

The Hodge Project

Contents

1	Introduction	17
1.1	What this project is	17
1.2	How the project is organised	17
1.3	A path to the Hodge conjecture	18
1.4	Sources and status	18
1.5	Conventions of the text	19
1.6	Mathematical conventions	19
I	Foundations	21
2	Mathematical Logic	22
2.1	Introduction	22
2.2	Propositional logic	22
2.3	First-order languages	24
2.4	Structures and satisfaction	26
2.5	Formal deductions	27
2.6	The completeness theorem	31
2.7	Compactness and the Löwenheim–Skolem theorem	32
2.8	The incompleteness theorems	33
2.9	Exercises	34
3	Set Theory	35
3.1	Introduction	35
3.2	The axioms	35
3.3	Relations and functions	36
3.4	The natural numbers	38
3.5	Ordinals	39
3.6	Cardinality	41
3.7	The axiom of choice	43
3.8	Universes	45
3.9	Exercises	46
4	Number Systems	47
4.1	Introduction	47
4.2	Arithmetic of the natural numbers	47
4.3	Commutative rings and fields	48
4.4	The integers	49
4.5	Divisibility and prime numbers	50
4.6	The rational numbers	51

4.7	Complete ordered fields	52
4.8	The real numbers	53
4.9	The complex numbers	54
4.10	Exercises	55
5	Categories	57
5.1	Introduction	57
5.2	Categories	57
5.3	Isomorphisms, monomorphisms and epimorphisms	59
5.4	Functors	60
5.5	Natural transformations	61
5.6	The Yoneda lemma	62
5.7	Equivalences of categories	64
5.8	Limits	65
5.9	Colimits	66
5.10	Adjoint functors	67
5.11	Filtered colimits	68
5.12	Additive categories	69
5.13	Exercises	70
II	Algebra	72
6	Groups	73
6.1	Introduction	73
6.2	Groups and homomorphisms	73
6.3	Cosets and Lagrange's theorem	75
6.4	Normal subgroups and quotients	76
6.5	Products	77
6.6	Group actions	78
6.7	The Sylow theorems	79
6.8	Symmetric and alternating groups	79
6.9	Solvable and nilpotent groups	81
6.10	Free groups and presentations	82
6.11	Abelian groups	83
6.12	Exercises	85
7	Rings and Modules	86
7.1	Introduction	86
7.2	Rings	86
7.3	Ideals and quotient rings	87
7.4	Prime and maximal ideals	87
7.5	Factorisation in integral domains	88
7.6	Polynomial rings	89
7.7	Modules	91
7.8	Vector spaces and rank	93
7.9	Tensor products	93
7.10	Algebras and graded rings	95
7.11	Modules over a principal ideal domain	95
7.12	Exercises	96

8	Fields and Galois Theory	98
8.1	Introduction	98
8.2	Field extensions	98
8.3	Splitting fields and algebraic closures	99
8.4	Separability	100
8.5	Galois extensions	101
8.6	Finite fields	102
8.7	The fundamental theorem of algebra	103
8.8	Cyclotomic fields	104
8.9	Solvability by radicals	104
8.10	Transcendence bases	106
8.11	Exercises	107
9	Linear Algebra	108
9.1	Introduction	108
9.2	Linear maps and dimension	108
9.3	Matrices	109
9.4	Duality	109
9.5	Determinants	110
9.6	Eigenvalues	111
9.7	Canonical forms	112
9.8	Bilinear forms	112
9.9	Inner product spaces and the spectral theorem	113
9.10	Complex structures	115
9.11	Exercises	116
10	Multilinear Algebra	117
10.1	Introduction	117
10.2	Tensor products of vector spaces	117
10.3	The tensor, symmetric and exterior algebras	118
10.4	Alternating forms	119
10.5	Orientations and the Hodge star	120
10.6	Forms of type (p, q)	121
10.7	Exercises	122
11	Representations of Groups and Lie Algebras	123
11.1	Introduction	123
11.2	Representations of groups	123
11.3	Characters of finite groups	124
11.4	Lie algebras	126
11.5	Representations of $\mathfrak{sl}_2$	126
11.6	Exercises	128
12	Commutative Algebra	130
12.1	Introduction	130
12.2	Prime ideals and radicals	130
12.3	Localisation	131
12.4	Nakayama's lemma	132
12.5	Noetherian and Artinian rings	132
12.6	Integral extensions	133
12.7	The Nullstellensatz	134

12.8	Primary decomposition	135
12.9	Dimension theory	136
12.10	Completions	137
12.11	Regular local rings and Dedekind domains	138
12.12	Kähler differentials	139
12.13	Exercises	139
13	Homological Algebra	140
13.1	Introduction	140
13.2	Abelian categories	140
13.3	Complexes	141
13.4	The long exact sequence	141
13.5	Projective and injective objects	143
13.6	Derived functors	143
13.7	Double complexes	145
13.8	Ext and Tor	145
13.9	Künneth and universal coefficient theorems	147
13.10	Exercises	148
14	Spectral Sequences	149
14.1	Introduction	149
14.2	Spectral sequences	149
14.3	The spectral sequence of a filtered complex	150
14.4	Double complexes	152
14.5	Exact couples	152
14.6	The Grothendieck spectral sequence	152
14.7	Exercises	153
15	Derived Categories	154
15.1	Introduction	154
15.2	The homotopy category	154
15.3	The derived category	155
15.4	Injective resolutions of complexes	156
15.5	t -structures	158
15.6	Filtered complexes and strictness	159
15.7	Exercises	159
III	Topology	161
16	General Topology	162
16.1	Introduction	162
16.2	Topological spaces and continuous maps	162
16.3	Metric spaces	164
16.4	Subspaces, products and quotients	166
16.5	Connectedness	167
16.6	Compactness	169
16.7	Complete and compact metric spaces	171
16.8	Separation axioms, Urysohn and Tietze	173
16.9	Local compactness and proper maps	175
16.10	Paracompactness and partitions of unity	176

16.11 Exercises	178
17 Homotopy and Covering Spaces	180
17.1 Introduction	180
17.2 Homotopy	180
17.3 The fundamental group	181
17.4 Covering spaces and the circle	182
17.5 The Seifert–van Kampen theorem	184
17.6 Classification of covering spaces	185
17.7 CW complexes	187
17.8 Higher homotopy groups and fibrations	188
17.9 Exercises	189
18 Topological Manifolds and Group Actions	191
18.1 Introduction	191
18.2 Topological groups	191
18.3 Group actions and orbit spaces	192
18.4 Covering space actions and quotients	194
18.5 Topological manifolds	195
18.6 Constructions of manifolds	197
18.7 Gluing	198
18.8 Exercises	199
19 Singular Homology	201
19.1 Introduction	201
19.2 Singular chains	201
19.3 Homotopy invariance	202
19.4 Relative homology and exact sequences	202
19.5 Excision and Mayer–Vietoris	203
19.6 Spheres, good pairs and degree	204
19.7 Cellular homology	206
19.8 Homology with coefficients	207
19.9 The first homology group	208
19.10 Exercises	208
20 Singular Cohomology	210
20.1 Introduction	210
20.2 Cochains and cohomology	210
20.3 The cup product	212
20.4 The Künneth formula	213
20.5 Cohomology rings of projective spaces	213
20.6 Cup length and coverings by contractible sets	214
20.7 The cap product	215
20.8 Exercises	215
21 Topological Manifolds and Poincaré Duality	217
21.1 Introduction	217
21.2 Topological manifolds	217
21.3 Orientations	218
21.4 The fundamental class	218
21.5 Cohomology with compact supports	220

21.6	Poincaré duality	220
21.7	Other dualities	222
21.8	Borel–Moore homology and fundamental classes	223
21.9	Exercises	223
22	Cohomology Operations	225
22.1	Introduction	225
22.2	Eilenberg–MacLane spaces	225
22.3	Cohomology operations	226
22.4	Steenrod squares	226
22.5	Cohomology of Eilenberg–MacLane spaces	228
22.6	Complex cobordism and algebraic cycles	228
22.7	Exercises	228
IV	Sheaves	230
23	Sheaves on Topological Spaces	231
23.1	Introduction	231
23.2	Presheaves and sheaves	231
23.3	Sheafification	233
23.4	Direct and inverse images	234
23.5	Locally constant sheaves and local systems	235
23.6	Ringed spaces and sheaves of modules	236
23.7	Exercises	237
24	Sheaf Cohomology	238
24.1	Introduction	238
24.2	Cohomology as a derived functor	238
24.3	Flasque sheaves	239
24.4	Soft and fine sheaves	239
24.5	Čech cohomology	241
24.6	Comparison with singular cohomology	242
24.7	Hypercohomology and the Leray spectral sequence	242
24.8	Cohomology with compact supports	243
24.9	Exercises	243
V	Analysis	245
25	Real Analysis	246
25.1	Introduction	246
25.2	Limits of sequences	246
25.3	Limits of functions	250
25.4	Continuous functions of a real variable	252
25.5	Upper and lower limits, and series	254
25.6	Differentiation in one variable	256
25.7	The exponential and trigonometric functions	259
25.8	The Riemann integral	261
25.9	Sequences of functions	262
25.10	Derivatives in several variables	264

25.11	The inverse and implicit function theorems	265
25.12	Ordinary differential equations	267
25.13	Stone–Weierstrass and Arzelà–Ascoli	268
25.14	Exercises	269
26	Measure and Integration	271
26.1	Introduction	271
26.2	Measures	271
26.3	Outer measures and Lebesgue measure	272
26.4	The integral	273
26.5	Lebesgue spaces	274
26.6	Modes of convergence	275
26.7	Product measures and Fubini’s theorem	276
26.8	Change of variables	277
26.9	The Radon–Nikodym and Riesz representation theorems	278
26.10	Exercises	278
27	Complex Analysis	280
27.1	Introduction	280
27.2	Holomorphic functions	280
27.3	Cauchy’s theorem	281
27.4	The Cauchy integral formula	282
27.5	Singularities, Laurent series and residues	283
27.6	Conformal maps and the Riemann mapping theorem	286
27.7	Exercises	287
28	Functional Analysis	289
28.1	Introduction	289
28.2	Banach spaces and bounded operators	289
28.3	Hilbert spaces	289
28.4	The Hahn–Banach and Baire theorems	291
28.5	Compact operators	292
28.6	Weak topologies	293
28.7	Exercises	294
29	Several Complex Variables	295
29.1	Introduction	295
29.2	Holomorphic functions of several variables	295
29.3	The Weierstrass theorems	297
29.4	The Dolbeault–Grothendieck lemma	299
29.5	Analytic sets	300
29.6	Exercises	300
30	Distributions and Sobolev Spaces	302
30.1	Introduction	302
30.2	Distributions	302
30.3	The Fourier transform	303
30.4	Sobolev spaces	304
30.5	Elliptic estimates	305
30.6	Exercises	305

VI Differential Geometry	307
31 Smooth Manifolds	308
31.1 Introduction	308
31.2 Manifolds and smooth maps	308
31.3 Tangent vectors	309
31.4 Submanifolds	310
31.5 Constructions of smooth manifolds	312
31.6 Partitions of unity	313
31.7 Vector fields and flows	314
31.8 Exercises	315
32 Lie Groups	316
32.1 Introduction	316
32.2 Lie groups and their Lie algebras	316
32.3 The exponential map	317
32.4 Subgroups and homogeneous spaces	318
32.5 Compact Lie groups	320
32.6 Exercises	320
33 Vector Bundles	322
33.1 Introduction	322
33.2 Vector bundles and cocycles	322
33.3 Operations on bundles	323
33.4 Line bundles and the first Chern class	325
33.5 Exercises	326
34 Differential Forms and Integration	327
34.1 Introduction	327
34.2 Differential forms	327
34.3 Orientations	329
34.4 Integration and Stokes' theorem	329
34.5 Exercises	331
35 de Rham Cohomology	332
35.1 Introduction	332
35.2 The de Rham complex	332
35.3 The de Rham theorem	333
35.4 Currents	336
35.5 Exercises	337
36 Connections and Curvature	338
36.1 Introduction	338
36.2 Connections	338
36.3 Parallel transport, holonomy and flatness	339
36.4 Metric connections	341
36.5 Principal bundles	341
36.6 Exercises	342

37 Characteristic Classes	343
37.1 Introduction	343
37.2 Chern–Weil theory	343
37.3 Chern classes: axioms and the splitting principle	345
37.4 The Thom isomorphism in de Rham cohomology	345
37.5 Euler, Pontryagin and Stiefel–Whitney classes	348
37.6 Exercises	349
38 Riemannian Geometry	350
38.1 Introduction	350
38.2 Metrics and the Levi-Civita connection	350
38.3 Covariant derivatives along curves	351
38.4 The exponential map and minimising geodesics	352
38.5 Parallel transport and holonomy	355
38.6 Curvature	356
38.7 The Hodge star and the Laplacian	359
38.8 Exercises	360
39 Elliptic Operators on Compact Manifolds	362
39.1 Introduction	362
39.2 Differential operators and symbols	362
39.3 Sobolev spaces of sections	363
39.4 Elliptic estimates and regularity	364
39.5 Fredholm theory and the spectral decomposition	364
39.6 Exercises	366
40 Harmonic Forms and the Hodge Theorem	367
40.1 Introduction	367
40.2 The Hodge decomposition	367
40.3 Applications	369
40.4 Exercises	370
VII Complex Geometry	371
41 Complex Manifolds	372
41.1 Introduction	372
41.2 Complex manifolds	372
41.3 Almost complex structures and forms of type (p, q)	374
41.4 Analytic subvarieties and blow-ups	375
41.5 Exercises	376
42 Holomorphic Vector Bundles	378
42.1 Introduction	378
42.2 Holomorphic bundles and sheaves	378
42.3 Divisors	380
42.4 Hermitian metrics and the Chern connection	381
42.5 The canonical bundle and adjunction	382
42.6 Exercises	382

43 Dolbeault Cohomology	384
43.1 Introduction	384
43.2 The Dolbeault theorem	384
43.3 Harmonic theory for $\bar{\partial}$	386
43.4 Examples and the Frölicher inequality	387
43.5 Exercises	388
44 Kähler Manifolds	389
44.1 Introduction	389
44.2 Hermitian metrics and the fundamental form	389
44.3 The Kähler condition	390
44.4 Examples	391
44.5 Linear algebra of the Lefschetz operator	393
44.6 Exercises	394
45 The Hodge Decomposition	396
45.1 Introduction	396
45.2 Conventions	396
45.3 The Kähler identities	397
45.4 Comparison of Laplacians	398
45.5 The Hodge decomposition	399
45.6 The $\partial\bar{\partial}$ -lemma	401
45.7 Examples	402
45.8 Exercises	403
46 Hard Lefschetz and the Hodge–Riemann Relations	405
46.1 Introduction	405
46.2 The $\mathfrak{sl}_2$ -action on cohomology	405
46.3 Pointwise Hodge–Riemann relations	407
46.4 The Hodge–Riemann bilinear relations	408
46.5 The Hodge index theorem	409
46.6 Further remarks and counterexamples	410
46.7 Exercises	410
47 Vanishing and Embedding Theorems	412
47.1 Introduction	412
47.2 Positive line bundles	412
47.3 The Bochner–Kodaira–Nakano identity	413
47.4 The Kodaira–Nakano vanishing theorem	415
47.5 The Lefschetz hyperplane theorem	415
47.6 The Kodaira embedding theorem	416
47.7 Exercises	419
48 Calabi–Yau Manifolds	421
48.1 Introduction	421
48.2 The Ricci form	421
48.3 The Calabi conjecture	422
48.4 Calabi–Yau manifolds and their Hodge numbers	422
48.5 Structure and symmetries	424
48.6 Exercises	425

VIII Algebraic Geometry	426
49 Algebraic Varieties	427
49.1 Introduction	427
49.2 Affine algebraic sets and the Zariski topology	427
49.3 Regular functions and morphisms	428
49.4 Projective varieties	430
49.5 Dimension	432
49.6 Tangent spaces and smoothness	433
49.7 Rational maps and birational equivalence	434
49.8 Blow-ups	435
49.9 Exercises	436
50 Schemes	437
50.1 Introduction	437
50.2 The spectrum of a ring	437
50.3 Schemes and their properties	438
50.4 Fibre products	440
50.5 Separated and proper morphisms	440
50.6 Projective schemes	441
50.7 Varieties as schemes	442
50.8 Flat and smooth morphisms	442
50.9 Base change and varieties over non-closed fields	444
50.10 Exercises	444
51 Quasi-coherent and Coherent Sheaves	446
51.1 Introduction	446
51.2 The sheaf associated to a module	446
51.3 Quasi-coherent sheaves	447
51.4 Locally free sheaves and the Picard group	449
51.5 Twisting sheaves on projective schemes	449
51.6 Sheaves of differentials	450
51.7 Exercises	451
52 Cohomology of Coherent Sheaves	453
52.1 Introduction	453
52.2 Vanishing on affine schemes	453
52.3 The cohomology of projective space	455
52.4 Finiteness and Serre vanishing	455
52.5 Euler characteristic and Hilbert polynomial	456
52.6 Serre duality	457
52.7 Semicontinuity and base change	457
52.8 Exercises	458
53 Divisors and Line Bundles	459
53.1 Introduction	459
53.2 Weil divisors	459
53.3 Cartier divisors and invertible sheaves	460
53.4 Linear systems and morphisms to projective space	461
53.5 Ample line bundles	461
53.6 Intersection numbers on surfaces	462

53.7	The Néron–Severi group	463
53.8	Exercises	464
54	Analytic Spaces and GAGA	465
54.1	Introduction	465
54.2	Complex analytic spaces	465
54.3	The analytification functor	465
54.4	Chow’s theorem and GAGA	466
54.5	Algebraic de Rham cohomology	468
54.6	Smooth projective varieties as Kähler manifolds	469
54.7	Exercises	469
55	Algebraic Curves and Riemann Surfaces	470
55.1	Introduction	470
55.2	Curves and their function fields	470
55.3	The Riemann–Roch theorem	471
55.4	The Riemann–Hurwitz formula	472
55.5	Hyperelliptic curves and the canonical embedding	473
55.6	The Jacobian and the Abel–Jacobi theorem	474
55.7	Exercises	474
56	Linear Algebraic Groups	476
56.1	Introduction	476
56.2	Affine group schemes	476
56.3	Representations and Tannakian categories	477
56.4	Tori, characters and the Deligne torus	478
56.5	The Lie algebra	479
56.6	Reductive groups	479
56.7	Weil restriction of scalars	480
56.8	Exercises	480
57	Abelian Varieties	482
57.1	Introduction	482
57.2	Complex tori and their cohomology	482
57.3	Line bundles and the Riemann conditions	483
57.4	The dual abelian variety	485
57.5	Jacobians	485
57.6	Endomorphisms and complex multiplication	485
57.7	Exercises	486
58	Algebraic Cycles and Chow Groups	488
58.1	Introduction	488
58.2	Cycles and rational equivalence	488
58.3	Functoriality	490
58.4	Chern classes and the intersection product	490
58.5	Correspondences	491
58.6	Adequate equivalence relations	491
58.7	Riemann–Roch	492
58.8	Exercises	493

59 The Cycle Class Map	494
59.1 Introduction	494
59.2 Fundamental classes	494
59.3 The cycle class map and its properties	495
59.4 Cycle classes are Hodge classes	496
59.5 Cycles and coherent sheaves	497
59.6 Exercises	497
60 Index Theorems and Riemann–Roch	498
60.1 Introduction	498
60.2 The index of an elliptic operator	498
60.3 The Atiyah–Singer index theorem	499
60.4 Riemann–Roch: direct verifications	500
60.5 Exercises	503
61 Étale Cohomology	504
61.1 Introduction	504
61.2 Étale morphisms and the étale site	504
61.3 The Kummer sequence	505
61.4 ℓ -adic cohomology	506
61.5 Galois actions and the Tate conjecture	507
61.6 Exercises	507
IX Hodge Theory	508
62 Hodge Structures	509
62.1 Introduction	509
62.2 Pure Hodge structures	509
62.3 Morphisms and the category of Hodge structures	510
62.4 Hodge structures as representations	511
62.5 Polarisation	511
62.6 Hodge classes	512
62.7 Weight one and complex tori	513
62.8 Exercises	514
63 Mumford–Tate Groups	515
63.1 Introduction	515
63.2 Definition and first properties	515
63.3 Reductivity and Hodge classes	516
63.4 Examples	517
63.5 Exercises	518
64 The Hodge Conjecture	520
64.1 Introduction	520
64.2 Algebraic classes and Hodge classes	520
64.3 First reductions and elementary cases	521
64.4 History and two corrections	522
64.5 Why one might believe it, and why it is hard	523
64.6 Exercises	524

65 The Lefschetz (1,1) Theorem	525
65.1 Introduction	525
65.2 The theorem for compact Kähler manifolds	525
65.3 The projective case: classes of divisors	527
65.4 Why the proof does not generalise	528
65.5 Exercises	528
66 Variations of Hodge Structure	530
66.1 Introduction	530
66.2 Ehresmann's theorem and the Gauss–Manin connection	530
66.3 Period domains and period maps	531
66.4 Global invariants and the theorem of the fixed part	533
66.5 Hodge loci	533
66.6 Exercises	534
67 Intermediate Jacobians and Normal Functions	535
67.1 Introduction	535
67.2 Intermediate Jacobians	535
67.3 The Abel–Jacobi map	536
67.4 Deligne cohomology	537
67.5 Griffiths' theorem	538
67.6 Normal functions	538
67.7 Exercises	539
68 Known Cases and Evidence	540
68.1 Introduction	540
68.2 Cases from general theory	540
68.3 Abelian varieties	541
68.4 Hypersurfaces and complete intersections	542
68.5 Uniruled varieties and the decomposition of the diagonal	542
68.6 K3 surfaces, hyper-Kähler varieties and Kuga–Satake	543
68.7 The evidence assessed	543
68.8 Exercises	543
69 Variants That Fail	545
69.1 Introduction	545
69.2 The integral Hodge conjecture	545
69.3 The Kähler counterexample	547
69.4 Hodge's general conjecture	547
69.5 The variational Hodge conjecture	547
69.6 A diagnostic list	548
69.7 Exercises	548
70 Mixed Hodge Structures	549
70.1 Introduction	549
70.2 Mixed Hodge structures	549
70.3 The Deligne splitting and strictness	550
70.4 The logarithmic de Rham complex	552
70.5 Singular varieties	553
70.6 Applications of weights	554
70.7 Exercises	556

71 Absolute Hodge Classes	558
71.1 Introduction	558
71.2 Conjugate varieties and comparison isomorphisms	558
71.3 Absolute Hodge classes	559
71.4 Operations on absolute Hodge classes	560
71.5 Galois action and fields of definition	561
71.6 Principle B	562
71.7 Deligne's theorem for abelian varieties	563
71.8 Fields of definition of Hodge loci	564
71.9 Exercises	564
72 The Tate Conjecture	566
72.1 Introduction	566
72.2 The statement	566
72.3 Divisors on abelian varieties	567
72.4 K3 surfaces and other known cases	568
72.5 The Mumford–Tate conjecture	569
72.6 Exercises	570
73 The Standard Conjectures and Motives	572
73.1 Introduction	572
73.2 Correspondences in cohomology	572
73.3 The standard conjectures	573
73.4 Pure motives	576
73.5 The Bloch–Beilinson conjectures	578
73.6 Exercises	579
X Miscellany	581
Bibliography	583

Chapter 1

Introduction

1.1 What this project is

The Hodge Project is a cumulative reference whose goal is to develop, with complete proofs, all the mathematics needed to state, understand and work with the Hodge conjecture, together with the theory that surrounds it.

Informally, the Hodge conjecture says the following: on a smooth complex projective variety X , every rational cohomology class of type (p, p) is a rational linear combination of the classes of algebraic subvarieties of X . A precise statement is given in Chapter 64. Each word of the informal statement hides a substantial theory: rational cohomology classes come from algebraic topology (Chapter 20), the type (p, p) comes from the Hodge decomposition of a compact Kähler manifold (Chapter 45), which rests on the analysis of elliptic operators (Chapter 39), and the classes of subvarieties come from algebraic geometry (Chapters 49 and 59). This project builds all of it, starting from logic and set theory.

The organisation is modelled on the Stacks project: the text is written in a strict logical order, every result has a permanent tag, and the whole project is a single document that can be read as a book or browsed as a website.

1.2 How the project is organised

The chapters are grouped into parts.

- Part I, *Foundations*: logic, set theory, the number systems and category theory.
- Part II, *Algebra*: from groups to derived categories.
- Part III, *Topology*: general topology, homotopy theory, singular homology and cohomology, Poincaré duality.
- Part IV, *Sheaves*: sheaves and sheaf cohomology.
- Part V, *Analysis*: real, complex and functional analysis, several complex variables and Sobolev spaces.
- Part VI, *Differential Geometry*: from smooth manifolds to the Hodge theorem for harmonic forms.
- Part VII, *Complex Geometry*: complex and Kähler manifolds, the Hodge decomposition, the Lefschetz theorems, the Kodaira vanishing and embedding theorems, and Calabi–Yau manifolds.

- Part VIII, *Algebraic Geometry*: varieties, schemes, coherent cohomology, GAGA, curves, algebraic groups, abelian varieties, algebraic cycles, the cycle class map and étale cohomology.
- Part IX, *Hodge Theory*: Hodge structures and Mumford–Tate groups, the Hodge conjecture, the evidence for it, its failed variants, mixed Hodge structures, absolute Hodge classes, and the Tate and standard conjectures around it.

The order of the chapters is the logical order of the project, not a recommended reading order. A reader interested mainly in the Hodge conjecture can start from Part IX and follow references backwards as needed.

1.3 A path to the Hodge conjecture

The following chapters form the spine of the project; everything else supports them or grows out of them.

1. *Cohomology*. Singular cohomology and Poincaré duality (Chapters 20 and 21), sheaf cohomology (Chapter 24) and de Rham’s theorem (Chapter 35) give the rational cohomology of a manifold and its description by differential forms.
2. *Harmonic theory*. Elliptic operators (Chapter 39) give the Hodge theorem: every de Rham class has a unique harmonic representative (Chapter 40).
3. *Kähler geometry*. On a compact Kähler manifold (Chapter 44) the Laplacian respects the bidegree of forms, which gives the Hodge decomposition $H^k = \bigoplus_{p+q=k} H^{p,q}$ (Chapter 45), hard Lefschetz and the Hodge–Riemann relations (Chapter 46).
4. *Algebraic geometry*. Projective varieties are Kähler, and GAGA identifies analytic and algebraic objects on them (Chapter 54). Subvarieties have cohomology classes, which are rational and of type (p, p) (Chapter 59).
5. *The conjecture*. Hodge structures (Chapter 62) isolate the linear algebra, and the Hodge conjecture asserts the converse of the last statement (Chapter 64). It is a theorem for divisors (Chapter 65), it is known in further cases (Chapter 68), and its natural strengthenings are false (Chapter 69).
6. *The surrounding theory*. Variations of Hodge structure and normal functions (Chapters 66 and 67), mixed Hodge structures (Chapter 70), absolute Hodge classes (Chapter 71), and the Tate and standard conjectures (Chapters 72 and 73) place the conjecture in its wider landscape.

1.4 Sources and status

The mathematics in this project is standard, and each chapter lists the references it follows; the bibliography collects them. The chapters of Part IX are based in part on the expository text *The Hodge Conjecture* (2026), whose claims were checked against the original literature and corrected where necessary. Where a theorem is quoted rather than proved, the text says so and gives a reference (Remark 1.5.3). All chapters are written; none has yet been checked by an independent reader, which is the condition for the status *checked* (Remark 1.5.6).

1.5 Conventions of the text

Notation 1.5.1. All numbered items of a section share a single counter: Lemma 5.2.3 is the third numbered item of Section 5.2, and it is followed by Example 5.2.4. Chapters are numbered consecutively through all the parts.

Notation 1.5.2. The text uses the following kinds of numbered items.

- **Definition:** introduces a notion.
- **Theorem, Proposition, Lemma, Corollary:** proved statements, in roughly decreasing order of importance.
- **Conjecture:** a statement that is not known to be true.
- **Example:** an instance of a definition or a result.
- **Counterexample:** shows that a hypothesis cannot be dropped, or that a plausible statement is false.
- **Remark:** context, motivation or intuition. Remarks are never used in proofs.
- **Notation:** fixes notation or a convention.
- **Exercise:** a statement for the reader to prove, sometimes followed by a solution.

Remark 1.5.3. A result followed by *Proof* has a complete proof that uses only earlier material. A result followed by *Sketch of proof* has an argument whose main ideas are given but whose omitted verifications are named. A result followed by no proof is quoted from the literature, with a reference. Sketches and quotations are temporary: the aim of the project is to replace all of them with complete proofs.

Remark 1.5.4. The project has no forward references: a proof may only use definitions and results that appear before it. Examples, counterexamples, remarks and exercises may mention notions that are defined later (for instance, the category of topological spaces appears as an example in Chapter 5), but such mentions are never used in a proof. The only exception is this introduction, which refers forward freely.

Remark 1.5.5. Every labelled item (chapter, section, definition, result, example, and so on) carries a permanent four-character *tag*, shown next to it on the website. The number of an item changes whenever material is inserted before it; its tag never changes. To refer to an item from outside the project, cite its tag.

Remark 1.5.6. Each chapter has a status. *Planned:* only an outline exists. *Draft:* the chapter is being written, and its proofs have not been checked. *Complete:* all planned material is written, and the proofs await independent checking. *Checked:* every proof in the chapter has been checked by a reader other than its author.

1.6 Mathematical conventions

Notation 1.6.1. We work in ZFC set theory together with the assumption that a Grothendieck universe exists, and we fix such a universe $\mathcal{U}$ once and for all. A set is *small* if it is an element of $\mathcal{U}$. Chapter 3 develops these notions; Chapter 5 explains why they are needed.

Notation 1.6.2. The natural numbers include zero: $\mathbb{N} = \{0, 1, 2, \dots\}$. A real number x is *positive* if $x > 0$ and *non-negative* if $x \geq 0$.

Notation 1.6.3. The composition of maps $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ is written $g \circ f$ or gf . The identity map of X is written id_X .

Notation 1.6.4. A ring is associative and has an identity element 1, and ring homomorphisms preserve identity elements. Rings are not assumed to be commutative, except from Chapter 12 onwards, where “ring” means “commutative ring” unless stated otherwise.

Part I

Foundations

Chapter 2

Mathematical Logic

2.1 Introduction

Mathematical logic studies the language in which mathematics is written: what a formula is, what it means for a formula to be true in a structure, and what it means to prove it. The main result of this chapter is Gödel's completeness theorem (Theorem 2.6.1 and Corollary 2.6.2): a statement follows from a set of axioms in every structure if and only if it can be formally deduced from them. Its consequences, the compactness and Löwenheim–Skolem theorems, are the first tools of model theory. We end with the statements of Gödel's incompleteness theorems, which limit what any axiomatic foundation, including the set theory of Chapter 3, can prove about itself.

We follow the approach of [End01]; other references are [Sho67] and [vD13], and [CK90] for model theory.

Prerequisites. None.

Remark 2.1.1. This chapter studies formal languages *from the outside*. The reasoning we use about formulas, structures and deductions (induction on the natural numbers, finite sequences, sets, countability) is ordinary informal mathematics, sometimes called the *metatheory*. In Chapter 3, set theory is itself formulated as a first-order theory in the sense of this chapter. There is no vicious circle: informal mathematics is used to describe formal systems, and one of these formal systems is then used as the official foundation of the rest of the project. The proofs in this chapter use only elementary facts about finite and countable sets, and in particular never the axiom of choice.

2.2 Propositional logic

Definition 2.2.1. Let P be a set, whose elements we call *propositional variables*, and let $\neg$, $\rightarrow$, $($ and $)$ be four symbols not in P . A *string* is a finite sequence of symbols. The set of *propositional formulas* over P is the smallest set of strings such that

1. every $p \in P$, viewed as a string of length one, is a formula;
2. if φ is a formula, then $\neg\varphi$ is a formula;
3. if φ and ψ are formulas, then $(\varphi \rightarrow \psi)$ is a formula.

Here $\neg\varphi$ denotes the string consisting of $\neg$ followed by the symbols of φ , and similarly for $(\varphi \rightarrow \psi)$.

Remark 2.2.2. Every formula is obtained from propositional variables by finitely many applications of the two operations, since the set of such strings contains P and is closed under the operations. Hence statements about all formulas can be proved by *induction on formulas*: it suffices to prove them for propositional variables and to show that they are preserved by the two operations.

Notation 2.2.3. We use the abbreviations $(\varphi \wedge \psi) := \neg(\varphi \rightarrow \neg\psi)$, $(\varphi \vee \psi) := (\neg\varphi \rightarrow \psi)$ and $(\varphi \leftrightarrow \psi) := ((\varphi \rightarrow \psi) \wedge (\psi \rightarrow \varphi))$, and we omit outer parentheses when no confusion is possible.

Lemma 2.2.4. *Let φ be a propositional formula. Then φ contains as many symbols (as symbols). Moreover, every proper initial segment of φ either consists only of symbols $\neg$ (possibly none), or contains strictly more symbols (than symbols).*

Proof. For a string u let $b(u)$ be the number of symbols (in u minus the number of symbols) in u . We argue by induction on φ .

If $\varphi = p$ is a variable, then $b(p) = 0$ and the only proper initial segment is empty.

If $\varphi = \neg\psi$, then $b(\varphi) = b(\psi) = 0$. A proper initial segment of φ is empty or of the form $\neg u$ with u a proper initial segment of ψ . By induction, u consists of symbols $\neg$ or $b(u) > 0$, and the same holds for $\neg u$.

If $\varphi = (\psi \rightarrow \chi)$, then $b(\varphi) = 1 + b(\psi) + b(\chi) - 1 = 0$. A nonempty proper initial segment of φ has one of the forms $(u$ with u an initial segment of ψ , or $(\psi \rightarrow u$ with u an initial segment of χ (possibly empty or equal to χ). By induction every initial segment u of ψ or χ satisfies $b(u) \geq 0$, so these segments have $b \geq 1$. $\square$

Theorem 2.2.5 (Unique readability). *Every propositional formula φ satisfies exactly one of the following: φ is a propositional variable; $\varphi = \neg\psi$ for a formula ψ ; $\varphi = (\psi \rightarrow \chi)$ for formulas ψ and χ . In the second case ψ is uniquely determined, and in the third case ψ and χ are uniquely determined.*

Proof. The three cases are distinguished by the first symbol of φ , and in the second case ψ is φ with its first symbol removed. Suppose $(\psi \rightarrow \chi) = (\psi' \rightarrow \chi')$ with $\psi \neq \psi'$. After the first symbol the two strings agree, so one of ψ, ψ' is a proper initial segment of the other, say ψ of ψ' . By Lemma 2.2.4, $b(\psi) = 0$, so ψ must consist only of symbols $\neg$. But every formula contains at least one propositional variable (by induction on formulas), a contradiction. Hence $\psi = \psi'$, and then $\chi = \chi'$. $\square$

Definition 2.2.6. A *truth assignment* is a map $v: P \rightarrow \{0, 1\}$. By Theorem 2.2.5, there is a unique map $\bar{v}$ from the set of formulas to $\{0, 1\}$, defined by recursion on formulas, such that $\bar{v}(p) = v(p)$ for $p \in P$, $\bar{v}(\neg\varphi) = 1 - \bar{v}(\varphi)$, and $\bar{v}(\varphi \rightarrow \psi) = 0$ if and only if $\bar{v}(\varphi) = 1$ and $\bar{v}(\psi) = 0$. We say that v *satisfies* φ if $\bar{v}(\varphi) = 1$, and that v satisfies a set of formulas Σ if it satisfies every element of Σ .

Definition 2.2.7. A formula φ is a *tautology* if every truth assignment satisfies it. A set Σ of formulas *tautologically implies* φ , written $\Sigma \models \varphi$, if every truth assignment satisfying Σ satisfies φ . A set of formulas is *satisfiable* if some truth assignment satisfies it.

Lemma 2.2.8. *For every truth assignment v : $\bar{v}(\varphi \wedge \psi) = 1$ if and only if $\bar{v}(\varphi) = \bar{v}(\psi) = 1$; $\bar{v}(\varphi \vee \psi) = 1$ if and only if $\bar{v}(\varphi) = 1$ or $\bar{v}(\psi) = 1$; and $\bar{v}(\varphi \leftrightarrow \psi) = 1$ if and only if $\bar{v}(\varphi) = \bar{v}(\psi)$.*

Proof. $\bar{v}(\neg(\varphi \rightarrow \neg\psi)) = 1$ if and only if $\bar{v}(\varphi \rightarrow \neg\psi) = 0$, that is, $\bar{v}(\varphi) = 1$ and $\bar{v}(\neg\psi) = 0$. Next, $\bar{v}(\neg\varphi \rightarrow \psi) = 0$ if and only if $\bar{v}(\varphi) = 0$ and $\bar{v}(\psi) = 0$. The last statement follows from the first and the definition of $\bar{v}$ on implications. $\square$

Example 2.2.9. The following are tautologies, as one checks by computing $\bar{v}$ for the finitely many possible values of v on the variables involved: $\varphi \vee \neg\varphi$ (excluded middle); $\neg\neg\varphi \leftrightarrow \varphi$; $((\varphi \rightarrow \psi) \rightarrow \varphi) \rightarrow \varphi$ (Peirce's law); $\neg(\varphi \wedge \psi) \leftrightarrow (\neg\varphi \vee \neg\psi)$ (De Morgan); $(\varphi \rightarrow (\psi \rightarrow \chi)) \rightarrow ((\varphi \rightarrow \psi) \rightarrow (\varphi \rightarrow \chi))$. By contrast, $(\varphi \rightarrow \psi) \rightarrow (\psi \rightarrow \varphi)$ is not a tautology when φ and ψ are distinct variables.

Proposition 2.2.10. *Let $p_1, \dots, p_n$ be distinct propositional variables, $n \geq 1$, and let $F: \{0, 1\}^n \rightarrow \{0, 1\}$ be any function. There is a formula φ whose variables are among $p_1, \dots, p_n$ such that $\bar{v}(\varphi) = F(v(p_1), \dots, v(p_n))$ for every truth assignment v .*

Proof. For $\varepsilon \in \{0, 1\}^n$ let $\gamma_\varepsilon = \ell_1 \wedge \dots \wedge \ell_n$, where $\ell_i = p_i$ if $\varepsilon_i = 1$ and $\ell_i = \neg p_i$ if $\varepsilon_i = 0$ (with parentheses grouped to the left). By Lemma 2.2.8, $\bar{v}(\gamma_\varepsilon) = 1$ if and only if $v(p_i) = \varepsilon_i$ for all i . If F is identically 0, take $\varphi = p_1 \wedge \neg p_1$. Otherwise let φ be the disjunction of the γ_ε over all ε with $F(\varepsilon) = 1$. Then $\bar{v}(\varphi) = 1$ if and only if $(v(p_1), \dots, v(p_n))$ is one of these ε , that is, if and only if $F(v(p_1), \dots, v(p_n)) = 1$. $\square$

Counterexample 2.2.11. Proposition 2.2.10 fails if only $\wedge$ and $\vee$ may be used, taken as primitive connectives with the truth tables of Lemma 2.2.8. Indeed, by induction on such formulas, if $v(p) \leq w(p)$ for every variable p then $\bar{v}(\varphi) \leq \bar{w}(\varphi)$. So no such formula computes the function $F(x) = 1 - x$.

Theorem 2.2.12 (Compactness, propositional case). *Let P be countable and let Σ be a set of formulas over P . If every finite subset of Σ is satisfiable, then Σ is satisfiable.*

Proof. Call a set of formulas *finitely satisfiable* if all its finite subsets are satisfiable. The formulas are finite strings over a countable set of symbols, so they can be enumerated as $\varphi_0, \varphi_1, \dots$. Put $\Delta_0 = \Sigma$, and let $\Delta_{n+1} = \Delta_n \cup \{\varphi_n\}$ if this set is finitely satisfiable, and $\Delta_{n+1} = \Delta_n \cup \{\neg\varphi_n\}$ otherwise.

Each Δ_n is finitely satisfiable. Suppose Δ_n is, but neither $\Delta_n \cup \{\varphi_n\}$ nor $\Delta_n \cup \{\neg\varphi_n\}$ is. Then there are finite $\Delta, \Delta' \subseteq \Delta_n$ such that $\Delta \cup \{\varphi_n\}$ and $\Delta' \cup \{\neg\varphi_n\}$ are not satisfiable. A truth assignment v satisfying $\Delta \cup \Delta'$ satisfies φ_n or $\neg\varphi_n$, hence one of the two sets; contradiction.

Let $\Delta = \bigcup_n \Delta_n$. Every finite subset of Δ lies in some Δ_n , so Δ is finitely satisfiable; in particular Δ never contains both ψ and $\neg\psi$. For every formula ψ , exactly one of $\psi, \neg\psi$ lies in Δ . Define $v(p) = 1$ if and only if $p \in \Delta$. We show by induction on ψ that $\bar{v}(\psi) = 1$ if and only if $\psi \in \Delta$. For variables this is the definition. For $\neg\psi$: $\bar{v}(\neg\psi) = 1$ iff $\psi \notin \Delta$ iff $\neg\psi \in \Delta$. For $\psi \rightarrow \chi$: if $(\psi \rightarrow \chi) \in \Delta$, $\psi \in \Delta$ and $\chi \notin \Delta$, then $\neg\chi \in \Delta$ and the finite set $\{\psi \rightarrow \chi, \psi, \neg\chi\}$ is not satisfiable. If $(\psi \rightarrow \chi) \notin \Delta$, then $\neg(\psi \rightarrow \chi) \in \Delta$; if moreover $\psi \notin \Delta$ or $\chi \in \Delta$, then Δ contains $\{\neg(\psi \rightarrow \chi), \neg\psi\}$ or $\{\neg(\psi \rightarrow \chi), \chi\}$, neither of which is satisfiable. So $(\psi \rightarrow \chi) \in \Delta$ if and only if $\psi \in \Delta$ or $\chi \in \Delta$, which by induction is equivalent to $\bar{v}(\psi \rightarrow \chi) = 1$. Hence v satisfies $\Delta \supseteq \Sigma$. $\square$

Remark 2.2.13. The theorem holds for arbitrary sets P , but the proof then needs a form of the axiom of choice (for instance Zorn's lemma, Chapter 3) to replace the enumeration of all formulas.

2.3 First-order languages

Definition 2.3.1. A *first-order language* $\mathcal{L}$ consists of a set of *constant symbols*, and for each $n \geq 1$ a set of *n -ary function symbols* and a set of *n -ary relation symbols*, all these sets being pairwise disjoint. The language is *countable* if the union of these sets is countable. Every language also has the *logical symbols*: a countably infinite set of *variables* $v_0, v_1, v_2, \dots$, disjoint from the other symbols, and the symbols $=, \neg, \rightarrow, \forall$.

Definition 2.3.2. Let $\mathcal{L}$ be a language. The set of *terms* is the smallest set that contains the variables and the constant symbols and contains the tuple $(f, t_1, \dots, t_n)$ whenever f is an n -ary function symbol and $t_1, \dots, t_n$ are terms. We write $f(t_1, \dots, t_n)$ for this tuple.

The *atomic formulas* are the tuples $(=, t_1, t_2)$ and $(R, t_1, \dots, t_n)$, where the t_i are terms and R is an n -ary relation symbol; we write $t_1 = t_2$ and $R(t_1, \dots, t_n)$. The set of *formulas* is the smallest set that contains the atomic formulas and contains $(\neg, \varphi)$, $(\rightarrow, \varphi, \psi)$ and $(\forall, x, \varphi)$ whenever φ, ψ are formulas and x is a variable. We write $\neg\varphi$, $(\varphi \rightarrow \psi)$ and $\forall x \varphi$ for these tuples.

Remark 2.3.3. We assume, as we may, that no symbol is itself a finite tuple of the above shapes. Since two tuples are equal only if they have the same length and the same entries, every term and every formula decomposes *uniquely* into its immediate constituents. This is the analogue of Theorem 2.2.5, obtained here for free: we have defined formulas as trees rather than strings. As in Remark 2.2.2, we can prove statements by induction on terms and formulas, and define functions on them by recursion. The *complexity* $c(\varphi)$ of a formula is the number of occurrences of $\neg$, $\rightarrow$ and $\forall$ in it: $c = 0$ on atomic formulas, $c(\neg\varphi) = c(\varphi) + 1$, $c(\varphi \rightarrow \psi) = c(\varphi) + c(\psi) + 1$ and $c(\forall x \varphi) = c(\varphi) + 1$.

In practice we write formulas as usual, for instance $x \cdot y = y \cdot x$ instead of $(=, (\cdot, x, y), (\cdot, y, x))$, and we use variables named $x, y, z, \dots$ for arbitrary distinct variables.

Notation 2.3.4. Besides the abbreviations of Notation 2.2.3, we write $\exists x \varphi$ for $\neg \forall x \neg \varphi$ and $t \neq s$ for $\neg(t = s)$.

Example 2.3.5. 1. The language of groups has a constant symbol e , a unary function symbol $^{-1}$ and a binary function symbol $\cdot$.

2. The language of rings has constant symbols $0, 1$ and binary function symbols $+$, $\cdot$ and a unary function symbol $-$.

3. The language of orders has a single binary relation symbol $<$.

4. The language of set theory has a single binary relation symbol $\in$.

5. The language of arithmetic $\mathcal{L}_A$ has a constant symbol 0 , a unary function symbol S and binary function symbols $+$ and $\cdot$.

All these languages are countable.

Definition 2.3.6. The set $\text{Var}(t)$ of variables of a term t is defined as usual. The set $\text{FV}(\varphi)$ of *free variables* of a formula is defined by recursion: $\text{FV}(t_1 = t_2) = \text{Var}(t_1) \cup \text{Var}(t_2)$, $\text{FV}(R(t_1, \dots, t_n)) = \bigcup_i \text{Var}(t_i)$, $\text{FV}(\neg\varphi) = \text{FV}(\varphi)$, $\text{FV}(\varphi \rightarrow \psi) = \text{FV}(\varphi) \cup \text{FV}(\psi)$ and $\text{FV}(\forall x \varphi) = \text{FV}(\varphi) \setminus \{x\}$. A *sentence* is a formula without free variables. A variable *occurs* in φ if it appears anywhere in the tree φ , free or not.

Definition 2.3.7. Let x be a variable and t a term. For a term u , the term $u[t/x]$ is obtained by replacing every occurrence of x in u by t . For formulas, $\varphi[t/x]$ is defined by recursion: $(t_1 = t_2)[t/x] = (t_1[t/x] = t_2[t/x])$ and similarly for $R(t_1, \dots, t_n)$; $(\neg\varphi)[t/x] = \neg(\varphi[t/x])$; $(\varphi \rightarrow \psi)[t/x] = (\varphi[t/x] \rightarrow \psi[t/x])$; $(\forall y \varphi)[t/x] = \forall y \varphi$ if $y = x$, and $(\forall y \varphi)[t/x] = \forall y (\varphi[t/x])$ if $y \neq x$.

The term t is *substitutable* for x in φ if: φ is atomic; or $\varphi = \neg\psi$ and t is substitutable for x in ψ ; or $\varphi = (\psi \rightarrow \chi)$ and t is substitutable for x in ψ and in χ ; or $\varphi = \forall y \psi$ and either $x \notin \text{FV}(\varphi)$, or y does not occur in t and t is substitutable for x in ψ .

Remark 2.3.8. Two facts follow at once by induction: if $x \notin \text{FV}(\varphi)$ then $\varphi[t/x] = \varphi$; and substitution does not change the complexity, $c(\varphi[t/x]) = c(\varphi)$. A constant symbol, or more generally a term without variables, is substitutable for every variable in every formula.

Lemma 2.3.9. *Let x and y be variables, and suppose y does not occur in the formula φ . Then y is substitutable for x in φ , x is substitutable for y in $\varphi[y/x]$, and $\varphi[y/x][x/y] = \varphi$.*

Proof. For terms u not containing y , clearly $u[y/x][x/y] = u$. This gives the claims for atomic formulas, and the cases $\neg$ and $\rightarrow$ follow by induction. Let $\varphi = \forall z \psi$; then $z \neq y$ and y does not occur in ψ . The variable z does not occur in the term y , so by induction y is substitutable for x in φ . If $z = x$, then $\varphi[y/x] = \varphi$, in which y is not free, so x is substitutable for y in it and $\varphi[x/y] = \varphi$. If $z \neq x$, then $\varphi[y/x] = \forall z \psi[y/x]$; since z does not occur in the term x , the induction hypothesis shows that x is substitutable for y in $\varphi[y/x]$, and $\varphi[y/x][x/y] = \forall z \psi[y/x][x/y] = \forall z \psi$. $\square$

2.4 Structures and satisfaction

Definition 2.4.1. A structure $\mathcal{A}$ for a language $\mathcal{L}$ consists of a nonempty set A , its *universe*, together with an element $c^{\mathcal{A}} \in A$ for every constant symbol c , a map $f^{\mathcal{A}}: A^n \rightarrow A$ for every n -ary function symbol f , and a subset $R^{\mathcal{A}} \subseteq A^n$ for every n -ary relation symbol R . An *assignment* in $\mathcal{A}$ is a map s from the set of variables to A . For an assignment s , a variable x and $a \in A$, we write $s[x \mapsto a]$ for the assignment that agrees with s except that it sends x to a .

Definition 2.4.2. Let $\mathcal{A}$ be a structure and s an assignment. The *value* $t^{\mathcal{A}}[s] \in A$ of a term is defined by $x^{\mathcal{A}}[s] = s(x)$, $c^{\mathcal{A}}[s] = c^{\mathcal{A}}$ and $f(t_1, \dots, t_n)^{\mathcal{A}}[s] = f^{\mathcal{A}}(t_1^{\mathcal{A}}[s], \dots, t_n^{\mathcal{A}}[s])$. The relation $\mathcal{A} \models \varphi[s]$, read “ $\mathcal{A}$ satisfies φ under s ”, is defined by recursion on φ :

1. $\mathcal{A} \models (t_1 = t_2)[s]$ if and only if $t_1^{\mathcal{A}}[s] = t_2^{\mathcal{A}}[s]$;
2. $\mathcal{A} \models R(t_1, \dots, t_n)[s]$ if and only if $(t_1^{\mathcal{A}}[s], \dots, t_n^{\mathcal{A}}[s]) \in R^{\mathcal{A}}$;
3. $\mathcal{A} \models \neg \varphi[s]$ if and only if $\mathcal{A} \not\models \varphi[s]$;
4. $\mathcal{A} \models (\varphi \rightarrow \psi)[s]$ if and only if $\mathcal{A} \not\models \varphi[s]$ or $\mathcal{A} \models \psi[s]$;
5. $\mathcal{A} \models \forall x \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$ for every $a \in A$.

Remark 2.4.3. With the abbreviations of Notation 2.3.4, $\mathcal{A} \models \exists x \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$ for some $a \in A$, and the connectives $\wedge$, $\vee$, $\leftrightarrow$ have their usual meaning, by the same computation as in Lemma 2.2.8.

Lemma 2.4.4 (Coincidence lemma). *Let s and s' be assignments in $\mathcal{A}$. If they agree on $\text{Var}(t)$, then $t^{\mathcal{A}}[s] = t^{\mathcal{A}}[s']$. If they agree on $\text{FV}(\varphi)$, then $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{A} \models \varphi[s']$.*

Proof. The statement for terms follows by induction on terms, and implies the statement for atomic formulas; the cases $\neg$ and $\rightarrow$ follow by induction. If $\varphi = \forall x \psi$ and s, s' agree on $\text{FV}(\varphi) = \text{FV}(\psi) \setminus \{x\}$, then for each $a \in A$ the assignments $s[x \mapsto a]$ and $s'[x \mapsto a]$ agree on $\text{FV}(\psi)$, and the claim follows from the induction hypothesis. $\square$

Notation 2.4.5. By Lemma 2.4.4, whether $\mathcal{A} \models \sigma[s]$ holds for a sentence σ does not depend on s ; we then write $\mathcal{A} \models \sigma$ and say that σ is *true* in $\mathcal{A}$.

Lemma 2.4.6 (Substitution lemma). *Let t be substitutable for x in φ , let s be an assignment in $\mathcal{A}$, and put $a = t^{\mathcal{A}}[s]$. Then $\mathcal{A} \models \varphi[t/x][s]$ if and only if $\mathcal{A} \models \varphi[s[x \mapsto a]]$. Also $u[t/x]^{\mathcal{A}}[s] = u^{\mathcal{A}}[s[x \mapsto a]]$ for every term u .*

Proof. The statement for terms follows by induction on u and gives the atomic case; the cases $\neg$ and $\rightarrow$ follow by induction. Let $\varphi = \forall y \psi$.

If $x \notin \text{FV}(\varphi)$, then $\varphi[t/x] = \varphi$ (Remark 2.3.8), and s and $s[x \mapsto a]$ agree on $\text{FV}(\varphi)$, so the claim follows from Lemma 2.4.4.

Otherwise $y \neq x$, y does not occur in t , and t is substitutable for x in ψ . Then $\varphi[t/x] = \forall y \psi[t/x]$. For $b \in A$, since y does not occur in t , we have $t^{\mathcal{A}}[s[y \mapsto b]] = a$, so by induction $\mathcal{A} \models \psi[t/x][s[y \mapsto b]]$ if and only if $\mathcal{A} \models \psi[s[y \mapsto b][x \mapsto a]]$. Since $x \neq y$, the assignment $s[y \mapsto b][x \mapsto a]$ equals $s[x \mapsto a][y \mapsto b]$. Quantifying over all $b \in A$ gives the claim. $\square$

Counterexample 2.4.7. The hypothesis that t is substitutable cannot be dropped. Let φ be $\exists y (x \neq y)$ and $t = y$. In a structure with at least two elements, φ is true under every assignment, but $\varphi[y/x] = \exists y (y \neq y)$ is false: the variable y of t has been “captured” by the quantifier $\exists y$.

Definition 2.4.8. Let Γ be a set of formulas. A structure $\mathcal{A}$ with an assignment s *satisfies* Γ if $\mathcal{A} \models \gamma[s]$ for all $\gamma \in \Gamma$. The set Γ *logically implies* φ , written $\Gamma \models \varphi$, if every $(\mathcal{A}, s)$ satisfying Γ satisfies φ ; φ is *valid* if $\emptyset \models \varphi$. A *theory* is a set of sentences. A *model* of a theory T is a structure in which all sentences of T are true. For a structure $\mathcal{A}$, the *theory of* $\mathcal{A}$, written $\text{Th}(\mathcal{A})$, is the set of sentences true in $\mathcal{A}$.

Example 2.4.9. Let T_{grp} consist of the sentences $\forall x \forall y \forall z ((x \cdot y) \cdot z = x \cdot (y \cdot z))$, $\forall x (x \cdot e = x \wedge e \cdot x = x)$ and $\forall x (x \cdot x^{-1} = e \wedge x^{-1} \cdot x = e)$ in the language of groups. Its models are exactly the groups (with the inverse map and the identity element as interpretations of $^{-1}$ and e). Adding $\forall x \forall y (x \cdot y = y \cdot x)$ gives a theory whose models are the abelian groups.

Definition 2.4.10. An *isomorphism* of $\mathcal{L}$ -structures $h: \mathcal{A} \rightarrow \mathcal{B}$ is a bijection $h: A \rightarrow B$ with $h(c^{\mathcal{A}}) = c^{\mathcal{B}}$, $h(f^{\mathcal{A}}(a_1, \dots, a_n)) = f^{\mathcal{B}}(h(a_1), \dots, h(a_n))$ and $(a_1, \dots, a_n) \in R^{\mathcal{A}}$ if and only if $(h(a_1), \dots, h(a_n)) \in R^{\mathcal{B}}$, for all symbols and all $a_i \in A$.

Proposition 2.4.11. Let $h: \mathcal{A} \rightarrow \mathcal{B}$ be an isomorphism and s an assignment in $\mathcal{A}$. Then $\mathcal{A} \models \varphi[s]$ if and only if $\mathcal{B} \models \varphi[h \circ s]$. In particular, isomorphic structures have the same theory.

Proof. By induction on terms, $h(t^{\mathcal{A}}[s]) = t^{\mathcal{B}}[h \circ s]$. Since h is injective and preserves relations in both directions, the atomic case follows; the cases $\neg$, $\rightarrow$ follow by induction. For $\forall x \psi$, note that $h \circ (s[x \mapsto a]) = (h \circ s)[x \mapsto h(a)]$ and that $a \mapsto h(a)$ is a bijection $A \rightarrow B$, so quantifying over $a \in A$ is the same as quantifying over $b \in B$. $\square$

2.5 Formal deductions

We now define formal proofs. The system below, a Hilbert-style system with modus ponens as its only rule, is the one of [End01].

Definition 2.5.1. A formula is *prime* if it is atomic or of the form $\forall x \varphi$. Every formula is built from prime formulas using $\neg$ and $\rightarrow$ alone. A *truth assignment* for $\mathcal{L}$ is a map v from the set of prime formulas to $\{0, 1\}$; it extends to $\bar{v}$ on all formulas as in Definition 2.2.6. A formula is a *tautology* if $\bar{v}(\varphi) = 1$ for all v , and a set Σ *tautologically implies* φ if $\bar{v}(\varphi) = 1$ for all v with $\bar{v}(\Sigma) = \{1\}$.

Definition 2.5.2. A *generalization* of a formula α is a formula $\forall x_1 \cdots \forall x_n \alpha$ with $n \geq 0$. The *logical axioms* are all generalizations of formulas of the following forms, where x, y, z are variables, t is a term and α, β are formulas:

- (A1) tautologies;
- (A2) $\forall x \alpha \rightarrow \alpha[t/x]$, where t is substitutable for x in α ;
- (A3) $\forall x (\alpha \rightarrow \beta) \rightarrow (\forall x \alpha \rightarrow \forall x \beta)$;
- (A4) $\alpha \rightarrow \forall x \alpha$, where $x \notin \text{FV}(\alpha)$;
- (A5) $x = x$;
- (A6) $x = y \rightarrow (\alpha[x/z] \rightarrow \alpha[y/z])$, where α is atomic.

Definition 2.5.3. Let Γ be a set of formulas. A *deduction* from Γ is a finite sequence $(\alpha_0, \dots, \alpha_n)$ of formulas such that each α_i is a logical axiom, or belongs to Γ , or is obtained by *modus ponens* from earlier formulas, that is, there are $j, k < i$ with $\alpha_k = (\alpha_j \rightarrow \alpha_i)$. We write $\Gamma \vdash \varphi$, and say that φ is a *theorem* of Γ , if there is a deduction from Γ whose last formula is φ . The set Γ is *inconsistent* if there is a formula β with $\Gamma \vdash \beta$ and $\Gamma \vdash \neg\beta$, and *consistent* otherwise.

Remark 2.5.4. A deduction is finite, so it uses only finitely many formulas of Γ . Hence $\Gamma \vdash \varphi$ if and only if $\Gamma_0 \vdash \varphi$ for some finite $\Gamma_0 \subseteq \Gamma$, and Γ is consistent if and only if all its finite subsets are. Whether $\Gamma \vdash \varphi$ holds could depend on the language in which deductions are carried out; see Lemma 2.5.12 below.

Lemma 2.5.5 (Rule T). *If $\Gamma \vdash \alpha_1, \dots, \Gamma \vdash \alpha_n$ and $\{\alpha_1, \dots, \alpha_n\}$ tautologically implies β , then $\Gamma \vdash \beta$. In particular, if Γ is inconsistent, then $\Gamma \vdash \beta$ for every formula β .*

Proof. The formula $\alpha_1 \rightarrow (\alpha_2 \rightarrow \dots (\alpha_n \rightarrow \beta) \dots)$ is a tautology, hence a logical axiom. Applying modus ponens n times gives $\Gamma \vdash \beta$. For the second claim, $\{\gamma, \neg\gamma\}$ tautologically implies every formula. $\square$

Theorem 2.5.6 (Soundness). *If $\Gamma \vdash \varphi$, then $\Gamma \models \varphi$. In particular, a set of formulas that is satisfied by some structure and assignment is consistent.*

Proof. We first show that every logical axiom is valid. If α is valid, so is every generalization of α , by the definition of satisfaction. Fix $\mathcal{A}$ and s . (A1) Let $v(\pi) = 1$ if and only if $\mathcal{A} \models \pi[s]$, for prime π . By induction, $\bar{v}(\alpha) = 1$ if and only if $\mathcal{A} \models \alpha[s]$; so tautologies are satisfied. (A2) If $\mathcal{A} \models \forall x \alpha[s]$ then $\mathcal{A} \models \alpha[s[x \mapsto t^{\mathcal{A}}[s]]]$, which by Lemma 2.4.6 means $\mathcal{A} \models \alpha[t/x][s]$. (A3) is immediate from the definitions. (A4) If $x \notin \text{FV}(\alpha)$, then s and $s[x \mapsto a]$ agree on $\text{FV}(\alpha)$, so this follows from Lemma 2.4.4. (A5) is clear. (A6) If $s(x) = s(y) = a$, then by Lemma 2.4.6 both $\alpha[x/z]$ and $\alpha[y/z]$ are satisfied under s if and only if α is satisfied under $s[z \mapsto a]$.

Now let $(\alpha_0, \dots, \alpha_n)$ be a deduction from Γ and let $(\mathcal{A}, s)$ satisfy Γ . By induction on i , $\mathcal{A} \models \alpha_i[s]$: this holds for axioms and elements of Γ , and if $\mathcal{A} \models \alpha_j[s]$ and $\mathcal{A} \models (\alpha_j \rightarrow \alpha_i)[s]$ then $\mathcal{A} \models \alpha_i[s]$. For the last claim, no $(\mathcal{A}, s)$ satisfies both β and $\neg\beta$. $\square$

Theorem 2.5.7 (Generalization theorem). *If $\Gamma \vdash \varphi$ and x is not free in any formula of Γ , then $\Gamma \vdash \forall x \varphi$.*

Proof. We show by induction on i that $\Gamma \vdash \forall x \alpha_i$ for every formula α_i of a deduction $(\alpha_0, \dots, \alpha_n)$ from Γ . If α_i is a logical axiom, so is its generalization $\forall x \alpha_i$. If $\alpha_i \in \Gamma$, then $x \notin \text{FV}(\alpha_i)$, so $\alpha_i \rightarrow \forall x \alpha_i$ is an instance of (A4), and modus ponens gives $\Gamma \vdash \forall x \alpha_i$. If α_i is obtained from α_j and $\alpha_j \rightarrow \alpha_i$ by modus ponens, then by induction $\Gamma \vdash \forall x \alpha_j$ and $\Gamma \vdash \forall x (\alpha_j \rightarrow \alpha_i)$; using the instance $\forall x (\alpha_j \rightarrow \alpha_i) \rightarrow (\forall x \alpha_j \rightarrow \forall x \alpha_i)$ of (A3) and modus ponens twice, $\Gamma \vdash \forall x \alpha_i$. $\square$

Theorem 2.5.8 (Deduction theorem). *$\Gamma \cup \{\gamma\} \vdash \varphi$ if and only if $\Gamma \vdash \gamma \rightarrow \varphi$.*

Proof. If $\Gamma \vdash \gamma \rightarrow \varphi$, then $\Gamma \cup \{\gamma\}$ proves both $\gamma \rightarrow \varphi$ and γ , hence φ . Conversely, let $(\alpha_0, \dots, \alpha_n)$ be a deduction from $\Gamma \cup \{\gamma\}$; we show $\Gamma \vdash \gamma \rightarrow \alpha_i$ by induction on i . If $\alpha_i = \gamma$, then $\gamma \rightarrow \gamma$ is a tautology. If α_i is a logical axiom or lies in Γ , then $\Gamma \vdash \alpha_i$, and α_i tautologically implies $\gamma \rightarrow \alpha_i$; apply Lemma 2.5.5. If α_i comes from α_j and $\alpha_j \rightarrow \alpha_i$ by modus ponens, then by induction Γ proves $\gamma \rightarrow \alpha_j$ and $\gamma \rightarrow (\alpha_j \rightarrow \alpha_i)$, and these two formulas tautologically imply $\gamma \rightarrow \alpha_i$. $\square$

Corollary 2.5.9 (Reductio ad absurdum). *If $\Gamma \cup \{\neg\varphi\}$ is inconsistent, then $\Gamma \vdash \varphi$. If $\Gamma \cup \{\varphi\}$ is inconsistent, then $\Gamma \vdash \neg\varphi$.*

Proof. If $\Gamma \cup \{\neg\varphi\}$ proves β and $\neg\beta$, then by Theorem 2.5.8 Γ proves $\neg\varphi \rightarrow \beta$ and $\neg\varphi \rightarrow \neg\beta$, which tautologically imply φ . The second statement is proved in the same way, using that $\{\varphi \rightarrow \beta, \varphi \rightarrow \neg\beta\}$ tautologically implies $\neg\varphi$. $\square$

For a constant symbol c and a variable y , let $\alpha[y/c]$ denote the result of replacing every occurrence of c in α by y (defined by recursion like substitution, but also inside the scope of quantifiers).

Lemma 2.5.10. *Let $(\alpha_0, \dots, \alpha_n)$ be a deduction from Γ , let c be a constant symbol that does not occur in any formula of Γ , and let y be a variable that does not occur in any α_i . Then $(\alpha_0[y/c], \dots, \alpha_n[y/c])$ is a deduction from Γ .*

Proof. Elements of Γ are unchanged, and replacement commutes with $\rightarrow$, so applications of modus ponens remain applications of modus ponens. It remains to see that the replacement of a logical axiom α is a logical axiom. Replacement commutes with generalization, so we may assume α is of one of the forms (A1)–(A6). Note that y does not occur in α .

(A1) Let w be a truth assignment, and define $v(\pi) = w(\pi[y/c])$ for prime π ; replacement sends prime formulas to prime formulas and commutes with $\neg$ and $\rightarrow$, so $\bar{v}(\alpha) = \bar{w}(\alpha[y/c])$. Hence if α is a tautology so is $\alpha[y/c]$. (A2) By induction on formulas, $\beta[t/x][y/c] = \beta[y/c][t[y/c]/x]$. The term $t[y/c]$ is substitutable for x in $\beta[y/c]$: the only variable it may contain besides those of t is y , and y is not quantified anywhere in β . (A3), (A5) are clear, and (A4) is preserved because the variable x occurs in α , so $x \neq y$, and replacing a constant by y does not make x free. (A6) Write the axiom as $x = x' \rightarrow (\beta[x/z] \rightarrow \beta[x'/z])$ with β atomic; renaming z if necessary we may assume $z \neq y$. Then its replacement is $x = x' \rightarrow (\beta[y/c][x/z] \rightarrow \beta[y/c][x'/z])$, again an instance of (A6). $\square$

Corollary 2.5.11 (Generalization on constants). *Suppose $\Gamma \vdash \varphi[c/x]$, where the constant symbol c occurs neither in Γ nor in φ . Then $\Gamma \vdash \forall x \varphi$.*

Proof. Choose a deduction of $\varphi[c/x]$ from Γ , and a variable $y \neq x$ that occurs neither in this deduction nor in φ . Since c does not occur in φ , we have $\varphi[c/x][y/c] = \varphi[y/x]$. By Lemma 2.5.10, there is a deduction of $\varphi[y/x]$ from a finite subset $\Gamma_0 \subseteq \Gamma$ in whose formulas y does not occur. By Theorem 2.5.7, $\Gamma_0 \vdash \forall y \varphi[y/x]$.

By Lemma 2.3.9, x is substitutable for y in $\varphi[y/x]$ and $\varphi[y/x][x/y] = \varphi$, so $\forall y \varphi[y/x] \rightarrow \varphi$ is an instance of (A2). Hence $\{\forall y \varphi[y/x]\} \vdash \varphi$. The variable x is not free in $\forall y \varphi[y/x]$, so Theorem 2.5.7 and Theorem 2.5.8 give $\vdash \forall y \varphi[y/x] \rightarrow \forall x \varphi$. By modus ponens, $\Gamma \vdash \forall x \varphi$. $\square$

Lemma 2.5.12. *Let $\mathcal{L}'$ be obtained from $\mathcal{L}$ by adding new constant symbols. If $\Gamma \cup \{\varphi\}$ is a set of $\mathcal{L}$ -formulas and $\Gamma \vdash \varphi$ by a deduction in $\mathcal{L}'$, then $\Gamma \vdash \varphi$ by a deduction in $\mathcal{L}$. In particular, a set of $\mathcal{L}$ -formulas that is consistent in $\mathcal{L}$ is consistent in $\mathcal{L}'$.*

Proof. A deduction in $\mathcal{L}'$ contains finitely many new constants. Replace them one at a time by distinct variables not occurring in the deduction. By Lemma 2.5.10 (applied in $\mathcal{L}'$), the result is a deduction from Γ of φ , which is unchanged because it contains no new constants.

All its formulas are $\mathcal{L}$ -formulas, and an $\mathcal{L}$ -formula that is a logical axiom of $\mathcal{L}'$ is a logical axiom of $\mathcal{L}$ (for (A1), every prime $\mathcal{L}$ -formula is a prime $\mathcal{L}'$ -formula). The last statement follows by applying the first to $\varphi = (x = x)$ and $\varphi = \neg(x = x)$, using Lemma 2.5.5. $\square$

Lemma 2.5.13 (Equality). *For all terms $t, t', t'', t_1, \dots, t_n, s_1, \dots, s_n$:*

1. $\vdash t = t$;
2. $\{t = t'\} \vdash t' = t$;
3. $\{t = t', t' = t''\} \vdash t = t''$;
4. $\{t_1 = s_1, \dots, t_n = s_n\} \vdash f(t_1, \dots, t_n) = f(s_1, \dots, s_n)$ for every n -ary function symbol f ;
5. $\{t_1 = s_1, \dots, t_n = s_n, R(t_1, \dots, t_n)\} \vdash R(s_1, \dots, s_n)$ for every n -ary relation symbol R .

Proof. We use the following observation. If $\vdash \beta$ where β has no quantifiers, and $x_1, \dots, x_k$ are distinct variables not occurring in the terms $u_1, \dots, u_k$, then $\vdash \beta[u_1/x_1] \cdots [u_k/x_k]$. Indeed, by Theorem 2.5.7, $\vdash \forall x_1 \cdots \forall x_k \beta$, and we remove the quantifiers one at a time with (A2): each term u_i is substitutable because it contains none of the quantified variables and the remaining formula has no other quantifiers. In each part below, the variables used are chosen distinct and not occurring in the terms of the statement.

(1) Apply the observation to the axiom $x = x$. (2) Let α be $z = x$. The instance $x = y \rightarrow (x = x \rightarrow y = x)$ of (A6), with $x := t$ and $y := t'$, together with (1) gives the claim. (3) Let α be $z = w$. The instance $x = y \rightarrow (x = w \rightarrow y = w)$ of (A6), with $x := t', y := t, w := t''$, gives $\vdash t' = t \rightarrow (t' = t'' \rightarrow t = t'')$; combine with (2). (4) For $0 \leq k \leq n$ let $a_k = f(y_1, \dots, y_k, x_{k+1}, \dots, x_n)$. With α the atomic formula $f(x_1, \dots, x_n) = f(y_1, \dots, y_{k-1}, z, x_{k+1}, \dots, x_n)$, axiom (A6) gives $\vdash x_k = y_k \rightarrow (f(x_1, \dots, x_n) = a_{k-1} \rightarrow f(x_1, \dots, x_n) = a_k)$. Starting from $f(x_1, \dots, x_n) = a_0$, which holds by (1), we obtain $\{x_1 = y_1, \dots, x_n = y_n\} \vdash f(x_1, \dots, x_n) = a_n$. By the deduction theorem this becomes a deducible quantifier-free implication, and the observation with $x_i := t_i, y_i := s_i$ gives the claim. (5) is proved in the same way as (4), with α the atomic formula $R(y_1, \dots, y_{k-1}, z, x_{k+1}, \dots, x_n)$. $\square$

Lemma 2.5.14 (Alphabetic variants). *Let φ be a formula, x a variable and t a term. There is a formula φ' with $c(\varphi') = c(\varphi)$ such that t is substitutable for x in φ' , $\vdash \varphi \rightarrow \varphi'$ and $\vdash \varphi' \rightarrow \varphi$.*

Proof. By induction on φ . For atomic φ take $\varphi' = \varphi$. For $\neg\psi$ and $\psi \rightarrow \chi$, take $\neg\psi'$ and $\psi' \rightarrow \chi'$ and use Lemma 2.5.5. Let $\varphi = \forall y \psi$, and let ψ' be given by induction. Choose a variable z that does not occur in ψ' or in t and is different from x , and put $\varphi' = \forall z \psi'[z/y]$.

The term t is substitutable for x in φ' . If $x = y$, then x is not free in φ' . If $x \neq y$, then z does not occur in t , and t is substitutable for x in $\psi'[z/y]$ because it is substitutable in ψ' and $\psi'[z/y]$ has the same quantifiers as ψ' in the same places.

From $\vdash \psi \rightarrow \psi'$, Theorem 2.5.7 and (A3) give $\vdash \forall y \psi \rightarrow \forall y \psi'$, and similarly $\vdash \forall y \psi' \rightarrow \forall y \psi$. By Lemma 2.3.9, $\forall y \psi' \rightarrow \psi'[z/y]$ and $\forall z \psi'[z/y] \rightarrow \psi'$ are instances of (A2). The variable z is not free in $\forall y \psi'$, and y is not free in $\forall z \psi'[z/y]$; so by Theorems 2.5.7 and 2.5.8, $\vdash \forall y \psi' \rightarrow \forall z \psi'[z/y]$ and $\vdash \forall z \psi'[z/y] \rightarrow \forall y \psi'$. Combining these implications with Lemma 2.5.5 gives the claim. Finally $c(\varphi') = c(\psi') + 1 = c(\varphi)$. $\square$

Example 2.5.15. Let us check that $\vdash \forall x \forall y \varphi \rightarrow \forall y \forall x \varphi$. By (A2) twice (a variable is substitutable for itself), $\{\forall x \forall y \varphi\} \vdash \varphi$. Neither x nor y is free in $\forall x \forall y \varphi$, so the generalization theorem gives $\{\forall x \forall y \varphi\} \vdash \forall y \forall x \varphi$, and the deduction theorem gives the claim. In practice, one rarely writes deductions: by Corollary 2.6.2 below it suffices to check that the implication is valid.

2.6 The completeness theorem

Theorem 2.6.1 (Gödel, Henkin). *Let $\mathcal{L}$ be a countable language and Γ a consistent set of $\mathcal{L}$ -formulas. Then there are an $\mathcal{L}$ -structure $\mathcal{A}$ with a countable universe and an assignment s such that $(\mathcal{A}, s)$ satisfies Γ .*

Proof. This is the method of [Hen49]; the theorem is due to [G30].

Step 1: new constants. Let $\mathcal{L}'$ be $\mathcal{L}$ together with new constant symbols $c_0, c_1, \dots$. It is still countable, and Γ is consistent in $\mathcal{L}'$ by Lemma 2.5.12. From now on all formulas and deductions are in $\mathcal{L}'$.

Step 2: witnesses. Enumerate all pairs of a formula and a variable as $(\varphi_1, x_1), (\varphi_2, x_2), \dots$. Define inductively

$$\theta_n = (\neg \forall x_n \varphi_n \rightarrow \neg \varphi_n[d_n/x_n]),$$

where d_n is the first of the new constants c_i that occurs neither in φ_n nor in $\theta_1, \dots, \theta_{n-1}$. Let $\Theta = \{\theta_1, \theta_2, \dots\}$. We claim that $\Gamma \cup \Theta$ is consistent. Otherwise, by Remark 2.5.4, there is a least m such that $\Gamma \cup \{\theta_1, \dots, \theta_m\}$ is inconsistent, and $m \geq 1$. By Corollary 2.5.9, $\Gamma \cup \{\theta_1, \dots, \theta_{m-1}\} \vdash \neg \theta_m$. The formula $\neg \theta_m$ tautologically implies both $\neg \forall x_m \varphi_m$ and $\varphi_m[d_m/x_m]$. The constant d_m occurs neither in Γ (whose formulas are in $\mathcal{L}$), nor in $\theta_1, \dots, \theta_{m-1}$, nor in φ_m ; so Corollary 2.5.11 gives $\Gamma \cup \{\theta_1, \dots, \theta_{m-1}\} \vdash \forall x_m \varphi_m$. This set is therefore inconsistent, contradicting the minimality of m .

Step 3: a maximal consistent set. Enumerate all formulas as $\psi_0, \psi_1, \dots$. Put $\Delta_0 = \Gamma \cup \Theta$, and $\Delta_{n+1} = \Delta_n \cup \{\psi_n\}$ if this set is consistent, $\Delta_{n+1} = \Delta_n \cup \{\neg \psi_n\}$ otherwise. If Δ_n is consistent, so is Δ_{n+1} : if both $\Delta_n \cup \{\psi_n\}$ and $\Delta_n \cup \{\neg \psi_n\}$ were inconsistent, then Δ_n would prove $\neg \psi_n$ and ψ_n by Corollary 2.5.9. Let $\Delta = \bigcup_n \Delta_n$. It is consistent (a deduction uses finitely many formulas), and for every formula ψ either $\psi \in \Delta$ or $\neg \psi \in \Delta$, but not both. Moreover, if $\Delta \vdash \psi$ then $\psi \in \Delta$: otherwise $\neg \psi \in \Delta$ and Δ would be inconsistent.

Step 4: the term structure. Let T be the set of terms of $\mathcal{L}'$; it is countable. For $t, t' \in T$ write $t \sim t'$ if $(t = t') \in \Delta$. Since Δ contains every formula it proves, Lemma 2.5.13 shows that $\sim$ is an equivalence relation and that

- if $t_i \sim s_i$ for all i , then $f(t_1, \dots, t_n) \sim f(s_1, \dots, s_n)$, and $R(t_1, \dots, t_n) \in \Delta$ if and only if $R(s_1, \dots, s_n) \in \Delta$.

Let $A = T/\sim$ and write $[t]$ for the class of t . Define an $\mathcal{L}'$ -structure $\mathcal{A}$ with universe A by $c^{\mathcal{A}} = [c]$, $f^{\mathcal{A}}([t_1], \dots, [t_n]) = [f(t_1, \dots, t_n)]$ and $([t_1], \dots, [t_n]) \in R^{\mathcal{A}}$ if and only if $R(t_1, \dots, t_n) \in \Delta$; these are well defined by the displayed property. Let $s(x) = [x]$. By induction on terms, $t^{\mathcal{A}}[s] = [t]$ for every term t .

Step 5: the truth lemma. We show that for every formula φ ,

$$\mathcal{A} \models \varphi[s] \iff \varphi \in \Delta,$$

by induction on the complexity $c(\varphi)$. For $t_1 = t_2$ both sides say $[t_1] = [t_2]$; for $R(t_1, \dots, t_n)$ both sides say $R(t_1, \dots, t_n) \in \Delta$. For $\neg \psi$: $\mathcal{A} \models \neg \psi[s]$ iff $\psi \notin \Delta$ iff $\neg \psi \in \Delta$. For $\psi \rightarrow \chi$: if $(\psi \rightarrow \chi) \in \Delta$ and $\psi \in \Delta$, then $\Delta \vdash \chi$, so $\chi \in \Delta$; conversely, if $\psi \notin \Delta$ then $\neg \psi \in \Delta$, and if $\chi \in \Delta$, in either case Δ proves $\psi \rightarrow \chi$ by Lemma 2.5.5. Hence $(\psi \rightarrow \chi) \in \Delta$ iff $\psi \notin \Delta$ or $\chi \in \Delta$, iff $\mathcal{A} \models (\psi \rightarrow \chi)[s]$.

Let $\varphi = \forall x \psi$. Suppose $\varphi \notin \Delta$. The pair (ψ, x) is (φ_n, x_n) for some n , and $\neg \forall x \psi \in \Delta$ and $\theta_n \in \Delta$, so $\neg \psi[d_n/x] \in \Delta$ and $\psi[d_n/x] \notin \Delta$. Since $c(\psi[d_n/x]) = c(\psi) < c(\varphi)$, the induction hypothesis gives $\mathcal{A} \not\models \psi[d_n/x][s]$, which by Lemma 2.4.6 means $\mathcal{A} \not\models \psi[s[x \mapsto [d_n]]]$. Hence $\mathcal{A} \not\models \varphi[s]$.

Suppose $\varphi \in \Delta$, and let $a \in A$, say $a = [t]$. Let ψ' be as in Lemma 2.5.14 for ψ, x and t . From $\vdash \psi \rightarrow \psi'$ we get $\vdash \forall x \psi \rightarrow \forall x \psi'$ (generalization and (A3)), so $\forall x \psi' \in \Delta$,

and by (A2) $\psi'[t/x] \in \Delta$. Since $c(\psi'[t/x]) = c(\psi) < c(\varphi)$, the induction hypothesis gives $\mathcal{A} \models \psi'[t/x][s]$, that is, $\mathcal{A} \models \psi'[s[x \mapsto a]]$ by Lemma 2.4.6. As $\vdash \psi' \rightarrow \psi$, Theorem 2.5.6 gives $\mathcal{A} \models \psi[s[x \mapsto a]]$. Since a was arbitrary, $\mathcal{A} \models \varphi[s]$.

Step 6: conclusion. Since $\Gamma \subseteq \Delta$, the pair $(\mathcal{A}, s)$ satisfies Γ . Forgetting the interpretations of the new constants gives an $\mathcal{L}$ -structure which, with the same assignment, still satisfies Γ , because the satisfaction of an $\mathcal{L}$ -formula only involves the symbols of $\mathcal{L}$. Its universe A is countable, as a quotient of T . $\square$

Corollary 2.6.2 (Completeness theorem). *Let $\mathcal{L}$ be countable. Then $\Gamma \models \varphi$ if and only if $\Gamma \vdash \varphi$.*

Proof. One direction is Theorem 2.5.6. Suppose $\Gamma \not\models \varphi$. By Corollary 2.5.9, $\Gamma \cup \{\neg\varphi\}$ is consistent, so by Theorem 2.6.1 it is satisfied by some $(\mathcal{A}, s)$. Then $(\mathcal{A}, s)$ satisfies Γ but not φ , so $\Gamma \not\models \varphi$. $\square$

Remark 2.6.3. The completeness theorem holds for languages of any cardinality, and the structure can then be taken of cardinality at most that of the language plus $\aleph_0$. The proof is the same, except that the enumerations of Steps 2 and 3 are replaced by transfinite ones, which requires the well-ordering theorem (Chapter 3); see [CK90, Section 2.2].

2.7 Compactness and the Löwenheim–Skolem theorem

Theorem 2.7.1 (Compactness theorem). *Let $\mathcal{L}$ be countable and Γ a set of formulas. If every finite subset of Γ is satisfiable, then Γ is satisfiable.*

Proof. If Γ is not satisfiable, it is inconsistent by Theorem 2.6.1, so some finite subset Γ_0 is inconsistent (Remark 2.5.4), and Γ_0 is not satisfiable by Theorem 2.5.6. $\square$

Theorem 2.7.2 (Löwenheim–Skolem). *Let $\mathcal{L}$ be countable. Every satisfiable set of $\mathcal{L}$ -formulas is satisfied by a structure with a countable (finite or countably infinite) universe. In particular, a theory with a model has a countable model.*

Proof. A satisfiable set is consistent by Theorem 2.5.6, and Theorem 2.6.1 produces a countable structure. $\square$

Proposition 2.7.3. *Let $\mathcal{L}$ be countable and T a theory that has finite models of arbitrarily large size. Then T has an infinite model.*

Proof. For $n \geq 2$ let λ_n be the sentence $\exists x_1 \cdots \exists x_n \bigwedge_{1 \leq i < j \leq n} x_i \neq x_j$, which is true in a structure if and only if it has at least n elements. Every finite subset of $T \cup \{\lambda_n : n \geq 2\}$ is contained in $T \cup \{\lambda_2, \dots, \lambda_N\}$ for some N , which is satisfied by a finite model of T with at least N elements. By Theorem 2.7.1, $T \cup \{\lambda_n : n \geq 2\}$ has a model, which is an infinite model of T . $\square$

Counterexample 2.7.4. “Being finite” cannot be expressed in first-order logic: in any countable language there is no theory whose models are exactly the finite structures. Indeed, such a theory would have finite models of every size (in any language, every nonempty finite set carries a structure), hence an infinite model by Proposition 2.7.3.

Example 2.7.5. Let $\mathcal{N}$ be the structure $(\mathbb{N}, 0, S, +, \cdot)$ for the language of arithmetic, with $S(n) = n + 1$, and for $n \in \mathbb{N}$ let $\underline{n} = S(S(\cdots S(0) \cdots))$ with n symbols S . Add a new constant symbol c and consider $\text{Th}(\mathcal{N}) \cup \{c \neq \underline{n} : n \in \mathbb{N}\}$. Every finite subset is satisfied by $\mathcal{N}$, interpreting c as a large enough number. By compactness there is a model $\mathcal{M}$ of $\text{Th}(\mathcal{N})$ with an element $c^{\mathcal{M}}$ different from the values of all $\underline{n}$. Such a model is not isomorphic to $\mathcal{N}$ (in $\mathcal{N}$ every element is the value of some $\underline{n}$, and isomorphisms preserve the values of closed terms), although it satisfies exactly the same sentences. So first-order sentences cannot characterise $\mathcal{N}$ up to isomorphism.

Remark 2.7.6. Applied to set theory (Chapter 3), the Löwenheim–Skolem theorem says that if ZFC has a model, it has a countable model $\mathcal{M}$, even though ZFC proves that uncountable sets exist. There is no contradiction: an element of $\mathcal{M}$ that $\mathcal{M}$ believes to be uncountable has only countably many $\mathcal{M}$ -elements, but no bijection with $\mathbb{N}$ exists *inside* $\mathcal{M}$. This is known as Skolem’s paradox.

2.8 The incompleteness theorems

The theorems of this section are quoted without proof: their proofs require a precise theory of computability, which is not developed in this project. Complete proofs can be found in [Smi13] and [Sho67, Chapter 6].

Definition 2.8.1. *Robinson arithmetic* Q is the theory in the language of arithmetic consisting of the sentences

$$\begin{array}{ll} \forall x (S(x) \neq 0), & \forall x \forall y (S(x) = S(y) \rightarrow x = y), \\ \forall x (x \neq 0 \rightarrow \exists y (x = S(y))), & \forall x (x + 0 = x), \\ \forall x \forall y (x + S(y) = S(x + y)), & \forall x (x \cdot 0 = 0), \\ \forall x \forall y (x \cdot S(y) = x \cdot y + x). \end{array}$$

Peano arithmetic PA consists of Q together with, for every formula φ whose free variables are among $x, y_1, \dots, y_k$, the induction axiom $\forall y_1 \dots \forall y_k ((\varphi[0/x] \wedge \forall x (\varphi \rightarrow \varphi[S(x)/x])) \rightarrow \forall x \varphi)$.

Definition 2.8.2. A theory T is *complete* if for every sentence σ , either $T \vdash \sigma$ or $T \vdash \neg\sigma$. It is *computably axiomatized* if there is an algorithm deciding, for any sentence, whether it belongs to T .

Example 2.8.3. For every structure $\mathcal{A}$, the theory $\text{Th}(\mathcal{A})$ is complete and consistent. For the structure $\mathcal{N}$ of Example 2.7.5, Theorem 2.8.4 implies that $\text{Th}(\mathcal{N})$ is not computably axiomatized.

Theorem 2.8.4 (First incompleteness theorem). *Let T be a consistent, computably axiomatized theory in the language of arithmetic with $T \vdash \sigma$ for every $\sigma \in Q$. Then T is not complete.*

This is due to [G31], under a slightly stronger hypothesis than consistency, and in this form to [Ros36]; the theory Q is from [TMR53].

Theorem 2.8.5 (Second incompleteness theorem). *Let T be a consistent, computably axiomatized theory in the language of arithmetic with $T \vdash \sigma$ for every $\sigma \in PA$. Then $T \not\vdash \text{Con}_T$, where Con_T is the arithmetic sentence which expresses, via a Gödel numbering of formulas, that T is consistent.*

See [G31] and [Smi13].

Remark 2.8.6. Both theorems apply, with the same proofs, to any consistent computably axiomatized theory in which arithmetic can be interpreted, such as ZFC: if ZFC is consistent, it does not prove its own consistency [Kun11]. This affects the foundations of this project. In Chapter 3 we assume the existence of a Grothendieck universe, and a universe is a model of ZFC; so ZFC together with this assumption proves Con_{ZFC} . By the second incompleteness theorem, the existence of a universe can therefore not be proved in ZFC (if ZFC is consistent): it is a genuinely new axiom, of modest strength.

2.9 Exercises

Exercise 2.9.1. Show that every function $\{0, 1\}^n \rightarrow \{0, 1\}$ with $n \geq 1$ is computed by a formula built from variables using only $\neg$ and $\wedge$ (as primitive connectives with their usual truth tables). Show that a formula built from variables using only $\wedge$, $\vee$ and $\rightarrow$ is satisfied by the truth assignment that is constantly 1, and deduce that these three connectives do not suffice to express negation.

Solution. By Lemma 2.2.8, $\varphi \vee \psi$ has the same truth table as $\neg(\neg\varphi \wedge \neg\psi)$ and $\varphi \rightarrow \psi$ the same as $\neg(\varphi \wedge \neg\psi)$; so the formula of Proposition 2.2.10 can be rewritten using $\neg$ and $\wedge$. For the second part, argue by induction: if v is constantly 1, then $\bar{v}$ is 1 on variables, and $1 \wedge 1$, $1 \vee 1$ and $1 \rightarrow 1$ all have value 1. Negation of a variable has value 0 under v , so it is not expressible. $\square$

Exercise 2.9.2. Show that $\exists x \forall y \varphi \rightarrow \forall y \exists x \varphi$ is valid for every formula φ , and give a formula φ in the language of orders and a structure in which $\forall y \exists x \varphi \rightarrow \exists x \forall y \varphi$ is false.

Exercise 2.9.3. A *well-order* is a total order in which every nonempty subset has a least element. Show that there is no theory T in the language of orders whose models are exactly the well-ordered sets. (Hint: add constants $c_0, c_1, \dots$ and the sentences $c_{n+1} < c_n$.)

Exercise 2.9.4. A group is a *torsion group* if every element has finite order. Show that there is no theory in the language of groups whose models are exactly the torsion groups. Show, on the other hand, that for every $n \geq 1$ there is a single sentence whose models among groups are exactly the groups satisfying $g^n = e$ for all g .

Exercise 2.9.5. Let T_0 be the theory of fields in the language of rings, and for a prime p let χ_p be the sentence $\underline{p} \neq 0$, where $\underline{p}$ is $1 + 1 + \dots + 1$ with p summands. The models of $T_0 \cup \{\chi_p : p \text{ prime}\}$ are the fields of characteristic zero. Show that there is no single sentence σ such that the models of $T_0 \cup \{\sigma\}$ are exactly the fields of characteristic zero. Deduce that a sentence true in all fields of characteristic zero is true in all fields of sufficiently large prime characteristic.

Exercise 2.9.6. Let T be a theory and σ a sentence with $T \models \sigma$. Show, without using deductions, that some finite subset $T_0 \subseteq T$ satisfies $T_0 \models \sigma$ (for $\mathcal{L}$ countable).

Solution. Otherwise, for every finite $T_0 \subseteq T$ the set $T_0 \cup \{\neg\sigma\}$ has a model. Then every finite subset of $T \cup \{\neg\sigma\}$ is satisfiable, so by Theorem 2.7.1 the whole set has a model, which is a model of T in which σ is false, contradicting $T \models \sigma$. $\square$

Chapter 3

Set Theory

3.1 Introduction

This chapter sets up the foundations of the project: Zermelo–Fraenkel set theory with the axiom of choice (ZFC), formulated as a first-order theory in the sense of Chapter 2. We develop the basic constructions (relations, functions, quotients), the natural numbers, ordinals and cardinals, prove the equivalence of the axiom of choice with Zorn’s lemma and the well-ordering theorem, and discuss Grothendieck universes, which provide the conventions on size used throughout the project (Notation 1.6.1).

Standard references are [Jec03] and [Kun11]; [Hal60] is an informal introduction.

Prerequisites. Chapter 2 (Mathematical Logic).

Remark 3.1.1. Set theory is the first-order theory, in the language with one binary relation symbol $\in$ (Example 2.3.5), whose axioms are listed in Definition 3.2.1. Its models have a universe whose elements are called *sets*. We write statements in ordinary mathematical language, but each of them can be translated into a formula of this language, and each “construction” is justified by the axioms. A *class* is given by a formula $\varphi(x)$, possibly with parameters, and we write $\{x : \varphi(x)\}$ for it; a class need not be a set. Statements about classes are abbreviations of statements about the corresponding formulas.

3.2 The axioms

We use the abbreviations $x \subseteq y$ for $\forall z (z \in x \rightarrow z \in y)$, $\forall z \in x \varphi$ for $\forall z (z \in x \rightarrow \varphi)$ and $\exists z \in x \varphi$ for $\exists z (z \in x \wedge \varphi)$.

Definition 3.2.1. The axioms of *Zermelo–Fraenkel set theory* ZF are the following sentences.

1. (Extensionality) $\forall x \forall y (\forall z (z \in x \leftrightarrow z \in y) \rightarrow x = y)$.
2. (Foundation) $\forall x (\exists y (y \in x) \rightarrow \exists y \in x \neg \exists z (z \in x \wedge z \in y))$.
3. (Separation) For every formula φ in which y is not free: $\forall w_1 \cdots \forall w_n \forall x \exists y \forall z (z \in y \leftrightarrow (z \in x \wedge \varphi))$, where $w_1, \dots, w_n$ are the other free variables of φ .
4. (Pairing) $\forall x \forall y \exists z (x \in z \wedge y \in z)$.
5. (Union) $\forall F \exists A \forall Y \forall x ((x \in Y \wedge Y \in F) \rightarrow x \in A)$.
6. (Replacement) For every formula φ in which B is not free: $\forall w_1 \cdots \forall w_n \forall A (\forall x \in A \exists! y \varphi \rightarrow \exists B \forall x \in A \exists y \in B \varphi)$, where $\exists! y \varphi$ abbreviates “there is exactly one y with φ ” and $w_1, \dots, w_n$ are the other free variables of φ .

7. (Infinity) $\exists x (\exists e \in x \forall z (z \notin e) \wedge \forall y \in x \exists s \in x \forall z (z \in s \leftrightarrow (z \in y \vee z = y)))$.
8. (Power set) $\forall x \exists y \forall z (z \subseteq x \rightarrow z \in y)$.

The theory ZFC consists of ZF together with the axiom of choice, Definition 3.7.1. The axioms of separation and replacement are *schemas*: they consist of one axiom for each formula φ .

Remark 3.2.2. Extensionality says that a set is determined by its elements. Separation says that for a set x and a property φ , the elements of x with the property form a set, written $\{z \in x : \varphi(z)\}$. Pairing, union and power set provide sets containing the pair $\{x, y\}$, the union of the elements of F , and the subsets of x ; by separation and extensionality there are then unique sets with exactly these elements. Replacement says that the image of a set under a class function is contained in a set. Infinity provides a set containing $\emptyset$ and closed under $y \mapsto y \cup \{y\}$. Foundation says that every nonempty set has an $\in$ -minimal element.

Lemma 3.2.3. 1. *There is a unique set $\emptyset$ with no elements.*

2. *For all sets x, y there are unique sets $\{x, y\}$, $\bigcup x$ and $\mathcal{P}(x)$ with, respectively, the elements x and y ; the elements of elements of x ; the subsets of x .*
3. *For all sets x, y there are unique sets $x \cup y$, $x \cap y$ and $x \setminus y$ with the usual elements. If F is a nonempty set, there is a unique set $\bigcap F$ whose elements are the sets belonging to every element of F .*

Proof. Uniqueness always follows from extensionality. (1) Some set x exists, because $\exists x (x = x)$ is valid (structures are nonempty); apply separation with the formula $z \neq z$. (2) Apply separation to the sets given by pairing, union and power set, with the formulas $z = x \vee z = y$, $\exists Y \in x (z \in Y)$ and $z \subseteq x$. (3) Put $x \cup y = \bigcup \{x, y\}$, and use separation on x for $x \cap y$ and $x \setminus y$. If $X \in F$, put $\bigcap F = \{z \in X : \forall Y \in F (z \in Y)\}$. $\square$

Proposition 3.2.4 (Russell). *There is no set of which every set is an element.*

Proof. Suppose V is such a set and let $R = \{x \in V : x \notin x\}$. Then $R \in V$, so $R \in R$ if and only if $R \notin R$, a contradiction. $\square$

Remark 3.2.5. So the class $\{x : x = x\}$ of all sets is not a set; such classes are called *proper classes*. Proposition 3.2.4 holds even without foundation. With foundation it is also immediate from Lemma 3.2.6.

Lemma 3.2.6. *There is no set x with $x \in x$, and there are no sets x, y with $x \in y$ and $y \in x$.*

Proof. If $x \in x$, the set $\{x\}$ has no $\in$ -minimal element: its only element x satisfies $x \in x \cap \{x\}$. If $x \in y \in x$, the set $\{x, y\}$ has no $\in$ -minimal element, since $y \in x \cap \{x, y\}$ and $x \in y \cap \{x, y\}$. $\square$

3.3 Relations and functions

Definition 3.3.1. The *ordered pair* of a and b is $(a, b) = \{\{a\}, \{a, b\}\}$.

Lemma 3.3.2. $(a, b) = (c, d)$ if and only if $a = c$ and $b = d$.

Proof. Suppose $(a, b) = (c, d)$. If $a = b$, then $(a, b) = \{\{a\}\}$, so $\{c\} = \{c, d\} = \{a\}$ and $a = b = c = d$. Similarly if $c = d$. Assume $a \neq b$ and $c \neq d$. Then $\{a\}$ is the only one-element set in (a, b) , and $\{c\}$ the only one in (c, d) , so $\{a\} = \{c\}$ and $a = c$. The two-element sets $\{a, b\}$ and $\{c, d\} = \{a, d\}$ are then equal, and since $b \neq a$ we get $b = d$. The converse is clear. $\square$

Definition 3.3.3. The *cartesian product* of sets A and B is $A \times B = \{p \in \mathcal{P}(\mathcal{P}(A \cup B)) : \exists a \in A \exists b \in B p = (a, b)\}$. It is a set by separation, and its elements are exactly the pairs (a, b) with $a \in A, b \in B$, because $\{a\}$ and $\{a, b\}$ are subsets of $A \cup B$.

Definition 3.3.4. A *relation* from A to B is a subset $R \subseteq A \times B$; we write aRb for $(a, b) \in R$. A *function* (or *map*) $f: A \rightarrow B$ is a relation $f \subseteq A \times B$ such that for every $a \in A$ there is exactly one $b \in B$ with $(a, b) \in f$, written $b = f(a)$. (Formally, a function records its target: it is the triple (A, B, f) ; we shall not insist on this.) For $X \subseteq A$ and $Y \subseteq B$, the *image* is $f[X] = \{f(a) : a \in X\}$ and the *preimage* is $f^{-1}[Y] = \{a \in A : f(a) \in Y\}$. The function is *injective*, *surjective* or *bijective* if it is so as a map in the usual sense. The composition $g \circ f$ of $f: A \rightarrow B$ and $g: B \rightarrow C$ is $\{(a, c) \in A \times C : \exists b ((a, b) \in f \wedge (b, c) \in g)\}$. The set of all functions from A to B is $B^A = \{f \in \mathcal{P}(A \times B) : f \text{ is a function } A \rightarrow B\}$.

Definition 3.3.5. A *family* $(x_i)_{i \in I}$ is a function $i \mapsto x_i$ with domain I . Its union is $\bigcup_{i \in I} x_i = \bigcup \{x_i : i \in I\}$, which is a set: the image of I under the family is a set by replacement and separation. Its *product* is the set $\prod_{i \in I} x_i = \{f \in (\bigcup_{i \in I} x_i)^I : f(i) \in x_i \text{ for all } i \in I\}$.

Lemma 3.3.6. A function $f: A \rightarrow B$ is bijective if and only if there is a function $g: B \rightarrow A$ with $g \circ f = \text{id}_A$ and $f \circ g = \text{id}_B$; such a g is unique.

Proof. If f is bijective, $g = \{(b, a) : (a, b) \in f\}$ is a function $B \rightarrow A$ with the two properties. Conversely, $g \circ f = \text{id}_A$ implies that f is injective and $f \circ g = \text{id}_B$ that f is surjective. If g' also has the properties, then $g' = g' \circ f \circ g = g$. $\square$

Definition 3.3.7. An *equivalence relation* on A is a relation $\sim$ from A to A that is reflexive, symmetric and transitive. The *class* of $a \in A$ is $[a] = \{x \in A : x \sim a\}$, and the *quotient* is the set $A/\sim = \{X \in \mathcal{P}(A) : \exists a \in A X = [a]\}$, with the *projection* $\pi: A \rightarrow A/\sim, a \mapsto [a]$. A *partition* of A is a set of nonempty, pairwise disjoint subsets of A whose union is A .

Proposition 3.3.8. Let A be a set. The map $\sim \mapsto A/\sim$ is a bijection from the set of equivalence relations on A to the set of partitions of A . Its inverse sends a partition Π to the relation “ x and y lie in the same element of Π ”.

Proof. If $\sim$ is an equivalence relation, the classes are nonempty ($a \in [a]$) and cover A . If $z \in [a] \cap [b]$, then $z \sim a$ and $z \sim b$, so $a \sim b$ by symmetry and transitivity, and then $[a] = [b]$. So $A/\sim$ is a partition. Conversely, the relation defined by a partition Π is clearly an equivalence relation whose classes are the elements of Π , and $x \sim y$ if and only if x and y lie in the same class of $\sim$. So the two constructions are inverse to each other. $\square$

Proposition 3.3.9. Let $\sim$ be an equivalence relation on A and $f: A \rightarrow B$ a map such that $a \sim a'$ implies $f(a) = f(a')$. There is a unique map $\bar{f}: A/\sim \rightarrow B$ with $\bar{f} \circ \pi = f$.

Proof. Uniqueness holds because π is surjective. Put $\bar{f} = \{(X, b) \in (A/\sim) \times B : \exists a \in X f(a) = b\}$. Every class X is nonempty, and all its elements have the same image under f , so $\bar{f}$ is a function, and $\bar{f}([a]) = f(a)$. $\square$

Definition 3.3.10. A *partial order* on A is a relation $\leq$ that is reflexive, antisymmetric and transitive; it is *total* if $x \leq y$ or $y \leq x$ for all x, y . We write $x < y$ for $x \leq y \wedge x \neq y$. A *chain* is a totally ordered subset. For $X \subseteq A$, an *upper bound* of X is an $a \in A$ with $x \leq a$ for all $x \in X$; a *least upper bound* or *supremum* is an upper bound below all upper bounds. An element m is *maximal* if there is no x with $m < x$, and *greatest* if $x \leq m$ for all x . A *well-order* is a total order in which every nonempty subset has a least element. An *isomorphism* of ordered sets is a bijection f with $x \leq y \iff f(x) \leq f(y)$. For $a \in A$, the *initial segment* determined by a is $A_{<a} = \{x \in A : x < a\}$.

Counterexample 3.3.11. A maximal element need not be greatest, and a greatest element need not exist. In $\{\{0\}, \{1\}\}$, ordered by inclusion, both elements are maximal and there is no greatest element.

Lemma 3.3.12. Let W be a well-ordered set and $f: W \rightarrow W$ a strictly increasing map. Then $f(w) \geq w$ for all $w \in W$. Consequently, W is not isomorphic to any initial segment $W_{<a}$, and the only isomorphism $W \rightarrow W$ is the identity. If two well-ordered sets are isomorphic, the isomorphism is unique.

Proof. If $\{w : f(w) < w\}$ is nonempty, let w be its least element. Then $f(w) < w$, so $f(f(w)) < f(w)$, and $f(w)$ is a smaller element of the same set; contradiction. An isomorphism $f: W \rightarrow W_{<a}$ would give a strictly increasing $W \rightarrow W$ with $f(a) < a$. If $f, g: W \rightarrow W'$ are isomorphisms, $g^{-1} \circ f$ and $f^{-1} \circ g$ are strictly increasing, so $f(w) \geq g(w)$ and $g(w) \geq f(w)$ for all w . $\square$

3.4 The natural numbers

Definition 3.4.1. The successor of a set x is $S(x) = x \cup \{x\}$. A set I is *inductive* if $\emptyset \in I$ and $S(y) \in I$ for all $y \in I$. By the axiom of infinity there is an inductive set I_0 , and we define

$$\omega = \{n \in I_0 : n \in I \text{ for every inductive set } I\}.$$

The elements of ω are the *natural numbers*; we write $\mathbb{N} = \omega$, $0 = \emptyset$, $1 = S(0)$, $2 = S(1)$, and so on.

Theorem 3.4.2 (Induction). *The set ω is inductive and is contained in every inductive set; it does not depend on the choice of I_0 . If $X \subseteq \omega$ contains 0 and $S(n) \in X$ for all $n \in X$, then $X = \omega$.*

Proof. Every inductive set contains $\emptyset$ and is closed under S , so the intersection ω has the same two properties; it is contained in every inductive set by definition. If I_1 is another inductive set, the set defined using I_1 is contained in I_0 and has the same elements. A set X as in the statement is inductive, so $\omega \subseteq X$. $\square$

Lemma 3.4.3. *Every element of a natural number is a natural number, and every natural number n is transitive: if $y \in x \in n$ then $y \in n$.*

Proof. Let X be the set of $n \in \omega$ such that $n \subseteq \omega$ and n is transitive. Clearly $0 \in X$. If $n \in X$, then $S(n) = n \cup \{n\} \subseteq \omega$; and if $y \in x \in S(n)$, then either $x \in n$, so $y \in n$, or $x = n$, so $y \in n$; in both cases $y \in S(n)$. By Theorem 3.4.2, $X = \omega$. $\square$

Proposition 3.4.4 (Peano axioms). *For all $m, n \in \omega$: $S(n) \neq 0$, and $S(m) = S(n)$ implies $m = n$.*

Proof. $n \in S(n)$, so $S(n) \neq \emptyset$. Suppose $S(m) = S(n)$ and $m \neq n$. Since $m \in S(n)$, we get $m \in n$; similarly $n \in m$. This contradicts Lemma 3.2.6. $\square$

Theorem 3.4.5 (Recursion). *Let A be a set, $a \in A$ and $g: \omega \times A \rightarrow A$ a map. There is a unique map $f: \omega \rightarrow A$ with $f(0) = a$ and $f(S(n)) = g(n, f(n))$ for all $n \in \omega$.*

Proof. Call a map h *acceptable* if its domain is $S(k)$ for some $k \in \omega$, its values lie in A , $h(0) = a$, and $h(S(i)) = g(i, h(i))$ whenever $S(i) \in \text{dom}(h)$. Acceptable maps are elements of $\mathcal{P}(\omega \times A)$, so they form a set H .

If $h, h' \in H$ and i lies in both domains, then $h(i) = h'(i)$. Indeed, let X be the set of $i \in \omega$ such that this holds whenever $i \in \text{dom}(h) \cap \text{dom}(h')$. Then $0 \in X$. If $i \in X$ and $S(i)$

lies in both domains, then so does i , because domains are transitive (Lemma 3.4.3), and $h(S(i)) = g(i, h(i)) = g(i, h'(i)) = h'(S(i))$. So $X = \omega$.

Hence $f = \bigcup H$ is a function with values in A . Every $n \in \omega$ lies in its domain: by induction there is an acceptable h_n with domain $S(n)$, since $\{(0, a)\}$ is acceptable with domain $S(0)$, and if h is acceptable with domain $S(n)$ then $h \cup \{(S(n), g(n, h(n)))\}$ is acceptable with domain $S(S(n))$. Then $f(0) = a$, and $f(S(n)) = h_{S(n)}(S(n)) = g(n, h_{S(n)}(n)) = g(n, f(n))$. If f' is another such map, induction shows $f(n) = f'(n)$ for all n . $\square$

Example 3.4.6. By Theorem 3.4.5, for each $m \in \omega$ there are unique maps $n \mapsto m + n$ and $n \mapsto m \cdot n$ with $m + 0 = m$, $m + S(n) = S(m + n)$, $m \cdot 0 = 0$ and $m \cdot S(n) = m \cdot n + m$. So $(\omega, 0, S, +, \cdot)$ is a model of Robinson arithmetic Q (Definition 2.8.1). The laws of arithmetic are proved in Chapter 4.

3.5 Ordinals

Definition 3.5.1. A set x is *transitive* if every element of x is a subset of x . An *ordinal* is a transitive set α such that the relation $\{(\beta, \gamma) \in \alpha \times \alpha : \beta \in \gamma\}$ is a strict well-order on α : it is transitive, for all $\beta, \gamma \in \alpha$ exactly one of $\beta \in \gamma$, $\beta = \gamma$, $\gamma \in \beta$ holds, and every nonempty subset of α has an $\in$ -least element. For ordinals we write $\beta < \gamma$ for $\beta \in \gamma$. The class of ordinals is denoted Ord .

Lemma 3.5.2. *Every element of an ordinal is an ordinal, and $\alpha \notin \alpha$ for every ordinal α .*

Proof. Let $\beta \in \alpha$. Then $\beta \subseteq \alpha$, so $\in$ is a strict well-order on β . If $\delta \in \gamma \in \beta$, then γ and δ lie in α , and transitivity of the order on α gives $\delta \in \beta$; so β is transitive. The last statement is Lemma 3.2.6. $\square$

Lemma 3.5.3. *If α and β are ordinals with $\alpha \subsetneq \beta$, then $\alpha \in \beta$.*

Proof. Let γ be the least element of $\beta \setminus \alpha$. If $\delta \in \gamma$, then $\delta \in \beta$ and $\delta < \gamma$, so $\delta \in \alpha$ by minimality; hence $\gamma \subseteq \alpha$. If $\delta \in \alpha$, then $\delta \in \beta$; $\delta = \gamma$ is impossible since $\gamma \notin \alpha$, and $\gamma \in \delta$ is impossible since it would give $\gamma \in \alpha$ by transitivity of α ; so $\delta \in \gamma$. Hence $\alpha = \gamma \in \beta$. $\square$

Theorem 3.5.4. *For all ordinals α, β , exactly one of $\alpha < \beta$, $\alpha = \beta$, $\beta < \alpha$ holds. Every nonempty set X of ordinals has a least element, namely $\bigcap X$, and $\sup X = \bigcup X$ is an ordinal which is the least ordinal $\geq$ every element of X .*

Proof. The set $\gamma = \alpha \cap \beta$ is transitive and a subset of α , hence an ordinal. By Lemma 3.5.3, $\gamma = \alpha$ or $\gamma \in \alpha$, and $\gamma = \beta$ or $\gamma \in \beta$. If $\gamma \in \alpha$ and $\gamma \in \beta$, then $\gamma \in \gamma$, contradicting Lemma 3.5.2. So $\alpha \subseteq \beta$ or $\beta \subseteq \alpha$, and the first claim follows from Lemma 3.5.3; at most one of the three holds by Lemma 3.2.6.

Let $\alpha \in X$. If no element of X is smaller than α , then α is least. Otherwise $\alpha \cap X$ is a nonempty subset of α and has a least element, which is least in X . By the first part, the least element is contained in every element of X , so it equals $\bigcap X$. Finally $\bigcup X$ is a transitive set of ordinals; by the first part $\in$ is a strict total order on it, and every nonempty subset has a least element; so it is an ordinal. It contains each $\alpha \in X$ as a subset, and any ordinal containing all $\alpha \in X$ as subsets contains $\bigcup X$. $\square$

Proposition 3.5.5 (Burali-Forti). *There is no set containing all ordinals.*

Proof. If X were such a set, $\Omega = \{x \in X : x \text{ is an ordinal}\}$ would be a transitive set (Lemma 3.5.2) well-ordered by $\in$ (Theorem 3.5.4), hence an ordinal, so $\Omega \in \Omega$. $\square$

Definition 3.5.6. For an ordinal α , the set $\alpha + 1 = S(\alpha)$ is an ordinal, the least ordinal greater than α . An ordinal is a *successor ordinal* if it is of the form $\alpha + 1$, and a *limit ordinal* if it is neither 0 nor a successor ordinal.

Proposition 3.5.7. Every natural number is an ordinal, ω is an ordinal, and ω is the least limit ordinal. On ω the order $m < n$ (that is, $m \in n$) is a well-order.

Proof. $\emptyset$ is an ordinal, and if α is an ordinal so is $S(\alpha)$ (it is transitive, and its elements are the ordinals in α together with α). By induction every natural number is an ordinal. By Lemma 3.4.3, ω is a transitive set of ordinals, hence an ordinal by Theorem 3.5.4. It is not 0, and not a successor, since $S(n) \in \omega$ for $n \in \omega$. If λ is a limit ordinal, then $0 \in \lambda$, and for $n \in \lambda$ we have $S(n) \leq \lambda$ and $S(n) \neq \lambda$, so $S(n) \in \lambda$; thus λ is inductive and $\omega \subseteq \lambda$. $\square$

Theorem 3.5.8 (Transfinite induction). Let C be a class. Suppose that for every ordinal α , if all ordinals $\beta < \alpha$ belong to C , then α belongs to C . Then every ordinal belongs to C .

Proof. Otherwise let α be an ordinal not in C . The set $\{\beta \in \alpha + 1 : \beta \notin C\}$ is nonempty, and its least element β satisfies the hypothesis, so $\beta \in C$; contradiction. $\square$

Theorem 3.5.9 (Transfinite recursion). Let G be a class function, that is, a formula $\psi(x, y)$ such that for every x there is exactly one y with $\psi(x, y)$, written $y = G(x)$. There is a unique class function F defined on all ordinals such that $F(\alpha) = G(F|_\alpha)$ for every ordinal α , where $F|_\alpha = \{(\beta, F(\beta)) : \beta < \alpha\}$.

Proof. As usual for class functions, the theorem is a schema: for each ψ we write down a formula defining F and prove its properties. Call a function f an α -attempt if its domain is the ordinal α and $f(\beta) = G(f|_\beta)$ for all $\beta < \alpha$. By transfinite induction on β , two attempts agree on the intersection of their domains.

We show by transfinite induction that an α -attempt exists for every α . Suppose that for each $\beta < \alpha$ there is a β -attempt f_β , which is unique. By replacement, $\{f_\beta : \beta < \alpha\}$ is a set. If $\alpha = \beta + 1$, then $f_\beta \cup \{(\beta, G(f_\beta))\}$ is an α -attempt. If α is 0 or a limit ordinal, then $\bigcup_{\beta < \alpha} f_\beta$ is a function (the f_β agree on common domains) with domain $\bigcup_{\beta < \alpha} \beta = \alpha$, and it is an α -attempt.

Define $F(\alpha) = G(f)$, where f is the α -attempt; then $F|_\alpha = f$ and $F(\alpha) = G(F|_\alpha)$. Uniqueness of F follows by transfinite induction. $\square$

Theorem 3.5.10. Every well-ordered set $(W, <)$ is isomorphic to a unique ordinal, by a unique isomorphism.

Proof. Uniqueness follows from Lemma 3.3.12: two isomorphic ordinals are equal, since otherwise one is an initial segment of the other. Let A be the set of $w \in W$ such that $W_{<w}$ is isomorphic to some ordinal, necessarily unique, which we call α_w . By replacement, $\alpha = \{\alpha_w : w \in A\}$ is a set. If $v < w \in A$, the isomorphism $W_{<w} \rightarrow \alpha_w$ restricts to an isomorphism from $W_{<v}$ to an initial segment of α_w , which is an ordinal; so $v \in A$ and $\alpha_v < \alpha_w$. Hence A is an initial segment of W or equal to W , α is a transitive set of ordinals, hence an ordinal, and $w \mapsto \alpha_w$ is an isomorphism $A \rightarrow \alpha$. If $A \neq W$, let w be the least element of $W \setminus A$; then $A = W_{<w}$, so $w \in A$, a contradiction. $\square$

Definition 3.5.11. By transfinite recursion, define sets V_α for all ordinals by $V_0 = \emptyset$, $V_{\alpha+1} = \mathcal{P}(V_\alpha)$ and $V_\lambda = \bigcup_{\alpha < \lambda} V_\alpha$ for limit ordinals λ .

Lemma 3.5.12. Every V_α is transitive, and $V_\beta \subseteq V_\alpha$ for $\beta \leq \alpha$. A set x lies in $V_{\alpha+1}$ if and only if $x \subseteq V_\alpha$.

Proof. The last statement is the definition. We prove the first two statements together by transfinite induction on α . For $\alpha = 0$ they are clear, and for limit α they follow from the induction hypothesis. Let $\alpha = \gamma + 1$. If $x \in V_\alpha$ and $y \in x$, then $y \in V_\gamma$, so $y \subseteq V_\gamma$ by transitivity of V_γ , that is, $y \in V_\alpha$. Also $V_\gamma \subseteq V_\alpha$, because each element of V_γ is a subset of V_γ ; combined with the induction hypothesis this gives $V_\beta \subseteq V_\alpha$ for all $\beta \leq \alpha$. $\square$

Lemma 3.5.13. *Every set x is a subset of a transitive set.*

Proof. Apply Theorem 3.5.9 to the class function G with $G(f) = \bigcup f(n)$ if f is a function whose domain is $S(n)$ for some $n \in \omega$, and $G(f) = x$ otherwise. The resulting F satisfies $F(0) = x$ and $F(n+1) = \bigcup F(n)$ for $n \in \omega$. By replacement, $\{F(n) : n \in \omega\}$ is a set; let T be its union. Then $x \subseteq T$, and if $z \in y \in T$, say $y \in F(n)$, then $z \in F(n+1) \subseteq T$. $\square$

Theorem 3.5.14. *Every set belongs to V_α for some ordinal α .*

Proof. Suppose x belongs to no V_α . Let T be a transitive set containing $\{x\}$ as a subset (Lemma 3.5.13) and let $Y = \{y \in T : y \notin V_\alpha \text{ for all } \alpha\}$; then $x \in Y$. By foundation there is $y \in Y$ with $y \cap Y = \emptyset$. Each $z \in y$ lies in T but not in Y , so there is a least ordinal α_z with $z \in V_{\alpha_z}$. By replacement, $\{\alpha_z : z \in y\}$ is a set; let α be its supremum. Then $y \subseteq V_\alpha$ by Lemma 3.5.12, so $y \in V_{\alpha+1}$, a contradiction. $\square$

Definition 3.5.15. The *rank* $\text{rk}(x)$ of a set x is the least ordinal α with $x \subseteq V_\alpha$.

3.6 Cardinality

Definition 3.6.1. Sets A and B are *equinumerous*, written $A \approx B$, if there is a bijection $A \rightarrow B$. We write $A \leq B$ if there is an injection $A \rightarrow B$, and $A < B$ if $A \leq B$ and not $A \approx B$.

Theorem 3.6.2 (Cantor–Schröder–Bernstein). *If $A \leq B$ and $B \leq A$, then $A \approx B$.*

Proof. Let $f: A \rightarrow B$ and $g: B \rightarrow A$ be injections. Define $C_0 = A \setminus g[B]$ and $C_{n+1} = g[f[C_n]]$ by recursion, and let $C = \bigcup_n C_n$. Define $h: A \rightarrow B$ by $h(a) = f(a)$ if $a \in C$, and $h(a) = g^{-1}(a)$ if $a \notin C$; this makes sense because $a \notin C$ implies $a \in g[B]$, and g is injective.

h is surjective: let $b \in B$. If $g(b) \notin C$, then $h(g(b)) = b$. If $g(b) \in C$, then $g(b) \notin C_0$, so $g(b) \in C_{n+1}$ for some n , that is, $g(b) = g(f(c))$ with $c \in C_n$; then $b = f(c) = h(c)$.

h is injective: f and g^{-1} are injective, so it suffices to exclude $f(a) = g^{-1}(a')$ with $a \in C$ and $a' \notin C$. But then $a' = g(f(a)) \in C$ since $a \in C_n$ implies $g(f(a)) \in C_{n+1}$. $\square$

Theorem 3.6.3 (Cantor). *For every set A there is no surjection $A \rightarrow \mathcal{P}(A)$. In particular $A < \mathcal{P}(A)$.*

Proof. Let $f: A \rightarrow \mathcal{P}(A)$ and $D = \{a \in A : a \notin f(a)\}$. If $D = f(d)$, then $d \in D$ if and only if $d \notin D$. So f is not surjective. The map $a \mapsto \{a\}$ is an injection $A \rightarrow \mathcal{P}(A)$. $\square$

Definition 3.6.4. A set is *finite* if it is equinumerous with some $n \in \omega$, and *infinite* otherwise. It is *countable* if $A \leq \omega$, and *countably infinite* if $A \approx \omega$.

Proposition 3.6.5 (Pigeonhole principle). *For $n \in \omega$ there is no injection $n+1 \rightarrow n$. Consequently, if $m \neq n$ are natural numbers then $m \not\approx n$, and ω is infinite.*

Proof. For $n = 0$ there is no map from $1 = \{0\}$ to $\emptyset$. Suppose there is no injection $n+1 \rightarrow n$, and let $f: n+2 \rightarrow n+1$ be injective. Let τ be the bijection of $n+1$ exchanging n and $f(n+1)$. Then $\tau \circ f$ is injective and sends $n+1$ to n , so it restricts to an injection $n+1 \rightarrow (n+1) \setminus \{n\} = n$, a contradiction. If $m < n$, then $m+1 \subseteq n$, so an injection $n \rightarrow m$ would give one $m+1 \rightarrow m$. Finally $\omega \approx n$ would give an injection $n+1 \rightarrow n$. $\square$

Proposition 3.6.6. *Every subset of a natural number is finite, and every subset of ω is finite or countably infinite. A set is countable if and only if it is finite or countably infinite.*

Proof. By induction on n : subsets of 0 are finite, and a subset X of $n+1$ is $X \cap n$ or $(X \cap n) \cup \{n\}$; if $X \cap n \approx k$ then $(X \cap n) \cup \{n\} \approx k+1$.

Let $X \subseteq \omega$ be infinite. Then X has no upper bound in ω (otherwise $X \subseteq m$ for some m , and X would be finite). Define $e(0) = \min X$ and $e(n+1) = \min\{x \in X : x > e(n)\}$ by recursion. Then e is strictly increasing, hence injective, and $e(n) \geq n$ by induction. Given $x \in X$, let n be least with $e(n) \geq x$. If $n = 0$ then $x \leq e(0) = \min X \leq x$. If $n > 0$ then $e(n-1) < x$, so $e(n) \leq x$ by definition of $e(n)$. In both cases $x = e(n)$, so $e: \omega \rightarrow X$ is a bijection. The last statement follows: a set injecting into ω is equinumerous with a subset of ω . $\square$

Definition 3.6.7. An ordinal κ is a *cardinal* if no $\beta < \kappa$ is equinumerous with κ . If A is equinumerous with some ordinal (for instance if A can be well-ordered, by Theorem 3.5.10), its *cardinality* $|A|$ is the least ordinal equinumerous with A ; it is a cardinal.

Example 3.6.8. By Proposition 3.6.5, every natural number and ω are cardinals, and a finite set A has $|A| \in \omega$, its number of elements. The ordinal $\omega+1$ is not a cardinal: $n \mapsto n+1$ for $n \in \omega$ and $\omega \mapsto 0$ define a bijection $\omega+1 \rightarrow \omega$.

Theorem 3.6.9 (Hartogs). *For every set A there is an ordinal $h(A)$ that does not inject into A ; the least such ordinal is a cardinal. This uses no form of the axiom of choice.*

Proof. Let $\mathcal{W}$ be the set of pairs $(X, R) \in \mathcal{P}(A) \times \mathcal{P}(A \times A)$ such that R is a well-order on X . By Theorem 3.5.10 and replacement, the order types of these well-orders form a set H . An ordinal α injects into A if and only if $\alpha \in H$ (transport the order of α to the image). If $\alpha \in H$ and $\beta < \alpha$, then $\beta \in H$; so H is a transitive set of ordinals, hence an ordinal, and $H \notin H$. Put $h(A) = H$; it is the least ordinal not injecting into A . If some $\beta < H$ were equinumerous with H , then H would inject into A through β . $\square$

Definition 3.6.10. By transfinite recursion let $\omega_0 = \omega$, $\omega_{\alpha+1} = h(\omega_\alpha)$ and $\omega_\lambda = \sup_{\alpha < \lambda} \omega_\alpha$ for limit λ . We also write $\aleph_\alpha$ for ω_α when we think of it as a cardinality.

Proposition 3.6.11. *The ω_α are cardinals, $\alpha < \beta$ implies $\omega_\alpha < \omega_\beta$, and every infinite cardinal equals ω_α for a unique α .*

Proof. Strict monotonicity is clear from the definition, because $\omega_\alpha < h(\omega_\alpha)$. By induction $\alpha \leq \omega_\alpha$. A supremum λ of cardinals is a cardinal: if $\beta < \lambda$ then $\beta < \kappa \leq \lambda$ for one of the cardinals κ , and $\beta \approx \lambda$ would give $\kappa \leq \beta$ and hence $\kappa \approx \beta$ by Theorem 3.6.2. With Theorem 3.6.9, all ω_α are cardinals.

Let κ be an infinite cardinal, so $\omega \leq \kappa$, and let α be least with $\kappa \leq \omega_\alpha$ (it exists since $\kappa \leq \omega_\kappa$). If $\alpha = 0$, then $\kappa = \omega$. If α is a limit and $\kappa < \omega_\alpha$, then $\kappa < \omega_\beta$ for some $\beta < \alpha$, contradicting minimality. If $\alpha = \beta+1$ and $\kappa < \omega_\alpha = h(\omega_\beta)$, then κ injects into ω_β , so $\kappa \leq \omega_\beta \leq \kappa$ and $\kappa \approx \omega_\beta$ by Theorem 3.6.2; as both are cardinals, $\kappa = \omega_\beta$, contradicting the choice of α . $\square$

Theorem 3.6.12 (Hessenberg). *For every infinite cardinal κ , $\kappa \times \kappa \approx \kappa$.*

Proof. Order pairs of ordinals by $(\alpha, \beta) \triangleleft (\gamma, \delta)$ if $\max(\alpha, \beta) < \max(\gamma, \delta)$, or the maxima are equal and $\alpha < \gamma$, or the maxima are equal, $\alpha = \gamma$ and $\beta < \delta$. On every set of pairs this is a strict total order, and every nonempty set of pairs has a least element (minimize the maximum, then the first coordinate, then the second). For an ordinal κ , the set $\kappa \times \kappa$ contains, with any pair, all $\triangleleft$ -smaller pairs.

Suppose the theorem fails and let κ be the least infinite cardinal with $\kappa \times \kappa \neq \kappa$. Let θ be the order type of $(\kappa \times \kappa, \triangleleft)$ (Theorem 3.5.10). If $\theta \leq \kappa$, then $\kappa \times \kappa \leq \kappa$, and since $\kappa \leq \kappa \times \kappa$ via $\xi \mapsto (\xi, 0)$, Theorem 3.6.2 gives a contradiction. So $\kappa < \theta$, and there is a pair $(\xi, \eta) \in \kappa \times \kappa$ whose set of $\triangleleft$ -predecessors P has order type κ . Let $\delta = \max(\xi, \eta) + 1$. An infinite cardinal is a limit ordinal: for an infinite ordinal α , the map $\alpha + 1 \rightarrow \alpha$ sending α to 0, each $n \in \omega$ to $n + 1$, and fixing all other elements, is a bijection, so $\alpha + 1$ is not a cardinal. Hence $\delta < \kappa$, and $P \subseteq \delta \times \delta$. If δ is finite, so is $\delta \times \delta$, contradicting $P \approx \kappa$. Otherwise $\mu = |\delta|$ is an infinite cardinal with $\mu \leq \delta < \kappa$, and by minimality of κ , $\kappa \approx P \leq \delta \times \delta \approx \mu \times \mu \approx \mu$. Then $\kappa \leq \mu \leq \kappa$, so $\kappa \approx \mu$, contradicting that κ is a cardinal. $\square$

Corollary 3.6.13. $\omega \times \omega$ is countably infinite, and so is ω^n for every $n \geq 1$. The set of finite sequences of elements of a countable set is countable.

Proof. The first statement is Theorem 3.6.12; fix a bijection $p: \omega \times \omega \rightarrow \omega$. By recursion on n define bijections $b_n: \omega^n \rightarrow \omega$ for $n \geq 1$ by $b_1(s) = s(0)$ and $b_{n+1}(s) = p(b_n(s|_n), s(n))$. It suffices to treat sequences of natural numbers, since an injection $X \rightarrow \omega$ induces an injection on sequences. The map sending a sequence s of length $n \geq 1$ to $p(n, b_n(s))$, and the empty sequence to $p(0, 0)$, is an injection into ω . $\square$

3.7 The axiom of choice

Definition 3.7.1. A *choice function* for a set F of nonempty sets is a function $c: F \rightarrow \bigcup F$ with $c(X) \in X$ for all $X \in F$. The *axiom of choice* (AC) states that every set of nonempty sets has a choice function.

Theorem 3.7.2. In ZF, the following statements are equivalent.

1. The axiom of choice.
2. (Well-ordering theorem) Every set admits a well-order.
3. (Zorn's lemma) Let $(P, \leq)$ be a partially ordered set in which every chain has an upper bound. Then P has a maximal element.

Proof. (1) $\Rightarrow$ (2). Let A be a set and c a choice function for $\mathcal{P}(A) \setminus \{\emptyset\}$. Fix some $\star \notin A$. By transfinite recursion let $F(\alpha) = c(A \setminus F[\alpha])$ if $A \setminus F[\alpha] \neq \emptyset$, and $F(\alpha) = \star$ otherwise, where $F[\alpha] = \{F(\beta) : \beta < \alpha\}$. If $F(\beta) = \star$, then $A \subseteq F[\beta]$, so $F(\alpha) = \star$ for all $\alpha > \beta$. Hence if $F(\alpha) \neq \star$, all $F(\beta)$ with $\beta \leq \alpha$ lie in A , and $F(\alpha) \notin F[\alpha]$; so these values are distinct. If $F(\alpha) \neq \star$ for all $\alpha < h(A)$, then F would inject $h(A)$ into A , contradicting Theorem 3.6.9. So there is a least α with $F(\alpha) = \star$, and then $F|_\alpha: \alpha \rightarrow A$ is a bijection; transport the order of α .

(2) $\Rightarrow$ (1). Well-order $\bigcup F$ and let $c(X)$ be the least element of X .

(1) $\Rightarrow$ (3). Suppose P has no maximal element. Then every chain C has a *strict* upper bound: an upper bound u of C is not maximal, so there is $v > u$. Let c be a choice function for $\mathcal{P}(P) \setminus \{\emptyset\}$ and $\star \notin P$. Define by transfinite recursion $F(\alpha) = c(S_\alpha)$, where S_α is the set of strict upper bounds of $F[\alpha]$, if $F[\alpha] \subseteq P$ is a chain and S_α is nonempty, and $F(\alpha) = \star$ otherwise. By induction on α , $F[\alpha]$ is a chain in P on which F is strictly increasing: if this holds for all $\beta < \alpha$ then $F[\beta]$ is a chain, so $S_\beta \neq \emptyset$ and $F(\beta)$ is greater than all $F(\gamma)$, $\gamma < \beta$. Hence F injects $h(P)$ into P , contradicting Theorem 3.6.9.

(3) $\Rightarrow$ (1). Let F be a set of nonempty sets. Let P be the set of functions c with $\text{dom}(c) \subseteq F$ and $c(X) \in X$ for all $X \in \text{dom}(c)$, ordered by inclusion. The union of a chain in P is in P and is an upper bound. Let c be maximal. If $X \in F$ were not in the domain of c , choose $x \in X$; then $c \cup \{(X, x)\}$ would be larger. So c is a choice function for F . $\square$

Remark 3.7.3. The well-ordering theorem was proved from the axiom of choice by Zermelo; see [Zer08]. Zorn's lemma is from [Zor35]. It is the form of choice used most often in algebra, for instance to show that every vector space has a basis or that every nonzero ring has a maximal ideal. Gödel showed that if ZF is consistent then so is ZFC [G38], and Cohen that the negation of AC is also consistent with ZF [Coh63]. From now on we work in ZFC.

Proposition 3.7.4. *Every set A has a cardinality $|A|$. For all sets, $A \approx B$ if and only if $|A| = |B|$, and $A \leq B$ if and only if $|A| \leq |B|$. In particular any two sets are comparable: $A \leq B$ or $B \leq A$.*

Proof. By Theorem 3.7.2, every set can be well-ordered, so $|A|$ is defined. The first equivalence is clear. If $|A| \leq |B|$, then $|A| \subseteq |B|$ and $A \approx |A| \leq |B| \approx B$. Conversely, if $A \leq B$ and $|B| < |A|$, then $|A| \leq |B| \subseteq |A|$ and Theorem 3.6.2 would give $|A| \approx |B|$, contradicting that $|A|$ is a cardinal. The last statement follows from Theorem 3.5.4. $\square$

Proposition 3.7.5. *Let κ be an infinite cardinal and $(X_i)_{i \in I}$ a family with $|I| \leq \kappa$ and $|X_i| \leq \kappa$ for all i . Then $|\bigcup_{i \in I} X_i| \leq \kappa$. In particular a countable union of countable sets is countable.*

Proof. We may assume that all X_i are nonempty. By the axiom of choice there is a family $(g_i)_{i \in I}$ of surjections $g_i: \kappa \rightarrow X_i$ (choose one from each nonempty set of surjections). The map $(i, \xi) \mapsto g_i(\xi)$ is a surjection $q: I \times \kappa \rightarrow \bigcup_i X_i$. Since $I \times \kappa \leq \kappa \times \kappa \approx \kappa$ by Theorem 3.6.12, there is an injection $j: I \times \kappa \rightarrow \kappa$. Sending $u \in \bigcup_i X_i$ to the least element of $j[q^{-1}[\{u\}]]$ is an injection $\bigcup_i X_i \rightarrow \kappa$, because the sets $q^{-1}[\{u\}]$ are nonempty and pairwise disjoint. $\square$

Corollary 3.7.6. *If A and B are nonempty and at least one is infinite, then $|A \times B| = |A \sqcup B| = \max(|A|, |B|)$, where $A \sqcup B = (\{0\} \times A) \cup (\{1\} \times B)$.*

Proof. Let $\kappa = \max(|A|, |B|)$, an infinite cardinal, and say $|A| = \kappa$. By Proposition 3.7.4 and Theorem 3.6.12, $A \times B \leq \kappa \times \kappa \approx \kappa$ and $A \sqcup B \leq 2 \times \kappa \leq \kappa \times \kappa \approx \kappa$. Conversely A injects into $A \sqcup B$, and into $A \times B$ via $a \mapsto (a, b_0)$ for a fixed $b_0 \in B$. Conclude with Theorem 3.6.2. $\square$

Remark 3.7.7. We write 2^κ for $|\mathcal{P}(\kappa)|$; by Theorem 3.6.3, $2^\kappa > \kappa$. The *continuum hypothesis* (CH) asserts $2^{\aleph_0} = \aleph_1$. It is independent of ZFC: it can be neither proved [Coh63] nor refuted [G38]. In Chapter 4 we show $|\mathbb{R}| = 2^{\aleph_0}$.

Definition 3.7.8. Let α be a limit ordinal. A map $f: \beta \rightarrow \alpha$ is *cofinal* if its image is unbounded in α . The *cofinality* $\text{cf}(\alpha)$ is the least ordinal β admitting a cofinal map $\beta \rightarrow \alpha$. An infinite cardinal κ is *regular* if $\text{cf}(\kappa) = \kappa$, and *singular* otherwise.

Lemma 3.7.9. *Let κ be a regular cardinal and $X \subseteq \kappa$ a set with $|X| < \kappa$. Then $\sup X < \kappa$.*

Proof. Let $\mu = |X| < \kappa$ and let $f: \mu \rightarrow X$ be a bijection. If $\sup X = \kappa$, then f is a cofinal map $\mu \rightarrow \kappa$, so $\text{cf}(\kappa) \leq \mu < \kappa$. $\square$

Proposition 3.7.10. *The cardinal ω is regular, and every infinite successor cardinal $\omega_{\alpha+1}$ is regular.*

Proof. A map from a finite ordinal n to ω has a largest value, so it is not cofinal. Let $\kappa = \omega_{\alpha+1}$ and suppose $f: \beta \rightarrow \kappa$ is cofinal with $\beta < \kappa$. Then $|\beta| \leq \omega_\alpha$, every $f(\xi) < \kappa$ has $|f(\xi)| \leq \omega_\alpha$, and $\kappa = \bigcup_{\xi < \beta} f(\xi)$ because κ is a limit ordinal. By Proposition 3.7.5, $|\kappa| \leq \omega_\alpha$, a contradiction. $\square$

Counterexample 3.7.11. Not every infinite cardinal is regular. The map $n \mapsto \omega_n$ is cofinal from ω to $\omega_\omega = \sup_n \omega_n$, so $\text{cf}(\omega_\omega) = \omega < \omega_\omega$.

Definition 3.7.12. A cardinal κ is *strongly inaccessible* if it is uncountable, regular, and $2^\lambda < \kappa$ for every cardinal $\lambda < \kappa$.

3.8 Universes

Definition 3.8.1. A Grothendieck universe is a set $\mathcal{U}$ such that

1. if $x \in \mathcal{U}$ and $y \in x$, then $y \in \mathcal{U}$;
2. if $x, y \in \mathcal{U}$, then $\{x, y\} \in \mathcal{U}$;
3. if $x \in \mathcal{U}$, then $\mathcal{P}(x) \in \mathcal{U}$;
4. if $I \in \mathcal{U}$ and $(x_i)_{i \in I}$ is a family with $x_i \in \mathcal{U}$ for all i , then $\bigcup_{i \in I} x_i \in \mathcal{U}$.

This is the definition of [AGV72, Exposé I, Appendice].

Lemma 3.8.2. Let $\mathcal{U}$ be a nonempty universe, $A, B, x, y \in \mathcal{U}$ and $I \in \mathcal{U}$.

1. $\emptyset \in \mathcal{U}$, and every subset of an element of $\mathcal{U}$ is in $\mathcal{U}$.
2. $\{x\}, (x, y), x \cup y, A \times B$ and B^A are in $\mathcal{U}$.
3. If $x_i \in \mathcal{U}$ for all $i \in I$, then $\prod_{i \in I} x_i \in \mathcal{U}$; and if $f: I \rightarrow \mathcal{U}$ is a map, the image $f[I]$ is in $\mathcal{U}$.
4. If $\sim$ is an equivalence relation on A , then $A/\sim \in \mathcal{U}$.

Proof. (1) If $y \subseteq x \in \mathcal{U}$, then $y \in \mathcal{P}(x) \in \mathcal{U}$, so $y \in \mathcal{U}$. Taking any $x \in \mathcal{U}$ and $y = \emptyset$ gives $\emptyset \in \mathcal{U}$. (2) $\{x\} = \{x, x\}$ and $(x, y) = \{\{x\}, \{x, y\}\}$. Since $\{A, B\} \in \mathcal{U}$, the union of the family A, B indexed by it is $A \cup B \in \mathcal{U}$. Then $A \times B \subseteq \mathcal{P}(\mathcal{P}(A \cup B))$ and $B^A \subseteq \mathcal{P}(A \times B)$. (3) $\bigcup_i x_i \in \mathcal{U}$ and $\prod_i x_i \subseteq (\bigcup_i x_i)^I$. The image is $f[I] = \bigcup_{i \in I} \{f(i)\}$. (4) $A/\sim \subseteq \mathcal{P}(A)$. $\square$

Theorem 3.8.3. If κ is a strongly inaccessible cardinal, then V_κ is a Grothendieck universe and $\omega \in V_\kappa$.

Proof. First, $|V_\alpha| < \kappa$ for all $\alpha < \kappa$, by transfinite induction: $V_0 = \emptyset$; $|V_{\alpha+1}| = 2^{|V_\alpha|} < \kappa$ by strong inaccessibility; and for a limit $\lambda < \kappa$, the cardinals $|V_\alpha|$, $\alpha < \lambda$, form a set of fewer than κ ordinals below κ , so their supremum μ is $< \kappa$ by Lemma 3.7.9, and $|V_\lambda| \leq \max(|\lambda|, \mu, \omega) < \kappa$ by Proposition 3.7.5.

Since κ is a limit ordinal (being an infinite cardinal), every $x \in V_\kappa$ lies in some $V_{\alpha+1}$ with $\alpha < \kappa$, so $x \subseteq V_\alpha$. We check the axioms. (1) holds because V_κ is transitive (Lemma 3.5.12). (2) If $x \subseteq V_\alpha$ and $y \subseteq V_\beta$ with $\alpha \leq \beta < \kappa$, then $x, y \in V_{\beta+1}$ and $\{x, y\} \in V_{\beta+2}$. (3) If $x \subseteq V_\alpha$ then $\mathcal{P}(x) \subseteq \mathcal{P}(V_\alpha) = V_{\alpha+1}$, so $\mathcal{P}(x) \in V_{\alpha+2}$. (4) Let $I \in V_\kappa$, so $|I| < \kappa$, and for $i \in I$ let $\alpha_i < \kappa$ be least with $x_i \subseteq V_{\alpha_i}$. The set $\{\alpha_i : i \in I\}$ has fewer than κ elements, so its supremum α is $< \kappa$ by Lemma 3.7.9. Then $\bigcup_i x_i \subseteq V_\alpha$, hence lies in $V_{\alpha+1}$. Finally, $\omega \subseteq V_\omega$, so $\omega \in V_{\omega+1}$, and $\omega + 1 < \kappa$ as κ is uncountable. $\square$

Theorem 3.8.4. A set $\mathcal{U}$ is a Grothendieck universe with $\omega \in \mathcal{U}$ if and only if $\mathcal{U} = V_\kappa$ for a strongly inaccessible cardinal κ .

One direction is Theorem 3.8.3; for the other, see [AGV72, Exposé I, Appendice].

Remark 3.8.5. If κ is strongly inaccessible, the structure $(V_\kappa, \in)$ is a model of ZFC [Jec03, Chapter 12]. Hence ZFC cannot prove that a universe containing ω exists, if ZFC is consistent (Remark 2.8.6). Following [AGV72], we therefore add to ZFC the axiom that there is a Grothendieck universe containing ω , and we fix one, $\mathcal{U}$, as in Notation 1.6.1. Only this single universe is needed in this project.

Remark 3.8.6. A set is *small* if it is an element of $\mathcal{U}$. By Lemma 3.8.2, the small sets are closed under all the constructions of this chapter applied to small sets: subsets, pairs, products, sets of maps, quotients, power sets, and unions and products of families indexed by small sets. Since $\omega \in \mathcal{U}$, the number systems $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$ and $\mathbb{C}$ constructed in Chapter 4 are small, and so is every object built from them by these constructions. A set that is merely equinumerous with a small set need not be small (it may even fail to be a subset of $\mathcal{U}$), but it is in bijection with one, which is all that matters in practice.

3.9 Exercises

Exercise 3.9.1. Let A, B, C be sets. Show that $f \mapsto (a \mapsto (b \mapsto f(a, b)))$ is a bijection $C^{A \times B} \rightarrow (C^B)^A$, and that $(A \times B) \times C \approx A \times (B \times C)$.

Exercise 3.9.2. Show that the axiom of choice is equivalent (in ZF) to the statement that every surjection $f: A \rightarrow B$ has a *section*, that is, a map $s: B \rightarrow A$ with $f \circ s = \text{id}_B$.

Solution. Given AC and a surjection f , choose $s(b)$ from the nonempty set $f^{-1}[\{b\}]$. Conversely, given a set F of nonempty sets, let $A = \{(X, x) : X \in F, x \in X\}$ and let $f: A \rightarrow F$ be the first projection, which is surjective. If s is a section, then $X \mapsto (\text{second coordinate of } s(X))$ is a choice function. $\square$

Exercise 3.9.3. Using the axiom of choice, show that every infinite set has a countably infinite subset.

Exercise 3.9.4. Show that an ordinal α is a limit ordinal if and only if $\alpha \neq 0$ and $\alpha = \bigcup \alpha$.

Exercise 3.9.5. Show that V_ω is a Grothendieck universe which does not contain ω , and that every element of V_ω is a finite set. Show that $\emptyset$ is also a universe. (So the condition $\omega \in \mathcal{U}$ in Remark 3.8.5 is not automatic.)

Exercise 3.9.6. Show that ω^ω (the set of maps $\omega \rightarrow \omega$) and $\mathcal{P}(\omega)$ are equinumerous.

Exercise 3.9.7. Show that for every set A of sets there is a set not equinumerous with a subset of any element of A . Deduce that the class of all cardinals is not a set.

Chapter 4

Number Systems

4.1 Introduction

Starting from the natural numbers of Chapter 3, we construct the integers, the rational numbers, the real numbers and the complex numbers, and establish the properties used everywhere else: the laws of arithmetic, unique factorisation of integers, the least upper bound property of $\mathbb{R}$ and its uniqueness, and the basic properties of $\mathbb{C}$. Along the way we introduce the vocabulary of commutative rings, fields and ordered fields; general rings are studied in Chapter 7.

Classical references are [Lan30], which carries out these constructions in full detail, [Ded72] for the real numbers, and [Rud76, Chapter 1].

Prerequisites. Chapter 3 (Set Theory).

4.2 Arithmetic of the natural numbers

Recall from Example 3.4.6 that addition and multiplication on $\mathbb{N} = \omega$ are defined by $m + 0 = m$, $m + S(n) = S(m + n)$, $m \cdot 0 = 0$ and $m \cdot S(n) = m \cdot n + m$. We have $m + 1 = S(m)$, since $m + S(0) = S(m + 0)$. All inductions below are instances of Theorem 3.4.2.

Proposition 4.2.1. *For all $m, n, k \in \mathbb{N}$:*

1. $(m + n) + k = m + (n + k)$;
2. $m + n = n + m$;
3. if $m + k = n + k$, then $m = n$;
4. if $m + n = 0$, then $m = n = 0$.

Proof. (1) By induction on k : for $k = 0$ both sides are $m + n$, and $(m + n) + S(k) = S((m + n) + k) = S(m + (n + k)) = m + S(n + k) = m + (n + S(k))$.

(2) First, $0 + n = n$ by induction on n . Next, $S(m) + n = S(m + n)$ by induction on n : $S(m) + 0 = S(m) = S(m + 0)$ and $S(m) + S(n) = S(S(m) + n) = S(S(m + n)) = S(m + S(n))$. Now induct on n : $m + 0 = m = 0 + m$, and $m + S(n) = S(m + n) = S(n + m) = S(n) + m$.

(3) By induction on k , using that S is injective (Proposition 3.4.4): $m + S(k) = n + S(k)$ means $S(m + k) = S(n + k)$.

(4) If $n = S(n')$ then $m + n = S(m + n') \neq 0$. So $n = 0$, and then $m = 0$. $\square$

Proposition 4.2.2. *For all $m, n, k \in \mathbb{N}$:*

1. $m \cdot (n + k) = m \cdot n + m \cdot k$;

2. $(m \cdot n) \cdot k = m \cdot (n \cdot k)$;
3. $m \cdot n = n \cdot m$ and $1 \cdot n = n$;
4. if $m \cdot n = 0$, then $m = 0$ or $n = 0$.

Proof. (1) By induction on k : $m(n + 0) = mn = mn + 0$, and $m(n + S(k)) = m \cdot S(n + k) = m(n + k) + m = (mn + mk) + m = mn + m \cdot S(k)$.

(2) By induction on k : $(mn) \cdot 0 = 0 = m(n \cdot 0)$, and, using (1), $(mn) \cdot S(k) = (mn)k + mn = m(nk) + mn = m(nk + n) = m(n \cdot S(k))$.

(3) By induction, $0 \cdot n = 0$ and $S(m) \cdot n = mn + n$; for the second, $S(m) \cdot S(n) = S(m)n + S(m) = (mn + n) + S(m)$ and $m \cdot S(n) + S(n) = (mn + m) + S(n)$, and both equal $S(mn + m + n)$ by Proposition 4.2.1. Then $mn = nm$ follows by induction on n , and $1 \cdot n = 0 \cdot n + n = n$.

(4) If $m = S(m')$ and $n = S(n')$, then $mn = mn' + m = S(mn' + m')$, which is not 0. $\square$

Proposition 4.2.3. For $m, n \in \mathbb{N}$, the following are equivalent: $m \leq n$ in the order of ordinals (Proposition 3.5.7); there is $k \in \mathbb{N}$ with $m + k = n$. This k is unique. Moreover, for all $m, n, k \in \mathbb{N}$: if $m \leq n$ then $m + k \leq n + k$ and $mk \leq nk$; and if $mk = nk$ with $k \neq 0$ then $m = n$.

Proof. By induction on k , $m \leq m + k$: indeed $m + S(k) = S(m + k)$ contains $m + k$. Conversely we show by induction on n that $m \leq n$ implies $m + k = n$ for some k . For $n = 0$, $m = 0$ and $k = 0$. If $m \leq S(n)$, then $m = S(n)$ (take $k = 0$) or $m \leq n$, in which case $m + k = n$ for some k and $m + S(k) = S(n)$. Uniqueness of k is Proposition 4.2.1(3).

If $m + j = n$, then $(m + k) + j = n + k$ and $mk + jk = nk$, which gives the monotonicity statements. If $mk = nk$ with $k \neq 0$ and, say, $m < n$, write $n = m + j$ with $j \neq 0$; then $mk + jk = mk + 0$, so $jk = 0$, contradicting Proposition 4.2.2(4). $\square$

Proposition 4.2.4 (Division with remainder). Let $m, n \in \mathbb{N}$ with $n \geq 1$. There are unique $q, r \in \mathbb{N}$ with $m = qn + r$ and $r < n$.

Proof. Existence, by induction on m : for $m = 0$ take $q = r = 0$. If $m = qn + r$ with $r < n$, then $m + 1 = qn + (r + 1)$, and either $r + 1 < n$, or $r + 1 = n$ and $m + 1 = (q + 1)n + 0$. Uniqueness: if $qn + r = q'n + r'$ with $r, r' < n$ and $q < q'$, then $q'n + r' \geq (q + 1)n = qn + n > qn + r$, a contradiction; so $q = q'$, and $r = r'$ by cancellation. $\square$

4.3 Commutative rings and fields

Definition 4.3.1. A *commutative ring* is a set R with elements $0, 1 \in R$ and maps $+, \cdot: R \times R \rightarrow R$, the *addition* $(a, b) \mapsto a + b$ and the *multiplication* $(a, b) \mapsto a \cdot b$, usually written ab , such that for all $a, b, c \in R$: $(a + b) + c = a + (b + c)$; $a + b = b + a$; $a + 0 = a$; there is $a' \in R$ with $a + a' = 0$; $(ab)c = a(bc)$; $ab = ba$; $1 \cdot a = a$; and $a(b + c) = ab + ac$. A *homomorphism* of commutative rings is a map f with $f(a + b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$ and $f(1) = 1$. A commutative ring is an *integral domain* if $1 \neq 0$ and $ab = 0$ implies $a = 0$ or $b = 0$; it is a *field* if $1 \neq 0$ and every $a \neq 0$ has an element a^{-1} with $aa^{-1} = 1$.

Lemma 4.3.2. In a commutative ring R : the element a' with $a + a' = 0$ is unique, and we write it $-a$; $0 \cdot a = 0$; $(-a)b = -(ab)$; $(-a)(-b) = ab$. A field is an integral domain, and in a field a^{-1} is unique.

Proof. If $a + a' = 0 = a + a''$, then $a' = a' + (a + a'') = (a' + a) + a'' = a''$. Next $0 \cdot a = (0 + 0)a = 0 \cdot a + 0 \cdot a$, and adding $-(0 \cdot a)$ gives $0 \cdot a = 0$. Then $ab + (-a)b = (a + (-a))b = 0$, so $(-a)b = -(ab)$, and $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. In a field, if $ab = 0$ and $a \neq 0$ then $b = a^{-1}ab = 0$. Uniqueness of inverses is proved like uniqueness of $-a$. $\square$

Definition 4.3.3. An *ordered ring* is a commutative ring R with a total order $\leq$ such that $a \leq b$ implies $a + c \leq b + c$, and $0 \leq a, 0 \leq b$ imply $0 \leq ab$. An *ordered field* is an ordered ring that is a field. An element a is *positive* if $a > 0$. The *absolute value* is $|a| = a$ if $a \geq 0$ and $|a| = -a$ otherwise.

Lemma 4.3.4. Let R be an ordered ring with $1 \neq 0$ and $a, b, c \in R$.

1. $a < b$ if and only if $-b < -a$; and $a^2 \geq 0$, so $1 > 0$.
2. If $a \leq b$ and $c \geq 0$, then $ac \leq bc$.
3. $|ab| = |a||b|$ and $|a + b| \leq |a| + |b|$.
4. The elements $n \cdot 1 = 1 + \cdots + 1$ (n summands) are positive for $n \geq 1$; in particular they are pairwise distinct.
5. If R is a field and $0 < a < b$, then $0 < b^{-1} < a^{-1}$.

Proof. (1) Add $-a - b$ to $a < b$. If $a \geq 0$ then $a^2 \geq 0$; if $a < 0$ then $-a > 0$ and $a^2 = (-a)(-a) \geq 0$. Then $1 = 1^2 \geq 0$ and $1 \neq 0$. (2) $b - a \geq 0$, so $(b - a)c \geq 0$. (3) Both sides of the first equation are $\pm ab$ with the sign making them non-negative. For the second, $-|a| \leq a \leq |a|$ and similarly for b , so $-(|a| + |b|) \leq a + b \leq |a| + |b|$. (4) By induction, $(n + 1) \cdot 1 = n \cdot 1 + 1 > n \cdot 1$. (5) $a^{-1} = a(a^{-1})^2 \geq 0$ and $a^{-1} \neq 0$; similarly $b^{-1} > 0$. Multiplying $a < b$ by $a^{-1}b^{-1} > 0$ gives $b^{-1} < a^{-1}$. $\square$

4.4 The integers

Definition 4.4.1. On $\mathbb{N} \times \mathbb{N}$ let $(a, b) \sim (c, d)$ if $a + d = b + c$. This is an equivalence relation (transitivity: from $a + d = b + c$ and $c + f = d + e$ we get $a + f + (c + d) = b + e + (c + d)$, and we cancel). The set of *integers* is $\mathbb{Z} = (\mathbb{N} \times \mathbb{N})/\sim$; we write $[a, b]$ for the class of (a, b) . Define

$$[a, b] + [c, d] = [a + c, b + d], \quad [a, b] \cdot [c, d] = [ac + bd, ad + bc],$$

$0 = [0, 0]$, $1 = [1, 0]$, and $[a, b] \leq [c, d]$ if $a + d \leq b + c$.

Theorem 4.4.2. The operations and the order on $\mathbb{Z}$ are well defined, and make $\mathbb{Z}$ an ordered integral domain. The map $\iota: \mathbb{N} \rightarrow \mathbb{Z}$, $n \mapsto [n, 0]$, is injective and preserves 0, 1, addition, multiplication and order. Every integer is of the form $\iota(n)$ or $-\iota(n)$ with $n \in \mathbb{N}$. Every nonempty set of integers that is bounded below has a least element.

Proof. If $(a, b) \sim (a', b')$, that is, $a + b' = b + a'$, then $(a + c) + (b' + d) = (b + d) + (a' + c)$, so addition is well defined. For multiplication, multiplying $a + b' = b + a'$ by c and $b + a' = a + b'$ by d and adding gives $ac + bd + a'd + b'c = ad + bc + a'c + b'd$, which says $(ac + bd, ad + bc) \sim (a'c + b'd, a'd + b'c)$; by symmetry of the formula the same holds in the second argument. For the order, $a + d \leq b + c$ and $a + b' = b + a'$ give $a' + d + (a + b') \leq b' + c + (a + b')$, and we cancel.

The ring axioms reduce to identities in $\mathbb{N}$. For instance, $[b, a]$ is an additive inverse of $[a, b]$, and distributivity follows from $[a, b]([c, d] + [e, f]) = [a(c + e) + b(d + f), a(d + f) + b(c + e)]$, whose entries are those of $[ac + bd, ad + bc] + [ae + bf, af + be]$ by Propositions 4.2.1 and 4.2.2. The other axioms are verified in the same way.

If $a \geq b$, write $a = b + n$; then $[a, b] = [n, 0]$. If $a < b$, then $[a, b] = [0, n] = -[n, 0]$ with $b = a + n$. So every integer is $\iota(n)$ or $-\iota(n)$. The map ι preserves the operations, and $[m, 0] \leq [n, 0]$ if and only if $m \leq n$, so it is injective. The relation $\leq$ is a total order because

$\leq$ is one on $\mathbb{N}$. It is compatible with addition, and if $x, y \geq 0$ then $x = \iota(m)$, $y = \iota(n)$ and $xy = \iota(mn) \geq 0$. If $xy = 0$ with $x = \pm\iota(m)$, $y = \pm\iota(n)$, then $\iota(mn) = 0$, so $m = 0$ or $n = 0$.

Finally, if $X \subseteq \mathbb{Z}$ is nonempty and bounded below by c , then $\{x - c : x \in X\}$ is a nonempty subset of $\iota(\mathbb{N})$, which has a least element by Proposition 3.5.7. $\square$

Notation 4.4.3. We identify $n \in \mathbb{N}$ with $\iota(n) \in \mathbb{Z}$, so that $\mathbb{N} \subseteq \mathbb{Z}$. For a commutative ring R , $n \in \mathbb{N}$ and $a \in R$, we write na for $a + \cdots + a$ (n summands) and $(-n)a = -(na)$.

4.5 Divisibility and prime numbers

Definition 4.5.1. Let $a, b \in \mathbb{Z}$. We say a divides b , written $a \mid b$, if $b = ac$ for some $c \in \mathbb{Z}$. An integer $p \geq 2$ is *prime* if its only positive divisors are 1 and p . A *greatest common divisor* of a and b , not both zero, is a positive common divisor d divisible by every common divisor.

Theorem 4.5.2 (Division with remainder). *Let $a, b \in \mathbb{Z}$ with $b \neq 0$. There are unique $q, r \in \mathbb{Z}$ with $a = qb + r$ and $0 \leq r < |b|$.*

Proof. Uniqueness: if $qb + r = q'b + r'$ then $(q - q')b = r' - r$ and $|r' - r| < |b|$, so $q = q'$ and $r = r'$. Existence: the set $\{a - kb : k \in \mathbb{Z}\} \cap \mathbb{N}$ is nonempty (take $k = -|a|$ if $b > 0$ and $k = |a|$ if $b < 0$, for which $a - kb = a + |a||b| \geq 0$), so it has a least element $r = a - qb$. If $r \geq |b|$, then $r - |b| = a - (q \pm 1)b$ is a smaller element. $\square$

Theorem 4.5.3 (Bézout). *Let $a, b \in \mathbb{Z}$ be not both zero, and let d be the least positive element of $\{ax + by : x, y \in \mathbb{Z}\}$. Then d is a greatest common divisor of a and b , and it is the only one. We write $d = \gcd(a, b)$.*

Proof. The set contains $a^2 + b^2 > 0$, so d exists; write $d = ax_0 + by_0$. Divide: $a = qd + r$ with $0 \leq r < d$. Then $r = a(1 - qx_0) + b(-qy_0)$ lies in the set, so $r = 0$ by minimality, and $d \mid a$; similarly $d \mid b$. Every common divisor of a and b divides $ax_0 + by_0 = d$. If d' is another greatest common divisor, then $d \mid d'$ and $d' \mid d$, and both are positive, so $d = d'$. $\square$

Lemma 4.5.4 (Euclid). *Let p be prime and $a, b \in \mathbb{Z}$. If $p \mid ab$, then $p \mid a$ or $p \mid b$.*

Proof. If $p \nmid a$, then $\gcd(p, a) = 1$, since it is a positive divisor of p different from p . By Theorem 4.5.3, $1 = px + ay$, so $b = pbx + aby$ is divisible by p . $\square$

Theorem 4.5.5 (Fundamental theorem of arithmetic). *Every integer $n \geq 2$ is a product $n = p_1 p_2 \cdots p_k$ of primes, and if $n = q_1 \cdots q_l$ is another such product, then $k = l$ and the q_j are a permutation of the p_i .*

Proof. Existence: if some $n \geq 2$ is not a product of primes, take the least such n . It is not prime, so $n = ab$ with $2 \leq a, b < n$, and a, b are products of primes, hence so is n . Uniqueness, by induction on k : p_k divides $q_1 \cdots q_l$, so by Lemma 4.5.4 (and induction on l) it divides some q_j , and then $p_k = q_j$ since q_j is prime. Cancelling gives two factorisations of n/p_k and the induction hypothesis applies (if $k = 1$, then $n = p_1$ is prime and $l = 1$). $\square$

Theorem 4.5.6 (Euclid). *There are infinitely many prime numbers.*

Proof. Let $p_1, \dots, p_k$ be primes and $N = p_1 \cdots p_k + 1 \geq 2$. By Theorem 4.5.5, some prime p divides N . If $p = p_i$, then p divides $N - p_1 \cdots p_k = 1$, which is impossible. So no finite set of primes contains all primes. $\square$

Example 4.5.7. Repeated division with remainder computes greatest common divisors, since $\gcd(a, b) = \gcd(b, r)$ when $a = qb + r$. For $a = 252$, $b = 198$: $252 = 1 \cdot 198 + 54$, $198 = 3 \cdot 54 + 36$, $54 = 1 \cdot 36 + 18$, $36 = 2 \cdot 18$. So $\gcd(252, 198) = 18$, and going backwards, $18 = 54 - 36 = 4 \cdot 54 - 198 = 4 \cdot 252 - 5 \cdot 198$.

Remark 4.5.8. Unique factorisation is a special property of $\mathbb{Z}$. In the ring $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$ one has $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$, and none of these four factors can be written as a product of two non-units. Such phenomena are studied in Chapters 7 and 12.

4.6 The rational numbers

Definition 4.6.1. Let $P = \{(a, b) \in \mathbb{Z} \times \mathbb{Z} : b > 0\}$ and $(a, b) \sim (c, d)$ if $ad = bc$. This is an equivalence relation (for transitivity, $ad = bc$ and $cf = de$ give $adf = bcf = bde$, and we cancel $d \neq 0$ in the integral domain $\mathbb{Z}$). The set of *rational numbers* is $\mathbb{Q} = P/\sim$, and a/b denotes the class of (a, b) . Define

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \quad \frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}, \quad \frac{a}{b} \leq \frac{c}{d} \iff ad \leq bc.$$

Theorem 4.6.2. The operations and the order on $\mathbb{Q}$ are well defined and make $\mathbb{Q}$ an ordered field. The map $\mathbb{Z} \rightarrow \mathbb{Q}$, $a \mapsto a/1$, is an injective ring homomorphism preserving the order, and $\mathbb{Q}$ is Archimedean: for every $x \in \mathbb{Q}$ there is $n \in \mathbb{N}$ with $x < n$.

Proof. Well-definedness: if $ab' = a'b$, then $(ad + bc)(b'd) = (a'd + b'c)(bd)$ and $(ac)(b'd) = (a'c)(bd)$, and $ad \leq bc$ is equivalent to $ab'd \leq bb'c$ (multiply by $b' > 0$), that is to $a'd \leq b'c$ (cancel $b > 0$). The field axioms are identities in $\mathbb{Z}$ as in Theorem 4.4.2; the zero is $0/1$, the unit $1/1$, $-(a/b) = (-a)/b$, and for $a \neq 0$ the inverse of a/b is b/a if $a > 0$ and $(-b)/(-a)$ if $a < 0$. The order is total, and compatible with the operations: $a/b \geq 0$ means $a \geq 0$, and a product or sum of such fractions again has non-negative numerator. The map $a \mapsto a/1$ is clearly an injective order-preserving homomorphism. Finally, for $x = a/b$ we have $x \leq |a| < |a| + 1$ because $b \geq 1$. $\square$

Proposition 4.6.3. Let K be an ordered field. There is a unique ring homomorphism $\mathbb{Q} \rightarrow K$; it is injective and order-preserving. We use it to regard $\mathbb{Q}$ as a subfield of K .

Proof. By Lemma 4.3.4(4) the map $\mathbb{N} \rightarrow K$, $n \mapsto n \cdot 1$, is injective, and it preserves sums and products (by induction). Extend it to $\mathbb{Z}$ by $-n \mapsto -(n \cdot 1)$; this is an injective order-preserving ring homomorphism j . Define $f(a/b) = j(a)j(b)^{-1}$; this is well defined since $ad = bc$ gives $j(a)j(d) = j(b)j(c)$, and it is a ring homomorphism by the formulas of Definition 4.6.1. It preserves order: for $b, d > 0$, $a/b \leq c/d$ iff $ad \leq bc$ iff $j(a)j(d) \leq j(b)j(c)$ iff $j(a)j(b)^{-1} \leq j(c)j(d)^{-1}$, by Lemma 4.3.4(5). Hence it is injective. Uniqueness: a ring homomorphism g satisfies $g(1) = 1$, hence $g(n) = n \cdot 1$ and $g(-n) = -g(n)$, and $g(b)g(a/b) = g(a)$, so $g(a/b) = j(a)j(b)^{-1}$. $\square$

Proposition 4.6.4. There is no $x \in \mathbb{Q}$ with $x^2 = 2$.

Proof. Suppose $(a/b)^2 = 2$ with $a, b > 0$, so $a^2 = 2b^2$. In the factorisations of Theorem 4.5.5, the prime 2 occurs an even number of times in a^2 and an odd number of times in $2b^2$, contradicting uniqueness. $\square$

Counterexample 4.6.5. The set $A = \{x \in \mathbb{Q} : x \leq 0 \text{ or } x^2 < 2\}$ is nonempty and bounded above (by 2), but has no least upper bound in $\mathbb{Q}$. Indeed, for $p \in \mathbb{Q}$ with $p > 0$ put $q = p - \frac{p^2-2}{p+2} = \frac{2p+2}{p+2}$;

then $q^2 - 2 = \frac{2(p^2-2)}{(p+2)^2}$. If $p \in A$ and $p > 0$, then $p^2 < 2$, so $q > p$ and $q^2 < 2$: A has no largest element. If $p > 0$ is an upper bound of A , then $p^2 > 2$ by Proposition 4.6.4 (and $p^2 < 2$ is impossible since then $p \in A$ is largest), so $q < p$ and $q^2 > 2$, and q is a smaller upper bound. Hence no upper bound is least.

4.7 Complete ordered fields

Definition 4.7.1. An ordered field K is *complete* if every nonempty subset of K that is bounded above has a least upper bound (supremum) in K . It is *Archimedean* if for every $x \in K$ there is $n \in \mathbb{N}$ with $x < n$.

Lemma 4.7.2. In a complete ordered field every nonempty subset bounded below has a greatest lower bound (infimum), namely $\inf A = -\sup(-A)$. If A, B are nonempty and bounded above, then $\sup(A + B) = \sup A + \sup B$, where $A + B = \{a + b : a \in A, b \in B\}$; if moreover A and B consist of positive elements, then $\sup(AB) = \sup A \cdot \sup B$.

Proof. The first claim is clear since $x \mapsto -x$ reverses the order. Let $s = \sup A$, $t = \sup B$. Clearly $s + t$ is an upper bound of $A + B$. If $u < s + t$, then $u - t < s$, so there is $a \in A$ with $a > u - t$; then $u - a < t$ and there is $b \in B$ with $b > u - a$, so $a + b > u$. For products, st is an upper bound of AB by Lemma 4.3.4(2). If $0 < u < st$, then $u/t < s$, so some $a \in A$ has $a > u/t$, and then $u/a < t$ and some $b \in B$ has $b > u/a$, so $ab > u$. Upper bounds $u \leq 0$ are not smaller than $st > 0$. $\square$

Proposition 4.7.3. A complete ordered field is Archimedean. In an Archimedean ordered field, $\mathbb{Q}$ is dense: for all $x < y$ there is $q \in \mathbb{Q}$ with $x < q < y$.

Proof. If $\mathbb{N} \subseteq K$ were bounded above, let $s = \sup \mathbb{N}$. Then $s - 1$ is not an upper bound, so $n > s - 1$ for some $n \in \mathbb{N}$, and $n + 1 > s$; contradiction.

Let K be Archimedean and $x < y$. Choose $n \in \mathbb{N}$ with $n > (y - x)^{-1}$, so $ny > nx + 1$. The set of integers m with $m > nx$ is nonempty and bounded below (both by the Archimedean property applied to $\pm nx$), so it has a least element m by Theorem 4.4.2. Then $m - 1 \leq nx$, so $nx < m \leq nx + 1 < ny$, and $q = m/n$ works. $\square$

Proposition 4.7.4. Let K be a complete ordered field, $x \in K$ with $x \geq 0$, and $n \geq 1$. There is a unique $y \in K$ with $y \geq 0$ and $y^n = x$. We write $y = \sqrt[n]{x}$, or $\sqrt{x}$ if $n = 2$. In particular the non-negative elements of K are exactly the squares.

Proof. Uniqueness: if $0 \leq y_1 < y_2$ then $y_1^n < y_2^n$. For existence we may assume $x > 0$ and $n \geq 2$ (for $n = 1$ take $y = x$). For $0 < a < b$ we have $b^n - a^n = (b - a)(b^{n-1} + b^{n-2}a + \dots + a^{n-1}) < (b - a)nb^{n-1}$. Let $E = \{t \in K : t \geq 0, t^n < x\}$. It contains 0, and is bounded above by $1 + x$ (if $t > 1 + x$ then $t^n \geq t > x$). Let $y = \sup E$; then $y > 0$, since $t = x/(1 + x)$ lies in E .

Suppose $y^n < x$. Choose h with $0 < h < 1$ and $h < (x - y^n)/(n(y + 1)^{n-1})$. Then $(y + h)^n - y^n < hn(y + h)^{n-1} < hn(y + 1)^{n-1} < x - y^n$, so $y + h \in E$, contradicting that y is an upper bound.

Suppose $y^n > x$. Let $k = (y^n - x)/(ny^{n-1})$; then $0 < k < y$. For $t \geq y - k$ we have $y^n - t^n \leq y^n - (y - k)^n < kny^{n-1} = y^n - x$, so $t^n > x$ and $t \notin E$. Hence $y - k$ is an upper bound of E , contradicting that y is least. So $y^n = x$. $\square$

Definition 4.7.5. Let K be an ordered field. A sequence $(x_n)_{n \in \mathbb{N}}$ in K converges to $x \in K$ if for every $\varepsilon > 0$ in K there is N with $|x_n - x| < \varepsilon$ for all $n \geq N$. It is a *Cauchy sequence* if for every $\varepsilon > 0$ there is N with $|x_n - x_m| < \varepsilon$ for all $m, n \geq N$.

Proposition 4.7.6. *In a complete ordered field every Cauchy sequence converges.*

Proof. Let (x_n) be Cauchy. Choosing N for $\varepsilon = 1$, all terms lie between $\min(x_0, \dots, x_N) - 1$ and $\max(x_0, \dots, x_N) + 1$, so the sequence is bounded. Let $s_n = \sup\{x_k : k \geq n\}$; this sequence is decreasing and bounded below, and we let $x = \inf_n s_n$. Given $\varepsilon > 0$, choose N with $|x_n - x_m| < \varepsilon$ for $m, n \geq N$. For $n \geq N$, every x_k with $k \geq n$ lies in $(x_n - \varepsilon, x_n + \varepsilon)$, so $x_n - \varepsilon \leq s_m \leq x_n + \varepsilon$ for all $m \geq n$; since the s_m decrease, $x = \inf_{m \geq n} s_m$ lies in $[x_n - \varepsilon, x_n + \varepsilon]$. Hence $|x_n - x| \leq \varepsilon$ for $n \geq N$. $\square$

Theorem 4.7.7. *Let K and L be complete ordered fields. There is exactly one ring homomorphism $K \rightarrow L$, and it is an order-preserving isomorphism of fields.*

Proof. Regard $\mathbb{Q} \subseteq K$ and $\mathbb{Q} \subseteq L$ by Proposition 4.6.3. For $x \in K$ let $D(x) = \{q \in \mathbb{Q} : q < x\}$, which is nonempty and bounded above in $\mathbb{Q}$ by Proposition 4.7.3. Define $\varphi(x) = \sup_L D(x)$.

By density, $x = \sup_K D(x)$, and $x < y$ implies that $D(x) \subsetneq D(y)$ contains two rationals between x and y , so $\varphi(x) < \varphi(y)$. For $q \in \mathbb{Q}$, $\varphi(q) = q$. Every $y \in L$ is in the image: with $D_L(y) = \{q : q < y\}$ and $x = \sup_K D_L(y)$ we get $D(x) = D_L(y)$ by density in both fields, so $\varphi(x) = y$. So φ is an order-preserving bijection.

By density, $D(x + y) = D(x) + D(y)$, so Lemma 4.7.2 gives $\varphi(x + y) = \varphi(x) + \varphi(y)$. For $x, y > 0$, the positive elements of $D(xy)$ are the products of positive elements of $D(x)$ and $D(y)$, so $\varphi(xy) = \varphi(x)\varphi(y)$. Since $\varphi(-x) = -\varphi(x)$ (from additivity) and $\varphi(0) = 0$, multiplicativity follows for all signs. Hence φ is an isomorphism.

Let $\psi : K \rightarrow L$ be any ring homomorphism. By Proposition 4.7.4, $x \geq 0$ implies $x = y^2$ and so $\psi(x) = \psi(y)^2 \geq 0$; hence ψ is order-preserving. Also ψ is the identity on $\mathbb{Q}$ by Proposition 4.6.3. For $q \in D(x)$ we get $q = \psi(q) \leq \psi(x)$, so $\varphi(x) \leq \psi(x)$; applying this to $-x$ gives $\varphi(x) \geq \psi(x)$. So $\psi = \varphi$. $\square$

4.8 The real numbers

Definition 4.8.1. Let C be the set of Cauchy sequences in $\mathbb{Q}$ and $\mathcal{N} \subseteq C$ the set of sequences converging to 0. For $x, y \in C$ write $x \sim y$ if $x - y \in \mathcal{N}$. The set of *real numbers* is $\mathbb{R} = C/\sim$. Operations are defined termwise: $[x] + [y] = [(x_n + y_n)_n]$ and $[x] \cdot [y] = [(x_n y_n)_n]$. A real number $[x]$ is *positive* if there are $\varepsilon \in \mathbb{Q}$, $\varepsilon > 0$, and N with $x_n > \varepsilon$ for all $n \geq N$, and $[x] < [y]$ means that $[y] - [x]$ is positive. A rational number q is identified with the class of the constant sequence $(q)_n$.

- Lemma 4.8.2.**
1. *Every $x \in C$ is bounded, and C is closed under termwise sums and products.*
 2. *$\mathcal{N}$ is closed under sums, and $xz \in \mathcal{N}$ for $x \in C$ and $z \in \mathcal{N}$. In particular $\sim$ is an equivalence relation compatible with sums and products.*
 3. *If $x \in C \setminus \mathcal{N}$, there are a rational $\varepsilon > 0$ and N such that either $x_n > \varepsilon$ for all $n \geq N$, or $x_n < -\varepsilon$ for all $n \geq N$.*

Proof. (1) Boundedness is proved as in Proposition 4.7.6. If $|x_n|, |y_n| \leq M$, then $|x_n y_n - x_m y_m| \leq M|y_n - y_m| + M|x_n - x_m|$, and similarly for sums. (2) If $|x_n| \leq M$ and $z \rightarrow 0$, then $|x_n z_n| \leq M|z_n| \rightarrow 0$. Sums are similar. If $x \sim x'$ and $y \sim y'$, then $xy - x'y' = x(y - y') + y'(x - x') \in \mathcal{N}$. (3) Since $x \notin \mathcal{N}$, there is a rational $\delta > 0$ such that $|x_n| \geq \delta$ for infinitely many n . Choose N with $|x_n - x_m| < \delta/2$ for $m, n \geq N$, and $m \geq N$ with $|x_m| \geq \delta$. If $x_m \geq \delta$ then $x_n > \delta/2$ for all $n \geq N$; if $x_m \leq -\delta$ then $x_n < -\delta/2$ for all $n \geq N$. Take $\varepsilon = \delta/2$. $\square$

Theorem 4.8.3. *$\mathbb{R}$ is a complete ordered field, and $\mathbb{Q} \rightarrow \mathbb{R}$ is an injective order-preserving ring homomorphism. By Theorem 4.7.7, $\mathbb{R}$ is the unique complete ordered field up to unique isomorphism.*

Proof. By Lemma 4.8.2, the operations are well defined, and the ring axioms hold termwise. If $[x] \neq 0$, then by part (3) of the lemma there is N with $|x_n| > \varepsilon$ for $n \geq N$. Let $y_n = x_n^{-1}$ for $n \geq N$ and $y_n = 0$ otherwise; since $|y_n - y_m| = |x_m - x_n|/|x_n x_m| \leq \varepsilon^{-2}|x_m - x_n|$ for $m, n \geq N$, y is Cauchy, and $xy - 1$ is eventually 0. So $\mathbb{R}$ is a field.

Positivity is well defined: if $x_n > \varepsilon$ for $n \geq N$ and $z \in \mathcal{N}$, then $x_n + z_n > \varepsilon/2$ for large n . By part (3), every nonzero real number is either positive or its negative is positive, and not both; sums and products of positive numbers are positive. It follows that $<$ is a strict total order compatible with the ring operations, so $\mathbb{R}$ is an ordered field. The embedding of $\mathbb{Q}$ is clearly an injective order-preserving homomorphism. $\mathbb{R}$ is Archimedean: if $|x_n| \leq M \in \mathbb{N}$ for all n , then $[x] < M + 1$.

Convergence of rational approximations. Let $x \in \mathcal{C}$. For rational $\varepsilon > 0$ choose N with $|x_n - x_m| < \varepsilon/2$ for $m, n \geq N$; then for $n \geq N$ the sequence $(\varepsilon - |x_n - x_m|)_m$ is eventually $> \varepsilon/2$, so $||x] - x_n| < \varepsilon$ in $\mathbb{R}$. Thus the rational sequence (x_n) converges to $[x]$ in $\mathbb{R}$; by density of $\mathbb{Q}$ this holds for every real $\varepsilon > 0$.

Completeness. Let $A \subseteq \mathbb{R}$ be nonempty and bounded above. Let u_0 be the least integer that is an upper bound of A , and a_0 an integer smaller than some element of A ; both exist by the Archimedean property and Theorem 4.4.2, and $a_0 < u_0$. Define rationals $a_n < u_n$ by recursion: let $m = (a_n + u_n)/2$; if m is an upper bound of A , put $(a_{n+1}, u_{n+1}) = (a_n, m)$, and otherwise $(a_{n+1}, u_{n+1}) = (m, u_n)$. Then every u_n is an upper bound, no a_n is, the sequences are monotone, and $u_n - a_n = (u_0 - a_0)/2^n \leq (u_0 - a_0)/n$ for $n \geq 1$. Hence (u_n) and (a_n) are Cauchy sequences in $\mathbb{Q}$ with the same class $s \in \mathbb{R}$, and both converge to s in $\mathbb{R}$. If $a \in A$, then $a \leq u_n$ for all n , hence $a \leq s$. If $t < s$, then $a_n > t$ for some n , and since a_n is not an upper bound, neither is t . So $s = \sup A$. $\square$

Remark 4.8.4. Dedekind's construction [Ded72] defines a real number as a *cut*: a nonempty proper subset $\alpha \subsetneq \mathbb{Q}$ which is downward closed and has no largest element. The order is inclusion and the supremum of a bounded set of cuts is their union. It also produces a complete ordered field (Exercise 4.10.1), which by Theorem 4.7.7 is canonically isomorphic to the one constructed above. See [Rud76, Appendix to Chapter 1].

Notation 4.8.5. For $a \leq b$ in $\mathbb{R}$ we write $[a, b] = \{x : a \leq x \leq b\}$, $(a, b) = \{x : a < x < b\}$, and similarly $[a, b)$, $(a, b]$, $[a, \infty)$ and so on. We use the *extended real line* $[-\infty, \infty]$, obtained by adding a least and a greatest element, where convenient.

Proposition 4.8.6. $|\mathbb{R}| = 2^{\aleph_0}$. In particular $\mathbb{R}$ is uncountable.

Proof. The map $x \mapsto \{q \in \mathbb{Q} : q < x\}$ is injective by density of $\mathbb{Q}$, so $\mathbb{R} \leq \mathcal{P}(\mathbb{Q})$. Since $\mathbb{Q}$ is countable (it injects into $\mathbb{Z} \times \mathbb{Z}$, which injects into $\mathbb{N} \times \mathbb{N}$, which is countable by Corollary 3.6.13), $\mathcal{P}(\mathbb{Q}) \leq \mathcal{P}(\mathbb{N})$.

Conversely, for $S \subseteq \mathbb{N}$ let $f(S) = \sup\{\sum_{n \in S, n < N} 2 \cdot 3^{-(n+1)} : N \in \mathbb{N}\}$; the partial sums are at most $\sum_{n < N} 2 \cdot 3^{-(n+1)} = 1 - 3^{-N} < 1$. Let $S \neq T$ and let n be the least element of the symmetric difference, say $n \in S \setminus T$. The sums over S and T agree below n , the term $2 \cdot 3^{-(n+1)}$ occurs only for S , and the terms of T above n add up to at most $\sum_{k=n+1}^M 2 \cdot 3^{-(k+1)} < 3^{-(n+1)}$. Hence $f(S) - f(T) \geq 2 \cdot 3^{-(n+1)} - 3^{-(n+1)} > 0$, so f is injective. By Theorem 3.6.2, $\mathbb{R} \approx \mathcal{P}(\mathbb{N})$, and $\mathbb{R}$ is uncountable by Theorem 3.6.3. $\square$

4.9 The complex numbers

Definition 4.9.1. The set of *complex numbers* is $\mathbb{C} = \mathbb{R} \times \mathbb{R}$ with

$$(a, b) + (c, d) = (a + c, b + d), \quad (a, b)(c, d) = (ac - bd, ad + bc).$$

We identify $a \in \mathbb{R}$ with $(a, 0)$ and write $i = (0, 1)$, so that $(a, b) = a + bi$. For $z = a + bi$, the real part is $\operatorname{Re} z = a$, the imaginary part is $\operatorname{Im} z = b$, the conjugate is $\bar{z} = a - bi$, and the absolute value is $|z| = \sqrt{a^2 + b^2}$.

Theorem 4.9.2. $\mathbb{C}$ is a field containing $\mathbb{R}$ as a subfield, and $i^2 = -1$. Every $z \in \mathbb{C}$ can be written uniquely as $a + bi$ with $a, b \in \mathbb{R}$. Complex conjugation is a field automorphism of $\mathbb{C}$ that is the identity exactly on $\mathbb{R}$, and for all $z, w \in \mathbb{C}$: $z\bar{z} = |z|^2$, $|zw| = |z||w|$, $|\operatorname{Re} z| \leq |z|$ and $|z + w| \leq |z| + |w|$.

Proof. The ring axioms are direct computations; for instance the product is associative because both $((a, b)(c, d))(e, f)$ and $(a, b)((c, d)(e, f))$ equal $(ace - bde - adf - bcf, acf - bdf + ade + bce)$. The unit is $(1, 0)$, and $(0, 1)(0, 1) = (-1, 0)$. If $z = a + bi \neq 0$, then $a^2 + b^2 > 0$ and $z \cdot (a - bi)/(a^2 + b^2) = 1$. The map $a \mapsto (a, 0)$ is an injective ring homomorphism. Conjugation preserves sums, and $(a + bi)(c + di) = (ac - bd) - (ad + bc)i = (a - bi)(c - di)$; it is its own inverse and fixes $a + bi$ iff $b = 0$.

$z\bar{z} = a^2 + b^2 = |z|^2$. Then $|zw|^2 = zw\bar{z}\bar{w} = |z|^2|w|^2$, and both $|zw|$ and $|z||w|$ are non-negative, so they are equal by the uniqueness in Proposition 4.7.4. Clearly $|\operatorname{Re} z| = |a| \leq \sqrt{a^2 + b^2}$. Finally $|z + w|^2 = |z|^2 + 2\operatorname{Re}(z\bar{w}) + |w|^2 \leq |z|^2 + 2|z||w| + |w|^2 = (|z| + |w|)^2$. $\square$

Proposition 4.9.3. Every complex number has a square root: for $z = a + bi$, the number

$$w = \sqrt{\frac{|z|+a}{2}} + \sigma\sqrt{\frac{|z|-a}{2}}i, \quad \sigma = 1 \text{ if } b \geq 0, \sigma = -1 \text{ if } b < 0,$$

satisfies $w^2 = z$. Consequently every polynomial equation $z^2 + pz + q = 0$ with $p, q \in \mathbb{C}$ has a solution in $\mathbb{C}$.

Proof. Both radicands are non-negative since $|a| \leq |z|$. We have $w^2 = \frac{|z|+a}{2} - \frac{|z|-a}{2} + 2\sigma\sqrt{\frac{|z|^2-a^2}{4}}i = a + \sigma|b|i = a + bi$. For the equation, take $z = (-p + w)/2$ where $w^2 = p^2 - 4q$. $\square$

Counterexample 4.9.4. There is no order on $\mathbb{C}$ making it an ordered field. Indeed, in an ordered field squares are non-negative and $-1 < 0$ (Lemma 4.3.4), but $-1 = i^2$.

Remark 4.9.5. The fundamental theorem of algebra, which says that every nonconstant complex polynomial has a root, is proved algebraically in Chapter 8 and again analytically in Chapter 27. The polar form $z = |z|e^{i\theta}$ requires the exponential function, which is defined in Chapter 25.

4.10 Exercises

Exercise 4.10.1. Carry out Dedekind's construction of Remark 4.8.4: define the sum of cuts α, β as $\{a + b : a \in \alpha, b \in \beta\}$, define products first for cuts containing positive rationals, and show that the cuts form a complete ordered field.

Exercise 4.10.2. Show that the only ring homomorphism $\mathbb{R} \rightarrow \mathbb{R}$ is the identity.

Solution. This is the uniqueness statement of Theorem 4.7.7 with $K = L = \mathbb{R}$. $\square$

Exercise 4.10.3. Show that the only ring homomorphisms $\mathbb{C} \rightarrow \mathbb{C}$ that are the identity on $\mathbb{R}$ are the identity and complex conjugation. (It is a much deeper fact, relevant to Chapter 71, that $\mathbb{C}$ has many other field automorphisms, which are not the identity on $\mathbb{R}$.)

Solution. Such a homomorphism σ is determined by $\sigma(i)$, because $\sigma(a + bi) = a + b\sigma(i)$. Since $\sigma(i)^2 = \sigma(-1) = -1$, and the only solutions of $w^2 = -1$ are $\pm i$ (if $w^2 = i^2$ then $(w - i)(w + i) = 0$), we get $\sigma(i) = \pm i$. $\square$

Exercise 4.10.4. Let $n \geq 1$ and $m \geq 0$ be integers. Show that $\sqrt[n]{m}$ is rational if and only if m is the n -th power of an integer.

Exercise 4.10.5. Let R be a commutative ring. Show that there is exactly one ring homomorphism $\mathbb{Z} \rightarrow R$.

Exercise 4.10.6. A complex number is *algebraic* if it is a root of a nonzero polynomial with integer coefficients. Show that the set of algebraic numbers is countable, and deduce that there are complex (even real) numbers that are not algebraic.

Exercise 4.10.7. Show that for every $x \in [0, 1)$ there is a unique sequence $(d_n)_{n \geq 1}$ with $d_n \in \{0, 1, \dots, 9\}$, not eventually equal to 9, such that $x = \sup_N \sum_{n=1}^N d_n 10^{-n}$.

Chapter 5

Categories

5.1 Introduction

Category theory is the language in which most of this project is written. This chapter introduces categories, functors and natural transformations, proves the Yoneda lemma, and develops equivalences, limits and colimits, adjoint functors, filtered colimits and additive categories. Abelian categories are treated in Chapter 13.

Categories, functors and natural transformations were introduced in [EML45], and adjoint functors in [Kan58]. Standard references are [ML98] and [Rie16]; [KS06] and the chapter on categories of [The] treat set-theoretic questions with universes, as we do.

Prerequisites. Chapter 3 (Set Theory), in particular the axiom of choice and Grothendieck universes.

Throughout, $\mathcal{U}$ is the universe fixed in Notation 1.6.1, and a set is *small* if it is an element of $\mathcal{U}$.

5.2 Categories

Definition 5.2.1. A *category* C consists of

1. a set $\text{Ob}(C)$, whose elements are called the *objects* of C ;
2. for all objects X, Y , a set $\text{Hom}_C(X, Y)$, whose elements are called *morphisms* from X to Y ;
3. for every object X , an element $\text{id}_X \in \text{Hom}_C(X, X)$, the *identity* of X ;
4. for all objects X, Y, Z , a map

$$\text{Hom}_C(Y, Z) \times \text{Hom}_C(X, Y) \longrightarrow \text{Hom}_C(X, Z), \quad (g, f) \longmapsto g \circ f,$$

called *composition*,

such that the following hold:

1. (associativity) $h \circ (g \circ f) = (h \circ g) \circ f$ for all $f \in \text{Hom}_C(X, Y)$, $g \in \text{Hom}_C(Y, Z)$ and $h \in \text{Hom}_C(Z, W)$;
2. (identities) $f \circ \text{id}_X = f = \text{id}_Y \circ f$ for all $f \in \text{Hom}_C(X, Y)$;
3. the sets $\text{Hom}_C(X, Y)$ are pairwise disjoint.

Notation 5.2.2. We write $X \in C$ instead of $X \in \text{Ob}(C)$, and $f: X \rightarrow Y$ instead of $f \in \text{Hom}_C(X, Y)$. By the last axiom of Definition 5.2.1, a morphism f determines its *source* X and its *target* Y . We often write gf for $g \circ f$, and $\text{Hom}(X, Y)$ when the category is clear.

Definition 5.2.3. A category C is *locally small* if $\text{Hom}_C(X, Y)$ is a small set for all $X, Y \in C$. It is *small* if it is locally small and $\text{Ob}(C)$ is a small set.

Remark 5.2.4. There is no set of all sets, so “the category of all sets” cannot be a category in the sense of Definition 5.2.1. The universe $\mathcal{U}$ solves this problem: it is a set, it is closed under all the usual constructions of set theory (pairs, products, power sets, sets of maps, unions indexed by small sets), and it contains every set that appears in ordinary mathematics. So we can form the category of all *small* sets without leaving ZFC.

Example 5.2.5. The category Set has as objects the small sets, that is, $\text{Ob}(\text{Set}) = \mathcal{U}$. For small sets X and Y , the set $\text{Hom}_{\text{Set}}(X, Y)$ is the set of maps from X to Y , composition is composition of maps, and id_X is the identity map. Since $\mathcal{U}$ is closed under forming sets of maps, Set is locally small. It is not small, because $\mathcal{U} \notin \mathcal{U}$.

(Strictly speaking, a map $f: X \rightarrow Y$ should be recorded together with its target Y , so that the last axiom of Definition 5.2.1 holds. We do this silently here and in similar examples.)

Example 5.2.6. In the same way there are locally small categories of small groups Grp , of small abelian groups Ab , of small rings Ring , of small vector spaces over a field k , written Vect_k , and of small topological spaces Top . In each case the morphisms are the maps that preserve the relevant structure (group homomorphisms, ring homomorphisms, linear maps, continuous maps), and composition is composition of maps. These notions are defined in later chapters.

Definition 5.2.7. A *monoid* is a set M with a map $M \times M \rightarrow M$, written $(a, b) \mapsto a \cdot b$ or simply ab , which is associative, together with an element $e \in M$ such that $ea = a = ae$ for all $a \in M$. (For a monoid given by addition, such as $(\mathbb{N}, +)$, we write $a + b$ and 0 instead.) A *homomorphism of monoids* $\varphi: M \rightarrow N$ is a map with $\varphi(ab) = \varphi(a)\varphi(b)$ for all $a, b \in M$ and $\varphi(e) = e$. The small monoids and their homomorphisms form a locally small category Mon .

Example 5.2.8. Let M be a monoid. There is a category BM with a single object $*$, with $\text{Hom}_{BM}(*, *) = M$, with composition given by the multiplication of M and with $\text{id}_* = e$. The axioms of a category are exactly the axioms of a monoid. Conversely, if C is a category and $X \in C$, then $\text{End}(X) = \text{Hom}_C(X, X)$ is a monoid under composition.

Example 5.2.9. Let P be a set with a relation $\leq$ that is reflexive and transitive (a *preorder*). There is a category, also written P , whose objects are the elements of P , with $\text{Hom}_P(x, y) = \{(x, y)\}$ if $x \leq y$ and $\text{Hom}_P(x, y) = \emptyset$ otherwise. Composition is forced, because every set of morphisms has at most one element; it exists by transitivity, and the identities exist by reflexivity.

Example 5.2.10. Let S be a set. The *discrete category* on S has objects the elements of S , with $\text{Hom}(s, s) = \{\text{id}_s\}$ and $\text{Hom}(s, t) = \emptyset$ for $s \neq t$. It is the category associated with the preorder “ $s \leq t$ if and only if $s = t$ ”.

Definition 5.2.11. Let C be a category. The *opposite category* C^{op} has the same objects as C and $\text{Hom}_{C^{\text{op}}}(X, Y) = \text{Hom}_C(Y, X)$, with the same identities and with composition $g \circ_{\text{op}} f = f \circ g$.

Remark 5.2.12. The axioms of a category are symmetric under reversing all morphisms, and $(C^{\text{op}})^{\text{op}} = C$. So every statement proved for all categories implies a *dual* statement, obtained by applying it to opposite categories.

Definition 5.2.13. A subcategory $\mathcal{D}$ of a category C consists of a subset $\text{Ob}(\mathcal{D}) \subseteq \text{Ob}(C)$ and subsets $\text{Hom}_{\mathcal{D}}(X, Y) \subseteq \text{Hom}_C(X, Y)$ for $X, Y \in \mathcal{D}$, which contain the identities and are closed under composition. It is a *full* subcategory if $\text{Hom}_{\mathcal{D}}(X, Y) = \text{Hom}_C(X, Y)$ for all $X, Y \in \mathcal{D}$.

Example 5.2.14. The category Ab is a full subcategory of Grp . The category whose objects are small sets and whose morphisms are the bijections is a subcategory of Set that is not full.

5.3 Isomorphisms, monomorphisms and epimorphisms

Definition 5.3.1. A morphism $f: X \rightarrow Y$ in a category C is an *isomorphism* if there is a morphism $g: Y \rightarrow X$ with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$. Two objects X and Y are *isomorphic*, written $X \cong Y$, if there is an isomorphism $X \rightarrow Y$.

Lemma 5.3.2. Let $f: X \rightarrow Y$ be an isomorphism. There is exactly one morphism $g: Y \rightarrow X$ with $g \circ f = \text{id}_X$ and $f \circ g = \text{id}_Y$.

Proof. Suppose g and g' both have the property. Then $g = g \circ \text{id}_Y = g \circ (f \circ g') = (g \circ f) \circ g' = \text{id}_X \circ g' = g'$. $\square$

Notation 5.3.3. The morphism of Lemma 5.3.2 is the *inverse* of f , written f^{-1} .

Definition 5.3.4. A morphism $f: X \rightarrow Y$ is a *monomorphism* if for every object W and all $g_1, g_2: W \rightarrow X$, the equality $f \circ g_1 = f \circ g_2$ implies $g_1 = g_2$. It is an *epimorphism* if for every object Z and all $h_1, h_2: Y \rightarrow Z$, the equality $h_1 \circ f = h_2 \circ f$ implies $h_1 = h_2$.

Lemma 5.3.5. Let C be a category.

1. Every isomorphism is both a monomorphism and an epimorphism.
2. A composition of monomorphisms is a monomorphism, and a composition of epimorphisms is an epimorphism.
3. If $g \circ f$ is a monomorphism, then f is a monomorphism. If $g \circ f$ is an epimorphism, then g is an epimorphism.

Proof. By Remark 5.2.12, a monomorphism in C is the same thing as an epimorphism in C^{op} , so in each part it suffices to prove the statement about monomorphisms.

(1) Let f be an isomorphism and suppose $f \circ g_1 = f \circ g_2$. Composing with f^{-1} on the left gives $g_1 = g_2$.

(2) Let $f: X \rightarrow Y$ and $f': Y \rightarrow Z$ be monomorphisms and suppose $f' \circ f \circ g_1 = f' \circ f \circ g_2$. Since f' is a monomorphism, $f \circ g_1 = f \circ g_2$, and since f is a monomorphism, $g_1 = g_2$.

(3) Suppose $f \circ g_1 = f \circ g_2$. Then $(g \circ f) \circ g_1 = (g \circ f) \circ g_2$, so $g_1 = g_2$. $\square$

Lemma 5.3.6. In the category Set , a map is a monomorphism if and only if it is injective, and an epimorphism if and only if it is surjective.

Proof. Let $f: X \rightarrow Y$ be a map of small sets.

Suppose f is injective and $f \circ g_1 = f \circ g_2$ for maps $g_1, g_2: W \rightarrow X$. For every $w \in W$ we have $f(g_1(w)) = f(g_2(w))$, hence $g_1(w) = g_2(w)$; so $g_1 = g_2$. Conversely, suppose f is a monomorphism and $f(x_1) = f(x_2)$. Let $W = \{0\}$, which is a small set, and let $g_i: W \rightarrow X$ be the map with $g_i(0) = x_i$. Then $f \circ g_1 = f \circ g_2$, so $g_1 = g_2$, that is, $x_1 = x_2$.

Suppose f is surjective and $h_1 \circ f = h_2 \circ f$ for maps $h_1, h_2: Y \rightarrow Z$. Every $y \in Y$ is of the form $f(x)$, so $h_1(y) = h_1(f(x)) = h_2(f(x)) = h_2(y)$; hence $h_1 = h_2$. Conversely, suppose f is

an epimorphism. Let $Z = \{0, 1\}$, let $h_1: Y \rightarrow Z$ be the map with $h_1(y) = 1$ if $y \in f(X)$ and $h_1(y) = 0$ otherwise, and let h_2 be the constant map with value 1. Then $h_1 \circ f = h_2 \circ f$, so $h_1 = h_2$, which says exactly that $f(X) = Y$. $\square$

Counterexample 5.3.7. A morphism that is both a monomorphism and an epimorphism need not be an isomorphism. Let $P = \{0, 1\}$ with the usual order $0 \leq 1$, viewed as a category as in Example 5.2.9, and let $u: 0 \rightarrow 1$ be the unique morphism. Every set of morphisms of P has at most one element, so any two parallel morphisms are equal; hence every morphism of P is a monomorphism and an epimorphism. But u is not an isomorphism, because $\text{Hom}_P(1, 0) = \emptyset$.

Counterexample 5.3.8. The same phenomenon occurs in categories of algebraic objects. Let $i: \mathbb{N} \rightarrow \mathbb{Z}$ be the inclusion, a homomorphism of monoids under addition. Then i is a monomorphism and an epimorphism in Mon , but not an isomorphism.

Indeed, if $u, v: M \rightarrow \mathbb{N}$ are homomorphisms with $i \circ u = i \circ v$, then $u = v$ because i is injective. Next, let $f, g: \mathbb{Z} \rightarrow M$ be homomorphisms with $f \circ i = g \circ i$, that is, $f(n) = g(n)$ for all $n \in \mathbb{N}$. Writing the operation of M multiplicatively, for every $n \in \mathbb{N}$ we have

$$f(-n) = f(-n) g(n+(-n)) = f(-n) g(n) g(-n) = f(-n) f(n) g(-n) = f(-n+n) g(-n) = g(-n),$$

where we used $g(0) = e$ and $f(0) = e$. Hence $f = g$. Finally, an isomorphism of monoids is in particular a bijection, and i is not surjective.

Remark 5.3.9. Counterexample 5.3.8 shows that being an epimorphism is not the same as being surjective, even in categories whose morphisms are maps. Whether a morphism is an epimorphism depends on the whole category, not only on the morphism itself.

5.4 Functors

Definition 5.4.1. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F: \mathcal{C} \rightarrow \mathcal{D}$ consists of a map $F: \text{Ob}(\mathcal{C}) \rightarrow \text{Ob}(\mathcal{D})$ and, for all $X, Y \in \mathcal{C}$, a map $F: \text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$, such that $F(\text{id}_X) = \text{id}_{F(X)}$ for all X and $F(g \circ f) = F(g) \circ F(f)$ for all composable morphisms f and g . A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is a functor $\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}$.

Remark 5.4.2. Unwinding the definitions, a contravariant functor F from $\mathcal{C}$ to $\mathcal{D}$ assigns to each $f: X \rightarrow Y$ in $\mathcal{C}$ a morphism $F(f): F(Y) \rightarrow F(X)$, and $F(g \circ f) = F(f) \circ F(g)$.

Example 5.4.3. Some functors.

1. Every category $\mathcal{C}$ has an identity functor $\text{id}_{\mathcal{C}}$. If $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$ are functors, then so is their composition $G \circ F$, defined on objects and on morphisms by composing maps.
2. Forgetful functors, such as $\text{Grp} \rightarrow \text{Set}$, which send a group to its underlying set and a homomorphism to its underlying map.
3. The covariant power set functor $\mathcal{P}: \text{Set} \rightarrow \text{Set}$ sends X to its power set $\mathcal{P}(X)$ and $f: X \rightarrow Y$ to the map $A \mapsto f(A)$. It is a functor because $(g \circ f)(A) = g(f(A))$.
4. The contravariant power set functor $\mathcal{P}^*: \text{Set}^{\text{op}} \rightarrow \text{Set}$ sends X to $\mathcal{P}(X)$ and $f: X \rightarrow Y$ to the map $\mathcal{P}(Y) \rightarrow \mathcal{P}(X)$, $B \mapsto f^{-1}(B)$. It is a functor because $(g \circ f)^{-1}(C) = f^{-1}(g^{-1}(C))$.
5. If P and Q are preorders, viewed as categories, a functor $P \rightarrow Q$ is the same thing as a map $\varphi: P \rightarrow Q$ with $\varphi(x) \leq \varphi(y)$ whenever $x \leq y$.

6. If M and N are monoids, a functor $BM \rightarrow BN$ is the same thing as a homomorphism of monoids $M \rightarrow N$.

Definition 5.4.4. Let C be a locally small category and $X \in C$. The functor $h_X: C^{\text{op}} \rightarrow \text{Set}$ sends Y to $\text{Hom}_C(Y, X)$ and a morphism $f: Y' \rightarrow Y$ of C to the map

$$h_X(f): \text{Hom}_C(Y, X) \rightarrow \text{Hom}_C(Y', X), \quad g \mapsto g \circ f.$$

Dually, the functor $h^X: C \rightarrow \text{Set}$ sends Y to $\text{Hom}_C(X, Y)$ and $f: Y \rightarrow Y'$ to $g \mapsto f \circ g$.

Lemma 5.4.5. Let $F: C \rightarrow \mathcal{D}$ be a functor. If f is an isomorphism in C , then $F(f)$ is an isomorphism in $\mathcal{D}$, with inverse $F(f^{-1})$.

Proof. If $f: X \rightarrow Y$, then $F(f^{-1}) \circ F(f) = F(f^{-1} \circ f) = F(\text{id}_X) = \text{id}_{F(X)}$, and similarly $F(f) \circ F(f^{-1}) = \text{id}_{F(Y)}$. $\square$

Definition 5.4.6. A functor $F: C \rightarrow \mathcal{D}$ is *faithful* (respectively *full*) if for all $X, Y \in C$ the map $\text{Hom}_C(X, Y) \rightarrow \text{Hom}_{\mathcal{D}}(F(X), F(Y))$ is injective (respectively surjective). It is *fully faithful* if it is both full and faithful. It is *essentially surjective* if every object of $\mathcal{D}$ is isomorphic to an object of the form $F(X)$.

Lemma 5.4.7. Let $F: C \rightarrow \mathcal{D}$ be a fully faithful functor and $f: X \rightarrow Y$ a morphism of C . If $F(f)$ is an isomorphism, then f is an isomorphism. In particular, if $F(X) \cong F(Y)$ then $X \cong Y$.

Proof. Let $h: F(Y) \rightarrow F(X)$ be the inverse of $F(f)$. Since F is full, there is $g: Y \rightarrow X$ with $F(g) = h$. Then $F(g \circ f) = h \circ F(f) = \text{id}_{F(X)} = F(\text{id}_X)$, and since F is faithful, $g \circ f = \text{id}_X$. In the same way $f \circ g = \text{id}_Y$.

For the second statement, let $\varphi: F(X) \rightarrow F(Y)$ be an isomorphism. Since F is full, $\varphi = F(f)$ for some $f: X \rightarrow Y$, and f is an isomorphism by the first statement. $\square$

Counterexample 5.4.8. Faithfulness alone does not suffice in Lemma 5.4.7. Let P be the category $0 \leq 1$ of Counterexample 5.3.7, let $\mathbf{1}$ be the category with one object and one morphism, and let $F: P \rightarrow \mathbf{1}$ be the unique functor. Every set of morphisms of P has at most one element, so F is faithful. The morphism $u: 0 \rightarrow 1$ is not an isomorphism, but $F(u)$ is an identity, hence an isomorphism. (The functor F is not full: $\text{Hom}_P(1, 0)$ is empty.)

Remark 5.4.9. A more natural instance of the same phenomenon is the forgetful functor $\text{Top} \rightarrow \text{Set}$, which is faithful. The identity map of $\mathbb{R}$, viewed as a map from $\mathbb{R}$ with the discrete topology to $\mathbb{R}$ with the Euclidean topology, is a continuous bijection, so its image in Set is an isomorphism; but it is not a homeomorphism.

5.5 Natural transformations

Definition 5.5.1. Let $F, G: C \rightarrow \mathcal{D}$ be functors. A *natural transformation* $\eta: F \Rightarrow G$ is a family of morphisms $\eta_X: F(X) \rightarrow G(X)$, one for each $X \in C$, such that $G(f) \circ \eta_X = \eta_Y \circ F(f)$ for every morphism $f: X \rightarrow Y$ of C ; that is, such that the square

$$\begin{array}{ccc} F(X) & \xrightarrow{\eta_X} & G(X) \\ F(f) \downarrow & & \downarrow G(f) \\ F(Y) & \xrightarrow{\eta_Y} & G(Y) \end{array}$$

commutes. It is a *natural isomorphism* if every η_X is an isomorphism. We write $\text{Nat}(F, G)$ for the set of natural transformations from F to G .

Lemma 5.5.2. *Let $\eta: F \Rightarrow G$ be a natural isomorphism. Then the morphisms $\eta_X^{-1}: G(X) \rightarrow F(X)$ form a natural transformation $\eta^{-1}: G \Rightarrow F$.*

Proof. Let $f: X \rightarrow Y$. Composing the equality $G(f) \circ \eta_X = \eta_Y \circ F(f)$ with η_Y^{-1} on the left and with η_X^{-1} on the right gives $\eta_Y^{-1} \circ G(f) = F(f) \circ \eta_X^{-1}$, which is the naturality of η^{-1} . $\square$

Definition 5.5.3. Let $\mathcal{C}$ and $\mathcal{D}$ be categories. If $\eta: F \Rightarrow G$ and $\theta: G \Rightarrow H$ are natural transformations, their composition $\theta \circ \eta: F \Rightarrow H$ is given by $(\theta \circ \eta)_X = \theta_X \circ \eta_X$. The functor category $\text{Fun}(\mathcal{C}, \mathcal{D})$ has as objects the functors $\mathcal{C} \rightarrow \mathcal{D}$, as morphisms the natural transformations, and this composition; the identity of F is the family $(\text{id}_{F(X)})_X$. (Formally, a natural transformation $\eta: F \Rightarrow G$ is the triple $(F, G, (\eta_X)_X)$, so that the sets $\text{Nat}(F, G)$ are pairwise disjoint.)

Lemma 5.5.4. *Definition 5.5.3 defines a category. A morphism η of $\text{Fun}(\mathcal{C}, \mathcal{D})$ is an isomorphism if and only if it is a natural isomorphism. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, then $\text{Fun}(\mathcal{C}, \mathcal{D})$ is locally small.*

Proof. If η and θ are natural, then for $f: X \rightarrow Y$ we have $H(f) \circ \theta_X \circ \eta_X = \theta_Y \circ G(f) \circ \eta_X = \theta_Y \circ \eta_Y \circ F(f)$, so $\theta \circ \eta$ is natural. The family of identities is clearly natural. Associativity and the identity axioms hold componentwise, because they hold in $\mathcal{D}$.

If η is an isomorphism with inverse θ , then $\theta_X \circ \eta_X = \text{id}_{F(X)}$ and $\eta_X \circ \theta_X = \text{id}_{G(X)}$ for every X , so each η_X is an isomorphism. Conversely, if η is a natural isomorphism, then η^{-1} of Lemma 5.5.2 is a morphism of $\text{Fun}(\mathcal{C}, \mathcal{D})$ that is inverse to η .

Finally, $\text{Nat}(F, G)$ is a subset of the product $\prod_{X \in \text{Ob}(\mathcal{C})} \text{Hom}_{\mathcal{D}}(F(X), G(X))$. If $\mathcal{C}$ is small and $\mathcal{D}$ is locally small, this is a product of small sets indexed by a small set, hence small, and so is every subset of it. $\square$

Example 5.5.5. Let k be a field. For a vector space V over k , let V^* be its dual and $\text{ev}_V: V \rightarrow V^{**}$ the map sending v to the linear form $\varphi \mapsto \varphi(v)$. The double dual is a functor $\text{Vect}_k \rightarrow \text{Vect}_k$, sending $f: V \rightarrow W$ to $f^{**}: \alpha \mapsto (\psi \mapsto \alpha(\psi \circ f))$, and the maps ev_V form a natural transformation $\text{ev}: \text{id} \Rightarrow (-)^{**}$: for $v \in V$ and $\psi \in W^*$,

$$f^{**}(\text{ev}_V(v))(\psi) = \text{ev}_V(v)(\psi \circ f) = \psi(f(v)) = \text{ev}_W(f(v))(\psi).$$

On the full subcategory of finite-dimensional vector spaces it is a natural isomorphism, since there each ev_V is an isomorphism.

Remark 5.5.6. A finite-dimensional vector space V is also isomorphic to its dual V^* , but an isomorphism $V \rightarrow V^*$ depends on a choice, such as a basis. The notion of a natural transformation was introduced to make precise the difference between “canonical” isomorphisms, such as $V \cong V^{**}$, and isomorphisms that depend on choices.

Example 5.5.7. Let $n \geq 1$. For a commutative ring R , the determinant is a group homomorphism $\det_R: \text{GL}_n(R) \rightarrow R^\times$. Both GL_n and $(-)^{\times}$ are functors from commutative rings to groups, and $\det$ is a natural transformation between them: if $\varphi: R \rightarrow S$ is a ring homomorphism and $A \in \text{GL}_n(R)$, then $\det_S(\varphi(A)) = \varphi(\det_R(A))$, because the determinant is a polynomial with integer coefficients in the entries of the matrix.

5.6 The Yoneda lemma

In this section $\mathcal{C}$ is a locally small category, so that the functors $h_X: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ of Definition 5.4.4 are defined.

Theorem 5.6.1 (Yoneda lemma). *Let $X \in \mathcal{C}$ and let $F: \mathcal{C}^{\text{op}} \rightarrow \text{Set}$ be a functor. The map*

$$\Phi: \text{Nat}(h_X, F) \longrightarrow F(X), \quad \eta \longmapsto \eta_X(\text{id}_X)$$

is a bijection. Its inverse sends $\xi \in F(X)$ to the natural transformation η^ξ given by $\eta_Y^\xi(g) = F(g)(\xi)$ for $Y \in \mathcal{C}$ and $g \in \text{Hom}_{\mathcal{C}}(Y, X)$.

Proof. First, η^ξ is natural. Let $f: Y' \rightarrow Y$ be a morphism of $\mathcal{C}$ and $g \in \text{Hom}_{\mathcal{C}}(Y, X)$. Using that F is contravariant (Remark 5.4.2),

$$\eta_{Y'}^\xi(h_X(f)(g)) = \eta_{Y'}^\xi(g \circ f) = F(g \circ f)(\xi) = F(f)(F(g)(\xi)) = F(f)(\eta_Y^\xi(g)),$$

which is the naturality condition for f .

Second, $\Phi(\eta^\xi) = \eta_X^\xi(\text{id}_X) = F(\text{id}_X)(\xi) = \xi$.

Third, let $\eta: h_X \Rightarrow F$ be natural and $g: Y \rightarrow X$. The naturality of η with respect to g gives $\eta_Y(h_X(g)(\text{id}_X)) = F(g)(\eta_X(\text{id}_X))$. Since $h_X(g)(\text{id}_X) = \text{id}_X \circ g = g$, this says $\eta_Y(g) = F(g)(\Phi(\eta)) = \eta_Y^{\Phi(\eta)}(g)$. Hence $\eta = \eta^{\Phi(\eta)}$.

The second and third steps show that $\xi \mapsto \eta^\xi$ is inverse to Φ . $\square$

Remark 5.6.2. The Yoneda lemma says that a natural transformation out of h_X is determined by what it does to the single element id_X , and that this element can be chosen freely. Informally: an object is determined by the morphisms into it.

Proposition 5.6.3. *The bijection Φ of Theorem 5.6.1 is natural in F and in X :*

1. *if $\theta: F \Rightarrow F'$ is a natural transformation, then $\Phi(\theta \circ \eta) = \theta_X(\Phi(\eta))$ for all $\eta \in \text{Nat}(h_X, F)$;*
2. *if $a: X' \rightarrow X$ is a morphism and $h_a: h_{X'} \Rightarrow h_X$ is the natural transformation given by $(h_a)_Y(g) = a \circ g$, then $\Phi(\eta \circ h_a) = F(a)(\Phi(\eta))$ for all $\eta \in \text{Nat}(h_X, F)$.*

Proof. We first check that h_a is natural: for $f: Y' \rightarrow Y$ and $g: Y \rightarrow X'$ we have $a \circ (g \circ f) = (a \circ g) \circ f$, that is, $(h_a)_{Y'}(h_{X'}(f)(g)) = h_X(f)((h_a)_Y(g))$.

(1) $\Phi(\theta \circ \eta) = \theta_X(\eta_X(\text{id}_X)) = \theta_X(\Phi(\eta))$.

(2) $\Phi(\eta \circ h_a) = \eta_{X'}((h_a)_{X'}(\text{id}_{X'})) = \eta_{X'}(a)$. By the third step of the proof of Theorem 5.6.1, $\eta_{X'}(a) = F(a)(\Phi(\eta))$. $\square$

Corollary 5.6.4. *For all $X, X' \in \mathcal{C}$, the map*

$$\text{Hom}_{\mathcal{C}}(X', X) \longrightarrow \text{Nat}(h_{X'}, h_X), \quad a \longmapsto h_a$$

is a bijection. Consequently, $X \mapsto h_X, a \mapsto h_a$ is a fully faithful functor $\mathcal{C} \rightarrow \text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$, called the Yoneda embedding.

Proof. Apply Theorem 5.6.1 with X' in place of X and $F = h_X$. Then $\Phi(h_a) = (h_a)_{X'}(\text{id}_{X'}) = a$, so the map $a \mapsto h_a$ is a right inverse of the bijection Φ , hence equal to its inverse and bijective. The identities $h_{\text{id}_X} = \text{id}_{h_X}$ and $h_{a \circ b} = h_a \circ h_b$ follow from the identity axiom and from associativity in $\mathcal{C}$, so $X \mapsto h_X$ is a functor, and it is fully faithful by the first statement. $\square$

Corollary 5.6.5. *Let $X, X' \in \mathcal{C}$. If there is a natural isomorphism $h_{X'} \Rightarrow h_X$, then $X' \cong X$.*

Proof. By Lemma 5.5.4, a natural isomorphism $h_{X'} \Rightarrow h_X$ is an isomorphism in $\text{Fun}(\mathcal{C}^{\text{op}}, \text{Set})$. The Yoneda embedding is fully faithful by Corollary 5.6.4, so the claim follows from Lemma 5.4.7. $\square$

Definition 5.6.6. A functor $F: C^{\text{op}} \rightarrow \mathbf{Set}$ is *representable* if there exist $X \in C$ and a natural isomorphism $h_X \Rightarrow F$. By Theorem 5.6.1, such a natural isomorphism is η^ξ for a unique $\xi \in F(X)$; the pair (X, ξ) is said to *represent* F .

Example 5.6.7. The contravariant power set functor $\mathcal{P}^*$ of Example 5.4.3 is represented by the pair $(\{0, 1\}, \{1\})$. Indeed, for a small set S the map $\eta_S^{\{1\}}$ sends $\chi: S \rightarrow \{0, 1\}$ to $\mathcal{P}^*(\chi)(\{1\}) = \chi^{-1}(1)$, and this is a bijection $\text{Hom}_{\mathbf{Set}}(S, \{0, 1\}) \rightarrow \mathcal{P}(S)$: its inverse sends a subset $A \subseteq S$ to its indicator function.

5.7 Equivalences of categories

Definition 5.7.1. A functor $F: C \rightarrow \mathcal{D}$ is an *equivalence of categories* if there are a functor $G: \mathcal{D} \rightarrow C$ and natural isomorphisms $G \circ F \Rightarrow \text{id}_C$ and $F \circ G \Rightarrow \text{id}_{\mathcal{D}}$; such a G is a *quasi-inverse* of F . It is an *isomorphism of categories* if moreover $G \circ F = \text{id}_C$ and $F \circ G = \text{id}_{\mathcal{D}}$. Two categories are *equivalent* if there is an equivalence between them.

Lemma 5.7.2. *An equivalence of categories is fully faithful and essentially surjective.*

Proof. Let $G, \eta: GF \Rightarrow \text{id}$ and $\varepsilon: FG \Rightarrow \text{id}$ be as in Definition 5.7.1. Every $Y \in \mathcal{D}$ is isomorphic to $F(G(Y))$ via ε_Y , so F is essentially surjective. For $f: X \rightarrow X'$, naturality of η gives $f = \eta_{X'} \circ GF(f) \circ \eta_X^{-1}$; so $F(f)$ determines f and F is faithful. By symmetry G is faithful. Let $g: F(X) \rightarrow F(X')$ and put $f = \eta_{X'} \circ G(g) \circ \eta_X^{-1}$. By the same formula applied to f , $GF(f) = \eta_{X'}^{-1} \circ f \circ \eta_X = G(g)$, and since G is faithful, $F(f) = g$. So F is full. $\square$

Theorem 5.7.3. *A functor $F: C \rightarrow \mathcal{D}$ is an equivalence of categories if and only if it is fully faithful and essentially surjective.*

Proof. One direction is Lemma 5.7.2. Conversely, for each $Y \in \mathcal{D}$ the set of pairs (X, e) with $X \in C$ and $e: F(X) \rightarrow Y$ an isomorphism is nonempty. By the axiom of choice (Definition 3.7.1) we may choose such a pair $(G(Y), \varepsilon_Y)$ for every Y . For $g: Y \rightarrow Y'$, let $G(g)$ be the unique morphism $G(Y) \rightarrow G(Y')$ with $F(G(g)) = \varepsilon_{Y'}^{-1} \circ g \circ \varepsilon_Y$; it exists and is unique because F is fully faithful. By uniqueness, G preserves identities and composition, and by construction $\varepsilon: FG \Rightarrow \text{id}_{\mathcal{D}}$ is a natural isomorphism.

For $X \in C$ let $\eta_X: GF(X) \rightarrow X$ be the unique morphism with $F(\eta_X) = \varepsilon_{F(X)}$. It is an isomorphism by Lemma 5.4.7. For $f: X \rightarrow X'$ we have $F(f \circ \eta_X) = F(f) \circ \varepsilon_{F(X)}$ and $F(\eta_{X'} \circ GF(f)) = \varepsilon_{F(X')} \circ \varepsilon_{F(X')}^{-1} \circ F(f) \circ \varepsilon_{F(X)}$, which are equal; as F is faithful, η is natural. $\square$

Example 5.7.4. Let k be a field, and let Mat_k be the category whose objects are the natural numbers, whose morphisms $n \rightarrow m$ are the $m \times n$ matrices with entries in k , and whose composition is matrix multiplication. The functor from Mat_k to the category of finite-dimensional vector spaces sending n to k^n and a matrix to the corresponding linear map is fully faithful, and essentially surjective because every finite-dimensional vector space has a basis. So it is an equivalence: linear algebra of finite-dimensional spaces is “the same” as matrix algebra.

Counterexample 5.7.5. Equivalent categories need not be isomorphic. Let FinSet be the category of finite small sets and maps, and $\mathcal{N}$ its full subcategory whose objects are the natural numbers. Every finite set is isomorphic to a natural number (Definition 3.6.4), so the inclusion $\mathcal{N} \rightarrow \text{FinSet}$ is fully faithful and essentially surjective, hence an equivalence. But an isomorphism of categories is a bijection on objects, and $\text{Ob}(\mathcal{N}) = \omega$ is countable while $\text{Ob}(\text{FinSet})$ contains the singletons $\{x\}$ for all $x \in \mathcal{U}$, and $\mathcal{U}$ is uncountable since it contains the uncountable set $\mathbb{R}$ as a subset.

5.8 Limits

Definition 5.8.1. Let $\mathcal{I}$ be a small category and $F: \mathcal{I} \rightarrow \mathcal{C}$ a functor, called a *diagram* of shape $\mathcal{I}$. A *cone* over F is an object L together with morphisms $\lambda_i: L \rightarrow F(i)$ for all $i \in \mathcal{I}$ such that $F(u) \circ \lambda_i = \lambda_j$ for every $u: i \rightarrow j$. A *limit* of F is a cone (L, λ) such that for every cone (M, μ) there is a unique morphism $m: M \rightarrow L$ with $\lambda_i \circ m = \mu_i$ for all i . We write $L = \lim F$ or $\lim_{i \in \mathcal{I}} F(i)$.

Lemma 5.8.2. If (L, λ) and (L', λ') are limits of F , there is a unique morphism $m: L' \rightarrow L$ with $\lambda_i \circ m = \lambda'_i$ for all i , and it is an isomorphism.

Proof. Existence and uniqueness of m is the definition. Let $m': L \rightarrow L'$ be the morphism given by the universal property of L' . Then $m \circ m'$ and id_L are both morphisms of cones from (L, λ) to itself, so they are equal by uniqueness; similarly $m' \circ m = \text{id}_{L'}$. $\square$

Example 5.8.3. 1. If $\mathcal{I}$ is empty, a limit is a *terminal object*: an object T such that every object has exactly one morphism to T .

2. If $\mathcal{I}$ is a discrete category, a limit of $(X_i)_{i \in \mathcal{I}}$ is a *product* $\prod_i X_i$ with projections p_i .

3. If $\mathcal{I}$ has two objects $0, 1$ and two non-identity morphisms $u, v: 0 \rightarrow 1$, a diagram is a pair of parallel morphisms $f, g: X \rightarrow Y$, and a limit is an *equaliser*: a morphism $e: E \rightarrow X$ with $fe = ge$ through which every h with $fh = gh$ factors uniquely.

4. If $\mathcal{I}$ is $1 \rightarrow 0 \leftarrow 2$, a diagram is a pair $f: X \rightarrow Z, g: Y \rightarrow Z$, and a limit is a *fibre product* $X \times_Z Y$.

5. If $\mathcal{I}$ is the poset $\mathbb{N}$ with the opposite order, a limit of a sequence $\cdots \rightarrow X_2 \rightarrow X_1 \rightarrow X_0$ is an *inverse limit*.

Definition 5.8.4. A category *has limits* (respectively *finite limits*) if every diagram $F: \mathcal{I} \rightarrow \mathcal{C}$ with $\mathcal{I}$ small (respectively with finitely many morphisms) has a limit. A functor $G: \mathcal{C} \rightarrow \mathcal{D}$ *preserves* the limit (L, λ) of F if $(G(L), G(\lambda))$ is a limit of $G \circ F$.

Proposition 5.8.5. *Set has limits. For $F: \mathcal{I} \rightarrow \text{Set}$ with $\mathcal{I}$ small, a limit is*

$$\lim F = \left\{ (x_i) \in \prod_{i \in \text{Ob}(\mathcal{I})} F(i) : F(u)(x_i) = x_j \text{ for all } u: i \rightarrow j \right\}$$

with the projections $\lambda_i((x_k)_k) = x_i$.

Proof. The set $\lim F$ is small by Lemma 3.8.2, and it is a cone by construction. If (M, μ) is a cone, the map $m(y) = (\mu_i(y))_i$ lands in $\lim F$, satisfies $\lambda_i \circ m = \mu_i$, and is the only such map. $\square$

Theorem 5.8.6. *If $\mathcal{C}$ has products indexed by small sets and equalisers, then it has limits. If $\mathcal{C}$ has a terminal object, binary products and equalisers, then it has finite limits.*

Proof. Let $F: \mathcal{I} \rightarrow \mathcal{C}$ be a diagram. Let $P = \prod_i F(i)$ over the objects of $\mathcal{I}$, and $Q = \prod_u F(t(u))$ over all morphisms $u: s(u) \rightarrow t(u)$ of $\mathcal{I}$, with projections p_i and q_u . Define $a, b: P \rightarrow Q$ by $q_u \circ a = F(u) \circ p_{s(u)}$ and $q_u \circ b = p_{t(u)}$. Let $e: L \rightarrow P$ be an equaliser of a and b , and $\lambda_i = p_i \circ e$. The equation $ae = be$ says exactly that $F(u) \circ \lambda_{s(u)} = \lambda_{t(u)}$ for all u , so (L, λ) is a cone. If (M, μ) is a cone, the morphism $\mu: M \rightarrow P$ with $p_i \circ \mu = \mu_i$ satisfies $a\mu = b\mu$, so it factors uniquely as $\mu = e \circ m$; and a morphism m satisfies $\lambda_i \circ m = \mu_i$ for all i if and only if $e \circ m = \mu$. So (L, λ) is a limit. If $\mathcal{I}$ is finite, only finite products are needed, and these can be built from binary products and a terminal object (the empty product) by induction. $\square$

Proposition 5.8.7. *Let C be locally small, $F: \mathcal{I} \rightarrow C$ a diagram and (L, λ) a cone over F . Then (L, λ) is a limit if and only if for every $X \in C$ the map*

$$\mathrm{Hom}_C(X, L) \longrightarrow \lim_{i \in \mathcal{I}} \mathrm{Hom}_C(X, F(i)), \quad m \longmapsto (\lambda_i \circ m)_i$$

*is a bijection, where the right-hand side is the limit in **Set** of Proposition 5.8.5. In particular the functors $h^X = \mathrm{Hom}_C(X, -)$ preserve all limits.*

Proof. An element of the right-hand side is exactly a cone with vertex X , so bijectivity for all X is the definition of a limit. $\square$

Proposition 5.8.8. *Let $\mathcal{A}$ be a category and suppose that C has limits of shape $\mathcal{I}$. Then $\mathrm{Fun}(\mathcal{A}, C)$ has limits of shape $\mathcal{I}$, computed pointwise: for $F: \mathcal{I} \rightarrow \mathrm{Fun}(\mathcal{A}, C)$ there is a limit L with $L(a) = \lim_i F(i)(a)$ for all $a \in \mathcal{A}$.*

Proof. For each a choose a limit $(L(a), \lambda^a)$ of $i \mapsto F(i)(a)$. For $f: a \rightarrow a'$, the morphisms $F(i)(f) \circ \lambda_i^a$ form a cone over $i \mapsto F(i)(a')$ (by naturality of the $F(u)$), so there is a unique $L(f): L(a) \rightarrow L(a')$ with $\lambda_i^{a'} \circ L(f) = F(i)(f) \circ \lambda_i^a$. Uniqueness shows that L is a functor and that the $\lambda_i = (\lambda_i^a)_a$ are natural transformations $L \Rightarrow F(i)$ forming a cone. If (M, μ) is a cone in the functor category, then for each a there is a unique m_a with $\lambda_i^a \circ m_a = \mu_{i,a}$, and uniqueness again shows that $m = (m_a)_a$ is natural; it is the unique morphism of cones. $\square$

Counterexample 5.8.9. The category **Field** of small fields and ring homomorphisms does not have binary products. First, a ring homomorphism $f: K \rightarrow K'$ between fields is injective: if $f(a) = 0$ with $a \neq 0$ then $1 = f(a)f(a^{-1}) = 0$. Let $\mathbb{F}_2 = \{0, 1\}$ be the field with $1 + 1 = 0$. If P were a product of $\mathbb{Q}$ and $\mathbb{F}_2$, there would be homomorphisms $P \rightarrow \mathbb{Q}$ and $P \rightarrow \mathbb{F}_2$. The second one sends $1 + 1$ to 0 , so $1 + 1 = 0$ in P by injectivity; but then the first one sends 0 to $2 \neq 0$.

5.9 Colimits

Definition 5.9.1. A *cocone* under a diagram $F: \mathcal{I} \rightarrow C$ is an object C with morphisms $\gamma_i: F(i) \rightarrow C$ such that $\gamma_j \circ F(u) = \gamma_i$ for all $u: i \rightarrow j$. A *colimit* is a cocone (C, γ) such that for every cocone (D, δ) there is a unique $m: C \rightarrow D$ with $m \circ \gamma_i = \delta_i$ for all i . We write $C = \mathrm{colim} F$.

Remark 5.9.2. A colimit of F is the same thing as a limit of the corresponding diagram $\mathcal{I}^{\mathrm{op}} \rightarrow C^{\mathrm{op}}$. By duality (Remark 5.2.12), all results of Section 5.8 have colimit versions: colimits are unique up to unique isomorphism, colimits can be built from coproducts and coequalisers, and (C, γ) is a colimit if and only if $\mathrm{Hom}(C, X) \rightarrow \lim_i \mathrm{Hom}(F(i), X)$ is bijective for all X . The colimit analogues of the examples in Example 5.8.3 are *initial objects*, *coproducts* $\coprod_i X_i$, *coequalisers*, *pushouts* $X \sqcup_Z Y$ and *direct limits*.

Proposition 5.9.3. ***Set** has colimits. For $F: \mathcal{I} \rightarrow \mathbf{Set}$ with $\mathcal{I}$ small, let $\sim$ be the smallest equivalence relation on $\coprod_i F(i) = \{(i, x) : i \in \mathrm{Ob}(\mathcal{I}), x \in F(i)\}$ with $(i, x) \sim (j, F(u)(x))$ for all $u: i \rightarrow j$ and $x \in F(i)$. Then $\mathrm{colim} F = (\coprod_i F(i))/\sim$, with $\gamma_i(x) = [(i, x)]$.*

Proof. The smallest equivalence relation containing a relation R exists: it is the intersection of all equivalence relations containing R , a subset of the square of the set. The maps γ_i form a cocone by construction. A cocone (D, δ) gives a map $\coprod_i F(i) \rightarrow D$, $(i, x) \mapsto \delta_i(x)$, whose fibres define an equivalence relation containing the generating relation; so it factors uniquely through the quotient (Proposition 3.3.9). $\square$

Example 5.9.4. In $\mathbf{Set}$, the coproduct is the disjoint union, and the initial object is $\emptyset$. In the category of abelian groups the coproduct of a family is its direct sum, and in the category of groups it is the free product; these are constructed in Chapter 6.

5.10 Adjoint functors

In this section all categories are locally small.

Definition 5.10.1. An *adjunction* between functors $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{C}$ is a family of bijections

$$\varphi_{X,Y}: \text{Hom}_{\mathcal{D}}(F(X), Y) \longrightarrow \text{Hom}_{\mathcal{C}}(X, G(Y))$$

for $X \in \mathcal{C}$, $Y \in \mathcal{D}$, natural in X and Y : for $f: X' \rightarrow X$, $g: F(X) \rightarrow Y$ and $h: Y \rightarrow Y'$, $\varphi(g \circ F(f)) = \varphi(g) \circ f$ and $\varphi(h \circ g) = G(h) \circ \varphi(g)$. We say F is *left adjoint* to G , and G *right adjoint* to F , written $F \dashv G$. The *unit* and *counit* are $\eta_X = \varphi(\text{id}_{F(X)}): X \rightarrow GF(X)$ and $\varepsilon_Y = \varphi^{-1}(\text{id}_{G(Y)}): FG(Y) \rightarrow Y$.

Proposition 5.10.2. Let $F \dashv G$. Then $\eta: \text{id}_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon: FG \Rightarrow \text{id}_{\mathcal{D}}$ are natural, $\varphi(g) = G(g) \circ \eta_X$, $\varphi^{-1}(f) = \varepsilon_Y \circ F(f)$, and the triangle identities

$$\varepsilon_{F(X)} \circ F(\eta_X) = \text{id}_{F(X)}, \quad G(\varepsilon_Y) \circ \eta_{G(Y)} = \text{id}_{G(Y)}$$

hold. Conversely, if $\eta: \text{id}_{\mathcal{C}} \Rightarrow GF$ and $\varepsilon: FG \Rightarrow \text{id}_{\mathcal{D}}$ are natural transformations satisfying the triangle identities, then $\varphi(g) = G(g) \circ \eta_X$ defines an adjunction with unit η and counit ε .

Proof. By naturality in Y , $\varphi(g) = \varphi(g \circ \text{id}_{F(X)}) = G(g) \circ \varphi(\text{id}_{F(X)}) = G(g) \circ \eta_X$. By naturality in X , for $f: X \rightarrow G(Y)$, $\varphi^{-1}(f) = \varphi^{-1}(\text{id}_{G(Y)} \circ f) = \varphi^{-1}(\text{id}_{G(Y)} \circ F(f)) = \varepsilon_Y \circ F(f)$. Naturality of η : for $f: X' \rightarrow X$, $GF(f) \circ \eta_{X'} = \varphi(F(f)) = \varphi(\text{id}_{F(X)} \circ F(f)) = \eta_X \circ f$; naturality of ε is dual. The triangle identities are $\varphi^{-1}(\eta_X) = \text{id}_{F(X)}$ and $\varphi(\varepsilon_Y) = \text{id}_{G(Y)}$.

Conversely, define $\varphi(g) = G(g) \circ \eta_X$ and $\psi(f) = \varepsilon_Y \circ F(f)$. Then $\psi(\varphi(g)) = \varepsilon_Y \circ FG(g) \circ G(\eta_X) = g \circ \varepsilon_{F(X)} \circ F(\eta_X) = g$ by naturality of ε and the first triangle identity, and similarly $\varphi(\psi(f)) = f$. Naturality of φ in X follows from naturality of η , and in Y from functoriality of G . $\square$

Proposition 5.10.3. If G and G' are both right adjoint to F , there is a unique natural isomorphism $\theta: G \Rightarrow G'$ compatible with the adjunction bijections, that is, with $\varphi'_{X,Y} = \text{Hom}(X, \theta_Y) \circ \varphi_{X,Y}$.

Proof. For fixed Y , the bijections $\varphi'_{X,Y} \circ \varphi_{X,Y}^{-1}: \text{Hom}(X, G(Y)) \rightarrow \text{Hom}(X, G'(Y))$ are natural in X , so they form a natural isomorphism $h_{G(Y)}: G(Y) \rightarrow G'(Y)$. By Corollary 5.6.4 it is h_{θ_Y} for a unique $\theta_Y: G(Y) \rightarrow G'(Y)$, which is an isomorphism because the Yoneda embedding is fully faithful (Lemma 5.4.7). Naturality in Y follows from naturality of φ and φ' in Y and faithfulness of the Yoneda embedding. $\square$

Theorem 5.10.4. A right adjoint preserves all limits that exist, and a left adjoint preserves all colimits that exist.

Proof. Let $F \dashv G$ and let (L, λ) be a limit of $D: \mathcal{I} \rightarrow \mathcal{D}$. For every $X \in \mathcal{C}$ we have bijections, natural in X ,

$$\text{Hom}(X, G(L)) \cong \text{Hom}(F(X), L) \cong \lim_i \text{Hom}(F(X), D(i)) \cong \lim_i \text{Hom}(X, G(D(i))),$$

by Proposition 5.8.7 in the middle. By naturality of φ in Y , the composite sends m to $(G(\lambda_i) \circ m)_i$. So $(G(L), G(\lambda))$ is a limit by Proposition 5.8.7. The statement about left adjoints follows by duality. $\square$

Example 5.10.5. 1. *Free monoids.* For a small set S , let $W(S)$ be the set of finite sequences of elements of S , a monoid under concatenation with the empty sequence as unit. A map $f: S \rightarrow M$ to a monoid extends uniquely to a homomorphism $W(S) \rightarrow M$, namely $(s_1, \dots, s_n) \mapsto f(s_1) \cdots f(s_n)$. This gives an adjunction $W \dashv U$, where $U: \mathbf{Mon} \rightarrow \mathbf{Set}$ is the forgetful functor. Similarly, free groups, free abelian groups and free modules are left adjoints of forgetful functors (Chapters 6 and 7).

2. *Products and maps.* For a small set B , the functor $- \times B$ on $\mathbf{Set}$ is left adjoint to $(-)^B$: the bijection $\text{Hom}(A \times B, C) \cong \text{Hom}(A, C^B)$ of Exercise 3.9.1 is natural in A and C .

3. *Topology.* The forgetful functor $\mathbf{Top} \rightarrow \mathbf{Set}$ has a left adjoint (discrete topology) and a right adjoint (indiscrete topology).

Counterexample 5.10.6. The forgetful functor $U: \mathbf{Field} \rightarrow \mathbf{Set}$ has no left adjoint. A left adjoint L would preserve colimits by Theorem 5.10.4, in particular initial objects, so $L(\emptyset)$ would be an initial field. But an initial field would admit homomorphisms to $\mathbb{Q}$ and to $\mathbb{F}_2$, which is impossible by the argument of Counterexample 5.8.9.

5.11 Filtered colimits

Definition 5.11.1. A category $\mathcal{I}$ is *filtered* if it is nonempty, for all objects i, j there are an object k and morphisms $i \rightarrow k$ and $j \rightarrow k$, and for all morphisms $u, v: i \rightarrow j$ there is a morphism $w: j \rightarrow k$ with $w \circ u = w \circ v$. A *filtered colimit* is a colimit of a diagram whose shape is a small filtered category.

Example 5.11.2. A preordered set I is *directed* if it is nonempty and every two elements have a common upper bound. Viewed as a category (Example 5.2.9), a directed set is filtered, the third condition being automatic. For example $\mathbb{N}$ is directed, and the set of open neighbourhoods of a point in a topological space, ordered by reverse inclusion, is directed; colimits over it define stalks of sheaves (Chapter 23).

Lemma 5.11.3. Let $\mathcal{I}$ be filtered.

1. For finitely many objects $i_1, \dots, i_n$ there is an object k with morphisms $i_m \rightarrow k$.
2. For finitely many morphisms $u_1, \dots, u_n: i \rightarrow j$ there is $w: j \rightarrow k$ with $wu_1 = wu_2 = \dots = wu_n$.
3. For finitely many morphisms $u_m: i \rightarrow j_m$, $1 \leq m \leq n$, there are an object k and morphisms $w_m: j_m \rightarrow k$ with $w_1u_1 = \dots = w_nu_n$.

Proof. (1) and (2) follow by induction on n from the definition; for (2), if $wu_1 = \dots = wu_{n-1}$ with $w: j \rightarrow k$, choose $w': k \rightarrow k'$ with $w'wu_{n-1} = w'wu_n$. (3) By (1) choose k' and $v_m: j_m \rightarrow k'$. The morphisms $v_mu_m: i \rightarrow k'$ are parallel, so by (2) there is $w: k' \rightarrow k$ with all wv_mu_m equal; put $w_m = wv_m$. $\square$

Proposition 5.11.4. Let $\mathcal{I}$ be a small filtered category and $F: \mathcal{I} \rightarrow \mathbf{Set}$. On $\coprod_i F(i)$, the relation

$$(i, x) \approx (j, y) \iff \text{there are } u: i \rightarrow k \text{ and } v: j \rightarrow k \text{ with } F(u)(x) = F(v)(y)$$

is an equivalence relation, and $\text{colim } F = (\coprod_i F(i)) / \approx$. In particular every element of $\text{colim } F$ is of the form $\gamma_i(x)$, and $\gamma_i(x) = \gamma_i(x')$ if and only if $F(u)(x) = F(u)(x')$ for some $u: i \rightarrow k$.

Proof. Reflexivity and symmetry are clear. For transitivity, let $F(u)(x) = F(v)(y)$ with $u: i \rightarrow k$, $v: j \rightarrow k$, and $F(v')(y) = F(w')(z)$ with $v': j \rightarrow k'$, $w': l \rightarrow k'$. By Lemma 5.11.3(3) applied to v and v' , there are $a: k \rightarrow m$ and $b: k' \rightarrow m$ with $av = bv'$. Then $F(au)(x) = F(av)(y) = F(bv')(y) = F(bw')(z)$. The relation $\approx$ contains the generating relation of Proposition 5.9.3 (take $v = \text{id}$), and it is contained in every equivalence relation containing it, since $(i, x) \sim (k, F(u)(x)) = (k, F(v)(y)) \sim (j, y)$. So it is the smallest one. For the last statement, if $F(u)(x) = F(v)(x')$ with $u, v: i \rightarrow k$, choose w with $wu = wv$ by filteredness; then $F(wu)(x) = F(wv)(x')$. $\square$

Theorem 5.11.5. *Let $\mathcal{I}$ be a small filtered category, $\mathcal{J}$ a category with finitely many morphisms, and $F: \mathcal{I} \times \mathcal{J} \rightarrow \mathbf{Set}$ a functor. Then the canonical map*

$$\kappa: \operatorname{colim}_{i \in \mathcal{I}} \lim_{j \in \mathcal{J}} F(i, j) \longrightarrow \lim_{j \in \mathcal{J}} \operatorname{colim}_{i \in \mathcal{I}} F(i, j)$$

is a bijection. That is, in $\mathbf{Set}$ filtered colimits commute with finite limits.

Proof. We use the descriptions of Propositions 5.8.5 and 5.11.4. An element of the left side is represented by some i and a family $(x_j)_j$ with $x_j \in F(i, j)$ compatible with all morphisms of $\mathcal{J}$; the map κ sends it to the family of classes $[(i, x_j)]_j$.

Injectivity. Let two elements of the left side be represented by $(x_j) \in \lim_j F(i, j)$ and $(y_j) \in \lim_j F(i', j)$. By Lemma 5.11.3(1) and the functoriality of the limit in i , we may assume $i = i'$. Suppose $[(i, x_j)] = [(i, y_j)]$ for all j . By Proposition 5.11.4, for each of the finitely many objects j there is $u_j: i \rightarrow k_j$ with $F(u_j, j)(x_j) = F(u_j, j)(y_j)$. By Lemma 5.11.3(3) there are $w_j: k_j \rightarrow k$ with all $w_j u_j$ equal to one $u: i \rightarrow k$. Then $F(u, j)(x_j) = F(u, j)(y_j)$ for all j , so the two families have the same image in $\lim_j F(k, j)$ and define the same element of the colimit.

Surjectivity. Let $(\xi_j)_j$ be an element of the right side. Each ξ_j is represented by some $x_j \in F(i_j, j)$. Using Lemma 5.11.3(1) and replacing each x_j by its image, we may assume all i_j equal to one i . For each morphism $a: j \rightarrow j'$ of $\mathcal{J}$, compatibility of (ξ_j) says that $F(i, a)(x_j)$ and $x_{j'}$ have the same class in $\operatorname{colim}_i F(i, j')$, so there is $u_a: i \rightarrow k_a$ with $F(u_a, j')(F(i, a)(x_j)) = F(u_a, j')(x_{j'})$. There are finitely many a ; by Lemma 5.11.3(3) we find $w_a: k_a \rightarrow k$ with all $w_a u_a$ equal to one $u: i \rightarrow k$. Put $y_j = F(u, j)(x_j) \in F(k, j)$. For every $a: j \rightarrow j'$, $F(k, a)(y_j) = F(u, j')(F(i, a)(x_j)) = F(w_a, j')F(u_a, j')(F(i, a)(x_j)) = F(u, j')(x_{j'}) = y_{j'}$, using that F is a functor on $\mathcal{I} \times \mathcal{J}$. So (y_j) lies in $\lim_j F(k, j)$, and its class maps to $[(k, y_j)]_j = (\xi_j)_j$. $\square$

Counterexample 5.11.6. Filtered colimits do not commute with infinite products in $\mathbf{Set}$. Let $\mathcal{I} = \mathbb{N}$, let $\mathcal{J}$ be the discrete category on $\mathbb{N}$, and let $F(n, j) = \{0, 1, \dots, n\}$ with the inclusions as transition maps. Then $\operatorname{colim}_n F(n, j) = \mathbb{N}$ for every j , so the right side is the set of all sequences of natural numbers, whereas the left side $\operatorname{colim}_n \prod_j \{0, \dots, n\}$ consists of the bounded sequences. The sequence $j \mapsto j$ is not in the image of κ .

Counterexample 5.11.7. Colimits that are not filtered do not commute with finite limits either. Take $\mathcal{I}$ discrete with two objects and $\mathcal{J}$ discrete with two objects, and $F(i, j) = \{0\}$ for all i, j . The left side is a coproduct of two products of one-element sets, with 2 elements, whereas the right side is a product of two coproducts of two one-element sets, with 4 elements.

5.12 Additive categories

Definition 5.12.1. A category $\mathcal{A}$ is *preadditive* if every set $\operatorname{Hom}(X, Y)$ has the structure of an abelian group, written additively, such that composition is bilinear: $(f + f') \circ g = f \circ g + f' \circ g$ and $h \circ (f + f') = h \circ f + h \circ f'$. A *zero object* is an object that is both initial and terminal.

A preadditive category is *additive* if it has a zero object and binary products. A functor between preadditive categories is *additive* if the maps $\text{Hom}(X, Y) \rightarrow \text{Hom}(F(X), F(Y))$ are group homomorphisms.

Lemma 5.12.2. *Let $\mathcal{A}$ be preadditive and X_1, X_2, P objects with morphisms $i_m: X_m \rightarrow P$ and $p_m: P \rightarrow X_m$ such that*

$$p_1 i_1 = \text{id}_{X_1}, \quad p_2 i_2 = \text{id}_{X_2}, \quad p_2 i_1 = 0, \quad p_1 i_2 = 0, \quad i_1 p_1 + i_2 p_2 = \text{id}_P.$$

Then (P, p_1, p_2) is a product and (P, i_1, i_2) is a coproduct of X_1 and X_2 . Conversely, if $\mathcal{A}$ has a zero object and (P, p_1, p_2) is a product of X_1 and X_2 , there are unique i_1, i_2 satisfying these equations.

Proof. Given $f_m: Y \rightarrow X_m$, the morphism $f = i_1 f_1 + i_2 f_2$ satisfies $p_m f = f_m$. If g also does, then $g = (i_1 p_1 + i_2 p_2)g = i_1 f_1 + i_2 f_2$. So P is a product; the coproduct statement is dual. Conversely, let $i_1: X_1 \rightarrow P$ be the unique morphism with $p_1 i_1 = \text{id}$ and $p_2 i_1 = 0$, and similarly i_2 (the zero morphism $0 \in \text{Hom}(X, Y)$ exists since Hom sets are groups). Then $p_m(i_1 p_1 + i_2 p_2) = p_m$ for $m = 1, 2$, so $i_1 p_1 + i_2 p_2 = \text{id}_P$ by the uniqueness in the universal property of the product. $\square$

Definition 5.12.3. An object P with morphisms as in Lemma 5.12.2 is a *biproduct* or *direct sum* of X_1 and X_2 , written $X_1 \oplus X_2$. In a preadditive category with a zero object, the *kernel* of $f: X \rightarrow Y$ is an equaliser of f and 0 , and its *cokernel* is a coequaliser of f and 0 .

Proposition 5.12.4. *Additive functors preserve biproducts and zero objects.*

Proof. The equations of Lemma 5.12.2 involve only composition, sums, identities and zero morphisms, all preserved by an additive functor F , so F sends a biproduct to a biproduct. An object Z is a zero object if and only if $\text{id}_Z = 0$ (then every $\text{Hom}(Z, Y)$ and $\text{Hom}(Y, Z)$ is the zero group), and this is preserved by F . $\square$

Example 5.12.5. The categories of abelian groups, of modules over a ring and of vector spaces over a field are additive, with the direct sum as biproduct (Chapter 7).

Counterexample 5.12.6. The category Set admits no preadditive structure: in a preadditive category every $\text{Hom}(X, Y)$ is nonempty, since it contains 0 , but $\text{Hom}_{\text{Set}}(\{0\}, \emptyset) = \emptyset$.

5.13 Exercises

Exercise 5.13.1. Show that a composition of faithful functors is faithful, and that a composition of full functors is full.

Solution. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ and $G: \mathcal{D} \rightarrow \mathcal{E}$. The map $\text{Hom}_{\mathcal{C}}(X, Y) \rightarrow \text{Hom}_{\mathcal{E}}(GF(X), GF(Y))$ induced by $G \circ F$ is the composition of the maps induced by F and by G . A composition of injective maps is injective, and a composition of surjective maps is surjective. $\square$

Exercise 5.13.2. Let $f: X \rightarrow Y$ be a morphism in a category. Show that if there is $r: Y \rightarrow X$ with $r \circ f = \text{id}_X$, then f is a monomorphism, and if there is $s: Y \rightarrow X$ with $f \circ s = \text{id}_Y$, then f is an epimorphism. Show that in Set every monomorphism with non-empty source has such an r . Show that the statement “every epimorphism in Set has such an s ” is equivalent to the axiom of choice (for small sets).

Exercise 5.13.3. Let P and Q be preorders, viewed as categories, and let $F, G: P \rightarrow Q$ be functors. Show that there is a natural transformation $F \Rightarrow G$ if and only if $F(x) \leq G(x)$ for all $x \in P$, and that it is then unique.

Solution. Every set of morphisms of Q has at most one element, so a family $(\eta_x: F(x) \rightarrow G(x))_x$ exists if and only if $F(x) \leq G(x)$ for all x , it is unique if it exists, and every square of morphisms of Q commutes. $\square$

Exercise 5.13.4. State and prove the analogue of Theorem 5.6.1 for functors $F: C \rightarrow \mathbf{Set}$ and the functors h^X of Definition 5.4.4. (Hint: apply the theorem to C^{op} .)

Exercise 5.13.5. Suppose (X, ξ) and (X', ξ') both represent a functor $F: C^{\text{op}} \rightarrow \mathbf{Set}$. Show that there is a unique morphism $a: X \rightarrow X'$ with $F(a)(\xi') = \xi$, and that a is an isomorphism.

Exercise 5.13.6. Show that the inclusion $\mathbb{Z} \rightarrow \mathbb{Q}$ is an epimorphism in the category of rings, although it is not surjective.

Exercise 5.13.7. Show that in $\mathbf{Set}$ a morphism that is both a monomorphism and an epimorphism is an isomorphism, and that the same holds in the category of groups for monomorphisms that are surjective.

Exercise 5.13.8. Let P be a partially ordered set viewed as a category. Show that a product of a family of elements is a greatest lower bound, a coproduct is a least upper bound, and that every diagram in P has a limit if and only if every subset of P has a greatest lower bound.

Solution. A cone over a family (x_i) with vertex y exists if and only if $y \leq x_i$ for all i , and it is unique; a limit is a lower bound y such that every lower bound z satisfies $z \leq y$. Coproducts are dual. For a general diagram $F: I \rightarrow P$ the cone condition only involves the objects, so a limit of F is a greatest lower bound of $\{F(i)\}$. $\square$

Exercise 5.13.9. Let $F \dashv G$ with unit η and counit ε . Show that F is fully faithful if and only if η is a natural isomorphism, and that F is an equivalence if and only if both η and ε are natural isomorphisms. Conversely, show that every equivalence is part of an adjunction whose unit and counit are isomorphisms.

Exercise 5.13.10. Let I, J be small and $F: I \times J \rightarrow C$, where C has limits. Show that $\lim_i \lim_j F(i, j) \cong \lim_{(i, j)} F(i, j) \cong \lim_j \lim_i F(i, j)$.

Exercise 5.13.11. Show that a functor $G: \mathcal{D} \rightarrow \mathbf{Set}$ is representable (in the covariant sense of Exercise 5.13.4) if and only if it has a left adjoint F with $F(\{0\}) \cong X$ for some X , provided $\mathcal{D}$ has coproducts indexed by small sets. (Hint: $F(S) = \coprod_{s \in S} X$.)

Exercise 5.13.12. In an additive category, show that $f + g$ equals the composite $X \rightarrow X \oplus X \rightarrow Y \oplus Y \rightarrow Y$ of the diagonal, $f \oplus g$ and the codiagonal. Deduce that the abelian group structure on Hom sets of an additive category is determined by the category itself.

Part II

Algebra

Chapter 6

Groups

6.1 Introduction

Groups describe symmetry. This chapter develops the basic theory: subgroups, quotients and the isomorphism theorems, group actions and the Sylow theorems, symmetric and alternating groups, products, solvable and nilpotent groups, free groups and presentations, and the structure of finitely generated abelian groups. Groups appear throughout the project: as fundamental groups in topology (Chapter 17), as Galois groups (Chapter 8), as structure groups of vector bundles, and as Mumford–Tate groups of Hodge structures (Chapter 63).

References are [Lan02], [Rot95] and [DF04]; for free groups and amalgamated products see [Ser80] and [LS77].

Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems) and Chapter 5 (Categories).

6.2 Groups and homomorphisms

Definition 6.2.1. A *group* is a set G together with a map $G \times G \rightarrow G$, called the *group law* and written $(a, b) \mapsto a \cdot b$, and an element $e \in G$, such that for all $a, b, c \in G$:

1. (associativity) $(a \cdot b) \cdot c = a \cdot (b \cdot c)$;
2. (neutral element) $e \cdot a = a \cdot e = a$;
3. (inverses) there is an element $a' \in G$ with $a \cdot a' = a' \cdot a = e$.

The group is *abelian* if $a \cdot b = b \cdot a$ for all $a, b \in G$. The *order* of G is its cardinality $|G|$.

Notation 6.2.2. Two notations are used for the group law.

- *Multiplicative notation*, used for all groups: the group law is written $a \cdot b$ or simply ab , the neutral element e (or 1), the inverse of a is written a^{-1} (it is unique by Lemma 6.2.3), and a^n denotes the n -th power (Lemma 6.2.9).
- *Additive notation*, used only for abelian groups: the group law is written $a + b$, the neutral element 0, the inverse $-a$, and na the n -th multiple.

Every statement in multiplicative notation applies to abelian groups written additively through the dictionary $ab \leftrightarrow a + b$, $e \leftrightarrow 0$, $a^{-1} \leftrightarrow -a$, $a^n \leftrightarrow na$. General results in this chapter are stated multiplicatively. For groups arising from a familiar structure we keep the familiar symbol: $+$ for $(\mathbb{Z}, +)$, multiplication for $\mathbb{Q}^\times$ or for invertible matrices, and composition

$\circ$ (written as juxtaposition) for groups of bijections. The symbol $*$ never denotes a group law; $G_1 * G_2$ is the free product of Definition 6.10.6. When the group law has to be named, we write $(G, \cdot)$ or $(G, +)$.

Lemma 6.2.3. *In a group the neutral element and inverses are unique, $(ab)^{-1} = b^{-1}a^{-1}$, $(a^{-1})^{-1} = a$, and $ab = ac$ or $ba = ca$ implies $b = c$.*

Proof. If e' is also neutral, $e = ee' = e'$. If b and c are inverses of a , then $b = b(ac) = (ba)c = c$; we write a^{-1} for the inverse. The element $b^{-1}a^{-1}$ is inverse to ab , and a is inverse to a^{-1} . Cancellation follows by multiplying with a^{-1} . $\square$

Example 6.2.4. 1. $(\mathbb{Z}, +)$, $(\mathbb{Q}, +)$, $(\mathbb{R}, +)$ and $(\mathbb{C}, +)$ are abelian groups, as are $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$, $\mathbb{R}^\times$ and $\mathbb{C}^\times$ under multiplication. More generally, for a commutative ring R the invertible elements form a group $R^\times$.

2. For a set X , the bijections $X \rightarrow X$ form the *symmetric group* $\text{Sym}(X)$ under composition. We write $S_n = \text{Sym}(\{1, \dots, n\})$.

3. For a field k , the invertible $n \times n$ matrices form the group $\text{GL}_n(k)$ (Chapter 9).

4. The *trivial group* has one element.

5. A group with one object as a category (Example 5.2.8) is a category with one object in which every morphism is an isomorphism.

Definition 6.2.5. A *homomorphism* of groups is a map $f: G \rightarrow H$ with $f(ab) = f(a)f(b)$ for all $a, b \in G$. Its *kernel* is $\ker f = \{a \in G : f(a) = e\}$ and its *image* is $\text{im } f = f[G]$. An *isomorphism* is a bijective homomorphism; an *automorphism* of G is an isomorphism $G \rightarrow G$, and they form a group $\text{Aut}(G)$. Small groups and homomorphisms form a locally small category Grp , with the full subcategory Ab of abelian groups.

Lemma 6.2.6. *A homomorphism f satisfies $f(e) = e$ and $f(a^{-1}) = f(a)^{-1}$; it is injective if and only if $\ker f = \{e\}$; and the inverse of a bijective homomorphism is a homomorphism.*

Proof. $f(e) = f(e)f(e)$ gives $f(e) = e$ by cancellation, and then $f(a)f(a^{-1}) = f(e) = e$. If $\ker f$ is trivial and $f(a) = f(b)$, then $f(ab^{-1}) = e$, so $a = b$. If f is bijective with inverse g , then $f(g(x)g(y)) = xy$, so $g(xy) = g(x)g(y)$. $\square$

Definition 6.2.7. A *subgroup* of G is a subset H containing e with $ab \in H$ and $a^{-1} \in H$ for all $a, b \in H$; it is a group with the restricted operation. We write $H \leq G$. For a subset $S \subseteq G$, the *subgroup generated by S* , written $\langle S \rangle$, is the intersection of all subgroups containing S .

Lemma 6.2.8. *The subgroup $\langle S \rangle$ consists of all finite products $s_1^{\epsilon_1} \cdots s_k^{\epsilon_k}$ with $k \geq 0$, $s_i \in S$ and $\epsilon_i \in \{1, -1\}$ (the empty product being e). Kernels and images of homomorphisms are subgroups.*

Proof. The set of such products contains S and is closed under products and inverses (Lemma 6.2.3), so it contains $\langle S \rangle$; and every subgroup containing S contains all such products. The second statement follows from Lemma 6.2.6. $\square$

Lemma 6.2.9. *Let G be a group and $g \in G$. There is a unique homomorphism $\mathbb{Z} \rightarrow G$ sending 1 to g ; we write $n \mapsto g^n$. Thus $g^{m+n} = g^m g^n$ and $(g^m)^n = g^{mn}$ for all $m, n \in \mathbb{Z}$.*

Proof. Uniqueness is clear since 1 generates $\mathbb{Z}$. For existence define g^n for $n \in \mathbb{N}$ by recursion ($g^0 = e$, $g^{n+1} = g^n g$) and put $g^{-n} = (g^{-1})^n$. One checks $g^{m+1} = g^m g$ for all $m \in \mathbb{Z}$ (for $m = -n - 1 < 0$ this reads $(g^{-1})^n = (g^{-1})^{n+1} g$, which holds since $g^{-1}g = e$), and then $g^{m+n} = g^m g^n$ follows by induction on $n \geq 0$ and on $-n \geq 0$. The last formula holds because $n \mapsto g^{mn}$ and $n \mapsto (g^m)^n$ are both homomorphisms sending 1 to g^m . $\square$

Proposition 6.2.10. Every subgroup of $\mathbb{Z}$ is of the form $n\mathbb{Z} = \{nk : k \in \mathbb{Z}\}$ for a unique $n \in \mathbb{N}$.

Proof. Let $H \leq \mathbb{Z}$. If $H = \{0\}$ take $n = 0$. Otherwise H contains a positive element, and we let n be the least one. Then $n\mathbb{Z} \subseteq H$, and for $h \in H$, division with remainder (Theorem 4.5.2) gives $h = qn + r$ with $0 \leq r < n$ and $r = h - qn \in H$, so $r = 0$. The number n is the least positive element of $n\mathbb{Z}$, hence unique. $\square$

Definition 6.2.11. The order of $g \in G$ is the least $n \geq 1$ with $g^n = e$ if it exists, and ∞ otherwise. By Proposition 6.2.10, the kernel of $n \mapsto g^n$ is $\text{ord}(g)\mathbb{Z}$ (with $\infty\mathbb{Z} := \{0\}$). A group is *cyclic* if it is generated by one element.

6.3 Cosets and Lagrange's theorem

Definition 6.3.1. Let $H \leq G$. A *left coset* of H is a set $gH = \{gh : h \in H\}$, and a *right coset* is a set Hg . The set of left cosets is G/H , and the *index* of H is $[G : H] = |G/H|$.

Lemma 6.3.2. The left cosets of H are the classes of the equivalence relation $a \sim b \iff a^{-1}b \in H$; hence they partition G . Each left coset is in bijection with H , and $gH \mapsto Hg^{-1}$ is a bijection from left to right cosets.

Proof. The relation is an equivalence relation because H is a subgroup, and the class of a is aH . The map $h \mapsto gh$ is a bijection $H \rightarrow gH$. Finally $aH = bH$ iff $a^{-1}b \in H$ iff $(a^{-1}b)^{-1} = b^{-1}a \in H$ iff $Ha^{-1} = Hb^{-1}$. $\square$

Theorem 6.3.3 (Lagrange). Let G be finite and $H \leq G$. Then $|G| = [G : H] \cdot |H|$. In particular the order of H divides the order of G , and the order of every element divides $|G|$, so $g^{|G|} = e$ for all $g \in G$.

Proof. By Lemma 6.3.2, G is the disjoint union of $[G : H]$ sets of size $|H|$. The subgroup $\langle g \rangle$ has order $\text{ord}(g)$, since $g^0, \dots, g^{\text{ord}(g)-1}$ are distinct and every power g^n equals g^r with r the remainder of n . $\square$

Corollary 6.3.4. A group of prime order p is cyclic, generated by any element $\neq e$.

Proof. If $g \neq e$, then $\text{ord}(g) > 1$ divides p , so it equals p , and $\langle g \rangle = G$. $\square$

Corollary 6.3.5 (Euler, Fermat, Wilson). Let $n \geq 2$, and write $a \equiv b \pmod{n}$ if $n \mid a - b$. Let U_n be the set of residue classes modulo n of the integers a with $\gcd(a, n) = 1$, and $\varphi(n) = |U_n|$ the number of $0 \leq a < n$ with $\gcd(a, n) = 1$.

1. Multiplication of representatives makes U_n an abelian group, with identity the class of 1.
2. (Euler) $a^{\varphi(n)} \equiv 1 \pmod{n}$ whenever $\gcd(a, n) = 1$.
3. (Fermat) If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$; and $a^p \equiv a \pmod{p}$ for every integer a .
4. (Wilson) If p is prime, then $(p-1)! \equiv -1 \pmod{p}$.

Proof. (1) If $a \equiv a'$ and $b \equiv b'$, then $ab - a'b' = a(b - b') + b'(a - a')$ is divisible by n , so the product of classes is well defined; it is associative and commutative because multiplication of integers is. If $\gcd(a, n) = \gcd(b, n) = 1$ then $\gcd(ab, n) = 1$: a prime dividing n and ab divides a or b (Lemma 4.5.4). By Bézout (Theorem 4.5.3), $\gcd(a, n) = 1$ gives $ax + ny = 1$, so the class of x , which is again coprime to n , is an inverse of the class of a .

(2) By Theorem 6.3.3 applied to the group U_n of order $\varphi(n)$.

(3) For $n = p$ prime, every a with $0 < a < p$ is coprime to p , so $\varphi(p) = p - 1$ and (2) applies. Multiplying by a gives $a^p \equiv a$, which also holds trivially when $p \mid a$.

(4) An element x of U_p is its own inverse exactly when $p \mid x^2 - 1 = (x - 1)(x + 1)$, that is, by Lemma 4.5.4, when $x \equiv \pm 1$. The remaining elements of U_p fall into pairs $\{x, x^{-1}\}$ with product 1. So the product of all elements of U_p , which is the class of $(p - 1)!$, is the class of $1 \cdot (-1) = -1$ (for $p = 2$, $1 \equiv -1$ and the statement is clear). $\square$

Example 6.3.6. $\varphi(12) = 4$, with $U_{12} = \{1, 5, 7, 11\}$, and every element squares to 1; so U_{12} is not cyclic, although U_{12} has order 4 and $U_5 = \{1, 2, 3, 4\}$, also of order 4, is cyclic, generated by 2 ($2^2 = 4$, $2^3 = 8 \equiv 3$, $2^4 = 16 \equiv 1$). Fermat's little theorem gives, for instance, $2^{100} = (2^{10})^{10} \equiv 1 \pmod{11}$. Its converse fails: $561 = 3 \cdot 11 \cdot 17$ is not prime, yet $a^{560} \equiv 1 \pmod{561}$ for all a coprime to 561, because 2, 10 and 16 divide 560, so Fermat's theorem gives the congruence modulo 3, 11 and 17, hence modulo their product, the three primes being distinct (Theorem 4.5.5). Such numbers are called *Carmichael numbers*.

Proposition 6.3.7. *If $K \leq H \leq G$ and the indices are finite, then $[G : K] = [G : H][H : K]$.*

Proof. Let (g_i) represent the left cosets of H in G and (h_j) those of K in H . Then the $g_i h_j K$ are the left cosets of K in G , and they are distinct: $G = \bigsqcup_i g_i H = \bigsqcup_i \bigsqcup_j g_i h_j K$. $\square$

6.4 Normal subgroups and quotients

Definition 6.4.1. A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for all $g \in G$, equivalently $gN = Ng$ for all g . A group $G \neq \{e\}$ is *simple* if its only normal subgroups are $\{e\}$ and G .

Lemma 6.4.2. *Kernels of homomorphisms are normal. Every subgroup of an abelian group is normal. A subgroup of index 2 is normal.*

Proof. If $f(n) = e$ then $f(gng^{-1}) = f(g)f(g)^{-1} = e$. The second claim is clear. If $[G : H] = 2$, then for $g \notin H$ both gH and Hg are the complement of H . $\square$

Theorem 6.4.3. *Let $N \trianglelefteq G$. The rule $(aN)(bN) = abN$ is a well-defined operation making G/N a group, the quotient group, and the projection $\pi: G \rightarrow G/N$ is a surjective homomorphism with kernel N . If $f: G \rightarrow H$ is a homomorphism with $N \subseteq \ker f$, there is a unique homomorphism $\bar{f}: G/N \rightarrow H$ with $\bar{f} \circ \pi = f$.*

Proof. If $aN = a'N$ and $bN = b'N$, write $a' = an$, $b' = bm$. Then $a'b' = anbm = ab(b^{-1}nb)m \in abN$ since N is normal. The group axioms follow from those of G , with neutral element N and $(aN)^{-1} = a^{-1}N$. Clearly π is a surjective homomorphism, and $\pi(a) = N$ iff $a \in N$. If $N \subseteq \ker f$ then f is constant on cosets, and $\bar{f}$ exists and is unique by Proposition 3.3.9; it is a homomorphism because π is surjective. $\square$

Theorem 6.4.4 (Isomorphism theorems). 1. *If $f: G \rightarrow H$ is a homomorphism, $\bar{f}: G/\ker f \rightarrow \text{im } f$ is an isomorphism.*

2. *If $H \leq G$ and $N \trianglelefteq G$, then HN is a subgroup, $N \trianglelefteq HN$, $H \cap N \trianglelefteq H$, and $H/(H \cap N) \cong HN/N$.*

3. *If $N \leq M$ are normal subgroups of G , then $M/N \trianglelefteq G/N$ and $(G/N)/(M/N) \cong G/M$.*

4. (Correspondence theorem) *If $N \trianglelefteq G$, then $K \mapsto K/N$ is a bijection from the set of subgroups of G containing N to the set of subgroups of G/N , preserving inclusions, normality and indices.*

Proof. (1) $\bar{f}$ is surjective onto $\text{im } f$ and has trivial kernel. (2) $HN = NH$ because $hN = Nh$, so HN is closed under products and inverses. The composite $H \rightarrow HN \rightarrow HN/N$ is surjective, since $hnN = hN$, with kernel $H \cap N$; apply (1). (3) The map $G/N \rightarrow G/M$, $gN \mapsto gM$, is a well-defined surjective homomorphism (Theorem 6.4.3) with kernel M/N ; apply (1). (4) The inverse map sends $\bar{K} \leq G/N$ to $\pi^{-1}[\bar{K}]$; since π is surjective, $\pi[\pi^{-1}[\bar{K}]] = \bar{K}$, and $\pi^{-1}[\pi[K]] = KN = K$ when $N \subseteq K$. Normality is preserved because π is surjective, and $gK \mapsto (gN)(K/N)$ is a bijection $G/K \rightarrow (G/N)/(K/N)$. $\square$

Example 6.4.5. For $n \geq 1$, the quotient $\mathbb{Z}/n\mathbb{Z}$ is a cyclic group of order n , with elements the classes of $0, 1, \dots, n-1$. If $G = \langle g \rangle$ is cyclic, the homomorphism $\mathbb{Z} \rightarrow G$, $k \mapsto g^k$ is surjective with kernel $\text{ord}(g)\mathbb{Z}$, so $G \cong \mathbb{Z}$ if g has infinite order and $G \cong \mathbb{Z}/n\mathbb{Z}$ if $\text{ord}(g) = n$. Thus a cyclic group is determined up to isomorphism by its order. By the correspondence theorem the subgroups of $\mathbb{Z}/n\mathbb{Z}$ are the $d\mathbb{Z}/n\mathbb{Z}$ for $d \mid n$, one for each divisor.

Definition 6.4.6. The *center* of G is $Z(G) = \{z \in G : zg = gz \text{ for all } g\}$. For $g \in G$, *conjugation* $c_g(x) = gxg^{-1}$ is an automorphism; automorphisms of this form are *inner*. The *centraliser* of x is $C_G(x) = \{g : gx = xg\}$ and the *normaliser* of $H \leq G$ is $N_G(H) = \{g : gHg^{-1} = H\}$.

Lemma 6.4.7. $Z(G)$ is a normal subgroup, and $g \mapsto c_g$ is a homomorphism $G \rightarrow \text{Aut}(G)$ with kernel $Z(G)$. If $G/Z(G)$ is cyclic, then G is abelian.

Proof. $c_{gh} = c_g \circ c_h$, and $c_g = \text{id}$ iff $g \in Z(G)$. If $G/Z(G)$ is generated by $gZ(G)$, every element of G is $g^k z$ with $z \in Z(G)$, and such elements commute with each other. $\square$

6.5 Products

Proposition 6.5.1. For a family of groups $(G_i)_{i \in I}$, the set $\prod_i G_i$ with the componentwise operation is a group, and with the projections it is a product in Grp (Example 5.8.3). Let $N, M \trianglelefteq G$ with $N \cap M = \{e\}$ and $NM = G$. Then $nm = mn$ for $n \in N$, $m \in M$, and $N \times M \rightarrow G$, $(n, m) \mapsto nm$, is an isomorphism.

Proof. The first statement is clear. For $n \in N$ and $m \in M$, the commutator $nmn^{-1}m^{-1}$ lies in N (as $mn^{-1}m^{-1} \in N$) and in M (as $nmn^{-1} \in M$), hence is e . So the map is a homomorphism; it is surjective since $NM = G$, and injective since $nm = e$ implies $n = m^{-1} \in N \cap M$. $\square$

Definition 6.5.2. Let N and H be groups and $\varphi: H \rightarrow \text{Aut}(N)$, $h \mapsto \varphi_h$, a homomorphism. The *semidirect product* $N \rtimes_{\varphi} H$ is the set $N \times H$ with the operation $(n, h)(n', h') = (n\varphi_h(n'), hh')$.

Proposition 6.5.3. $N \rtimes_{\varphi} H$ is a group, in which $N \times \{e\}$ is a normal subgroup isomorphic to N , $\{e\} \times H$ is a subgroup isomorphic to H , and $(e, h)(n, e)(e, h)^{-1} = (\varphi_h(n), e)$. Conversely, if G is a group with $N \trianglelefteq G$ and $H \leq G$ such that $N \cap H = \{e\}$ and $NH = G$, then $(n, h) \mapsto nh$ is an isomorphism $N \rtimes_{\varphi} H \rightarrow G$, where $\varphi_h(n) = hnh^{-1}$.

Proof. Associativity: $((n, h)(n', h'))(n'', h'') = (n\varphi_h(n')\varphi_{hh'}(n''), hh'h'')$ and $(n, h)((n', h')(n'', h'')) = (n\varphi_h(n'\varphi_{h'}(n'')), hh'h'')$ agree since φ is a homomorphism into $\text{Aut}(N)$. The neutral element is (e, e) and $(n, h)^{-1} = (\varphi_{h^{-1}}(n^{-1}), h^{-1})$. The projection $(n, h) \mapsto h$ is a homomorphism with kernel $N \times \{e\}$, and the conjugation formula is a direct computation. For the converse, every $g \in G$ is uniquely nh (if $nh = n'h'$ then $n'^{-1}n = h'h^{-1} \in N \cap H$), and $nh \cdot n'h' = n(hn'h^{-1})hh'$, which is the product rule. $\square$

Example 6.5.4. For $n \geq 3$ let $\varphi: \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Aut}(\mathbb{Z}/n\mathbb{Z})$ send the generator to $x \mapsto -x$. The group $D_n = \mathbb{Z}/n\mathbb{Z} \rtimes_{\varphi} \mathbb{Z}/2\mathbb{Z}$ of order $2n$ is the *dihedral group*; it is the group of symmetries of a regular n -gon (rotations and reflections). It is not abelian, since $(0, 1)(1, 0) = (-1, 1) \neq (1, 1) = (1, 0)(0, 1)$.

6.6 Group actions

Definition 6.6.1. An *action* of a group G on a set X is a map $G \times X \rightarrow X$, $(g, x) \mapsto gx$, with $ex = x$ and $g(hx) = (gh)x$; equivalently a homomorphism $G \rightarrow \text{Sym}(X)$. The *orbit* of x is $Gx = \{gx : g \in G\}$, its *stabiliser* is $G_x = \{g : gx = x\}$, and $X^G = \{x : gx = x \text{ for all } g\}$ is the set of *fixed points*. The action is *transitive* if there is only one orbit, and *free* if all stabilisers are trivial.

Theorem 6.6.2 (Orbit–stabiliser). *For an action of G on X and $x \in X$, the stabiliser G_x is a subgroup and $gG_x \mapsto gx$ is a bijection $G/G_x \rightarrow Gx$. In particular $|Gx| = [G : G_x]$, and the orbits partition X .*

Proof. $gG_x = hG_x$ iff $h^{-1}g \in G_x$ iff $gx = hx$, so the map is well defined and injective; it is surjective by definition. The relation “ $y \in Gx$ ” is an equivalence relation since G is a group. $\square$

Example 6.6.3. 1. G acts on itself by left multiplication, freely and transitively. This gives an injective homomorphism $G \rightarrow \text{Sym}(G)$ (*Cayley’s theorem*): every group is isomorphic to a group of permutations.

2. G acts on itself by conjugation; the orbits are the *conjugacy classes* and $G_x = C_G(x)$.
3. G acts on the set of its subgroups by conjugation, with stabiliser $N_G(H)$; so the number of conjugates of H is $[G : N_G(H)]$.
4. If $H \leq G$, then G acts transitively on G/H by $g(aH) = gaH$.

Corollary 6.6.4 (Class equation). *Let G be finite and let $x_1, \dots, x_r$ represent the conjugacy classes with more than one element. Then $|G| = |Z(G)| + \sum_{i=1}^r [G : C_G(x_i)]$.*

Proof. The conjugacy classes partition G ; the one-element classes are the elements of $Z(G)$, and the class of x_i has $[G : C_G(x_i)]$ elements by Theorem 6.6.2. $\square$

Definition 6.6.5. Let p be a prime. A p -group is a finite group whose order is a power of p .

Lemma 6.6.6. *Let a p -group P act on a finite set X . Then $|X| \equiv |X^P| \pmod{p}$.*

Proof. Each orbit that is not a fixed point has size $[P : P_x] > 1$, a divisor of $|P|$, hence divisible by p . $\square$

Proposition 6.6.7. *A nontrivial p -group has nontrivial center. Every group of order p^2 is abelian.*

Proof. Apply Lemma 6.6.6 to the conjugation action: $|Z(G)| \equiv |G| \equiv 0 \pmod{p}$ and $e \in Z(G)$, so $|Z(G)| \geq p$. If $|G| = p^2$, then $|G/Z(G)| \in \{1, p\}$, so $G/Z(G)$ is cyclic (Corollary 6.3.4), and G is abelian by Lemma 6.4.7. $\square$

Theorem 6.6.8 (Cauchy). *If a prime p divides the order of a finite group G , then G has an element of order p .*

Proof. Let $X = \{(g_1, \dots, g_p) \in G^p : g_1 g_2 \cdots g_p = e\}$. Since g_p is determined by the others, $|X| = |G|^{p-1} \equiv 0 \pmod{p}$. The group $\mathbb{Z}/p\mathbb{Z}$ acts on X by cyclic rotation: if $g_1 \cdots g_p = e$ then $g_2 \cdots g_p g_1 = g_1^{-1}(g_1 \cdots g_p)g_1 = e$. The fixed points are the tuples $(g, \dots, g)$ with $g^p = e$. By Lemma 6.6.6 their number is divisible by p , and $(e, \dots, e)$ is one of them, so there is a fixed point with $g \neq e$. $\square$

6.7 The Sylow theorems

Definition 6.7.1. Let G be finite and $|G| = p^a m$ with p prime and $p \nmid m$. A Sylow p -subgroup of G is a subgroup of order p^a . We write $\text{Syl}_p(G)$ for the set of them and $n_p = |\text{Syl}_p(G)|$.

Theorem 6.7.2 (Sylow). Let G be a finite group, p a prime and $|G| = p^a m$ with $p \nmid m$.

1. Sylow p -subgroups exist.
2. Every p -subgroup of G is contained in a Sylow p -subgroup, and any two Sylow p -subgroups are conjugate.
3. n_p divides m and $n_p \equiv 1 \pmod{p}$.

Proof. (1) By induction on $|G|$; if $a = 0$ there is nothing to prove. Suppose $p \mid |Z(G)|$. By Theorem 6.6.8 there is $z \in Z(G)$ of order p ; $N = \langle z \rangle$ is normal, and G/N has order $p^{a-1}m$, so by induction it has a subgroup of order p^{a-1} , whose preimage in G has order p^a . Suppose $p \nmid |Z(G)|$. In the class equation (Corollary 6.6.4) the left side is divisible by p , so some $[G : C_G(x_i)]$ is not divisible by p . Then $C_G(x_i)$ is a proper subgroup of order divisible by p^a , and by induction it contains a subgroup of order p^a .

(2) Let P be a Sylow p -subgroup and Q a p -subgroup. Let Q act on G/P by left multiplication. Since $|G/P| = m \not\equiv 0 \pmod{p}$, Lemma 6.6.6 gives a fixed point gP . Then $QgP = gP$, so $g^{-1}Qg \subseteq P$ and $Q \subseteq gPg^{-1}$, which is a Sylow p -subgroup. If Q is itself Sylow, then $Q = gPg^{-1}$ by comparing orders.

(3) By (2), G acts transitively by conjugation on $\text{Syl}_p(G)$, so $n_p = [G : N_G(P)]$, which divides $[G : P] = m$ since $P \leq N_G(P)$ (Proposition 6.3.7). Let P act on $\text{Syl}_p(G)$ by conjugation. If Q is a fixed point, then $P \leq N_G(Q)$, so P and Q are Sylow p -subgroups of $N_G(Q)$, hence conjugate in $N_G(Q)$ by (2); as Q is normal in $N_G(Q)$, $P = Q$. So P is the only fixed point, and $n_p \equiv 1 \pmod{p}$ by Lemma 6.6.6. $\square$

Corollary 6.7.3. A Sylow p -subgroup is normal if and only if $n_p = 1$.

Proof. By Theorem 6.7.2(2) the Sylow p -subgroups are the conjugates of one of them. $\square$

Example 6.7.4. Every group G of order 15 is cyclic. Indeed n_3 divides 5 and is 1 modulo 3, so $n_3 = 1$; similarly $n_5 = 1$. The Sylow subgroups P_3, P_5 are normal, cyclic of prime order, and $P_3 \cap P_5 = \{e\}$ by Lagrange. Then $|P_3 P_5| = 15$ (the map $P_3 \times P_5 \rightarrow G$ is injective), so $G \cong \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/5\mathbb{Z}$ by Proposition 6.5.1, and the element $(1, 1)$ has order 15.

6.8 Symmetric and alternating groups

Definition 6.8.1. Let $n \geq 1$. For distinct $a_1, \dots, a_k \in \{1, \dots, n\}$, the cycle $(a_1 a_2 \cdots a_k) \in S_n$ sends a_i to a_{i+1} for $i < k$, a_k to a_1 , and fixes all other points. A cycle of length 2 is a *transposition*. Two cycles are *disjoint* if they move disjoint sets of points. Products in S_n are compositions, applied from right to left.

Proposition 6.8.2. Every $\sigma \in S_n$ is a product of disjoint cycles of length ≥ 2 , uniquely up to order; disjoint cycles commute. Every element of S_n is a product of transpositions. For $\tau \in S_n$, $\tau(a_1 \cdots a_k)\tau^{-1} = (\tau(a_1) \cdots \tau(a_k))$, so two permutations are conjugate if and only if they have the same cycle lengths.

Proof. The orbits of $\langle \sigma \rangle$ on $\{1, \dots, n\}$ partition it; on an orbit $\{a, \sigma(a), \dots, \sigma^{k-1}(a)\}$ of size k , σ acts as the cycle $(a \sigma(a) \dots \sigma^{k-1}(a))$, and σ is the product of these cycles for the orbits of size ≥ 2 . Conversely the cycles in any such decomposition must be the orbits, which gives uniqueness. Disjoint cycles commute since they act on disjoint sets. We have $(a_1 \dots a_k) = (a_1 a_k)(a_1 a_{k-1}) \dots (a_1 a_2)$, as one checks by evaluating both sides. The conjugation formula follows by evaluating both sides at $\tau(a_i)$ and at points not of this form. If σ and σ' have the same cycle lengths, a bijection matching their cycles gives τ with $\tau\sigma\tau^{-1} = \sigma'$. $\square$

Proposition 6.8.3. *There is a unique homomorphism $\text{sgn}: S_n \rightarrow \{1, -1\}$ sending every transposition to -1 (for $n \geq 2$). It is given by $\text{sgn}(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}$.*

Proof. For $\sigma \in S_n$ let $s(\sigma)$ be the product in the statement, a rational number. The factors $\sigma(j) - \sigma(i)$ run, up to sign, over all differences $j - i$ with $i < j$, so $s(\sigma) = \pm 1$. For $\sigma, \tau \in S_n$,

$$s(\sigma\tau) = \prod_{i < j} \frac{\sigma(\tau(j)) - \sigma(\tau(i))}{\tau(j) - \tau(i)} \cdot \prod_{i < j} \frac{\tau(j) - \tau(i)}{j - i}.$$

The expression $\frac{\sigma(b) - \sigma(a)}{b - a}$ is symmetric in a and b , and $\{\tau(i), \tau(j)\}$ runs over all two-element subsets as $\{i, j\}$ does; so the first product is $s(\sigma)$, and s is a homomorphism. For the transposition $(1 2)$, the only factor that is negative is the one with $(i, j) = (1, 2)$, so $s((1 2)) = -1$. All transpositions are conjugate by Proposition 6.8.2, hence have sign -1 . Uniqueness holds because transpositions generate S_n . $\square$

Definition 6.8.4. The alternating group A_n is the kernel of sgn . Permutations in A_n are even. For $n \geq 2$, $[S_n : A_n] = 2$.

Lemma 6.8.5. *For $n \geq 3$, the group A_n is generated by the 3-cycles. For $n \geq 5$, any two 3-cycles are conjugate in A_n .*

Proof. A 3-cycle is even, being a product of two transpositions. Every even permutation is a product of an even number of transpositions, and for distinct a, b, c, d we have $(a b)(a c) = (a c b)$ and $(a b)(c d) = (a b c)(b c d)$, as one checks by evaluation. For the second claim, two 3-cycles $(a b c)$ and $(a' b' c')$ are conjugate by some $\tau \in S_n$. If τ is odd, choose $d, e \notin \{a, b, c\}$ distinct (possible since $n \geq 5$); then $\tau(d e)$ is even and also conjugates $(a b c)$ to $(a' b' c')$, because $(d e)$ commutes with $(a b c)$. $\square$

Theorem 6.8.6. *For $n \geq 5$ the group A_n is simple.*

Proof. Let $N \trianglelefteq A_n$ with $N \neq \{e\}$. By Lemma 6.8.5 it suffices to show that N contains a 3-cycle. Choose $\sigma \in N \setminus \{e\}$ with the largest possible number of fixed points. For $\rho \in A_n$, the element $\tau = (\rho\sigma\rho^{-1})\sigma^{-1}$ lies in N .

Suppose σ moves exactly 4 points. Being even, it is then of the form $(a b)(c d)$. Choose $e \notin \{a, b, c, d\}$ and $\rho = (c d e)$. Then $\rho\sigma\rho^{-1} = (a b)(d e)$ and $\tau = (a b)(d e)(a b)(c d) = (d e)(c d) = (c e d)$, a 3-cycle in N .

Suppose σ moves at least 5 points. If σ has a cycle of length ≥ 3 , write it as $(a b c \dots)$ and choose further points d, e moved by σ , distinct from a, b, c . Otherwise σ is a product of at least three disjoint transpositions, and we write $\sigma = (a b)(c d)(e f) \dots$. In both cases let $\rho = (c d e)$. Every point fixed by σ is fixed by ρ , hence by τ ; and $\tau(\sigma(x)) = \rho\sigma\rho^{-1}(x)$. In the first case $\tau(b) = \tau(\sigma(a)) = \rho(\sigma(a)) = \rho(b) = b$, and $\tau(c) = \tau(\sigma(b)) = \rho(\sigma(b)) = \rho(c) = d \neq c$. In the second case $\tau(b) = \rho\sigma(a) = b$, $\tau(a) = \rho\sigma(b) = a$, and $\tau(d) = \tau(\sigma(c)) = \rho\sigma\rho^{-1}(c) = \rho(\sigma(e)) = \rho(f) = f \neq d$. In both cases $\tau \neq e$, τ fixes all points fixed by σ and at least one point moved by σ , contradicting the choice of σ . So this case does not occur. The remaining possibility is that σ moves exactly 3 points (an even permutation cannot move exactly 2), and then σ is a 3-cycle. In all cases N contains a 3-cycle. $\square$

Counterexample 6.8.7. The converse of Lagrange's theorem is false: A_4 has order 12 but no subgroup of order 6. Such a subgroup H would have index 2, so $g^2 \in H$ for all $g \in A_4$ (as G/H has order 2). But every 3-cycle is a square: $(a b c) = (a c b)^2$. So H would contain the eight 3-cycles of A_4 .

Counterexample 6.8.8. Normality is not transitive. In A_4 , the subgroup $V = \{e, (12)(34), (13)(24), (14)(23)\}$ is normal, being a union of conjugacy classes by Proposition 6.8.2. Its subgroup $K = \{e, (12)(34)\}$ is normal in the abelian group V . But $(123)(12)(34)(123)^{-1} = (23)(14) \notin K$, so K is not normal in A_4 .

6.9 Solvable and nilpotent groups

Definition 6.9.1. The *commutator* of $a, b \in G$ is $[a, b] = aba^{-1}b^{-1}$. For subgroups H, K , $[H, K]$ is the subgroup generated by the $[h, k]$. The *derived subgroup* is $G' = [G, G]$, and the *derived series* is $G^{(0)} = G$, $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. The group G is *solvable* if $G^{(n)} = \{e\}$ for some n .

Proposition 6.9.2. $[G, G]$ is a normal subgroup, $G^{\text{ab}} = G/[G, G]$ is abelian, and a normal subgroup N has abelian quotient G/N if and only if $[G, G] \subseteq N$. Every homomorphism from G to an abelian group factors uniquely through G^{ab} ; thus $G \mapsto G^{\text{ab}}$ is left adjoint to the inclusion $\text{Ab} \rightarrow \text{Grp}$.

Proof. $g[a, b]g^{-1} = [gag^{-1}, gbg^{-1}]$, so $[G, G]$ is normal. The quotient G/N is abelian iff $abN = baN$ for all a, b , iff all commutators lie in N . A homomorphism f to an abelian group kills all commutators, so it factors uniquely by Theorem 6.4.3. The last statement is the resulting natural bijection $\text{Hom}_{\text{Grp}}(G, A) \cong \text{Hom}_{\text{Ab}}(G^{\text{ab}}, A)$. $\square$

Proposition 6.9.3. 1. G is solvable if and only if there is a chain $\{e\} = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = G$ with all G_i/G_{i+1} abelian.

2. Subgroups and quotients of solvable groups are solvable. If $N \trianglelefteq G$ and both N and G/N are solvable, then G is solvable.

3. p -groups are solvable, and S_n is not solvable for $n \geq 5$.

Proof. (1) The derived series is such a chain. Conversely, given a chain, induction shows $G^{(i)} \subseteq G_i$, because G_i/G_{i+1} abelian gives $[G_i, G_i] \subseteq G_{i+1}$ (Proposition 6.9.2). (2) For $H \leq G$, $H^{(i)} \subseteq G^{(i)}$. For a surjection $f: G \rightarrow Q$, $f[G^{(i)}] = Q^{(i)}$. If $(G/N)^{(n)}$ is trivial then $G^{(n)} \subseteq N$, and then $G^{(n+m)} \subseteq N^{(m)}$ is trivial for large m . (3) By induction on $|G|$, using (2) and Proposition 6.6.7: $Z(G)$ is abelian and nontrivial, and $G/Z(G)$ is a smaller p -group. For $n \geq 5$, A_n is simple and not abelian, so $[A_n, A_n] = A_n$ and A_n , hence S_n , is not solvable. $\square$

Example 6.9.4. S_4 is solvable: $\{e\} \trianglelefteq V \trianglelefteq A_4 \trianglelefteq S_4$, with V as in Counterexample 6.8.8, has quotients of orders 4, 3 and 2, which are abelian by Proposition 6.6.7. The failure of solvability of S_n for $n \geq 5$ is the group-theoretic reason why polynomial equations of degree ≥ 5 cannot in general be solved by radicals (Chapter 8).

Definition 6.9.5. The *upper central series* of G is defined by $Z_0 = \{e\}$ and $Z_{i+1}/Z_i = Z(G/Z_i)$ (using the correspondence theorem). The group is *nilpotent* if $Z_n = G$ for some n .

Proposition 6.9.6. p -groups are nilpotent, and nilpotent groups are solvable. The group S_3 is solvable but not nilpotent.

Proof. If G is a p -group and $Z_i \neq G$, then G/Z_i is a nontrivial p -group, so $Z_{i+1} \supsetneq Z_i$ by Proposition 6.6.7; hence $Z_n = G$ eventually. The chain $Z_0 \trianglelefteq Z_1 \trianglelefteq \cdots$ has abelian quotients, so nilpotent groups are solvable by Proposition 6.9.3. S_3 is solvable via $\{e\} \trianglelefteq A_3 \trianglelefteq S_3$, but $Z(S_3) = \{e\}$ (a non-identity permutation of $\{1, 2, 3\}$ does not commute with every transposition), so $Z_i = \{e\}$ for all i . $\square$

Definition 6.9.7. A *composition series* of a group G is a chain $\{e\} = G_n \trianglelefteq G_{n-1} \trianglelefteq \cdots \trianglelefteq G_0 = G$ in which every factor G_i/G_{i+1} is simple; the factors are the *composition factors* and n is the *length*.

Theorem 6.9.8 (Jordan–Hölder). *Every finite group has a composition series, and any two composition series of a finite group have the same length and the same composition factors up to isomorphism and permutation.*

Proof. Existence: by induction on $|G|$; if $G \neq \{e\}$, a normal subgroup $N \neq G$ of largest order has G/N simple by the correspondence theorem, and N has a composition series.

Uniqueness, by induction on $|G|$. Let (G_i) and (H_j) be composition series of lengths n and m . If $G_1 = H_1$, apply induction to G_1 . Otherwise $H_1 \not\subseteq G_1$: if $H_1 \subsetneq G_1$, then G_1/H_1 would be a nontrivial proper normal subgroup of the simple group G/H_1 . So G_1H_1 is a normal subgroup strictly containing G_1 , and $G_1H_1 = G$ because G/G_1 is simple. Let $K = G_1 \cap H_1$. By Theorem 6.4.4(2), $G_1/K \cong G/H_1$ and $H_1/K \cong G/G_1$, both simple. Choose a composition series of K , of length l . Then $G_1 \supseteq K \supseteq \cdots$ is a composition series of G_1 of length $l + 1$; by induction applied to G_1 it has the same length and factors as $G_1 \supseteq G_2 \supseteq \cdots$, so $n - 1 = l + 1$. Similarly $m - 1 = l + 1$. The factors of (G_i) are G/G_1 , $G_1/K \cong G/H_1$ and those of K ; the factors of (H_j) are G/H_1 , $H_1/K \cong G/G_1$ and those of K . These agree up to permutation. $\square$

6.10 Free groups and presentations

Definition 6.10.1. Let S be a set and $A = S \times \{1, -1\}$, whose elements we write s and s^{-1} ; for $x = s^{\pm 1}$ write $x^{-1} = s^{\mp 1}$. A *word* is a finite sequence $x_1 \cdots x_k$ of elements of A , and it is *reduced* if no two adjacent letters are of the form xx^{-1} . Let $F(S)$ be the set of reduced words.

Theorem 6.10.2. *There is a unique group structure on $F(S)$ in which the product of reduced words u and v is obtained from the concatenation uv by repeatedly deleting adjacent pairs xx^{-1} . With the map $\iota: S \rightarrow F(S)$ sending s to the one-letter word, $F(S)$ is the free group on S : for every group G and map $f: S \rightarrow G$ there is a unique homomorphism $\bar{f}: F(S) \rightarrow G$ with $\bar{f} \circ \iota = f$. So F is left adjoint to the forgetful functor $\text{Grp} \rightarrow \text{Set}$.*

Proof. For $x \in A$ define $L_x: F(S) \rightarrow F(S)$ by $L_x(w) = xw$ if w does not begin with x^{-1} , and $L_x(x^{-1}w') = w'$. Then $L_x L_{x^{-1}} = \text{id}$ for all x , so each L_x is a bijection, and $L_{x^{-1}} = L_x^{-1}$. Let $\Phi \leq \text{Sym}(F(S))$ be the subgroup generated by the L_x , and $\epsilon: \Phi \rightarrow F(S)$, $\pi \mapsto \pi(\emptyset)$, where $\emptyset$ is the empty word. By Lemma 6.2.8, every $\pi \in \Phi$ is a product $L_{y_1} \cdots L_{y_m}$; deleting adjacent factors $L_x L_{x^{-1}} = \text{id}$ we may assume $y_1 \cdots y_m$ is reduced, and then $\pi(\emptyset) = y_1 \cdots y_m$. Hence π is determined by $\pi(\emptyset)$, and every reduced word $w = x_1 \cdots x_k$ equals $L_{x_1} \cdots L_{x_k}(\emptyset)$. So ϵ is a bijection, and we transport the group structure of Φ to $F(S)$. The product of u and v is then $L_{u_1} \cdots L_{u_k}(v)$, which is computed by the deletion process of the statement.

For $f: S \rightarrow G$ extend f to A by $f(s^{-1}) = f(s)^{-1}$ and define $\bar{f}(x_1 \cdots x_k) = f(x_1) \cdots f(x_k)$. The value of $f(x_1) \cdots f(x_k)$ on an arbitrary word does not change when an adjacent pair xx^{-1} is deleted, and the product in $F(S)$ is obtained from concatenation by such deletions; so $\bar{f}$ is a homomorphism. It is unique because $\iota(S)$ generates $F(S)$. $\square$

Definition 6.10.3. Let S be a set and $R \subseteq F(S)$. The group with *presentation* $\langle S \mid R \rangle$ is $F(S)/N$, where N is the smallest normal subgroup containing R (the intersection of all normal subgroups containing R). Every group G has a presentation: take $S = G$ and R the kernel of $F(G) \rightarrow G$.

Proposition 6.10.4. *Let $G = \langle S \mid R \rangle$ and let $f: S \rightarrow H$ be a map to a group such that $\bar{f}(r) = e$ for all $r \in R$. Then there is a unique homomorphism $G \rightarrow H$ sending the class of each s to $f(s)$.*

Proof. $\ker \bar{f}$ is a normal subgroup containing R , hence containing N ; apply Theorem 6.4.3. Uniqueness holds because the classes of S generate G . $\square$

- Example 6.10.5.** 1. $\langle a \mid a^n \rangle \cong \mathbb{Z}/n\mathbb{Z}$: the homomorphism $F(\{a\}) \rightarrow \mathbb{Z}/n\mathbb{Z}, a \mapsto 1$, kills a^n and so induces a surjection $\langle a \mid a^n \rangle \rightarrow \mathbb{Z}/n\mathbb{Z}$; the source is generated by the class of a , which has order dividing n , so it has at most n elements.
2. $\langle a, b \mid aba^{-1}b^{-1} \rangle \cong \mathbb{Z}^2$: the map $a \mapsto (1, 0), b \mapsto (0, 1)$ induces a surjection, and a homomorphism $\mathbb{Z}^2 \rightarrow \langle a, b \mid aba^{-1}b^{-1} \rangle$ inverse to it exists because the classes of a and b commute.
3. $D_n \cong \langle r, s \mid r^n, s^2, srsr \rangle$, by a similar argument.

Definition 6.10.6. Let $\alpha_1: H \rightarrow G_1$ and $\alpha_2: H \rightarrow G_2$ be homomorphisms. The *amalgamated free product* $G_1 *_H G_2$ is the group with generators the disjoint union of the sets G_1 and G_2 and relations xyz^{-1} whenever $xy = z$ holds in G_1 or in G_2 , and $\alpha_1(h)\alpha_2(h)^{-1}$ for $h \in H$. If H is trivial, it is the *free product* $G_1 * G_2$.

Proposition 6.10.7. With the evident homomorphisms $G_i \rightarrow G_1 *_H G_2$, the amalgamated free product is a pushout of $G_1 \leftarrow H \rightarrow G_2$ in $\mathbf{Grp}$. In particular $G_1 * G_2$ is a coproduct, and $\mathbf{Grp}$ has all colimits.

Proof. The relations of the first kind make the maps $G_i \rightarrow G_1 *_H G_2$ homomorphisms, and those of the second kind make the square commute. Given homomorphisms $f_i: G_i \rightarrow K$ with $f_1\alpha_1 = f_2\alpha_2$, the map on generators given by f_1 and f_2 kills all relations, so it induces a homomorphism by Proposition 6.10.4, which is unique since the images of G_1 and G_2 generate. Coproducts of arbitrary families are constructed in the same way, and coequalisers are quotients by normal subgroups; so $\mathbf{Grp}$ has all colimits by the dual of Theorem 5.8.6. $\square$

Remark 6.10.8. If α_1 and α_2 are injective, the maps $G_i \rightarrow G_1 *_H G_2$ are injective, and every element has a normal form as an alternating product of coset representatives [Ser80, Chapter I, Theorem 1], [LS77, Chapter IV]. For example $\mathrm{SL}_2(\mathbb{Z}) \cong \mathbb{Z}/4\mathbb{Z} *__{\mathbb{Z}/2\mathbb{Z}} \mathbb{Z}/6\mathbb{Z}$ [Ser80, Chapter I, §4]. Amalgamated free products compute fundamental groups of unions of spaces via the Seifert–van Kampen theorem (Chapter 17).

6.11 Abelian groups

In this section groups are abelian and written additively. For $n \in \mathbb{Z}$ and $a \in A$ we write na for the n -th power of Lemma 6.2.9.

Definition 6.11.1. For a set S , let $\mathbb{Z}^{(S)}$ be the set of maps $S \rightarrow \mathbb{Z}$ that vanish outside a finite set, with pointwise addition, and e_s the map that is 1 at s and 0 elsewhere. An abelian group is *free* if it is isomorphic to some $\mathbb{Z}^{(S)}$, and a family (a_s) in A is a *basis* if the homomorphism $\mathbb{Z}^{(S)} \rightarrow A, e_s \mapsto a_s$, is an isomorphism. The *torsion subgroup* of A is $A_{\mathrm{tors}} = \{a : na = 0 \text{ for some } n \geq 1\}$, and A is *torsion-free* if $A_{\mathrm{tors}} = 0$.

Proposition 6.11.2. For every abelian group A and map $f: S \rightarrow A$ there is a unique homomorphism $\mathbb{Z}^{(S)} \rightarrow A$ with $e_s \mapsto f(s)$, namely $\sum_s n_s e_s \mapsto \sum_s n_s f(s)$. If S is finite with n elements, $\mathbb{Z}^{(S)} \cong \mathbb{Z}^n$, and $\mathbb{Z}^n \cong \mathbb{Z}^m$ implies $n = m$.

Proof. The formula defines a homomorphism, and it is forced. If $\mathbb{Z}^n \cong \mathbb{Z}^m$, then $\mathbb{Z}^n/2\mathbb{Z}^n \cong \mathbb{Z}^m/2\mathbb{Z}^m$, and these groups have 2^n and 2^m elements. $\square$

Definition 6.11.3. The *rank* of a free abelian group $\mathbb{Z}^n$ is n ; it is well defined by Proposition 6.11.2.

Proposition 6.11.4. *Every subgroup of $\mathbb{Z}^n$ is free of rank at most n .*

Proof. By induction on n ; for $n = 0$ it is clear. Let $H \leq \mathbb{Z}^n$ and $p: \mathbb{Z}^n \rightarrow \mathbb{Z}$ the last coordinate. Then $H \cap \ker p$ is a subgroup of $\ker p \cong \mathbb{Z}^{n-1}$, hence free of rank $\leq n - 1$ with some basis $(h_1, \dots, h_k)$. By Proposition 6.2.10, $p[H] = d\mathbb{Z}$. If $d = 0$, $H = H \cap \ker p$. Otherwise choose $h \in H$ with $p(h) = d$; every $x \in H$ is uniquely $x = (x - qh) + qh$ with $q = p(x)/d$ and $x - qh \in H \cap \ker p$, so $(h_1, \dots, h_k, h)$ is a basis of H . $\square$

Theorem 6.11.5 (Smith normal form over $\mathbb{Z}$). *Let M be an $m \times n$ matrix with integer entries. There are invertible integer matrices P and Q (with integer inverses) such that PMQ is diagonal, with diagonal entries $d_1, \dots, d_r, 0, \dots, 0$ where $d_i \geq 1$ and $d_1 \mid d_2 \mid \dots \mid d_r$.*

Proof. Multiplying by invertible matrices on the left and right performs the elementary operations: swapping two rows or columns, multiplying a row or column by -1 , and adding an integer multiple of one row (column) to another. We argue by induction on $m + n$; if $M = 0$ there is nothing to do. Among all matrices obtainable from M by elementary operations, choose one, M' , whose smallest nonzero entry in absolute value, δ , is as small as possible; move it to position $(1, 1)$ and make it positive. Every entry a of the first row or column is divisible by δ : otherwise subtracting a multiple of the first column (row) would produce the remainder of a modulo δ , a smaller nonzero entry. So we can clear the first row and column. Every remaining entry a is also divisible by δ : otherwise adding its row to the first row and then subtracting a multiple of the first column would create a smaller nonzero entry. Apply induction to the remaining $(m - 1) \times (n - 1)$ block; the operations on it do not affect divisibility of its entries by δ , so $d_1 = \delta$ divides all later d_i . $\square$

Theorem 6.11.6 (Structure of finitely generated abelian groups). *Let A be a finitely generated abelian group. There are unique integers $r \geq 0$ and $d_1, \dots, d_k \geq 2$ with $d_1 \mid d_2 \mid \dots \mid d_k$ such that*

$$A \cong \mathbb{Z}^r \oplus \mathbb{Z}/d_1\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/d_k\mathbb{Z}.$$

Here $A_{\text{tors}} \cong \bigoplus_i \mathbb{Z}/d_i\mathbb{Z}$, and $A/A_{\text{tors}} \cong \mathbb{Z}^r$.

Proof. Existence. Choose generators $a_1, \dots, a_n$, giving a surjection $\pi: \mathbb{Z}^n \rightarrow A$. By Proposition 6.11.4, $\ker \pi$ is free with a basis $v_1, \dots, v_m$, $m \leq n$. Let M be the $n \times m$ matrix with columns v_j . Changing the basis of $\mathbb{Z}^n$ corresponds to multiplying M by an invertible matrix on the left, and changing the basis of $\ker \pi$ to multiplying on the right. By Theorem 6.11.5 there are bases $f_1, \dots, f_n$ of $\mathbb{Z}^n$ and $w_1, \dots, w_m$ of $\ker \pi$ with $w_i = d_i f_i$ for $i \leq s$ and $w_i = 0$ for $i > s$; since the w_i are a basis, $s = m$ and all $d_i \geq 1$. Hence $A \cong \mathbb{Z}^n / \ker \pi \cong \bigoplus_{i \leq m} \mathbb{Z}/d_i\mathbb{Z} \oplus \mathbb{Z}^{n-m}$, and we discard the factors with $d_i = 1$.

Uniqueness. In such a decomposition the torsion subgroup is $T = \bigoplus \mathbb{Z}/d_i\mathbb{Z}$ and $A/T \cong \mathbb{Z}^r$, so r is determined by Proposition 6.11.2. For a prime p and $j \geq 0$, $p^j(\mathbb{Z}/d\mathbb{Z})/p^{j+1}(\mathbb{Z}/d\mathbb{Z})$ has order p if $p^{j+1} \mid d$ and order 1 otherwise. Hence $|p^j T / p^{j+1} T| = p^{t(p,j)}$, where $t(p, j)$ is the number of i with $p^{j+1} \mid d_i$. By the divisibility chain, these are the last $t(p, j)$ indices. So the numbers $t(p, j)$, which depend only on T , determine the exponent of p in every d_i , provided k is known; and $k = \max_p t(p, 0)$, since a prime dividing d_1 divides all d_i . $\square$

Corollary 6.11.7. *If $\gcd(m, n) = 1$, then $\mathbb{Z}/mn\mathbb{Z} \cong \mathbb{Z}/m\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$. Every finite abelian group is a direct sum of cyclic groups of prime power order.*

Proof. The element $(1, 1)$ of $\mathbb{Z}/m\mathbb{Z} \oplus \mathbb{Z}/n\mathbb{Z}$ has order $\text{lcm}(m, n) = mn$, so it generates. Apply this to the prime factorisations of the d_i . $\square$

Counterexample 6.11.8. The group $(\mathbb{Q}, +)$ is torsion-free but not free, and not finitely generated. Any two elements $a/b, c/d$ satisfy the nontrivial relation $bc \cdot (a/b) - ad \cdot (c/d) = 0$, so a basis would have one element, and $\mathbb{Q} \cong \mathbb{Z}$ is impossible since $\mathbb{Q}$ is divisible ($x = 2 \cdot (x/2)$) while $1 \in \mathbb{Z}$ is not twice anything. A finitely generated torsion-free group would be free by Theorem 6.11.6.

6.12 Exercises

Exercise 6.12.1. Let a finite group G act on a finite set X . Show that the number of orbits is $\frac{1}{|G|} \sum_{g \in G} |X^g|$, where $X^g = \{x : gx = x\}$.

Solution. Count the set $\{(g, x) : gx = x\}$ in two ways: it has $\sum_g |X^g|$ elements, and also $\sum_x |G_x| = \sum_x |G|/|Gx|$. Each orbit O contributes $\sum_{x \in O} |G|/|O| = |G|$ to the last sum. $\square$

Exercise 6.12.2. Show that every group of order at most 5 is abelian, and that S_3 is the smallest non-abelian group.

Exercise 6.12.3. Show that $[S_n, S_n] = A_n$ for $n \geq 2$ and $S_n^{\text{ab}} \cong \mathbb{Z}/2\mathbb{Z}$.

Exercise 6.12.4. (Frattini argument.) Let $N \trianglelefteq G$ be finite and P a Sylow p -subgroup of N . Show that $G = N \cdot N_G(P)$.

Exercise 6.12.5. Let $p < q$ be primes with $p \nmid q - 1$. Show that every group of order pq is cyclic.

Exercise 6.12.6. Show that $F(S) \cong F(T)$ implies $|S| = |T|$ for finite sets S, T . (Hint: count homomorphisms to $\mathbb{Z}/2\mathbb{Z}$.)

Exercise 6.12.7. Show that the free group on two generators is not solvable. (Hint: construct a surjection onto S_5 .)

Exercise 6.12.8. Let $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ with $i^2 = j^2 = k^2 = ijk = -1$. Show that every subgroup of Q_8 is normal but Q_8 is not abelian, and that Q_8 is not a semidirect product of two proper subgroups.

Chapter 7

Rings and Modules

7.1 Introduction

This chapter develops the basic theory of rings and modules: ideals and quotients, prime and maximal ideals, factorisation in integral domains, polynomial rings, modules and exact sequences, tensor products, algebras, and the structure of finitely generated modules over a principal ideal domain. Commutative rings were introduced in Section 4.3; the deeper theory of commutative rings is in Chapter 12.

References are [Lan02], [DF04] and, for modules over commutative rings, [AM69].

Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems), Chapter 5 (Categories) and Chapter 6 (Groups).

7.2 Rings

Definition 7.2.1. A *ring* is a set R with two operations, addition $(a, b) \mapsto a + b$ and multiplication $(a, b) \mapsto a \cdot b$ (usually written ab), and elements $0, 1$ such that $(R, +, 0)$ is an abelian group (in additive notation, Notation 6.2.2), multiplication is associative with neutral element 1 , and both distributive laws $a(b + c) = ab + ac$ and $(a + b)c = ac + bc$ hold. A *homomorphism* of rings is a map preserving $+$, $\cdot$ and 1 . A *subring* is a subset containing 0 and 1 and closed under $+$, $-$ and $\cdot$. An element u is a *unit* if it has a two-sided inverse; the units form a group $R^\times$. A *division ring* is a ring with $1 \neq 0$ in which every nonzero element is a unit; a commutative division ring is a field. The rings form a category $\mathbf{Ring}$, and the commutative rings a full subcategory $\mathbf{CRing}$.

Remark 7.2.2. The computations of Lemma 4.3.2 hold in every ring. If $1 = 0$ then $a = 1 \cdot a = 0 \cdot a = 0$ for all a , so $R = \{0\}$ is the *zero ring*; it is the terminal object of $\mathbf{Ring}$, and $\mathbb{Z}$ is the initial object (Exercise 4.10.5).

Example 7.2.3. 1. $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$, and $\mathbb{Z}/n\mathbb{Z}$ with the multiplication $[a][b] = [ab]$, which is well defined since $a - a' \in n\mathbb{Z}$ implies $ab - a'b \in n\mathbb{Z}$.

2. For a ring R , the $n \times n$ matrices $M_n(R)$ form a ring under matrix addition and multiplication; it is not commutative for $n \geq 2$ if $1 \neq 0$.
3. For an abelian group A , the endomorphisms $\text{End}(A)$ form a ring under pointwise addition and composition.
4. The product $\prod_i R_i$ of rings with componentwise operations is a product in $\mathbf{Ring}$.

5. The *quaternions* $\mathbb{H}$ are the matrices $\begin{pmatrix} z & -\bar{w} \\ w & \bar{z} \end{pmatrix}$ with $z, w \in \mathbb{C}$. They form a subring of $M_2(\mathbb{C})$, and a division ring: the determinant of such a matrix is $|z|^2 + |w|^2$, and if it is nonzero the inverse $\frac{1}{|z|^2 + |w|^2} \begin{pmatrix} \bar{z} & \bar{w} \\ -w & z \end{pmatrix}$ is again of this form. With $i = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$, $j = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $k = ij$ we get $i^2 = j^2 = k^2 = ijk = -1$, so $\mathbb{H}$ is not commutative.

7.3 Ideals and quotient rings

Definition 7.3.1. A *left ideal* of a ring R is an additive subgroup I with $ra \in I$ for $r \in R$, $a \in I$; right ideals are defined similarly, and a (two-sided) *ideal* is both. For a commutative ring these notions agree. The ideal generated by a subset S of a commutative ring is $(S) = \{r_1 s_1 + \cdots + r_n s_n : r_i \in R, s_i \in S\}$; an ideal of the form (a) is *principal*. For ideals I, J of a commutative ring, $I + J$, $I \cap J$ and the product IJ , generated by the products ab with $a \in I$, $b \in J$, are ideals, and $IJ \subseteq I \cap J$.

Theorem 7.3.2. Let I be an ideal of R . The quotient group R/I is a ring with $(a + I)(b + I) = ab + I$, and $\pi: R \rightarrow R/I$ is a surjective ring homomorphism with kernel I . Every ring homomorphism $f: R \rightarrow S$ with $I \subseteq \ker f$ factors uniquely as $\bar{f} \circ \pi$. Kernels of ring homomorphisms are ideals, and $R/\ker f \cong \text{im } f$. The ideals of R/I correspond to the ideals of R containing I via $J \mapsto J/I$, and $(R/I)/(J/I) \cong R/J$.

Proof. If $a - a', b - b' \in I$, then $ab - a'b' = a(b - b') + (a - a')b' \in I$, so the product is well defined; the ring axioms follow from those of R . The remaining statements follow from Theorems 6.4.3 and 6.4.4 applied to the additive groups, together with the observation that the maps constructed there preserve products and that preimages and images of ideals under surjective ring homomorphisms are ideals. $\square$

Theorem 7.3.3 (Chinese remainder theorem). Let R be a commutative ring and $I_1, \dots, I_n$ ideals with $I_i + I_j = R$ for $i \neq j$. Then $I_1 \cap \cdots \cap I_n = I_1 \cdots I_n$, and the map $R \rightarrow \prod_i R/I_i$ induces an isomorphism $R/(I_1 \cap \cdots \cap I_n) \cong \prod_i R/I_i$.

Proof. For $n = 2$: write $1 = a + b$ with $a \in I_1$, $b \in I_2$. If $x \in I_1 \cap I_2$ then $x = xa + xb \in I_1 I_2$. Given $r_1, r_2 \in R$, the element $r_1 b + r_2 a$ is congruent to r_1 modulo I_1 and to r_2 modulo I_2 , so the map is surjective; its kernel is $I_1 \cap I_2$. For $n > 2$, note that $I_1 + I_2 \cdots I_n = R$: from $1 = a_j + b_j$ with $a_j \in I_1$, $b_j \in I_j$ we get $1 = \prod_{j \geq 2} (a_j + b_j) \in I_1 + I_2 \cdots I_n$. Now use induction. $\square$

Example 7.3.4. If $n = p_1^{e_1} \cdots p_k^{e_k}$ with distinct primes p_i , then $\mathbb{Z}/n\mathbb{Z} \cong \prod_i \mathbb{Z}/p_i^{e_i}\mathbb{Z}$ as rings, since $(p_i^{e_i}) + (p_j^{e_j}) = \mathbb{Z}$ by Theorem 4.5.3.

7.4 Prime and maximal ideals

In this section all rings are commutative.

Definition 7.4.1. An ideal $\mathfrak{p} \neq R$ is *prime* if $ab \in \mathfrak{p}$ implies $a \in \mathfrak{p}$ or $b \in \mathfrak{p}$. An ideal $\mathfrak{m} \neq R$ is *maximal* if there is no ideal J with $\mathfrak{m} \subsetneq J \subsetneq R$.

Proposition 7.4.2. An ideal I is prime if and only if R/I is an integral domain, and maximal if and only if R/I is a field. In particular maximal ideals are prime.

Proof. The first statement is a restatement of the definitions. A commutative ring $K \neq 0$ is a field if and only if its only ideals are 0 and K : if $a \neq 0$ then $(a) = K$ means a is a unit. By the correspondence of ideals in Theorem 7.3.2, R/I has only the ideals 0 and R/I exactly when I is maximal. $\square$

Theorem 7.4.3 (Krull). *Every ideal $I \neq R$ is contained in a maximal ideal. In particular every nonzero ring has a maximal ideal.*

Proof. Let P be the set of ideals J with $I \subseteq J \neq R$, ordered by inclusion. It contains I . The union of a nonempty chain in P is an ideal, and it does not contain 1, so it lies in P . By Zorn's lemma (Theorem 3.7.2) P has a maximal element, which is a maximal ideal. $\square$

Example 7.4.4. The ideals of $\mathbb{Z}$ are the $n\mathbb{Z}$ (Proposition 6.2.10). The prime ideals are 0 and $p\mathbb{Z}$ for p prime, by Lemma 4.5.4, and the maximal ideals are the $p\mathbb{Z}$. Hence $\mathbb{Z}/n\mathbb{Z}$ is a field if and only if n is prime; we write $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The prime ideal 0 is not maximal.

Definition 7.4.5. The *characteristic* of a ring R is the $n \geq 0$ with $\ker(\mathbb{Z} \rightarrow R) = n\mathbb{Z}$.

Lemma 7.4.6. *The characteristic of an integral domain is 0 or a prime number.*

Proof. The image of $\mathbb{Z} \rightarrow R$ is a subring of a domain, hence a domain, so the kernel is a prime ideal of $\mathbb{Z}$. $\square$

Proposition 7.4.7. *Let R be an integral domain. There is a field $\text{Frac}(R)$ and an injective homomorphism $R \rightarrow \text{Frac}(R)$ such that every injective homomorphism from R to a field K extends uniquely to a homomorphism $\text{Frac}(R) \rightarrow K$.*

Proof. Construct $\text{Frac}(R)$ as the set of classes a/b of pairs with $b \neq 0$ under $(a, b) \sim (c, d) \iff ad = bc$, with the operations of Definition 4.6.1; the verification of Theorem 4.6.2 only used that $\mathbb{Z}$ is an integral domain. Given $f: R \rightarrow K$ injective, $f(b) \neq 0$ for $b \neq 0$, and $a/b \mapsto f(a)f(b)^{-1}$ is the unique extension, as in Proposition 4.6.3. $\square$

7.5 Factorisation in integral domains

In this section R is an integral domain.

Definition 7.5.1. Elements a, b are *associates* if $a = ub$ with $u \in R^\times$. A nonzero nonunit a is *irreducible* if $a = bc$ implies that b or c is a unit, and *prime* if $a \mid bc$ implies $a \mid b$ or $a \mid c$. The domain R is a *unique factorisation domain* (UFD) if every nonzero nonunit is a product of irreducible elements, and this product is unique up to order and associates. It is a *principal ideal domain* (PID) if every ideal is principal, and a *Euclidean domain* if there is a map $N: R \setminus \{0\} \rightarrow \mathbb{N}$ such that for all a and $b \neq 0$ there are q, r with $a = qb + r$ and $r = 0$ or $N(r) < N(b)$.

Lemma 7.5.2. *A nonzero element p is prime if and only if (p) is a prime ideal. Prime elements are irreducible.*

Proof. The first statement is a restatement. If p is prime and $p = bc$, then $p \mid b$ or $p \mid c$, say $b = pd$; then $p = pdc$, so $dc = 1$ and c is a unit. $\square$

Lemma 7.5.3. *Suppose every nonzero nonunit of R is a product of irreducible elements. Then R is a UFD if and only if every irreducible element is prime.*

Proof. If irreducibles are prime, uniqueness follows as in the proof of Theorem 4.5.5: a prime dividing a product of irreducibles divides one of them, hence is associate to it, and we cancel. Conversely, let R be a UFD, p irreducible and $p \mid bc$, say $bc = pd$. Factoring b , c and d into irreducibles and comparing the two factorisations of bc shows that p is associate to an irreducible factor of b or of c (if b or c is a unit or zero the claim is immediate). $\square$

Theorem 7.5.4. *A Euclidean domain is a PID, and a PID is a UFD.*

Proof. Let R be Euclidean and $I \neq 0$ an ideal; choose $b \in I \setminus \{0\}$ with $N(b)$ minimal. For $a \in I$ write $a = qb + r$; then $r \in I$, so $r = 0$ and $I = (b)$.

Let R be a PID. The union of a chain of ideals $(a_1) \subseteq (a_2) \subseteq \cdots$ is an ideal (a) , and $a \in (a_n)$ for some n , so the chain is eventually constant. If some nonzero nonunit a were not a product of irreducibles, then a is not irreducible, so $a = bc$ with b, c nonunits, one of which, say a_1 , is not a product of irreducibles; and $(a) \subsetneq (a_1)$. Repeating gives a strictly increasing chain of principal ideals, a contradiction. (The repeated choices are made using the axiom of choice in the form of Theorem 3.4.5 applied to a choice function.) Finally, if p is irreducible then (p) is maximal: if $(p) \subseteq (c) \neq R$ then $p = cd$ with c a nonunit, so d is a unit and $(c) = (p)$. Hence (p) is prime by Proposition 7.4.2, and R is a UFD by Lemma 7.5.3. $\square$

Example 7.5.5. The ring $\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$ of *Gaussian integers* is Euclidean for $N(z) = |z|^2$. Let $a, b \in \mathbb{Z}[i]$, $b \neq 0$, and write $a/b = u + vi$ with $u, v \in \mathbb{Q}$. Choose integers m, n with $|u - m| \leq 1/2$ and $|v - n| \leq 1/2$, and put $q = m + ni$, $r = a - qb \in \mathbb{Z}[i]$. Then $N(r) = |b|^2|a/b - q|^2 \leq N(b)(1/4 + 1/4) < N(b)$.

Counterexample 7.5.6. The ring $R = \mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\} \subseteq \mathbb{C}$ is not a UFD. The norm $N(z) = |z|^2 = a^2 + 5b^2$ is multiplicative and takes values in $\mathbb{N}$, and the units are the elements of norm 1, namely ± 1 . No element has norm 2 or 3. Hence 2 (norm 4) is irreducible: a factorisation into nonunits would have factors of norm 2. But 2 divides $6 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ without dividing either factor, since $N(1 \pm \sqrt{-5}) = 6$ is not divisible by $N(2) = 4$. So 2 is irreducible but not prime.

7.6 Polynomial rings

Definition 7.6.1. Let R be a ring. The *polynomial ring* $R[x]$ is the set of sequences $(a_0, a_1, \dots)$ in R with only finitely many nonzero terms, with termwise addition and the product $(a_i)(b_j) = (c_k)$, $c_k = \sum_{i+j=k} a_i b_j$. We identify $r \in R$ with $(r, 0, 0, \dots)$ and write $x = (0, 1, 0, \dots)$, so that every polynomial is uniquely $f = \sum_{i=0}^n a_i x^i$. If $f \neq 0$, its *degree* $\deg f$ is the largest i with $a_i \neq 0$, and $a_{\deg f}$ is the *leading coefficient*; f is *monic* if the leading coefficient is 1. The ring $R[x_1, \dots, x_n]$ is defined inductively as $R[x_1, \dots, x_{n-1}][x_n]$, and the *power series ring* $R[[x]]$ is defined like $R[x]$ but with all sequences allowed.

Proposition 7.6.2. Let $\varphi: R \rightarrow S$ be a ring homomorphism and $s \in S$ an element commuting with all elements of $\varphi(R)$. There is a unique ring homomorphism $R[x] \rightarrow S$ extending φ and sending x to s , namely $\sum a_i x^i \mapsto \sum \varphi(a_i) s^i$. In particular, for R commutative, $R[x_1, \dots, x_n]$ is the free commutative R -algebra on $x_1, \dots, x_n$: homomorphisms to a commutative ring S extending φ correspond to n -tuples of elements of S .

Proof. The formula is forced, and it preserves products because s commutes with the $\varphi(a_i)$: $(\sum \varphi(a_i) s^i)(\sum \varphi(b_j) s^j) = \sum_k (\sum_{i+j=k} \varphi(a_i) \varphi(b_j)) s^k$. The last statement follows by induction on n . $\square$

Notation 7.6.3. For R commutative, $f \in R[x]$ and $a \in R$, we write $f(a)$ for the image of f under the homomorphism $R[x] \rightarrow R$ with $x \mapsto a$. An element a with $f(a) = 0$ is a *root* of f .

Proposition 7.6.4. *If R is an integral domain, then $\deg(fg) = \deg f + \deg g$ for nonzero f, g , so $R[x]$ is an integral domain and $R[x]^\times = R^\times$.*

Proof. The coefficient of $x^{\deg f + \deg g}$ in fg is the product of the leading coefficients, which is nonzero; all higher coefficients vanish. If $f g = 1$ then $\deg f = \deg g = 0$. $\square$

Theorem 7.6.5 (Division with remainder). *Let R be commutative and $g \in R[x]$ a polynomial whose leading coefficient is a unit. For every $f \in R[x]$ there are unique $q, r \in R[x]$ with $f = qg + r$ and $r = 0$ or $\deg r < \deg g$. Consequently, if k is a field then $k[x]$ is Euclidean with $N = \deg$, hence a PID and a UFD.*

Proof. Existence by induction on $\deg f$: if $f = 0$ or $\deg f < \deg g$ take $q = 0$. Otherwise, with a and b the leading coefficients of f and g , the polynomial $f - ab^{-1}x^{\deg f - \deg g}g$ has smaller degree or is zero. Uniqueness: if $qg + r = q'g + r'$ with $q \neq q'$, then $(q - q')g = r' - r$ has degree $\geq \deg g$, because the leading coefficient of g is a unit (so it is not a zero divisor), which is impossible. The last statement follows from Theorem 7.5.4. $\square$

Corollary 7.6.6. *Let R be commutative, $f \in R[x]$ and $a \in R$. Then $f(a) = 0$ if and only if $(x - a) \mid f$. If R is an integral domain and $f \neq 0$, then f has at most $\deg f$ roots in R .*

Proof. Divide: $f = q(x - a) + r$ with $r \in R$, and $r = f(a)$. If R is a domain and $a_1, \dots, a_m$ are distinct roots, induction shows $(x - a_1) \cdots (x - a_m) \mid f$: if $f = (x - a_1)g$ then $0 = f(a_j) = (a_j - a_1)g(a_j)$ gives $g(a_j) = 0$ for $j \geq 2$. So $m \leq \deg f$ by Proposition 7.6.4. $\square$

Counterexample 7.6.7. The hypothesis that R is a domain is needed: in $\mathbb{Z}/8\mathbb{Z}$ the polynomial $x^2 - 1$ of degree 2 has the four roots 1, 3, 5, 7.

Proposition 7.6.8. *Let R be commutative. A power series $f = \sum a_i x^i \in R[[x]]$ is a unit if and only if a_0 is a unit of R .*

Proof. If $fg = 1$ then $a_0 b_0 = 1$. Conversely, if a_0 is a unit define $b_0 = a_0^{-1}$ and recursively $b_k = -a_0^{-1} \sum_{i=1}^k a_i b_{k-i}$; then $\sum_{i+j=k} a_i b_j = 0$ for $k \geq 1$. $\square$

Definition 7.6.9. Let R be a UFD. A *greatest common divisor* of elements of R is a common divisor divisible by every common divisor; it exists (take the product of the common prime factors with their minimal multiplicities) and is unique up to associates. The *content* $c(f)$ of a nonzero $f \in R[x]$ is a greatest common divisor of its coefficients, and f is *primitive* if $c(f)$ is a unit.

Lemma 7.6.10 (Gauss). *Let R be a UFD. A product of primitive polynomials is primitive, and $c(fg) = c(f)c(g)$ up to associates.*

Proof. If f, g are primitive and a prime p divides all coefficients of fg , then the images of f and g in $(R/(p))[x]$ are nonzero but their product is zero; this is impossible because $R/(p)$ is a domain, hence so is $(R/(p))[x]$ (Proposition 7.6.4). In general write $f = c(f)f_0$ and $g = c(g)g_0$ with f_0, g_0 primitive; then $fg = c(f)c(g)f_0g_0$ with f_0g_0 primitive. $\square$

Theorem 7.6.11. *Let R be a UFD with fraction field K .*

1. *A primitive polynomial $f \in R[x]$ of positive degree is irreducible in $R[x]$ if and only if it is irreducible in $K[x]$.*
2. *$R[x]$ is a UFD, whose prime elements are the prime elements of R and the primitive polynomials of positive degree that are irreducible in $K[x]$.*

Consequently $\mathbb{Z}[x_1, \dots, x_n]$ and $k[x_1, \dots, x_n]$ for a field k are UFDs.

Proof. Every $h \in K[x]$, $h \neq 0$, can be written $h = ah_0$ with $a \in K^\times$ and $h_0 \in R[x]$ primitive (clear denominators, then divide by the content), and if also $h = a'h'_0$ then $h'_0 = uh_0$ with $u \in R^\times$: writing $a/a' = b/b'$ with $b, b' \in R$, $bh_0 = b'h'_0$ and comparing contents gives $b = b'u$.

(1) If $f = gh$ in $K[x]$ with $\deg g, \deg h > 0$, write $g = ag_0$, $h = bh_0$; then $f = abg_0h_0$ with g_0h_0 primitive (Lemma 7.6.10), so $ab \in R^\times$ by the uniqueness just shown, and $f = (abg_0)h_0$ is a factorisation in $R[x]$ into polynomials of positive degree. Conversely a factorisation in $R[x]$ of a primitive polynomial into nonunits has both factors of positive degree (a constant factor would divide the content), hence is a factorisation in $K[x]$.

(2) A prime p of R is prime in $R[x]$ since $R[x]/(p) \cong (R/(p))[x]$ is a domain. Let f be primitive of positive degree and irreducible in $K[x]$, and suppose $f \mid gh$ in $R[x]$ with $g, h \neq 0$. Since $K[x]$ is a PID, f is prime in $K[x]$, so say $g = fq$ with $q \in K[x]$. Write $q = aq_0$ with $a = c/c'$, $c, c' \in R$ and $q_0 \in R[x]$ primitive. Then $c'g = cfq_0$, and comparing contents using Lemma 7.6.10 gives $c = c'c(g)u$ with $u \in R^\times$, so $a \in R$ and $q \in R[x]$. Hence f is prime in $R[x]$.

Every irreducible element of $R[x]$ is of one of these two kinds: an irreducible of degree 0 is irreducible in R , hence prime in R ; an irreducible f of positive degree is primitive (otherwise $c(f)$ would be a nonunit factor), hence irreducible in $K[x]$ by (1). Every nonzero nonunit of $R[x]$ is a product of irreducibles: write $f = c(f)f_0$, factor $c(f)$ in R , and factor f_0 in $K[x]$ into irreducibles, which by the first paragraph and (1) can be taken primitive in $R[x]$, the remaining constant being a unit of R by Lemma 7.6.10. By Lemma 7.5.3, $R[x]$ is a UFD. The last statement follows by induction from $\mathbb{Z}$ and k being UFDs. $\square$

Proposition 7.6.12 (Eisenstein's criterion). *Let R be a UFD with fraction field K , $f = a_nx^n + \dots + a_0 \in R[x]$ with $n \geq 1$, and p a prime of R with $p \nmid a_n$, $p \mid a_i$ for $i < n$ and $p^2 \nmid a_0$. Then f is irreducible in $K[x]$.*

Proof. By Theorem 7.6.11(1) applied to $f/c(f)$ (note $p \nmid c(f)$) it suffices to exclude $f = gh$ in $R[x]$ with $\deg g, \deg h \geq 1$. Reducing modulo p gives $\bar{g}\bar{h} = \bar{a}_nx^n$ in $(R/(p))[x]$, which embeds in $L[x]$ for $L = \text{Frac}(R/(p))$. By unique factorisation in $L[x]$, $\bar{g} = bx^k$ and $\bar{h} = cx^{n-k}$ with $k = \deg g$ and $n - k = \deg h$ both positive. So p divides the constant terms of g and h , and $p^2 \mid a_0$, a contradiction. $\square$

Counterexample 7.6.13. A UFD need not be a PID: in $\mathbb{Z}[x]$ the ideal $(2, x)$ is not principal. If $(2, x) = (f)$, then $f \mid 2$, so $f = \pm 1$ or $f = \pm 2$. But $2 \nmid x$, and $1 \notin (2, x)$ since every element of $(2, x)$ has even constant term.

7.7 Modules

Definition 7.7.1. Let R be a ring. A (left) R -module is an abelian group M with a map $R \times M \rightarrow M$, $(r, m) \mapsto rm$, such that $r(m + m') = rm + rm'$, $(r + s)m = rm + sm$, $(rs)m = r(sm)$ and $1m = m$; equivalently, a ring homomorphism $R \rightarrow \text{End}(M)$. A homomorphism (or R -linear map) is an additive map with $f(rm) = rf(m)$. Submodules are subgroups closed under the action. The set $\text{Hom}_R(M, N)$ of homomorphisms is an abelian group, and an R -module if R is commutative. The small R -modules form a category $R\text{-Mod}$.

Example 7.7.2. 1. R is a module over itself, and its submodules are its left ideals.

2. A $\mathbb{Z}$ -module is the same thing as an abelian group, by Lemma 6.2.9.

3. A module over a field k is a vector space.

4. A $k[x]$ -module is a k -vector space V together with a k -linear map $T: V \rightarrow V$ (the action of x), by Proposition 7.6.2.

Proposition 7.7.3. *Quotients M/N by submodules are modules, and the isomorphism and correspondence theorems of Theorem 6.4.4 hold for modules. For a family $(M_i)_{i \in I}$, the product $\prod_i M_i$ and the direct sum $\bigoplus_i M_i = \{(m_i) \in \prod_i M_i : m_i = 0 \text{ for almost all } i\}$ are a product and a coproduct in $R\text{-Mod}$. The category $R\text{-Mod}$ is additive (Definition 5.12.1), has kernels and cokernels, and has all limits and colimits.*

Proof. The action on M/N is $r(m + N) = rm + N$, well defined since N is a submodule; the maps in the isomorphism theorems are R -linear. The universal properties of $\prod$ and $\bigoplus$ are verified componentwise; for $\bigoplus$, a family $f_i: M_i \rightarrow N$ gives $(m_i) \mapsto \sum_i f_i(m_i)$, a finite sum. Hom sets are abelian groups with bilinear composition, 0 is a zero object, and finite products exist, so the category is additive. The kernel of f is $\ker f \rightarrow M$ and the cokernel is $N \rightarrow N/\text{im } f$. Equalisers of f, g are kernels of $f - g$, so all limits exist by Theorem 5.8.6, and dually all colimits exist. $\square$

Definition 7.7.4. For a set S , the *free module* $R^{(S)} = \bigoplus_{s \in S} R$ has basis vectors e_s . A family $(m_s)_{s \in S}$ in M *generates* M if every element is a finite R -linear combination of the m_s , is *linearly independent* if $\sum r_s m_s = 0$ implies all $r_s = 0$, and is a *basis* if both hold. M is *finitely generated* if it has a finite generating family.

Proposition 7.7.5. *For every module M and family $(m_s)_{s \in S}$ in M there is a unique homomorphism $R^{(S)} \rightarrow M$ with $e_s \mapsto m_s$. It is surjective iff the family generates, injective iff it is linearly independent. Every module is a quotient of a free module, and a finitely generated module is a quotient of some R^n .*

Proof. The map is $(r_s) \mapsto \sum r_s m_s$. The other statements are immediate; take the family of all elements of M (or a finite generating family). $\square$

Definition 7.7.6. A sequence of homomorphisms $\cdots \rightarrow M_{i-1} \xrightarrow{f_{i-1}} M_i \xrightarrow{f_i} M_{i+1} \rightarrow \cdots$ is *exact* at M_i if $\text{im } f_{i-1} = \ker f_i$, and *exact* if it is exact everywhere. A *short exact sequence* is an exact sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$.

Proposition 7.7.7 (Splitting lemma). *For a short exact sequence $0 \rightarrow A \xrightarrow{f} B \xrightarrow{g} C \rightarrow 0$ the following are equivalent:*

1. *there is $r: B \rightarrow A$ with $rf = \text{id}_A$;*
2. *there is $s: C \rightarrow B$ with $gs = \text{id}_C$;*
3. *there is an isomorphism $h: B \rightarrow A \oplus C$ with $hf(a) = (a, 0)$ and $g = \text{pr}_C \circ h$.*

The sequence is then called split. If C is free, the sequence splits.

Proof. (1) $\Rightarrow$ (3): $h = (r, g)$ is injective (if $r(b) = 0$ and $g(b) = 0$ then $b = f(a)$ and $a = rf(a) = 0$) and surjective (given (a, c) , pick b with $g(b) = c$; then $b - fr(b) + f(a)$ maps to (a, c)). (2) $\Rightarrow$ (3): the map $A \oplus C \rightarrow B$, $(a, c) \mapsto f(a) + s(c)$, is bijective by a similar check; take its inverse. (3) $\Rightarrow$ (1), (2): take $r = \text{pr}_A \circ h$ and $s = h^{-1} \circ \iota_C$. If C has a basis (c_s) , choose preimages b_s and define $s(c_s) = b_s$ using Proposition 7.7.5. $\square$

Counterexample 7.7.8. The sequence $0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow 0$ of abelian groups is exact but not split, since $\mathbb{Z}$ has no element of order 2 and so cannot contain a copy of $\mathbb{Z}/2\mathbb{Z}$.

7.8 Vector spaces and rank

Lemma 7.8.1 (Exchange lemma). *Let k be a field and V a vector space generated by $v_1, \dots, v_n$. Then any $n + 1$ vectors $w_1, \dots, w_{n+1}$ in V are linearly dependent.*

Proof. Induction on n . For $n = 0$, $V = 0$ and $w_1 = 0$. For $n \geq 1$ write $w_j = \sum_{i=1}^n a_{ij}v_i$. If $a_{nj} = 0$ for all j , then $w_1, \dots, w_n$ lie in the span of $v_1, \dots, v_{n-1}$ and are dependent by induction. Otherwise, renumbering, let $a_{n,n+1} \neq 0$ and put $w'_j = w_j - a_{nj}a_{n,n+1}^{-1}w_{n+1}$ for $j \leq n$. These lie in the span of $v_1, \dots, v_{n-1}$, so by induction $\sum_{j \leq n} c_j w'_j = 0$ with some $c_j \neq 0$. Then $\sum_{j \leq n} c_j w_j - (\sum_{j \leq n} c_j a_{nj} a_{n,n+1}^{-1}) w_{n+1} = 0$ is a nontrivial relation. $\square$

Theorem 7.8.2. *Let R be a nonzero commutative ring. If $R^m \cong R^n$ as R -modules, then $m = n$. In particular any two bases of a finitely generated vector space have the same number of elements.*

Proof. For a field, a basis of n elements generates, so by Lemma 7.8.1 every linearly independent family has at most n elements; hence $k^m \cong k^n$ implies $m \leq n$ and $n \leq m$. For general R , choose a maximal ideal $\mathfrak{m}$ (Theorem 7.4.3). An isomorphism $R^m \cong R^n$ maps $\mathfrak{m}R^m$ onto $\mathfrak{m}R^n$, so it induces an isomorphism $(R/\mathfrak{m})^m \cong R^m/\mathfrak{m}R^m \cong R^n/\mathfrak{m}R^n \cong (R/\mathfrak{m})^n$ of vector spaces over the field $R/\mathfrak{m}$. $\square$

Definition 7.8.3. For R nonzero commutative, the *rank* of a free module R^n is n .

Theorem 7.8.4. *Let k be a field, V a vector space, $I \subseteq V$ a linearly independent subset and $S \supseteq I$ a generating subset. There is a basis B of V with $I \subseteq B \subseteq S$. In particular every vector space has a basis, every linearly independent subset extends to a basis, and every generating subset contains a basis.*

Proof. Let P be the set of linearly independent subsets J with $I \subseteq J \subseteq S$, ordered by inclusion. The union of a chain in P is linearly independent, since a linear relation involves finitely many vectors, which lie in one member of the chain. By Zorn's lemma (Theorem 3.7.2) there is a maximal $B \in P$. If some $s \in S$ were not a linear combination of B , then $B \cup \{s\}$ would be linearly independent, contradicting maximality. So every element of S , and hence of V , is a linear combination of B . $\square$

Theorem 7.8.5. *Any two bases of a vector space V have the same cardinality, called the dimension $\dim_k V$.*

Proof. Let B and C be bases. If B is finite with n elements, then by Lemma 7.8.1 every linearly independent set has at most n elements, so C is finite, and $|B| = |C|$ by Theorem 7.8.2. Suppose both are infinite. Each $c \in C$ is a linear combination of a finite subset $F_c \subseteq B$. Every $b \in B$ lies in some F_c : otherwise b would be a linear combination of $\bigcup_c F_c \subseteq B \setminus \{b\}$, contradicting independence. So $B = \bigcup_{c \in C} F_c$ and $|B| \leq |C|$ by Proposition 3.7.5. By symmetry $|C| \leq |B|$, and Theorem 3.6.2 gives $|B| = |C|$. $\square$

Counterexample 7.8.6. Theorem 7.8.2 fails for noncommutative rings. Let k be a field, $V = k^{(\mathbb{N})}$ and $R = \text{End}_k(V)$. Splitting $\mathbb{N}$ into even and odd numbers gives $V \cong V \oplus V$. As a left R -module (by post-composition), $R = \text{Hom}_k(V, V) \cong \text{Hom}_k(V \oplus V, V) \cong \text{Hom}_k(V, V) \oplus \text{Hom}_k(V, V) = R^2$.

7.9 Tensor products

In this section R is a commutative ring.

Definition 7.9.1. Let M, N, P be R -modules. A map $\beta: M \times N \rightarrow P$ is *bilinear* if it is R -linear in each variable. A *tensor product* of M and N is a module $M \otimes_R N$ with a bilinear map $\otimes: M \times N \rightarrow M \otimes_R N$, $(m, n) \mapsto m \otimes n$, such that every bilinear $\beta: M \times N \rightarrow P$ factors as $\beta = \tilde{\beta} \circ \otimes$ for a unique linear $\tilde{\beta}$.

Proposition 7.9.2. *Tensor products exist and are unique up to unique isomorphism. The elements $m \otimes n$ generate $M \otimes_R N$.*

Proof. Uniqueness is the usual argument for universal properties. Let $F = R^{(M \times N)}$ and $Q \subseteq F$ the submodule generated by the elements $e_{(m+m', n)} - e_{(m, n)} - e_{(m', n)}$, $e_{(m, n+n')} - e_{(m, n)} - e_{(m, n')}$, $e_{(rm, n)} - re_{(m, n)}$ and $e_{(m, rn)} - re_{(m, n)}$. Put $M \otimes_R N = F/Q$ and $m \otimes n = e_{(m, n)} + Q$. A bilinear β induces a linear map $F \rightarrow P$, $e_{(m, n)} \mapsto \beta(m, n)$, vanishing on Q , hence a unique $\tilde{\beta}$. The elements $m \otimes n$ generate since the $e_{(m, n)}$ generate F . $\square$

Proposition 7.9.3. *There are natural isomorphisms $R \otimes_R M \cong M$, $M \otimes N \cong N \otimes M$, $(M \otimes N) \otimes P \cong M \otimes (N \otimes P)$ and $(\bigoplus_i M_i) \otimes N \cong \bigoplus_i (M_i \otimes N)$. In particular $R^m \otimes R^n \cong R^{mn}$. For an ideal I , $(R/I) \otimes_R M \cong M/IM$.*

Proof. Each isomorphism is obtained by constructing maps in both directions from bilinear maps and checking that the composites fix generators $m \otimes n$. For instance $(r, m) \mapsto rm$ induces $R \otimes M \rightarrow M$, inverse to $m \mapsto 1 \otimes m$. For associativity, for fixed p the map $(m, n) \mapsto m \otimes (n \otimes p)$ is bilinear, giving $(m \otimes n) \mapsto m \otimes (n \otimes p)$; this is bilinear in $(m \otimes n, p)$, and an inverse is constructed symmetrically. For direct sums, $((m_i), n) \mapsto (m_i \otimes n)$ is bilinear, and the inverse is induced by the maps $M_i \otimes N \rightarrow (\bigoplus M_i) \otimes N$. Finally, $(\bar{r}, m) \mapsto \bar{r}m$ induces $R/I \otimes M \rightarrow M/IM$, with inverse $\bar{m} \mapsto \bar{1} \otimes m$, well defined since $1 \otimes am = \bar{a} \otimes m = 0$ for $a \in I$. $\square$

Theorem 7.9.4 (Tensor–hom adjunction). *For R -modules M, N, P there is a bijection, natural in all three variables,*

$$\text{Hom}_R(M \otimes_R N, P) \cong \text{Hom}_R(M, \text{Hom}_R(N, P)), \quad f \mapsto (m \mapsto (n \mapsto f(m \otimes n))).$$

So $- \otimes_R N$ is left adjoint to $\text{Hom}_R(N, -)$. Consequently $- \otimes_R N$ preserves colimits, and for every exact sequence $A \rightarrow B \rightarrow C \rightarrow 0$ the sequence $A \otimes N \rightarrow B \otimes N \rightarrow C \otimes N \rightarrow 0$ is exact.

Proof. Both sides are in bijection with bilinear maps $M \times N \rightarrow P$, naturally. The functor $- \otimes_R N$ preserves colimits by Theorem 5.10.4. A sequence $A \xrightarrow{u} B \xrightarrow{v} C \rightarrow 0$ is exact if and only if v is a cokernel of u , that is, a coequaliser of u and 0 ; this is preserved. $\square$

Example 7.9.5. $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}/d\mathbb{Z}$ with $d = \gcd(m, n)$, since by Proposition 7.9.3 it is $(\mathbb{Z}/n\mathbb{Z})/m(\mathbb{Z}/n\mathbb{Z}) = \mathbb{Z}/(m, n) = \mathbb{Z}/d\mathbb{Z}$. In particular $\mathbb{Z}/2\mathbb{Z} \otimes \mathbb{Z}/3\mathbb{Z} = 0$, and $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} = 0$ since $x \otimes \bar{1} = nx/n \otimes \bar{1} = x/n \otimes \bar{n} = 0$.

Counterexample 7.9.6. Tensoring does not preserve injectivity. The map $2: \mathbb{Z} \rightarrow \mathbb{Z}$ is injective, but tensoring with $\mathbb{Z}/2\mathbb{Z}$ gives the zero map $\mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$. Similarly $\text{Hom}_R(N, -)$ does not preserve surjectivity: $\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z}$ is surjective but $\text{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}) = 0 \rightarrow \text{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ is not.

Definition 7.9.7. An R -module N is *flat* if $- \otimes_R N$ preserves injective homomorphisms. Free modules are flat, by Proposition 7.9.3.

Proposition 7.9.8 (Extension of scalars). *Let $\varphi: R \rightarrow S$ be a homomorphism of commutative rings. For an R -module M , $S \otimes_R M$ is an S -module via $s(s' \otimes m) = ss' \otimes m$, and for every S -module N (an R -module via φ) there is a natural bijection $\text{Hom}_S(S \otimes_R M, N) \cong \text{Hom}_R(M, N)$, $f \mapsto (m \mapsto f(1 \otimes m))$.*

Proof. For $s \in S$, multiplication by s on S is R -linear, so it induces an endomorphism of $S \otimes_R M$; this gives the module structure. Given an R -linear $g: M \rightarrow N$, the map $(s, m) \mapsto sg(m)$ is R -bilinear and induces an S -linear map $S \otimes M \rightarrow N$; the two constructions are inverse because $1 \otimes m$ generate $S \otimes M$ as an S -module. $\square$

7.10 Algebras and graded rings

Definition 7.10.1. Let R be commutative. An R -algebra is a ring A with a ring homomorphism $R \rightarrow A$ whose image lies in the center of A . Homomorphisms of R -algebras are ring homomorphisms compatible with the maps from R . A *graded ring* is a ring $A = \bigoplus_{n \in \mathbb{Z}} A_n$ (as abelian group) with $A_m A_n \subseteq A_{m+n}$; elements of A_n are *homogeneous of degree n* . A *graded module* over A is a module $M = \bigoplus_n M_n$ with $A_m M_n \subseteq M_{m+n}$. An ideal of a graded ring is *homogeneous* if it is generated by homogeneous elements.

Proposition 7.10.2. Let A and B be commutative R -algebras. There is a unique ring structure on $A \otimes_R B$ with $(a \otimes b)(a' \otimes b') = aa' \otimes bb'$, and with the maps $a \mapsto a \otimes 1$ and $b \mapsto 1 \otimes b$ it is a coproduct of A and B in the category of commutative R -algebras.

Proof. For fixed (a', b') the map $(a, b) \mapsto aa' \otimes bb'$ is bilinear, inducing $\mu_{a', b'}: A \otimes B \rightarrow A \otimes B$, and $(a', b') \mapsto \mu_{a', b'}$ is bilinear, inducing a bilinear multiplication on $A \otimes B$ with the required formula. Associativity, commutativity and the unit $1 \otimes 1$ hold on generators, hence everywhere by bilinearity. Given algebra homomorphisms $f: A \rightarrow C$, $g: B \rightarrow C$ to a commutative algebra, the bilinear map $(a, b) \mapsto f(a)g(b)$ induces a linear map $A \otimes B \rightarrow C$, which is multiplicative on generators and hence a ring homomorphism; it is unique since the elements $a \otimes b = (a \otimes 1)(1 \otimes b)$ generate. $\square$

Example 7.10.3. The polynomial ring $R[x_1, \dots, x_n]$ is graded by total degree, with A_d the span of the monomials of degree d . By Proposition 7.10.2 and Proposition 7.6.2, $R[x] \otimes_R R[y] \cong R[x, y]$. Graded rings of this kind are the homogeneous coordinate rings of projective varieties (Chapter 49).

Lemma 7.10.4. An ideal I of a graded commutative ring A is homogeneous if and only if, for every $f \in I$, all homogeneous components of f lie in I .

Proof. If I is generated by homogeneous g_j and $f = \sum_j a_j g_j \in I$, then the degree- n component of f is $\sum_j (a_j)_{n-\deg g_j} g_j \in I$. Conversely, if I contains the components of its elements, it is generated by these components. $\square$

7.11 Modules over a principal ideal domain

In this section R is a principal ideal domain.

Proposition 7.11.1. Every submodule of R^n is free of rank at most n .

Proof. The proof of Proposition 6.11.4 applies verbatim, using that the image of a submodule under the last coordinate is an ideal (d) of R . $\square$

Theorem 7.11.2 (Smith normal form). Let M be an $m \times n$ matrix over R . There are invertible matrices P, Q over R such that PMQ is diagonal with entries $d_1, \dots, d_r, 0, \dots, 0$, where $d_i \neq 0$ and $d_1 \mid d_2 \mid \dots \mid d_r$.

Proof. Since R is a UFD (Theorem 7.5.4), each nonzero $a \in R$ has a length $\ell(a)$, the number of prime factors counted with multiplicity; if $d \mid a$ and d is not associate to a , then $\ell(d) < \ell(a)$. Besides row and column swaps and adding multiples of one row (column) to another, we use the following operation: for entries a, b in the same row, in columns i and j , write $d = \gcd(a, b) = xa + yb$ (possible since $(a, b) = (d)$) and replace columns i, j by $x \text{col}_i + y \text{col}_j$ and $-(b/d)\text{col}_i + (a/d)\text{col}_j$. This is multiplication by an invertible matrix (its 2×2 block has determinant 1), and it replaces a, b by $d, 0$. The same operation exists for rows.

Now argue as in Theorem 6.11.5, with ℓ in place of the absolute value. Among all matrices obtained from M by these operations choose one whose nonzero entry of least length is as short as possible, and move it to position $(1, 1)$. If it did not divide an entry in the first row or column, the operation above would produce a proper divisor, of smaller length. So it divides these entries, and we clear the first row and column. If it did not divide some other entry, adding that row to the first row and applying the operation would again produce a shorter entry. Apply induction to the remaining block. $\square$

Theorem 7.11.3 (Structure theorem). *Let M be a finitely generated R -module. There are $r \geq 0$ and nonzero nonunits $d_1 \mid d_2 \mid \cdots \mid d_k$ in R with*

$$M \cong R^r \oplus R/(d_1) \oplus \cdots \oplus R/(d_k),$$

and r and the ideals $(d_1), \dots, (d_k)$ are uniquely determined by M .

Proof. Existence is proved exactly as in Theorem 6.11.6, using Proposition 7.11.1 and Theorem 7.11.2. For uniqueness, the torsion submodule $T = \{m : am = 0 \text{ for some } a \neq 0\}$ is $\bigoplus R/(d_i)$, and $M/T \cong R^r$, so r is determined by Theorem 7.8.2. Let p be a prime of R and $j \geq 0$. For $d \neq 0$ with $v = v_p(d)$, the $R/(p)$ -vector space $p^j(R/(d))/p^{j+1}(R/(d))$ is one-dimensional if $v \geq j + 1$ and zero otherwise: indeed $p^j(R/(d)) = (p^{\min(j,v)})(d)$. Hence $\dim_{R/(p)} p^j T / p^{j+1} T$ is the number of i with $p^{j+1} \mid d_i$, which by Theorem 7.8.2 depends only on T . As in the proof of Theorem 6.11.6, these numbers determine k and the p -adic valuations of all d_i , hence the d_i up to units. $\square$

Corollary 7.11.4. *Every finitely generated torsion R -module is a direct sum of modules $R/(p^e)$ with p prime and $e \geq 1$, and the multiset of the ideals (p^e) is uniquely determined.*

Proof. Apply Theorem 7.3.3 to $R/(d_i)$ with the prime factorisation of d_i ; uniqueness follows from the uniqueness in Theorem 7.11.3, since the d_i are recovered from the prime powers by taking, for each p , the largest powers for d_k , the next largest for d_{k-1} , and so on. $\square$

7.12 Exercises

Exercise 7.12.1. Let k be a field. Show that $M_n(k)$ has no two-sided ideals other than 0 and $M_n(k)$, although it has many left ideals for $n \geq 2$.

Exercise 7.12.2. Show that $\mathbb{Z}[x]/(x^2 + 1) \cong \mathbb{Z}[i]$, and that $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.

Exercise 7.12.3. Show that $(\mathbb{Z}/n\mathbb{Z})^\times$ consists of the classes of integers coprime to n , and deduce Fermat's little theorem: $a^{p-1} \equiv 1 \pmod{p}$ for p prime and $p \nmid a$.

Solution. If $\gcd(a, n) = 1$ then $ax + ny = 1$ by Theorem 4.5.3, so $[a][x] = 1$. If $d = \gcd(a, n) > 1$ then $[a][n/d] = 0$ with $[n/d] \neq 0$, so $[a]$ is a zero divisor and not a unit. For Fermat, apply Theorem 6.3.3 to the group $\mathbb{F}_p^\times$ of order $p - 1$. $\square$

Exercise 7.12.4. Let p be prime. Show that $x^{p-1} + x^{p-2} + \cdots + 1$ is irreducible in $\mathbb{Q}[x]$ by applying Proposition 7.6.12 to its image under $x \mapsto x + 1$.

Exercise 7.12.5. Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and show that $\mathbb{Q}$ is a flat $\mathbb{Z}$ -module which is not free.

Exercise 7.12.6. Let k be a field. Show that $k[x, y]$ is a UFD but not a PID.

Exercise 7.12.7. Let k be a field, V a finite-dimensional vector space and $T: V \rightarrow V$ linear. Using Example 7.7.2 and Theorem 7.11.3, show that $V \cong \bigoplus_i k[x]/(f_i)$ as $k[x]$ -modules for monic $f_1 \mid \cdots \mid f_k$, and that f_k is the monic polynomial of least degree with $f_k(T) = 0$.

Chapter 8

Fields and Galois Theory

8.1 Introduction

This chapter studies field extensions: algebraic elements, splitting fields and algebraic closures, separability, the fundamental theorem of Galois theory, finite fields, cyclotomic fields and solvability by radicals. We give Artin's algebraic proof of the fundamental theorem of algebra, and we study transcendence bases, which show that $\mathbb{C}$ has an enormous group of automorphisms. These automorphisms reappear in the theory of absolute Hodge classes (Chapter 71).

References are [Lan02], [Art42] and [Ste15].

Prerequisites. Chapter 4 (Number Systems), Chapter 6 (Groups) and Chapter 7 (Rings and Modules).

8.2 Field extensions

Definition 8.2.1. A *field extension* L/K is a field L together with a subfield K (more generally, a homomorphism of fields $K \rightarrow L$, which is injective by Counterexample 5.8.9). Then L is a K -vector space, and its dimension $[L : K] = \dim_K L$ is the *degree* of the extension; the extension is *finite* if the degree is finite. For a subset $S \subseteq L$, $K(S)$ denotes the smallest subfield of L containing K and S , and $K[S]$ the smallest subring. A *K -homomorphism* between extensions of K is a field homomorphism that is the identity on K ; $\text{Aut}(L/K)$ is the group of K -automorphisms of L .

Theorem 8.2.2 (Tower law). *Let $K \subseteq L \subseteq M$ be fields. If $(l_i)_{i \in I}$ is a basis of L over K and $(m_j)_{j \in J}$ a basis of M over L , then $(l_i m_j)_{(i,j) \in I \times J}$ is a basis of M over K . In particular $[M : K] = [M : L][L : K]$.*

Proof. Every $x \in M$ is $\sum_j a_j m_j$ with $a_j \in L$, and each $a_j = \sum_i c_{ij} l_i$ with $c_{ij} \in K$, so the $l_i m_j$ generate. If $\sum_{i,j} c_{ij} l_i m_j = 0$, then $\sum_j (\sum_i c_{ij} l_i) m_j = 0$ gives $\sum_i c_{ij} l_i = 0$ for all j , hence all $c_{ij} = 0$. $\square$

Definition 8.2.3. Let L/K be an extension. An element $\alpha \in L$ is *algebraic* over K if $f(\alpha) = 0$ for some nonzero $f \in K[x]$, and *transcendental* otherwise. The extension is *algebraic* if all elements of L are algebraic over K .

Proposition 8.2.4. *Let $\alpha \in L$ be algebraic over K . The kernel of the evaluation homomorphism $K[x] \rightarrow L, f \mapsto f(\alpha)$, is generated by a unique monic irreducible polynomial m_α , the minimal polynomial of α . The evaluation induces an isomorphism $K[x]/(m_\alpha) \cong K[\alpha]$, so $K[\alpha] = K(\alpha)$ is a field with basis $1, \alpha, \dots, \alpha^{d-1}$ over K , where $d = \deg m_\alpha = [K(\alpha) : K]$.*

Proof. The kernel is a nonzero ideal of the PID $K[x]$ (Theorem 7.6.5), hence generated by a unique monic polynomial. The image $K[\alpha]$ is a subring of a field, hence a domain, so the kernel is prime, and the generator is irreducible. Nonzero prime ideals of a PID are maximal (proof of Theorem 7.5.4), so $K[\alpha] \cong K[x]/(m_\alpha)$ is a field (Proposition 7.4.2). By division with remainder, the classes of $1, x, \dots, x^{d-1}$ form a basis of $K[x]/(m_\alpha)$. $\square$

Proposition 8.2.5. 1. An element α is algebraic over K if and only if $[K(\alpha) : K] < \infty$. Every finite extension is algebraic.

2. If $\alpha_1, \dots, \alpha_n$ are algebraic over K , then $K(\alpha_1, \dots, \alpha_n)/K$ is finite. The elements of L algebraic over K form a subfield of L .

3. If L/K and M/L are algebraic, then M/K is algebraic.

Proof. (1) If $[L : K] = n$ is finite and $\alpha \in L$, the $n+1$ elements $1, \alpha, \dots, \alpha^n$ are linearly dependent, which gives a nonzero polynomial with root α . The rest follows from Proposition 8.2.4. (2) Each α_i is algebraic over $K(\alpha_1, \dots, \alpha_{i-1})$, so the claim follows from the tower law. If α, β are algebraic, then $\alpha \pm \beta, \alpha\beta$ and α^{-1} lie in the finite extension $K(\alpha, \beta)$ and are algebraic by (1). (3) Let $\alpha \in M$ be a root of $c_n x^n + \dots + c_0$ with $c_i \in L$ not all zero. Then α is algebraic over $K(c_0, \dots, c_n)$, which is finite over K by (2); so $K(c_0, \dots, c_n, \alpha)$ is finite over K and α is algebraic by (1). $\square$

Example 8.2.6. $[\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] = 2$, since $x^2 - 2$ is irreducible by Proposition 4.6.4. The real number $\sqrt[3]{2}$ has minimal polynomial $x^3 - 2$, irreducible by Eisenstein's criterion (Proposition 7.6.12), so $[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$. Hence $\sqrt[3]{2} \notin \mathbb{Q}(\sqrt{2})$, by the tower law. Transcendental numbers exist by Exercise 4.10.6.

Proposition 8.2.7 (Kronecker). Let $f \in K[x]$ be irreducible. Then $L = K[x]/(f)$ is an extension of K of degree $\deg f$ in which the class of x is a root of f .

Proof. (f) is maximal, so L is a field; the composite $K \rightarrow K[x] \rightarrow L$ is injective, and $f(\bar{x}) = \bar{f}(x) = 0$. The degree is computed as in Proposition 8.2.4. $\square$

8.3 Splitting fields and algebraic closures

Definition 8.3.1. Let $f \in K[x]$ be nonconstant. An extension L/K is a *splitting field* of f if $f = c(x - \alpha_1) \cdots (x - \alpha_n)$ in $L[x]$ and $L = K(\alpha_1, \dots, \alpha_n)$. A field is *algebraically closed* if every nonconstant polynomial over it has a root, and an *algebraic closure* of K is an algebraic extension that is algebraically closed.

Proposition 8.3.2. Every nonconstant $f \in K[x]$ has a splitting field L , with $[L : K] \leq (\deg f)!$.

Proof. By induction on $n = \deg f$. Let g be an irreducible factor of f and $K_1 = K[x]/(g)$, in which f has a root α_1 and $[K_1 : K] \leq n$. Over K_1 , $f = (x - \alpha_1)f_1$ with $\deg f_1 = n - 1$; a splitting field of f_1 over K_1 is a splitting field of f over K of degree at most $n \cdot (n - 1)!$. $\square$

Theorem 8.3.3 (Extension of homomorphisms). Let $L = K(\alpha_1, \dots, \alpha_r)$ be a finite extension and $\tau : K \rightarrow M$ a field homomorphism. The number of homomorphisms $L \rightarrow M$ extending τ is at most $[L : K]$. For $i = 1, \dots, r$ let g_i be the minimal polynomial of α_i over $K(\alpha_1, \dots, \alpha_{i-1})$. If for every extension τ' of τ to $K(\alpha_1, \dots, \alpha_{i-1})$ the polynomial $g_i^{\tau'}$ splits into distinct linear factors in $M[x]$, then there are exactly $[L : K]$ extensions.

Proof. By induction on r , using the tower law, it suffices to treat $r = 1$: $L = K(\alpha)$ with minimal polynomial g . An extension σ of τ is determined by $\sigma(\alpha)$, which is a root of g^τ (the polynomial obtained by applying τ to the coefficients). Conversely, for each root $\beta \in M$ of g^τ , the homomorphism $K[x] \rightarrow M$ extending τ with $x \mapsto \beta$ kills g and so induces an extension $K(\alpha) \cong K[x]/(g) \rightarrow M$. Hence the extensions correspond to the roots of g^τ in M , of which there are at most $\deg g = [K(\alpha) : K]$ by Corollary 7.6.6, with equality if g^τ splits into distinct linear factors. $\square$

Corollary 8.3.4. *Let L and L' be splitting fields of $f \in K[x]$. Then there is a K -isomorphism $L \rightarrow L'$. For every finite extension L/K , $|\text{Aut}(L/K)| \leq [L : K]$.*

Proof. Write $L = K(\alpha_1, \dots, \alpha_n)$ with the α_i the roots of f . At each stage the minimal polynomial g_i divides f , and f splits in L' , so $g_i^{\tau'}$ has a root in L' and the proof of Theorem 8.3.3 gives at least one extension of the inclusion $K \rightarrow L'$ to $\sigma: L \rightarrow L'$. It is injective, and its image contains all roots of f in L' (it contains n roots counted with multiplicity), so it is L' . The second statement is the first part of Theorem 8.3.3 with $M = L$. $\square$

Theorem 8.3.5. *Every field K has an algebraic closure $\bar{K}$. If L/K is algebraic and $\tau: K \rightarrow M$ is a homomorphism into an algebraically closed field, then τ extends to $L \rightarrow M$. Any two algebraic closures of K are K -isomorphic.*

Proof. Existence (Artin). Let $\mathcal{F}$ be the set of monic nonconstant polynomials in $K[x]$, let $A = K[x_f : f \in \mathcal{F}]$ be the polynomial ring in variables indexed by $\mathcal{F}$ (the union of the polynomial rings in finitely many of them), and I the ideal generated by the $f(x_f)$. Then $I \neq A$: otherwise $1 = \sum_{i=1}^n g_i f_i(x_{f_i})$ for finitely many f_i ; in a splitting field of $f_1 \cdots f_n$ choose roots β_i of f_i , and apply the homomorphism $x_{f_i} \mapsto \beta_i$, other variables $\mapsto 0$, to get $1 = 0$. Let $\mathfrak{m} \supseteq I$ be maximal (Theorem 7.4.3) and $E_1 = A/\mathfrak{m}$, a field extension of K in which every $f \in \mathcal{F}$ has a root. Repeating, we get $K = E_0 \subseteq E_1 \subseteq E_2 \subseteq \cdots$ where every nonconstant polynomial over E_n has a root in E_{n+1} . The union E is algebraically closed, since a polynomial over E has coefficients in some E_n . The elements of E algebraic over K form a field $\bar{K}$ (Proposition 8.2.5). If $g \in \bar{K}[x]$ is nonconstant, a root $\alpha \in E$ is algebraic over $\bar{K}$, hence over K , so $\alpha \in \bar{K}$. Thus $\bar{K}$ is an algebraic closure.

Extension. Consider pairs (F, σ) with $K \subseteq F \subseteq L$ a subfield and $\sigma: F \rightarrow M$ extending τ , ordered by extension. Chains have upper bounds (take unions), so by Zorn's lemma there is a maximal (F, σ) . If $\alpha \in L \setminus F$, its minimal polynomial over F maps to a polynomial with a root in M , and Theorem 8.3.3 extends σ to $F(\alpha)$; contradiction. So $F = L$.

Uniqueness. If $\bar{K}$ and $\bar{K}'$ are algebraic closures, extend the inclusion $K \rightarrow \bar{K}'$ to $\sigma: \bar{K} \rightarrow \bar{K}'$. Its image is an algebraically closed field over which $\bar{K}'$ is algebraic, so every element of $\bar{K}'$ is a root of a polynomial that splits over $\sigma(\bar{K})$, and σ is surjective. $\square$

8.4 Separability

Definition 8.4.1. A nonzero polynomial $f \in K[x]$ is *separable* if it has no repeated root in a splitting field. An algebraic element is *separable* if its minimal polynomial is, and an algebraic extension is separable if all its elements are. The *derivative* of $f = \sum a_i x^i$ is $f' = \sum i a_i x^{i-1}$; it satisfies $(fg)' = f'g + fg'$.

Proposition 8.4.2. *A nonzero polynomial f is separable if and only if $\gcd(f, f') = 1$ in $K[x]$. An irreducible polynomial is separable if and only if $f' \neq 0$. If $\text{char } K = 0$, every irreducible polynomial is separable; if $\text{char } K = p > 0$, an irreducible f is inseparable if and only if $f = g(x^p)$ for some $g \in K[x]$.*

Proof. In a splitting field, if $f = (x - \alpha)^2 h$ then $f'(\alpha) = 0$; if $f = (x - \alpha)h$ with $h(\alpha) \neq 0$ then $f'(\alpha) = h(\alpha) \neq 0$. So f has a repeated root iff f and f' have a common root in a splitting field, iff they have a common factor of positive degree in $K[x]$ (a common root α makes m_α a common factor; conversely a common factor has a root in the splitting field). If f is irreducible and $f' \neq 0$, the gcd divides f and has degree $< \deg f$, so it is 1. If $f' = 0$ then $\gcd(f, f') = f$. Finally $f' = 0$ iff $ia_i = 0$ for all i , which in characteristic 0 means f is constant, and in characteristic p means $a_i = 0$ unless $p \mid i$. $\square$

Definition 8.4.3. A field K is *perfect* if it has characteristic 0, or characteristic p and the Frobenius $x \mapsto x^p$ is surjective. (The Frobenius is always an injective ring homomorphism in characteristic p , since p divides the binomial coefficients $\binom{p}{i}$ for $0 < i < p$.)

Proposition 8.4.4. *Over a perfect field every irreducible polynomial is separable, so every algebraic extension is separable. Finite fields are perfect.*

Proof. If $\text{char } K = p$ and $f = \sum b_i x^{pi}$ with $b_i = c_i^p$, then $f = (\sum c_i x^i)^p$ is not irreducible. A finite field is perfect because an injective map from a finite set to itself is surjective. $\square$

Counterexample 8.4.5. Let $K = \mathbb{F}_p(t)$ be the field of fractions of $\mathbb{F}_p[t]$. The polynomial $f = x^p - t$ is irreducible, by Eisenstein's criterion for the prime t of the UFD $\mathbb{F}_p[t]$, but $f' = 0$. In a splitting field, with $\beta^p = t$, we have $f = (x - \beta)^p$: a single root of multiplicity p . So K is not perfect and $K(\beta)/K$ is not separable.

8.5 Galois extensions

Definition 8.5.1. An algebraic extension L/K is *normal* if every irreducible polynomial in $K[x]$ with a root in L splits into linear factors in $L[x]$. For a group G of automorphisms of a field L , the *fixed field* is $L^G = \{x \in L : \sigma(x) = x \text{ for all } \sigma \in G\}$.

Lemma 8.5.2 (Artin). *Let G be a finite group of automorphisms of a field L . Then $[L : L^G] \leq |G|$.*

Proof. Let $K = L^G$, $G = \{\sigma_1, \dots, \sigma_n\}$ with $\sigma_1 = \text{id}$, and let $\alpha_1, \dots, \alpha_m \in L$ with $m > n$. The m vectors $(\sigma_i(\alpha_j))_i \in L^n$ are linearly dependent over L by Lemma 7.8.1, so the system $\sum_j \sigma_i(\alpha_j)x_j = 0$, $i = 1, \dots, n$, has a nonzero solution $(x_j) \in L^m$. Choose one with the fewest nonzero entries and, scaling, with $x_{j_0} = 1$ for some j_0 . For $\tau \in G$, applying τ gives $\sum_j \tau\sigma_i(\alpha_j)\tau(x_j) = 0$ for all i , and since $\tau\sigma_i$ runs over G as i does, $(\tau(x_j))_j$ is also a solution. The difference $(x_j - \tau(x_j))_j$ is a solution with fewer nonzero entries (the j_0 entry vanishes), so it is zero. Hence all $x_j \in K$, and the equation for $\sigma_1 = \text{id}$ is a nontrivial K -linear relation $\sum_j \alpha_j x_j = 0$. So any $m > n$ elements are linearly dependent over K . $\square$

Theorem 8.5.3. *For a finite extension L/K with $G = \text{Aut}(L/K)$, the following are equivalent:*

1. $|G| = [L : K]$;
2. $L^G = K$;
3. L is a splitting field over K of a separable polynomial;
4. L/K is normal and separable.

Such an extension is called *Galois*, and $\text{Gal}(L/K) = G$ its Galois group. Moreover, if G is any finite group of automorphisms of a field L , then L/L^G is Galois with group G .

Proof. For the last statement: $G \subseteq \text{Aut}(L/L^G)$, so by Lemma 8.5.2 and Corollary 8.3.4, $[L : L^G] \leq |G| \leq |\text{Aut}(L/L^G)| \leq [L : L^G]$.

(1) $\Rightarrow$ (2): $K \subseteq L^G$ and by the last statement $[L : L^G] = |G| = [L : K]$, so $L^G = K$ by the tower law.

(2) $\Rightarrow$ (4): Let $\alpha \in L$ and let $\alpha_1, \dots, \alpha_r$ be the distinct elements of the orbit $G\alpha$. The polynomial $g = \prod_i (x - \alpha_i)$ has coefficients fixed by G , hence in K , and $g(\alpha) = 0$, so $m_\alpha \mid g$. Every $\alpha_i = \sigma(\alpha)$ is a root of m_α , so $g \mid m_\alpha$. Thus $m_\alpha = g$ splits in L with distinct roots.

(4) $\Rightarrow$ (3): Write $L = K(\beta_1, \dots, \beta_s)$ and let f be the product of the distinct minimal polynomials of the β_j . It is separable (distinct monic irreducible polynomials have no common root), it splits in L by normality, and its roots generate L .

(3) $\Rightarrow$ (1): Apply Theorem 8.3.3 with $M = L$ and τ the inclusion $K \rightarrow L$: at every stage the minimal polynomial divides f , which splits into distinct linear factors in L . So there are exactly $[L : K]$ K -homomorphisms $L \rightarrow L$. Each is an injective K -linear map of the finite-dimensional K -vector space L to itself, hence bijective by Theorem 7.8.5 (its image is a subspace of the same finite dimension). So $|G| = [L : K]$. $\square$

Theorem 8.5.4 (Fundamental theorem of Galois theory). *Let L/K be a finite Galois extension with group G . The maps $H \mapsto L^H$ and $F \mapsto \text{Gal}(L/F)$ are mutually inverse, inclusion-reversing bijections between subgroups $H \leq G$ and intermediate fields $K \subseteq F \subseteq L$. Moreover $[L : L^H] = |H|$ and $[L^H : K] = [G : H]$. An intermediate field F is Galois over K if and only if $\text{Gal}(L/F)$ is normal in G , and then restriction induces an isomorphism $G/\text{Gal}(L/F) \cong \text{Gal}(F/K)$.*

Proof. If L is the splitting field of a separable $f \in K[x]$, it is also the splitting field of f over F , so L/F is Galois and $L^{\text{Gal}(L/F)} = F$ by Theorem 8.5.3. For $H \leq G$, the last statement of that theorem gives $\text{Gal}(L/L^H) = H$ and $[L : L^H] = |H|$; then $[L^H : K] = [L : K]/|H| = [G : H]$ by the tower law. The maps clearly reverse inclusions.

For $\sigma \in G$, $\text{Gal}(L/\sigma(F)) = \sigma \text{Gal}(L/F) \sigma^{-1}$. If F/K is Galois, it is normal, so σ maps each $\alpha \in F$ to a root of m_α , which lies in F ; thus $\sigma(F) = F$ and $\text{Gal}(L/F)$ is normal. Conversely, if $N = \text{Gal}(L/F)$ is normal, then $\sigma(F) = L^{\sigma N \sigma^{-1}} = F$ for all σ , and restriction is a homomorphism $G \rightarrow \text{Aut}(F/K)$ with kernel N . Its image has $[G : N] = [F : K]$ elements and lies in a group with at most $[F : K]$ elements (Corollary 8.3.4), so $|\text{Aut}(F/K)| = [F : K]$, F/K is Galois and the restriction map is surjective. $\square$

Example 8.5.5. The field $L = \mathbb{Q}(\sqrt{2}, \sqrt{3})$ is the splitting field of $(x^2 - 2)(x^2 - 3)$, of degree 4 over $\mathbb{Q}$: if $\sqrt{3} = a + b\sqrt{2}$ with $a, b \in \mathbb{Q}$, squaring gives $2ab = 0$, and neither $a = 0$ nor $b = 0$ is possible by the argument of Proposition 4.6.4. Its Galois group consists of the four automorphisms $\sqrt{2} \mapsto \pm\sqrt{2}$, $\sqrt{3} \mapsto \pm\sqrt{3}$, so it is isomorphic to $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$. It has three subgroups of order 2, corresponding to the three quadratic subfields $\mathbb{Q}(\sqrt{2})$, $\mathbb{Q}(\sqrt{3})$ and $\mathbb{Q}(\sqrt{6})$.

Counterexample 8.5.6. The extension $\mathbb{Q}(\sqrt[3]{2})/\mathbb{Q}$ of degree 3 is not Galois. A $\mathbb{Q}$ -automorphism must send $\sqrt[3]{2}$ to a cube root of 2 in $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$, and the only real cube root of 2 is $\sqrt[3]{2}$ (Proposition 4.7.4); so the automorphism group is trivial. The polynomial $x^3 - 2$ has a root in $\mathbb{Q}(\sqrt[3]{2})$ but does not split there, so the extension is not normal.

8.6 Finite fields

Lemma 8.6.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

Proof. Let G be such a subgroup. By Theorem 6.11.6, $G \cong \mathbb{Z}/d_1\mathbb{Z} \oplus \dots \oplus \mathbb{Z}/d_k\mathbb{Z}$ with $d_1 \mid \dots \mid d_k$, so every element satisfies $x^{d_k} = 1$. The polynomial $x^{d_k} - 1$ has at most d_k roots, so $|G| \leq d_k$ and $k \leq 1$. $\square$

- Theorem 8.6.2.** 1. A finite field has $q = p^n$ elements, where p is its characteristic and n its degree over $\mathbb{F}_p$.
2. For every prime power $q = p^n$ there is a field $\mathbb{F}_q$ with q elements, unique up to isomorphism: a splitting field of $x^q - x$ over $\mathbb{F}_p$, consisting of the roots of this polynomial.
3. $\mathbb{F}_{q^n}/\mathbb{F}_q$ is Galois with cyclic group of order n , generated by the Frobenius $x \mapsto x^q$. The subfields of $\mathbb{F}_{p^n}$ are the $\mathbb{F}_{p^m}$ with $m \mid n$.

Proof. (1) The characteristic is prime by Lemma 7.4.6, and F is a vector space of some finite dimension n over $\mathbb{F}_p$. (2) Let L be a splitting field of $f = x^q - x$. Since $f' = -1$, f has q distinct roots. The set of roots is closed under sums, products and inverses because $x \mapsto x^q$ is a ring homomorphism (a power of the Frobenius), so it is a subfield, and it equals L . If F has q elements, then $F^\times$ has order $q - 1$, so $x^q = x$ for all $x \in F$ by Theorem 6.3.3; hence F is a splitting field of f and $F \cong L$ by Corollary 8.3.4. (3) $\mathbb{F}_{q^n}$ is the splitting field of the separable polynomial $x^{q^n} - x$ over $\mathbb{F}_q$, hence Galois of degree n . The Frobenius $\phi(x) = x^q$ fixes $\mathbb{F}_q$, and $\phi^k = \text{id}$ means all q^n elements are roots of $x^{q^k} - x$, which requires $k \geq n$. So ϕ has order n and generates the Galois group. By the fundamental theorem, subfields correspond to subgroups of $\mathbb{Z}/n\mathbb{Z}$, one for each divisor. $\square$

Theorem 8.6.3 (Primitive element theorem). Every finite separable extension L/K is simple: $L = K(\gamma)$ for some $\gamma \in L$.

Proof. If K is finite, so is L , and a generator γ of the cyclic group $L^\times$ (Lemma 8.6.1) works. Let K be infinite. By induction on the number of generators it suffices to treat $L = K(\alpha, \beta)$. Let M be a splitting field of $m_\alpha m_\beta$ over L . The minimal polynomials over intermediate fields divide m_α or m_β , which are separable, so by Theorem 8.3.3 there are exactly $n = [L : K]$ distinct K -homomorphisms $\sigma_1, \dots, \sigma_n: L \rightarrow M$. The polynomial $P(t) = \prod_{i < j} ((\sigma_i(\alpha) - \sigma_j(\alpha)) + t(\sigma_i(\beta) - \sigma_j(\beta))) \in M[t]$ is nonzero, because $\sigma_i \neq \sigma_j$ differ on α or on β . It has finitely many roots, so there is $c \in K$ with $P(c) \neq 0$. Then $\gamma = \alpha + c\beta$ has n distinct images $\sigma_i(\gamma)$. The restrictions of the σ_i to $K(\gamma)$ are therefore distinct, so $[K(\gamma) : K] \geq n$ by Theorem 8.3.3, and $K(\gamma) = L$. $\square$

8.7 The fundamental theorem of algebra

Lemma 8.7.1. Every polynomial of odd degree in $\mathbb{R}[x]$ has a root in $\mathbb{R}$.

Proof. We may assume $p = x^n + a_{n-1}x^{n-1} + \dots + a_0$ is monic, with n odd. Let $M = 1 + \sum_i |a_i|$. For $|x| \geq M$ we have $|a_{n-1}x^{n-1} + \dots + a_0| \leq (M-1)|x|^{n-1} < |x|^n$, so $p(x)$ has the sign of x^n , that is, of x . On $[-M, M]$, the identity $x^i - y^i = (x - y) \sum_k x^k y^{i-1-k}$ gives $|p(x) - p(y)| \leq C|x - y|$ with $C = \sum_{i=1}^n i|a_i|M^{i-1}$ (with $a_n = 1$). Let $S = \{x \in [-M, M] : p(x) < 0\}$, which contains $-M$, and $c = \sup S$. If $p(c) < 0$, then $c < M$ and $p < 0$ on $[c, c + |p(c)|/C] \cap [-M, M]$, contradicting $c = \sup S$. If $p(c) > 0$, then $c > -M$ and $p > 0$ on $(c - p(c)/C, c]$, so $\sup S \leq c - p(c)/C$, again a contradiction. Hence $p(c) = 0$. $\square$

Lemma 8.7.2. Every group of order p^n with $n \geq 1$ has a subgroup of index p .

Proof. By induction on n . By Proposition 6.6.7 and Theorem 6.6.8 there is $z \in Z(G)$ of order p . If $n = 1$ take $\{e\}$. Otherwise $G/\langle z \rangle$ has order p^{n-1} and, by induction, a subgroup of index p , whose preimage in G has index p . $\square$

Theorem 8.7.3 (Fundamental theorem of algebra). The field $\mathbb{C}$ is algebraically closed.

Proof. Let $f \in \mathbb{C}[x]$ be nonconstant, and let $\bar{f}$ be obtained by conjugating the coefficients. Then $g = (x^2 + 1)f\bar{f}$ lies in $\mathbb{R}[x]$. Let L be a splitting field of g over $\mathbb{C}$; it is also a splitting field of g over $\mathbb{R}$ (since $\mathbb{C} = \mathbb{R}(i)$ and i is a root of g), hence Galois over $\mathbb{R}$ as $\text{char } \mathbb{R} = 0$. Let $G = \text{Gal}(L/\mathbb{R})$ and let P be a Sylow 2-subgroup. The fixed field L^P has odd degree $[G : P]$ over $\mathbb{R}$; by Theorem 8.6.3 it is $\mathbb{R}(\gamma)$, and the minimal polynomial of γ is irreducible of odd degree, hence of degree 1 by Lemma 8.7.1. So $P = G$ is a 2-group, and so is $H = \text{Gal}(L/\mathbb{C})$. If $H \neq \{e\}$, it has a subgroup of index 2 (Lemma 8.7.2), whose fixed field is an extension of $\mathbb{C}$ of degree 2, generated by a root of an irreducible quadratic polynomial over $\mathbb{C}$. This contradicts Proposition 4.9.3. So $L = \mathbb{C}$, and f has its roots in $\mathbb{C}$. $\square$

Corollary 8.7.4. *Every nonconstant polynomial over $\mathbb{C}$ is a product of linear factors, and $\mathbb{C}$ is an algebraic closure of $\mathbb{R}$. The irreducible polynomials in $\mathbb{R}[x]$ are the linear ones and the quadratic ones without real roots.*

Proof. The first statement follows by induction on the degree. Since $[\mathbb{C} : \mathbb{R}] = 2$, the extension is algebraic. An irreducible real polynomial of degree ≥ 2 has a non-real root α ; then $\bar{\alpha}$ is also a root, and $(x - \alpha)(x - \bar{\alpha}) \in \mathbb{R}[x]$ divides it. $\square$

8.8 Cyclotomic fields

Definition 8.8.1. For $n \geq 1$, the n -th roots of unity in $\mathbb{C}$ form a group μ_n of order n (the polynomial $x^n - 1$ has derivative nx^{n-1} , so its n roots are distinct), cyclic by Lemma 8.6.1. Its generators are the *primitive n -th roots of unity*. The *n -th cyclotomic polynomial* is $\Phi_n = \prod_{\zeta}(x - \zeta)$ over the primitive n -th roots of unity ζ ; it has degree $\varphi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$.

Theorem 8.8.2. Φ_n lies in $\mathbb{Z}[x]$ and is irreducible over $\mathbb{Q}$. The extension $\mathbb{Q}(\zeta)/\mathbb{Q}$ for a primitive n -th root of unity ζ is Galois, and $\sigma \mapsto a_\sigma$, where $\sigma(\zeta) = \zeta^{a_\sigma}$, is an isomorphism $\text{Gal}(\mathbb{Q}(\zeta)/\mathbb{Q}) \cong (\mathbb{Z}/n\mathbb{Z})^\times$.

Proof. Every n -th root of unity is a primitive d -th root of unity for exactly one $d \mid n$, so $x^n - 1 = \prod_{d \mid n} \Phi_d$. By induction on n , Φ_n is the quotient of $x^n - 1$ by a monic polynomial in $\mathbb{Z}[x]$, hence lies in $\mathbb{Z}[x]$ by division with remainder.

Let f be the minimal polynomial of ζ over $\mathbb{Q}$. Since $f \mid \Phi_n$ in $\mathbb{Q}[x]$, write $\Phi_n = fg$; by Lemma 7.6.10 (and since f, g are monic) $f, g \in \mathbb{Z}[x]$. Let p be a prime with $p \nmid n$. We claim $f(\zeta^p) = 0$. Otherwise $g(\zeta^p) = 0$, so $f(x) \mid g(x^p)$ in $\mathbb{Z}[x]$. Reducing modulo p and using $\bar{g}(x^p) = \bar{g}(x)^p$ in $\mathbb{F}_p[x]$, we find that $\bar{f}$ and $\bar{g}$ have a common irreducible factor, so $\bar{\Phi}_n = \bar{f}\bar{g}$, and hence $x^n - 1$, has a repeated root in a splitting field over $\mathbb{F}_p$. But the derivative nx^{n-1} is coprime to $x^n - 1$ in $\mathbb{F}_p[x]$, a contradiction. Every primitive root of unity is ζ^a with $\gcd(a, n) = 1$, and writing a as a product of primes not dividing n , repeated application of the claim shows that ζ^a is a root of f . So $f = \Phi_n$.

$\mathbb{Q}(\zeta)$ contains all n -th roots of unity (powers of ζ), so it is the splitting field of $x^n - 1$, hence Galois. Each σ maps ζ to a primitive root ζ^{a_σ} , and $\sigma \mapsto a_\sigma$ is an injective homomorphism into $(\mathbb{Z}/n\mathbb{Z})^\times$, surjective since $|\text{Gal}| = \deg \Phi_n = \varphi(n)$. $\square$

8.9 Solvability by radicals

In this section all fields have characteristic 0.

Definition 8.9.1. A *radical extension* of K is an extension E/K with a chain $K = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_r = E$ where $E_{i+1} = E_i(a_i)$ and $a_i^{n_i} \in E_i$ for some $n_i \geq 1$. A polynomial $f \in K[x]$ is *solvable by radicals* if a splitting field of f is contained in a radical extension of K .

Lemma 8.9.2. *Let K contain a primitive n -th root of unity ω and let b in an extension satisfy $b^n = a \in K$. Then $K(b)/K$ is Galois with cyclic Galois group. For any K and $m \geq 1$, the splitting field of $x^m - 1$ over K is Galois with abelian Galois group.*

Proof. If $b = 0$ there is nothing to prove. Otherwise the roots of $x^n - a$ are the distinct $\omega^i b$, all in $K(b)$, so $K(b)$ is a splitting field of a separable polynomial. Each σ satisfies $\sigma(b) = \omega^{k_\sigma} b$, and since σ fixes ω , $\sigma \mapsto \omega^{k_\sigma}$ is an injective homomorphism into the cyclic group generated by ω . For the second statement, the roots of $x^m - 1$ form a cyclic group generated by some ζ (Lemma 8.6.1; they are distinct in characteristic 0), and $\sigma(\zeta) = \zeta^{a_\sigma}$ defines an injective homomorphism into the abelian group $(\mathbb{Z}/m\mathbb{Z})^\times$. $\square$

Theorem 8.9.3 (Galois). *If $f \in K[x]$ is solvable by radicals, then the Galois group of a splitting field of f over K is solvable.*

Proof. Let $L \subseteq E$ be a splitting field of f inside a radical extension with data a_i, n_i as in Definition 8.9.1. Let $m = n_0 \cdots n_{r-1}$, $h = (x^m - 1) \prod_i m_{a_i}$, and let N be a splitting field of h over E (Proposition 8.3.2). Since E is generated over K by roots of h , N is also a splitting field of h over K . Then N/K is Galois with group G , and N is generated over K by a primitive m -th root of unity ζ and the elements $\sigma(a_i)$, $\sigma \in G$, since the roots of m_{a_i} are the $\sigma(a_i)$ (proof of Theorem 8.5.3). Adjoin these generators in the order ζ , then all $\sigma(a_0)$, then all $\sigma(a_1)$, and so on, obtaining a chain $K = F_0 \subseteq F_1 = K(\zeta) \subseteq F_2 \subseteq \cdots \subseteq F_s = N$. Each $F_{j+1} = F_j(b)$ with F_1/F_0 abelian Galois, and for $j \geq 1$, $b = \sigma(a_i)$ with $b^{n_i} = \sigma(a_i^{n_i})$ in F_j (because $a_i^{n_i} \in E_i = K(a_0, \dots, a_{i-1})$) and F_j containing a primitive n_i -th root of unity. By Lemma 8.9.2 each F_{j+1}/F_j is Galois with abelian group. By the fundamental theorem, $G_j = \text{Gal}(N/F_j)$ satisfies $G_{j+1} \trianglelefteq G_j$ and $G_j/G_{j+1} \cong \text{Gal}(F_{j+1}/F_j)$ is abelian; so G is solvable (Proposition 6.9.3), and so is its quotient $\text{Gal}(L/K) \cong G/\text{Gal}(N/L)$. $\square$

Remark 8.9.4. The converse also holds: if the Galois group is solvable, then f is solvable by radicals. The proof uses Hilbert's Theorem 90 to show that cyclic extensions of fields containing enough roots of unity are generated by radicals; see [Lan02, Chapter VI, §7].

Lemma 8.9.5. *Let p be prime and $G \leq S_p$ a subgroup acting transitively on $\{1, \dots, p\}$ and containing a transposition. Then $G = S_p$.*

Proof. The orbit has size p , so p divides $|G|$ (Theorem 6.6.2) and G contains an element of order p (Theorem 6.6.8), necessarily a p -cycle σ . Let $(a\ b) \in G$. Some power of σ sends a to b and is again a p -cycle; renumbering, we may assume it is $(1\ 2 \cdots p)$ and the transposition is $(1\ 2)$. Conjugating $(1\ 2)$ by powers of the p -cycle gives all $(i\ i+1)$, and these generate S_p since $(i\ j) = (j-1\ j)(i\ j-1)(j-1\ j)$ recursively. $\square$

Example 8.9.6. The polynomial $f = x^5 - 4x + 2$ is not solvable by radicals over $\mathbb{Q}$. It is irreducible by Eisenstein's criterion at $p = 2$. Its roots in $\mathbb{C}$ (Theorem 8.7.3) satisfy $\sum \alpha_i = 0$ and $\sum_{i < j} \alpha_i \alpha_j = 0$ (the coefficients of x^4 and x^3 vanish), so $\sum \alpha_i^2 = 0$; hence not all roots are real. Since $f(-2) < 0 < f(0)$, $f(1) < 0 < f(2)$, the proof of Lemma 8.7.1 gives real roots in $(-2, 0)$, $(0, 1)$ and $(1, 2)$. Non-real roots come in conjugate pairs, so there are exactly 3 real and 2 non-real roots. Complex conjugation restricts to an element of the Galois group G of the splitting field in $\mathbb{C}$ that exchanges the two non-real roots and fixes the others: a transposition. G acts transitively on the roots (they are roots of one irreducible polynomial, and Theorem 8.3.3 moves any root to any other). By Lemma 8.9.5, $G \cong S_5$, which is not solvable (Proposition 6.9.3).

8.10 Transcendence bases

Definition 8.10.1. Let L/K be an extension. A subset $S \subseteq L$ is *algebraically independent* over K if for all distinct $s_1, \dots, s_n \in S$ the homomorphism $K[x_1, \dots, x_n] \rightarrow L, x_i \mapsto s_i$, is injective. A *transcendence basis* is an algebraically independent set S such that L is algebraic over $K(S)$. An extension is *purely transcendental* if $L = K(S)$ for an algebraically independent S .

Lemma 8.10.2. Let $S \subseteq L$ be algebraically independent over K and $\alpha \in L$. Then $S \cup \{\alpha\}$ is algebraically independent if and only if $\alpha \notin K(S)$ and α is transcendental over $K(S)$.

Proof. A nonzero polynomial relation $P(s_1, \dots, s_n, \alpha) = 0$ with $P \notin K[x_1, \dots, x_n]$ is, after collecting powers of α , a nonzero polynomial over $K[s_1, \dots, s_n] \cong K[x_1, \dots, x_n]$ with root α ; and conversely a nonzero polynomial over $K(S)$ with root α can be cleared of denominators to give such a relation. $\square$

Theorem 8.10.3. Every extension L/K has a transcendence basis, and any two have the same cardinality, the transcendence degree $\text{trdeg}_K L$. More precisely, if $T \subseteq L$ is such that L is algebraic over $K(T)$ and $S_0 \subseteq T$ is algebraically independent, then there is a transcendence basis S with $S_0 \subseteq S \subseteq T$.

Proof. By Zorn's lemma choose a maximal algebraically independent subset S with $S_0 \subseteq S \subseteq T$. By Lemma 8.10.2 every $t \in T$ is algebraic over $K(S)$, so $K(T)$ is algebraic over $K(S)$ and L is algebraic over $K(S)$ by Proposition 8.2.5.

We use the following fact: if F is a field, s is algebraic over $F(t)$ but not over F , then t is algebraic over $F(s)$. Indeed, clearing denominators gives a nonzero $P = \sum_k P_k(X)Y^k \in F[X, Y]$ with $P(s, t) = 0$; not all $P_k(s)$ vanish, since otherwise all $P_k = 0$ as s is transcendental over F ; so t is a root of the nonzero polynomial $\sum_k P_k(s)Y^k$.

Let S and T be transcendence bases. Suppose first $T = \{t_1, \dots, t_n\}$ is finite and let $s_1, \dots, s_m \in S$ be distinct. Suppose that after $i < m$ steps L is algebraic over $K(s_1, \dots, s_i, T_i)$ with $T_i \subseteq T$ of size $n - i$. Then s_{i+1} is algebraic over this field; let $T' \subseteq T_i$ be minimal such that s_{i+1} is algebraic over $K(s_1, \dots, s_i, T')$. Since s_{i+1} is not algebraic over $K(s_1, \dots, s_i)$, T' is nonempty; pick $t \in T'$ and let $F = K(s_1, \dots, s_i, T' \setminus \{t\})$. By the fact above, t is algebraic over $F(s_{i+1})$, so with $T_{i+1} = T_i \setminus \{t\}$ the field $K(s_1, \dots, s_i, T_i)$ is algebraic over $K(s_1, \dots, s_{i+1}, T_{i+1})$, and so is L . In particular $T_i \neq \emptyset$ at every step $i < m$, so $m \leq n$. Hence S is finite with $|S| \leq |T|$, and by symmetry $|S| = |T|$.

If both are infinite, each $t \in T$ is algebraic over $K(S_t)$ for a finite $S_t \subseteq S$, and $S' = \bigcup_t S_t$ satisfies $|S'| \leq |T|$ by Proposition 3.7.5. Then L is algebraic over $K(S')$, so every $s \in S$ is algebraic over $K(S')$, which forces $S' = S$ by Lemma 8.10.2. So $|S| \leq |T|$, and by symmetry the cardinalities agree. $\square$

Theorem 8.10.4. Let Ω and Ω' be algebraically closed fields of the same characteristic, with prime fields $P \subseteq \Omega$ and $P' \subseteq \Omega'$, and $\text{trdeg}_P \Omega = \text{trdeg}_{P'} \Omega'$. Then every bijection between transcendence bases $B \rightarrow B'$ extends to an isomorphism $\Omega \rightarrow \Omega'$.

Proof. The prime fields are isomorphic ($\mathbb{Q}$ or $\mathbb{F}_p$). A bijection $B \rightarrow B'$ induces an isomorphism of polynomial rings $P[B] \cong P'[B']$ and hence of their fields of fractions $P(B) \cong P'(B')$. The field Ω is an algebraic closure of $P(B)$ and Ω' one of $P'(B')$, so by Theorem 8.3.5 the isomorphism extends to an isomorphism $\Omega \rightarrow \Omega'$. $\square$

Corollary 8.10.5. $\text{trdeg}_{\mathbb{Q}} \mathbb{C} = 2^{\aleph_0}$. If $\alpha, \beta \in \mathbb{C}$ are both transcendental over $\mathbb{Q}$, there is a field automorphism σ of $\mathbb{C}$ with $\sigma(\alpha) = \beta$. Every automorphism of the algebraic closure $\bar{\mathbb{Q}}$ of $\mathbb{Q}$ in $\mathbb{C}$ extends to an automorphism of $\mathbb{C}$.

Proof. If B is a transcendence basis, $\mathbb{C}$ is algebraic over $\mathbb{Q}(B)$, and a field algebraic over a field of infinite cardinality κ has cardinality κ (each element is a root of one of κ polynomials, each with finitely many roots); $\mathbb{Q}(B)$ has cardinality $\max(|B|, \aleph_0)$. Since $|\mathbb{C}| = 2^{\aleph_0}$ (Proposition 4.8.6) is uncountable, $|B| = 2^{\aleph_0}$. Extend $\{\alpha\}$ and $\{\beta\}$ to transcendence bases B, B' (Theorem 8.10.3 with $T = \mathbb{C}$ and $S_0 = \{\alpha\}$, respectively $S_0 = \{\beta\}$); both have cardinality $2^{\aleph_0}$, and removing α and β leaves sets of the same cardinality, so there is a bijection $B \rightarrow B'$ with $\alpha \mapsto \beta$. Apply Theorem 8.10.4. For the last statement, choose a transcendence basis B of $\mathbb{C}$ over $\bar{\mathbb{Q}}$ (it is also one over $\mathbb{Q}$), extend the automorphism to $\bar{\mathbb{Q}}(B)$ by fixing B , and then to $\mathbb{C}$ by Theorem 8.3.5. $\square$

Remark 8.10.6. The automorphisms of Corollary 8.10.5 are constructed with the axiom of choice, and apart from the identity and complex conjugation they are not continuous (Exercise 4.10.3 shows that the only ones fixing $\mathbb{R}$ are these two). They nevertheless act on algebraic varieties defined over $\mathbb{C}$ by acting on coefficients, and the question of how they interact with Hodge theory is the subject of Chapter 71.

Remark 8.10.7. For an algebraic extension L/K that is normal and separable but infinite, $\text{Gal}(L/K)$ is the inverse limit of the finite groups $\text{Gal}(F/K)$ over finite Galois subextensions F , and the fundamental theorem holds for closed subgroups with respect to the resulting profinite topology; see [Lan02, Chapter VI, §14]. Topological groups are introduced in Chapter 16.

8.11 Exercises

Exercise 8.11.1. Show that the splitting field of $x^3 - 2$ over $\mathbb{Q}$ is $\mathbb{Q}(\sqrt[3]{2}, \omega)$ with $\omega = (-1 + \sqrt{-3})/2$, that it has degree 6, and that its Galois group is isomorphic to S_3 . Describe all intermediate fields.

Exercise 8.11.2. Show that if $\alpha \in \mathbb{R}$ lies in a tower $\mathbb{Q} = K_0 \subseteq \cdots \subseteq K_r$ with $[K_{i+1} : K_i] = 2$, then $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ is a power of 2. Deduce that $\sqrt[3]{2}$ is not in any such tower (impossibility of doubling the cube).

Exercise 8.11.3. Show that for every $n \geq 1$ there is an irreducible polynomial of degree n over $\mathbb{F}_p$, and that $x^{p^n} - x$ is the product of all monic irreducible polynomials over $\mathbb{F}_p$ of degree dividing n .

Exercise 8.11.4. Let G be a group and $\chi_1, \dots, \chi_n : G \rightarrow L^\times$ distinct homomorphisms into the multiplicative group of a field. Show that the χ_i are linearly independent over L as maps $G \rightarrow L$.

Exercise 8.11.5. Let k be a field and $G \leq \text{Aut}(k(t)/k)$ the group of order 2 generated by $t \mapsto 1/t$. Show that $k(t)^G = k(t + 1/t)$.

Exercise 8.11.6. Show that $\bar{\mathbb{Q}}$ is countable and that $\text{trdeg}_{\bar{\mathbb{Q}}} \mathbb{C} = 2^{\aleph_0}$.

Chapter 9

Linear Algebra

9.1 Introduction

This chapter develops linear algebra over a field: linear maps and matrices, duality, determinants, eigenvalues and canonical forms, bilinear forms, inner product spaces and the spectral theorem, and complex structures on real vector spaces. The last topic is the linear algebra underlying Hodge theory: the decomposition $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ of a real vector space with a complex structure is the pointwise version of the decomposition of forms on a complex manifold (Chapter 41) and of the Hodge decomposition (Chapter 45).

Bases and dimension were introduced in Section 7.8. References are [Hal74], [Axl15] and [Lan02, Chapters XIII–XV].

Prerequisites. Chapter 6 (Groups), Chapter 7 (Rings and Modules) and Chapter 8 (Fields and Galois Theory).

Throughout, k is a field and vector spaces are k -vector spaces, unless stated otherwise.

9.2 Linear maps and dimension

Proposition 9.2.1. *Let V be a vector space and $W \subseteq V$ a subspace.*

1. *W has a complement: a subspace U with $V = W \oplus U$ (that is, $W + U = V$ and $W \cap U = 0$). Then $U \cong V/W$ and $\dim V = \dim W + \dim V/W$.*
2. *$\dim W \leq \dim V$, and if $\dim V$ is finite and $\dim W = \dim V$ then $W = V$.*
3. *If $U, W \subseteq V$ are finite-dimensional, $\dim(U + W) + \dim(U \cap W) = \dim U + \dim W$.*

Proof. (1) Extend a basis B_W of W to a basis B of V (Theorem 7.8.4) and let U be spanned by $B \setminus B_W$. The projection $V \rightarrow V/W$ restricts to an isomorphism $U \rightarrow V/W$, and $B = B_W \sqcup (B \setminus B_W)$ gives the formula (as cardinals). (2) The first claim follows from (1). If $\dim W = \dim V < \infty$, a basis of W has $\dim V$ elements and extends to a basis of V , which by Theorem 7.8.5 has no further elements. (3) Extend a basis of $U \cap W$ to bases of U and of W ; the union of these bases is a basis of $U + W$. $\square$

Theorem 9.2.2 (Rank–nullity). *For a linear map $f: V \rightarrow W$, $V/\ker f \cong \operatorname{im} f$, so $\dim V = \dim \ker f + \dim \operatorname{im} f$. If $\dim V = \dim W < \infty$, then f is injective if and only if it is surjective.*

Proof. The isomorphism is the first isomorphism theorem (Proposition 7.7.3); combine with Proposition 9.2.1(1). If $\dim V = \dim W = n$, then f is injective iff $\dim \operatorname{im} f = n$ iff $\operatorname{im} f = W$ by Proposition 9.2.1(2). $\square$

Counterexample 9.2.3. The last statement fails in infinite dimension. On $V = k^{(\mathbb{N})}$ the shift $e_n \mapsto e_{n+1}$ is injective but not surjective, and $e_0 \mapsto 0, e_{n+1} \mapsto e_n$ is surjective but not injective.

9.3 Matrices

Definition 9.3.1. Let V and W have ordered bases $\mathcal{B} = (v_1, \dots, v_n)$ and $\mathcal{C} = (w_1, \dots, w_m)$. The *matrix* of a linear map $f: V \rightarrow W$ is the $m \times n$ matrix $[f]_{\mathcal{C}, \mathcal{B}} = (a_{ij})$ with $f(v_j) = \sum_i a_{ij} w_i$: its j -th column contains the coordinates of $f(v_j)$.

Proposition 9.3.2. The map $f \mapsto [f]_{\mathcal{C}, \mathcal{B}}$ is an isomorphism of vector spaces $\text{Hom}_k(V, W) \rightarrow M_{m \times n}(k)$, so $\dim \text{Hom}_k(V, W) = mn$. Composition corresponds to matrix multiplication: $[g \circ f]_{\mathcal{D}, \mathcal{B}} = [g]_{\mathcal{D}, \mathcal{C}} [f]_{\mathcal{C}, \mathcal{B}}$. If $\mathcal{B}', \mathcal{C}'$ are other bases and $P = [\text{id}]_{\mathcal{B}, \mathcal{B}'}$, $Q = [\text{id}]_{\mathcal{C}, \mathcal{C}'}$, then $[f]_{\mathcal{C}', \mathcal{B}'} = Q^{-1} [f]_{\mathcal{C}, \mathcal{B}} P$. In particular the matrices of an endomorphism in different bases are similar: $A' = P^{-1} A P$.

Proof. A linear map is determined by the images of a basis, which can be arbitrary (Proposition 7.7.5); this gives bijectivity, and linearity is clear. If $f(v_j) = \sum_i a_{ij} w_i$ and $g(w_i) = \sum_l b_{li} u_l$, then $g(f(v_j)) = \sum_l (\sum_i b_{li} a_{ij}) u_l$. The change of basis formula follows by writing $f = \text{id}_W \circ f \circ \text{id}_V$ and using that $[\text{id}]_{\mathcal{C}', \mathcal{C}} = Q^{-1}$. $\square$

Definition 9.3.3. The *rank* of a linear map is $\dim \text{im } f$. The *column rank* of a matrix $A \in M_{m \times n}(k)$ is the dimension of the span of its columns in k^m , that is, the rank of the map $k^n \rightarrow k^m$ it defines, and its *row rank* is the column rank of the transpose A^T .

9.4 Duality

Definition 9.4.1. The *dual* of V is $V^* = \text{Hom}_k(V, k)$. For $f: V \rightarrow W$, the *transpose* is $f^*: W^* \rightarrow V^*$, $\varphi \mapsto \varphi \circ f$. For a subspace $W \subseteq V$, the *annihilator* is $W^\perp = \{\varphi \in V^* : \varphi|_W = 0\}$. If $(v_1, \dots, v_n)$ is a basis of V , the *dual basis* $(v_1^*, \dots, v_n^*)$ is given by $v_i^*(v_j) = \delta_{ij}$.

Proposition 9.4.2. Let V, W be finite-dimensional.

1. The dual basis is a basis of V^* ; so $\dim V^* = \dim V$.
2. The map $\text{ev}: V \rightarrow V^{**}$ of Example 5.5.5 is an isomorphism.
3. In dual bases, the matrix of f^* is the transpose of the matrix of f .
4. $\dim W^\perp = \dim V - \dim W$ for $W \subseteq V$, and $(\text{im } f)^\perp = \ker f^*$.
5. $\text{rank } f^* = \text{rank } f$. Hence row rank equals column rank for every matrix.

Proof. (1) $\varphi = \sum_i \varphi(v_i) v_i^*$ for $\varphi \in V^*$, and the coefficients are determined by evaluating on the v_j . (2) ev is injective: if $v \neq 0$, extend v to a basis and take the corresponding dual basis element. Both sides have dimension n ; apply Theorem 9.2.2. (3) If $f(v_j) = \sum_i a_{ij} w_i$ then $f^*(w_i^*)(v_j) = a_{ij}$, so $f^*(w_i^*) = \sum_j a_{ij} v_j^*$. (4) Extend a basis $v_1, \dots, v_r$ of W to a basis of V ; then $W^\perp$ has basis $v_{r+1}^*, \dots, v_n^*$. And $\varphi \in \ker f^*$ iff $\varphi \circ f = 0$ iff φ vanishes on $\text{im } f$. (5) By (4) and rank-nullity for f^* , $\text{rank } f^* = \dim W - \dim \ker f^* = \dim W - \dim(\text{im } f)^\perp = \dim \text{im } f$. Apply (3). $\square$

Counterexample 9.4.3. In infinite dimension $\text{ev}: V \rightarrow V^{**}$ is injective (by the same argument) but not surjective. Let $k = \mathbb{Q}$ and $V = \mathbb{Q}^{(\mathbb{N})}$, which is countable. Then $V^* \cong \mathbb{Q}^{\mathbb{N}}$, which is uncountable (it contains the sequences with values in $\{0, 1\}$). Choosing a basis of V^* gives an injection $V^* \rightarrow V^{**}$ (send the basis to its dual family), so V^{**} is uncountable, and cannot be the image of the countable set V .

9.5 Determinants

In this section R is a commutative ring. For $A \in M_n(R)$ we write $A = (a_1 | \cdots | a_n)$ with columns $a_j \in R^n$.

Definition 9.5.1. A map $D: (R^n)^n \rightarrow R$ is *multilinear* if it is R -linear in each argument, and *alternating* if $D(a_1, \dots, a_n) = 0$ whenever $a_i = a_j$ for some $i \neq j$. The *determinant* of $A \in M_n(R)$ is

$$\det A = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) a_{\sigma(1)1} a_{\sigma(2)2} \cdots a_{\sigma(n)n},$$

where sgn is the sign of Proposition 6.8.3.

Theorem 9.5.2. The determinant, as a function of the columns, is multilinear and alternating with $\det I = 1$. Every multilinear alternating $D: (R^n)^n \rightarrow R$ satisfies $D(a_1, \dots, a_n) = \det(a_1 | \cdots | a_n) \cdot D(e_1, \dots, e_n)$.

Proof. Multilinearity is clear, and $\det I = 1$ since only $\sigma = \operatorname{id}$ contributes. If $a_i = a_j$ with $i \neq j$, let $\tau = (i j)$. The map $\sigma \mapsto \sigma\tau$ is a bijection from even to odd permutations, and the terms for σ and $\sigma\tau$ have the same product of entries (as columns i and j agree) and opposite signs; so the sum vanishes.

Let D be multilinear and alternating. First, D changes sign when two arguments are exchanged: expand $0 = D(\dots, a+b, \dots, a+b, \dots)$. Hence $D(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \operatorname{sgn}(\sigma) D(e_1, \dots, e_n)$ for $\sigma \in S_n$, writing σ as a product of transpositions. Writing $a_j = \sum_i a_{ij} e_i$ and expanding by multilinearity, $D(a_1, \dots, a_n) = \sum_{\phi} a_{\phi(1)1} \cdots a_{\phi(n)n} D(e_{\phi(1)}, \dots, e_{\phi(n)})$ over all maps $\phi: \{1, \dots, n\} \rightarrow \{1, \dots, n\}$. The terms with ϕ not injective vanish, and the others give $\det(a_1 | \cdots | a_n) D(e_1, \dots, e_n)$. $\square$

Proposition 9.5.3. For $A, B \in M_n(R)$:

1. $\det(AB) = \det A \det B$;
2. $\det A^T = \det A$, so $\det$ is also multilinear and alternating in the rows;
3. (Laplace expansion) for each j , $\det A = \sum_i (-1)^{i+j} a_{ij} \det A_{ij}$, where A_{ij} is obtained by deleting row i and column j ;
4. the adjugate $\operatorname{adj} A$, with (i, j) entry $(-1)^{i+j} \det A_{ji}$, satisfies $A \operatorname{adj} A = \operatorname{adj} A A = \det(A)I$;
5. A is invertible in $M_n(R)$ if and only if $\det A \in R^\times$.

Proof. (1) The map $(b_1, \dots, b_n) \mapsto \det(Ab_1 | \cdots | Ab_n)$ is multilinear and alternating, with value $\det A$ on $(e_1, \dots, e_n)$; apply Theorem 9.5.2. (2) Reindex the sum by $\sigma \mapsto \sigma^{-1}$, using $\operatorname{sgn}(\sigma^{-1}) = \operatorname{sgn}(\sigma)$. (3) Both sides are multilinear and alternating in the columns other than j , and linear in column j . Writing $a_j = \sum_i a_{ij} e_i$, it suffices to treat $a_j = e_i$. Moving row i to the top and column j to the front by $i-1$ and $j-1$ adjacent exchanges changes the determinant by $(-1)^{i+j}$ and gives a matrix with first column e_1 , whose determinant is $\det A_{ij}$ by the Leibniz formula (only permutations fixing 1 contribute). (4) The (j, j') entry of $\operatorname{adj}(A)A$ is $\sum_i (-1)^{i+j} \det A_{ij} a_{ij'}$, which by (3) is the determinant of A with column j replaced by column j' : this is $\det A$ if $j = j'$ and 0 otherwise. The other product is handled with (2). (5) If $AB = I$ then $\det A \det B = 1$. Conversely $(\det A)^{-1} \operatorname{adj} A$ is an inverse by (4). $\square$

Definition 9.5.4. The determinant of an endomorphism f of a finite-dimensional vector space (or of a finitely generated free module over R) is the determinant of its matrix in any basis; it is well defined since $\det(P^{-1}AP) = \det A$. The group $\operatorname{GL}(V)$ of invertible endomorphisms contains $\operatorname{SL}(V) = \{f : \det f = 1\}$, and $\operatorname{GL}_n(R) = M_n(R)^\times$.

9.6 Eigenvalues

In this section V is a vector space of finite dimension n and $f: V \rightarrow V$ is linear.

Definition 9.6.1. A scalar $\lambda \in k$ is an *eigenvalue* of f if $\ker(f - \lambda) \neq 0$; the nonzero elements of the *eigenspace* $V_\lambda = \ker(f - \lambda)$ are *eigenvectors*. The *characteristic polynomial* is $\chi_f(x) = \det(xI - A) \in k[x]$ for the matrix A of f in any basis; it is monic of degree n and independent of the basis. The *trace* of f is $\operatorname{tr} f = \sum_i a_{ii}$. The map f is *diagonalisable* if V has a basis of eigenvectors.

Proposition 9.6.2. 1. The eigenvalues of f are the roots of χ_f in k .

2. $\chi_f(x) = x^n - (\operatorname{tr} f)x^{n-1} + \cdots + (-1)^n \det f$; in particular the trace is independent of the basis.

3. Eigenvectors with pairwise distinct eigenvalues are linearly independent.

Proof. (1) λ is an eigenvalue iff $\lambda - f$ is not injective iff $\det(\lambda I - A) = 0$, by Theorem 9.2.2 and Proposition 9.5.3(5). (2) In the Leibniz expansion of $\det(xI - A)$ only $\sigma = \operatorname{id}$ contributes to the coefficients of x^n and x^{n-1} , since any other permutation moves at least two indices; evaluate at $x = 0$ for the constant term. (3) Suppose $\sum_{i=1}^m c_i v_i = 0$ with m minimal and all $c_i \neq 0$. Applying $f - \lambda_m$ gives $\sum_{i < m} c_i (\lambda_i - \lambda_m) v_i = 0$, a shorter relation with nonzero coefficients. $\square$

Theorem 9.6.3 (Cayley–Hamilton). $\chi_f(f) = 0$.

Proof. Let A be the matrix of f . By Proposition 9.5.3(4) over $R = k[x]$, $(xI - A) \operatorname{adj}(xI - A) = \chi_f(x)I$. The entries of $\operatorname{adj}(xI - A)$ have degree $< n$, so we can write $\operatorname{adj}(xI - A) = \sum_{i=0}^{n-1} B_i x^i$ with $B_i \in M_n(k)$; write $\chi_f = \sum_{i=0}^n c_i x^i$. Comparing coefficients of x^i : $B_{i-1} - AB_i = c_i I$ for $0 \leq i \leq n$, with $B_{-1} = B_n = 0$. Multiplying the i -th equation by A^i on the left and summing, the left sides telescope to 0, so $\sum_i c_i A^i = 0$. $\square$

Definition 9.6.4. The *minimal polynomial* m_f is the monic generator of the ideal $\{p \in k[x] : p(f) = 0\}$, which is nonzero by Theorem 9.6.3. So $m_f \mid \chi_f$.

Lemma 9.6.5 (Kernel lemma). If $p, q \in k[x]$ are coprime, then $\ker(pq)(f) = \ker p(f) \oplus \ker q(f)$.

Proof. Write $1 = ap + bq$. For $v \in \ker(pq)(f)$, $v = a(f)p(f)v + b(f)q(f)v$, where the first term lies in $\ker q(f)$ (since $q(f)a(f)p(f)v = a(f)(pq)(f)v = 0$) and the second in $\ker p(f)$. If $v \in \ker p(f) \cap \ker q(f)$ then $v = a(f)p(f)v + b(f)q(f)v = 0$. $\square$

Theorem 9.6.6. f is diagonalisable if and only if m_f is a product of distinct monic linear factors. In that case $V = \bigoplus_\lambda V_\lambda$ over the eigenvalues. Commuting diagonalisable endomorphisms are simultaneously diagonalisable.

Proof. If V has a basis of eigenvectors with distinct eigenvalues $\lambda_1, \dots, \lambda_r$, then $\prod_i (f - \lambda_i) = 0$, so m_f divides $\prod_i (x - \lambda_i)$. Conversely if $m_f = \prod_i (x - \lambda_i)$ with distinct λ_i , the kernel lemma gives $V = \ker m_f(f) = \bigoplus_i \ker(f - \lambda_i)$. If g commutes with f , it preserves each V_λ , and its restriction to V_λ has minimal polynomial dividing m_g , hence is diagonalisable; choose eigenbases of these restrictions. $\square$

Counterexample 9.6.7. The rotation $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ has characteristic polynomial $x^2 + 1$ and no eigenvalue over $\mathbb{R}$, although it is diagonalisable over $\mathbb{C}$. The matrix $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ has $m_N = x^2$ and is not diagonalisable over any field: its only eigenvalue is 0 and $\ker N$ is one-dimensional.

9.7 Canonical forms

Let $f: V \rightarrow V$ be as in the previous section. By Example 7.7.2, V is a $k[x]$ -module V_f with x acting as f ; it is finitely generated and torsion, being annihilated by χ_f .

Definition 9.7.1. The *companion matrix* of a monic $g = x^d + c_{d-1}x^{d-1} + \cdots + c_0$ is the $d \times d$ matrix $C(g)$ with 1's below the diagonal, last column $(-c_0, \dots, -c_{d-1})^T$ and zeros elsewhere; it is the matrix of multiplication by x on $k[x]/(g)$ in the basis $1, x, \dots, x^{d-1}$. The *Jordan block* $J_e(\lambda)$ is the $e \times e$ matrix with λ on the diagonal, 1 directly above the diagonal and zeros elsewhere.

Lemma 9.7.2. The minimal and characteristic polynomials of $C(g)$ both equal g . The minimal and characteristic polynomials of $J_e(\lambda)$ both equal $(x - \lambda)^e$.

Proof. On $k[x]/(g)$ a polynomial p acts as zero iff $p \cdot 1 = p$ lies in (g) , so the minimal polynomial is g . The characteristic polynomial is monic of degree d and divisible by g (Definition 9.6.4), hence equal to g . The matrix $J_e(\lambda)$ is the matrix of x on $k[x]/((x - \lambda)^e)$ in the basis $(x - \lambda)^{e-1}, \dots, (x - \lambda), 1$, and the same argument applies. $\square$

Theorem 9.7.3 (Rational canonical form). *There are unique monic polynomials $f_1 \mid f_2 \mid \cdots \mid f_s$ of positive degree such that f has, in a suitable basis, the block diagonal matrix $C(f_1) \oplus \cdots \oplus C(f_s)$. Moreover $m_f = f_s$ and $\chi_f = f_1 \cdots f_s$. Two endomorphisms (or matrices) are similar if and only if they have the same f_i , the invariant factors.*

Proof. By Theorem 7.11.3, $V_f \cong \bigoplus_i k[x]/(f_i)$ with $f_1 \mid \cdots \mid f_s$ monic of positive degree (there is no free part since V_f is torsion), uniquely determined. Bases of the summands as in Definition 9.7.1 give the block form. An isomorphism $V_f \cong V_g$ of $k[x]$ -modules is the same as a linear isomorphism h with $hf = gh$, so similarity corresponds to isomorphism of modules. The polynomial f_s kills every summand and is the minimal polynomial of the last one, so $m_f = f_s$; the characteristic polynomial of a block matrix is the product of those of the blocks, which gives χ_f by Lemma 9.7.2. $\square$

Theorem 9.7.4 (Jordan normal form). *Suppose χ_f splits into linear factors over k (for instance, k algebraically closed). Then f has, in a suitable basis, a block diagonal matrix $J_{e_1}(\lambda_1) \oplus \cdots \oplus J_{e_r}(\lambda_r)$, and the multiset of pairs (λ_i, e_i) is uniquely determined by f .*

Proof. By Corollary 7.11.4, $V_f \cong \bigoplus_i k[x]/(p_i^{e_i})$ with p_i monic irreducible, uniquely. Each p_i divides χ_f , hence is $x - \lambda_i$. Use the bases of Lemma 9.7.2. $\square$

Example 9.7.5. A nilpotent endomorphism ($f^m = 0$ for some m) over any field has $\chi_f = x^n$ (its only possible eigenvalue is 0, and $m_f \mid x^m$), so it is similar to a direct sum of blocks $J_e(0)$; the sizes of the blocks form a partition of n , and they classify nilpotent endomorphisms up to similarity. Such nilpotent operators, and their Jordan decompositions, organise the limiting behaviour of variations of Hodge structure (Chapter 66).

9.8 Bilinear forms

Definition 9.8.1. A *bilinear form* on V is a bilinear map $B: V \times V \rightarrow k$. It is *symmetric* if $B(v, w) = B(w, v)$, *alternating* if $B(v, v) = 0$ for all v , and *nondegenerate* if $v \mapsto B(v, -)$ is an isomorphism $V \rightarrow V^*$ (for V finite-dimensional: if $B(v, w) = 0$ for all w implies $v = 0$). Its *Gram matrix* in a basis (v_i) is $(B(v_i, v_j))$. For a subspace W , $W^{\perp_B} = \{v : B(v, w) = 0 \text{ for all } w \in W\}$. A *quadratic form* is $q(v) = B(v, v)$ for a symmetric B .

Lemma 9.8.2. *Let B be symmetric or alternating on a finite-dimensional V , and $W \subseteq V$ a subspace on which B is nondegenerate. Then $V = W \oplus W^{\perp_B}$. An alternating form B satisfies $B(v, w) = -B(w, v)$.*

Proof. $W \cap W^{\perp_B} = 0$ by nondegeneracy on W . Given $v \in V$, the functional $B(v, -)|_W$ equals $B(w, -)|_W$ for a unique $w \in W$, and then $v - w \in W^{\perp_B}$ (for alternating forms use $B(-, w) = -B(w, -)$). The last statement follows from expanding $B(v + w, v + w) = 0$. $\square$

Theorem 9.8.3. *Let $\text{char } k \neq 2$ and B symmetric on a finite-dimensional V . Then V has a basis (v_i) with $B(v_i, v_j) = 0$ for $i \neq j$.*

Proof. By induction on $\dim V$. If $B = 0$ any basis works. Otherwise some $q(v) \neq 0$: if $B(v, w) \neq 0$ then by the polarisation identity $2B(v, w) = q(v + w) - q(v) - q(w)$ one of $q(v + w)$, $q(v)$, $q(w)$ is nonzero. Then B is nondegenerate on kv , so $V = kv \oplus (kv)^{\perp_B}$ by Lemma 9.8.2; apply induction to $(kv)^{\perp_B}$. $\square$

Counterexample 9.8.4. The hypothesis $\text{char } k \neq 2$ is needed. On $\mathbb{F}_2^2$ the form $B((a, b), (c, d)) = ad + bc$ is symmetric and nondegenerate, but alternating ($B(v, v) = 2ab = 0$). A basis with $B(v_i, v_j) = 0$ for $i \neq j$ would make B zero.

Theorem 9.8.5 (Sylvester's law of inertia). *Let B be a symmetric bilinear form on a real vector space V of dimension n . There is a basis in which the Gram matrix is diagonal with p entries 1, q entries -1 and $n - p - q$ entries 0, and the numbers p and q do not depend on the basis. The pair (p, q) is the signature of B .*

Proof. Take an orthogonal basis (Theorem 9.8.3) and rescale each v_i with $q(v_i) \neq 0$ by $|q(v_i)|^{-1/2}$ (Proposition 4.7.4). For uniqueness, p is the largest dimension of a subspace on which q is positive definite: q is positive definite on the span P of the v_i with $q(v_i) = 1$, and if U is a subspace on which q is positive definite, then U meets the span N of the other basis vectors, on which $q \leq 0$, only in 0; so $\dim U \leq n - \dim N = p$. Similarly for q . $\square$

Theorem 9.8.6. *Let ω be a nondegenerate alternating form on a finite-dimensional V . Then $\dim V = 2m$ is even, and V has a symplectic basis $e_1, \dots, e_m, f_1, \dots, f_m$ with $\omega(e_i, f_j) = \delta_{ij}$ and $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$.*

Proof. If $V \neq 0$, choose $e_1 \neq 0$, and f_1 with $\omega(e_1, f_1) = 1$ by nondegeneracy. Then e_1, f_1 are independent (as $\omega(e_1, e_1) = 0$), and ω is nondegenerate on their span W , whose Gram matrix is $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. By Lemma 9.8.2, $V = W \oplus W^{\perp_\omega}$, and ω is nondegenerate on $W^{\perp_\omega}$. Apply induction. $\square$

9.9 Inner product spaces and the spectral theorem

Definition 9.9.1. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$. An *inner product* on a $\mathbb{K}$ -vector space V is a map $\langle -, - \rangle: V \times V \rightarrow \mathbb{K}$, linear in the first variable, with $\langle w, v \rangle = \overline{\langle v, w \rangle}$ and $\langle v, v \rangle > 0$ for $v \neq 0$. (Over $\mathbb{C}$ it is then conjugate-linear in the second variable; such a form is also called a *Hermitian inner product*.) We write $\|v\| = \sqrt{\langle v, v \rangle}$. A family is *orthonormal* if $\langle v_i, v_j \rangle = \delta_{ij}$.

Proposition 9.9.2. $|\langle v, w \rangle| \leq \|v\| \|w\|$, and $\|v + w\| \leq \|v\| + \|w\|$.

Proof. If $w = 0$ this is clear. Otherwise let $u = v - \frac{\langle v, w \rangle}{\|w\|^2} w$; then $\langle u, w \rangle = 0$ and $0 \leq \|u\|^2 = \|v\|^2 - |\langle v, w \rangle|^2 / \|w\|^2$. For the triangle inequality, $\|v + w\|^2 = \|v\|^2 + 2 \operatorname{Re} \langle v, w \rangle + \|w\|^2 \leq (\|v\| + \|w\|)^2$. $\square$

Theorem 9.9.3 (Gram–Schmidt). *If $v_1, v_2, \dots$ is a finite or countable linearly independent family, there is an orthonormal family $u_1, u_2, \dots$ with $\text{span}(u_1, \dots, u_j) = \text{span}(v_1, \dots, v_j)$ for all j . Every finite-dimensional inner product space has an orthonormal basis, and for every finite-dimensional subspace W , $V = W \oplus W^\perp$, where $W^\perp = \{v : \langle v, w \rangle = 0 \text{ for all } w \in W\}$.*

Proof. Define recursively $u'_j = v_j - \sum_{i < j} \langle v_j, u_i \rangle u_i$, which is nonzero since $v_j \notin \text{span}(u_1, \dots, u_{j-1})$, and $u_j = u'_j / \|u'_j\|$. For the last claim, with an orthonormal basis (u_i) of W , write $v = \sum_i \langle v, u_i \rangle u_i + (v - \sum_i \langle v, u_i \rangle u_i)$; the second term lies in $W^\perp$, and $W \cap W^\perp = 0$ by positivity. $\square$

Definition 9.9.4. Let V be a finite-dimensional inner product space and $f : V \rightarrow V$ linear. The *adjoint* $f^\dagger$ is the unique linear map with $\langle f(v), w \rangle = \langle v, f^\dagger(w) \rangle$ for all v, w . The map f is *self-adjoint* if $f^\dagger = f$, *normal* if $ff^\dagger = f^\dagger f$, and *unitary* (or *orthogonal*, if $\mathbb{K} = \mathbb{R}$) if $f^\dagger f = \text{id}$. The unitary operators of $\mathbb{C}^n$ form the *unitary group* $U(n)$, and the orthogonal operators of $\mathbb{R}^n$ the *orthogonal group* $O(n)$.

Lemma 9.9.5. *The adjoint exists and is unique. In an orthonormal basis its matrix is the conjugate transpose of the matrix of f . If W is f -invariant then $W^\perp$ is $f^\dagger$ -invariant.*

Proof. With an orthonormal basis (u_i) , the map $f^\dagger(w) = \sum_i \langle w, f(u_i) \rangle u_i$ works, and a linear map is determined by the values $\langle v, f^\dagger(w) \rangle$. Its matrix entries are $\langle f^\dagger(u_j), u_i \rangle = \overline{\langle f(u_i), u_j \rangle}$. If $f(W) \subseteq W$ and $v \in W^\perp$, then $\langle f^\dagger(v), w \rangle = \langle v, f(w) \rangle = 0$ for $w \in W$. $\square$

Theorem 9.9.6 (Spectral theorem). *Let V be a finite-dimensional inner product space.*

1. *If $\mathbb{K} = \mathbb{C}$ and f is normal, then V has an orthonormal basis of eigenvectors of f .*
2. *If f is self-adjoint (over $\mathbb{R}$ or $\mathbb{C}$), then all its eigenvalues are real and V has an orthonormal basis of eigenvectors of f .*

Proof. For self-adjoint f and an eigenvector v with eigenvalue λ , $\lambda \|v\|^2 = \langle f v, v \rangle = \langle v, f v \rangle = \bar{\lambda} \|v\|^2$, so $\lambda \in \mathbb{R}$.

(1) By induction on $\dim V$. By Theorem 8.7.3, χ_f has a root, so f has an eigenvector v . Since f and $f^\dagger$ commute, $f^\dagger$ preserves the eigenspace V_λ , and its restriction there has an eigenvector u ; then u is an eigenvector of both f and $f^\dagger$. Then $W = \mathbb{C}u$ is invariant under both, so $W^\perp$ is invariant under $f^\dagger$ and $(f^\dagger)^\dagger = f$ (Lemma 9.9.5), and the restriction of f to $W^\perp$ is normal. Apply induction and normalise u .

(2) Over $\mathbb{C}$ this follows from (1). Over $\mathbb{R}$, choose an orthonormal basis and let A be the real symmetric matrix of f . As a complex matrix acting on $\mathbb{C}^n$ with the standard inner product, A is self-adjoint, so it has an eigenvalue, which is real, and hence a real root λ of χ_f ; so f has an eigenvector $u \in V$. The line $\mathbb{R}u$ and its orthogonal complement are invariant under $f = f^\dagger$, and we conclude by induction. $\square$

Counterexample 9.9.7. Normality is necessary in (1): the matrix N of Counterexample 9.6.7 is not diagonalisable at all. Over $\mathbb{R}$, normal is not enough: the rotation of that counterexample is orthogonal, hence normal, but has no real eigenvector.

Corollary 9.9.8. *A real symmetric (or complex Hermitian) matrix A defines an inner product $\langle v, w \rangle = w^* A v$ if and only if all its eigenvalues are positive. Every inner product on $\mathbb{R}^n$ (or $\mathbb{C}^n$) is of this form.*

Proof. Diagonalise $A = U D U^*$ with U unitary (Theorem 9.9.6); then $w^* A v = (U^* w)^* D (U^* v)$, which is positive definite iff all diagonal entries of D are positive. The matrix of an inner product in the standard basis is Hermitian. $\square$

Proposition 9.9.9 (Singular value decomposition). *Let $f: V \rightarrow W$ be a linear map between finite-dimensional inner product spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, of rank r . There are orthonormal bases $(v_1, \dots, v_n)$ of V and $(w_1, \dots, w_m)$ of W and real numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$ with $f(v_i) = \sigma_i w_i$ for $i \leq r$ and $f(v_i) = 0$ for $i > r$. The σ_i^2 are the nonzero eigenvalues of $f^\dagger f$, so the singular values σ_i are determined by f ; the largest one is the operator norm $\|f\| = \sup_{\|v\|=1} \|f(v)\|$.*

Proof. The endomorphism $f^\dagger f$ of V is self-adjoint and $\langle f^\dagger f v, v \rangle = \|f(v)\|^2 \geq 0$, so by Theorem 9.9.6 there is an orthonormal basis (v_i) of V with $f^\dagger f v_i = \lambda_i v_i$ and $\lambda_i \geq 0$; order it so that $\lambda_1 \geq \lambda_2 \geq \dots$. Then $\langle f(v_i), f(v_j) \rangle = \langle f^\dagger f v_i, v_j \rangle = \lambda_i \delta_{ij}$, so the $f(v_i)$ are pairwise orthogonal, $f(v_i) = 0$ exactly when $\lambda_i = 0$, and the number of i with $\lambda_i > 0$ is the rank r . Put $\sigma_i = \sqrt{\lambda_i}$ and $w_i = f(v_i)/\sigma_i$ for $i \leq r$; these are orthonormal, and we extend them to an orthonormal basis of W (Theorem 9.9.3). For $v = \sum_i c_i v_i$ of norm one, $\|f(v)\|^2 = \sum_{i \leq r} \sigma_i^2 |c_i|^2 \leq \sigma_1^2$, with equality at $v = v_1$. $\square$

Example 9.9.10. For $f(x, y) = (x + y, 0)$ on $\mathbb{R}^2$, $f^\dagger f$ has matrix $\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ with eigenvalues 2 and 0 and eigenvectors $(1, 1)/\sqrt{2}$ and $(1, -1)/\sqrt{2}$. So $\sigma_1 = \sqrt{2}$, $w_1 = (1, 0)$, and $\|f\| = \sqrt{2}$, although the eigenvalues of f itself are 1 and 0: for non-normal maps the operator norm is not the largest absolute value of an eigenvalue.

9.10 Complex structures

Definition 9.10.1. Let V be a real vector space. Its *complexification* is $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$, a complex vector space with the *complex conjugation* $\bar{v} \otimes \bar{z} = v \otimes z$, which is $\mathbb{R}$ -linear and conjugate-linear over $\mathbb{C}$. A *complex structure* on V is an $\mathbb{R}$ -linear map $J: V \rightarrow V$ with $J^2 = -\text{id}$.

Proposition 9.10.2. *Let J be a complex structure on a finite-dimensional real vector space V .*

1. *V becomes a complex vector space with $(a + bi)v = av + bJv$; hence $\dim_{\mathbb{R}} V = 2 \dim_{\mathbb{C}} V$ is even.*
2. *Extend J $\mathbb{C}$ -linearly to $V_{\mathbb{C}}$. Then $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$, where $V^{1,0} = \ker(J - i)$ and $V^{0,1} = \ker(J + i)$, and $\overline{V^{1,0}} = V^{0,1}$.*
3. *The map $V \rightarrow V^{1,0}$, $v \mapsto \frac{1}{2}(v - iJv)$, is an isomorphism of complex vector spaces from (V, J) to $V^{1,0}$, and $v \mapsto \frac{1}{2}(v + iJv)$ is a conjugate-linear isomorphism $V \rightarrow V^{0,1}$.*

Proof. (1) The axioms of a module over $\mathbb{C} = \mathbb{R}[x]/(x^2 + 1)$ follow from $J^2 = -1$ (Example 7.7.2 and Exercise 7.12.2); a complex basis (v_j) gives the real basis (v_j, Jv_j) . (2) The polynomial $x^2 + 1 = (x - i)(x + i)$ kills J on $V_{\mathbb{C}}$, and the kernel lemma (Lemma 9.6.5) gives the decomposition. Conjugation commutes with J (which is real), so it exchanges the $\pm i$ -eigenspaces. (3) $J(v - iJv) = Jv + iv = i(v - iJv)$, so the image lies in $V^{1,0}$, and J on V corresponds to multiplication by i . The map is injective because v is recovered as twice the real part, and both sides have complex dimension $\frac{1}{2} \dim_{\mathbb{R}} V$. The second map is the conjugate of the first. $\square$

Proposition 9.10.3. *Let J be a complex structure on a real inner product space (V, g) with $g(Jv, Jw) = g(v, w)$. Put $\omega(v, w) = g(Jv, w)$ and $h = g - i\omega$. Then ω is a nondegenerate alternating form with $\omega(Jv, Jw) = \omega(v, w)$, and h is a Hermitian inner product on the complex vector space (V, J) , whose real part is g . Conversely, g and J determine ω , and ω and J determine $g(v, w) = \omega(v, Jw)$.*

Proof. From J -invariance, $g(Jv, w) = g(J^2v, Jw) = -g(v, Jw)$. Hence $\omega(w, v) = g(Jw, v) = -g(w, Jv) = -\omega(v, w)$, so ω is alternating; $\omega(Jv, Jw) = g(J(Jv), Jw) = g(Jv, w) = \omega(v, w)$ by J -invariance of g ; and ω is nondegenerate because $\omega(v, Jv) = g(Jv, Jv) > 0$ for $v \neq 0$. For h : $h(v, w) = g(v, w) + ig(v, Jw)$. Then $h(Jv, w) = g(Jv, w) + ig(Jv, Jw) = -g(v, Jw) + ig(v, w) = ih(v, w)$, so h is complex linear in the first variable; $h(w, v) = g(v, w) - ig(v, Jw) = \overline{h(v, w)}$; and $h(v, v) = g(v, v) > 0$ for $v \neq 0$ since $g(v, Jv) = -g(Jv, v) = -g(v, Jv)$ vanishes. Finally $\omega(v, Jw) = g(Jv, Jw) = g(v, w)$. $\square$

Example 9.10.4. On $\mathbb{R}^{2n}$ with coordinates $(x_1, y_1, \dots, x_n, y_n)$, the standard complex structure is $J(\partial_{x_j}) = \partial_{y_j}$, $J(\partial_{y_j}) = -\partial_{x_j}$ (writing $\partial_{x_j}, \partial_{y_j}$ for the standard basis vectors), which identifies $(\mathbb{R}^{2n}, J)$ with $\mathbb{C}^n$ via $x + iy$. The standard inner product is J -invariant, the associated form satisfies $\omega(\partial_{x_j}, \partial_{y_j}) = g(\partial_{y_j}, \partial_{y_j}) = 1$, so the $\partial_{x_j}, \partial_{y_j}$ form a symplectic basis, and h is the standard Hermitian inner product of $\mathbb{C}^n$. The vectors $\partial_{z_j} = \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ span $V^{1,0}$. These formulas reappear as the local model of Kähler manifolds (Chapter 44).

9.11 Exercises

Exercise 9.11.1. Show that $\text{tr}(AB) = \text{tr}(BA)$ for $A \in M_{m \times n}(k)$, $B \in M_{n \times m}(k)$, and deduce that similar matrices have the same trace. Show that if $\chi_f = \prod_i (x - \lambda_i)$ then $\text{tr } f = \sum \lambda_i$ and $\det f = \prod \lambda_i$.

Exercise 9.11.2. Suppose χ_f splits. Show that the number of Jordan blocks $J_e(\lambda)$ with $e \geq j$ in the Jordan form of f is $\dim \ker(f - \lambda)^j - \dim \ker(f - \lambda)^{j-1}$.

Exercise 9.11.3. Let V be a finite-dimensional complex vector space and $\mathcal{F}$ a set of pairwise commuting endomorphisms. Show that there is a basis in which all elements of $\mathcal{F}$ are upper triangular.

Exercise 9.11.4. Show that every invertible complex matrix is uniquely $A = UP$ with U unitary and P Hermitian with positive eigenvalues.

Exercise 9.11.5. Let ω be a nondegenerate alternating form on V of dimension $2m$ and $\text{Sp}(V, \omega)$ the group of $f \in \text{GL}(V)$ with $\omega(fv, fw) = \omega(v, w)$. Show that $\det f = \pm 1$ for $f \in \text{Sp}(V, \omega)$. (It is in fact always 1; this can be shown using the Pfaffian, Chapter 10.)

Exercise 9.11.6. Show that the set of complex structures J on $\mathbb{R}^2$ compatible with the standard inner product and with $\omega(e_1, e_2) > 0$ has exactly one element.

Chapter 10

Multilinear Algebra

10.1 Introduction

This chapter constructs the tensor, symmetric and exterior algebras of a vector space and studies the structures on them that underlie differential forms: alternating forms and interior products, orientations, the Hodge star operator of an inner product space, and the decomposition of the exterior algebra of a real vector space with a complex structure into forms of type (p, q) . These are the pointwise versions of the constructions of Chapters 34, 38 and 41.

References are [Lan02, Chapters XVI and XIX] and [Gre78].

Prerequisites. Chapter 7 (Rings and Modules) and Chapter 9 (Linear Algebra).

Throughout, k is a field, vector spaces are over k and $\otimes = \otimes_k$. Tensor products were defined in Section 7.9.

10.2 Tensor products of vector spaces

Proposition 10.2.1. *Let $(v_i)_{i \in I}$ be a basis of V and $(w_j)_{j \in J}$ a basis of W . Then $(v_i \otimes w_j)$ is a basis of $V \otimes W$; in particular $\dim(V \otimes W) = \dim V \cdot \dim W$. For vector spaces $V_1, \dots, V_n, U$, multilinear maps $V_1 \times \dots \times V_n \rightarrow U$ correspond bijectively to linear maps $V_1 \otimes \dots \otimes V_n \rightarrow U$.*

Proof. $V \cong k^{(I)}$ and $W \cong k^{(J)}$, and by Proposition 7.9.3 $k^{(I)} \otimes k^{(J)} \cong \bigoplus_i (k \otimes k^{(J)}) \cong k^{(I \times J)}$, with $e_i \otimes e_j \mapsto e_{(i,j)}$. The second statement follows by induction from the definition of the tensor product and the associativity isomorphism. $\square$

Proposition 10.2.2. *The linear map $V^* \otimes W \rightarrow \text{Hom}_k(V, W)$, $\varphi \otimes w \mapsto (v \mapsto \varphi(v)w)$, is injective, with image the maps of finite rank; it is an isomorphism if V or W is finite-dimensional. For finite-dimensional V, W the map $V^* \otimes W^* \rightarrow (V \otimes W)^*$, $\varphi \otimes \psi \mapsto (v \otimes w \mapsto \varphi(v)\psi(w))$, is an isomorphism. Under $V^* \otimes V \cong \text{End}(V)$, the contraction $\varphi \otimes v \mapsto \varphi(v)$ is the trace.*

Proof. Let (w_j) be a basis of W . An element of $V^* \otimes W$ is uniquely $\sum_j \varphi_j \otimes w_j$ with finitely many nonzero φ_j (Proposition 10.2.1 applied to a basis of V^*); it maps to $v \mapsto \sum_j \varphi_j(v)w_j$, which is zero only if all $\varphi_j = 0$. The image consists of maps with image in a finite-dimensional subspace, and every such map is of this form (choose a basis of W extending a basis of a finite-dimensional subspace containing the image). If V is finite-dimensional, every map has finite rank; if W is, likewise. For the second isomorphism, compare dual bases. Finally, with a basis (e_i) of V , $e_i^* \otimes e_j$ corresponds to the matrix unit E_{ji} , whose trace is $\delta_{ij} = e_i^*(e_j)$. $\square$

Counterexample 10.2.3. Not every element of $V \otimes W$ is of the form $v \otimes w$. Under $k^2 \otimes k^2 \cong M_2(k)$ (sending $e_i \otimes e_j$ to the matrix unit E_{ij}), the element $v \otimes w$ corresponds to a matrix of rank at most 1, while $e_1 \otimes e_1 + e_2 \otimes e_2$ corresponds to the identity matrix, of rank 2.

10.3 The tensor, symmetric and exterior algebras

Definition 10.3.1. The *tensor algebra* of V is the graded k -algebra $T(V) = \bigoplus_{n \geq 0} T^n(V)$, with $T^0(V) = k$, $T^n(V) = V^{\otimes n}$, and multiplication induced by $(v_1 \otimes \cdots \otimes v_p)(w_1 \otimes \cdots \otimes w_q) = v_1 \otimes \cdots \otimes v_p \otimes w_1 \otimes \cdots \otimes w_q$. The *symmetric algebra* is $S(V) = T(V)/I_S$, where I_S is the two-sided ideal generated by the elements $v \otimes w - w \otimes v$, and the *exterior algebra* is $\Lambda(V) = T(V)/I_\Lambda$, where I_Λ is generated by the elements $v \otimes v$. The ideals are homogeneous, so both quotients are graded: $S(V) = \bigoplus_n S^n(V)$ and $\Lambda(V) = \bigoplus_n \Lambda^n(V)$. We write $v_1 \cdots v_n$ for the image of $v_1 \otimes \cdots \otimes v_n$ in $S(V)$ and $v_1 \wedge \cdots \wedge v_n$ for its image in $\Lambda(V)$.

Proposition 10.3.2. Let A be an associative k -algebra and $f: V \rightarrow A$ linear.

1. f extends uniquely to an algebra homomorphism $T(V) \rightarrow A$.
2. If A is commutative, it extends uniquely to $S(V) \rightarrow A$.
3. If $f(v)^2 = 0$ for all $v \in V$, it extends uniquely to $\Lambda(V) \rightarrow A$.

Moreover, n -linear maps $V^n \rightarrow U$ that are symmetric (respectively alternating) correspond to linear maps $S^n(V) \rightarrow U$ (respectively $\Lambda^n(V) \rightarrow U$).

Proof. (1) The n -linear map $(v_1, \dots, v_n) \mapsto f(v_1) \cdots f(v_n)$ induces $T^n(V) \rightarrow A$, and the sum of these is an algebra homomorphism; it is unique since V generates $T(V)$. (2), (3) The homomorphism of (1) kills the generators of I_S , respectively I_Λ .

For the last statement, the degree- n part of I_Λ is spanned by the tensors $x \otimes v \otimes v \otimes y$ with x, y pure tensors, and similarly for I_S . A multilinear $\beta: V^n \rightarrow U$ vanishes on the tensors with two equal adjacent factors if and only if it is alternating: expanding $\beta(\dots, v+w, v+w, \dots) = 0$ shows it changes sign under adjacent transpositions, hence under all transpositions up to sign, so it vanishes whenever any two arguments are equal. The symmetric case is analogous. $\square$

Proposition 10.3.3. 1. In $\Lambda(V)$, $v \wedge w = -w \wedge v$ for $v, w \in V$, and $\alpha \wedge \beta = (-1)^{pq} \beta \wedge \alpha$ for $\alpha \in \Lambda^p(V)$, $\beta \in \Lambda^q(V)$.

2. $S(V)$ is commutative. If $(e_1, \dots, e_n)$ is a basis of V , then $S(V) \cong k[x_1, \dots, x_n]$ with $e_i \mapsto x_i$, and $S^d(V)$ corresponds to the homogeneous polynomials of degree d .
3. If $(e_1, \dots, e_n)$ is a basis of V , the elements $e_I = e_{i_1} \wedge \cdots \wedge e_{i_p}$, for $I = \{i_1 < \cdots < i_p\} \subseteq \{1, \dots, n\}$, form a basis of $\Lambda^p(V)$. So $\dim \Lambda^p(V) = \binom{n}{p}$, $\Lambda^p(V) = 0$ for $p > n$, and $\dim \Lambda(V) = 2^n$.
4. Vectors $v_1, \dots, v_p$ are linearly independent if and only if $v_1 \wedge \cdots \wedge v_p \neq 0$.
5. For $f: V \rightarrow W$ linear there are induced maps $\Lambda^p f: \Lambda^p V \rightarrow \Lambda^p W$, $v_1 \wedge \cdots \wedge v_p \mapsto f(v_1) \wedge \cdots \wedge f(v_p)$, and if $\dim V = n$, $\Lambda^n f$ is multiplication by $\det f$ for $f: V \rightarrow V$.

Proof. (1) $0 = (v+w) \wedge (v+w) = v \wedge w + w \wedge v$. Moving each of the q factors of a decomposable β past the p factors of α gives pq sign changes.

(2) Commutativity holds on generators. The two universal properties (Proposition 10.3.2(2) and Proposition 7.6.2) give inverse homomorphisms.

(3) By (1) and $e_i \wedge e_i = 0$, the e_I span $\Lambda^p(V)$. For independence, fix I and consider the map $V^p \rightarrow k$, $(v_1, \dots, v_p) \mapsto \det(e_{i_a}^*(v_b))_{a,b}$. It is multilinear and alternating by Theorem 9.5.2, so it induces $\varphi_I: \Lambda^p(V) \rightarrow k$, and $\varphi_I(e_J) = \delta_{IJ}$. A relation $\sum_J c_J e_J = 0$ gives $c_I = 0$ after applying φ_I .

(4) If $v_1 = \sum_{i \geq 2} c_i v_i$, the product vanishes by (1) and $v_i \wedge v_i = 0$. If they are independent, extend them to a basis and use (3).

(5) The map exists by the universal property. For $V = W$ of dimension n with basis (e_i) , $(v_1, \dots, v_n) \mapsto$ the coefficient of $f(v_1) \wedge \dots \wedge f(v_n)$ on $e_1 \wedge \dots \wedge e_n$ is multilinear alternating; by Theorem 9.5.2 applied to the columns of the matrix of f , $f(e_1) \wedge \dots \wedge f(e_n) = \det(f) e_1 \wedge \dots \wedge e_n$. $\square$

Proposition 10.3.4. *Let $f: V \rightarrow W$ be linear with matrix $A = (a_{ij})$ in bases $(e_1, \dots, e_m)$ of V and $(e'_1, \dots, e'_n)$ of W . For subsets $I \subseteq \{1, \dots, n\}$, $J \subseteq \{1, \dots, m\}$ of size p , let $A_{I,J}$ be the $p \times p$ submatrix with rows I and columns J .*

1. *The matrix of $\Lambda^p f$ in the bases (e_J) and (e'_I) is $(\det A_{I,J})_{I,J}$: $\Lambda^p f(e_J) = \sum_I \det(A_{I,J}) e'_I$.*
2. *(Cauchy–Binet) For an $n \times m$ matrix A and an $m \times n$ matrix B , $\det(AB) = \sum_S \det A_{\{1..n\},S} \det B_{S,\{1..n\}}$, the sum over the subsets $S \subseteq \{1, \dots, m\}$ of size n ; in particular $\det(AB) = 0$ if $m < n$, and $\det(AB) = \det A \det B$ if $m = n$.*
3. *The rank of f is the largest p with $\Lambda^p f \neq 0$, that is, the largest size of a nonzero minor of A .*

Proof. (1) $\Lambda^p f(e_J) = f(e_{j_1}) \wedge \dots \wedge f(e_{j_p})$, and its coefficient on e'_I is obtained by applying the functional φ_I of the proof of Proposition 10.3.3(3), which gives $\det(e'^*_a(f(e_{j_b})))_{a,b} = \det(a_{ia_{j_b}}) = \det A_{I,J}$.

(2) By functoriality $\Lambda^n(AB) = \Lambda^n A \circ \Lambda^n B$ on $\Lambda^n(k^n)$, which is one-dimensional with basis $e_{\{1..n\}}$, and $\Lambda^n(AB)$ is multiplication by $\det(AB)$ (Proposition 10.3.3(5)). By (1), $\Lambda^n B(e_{\{1..n\}}) = \sum_S \det B_{S,\{1..n\}} e_S$ and $\Lambda^n A(e_S) = \det A_{\{1..n\},S} e_{\{1..n\}}$. If $m < n$ there are no such S .

(3) The image of $\Lambda^p f$ is spanned by the products $f(v_1) \wedge \dots \wedge f(v_p)$, so it is the image of $\Lambda^p(\text{im } f) \rightarrow \Lambda^p W$, which is injective because a basis of $\text{im } f$ extends to a basis of W and Proposition 10.3.3(3) applies to both. So $\Lambda^p f \neq 0$ if and only if $\Lambda^p(\text{im } f) \neq 0$, that is, $p \leq \text{rank } f$. Combine with (1). $\square$

Counterexample 10.3.5. In characteristic 2 symmetric and alternating notions separate. Let $k = \mathbb{F}_2$ and $V = k$. The bilinear form $B(x, y) = xy$ satisfies $B(y, x) = B(x, y) = -B(x, y)$, so it is both symmetric and antisymmetric, but it is not alternating, since $B(1, 1) = 1$; so it does not factor through $\Lambda^2 V = 0$. Similarly the symmetrisation $S^2 V \rightarrow V \otimes V$, $vw \mapsto v \otimes w + w \otimes v$, sends the generator e^2 of $S^2 V$ to $2e \otimes e = 0$, so it is not injective. Factorials cause the same phenomena elsewhere: in Proposition 10.4.6 the characteristic enters through the factor $m!$ in $\omega^m = m! e_1^* \wedge f_1^* \wedge \dots \wedge e_m^* \wedge f_m^*$, which vanishes in characteristic $p \leq m$.

10.4 Alternating forms

In this section V has finite dimension n .

Proposition 10.4.1. *The pairing $\Lambda^p(V^*) \times \Lambda^p(V) \rightarrow k$ determined by $\langle \varphi_1 \wedge \dots \wedge \varphi_p, v_1 \wedge \dots \wedge v_p \rangle = \det(\varphi_i(v_j))$ is well defined and perfect. Hence $\Lambda^p(V^*) \cong \Lambda^p(V)^*$, which by Proposition 10.3.2 is the space $\text{Alt}^p(V)$ of alternating p -linear forms on V .*

Proof. The determinant is multilinear and alternating both in the φ_i and in the v_j (using Proposition 9.5.3(2)), so it induces a bilinear pairing. For a basis (e_i) with dual basis (e_i^*) , $\langle e_i^*, e_j \rangle = \delta_{ij}$, so the pairing is perfect. $\square$

Notation 10.4.2. We identify $\Lambda^p(V^*)$ with $\text{Alt}^p(V)$ via Proposition 10.4.1: an element $\alpha \in \Lambda^p(V^*)$ is evaluated on vectors as $\alpha(v_1, \dots, v_p) = \langle \alpha, v_1 \wedge \dots \wedge v_p \rangle$. With this convention, $(\varphi_1 \wedge \dots \wedge \varphi_p)(v_1, \dots, v_p) = \det(\varphi_i(v_j))$, without a factor $1/p!$.

Proposition 10.4.3. For $\alpha \in \Lambda^p(V^*)$ and $\beta \in \Lambda^q(V^*)$,

$$(\alpha \wedge \beta)(v_1, \dots, v_{p+q}) = \sum_{\sigma} \text{sgn}(\sigma) \alpha(v_{\sigma(1)}, \dots, v_{\sigma(p)}) \beta(v_{\sigma(p+1)}, \dots, v_{\sigma(p+q)}),$$

the sum running over the shuffles: permutations $\sigma \in S_{p+q}$ with $\sigma(1) < \dots < \sigma(p)$ and $\sigma(p+1) < \dots < \sigma(p+q)$.

Proof. Both sides are bilinear in (α, β) , so we may take $\alpha = \varphi_1 \wedge \dots \wedge \varphi_p$ and $\beta = \varphi_{p+1} \wedge \dots \wedge \varphi_{p+q}$. The left side is $\det M$ with $M = (\varphi_i(v_j))$. In the Leibniz formula $\det M = \sum_{\pi} \text{sgn}(\pi) \prod_j M_{\pi(j)j}$, group the permutations π according to the set $A = \{j : \pi(j) \leq p\}$ of columns sent to the first p rows. For fixed $A = \{a_1 < \dots < a_p\}$ with complement $\{b_1 < \dots < b_q\}$, let σ be the shuffle with $\sigma(i) = a_i$ and $\sigma(p+i) = b_i$. Every such π is uniquely $\pi = (\pi_1 \times \pi_2) \circ \sigma^{-1}$ with $\pi_1 \in S_p$ acting on $\{1, \dots, p\}$ and π_2 on $\{p+1, \dots, p+q\}$, and $\text{sgn } \pi = \text{sgn } \pi_1 \text{sgn } \pi_2 \text{sgn } \sigma$. Summing over π_1 and π_2 gives $\text{sgn}(\sigma) \det(\varphi_i(v_{a_j})) \det(\varphi_{p+i}(v_{b_j}))$, which is the term of σ on the right. $\square$

Definition 10.4.4. For $v \in V$ and $\alpha \in \Lambda^p(V^*)$ with $p \geq 1$, the *interior product* $\iota_v \alpha \in \Lambda^{p-1}(V^*)$ is $(\iota_v \alpha)(v_2, \dots, v_p) = \alpha(v, v_2, \dots, v_p)$; for $p = 0$, $\iota_v \alpha = 0$.

Proposition 10.4.5. $\iota_v(\alpha \wedge \beta) = \iota_v \alpha \wedge \beta + (-1)^p \alpha \wedge \iota_v \beta$ for $\alpha \in \Lambda^p(V^*)$, and $\iota_v \iota_w = -\iota_w \iota_v$.

Proof. Expanding the determinant along the first column (Laplace expansion) gives $\iota_v(\varphi_1 \wedge \dots \wedge \varphi_p) = \sum_{i=1}^p (-1)^{i-1} \varphi_i(v) \varphi_1 \wedge \dots \wedge \widehat{\varphi_i} \wedge \dots \wedge \varphi_p$. Applying this formula to a product of decomposable forms and splitting the sum according to whether i is among the first p factors gives the first identity. The second holds because $\alpha(w, v, \dots) = -\alpha(v, w, \dots)$. $\square$

Proposition 10.4.6. Suppose $\text{char } k = 0$, and let $\omega \in \Lambda^2(V^*)$ be nondegenerate, $\dim V = 2m$. Then $\omega^m = \omega \wedge \dots \wedge \omega \neq 0$, and every $f \in \text{GL}(V)$ with $\omega(fv, fw) = \omega(v, w)$ has $\det f = 1$.

Proof. With a symplectic basis (Theorem 9.8.6) and its dual basis, $\omega = \sum_i e_i^* \wedge f_i^*$, as both sides take the same values on pairs of basis vectors. The 2-forms $e_i^* \wedge f_i^*$ commute and square to zero, so $\omega^m = m! e_1^* \wedge f_1^* \wedge \dots \wedge e_m^* \wedge f_m^* \neq 0$. The map $f^*: V^* \rightarrow V^*$, $\varphi \mapsto \varphi \circ f$, satisfies $(\Lambda^2 f^*)(\omega) = \omega$, hence $(\Lambda^{2m} f^*)(\omega^m) = \omega^m$; but $\Lambda^{2m} f^*$ is multiplication by $\det f^* = \det f$ (Proposition 10.3.3(5) and Proposition 9.4.2(3)). $\square$

10.5 Orientations and the Hodge star

Definition 10.5.1. Let V be a real vector space of dimension $n \geq 1$. Two ordered bases are *equally oriented* if the change of basis matrix has positive determinant. This is an equivalence relation with two classes, the *orientations* of V ; equivalently, an orientation is one of the two components $\{c e_1 \wedge \dots \wedge e_n : c > 0\}$, $\{c e_1 \wedge \dots \wedge e_n : c < 0\}$ of $\Lambda^n V \setminus \{0\}$.

Proposition 10.5.2. Let (V, g) be a real inner product space of dimension n . There is a unique inner product on $\Lambda^p(V)$ with $\langle v_1 \wedge \dots \wedge v_p, w_1 \wedge \dots \wedge w_p \rangle = \det(g(v_i, w_j))$. If (e_i) is orthonormal, the e_i are orthonormal. If V is oriented, the element $\text{vol} = e_1 \wedge \dots \wedge e_n$ does not depend on the choice of an oriented orthonormal basis; it is the volume element.

Proof. The formula is multilinear and alternating in the v_i and in the w_j , so it induces a bilinear form, which is symmetric by Proposition 9.5.3(2). On an orthonormal basis $\langle e_i, e_j \rangle = \delta_{ij}$, so the form is positive definite. If (f_i) is another oriented orthonormal basis, $f_j = \sum_i a_{ij} e_i$ with $A \in \text{O}(n)$ and $\det A > 0$, hence $\det A = 1$, and $f_1 \wedge \dots \wedge f_n = \det(A) e_1 \wedge \dots \wedge e_n$. $\square$

Theorem 10.5.3. *Let (V, g) be an oriented real inner product space of dimension n . There is a unique linear map $*$: $\Lambda^p(V) \rightarrow \Lambda^{n-p}(V)$ with*

$$\alpha \wedge * \beta = \langle \alpha, \beta \rangle \text{vol} \quad \text{for all } \alpha, \beta \in \Lambda^p(V).$$

*For an oriented orthonormal basis, $*e_I = \varepsilon_I e_{I^c}$ where $e_I \wedge e_{I^c} = \varepsilon_I \text{vol}$, $\varepsilon_I = \pm 1$. The map $*$ is an isometry, $** = (-1)^{p(n-p)}$ on $\Lambda^p(V)$, and $\alpha \wedge * \beta = \beta \wedge * \alpha$.*

Proof. The pairing $\Lambda^p(V) \times \Lambda^{n-p}(V) \rightarrow \Lambda^n(V) = \mathbb{R} \text{vol}$ is perfect, since $e_I \wedge e_J$ is $\pm \text{vol}$ if $J = I^c$ and 0 otherwise. Hence for each β there is a unique $*\beta$ representing the functional $\alpha \mapsto \langle \alpha, \beta \rangle$, and it depends linearly on β . The formula: $e_J \wedge \varepsilon_I e_{I^c}$ is $\varepsilon_I^2 \text{vol} = \text{vol}$ if $J = I$ and 0 otherwise, which is $\langle e_J, e_I \rangle \text{vol}$. So $*$ maps the orthonormal basis (e_I) to $(\pm e_{I^c})$, hence is an isometry. Next, $**e_I = \varepsilon_I \varepsilon_{I^c} e_I$, and $e_{I^c} \wedge e_I = (-1)^{p(n-p)} e_I \wedge e_{I^c}$ by Proposition 10.3.3(1), so $\varepsilon_{I^c} = (-1)^{p(n-p)} \varepsilon_I$. The last identity follows from the symmetry of the inner product. $\square$

Example 10.5.4. On $\mathbb{R}^3$ with the standard orientation, $*e_1 = e_2 \wedge e_3$, $*e_2 = e_3 \wedge e_1$, $*e_3 = e_1 \wedge e_2$, $*1 = \text{vol}$ and $*\text{vol} = 1$. The map $(v, w) \mapsto *(v \wedge w)$ is the cross product. On $\mathbb{R}^4$, $*$ is an involution of Λ^2 (since $(-1)^{2 \cdot 2} = 1$), which splits into ± 1 -eigenspaces of dimension 3: the self-dual and anti-self-dual 2-vectors.

10.6 Forms of type (p, q)

Let V be a real vector space of dimension $2n$ with a complex structure J and $W = V^* = \text{Hom}_{\mathbb{R}}(V, \mathbb{R})$. Then $W_{\mathbb{C}} = W \otimes_{\mathbb{R}} \mathbb{C}$ is identified with $\text{Hom}_{\mathbb{R}}(V, \mathbb{C})$, and $\Lambda^k(W_{\mathbb{C}})$ (exterior power over $\mathbb{C}$) with $\Lambda^k(W) \otimes_{\mathbb{R}} \mathbb{C}$, the complex-valued alternating k -forms on V .

Definition 10.6.1. Let $W^{1,0} \subseteq W_{\mathbb{C}}$ be the space of $\mathbb{R}$ -linear maps $\varphi: V \rightarrow \mathbb{C}$ with $\varphi(Jv) = i\varphi(v)$, that is, the $\mathbb{C}$ -linear maps $(V, J) \rightarrow \mathbb{C}$, and $W^{0,1} = \overline{W^{1,0}}$, the maps with $\varphi(Jv) = -i\varphi(v)$. For $p, q \geq 0$ let $\Lambda^{p,q} \subseteq \Lambda^{p+q}(W_{\mathbb{C}})$ be the span of the products $\alpha \wedge \beta$ with $\alpha \in \Lambda^p W^{1,0}$ and $\beta \in \Lambda^q W^{0,1}$. Its elements are the forms of type (p, q) .

Proposition 10.6.2. 1. $W_{\mathbb{C}} = W^{1,0} \oplus W^{0,1}$, and both summands have complex dimension n .

$$2. \Lambda^k(W_{\mathbb{C}}) = \bigoplus_{p+q=k} \Lambda^{p,q}, \text{ with } \dim_{\mathbb{C}} \Lambda^{p,q} = \binom{n}{p} \binom{n}{q}, \text{ and } \overline{\Lambda^{p,q}} = \Lambda^{q,p}.$$

3. A real 2-form $\alpha \in \Lambda^2(W)$ lies in $\Lambda^{1,1}$ if and only if $\alpha(Jv, Jw) = \alpha(v, w)$ for all v, w .

Proof. (1) The map $J^*: W \rightarrow W$, $\varphi \mapsto \varphi \circ J$, is a complex structure on W , and $W^{1,0}, W^{0,1}$ are the $\pm i$ -eigenspaces of its complexification; apply Proposition 9.10.2. (2) Choose a basis $\zeta_1, \dots, \zeta_n$ of $W^{1,0}$; then $\bar{\zeta}_1, \dots, \bar{\zeta}_n$ is a basis of $W^{0,1}$ and together they form a basis of $W_{\mathbb{C}}$. By Proposition 10.3.3(3) the products $\zeta_I \wedge \bar{\zeta}_{I'}$ with $|I| + |I'| = k$ form a basis of $\Lambda^k(W_{\mathbb{C}})$ (after reordering factors, which only changes signs), and those with $|I| = p, |I'| = q$ span $\Lambda^{p,q}$. Complex conjugation, extended to forms, maps $\zeta_I \wedge \bar{\zeta}_{I'}$ to $\bar{\zeta}_I \wedge \zeta_{I'}$. (3) The map $\alpha \mapsto J^* \alpha$, $(J^* \alpha)(v, w) = \alpha(Jv, Jw)$, equals Λ^2 of J^* on $W_{\mathbb{C}}$. On $\Lambda^{2,0}$ it acts by $i \cdot i = -1$, on $\Lambda^{0,2}$ by $(-i)(-i) = -1$, and on $\Lambda^{1,1}$ by $i(-i) = 1$. So $\Lambda^{1,1}$ is exactly the fixed space of J^* on $\Lambda^2(W_{\mathbb{C}})$. $\square$

Example 10.6.3. On $\mathbb{R}^{2n} = \mathbb{C}^n$ with the standard structure (Example 9.10.4) let dx_j, dy_j be the dual basis and $dz_j = dx_j + i dy_j$, $d\bar{z}_j = dx_j - i dy_j$. Then $dz_j \in W^{1,0}$, since $dz_j(J\partial_{x_j}) = dz_j(\partial_{y_j}) = i = i dz_j(\partial_{x_j})$ and similarly on ∂_{y_j} . The fundamental form of the standard inner product is

$$\omega = \sum_j dx_j \wedge dy_j = \frac{i}{2} \sum_j dz_j \wedge d\bar{z}_j,$$

because $dz_j \wedge d\bar{z}_j = -2i dx_j \wedge dy_j$. It is a real form of type $(1,1)$, in accordance with Proposition 10.6.2(3) and Proposition 9.10.3.

10.7 Exercises

Exercise 10.7.1. Show that $\dim S^d(k^n) = \binom{n+d-1}{d}$.

Exercise 10.7.2. Let $v \in V$ be nonzero and $\alpha \in \Lambda^p(V)$. Show that $v \wedge \alpha = 0$ if and only if $\alpha = v \wedge \beta$ for some $\beta \in \Lambda^{p-1}(V)$.

Exercise 10.7.3. Let $\text{char } k \neq 2$. Show that $\omega \in \Lambda^2(V)$ is decomposable, $\omega = v \wedge w$, if and only if $\omega \wedge \omega = 0$.

Exercise 10.7.4. In the situation of Theorem 10.5.3, show that $\alpha \wedge * \alpha = \|\alpha\|^2 \text{vol}$, and compute $*$ on $\Lambda^1(\mathbb{R}^2)$: it is the rotation by 90 degrees.

Solution. The first identity is the definition with $\beta = \alpha$. On $\mathbb{R}^2$, $*e_1 = e_2$ and $*e_2 = -e_1$, since $e_1 \wedge e_2 = \text{vol}$ and $e_2 \wedge (-e_1) = \text{vol}$. This is the standard complex structure J . $\square$

Exercise 10.7.5. Let $V = \mathbb{C}^n$. Show that the real (k, k) -forms, that is, $\alpha \in \Lambda^{k,k}$ with $\bar{\alpha} = \alpha$, form a real vector space of dimension $\binom{n}{k}^2$.

Exercise 10.7.6. With notation as in Section 10.6, let C act on $\Lambda^{p,q}$ as multiplication by i^{p-q} . Show that C commutes with complex conjugation, so preserves real forms, and that C is the map induced on $\Lambda(W_{\mathbb{C}})$ by J^* .

Chapter 11

Representations of Groups and Lie Algebras

11.1 Introduction

A representation of a group or a Lie algebra is an action by linear maps. This chapter develops the basic theory: Schur's lemma, Maschke's theorem and character theory for finite groups, and the complete classification of the finite-dimensional representations of the Lie algebra $\mathfrak{sl}_2(\mathbb{C})$. The last result is used repeatedly in Hodge theory: on the cohomology of a compact Kähler manifold the Lefschetz operator, its adjoint and the degree operator span a copy of $\mathfrak{sl}_2$, and the hard Lefschetz theorem and the primitive decomposition are consequences of the structure of $\mathfrak{sl}_2$ -representations (Chapters 44 and 46).

References are [Ser77], [FH91] and [Hum72].

Prerequisites. Chapter 6 (Groups), Chapter 8 (Fields and Galois Theory), Chapter 9 (Linear Algebra) and Chapter 10 (Multilinear Algebra).

Throughout, k is a field and representations are on k -vector spaces.

11.2 Representations of groups

Definition 11.2.1. A *representation* of a group G is a vector space V with a homomorphism $\rho: G \rightarrow \mathrm{GL}(V)$; we write gv for $\rho(g)v$. A *homomorphism* of representations (or G -equivariant map) is a linear map f with $f(gv) = g f(v)$; the space of such maps is $\mathrm{Hom}_G(V, W)$. A *subrepresentation* is a G -stable subspace. V is *irreducible* if $V \neq 0$ and its only subrepresentations are 0 and V , and *completely reducible* (or *semisimple*) if it is a direct sum of irreducible subrepresentations. The *invariants* are $V^G = \{v : gv = v \text{ for all } g\}$.

Example 11.2.2. Given representations V, W : quotients V/U by subrepresentations, $V \oplus W$, $V \otimes W$ with $g(v \otimes w) = gv \otimes gw$, the dual V^* with $(g\varphi)(v) = \varphi(g^{-1}v)$, and $\mathrm{Hom}_k(V, W)$ with $(gf)(v) = gf(g^{-1}v)$ are representations, and $\mathrm{Hom}_k(V, W)^G = \mathrm{Hom}_G(V, W)$. The *trivial representation* is k with $gv = v$. If G acts on a finite set X , the *permutation representation* is k^X with $ge_x = e_{gx}$; for $X = G$ with the left action this is the *regular representation*.

Lemma 11.2.3 (Schur). *Let V, W be irreducible representations. Every $f \in \mathrm{Hom}_G(V, W)$ is zero or an isomorphism. If k is algebraically closed and V is finite-dimensional, then $\mathrm{Hom}_G(V, V) = k \cdot \mathrm{id}$.*

Proof. $\ker f$ and $\mathrm{im} f$ are subrepresentations, so $\ker f \in \{0, V\}$ and $\mathrm{im} f \in \{0, W\}$. If k is algebraically closed and $f \in \mathrm{Hom}_G(V, V)$, f has an eigenvalue λ , and $f - \lambda \in \mathrm{Hom}_G(V, V)$ is not an isomorphism, hence zero. $\square$

Theorem 11.2.4 (Maschke). *Let G be finite with $\text{char } k \nmid |G|$. Every subrepresentation U of a representation V has a G -stable complement. Hence every finite-dimensional representation is completely reducible.*

Proof. Choose a linear projection $\pi: V \rightarrow U$ (a linear map with $\pi|_U = \text{id}$), using a complement of U (Proposition 9.2.1), and put $\pi_0 = \frac{1}{|G|} \sum_{g \in G} g\pi g^{-1}$. It is G -equivariant, has image in U , and restricts to the identity on U ; so $\ker \pi_0$ is a G -stable complement of U . The second statement follows by induction on the dimension. $\square$

Counterexample 11.2.5. Both hypotheses of Maschke's theorem are needed. The group $\mathbb{Z}$ acts on $\mathbb{C}^2$ by $n \mapsto \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$; the line $\mathbb{C}e_1$ is stable, but it has no stable complement, since a stable line would be spanned by an eigenvector and e_1 spans the only eigenline. Similarly $\mathbb{Z}/p\mathbb{Z}$ acts on $\mathbb{F}_p^2$ by the same matrices (with n read modulo p), and the representation is not completely reducible although $\mathbb{Z}/p\mathbb{Z}$ is finite; here p divides the order of the group.

11.3 Characters of finite groups

In this section G is finite, $k = \mathbb{C}$, and representations are finite-dimensional.

Definition 11.3.1. The *character* of a representation V is $\chi_V: G \rightarrow \mathbb{C}$, $\chi_V(g) = \text{tr}(g|_V)$. A *class function* is a function $G \rightarrow \mathbb{C}$ constant on conjugacy classes. On class functions put the Hermitian inner product $\langle \phi, \psi \rangle = \frac{1}{|G|} \sum_{g \in G} \phi(g) \overline{\psi(g)}$.

Proposition 11.3.2. *Characters are class functions, $\chi_V(e) = \dim V$, $\chi_{V \oplus W} = \chi_V + \chi_W$, $\chi_{V \otimes W} = \chi_V \chi_W$, and $\chi_{V^*}(g) = \overline{\chi_V(g)} = \chi_V(g^{-1})$. Moreover $\dim V^G = \frac{1}{|G|} \sum_g \chi_V(g)$.*

Proof. The trace is invariant under conjugation, additive on direct sums and multiplicative on tensor products (compute in bases). Each g has finite order, so $g|_V$ satisfies $x^n - 1$ and is diagonalisable with roots of unity λ_i as eigenvalues (Theorem 9.6.6); then $\chi_{V^*}(g) = \sum \lambda_i^{-1} = \sum \bar{\lambda}_i$. Finally $p = \frac{1}{|G|} \sum_g g$ is a projection onto V^G (it is the identity on V^G and $hp = p$ for all h), and the trace of a projection is the dimension of its image. $\square$

Theorem 11.3.3 (Orthogonality). *For representations V, W , $\langle \chi_W, \chi_V \rangle = \dim \text{Hom}_G(V, W)$. Consequently the characters of irreducible representations are orthonormal, a representation is determined up to isomorphism by its character, and V is irreducible if and only if $\langle \chi_V, \chi_V \rangle = 1$.*

Proof. $\text{Hom}_G(V, W) = \text{Hom}_{\mathbb{C}}(V, W)^G$, and $\text{Hom}_{\mathbb{C}}(V, W) \cong V^* \otimes W$ as representations, so by Proposition 11.3.2 its dimension is $\frac{1}{|G|} \sum_g \overline{\chi_V(g)} \chi_W(g)$. For irreducible V, W , Schur's lemma gives $\dim \text{Hom}_G(V, W) = 1$ if $V \cong W$ and 0 otherwise. By Maschke's theorem $V \cong \bigoplus_i V_i^{m_i}$ with distinct irreducible V_i , and $m_i = \langle \chi_V, \chi_{V_i} \rangle$ is determined by χ_V ; also $\langle \chi_V, \chi_V \rangle = \sum m_i^2$. $\square$

Theorem 11.3.4. *The characters of the irreducible representations of G form an orthonormal basis of the space of class functions. In particular the number of isomorphism classes of irreducible representations equals the number of conjugacy classes, and if $V_1, \dots, V_r$ are the irreducible representations, then $\sum_i (\dim V_i)^2 = |G|$.*

Proof. The regular representation has character $|G|$ at e and 0 elsewhere, so by Theorem 11.3.3 it contains each V_i with multiplicity $\langle \chi_{\text{reg}}, \chi_{V_i} \rangle = \dim V_i$; comparing dimensions gives $\sum (\dim V_i)^2 = |G|$. In particular there are finitely many irreducible representations up to isomorphism.

It remains to show that a class function ϕ orthogonal to all χ_{V_i} is zero. For a representation V let $T_V = \sum_g \overline{\phi(g)} g|_V$. Since ϕ is a class function, T_V commutes with all $h|_V$. If V is irreducible, $T_V = \lambda \text{id}$ by Schur's lemma, and taking traces, $\lambda \dim V = \sum_g \overline{\phi(g)} \chi_V(g) = |G| \langle \chi_V, \phi \rangle = 0$. So $T_V = 0$ for every irreducible V , hence for every V by Maschke's theorem, in particular for the regular representation; but there $T(e_e) = \sum_g \overline{\phi(g)} e_g$, so $\phi = 0$. $\square$

Example 11.3.5. S_3 has three conjugacy classes: $\{e\}$, the three transpositions and the two 3-cycles. Its irreducible representations are the trivial representation, the sign representation and the two-dimensional representation $U = \{x \in \mathbb{C}^3 : x_1 + x_2 + x_3 = 0\}$ (a subrepresentation of the permutation representation). Their characters take the values $(1, 1, 1)$, $(1, -1, 1)$ and $(2, 0, -1)$ on the three classes. One checks orthonormality directly: for instance $\langle \chi_U, \chi_U \rangle = \frac{1}{6}(4 + 0 + 2 \cdot 1) = 1$.

Proposition 11.3.6. *Let G be a finite abelian group. Every irreducible complex representation of G is one-dimensional, given by a homomorphism $\chi : G \rightarrow \mathbb{C}^\times$, and there are exactly $|G|$ of them. For $G = \mathbb{Z}/n\mathbb{Z}$ they are $\chi_j(k) = e^{2\pi i j k / n}$, $0 \leq j < n$.*

Proof. Let V be irreducible. Each g commutes with the action of G , so by Schur's lemma (Lemma 11.2.3) it acts by a scalar; then every line in V is a subrepresentation, and $\dim V = 1$. The conjugacy classes of an abelian group are its elements, so by Theorem 11.3.4 there are $|G|$ irreducible representations. For $\mathbb{Z}/n\mathbb{Z}$, a homomorphism to $\mathbb{C}^\times$ is determined by the image ζ of 1, with $\zeta^n = 1$, and the n choices give the χ_j . $\square$

Counterexample 11.3.7. The second statement of Schur's lemma needs an algebraically closed field. The group $\mathbb{Z}/4\mathbb{Z}$ acting on $\mathbb{R}^2$ by rotation by $\pi/2$, with generator acting by $J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$, is irreducible over $\mathbb{R}$, since J has no real eigenvector; but its endomorphisms, the matrices commuting with J , are the $aI + bJ$, a field isomorphic to $\mathbb{C}$, not $\mathbb{R} \cdot \text{id}$. Over $\mathbb{C}$ the representation splits as $\chi_1 \oplus \chi_3$ in the notation of Proposition 11.3.6.

Definition 11.3.8. Let $H \subseteq G$ be a subgroup of a finite group and W a representation of H . The *induced representation* is $\text{Ind}_H^G W = \{f : G \rightarrow W : f(hx) = hf(x) \text{ for } h \in H, x \in G\}$, with $(gf)(x) = f(xg)$. For a representation V of G , $\text{Res}_H^G V$ is V viewed as a representation of H .

Proposition 11.3.9 (Frobenius reciprocity). *In the situation of Definition 11.3.8:*

1. $\text{Hom}_G(V, \text{Ind}_H^G W) \cong \text{Hom}_H(\text{Res}_H^G V, W)$, via $\varphi \mapsto (v \mapsto \varphi(v)(e))$;
2. $\dim \text{Ind}_H^G W = [G : H] \dim W$, and

$$\chi_{\text{Ind } W}(g) = \frac{1}{|H|} \sum_{x \in G, xgx^{-1} \in H} \chi_W(xgx^{-1});$$

3. $\langle \chi_{\text{Ind } W}, \chi_V \rangle_G = \langle \chi_W, \chi_{\text{Res } V} \rangle_H$.

Proof. (1) If φ is G -equivariant, then $\psi(v) = \varphi(v)(e)$ satisfies $\psi(hv) = \varphi(hv)(e) = (h\varphi(v))(e) = \varphi(v)(h) = h\varphi(v)(e) = h\psi(v)$. Conversely, for $\psi \in \text{Hom}_H(V, W)$ put $\varphi(v)(x) = \psi(xv)$; then $\varphi(v)(hx) = \psi(hxv) = h\varphi(v)(x)$, so $\varphi(v) \in \text{Ind } W$, and $\varphi(gv)(x) = \psi(xgv) = \varphi(v)(xg) = (g\varphi(v))(x)$. The two constructions are inverse, since $\varphi(v)(x) = (x\varphi(v))(e) = \varphi(xv)(e)$.

(2) Choose representatives $x_1, \dots, x_m$ of the cosets Hx , $m = [G : H]$. An element f is determined by the values $f(x_i)$, which are arbitrary, so $f \mapsto (f(x_i))_i$ is an isomorphism with W^m . For $g \in G$ write $x_i g = h_i x_{\sigma(i)}$ with $h_i \in H$ and σ a permutation; then $(gf)(x_i) = f(x_i g) = h_i f(x_{\sigma(i)})$. So in the decomposition W^m , g maps the $\sigma(i)$ -th summand to the i -th by

h_i , and only the indices with $\sigma(i) = i$ contribute to the trace, with $h_i = x_i g x_i^{-1} \in H$. Hence $\chi_{\text{Ind } W}(g) = \sum_{i: x_i g x_i^{-1} \in H} \chi_W(x_i g x_i^{-1})$, and since χ_W is a class function on H , each coset Hx_i contributes $|H|$ equal terms to the sum in the statement.

(3) By Theorem 11.3.3, both sides are dimensions of the spaces in (1): the left side is $\dim \text{Hom}_G(V, \text{Ind}_H^G W)$ and the right side is $\dim \text{Hom}_H(\text{Res}_H^G V, W)$. $\square$

Counterexample 11.3.10. The character table does not determine the group. Let G be a group of order 8 whose centre Z has order 2 and with $G/Z \cong (\mathbb{Z}/2\mathbb{Z})^2$; the dihedral group D_4 of symmetries of a square and the quaternion group $Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}$ are examples. For $x \notin Z$ the centraliser contains $\langle x \rangle Z$, of order 4, and is not G , so the class of x is $\{x, zx\}$, $Z = \{e, z\}$: there are five classes. For $x \notin Z$, $x^2 \in Z$ and every conjugate of x lies in xZ , because G/Z is abelian of exponent 2. The commutator subgroup is Z (G/Z is abelian and G is not), so the one-dimensional characters are the four characters of G/Z , equal to 1 at z ; at each non-identity element of G/Z two of them take the value 1 and two the value -1 ; by Theorem 11.3.4 the fifth irreducible character χ has degree 2 ($4 + 2^2 = 8$), and comparing with the regular character, $2\chi = \chi_{\text{reg}} - \sum \chi_{1\text{-dim}}$ gives $\chi(z) = -2$ and $\chi(x) = 0$ for $x \notin Z$. So the character tables of D_4 and Q_8 coincide, but the groups are not isomorphic: D_4 has five elements of order 2 and Q_8 only one.

11.4 Lie algebras

Definition 11.4.1. A Lie algebra over k is a vector space $\mathfrak{g}$ with a bilinear map $[-, -]: \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, the *bracket*, with $[x, x] = 0$ and the *Jacobi identity* $[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0$. A *homomorphism* of Lie algebras is a linear map preserving brackets, an *ideal* is a subspace $\mathfrak{a}$ with $[\mathfrak{g}, \mathfrak{a}] \subseteq \mathfrak{a}$, and $\mathfrak{g}$ is *abelian* if its bracket is zero.

Example 11.4.2. 1. For an associative algebra A , the commutator $[x, y] = xy - yx$ makes A a Lie algebra. For $A = \text{End}(V)$ this is $\mathfrak{gl}(V)$, and for $A = M_n(k)$ it is $\mathfrak{gl}_n(k)$.

2. $\mathfrak{sl}_n(k) = \{x \in \mathfrak{gl}_n(k) : \text{tr } x = 0\}$ is a Lie subalgebra, since $\text{tr}[x, y] = 0$ (Exercise 9.11.1).

3. The skew-symmetric matrices form $\mathfrak{so}_n(k)$, and more generally the x with $B(xv, w) + B(v, xw) = 0$ for a bilinear form B form a Lie subalgebra of $\mathfrak{gl}(V)$.

Definition 11.4.3. A *representation* of a Lie algebra $\mathfrak{g}$ is a vector space V with a Lie algebra homomorphism $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$; we write xv for the action, so $[x, y]v = x(yv) - y(xv)$. Subrepresentations, irreducibility, homomorphisms and direct sums are defined as for groups. The tensor product $V \otimes W$ is a representation with $x(v \otimes w) = xv \otimes w + v \otimes xw$, and the dual with $(x\varphi)(v) = -\varphi(xv)$. The *adjoint representation* is $\mathfrak{g}$ itself with $x \cdot y = [x, y]$, a representation by the Jacobi identity.

Remark 11.4.4. Lie algebras arise as tangent spaces of Lie groups at the identity (Chapter 32), and representations of a Lie group differentiate to representations of its Lie algebra; the formula for the tensor product is the derivative of $g(v \otimes w) = gv \otimes gw$.

11.5 Representations of $\mathfrak{sl}_2$

In this section $k = \mathbb{C}$ and $\mathfrak{sl}_2 = \mathfrak{sl}_2(\mathbb{C})$, with basis

$$e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

satisfying $[h, e] = 2e$, $[h, f] = -2f$ and $[e, f] = h$. All representations are finite-dimensional.

Lemma 11.5.1. *Let V be a representation and for $\lambda \in \mathbb{C}$ let $V_\lambda = \ker(h - \lambda)$ and $V^{(\lambda)} = \bigcup_N \ker(h - \lambda)^N$ (the generalised eigenspace). Then e maps V_λ to $V_{\lambda+2}$ and $V^{(\lambda)}$ to $V^{(\lambda+2)}$, and f maps them to $V_{\lambda-2}$ and $V^{(\lambda-2)}$. Moreover, for $k \geq 1$,*

$$ef^k = f^k e + k f^{k-1}(h - k + 1) \quad \text{as operators on } V.$$

Proof. From $he = eh + 2e$ we get $(h - \lambda - 2)e = e(h - \lambda)$, hence $(h - \lambda - 2)^N e = e(h - \lambda)^N$; similarly for f . The identity holds for $k = 1$ since $ef = fe + h$. If it holds for k , then using $hf = fh - 2f$, $ef^{k+1} = f^k ef + k f^{k-1}(h - k + 1)f = f^k(fe + h) + k f^k(h - k - 1) = f^{k+1}e + (k + 1)f^k(h - k)$. $\square$

Theorem 11.5.2. *For every $m \geq 0$ there is an irreducible representation $V(m)$ of dimension $m + 1$ with a basis $v_0, \dots, v_m$ such that*

$$hv_j = (m - 2j)v_j, \quad fv_j = (j + 1)v_{j+1}, \quad ev_j = (m - j + 1)v_{j-1},$$

with $v_{-1} = v_{m+1} = 0$. Every irreducible representation is isomorphic to exactly one $V(m)$. More precisely, if V is any representation and $v_0 \neq 0$ satisfies $ev_0 = 0$ and $hv_0 = \lambda v_0$, then $\lambda = m$ is a non-negative integer and the vectors $v_j = f^j v_0 / j!$ span a subrepresentation isomorphic to $V(m)$.

Proof. The formulas define a representation: it suffices to check the three bracket relations on each v_j . For instance $(ef - fe)v_j = (j + 1)(m - j)v_j - j(m - j + 1)v_j = (m - 2j)v_j = hv_j$, and $[h, e] = 2e$, $[h, f] = -2f$ hold because e raises and f lowers the h -eigenvalue by 2. It is irreducible: a nonzero subrepresentation is h -stable, so it contains some v_j (the eigenvalues $m - 2j$ are distinct), then v_0 by applying e (the coefficients $m - j + 1$ are nonzero for $1 \leq j \leq m$), then all v_j by applying f .

Now let v_0 be as in the last statement and $v_j = f^j v_0 / j!$. By Lemma 11.5.1, $hv_j = (\lambda - 2j)v_j$, and $fv_j = (j + 1)v_{j+1}$ by definition. By the identity of the lemma, $ev_j = \frac{1}{j!} j f^{j-1}(h - j + 1)v_0 = (\lambda - j + 1)v_{j-1}$. The nonzero v_j have distinct h -eigenvalues, hence are linearly independent, so there is a largest m with $v_m \neq 0$, and then $v_{m+1} = 0$. Hence $0 = ev_{m+1} = (\lambda - m)v_m$, so $\lambda = m$, and the span of $v_0, \dots, v_m$ is a subrepresentation isomorphic to $V(m)$.

Every representation $V \neq 0$ contains such a v_0 : h has an eigenvector w , with eigenvalue μ , and among $w, ew, e^2w, \dots$ (which lie in the distinct eigenspaces $V_{\mu+2i}$ or vanish) there is a last nonzero one. If V is irreducible, the subrepresentation generated by v_0 is all of V , so $V \cong V(m)$ with $m = \dim V - 1$. $\square$

Theorem 11.5.3. *Every finite-dimensional representation of $\mathfrak{sl}_2$ is completely reducible, and h acts diagonalisably with integer eigenvalues.*

Proof. Let $C = ef + fe + \frac{1}{2}h^2$ (the Casimir operator). A direct computation with the relations shows that C commutes with e, f and h ; for example $[C, e] = e[f, e] + [f, e]e + \frac{1}{2}(h[h, e] + [h, e]h) = -eh - he + eh + he = 0$. On $V(m)$, C is scalar by Lemma 11.2.3, and evaluating on v_0 gives $C = m + \frac{1}{2}m^2 = \frac{1}{2}m(m + 2)$ (using $efv_0 = mv_0$ and $ev_0 = 0$). These scalars are distinct for different $m \geq 0$.

Since C commutes with the action, V is the direct sum of the generalised eigenspaces of C , each of which is a subrepresentation (Lemma 9.6.5). By choosing an irreducible subrepresentation and passing to quotients repeatedly, every representation has a chain of subrepresentations with irreducible successive quotients, each isomorphic to some $V(m)$ by Theorem 11.5.2; C acts on the quotient $V(m)$ by $\frac{1}{2}m(m + 2)$. Hence on the generalised c -eigenspace, all these quotients are isomorphic to the same $V(m)$ (with $c = \frac{1}{2}m(m + 2)$). It therefore suffices to prove: if V has such a chain $0 = F_0 \subsetneq F_1 \subsetneq \dots \subsetneq F_r = V$ with $F_i/F_{i-1} \cong V(m)$ for all i , then $V \cong V(m)^r$.

The eigenvalues of h on V are among $m, m - 2, \dots, -m$, so $V^{(m+2)} = 0$ and e kills $V^{(m)}$ (Lemma 11.5.1). Also $\dim V^{(m)} = r$, since each quotient contributes one dimension. The map

$f^m : V^{(m)} \rightarrow V^{(-m)}$ is injective: if $u \in V^{(m)}$ is nonzero, let i be least with $u \in F_i$; the image of u in $F_i/F_{i-1} \cong V(m)$ is a nonzero vector of weight m , on which f^m is nonzero, so $f^m u \notin F_{i-1}$. Now $f^{m+1}u \in V^{(-m-2)} = 0$, and by Lemma 11.5.1, $0 = e f^{m+1}u = f^{m+1}eu + (m+1)f^m(h-m)u = (m+1)f^m(h-m)u$. Since $(h-m)u \in V^{(m)}$ and f^m is injective there, $(h-m)u = 0$. So $h = m$ on $V^{(m)}$.

Let $u_1, \dots, u_r$ be a basis of $V^{(m)} = V_m$. Each u_i satisfies $eu_i = 0$ and $hu_i = mu_i$, so it generates a subrepresentation $W_i \cong V(m)$ by Theorem 11.5.2. The kernel K of $\bigoplus_i W_i \rightarrow V$ is a subrepresentation whose weight- m part is zero, since the u_i are independent. If $K \neq 0$, the last paragraph of the proof of Theorem 11.5.2, applied to K , gives $v_0 \in K$ with $v_0 \neq 0$ and $ev_0 = 0$. But in $V(m)$ the kernel of e is $\mathbb{C}v_0$, since $e(\sum_j c_j v_j) = \sum_{j \geq 1} c_j(m-j+1)v_{j-1}$ vanishes only if $c_j = 0$ for $j \geq 1$; so in $\bigoplus_i W_i \cong V(m)^r$ the kernel of e is the weight- m space, and v_0 would be a nonzero element of K of weight m . Hence $K = 0$, the sum is direct, and by dimension $(r(m+1) = \dim V)$ it is all of V . In each W_i , h is diagonal with integer eigenvalues. $\square$

Corollary 11.5.4. *Let V be a finite-dimensional representation of $\mathfrak{sl}_2$, with weight spaces $V_\lambda = \ker(h - \lambda)$, $\lambda \in \mathbb{Z}$. For every $j \geq 0$ the map $f^j : V_j \rightarrow V_{-j}$ is an isomorphism. Let $P_j = V_j \cap \ker e$ (the primitive vectors of weight j). Then*

$$V = \bigoplus_{j \geq 0} \bigoplus_{i=0}^j f^i P_j,$$

and f^i is injective on P_j for $0 \leq i \leq j$.

Proof. By Theorem 11.5.3 it suffices to treat $V = V(m)$, where $V_{m-2i} = \mathbb{C}v_i$, f^j maps $v_{(m-j)/2}$ to a nonzero multiple of $v_{(m+j)/2}$ when $m-j$ is even and non-negative, $P_m = \mathbb{C}v_0$, $P_j = 0$ for $j \neq m$, and $f^i P_m = \mathbb{C}v_i$. $\square$

Remark 11.5.5. Theorem 11.5.3 is a special case of Weyl's theorem: every finite-dimensional representation of a semisimple Lie algebra over $\mathbb{C}$ is completely reducible. The irreducible representations are classified by their highest weights, and the semisimple Lie algebras themselves by Dynkin diagrams. See [Hum72] and [FH91].

11.6 Exercises

Exercise 11.6.1. Let $\mathfrak{sl}_2$ act on $\mathbb{C}^2$ by matrix multiplication. Show that $S^m(\mathbb{C}^2) \cong V(m)$.

Exercise 11.6.2. Show that $V(m) \otimes V(n) \cong \bigoplus_{i=0}^{\min(m,n)} V(m+n-2i)$. (Hint: count weights.)

Solution. The vectors $v_a \otimes w_b$ form a basis of weight vectors, of weight $(m-2a) + (n-2b)$. So the weight $l = m+n-2s$ has multiplicity $c(s) = |\{(a,b) : 0 \leq a \leq m, 0 \leq b \leq n, a+b=s\}|$, and for $0 \leq s \leq (m+n)/2$ one finds $c(s) = \min(s, m, n) + 1$. By Theorem 11.5.3 the representation is a direct sum of $V(l)$'s, and the multiplicity of $V(l)$, $l \geq 0$, is $\dim V_l - \dim V_{l+2} = c(s) - c(s-1)$ (with $c(-1) = 0$), which is 1 for $s \leq \min(m, n)$ and 0 otherwise. $\square$

Exercise 11.6.3. Determine the character table of S_4 .

Exercise 11.6.4. Show that a finite group is abelian if and only if all its irreducible complex representations are one-dimensional. (One direction is Proposition 11.3.6; for the other use Theorem 11.3.4.)

Exercise 11.6.5. Let $V = \bigoplus_{k=0}^{2n} V^k$ be a graded finite-dimensional vector space and $L: V^k \rightarrow V^{k+2}$ a linear map. Suppose there is $\Lambda: V^k \rightarrow V^{k-2}$ such that, with $H = (k - n)\text{id}$ on V^k , the relations $[H, L] = 2L$, $[H, \Lambda] = -2\Lambda$, $[L, \Lambda] = H$ hold. Show that $L^j: V^{n-j} \rightarrow V^{n+j}$ is an isomorphism for all $j \geq 0$. (This is the algebraic form of the hard Lefschetz theorem.)

Chapter 12

Commutative Algebra

12.1 Introduction

Commutative algebra is the local theory of algebraic geometry. This chapter develops the tools used in Part VIII and in the local theory of analytic functions (Chapter 29): localisation, Nakayama's lemma, Noetherian and Artinian rings, integral extensions and Noether normalisation, the Nullstellensatz, primary decomposition, dimension theory, regular local rings and Dedekind domains, completions, and Kähler differentials.

References are [AM69], [Mat86] and [Eis95].

Prerequisites. Chapter 7 (Rings and Modules), Chapter 8 (Fields and Galois Theory) and Chapter 9 (Linear Algebra).

In accordance with Notation 1.6.4, from now on all rings are commutative with 1.

12.2 Prime ideals and radicals

Definition 12.2.1. The *spectrum* $\text{Spec } R$ is the set of prime ideals of R . For an ideal I let $V(I) = \{\mathfrak{p} \in \text{Spec } R : I \subseteq \mathfrak{p}\}$, and for $f \in R$ let $D(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$. The *nilradical* of R is the set of nilpotent elements, the *Jacobson radical* $J(R)$ is the intersection of all maximal ideals, and the *radical* of an ideal is $\sqrt{I} = \{f : f^n \in I \text{ for some } n\}$.

Proposition 12.2.2. 1. The nilradical is the intersection of all prime ideals, and $\sqrt{I}$ is the intersection of all primes containing I .

2. More precisely, if $S \subseteq R$ is closed under multiplication, $1 \in S$, and I is an ideal with $I \cap S = \emptyset$, then there is a prime $\mathfrak{p} \supseteq I$ with $\mathfrak{p} \cap S = \emptyset$.

3. $x \in J(R)$ if and only if $1 - xy$ is a unit for all $y \in R$.

Proof. (2) The set of ideals containing I and disjoint from S is nonempty and closed under unions of chains, so it has a maximal element $\mathfrak{p}$ by Zorn's lemma. If $a, b \notin \mathfrak{p}$, then $\mathfrak{p} + (a)$ and $\mathfrak{p} + (b)$ meet S , say $s = p_1 + ra$ and $t = p_2 + r'b$; then $st \in \mathfrak{p} + (ab)$ lies in S , so $ab \notin \mathfrak{p}$. Also $1 \in S$, so $\mathfrak{p} \neq R$. (1) A nilpotent element lies in every prime. If f is not nilpotent, apply (2) with $S = \{f^n\}$ and $I = 0$. For $\sqrt{I}$, apply this to R/I . (3) If $x \in J(R)$ and $1 - xy$ is not a unit, it lies in a maximal ideal $\mathfrak{m}$ (Theorem 7.4.3), which also contains xy , hence 1. Conversely if $x \notin \mathfrak{m}$ for a maximal $\mathfrak{m}$, then $\mathfrak{m} + (x) = R$, so $1 = u + xy$ with $u \in \mathfrak{m}$, and $1 - xy = u$ is not a unit. $\square$

Lemma 12.2.3 (Prime avoidance). If an ideal I is contained in the union of prime ideals $\mathfrak{p}_1, \dots, \mathfrak{p}_n$, then $I \subseteq \mathfrak{p}_i$ for some i .

Proof. By induction on n ; suppose $I \not\subseteq \bigcup_{j \neq i} \mathfrak{p}_j$ for each i (otherwise use induction), and choose $x_i \in I$ with $x_i \notin \mathfrak{p}_j$ for $j \neq i$. Then $x_i \in \mathfrak{p}_i$ for all i . The element $y = x_1 + x_2 \cdots x_n$ lies in I but in no $\mathfrak{p}_i$: for $i = 1$, $x_1 \in \mathfrak{p}_1$ and $x_2 \cdots x_n \notin \mathfrak{p}_1$; for $i \geq 2$, $x_1 \notin \mathfrak{p}_i$ while $x_2 \cdots x_n \in \mathfrak{p}_i$. This contradicts $I \subseteq \mathfrak{p}_1 \cup \cdots \cup \mathfrak{p}_n$. $\square$

12.3 Localisation

Definition 12.3.1. A subset $S \subseteq R$ is *multiplicative* if $1 \in S$ and $st \in S$ for $s, t \in S$. For an R -module M , the *localisation* $S^{-1}M$ is the set of equivalence classes m/s of pairs $(m, s) \in M \times S$ under $(m, s) \sim (m', s') \iff t(s'm - sm') = 0$ for some $t \in S$, with $m/s + m'/s' = (s'm + sm')/ss'$. For $M = R$ it is a ring with $(r/s)(r'/s') = rr'/ss'$, and $S^{-1}R$ is an $S^{-1}R$ -module. For $f \in R$ we write R_f for $S^{-1}R$ with $S = \{f^n\}$, and for a prime $\mathfrak{p}$ we write $R_{\mathfrak{p}}$, $M_{\mathfrak{p}}$ for $S = R \setminus \mathfrak{p}$. A ring is *local* if it has exactly one maximal ideal.

Proposition 12.3.2. 1. $\sim$ is an equivalence relation and the operations are well defined. The map $R \rightarrow S^{-1}R$, $r \mapsto r/1$, is a ring homomorphism sending S to units, and every homomorphism $\varphi: R \rightarrow B$ sending S to units factors uniquely through it.

2. If $M' \rightarrow M \rightarrow M''$ is exact, so is $S^{-1}M' \rightarrow S^{-1}M \rightarrow S^{-1}M''$. There is a natural isomorphism $S^{-1}R \otimes_R M \cong S^{-1}M$; hence $S^{-1}R$ is a flat R -module.

3. The primes of $S^{-1}R$ are the $S^{-1}\mathfrak{p}$ for primes $\mathfrak{p}$ of R with $\mathfrak{p} \cap S = \emptyset$, and this is an inclusion-preserving bijection with inverse $\mathfrak{q} \mapsto$ its preimage in R . In particular $R_{\mathfrak{p}}$ is local with maximal ideal $\mathfrak{p}R_{\mathfrak{p}}$.

4. A ring is local if and only if its nonunits form an ideal.

Proof. (1) Transitivity: from $t(s'm - sm') = 0$ and $t'(s''m' - s'm'') = 0$, multiplying by $t's''$ and ts and adding gives $tt's'(s''m - sm'') = 0$. The other verifications are similar. Given φ , the map $r/s \mapsto \varphi(r)\varphi(s)^{-1}$ is well defined since $t(s'r - sr') = 0$ implies $\varphi(s'r) = \varphi(sr')$, and it is forced. (2) Let $f: M' \rightarrow M$, $g: M \rightarrow M''$ with $\text{im } f = \ker g$. If $g(m)/s = 0$, then $tg(m) = 0$ for some t , so $tm = f(m')$ and $m/s = f(m')/ts$. The map $S^{-1}R \times M \rightarrow S^{-1}M$, $(r/s, m) \mapsto rm/s$, is bilinear, and $m/s \mapsto (1/s) \otimes m$ is a well-defined inverse of the induced map: if $t(s'm - sm') = 0$ then $(1/s) \otimes m = (1/tss') \otimes ts'm = (1/tss') \otimes tsm' = (1/s') \otimes m'$. So $S^{-1}R \otimes -$ is exact, that is, $S^{-1}R$ is flat. (3) If $\mathfrak{q}$ is a prime of $S^{-1}R$, its preimage $\mathfrak{p}$ is prime and disjoint from S , and $\mathfrak{q} = S^{-1}\mathfrak{p}$ since $r/s \in \mathfrak{q}$ iff $r/1 \in \mathfrak{q}$. Conversely, for $\mathfrak{p}$ disjoint from S , $S^{-1}\mathfrak{p}$ is a proper ideal, it is prime because $(r/s)(r'/s') \in S^{-1}\mathfrak{p}$ gives $trr' \in \mathfrak{p}$ for some $t \in S$, and its preimage is $\mathfrak{p}$ by the same argument. For $R_{\mathfrak{p}}$, the primes correspond to primes contained in $\mathfrak{p}$. (4) If the nonunits form an ideal, it is the unique maximal ideal. If R is local with maximal ideal $\mathfrak{m}$, every nonunit lies in a maximal ideal, hence in $\mathfrak{m}$. $\square$

Proposition 12.3.3. For an R -module M the following are equivalent: $M = 0$; $M_{\mathfrak{p}} = 0$ for all primes $\mathfrak{p}$; $M_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$. A homomorphism $M \rightarrow N$ is injective (surjective) if and only if all $M_{\mathfrak{m}} \rightarrow N_{\mathfrak{m}}$ are.

Proof. If $x \in M$ is nonzero, its annihilator is a proper ideal, contained in a maximal ideal $\mathfrak{m}$, and then $x/1 \neq 0$ in $M_{\mathfrak{m}}$. The second statement follows by applying the first to kernels and cokernels, which commute with localisation by Proposition 12.3.2(2). $\square$

Example 12.3.4. $\mathbb{Z}_{(p)}$ is the ring of rationals whose denominators are prime to p , a local ring with maximal ideal $p\mathbb{Z}_{(p)}$; $\mathbb{Z}_{(0)} = \mathbb{Q}$; and the localisation of $\mathbb{Z}$ at $f = p$ is $\mathbb{Z}[1/p]$, the ring of rationals whose denominators are powers of p (we avoid the notation $\mathbb{Z}_p$ for it, which is reserved for the p -adic integers). Localisation can kill elements: in $(\mathbb{Z}/6\mathbb{Z})_{(2)}$ the element 3 is a unit and $2 \cdot 3 = 0$, so $2/1 = 0$ and $(\mathbb{Z}/6\mathbb{Z})_{(2)} \cong \mathbb{Z}/2\mathbb{Z}$. If f is nilpotent, $R_f = 0$.

12.4 Nakayama's lemma

Lemma 12.4.1 (Determinant trick). *Let M be an R -module generated by n elements, I an ideal and $\varphi: M \rightarrow M$ an R -linear map with $\varphi(M) \subseteq IM$. Then $\varphi^n + a_1\varphi^{n-1} + \cdots + a_n = 0$ for some $a_i \in I$.*

Proof. Let $x_1, \dots, x_n$ generate M and write $\varphi(x_i) = \sum_j a_{ij}x_j$ with $a_{ij} \in I$. View M as a module over the commutative ring $R[\varphi]$ of endomorphisms. The matrix $T = \varphi I - (a_{ij})$ over $R[\varphi]$ kills the column vector (x_j) , so by Proposition 9.5.3(4) $\det(T)x_j = \operatorname{adj}(T)T(x) = 0$ for all j , that is, $\det T = 0$ on M . Expanding the determinant gives the claim. $\square$

Theorem 12.4.2 (Nakayama's lemma). *Let M be a finitely generated R -module and I an ideal with $IM = M$. Then there is $a \in 1 + I$ with $aM = 0$. If $I \subseteq J(R)$, then $M = 0$.*

Proof. Apply Lemma 12.4.1 to $\varphi = \operatorname{id}$: $a = 1 + a_1 + \cdots + a_n$ kills M . If $I \subseteq J(R)$, a is a unit by Proposition 12.2.2(3). $\square$

Corollary 12.4.3. *Let $(R, \mathfrak{m}, k)$ be a local ring and M a finitely generated module. Elements $x_1, \dots, x_n \in M$ generate M if and only if their images generate the k -vector space $M/\mathfrak{m}M$. Every minimal generating set has $\dim_k M/\mathfrak{m}M$ elements.*

Proof. Let N be the submodule generated by the x_i . If their images generate $M/\mathfrak{m}M$ then $M = N + \mathfrak{m}M$, so $\mathfrak{m}(M/N) = M/N$ and $M/N = 0$ by Theorem 12.4.2. The last statement follows because a generating set of M maps to a spanning set of $M/\mathfrak{m}M$, and a minimal one to a basis. $\square$

12.5 Noetherian and Artinian rings

Definition 12.5.1. A module is *Noetherian* if every increasing chain of submodules is eventually constant, and *Artinian* if every decreasing chain is. A ring is Noetherian (Artinian) if it is so as a module over itself.

Proposition 12.5.2. 1. *M is Noetherian if and only if every submodule is finitely generated, if and only if every nonempty set of submodules has a maximal element.*

2. *If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, then M is Noetherian (Artinian) if and only if M' and M'' are.*

3. *Over a Noetherian ring, finitely generated modules are Noetherian; quotients and localisations of Noetherian rings are Noetherian.*

Proof. (1) If N is not finitely generated, choose $x_1, x_2, \dots \in N$ with $x_{n+1} \notin (x_1, \dots, x_n)$, a strictly increasing chain (using dependent choice). If every submodule is finitely generated, the union of a chain is generated by finitely many elements, which lie in one member. A nonempty set without maximal element contains a strictly increasing chain, and conversely a strictly increasing chain has no maximal element. (2) A chain in M gives chains in M' and M'' ; if both stabilise, so does the original, because $N_1 \subseteq N_2$ with $N_1 \cap M' = N_2 \cap M'$ and equal images in M'' implies $N_1 = N_2$. The converse is clear. (3) R^n is Noetherian by (2) and induction, and finitely generated modules are its quotients. Ideals of $S^{-1}R$ are of the form $S^{-1}I$ (by the argument of Proposition 12.3.2(3)), so they are finitely generated. $\square$

Theorem 12.5.3 (Hilbert's basis theorem). *If R is Noetherian, so is $R[x]$. Consequently every finitely generated R -algebra is Noetherian; in particular $k[x_1, \dots, x_n]$ and $\mathbb{Z}[x_1, \dots, x_n]$ are Noetherian.*

Proof. Let $I \subseteq R[x]$ be an ideal and $L_d \subseteq R$ the set of leading coefficients of elements of I of degree d , together with 0. These are ideals with $L_d \subseteq L_{d+1}$ (multiply by x), so $L_d = L_{d_0}$ for $d \geq d_0$. For each $d \leq d_0$ choose finitely many $f_{d,j} \in I$ of degree d whose leading coefficients generate L_d . Let I' be the ideal they generate. If $f \in I$ has degree d , some combination $g = \sum_j c_j x^{d-\min(d,d_0)} f_{\min(d,d_0),j}$ has the same degree and leading coefficient, so $f - g \in I$ has smaller degree; by induction on the degree $f \in I'$. So $I = I'$. The last statement follows from quotients (Proposition 12.5.2). $\square$

Counterexample 12.5.4. The polynomial ring $k[x_1, x_2, \dots]$ in infinitely many variables is not Noetherian: $(x_1) \subsetneq (x_1, x_2) \subsetneq \dots$. It is a subring of its fraction field, which is Noetherian; so subrings of Noetherian rings need not be Noetherian.

Lemma 12.5.5. *Let R be a ring and $\mathfrak{m}_1, \dots, \mathfrak{m}_n$ maximal ideals (not necessarily distinct) with $\mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Then R is Noetherian if and only if it is Artinian.*

Proof. Consider the chain $R \supseteq \mathfrak{m}_1 \supseteq \mathfrak{m}_1 \mathfrak{m}_2 \supseteq \dots \supseteq \mathfrak{m}_1 \cdots \mathfrak{m}_n = 0$. Each quotient is a vector space over the field $R/\mathfrak{m}_i$, and for a vector space the ascending and the descending chain conditions are both equivalent to finite dimension. By Proposition 12.5.2(2) applied along the chain, R satisfies ACC iff all quotients are finite-dimensional iff R satisfies DCC. $\square$

Theorem 12.5.6. *An Artinian ring R is Noetherian, every prime ideal of R is maximal, R has finitely many maximal ideals, its nilradical is nilpotent, and R is a finite product of local Artinian rings. (For the converse see Corollary 12.8.5.)*

Proof. Let R be Artinian. If $\mathfrak{p}$ is prime, $R/\mathfrak{p}$ is an Artinian domain; for $x \neq 0$ the chain $(x) \supseteq (x^2) \supseteq \dots$ stabilises, $x^n = x^{n+1}y$, so $xy = 1$; hence $\mathfrak{p}$ is maximal. Among finite intersections of maximal ideals choose a minimal one, $\mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n$; every maximal $\mathfrak{m}$ contains it, hence contains $\mathfrak{m}_1 \cdots \mathfrak{m}_n$ and so some $\mathfrak{m}_i$, and $\mathfrak{m} = \mathfrak{m}_i$. The nilradical $N = \mathfrak{m}_1 \cap \dots \cap \mathfrak{m}_n$ satisfies $N^k = N^{k+1} = \dots =: I$ for some k . If $I \neq 0$, choose an ideal J minimal with $IJ \neq 0$; there is $x \in J$ with $xI \neq 0$, so $J = (x)$ by minimality, and $(xI)I = xI \neq 0$ with $xI \subseteq (x)$ gives $xI = (x)$, so $x = xy$ with $y \in I$, hence $x = xy^m = 0$ as y is nilpotent; contradiction. So N is nilpotent, $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k \subseteq N^k = 0$, and R is Noetherian by Lemma 12.5.5.

Finally, the ideals $\mathfrak{m}_i^k$ are pairwise comaximal and their product is 0, so $R \cong \prod_i R/\mathfrak{m}_i^k$ by Theorem 7.3.3, and $R/\mathfrak{m}_i^k$ is local with maximal ideal $\mathfrak{m}_i/\mathfrak{m}_i^k$. $\square$

12.6 Integral extensions

Definition 12.6.1. Let $A \subseteq B$ be rings. An element $b \in B$ is *integral over A* if it is a root of a monic polynomial with coefficients in A . B is *integral over A* if all its elements are, and *finite over A* if it is a finitely generated A -module. The *integral closure* of A in B is the set of elements integral over A . A domain is *normal* (or *integrally closed*) if it equals its integral closure in its fraction field.

Proposition 12.6.2. 1. *b is integral over A if and only if $A[b]$ is a finite A -module, if and only if b lies in a subring $C \subseteq B$ that is a finite A -module.*

2. *The integral closure of A in B is a subring. If $A \subseteq B \subseteq C$ with C integral over B and B integral over A , then C is integral over A .*

3. *If $A \subseteq B$ are domains and B is integral over A , then B is a field if and only if A is.*

4. *A unique factorisation domain is normal.*

Proof. (1) A monic equation of degree n shows that $1, b, \dots, b^{n-1}$ generate $A[b]$. If $b \in C$ with C finite, apply Lemma 12.4.1 to multiplication by b on C with $I = A$; the resulting polynomial kills $1 \in C$, so it vanishes at b . (2) If b, b' are integral, $A[b, b']$ is finite over A , so $b \pm b'$ and bb' are integral by (1). If $c \in C$ satisfies a monic equation with coefficients $b_1, \dots, b_n \in B$, then $A[b_1, \dots, b_n, c]$ is finite over A by the tower of finite extensions, so c is integral over A . (3) If A is a field and $0 \neq b \in B$ satisfies $b^n + a_1 b^{n-1} + \dots + a_n = 0$ with n minimal, then $a_n \neq 0$ (as B is a domain), and $b^{-1} = -a_n^{-1}(b^{n-1} + \dots + a_{n-1}) \in B$. If B is a field and $0 \neq a \in A$, then $a^{-1} \in B$ is integral over A : $a^{-m} + c_1 a^{-m+1} + \dots + c_m = 0$, so $a^{-1} = -(c_1 + c_2 a + \dots + c_m a^{m-1}) \in A$. (4) If $x = r/s$ in lowest terms satisfies a monic equation of degree n , then $r^n = -s(a_1 r^{n-1} + \dots + a_n s^{n-1})$, so every prime factor of s divides r ; hence s is a unit. $\square$

Theorem 12.6.3 (Lying over and going up). *Let $A \subseteq B$ be an integral extension and $\mathfrak{p}$ a prime of A .*

1. *There is a prime $\mathfrak{q}$ of B with $\mathfrak{q} \cap A = \mathfrak{p}$.*
2. *If $\mathfrak{q} \subseteq \mathfrak{q}'$ are primes of B with $\mathfrak{q} \cap A = \mathfrak{q}' \cap A$, then $\mathfrak{q} = \mathfrak{q}'$ (incomparability). A prime $\mathfrak{q}$ is maximal if and only if $\mathfrak{q} \cap A$ is maximal.*
3. *If $\mathfrak{p} \subseteq \mathfrak{p}'$ are primes of A and $\mathfrak{q} \cap A = \mathfrak{p}$, there is $\mathfrak{q}' \supseteq \mathfrak{q}$ with $\mathfrak{q}' \cap A = \mathfrak{p}'$ (going up).*

Proof. For a prime $\mathfrak{q}$ of B over $\mathfrak{p}$, the extension $A/\mathfrak{p} \subseteq B/\mathfrak{q}$ is integral, so by Proposition 12.6.2(3) $\mathfrak{q}$ is maximal iff $\mathfrak{p}$ is. This gives the second part of (2).

(1) Let $S = A \setminus \mathfrak{p}$. Then $A_{\mathfrak{p}} \subseteq S^{-1}B$ is integral and $S^{-1}B \neq 0$. Let $\mathfrak{n}$ be a maximal ideal of $S^{-1}B$. Then $\mathfrak{n} \cap A_{\mathfrak{p}}$ is maximal by the first remark, hence equals $\mathfrak{p}A_{\mathfrak{p}}$, and the preimage $\mathfrak{q}$ of $\mathfrak{n}$ in B satisfies $\mathfrak{q} \cap A = \mathfrak{p}$. (2) Localising at $S = A \setminus \mathfrak{p}$ as in (1), $S^{-1}\mathfrak{q} \subseteq S^{-1}\mathfrak{q}'$ both lie over the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$, hence are maximal, hence equal, and so $\mathfrak{q} = \mathfrak{q}'$. (3) Apply (1) to the integral extension $A/\mathfrak{p} \subseteq B/\mathfrak{q}$ and the prime $\mathfrak{p}'/\mathfrak{p}$. $\square$

Remark 12.6.4. If moreover A is a normal domain and B is a domain, the *going down* theorem holds: given $\mathfrak{p}' \subseteq \mathfrak{p}$ and $\mathfrak{q}$ over $\mathfrak{p}$, there is $\mathfrak{q}' \subseteq \mathfrak{q}$ over $\mathfrak{p}'$ [AM69, Theorem 5.16].

Theorem 12.6.5 (Noether normalisation). *Let A be a finitely generated algebra over a field k . There are elements $y_1, \dots, y_d \in A$, algebraically independent over k , such that A is finite over $k[y_1, \dots, y_d]$.*

Proof. By induction on the number n of generators $x_1, \dots, x_n$. If they are algebraically independent, take $y_i = x_i$. Otherwise let $F \neq 0$ be a polynomial with $F(x_1, \dots, x_n) = 0$, and let r be larger than all exponents occurring in F . Put $z_i = x_i - x_n^{r_i}$ for $i < n$. Substituting $x_i = z_i + x_n^{r_i}$, a monomial $c x_1^{a_1} \dots x_n^{a_n}$ of F becomes $c x_n^{a_n + a_1 r + \dots + a_{n-1} r^{n-1}}$ plus terms of lower degree in x_n with coefficients in $k[z_1, \dots, z_{n-1}]$. The exponents $a_n + a_1 r + \dots$ of distinct monomials are distinct (base r expansions), so exactly one monomial produces the highest power of x_n . Dividing by its coefficient, x_n satisfies a monic equation over $k[z_1, \dots, z_{n-1}]$. So $A = k[z_1, \dots, z_{n-1}][x_n]$ is finite over $k[z_1, \dots, z_{n-1}]$, which by induction is finite over a polynomial ring $k[y_1, \dots, y_d]$; conclude with Proposition 12.6.2(2). $\square$

12.7 The Nullstellensatz

Theorem 12.7.1 (Zariski's lemma). *Let k be a field and L a field that is finitely generated as a k -algebra. Then L is a finite extension of k . Consequently, for every finitely generated k -algebra A and maximal ideal $\mathfrak{m}$, $A/\mathfrak{m}$ is finite over k .*

Proof. By Theorem 12.6.5, L is finite over some $k[y_1, \dots, y_d]$, which is then a field by Proposition 12.6.2(3). A polynomial ring in $d \geq 1$ variables is not a field, so $d = 0$. $\square$

Theorem 12.7.2 (Hilbert's Nullstellensatz). *Let k be algebraically closed. For $I \subseteq k[x_1, \dots, x_n]$ let $Z(I) = \{a \in k^n : f(a) = 0 \text{ for all } f \in I\}$.*

1. *The maximal ideals of $k[x_1, \dots, x_n]$ are the ideals $(x_1 - a_1, \dots, x_n - a_n)$, $a \in k^n$.*
2. *If $I \neq (1)$ then $Z(I) \neq \emptyset$.*
3. *If f vanishes on $Z(I)$, then $f \in \sqrt{I}$.*

Proof. (1) The ideal $(x_1 - a_1, \dots, x_n - a_n)$ is the kernel of evaluation at a , which is surjective onto k , so it is maximal. If $\mathfrak{m}$ is maximal, $k[x]/\mathfrak{m}$ is a finite extension of k by Theorem 12.7.1, hence equals k ; so $x_i \equiv a_i$ modulo $\mathfrak{m}$ for some $a_i \in k$, and $\mathfrak{m}$ contains, hence equals, $(x - a)$. (2) A proper ideal lies in a maximal ideal $(x - a)$, and then $a \in Z(I)$. (3) (Rabinowitsch) We may assume $f \neq 0$. In $k[x_1, \dots, x_n, y]$ the ideal $J = Ik[x, y] + (1 - yf)$ has no zeros, since at a zero (a, b) we would have $a \in Z(I)$, so $f(a) = 0$ and $1 - bf(a) = 1$. By (2), $1 = \sum_i g_i h_i + g(1 - yf)$ with $h_i \in I$. Substituting $y = 1/f$ in $k(x_1, \dots, x_n)$ and multiplying by a high power f^N gives $f^N = \sum_i g'_i h_i \in I$. $\square$

Counterexample 12.7.3. Algebraic closedness is needed: over $\mathbb{R}$, the ideal $(x^2 + 1) \subsetneq \mathbb{R}[x]$ is maximal but has no zeros in $\mathbb{R}$, and $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$ is a finite extension of $\mathbb{R}$ different from $\mathbb{R}$.

Corollary 12.7.4. *In a finitely generated algebra A over a field, the nilradical equals the Jacobson radical.*

Proof. Let $f \in A$ be non-nilpotent. Then $A_f \neq 0$ has a maximal ideal $\mathfrak{n}$, and $A_f \cong A[t]/(tf - 1)$ is finitely generated, so $A_f/\mathfrak{n}$ is a finite field extension of k . The preimage $\mathfrak{m}$ of $\mathfrak{n}$ in A does not contain f , and $k \subseteq A/\mathfrak{m} \subseteq A_f/\mathfrak{n}$, so $A/\mathfrak{m}$ is a domain finite-dimensional over k , hence a field (Proposition 12.6.2(3)). So $f \notin J(A)$. $\square$

12.8 Primary decomposition

In this section R is Noetherian.

Definition 12.8.1. Let M be an R -module. A prime $\mathfrak{p}$ is *associated* to M if $\mathfrak{p} = \text{Ann}(x)$ for some $x \in M$; we write $\text{Ass}(M)$. An ideal $\mathfrak{q} \neq R$ is *primary* if $xy \in \mathfrak{q}$ and $x \notin \mathfrak{q}$ imply $y^n \in \mathfrak{q}$ for some n . Then $\mathfrak{p} = \sqrt{\mathfrak{q}}$ is prime, and $\mathfrak{q}$ is called $\mathfrak{p}$ -primary.

Proposition 12.8.2. *Let M be a finitely generated R -module.*

1. *Every maximal element of $\{\text{Ann}(x) : 0 \neq x \in M\}$ is a prime ideal; so $M \neq 0$ implies $\text{Ass}(M) \neq \emptyset$.*
2. *The set of zero divisors on M is $\bigcup_{\mathfrak{p} \in \text{Ass}(M)} \mathfrak{p}$.*
3. *There is a chain $0 = M_0 \subseteq \dots \subseteq M_n = M$ with $M_i/M_{i-1} \cong R/\mathfrak{p}_i$ for primes $\mathfrak{p}_i$, and $\text{Ass}(M) \subseteq \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$; in particular $\text{Ass}(M)$ is finite.*

Proof. (1) Let $I = \text{Ann}(x)$ be maximal and $ab \in I$ with $b \notin I$. Then $bx \neq 0$ and $\text{Ann}(bx) \supseteq I + (a)$, so $a \in I$ by maximality. Maximal elements exist by Proposition 12.5.2(1). (2) Elements of associated primes are clearly zero divisors. If $ax = 0$ with $x \neq 0$, choose $z \in Rx$, $z \neq 0$, with $\text{Ann}(z)$ maximal among the annihilators of nonzero elements of Rx . The argument of (1), applied inside Rx , shows that $\text{Ann}(z)$ is prime, and it contains $\text{Ann}(x) \ni a$. (3) By (1), choose $M_1 = Rx_1 \cong R/\mathfrak{p}_1$, then apply (1) to M/M_1 , and so on; the chain stops since M is Noetherian. If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact then $\text{Ass}(M) \subseteq \text{Ass}(M') \cup \text{Ass}(M'')$: if $\text{Ann}(x) = \mathfrak{p}$ and $Rx \cap M' \neq 0$, a nonzero $yx \in M'$ has annihilator $\mathfrak{p}$ (as $R/\mathfrak{p}$ is a domain); otherwise Rx maps isomorphically into M'' . Finally $\text{Ass}(R/\mathfrak{p}) = \{\mathfrak{p}\}$ since $R/\mathfrak{p}$ is a domain. $\square$

Theorem 12.8.3 (Lasker–Noether). *Every ideal $I \neq R$ of a Noetherian ring is a finite intersection $I = q_1 \cap \cdots \cap q_r$ of primary ideals. One can choose the decomposition with the primes $\sqrt{q_i}$ distinct and no q_i superfluous; the set of these primes is then $\text{Ass}(R/I)$, independent of the decomposition. The minimal elements of $\text{Ass}(R/I)$ are the minimal primes containing I .*

Proof. Call I *irreducible* if $I = J \cap J'$ implies $I = J$ or $I = J'$. If some ideal is not a finite intersection of irreducible ideals, a maximal such ideal is reducible, a contradiction; so every ideal is a finite intersection of irreducibles. An irreducible ideal I is primary: passing to R/I , suppose $xy = 0$ with $x \neq 0$. The chain $\text{Ann}(y) \subseteq \text{Ann}(y^2) \subseteq \cdots$ stabilises at n ; then $(x) \cap (y^n) = 0$ (if $ax = by^n$ then $axy = 0 = by^{n+1}$, so $by^n = 0$), and irreducibility of 0 gives $y^n = 0$.

If q is $\mathfrak{p}$ -primary, then $\text{Ass}(R/q) = \{\mathfrak{p}\}$: if $\text{Ann}(\bar{x}) = \mathfrak{p}'$ with $x \notin q$, then $\mathfrak{p}'x \subseteq q$ gives $\mathfrak{p}' \subseteq \mathfrak{p}$, and every $a \in \mathfrak{p}$ has $a^n \bar{x} = 0$, so $a \in \mathfrak{p}'$. A finite intersection of $\mathfrak{p}$ -primary ideals is $\mathfrak{p}$ -primary, so we may merge components with the same prime, and delete superfluous ones. Then $R/I \hookrightarrow \bigoplus_i R/q_i$ gives $\text{Ass}(R/I) \subseteq \{\mathfrak{p}_i\}$, by the argument in Proposition 12.8.2(3). Conversely, if q_i is not superfluous, choose $x \in \bigcap_{j \neq i} q_j \setminus q_i$; then $\bar{x} \in R/I$ generates a submodule isomorphic to a submodule of R/q_i , whose associated primes are $\{\mathfrak{p}_i\}$, so $\mathfrak{p}_i \in \text{Ass}(R/I)$. For the last statement, a prime contains $I = \bigcap q_i$ iff it contains some q_i iff it contains some $\mathfrak{p}_i$. $\square$

Example 12.8.4. In $k[x, y]$, $(x^2, xy) = (x) \cap (x^2, y)$, where (x) is prime and (x^2, y) is (x, y) -primary. The associated primes are (x) and the *embedded* prime $(x, y) \supsetneq (x)$. Geometrically, the ideal defines the line $x = 0$ with an extra infinitesimal structure at the origin.

Corollary 12.8.5. *A Noetherian ring in which every prime ideal is maximal is Artinian.*

Proof. By Theorem 12.8.3 applied to $I = 0$, the ring has finitely many minimal primes; since all primes are maximal, these are all the primes, say $\mathfrak{m}_1, \dots, \mathfrak{m}_n$. Their intersection is the nilradical (Proposition 12.2.2), which is finitely generated, hence nilpotent: $(\mathfrak{m}_1 \cap \cdots \cap \mathfrak{m}_n)^k = 0$. Then $(\mathfrak{m}_1 \cdots \mathfrak{m}_n)^k = 0$, and the ring is Artinian by Lemma 12.5.5. $\square$

12.9 Dimension theory

Definition 12.9.1. The *Krull dimension* $\dim R$ is the supremum of the lengths n of chains of primes $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n$. The *height* of a prime is $\text{ht } \mathfrak{p} = \dim R_{\mathfrak{p}}$, the supremum of lengths of chains ending at $\mathfrak{p}$.

Proposition 12.9.2. *If B is integral over A , then $\dim B = \dim A$.*

Proof. A chain of primes of B contracts to a chain in A with strict inclusions by incomparability, and a chain in A lifts by lying over and going up (Theorem 12.6.3). $\square$

Theorem 12.9.3. *Let A be a finitely generated k -algebra that is a domain. Then $\dim A = \text{trdeg}_k \text{Frac}(A)$. In particular $\dim k[x_1, \dots, x_n] = n$.*

Proof. Let $t = \text{trdeg}_k \text{Frac}(A)$. By Theorem 12.6.5, A is finite over a polynomial ring $k[y_1, \dots, y_d]$; then $\text{Frac}(A)$ is algebraic over $k(y_1, \dots, y_d)$, so $d = t$, and $\dim A = \dim k[y_1, \dots, y_t]$ by Proposition 12.9.2. The chain $0 \subsetneq (y_1) \subsetneq (y_1, y_2) \subsetneq \cdots \subsetneq (y_1, \dots, y_t)$ consists of primes (the quotients are polynomial rings), so $\dim \geq t$. We show $\dim A \leq t$ for all such A by induction on t ; by the above we may assume $A = k[y_1, \dots, y_t]$. Let $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$ be a chain. The domain $A/\mathfrak{p}_1$ is generated by the images of the y_i , which satisfy a nontrivial polynomial relation (any nonzero element of $\mathfrak{p}_1$), so its fraction field has transcendence degree $< t$ (Theorem 8.10.3). By induction the chain $\mathfrak{p}_1/\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m/\mathfrak{p}_1$ has length $m - 1 \leq t - 1$. $\square$

Lemma 12.9.4. *A Noetherian local ring $(R, \mathfrak{m})$ in which $\mathfrak{m}$ is the only prime is Artinian, and $\mathfrak{m}^n = 0$ for some n .*

Proof. The nilradical is $\mathfrak{m}$ (Proposition 12.2.2), which is finitely generated, so $\mathfrak{m}^n = 0$; apply Lemma 12.5.5. $\square$

Theorem 12.9.5 (Krull's principal ideal theorem). *Let R be Noetherian, $x \in R$, and $\mathfrak{p}$ a prime minimal among primes containing x . Then $\text{ht } \mathfrak{p} \leq 1$. More generally, if $\mathfrak{p}$ is minimal over $(x_1, \dots, x_r)$, then $\text{ht } \mathfrak{p} \leq r$.*

Proof. Localising at $\mathfrak{p}$ we may assume $(R, \mathfrak{p})$ local with $\mathfrak{p}$ minimal over (x) . Let $\mathfrak{q} \subsetneq \mathfrak{p}$ be prime; we show $\text{ht } \mathfrak{q} = 0$. For $n \geq 1$ let $\mathfrak{q}^{(n)} = \mathfrak{q}^n R_{\mathfrak{q}} \cap R$. The ring $R/(x)$ has $\mathfrak{p}/(x)$ as its only prime, so it is Artinian (Lemma 12.9.4), and the chain $\mathfrak{q}^{(n)} + (x)$ stabilises: for large n , $\mathfrak{q}^{(n)} \subseteq \mathfrak{q}^{(n+1)} + (x)$. For $a \in \mathfrak{q}^{(n)}$ write $a = b + rx$ with $b \in \mathfrak{q}^{(n+1)}$; then $rx \in \mathfrak{q}^{(n)}$, and since $x \notin \mathfrak{q}$ (by minimality of $\mathfrak{p}$) and x is a unit in $R_{\mathfrak{q}}$, $r \in \mathfrak{q}^{(n)}$. Hence $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)} + x\mathfrak{q}^{(n)}$, and Nakayama's lemma gives $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$. Localising, $\mathfrak{q}^n R_{\mathfrak{q}} = \mathfrak{q}^{n+1} R_{\mathfrak{q}}$, and Nakayama in $R_{\mathfrak{q}}$ gives $\mathfrak{q}^n R_{\mathfrak{q}} = 0$. So every prime of $R_{\mathfrak{q}}$, which contains $0 = (\mathfrak{q} R_{\mathfrak{q}})^n$, equals $\mathfrak{q} R_{\mathfrak{q}}$, and $\text{ht } \mathfrak{q} = 0$.

For the general statement, argue by induction on r , again with $(R, \mathfrak{p})$ local. Let $\mathfrak{q} \subsetneq \mathfrak{p}$ be a prime such that there is no prime strictly between them; it suffices to show $\text{ht } \mathfrak{q} \leq r - 1$. As $\mathfrak{p}$ is minimal over $(x_1, \dots, x_r)$, some x_i , say x_r , is not in $\mathfrak{q}$. Then $\mathfrak{p}$ is minimal over $\mathfrak{q} + (x_r)$, so $R/(\mathfrak{q} + (x_r))$ is Artinian and $x_i^N \in \mathfrak{q} + (x_r)$ for $i < r$: write $x_i^N = y_i + a_i x_r$ with $y_i \in \mathfrak{q}$. Every prime containing $(y_1, \dots, y_{r-1}, x_r)$ contains all x_i , so $\mathfrak{p}$ is minimal over $(y_1, \dots, y_{r-1}, x_r)$, hence $\mathfrak{p}/I$ is minimal over $(\bar{x}_r)$ in R/I , $I = (y_1, \dots, y_{r-1})$. By the case $r = 1$, $\text{ht}(\mathfrak{p}/I) \leq 1$, so $\mathfrak{q}/I$ is a minimal prime of R/I , that is, $\mathfrak{q}$ is minimal over I , and by induction $\text{ht } \mathfrak{q} \leq r - 1$. $\square$

Corollary 12.9.6. *A Noetherian local ring $(R, \mathfrak{m}, k)$ has $\dim R \leq \dim_k \mathfrak{m}/\mathfrak{m}^2 < \infty$.*

Proof. By Corollary 12.4.3, $\mathfrak{m}$ is generated by $\dim_k \mathfrak{m}/\mathfrak{m}^2$ elements, and $\text{ht } \mathfrak{m} = \dim R$. $\square$

12.10 Completions

Lemma 12.10.1 (Artin–Rees). *Let R be Noetherian, I an ideal, M a finitely generated module and $N \subseteq M$ a submodule. There is c such that $I^n M \cap N = I^{n-c}(I^c M \cap N)$ for all $n \geq c$.*

Proof. The Rees algebra $R^* = \bigoplus_n I^n$ is a finitely generated R -algebra (generated by generators of I in degree 1), hence Noetherian, and $M^* = \bigoplus_n I^n M$ is a finitely generated graded R^* -module. Its graded submodule $N^* = \bigoplus_n (I^n M \cap N)$ is therefore finitely generated, by homogeneous elements of degrees at most some c . For $n \geq c$, this means $I^n M \cap N = \sum_{j \leq c} I^{n-j}(I^j M \cap N) \subseteq I^{n-c}(I^c M \cap N)$, and the reverse inclusion is clear. $\square$

Theorem 12.10.2 (Krull's intersection theorem). *Let R be Noetherian, I an ideal and M finitely generated, and $N = \bigcap_n I^n M$. Then $IN = N$, so $aN = 0$ for some $a \in 1 + I$. If $I \subseteq J(R)$, for instance if R is local and $I \neq R$, then $\bigcap_n I^n M = 0$.*

Proof. By Lemma 12.10.1, $N = I^{c+1} M \cap N = I(I^c M \cap N) = IN$. Apply Theorem 12.4.2. $\square$

Definition 12.10.3. The I -adic completion of a module M is $\hat{M} = \varprojlim_n M/I^n M$, the inverse limit (Example 5.8.3) of the quotients. For $M = R$ it is a ring $\hat{R}$, and there is a natural map $M \rightarrow \hat{M}$, injective for R Noetherian local, $I = \mathfrak{m}$ and M finitely generated, by Theorem 12.10.2.

Example 12.10.4. The completion of $k[x_1, \dots, x_n]$ at $(x_1, \dots, x_n)$ is the power series ring $k[[x_1, \dots, x_n]]$, and the completion of $\mathbb{Z}$ at (p) is the ring $\mathbb{Z}_p$ of p -adic integers. For R Noetherian, $\hat{R}$ is Noetherian and flat over R , and completion is exact on finitely generated modules [AM69, Chapter 10].

12.11 Regular local rings and Dedekind domains

Definition 12.11.1. A Noetherian local ring $(R, \mathfrak{m}, k)$ is *regular* if $\dim R = \dim_k \mathfrak{m}/\mathfrak{m}^2$. A *discrete valuation ring* (DVR) is a domain R with a surjective map $v: \text{Frac}(R)^\times \rightarrow \mathbb{Z}$ such that $v(xy) = v(x) + v(y)$, $v(x + y) \geq \min(v(x), v(y))$, and $R = \{0\} \cup \{x : v(x) \geq 0\}$. A *Dedekind domain* is a Noetherian normal domain of dimension at most 1.

Theorem 12.11.2. For a Noetherian local domain $(R, \mathfrak{m}, k)$ of dimension 1, the following are equivalent: R is a DVR; R is normal; $\mathfrak{m}$ is principal; R is regular.

Proof. DVR $\Rightarrow$ normal: if $x \notin R$ satisfies a monic equation $x^n + a_1x^{n-1} + \cdots + a_n = 0$, then $v(x) < 0$ and $v(x^n)$ is smaller than the valuation of every other term, a contradiction.

Normal $\Rightarrow \mathfrak{m}$ principal: the only primes are 0 and $\mathfrak{m}$, so for $0 \neq a \in \mathfrak{m}$, $R/(a)$ is Artinian and $\mathfrak{m}^n \subseteq (a)$, $\mathfrak{m}^{n-1} \not\subseteq (a)$ for some $n \geq 1$. Choose $b \in \mathfrak{m}^{n-1} \setminus (a)$ and let $x = a/b \in \text{Frac}(R)$. Then $x^{-1} \notin R$ and $x^{-1}\mathfrak{m} \subseteq R$ (since $b\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (a)$). If $x^{-1}\mathfrak{m} \subseteq \mathfrak{m}$, apply Lemma 12.4.1 to multiplication by x^{-1} on the finitely generated module $\mathfrak{m}$ (with $I = R$): the resulting monic polynomial in x^{-1} kills the nonzero elements of $\mathfrak{m} \subseteq \text{Frac}(R)$, hence vanishes at x^{-1} , so x^{-1} is integral over R , contradicting normality. So $x^{-1}\mathfrak{m}$ is an ideal not contained in $\mathfrak{m}$, that is, $x^{-1}\mathfrak{m} = R$, and $\mathfrak{m} = xR$.

$\mathfrak{m}$ principal $\Rightarrow$ DVR: let $\mathfrak{m} = (t)$. By Krull's intersection theorem (Theorem 12.10.2 below), $\bigcap_n \mathfrak{m}^n = 0$, so every $0 \neq r \in R$ is $r = ut^n$ with u a unit and n unique; put $v(ut^n/ut'^n) = n - n'$.

$\mathfrak{m}$ principal $\Leftrightarrow$ regular: $\dim_k \mathfrak{m}/\mathfrak{m}^2 \geq \dim R = 1$ by Corollary 12.9.6, and it equals 1 iff $\mathfrak{m}$ is principal by Corollary 12.4.3. $\square$

Remark 12.11.3. Every regular local ring is a domain, and in fact a UFD [Mat86, Theorems 14.3 and 20.3]. The local rings $k[x_1, \dots, x_n]_{(x_1, \dots, x_n)}$ are regular of dimension n : the images of the x_i form a basis of $\mathfrak{m}/\mathfrak{m}^2$, and the dimension is at least n because of the chain $0 \subsetneq (x_1) \subsetneq \cdots$, and at most n by Corollary 12.9.6. Regularity is the algebraic counterpart of smoothness (Chapter 49).

Theorem 12.11.4. In a Dedekind domain R every nonzero ideal I is uniquely a product $\mathfrak{p}_1^{n_1} \cdots \mathfrak{p}_r^{n_r}$ of distinct nonzero primes.

Proof. If $I = R$ take the empty product. Otherwise let $I = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_r$ be a primary decomposition with distinct primes $\mathfrak{p}_i = \sqrt{\mathfrak{q}_i}$, which are nonzero, hence maximal. The $\mathfrak{q}_i$ are pairwise comaximal (their radicals are distinct maximal ideals), so $I = \mathfrak{q}_1 \cdots \mathfrak{q}_r$ by Theorem 7.3.3. Each $R_{\mathfrak{p}_i}$ is a Noetherian local domain of dimension 1, and it is normal: if $x \in \text{Frac}(R)$ satisfies a monic equation with coefficients $a_j/s_j \in R_{\mathfrak{p}_i}$, then with $s = \prod_j s_j$ the element sx satisfies a monic equation over R , so $sx \in R$ and $x \in R_{\mathfrak{p}_i}$. By Theorem 12.11.2 it is a DVR, so $\mathfrak{q}_i R_{\mathfrak{p}_i} = \mathfrak{p}_i^{n_i} R_{\mathfrak{p}_i}$ for some n_i . Both $\mathfrak{q}_i$ and $\mathfrak{p}_i^{n_i}$ are $\mathfrak{p}_i$ -primary (an ideal whose radical is a maximal ideal $\mathfrak{m}$ is $\mathfrak{m}$ -primary, because in the quotient every element outside $\mathfrak{m}$ is a unit), and a $\mathfrak{p}$ -primary ideal $\mathfrak{q}$ is the preimage of $\mathfrak{q}R_{\mathfrak{p}}$ (if $sx \in \mathfrak{q}$ with $s \notin \mathfrak{p}$ then $x \in \mathfrak{q}$); so $\mathfrak{q}_i = \mathfrak{p}_i^{n_i}$. Uniqueness: localising a factorisation at $\mathfrak{p}$ gives $IR_{\mathfrak{p}} = \mathfrak{p}^n R_{\mathfrak{p}}$ with n the exponent of $\mathfrak{p}$, which determines n as $v_{\mathfrak{p}}$ of a generator. $\square$

Example 12.11.5. The ring $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain (it is the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{-5})$, see Exercise 12.13.2). Although $6 = 2 \cdot 3 = (1 + \sqrt{-5})(1 - \sqrt{-5})$ has two factorisations into irreducibles (Counterexample 7.5.6), the ideal (6) factors uniquely into primes: $(2) = \mathfrak{p}^2$ with $\mathfrak{p} = (2, 1 + \sqrt{-5})$, and $(3) = \mathfrak{r}\bar{\mathfrak{r}}$ with $\mathfrak{r} = (3, 1 + \sqrt{-5})$, $\bar{\mathfrak{r}} = (3, 1 - \sqrt{-5})$, while $(1 \pm \sqrt{-5}) = \mathfrak{p}\bar{\mathfrak{r}}$.

12.12 Kähler differentials

Definition 12.12.1. Let $A \rightarrow B$ be a ring homomorphism and M a B -module. An A -derivation $D: B \rightarrow M$ is an A -linear map with $D(bb') = bD(b') + b'D(b)$. The *module of Kähler differentials* $\Omega_{B/A}$ is the free B -module on symbols db , $b \in B$, modulo the relations $d(b + b') - db - db'$, $d(bb') - b db' - b' db$ and da for a in the image of A ; the map $d: B \rightarrow \Omega_{B/A}$ is an A -derivation.

Proposition 12.12.2. 1. $\text{Hom}_B(\Omega_{B/A}, M) \cong \text{Der}_A(B, M)$, $\varphi \mapsto \varphi \circ d$, naturally in M .

2. $\Omega_{A[x_1, \dots, x_n]/A}$ is free with basis $dx_1, \dots, dx_n$, and $df = \sum_i (\partial f / \partial x_i) dx_i$.

3. For $A \rightarrow B \rightarrow C$ there is an exact sequence $C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow \Omega_{C/B} \rightarrow 0$.

4. If $C = B/I$, there is an exact sequence $I/I^2 \xrightarrow{\delta} C \otimes_B \Omega_{B/A} \rightarrow \Omega_{C/A} \rightarrow 0$ with $\delta(\bar{f}) = 1 \otimes df$.

5. $S^{-1}\Omega_{B/A} \cong \Omega_{S^{-1}B/A}$ for a multiplicative $S \subseteq B$.

Proof. (1) is the universal property of the construction. (2) The formal partial derivatives define a derivation $A[x] \rightarrow \bigoplus A[x]e_i$, $f \mapsto \sum_i (\partial f / \partial x_i)e_i$; a derivation is determined by its values on the x_i , which can be arbitrary. By (1), Ω is free on the dx_i . (3) and (4): a sequence $X \rightarrow Y \rightarrow Z \rightarrow 0$ of C -modules is exact if and only if $0 \rightarrow \text{Hom}_C(Z, M) \rightarrow \text{Hom}_C(Y, M) \rightarrow \text{Hom}_C(X, M)$ is exact for every C -module M (for the nontrivial direction take $M = Z/\text{im}(Y)$ and $M = \text{coker}(X \rightarrow Y)$). By (1), $\text{Hom}_C(C \otimes_B \Omega_{B/A}, M) = \text{Der}_A(B, M)$ and $\text{Hom}_C(I/I^2, M) = \text{Hom}_B(I, M)$, so it suffices to show that for every C -module M the sequences $0 \rightarrow \text{Der}_B(C, M) \rightarrow \text{Der}_A(C, M) \rightarrow \text{Der}_A(B, M)$ and $0 \rightarrow \text{Der}_A(C, M) \rightarrow \text{Der}_A(B, M) \rightarrow \text{Hom}_B(I, M)$ are exact, which is immediate: a derivation of C over A vanishing on B is a B -derivation; and an A -derivation of B into a C -module that vanishes on I factors through $C = B/I$ (it automatically vanishes on I^2). (5) Derivations $S^{-1}B \rightarrow M$ correspond to derivations $B \rightarrow M$ into $S^{-1}B$ -modules by the quotient rule $D(b/s) = (sDb - bDs)/s^2$. $\square$

Example 12.12.3. For $B = k[x, y]/(y^2 - x^3)$, Proposition 12.12.2(4) gives $\Omega_{B/k} = (B dx \oplus B dy)/(2y dy - 3x^2 dx)$. At the origin, $\Omega_{B/k} \otimes k$ has dimension 2 although B has dimension 1; this detects the singularity of the cusp. For a finite separable field extension L/K one has $\Omega_{L/K} = 0$: if $L = K(\gamma)$ with minimal polynomial f , then $0 = d(f(\gamma)) = f'(\gamma) d\gamma$ and $f'(\gamma) \neq 0$ (Theorem 8.6.3 and Proposition 8.4.2).

12.13 Exercises

Exercise 12.13.1. Describe $\text{Spec } \mathbb{Z}$, $\text{Spec } \mathbb{Z}[x]$ (in terms of prime numbers and irreducible polynomials) and $\text{Spec } k[x, y]$ for k algebraically closed.

Exercise 12.13.2. Let $d \neq 0, 1$ be a squarefree integer with $d \equiv 2, 3 \pmod{4}$. Show that the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(\sqrt{d})$ is $\mathbb{Z}[\sqrt{d}]$, and deduce that $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain.

Exercise 12.13.3. Let $(R, \mathfrak{m})$ be a local ring and M a finitely generated R -module that is a direct summand of a free module. Show that M is free.

Exercise 12.13.4. Let $K = \mathbb{F}_p(t)$ and $L = K(t^{1/p})$. Show that $\Omega_{L/K}$ is one-dimensional over L .

Exercise 12.13.5. (Hensel's lemma.) Let $f \in \mathbb{Z}_p[x]$ and $a \in \mathbb{Z}_p$ with $f(a) \equiv 0 \pmod{p}$ and $f'(a) \not\equiv 0 \pmod{p}$. Show that there is a unique $b \in \mathbb{Z}_p$ with $f(b) = 0$ and $b \equiv a \pmod{p}$.

Exercise 12.13.6. Let k be a field and $f \in k[x_1, \dots, x_n]$ irreducible. Show that $\dim k[x_1, \dots, x_n]/(f) = n - 1$.

Chapter 13

Homological Algebra

13.1 Introduction

Homological algebra measures the failure of exactness. Its basic objects are complexes and their cohomology; its basic tools are the long exact sequence and derived functors. Everything in this project that is called a cohomology theory (singular cohomology in Chapter 20, sheaf cohomology in Chapter 24, de Rham and Dolbeault cohomology in Chapters 35 and 43) is built with the machinery of this chapter.

We work in abelian categories, but prove the diagram lemmas for modules; by the Freyd–Mitchell embedding theorem they then hold in every abelian category (Remark 13.2.5). References are [Wei94], [CE56] and [GM03].

Prerequisites. Chapter 5 (Categories) and Chapter 7 (Rings and Modules).

13.2 Abelian categories

Definition 13.2.1. An additive category $\mathcal{A}$ (Definition 5.12.1) is *abelian* if every morphism has a kernel and a cokernel (Definition 5.12.3), every monomorphism is a kernel of its cokernel, and every epimorphism is a cokernel of its kernel. For $f: X \rightarrow Y$ the *image* is $\text{im } f = \ker(\text{coker } f)$. A sequence $X \xrightarrow{f} Y \xrightarrow{g} Z$ with $gf = 0$ is *exact* at Y if the induced morphism $\text{im } f \rightarrow \ker g$ is an isomorphism. An additive functor between abelian categories is *left exact* if it sends exact sequences $0 \rightarrow X \rightarrow Y \rightarrow Z$ to exact sequences, *right exact* if it sends exact $X \rightarrow Y \rightarrow Z \rightarrow 0$ to exact sequences, and *exact* if both.

Example 13.2.2. For a ring R , the category $R\text{-Mod}$ is abelian (Proposition 7.7.3): kernels and cokernels are the usual ones, an injective map is the kernel of the projection onto its cokernel, and a surjective map is the cokernel of the inclusion of its kernel. In particular $\mathbf{Ab}$ is abelian. The category of sheaves of abelian groups on a topological space is abelian (Chapter 23). By Theorem 7.9.4, $-\otimes_R N$ is right exact, and $\text{Hom}_R(N, -)$ is left exact.

Lemma 13.2.3. *In an abelian category a morphism that is both a monomorphism and an epimorphism is an isomorphism.*

Proof. Let $f: X \rightarrow Y$ be both. Its cokernel is $Y \rightarrow 0$, since f is an epimorphism; as f is a monomorphism, it is a kernel of $Y \rightarrow 0$. But id_Y is also a kernel of $Y \rightarrow 0$, so f is an isomorphism by uniqueness of kernels (Lemma 5.8.2). $\square$

Counterexample 13.2.4. The category $\mathcal{F}$ of finitely generated free abelian groups is additive and has kernels (Proposition 6.11.4), but it is not abelian. The morphism $2: \mathbb{Z} \rightarrow \mathbb{Z}$ is a

monomorphism, and it is an epimorphism in $\mathcal{F}$: if $g \circ 2 = 0$ for $g: \mathbb{Z} \rightarrow \mathbb{Z}^n$, then $2g(1) = 0$, so $g = 0$. It is not an isomorphism, contradicting Lemma 13.2.3.

Remark 13.2.5. By the Freyd–Mitchell embedding theorem, every small abelian category admits a fully faithful exact functor into $R\text{-Mod}$ for some ring R [Wei94, Theorem 1.6.1]. Every statement about a finite diagram in an abelian category $\mathcal{A}$ that asserts exactness, or the existence of morphisms constructed from kernels, cokernels and compositions, involves only a small abelian subcategory of $\mathcal{A}$; hence it suffices to prove it for modules, where one may argue with elements (“diagram chasing”). We use this principle for Lemmas 13.4.1 and 13.4.2, and wherever we prove a statement about abelian categories by a chase.

13.3 Complexes

Definition 13.3.1. A *complex* (or *cochain complex*) $A^\bullet$ in an abelian category $\mathcal{A}$ is a family of objects A^n , $n \in \mathbb{Z}$, with morphisms $d^n: A^n \rightarrow A^{n+1}$ such that $d^{n+1}d^n = 0$. Its *cocycles*, *coboundaries* and *cohomology* are $Z^n = \ker d^n$, $B^n = \text{im } d^{n-1}$ and $H^n(A^\bullet) = Z^n/B^n$. A *morphism of complexes* $f: A^\bullet \rightarrow B^\bullet$ is a family $f^n: A^n \rightarrow B^n$ with $df = f d$; it induces $H^n(f)$. It is a *quasi-isomorphism* if all $H^n(f)$ are isomorphisms. A complex is *acyclic* (or *exact*) if all H^n vanish, and *bounded below* if $A^n = 0$ for $n \ll 0$. The complexes form an abelian category $\text{Ch}(\mathcal{A})$, with kernels and cokernels computed degreewise.

Notation 13.3.2. A *chain complex* $C_\bullet$ has differentials $d_n: C_n \rightarrow C_{n-1}$ and *homology* $H_n = \ker d_n / \text{im } d_{n+1}$. Setting $C^n = C_{-n}$ turns it into a complex, with $H^n = H_{-n}$; all statements about complexes apply to chain complexes in this way. For a complex $A^\bullet$ and $k \in \mathbb{Z}$, the *shift* $A[k]$ has $A[k]^n = A^{n+k}$ and differential $(-1)^k d$.

Definition 13.3.3. Two morphisms $f, g: A^\bullet \rightarrow B^\bullet$ are *homotopic*, $f \simeq g$, if there are morphisms $h^n: A^n \rightarrow B^{n-1}$ with $f^n - g^n = d^{n-1}h^n + h^{n+1}d^n$. A morphism f is a *homotopy equivalence* if there is g with $gf \simeq \text{id}$ and $fg \simeq \text{id}$.

Lemma 13.3.4. *Homotopic morphisms induce the same maps on cohomology; homotopy equivalences are quasi-isomorphisms. Additive functors preserve homotopies.*

Proof. For a cocycle z , $f(z) - g(z) = dh(z) + hd(z) = dh(z)$ is a coboundary. The second statement follows, and the third because the defining identity is preserved by additive functors. $\square$

Definition 13.3.5. The *mapping cone* of $f: A^\bullet \rightarrow B^\bullet$ is the complex $\text{Cone}(f)$ with $\text{Cone}(f)^n = A^{n+1} \oplus B^n$ and $d(a, b) = (-da, f(a) + db)$.

13.4 The long exact sequence

Lemma 13.4.1 (Snake lemma). *Consider a commutative diagram of modules with exact rows*

$$\begin{array}{ccccccc} A' & \xrightarrow{i'} & B' & \xrightarrow{p'} & C' & \longrightarrow & 0 \\ & \downarrow f & \downarrow g & & \downarrow h & & \\ 0 & \longrightarrow & A & \xrightarrow{i} & B & \xrightarrow{p} & C \end{array}$$

There is a homomorphism $\delta: \ker h \rightarrow \text{coker } f$ such that the sequence

$$\ker f \rightarrow \ker g \rightarrow \ker h \xrightarrow{\delta} \text{coker } f \rightarrow \text{coker } g \rightarrow \text{coker } h$$

is exact. If i' is injective so is $\ker f \rightarrow \ker g$, and if p is surjective so is $\text{coker } g \rightarrow \text{coker } h$.

Proof. Definition of δ . Let $c' \in \ker h$. Choose $b' \in B'$ with $p'(b') = c'$. Then $p(g(b')) = h(c') = 0$, so $g(b') = i(a)$ for a unique $a \in A$; put $\delta(c') = [a]$. If b' is replaced by $b' + i'(a')$, then a is replaced by $a + f(a')$, with the same class; so δ is well defined, and it is clearly a homomorphism.

Exactness at $\ker g$. The composite is zero. If $b' \in \ker g$ and $p'(b') = 0$, then $b' = i'(a')$ and $i(f(a')) = g(b') = 0$, so $f(a') = 0$.

Exactness at $\ker h$. If $c' = p'(b')$ with $g(b') = 0$, then $a = 0$ and $\delta(c') = 0$. Conversely if $\delta(c') = 0$, then $a = f(a'')$, and $b' - i'(a'')$ is a preimage of c' with $g(b' - i'(a'')) = i(a) - i(f(a'')) = 0$.

Exactness at $\text{coker } f$. The image of $\delta(c')$ in $\text{coker } g$ is the class of $i(a) = g(b')$, which is zero. Conversely, if $i(a) = g(b')$ for some b' , then $c' = p'(b')$ satisfies $h(c') = p(g(b')) = pi(a) = 0$ and $\delta(c') = [a]$.

Exactness at $\text{coker } g$. The composite is zero. If $b \in B$ maps to zero in $\text{coker } h$, then $p(b) = h(p'(b'))$ for some b' , so $p(b - g(b')) = 0$ and $b - g(b') = i(a)$; hence $[b]$ is the image of $[a]$.

The last two statements are clear. $\square$

Lemma 13.4.2 (Five lemma). *Consider a commutative diagram of modules with exact rows $A_1 \rightarrow A_2 \rightarrow A_3 \rightarrow A_4 \rightarrow A_5$ and $B_1 \rightarrow B_2 \rightarrow B_3 \rightarrow B_4 \rightarrow B_5$ and vertical maps $f_1, \dots, f_5$. If f_2 and f_4 are isomorphisms, f_1 is surjective and f_5 is injective, then f_3 is an isomorphism.*

Proof. Write $\alpha_i: A_i \rightarrow A_{i+1}$ and $\beta_i: B_i \rightarrow B_{i+1}$. *Injectivity:* let $f_3(x) = 0$. Then $f_4\alpha_3(x) = 0$, so $\alpha_3(x) = 0$ and $x = \alpha_2(y)$. Now $\beta_2f_2(y) = 0$, so $f_2(y) = \beta_1(z)$ and $z = f_1(w)$. Then $f_2(y - \alpha_1(w)) = 0$, so $y = \alpha_1(w)$ and $x = \alpha_2\alpha_1(w) = 0$. *Surjectivity:* let $u \in B_3$. Choose $x \in A_4$ with $f_4(x) = \beta_3(u)$. Then $f_5\alpha_4(x) = \beta_4\beta_3(u) = 0$, so $\alpha_4(x) = 0$ and $x = \alpha_3(y)$. Then $\beta_3(u - f_3(y)) = 0$, so $u - f_3(y) = \beta_2(v)$ with $v = f_2(t)$, and $u = f_3(y + \alpha_2(t))$. $\square$

Theorem 13.4.3 (Long exact sequence). *A short exact sequence $0 \rightarrow A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \rightarrow 0$ of complexes in an abelian category induces a long exact sequence*

$$\cdots \rightarrow H^n(A) \rightarrow H^n(B) \rightarrow H^n(C) \xrightarrow{\delta} H^{n+1}(A) \rightarrow H^{n+1}(B) \rightarrow \cdots,$$

natural in the short exact sequence. For $c \in Z^n(C)$ (in the case of modules), $\delta[c]$ is the class of a where $f(a) = d(b)$ and $g(b) = c$.

Proof. Applying the snake lemma to the vertical maps d^{n-1} between the short exact sequences in degrees $n-1$ and n , the cokernel sequence $A^n/B^n(A) \rightarrow B^n/B^n(B) \rightarrow C^n/B^n(C) \rightarrow 0$ is exact; applying it to d^{n+1} , the kernel sequence $0 \rightarrow Z^{n+1}(A) \rightarrow Z^{n+1}(B) \rightarrow Z^{n+1}(C)$ is exact. The differential d^n induces vertical maps from the first of these rows to the second, whose kernels are the H^n and whose cokernels are the H^{n+1} . Apply Lemma 13.4.1. Naturality follows from the explicit formula for δ . $\square$

Proposition 13.4.4. *For $f: A^\bullet \rightarrow B^\bullet$ there is a short exact sequence $0 \rightarrow B^\bullet \rightarrow \text{Cone}(f) \rightarrow A[1]^\bullet \rightarrow 0$ whose connecting map $H^n(A[1]) = H^{n+1}(A) \rightarrow H^{n+1}(B)$ is $H^{n+1}(f)$. In particular f is a quasi-isomorphism if and only if $\text{Cone}(f)$ is acyclic.*

Proof. The maps $b \mapsto (0, b)$ and $(a, b) \mapsto a$ are morphisms of complexes (with the sign convention of Notation 13.3.2 for $A[1]$). For a cocycle $a \in A^{n+1}$, the element $(a, 0)$ lifts it, and $d(a, 0) = (-da, f(a)) = (0, f(a))$; so $\delta[a] = [f(a)]$. The last statement follows from the long exact sequence. $\square$

13.5 Projective and injective objects

Definition 13.5.1. An object P of an abelian category is *projective* if $\text{Hom}(P, -)$ is exact, that is, every morphism $P \rightarrow Y$ lifts along every epimorphism $X \rightarrow Y$. An object I is *injective* if $\text{Hom}(-, I)$ is exact, that is, every morphism $X \rightarrow I$ extends along every monomorphism $X \rightarrow Y$. The category has *enough projectives (injectives)* if every object is a quotient of a projective (a subobject of an injective).

Proposition 13.5.2. *An R -module is projective if and only if it is a direct summand of a free module. $R\text{-Mod}$ has enough projectives.*

Proof. Free modules are projective by Proposition 7.7.5 (choose lifts of the images of a basis), and direct summands of projectives are projective. If P is projective, choose a surjection $F \rightarrow P$ from a free module; the identity of P lifts to a section, so P is a direct summand of F by Proposition 7.7.7. Every module is a quotient of a free module. $\square$

Proposition 13.5.3 (Baer's criterion). *An R -module I is injective if and only if every homomorphism $J \rightarrow I$ from a left ideal $J \subseteq R$ extends to R .*

Proof. Necessity is clear. Let $N \subseteq M$ and $\varphi: N \rightarrow I$. By Zorn's lemma choose a maximal extension $\psi: N' \rightarrow I$ with $N \subseteq N' \subseteq M$. If $m \in M \setminus N'$, let $J = \{r : rm \in N'\}$ and extend $r \mapsto \psi(rm)$ to $\theta: R \rightarrow I$. Then $n' + rm \mapsto \psi(n') + \theta(r)$ is a well-defined extension of ψ to $N' + Rm$ (if $n' + rm = 0$ then $r \in J$ and $\theta(r) = \psi(rm) = -\psi(n')$), contradicting maximality. $\square$

Theorem 13.5.4. *$R\text{-Mod}$ has enough injectives. An abelian group D is injective if and only if it is divisible: for every $x \in D$ and $n \geq 1$ there is y with $ny = x$.*

Proof. By Baer's criterion and Proposition 6.2.10, D is injective iff every homomorphism $n\mathbb{Z} \rightarrow D$ extends to $\mathbb{Z}$, iff divisible. Every abelian group M embeds in a divisible group: write $M = F/K$ with $F = \mathbb{Z}^{(S)}$ free; then $M \subseteq \mathbb{Q}^{(S)}/K$, which is divisible.

Let M be an R -module and $M \subseteq D$ an embedding of abelian groups with D divisible. Make $\text{Hom}_{\mathbb{Z}}(R, D)$ an R -module by $(r\varphi)(s) = \varphi(sr)$. For every R -module N there is a natural bijection $\text{Hom}_R(N, \text{Hom}_{\mathbb{Z}}(R, D)) \cong \text{Hom}_{\mathbb{Z}}(N, D)$, $\psi \mapsto (n \mapsto \psi(n)(1))$, with inverse $\chi \mapsto (n \mapsto (s \mapsto \chi(sn)))$. The right side is exact in N since D is injective, so $\text{Hom}_{\mathbb{Z}}(R, D)$ is an injective R -module. The map $M \rightarrow \text{Hom}_{\mathbb{Z}}(R, D)$, $m \mapsto (s \mapsto sm)$, is R -linear and injective. $\square$

13.6 Derived functors

In this section $\mathcal{A}$ and $\mathcal{B}$ are abelian categories.

Definition 13.6.1. An *injective resolution* of an object X is an exact sequence $0 \rightarrow X \rightarrow I^0 \rightarrow I^1 \rightarrow \cdots$ with all I^n injective; we regard $I^\bullet$ as a complex with a quasi-isomorphism $X \rightarrow I^\bullet$ (X in degree 0). Projective resolutions $\cdots \rightarrow P_1 \rightarrow P_0 \rightarrow X \rightarrow 0$ are defined dually. If $\mathcal{A}$ has enough injectives, every object has an injective resolution: embed X in I^0 , embed I^0/X in I^1 , and so on.

Theorem 13.6.2 (Comparison theorem). *Let $0 \rightarrow X \rightarrow E^\bullet$ be exact with $E^n = 0$ for $n < 0$, and let $0 \rightarrow Y \rightarrow I^\bullet$ be a complex with all I^n injective, $I^n = 0$ for $n < 0$. Every morphism $\varphi: X \rightarrow Y$ extends to a morphism of complexes $E^\bullet \rightarrow I^\bullet$, and any two extensions are homotopic.*

Proof. Write $\varepsilon: X \rightarrow E^0$ and $\eta: Y \rightarrow I^0$. The morphism $\eta\varphi$ extends along the monomorphism ε to $f^0: E^0 \rightarrow I^0$. Suppose $f^0, \dots, f^n$ are constructed commuting with the differentials (and with ε, η). Then df^n vanishes on the image of $E^{n-1} \rightarrow E^n$ (or of X when $n = 0$), so it factors through $E^n/\text{im}(E^{n-1}) \hookrightarrow E^{n+1}$, and extends along this monomorphism to $f^{n+1}: E^{n+1} \rightarrow I^{n+1}$ by injectivity.

If f, g both extend φ , let $u = f - g$, which extends 0, so $u^0\varepsilon = 0$. Put $h^0 = 0: E^0 \rightarrow I^{-1} = 0$. Suppose $h^0, \dots, h^n$ are constructed with $u^k = dh^k + h^{k+1}d$ for all $k < n$. For $n \geq 1$,

$$(u^n - dh^n)d = u^n d - dh^n d = du^{n-1} - d(u^{n-1} - dh^{n-1}) = ddh^{n-1} = 0,$$

using that u commutes with the differentials and the identity for $k = n - 1$; for $n = 0$, $(u^0 - dh^0)\varepsilon = u^0\varepsilon = 0$. So $u^n - dh^n$ vanishes on the image of $E^{n-1} \rightarrow E^n$ (of $X \rightarrow E^0$ if $n = 0$), factors through the monomorphism $E^n/\text{im} \hookrightarrow E^{n+1}$, and extends to $h^{n+1}: E^{n+1} \rightarrow I^n$ with $h^{n+1}d = u^n - dh^n$. $\square$

Lemma 13.6.3 (Horseshoe lemma). *Let $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ be exact, and let $X' \rightarrow I'^\bullet$, $X'' \rightarrow I''^\bullet$ be injective resolutions. There is an injective resolution $X \rightarrow I^\bullet$ with $I^n = I'^n \oplus I''^n$ such that $0 \rightarrow I'^\bullet \rightarrow I^\bullet \rightarrow I''^\bullet \rightarrow 0$ (with the inclusion and projection in each degree) is a short exact sequence of complexes compatible with the given sequence.*

Proof. Let $\varepsilon': X' \rightarrow I'^0$, $\varepsilon'': X'' \rightarrow I''^0$, and $\iota: X' \rightarrow X$, $\pi: X \rightarrow X''$. Extend ε' along ι to $\lambda: X \rightarrow I'^0$ and define $\varepsilon = (\lambda, \varepsilon''\pi): X \rightarrow I'^0 \oplus I''^0$. The diagram $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ over $0 \rightarrow I'^0 \rightarrow I^0 \rightarrow I''^0 \rightarrow 0$ commutes, and the snake lemma shows that ε is a monomorphism and that $0 \rightarrow \text{coker } \varepsilon' \rightarrow \text{coker } \varepsilon \rightarrow \text{coker } \varepsilon'' \rightarrow 0$ is exact. Apply the same construction to this sequence and the resolutions $I'^{\geq 1}, I''^{\geq 1}$ of the cokernels, and continue by induction. $\square$

Definition 13.6.4. Let $\mathcal{A}$ have enough injectives and $F: \mathcal{A} \rightarrow \mathcal{B}$ be a left exact functor. For each X choose an injective resolution $X \rightarrow I_X^\bullet$ and define the *right derived functors* $R^n F(X) = H^n(F(I_X^\bullet))$. A morphism $X \rightarrow Y$ induces a morphism of resolutions, unique up to homotopy (Theorem 13.6.2), hence a well-defined morphism $R^n F(X) \rightarrow R^n F(Y)$ by Lemma 13.3.4. Left derived functors $L_n F$ of a right exact functor on a category with enough projectives are defined dually using projective resolutions.

Theorem 13.6.5. 1. *The $R^n F$ are additive functors, independent of the choice of resolutions up to canonical isomorphism, with $R^n F = 0$ for $n < 0$ and $R^0 F \cong F$.*

2. *$R^n F(I) = 0$ for $n > 0$ and I injective.*

3. *Every short exact sequence $0 \rightarrow X' \rightarrow X \rightarrow X'' \rightarrow 0$ induces a long exact sequence $0 \rightarrow F(X') \rightarrow F(X) \rightarrow F(X'') \rightarrow R^1 F(X') \rightarrow R^1 F(X) \rightarrow \dots$, natural in the short exact sequence.*

Proof. (1) Two resolutions of X are related by morphisms extending id_X in both directions, whose composites are homotopic to the identities by the uniqueness part of Theorem 13.6.2; applying F gives homotopy equivalences and canonical isomorphisms on cohomology. Functoriality and additivity follow from uniqueness up to homotopy. $R^0 F(X) = \ker(F(I^0) \rightarrow F(I^1)) = F(\ker(I^0 \rightarrow I^1)) = F(X)$ by left exactness. (2) Use the resolution $0 \rightarrow I \rightarrow I \rightarrow 0$. (3) Choose resolutions of X' and X'' and form the resolution of X given by Lemma 13.6.3. The sequence $0 \rightarrow I'^\bullet \rightarrow I^\bullet \rightarrow I''^\bullet \rightarrow 0$ is split in every degree, so it stays exact after applying the additive functor F . Apply Theorem 13.4.3. Naturality follows from the comparison theorem. $\square$

Proposition 13.6.6. *Call J F -acyclic if $R^n F(J) = 0$ for all $n > 0$. If $0 \rightarrow X \rightarrow J^0 \rightarrow J^1 \rightarrow \dots$ is exact with all J^n F -acyclic, then $R^n F(X) \cong H^n(F(J^\bullet))$ for all n .*

Proof. Let $K = \text{coker}(X \rightarrow J^0) = \ker(J^1 \rightarrow J^2)$, so that $0 \rightarrow X \rightarrow J^0 \rightarrow K \rightarrow 0$ is exact and $K \rightarrow J^{\geq 1}$ is an F -acyclic resolution. The long exact sequence and acyclicity of J^0 give $R^n F(X) \cong R^{n-1} F(K)$ for $n \geq 2$, and $R^1 F(X) \cong \text{coker}(F(J^0) \rightarrow F(K))$. By left exactness $F(K) = \ker(F(J^1) \rightarrow F(J^2))$, so $R^1 F(X) \cong H^1(F(J^\bullet))$. For $n \geq 2$ argue by induction on n , for all objects and resolutions simultaneously: $K \rightarrow J^1 \rightarrow J^2 \rightarrow \dots$ is an F -acyclic resolution of K , with J^1 in degree 0, so $R^n F(X) \cong R^{n-1} F(K) \cong H^{n-1}(F(J^1 \rightarrow F(J^2) \rightarrow \dots)) = H^n(F(J^\bullet))$. The case $n = 0$ is left exactness. $\square$

13.7 Double complexes

Definition 13.7.1. A double complex $C^{\bullet, \bullet}$ consists of objects $C^{p,q}$ with morphisms $d_h: C^{p,q} \rightarrow C^{p+1,q}$ and $d_v: C^{p,q} \rightarrow C^{p,q+1}$ such that $d_h^2 = 0$, $d_v^2 = 0$ and $d_h d_v = d_v d_h$. It is *first quadrant* if $C^{p,q} = 0$ unless $p, q \geq 0$. Its *total complex* is $\text{Tot}^n = \bigoplus_{p+q=n} C^{p,q}$ with $d = d_h + (-1)^p d_v$ on $C^{p,q}$; then $d^2 = 0$.

Lemma 13.7.2. *Let C be a first-quadrant double complex of modules whose rows are exact in positive degrees: for each q , the complex $C^{0,q} \rightarrow C^{1,q} \rightarrow \dots$ is exact at $C^{p,q}$ for all $p \geq 1$. Let $K^q = \ker(d_h: C^{0,q} \rightarrow C^{1,q})$, a complex with differential d_v . Then the inclusion $K^\bullet \rightarrow \text{Tot}^\bullet(C)$ is a quasi-isomorphism.*

Proof. The inclusion is a morphism of complexes, since $d = d_v$ on K . *Surjectivity on cohomology.* Let $x = (x^{0,n}, x^{1,n-1}, \dots, x^{n,0})$ be a cocycle of degree n . Let p be largest with $x^{p,n-p} \neq 0$. If $p \geq 1$, the component of dx in $C^{p+1,n-p}$ is $d_h x^{p,n-p} = 0$ (the other contribution comes from $x^{p+1,n-p-1} = 0$), so $x^{p,n-p} = d_h y$ with $y \in C^{p-1,n-p}$, and $x - dy$ is a cohomologous cocycle whose components vanish in positions $\geq p$. Repeating, we reach a cocycle concentrated in $C^{0,n}$; the vanishing of its $C^{1,n}$ -component of d means it lies in K^n , and it is a d_v -cocycle.

Injectivity on cohomology. Let $k \in K^n$ with $k = dy$ for $y \in \text{Tot}^{n-1}$. As above, if $p \geq 1$ is largest with $y^{p,n-1-p} \neq 0$, the component of dy in $C^{p+1,n-1-p}$ is $d_h y^{p,n-1-p}$, which is zero because $dy = k$ lives in $C^{0,n}$ (here $p+1 \geq 2 > 0$); so $y^{p,n-1-p} = d_h z$ and we replace y by $y - dz$, which does not change dy . We reach $y \in C^{0,n-1}$ with $d_h y$ equal to the $C^{1,n-1}$ -component of k , which is 0; so $y \in K^{n-1}$ and $k = d_v y$. $\square$

13.8 Ext and Tor

In this section R is a ring and modules are left R -modules; for Tor we assume R commutative.

Definition 13.8.1. $\text{Ext}_R^n(M, N) = R^n \text{Hom}_R(M, -)(N)$, computed with an injective resolution of N , and $\text{Tor}_n^R(M, N) = L_n(M \otimes_R -)(N)$, computed with a projective resolution of N .

Theorem 13.8.2 (Balance). *If $P_\bullet \rightarrow M$ is a projective resolution, then $\text{Ext}_R^n(M, N) \cong H^n(\text{Hom}_R(P_\bullet, N))$. Similarly $\text{Tor}_n^R(M, N) \cong H_n(P_\bullet \otimes_R N)$, and $\text{Tor}_n^R(M, N) \cong \text{Tor}_n^R(N, M)$. Consequently $\text{Ext}^n(-, N)$ are the right derived functors of the contravariant left exact functor $\text{Hom}_R(-, N)$, and there are long exact sequences in both variables.*

Proof. Let $N \rightarrow I^\bullet$ be an injective resolution and $C^{p,q} = \text{Hom}_R(P_p, I^q)$, a first-quadrant double complex. For fixed q the row $\text{Hom}_R(P_\bullet, I^q)$ is exact in positive degrees with kernel $\text{Hom}_R(M, I^q)$ in degree 0, because $\text{Hom}_R(-, I^q)$ is exact and $P_\bullet \rightarrow M \rightarrow 0$ is exact. For fixed p the column $\text{Hom}_R(P_p, I^\bullet)$ is exact in positive degrees with kernel $\text{Hom}_R(P_p, N)$, because

P_p is projective. By Lemma 13.7.2, applied to C and to its transpose, both $\text{Hom}_R(M, I^\bullet)$ and $\text{Hom}_R(P_\bullet, N)$ are quasi-isomorphic to $\text{Tot}(C)$. The same argument with $P_p \otimes Q_q$ for projective resolutions P of M and Q of N (projective modules are flat, being summands of free modules) gives the statements for Tor , and symmetry follows from $P_p \otimes Q_q \cong Q_q \otimes P_p$. The long exact sequence in the first variable follows from the horseshoe lemma for projective resolutions and Theorem 13.4.3. $\square$

Example 13.8.3. Using the projective resolution $0 \rightarrow \mathbb{Z} \xrightarrow{n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$, for every abelian group A : $\text{Hom}(\mathbb{Z}/n\mathbb{Z}, A) = A[n] = \{a : na = 0\}$, $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, A) = A/nA$, $\mathbb{Z}/n\mathbb{Z} \otimes A = A/nA$ and $\text{Tor}_{\mathbb{Z}}^2(\mathbb{Z}/n\mathbb{Z}, A) = A[n]$; all higher groups vanish. For instance $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/n\mathbb{Z}$, although $\text{Hom}(\mathbb{Z}/n\mathbb{Z}, \mathbb{Z}) = 0$.

Proposition 13.8.4. *A module N is flat if and only if $\text{Tor}_1^R(M, N) = 0$ for all M ; then $\text{Tor}_n^R(M, N) = 0$ for all $n \geq 1$. If R is Noetherian local with residue field k and N is finitely generated, then N is flat if and only if it is free, if and only if $\text{Tor}_1^R(k, N) = 0$.*

Proof. If $0 \rightarrow M' \rightarrow M \rightarrow M'' \rightarrow 0$ is exact, the long exact sequence $\text{Tor}_1(M'', N) \rightarrow M' \otimes N \rightarrow M \otimes N$ shows that vanishing of Tor_1 gives injectivity; conversely if N is flat, computing Tor with a projective resolution of M tensored with N (Theorem 13.8.2) gives an exact complex. For the second statement, free implies flat. Suppose $\text{Tor}_1(k, N) = 0$. Choose a minimal set of generators, giving $0 \rightarrow K \rightarrow R^n \rightarrow N \rightarrow 0$ with $n = \dim_k N/\mathfrak{m}N$ (Corollary 12.4.3). Tensoring with k gives the exact sequence $0 \rightarrow K \otimes k \rightarrow k^n \rightarrow N \otimes k \rightarrow 0$, in which the last map is an isomorphism of n -dimensional spaces; so $K/\mathfrak{m}K = 0$, and $K = 0$ by Nakayama's lemma, as K is finitely generated (R is Noetherian). $\square$

Proposition 13.8.5. *If R is a principal ideal domain, every submodule of a free R -module is free. Hence every R -module has a projective resolution of length at most 1, and $\text{Ext}_R^n = \text{Tor}_n^R = 0$ for $n \geq 2$.*

Proof. Let F be free with basis $(e_i)_{i \in I}$, where I is well-ordered (Theorem 3.7.2), and $M \subseteq F$. For $i \in I$ let $F_{\leq i}$ and $F_{< i}$ be spanned by the e_j with $j \leq i$, respectively $j < i$, let $M_i = M \cap F_{\leq i}$, and let $\pi_i: M_i \rightarrow R$ be the e_i -coordinate. Its image is an ideal (a_i) ; if $a_i \neq 0$ choose $m_i \in M_i$ with $\pi_i(m_i) = a_i$. The m_i are linearly independent: in a relation $\sum c_i m_i = 0$, the largest i with $c_i \neq 0$ gives $c_i a_i = 0$ on the e_i -coordinate. They generate M : otherwise let i be least such that M_i contains an element m not in their span. Then $\pi_i(m) = r a_i$, and $m - r m_i$ (or m itself if $a_i = 0$) lies in $M \cap F_{< i}$; having finite support, it lies in some M_j with $j < i$, hence in the span, and so does m . For the last statement, write M as a quotient of a free module; the kernel is free. $\square$

Proposition 13.8.6. *Let M, N be R -modules. There is a bijection between $\text{Ext}_R^1(M, N)$ and the set of equivalence classes of short exact sequences $0 \rightarrow N \rightarrow E \rightarrow M \rightarrow 0$, where two sequences are equivalent if there is an isomorphism of the middle terms compatible with the given maps; the zero element corresponds to the split sequence.*

Proof. Choose $0 \rightarrow K \xrightarrow{j} P \rightarrow M \rightarrow 0$ with P projective. By Theorem 13.8.2 and the long exact sequence, $\text{Ext}_R^1(M, N) = \text{coker}(\text{Hom}(P, N) \rightarrow \text{Hom}(K, N))$. Given an extension $0 \rightarrow N \rightarrow E \xrightarrow{q} M \rightarrow 0$, lift $P \rightarrow M$ to $\beta: P \rightarrow E$; then βj lands in N , and its class $\theta(E)$ does not depend on β (two lifts differ by a map $P \rightarrow N$). Conversely, given $\alpha: K \rightarrow N$, let E_α be the pushout of $N \xleftarrow{\alpha} K \xrightarrow{j} P$, that is, $(N \oplus P)/\{(\alpha(k), -j(k)) : k \in K\}$; it sits in an exact sequence $0 \rightarrow N \rightarrow E_\alpha \rightarrow M \rightarrow 0$, and α and $\alpha + \gamma j$ ($\gamma: P \rightarrow N$) give equivalent extensions via $E_\alpha \rightarrow E_{\alpha+\gamma j}$, $(n, p) \mapsto (n - \gamma(p), p)$, which maps the relation $(\alpha(k), -j(k))$ to $((\alpha + \gamma j)(k), -j(k))$. The lift $p \mapsto (0, p)$ of $P \rightarrow M$ to E_α restricts on K to $k \mapsto (0, j(k)) = (\alpha(k), 0)$, so $\theta(E_\alpha)$ is the class of α . For an extension E with lift β , the map $E_{\beta j} \rightarrow E$, $(n, p) \mapsto n + \beta(p)$, is well

defined and an equivalence by the five lemma. Hence the constructions are inverse. The split extension admits a lift with $\beta j = 0$. $\square$

Example 13.8.7. Let p be a prime. By Proposition 13.8.6, computed with the resolution $0 \rightarrow p\mathbb{Z} \xrightarrow{j} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0$: every $f: \mathbb{Z} \rightarrow \mathbb{Z}/p$ vanishes on $p\mathbb{Z}$, so $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/p, \mathbb{Z}/p) = \text{Hom}(p\mathbb{Z}, \mathbb{Z}/p) \cong \mathbb{Z}/p$, via $\alpha \mapsto a = \alpha(p)$. For $a = 0$ the extension is split, $E = \mathbb{Z}/p \oplus \mathbb{Z}/p$. For $a \neq 0$ the pushout is $E_\alpha = (\mathbb{Z}/p \oplus \mathbb{Z}) / \langle (a, -p) \rangle$, in which $(0, 1)$ has order p^2 : if $m(0, 1) = k(a, -p)$, then $ka \equiv 0 \pmod{p}$ forces $p \mid k$ and $p^2 \mid m$. So $E_\alpha \cong \mathbb{Z}/p^2$, with $\mathbb{Z}/p$ embedded by $n \mapsto na^{-1}p$. The $p-1$ nonzero classes thus give pairwise inequivalent extensions whose middle terms are all isomorphic to $\mathbb{Z}/p^2$: equivalence of extensions remembers the maps, not only the middle group.

Counterexample 13.8.8. Projective, flat and injective are different conditions, already for abelian groups.

1. $\mathbb{Q}$ is flat, since $\mathbb{Q} = S^{-1}\mathbb{Z}$ with $S = \mathbb{Z} \setminus \{0\}$ (Proposition 12.3.2(2)), and injective, since it is divisible (Theorem 13.5.4). It is not projective: a projective abelian group is a direct summand, hence a subgroup, of a free group, hence free (Propositions 13.5.2 and 13.8.5); but $\text{Hom}(\mathbb{Q}, \mathbb{Z}) = 0$, because the image of 1 under a homomorphism $\mathbb{Q} \rightarrow \mathbb{Z}$ is divisible by every $n \geq 1$ in $\mathbb{Z}$, while a nonzero free group has a nonzero coordinate map to $\mathbb{Z}$.
2. $\mathbb{Z}$ is projective, hence flat, but not injective, as it is not divisible.
3. $\mathbb{Z}/n$ for $n \geq 2$ is none of the three: it is not flat, since $\text{Tor}_1(\mathbb{Z}/n, \mathbb{Z}/n) = \mathbb{Z}/n \neq 0$ (Example 13.8.3 and Proposition 13.8.4), hence not projective, and it is not divisible, hence not injective.

13.9 Künneth and universal coefficient theorems

In this section R is a principal ideal domain and we use chain complexes of R -modules. The tensor product of chain complexes is $(C \otimes D)_n = \bigoplus_{p+q=n} C_p \otimes D_q$ with $d(x \otimes y) = dx \otimes y + (-1)^p x \otimes dy$ for $x \in C_p$, and $\text{Hom}(C, M)$ is the cochain complex $\text{Hom}(C_n, M)$ with $(d\varphi)(x) = (-1)^{n+1} \varphi(dx)$ for $\varphi \in \text{Hom}(C_n, M)$.

Theorem 13.9.1 (Künneth formula). *Let C be a chain complex of free R -modules and D any chain complex. For every n there is a natural short exact sequence*

$$0 \rightarrow \bigoplus_{p+q=n} H_p(C) \otimes H_q(D) \rightarrow H_n(C \otimes D) \rightarrow \bigoplus_{p+q=n-1} \text{Tor}_1^R(H_p(C), H_q(D)) \rightarrow 0,$$

where the first map sends $[x] \otimes [y]$ to $[x \otimes y]$. If D is also a complex of free modules, the sequence splits (non-naturally).

Proof. Let $Z_p = \ker d_p$ and $B_p = \text{im } d_{p+1} \subseteq C_p$; they are free by Proposition 13.8.5, hence flat. Regard Z and B as complexes with zero differential, and let B' be the complex with $B'_p = B_{p-1}$ and zero differential. The sequences $0 \rightarrow Z_p \rightarrow C_p \xrightarrow{d} B_{p-1} \rightarrow 0$ form a short exact sequence of complexes $0 \rightarrow Z \rightarrow C \rightarrow B' \rightarrow 0$, split in each degree since B_{p-1} is free; so it stays exact after tensoring with D . As Z and B are flat with zero differential, $H_n(Z \otimes D) = \bigoplus_p Z_p \otimes H_q(D)$ and similarly for B . The long exact sequence reads

$$\cdots \rightarrow \bigoplus_{p+q=n} B_p \otimes H_q(D) \xrightarrow{\partial} \bigoplus_{p+q=n} Z_p \otimes H_q(D) \rightarrow H_n(C \otimes D) \rightarrow \bigoplus_{p+q=n-1} B_p \otimes H_q(D) \xrightarrow{\partial} \cdots,$$

and computing the connecting map with Theorem 13.4.3 shows that ∂ is induced by the inclusions $B_p \rightarrow Z_p$ (a class $b \otimes [y]$ lifts to $c \otimes y$ with $dc = b$, and $d(c \otimes y) = b \otimes y$). Hence $H_n(C \otimes D)$ is an extension of $\ker(\partial)$ in degree $n - 1$ by $\operatorname{coker}(\partial)$ in degree n . Since $0 \rightarrow B_p \rightarrow Z_p \rightarrow H_p(C) \rightarrow 0$ is a free resolution, $\operatorname{coker}(B_p \otimes H_q(D) \rightarrow Z_p \otimes H_q(D)) = H_p(C) \otimes H_q(D)$ and $\ker = \operatorname{Tor}_1(H_p(C), H_q(D))$, by right exactness of $\otimes$ and Theorem 13.8.2.

If D is free as well, choose retractions $r_C: C_p \rightarrow Z_p$ (which exist since B_{p-1} is free) and r_D . The composites $C_p \rightarrow Z_p \rightarrow H_p(C)$ form a morphism of complexes $C \rightarrow H(C)$ (to the complex with zero differential), since $r_C(dx) = dx \in B_p$ maps to 0; similarly $D \rightarrow H(D)$. Their tensor product induces $H_n(C \otimes D) \rightarrow \bigoplus H_p(C) \otimes H_q(D)$, which is a left inverse of the first map. $\square$

Theorem 13.9.2 (Universal coefficient theorem). *Let C be a chain complex of free R -modules and M an R -module. There is a natural short exact sequence*

$$0 \rightarrow \operatorname{Ext}_R^1(H_{n-1}(C), M) \rightarrow H^n(\operatorname{Hom}(C, M)) \rightarrow \operatorname{Hom}_R(H_n(C), M) \rightarrow 0,$$

where the second map sends the class of a cocycle φ to its restriction to cycles. It splits non-naturally.

Proof. Apply $\operatorname{Hom}(-, M)$ to the degreewise split sequence $0 \rightarrow Z \rightarrow C \rightarrow B' \rightarrow 0$ of the previous proof. As Z and B have zero differential, the long exact sequence becomes $\cdots \rightarrow \operatorname{Hom}(B_{n-1}, M) \rightarrow H^n(\operatorname{Hom}(C, M)) \rightarrow \operatorname{Hom}(Z_n, M) \xrightarrow{\partial} \operatorname{Hom}(B_n, M) \rightarrow \cdots$, where ∂ is restriction along $B_n \subseteq Z_n$ (up to sign). From the free resolution $0 \rightarrow B_n \rightarrow Z_n \rightarrow H_n(C) \rightarrow 0$ we get $\ker(\partial) = \operatorname{Hom}(H_n(C), M)$ and $\operatorname{coker}(\operatorname{Hom}(Z_{n-1}, M) \rightarrow \operatorname{Hom}(B_{n-1}, M)) = \operatorname{Ext}_R^1(H_{n-1}(C), M)$, which gives the sequence. A splitting is obtained from a retraction $r: C_n \rightarrow Z_n$: a homomorphism $H_n(C) \rightarrow M$ composed with $C_n \rightarrow Z_n \rightarrow H_n(C)$ is a cocycle. $\square$

Example 13.9.3. Let C be the complex $\cdots \rightarrow 0 \rightarrow \mathbb{Z} \xrightarrow{2} \mathbb{Z} \rightarrow 0$ in degrees 1 and 0, so $H_0 = \mathbb{Z}/2\mathbb{Z}$ and $H_1 = 0$. Then $\operatorname{Hom}(C, \mathbb{Z})$ is $\mathbb{Z} \xrightarrow{\pm 2} \mathbb{Z}$ in degrees 0 and 1, with $H^0 = 0$ and $H^1 = \mathbb{Z}/2\mathbb{Z}$. Indeed $\operatorname{Hom}(H_1, \mathbb{Z}) = 0$ and $\operatorname{Ext}_R^1(H_0, \mathbb{Z}) = \mathbb{Z}/2\mathbb{Z}$: torsion in homology shifts up one degree in cohomology. This is the algebraic reason why $H^2(\mathbb{RP}^2, \mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$ while $H_2(\mathbb{RP}^2, \mathbb{Z}) = 0$ (Chapter 20).

13.10 Exercises

Exercise 13.10.1. Prove the 3×3 lemma: in a commutative 3×3 diagram of modules with exact columns, if two of the rows are short exact sequences, so is the third.

Exercise 13.10.2. Show that $\operatorname{Ext}_\mathbb{Z}^1(\mathbb{Q}, \mathbb{Z}) \neq 0$. (Hint: use $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{Q}/\mathbb{Z} \rightarrow 0$.)

Exercise 13.10.3. Let R be a PID. Show that an R -module is flat if and only if it is torsion-free.

Exercise 13.10.4. Let $f: A^\bullet \rightarrow B^\bullet$ be a morphism of complexes. Show that f is homotopic to zero if and only if the sequence $0 \rightarrow B \rightarrow \operatorname{Cone}(f) \rightarrow A[1] \rightarrow 0$ splits as a sequence of complexes.

Exercise 13.10.5. Show that an abelian group is injective if and only if $\operatorname{Ext}_\mathbb{Z}^1(\mathbb{Z}/n\mathbb{Z}, A) = 0$ for all $n \geq 1$.

Solution. By Example 13.8.3, $\operatorname{Ext}_\mathbb{Z}^1(\mathbb{Z}/n\mathbb{Z}, A) = A/nA$, which vanishes for all n iff A is divisible, iff A is injective by Theorem 13.5.4. $\square$

Chapter 14

Spectral Sequences

14.1 Introduction

A spectral sequence is a bookkeeping device for computing the cohomology of a filtered complex from the cohomology of its graded pieces, in a sequence of successive approximations. This chapter constructs the spectral sequence of a filtered complex with complete proofs, derives the two spectral sequences of a double complex, and proves the Grothendieck spectral sequence of a composition of functors. Later chapters use the Leray spectral sequence (Chapter 24), the Frölicher spectral sequence (Chapter 43) and the weight and Hodge spectral sequences of mixed Hodge theory (Chapter 70).

References are [Wei94, Chapter 5] and [McC01]. As in Chapter 13, we argue with elements; by Remark 13.2.5 the results hold in every abelian category.

Prerequisites. Chapter 13 (Homological Algebra).

14.2 Spectral sequences

Definition 14.2.1. A (cohomological) *spectral sequence* starting at page r_0 consists of objects $E_r^{p,q}$ for $p, q \in \mathbb{Z}$ and $r \geq r_0$, morphisms $d_r: E_r^{p,q} \rightarrow E_r^{p+r, q-r+1}$ with $d_r \circ d_r = 0$, and isomorphisms $E_{r+1}^{p,q} \cong \ker(d_r)/\text{im}(d_r)$ at $E_r^{p,q}$. The object $E_r = \bigoplus_{p,q} E_r^{p,q}$ is the r -th page, and $p + q$ is the *total degree*. It is a *first-quadrant* spectral sequence if $E_{r_0}^{p,q} = 0$ unless $p, q \geq 0$.

Lemma 14.2.2. In a first-quadrant spectral sequence $E_r^{p,q} = 0$ unless $p, q \geq 0$, for all r , and $E_r^{p,q} = E_{r+1}^{p,q}$ whenever $r > \max(p, q + 1)$. The common value is denoted $E_\infty^{p,q}$.

Proof. The first claim holds because $E_{r+1}^{p,q}$ is a subquotient of $E_r^{p,q}$. For $r > p$ the source $E_r^{p-r, q+r-1}$ of the incoming differential has negative first index, and for $r > q + 1$ the target $E_r^{p+r, q-r+1}$ of the outgoing differential has negative second index; so both differentials vanish. $\square$

Definition 14.2.3. A first-quadrant spectral sequence *converges* to objects H^n , written $E_{r_0}^{p,q} \Rightarrow H^{p+q}$, if each H^n has a finite decreasing filtration $H^n = F^0 H^n \supseteq F^1 H^n \supseteq \dots \supseteq F^{n+1} H^n = 0$ with isomorphisms $E_\infty^{p,q} \cong F^p H^{p+q} / F^{p+1} H^{p+q}$. The spectral sequence *degenerates* at page r if $d_s = 0$ for all $s \geq r$, so that $E_r = E_\infty$.

Proposition 14.2.4. Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a convergent first-quadrant spectral sequence.

1. There are edge maps $E_2^{n,0} \rightarrow H^n$ and $H^n \rightarrow E_2^{0,n}$: the first is the composite $E_2^{n,0} \twoheadrightarrow E_\infty^{n,0} = F^n H^n \hookrightarrow H^n$, the second $H^n \twoheadrightarrow H^n / F^1 H^n = E_\infty^{0,n} \hookrightarrow E_2^{0,n}$.

2. (Five-term exact sequence) The sequence $0 \rightarrow E_2^{1,0} \rightarrow H^1 \rightarrow E_2^{0,1} \xrightarrow{d_2} E_2^{2,0} \rightarrow H^2$ is exact.
3. If $E_2^{p,q} = 0$ for $q \neq 0$, then $H^n \cong E_2^{n,0}$. If $E_2^{p,q} = 0$ for $p \neq 0$, then $H^n \cong E_2^{0,n}$.

Proof. (1) Since nothing is in negative degrees, $E_{r+1}^{n,0}$ is a quotient of $E_r^{n,0}$ (no outgoing differential lands in $q < 0$) and $E_{r+1}^{0,n}$ a subobject of $E_r^{0,n}$ (no incoming differential starts at $p < 0$). Also $F^{n+1}H^n = 0$, so $E_\infty^{n,0} = F^n H^n$, and $F^0 H^n = H^n$, so $E_\infty^{0,n} = H^n / F^1 H^n$. (2) In total degree 1, H^1 has the filtration $0 \subseteq F^1 H^1 \subseteq H^1$ with $F^1 H^1 = E_\infty^{1,0} = E_2^{1,0}$ (no differential reaches or leaves $E_r^{1,0}$ nontrivially: incoming would start at $p = 1 - r < 0$, outgoing lands at $q = 1 - r < 0$) and $H^1 / F^1 H^1 = E_\infty^{0,1} = \ker(d_2: E_2^{0,1} \rightarrow E_2^{2,0})$ (for $r \geq 3$ all differentials at $(0, 1)$ vanish). This gives exactness at $E_2^{1,0}$, H^1 and $E_2^{0,1}$. Next $E_\infty^{2,0} = E_3^{2,0} = E_2^{2,0} / \text{im}(d_2)$, and it injects into H^2 as $F^2 H^2$; so the kernel of $E_2^{2,0} \rightarrow H^2$ is $\text{im}(d_2)$. (3) All but one of the graded pieces $E_\infty^{p,n-p}$ of H^n vanish, and all differentials vanish because either the source or the target is zero. $\square$

Proposition 14.2.5. Let $E_2^{p,q} \Rightarrow H^{p+q}$ be a convergent first-quadrant spectral sequence with $E_2^{p,q} = 0$ unless $q \in \{0, s\}$, for some $s \geq 1$. Then there is a long exact sequence

$$\cdots \rightarrow E_2^{n,0} \rightarrow H^n \rightarrow E_2^{n-s,s} \xrightarrow{d_{s+1}} E_2^{n+1,0} \rightarrow H^{n+1} \rightarrow \cdots$$

Proof. The only possibly nonzero differentials are $d_{s+1}: E_{s+1}^{p,s} \rightarrow E_{s+1}^{p+s+1,0}$, and $E_{s+1} = E_2$. So $E_\infty^{n,0} = \text{coker}(d_{s+1}: E_2^{n-s-1,s} \rightarrow E_2^{n,0})$ and $E_\infty^{n-s,s} = \ker(d_{s+1}: E_2^{n-s,s} \rightarrow E_2^{n+1,0})$, and H^n is an extension $0 \rightarrow E_\infty^{n,0} \rightarrow H^n \rightarrow E_\infty^{n-s,s} \rightarrow 0$. Splicing these sequences gives the claim. $\square$

Proposition 14.2.6. Let E be a spectral sequence of finite-dimensional vector spaces over a field, with only finitely many nonzero $E_{r_0}^{p,q}$. Then $\chi(E_r) = \sum_{p,q} (-1)^{p+q} \dim E_r^{p,q}$ does not depend on r , and if $E_{r_0}^{p,q} \Rightarrow H^{p+q}$ converges, then $\chi(E_{r_0}) = \sum_n (-1)^n \dim H^n$. In particular $\dim H^n \leq \sum_{p+q=n} \dim E_r^{p,q}$ for every r .

Proof. Grading E_r by total degree $n = p + q$, the differential d_r raises n by one, so (E_r, d_r) is a finite complex of finite-dimensional spaces with cohomology E_{r+1} . For such a complex $C^\bullet$, rank-nullity gives $\dim C^n = \dim Z^n + \dim B^{n+1}$ and $\dim H^n = \dim Z^n - \dim B^n$, so $\sum_n (-1)^n \dim C^n = \sum_n (-1)^n \dim H^n$. Hence $\chi(E_{r+1}) = \chi(E_r)$. Only finitely many positions are nonzero, so the pages stabilise, and convergence gives $\dim H^n = \sum_p \dim E_\infty^{p,n-p}$. The inequality holds because each $E_\infty^{p,q}$ is a subquotient of $E_r^{p,q}$. $\square$

Counterexample 14.2.7. The limit of a spectral sequence determines the abutment only up to extensions. Let K and K' be the complexes concentrated in degree 0, with zero differential, given by $K^0 = \mathbb{Z}/4$ with the filtration $F^0 = \mathbb{Z}/4 \supseteq F^1 = 2\mathbb{Z}/4 \supseteq F^2 = 0$, and $K'^0 = \mathbb{Z}/2 \oplus \mathbb{Z}/2$ with $F^0 = K'^0 \supseteq F^1 = 0 \oplus \mathbb{Z}/2 \supseteq F^2 = 0$. In both cases all differentials vanish and the spectral sequence of the filtered complex (Section 14.3) has $E_0^{0,0} = \mathbb{Z}/2$ and $E_0^{1,-1} = \mathbb{Z}/2$ on every page. The spectral sequences are isomorphic, but the abutments $\mathbb{Z}/4$ and $\mathbb{Z}/2 \oplus \mathbb{Z}/2$ are not. Over a field this cannot happen, since every extension of vector spaces splits and $H^n \cong \bigoplus_p E_\infty^{p,n-p}$.

14.3 The spectral sequence of a filtered complex

Definition 14.3.1. A *filtered complex* is a complex $K^\bullet$ with subcomplexes $\cdots \supseteq F^p K \supseteq F^{p+1} K \supseteq \cdots$. The filtration is *finite in each degree* if for every n there are $a \leq b$ with $F^a K^n = K^n$ and $F^b K^n = 0$. The *associated graded complexes* are $\text{gr}^p K = F^p K / F^{p+1} K$, and the induced filtration on cohomology is $F^p H^n(K) = \text{im}(H^n(F^p K) \rightarrow H^n(K))$.

Definition 14.3.2. For a filtered complex K we abbreviate $F^p = F^p K$ and, for $r \geq 0$,

$$Z_r^p = F^p \cap d^{-1}(F^{p+r}), \quad B_r^p = F^p \cap d(F^{p-r+1}), \quad E_r^p = Z_r^p / (B_r^p + Z_{r-1}^{p+1}),$$

with the convention $Z_{-1}^{p+1} = F^{p+1}$. These are graded by the degree of K , and we write $E_r^{p,q}$ for the part of E_r^p in degree $p + q$. Let $d_r: E_r^p \rightarrow E_r^{p+r}$ be induced by d .

Lemma 14.3.3. The inclusions $B_r^p \subseteq Z_r^p$ and $Z_{r-1}^{p+1} \subseteq Z_r^p$ hold, so E_r^p is defined, and d_r is a well-defined morphism with $d_r^2 = 0$, of bidegree $(r, 1 - r)$. Moreover $E_0^p = \text{gr}^p K$ with d_0 induced by d , and $E_1^{p,q} = H^{p+q}(\text{gr}^p K)$.

Proof. Elements of B_r^p are boundaries, hence cycles; and $x \in F^{p+1}$ with $dx \in F^{p+r}$ lies in Z_r^p . If $x \in Z_r^p$ then $dx \in F^{p+r}$ and $d(dx) = 0$, so $dx \in Z_{r-1}^{p+r}$. We must check that d maps $B_r^p + Z_{r-1}^{p+1}$ into $B_r^{p+r} + Z_{r-1}^{p+r+1}$: $d(B_r^p) = 0$, and for $x \in Z_{r-1}^{p+1}$ we have $dx \in F^{p+r} \cap d(F^{p+1}) = B_r^{p+r}$. Clearly $d_r^2 = 0$. For $r = 0$: $Z_0^p = F^p$ and $B_0^p = d(F^{p+1}) \subseteq F^{p+1}$, so $E_0^p = F^p / F^{p+1}$. For $r = 1$: Z_1^p consists of the elements of F^p whose image in gr^p is a cocycle, $Z_0^{p+1} = F^{p+1}$ and $B_1^p = d(F^p)$, so $E_1^p = Z_1^p / (d(F^p) + F^{p+1})$ is the cohomology of $\text{gr}^p K$. $\square$

Theorem 14.3.4. For every filtered complex, the objects $E_r^{p,q}$ and differentials d_r , $r \geq 0$, form a spectral sequence: $H(E_r, d_r) \cong E_{r+1}$. If the filtration is finite in each degree, then for fixed (p, q) and all sufficiently large r , $E_r^{p,q} = E_\infty^{p,q}$ with

$$E_\infty^{p,q} \cong F^p H^{p+q}(K) / F^{p+1} H^{p+q}(K).$$

In particular, if $F^0 K = K$ and $F^{n+1} K^n = 0$ for all n , the spectral sequence is first-quadrant (from E_1 on, since $E_1^{p,q}$ vanishes for $p < 0$ or $q < 0$) and $E_1^{p,q} = H^{p+q}(\text{gr}^p K) \Rightarrow H^{p+q}(K)$.

Proof. Kernel of d_r at E_r^p . Let $x \in Z_r^p$ with $dx \in B_r^{p+r} + Z_{r-1}^{p+r+1}$, say $dx = dy + z$ with $y \in F^{p+1}$, $dy \in F^{p+r}$ and $z \in F^{p+r+1}$. Then $y \in Z_{r-1}^{p+1}$, and $x - y \in F^p$ with $d(x - y) = z \in F^{p+r+1}$, so $x - y \in Z_{r-1}^{p+1}$. Conversely, Z_{r+1}^p maps into the kernel, since $dx \in F^{p+r+1}$ and $d(dx) = 0$ give $dx \in Z_{r-1}^{p+r+1}$. Hence the kernel is $(Z_{r+1}^p + Z_{r-1}^{p+1}) / (B_r^p + Z_{r-1}^{p+1})$.

Image of d_r in E_r^p . It consists of the classes of dx with $x \in Z_r^{p-r}$, that is of the elements of $F^p \cap d(F^{p-r}) = B_{r+1}^p$. Since $B_r^p \subseteq B_{r+1}^p$, the image is $(B_{r+1}^p + Z_{r-1}^{p+1}) / (B_r^p + Z_{r-1}^{p+1})$.

Cohomology. Therefore $H(E_r)^p = (Z_{r+1}^p + Z_{r-1}^{p+1}) / (B_{r+1}^p + Z_{r-1}^{p+1})$. The map $Z_{r+1}^p \rightarrow H(E_r)^p$ is surjective, with kernel $Z_{r+1}^p \cap (B_{r+1}^p + Z_{r-1}^{p+1}) = B_{r+1}^p + (Z_{r+1}^p \cap Z_{r-1}^{p+1})$ (as $B_{r+1}^p \subseteq Z_{r+1}^p$). Now $Z_{r+1}^p \cap Z_{r-1}^{p+1} = F^{p+1} \cap d^{-1}(F^{p+r+1}) = Z_r^{p+1}$. So $H(E_r)^p \cong Z_{r+1}^p / (B_{r+1}^p + Z_r^{p+1}) = E_{r+1}^p$.

Convergence. Fix n and p and let Z, B denote the cocycles and coboundaries of K . If the filtration is finite in degrees $n - 1, n$ and $n + 1$, then for large r we have $F^{p+r} K^{n+1} = 0$ and $F^{p-r+1} K^{n-1} = K^{n-1}$, so in degree n : $Z_r^p = F^p \cap Z$, $Z_{r-1}^{p+1} = F^{p+1} \cap Z$ and $B_r^p = F^p \cap B$. Hence $E_\infty^p = (F^p \cap Z) / ((F^p \cap B) + (F^{p+1} \cap Z))$. On the other hand $F^p H^n = (F^p \cap Z + B) / B$, and

$$\frac{F^p H^n}{F^{p+1} H^n} \cong \frac{F^p \cap Z}{(F^p \cap Z) \cap (F^{p+1} \cap Z + B)} = \frac{F^p \cap Z}{F^{p+1} \cap Z + F^p \cap B},$$

using the modular law. For the last statement, $\text{gr}^p K = 0$ for $p < 0$ and $\text{gr}^p K^n = 0$ for $p > n$, so $E_1^{p,q} = 0$ unless $p \geq 0$ and $q \geq 0$, and $F^p H^n$ runs from H^n to 0. $\square$

14.4 Double complexes

Theorem 14.4.1. *Let C be a first-quadrant double complex (Definition 13.7.1) and $T = \text{Tot}(C)$. There are two convergent first-quadrant spectral sequences*

$${}^I E_2^{p,q} = H_h^p(H_v^q(C)) \Rightarrow H^{p+q}(T), \quad {}^{II} E_2^{p,q} = H_v^p(H_h^q(C)) \Rightarrow H^{p+q}(T),$$

where $H_v^q(C)$ is the cohomology of the columns (a complex in p via d_h), and $H_h^q(C)$ that of the rows (a complex in the other index via d_v).

Proof. Filter T by columns: ${}^I F^p T^n = \bigoplus_{i \geq p} C^{i,n-i}$. These are subcomplexes, $F^0 T = T$ and $F^{n+1} T^n = 0$. The graded piece $\text{gr}^p T$ is the column $C^{p,\bullet}$ (shifted by p) with differential $\pm d_v$, so ${}^I E_1^{p,q} = H_v^q(C^{p,\bullet})$ by Theorem 14.3.4. The differential d_1 is the connecting map of $0 \rightarrow \text{gr}^{p+1} \rightarrow F^p/F^{p+2} \rightarrow \text{gr}^p \rightarrow 0$; a cocycle $x \in C^{p,q}$ of d_v lifts to itself, and its total differential is $d_h x \in C^{p+1,q}$, so $d_1 = H_v^q(d_h)$ and ${}^I E_2^{p,q} = H_h^p(H_v^q(C))$. Convergence follows from Theorem 14.3.4. Filtering by rows, ${}^{II} F^p T^n = \bigoplus_{j \geq p} C^{n-j,j}$, gives the other spectral sequence in the same way. $\square$

Example 14.4.2. If the rows of C are exact in positive degrees, with kernel complex $K^\bullet$ in column 0, then ${}^{II} E_1^{p,q} = H_h^q(C^{\bullet,p})$ vanishes for $q > 0$ and equals K^p for $q = 0$. So ${}^{II} E_2^{p,0} = H^p(K)$, all other terms vanish, and $H^n(T) \cong H^n(K)$ by Proposition 14.2.4(3). This recovers Lemma 13.7.2.

14.5 Exact couples

Definition 14.5.1. An *exact couple* consists of objects D, E and morphisms $i: D \rightarrow D$, $j: D \rightarrow E$, $k: E \rightarrow D$ such that the triangle $D \xrightarrow{i} D \xrightarrow{j} E \xrightarrow{k} D$ is exact at each vertex. Put $d = jk: E \rightarrow E$; then $d^2 = j(kj)k = 0$.

Proposition 14.5.2. *Given an exact couple, let $D' = i(D)$, $E' = \ker d / \text{im } d$, $i' = i|_{D'}$, $j'(i(x)) = [j(x)]$ and $k'([e]) = k(e)$. Then (D', E', i', j', k') is an exact couple, the derived couple. Iterating gives a spectral sequence $E^{(r)}$ with differentials $d^{(r)} = j^{(r)} k^{(r)}$.*

Proof. j' is well defined: if $i(x) = i(x')$ then $x - x' = k(e)$ and $j(x - x') = jk(e) = d(e)$; and $j(x)$ is a d -cocycle since $dj(x) = jk(j(x)) = 0$. k' is well defined: if $e \in \ker d$ then $jk(e) = 0$ so $k(e) \in \ker j = \text{im } i = D'$, and $k(d(e')) = jk(e') = 0$. Exactness at D' (between i' and j'): $j'(i(x)) = 0$ iff $j(x) = jk(e)$ iff $x - k(e) \in \ker j = \text{im } i$, iff $i(x) \in i(\text{im } i) + ik(E) = i(D')$ since $ik = 0$. Exactness at E' : $k'([e]) = 0$ iff $k(e) = 0$ iff $e = j(x)$, iff $[e] = j'(i(x))$. Exactness at the other D' (between k' and i'): $i'(i(x)) = 0$ iff $i(x) \in \ker i = \text{im } k$, iff $i(x) = k(e)$ for some e , which is then a d -cocycle since $jk(e) = ji(x) = 0$; that is, iff $i(x) \in \text{im } k'$. $\square$

Remark 14.5.3. A filtered complex gives an exact couple with $D = \bigoplus_{p,n} H^n(F^p K)$, $E = \bigoplus_{p,n} H^n(\text{gr}^p K)$, i induced by the inclusions $F^{p+1} \subseteq F^p$, and j, k from the long exact sequences of $0 \rightarrow F^{p+1} \rightarrow F^p \rightarrow \text{gr}^p \rightarrow 0$. The resulting spectral sequence is the one of Theorem 14.3.4, from E_1 on [Wei94, Section 5.9].

14.6 The Grothendieck spectral sequence

Definition 14.6.1. Let $A^\bullet$ be a complex with $A^n = 0$ for $n < 0$ in an abelian category with enough injectives. A *Cartan–Eilenberg resolution* of A is a first-quadrant double complex

$I^{\bullet,\bullet}$ of injectives with a morphism $A^p \rightarrow I^{p,0}$ for each p , such that for every p the columns give injective resolutions $A^p \rightarrow I^{p,\bullet}$, $Z^p(A) \rightarrow Z_h^p(I)^\bullet$, $B^p(A) \rightarrow B_h^p(I)^\bullet$ and $H^p(A) \rightarrow H_h^p(I)^\bullet$, where Z_h , B_h , H_h denote horizontal cocycles, coboundaries and cohomology.

Lemma 14.6.2. *Every such complex A has a Cartan–Eilenberg resolution.*

Proof. For each p choose injective resolutions $B^p(A) \rightarrow J_B^{p,\bullet}$ and $H^p(A) \rightarrow J_H^{p,\bullet}$. The horseshoe lemma (Lemma 13.6.3) applied to $0 \rightarrow B^p \rightarrow Z^p \rightarrow H^p \rightarrow 0$ gives a resolution $J_Z^{p,\bullet}$ of Z^p with $J_Z^{p,q} = J_B^{p,q} \oplus J_H^{p,q}$, and applied to $0 \rightarrow Z^p \rightarrow A^p \rightarrow B^{p+1} \rightarrow 0$ gives a resolution $I^{p,\bullet}$ of A^p with $I^{p,q} = J_Z^{p,q} \oplus J_B^{p+1,q}$. Define $d_h: I^{p,q} \rightarrow I^{p+1,q}$ as the composite $I^{p,q} \rightarrow J_B^{p+1,q} \rightarrow J_Z^{p+1,q} \rightarrow I^{p+1,q}$ of the projection and inclusions provided by the horseshoe lemma. Then $d_h^2 = 0$ (the projection $I^{p+1,q} \rightarrow J_B^{p+2,q}$ kills $J_Z^{p+1,q}$), d_h commutes with the vertical differentials since all maps involved are morphisms of complexes, and it covers $d: A^p \rightarrow A^{p+1}$. By construction the horizontal cocycles, coboundaries and cohomology in column p are $J_Z^{p,\bullet}$, $J_B^{p,\bullet}$ and $J_H^{p,\bullet}$. $\square$

Theorem 14.6.3 (Grothendieck spectral sequence). *Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories, $\mathcal{A}$ and $\mathcal{B}$ with enough injectives, and $F: \mathcal{A} \rightarrow \mathcal{B}$, $G: \mathcal{B} \rightarrow \mathcal{C}$ left exact functors such that F maps injective objects to G -acyclic objects. For every $X \in \mathcal{A}$ there is a convergent first-quadrant spectral sequence*

$$E_2^{p,q} = R^p G(R^q F(X)) \Rightarrow R^{p+q}(G \circ F)(X).$$

Proof. Let $X \rightarrow A^\bullet$ be an injective resolution in $\mathcal{A}$, and let $I^{\bullet,\bullet}$ be a Cartan–Eilenberg resolution of the complex $F(A^\bullet)$ in $\mathcal{B}$. Consider the double complex $G(I)$ and its total complex T .

First filtration (by columns). For each p , the column $I^{p,\bullet}$ is an injective resolution of the G -acyclic object $F(A^p)$, so $H_v^q(G(I^{p,\bullet})) = R^q G(F(A^p))$ is $G(F(A^p))$ for $q = 0$ and 0 for $q > 0$. So ${}^I E_2^{p,q} = 0$ for $q > 0$ and ${}^I E_2^{p,0} = H^p(GF(A^\bullet)) = R^p(GF)(X)$, and $H^n(T) \cong R^n(GF)(X)$ by Proposition 14.2.4(3).

Second filtration (by rows). Fix a row index j . In the row $I^{\bullet,j}$ the sequences $0 \rightarrow B_h^{q,j} \rightarrow Z_h^{q,j} \rightarrow H_h^{q,j} \rightarrow 0$ and $0 \rightarrow Z_h^{q,j} \rightarrow I^{q,j} \rightarrow B_h^{q+1,j} \rightarrow 0$ consist of injective objects (they are terms of injective resolutions), hence split; so G commutes with taking horizontal cocycles, coboundaries and cohomology of this row, and $H_h^q(G(I^{\bullet,j})) = G(H_h^{q,j}(I))$. For fixed q , the column $H_h^{q,\bullet}(I)$ is an injective resolution of $H^q(F(A^\bullet)) = R^q F(X)$. By Theorem 14.4.1, ${}^{II} E_2^{p,q} = H_v^p(H_h^q(G(I))) = H^p(G(H_h^{q,\bullet}(I))) = R^p G(R^q F(X))$, and it converges to $H^{p+q}(T) \cong R^{p+q}(GF)(X)$. $\square$

14.7 Exercises

Exercise 14.7.1. State and prove the analogue of Proposition 14.2.5 for a spectral sequence concentrated in two columns $p \in \{0, s\}$.

Exercise 14.7.2. Let $0 \rightarrow A^\bullet \rightarrow B^\bullet \rightarrow C^\bullet \rightarrow 0$ be a short exact sequence of bounded below complexes, viewed as a double complex with three columns. Using both spectral sequences of Theorem 14.4.1 (after shifting so that the double complex is first-quadrant), show that if two of A, B, C are acyclic, then so is the third.

Exercise 14.7.3. Use Theorem 14.4.1 to give a second proof of Theorem 13.8.2.

Exercise 14.7.4. Let $R \rightarrow S$ be a homomorphism of commutative rings, M an S -module and N an R -module. Construct a spectral sequence $\mathrm{Tor}_p^S(M, \mathrm{Tor}_q^R(S, N)) \Rightarrow \mathrm{Tor}_{p+q}^R(M, N)$ (homological indexing).

Chapter 15

Derived Categories

15.1 Introduction

The derived category of an abelian category is obtained from complexes by declaring quasi-isomorphisms to be isomorphisms. It is the natural home of derived functors: instead of a sequence of functors $R^n F$ one has a single functor RF on complexes, compositions of derived functors are derived functors of compositions, and hypercohomology of complexes (for instance of the de Rham complex of a manifold) is defined without choices. Filtered derived categories and strict morphisms, discussed at the end, are the framework of Deligne's mixed Hodge theory (Chapter 70).

Some foundational verifications (the octahedral axiom, and the construction of localisations by roofs) are quoted from [Wei94, Chapter 10], [GM03, Chapter III] and [KS06]; everything used about derived functors is proved.

Prerequisites. Chapter 5 (Categories), Chapter 13 (Homological Algebra) and Chapter 14 (Spectral Sequences).

Throughout, $\mathcal{A}$ is an abelian category, complexes are as in Definition 13.3.1, and we argue with elements as justified by Remark 13.2.5.

15.2 The homotopy category

Definition 15.2.1. The *homotopy category* $K(\mathcal{A})$ has the complexes as objects and homotopy classes of morphisms of complexes (Definition 13.3.3) as morphisms. $K^+(\mathcal{A})$ is its full subcategory of bounded below complexes. The *shift* $X \mapsto X[1]$ is an automorphism of $K(\mathcal{A})$. A *distinguished triangle* in $K(\mathcal{A})$ is a diagram $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ isomorphic (in $K(\mathcal{A})$) to one of the form $X \xrightarrow{f} Y \xrightarrow{i} \text{Cone}(f) \xrightarrow{p} X[1]$, with i and p the maps of Proposition 13.4.4.

Remark 15.2.2. Homotopy is compatible with sums and compositions, so $K(\mathcal{A})$ is an additive category, and cohomology $H^n: K(\mathcal{A}) \rightarrow \mathcal{A}$ is a well-defined functor by Lemma 13.3.4.

Definition 15.2.3. A *triangulated category* is an additive category $\mathcal{T}$ with an additive automorphism $X \mapsto X[1]$ and a class of *distinguished triangles* $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ such that:

(TR1) $X \xrightarrow{\text{id}} X \rightarrow 0 \rightarrow X[1]$ is distinguished; triangles isomorphic to distinguished ones are distinguished; every morphism $X \rightarrow Y$ is the first morphism of a distinguished triangle.

(TR2) $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ is distinguished if and only if $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$ is.

- (TR3) Given distinguished triangles $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ and $X' \rightarrow Y' \rightarrow Z' \rightarrow X'[1]$ and morphisms $u: X \rightarrow X'$, $v: Y \rightarrow Y'$ making the first square commute, there is $w: Z \rightarrow Z'$ making the whole diagram commute.
- (TR4) (Octahedral axiom) For composable $X \xrightarrow{f} Y \xrightarrow{g} Z$ and distinguished triangles on f , g and gf with third vertices Z' , X' , Y' , there is a distinguished triangle $Z' \rightarrow Y' \rightarrow X' \rightarrow Z'[1]$ compatible with the given morphisms.

An additive functor between triangulated categories commuting with the shifts and preserving distinguished triangles is *exact* (or *triangulated*).

Theorem 15.2.4. $K(\mathcal{A})$ and $K^+(\mathcal{A})$ are triangulated categories.

Sketch of proof. (TR1): $\text{Cone}(\text{id}_X)$ is contractible, with contracting homotopy $(a, b) \mapsto (b, 0)$, so it is isomorphic to 0 in $K(\mathcal{A})$; the other two parts hold by definition. (TR3): if h is a homotopy with $vf - f'u = dh + hd$, then $(a, b) \mapsto (u(a), v(b) + h(a))$ is a morphism of complexes $\text{Cone}(f) \rightarrow \text{Cone}(f')$ (a direct check using the formula for the differential of the cone in Definition 13.3.5) compatible with the maps of the triangles. The omitted verifications are (TR2), which amounts to an explicit homotopy equivalence $\text{Cone}(i) \simeq X[1]$ for $i: Y \rightarrow \text{Cone}(f)$, and the octahedral axiom (TR4); see [Wei94, Sections 10.1–10.2]. $\square$

Proposition 15.2.5. Let $X \xrightarrow{f} Y \xrightarrow{g} Z \xrightarrow{h} X[1]$ be a distinguished triangle in a triangulated category $\mathcal{T}$ and W an object. Then $gf = 0$, and the sequence $\text{Hom}(W, X) \rightarrow \text{Hom}(W, Y) \rightarrow \text{Hom}(W, Z)$ is exact; so by (TR2) there is a long exact sequence $\cdots \rightarrow \text{Hom}(W, X) \rightarrow \text{Hom}(W, Y) \rightarrow \text{Hom}(W, Z) \rightarrow \text{Hom}(W, X[1]) \rightarrow \cdots$. Similarly for $\text{Hom}(-, W)$. For $\mathcal{T} = K(\mathcal{A})$, the functor H^0 sends distinguished triangles to long exact sequences.

Proof. Apply (TR3) to the triangles $X \xrightarrow{\text{id}} X \rightarrow 0 \rightarrow X[1]$ and $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ with $u = \text{id}$, $v = f$: the resulting w satisfies $gf = w \circ 0 = 0$. Let $\varphi: W \rightarrow Y$ with $g\varphi = 0$. By (TR1) and (TR2), the rotated triangles $W \rightarrow 0 \rightarrow W[1] \xrightarrow{-\text{id}} W[1]$ and $Y \xrightarrow{g} Z \xrightarrow{h} X[1] \xrightarrow{-f[1]} Y[1]$ are distinguished. The square formed by $\varphi: W \rightarrow Y$ and $0: 0 \rightarrow Z$ commutes because $g\varphi = 0$, so (TR3) gives $w: W[1] \rightarrow X[1]$ with $(-f[1]) \circ w = \varphi[1] \circ (-\text{id})$. Then $\psi = w[-1]$ satisfies $f\psi = \varphi$. The dual statement is proved in the same way. For $K(\mathcal{A})$ the last claim follows from the long exact sequence of the cone (Proposition 13.4.4) and invariance under isomorphism. $\square$

15.3 The derived category

Definition 15.3.1. A class S of morphisms in a category C is a *multiplicative system* if it contains the identities and is closed under composition, and:

1. for $s: X' \rightarrow X$ in S and $f: Y \rightarrow X$ there are $t: Y' \rightarrow Y$ in S and $g: Y' \rightarrow X'$ with $ft = sg$, and dually;
2. if $f, g: X \rightarrow Y$ and $s: Y \rightarrow Y'$ in S satisfy $sf = sg$, then $ft = gt$ for some $t: X' \rightarrow X$ in S , and dually.

Theorem 15.3.2 (Localisation). Let S be a multiplicative system in C . There is a category $C[S^{-1}]$ with the same objects and a functor $Q: C \rightarrow C[S^{-1}]$ that is the identity on objects, sends S to isomorphisms, and is universal: every functor sending S to isomorphisms factors uniquely through Q . Morphisms $X \rightarrow Y$ in $C[S^{-1}]$ are equivalence classes of roofs $X \xleftarrow{s} X' \xrightarrow{f} Y$ with $s \in S$,

representing $Q(f)Q(s)^{-1}$, and for $f, g: X \rightarrow Y$ in $\mathcal{C}$ one has $Q(f) = Q(g)$ if and only if $ft = gt$ for some $t \in S$. If $\mathcal{C}$ is additive (triangulated) and S is compatible with the structure, then $\mathcal{C}[S^{-1}]$ is additive (triangulated), with distinguished triangles the images of those of $\mathcal{C}$.

This is proved in [GM03, Chapter III] and [KS06, Chapter 7]; for the triangulated case see [Wei94, Section 10.3].

Proposition 15.3.3. *The quasi-isomorphisms form a multiplicative system in $K(\mathcal{A})$ and in $K^+(\mathcal{A})$.*

Proof. Identities and composites of quasi-isomorphisms are quasi-isomorphisms. Note that a morphism is a quasi-isomorphism iff its cone is acyclic (Proposition 13.4.4), and by Proposition 15.2.5 the cohomology of the three vertices of a distinguished triangle forms a long exact sequence.

(1) Let $s: X' \rightarrow X$ be a quasi-isomorphism and $f: Y \rightarrow X$. Complete s to a triangle $X' \rightarrow X \xrightarrow{p} C \rightarrow X'[1]$, so C is acyclic, and complete $pf: Y \rightarrow C$ to a triangle $Y' \xrightarrow{t} Y \xrightarrow{pf} C \rightarrow Y'[1]$ (using (TR1) and (TR2)). Then t is a quasi-isomorphism since C is acyclic, and (TR3) applied to the rotated triangles gives $g: Y' \rightarrow X'$ with $sg = ft$. The dual statement is proved in the same way.

(2) Suppose $s(f - g) = 0$ with $s: Y \rightarrow Y'$ a quasi-isomorphism. Complete s to a triangle $C[-1] \xrightarrow{k} Y \xrightarrow{s} Y' \rightarrow C$ with C acyclic. By Proposition 15.2.5, $f - g = k\varphi$ for some $\varphi: X \rightarrow C[-1]$. Complete φ to a triangle $X' \xrightarrow{t} X \xrightarrow{\varphi} C[-1] \rightarrow X'[1]$; then t is a quasi-isomorphism and $(f - g)t = k\varphi t = 0$. Dually. $\square$

Definition 15.3.4. The *derived category* is $D(\mathcal{A}) = K(\mathcal{A})[\text{Qis}^{-1}]$, and $D^+(\mathcal{A}) = K^+(\mathcal{A})[\text{Qis}^{-1}]$. By Theorem 15.3.2 and Proposition 15.3.3 they are triangulated, H^n factors through them, and a morphism in $D(\mathcal{A})$ is an isomorphism if and only if it induces isomorphisms on all H^n .

Remark 15.3.5. The morphism classes of $D(\mathcal{A})$ need not be small sets for an arbitrary locally small $\mathcal{A}$, since roofs involve all complexes. For $\mathcal{A}$ with enough injectives this problem disappears for $D^+(\mathcal{A})$ by Corollary 15.4.2, which computes morphisms using a single complex of injectives.

15.4 Injective resolutions of complexes

Lemma 15.4.1. *Let X be an acyclic complex and I a bounded below complex of injective objects. Every morphism of complexes $f: X \rightarrow I$ is homotopic to zero.*

Proof. Let $I^n = 0$ for $n < n_0$; put $h^n = 0: X^n \rightarrow I^{n-1}$ for $n \leq n_0$. Suppose h^k is defined for $k \leq n$ with $f^{k-1} = dh^{k-1} + h^k d$ for $k \leq n$. Let $g = f^n - dh^n: X^n \rightarrow I^n$. Then $gd = df^{n-1} - dh^n d = df^{n-1} - d(f^{n-1} - dh^{n-1}) = 0$, so g vanishes on $\text{im}(X^{n-1} \rightarrow X^n) = \ker(X^n \rightarrow X^{n+1})$. Hence g factors through $X^n/\ker \cong \text{im}(X^n \rightarrow X^{n+1}) \subseteq X^{n+1}$ and extends, by injectivity of I^n , to $h^{n+1}: X^{n+1} \rightarrow I^n$ with $h^{n+1}d = g$, that is, $f^n = dh^n + h^{n+1}d$. $\square$

Corollary 15.4.2. *Let I be a bounded below complex of injectives. For every quasi-isomorphism $s: X' \rightarrow X$, composition with s is a bijection $\text{Hom}_{K(\mathcal{A})}(X, I) \rightarrow \text{Hom}_{K(\mathcal{A})}(X', I)$. Consequently $\text{Hom}_{D(\mathcal{A})}(X, I) = \text{Hom}_{K(\mathcal{A})}(X, I)$, and every quasi-isomorphism $I \rightarrow I'$ between bounded below complexes of injectives is a homotopy equivalence.*

Proof. Complete s to a distinguished triangle $X' \rightarrow X \rightarrow C \rightarrow X'[1]$ with C acyclic. By Lemma 15.4.1, $\text{Hom}_K(C, I) = \text{Hom}_K(C[-1], I) = 0$ ($C[-1]$ is also acyclic), and the long exact sequence of Proposition 15.2.5 gives the bijection. For the second statement, we use the description of morphisms in Theorem 15.3.2. The map $Q: \text{Hom}_K(X, I) \rightarrow \text{Hom}_D(X, I)$ is

surjective: a roof $X \xleftarrow{s} X' \xrightarrow{f} I$ has $f = f's$ for some $f': X \rightarrow I$, and then $Q(f)Q(s)^{-1} = Q(f')$. It is injective: if $Q(f_1) = Q(f_2)$, then $f_1 t = f_2 t$ for a quasi-isomorphism t , so $f_1 = f_2$ by the bijection just proved. For the last, a quasi-isomorphism $I \rightarrow I'$ is an isomorphism in $D(\mathcal{A})$, hence by the second statement an isomorphism in $K(\mathcal{A})$. $\square$

Proposition 15.4.3. *If $\mathcal{A}$ has enough injectives, every bounded below complex X admits a quasi-isomorphism $X \rightarrow I$ to a bounded below complex of injectives. Hence $K^+(\text{Inj } \mathcal{A}) \rightarrow D^+(\mathcal{A})$ is an equivalence of categories.*

Proof. Shifting, assume $X^n = 0$ for $n < 0$. Let $J^{\bullet,\bullet}$ be a Cartan–Eilenberg resolution of X (Lemma 14.6.2) and $I = \text{Tot}(J)$, a bounded below complex of injectives (each I^n is a finite sum). The augmentations $X^p \rightarrow J^{p,0}$ define a morphism $X \rightarrow I$. For each p the column $0 \rightarrow X^p \rightarrow J^{p,0} \rightarrow J^{p,1} \rightarrow \dots$ is exact, so by Lemma 13.7.2 (applied with the roles of rows and columns exchanged) the map $X \rightarrow I$ is a quasi-isomorphism. (The total complex of the transposed double complex is isomorphic to I by the signs $(-1)^{pq}$, and the compatibility of $X^p \rightarrow J^{p,0}$ with the differentials holds because the column differential kills the image of X^p .) The functor $K^+(\text{Inj } \mathcal{A}) \rightarrow D^+(\mathcal{A})$ is fully faithful by Corollary 15.4.2 and essentially surjective by the first statement. $\square$

Definition 15.4.4. Let $\mathcal{A}$ have enough injectives and $F: \mathcal{A} \rightarrow \mathcal{B}$ be an additive functor to an abelian category. For each bounded below complex X choose a quasi-isomorphism $X \rightarrow I_X$ as in Proposition 15.4.3. The *right derived functor* is $RF: D^+(\mathcal{A}) \rightarrow D^+(\mathcal{B})$, $RF(X) = F(I_X)$ (with F applied termwise), on morphisms via $\text{Hom}_D(X, Y) \cong \text{Hom}_K(I_X, I_Y)$. For $X \in D^+(\mathcal{A})$ and $n \in \mathbb{Z}$, $R^n F(X) = H^n(RF(X))$ is the n -th *hypercohomology* of F at X .

Proposition 15.4.5. *RF is an exact functor of triangulated categories, independent of the choices up to canonical isomorphism. If F is left exact and $X \in \mathcal{A}$ is viewed as a complex in degree 0, then $H^n(RF(X)) = R^n F(X)$ as in Definition 13.6.4. For $X, Y \in \mathcal{A}$ and $n \in \mathbb{Z}$, $\text{Hom}_D(\mathcal{A})(X, Y[n]) \cong R^n \text{Hom}(X, -)(Y)$; in particular $\text{Hom}_D(R\text{-Mod})(M, N[n]) \cong \text{Ext}_R^n(M, N)$ for R -modules M, N .*

Proof. Independence of the choices follows from uniqueness of I_X up to unique isomorphism in $K(\mathcal{A})$ (Corollary 15.4.2). For exactness, a distinguished triangle of $D^+(\mathcal{A})$ is isomorphic to one of the form $X \xrightarrow{g} Y \rightarrow \text{Cone}(g) \rightarrow X[1]$ with g a morphism of bounded below complexes. By Corollary 15.4.2 there is $g_I: I_X \rightarrow I_Y$ with $g_I \iota_X = \iota_Y g$ in $K(\mathcal{A})$, where ι denotes the resolutions, and (TR3) in $K(\mathcal{A})$ gives a morphism of triangles (ι_X, ι_Y, w) from the triangle of g to that of g_I . By Proposition 15.2.5, the five lemma (Lemma 13.4.2) shows that w is a quasi-isomorphism. So the triangle is isomorphic in $D(\mathcal{A})$ to $I_X \rightarrow I_Y \rightarrow \text{Cone}(g_I) \rightarrow I_X[1]$, whose vertices are bounded below complexes of injectives, and RF sends it to $F(I_X) \rightarrow F(I_Y) \rightarrow F(\text{Cone}(g_I)) \rightarrow F(I_X)[1]$. Since F is additive and applied termwise, $F(\text{Cone}(g_I)) = \text{Cone}(F(g_I))$ with the corresponding maps, so this triangle is distinguished.

For $X \in \mathcal{A}$, an injective resolution is a quasi-isomorphism $X \rightarrow I$, so $H^n(RF(X)) = H^n(F(I)) = R^n F(X)$. Finally $\text{Hom}_D(X, Y[n]) = \text{Hom}_K(X, I_Y[n])$ by Corollary 15.4.2. A morphism of complexes $X \rightarrow I_Y[n]$ is a morphism $\varphi: X \rightarrow I_Y^n$ with $d\varphi = 0$, that is, an element of $\ker(\text{Hom}(X, I_Y^n) \rightarrow \text{Hom}(X, I_Y^{n+1}))$, and it is homotopic to zero if and only if $\varphi = dh$ for some $h: X \rightarrow I_Y^{n-1}$. So $\text{Hom}_K(X, I_Y[n]) = H^n(\text{Hom}(X, I_Y)) = R^n \text{Hom}(X, -)(Y)$. For modules this is $\text{Ext}_R^n(M, N)$ by Definition 13.8.1, as $R\text{-Mod}$ has enough injectives (Theorem 13.5.4). $\square$

Lemma 15.4.6. *Let $\mathcal{B}$ have enough injectives, $G: \mathcal{B} \rightarrow \mathcal{C}$ be left exact, and Y a bounded below acyclic complex of G -acyclic objects (Proposition 13.6.6). Then $G(Y)$ is acyclic. Consequently, if Y is any bounded below complex of G -acyclic objects, then $RG(Y) \cong G(Y)$ in $D^+(\mathcal{C})$.*

Proof. Let $Z^n = \ker(Y^n \rightarrow Y^{n+1})$. Acyclicity gives short exact sequences $0 \rightarrow Z^n \rightarrow Y^n \rightarrow Z^{n+1} \rightarrow 0$, and $Z^n = 0$ for $n \ll 0$. By induction and the long exact sequence of Theorem 13.6.5, all Z^n are G -acyclic, so G preserves the exactness of these sequences ($R^1G(Z^n) = 0$). Since $G(Z^{n+1}) \rightarrow G(Y^{n+1})$ is a monomorphism, the kernel of $G(Y^n) \rightarrow G(Y^{n+1})$ is $G(Z^n)$, which is the image of $G(Y^{n-1}) \rightarrow G(Z^n) \rightarrow G(Y^n)$; so $G(Y)$ is acyclic. For the second statement, let $Y \rightarrow I$ be a quasi-isomorphism to a complex of injectives; its cone is a bounded below acyclic complex (Proposition 13.4.4) of G -acyclic objects (termwise $Y^{n+1} \oplus I^n$, and R^kG is additive). By the first statement $G(\text{Cone}) = \text{Cone}(G(Y) \rightarrow G(I))$ is acyclic, so $G(Y) \rightarrow G(I) = RG(Y)$ is a quasi-isomorphism. $\square$

Theorem 15.4.7. *Let $F: \mathcal{A} \rightarrow \mathcal{B}$ and $G: \mathcal{B} \rightarrow \mathcal{C}$ be left exact, with $\mathcal{A}$ and $\mathcal{B}$ having enough injectives, and suppose F sends injectives to G -acyclic objects. Then $R(G \circ F) \cong RG \circ RF$.*

Proof. For $X \in D^+(\mathcal{A})$ with resolution $X \rightarrow I_X$, $RF(X) = F(I_X)$ is a bounded below complex of G -acyclic objects, so $RG(RF(X)) \cong G(F(I_X)) = R(GF)(X)$ by Lemma 15.4.6. $\square$

Proposition 15.4.8 (Hypercohomology spectral sequences). *Let F be left exact, $\mathcal{A}$ with enough injectives, and X a complex with $X^n = 0$ for $n < 0$. There are convergent first-quadrant spectral sequences*

$$'E_1^{p,q} = R^qF(X^p) \Rightarrow R^{p+q}F(X), \quad ''E_2^{p,q} = R^pF(H^q(X)) \Rightarrow R^{p+q}F(X).$$

Proof. Let J be a Cartan–Eilenberg resolution of X ; by the proof of Proposition 15.4.3, $RF(X) = F(\text{Tot}(J)) = \text{Tot}(F(J))$. Filtering by columns as in the proof of Theorem 14.4.1, $'E_1^{p,q} = H_v^q(F(J^{p,\bullet})) = R^qF(X^p)$, since $J^{p,\bullet}$ is an injective resolution of X^p . Filtering by rows, the rows of J are split (second filtration in the proof of Theorem 14.6.3), so $H_h^q(F(J)) = F(H_h^q(J))$, and $H_h^q(J)^\bullet$ is an injective resolution of $H^q(X)$; hence $''E_2^{p,q} = H^p(F(H_h^q(J))) = R^pF(H^q(X))$. Convergence is Theorem 14.4.1. $\square$

15.5 t -structures

Definition 15.5.1. For a complex X and $n \in \mathbb{Z}$ the *truncations* are

$$\tau_{\leq n}X = (\cdots \rightarrow X^{n-1} \rightarrow Z^n(X) \rightarrow 0 \rightarrow \cdots), \quad \tau_{\geq n}X = (\cdots \rightarrow 0 \rightarrow X^n/B^n(X) \rightarrow X^{n+1} \rightarrow \cdots).$$

Proposition 15.5.2. *There are natural morphisms $\tau_{\leq n}X \rightarrow X \rightarrow \tau_{\geq n}X$, and $H^k(\tau_{\leq n}X) = H^k(X)$ for $k \leq n$ and 0 otherwise, while $H^k(\tau_{\geq n}X) = H^k(X)$ for $k \geq n$ and 0 otherwise. There is a distinguished triangle $\tau_{\leq n}X \rightarrow X \rightarrow \tau_{\geq n+1}X \rightarrow (\tau_{\leq n}X)[1]$ in $D(\mathcal{A})$.*

Proof. The cohomology computations are immediate from the definitions. The morphism $X \rightarrow \tau_{\geq n+1}X$ is surjective with kernel the complex $K = (\cdots \rightarrow X^{n-1} \rightarrow X^n \rightarrow B^{n+1}(X) \rightarrow 0 \rightarrow \cdots)$, and $\tau_{\leq n}X \rightarrow K$ is a quasi-isomorphism (both have cohomology $H^k(X)$ for $k \leq n$ and zero above, $H^{n+1}(K) = 0$ as $X^n \rightarrow B^{n+1}$ is onto). Finally, a short exact sequence of complexes $0 \rightarrow K \xrightarrow{\iota} X \xrightarrow{\pi} Q \rightarrow 0$ gives a distinguished triangle $K \rightarrow X \rightarrow Q \rightarrow K[1]$ in $D(\mathcal{A})$: the map $\text{Cone}(\iota) \rightarrow Q$, $(a, b) \mapsto \pi(b)$, is a surjective morphism of complexes whose kernel $K^{n+1} \oplus K^n$ is $\text{Cone}(\text{id}_K)$, which is contractible (proof of Theorem 15.2.4); by the long exact sequence (Theorem 13.4.3) it is a quasi-isomorphism, so the triangle of ι is isomorphic in $D(\mathcal{A})$ to $K \rightarrow X \rightarrow Q \rightarrow K[1]$. Apply this to $K \rightarrow X \rightarrow \tau_{\geq n+1}X$ and replace K by $\tau_{\leq n}X$. $\square$

Remark 15.5.3. The full subcategories $D^{\leq 0}$ and $D^{\geq 0}$ of complexes with cohomology in degrees ≤ 0 , respectively ≥ 0 , form the *standard t -structure* on $D(\mathcal{A})$: $\text{Hom}(D^{\leq 0}, D^{\geq 1}) = 0$, $D^{\leq 0}[1] \subseteq D^{\leq 0}$, and every object sits in a triangle as in Proposition 15.5.2. Its *heart* $D^{\leq 0} \cap D^{\geq 0}$ is

equivalent to $\mathcal{A}$. Abstract t -structures, and in particular the perverse t -structure on derived categories of sheaves, were introduced in [BBD82]; perverse sheaves and M. Saito's mixed Hodge modules are the modern setting for the Hodge theory of singular varieties.

15.6 Filtered complexes and strictness

Definition 15.6.1. A *filtered object* (A, F) is an object with a decreasing filtration $F^p A$, *finite* if $F^a A = A$ and $F^b A = 0$ for some a, b . A morphism $f: (A, F) \rightarrow (B, F)$ is *filtered* if $f(F^p A) \subseteq F^p B$, and *strict* if $f(F^p A) = f(A) \cap F^p B$ for all p . A morphism of filtered complexes is a *filtered quasi-isomorphism* if it induces quasi-isomorphisms on all gr^p .

Counterexample 15.6.2. Filtered morphisms need not be strict, and the category of filtered vector spaces is not abelian. Let $A = B = k$ with $F^0 A = A$, $F^1 A = 0$ and $F^0 B = F^1 B = B$, $F^2 B = 0$. The identity $A \rightarrow B$ is filtered and bijective, but not strict ($f(F^1 A) = 0 \neq F^1 B \cap B = k$), and it is not an isomorphism of filtered objects, because its inverse is not filtered. Its kernel and cokernel (with the induced filtrations) are zero, so it is a monomorphism and an epimorphism; by Lemma 13.2.3 the category is not abelian.

Proposition 15.6.3 (Deligne). *Let K be a complex with a filtration that is finite in each degree. The spectral sequence of Theorem 14.3.4 degenerates at E_1 (that is, $d_r = 0$ for all $r \geq 1$) if and only if every differential $d: K^n \rightarrow K^{n+1}$ is strict. In this case $H^n(\text{gr}^p K) \cong \text{gr}^p H^n(K)$.*

Proof. We use the notation of Definition 14.3.2; strictness of d means $F^p \cap d(K) = d(F^p)$.

Suppose d is strict. Let $x \in Z_r^p$ with $r \geq 1$. Then $dx \in F^{p+1} \cap d(K) = d(F^{p+1})$, so $dx = dy$ with $y \in F^{p+1}$. Since $dy = dx \in F^{p+r}$, $y \in Z_{r-1}^{p+1}$, so x and the cocycle $x - y$ have the same class in E_r^p . Hence every class in E_r^p is represented by a cocycle of K , and $d_r = 0$.

Suppose all d_r , $r \geq 1$, vanish. Fix a degree n . We show $F^p \cap d(K^{n-1}) \subseteq d(F^p K^{n-1})$ for all p by descending induction on p (for p large, $F^p K^n = 0$). Let $y = dx \in F^p$ with $x \in F^a K^{n-1}$; since the filtration is finite in degree $n - 1$, it suffices to treat all a , by descending induction on a . If $a \geq p$ there is nothing to show. If $a < p$, then $x \in Z_{p-a}^a$, and $d_{p-a}[x] = [dx] = 0$ in E_{p-a}^p , so $dx \in B_{p-a}^p + Z_{p-a-1}^{p+1}$: $dx = dx' + z$ with $x' \in F^{a+1}$, $dx' \in F^p$, and $z \in F^{p+1}$ with $z = d(x - x') \in d(K)$. By induction on a (as $x' \in F^{a+1}$), $dx' \in d(F^p)$; by induction on p , $z \in d(F^{p+1}) \subseteq d(F^p)$. So $y \in d(F^p)$. The last statement follows from $E_1 = E_\infty$ and Theorem 14.3.4. $\square$

Remark 15.6.4. Localising the homotopy category of filtered complexes at filtered quasi-isomorphisms gives the *filtered derived category* $DF(\mathcal{A})$. Deligne used it to construct mixed Hodge structures on the cohomology of algebraic varieties [Del71]; the degeneration of the Hodge-to-de Rham spectral sequence of a compact Kähler manifold (Chapter 45) is, by Proposition 15.6.3, the strictness of the de Rham differential with respect to the Hodge filtration.

15.7 Exercises

Exercise 15.7.1. Show that in a triangulated category, if (u, v, w) is a morphism of distinguished triangles and two of u, v, w are isomorphisms, so is the third.

Solution. Let (u, v, w) be a morphism from $X \rightarrow Y \rightarrow Z \rightarrow X[1]$ to $X' \rightarrow Y' \rightarrow Z' \rightarrow X'[1]$ with u, v isomorphisms, and let W be any object. Applying $\text{Hom}(W, -)$ and Proposition 15.2.5 gives two long exact sequences of abelian groups connected by the maps induced by u, v ,

$w, u[1], v[1]$. The five lemma (Lemma 13.4.2) shows that $w_*: \text{Hom}(W, Z) \rightarrow \text{Hom}(W, Z')$ is bijective for all W . Taking $W = Z'$ gives $s: Z' \rightarrow Z$ with $ws = \text{id}$; then $w(sw) = w$, and injectivity of w_* for $W = Z$ gives $sw = \text{id}$. The other cases follow by rotating the triangles with (TR2). $\square$

Exercise 15.7.2. Let k be a field. Show that every complex X of k -vector spaces is isomorphic in $D(\text{Vect}_k)$ to the complex $\bigoplus_n H^n(X)[-n]$ with zero differential.

Solution. Every subspace has a complement. Choose complements $X^n = Z^n \oplus C^n$ and $Z^n = B^n \oplus H^n$, where $H^n \cong H^n(X)$. The subcomplex $H = \bigoplus_n H^n$ has zero differential. Choose linear maps $r^n: X^n \rightarrow H^n$ that are the identity on H^n and vanish on $B^n \oplus C^n$. Then $r: X \rightarrow H$ is a morphism of complexes: $r(dx) = 0$ because $dx \in B^{n+1}$. It induces the identity on cohomology, so it is a quasi-isomorphism, hence an isomorphism in $D(\text{Vect}_k)$. $\square$

Exercise 15.7.3. Using Proposition 15.4.5, define the Yoneda product $\text{Ext}^m(Y, Z) \times \text{Ext}^n(X, Y) \rightarrow \text{Ext}^{m+n}(X, Z)$ as composition in $D(\mathcal{A})$ and show that it is associative.

Exercise 15.7.4. Show that $D(\mathcal{A})$ is not abelian unless every short exact sequence in $\mathcal{A}$ splits. (Hint: a monomorphism in a triangulated category is split.)

Solution. Let $f: X \rightarrow Y$ be a monomorphism in a triangulated category, and complete it to a triangle $Z[-1] \xrightarrow{h} X \xrightarrow{f} Y \rightarrow Z$. Then $fh = 0$, so $h = 0$. Applying Proposition 15.2.5 with $\text{Hom}(-, X)$ shows that $\text{Hom}(Y, X) \rightarrow \text{Hom}(X, X)$ is surjective (the next term maps through $h = 0$), so f has a retraction. Dually, epimorphisms have sections.

Suppose $D(\mathcal{A})$ is abelian and let $0 \rightarrow A \rightarrow B \xrightarrow{g} C \rightarrow 0$ be exact in $\mathcal{A}$. As in the proof of Proposition 15.5.2, it gives a distinguished triangle $A \rightarrow B \xrightarrow{g} C \xrightarrow{\delta} A[1]$. Factor $\delta = me$ with $e: C \rightarrow I$ an epimorphism and $m: I \rightarrow A[1]$ a monomorphism. Both split, so I is a direct summand of C and of $A[1]$ in $D(\mathcal{A})$. Cohomology is additive, so $H^k(I)$ vanishes for $k \neq 0$ (as a summand of C) and for $k \neq -1$ (as a summand of $A[1]$). Hence I is acyclic, $I \cong 0$ and $\delta = 0$. By Proposition 15.2.5, the sequence $\text{Hom}(C, B) \rightarrow \text{Hom}(C, C) \rightarrow \text{Hom}(C, A[1])$ is exact and $\text{id}_C \mapsto \delta = 0$, so $g\sigma = \text{id}_C$ for some $\sigma: C \rightarrow B$ in $D(\mathcal{A})$. Applying H^0 gives $g \circ H^0(\sigma) = \text{id}_C$ in $\mathcal{A}$, so the sequence splits. $\square$

Part III

Topology

Chapter 16

General Topology

16.1 Introduction

Topology axiomatises the notions of nearness and continuity. This chapter develops the point-set topology used in the rest of the project: metric spaces for analysis, quotients and compactness for geometry, and paracompactness and partitions of unity, which are the basic tool for passing from local to global constructions on manifolds. The main theorems are Tychonoff's theorem (Theorem 16.6.7), the characterisation of compact metric spaces (Theorem 16.7.4), the Baire category theorem (Theorem 16.7.7), Urysohn's lemma and the Tietze extension theorem (Theorems 16.8.4 and 16.8.5), and the existence of partitions of unity (Theorem 16.10.6). Standard references are [Mun00] and [Eng89].

Prerequisites. Chapter 3 (Set Theory), Chapter 4 (Number Systems), Chapter 5 (Categories) and Chapter 9 (Linear Algebra).

16.2 Topological spaces and continuous maps

Definition 16.2.1. A *topology* on a set X is a set $\mathcal{T}$ of subsets of X , called *open* sets, such that $\emptyset$ and X are open, every union of open sets is open, and every finite intersection of open sets is open. A *topological space* is a set with a topology. A subset $C \subseteq X$ is *closed* if $X \setminus C$ is open. A *neighbourhood* of a point x (or of a subset A) is a set containing an open set containing x (or A). If $\mathcal{T} \subseteq \mathcal{T}'$ are two topologies on X , then $\mathcal{T}'$ is *finer* and $\mathcal{T}$ is *coarser*.

Example 16.2.2. 1. The *discrete* topology $\mathcal{P}(X)$ and the *indiscrete* topology $\{\emptyset, X\}$ are the finest and the coarsest topologies on X .

2. The *cofinite* topology on X consists of $\emptyset$ and the complements of finite sets. The *cocountable* topology consists of $\emptyset$ and the complements of countable sets.

3. The *Sierpiński space* is $\{0, 1\}$ with open sets $\emptyset$, $\{1\}$ and $\{0, 1\}$.

Definition 16.2.3. Let A be a subset of a space X . Its *interior* $\text{int } A$ is the union of all open sets contained in A , its *closure* $\overline{A}$ is the intersection of all closed sets containing A , and its *boundary* is $\partial A = \overline{A} \setminus \text{int } A$. The set A is *dense* if $\overline{A} = X$.

Lemma 16.2.4. Let $A, B \subseteq X$. Then $\text{int } A$ is the largest open subset of A and $\overline{A}$ is the smallest closed set containing A ; $X \setminus \overline{A} = \text{int}(X \setminus A)$; a point x lies in $\overline{A}$ if and only if every open set containing x meets A ; and $\overline{A \cup B} = \overline{A} \cup \overline{B}$.

Proof. The first two statements hold because unions of open sets are open and intersections of closed sets are closed. Taking complements turns closed sets containing A into open sets contained in $X \setminus A$, which gives the third statement. So $x \notin \overline{A}$ if and only if x lies in an open set contained in $X \setminus A$, that is, an open set not meeting A . Finally $\overline{A} \cup \overline{B}$ is closed and contains $A \cup B$, so it contains $\overline{A \cup B}$; the reverse inclusion holds as $\overline{A}, \overline{B} \subseteq \overline{A \cup B}$. $\square$

Definition 16.2.5. A *basis* of a topology $\mathcal{T}$ is a subset $\mathcal{B} \subseteq \mathcal{T}$ such that every open set is a union of members of $\mathcal{B}$. A *subbasis* is a subset $\mathcal{S} \subseteq \mathcal{T}$ such that the finite intersections of members of $\mathcal{S}$ (the empty intersection being X) form a basis.

Lemma 16.2.6. 1. A set $\mathcal{B}$ of subsets of X is a basis of some topology if and only if $\mathcal{B}$ covers X and for all $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$ there is $B_3 \in \mathcal{B}$ with $x \in B_3 \subseteq B_1 \cap B_2$. This topology consists of the unions of members of $\mathcal{B}$.

2. For every set $\mathcal{S}$ of subsets of X there is a coarsest topology containing $\mathcal{S}$, the topology generated by $\mathcal{S}$, and $\mathcal{S}$ is a subbasis of it.

Proof. (1) If $\mathcal{B}$ is a basis, then X is a union of members and $B_1 \cap B_2$ is open, hence a union of members, which gives the two conditions. Conversely, let $\mathcal{T}$ be the set of unions of members of $\mathcal{B}$. It contains $\emptyset$ (the empty union) and X (first condition), and is closed under unions. If $U = \bigcup_i B_i$ and $V = \bigcup_j B'_j$, then $U \cap V = \bigcup_{i,j} B_i \cap B'_j$, and each $B_i \cap B'_j$ is a union of members of $\mathcal{B}$ by the second condition. By induction $\mathcal{T}$ is closed under finite intersections.

(2) The set $\mathcal{B}$ of finite intersections of members of $\mathcal{S}$ contains X and is closed under finite intersections, so it satisfies the conditions of (1). The topology it generates contains $\mathcal{S}$, and every topology containing $\mathcal{S}$ contains $\mathcal{B}$ and all unions of its members. $\square$

Definition 16.2.7. A map $f: X \rightarrow Y$ between spaces is *continuous* if $f^{-1}(V)$ is open for every open $V \subseteq Y$. It is a *homeomorphism* if it is bijective and both f and f^{-1} are continuous. It is *open* (*closed*) if it maps open (closed) sets to open (closed) sets.

Proposition 16.2.8. For a map $f: X \rightarrow Y$ the following are equivalent:

1. f is continuous;
2. $f^{-1}(C)$ is closed for every closed $C \subseteq Y$;
3. $f(\overline{A}) \subseteq \overline{f(A)}$ for every $A \subseteq X$;
4. for every $x \in X$ and every open $V \ni f(x)$ there is an open $U \ni x$ with $f(U) \subseteq V$;
5. $f^{-1}(S)$ is open for every member S of a subbasis of Y .

Compositions of continuous maps are continuous, so spaces and continuous maps form a category **Top**.

Proof. (1) $\Leftrightarrow$ (2) since $f^{-1}(Y \setminus C) = X \setminus f^{-1}(C)$. (2) $\Rightarrow$ (3): $f^{-1}(\overline{f(A)})$ is closed and contains A , hence contains $\overline{A}$. (3) $\Rightarrow$ (2): if C is closed and $A = f^{-1}(C)$, then $f(\overline{A}) \subseteq \overline{f(A)} \subseteq C$, so $\overline{A} \subseteq A$. (1) $\Rightarrow$ (4): take $U = f^{-1}(V)$. (4) $\Rightarrow$ (1): $f^{-1}(V)$ is the union of the open sets U provided by (4) for its points. (1) $\Leftrightarrow$ (5): preimages commute with unions and intersections, and every open set is a union of finite intersections of members of the subbasis. The last statement follows from $(g \circ f)^{-1}(W) = f^{-1}(g^{-1}(W))$. $\square$

Counterexample 16.2.9. A continuous bijection need not be a homeomorphism: the identity map from X with the discrete topology to X with the indiscrete topology is continuous, and its inverse is not if X has at least two points.

Definition 16.2.10. A space X is *Hausdorff* if any two distinct points have disjoint open neighbourhoods. A sequence $(x_n)_{n \in \mathbb{N}}$ in X *converges* to x if every open set containing x contains x_n for all sufficiently large n .

Lemma 16.2.11. *In a Hausdorff space every sequence has at most one limit, and every finite subset is closed.*

Proof. If $x \neq y$ are limits, take disjoint open $U \ni x$ and $V \ni y$; for large n , $x_n \in U \cap V$, a contradiction. For a point x , every $y \neq x$ has an open neighbourhood not containing x , so $X \setminus \{x\}$ is open; finite unions of closed sets are closed. $\square$

Counterexample 16.2.12. The cofinite topology on an infinite set X is not Hausdorff: two nonempty open sets have finite complements, and the union of these complements is not all of X , so they meet. In this space every injective sequence converges to every point, since an open set misses only finitely many of its terms.

16.3 Metric spaces

Definition 16.3.1. A *metric* on a set X is a map $d: X \times X \rightarrow \mathbb{R}$ such that for all $x, y, z \in X$: $d(x, y) \geq 0$ with equality if and only if $x = y$; $d(x, y) = d(y, x)$; and $d(x, z) \leq d(x, y) + d(y, z)$ (the *triangle inequality*). The *open ball* of radius $r > 0$ around x is $B(x, r) = \{y : d(x, y) < r\}$. A set A is *bounded* if $A \subseteq B(x, r)$ for some x and r .

Lemma 16.3.2. *The open balls of a metric space form a basis of a topology, the metric topology: U is open if and only if for every $x \in U$ there is $\varepsilon > 0$ with $B(x, \varepsilon) \subseteq U$. It is Hausdorff. A space whose topology is the metric topology of some metric is metrisable.*

Proof. If $x \in B(a, r) \cap B(b, s)$, let $\varepsilon = \min(r - d(a, x), s - d(b, x)) > 0$. By the triangle inequality $B(x, \varepsilon)$ is contained in both balls, so Lemma 16.2.6 applies; the description of open sets follows. If $x \neq y$ and $r = d(x, y)$, the balls $B(x, r/2)$ and $B(y, r/2)$ are disjoint by the triangle inequality. $\square$

Definition 16.3.3. Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, with absolute value $|\cdot|$. A *norm* on a $\mathbb{K}$ -vector space V is a map $\|\cdot\|: V \rightarrow \mathbb{R}$ such that $\|v\| = 0$ only for $v = 0$, $\|\lambda v\| = |\lambda| \|v\|$ and $\|v + w\| \leq \|v\| + \|w\|$ (so $\|v\| \geq 0$). Then $d(v, w) = \|v - w\|$ is a metric. On $\mathbb{K}^n$ we use the *Euclidean norm* $\|x\|_2 = (\sum_i |x_i|^2)^{1/2}$, the norm of the standard inner product (a norm by Proposition 9.9.2), and the *maximum norm* $\|x\|_\infty = \max_i |x_i|$.

Lemma 16.3.4. *If d, d' are metrics on X with $c d \leq d' \leq C d$ for constants $c, C > 0$, they define the same topology. In particular, since $\|x\|_\infty \leq \|x\|_2 \leq \sqrt{n} \|x\|_\infty$, the two norms on $\mathbb{K}^n$ define the same topology. On $\mathbb{R}$ it is the topology with basis the open intervals, the standard topology.*

Proof. $B_{d'}(x, c\varepsilon) \subseteq B_d(x, \varepsilon)$ and $B_d(x, \varepsilon/C) \subseteq B_{d'}(x, \varepsilon)$, so the criterion of Lemma 16.3.2 gives the same open sets. The norm inequalities hold because $\max_i |x_i|^2 \leq \sum_i |x_i|^2 \leq n \max_i |x_i|^2$. On $\mathbb{R}$ the balls are the bounded open intervals $(a - r, a + r)$, and every bounded open interval is such a ball. $\square$

Proposition 16.3.5. *Let $f: X \rightarrow Y$ be a map of metric spaces.*

1. *f is continuous if and only if for every x and $\varepsilon > 0$ there is $\delta > 0$ with $d(f(x), f(x')) < \varepsilon$ whenever $d(x, x') < \delta$.*
2. *For $A \subseteq X$, a point x lies in $\overline{A}$ if and only if it is the limit of a sequence in A .*

3. f is continuous if and only if $f(x_n) \rightarrow f(x)$ whenever $x_n \rightarrow x$.

Proof. (1) is Proposition 16.2.8(4) with balls. (2) If $x \in \overline{A}$, choose $a_n \in A \cap B(x, 1/(n+1))$ by Lemma 16.2.4; then $a_n \rightarrow x$. Conversely, if $a_n \rightarrow x$, every open set containing x contains some a_n . (3) If f is continuous and $x_n \rightarrow x$, every open $V \ni f(x)$ has open preimage containing x , hence containing x_n for large n . If f is not continuous at x , there are $\varepsilon > 0$ and x_n with $d(x, x_n) < 1/(n+1)$ and $d(f(x), f(x_n)) \geq \varepsilon$; then $x_n \rightarrow x$ but $f(x_n) \not\rightarrow f(x)$. $\square$

Counterexample 16.3.6. In general spaces sequences do not detect closures. Give $\mathbb{R}$ the cocountable topology. If $x_n \rightarrow x$, the set $C = \{x_n : x_n \neq x\}$ is countable, so $\mathbb{R} \setminus C$ is an open set containing x , and hence $x_n = x$ for large n . Now $A = \mathbb{R} \setminus \{0\}$ is uncountable (Proposition 4.8.6), so the only closed set containing it is $\mathbb{R}$ and $0 \in \overline{A}$; but no sequence in A converges to 0. (Nets or filters replace sequences in general [Eng89, Chapter 1].)

Proposition 16.3.7. Addition and multiplication $\mathbb{K} \times \mathbb{K} \rightarrow \mathbb{K}$ are continuous for the metric $\|(x, y) - (a, b)\|_\infty$, and so is $x \mapsto 1/x$ on $\mathbb{K} \setminus \{0\}$.

Proof. We have $|(x+y) - (a+b)| \leq |x-a| + |y-b|$, and if $|x-a|, |y-b| < \delta \leq 1$ then $|xy - ab| \leq |x||y-b| + |b||x-a| \leq (|a|+1+|b|)\delta$. If $|x-a| < |a|/2$ then $|x| > |a|/2$ and $|1/x - 1/a| = |x-a|/(|x||a|) \leq 2|x-a|/|a|^2$. In each case the criterion of Proposition 16.3.5(1) holds. $\square$

Proposition 16.3.8. Let X be a space, (Y, d) a metric space and $f_n: X \rightarrow Y$ continuous maps converging uniformly to f : for every $\varepsilon > 0$ there is N with $d(f_n(x), f(x)) < \varepsilon$ for all $n \geq N$ and all x . Then f is continuous.

Proof. Let $x \in X$ and $\varepsilon > 0$. Choose n with $d(f_n, f) < \varepsilon/3$ everywhere, and an open $U \ni x$ with $d(f_n(x), f_n(x')) < \varepsilon/3$ for $x' \in U$. Then $d(f(x), f(x')) < \varepsilon$ for $x' \in U$, and Proposition 16.2.8(4) applies. $\square$

Definition 16.3.9. A space is *first countable* if every point x has countably many open neighbourhoods such that every open set containing x contains one of them; *second countable* if its topology has a countable basis; *separable* if it has a countable dense subset; and *Lindelöf* if every open cover has a countable subcover.

Proposition 16.3.10. Metric spaces are first countable. Second countable spaces are separable and Lindelöf. A separable metric space is second countable. In particular $\mathbb{R}^n$ and $\mathbb{C}^n$ are second countable.

Proof. The balls $B(x, 1/(n+1))$ show first countability. Let $\mathcal{B}$ be a countable basis. Choosing a point in each nonempty member gives a countable set meeting every nonempty open set, hence dense by Lemma 16.2.4. Given an open cover $\mathcal{U}$, for each $B \in \mathcal{B}$ contained in some member of $\mathcal{U}$ choose one such member U_B ; these countably many sets cover, because a point $x \in U \in \mathcal{U}$ lies in some $B \in \mathcal{B}$ with $B \subseteq U$, and then $x \in U_B$. If D is a countable dense subset of a metric space, the balls $B(q, 1/n)$ with $q \in D, n \geq 1$ form a basis: if $x \in U$ with U open, choose n with $B(x, 2/n) \subseteq U$ and $q \in D \cap B(x, 1/n)$; then $x \in B(q, 1/n) \subseteq B(x, 2/n) \subseteq U$. Finally $\mathbb{Q}^n$ and $(\mathbb{Q} + i\mathbb{Q})^n$ are countable (Corollary 3.6.13) and dense, by density of $\mathbb{Q}$ in $\mathbb{R}$ (Proposition 4.7.3). $\square$

Counterexample 16.3.11 (The Sorgenfrey line). The half-open intervals $[a, b)$ form a basis of a topology on $\mathbb{R}$, the *Sorgenfrey line*, finer than the Euclidean topology. It is first countable (the sets $[x, x+1/n)$ form a neighbourhood basis at x) and separable ($\mathbb{Q}$ is dense, as every $[a, b)$ contains a rational number). It is not second countable: if $\mathcal{B}$ is a basis, then for each x some $B_x \in \mathcal{B}$ satisfies $x \in B_x \subseteq [x, x+1)$, so $x = \min B_x$, and $x \mapsto B_x$ is injective from the uncountable set $\mathbb{R}$ to $\mathcal{B}$. By Proposition 16.3.10, it is therefore not metrisable: separable metric spaces are second countable.

16.4 Subspaces, products and quotients

Definition 16.4.1. Let X be a set. For a family of maps $f_i: X \rightarrow X_i$ into spaces, the *initial topology* is the topology generated by the sets $f_i^{-1}(U)$, $U \subseteq X_i$ open. For a family of maps $g_i: X_i \rightarrow X$ from spaces, the *final topology* consists of the sets $U \subseteq X$ such that $g_i^{-1}(U)$ is open for all i .

Proposition 16.4.2. With the initial topology, a map $h: Z \rightarrow X$ from a space is continuous if and only if every $f_i \circ h$ is continuous. With the final topology, a map $h: X \rightarrow Z$ to a space is continuous if and only if every $h \circ g_i$ is continuous.

Proof. The final topology is a topology because preimages commute with unions and intersections. For the initial topology, $h^{-1}(f_i^{-1}(U)) = (f_i h)^{-1}(U)$ and Proposition 16.2.8(5) give the claim. For the final topology, $h^{-1}(W)$ is open if and only if $g_i^{-1}(h^{-1}(W)) = (h g_i)^{-1}(W)$ is open for all i . $\square$

Definition 16.4.3. The *subspace topology* on $A \subseteq X$ is the initial topology for the inclusion: its open sets are the sets $U \cap A$ with U open in X , and its closed sets are the sets $C \cap A$ with C closed in X . An *embedding* is a map $f: X \rightarrow Y$ that is a homeomorphism onto $f(X)$ with the subspace topology.

Lemma 16.4.4 (Pasting lemma). Let $f: X \rightarrow Y$ be a map and suppose $X = \bigcup_i A_i$, where either all A_i are open, or there are finitely many A_i and all are closed. If every restriction $f|_{A_i}$ is continuous, then f is continuous.

Proof. For $V \subseteq Y$ we have $f^{-1}(V) = \bigcup_i (f|_{A_i})^{-1}(V)$. In the first case, for V open each $(f|_{A_i})^{-1}(V)$ is open in the open set A_i , hence open in X . In the second case, for V closed each term is closed in the closed set A_i , hence closed in X , and a finite union of closed sets is closed. $\square$

Definition 16.4.5. The *product* of a family of spaces $(X_i)_{i \in I}$ is the set $\prod_i X_i$ with the initial topology for the projections $p_i: \prod_j X_j \rightarrow X_i$, the *product topology*. The *disjoint union* $\coprod_i X_i$ carries the final topology for the inclusions.

Proposition 16.4.6. 1. The sets $\prod_i U_i$, with $U_i \subseteq X_i$ open and $U_i = X_i$ for all but finitely many i , form a basis of the product topology.

2. $\prod_i X_i$ with the projections is a product in **Top**, and $\coprod_i X_i$ with the inclusions is a coproduct, in the sense of Example 5.8.3.
3. The projections are open maps. A product of Hausdorff spaces is Hausdorff, and so is a subspace of a Hausdorff space.
4. The product topology on $\mathbb{K}^n = \mathbb{K} \times \cdots \times \mathbb{K}$ is the topology of the Euclidean norm.
5. For every space X , sums and products of continuous functions $X \rightarrow \mathbb{K}$ are continuous, and so is $1/g$ for continuous $g: X \rightarrow \mathbb{K} \setminus \{0\}$.

Proof. (1) These sets are the finite intersections of the subbasic sets $p_i^{-1}(U_i)$. (2) By Proposition 16.4.2, a map $h: Z \rightarrow \prod_i X_i$ is continuous if and only if all its components $p_i h$ are; so continuous families $Z \rightarrow X_i$ correspond bijectively to continuous maps $Z \rightarrow \prod_i X_i$. Dually for disjoint unions. (3) The image of a basic open set under p_i is U_i , and images commute with unions. If $x \neq y$ in the product, then $x_i \neq y_i$ for some i , and preimages under p_i of disjoint neighbourhoods of x_i, y_i separate x, y . For subspaces intersect with A . (4) By Lemma 16.3.4 we may use $\|\cdot\|_\infty$, whose balls are products of balls; and every basic open set $\prod_i U_i$ contains,

around each of its points a , the $\|\cdot\|_\infty$ -ball of radius $\min_i \varepsilon_i$, where $B(a_i, \varepsilon_i) \subseteq U_i$. (5) If f, g are continuous, then $(f, g): X \rightarrow \mathbb{K}^2$ is continuous by (2), and $\mathbb{K}^2$ carries the metric topology of $\|\cdot\|_\infty$ by (4); compose with the maps of Proposition 16.3.7. $\square$

Counterexample 16.4.7. The *box topology* on $\prod_i X_i$, with basis all products of open sets, is finer than the product topology and does not have the universal property of the product. The map $\mathbb{R} \rightarrow \mathbb{R}^\mathbb{N}$, $t \mapsto (t, t, \dots)$, has continuous components, but the preimage of the box $\prod_n (-1/(n+1), 1/(n+1))$ is $\{0\}$, which is not open.

Definition 16.4.8. A surjective map $q: X \rightarrow Y$ from a space is a *quotient map* if Y carries the final topology: $V \subseteq Y$ is open if and only if $q^{-1}(V)$ is open. For an equivalence relation $\sim$ on X , the *quotient space* $X/\sim$ is the set of classes with the final topology for the projection. For $A \subseteq X$, X/A denotes the quotient identifying all points of A with each other.

Proposition 16.4.9. Let $q: X \rightarrow Y$ be a quotient map. A map $h: Y \rightarrow Z$ is continuous if and only if $h \circ q$ is continuous; hence every continuous map $g: X \rightarrow Z$ that is constant on the fibres of q factors uniquely as $g = \bar{g} \circ q$ with $\bar{g}$ continuous. A continuous surjection that is open or closed is a quotient map.

Proof. The first statement is Proposition 16.4.2; $\bar{g}$ is well defined as a map of sets because g is constant on fibres. If $f: X \rightarrow Y$ is a continuous open surjection and $f^{-1}(V)$ is open, then $V = f(f^{-1}(V))$ is open. The closed case is the same with closed sets, using $f^{-1}(Y \setminus V) = X \setminus f^{-1}(V)$. $\square$

- Example 16.4.10.**
1. The *unit sphere* $S^n = \{x \in \mathbb{R}^{n+1} : \|x\|_2 = 1\}$ is a subspace of $\mathbb{R}^{n+1}$, and the *real projective space* is $\mathbb{RP}^n = S^n/(x \sim -x)$.
 2. The *cylinder* and the *torus* are the quotients of $[0, 1]^2$ identifying $(0, t) \sim (1, t)$, respectively identifying in addition $(s, 0) \sim (s, 1)$. The *cone* on X is $(X \times [0, 1])/(X \times \{1\})$.
 3. Two spaces X, Y with a subspace $A \subseteq X$ and a continuous map $f: A \rightarrow Y$ give the *adjunction space* $X \cup_f Y = (X \sqcup Y)/(a \sim f(a))$, the space obtained by *attaching* X to Y along f . These are the building blocks of CW complexes (Chapter 17).

Counterexample 16.4.11. Quotients of Hausdorff spaces need not be Hausdorff.

1. Let $x \sim y$ on $\mathbb{R}$ if $x - y \in \mathbb{Q}$. If $V \subseteq \mathbb{R}/\sim$ is open and nonempty, then $W = q^{-1}(V)$ is a nonempty open set with $W + \mathbb{Q} = W$. Its complement is closed and invariant under translation by $\mathbb{Q}$; if it contained a point y it would contain the dense set $y + \mathbb{Q}$, hence be all of $\mathbb{R}$. So $W = \mathbb{R}$, and the quotient is indiscrete.
2. (The line with two origins.) Let $X = (\mathbb{R} \times \{1, 2\})/\sim$ with $(t, 1) \sim (t, 2)$ for $t \neq 0$. The projection q is open, since for U open $q^{-1}(q(U))$ is the union of U and the swap of $U \setminus (\{0\} \times \{1, 2\})$; so q restricts to homeomorphisms of each copy of $\mathbb{R}$ onto open subsets, and every point of X has an open neighbourhood homeomorphic to $\mathbb{R}$. But the two origins $q(0, 1)$ and $q(0, 2)$ have no disjoint neighbourhoods, since preimages of neighbourhoods contain $(-\varepsilon, \varepsilon) \times \{1\}$ and $(-\varepsilon, \varepsilon) \times \{2\}$, whose images share $q(\varepsilon/2, 1)$.

16.5 Connectedness

Definition 16.5.1. A space X is *connected* if it is nonempty and is not the union of two disjoint nonempty open subsets. A subset is connected if it is connected as a subspace.

Lemma 16.5.2. *For a nonempty space X the following are equivalent: X is connected; the only subsets of X that are both open and closed are $\emptyset$ and X ; every continuous map from X to the discrete space $\{0, 1\}$ is constant.*

Proof. A decomposition $X = U \sqcup V$ into disjoint open sets is the same as a set $U = X \setminus V$ that is open and closed, and the same as the continuous map equal to 0 on U and 1 on V . $\square$

Theorem 16.5.3. *A nonempty subset $I \subseteq \mathbb{R}$ is connected if and only if it is an interval: $a, b \in I$ and $a < c < b$ imply $c \in I$.*

Proof. If I is not an interval, with $a, b \in I$, $a < c < b$ and $c \notin I$, then $I \cap (-\infty, c)$ and $I \cap (c, \infty)$ disconnect I .

Let I be an interval and suppose $I = U \sqcup V$ with U, V nonempty and open in I . Choose $a \in U$ and $b \in V$; exchanging the roles of U and V if necessary, $a < b$. Then $[a, b] \subseteq I$. Let $s = \sup(U \cap [a, b])$, which exists by completeness of $\mathbb{R}$, and $s \in [a, b] \subseteq I$. Suppose $s \in U$. Then $s < b$, and since U is open in I there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap I \subseteq U$; the points of $(s, \min(s + \varepsilon, b))$ lie in $U \cap [a, b]$, contradicting the definition of s . Suppose $s \in V$. Then $s > a$, and there is $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap I \subseteq V$. So $U \cap [a, b] \subseteq [a, \max(a, s - \varepsilon)]$, whose supremum is smaller than s , again a contradiction. $\square$

Proposition 16.5.4. 1. *Continuous images of connected spaces are connected.*

2. *If (A_i) is a family of connected subsets of X with a common point, then $\bigcup_i A_i$ is connected.*

3. *If $A \subseteq X$ is connected and $A \subseteq B \subseteq \overline{A}$, then B is connected.*

4. *A product of connected spaces is connected.*

Proof. We use Lemma 16.5.2. (1) If $f: X \rightarrow Y$ is surjective and $h: Y \rightarrow \{0, 1\}$ is continuous, then hf is constant, so h is. (2) A continuous map $\bigcup_i A_i \rightarrow \{0, 1\}$ is constant on each A_i , with the value it takes at the common point. (3) A continuous $h: B \rightarrow \{0, 1\}$ is constant, say equal to c , on A ; then $h^{-1}(c)$ is closed in B and contains A , so it contains the closure of A in B , which is B . (4) For two factors, fix (x_0, y_0) . For each y , the set $(X \times \{y\}) \cup (\{x_0\} \times Y)$ is connected by (2), both pieces being homeomorphic to X or Y and sharing (x_0, y) . All these sets contain (x_0, y_0) and cover $X \times Y$, so (2) applies; by induction finite products are connected. For an arbitrary product, fix a point a and let D be the set of points that differ from a in finitely many coordinates. For each finite set F of indices, the points agreeing with a outside F form a subspace homeomorphic to $\prod_{i \in F} X_i$, which is connected and contains a ; so D is connected by (2). Every basic open set of Proposition 16.4.6(1) meets D , so D is dense and the product is connected by (3). $\square$

Corollary 16.5.5 (Intermediate value theorem). *If X is connected and $f: X \rightarrow \mathbb{R}$ is continuous, then $f(X)$ is an interval. In particular, a continuous $f: [a, b] \rightarrow \mathbb{R}$ takes every value between $f(a)$ and $f(b)$.*

Proof. Combine Theorem 16.5.3 and Proposition 16.5.4(1). $\square$

Definition 16.5.6. The *connected component* of $x \in X$ is the union of all connected subsets containing x . A *path* from x to y is a continuous map $\gamma: [0, 1] \rightarrow X$ with $\gamma(0) = x$ and $\gamma(1) = y$. The space X is *path-connected* if it is nonempty and any two points are joined by a path. It is *locally connected* (*locally path-connected*) if every neighbourhood of every point contains a connected (path-connected) open neighbourhood.

Proposition 16.5.7. 1. *The connected components are connected, closed, and partition X . In a locally connected space they are open.*

2. Path-connected spaces are connected. The relation “joined by a path” is an equivalence relation, whose classes are the path components.
3. In a locally path-connected space the path components are open, and they coincide with the connected components. In particular, a connected open subset of $\mathbb{R}^n$ is path-connected.

Proof. (1) The component C of x is connected by Proposition 16.5.4(2), and $\overline{C}$ is connected by (3), so $C = \overline{C}$. If two components meet, their union is connected, so they are equal. If X is locally connected and $y \in C$, a connected open neighbourhood of y lies in the component of y , which is C .

(2) If γ is a path from x to y and δ one from y to z , then $\gamma * \delta$, equal to $\gamma(2t)$ on $[0, 1/2]$ and to $\delta(2t - 1)$ on $[1/2, 1]$, is continuous by the pasting lemma (Lemma 16.4.4); reversing $t \mapsto 1 - t$ gives symmetry. If X is path-connected and $x \in X$, then X is the union of the images of paths starting at x , which are connected by Theorem 16.5.3 and Proposition 16.5.4(1), so X is connected by Proposition 16.5.4(2).

(3) If y lies in the path component P and U is a path-connected open neighbourhood of y , then $U \subseteq P$; so path components are open. Each connected component C is the disjoint union of the path components it contains, which are open; by connectedness there is only one. Open balls in $\mathbb{R}^n$ are path-connected (use straight segments), so open subsets of $\mathbb{R}^n$ are locally path-connected. $\square$

Counterexample 16.5.8 (Topologist’s sine curve). Connected does not imply path-connected. Let $f: (0, 1] \rightarrow [-1, 1]$ be the continuous function with $f(1/n) = (-1)^n$ for $n \geq 1$ that is affine on each interval $[1/(n+1), 1/n]$ (continuous by the pasting lemma applied on the open sets $(1/(n+1), 1/(n-1))$ and $(1/2, 1]$); classically one uses $\sin(1/x)$ instead. Let G be its graph and $T = G \cup (\{0\} \times [-1, 1]) \subseteq \mathbb{R}^2$. The map $x \mapsto (x, f(x))$ is continuous, so G is connected. By Corollary 16.5.5, for each $y \in [-1, 1]$ and n there is $x_n \in [1/(n+1), 1/n]$ with $f(x_n) = y$, and $(x_n, y) \rightarrow (0, y)$; so $G \subseteq T \subseteq \overline{G}$ and T is connected.

Suppose $\gamma = (x, y): [0, 1] \rightarrow T$ is a path from $(0, 0)$ to $(1, f(1))$. The set of t with $x(t) = 0$ is closed and contains 0; let t_0 be its largest element, so $t_0 < 1$ and $\gamma(t) \in G$ for $t > t_0$. By continuity there is $\delta > 0$ with $t_0 + \delta \leq 1$ and $|y(t) - y(t_0)| < 1/2$ for $t \in [t_0, t_0 + \delta]$. Let $u = x(t_0 + \delta) > 0$ and choose n with $1/n < u$. By Corollary 16.5.5 there are $t, t' \in (t_0, t_0 + \delta]$ with $x(t) = 1/n$ and $x(t') = 1/(n+1)$, so $y(t)$ and $y(t')$ are 1 and -1 in some order; both lie within $1/2$ of $y(t_0)$, which is impossible.

Example 16.5.9. $\mathbb{R}$ and $\mathbb{R}^2$ are not homeomorphic. A homeomorphism would restrict to a homeomorphism between $\mathbb{R} \setminus \{a\}$, which is not connected, and $\mathbb{R}^2 \setminus \{b\}$, which is path-connected: two points $p, q \neq b$ are joined by the segment $[p, q]$ if it avoids b , and otherwise by the segments $[p, r]$ and $[r, q]$ for any r not on the line through p and q . Similarly S^n is path-connected for $n \geq 1$, being a continuous image of $\mathbb{R}^{n+1} \setminus \{0\}$ under $x \mapsto x/\|x\|_2$.

16.6 Compactness

Definition 16.6.1. An open cover of a space X is a family of open sets whose union is X . The space X is compact if every open cover has a finite subcover. A subset is compact if it is compact as a subspace, that is, if every cover of it by open subsets of X has a finite subcover.

Proposition 16.6.2. 1. A closed subset of a compact space is compact.

2. In a Hausdorff space, disjoint compact subsets K, L have disjoint open neighbourhoods; in particular compact subsets are closed.

3. Continuous images of compact spaces are compact.
4. A continuous map from a compact space to a Hausdorff space is closed. If it is bijective, it is a homeomorphism; if it is injective, it is an embedding; if it is surjective, it is a quotient map.

Proof. (1) An open cover of $C \subseteq X$, together with $X \setminus C$, covers X . (2) First let $L = \{y\}$. For each $k \in K$ choose disjoint open $U_k \ni k$ and $V_k \ni y$; finitely many $U_{k_1}, \dots, U_{k_m}$ cover K , and $U = \bigcup_j U_{k_j}$ and $V = \bigcap_j V_{k_j}$ are disjoint. For general L , do this for each $y \in L$ and use a finite subcover of L by the resulting V 's in the same way. (3) Preimages of an open cover of $f(X)$ cover X . (4) Closed subsets are compact by (1), their images are compact by (3), hence closed by (2). A closed continuous bijection has continuous inverse; for an injection apply this to $X \rightarrow f(X)$, and for a surjection use Proposition 16.4.9. $\square$

Proposition 16.6.3. *If X and Y are compact, so is $X \times Y$.*

Proof. Let $\mathcal{W}$ be an open cover of $X \times Y$ and fix $y \in Y$. For each $x \in X$ choose $W_x \in \mathcal{W}$ containing (x, y) and a basic open set $U_x \times V_x$ with $(x, y) \in U_x \times V_x \subseteq W_x$. Finitely many $U_{x_1}, \dots, U_{x_m}$ cover X ; let $N_y = \bigcap_j V_{x_j}$. Then $X \times N_y$ is covered by $W_{x_1}, \dots, W_{x_m}$ (the tube lemma). Finitely many N_y cover Y , and the corresponding finitely many members of $\mathcal{W}$ cover $X \times Y$. $\square$

Theorem 16.6.4 (Heine–Borel). *Every closed bounded interval $[a, b] \subseteq \mathbb{R}$ is compact. A subset of $\mathbb{R}^n$ (or $\mathbb{C}^n = \mathbb{R}^{2n}$) is compact if and only if it is closed and bounded.*

Proof. Let $\mathcal{U}$ be an open cover of $[a, b]$ and S the set of $t \in [a, b]$ such that $[a, t]$ is covered by finitely many members. Then $a \in S$; let $s = \sup S$. Choose $U \in \mathcal{U}$ with $s \in U$ and $\varepsilon > 0$ with $(s - \varepsilon, s + \varepsilon) \cap [a, b] \subseteq U$, and $t \in S$ with $t > s - \varepsilon$. A finite cover of $[a, t]$ together with U covers $[a, t']$ for $t' = \min(s + \varepsilon/2, b)$, so $t' \in S$ and $t' \leq s$. Hence $s = b \in S$.

For the second statement, a compact $K \subseteq \mathbb{R}^n$ is closed by Proposition 16.6.2(2) and bounded, since the balls $B(0, m)$ cover it. A closed bounded set lies in some cube $[-R, R]^n$, which is compact as a finite product of compact spaces by Proposition 16.6.3 (with the product topology, which is the Euclidean one by Proposition 16.4.6(4)); so it is compact by Proposition 16.6.2(1). $\square$

Corollary 16.6.5 (Extreme value theorem). *A continuous function $f: X \rightarrow \mathbb{R}$ on a nonempty compact space attains a maximum and a minimum.*

Proof. $f(X)$ is compact, hence closed and bounded, so it contains its supremum and infimum, which lie in its closure. $\square$

Lemma 16.6.6 (Alexander subbasis lemma). *Let $\mathcal{S}$ be a subbasis of the topology of X . If every cover of X by members of $\mathcal{S}$ has a finite subcover, then X is compact.*

Proof. Suppose not. The set of open covers of X without finite subcover is nonempty, and ordered by inclusion every chain in it has an upper bound, its union: a finite subcover of the union would lie in one member of the chain. By Zorn's lemma (Theorem 3.7.2) there is a maximal open cover $\mathcal{C}$ without finite subcover. The members of $\mathcal{C} \cap \mathcal{S}$ do not cover X , since otherwise they would have a finite subcover. Choose x not covered by them and $U \in \mathcal{C}$ with $x \in U$. There are $S_1, \dots, S_k \in \mathcal{S}$ with $x \in S_1 \cap \dots \cap S_k \subseteq U$, and no S_j belongs to $\mathcal{C}$. By maximality, $\mathcal{C} \cup \{S_j\}$ has a finite subcover, so $X = S_j \cup Y_j$ with Y_j a finite union of members of $\mathcal{C}$. Then $X = (S_1 \cap \dots \cap S_k) \cup Y_1 \cup \dots \cup Y_k \subseteq U \cup Y_1 \cup \dots \cup Y_k$ is covered by finitely many members of $\mathcal{C}$, a contradiction. (If $k = 0$, then $U = X$ is itself a finite subcover.) $\square$

Theorem 16.6.7 (Tychonoff). *Every product of compact spaces is compact.*

Proof. Let $X = \prod_i X_i$. If some X_i is empty, so is X . Otherwise, by Lemma 16.6.6 it suffices to consider a cover $\mathcal{S}$ of X by subbasic sets $p_i^{-1}(U)$. For each i let $\mathcal{U}_i$ be the set of open $U \subseteq X_i$ with $p_i^{-1}(U) \in \mathcal{S}$. If for some i the family $\mathcal{U}_i$ covers X_i , then a finite subcover of $\mathcal{U}_i$ gives a finite subcover of $\mathcal{S}$. Otherwise choose, by the axiom of choice, $x_i \in X_i$ not covered by $\mathcal{U}_i$ for each i ; then the point $(x_i)_i$ is not covered by $\mathcal{S}$, a contradiction. $\square$

Remark 16.6.8. Tychonoff's theorem implies the axiom of choice over ZF [Kel50], so the use of choice in its proof cannot be avoided.

Example 16.6.9. The sphere S^n is compact by Theorem 16.6.4, and so is $\mathbb{R}P^n$, a continuous image of S^n . The quotient $[0, 4]/\{0, 4\}$ is homeomorphic to S^1 . Indeed, let $Q = \{x \in \mathbb{R}^2 : \|x\|_\infty = 1\}$ be the boundary of the square. The maps $x \mapsto x/\|x\|_2$ and $x \mapsto x/\|x\|_\infty$ are mutually inverse continuous maps $Q \rightarrow S^1$ and $S^1 \rightarrow Q$. The map $[0, 4] \rightarrow Q$ running affinely along the four sides, from $(1, -1)$ to $(1, 1)$, $(-1, 1)$, $(-1, -1)$ and back, is a continuous surjection identifying only 0 and 4; the induced continuous bijection $[0, 4]/\{0, 4\} \rightarrow Q$ from a compact space to a Hausdorff space is a homeomorphism by Proposition 16.6.2(4).

Counterexample 16.6.10. 1. An infinite set X with the metric $d(x, y) = 1$ for $x \neq y$ is closed and bounded in itself but not compact: its topology is discrete, and the cover by singletons has no finite subcover.

2. In the cofinite topology on $\mathbb{N}$ every subset is compact (any member of a cover contains all but finitely many points), and the set of even numbers is compact but not closed. Hausdorffness in Proposition 16.6.2(2) is needed.

16.7 Complete and compact metric spaces

Definition 16.7.1. A sequence (x_n) in a metric space is *Cauchy* if for every $\varepsilon > 0$ there is N with $d(x_m, x_n) < \varepsilon$ for $m, n \geq N$. The space is *complete* if every Cauchy sequence converges. It is *totally bounded* if for every $\varepsilon > 0$ it is covered by finitely many balls of radius ε . A *Banach space* is a normed space that is complete for the metric of its norm.

Proposition 16.7.2. 1. $\mathbb{R}^n$ and $\mathbb{C}^n$ are complete, and so are all their closed subsets.

2. A complete subspace of a metric space is closed; a closed subspace of a complete space is complete.

3. For a nonempty set X and a complete metric space (Y, d) , the set $B(X, Y)$ of maps with bounded image, with $d_\infty(f, g) = \sup_x d(f(x), g(x))$, is a complete metric space. If X is a space, the subset $C_b(X, Y)$ of continuous maps is closed, hence complete. In particular $C_b(X, \mathbb{K})$ with the norm $\|f\|_\infty = \sup |f|$ is a Banach space.

Proof. (2) If $A \subseteq X$ is complete and $x \in \bar{A}$, a sequence in A converging to x (Proposition 16.3.5) is Cauchy, so it converges in A , and by uniqueness of limits $x \in A$. If X is complete and A closed, a Cauchy sequence in A converges in X , and its limit lies in $\bar{A} = A$.

(1) With the maximum norm, a sequence is Cauchy if and only if each coordinate sequence is, and converges if and only if each coordinate does. Real Cauchy sequences converge by Proposition 4.7.6, and complex ones because real and imaginary parts are Cauchy. The second part follows from (2).

(3) For bounded f, g , $d_\infty(f, g)$ is finite by the triangle inequality, and d_∞ is a metric. Let (f_n) be Cauchy. For each x , $(f_n(x))$ is Cauchy in Y ; let $f(x)$ be its limit. Given $\varepsilon > 0$ choose N with $d(f_n(x), f_m(x)) < \varepsilon$ for all x and $m, n \geq N$. Since $|d(a, b) - d(a, c)| \leq d(b, c)$, letting $m \rightarrow \infty$ gives $d(f_n(x), f(x)) \leq \varepsilon$ for all x and $n \geq N$. So f is bounded and $f_n \rightarrow f$ in d_∞ . Uniform limits of continuous maps are continuous (Proposition 16.3.8), so $C_b(X, Y)$ is closed. $\square$

Lemma 16.7.3 (Lebesgue number lemma). *Let X be a metric space in which every sequence has a convergent subsequence, and $\mathcal{U}$ an open cover. There is $\delta > 0$ such that every ball of radius δ is contained in some member of $\mathcal{U}$.*

Proof. Otherwise choose x_n such that $B(x_n, 1/n)$ lies in no member of $\mathcal{U}$, and a subsequence $x_{n_k} \rightarrow x$. Let $x \in U \in \mathcal{U}$ and $B(x, 2\varepsilon) \subseteq U$. For k large, $d(x_{n_k}, x) < \varepsilon$ and $1/n_k < \varepsilon$, so $B(x_{n_k}, 1/n_k) \subseteq B(x, 2\varepsilon) \subseteq U$, a contradiction. $\square$

Theorem 16.7.4. *For a metric space X the following are equivalent:*

1. X is compact;
2. every sequence in X has a convergent subsequence (X is sequentially compact);
3. X is complete and totally bounded.

Proof. (1) $\Rightarrow$ (2): Let (x_n) have no convergent subsequence. Then every $x \in X$ has an open ball containing x_n for only finitely many n : otherwise we could choose $n_1 < n_2 < \dots$ with $d(x_{n_k}, x) < 1/k$. Finitely many such balls cover X , but together they contain x_n for only finitely many n , a contradiction.

(2) $\Rightarrow$ (3): A Cauchy sequence with a convergent subsequence converges to the same limit. If X were not totally bounded, for some $\varepsilon > 0$ we could choose inductively $x_n \notin \bigcup_{m < n} B(x_m, \varepsilon)$; this sequence has no Cauchy subsequence.

(3) $\Rightarrow$ (2): Let (x_n) be a sequence. Cover X by finitely many balls of radius 1; one of them contains x_n for infinitely many n . Among these indices, cover by balls of radius $1/2$ and choose again, and so on; this gives nested infinite index sets $N_1 \supseteq N_2 \supseteq \dots$ such that $d(x_m, x_n) < 2/k$ for $m, n \in N_k$. Choosing $n_k \in N_k$ increasing gives a Cauchy subsequence, which converges.

(2) $\Rightarrow$ (1): Let $\mathcal{U}$ be an open cover and δ a Lebesgue number (Lemma 16.7.3). By (2) $\Rightarrow$ (3), X is covered by finitely many balls of radius δ , each contained in a member of $\mathcal{U}$. $\square$

Corollary 16.7.5. *A continuous map $f: X \rightarrow Y$ from a compact metric space to a metric space is uniformly continuous: for every $\varepsilon > 0$ there is $\delta > 0$ such that $d(x, x') < \delta$ implies $d(f(x), f(x')) < \varepsilon$.*

Proof. Let δ be a Lebesgue number of the cover by the sets $f^{-1}(B(y, \varepsilon/2))$. If $d(x, x') < \delta$, then $x, x' \in B(x, \delta)$ lie in one such set, so $d(f(x), f(x')) < \varepsilon$. $\square$

Theorem 16.7.6 (Banach fixed point theorem). *Let X be a nonempty complete metric space and $T: X \rightarrow X$ a contraction: $d(Tx, Ty) \leq c d(x, y)$ for some $0 \leq c < 1$. Then T has a unique fixed point, and for every x_0 the sequence $x_{n+1} = Tx_n$ converges to it.*

Proof. By induction $d(x_{n+1}, x_n) \leq c^n d(x_1, x_0)$, so for $m > n$, $d(x_m, x_n) \leq \sum_{k=n}^{m-1} c^k d(x_1, x_0) = c^n \frac{1-c^{m-n}}{1-c} d(x_1, x_0) \leq \frac{c^n}{1-c} d(x_1, x_0)$. Since $c^n \rightarrow 0$ (if $c > 0$, write $1/c = 1 + h$ with $h > 0$; then $c^{-n} \geq 1 + nh$), the sequence is Cauchy; let x be its limit. A contraction is continuous, so $Tx = \lim Tx_n = \lim x_{n+1} = x$. If also $Ty = y$, then $d(x, y) \leq c d(x, y)$, so $x = y$. $\square$

Theorem 16.7.7 (Baire category theorem). *In a complete metric space X , a countable intersection of dense open sets is dense. Equivalently, a countable union of closed sets with empty interior has empty interior.*

Proof. Let $U_1, U_2, \dots$ be dense open and W nonempty open. Since $W \cap U_1$ is nonempty and open, it contains a ball $B(x_1, 2r_1)$ with $r_1 < 1$, and then the closed ball $\bar{B}(x_1, r_1) = \{y : d(x_1, y) \leq r_1\}$. Inductively, $B(x_n, r_n) \cap U_{n+1}$ is nonempty and open, so it contains $\bar{B}(x_{n+1}, r_{n+1})$ with $r_{n+1} < 1/(n+1)$. For $m \geq n$, $x_m \in B(x_n, r_n)$, so (x_n) is Cauchy; its limit x lies in

every closed set $\bar{B}(x_n, r_n)$, hence in $W \cap \bigcap_n U_n$. The second statement is the first for the complements. $\square$

Corollary 16.7.8. *A nonempty complete metric space in which no point is isolated (no singleton is open) is uncountable.*

Proof. If $X = \{x_1, x_2, \dots\}$, then X is the countable union of the closed sets $\{x_n\}$, which have empty interior, contradicting Theorem 16.7.7. $\square$

Counterexample 16.7.9. Completeness is not a topological property: $t \mapsto t/(1 - |t|)$ is a homeomorphism $(-1, 1) \rightarrow \mathbb{R}$ with inverse $s \mapsto s/(1 + |s|)$, but $(-1, 1)$ is not complete. And $\mathbb{Q}$, which is not complete, is a countable union of closed sets with empty interior, so the Baire category theorem fails for it.

16.8 Separation axioms, Urysohn and Tietze

Definition 16.8.1. A space X is T_1 if its points are closed. A T_1 space is *regular* if a point x and a closed set $C \not\ni x$ have disjoint open neighbourhoods, and *normal* if any two disjoint closed sets have disjoint open neighbourhoods. So $\text{normal} \Rightarrow \text{regular} \Rightarrow \text{Hausdorff} \Rightarrow T_1$.

Lemma 16.8.2. *A T_1 space X is normal if and only if for every closed C and open $U \supseteq C$ there is an open V with $C \subseteq V \subseteq \bar{V} \subseteq U$. It is regular if and only if this holds for C a point.*

Proof. If X is normal, separate C and $X \setminus U$ by disjoint open V and W ; then $\bar{V} \subseteq X \setminus W \subseteq U$. Conversely, for disjoint closed C, D choose V for $U = X \setminus D$; then V and $X \setminus \bar{V}$ separate them. The regular case is identical. $\square$

Proposition 16.8.3. *Metric spaces and compact Hausdorff spaces are normal.*

Proof. For a nonempty subset A of a metric space let $d(x, A) = \inf_{a \in A} d(x, a)$. Then $|d(x, A) - d(y, A)| \leq d(x, y)$, so $d(-, A)$ is continuous, and $d(x, A) = 0$ if and only if $x \in \bar{A}$. For disjoint nonempty closed A, B , the continuous function (Proposition 16.4.6(5)) $g(x) = d(x, A) - d(x, B)$ is negative on A and positive on B , so $\{g < 0\}$ and $\{g > 0\}$ separate them. For compact Hausdorff spaces use Propositions 16.6.2(1) and (2). $\square$

Theorem 16.8.4 (Urysohn's lemma). *Let A, B be disjoint closed subsets of a normal space X . There is a continuous $f: X \rightarrow [0, 1]$ with $f = 0$ on A and $f = 1$ on B .*

Proof. Let D be the set of dyadic rationals $k/2^n$ in $[0, 1]$. We construct open sets U_r , $r \in D$, with $A \subseteq U_0$, $U_1 = X \setminus B$ and $\bar{U}_r \subseteq U_s$ for $r < s$. Choose U_0 with $A \subseteq U_0 \subseteq \bar{U}_0 \subseteq U_1$ by Lemma 16.8.2. If U_r is defined for all r with denominator 2^n , then for $r = (2k+1)/2^{n+1}$ choose U_r with $\bar{U}_{k/2^n} \subseteq U_r \subseteq \bar{U}_r \subseteq U_{(k+1)/2^n}$. The required inclusions for $r < s$ with denominators dividing 2^{n+1} follow: if one of them is new, pass through the neighbouring old index, and if both are new, there is an old index strictly between them.

Define $f(x) = \inf(\{r \in D : x \in U_r\} \cup \{1\})$. Then $f = 0$ on A , and $f = 1$ on B since B meets no U_r . If $x \in \bar{U}_r$, then $x \in U_s$ for all $s > r$ in D , so $f(x) \leq r$ by density of D . If $x \notin U_r$, then $x \notin U_s$ for $s \leq r$, so $f(x) \geq r$. Hence for $a \in \mathbb{R}$,

$$f^{-1}([0, a)) = \bigcup_{r \in D, r < a} U_r \quad (a \leq 1), \quad f^{-1}((a, 1]) = \bigcup_{r \in D, r > a} (X \setminus \bar{U}_r) \quad (a \geq 0).$$

For the second equality, if $f(x) > a$ choose $r \in D$ with $a < r < f(x)$; then $x \notin \bar{U}_r$. These sets are open, and the intervals $[0, a)$, $(a, 1]$ form a subbasis of $[0, 1]$, so f is continuous. $\square$

Theorem 16.8.5 (Tietze extension theorem). *Let A be a closed subset of a normal space X . Every continuous $f: A \rightarrow [-1, 1]$ extends to a continuous $F: X \rightarrow [-1, 1]$, and every continuous $f: A \rightarrow \mathbb{R}$ extends to a continuous $F: X \rightarrow \mathbb{R}$.*

Proof. *Step 1.* If $g: A \rightarrow [-c, c]$ is continuous, there is a continuous $h: X \rightarrow [-c/3, c/3]$ with $|g - h| \leq 2c/3$ on A : the sets $g^{-1}([-c, -c/3])$ and $g^{-1}([c/3, c])$ are disjoint and closed in X , and Urysohn's lemma (rescaled) gives h equal to $-c/3$ on the first and $c/3$ on the second; check $|g - h| \leq 2c/3$ separately on the two sets and on the rest of A , where $|g| < c/3$.

Step 2. Put $g_0 = f$ and, given $g_n: A \rightarrow [-(2/3)^n, (2/3)^n]$, let h_n be as in Step 1 and $g_{n+1} = g_n - h_n|_A$. Then $|h_n| \leq \frac{1}{3}(\frac{2}{3})^n$, so $F = \sum_n h_n$ converges uniformly and is continuous by Proposition 16.3.8, with $|F| \leq \sum_n \frac{1}{3}(\frac{2}{3})^n = 1$. On A , $f - \sum_{n < N} h_n = g_N \rightarrow 0$, so $F|_A = f$.

Step 3. For $f: A \rightarrow \mathbb{R}$, compose with the homeomorphism $\mathbb{R} \rightarrow (-1, 1)$ of Counterexample 16.7.9 and extend to $F: X \rightarrow [-1, 1]$ by Step 2. The set $E = F^{-1}(\{-1, 1\})$ is closed and disjoint from A ; by Urysohn's lemma choose $\varphi: X \rightarrow [0, 1]$ with $\varphi = 0$ on E and $\varphi = 1$ on A . Then φF extends f and takes values in $(-1, 1)$; compose back. $\square$

Counterexample 16.8.6. Let $K = \{1/n : n \geq 1\}$ and give $\mathbb{R}$ the topology with basis the open intervals (a, b) and the sets $(a, b) \setminus K$ (the conditions of Lemma 16.2.6 are clear). It is finer than the standard topology, hence Hausdorff, and K is closed. Suppose $U \supseteq K$ and $V \ni 0$ are disjoint open sets. Since every interval around 0 meets K , V contains a set $(a, b) \setminus K$ with $a < 0 < b$. Choose n with $1/n < b$. The only basic sets containing $1/n$ are intervals, so $U \supseteq (c, d) \ni 1/n$, and $(c, d) \cap (a, b)$ contains points not in K , which lie in $U \cap V$. So this Hausdorff space is not regular.

Lemma 16.8.7 (Tychonoff). *A regular space in which every open cover has a countable subcover (a Lindelöf space), for instance a regular second countable space, is normal.*

Proof. Let A, B be disjoint closed sets. For $a \in A$, regularity and Lemma 16.8.2 give an open $U \ni a$ with $\overline{U} \cap B = \emptyset$. As A is closed in a Lindelöf space, it is Lindelöf, so countably many such sets $U_1, U_2, \dots$ cover A ; similarly open $V_1, V_2, \dots$ cover B with $\overline{V_k} \cap A = \emptyset$. Put $U'_n = U_n \setminus \bigcup_{k \leq n} \overline{V_k}$ and $V'_n = V_n \setminus \bigcup_{k \leq n} \overline{U_k}$, which are open, and $U' = \bigcup_n U'_n$, $V' = \bigcup_n V'_n$. Then $A \subseteq U'$, since a point of A lies in some U_n and in no $\overline{V_k}$; similarly $B \subseteq V'$. If $x \in U'_n \cap V'_m$ with $n \leq m$, then $x \in U_n$ but $x \notin \overline{U_n}$, a contradiction, and the case $n > m$ is symmetric. So $U' \cap V' = \emptyset$. Second countable spaces are Lindelöf by Proposition 16.3.10. $\square$

Theorem 16.8.8 (Urysohn metrisation theorem). *Every regular second countable space is metrisable: its topology is induced by a metric.*

Proof. Let X be regular with a countable basis (B_n) ; by Lemma 16.8.7 it is normal. For each pair (n, m) with $\overline{B_n} \subseteq B_m$ choose, by Urysohn's lemma (Theorem 16.8.4), a continuous $f_{n,m}: X \rightarrow [0, 1]$ equal to 1 on $\overline{B_n}$ and to 0 outside B_m . Enumerate these functions as $f_1, f_2, \dots$. They separate points from closed sets: if $x \in U$ with U open, choose m with $x \in B_m \subseteq U$, and by regularity and Lemma 16.8.2 an open V with $x \in V \subseteq \overline{V} \subseteq B_m$, and then n with $x \in B_n \subseteq V$; the corresponding f_k has $f_k(x) = 1$ and vanishes outside U .

Define $d(x, y) = \sum_k 2^{-k} |f_k(x) - f_k(y)|$. It is symmetric and satisfies the triangle inequality, and $d(x, y) > 0$ for $x \neq y$, because X is T_1 , so $X \setminus \{y\}$ is an open neighbourhood of x , and some f_k equals 1 at x and vanishes outside it, so $f_k(x) = 1 \neq 0 = f_k(y)$. Each $d(x, -)$ is continuous, being a uniformly convergent series of continuous functions (Proposition 16.3.8), so d -balls are open. Conversely, if $x \in U$ with U open and f_k is as above, then $d(x, y) < 2^{-k}$ implies $|f_k(y) - 1| < 1$, so $f_k(y) > 0$ and $y \in U$: the d -ball of radius 2^{-k} about x lies in U . So d induces the topology of X . $\square$

Remark 16.8.9. Topological manifolds are regular (indeed locally compact Hausdorff) and second countable, hence metrisable (Chapter 18). Regularity cannot be weakened to the Hausdorff property: the space of Counterexample 16.8.6 is second countable (a basis is given by the intervals with rational endpoints and their differences with K) and Hausdorff, but not regular, hence not metrisable, since metric spaces are normal (Proposition 16.8.3).

16.9 Local compactness and proper maps

Definition 16.9.1. A space is *locally compact* if every point has a compact neighbourhood. A continuous map $f: X \rightarrow Y$ is *proper* if $f^{-1}(K)$ is compact for every compact $K \subseteq Y$. The *support* of a function $f: X \rightarrow \mathbb{K}$ is $\text{supp } f = \overline{\{f \neq 0\}}$, and $C_c(X)$ denotes the continuous functions with compact support.

Definition 16.9.2. Let X be a Hausdorff space. The *one-point compactification* is $X^+ = X \sqcup \{\infty\}$, whose open sets are the open subsets of X and the sets $(X \setminus K) \cup \{\infty\}$ with $K \subseteq X$ compact.

Theorem 16.9.3. Let X be Hausdorff. Then X^+ is a compact space containing X as an open subspace, and X^+ is Hausdorff if and only if X is locally compact.

Proof. Compact subsets of X are closed (Proposition 16.6.2), so the sets $X \setminus K$ are open. The open sets form a topology: $\bigcup_i ((X \setminus K_i) \cup \{\infty\}) = (X \setminus \bigcap_i K_i) \cup \{\infty\}$ and $U \cup (X \setminus K) \cup \{\infty\} = (X \setminus (K \setminus U)) \cup \{\infty\}$, where $\bigcap_i K_i$ and $K \setminus U$ are closed subsets of a compact set, hence compact; and $U \cap ((X \setminus K) \cup \{\infty\}) = U \cap (X \setminus K)$ and $((X \setminus K) \cup \{\infty\}) \cap ((X \setminus L) \cup \{\infty\}) = (X \setminus (K \cup L)) \cup \{\infty\}$. The traces on X of the open sets are open in X , so X is an open subspace. Given an open cover of X^+ , a member $(X \setminus K) \cup \{\infty\}$ contains ∞ , and finitely many members cover K .

If X is locally compact, points of X are separated in X , and $x \in X$ with a compact neighbourhood K is separated from ∞ by $\text{int } K$ and $(X \setminus K) \cup \{\infty\}$. Conversely, if x and ∞ have disjoint neighbourhoods U and $(X \setminus K) \cup \{\infty\}$, then $U \subseteq K$, so K is a compact neighbourhood of x . $\square$

Example 16.9.4. $(\mathbb{R}^n)^+ \cong S^n$. Let $N = (0, \dots, 0, 1)$ and define $\tau: \mathbb{R}^n \rightarrow S^n$ by $\tau(y) = (2y, \|y\|^2 - 1)/(\|y\|^2 + 1)$. A direct computation shows $\|\tau(y)\| = 1$, that τ is a bijection onto $S^n \setminus \{N\}$ with inverse the *stereographic projection* $x \mapsto (x_1, \dots, x_n)/(1 - x_{n+1})$, and that $\|\tau(y) - N\| = 2/\sqrt{\|y\|^2 + 1}$. Extend τ by $\tau(\infty) = N$. It is continuous at ∞ : for $\varepsilon > 0$, τ maps the complement of a closed ball of large radius R , which is a compact set, into $B(N, \varepsilon)$. So $\tau: (\mathbb{R}^n)^+ \rightarrow S^n$ is a continuous bijection from a compact space to a Hausdorff space, hence a homeomorphism.

Lemma 16.9.5. Let X be locally compact Hausdorff, $K \subseteq X$ compact and $U \supseteq K$ open. There is an open V with compact closure such that $K \subseteq V \subseteq \overline{V} \subseteq U$, and there is $f \in C_c(X)$ with values in $[0, 1]$, $f = 1$ on K and $\text{supp } f \subseteq U$.

Proof. By Theorem 16.9.3 and Proposition 16.8.3, X^+ is normal. The set K is closed in X^+ , $X^+ \setminus U$ is closed, and they are disjoint. By Lemma 16.8.2 there is an open V with $K \subseteq V$ and $\overline{V} \subseteq U$ (closure in X^+); then $\overline{V}$ is a closed subset of X^+ not containing ∞ , so it is compact and equals the closure of V in X . Urysohn's lemma (Theorem 16.8.4) in X^+ gives a continuous $f: X^+ \rightarrow [0, 1]$ equal to 1 on K and 0 on $X^+ \setminus V$. Its restriction to X has support contained in the compact set $\overline{V} \subseteq U$, so the support is compact. $\square$

Proposition 16.9.6. Let $f: X \rightarrow Y$ be a continuous map to a locally compact Hausdorff space. If f is proper, then f is closed. A continuous map from a compact space to a Hausdorff space is proper.

Proof. Let $C \subseteq X$ be closed and $y \in \overline{f(C)}$. Choose a compact neighbourhood K of y . Then $C \cap f^{-1}(K)$ is compact, so $f(C \cap f^{-1}(K)) = f(C) \cap K$ is compact, hence closed. Every open $W \ni y$ gives an open set $W \cap \text{int } K \ni y$, which meets $f(C)$; so $y \in \overline{f(C) \cap K} = f(C) \cap K$. For the last statement, compact subsets of a Hausdorff space are closed, and closed subsets of compact spaces are compact. $\square$

Counterexample 16.9.7. $\mathbb{Q}$ is not locally compact. A compact neighbourhood K of 0 in $\mathbb{Q}$ would contain $[-\varepsilon, \varepsilon] \cap \mathbb{Q}$ for some $\varepsilon > 0$, which is closed in K and hence compact. But choosing an irrational $\alpha \in (0, \varepsilon)$, the open sets $(\mathbb{Q} \cap (-\infty, \alpha - 1/n)) \cup (\mathbb{Q} \cap (\alpha + 1/n, \infty))$, $n \geq 1$, form an increasing family that covers $[-\varepsilon, \varepsilon] \cap \mathbb{Q}$ without finite subcover.

16.10 Paracompactness and partitions of unity

Definition 16.10.1. A family $(A_i)_{i \in I}$ of subsets of X is *locally finite* if every point has a neighbourhood meeting A_i for only finitely many i . A *refinement* of a cover (U_i) is a cover (V_j) such that each V_j is contained in some U_i . A space is *paracompact* if every open cover has a locally finite open refinement.

Lemma 16.10.2. If (A_i) is locally finite, then so is $(\overline{A_i})$, and $\overline{\bigcup_i A_i} = \bigcup_i \overline{A_i}$. In particular a union of a locally finite family of closed sets is closed. If (f_i) are continuous functions $X \rightarrow \mathbb{K}$ whose supports form a locally finite family, then $\sum_i f_i$ is a well-defined continuous function.

Proof. An open set meets $\overline{A_i}$ if and only if it meets A_i . Let $x \in \overline{\bigcup_i A_i}$ and let W be an open neighbourhood of x meeting only $A_{i_1}, \dots, A_{i_k}$. Every open $W' \ni x$ satisfies $W' \cap W \cap \bigcup_i A_i \neq \emptyset$, so W' meets $A_{i_1} \cup \dots \cup A_{i_k}$; hence $x \in \overline{A_{i_1} \cup \dots \cup A_{i_k}} = \overline{A_{i_1}} \cup \dots \cup \overline{A_{i_k}}$ by Lemma 16.2.4. For the sum, each point has an open neighbourhood on which all but finitely many f_i vanish, and on it the sum is a finite sum of continuous functions (Proposition 16.4.6(5)); conclude by the pasting lemma. $\square$

Proposition 16.10.3. A paracompact Hausdorff space is normal.

Proof. Regularity. Let C be closed and $x \notin C$. For $c \in C$ choose disjoint open $U_c \ni c$ and $W_c \ni x$. Let $\mathcal{V}$ be a locally finite open refinement of the cover $\{U_c\} \cup \{X \setminus C\}$, and $\mathcal{V}'$ the members meeting C ; each is contained in some U_c , so its closure is contained in $X \setminus W_c$ and does not contain x . Then $V = \bigcup \mathcal{V}'$ is an open set containing C with $\overline{V} = \bigcup_{V' \in \mathcal{V}'} \overline{V'} \not\ni x$ by Lemma 16.10.2. So V and $X \setminus \overline{V}$ separate C and x .

Normality. The same argument, with x replaced by a closed set D disjoint from C and the sets U_c, W_c provided by regularity, separates C and D . $\square$

Lemma 16.10.4 (Shrinking lemma). Let X be normal and $(U_i)_{i \in I}$ an open cover that is point-finite (each point lies in finitely many U_i), for instance locally finite. There is an open cover $(V_i)_{i \in I}$ with $\overline{V_i} \subseteq U_i$ for all i .

Proof. Consider pairs $(J, (V_j)_{j \in J})$ with $J \subseteq I$, V_j open and $\overline{V_j} \subseteq U_j$, such that $(V_j)_{j \in J} \cup (U_i)_{i \notin J}$ covers X , ordered by extension. The pair $(\emptyset, ())$ is one. Given a chain, let J^* be the union of the index sets, with the V_j defined compatibly. Let $x \in X$ and F the finite nonempty set of i with $x \in U_i$. If $F \not\subseteq J^*$, x is covered by some U_i with $i \notin J^*$. Otherwise $F \subseteq J$ for some member (J, V) of the chain; there x is covered by some V_j , $j \in J$, since $x \in U_i$ implies $i \in F \subseteq J$. So the union is an upper bound, and Zorn's lemma gives a maximal pair (J, V) . If $i \notin J$, the set $C = X \setminus (\bigcup_{j \in J} V_j \cup \bigcup_{k \notin J, k \neq i} U_k)$ is closed and contained in U_i ; by Lemma 16.8.2 choose open V_i with $C \subseteq V_i \subseteq \overline{V_i} \subseteq U_i$, contradicting maximality. So $J = I$. $\square$

Definition 16.10.5. A *partition of unity* on X is a family of continuous functions $\psi_i: X \rightarrow [0, 1]$ whose supports form a locally finite family and with $\sum_i \psi_i = 1$. It is *subordinate* to an open cover $(U_i)_{i \in I}$ (with the same index set) if $\text{supp } \psi_i \subseteq U_i$ for all i .

Theorem 16.10.6. Let X be paracompact Hausdorff. Every open cover $(U_i)_{i \in I}$ has a subordinate partition of unity.

Proof. Let $(W_k)_{k \in K}$ be a locally finite open refinement, with $W_k \subseteq U_{\varphi(k)}$ for a chosen map $\varphi: K \rightarrow I$. The space is normal (Proposition 16.10.3), so by Lemma 16.10.4 there is an open cover (V_k) with $\overline{V_k} \subseteq W_k$, and by Lemma 16.8.2 open V'_k with $\overline{V_k} \subseteq V'_k \subseteq \overline{V'_k} \subseteq W_k$. By Urysohn's lemma choose $f_k: X \rightarrow [0, 1]$ equal to 1 on $\overline{V_k}$ and 0 outside V'_k , so $\text{supp } f_k \subseteq \overline{V'_k} \subseteq W_k$. By Lemma 16.10.2, $f = \sum_k f_k$ is continuous, and $f \geq 1$ because the V_k cover. The functions $g_k = f_k/f$ are continuous with supports in W_k and sum 1. Finally let $\psi_i = \sum_{\varphi(k)=i} g_k$. Its support is contained in $\bigcup_{\varphi(k)=i} \overline{V'_k}$, which is closed by Lemma 16.10.2 and contained in U_i ; the supports of the ψ_i are locally finite because each point has a neighbourhood meeting only finitely many W_k ; and $\sum_i \psi_i = \sum_k g_k = 1$. $\square$

Proposition 16.10.7. Let X be locally compact Hausdorff and second countable. There are compact sets $K_1 \subseteq K_2 \subseteq \dots$ with $K_n \subseteq \text{int } K_{n+1}$ and $\bigcup_n K_n = X$ (an exhaustion by compact sets), and X is paracompact.

Proof. Let $\mathcal{B}$ be a countable basis, and $\mathcal{B}'$ the members with compact closure. $\mathcal{B}'$ covers X : if x has a compact neighbourhood K and $x \in B \subseteq \text{int } K$ with $B \in \mathcal{B}$, then $\overline{B} \subseteq K$ is compact. Enumerate $\mathcal{B}' = \{B_1, B_2, \dots\}$ (with repetitions if finite; if $X = \emptyset$ there is nothing to prove). Let $K_1 = \overline{B_1}$, and given K_n choose $m_n \geq n + 1$ with $K_n \subseteq B_1 \cup \dots \cup B_{m_n}$ and let $K_{n+1} = \overline{B_1} \cup \dots \cup \overline{B_{m_n}}$. Then $K_n \subseteq \text{int } K_{n+1}$ and $B_n \subseteq K_{n+1}$.

Put $K_0 = K_{-1} = \emptyset$ and let $\mathcal{U}$ be an open cover. For $n \geq 1$ the compact set $L_n = K_n \setminus \text{int } K_{n-1}$ is contained in the open set $O_n = \text{int } K_{n+1} \setminus K_{n-2}$. Cover L_n by finitely many sets $U \cap O_n$ with $U \in \mathcal{U}$, and let $\mathcal{V}$ be the collection of all these sets for all n . It is an open refinement of $\mathcal{U}$, and it covers X because a point lies in L_n for the least n with $x \in K_n$. A point x lies in $\text{int } K_m$ for some m , and the sets of $\mathcal{V}$ coming from L_n lie in $X \setminus K_{n-2}$, so they meet $\text{int } K_m$ only if $n \leq m + 1$. Hence $\mathcal{V}$ is locally finite. $\square$

Theorem 16.10.8 (Stone). Every metric space is paracompact.

Proof. This is [Sto48]; we give the proof of [Rud69]. Let $(U_s)_{s \in S}$ be an open cover and well-order S (Theorem 3.7.2). For $n \geq 1$ define inductively $V_{s,n}$ as the union of the balls $B(x, 2^{-n})$ over all x such that: (i) s is the least index with $x \in U_s$; (ii) $x \notin V_{t,j}$ for all t and all $j < n$; (iii) $B(x, 3 \cdot 2^{-n}) \subseteq U_s$. Each $V_{s,n}$ is open and contained in U_s by (iii).

Cover. For $x \in X$ let s be least with $x \in U_s$ and n with $B(x, 3 \cdot 2^{-n}) \subseteq U_s$. Either $x \in V_{t,j}$ for some $j < n$, or x satisfies (i)–(iii) and $x \in V_{s,n}$.

Local finiteness. Let $x \in X$, let s be least such that $x \in V_{s,n}$ for some n , fix such n , and choose $j \geq 1$ with $B(x, 2^{-j}) \subseteq V_{s,n}$. We claim that $B(x, 2^{-n-j})$ meets only finitely many $V_{t,i}$. If $i \geq n + j$, a ball $B(y, 2^{-i})$ in $V_{t,i}$ has $y \notin V_{s,n}$ by (ii) (as $n < i$), so $d(x, y) \geq 2^{-j}$; since $2^{-i} + 2^{-n-j} \leq 2^{-j}$, it does not meet $B(x, 2^{-n-j})$. If $i < n + j$, then $B(x, 2^{-n-j})$ meets $V_{t,i}$ for at most one t : suppose it contains $p \in B(y, 2^{-i}) \subseteq V_{t,i}$ and $q \in B(z, 2^{-i}) \subseteq V_{t',i}$ with $t < t'$ and centres y, z as in the definition. By (iii) $B(y, 3 \cdot 2^{-i}) \subseteq U_t$, and by (i) $z \notin U_t$, so $d(y, z) \geq 3 \cdot 2^{-i}$. But $d(y, z) < 2^{-i} + 2 \cdot 2^{-n-j} + 2^{-i} \leq 3 \cdot 2^{-i}$, since $2^{-n-j} \leq 2^{-i-1}$. So only finitely many pairs (t, i) occur. $\square$

Remark 16.10.9. Topological manifolds (Hausdorff, second countable and locally homeomorphic to $\mathbb{R}^n$) are locally compact, hence paracompact by Proposition 16.10.7, and admit

partitions of unity; smooth partitions of unity are constructed in Chapter 31. Without second countability this fails: the long line is a Hausdorff space locally homeomorphic to $\mathbb{R}$ that is not paracompact [Eng89]. Paracompactness is also the natural hypothesis under which Čech cohomology computes sheaf cohomology (Chapter 24).

16.11 Exercises

Exercise 16.11.1. Show that X is Hausdorff if and only if the diagonal $\Delta = \{(x, x)\}$ is closed in $X \times X$. Deduce that if $f, g: Z \rightarrow X$ are continuous and X is Hausdorff, then $\{z : f(z) = g(z)\}$ is closed, and that the graph of a continuous map to a Hausdorff space is closed.

Solution. $(x, y) \notin \Delta$ has a basic neighbourhood $U \times V$ disjoint from Δ if and only if $U \cap V = \emptyset$. So $X \times X \setminus \Delta$ is open if and only if distinct points have disjoint neighbourhoods. The set $\{f = g\}$ is the preimage of Δ under the continuous map $(f, g): Z \rightarrow X \times X$, and the graph of $f: Y \rightarrow X$ is $\{(y, x) : f(y) = x\}$, the equaliser of $f \circ p_1$ and p_2 . $\square$

Exercise 16.11.2. Show that $\overline{\prod_i A_i} = \prod_i \overline{A_i}$ in a product of spaces.

Solution. A point x lies in the closure of $\prod_i A_i$ if and only if every basic open set $\prod_i U_i \ni x$ meets $\prod_i A_i$, that is, $U_i \cap A_i \neq \emptyset$ for the finitely many i with $U_i \neq X_i$. This holds for all such sets if and only if $x_i \in \overline{A_i}$ for every i . $\square$

Exercise 16.11.3. Show that $\mathbb{RP}^n$ is a compact Hausdorff space, and that the projection $S^n \rightarrow \mathbb{RP}^n$ is open.

Solution. If $U \subseteq S^n$ is open, $q^{-1}(q(U)) = U \cup (-U)$ is open, so q is open. Given $[x] \neq [y]$, we have $x \neq \pm y$; choose $\varepsilon < \min(\|x - y\|, \|x + y\|)/2$. The open sets $U = (B(x, \varepsilon) \cup B(-x, \varepsilon)) \cap S^n$ and $V = (B(y, \varepsilon) \cup B(-y, \varepsilon)) \cap S^n$ are invariant under $x \mapsto -x$ and disjoint, so $q(U)$ and $q(V)$ are disjoint open neighbourhoods. Compactness was shown in Example 16.6.9. $\square$

Exercise 16.11.4. Prove the Baire category theorem for locally compact Hausdorff spaces.

Solution. Let $U_1, U_2, \dots$ be dense open and W nonempty open. By Lemma 16.9.5, choose nonempty open sets V_n with compact closure and $\overline{V_1} \subseteq W \cap U_1$, $\overline{V_{n+1}} \subseteq V_n \cap U_{n+1}$. The compact sets $\overline{V_n}$ are nested and nonempty, so their intersection is nonempty: otherwise the open sets $\overline{V_1} \setminus \overline{V_n}$ would cover $\overline{V_1}$ without finite subcover. A point of the intersection lies in $W \cap \bigcap_n U_n$. $\square$

Exercise 16.11.5. Let X be a metric space. Show that $x \mapsto d(x, -) - d(x_0, -)$ is an isometric embedding of X into the Banach space $C_b(X, \mathbb{R})$, for a fixed $x_0 \in X$, and deduce that X has a *completion*: a complete metric space containing X as a dense subspace.

Solution. The function $\phi_x = d(x, -) - d(x_0, -)$ is bounded by $d(x, x_0)$ and continuous. By the triangle inequality $|\phi_x(z) - \phi_y(z)| \leq d(x, y)$ with equality at $z = y$, so $\|\phi_x - \phi_y\|_\infty = d(x, y)$. The closure of the image in the complete space $C_b(X, \mathbb{R})$ (Proposition 16.7.2) is complete and contains X as a dense subspace. $\square$

Exercise 16.11.6. Show that the one-point compactification of $\mathbb{N}$ (discrete) is homeomorphic to $\{0\} \cup \{1/n : n \geq 1\} \subseteq \mathbb{R}$, and that $(0, 1)^+$ is homeomorphic to S^1 .

Exercise 16.11.7. Let $f: X \rightarrow Y$ be a continuous map between locally compact Hausdorff spaces. Show that f is proper if and only if its extension $f^+: X^+ \rightarrow Y^+$ with $f^+(\infty) = \infty$ is continuous.

Exercise 16.11.8. Let $q: X \rightarrow Y$ be a quotient map and K a compact Hausdorff space. Show that $q \times \text{id}_K: X \times K \rightarrow Y \times K$ is a quotient map. (Hint: use the tube lemma from the proof of Proposition 16.6.3.)

Chapter 17

Homotopy and Covering Spaces

17.1 Introduction

Homotopy theory studies spaces up to continuous deformation. Its first invariant is the fundamental group, which we compute for the circle using covering spaces (Theorem 17.4.6) and for spaces glued from simpler pieces using the Seifert–van Kampen theorem (Theorem 17.5.1). Covering spaces of a reasonable space are classified by subgroups of its fundamental group (Theorem 17.6.6), in close analogy with Galois theory. We then introduce CW complexes and the higher homotopy groups. Our main references are [Hat02] and [Spa66].

Prerequisites. Chapter 16 (General Topology) and Chapter 6 (Groups).

Throughout, $I = [0, 1]$, $D^n = \{x \in \mathbb{R}^n : \|x\|_2 \leq 1\}$ is the closed unit ball and $S^{n-1} = \partial D^n$ the unit sphere. A *map* of spaces means a continuous map.

17.2 Homotopy

Definition 17.2.1. A *homotopy* between maps $f, g: X \rightarrow Y$ is a map $H: X \times I \rightarrow Y$ with $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$; we write $f \simeq g$ and $H_t = H(-, t)$. It is a *homotopy relative to* $A \subseteq X$ if $H_t(a) = f(a)$ for all $a \in A$ and $t \in I$. A map $f: X \rightarrow Y$ is a *homotopy equivalence* if there is $g: Y \rightarrow X$ with $gf \simeq \text{id}_X$ and $fg \simeq \text{id}_Y$; then X and Y are *homotopy equivalent*, $X \simeq Y$. A space is *contractible* if it is homotopy equivalent to a point. A subspace $A \subseteq X$ is a *retract* if there is a map $r: X \rightarrow A$ with $r|_A = \text{id}_A$, and a *deformation retract* if moreover $ir \simeq \text{id}_X$ relative to A , where $i: A \rightarrow X$ is the inclusion.

Lemma 17.2.2. *Homotopy (relative to A) is an equivalence relation on maps $X \rightarrow Y$, compatible with composition: if $f \simeq f'$ and $g \simeq g'$, then $gf \simeq g'f'$. Hence spaces and homotopy classes of maps form a category $\mathbf{hTop}$, and homotopy equivalences are its isomorphisms.*

Proof. Reflexivity: $H(x, t) = f(x)$. Symmetry: $H(x, 1 - t)$. Transitivity: if H goes from f to g and K from g to h , the map equal to $H(x, 2t)$ for $t \leq 1/2$ and $K(x, 2t - 1)$ for $t \geq 1/2$ is continuous by the pasting lemma (Lemma 16.4.4). If H goes from f to f' and K from g to g' , then $(x, t) \mapsto K(H(x, t), t)$ goes from gf to $g'f'$; it is continuous as the composite of $(x, t) \mapsto (H(x, t), t)$ and K . Relative homotopies are preserved by these constructions. $\square$

Example 17.2.3. 1. A convex subset $C \subseteq \mathbb{R}^n$ is contractible: for $c \in C$, the *straight-line homotopy* $H(x, t) = (1 - t)x + tc$ goes from id_C to the constant map, and it is continuous by Proposition 16.4.6(5). More generally, any two maps $f, g: X \rightarrow C$ are homotopic via $(1 - t)f + tg$, relative to the set where they agree.

2. S^{n-1} is a deformation retract of $\mathbb{R}^n \setminus \{0\}$ and of $D^n \setminus \{0\}$, via $H(x, t) = (1 - t)x + tx/\|x\|$, which never vanishes.

Lemma 17.2.4. *If $q: X \rightarrow Y$ is a quotient map and K is a compact Hausdorff space, then $q \times \text{id}_K: X \times K \rightarrow Y \times K$ is a quotient map. If $U \subseteq Y$ is open, then $q^{-1}(U) \rightarrow U$ is a quotient map.*

Proof. Let $W \subseteq Y \times K$ with $W' = (q \times \text{id})^{-1}(W)$ open, and let $(y_0, k_0) \in W$ with $y_0 = q(x_0)$. The set $\{k : (x_0, k) \in W'\}$ is open and contains k_0 ; since K is regular (Proposition 16.8.3), there is an open $N \ni k_0$ whose closure $\bar{N}$ lies in it. Let $U' = \{x \in X : \{x\} \times \bar{N} \subseteq W'\}$, so $x_0 \in U'$. The set U' is open: for $x \in U'$, each (x, k) with $k \in \bar{N}$ has a basic neighbourhood $M_k \times O_k \subseteq W'$, finitely many O_k cover the compact set $\bar{N}$, and the intersection M of the corresponding M_k satisfies $M \times \bar{N} \subseteq W'$. It is saturated: if $q(x) = q(x')$ and $x \in U'$, then (x', k) and (x, k) have the same image, and W' is a union of fibres of $q \times \text{id}$. So $q(U')$ is open, and $(y_0, k_0) \in q(U') \times N \subseteq W$. Hence W is open. For the second statement, if $V \subseteq U$ and $q^{-1}(V)$ is open in $q^{-1}(U)$, then it is open in X , so V is open in Y . $\square$

17.3 The fundamental group

Definition 17.3.1. For paths $\gamma, \delta: I \rightarrow X$ with $\gamma(1) = \delta(0)$, the *concatenation* $\gamma * \delta$ is $\gamma(2s)$ for $s \leq 1/2$ and $\delta(2s - 1)$ for $s \geq 1/2$ (first γ , then δ), and the *reverse* of γ is $\bar{\gamma}(s) = \gamma(1 - s)$. We write $\gamma \simeq \delta$ for homotopy relative to $\{0, 1\}$ and $[\gamma]$ for the class. A *loop* at x_0 is a path with $\gamma(0) = \gamma(1) = x_0$, and c_{x_0} denotes the constant loop.

Lemma 17.3.2. *Concatenation is compatible with $\simeq$, and for composable paths:*

1. $\gamma \circ \varphi \simeq \gamma$ for every map $\varphi: I \rightarrow I$ with $\varphi(0) = 0$ and $\varphi(1) = 1$;
2. $(\gamma * \delta) * \epsilon \simeq \gamma * (\delta * \epsilon)$, and $c_{\gamma(0)} * \gamma \simeq \gamma \simeq \gamma * c_{\gamma(1)}$;
3. $\gamma * \bar{\gamma} \simeq c_{\gamma(0)}$ and $\bar{\gamma} * \gamma \simeq c_{\gamma(1)}$.

Proof. If H and K are homotopies $\gamma \simeq \gamma'$ and $\delta \simeq \delta'$, then $H(-, t) * K(-, t)$ is a homotopy $\gamma * \delta \simeq \gamma' * \delta'$. (1) $H(s, t) = \gamma((1 - t)\varphi(s) + ts)$. (2) Each side is the other composed with a piecewise affine φ as in (1): for instance $(\gamma * \delta) * \epsilon = (\gamma * (\delta * \epsilon)) \circ \varphi$ with φ affine on $[0, \frac{1}{4}]$, $[\frac{1}{4}, \frac{1}{2}]$, $[\frac{1}{2}, 1]$ mapping them to $[0, \frac{1}{2}]$, $[\frac{1}{2}, \frac{3}{4}]$, $[\frac{3}{4}, 1]$. (3) $H(s, t) = \gamma(2s(1 - t))$ for $s \leq 1/2$ and $H(s, t) = \gamma((2 - 2s)(1 - t))$ for $s \geq 1/2$ is a homotopy from $\gamma * \bar{\gamma}$ to $c_{\gamma(0)}$; apply this to $\bar{\gamma}$ for the second statement. $\square$

Definition 17.3.3. The *fundamental group* $\pi_1(X, x_0)$ is the set of classes of loops at x_0 with the multiplication $[\gamma][\delta] = [\gamma * \delta]$, which is a group by Lemma 17.3.2, with identity $[c_{x_0}]$ and inverse $[\gamma]^{-1} = [\bar{\gamma}]$. (The group is written multiplicatively; note that $[\gamma][\delta]$ means “first γ ”.) A map $f: (X, x_0) \rightarrow (Y, y_0)$ of pointed spaces induces a homomorphism $f_*[\gamma] = [f \circ \gamma]$, and $(gf)_* = g_*f_*$, $\text{id}_* = \text{id}$. A path-connected space X is *simply connected* if $\pi_1(X, x_0)$ is trivial.

Lemma 17.3.4. *A path h from x_0 to x_1 induces an isomorphism $\beta_h: \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$, $[\gamma] \mapsto [\bar{h} * \gamma * h]$. If H is a homotopy from f to $g: X \rightarrow Y$ and $h(t) = H(x_0, t)$, then $g_* = \beta_h \circ f_*: \pi_1(X, x_0) \rightarrow \pi_1(Y, g(x_0))$. Consequently a homotopy equivalence induces isomorphisms on fundamental groups, and contractible spaces are simply connected.*

Proof. β_h is a homomorphism by Lemma 17.3.2, with inverse $\beta_{\bar{h}}$. For the second statement let γ be a loop at x_0 and $F(s, t) = H(\gamma(s), t)$ on the square I^2 . In the convex set I^2 , the path along the bottom and then the right side and the path along the left side and then the top

have the same endpoints, so they are homotopic relative to the endpoints (Example 17.2.3(1)). Composing with F gives $(f \circ \gamma) * h \simeq h * (g \circ \gamma)$, that is, $g_*[\gamma] = [h * (f \circ \gamma) * h]$.

Let f be a homotopy equivalence with inverse g . Since $gf \simeq \text{id}_X$, the composite $\pi_1(X, x_0) \xrightarrow{f_*} \pi_1(Y, f(x_0)) \xrightarrow{g_*} \pi_1(X, gf(x_0))$ is β_h , a bijection. Since $fg \simeq \text{id}_Y$, the composite $\pi_1(Y, f(x_0)) \xrightarrow{g_*} \pi_1(X, gf(x_0)) \xrightarrow{f_*} \pi_1(Y, fgf(x_0))$ is also a bijection. So this g_* is injective and surjective, hence bijective, and then $f_* = g_*^{-1}(g_* f_*)$ is bijective. $\square$

Proposition 17.3.5. $\pi_1(X \times Y, (x_0, y_0)) \cong \pi_1(X, x_0) \times \pi_1(Y, y_0)$ via the projections.

Proof. Loops and homotopies in $X \times Y$ are pairs of loops and homotopies in X and Y (Proposition 16.4.6(2)). $\square$

17.4 Covering spaces and the circle

Definition 17.4.1. A *covering map* is a surjective map $p: E \rightarrow B$ such that every $b \in B$ has an open neighbourhood U that is *evenly covered*: $p^{-1}(U)$ is a disjoint union of open sets (the *sheets* over U), each mapped homeomorphically onto U by p . A *lift* of a map $f: Y \rightarrow B$ is a map $\tilde{f}: Y \rightarrow E$ with $p\tilde{f} = f$.

Lemma 17.4.2 (Unique lifting). *Let $p: E \rightarrow B$ be a covering map and Y connected. Two lifts of $f: Y \rightarrow B$ that agree at one point are equal.*

Proof. Let $\tilde{f}_1, \tilde{f}_2$ be lifts, $y \in Y$, U an evenly covered neighbourhood of $f(y)$, and $\tilde{U}_1, \tilde{U}_2$ the sheets containing $\tilde{f}_1(y), \tilde{f}_2(y)$. On the open neighbourhood $N = \tilde{f}_1^{-1}(\tilde{U}_1) \cap \tilde{f}_2^{-1}(\tilde{U}_2)$ of y : if $\tilde{f}_1(y) = \tilde{f}_2(y)$, then $\tilde{U}_1 = \tilde{U}_2$ and, p being injective on it, $\tilde{f}_1 = \tilde{f}_2$ on N ; if $\tilde{f}_1(y) \neq \tilde{f}_2(y)$, then $\tilde{U}_1 \neq \tilde{U}_2$ (as p is injective on sheets), the sheets are disjoint and $\tilde{f}_1 \neq \tilde{f}_2$ on N . So the set where the lifts agree is open and closed, and nonempty, hence all of Y . $\square$

Proposition 17.4.3 (Homotopy lifting). *Let $p: E \rightarrow B$ be a covering map, $F: Y \times I \rightarrow B$ a homotopy and $\tilde{F}_0: Y \rightarrow E$ a lift of $F(-, 0)$. There is a unique lift $\tilde{F}: Y \times I \rightarrow E$ of F with $\tilde{F}(-, 0) = \tilde{F}_0$. In particular (for Y a point) every path in B has a unique lift with a given initial point.*

Proof. Fix $y_0 \in Y$. Each (y_0, t) has a neighbourhood $N_t \times J_t$ mapped by F into an evenly covered set. By compactness of I there are an open $N \ni y_0$ and a partition $0 = t_0 < t_1 < \dots < t_m = 1$ with $F(N \times [t_i, t_{i+1}]) \subseteq U_i$, U_i evenly covered (intersect finitely many N_t , and take a partition subordinate to a Lebesgue number of the cover (J_t) , Lemma 16.7.3). Suppose a lift $\tilde{F}$ has been constructed on $N \times [0, t_i]$ (for $i = 0$ this is $\tilde{F}_0$ on $N \times \{0\}$). Let $\tilde{U}_i$ be the sheet over U_i containing $\tilde{F}(y_0, t_i)$; replacing N by the open set $\{y \in N : \tilde{F}(y, t_i) \in \tilde{U}_i\}$, define $\tilde{F} = (p|_{\tilde{U}_i})^{-1} \circ F$ on $N \times [t_i, t_{i+1}]$; this agrees on $N \times \{t_i\}$ with the previous definition, and the pasting lemma gives a lift on $N \times [0, t_{i+1}]$. After m steps we have a lift on $N_{y_0} \times I$ for an open $N_{y_0} \ni y_0$.

For each y , the restrictions to $\{y\} \times I$ of two lifts agreeing at $(y, 0)$ coincide, by Lemma 17.4.2. Hence the lifts on the sets $N_{y_0} \times I$ agree on overlaps, glue to a continuous lift on $Y \times I$ (pasting lemma for the open cover), and this lift is unique. $\square$

Corollary 17.4.4. *Let $p: E \rightarrow B$ be a covering map, $e_0 \in E$ and $b_0 = p(e_0)$.*

1. *If $\gamma \simeq \gamma'$ are paths starting at b_0 , their lifts starting at e_0 have the same endpoint and are homotopic.*
2. *$p_*: \pi_1(E, e_0) \rightarrow \pi_1(B, b_0)$ is injective, and its image consists of the classes of loops whose lift starting at e_0 is a loop.*

3. The monodromy $e \cdot [\gamma] = \tilde{\gamma}_e(1)$, where $\tilde{\gamma}_e$ is the lift of γ starting at e , is a right action of $\pi_1(B, b_0)$ on the fibre $p^{-1}(b_0)$. If E is path-connected, the action is transitive and the stabiliser of e_0 is $p_*\pi_1(E, e_0)$; so the fibre is in bijection with the set of right cosets of $p_*\pi_1(E, e_0)$.

Proof. (1) Lift a homotopy F from γ to γ' with $\tilde{F}(-, 0) = \tilde{\gamma}$. The paths $t \mapsto \tilde{F}(0, t)$ and $t \mapsto \tilde{F}(1, t)$ lift constant paths, so they are constant by uniqueness; hence $\tilde{F}(-, 1)$ is the lift of γ' starting at e_0 and has the same endpoint as $\tilde{\gamma}$. (2) If $p \circ \tilde{\gamma} \simeq c_{b_0}$ for a loop $\tilde{\gamma}$, lifting the homotopy gives $\tilde{\gamma} \simeq c_{e_0}$ by (1). If the lift of γ is a loop $\tilde{\gamma}$, then $[\gamma] = p_*[\tilde{\gamma}]$; conversely the lift of $p \circ \tilde{\gamma}$ is $\tilde{\gamma}$, and (1) handles homotopic loops. (3) The action is well defined by (1), and the lift of $\gamma * \delta$ starting at e is $\tilde{\gamma}_e * \tilde{\delta}_{e \cdot [\gamma]}$, so $e \cdot ([\gamma][\delta]) = (e \cdot [\gamma]) \cdot [\delta]$. If η is a path from e_0 to e , then $e_0 \cdot [p \circ \eta] = e$. The stabiliser statement is (2). $\square$

Proposition 17.4.5. Let $Q = \{x \in \mathbb{R}^2 : \|x\|_\infty = 1\}$ and let $P: \mathbb{R} \rightarrow Q$ be the map of period 4 that on $[0, 4]$ runs affinely through the vertices $(1, -1), (1, 1), (-1, 1), (-1, -1), (1, -1)$ at the times $0, 1, 2, 3, 4$. Let $r: Q \rightarrow S^1$, $x \mapsto x/\|x\|_2$, and $p = r \circ P$. Then p is a covering map, $p(t) = p(t')$ if and only if $t - t' \in 4\mathbb{Z}$, and $p(t + 4) = p(t)$.

Proof. P is continuous by the pasting lemma (applied to the closed intervals $[k, k + 1]$ inside each bounded open interval), and r is a homeomorphism with inverse $x \mapsto x/\|x\|_\infty$ (Example 16.6.9). By construction P is injective on $[a, a + 4)$ for every a and has period 4, which gives the second statement. For $a \in \mathbb{R}$, the restriction $P|_{[a, a+4]}$ is a closed map (Proposition 16.6.2(4)). If $V \subseteq (a, a + 4)$ is open, then $P(V) = Q \setminus P([a, a + 4] \setminus V)$, because P is surjective, injective on $(a, a + 4)$ and identifies only a and $a + 4$; so $P(V)$ is open, and P maps $(a, a + 4)$ homeomorphically onto $Q \setminus \{P(a)\}$. Given $q \in Q$, choose a with $P(a) \neq q$; then $U = Q \setminus \{P(a)\}$ is an open neighbourhood of q with $P^{-1}(U) = \bigsqcup_{k \in \mathbb{Z}} (a + 4k, a + 4k + 4)$, each interval mapped homeomorphically onto U . Composing with r proves that p is a covering map. $\square$

Theorem 17.4.6. Let $b_0 = p(0)$ and $\omega(s) = p(4s)$. Then $\pi_1(S^1, b_0) \cong \mathbb{Z}$, with $[\omega]$ corresponding to 1. Hence $\pi_1(S^1, x) \cong \mathbb{Z}$ for every $x \in S^1$.

Proof. The space $\mathbb{R}$ is contractible, hence simply connected (Lemma 17.3.4), and path-connected. By Corollary 17.4.4(3), $[\gamma] \mapsto 0 \cdot [\gamma]$ is a bijection from $\pi_1(S^1, b_0)$ onto $p^{-1}(b_0) = 4\mathbb{Z}$. If the lifts of γ and δ starting at 0 end at $4m$ and $4n$, then $t \mapsto \tilde{\delta}_0(t) + 4m$ lifts δ (as p has period 4) and starts at $4m$, so the lift of $\gamma * \delta$ ends at $4(m + n)$. So $[\gamma] \mapsto \tilde{\gamma}_0(1)/4$ is an isomorphism onto $\mathbb{Z}$, and $\tilde{\omega}_0(s) = 4s$ ends at 4. The last statement follows from Lemma 17.3.4. $\square$

Remark 17.4.7. With the exponential function of Chapter 25, the usual covering is $\mathbb{R} \rightarrow S^1$, $t \mapsto (\cos 2\pi t, \sin 2\pi t)$. We used a piecewise affine model so as not to depend on analysis. A map $f: S^1 \rightarrow S^1$ induces multiplication by an integer $\deg f$ on $\pi_1(S^1) \cong \mathbb{Z}$ (after identifying basepoints by a path, which does not matter since $\mathbb{Z}$ is abelian), its *degree*.

Corollary 17.4.8 (Brouwer fixed point theorem in dimension 2). *There is no retraction $D^2 \rightarrow S^1$, and every map $f: D^2 \rightarrow D^2$ has a fixed point.*

Proof. If $r: D^2 \rightarrow S^1$ were a retraction with inclusion i , then $r_*i_* = \text{id}$ on $\pi_1(S^1) \cong \mathbb{Z}$ would factor through $\pi_1(D^2) = 0$. If f had no fixed point, let $u(x) = x - f(x) \neq 0$ and $r(x) = x + \lambda(x)u(x)$, where

$$\lambda(x) = \frac{-\langle x, u \rangle + \sqrt{\langle x, u \rangle^2 + \|u\|^2(1 - \|x\|^2)}}{\|u\|^2}$$

is the non-negative solution of $\|x + \lambda u\|^2 = 1$. The square root is continuous on $[0, \infty)$ since $|\sqrt{a} - \sqrt{b}| \leq \sqrt{|a - b|}$, so r is continuous, and $r(x) \in S^1$. If $\|x\| = 1$, then $\langle x, u \rangle = 1 - \langle x, f(x) \rangle \geq 0$ by Cauchy–Schwarz, so $\lambda(x) = 0$ and $r(x) = x$. This contradicts the first statement. $\square$

- Example 17.4.9.** 1. For $n \geq 1$ the map $S^1 \rightarrow S^1$, $p(t) \mapsto p(nt)$, is well defined and is a covering map with n sheets; it corresponds to the subgroup $n\mathbb{Z} \subseteq \mathbb{Z}$ in Theorem 17.6.6.
2. The projection $q: S^n \rightarrow \mathbb{RP}^n$ is a two-sheeted covering: for $x \in S^n$ the open hemisphere $W = \{y \in S^n : \langle x, y \rangle > 0\}$ is mapped injectively by q , $q^{-1}(q(W)) = W \sqcup (-W)$ is open, so $q(W)$ is open and $q|_W, q|_{-W}$ are homeomorphisms onto it (q is open, since $q^{-1}(q(V)) = V \cup (-V)$). For $n \geq 2$, S^n is simply connected (Corollary 17.5.2), so $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$ by Corollary 17.4.4(3).

17.5 The Seifert–van Kampen theorem

Theorem 17.5.1 (Seifert–van Kampen). *Let $X = U \cup V$ with U, V open and $U, V, U \cap V$ path-connected, and $x_0 \in U \cap V$. The homomorphism*

$$\Phi: \pi_1(U, x_0) *_{\pi_1(U \cap V, x_0)} \pi_1(V, x_0) \rightarrow \pi_1(X, x_0)$$

induced by the inclusions (Proposition 6.10.7) is an isomorphism.

Proof. Write G for the amalgamated product and ι_U, ι_V for the maps from $\pi_1(U), \pi_1(V)$ to G ; by Definition 6.10.6, G is generated by their images. For loops $\gamma_1, \dots, \gamma_k$ at x_0 let $\gamma_1 \cdots \gamma_k$ denote the loop running through γ_i on $[(i-1)/k, i/k]$; it is homotopic to the iterated concatenation by Lemma 17.3.2(1).

Surjectivity. Let γ be a loop at x_0 . By the Lebesgue number lemma there is a partition $0 = s_0 < \dots < s_m = 1$ such that each $\gamma([s_{i-1}, s_i])$ lies in a set $A_i \in \{U, V\}$; merging adjacent intervals with the same label, we may assume $A_i \neq A_{i+1}$, so $\gamma(s_i) \in U \cap V$ for $0 < i < m$. Choose paths g_i in $U \cap V$ from x_0 to $\gamma(s_i)$, with g_0, g_m constant. Then $\gamma \simeq \prod_i (g_{i-1} * \gamma|_{[s_{i-1}, s_i]} * \bar{g}_i)$ (with the restriction reparametrised to I), a product of loops in U or V .

Injectivity. Let $\gamma_1, \dots, \gamma_k$ and $\gamma'_1, \dots, \gamma'_l$ be loops at x_0 , each in U or V (label them A_i and A'_j), with $F: I \times I \rightarrow X$ a homotopy from $\gamma_1 \cdots \gamma_k$ to $\gamma'_1 \cdots \gamma'_l$ relative to $\{0, 1\}$. We show $\prod_i \iota_{A_i}[\gamma_i] = \prod_j \iota_{A'_j}[\gamma'_j]$ in G . This proves that $\ker \Phi$ is trivial: an element of the kernel can be written as $\prod_i \iota_{A_i}[\gamma_i]$, and $\gamma_1 \cdots \gamma_k$ is then homotopic to the constant loop, which we take as γ'_1 with $l = 1$.

Choose N divisible by k and l such that each small square R of the grid of mesh $1/N$ is mapped by F into U or V (Lebesgue number lemma), and fix a label $A_R \in \{U, V\}$ with $F(R) \subseteq A_R$. For each grid vertex v let S_v be the intersection of the labels of the squares containing v and, for v on the bottom (top) edge in the domain of γ_i (γ'_j), also of A_i (A'_j). Then S_v is U, V or $U \cap V$, so it is path-connected, and it contains $F(v)$. Choose a path g_v in S_v from x_0 to $F(v)$, constant if $F(v) = x_0$ and v lies on the boundary of the square. For an edge e from v to w let $\ell_e = g_v * F|_e * \bar{g}_w$, a loop in the intersection of the labels of the (one or two) squares containing e . Put $\phi(e) = \iota_{A_R}[\ell_e]$ for a square R containing e ; this does not depend on R , because if two squares with different labels contain e , then ℓ_e lies in $U \cap V$ and the two images agree in G by the amalgamation relations. For an edge path P let $\phi(P)$ be the product of the $\phi(e)$ along P .

For a square R with lower left vertex a , lower right b , upper left c and upper right d , all four loops ℓ_e of its edges lie in A_R , and $F|_R$ gives, in the convex set R , a homotopy relative to endpoints in A_R between $F|_{[a,b]} * F|_{[b,d]}$ and $F|_{[a,c]} * F|_{[c,d]}$. Hence $[\ell_{ab}][\ell_{bd}] = [g_a * F|_{[a,b]} * F|_{[b,d]} * \bar{g}_d] = [\ell_{ac}][\ell_{cd}]$ in $\pi_1(A_R)$. So replacing the two lower-right edges of a square by its two upper-left edges does not change $\phi(P)$, and moving square by square the path P_0 along the bottom and right sides of I^2 to the path P_1 along the left and top sides gives $\phi(P_0) = \phi(P_1)$. On the vertical sides F is constant at x_0 and all g_v are constant, so those edges contribute 1. The bottom edges in the domain of γ_i have loops ℓ_e in A_i ,

and their product in $\pi_1(A_i)$ is $[\gamma_i]$ (the paths g_v cancel, and g_v is constant at the ends of the domain); since each ℓ_e also lies in A_R , $\iota_{A_R}[\ell_e] = \iota_{A_i}[\ell_e]$. Hence $\phi(P_0) = \prod_i \iota_{A_i}[\gamma_i]$, and likewise $\phi(P_1) = \prod_j \iota_{A'_j}[\gamma'_j]$. $\square$

Corollary 17.5.2. *For $n \geq 2$ the sphere S^n is simply connected. If $X = U \cup V$ as in Theorem 17.5.1 with $U \cap V$ simply connected, then $\pi_1(X) \cong \pi_1(U) * \pi_1(V)$; if V is simply connected, then $\pi_1(U) \rightarrow \pi_1(X)$ is surjective with kernel the normal subgroup generated by the image of $\pi_1(U \cap V)$.*

Proof. Let $U = S^n \setminus \{N\}$ and $V = S^n \setminus \{-N\}$. By stereographic projection (Example 16.9.4), U and V are homeomorphic to $\mathbb{R}^n$, hence contractible, and $U \cap V$ is homeomorphic to $\mathbb{R}^n \setminus \{0\}$, which is path-connected for $n \geq 2$. So $\pi_1(S^n) \cong 1 *_{\pi_1(U \cap V)} 1 = 1$. The other statements follow from the presentation of the amalgamated product (Definition 6.10.6): the relations from $\pi_1(U \cap V)$ are trivial in the first case, and in the second they identify the image of each $h \in \pi_1(U \cap V)$ in $\pi_1(U)$ with 1. $\square$

Example 17.5.3. Let $X = S^1 \vee S^1$ be the wedge of two circles, the quotient of $S^1 \sqcup S^1$ identifying the two basepoints b_0 to a point x_0 . Let U (V) be the union of the first (second) circle with the second (first) circle minus the point $-b_0$. The open arc $S^1 \setminus \{-b_0\}$ deformation retracts onto b_0 (it is homeomorphic to an open interval by Proposition 17.4.5), so U and V deformation retract onto the circles (Lemma 17.2.4 ensures continuity of the homotopies on the quotient), and $U \cap V$ is contractible. By Corollary 17.5.2, $\pi_1(S^1 \vee S^1) \cong \mathbb{Z} * \mathbb{Z}$, the free group on two generators. It is not abelian, although $\pi_1(S^1 \times S^1) \cong \mathbb{Z}^2$ by Proposition 17.3.5.

Proposition 17.5.4 (Attaching cells). *Let X be path-connected, $n \geq 2$, $\varphi: S^{n-1} \rightarrow X$ a map, and $Y = X \cup_{\varphi} D^n$ the space obtained by attaching an n -cell (Example 16.4.10). Let $z_0 \in S^{n-1}$ and $x_0 = \varphi(z_0)$. The inclusion induces a surjection $\pi_1(X, x_0) \rightarrow \pi_1(Y, x_0)$. For $n = 2$ its kernel is the normal subgroup generated by $[\varphi \circ \omega]$, where ω is the loop of Theorem 17.4.6 based at z_0 ; for $n \geq 3$ it is an isomorphism.*

Proof. Let $q: X \sqcup D^n \rightarrow Y$ be the quotient map. For $C \subseteq X$ closed, $q^{-1}(q(C)) = C \sqcup \varphi^{-1}(C)$ is closed, so $q|_X$ is an embedding onto a closed subspace, which we identify with X ; similarly q maps the open ball $\text{int } D^n$ homeomorphically onto an open subset. Let $V = q(\text{int } D^n)$, which is contractible, and $U = Y \setminus \{q(0)\}$. On $q^{-1}(U) \times I = (X \times I) \sqcup ((D^n \setminus \{0\}) \times I)$ define H as the projection on $X \times I$ and as $q((1-t)z + tz/\|z\|)$ on the second part; it respects the identifications, so by Lemma 17.2.4 it induces a homotopy on $U \times I$, which is a deformation retraction of U onto X . The set $U \cap V$ is homeomorphic to $\text{int } D^n \setminus \{0\}$, which is path-connected and deformation retracts onto the sphere of radius $1/2$; so $\pi_1(U \cap V)$ is $\mathbb{Z}$ for $n = 2$ (generated by $c(s) = q(\omega(s)/2)$) and trivial for $n \geq 3$ (Corollary 17.5.2).

Work at the basepoint $y_1 = q(z_0/2) \in U \cap V$. By Corollary 17.5.2, $\pi_1(U, y_1) \rightarrow \pi_1(Y, y_1)$ is surjective with kernel the normal subgroup generated by the image of $\pi_1(U \cap V, y_1)$. Transport to x_0 along the path $\delta(t) = H(y_1, t)$ from y_1 to x_0 : the inclusion $X \rightarrow U$ is a homotopy equivalence, and by Lemma 17.3.4 applied to the homotopy H , the class $[c]$ corresponds to $[\bar{\delta} * c * \delta] = [H_1 \circ c] = [\varphi \circ \omega]$ in $\pi_1(X, x_0)$. Since the change-of-basepoint isomorphisms β_{δ} in U and Y are compatible with the inclusion, the statement follows. $\square$

17.6 Classification of covering spaces

In this section B is path-connected and locally path-connected. Then a covering space E is locally path-connected, being locally homeomorphic to B , and if E is connected it is path-connected (Proposition 16.5.7).

Proposition 17.6.1 (Lifting criterion). *Let $p: (E, e_0) \rightarrow (B, b_0)$ be a covering map, Y path-connected and locally path-connected, and $f: (Y, y_0) \rightarrow (B, b_0)$ a map. A lift $\tilde{f}: (Y, y_0) \rightarrow (E, e_0)$ exists if and only if $f_*\pi_1(Y, y_0) \subseteq p_*\pi_1(E, e_0)$, and it is then unique.*

Proof. Necessity follows from $f_* = p_*\tilde{f}_*$, and uniqueness from Lemma 17.4.2. Conversely, for $y \in Y$ choose a path η from y_0 to y and let $\tilde{f}(y)$ be the endpoint of the lift of $f \circ \eta$ starting at e_0 . If η' is another such path, the loop $h = (f \circ \eta') * (f \circ \bar{\eta})$ represents an element of $f_*\pi_1(Y) \subseteq p_*\pi_1(E)$, so its lift starting at e_0 is a loop (Corollary 17.4.4(2)). This lift is the lift of $f \circ \eta'$, ending at some e' , followed by the lift of $f \circ \bar{\eta}$ starting at e' , which therefore ends at e_0 ; reversing it gives the lift of $f \circ \eta$ starting at e_0 , which ends at e' . So $\tilde{f}$ is well defined, and $p\tilde{f} = f$.

Continuity: let $y \in Y$, U an evenly covered neighbourhood of $f(y)$ and $\tilde{U}$ the sheet containing $\tilde{f}(y)$. Choose a path-connected open $W \ni y$ with $f(W) \subseteq U$. For $y' \in W$, using $\eta * \sigma$ with σ a path in W from y to y' , the lift of $f \circ \sigma$ starting at $\tilde{f}(y)$ is $(p|_{\tilde{U}})^{-1} \circ f \circ \sigma$. So $\tilde{f} = (p|_{\tilde{U}})^{-1} \circ f$ on W . $\square$

Definition 17.6.2. Two coverings $p: E \rightarrow B$ and $p': E' \rightarrow B$ are *isomorphic* if there is a homeomorphism $\tau: E \rightarrow E'$ with $p'\tau = p$. The automorphisms of a covering form the group $\text{Deck}(p)$ of *deck transformations*. A covering with E path-connected is *normal* (or *Galois*) if $\text{Deck}(p)$ acts transitively on the fibres, and *universal* if E is simply connected.

Proposition 17.6.3. *Let $p: E \rightarrow B$ be a covering with E path-connected, $e_0 \in p^{-1}(b_0)$ and $H = p_*\pi_1(E, e_0)$.*

1. *For $e \in p^{-1}(b_0)$ there is a deck transformation τ with $\tau(e_0) = e$ if and only if $p_*\pi_1(E, e) = H$. For $e = e_0 \cdot [\gamma]$ one has $p_*\pi_1(E, e) = [\gamma]^{-1}H[\gamma]$.*
2. *The covering is normal if and only if H is a normal subgroup, and then $\text{Deck}(p) \cong \pi_1(B, b_0)/H$. In particular, for a universal covering, $\text{Deck}(p) \cong \pi_1(B, b_0)$.*

Proof. (1) If $p_*\pi_1(E, e) = H$, the lifting criterion gives maps $\tau, \tau': E \rightarrow E$ over B with $\tau(e_0) = e$ and $\tau'(e) = e_0$; then $\tau'\tau$ and $\tau\tau'$ are lifts of p fixing a point, hence identities. The converse is clear. If $\tilde{\gamma}$ is the lift of γ from e_0 to e , then $[\tilde{\alpha}] \mapsto [\tilde{\gamma} * \tilde{\alpha} * \tilde{\gamma}]$ is an isomorphism $\pi_1(E, e_0) \rightarrow \pi_1(E, e)$, whence the formula.

(2) The first statement follows from (1) and transitivity of monodromy (Corollary 17.4.4). A deck transformation commutes with monodromy, $\tau(e \cdot [\gamma]) = \tau(e) \cdot [\gamma]$, since $\tau \circ \tilde{\gamma}_e$ lifts γ from $\tau(e)$. For H normal let $\Psi[\gamma]$ be the deck transformation with $\Psi[\gamma](e_0) = e_0 \cdot [\gamma]$. Then $\Psi([\gamma][\delta])(e_0) = (e_0 \cdot [\gamma]) \cdot [\delta] = \Psi[\gamma](e_0 \cdot [\delta]) = \Psi[\gamma]\Psi[\delta](e_0)$, so Ψ is a homomorphism (deck transformations agreeing at a point are equal). It is surjective by (1), and its kernel is the stabiliser H of e_0 . $\square$

Definition 17.6.4. B is *semilocally simply connected* if every point b has a neighbourhood U such that $\pi_1(U, b) \rightarrow \pi_1(B, b)$ is trivial.

Theorem 17.6.5. *If B is path-connected, locally path-connected and semilocally simply connected, then B has a universal covering.*

Proof. Let $\tilde{B}$ be the set of classes $[\gamma]$ of paths starting at b_0 (up to homotopy relative to endpoints), and $p[\gamma] = \gamma(1)$. Let $\mathcal{U}$ be the set of path-connected open $U \subseteq B$ with $\pi_1(U) \rightarrow \pi_1(B)$ trivial (at any, hence every, basepoint in U). Every path-connected open subset of a member of $\mathcal{U}$ is in $\mathcal{U}$, so by the hypotheses $\mathcal{U}$ is a basis of B . For $U \in \mathcal{U}$ and $\gamma(1) \in U$ let $U_{[\gamma]} = \{[\gamma * \eta] : \eta \text{ a path in } U \text{ starting at } \gamma(1)\}$. Then $p: U_{[\gamma]} \rightarrow U$ is bijective: surjective as U is path-connected, and injective since for two paths η, η' in U with the same endpoints, $\eta * \bar{\eta}'$ is null-homotopic in B . If $[\gamma'] \in U_{[\gamma]}$ then $U_{[\gamma']} = U_{[\gamma]}$. Hence if

$[\gamma''] \in U_{[\gamma]} \cap V_{[\gamma']}$, choosing $W \in \mathcal{U}$ with $\gamma''(1) \in W \subseteq U \cap V$ gives $[\gamma''] \in W_{[\gamma'']} \subseteq U_{[\gamma]} \cap V_{[\gamma']}$; so the sets $U_{[\gamma]}$ form a basis of a topology on $\tilde{B}$ (Lemma 16.2.6).

For $V \subseteq U$ in $\mathcal{U}$, p maps the basic open sets $V_{[\gamma']} \subseteq U_{[\gamma']}$ onto V , and $p^{-1}(V) \cap U_{[\gamma]} = V_{[\gamma']}$ for $[\gamma'] \in U_{[\gamma]}$ with $\gamma'(1) \in V$. So $p|_{U_{[\gamma]}}$ is a homeomorphism onto U . The set $p^{-1}(U)$ is the union of the sets $U_{[\gamma]}$ with $\gamma(1) \in U$, and two of them are equal or disjoint. So p is a covering map.

For $[\gamma] \in \tilde{B}$, let $\gamma_t(s) = \gamma(ts)$. The path $t \mapsto [\gamma_t]$ is continuous: near t_0 , with $\gamma([t_0, t]) \subseteq U \in \mathcal{U}$, it equals $(p|_{U_{[\gamma_{t_0}]}})^{-1} \circ \gamma$. It is the lift of γ starting at $\tilde{b}_0 = [c_{b_0}]$ and ends at $[\gamma]$. So $\tilde{B}$ is path-connected, and a loop γ whose lift is a loop satisfies $[\gamma] = [c_{b_0}]$. By Corollary 17.4.4(2), $p_*\pi_1(\tilde{B}) = 1$, and as p_* is injective, $\tilde{B}$ is simply connected. $\square$

Theorem 17.6.6 (Classification of coverings). *Let B be path-connected, locally path-connected and semilocally simply connected, and $b_0 \in B$. Sending $p: (E, e_0) \rightarrow (B, b_0)$ to $p_*\pi_1(E, e_0)$ is a bijection between isomorphism classes of pointed coverings with E path-connected and subgroups of $\pi_1(B, b_0)$. Forgetting basepoints, isomorphism classes of coverings with E path-connected correspond to conjugacy classes of subgroups.*

Proof. Injectivity: if $p_*\pi_1(E, e_0) = p'_*\pi_1(E', e'_0)$, the lifting criterion gives pointed maps $E \rightarrow E'$ and $E' \rightarrow E$ over B , whose composites are identities by uniqueness of lifts.

Surjectivity: let $H \subseteq \pi_1(B, b_0)$ and $\tilde{B}$ be as in the proof of Theorem 17.6.5. Put $[\gamma] \sim [\gamma']$ if $\gamma(1) = \gamma'(1)$ and $[\gamma * \bar{\gamma}'] \in H$, an equivalence relation, and let $E_H = \tilde{B}/\sim$. If $[\gamma] \sim [\gamma']$ then $[\gamma * \eta] \sim [\gamma' * \eta]$, so the saturation of a basic set $U_{[\gamma]}$ is the union of the sets $U_{[\gamma']}$ with $[\gamma'] \sim [\gamma]$; hence the projection $\tilde{B} \rightarrow E_H$ is open, and the images of the sets $U_{[\gamma]}$ are open, pairwise equal or disjoint, and mapped bijectively (as distinct points of $U_{[\gamma]}$ have distinct endpoints), continuously and openly onto U . So $E_H \rightarrow B$ is a covering, and E_H is path-connected. The lift of a loop γ at b_0 starting at the class of $[c_{b_0}]$ is $t \mapsto$ class of $[\gamma_t]$, which is a loop if and only if $[\gamma] \in H$; by Corollary 17.4.4(2) the subgroup of E_H is H .

Changing the basepoint within the fibre replaces the subgroup by a conjugate, and every conjugate arises (Proposition 17.6.3); an isomorphism of unpointed coverings carries e_0 to some point of the fibre, which gives the last statement. $\square$

Counterexample 17.6.7. Semilocal simple connectivity is necessary: a space with a simply connected covering $p: E \rightarrow B$ is semilocally simply connected, because a loop in an evenly covered U lifts to a loop in a sheet, which is null-homotopic in E . The *Hawaiian earring*, the union of the circles of radius $1/n$ centred at $(1/n, 0)$ in $\mathbb{R}^2$, is path-connected and locally path-connected but not semilocally simply connected (every neighbourhood of the origin contains a whole circle, which is a retract of the earring and hence not null-homotopic in it), so it has no universal covering [Hat02].

17.7 CW complexes

Definition 17.7.1. A *CW complex* is a space X with an increasing sequence of subspaces $X^0 \subseteq X^1 \subseteq \dots$, its *skeleta*, such that X^0 is discrete, X^n is obtained from X^{n-1} by attaching a family of n -cells, $X^n = (X^{n-1} \sqcup \coprod_{\alpha} D_{\alpha}^n) / (z \sim \varphi_{\alpha}(z))$ for maps $\varphi_{\alpha}: S^{n-1} \rightarrow X^{n-1}$, and $X = \bigcup_n X^n$ carries the final topology for the inclusions of the X^n (a set is open if and only if its intersection with every X^n is open). The images e_{α}^n of the open balls are the *open cells*, the maps $\Phi_{\alpha}: D_{\alpha}^n \rightarrow X$ are the *characteristic maps*, and X is *finite* if there are finitely many cells and *finite-dimensional* if $X = X^n$ for some n . A *subcomplex* is a closed union of cells containing, with each cell, the image of its characteristic map; (X, A) is then a *CW pair*.

- Example 17.7.2.** 1. S^n has a CW structure with one 0-cell and one n -cell attached by the constant map, since $D^n/S^{n-1} \cong S^n$. It also has a structure with two cells in each dimension $0, \dots, n$ (the hemispheres), compatible with the antipodal map.
2. Passing to the quotient by the antipodal map in the second structure gives $\mathbb{RP}^n$ one cell in each dimension $0, \dots, n$, with $\mathbb{RP}^n = \mathbb{RP}^{n-1} \cup e^n$ attached along the double covering $S^{n-1} \rightarrow \mathbb{RP}^{n-1}$. Similarly $\mathbb{CP}^n = \mathbb{CP}^{n-1} \cup e^{2n}$ has one cell in each even dimension (Chapter 41).
3. A graph is a 1-dimensional CW complex. The torus is obtained from $S^1 \vee S^1$ by attaching a 2-cell along the loop $aba^{-1}b^{-1}$, and the closed orientable surface of genus g from a wedge of $2g$ circles along $a_1b_1a_1^{-1}b_1^{-1} \cdots a_gb_ga_g^{-1}b_g^{-1}$. By Proposition 17.5.4 and Example 17.5.3 (extended to $2g$ circles), its fundamental group is $\langle a_1, b_1, \dots, a_g, b_g \mid \prod_i a_i b_i a_i^{-1} b_i^{-1} \rangle$.

Theorem 17.7.3. *Let X be a CW complex.*

1. X is Hausdorff, and in fact paracompact, and it is locally contractible.
2. Every compact subset of X lies in a finite subcomplex.
3. (Homotopy extension property) If (X, A) is a CW pair, then $X \times \{0\} \cup A \times I$ is a deformation retract of $X \times I$; so every map $X \times \{0\} \cup A \times I \rightarrow Y$ extends to $X \times I$.
4. The inclusion $X^2 \rightarrow X$ induces an isomorphism on fundamental groups (at a 0-cell).

These are [Hat02, Appendix, and Propositions 0.16 and 1.26]. Part (4) follows from Proposition 17.5.4 applied to the finite subcomplexes containing a given loop or homotopy, using (2).

Theorem 17.7.4 (Cellular approximation). *Every map $f : X \rightarrow Y$ of CW complexes is homotopic to a cellular map ($f(X^n) \subseteq Y^n$ for all n), by a homotopy relative to any subcomplex on which f is already cellular.*

This is [Hat02, Theorem 4.8].

17.8 Higher homotopy groups and fibrations

Definition 17.8.1. For $n \geq 1$, $\pi_n(X, x_0)$ is the set of homotopy classes, relative to ∂I^n , of maps $f : (I^n, \partial I^n) \rightarrow (X, x_0)$, with $[f] + [g] = [f *_1 g]$, where $(f *_1 g)(s) = f(2s_1, s_2, \dots, s_n)$ for $s_1 \leq 1/2$ and $g(2s_1 - 1, s_2, \dots, s_n)$ for $s_1 \geq 1/2$. We write the operation additively for $n \geq 2$ and multiplicatively for $n = 1$, where it is the fundamental group. Also $\pi_0(X, x_0)$ is the pointed set of path components.

Proposition 17.8.2. $\pi_n(X, x_0)$ is a group, abelian for $n \geq 2$. A pointed map induces homomorphisms f_* , pointed homotopic maps induce the same homomorphisms, and $\pi_n(X \times Y) \cong \pi_n(X) \times \pi_n(Y)$.

Proof. The proofs of Lemma 17.3.2 apply verbatim in the first coordinate, the other coordinates being carried along. For $n \geq 2$ define $*_2$ in the same way using the second coordinate; it also induces a well-defined operation on π_n with the same identity e , the class of the constant map. For maps, $(a *_1 b) *_2 (c *_1 d) = (a *_2 c) *_1 (b *_2 d)$ holds exactly (both sides are a, b, c, d on the four quarters of $I^2 \times I^{n-2}$, rescaled). Hence on classes

$$a *_2 b = (a *_1 e) *_2 (e *_1 b) = (a *_2 e) *_1 (e *_2 b) = a *_1 b, \quad a *_1 b = (e *_1 a) *_2 (b *_1 e) = (e *_2 b) *_1 (a *_2 e) = b *_1 a$$

(the Eckmann–Hilton argument). The remaining statements are clear. $\square$

Proposition 17.8.3. *A covering map $p: E \rightarrow B$ induces isomorphisms $\pi_n(E, e_0) \rightarrow \pi_n(B, b_0)$ for $n \geq 2$. In particular $\pi_n(S^1) = 0$ and $\pi_n(\mathbb{RP}^m) \cong \pi_n(S^m)$ for $n \geq 2$.*

Proof. For $n \geq 2$, ∂I^n is homeomorphic to S^{n-1} (by $x \mapsto x/\|x\|_2$ after centring, as in Example 16.6.9), hence path-connected. Let $f: (I^n, \partial I^n) \rightarrow (B, b_0)$. The cube is simply connected and locally path-connected, so f has a lift $\tilde{f}$ with $\tilde{f}(0) = e_0$ (Proposition 17.6.1); $\tilde{f}(\partial I^n)$ is a connected subset of the fibre $p^{-1}(b_0)$, which is discrete, so $\tilde{f}(\partial I^n) = e_0$. So p_* is surjective. If $p \circ \tilde{f}$ is null-homotopic relative to ∂I^n , lift the homotopy (Proposition 17.4.3); the image of the connected set $\partial I^n \times I$ lies in the fibre, hence is e_0 , and at time 1 the lift of the constant map is constant. So p_* is injective. Apply this to $\mathbb{R} \rightarrow S^1$ ($\mathbb{R}$ is contractible) and $S^m \rightarrow \mathbb{RP}^m$. $\square$

Definition 17.8.4. A map $p: E \rightarrow B$ has the *homotopy lifting property* with respect to Y if for every homotopy $F: Y \times I \rightarrow B$ and lift $\tilde{F}_0$ of F_0 there is a lift $\tilde{F}$ of F with $\tilde{F}(-, 0) = \tilde{F}_0$ (not necessarily unique). It is a *Hurewicz fibration* if this holds for all Y , and a *Serre fibration* if it holds for all cubes $Y = I^k$. The fibre over b_0 is $F = p^{-1}(b_0)$. For a pair (X, A) with $x_0 \in A$ and $n \geq 1$, the *relative homotopy group* $\pi_n(X, A, x_0)$ is the set of homotopy classes of maps $(I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$, where $J^{n-1} = \partial I^n \setminus I^{n-1} \times \{0\}$; it is a group for $n \geq 2$.

Theorem 17.8.5 (Long exact sequences). 1. For a pair (X, A) there is a long exact sequence

$$\cdots \rightarrow \pi_n(A) \rightarrow \pi_n(X) \rightarrow \pi_n(X, A) \xrightarrow{\partial} \pi_{n-1}(A) \rightarrow \cdots \rightarrow \pi_0(X), \text{ with } \partial[f] = [f|_{I^{n-1} \times \{0\}}].$$

2. For a Serre fibration $p: E \rightarrow B$ with B path-connected, fibre F and $x_0 \in F$, the map $p_*: \pi_n(E, F, x_0) \rightarrow \pi_n(B, b_0)$ is an isomorphism for $n \geq 1$, which gives a long exact sequence

$$\cdots \rightarrow \pi_n(F) \rightarrow \pi_n(E) \rightarrow \pi_n(B) \rightarrow \pi_{n-1}(F) \rightarrow \cdots \rightarrow \pi_0(E) \rightarrow 0.$$

Sketch of proof. (1) Exactness at each term is a direct manipulation of maps of cubes, based on the *compression criterion*: a map $f: (I^n, \partial I^n, J^{n-1}) \rightarrow (X, A, x_0)$ represents 0 if and only if it is homotopic relative to ∂I^n to a map into A . (2) Surjectivity of p_* lifts a map $(I^n, \partial I^n) \rightarrow (B, b_0)$, viewed as a homotopy of maps $I^{n-1} \rightarrow B$, starting from the constant lift; injectivity lifts a homotopy in the same way, using the homotopy lifting property for $I^{n-1} \times I$ relative to J^{n-1} (which holds for Serre fibrations because $(I^n, J^{n-1}) \cong (I^n, I^{n-1} \times \{0\})$). Substituting into (1) for the pair (E, F) gives the sequence. The omitted verifications are in [Hat02, Theorems 4.3 and 4.41]. $\square$

Theorem 17.8.6 (Whitehead). *A map between connected CW complexes that induces isomorphisms on all homotopy groups π_n , $n \geq 1$, is a homotopy equivalence.*

This is [Hat02, Theorem 4.5].

Remark 17.8.7. Higher homotopy groups are hard to compute. By cellular approximation, $\pi_k(S^n) = 0$ for $k < n$ (a map $S^k \rightarrow S^n$ is homotopic to a cellular map, which misses the open n -cell and so factors through a point). The *Hopf fibration* $S^3 \rightarrow S^2 = \mathbb{CP}^1$, $(z_1, z_2) \mapsto [z_1 : z_2]$, has fibre S^1 , and its long exact sequence gives $\pi_3(S^2) \cong \pi_3(S^3) \cong \mathbb{Z}$, although $H_3(S^2) = 0$. Homology (Chapter 19) is much more computable, and the Hurewicz theorem relates the two.

17.9 Exercises

Exercise 17.9.1. Show that if $A \subseteq X$ is a retract, then $i_*: \pi_1(A, a) \rightarrow \pi_1(X, a)$ is injective, and if A is a deformation retract it is an isomorphism.

Solution. $r_*i_* = (ri)_* = \text{id}$, so i_* is injective. For a deformation retract, $ir \simeq \text{id}_X$ relative to A , so i is a homotopy equivalence and Lemma 17.3.4 applies (the path h is constant since $a \in A$ is fixed). $\square$

Exercise 17.9.2. Let G be a topological group (a group with continuous multiplication and inversion) with identity 1. Show that $\pi_1(G, 1)$ is abelian, and that the group law of π_1 coincides with $[\gamma][\delta] = [s \mapsto \gamma(s)\delta(s)]$.

Solution. Let $\gamma \cdot \delta$ denote the pointwise product. For loops, $(\gamma * \delta) \cdot (\gamma' * \delta') = (\gamma \cdot \gamma') * (\delta \cdot \delta')$ exactly, and the constant loop is a two-sided unit for both operations on classes. The Eckmann–Hilton argument of Proposition 17.8.2 shows that the two operations coincide and are commutative. $\square$

Exercise 17.9.3. Show that every map $S^n \rightarrow S^1$ with $n \geq 2$ is null-homotopic, and that every map $S^1 \rightarrow S^1$ of degree 0 is null-homotopic.

Solution. In both cases $f_*\pi_1 = 0$ (S^n is simply connected for $n \geq 2$; for degree 0 by definition), so f lifts to $\tilde{f}: S^n \rightarrow \mathbb{R}$ by the lifting criterion. Then $f = p \circ \tilde{f} \simeq p \circ c$ via $p \circ ((1-t)\tilde{f} + tc)$, where c is a constant. $\square$

Exercise 17.9.4. Show that the deck transformation group of $p: \mathbb{R} \rightarrow S^1$ consists of the translations $t \mapsto t + 4k$, $k \in \mathbb{Z}$, and deduce again $\pi_1(S^1) \cong \mathbb{Z}$ from Proposition 17.6.3.

Exercise 17.9.5. (Borsuk–Ulam in dimension 2.) Show that for every map $f: S^2 \rightarrow \mathbb{R}^2$ there is x with $f(x) = f(-x)$. (Hint: otherwise $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is an odd map $S^2 \rightarrow S^1$; restrict to the equator and compare lifts to $\mathbb{R}$ along a half circle.)

Exercise 17.9.6. Find all connected two-sheeted coverings of $S^1 \vee S^1$, and describe the covering corresponding to the kernel of $\mathbb{Z} * \mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

Chapter 18

Topological Manifolds and Group Actions

18.1 Introduction

A topological manifold is a space that looks locally like $\mathbb{R}^n$. The definition has three axioms: locally Euclidean, Hausdorff and second countable. This chapter explains why each is there, by counterexamples showing that none follows from the other two (Section 18.5), and develops the two main ways of producing manifolds from old ones: quotients by group actions (Sections 18.2 to 18.4) and gluing (Section 18.7). Smooth structures come in Chapter 31, and homological invariants such as the dimension and the orientation in Chapter 21. References: [Mun00], [Hat02], [Lee13].

Prerequisites. Chapter 16 (General Topology), Chapter 17 (Homotopy and Covering Spaces) and Chapter 6 (Groups).

Conventions. Groups are written multiplicatively, with identity element e , unless stated otherwise; $\mathbb{R}^n$ and $\mathbb{Z}^n$ are groups under addition. A *neighbourhood* of a point is an open set containing it.

18.2 Topological groups

Definition 18.2.1. A *topological group* is a group G with a topology such that the multiplication $G \times G \rightarrow G, (g, h) \mapsto gh$, and the inversion $G \rightarrow G, g \mapsto g^{-1}$, are continuous. A *homomorphism* of topological groups is a continuous group homomorphism.

- Example 18.2.2.**
1. $(\mathbb{R}^n, +)$, $(\mathbb{C}, +)$, the multiplicative groups $\mathbb{R}^\times$, $\mathbb{R}_{>0}$ and $\mathbb{C}^\times$, and the circle $S^1 = \{z \in \mathbb{C} : |z| = 1\}$, with the Euclidean topologies: the operations are polynomial or rational.
 2. $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, open subsets of the spaces of matrices $\mathbb{R}^{n^2}$ and $\mathbb{C}^{n^2}$ (as the complement of the zero set of $\det$); multiplication is polynomial in the entries and inversion is rational by Cramer's rule. Their subgroups SL_n , $\mathrm{O}(n) = \{A : A^T A = I\}$, $\mathrm{SO}(n)$, $\mathrm{U}(n) = \{A : \bar{A}^T A = I\}$ and $\mathrm{SU}(n)$, with the subspace topologies. $\mathrm{O}(n)$ and $\mathrm{U}(n)$ are compact, being closed and bounded (their columns are unit vectors).
 3. Any group with the discrete topology; products of topological groups.
 4. $(\mathbb{Q}, +)$ with the subspace topology, a topological group that is neither discrete nor locally compact.

Proposition 18.2.3. *Let G be a topological group.*

1. *For $g \in G$, left translation $h \mapsto gh$, right translation and inversion are homeomorphisms of G .*
2. *An open subgroup H is also closed.*
3. *Let U be a neighbourhood of e with $U = U^{-1}$. The subgroup generated by U is open. If G is connected, it is G .*
4. *The connected component G^0 of e is a closed normal subgroup.*
5. *G is Hausdorff if and only if $\{e\}$ is closed.*

Proof. (1) Left translation by g is continuous, as the composite of $h \mapsto (g, h)$ and multiplication, with continuous inverse left translation by g^{-1} ; similarly on the right, and inversion is its own inverse. (2) The complement of H is the union of the cosets gH , $g \notin H$, which are open by (1). (3) The subgroup H generated by U is the union of the sets $u_1 \cdots u_k U$ with $u_i \in U$, which are open by (1). By (2) it is also closed, and nonempty, so $H = G$ if G is connected. (4) $G^0 G^0$ is the image of the connected set $G^0 \times G^0$ under multiplication, hence connected, and it contains e ; so $G^0 G^0 \subseteq G^0$. Similarly $(G^0)^{-1} \subseteq G^0$ and $g G^0 g^{-1} \subseteq G^0$ for all g . Components are closed. (5) If G is Hausdorff, points are closed. Conversely, if $\{e\}$ is closed, so is the diagonal $\{(g, h) : gh^{-1} = e\}$, the preimage of $\{e\}$ under the continuous map $(g, h) \mapsto gh^{-1}$. If $g \neq h$, then (g, h) lies in the open complement of the diagonal, so some product neighbourhood $A \times B$ of (g, h) misses the diagonal, that is, $A \cap B = \emptyset$. $\square$

Counterexample 18.2.4. Any group with the indiscrete topology, whose only open sets are $\emptyset$ and G , is a topological group, and it is not Hausdorff if $|G| > 1$. So (5) is a genuine condition. More generally, for any normal subgroup N the quotient G/N with the quotient topology is a topological group, and it is Hausdorff if and only if N is closed (Exercise 18.8.1).

18.3 Group actions and orbit spaces

Definition 18.3.1. A (left) *action* of a topological group G on a space X is a continuous map $G \times X \rightarrow X$, $(g, x) \mapsto g \cdot x = gx$, with $ex = x$ and $g(hx) = (gh)x$. The *orbit* of x is $Gx = \{gx : g \in G\}$, the *stabiliser* is $G_x = \{g : gx = x\}$, and the *orbit space* X/G is the set of orbits with the quotient topology for the projection $\pi : X \rightarrow X/G$. The action is *free* if all stabilisers are trivial, and *transitive* if there is only one orbit.

Each $g \in G$ acts by a homeomorphism of X , with inverse the action of g^{-1} . The orbits partition X , and $gG_x g^{-1} = G_{gx}$.

Proposition 18.3.2. *Let G act on X , with $\pi : X \rightarrow X/G$.*

1. *π is open: for U open, $\pi^{-1}(\pi(U)) = \bigcup_{g \in G} gU$ is open.*
2. *X/G is Hausdorff if and only if the orbit relation $R = \{(x, gx) : x \in X, g \in G\}$ is closed in $X \times X$.*
3. *If X is second countable, so is X/G ; if X is compact, connected or path-connected, so is X/G .*
4. *If G is compact and X Hausdorff, then X/G is Hausdorff, π is closed, and for each x the map $G/G_x \rightarrow Gx$, $gG_x \mapsto gx$, is a homeomorphism.*

Proof. (1) Each gU is open. (2) Suppose R is closed and $\pi(x) \neq \pi(y)$. Then $(x, y) \notin R$, so there are open $U \ni x$ and $V \ni y$ with $(U \times V) \cap R = \emptyset$. The sets $\pi(U)$ and $\pi(V)$ are open by (1) and disjoint: if $\pi(u) = \pi(v)$ then $(u, v) \in R$. Conversely, if X/G is Hausdorff, its diagonal is closed (a point (a, b) with $a \neq b$ has the neighbourhood $A \times B$, for disjoint neighbourhoods $A \ni a, B \ni b$, which misses the diagonal), and R is its preimage under the continuous map $\pi \times \pi$. (3) The images of a countable basis form a countable basis, since π is open, continuous and surjective; continuous images of compact, connected or path-connected spaces have the same property.

(4) We first show that projections along a compact factor are closed: if K is compact, Y any space and $C \subseteq K \times Y$ closed, then $\text{pr}_Y(C)$ is closed. Indeed, if $y \notin \text{pr}_Y(C)$, each $k \in K$ has a product neighbourhood $A_k \times B_k$ of (k, y) disjoint from C ; finitely many $A_{k_1}, \dots, A_{k_r}$ cover K , and $B = \bigcap_j B_{k_j}$ is a neighbourhood of y with $(K \times B) \cap C = \emptyset$, so B misses $\text{pr}_Y(C)$. Now let $C \subseteq X$ be closed. Then $\pi^{-1}(\pi(C)) = GC$ is the projection to X of $\{(g, x) : g^{-1}x \in C\}$, which is closed in $G \times X$; so GC is closed, and π is a closed map. Next, as X is Hausdorff its diagonal Δ_X is closed (as in (2)), and R is the projection to $X \times X$ of the closed set $\{(g, x, y) : (gx, y) \in \Delta_X\} \subseteq G \times X \times X$; so R is closed and X/G is Hausdorff by (2). Finally $g \mapsto gx$ induces a continuous bijection from G/G_x , which is compact as the image of G , to the Hausdorff space Gx ; it is a homeomorphism by Proposition 16.6.2(4). $\square$

Example 18.3.3. 1. $\mathbb{Z}$ acts on $\mathbb{R}$ by translations. The map $t \mapsto e^{2\pi it}$ induces a continuous bijection $\mathbb{R}/\mathbb{Z} \rightarrow S^1$, which is a homeomorphism because $\mathbb{R}/\mathbb{Z}$ is compact, the image of $[0, 1]$. Likewise $\mathbb{R}^n/\mathbb{Z}^n \cong (S^1)^n = T^n$, the n -torus.

2. $\{\pm 1\}$ acts on S^n by $x \mapsto \pm x$, and $S^n/\{\pm 1\} = \mathbb{RP}^n$, the real projective space.

3. $\mathbb{R}^\times$ acts on $\mathbb{R}^{n+1} \setminus \{0\}$ by scalar multiplication, and the orbit space is again $\mathbb{RP}^n$; $\mathbb{C}^\times$ acting on $\mathbb{C}^{n+1} \setminus \{0\}$ gives the complex projective space $\mathbb{CP}^n$, and S^1 acting on $S^{2n+1} \subseteq \mathbb{C}^{n+1}$ gives $\mathbb{CP}^n$ as well (Proposition 18.6.3).

4. $O(n)$ acts transitively on S^{n-1} , with the stabiliser of e_n equal to $O(n-1)$ (embedded in the upper left corner). By Proposition 18.3.2(4), $S^{n-1} \cong O(n)/O(n-1)$; similarly $S^{2n-1} \cong U(n)/U(n-1)$.

Lemma 18.3.4 (Kronecker). *If $\alpha \in \mathbb{R}$ is irrational, the set $\{n\alpha - \lfloor n\alpha \rfloor : n \in \mathbb{Z}\}$ of fractional parts is dense in $[0, 1]$.*

Proof. Let $N \geq 1$. The $N+1$ fractional parts of $0, \alpha, \dots, N\alpha$ lie in $[0, 1)$, which is the union of the N intervals $[j/N, (j+1)/N)$, so two of them, of $k\alpha$ and $l\alpha$ with $k < l$, lie in the same interval. Then $m = l - k$ satisfies $0 < |m\alpha - p| < 1/N$ for some integer p ; the first inequality holds because α is irrational. Replacing m, p by $-m, -p$ if necessary, we may assume that $\beta = m\alpha - p$ satisfies $0 < \beta < 1/N$. The numbers $j\beta, j = 0, 1, 2, \dots$, increase in steps of $\beta < 1/N$, so every interval $[c, c + 1/N) \subseteq [0, 1)$ contains some $j\beta$; and for $0 \leq j\beta < 1$, $j\beta = (jm)\alpha - jp$ is the fractional part of $(jm)\alpha$. As N is arbitrary, the set is dense. $\square$

Counterexample 18.3.5. Orbit spaces can be very bad. Let α be irrational and let $\mathbb{Z}$ act on S^1 by $n \cdot z = e^{2\pi i n \alpha} z$. By Lemma 18.3.4 every orbit is dense. If $U \subseteq S^1$ is open, saturated and nonempty, then for every $y \in S^1$ the orbit of y meets U , so $y \in U$; hence $U = S^1$. The orbit space $S^1/\mathbb{Z}$ is therefore an uncountable set with the indiscrete topology: it is not Hausdorff, and every map from it to a Hausdorff space is constant. The action is free.

18.4 Covering space actions and quotients

From now on in this section G is a group acting on X by homeomorphisms, that is, a topological group with the discrete topology.

Definition 18.4.1. An action of a discrete group G on X is

1. a *covering space action* if every $x \in X$ has a neighbourhood U with $gU \cap U = \emptyset$ for all $g \neq e$;
2. *proper* if for every compact $K \subseteq X$ the set $\{g \in G : gK \cap K \neq \emptyset\}$ is finite.

A covering space action is free. An action of a finite group is proper.

Theorem 18.4.2. Let G act on X by a covering space action. Then $\pi : X \rightarrow X/G$ is a covering map. If X is path-connected, then π is a normal covering whose deck transformation group is G , and $\pi_1(X/G)/\pi_*\pi_1(X) \cong G$; in particular $\pi_1(X/G) \cong G$ if X is simply connected.

Proof. Let $x \in X$ and U as in Definition 18.4.1. Then $\pi^{-1}(\pi(U)) = \bigsqcup_{g \in G} gU$, a disjoint union of open sets: if $gU \cap hU \neq \emptyset$ then $h^{-1}gU \cap U \neq \emptyset$, so $g = h$. The restriction of π to gU is continuous, open (Proposition 18.3.2(1)) and injective, since $\pi(gu) = \pi(gu')$ means $gu' = hgu$ for some h , whence $g^{-1}hg = e$ and $u = u'$. So $\pi|_{gU}$ is a homeomorphism onto the open set $\pi(gU)$, and $\pi(U)$ is evenly covered.

Each g is a deck transformation, since $\pi(gx) = \pi(x)$, and G acts transitively on each fibre $\pi^{-1}(\pi(x)) = Gx$. If X is path-connected, a deck transformation is determined by the image of one point (Chapter 17), so every deck transformation τ equals the g with $gx = \tau(x)$, and $g \mapsto (x \mapsto gx)$ is an isomorphism $G \rightarrow \text{Deck}(\pi)$, injective because the action is free. The deck group acts transitively on the fibres, so the covering is normal, and Proposition 17.6.3(2) gives $\pi_1(X/G)/\pi_*\pi_1(X) \cong \text{Deck}(\pi) \cong G$. $\square$

Theorem 18.4.3. Let a discrete group G act freely and properly on a locally compact Hausdorff space X . Then the action is a covering space action and X/G is Hausdorff.

Proof. Let $x \in X$ and let K be a compact neighbourhood of x (containing an open neighbourhood of x). The set $F = \{g \neq e : gK \cap K \neq \emptyset\}$ is finite. For $g \in F$, $gx \neq x$ as the action is free, so there are disjoint open sets $V_g \ni x$ and $W_g \ni gx$. Let $U = \text{int } K \cap \bigcap_{g \in F} (V_g \cap g^{-1}W_g)$, an open neighbourhood of x . For $g \neq e$ not in F , $gU \cap U \subseteq gK \cap K = \emptyset$; for $g \in F$, $gU \subseteq W_g$ and $U \subseteq V_g$, which are disjoint. So the action is a covering space action.

For the Hausdorff property we show that the orbit relation R is closed (Proposition 18.3.2(2)). Let $(x, y) \notin R$ and choose compact neighbourhoods K_x of x and K_y of y . With $K = K_x \cup K_y$, the set $F = \{g : gK_x \cap K_y \neq \emptyset\}$ is contained in $\{g : gK \cap K \neq \emptyset\}$, hence finite. For $g \in F$, $gx \neq y$, so there are disjoint open $A_g \ni gx$ and $B_g \ni y$. Put $U = \text{int } K_x \cap \bigcap_{g \in F} g^{-1}A_g$ and $V = \text{int } K_y \cap \bigcap_{g \in F} B_g$. For every g , $gU \cap V = \emptyset$: if $g \notin F$ because $gK_x \cap K_y = \emptyset$, and if $g \in F$ because $gU \subseteq A_g$ and $V \subseteq B_g$. Hence $(U \times V) \cap R = \emptyset$. $\square$

Counterexample 18.4.4. A covering space action need not have a Hausdorff quotient. Let $\mathbb{Z}$ act on $X = \mathbb{R}^2 \setminus \{0\}$ by $n \cdot (x, y) = (2^n x, 2^{-n} y)$. It is a covering space action: a point with $x \neq 0$ has the neighbourhood $\{(x', y') : |x|/\sqrt{2} < |x'| < \sqrt{2}|x|\}$, which is moved off itself by every $n \neq 0$, and similarly if $y \neq 0$. But the points $p = (1, 0)$ and $q = (0, 1)$ lie in different orbits, while $(1, 2^{-k}) \rightarrow p$ and $(-k) \cdot (1, 2^{-k}) = (2^{-k}, 1) \rightarrow q$; so (p, q) lies in the closure of the orbit relation, which is not closed, and $X/\mathbb{Z}$ is not Hausdorff. The action is free but not proper: the compact set $K = \{(x, y) : \max(|x|, |y|) = 1\}$ meets its translates by all n .

Example 18.4.5. 1. $\mathbb{Z}^n$ acts freely and properly on $\mathbb{R}^n$ by translations; $T^n = \mathbb{R}^n/\mathbb{Z}^n$ and $\pi_1(T^n) \cong \mathbb{Z}^n$, as $\mathbb{R}^n$ is simply connected.

2. $\{\pm 1\}$ acts freely on S^n , so $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2$ for $n \geq 2$, since S^n is simply connected for $n \geq 2$ (Chapter 17).
3. (Lens spaces) Let $p \geq 2$ and q be coprime integers, $\zeta = e^{2\pi i/p}$. The group $\mathbb{Z}/p$ acts freely on $S^3 \subseteq \mathbb{C}^2$ by $k \cdot (z_1, z_2) = (\zeta^k z_1, \zeta^{qk} z_2)$: if $\zeta^k z_1 = z_1$ and $\zeta^{qk} z_2 = z_2$ with $(z_1, z_2) \neq 0$, then $\zeta^k = 1$ or $\zeta^{qk} = 1$, and either forces $p \mid k$ as q is prime to p . The quotient $L(p, q) = S^3/(\mathbb{Z}/p)$ is the *lens space*, with $\pi_1 \cong \mathbb{Z}/p$.
4. (Klein bottle) The transformations $a(x, y) = (x + \frac{1}{2}, -y)$ and $b(x, y) = (x, y + 1)$ of $\mathbb{R}^2$ generate a group G acting freely and properly, with $aba^{-1} = b^{-1}$; the quotient $\mathbb{R}^2/G$ is the Klein bottle, with $\pi_1 \cong G$, a non-abelian group.

18.5 Topological manifolds

Definition 18.5.1. Let $n \geq 0$. A space M is *locally Euclidean of dimension n* if every point has a neighbourhood homeomorphic to an open subset of $\mathbb{R}^n$ (equivalently, to $\mathbb{R}^n$ itself, since every point of an open subset of $\mathbb{R}^n$ has a ball neighbourhood, and balls are homeomorphic to $\mathbb{R}^n$). A *topological manifold of dimension n* , or *n -manifold*, is a space that is

1. locally Euclidean of dimension n ,
2. Hausdorff, and
3. second countable.

A *chart* is a homeomorphism $\varphi: U \rightarrow \varphi(U)$ from an open $U \subseteq M$ onto an open subset of $\mathbb{R}^n$.

That the dimension of a nonempty manifold is determined by the space is a theorem, proved with homology (Chapter 21). The next three counterexamples show that none of the three axioms follows from the other two.

Counterexample 18.5.2 (Not Hausdorff). The *line with two origins* L is the quotient of $\mathbb{R} \times \{1, 2\}$ by $(t, 1) \sim (t, 2)$ for $t \neq 0$ (Counterexample 16.4.11). The quotient map is open, the two copies of $\mathbb{R}$ map homeomorphically onto open subsets covering L , so L is locally Euclidean of dimension 1 and second countable. But the two origins $0_1, 0_2$ have no disjoint neighbourhoods: every neighbourhood of 0_i contains the images of $(-\varepsilon, \varepsilon) \setminus \{0\}$ for some $\varepsilon > 0$. In L the sequence $1/n$ converges to both origins, so limits are not unique.

Counterexample 18.5.3 (Not second countable). Let I be an uncountable set with the discrete topology, and $M = I \times \mathbb{R}$, a disjoint union of uncountably many lines. It is locally Euclidean of dimension 1 and Hausdorff. It is not second countable: a basis must contain, for each $i \in I$, a nonempty open subset of $\{i\} \times \mathbb{R}$, and these are pairwise distinct. Connected examples also exist: the *long line* and the *Prüfer surface* are connected, Hausdorff and locally Euclidean of dimensions 1 and 2 but not second countable [Mun00]. Without second countability, partitions of unity (Theorem 16.10.6) and embeddings in $\mathbb{R}^N$ can fail.

Lemma 18.5.4. Let $n \geq 2$, $W \subseteq \mathbb{R}^n$ open and connected, and $p \in W$. Then $W \setminus \{p\}$ is connected. For $n = 1$, removing a point from an open interval leaves exactly two components.

Proof. $\mathbb{R}^n \setminus \{0\}$ is path-connected for $n \geq 2$: two points a, b are joined by the segment $[a, b]$ unless 0 lies on it, and then by the segments $[a, c]$ and $[c, b]$ for any c not on the line through a and b . Hence every punctured ball $B \setminus \{p\}$, homeomorphic to $\mathbb{R}^n \setminus \{0\}$, is path-connected. Choose a ball B with $p \in B \subseteq W$ and let A be the set of points of $W \setminus \{p\}$ that can be joined to a point of $B \setminus \{p\}$ by a path in $W \setminus \{p\}$. Every point of $W \setminus \{p\}$ has a ball neighbourhood

in $W \setminus \{p\}$, which is path-connected; so A and its complement in $W \setminus \{p\}$ are open. If the complement were nonempty, then $W = (A \cup B) \cup (W \setminus (A \cup \{p\}))$ would be a decomposition into disjoint nonempty open sets ($A \cup B$ is open, and $B \setminus \{p\} \subseteq A$), contradicting connectedness. So $W \setminus \{p\} = A$ is path-connected. The case $n = 1$ is clear, as connected subsets of $\mathbb{R}$ are intervals (Theorem 16.5.3). $\square$

Counterexample 18.5.5 (Not locally Euclidean). The union $X = \{(x, y) \in \mathbb{R}^2 : xy = 0\}$ of the two coordinate axes is Hausdorff and second countable, being a subspace of $\mathbb{R}^2$, but not locally Euclidean at the origin. Suppose $\varphi: U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ were a chart at 0. Choose $\varepsilon > 0$ with $V = X \cap (-\varepsilon, \varepsilon)^2 \subseteq U$. Then V is a cross with four open arms, open in X and connected, and $V \setminus \{0\}$ has exactly four components, the arms. So $W = \varphi(V)$ is a connected open subset of $\mathbb{R}^n$, and $W \setminus \{\varphi(0)\}$, homeomorphic to $V \setminus \{0\}$, has four components. This contradicts Lemma 18.5.4 for $n \geq 1$, and $n = 0$ is impossible because $\{0\}$ is not open in X .

Similarly the closed half-line $[0, +\infty)$ is not locally Euclidean at 0. If φ were a chart at 0 with values in $\mathbb{R}^n$: $n = 0$ is impossible as before; for $n = 1$, a connected neighbourhood $[0, \varepsilon) \subseteq U$ would map to an open interval, which is disconnected by removing the image of 0, while $(0, \varepsilon)$ is connected; for $n \geq 2$, a small interval $J \subseteq (0, \varepsilon)$ would map to a connected open subset of $\mathbb{R}^n$ that stays connected after removing a point, while J does not. It is a manifold with boundary (Definition 18.5.9).

Proposition 18.5.6. *Let M be an n -manifold.*

1. *M is locally compact and locally path-connected. Its connected components are open, coincide with its path components, and are countable in number.*
2. *Every open subset of M is an n -manifold.*
3. *M is paracompact and admits partitions of unity subordinate to any open cover.*
4. *M has a countable cover by open sets B_i with homeomorphisms $B_i \cong \mathbb{R}^n$ (coordinate balls).*

Proof. (1) Each point has a neighbourhood homeomorphic to $\mathbb{R}^n$, and closed balls and open balls in $\mathbb{R}^n$ give compact and path-connected neighbourhoods. In a locally path-connected space, path components are open, hence also closed, and so they are the components. They are disjoint open sets, each containing an element of a countable basis, so there are countably many. (2) The three axioms pass to open subspaces. (3) By (1) and Hausdorffness M is locally compact Hausdorff and second countable, so Proposition 16.10.7 and Theorem 16.10.6 apply. (4) Let $\mathcal{B}$ be a countable basis and $\mathcal{B}'$ the set of $W \in \mathcal{B}$ contained in the domain of some chart; for each $W \in \mathcal{B}'$ fix a chart φ_W whose domain contains W . The sets $\varphi_W^{-1}(B)$, for $W \in \mathcal{B}'$ and B an open ball with rational centre and radius contained in $\varphi_W(W)$, are countably many coordinate balls. They cover M : a point x lies in a chart domain, hence in some $W \in \mathcal{B}$ inside it, so $W \in \mathcal{B}'$, and $\varphi_W(x)$ lies in a rational ball contained in the open set $\varphi_W(W)$. $\square$

Theorem 18.5.7. *The fundamental group of a connected manifold is countable.*

Proof. Let $\{B_i\}$ be a countable cover of M by coordinate balls (Proposition 18.5.6(4)); each B_i is simply connected, being homeomorphic to $\mathbb{R}^n$. For each pair i, j , the open set $B_i \cap B_j$ has countably many path components (by Proposition 18.5.6(1)); choose a point in each, and let S be the countable set of all chosen points, together with a base point p . For any two points $s, s' \in S$ lying in a common B_i , choose a path $\sigma_{s,s'}^i$ in B_i from s to s' ; these are countably many paths. We claim that every loop γ at p is homotopic to a finite concatenation of such paths; since there are countably many finite concatenations, this proves the theorem.

By the Lebesgue number lemma there is a partition $0 = t_0 < t_1 < \dots < t_k = 1$ with $\gamma([t_{m-1}, t_m]) \subseteq B_{i_m}$. We may take $p \in B_{i_1} \cap B_{i_k}$. For $0 < m < k$, the point $\gamma(t_m)$ lies in

$B_{i_m} \cap B_{i_{m+1}}$, in a path component containing some $s_m \in S$; choose a path τ_m in that component from $\gamma(t_m)$ to s_m , and let τ_0, τ_k be constant at $p = s_0 = s_k$. Then γ is homotopic to the concatenation over m of the paths $\bar{\tau}_{m-1} \cdot \gamma|_{[t_{m-1}, t_m]} \cdot \tau_m$, each of which runs in B_{i_m} from s_{m-1} to s_m . As B_{i_m} is simply connected, each is homotopic relative to its endpoints to $\sigma_{s_{m-1}, s_m}^{i_m}$. $\square$

Corollary 18.5.8. *If M is a connected n -manifold and $p: E \rightarrow M$ a covering map with E connected, then E is an n -manifold.*

Proof. E is locally homeomorphic to M , hence locally Euclidean. It is Hausdorff: two points in different fibres are separated by preimages of disjoint neighbourhoods in M , and two points in the same fibre lie in different sheets over an evenly covered neighbourhood. The fibres are in bijection with the cosets of $p_*\pi_1(E)$ in the countable group $\pi_1(M)$ (Chapter 17), hence countable. If $\{W_k\}$ is a countable basis of M consisting of evenly covered sets, then the sheets over the W_k form a countable family of open sets, and the sheets over basic open subsets of the W_k give a countable basis of E . $\square$

Definition 18.5.9. Let $\mathbb{H}^n = \{x \in \mathbb{R}^n : x_n \geq 0\}$. An n -manifold with boundary is a Hausdorff, second countable space in which every point has a neighbourhood homeomorphic to an open subset of $\mathbb{H}^n$. A point is an *interior point* if it has a neighbourhood homeomorphic to $\mathbb{R}^n$, and a *boundary point* otherwise; the boundary is written ∂M .

A point sent by a chart to a point with $x_n > 0$ is interior. That a point sent by some chart to $\{x_n = 0\}$ is not interior, so that ∂M is the set of such points and ∂M is an $(n-1)$ -manifold, uses invariance of domain (Chapter 21); for $n = 1$ it follows from Lemma 18.5.4.

Example 18.5.10. The closed disc D^n is an n -manifold with boundary S^{n-1} ; $[0, 1]$ has boundary $\{0, 1\}$; the *Möbius band*, the quotient of $[0, 1] \times [-1, 1]$ by $(0, t) \sim (1, -t)$, has boundary a single circle.

18.6 Constructions of manifolds

Proposition 18.6.1. 1. *Open subsets of n -manifolds and countable disjoint unions of n -manifolds are n -manifolds.*

2. *If M is an m -manifold and N an n -manifold, $M \times N$ is an $(m+n)$ -manifold.*

3. *If a discrete group G acts freely and properly on an n -manifold M , then M/G is an n -manifold, and $M \rightarrow M/G$ is a covering map.*

4. *Connected covering spaces of connected n -manifolds are n -manifolds.*

Proof. (1) Clear. (2) Products of charts are charts, $\mathbb{R}^m \times \mathbb{R}^n = \mathbb{R}^{m+n}$, and products of Hausdorff or second countable spaces are Hausdorff or second countable. (3) By Theorem 18.4.3 and Proposition 18.5.6(1) the quotient is Hausdorff and π is a covering map, so M/G is locally homeomorphic to M ; it is second countable by Proposition 18.3.2(3). (4) is Corollary 18.5.8. $\square$

Example 18.6.2. 1. $\mathbb{R}^n$; open subsets such as $\text{GL}_n(\mathbb{R}) \subseteq \mathbb{R}^{n^2}$.

2. The sphere $S^n \subseteq \mathbb{R}^{n+1}$, with the two stereographic charts (Example 16.9.4); it is Hausdorff and second countable as a subspace of $\mathbb{R}^{n+1}$.

3. The graph $\{(x, f(x))\}$ of a continuous map $f: U \rightarrow \mathbb{R}^k$, $U \subseteq \mathbb{R}^n$ open, is an n -manifold, homeomorphic to U by the projection.

4. The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, $\mathbb{RP}^n = S^n/\{\pm 1\}$, the lens spaces and the Klein bottle of Example 18.4.5, by Proposition 18.6.1(3).

Proposition 18.6.3. $\mathbb{CP}^n = (\mathbb{C}^{n+1} \setminus \{0\})/\mathbb{C}^\times$ is a compact $2n$ -manifold, homeomorphic to S^{2n+1}/S^1 . The sets $U_i = \{[z] : z_i \neq 0\}$ are open and $\varphi_i([z]) = (z_0/z_i, \dots, z_i/z_i, \dots, z_n/z_i)$ is a homeomorphism $U_i \rightarrow \mathbb{C}^n$. The same holds for $\mathbb{RP}^n$, with $\mathbb{R}$ in place of $\mathbb{C}$.

Proof. The inclusion $S^{2n+1} \rightarrow \mathbb{C}^{n+1} \setminus \{0\}$ induces a continuous map $S^{2n+1}/S^1 \rightarrow \mathbb{CP}^n$, which is bijective since every line meets the sphere in a circle, and has continuous inverse induced by $z \mapsto z/|z|$. By Proposition 18.3.2(3),(4) applied to the compact group S^1 , $\mathbb{CP}^n$ is compact, Hausdorff and second countable. The set U_i is open because its preimage $\{z_i \neq 0\}$ is. The map φ_i is well defined and continuous (it is induced by a continuous map constant on orbits), with continuous inverse $w \mapsto [w_0 : \dots : 1 : \dots : w_n]$, the 1 in position i . $\square$

18.7 Gluing

Lemma 18.7.1. Let X and Y be spaces, $U \subseteq X$ and $V \subseteq Y$ open, and $f: U \rightarrow V$ a homeomorphism. Let $Z = X \cup_f Y$ be the quotient of $X \sqcup Y$ by $u \sim f(u)$ for $u \in U$.

1. X and Y map homeomorphically onto open subsets of Z covering Z .
2. If X and Y are Hausdorff, then Z is Hausdorff if and only if the graph $\Gamma_f = \{(u, f(u))\}$ is closed in $X \times Y$.
3. If X and Y are n -manifolds and Γ_f is closed, then Z is an n -manifold.

Proof. (1) The saturation of an open $A \subseteq X$ is $A \sqcup f(A \cap U)$, which is open, so the quotient map q is open, and it is injective on X and on Y . (2) Two points of Z coming from X are separated by images of disjoint open subsets of X , and similarly for Y . The remaining case is $q(x) \neq q(y)$ with $x \in X \setminus U$, $y \in Y \setminus V$; then $(x, y) \notin \Gamma_f$. If Γ_f is closed, there are open $A \ni x$, $B \ni y$ with $(A \times B) \cap \Gamma_f = \emptyset$, that is $f(A \cap U) \cap B = \emptyset$, and $q(A)$, $q(B)$ are disjoint neighbourhoods. Conversely, if $(x, y) \in \bar{\Gamma}_f \setminus \Gamma_f$, then every pair of neighbourhoods $A \ni x$, $B \ni y$ has $f(A \cap U) \cap B \neq \emptyset$, so $q(A) \cap q(B) \neq \emptyset$; and $q(x) \neq q(y)$ (if $x \in U$ then $y \neq f(x)$ since $(x, y) \notin \Gamma_f$; if $x \notin U$ and $y \notin V$ they are not identified; and $x \notin U$, $y \in V$ is also not identified). (3) Locally Euclidean by (1), Hausdorff by (2), and second countable as the union of two open second countable subspaces. $\square$

Counterexample 18.7.2. The line with two origins is $\mathbb{R} \cup_f \mathbb{R}$ with $U = V = \mathbb{R} \setminus \{0\}$ and $f = \text{id}$. The graph of f is the diagonal minus the origin, whose closure in $\mathbb{R} \times \mathbb{R}$ contains $(0, 0)$; in accordance with Lemma 18.7.1(2), the result is not Hausdorff (Counterexample 18.5.2).

Definition 18.7.3. Let M and N be connected n -manifolds, $n \geq 2$, and let $\varphi: U \rightarrow \mathbb{R}^n$, $\psi: V \rightarrow \mathbb{R}^n$ be charts onto $\mathbb{R}^n$. Let $X = M \setminus \varphi^{-1}(\bar{B}_1)$ and $Y = N \setminus \psi^{-1}(\bar{B}_1)$, where $\bar{B}_r = \{|x| \leq r\}$, and let f be the homeomorphism from $\varphi^{-1}(\{1 < |x| < 2\}) \subseteq X$ onto $\psi^{-1}(\{1 < |y| < 2\}) \subseteq Y$ given in coordinates by $x \mapsto (3 - |x|)x/|x|$, which exchanges the two boundary spheres of the annulus. The *connected sum* is $M \# N = X \cup_f Y$.

Proposition 18.7.4. $M \# N$ is a connected n -manifold, compact if M and N are.

Proof. X and Y are n -manifolds by Proposition 18.6.1(1). We check that the graph of f is closed in $X \times Y$. Let $(x_k, f(x_k)) \rightarrow (x, y) \in X \times Y$ with $1 < |\varphi(x_k)| < 2$; sequences suffice, as $X \times Y$ is first countable. The points $\varphi(x_k)$ lie in the compact set $\{1 \leq |w| \leq 2\}$, so a subsequence converges to some w there, and then the corresponding subsequence of (x_k) converges to

$\varphi^{-1}(w)$; by uniqueness of limits in the Hausdorff space M , $x = \varphi^{-1}(w)$, and $|w| > 1$ because $x \in X$. In the same way $y = \psi^{-1}(w')$ with $|w'| > 1$ a limit point of $\psi(f(x_k))$, whose norms are $3 - |\varphi(x_k)|$; so $|w'| = 3 - |w|$, which forces $|w| < 2$. Hence x lies in the domain of f , and $y = f(x)$ by continuity of f . By Lemma 18.7.1, $M \# N$ is an n -manifold.

X is connected: the set $E = \varphi^{-1}(\{|w| > 1\})$ is connected, being homeomorphic to $S^{n-1} \times (1, \infty)$ with S^{n-1} path-connected for $n \geq 2$ (as the image of $\mathbb{R}^n \setminus \{0\}$ under $w \mapsto w/|w|$). If $X = A \sqcup A'$ with A, A' open, we may assume $E \subseteq A$; then $M = (A \cup \varphi^{-1}(B_2)) \sqcup A'$ is a decomposition into disjoint open sets, since the points of $\varphi^{-1}(B_2)$ in X lie in E ; as M is connected, $A' = \emptyset$. Similarly Y is connected, and the images of X and Y in $M \# N$ meet, so $M \# N$ is connected. If M and N are compact, the closed subsets $M \setminus \varphi^{-1}(B_{3/2}) \subseteq X$ and $N \setminus \psi^{-1}(B_{3/2}) \subseteq Y$ are compact, and their images cover $M \# N$, because a point of X with $1 < |\varphi| < 3/2$ is identified with a point of Y with $3/2 < |\psi| < 2$. $\square$

Remark 18.7.5. The homeomorphism type of $M \# N$ can depend on the choice of charts, through orientations: for orientable manifolds the two choices of relative orientation can give different results. For surfaces, and more generally up to these orientation choices, the result is independent of the charts, but the proof in the topological category uses deep results; in the smooth category it follows from the disc theorem of Palais. We shall only use connected sums of surfaces, where $T^2 \# T^2$ is the closed orientable surface of genus 2 and $\mathbb{RP}^2 \# \mathbb{RP}^2$ is the Klein bottle.

Theorem 18.7.6 (Classification of surfaces). *Every compact connected 2-manifold is homeomorphic to exactly one of: the sphere S^2 , a connected sum $T^2 \# \dots \# T^2$ of $g \geq 1$ tori, or a connected sum $\mathbb{RP}^2 \# \dots \# \mathbb{RP}^2$ of $k \geq 1$ projective planes.*

This is [Mun00, Chapter 12]. The surfaces in the list are distinguished by their fundamental groups (their abelianisations are $\mathbb{Z}^{2g}$ and $\mathbb{Z}^{k-1} \oplus \mathbb{Z}/2$), and every compact surface is homeomorphic to one obtained by identifying the edges of a polygon in pairs, which is the route to the proof. In dimension 1 the classification is simpler: every connected 1-manifold is homeomorphic to $\mathbb{R}$ or to S^1 [Mil97, Appendix].

18.8 Exercises

Exercise 18.8.1. Let N be a normal subgroup of a topological group G . Show that G/N with the quotient topology is a topological group, and that it is Hausdorff if and only if N is closed.

Solution. The quotient map $\pi: G \rightarrow G/N$ is open, by the argument of Proposition 18.3.2(1) for the action of N by right translation. So $\pi \times \pi$ is open and surjective, hence a quotient map, and multiplication $G/N \times G/N \rightarrow G/N$ is continuous because its composite with $\pi \times \pi$ is. Inversion is similar. By Proposition 18.2.3(5), G/N is Hausdorff if and only if $\{eN\}$ is closed, that is, if and only if $\pi^{-1}(eN) = N$ is closed (the complement of $\{eN\}$ is open if and only if its preimage $G \setminus N$ is). $\square$

Exercise 18.8.2. Show that a compact Hausdorff space that is locally Euclidean is second countable, hence a manifold.

Solution. Finitely many chart domains $U_1, \dots, U_k$ cover the space, and each is second countable, being homeomorphic to an open subset of $\mathbb{R}^n$. The union of countable bases of the U_i is a countable basis. $\square$

Exercise 18.8.3. Show that the Möbius band B of Example 18.5.10 is the quotient of $\mathbb{R} \times [-1, 1]$ by the free proper action of $\mathbb{Z}$ given by $n \cdot (x, t) = (x + n, (-1)^n t)$, and deduce that $B \setminus \partial B$ is a 2-manifold whose fundamental group is $\mathbb{Z}$.

Solution. Every orbit meets $[0, 1] \times [-1, 1]$, in one point or, for $x \in \mathbb{Z}$, in the two points $(0, t)$ and $(1, -t)$; the induced continuous bijection from the band to the orbit space is a homeomorphism since the band is compact and the orbit space Hausdorff (Theorem 18.4.3, which applies because $K \cap (K + n) = \emptyset$ for compact K and large $|n|$). Restricted to $\mathbb{R} \times (-1, 1)$, which is simply connected and a 2-manifold, Theorem 18.4.2 gives $\pi_1(B \setminus \partial B) \cong \mathbb{Z}$. $\square$

Exercise 18.8.4. Let $\mathbb{Z}$ act on $\mathbb{R}^n \setminus \{0\}$ by $k \cdot x = 2^k x$. Show that the action is free and proper, and that the quotient is homeomorphic to $S^{n-1} \times S^1$.

Solution. The map $\mathbb{R}^n \setminus \{0\} \rightarrow S^{n-1} \times \mathbb{R}$, $x \mapsto (x/|x|, \log_2 |x|)$, is a homeomorphism carrying the action to $k \cdot (u, s) = (u, s + k)$, which is free and proper on $S^{n-1} \times \mathbb{R}$ with quotient $S^{n-1} \times (\mathbb{R}/\mathbb{Z}) \cong S^{n-1} \times S^1$ (Example 18.3.3(1)). The action on $S^{n-1} \times \mathbb{R}$ is proper because a compact set lies in some $S^{n-1} \times [-r, r]$, which meets its translate by k only if $|k| \leq 2r$. For $n = 2$ the quotient is a torus. For $n \geq 3$, $\mathbb{R}^n \setminus \{0\} \cong S^{n-1} \times \mathbb{R}$ is simply connected, S^{n-1} being simply connected (Chapter 17), so the quotient has $\pi_1 \cong \mathbb{Z}$ by Theorem 18.4.2. $\square$

Chapter 19

Singular Homology

19.1 Introduction

Singular homology assigns to every space X abelian groups $H_n(X)$, functorial in X and invariant under homotopy. Unlike homotopy groups, they are computable: the long exact sequence of a pair, excision (Theorem 19.5.4) and the Mayer–Vietoris sequence reduce computations to simple pieces, and for CW complexes cellular homology (Theorem 19.7.3) computes them from the cells. We follow [Hat02, Chapter 2]; see also [Spa66].

Prerequisites. Chapter 17 (Homotopy and Covering Spaces) and Chapter 13 (Homological Algebra).

Abelian groups are written additively. Chain complexes are as in Notation 13.3.2, with differential ∂ of degree -1 .

19.2 Singular chains

Definition 19.2.1. The *standard n -simplex* is $\Delta^n = \{(t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_i t_i = 1\}$, with vertices the standard basis vectors $e_0, \dots, e_n$. For points $w_0, \dots, w_k$ of some $\mathbb{R}^N$ we write $[w_0, \dots, w_k]: \Delta^k \rightarrow \mathbb{R}^N$ for the affine map $(t_i) \mapsto \sum_i t_i w_i$. The *face maps* are $\delta^i = [e_0, \dots, \hat{e}_i, \dots, e_n]: \Delta^{n-1} \rightarrow \Delta^n$, where the hat means omission.

Definition 19.2.2. A *singular n -simplex* in a space X is a map $\sigma: \Delta^n \rightarrow X$. The group $C_n(X)$ of *singular n -chains* is the free abelian group (Definition 6.11.1) with basis the singular n -simplices; its elements are finite formal sums $\sum_i n_i \sigma_i$ with $n_i \in \mathbb{Z}$. For $n < 0$, $C_n(X) = 0$. The *boundary* $\partial: C_n(X) \rightarrow C_{n-1}(X)$ is the homomorphism with

$$\partial\sigma = \sum_{i=0}^n (-1)^i \sigma \circ \delta^i.$$

The *singular homology* of X is $H_n(X) = \ker(\partial: C_n \rightarrow C_{n-1})/\text{im}(\partial: C_{n+1} \rightarrow C_n)$. A map $f: X \rightarrow Y$ induces $f_\#: C_n(X) \rightarrow C_n(Y)$, $\sigma \mapsto f \circ \sigma$, and $f_*: H_n(X) \rightarrow H_n(Y)$.

Lemma 19.2.3. $\partial\partial = 0$, and $f_\#$ is a morphism of chain complexes. Hence H_n is a functor $\text{Top} \rightarrow \text{Ab}$.

Proof. For $i < j$ one has $\delta^j \delta^i = \delta^i \delta^{j-1}$ as maps $\Delta^{n-2} \rightarrow \Delta^n$: both are the affine map omitting e_i and e_j . So

$$\partial\partial\sigma = \sum_j \sum_i (-1)^{i+j} \sigma \delta^j \delta^i = \sum_{i < j} (-1)^{i+j} \sigma \delta^i \delta^{j-1} + \sum_{i \geq j} (-1)^{i+j} \sigma \delta^j \delta^i,$$

and substituting $(i, j-1) \mapsto (j', i')$ in the first sum shows that it is the negative of the second. Clearly $\partial f_\# = f_\# \partial$. $\square$

Proposition 19.2.4. 1. If (X_α) are the path components of X , then $H_n(X) \cong \bigoplus_\alpha H_n(X_\alpha)$.

2. If X is path-connected and nonempty, then $H_0(X) \cong \mathbb{Z}$, via $\sum n_i x_i \mapsto \sum n_i$.

3. For a point, $H_0 = \mathbb{Z}$ and $H_n = 0$ for $n \neq 0$.

Proof. (1) The image of a singular simplex is path-connected, so it lies in one X_α , and $C_\bullet(X) = \bigoplus_\alpha C_\bullet(X_\alpha)$ as chain complexes. (2) Singular 0-simplices are points and $\partial\sigma = \sigma(e_1) - \sigma(e_0)$ for a 1-simplex, a path. So the augmentation $\varepsilon(\sum n_i x_i) = \sum n_i$ vanishes on boundaries and is surjective. If $\varepsilon(\sum n_i x_i) = 0$, fix x_0 and paths γ_i from x_0 to x_i ; then $\sum n_i x_i = \sum n_i(x_i - x_0) = \partial(\sum n_i \gamma_i)$. (3) There is exactly one singular n -simplex σ_n for each n , and $\partial\sigma_n = \sum_{i=0}^n (-1)^i \sigma_{n-1}$ is 0 for n odd and σ_{n-1} for $n \geq 2$ even. So the complex is $\cdots \rightarrow \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \xrightarrow{1} \mathbb{Z} \xrightarrow{0} \mathbb{Z} \rightarrow 0$, with homology $\mathbb{Z}$ in degree 0 only. $\square$

Definition 19.2.5. The *reduced homology* $\tilde{H}_n(X)$ of a nonempty space is the homology of the augmented complex $\cdots \rightarrow C_1(X) \rightarrow C_0(X) \xrightarrow{\varepsilon} \mathbb{Z} \rightarrow 0$, with $\mathbb{Z}$ in degree -1 . So $\tilde{H}_n(X) = H_n(X)$ for $n > 0$, $H_0(X) \cong \tilde{H}_0(X) \oplus \mathbb{Z}$, and $\tilde{H}_n(\text{pt}) = 0$ for all n .

19.3 Homotopy invariance

Theorem 19.3.1. If $f \simeq g: X \rightarrow Y$, then $f_\#$ and $g_\#$ are chain homotopic, so $f_* = g_*: H_n(X) \rightarrow H_n(Y)$. Consequently homotopy equivalences induce isomorphisms, and contractible spaces have $\tilde{H}_\bullet = 0$.

Proof. Let $F: X \times I \rightarrow Y$ be a homotopy from f to g . In $\Delta^n \times I \subseteq \mathbb{R}^{n+2}$ let $v_i = (e_i, 0)$ and $w_i = (e_i, 1)$. Define the *prism operator* $P: C_n(X) \rightarrow C_{n+1}(Y)$ by

$$P(\sigma) = \sum_{i=0}^n (-1)^i F \circ (\sigma \times \text{id}_I) \circ [v_0, \dots, v_i, w_i, \dots, w_n].$$

Write $[\dots]$ for $F \circ (\sigma \times \text{id}) \circ [\dots]$. Then

$$\partial P(\sigma) = \sum_{j \leq i} (-1)^{i+j} [v_0, \dots, \widehat{v_j}, \dots, v_i, w_i, \dots, w_n] + \sum_{j \geq i} (-1)^{i+j+1} [v_0, \dots, v_i, w_i, \dots, \widehat{w_j}, \dots, w_n].$$

In the terms with $i = j$, the term $[v_0, \dots, v_{i-1}, w_i, \dots, w_n]$ of the first sum cancels the term with index $i-1$ of the second sum, for $1 \leq i \leq n$. What remains of them is $[w_0, \dots, w_n] = g \circ \sigma$ (first sum, $i = 0$) and $-[v_0, \dots, v_n] = -f \circ \sigma$ (second sum, $i = n$). Since $\sigma \circ \delta^j$ corresponds to the vertices with index j omitted,

$$P(\partial\sigma) = \sum_{i < j} (-1)^{i+j} [v_0, \dots, v_i, w_i, \dots, \widehat{w_j}, \dots, w_n] + \sum_{i > j} (-1)^{i-1+j} [v_0, \dots, \widehat{v_j}, \dots, v_i, w_i, \dots, w_n],$$

which is the negative of the terms with $i \neq j$ above. Hence $\partial P + P\partial = g_\# - f_\#$, and chain homotopic maps induce the same map on homology (Lemma 13.3.4). The rest follows from functoriality and Proposition 19.2.4(3). $\square$

19.4 Relative homology and exact sequences

Definition 19.4.1. For a subspace $A \subseteq X$, $C_n(X, A) = C_n(X)/C_n(A)$ with the induced boundary, and $H_n(X, A)$ is its homology, the *relative homology*. A map of pairs $f: (X, A) \rightarrow (Y, B)$ (a map with $f(A) \subseteq B$) induces $f_*: H_n(X, A) \rightarrow H_n(Y, B)$, and homotopic maps of pairs (through maps of pairs) induce the same maps, since the prism operator preserves chains in A .

Theorem 19.4.2. For $B \subseteq A \subseteq X$ there is a natural long exact sequence

$$\cdots \rightarrow H_n(A, B) \rightarrow H_n(X, B) \rightarrow H_n(X, A) \xrightarrow{\partial} H_{n-1}(A, B) \rightarrow \cdots \rightarrow H_0(X, A) \rightarrow 0.$$

For $B = \emptyset$ this is the long exact sequence of the pair (X, A) , where $\partial[c] = [\partial c]$ for a relative cycle c . The same holds with reduced homology when $B = \emptyset$ and $A \neq \emptyset$.

Proof. The sequence $0 \rightarrow C(A, B) \rightarrow C(X, B) \rightarrow C(X, A) \rightarrow 0$ of chain complexes is exact. Apply Theorem 13.4.3. For reduced homology use the augmented complexes, whose quotient is still $C(X, A)$. $\square$

Example 19.4.3. For $x_0 \in X$, the long exact sequence of (X, x_0) in reduced homology gives $H_n(X, x_0) \cong \tilde{H}_n(X)$. If A is a deformation retract of X , then $H_n(X, A) = 0$ for all n , by exactness of the long exact sequence, since $H_n(A) \rightarrow H_n(X)$ is an isomorphism for all n (Theorem 19.3.1).

19.5 Excision and Mayer-Vietoris

Definition 19.5.1. Let $\mathcal{U}$ be a family of subspaces of X whose interiors cover X . Then $C_n^{\mathcal{U}}(X) \subseteq C_n(X)$ is the subgroup generated by singular simplices with image in some $U \in \mathcal{U}$. It is a subcomplex.

Lemma 19.5.2. There are homomorphisms $S: C_n(X) \rightarrow C_n(X)$ and $T: C_n(X) \rightarrow C_{n+1}(X)$, natural in X ($f_{\#}S = S f_{\#}$, $f_{\#}T = T f_{\#}$), with $\partial S = S \partial$ and $\partial T + T \partial = 1 - S$, such that $S(\sigma)$ and $T(\sigma)$ are combinations of simplices with image in $\sigma(\Delta^n)$. Moreover $S^m(\sigma) = \sigma_{\#}(S^m \iota_n)$ with $\iota_n = [e_0, \dots, e_n]$, and $S^m \iota_n$ is a combination of affine simplices of diameter at most $(n/(n+1))^m \sqrt{2}$.

Proof. Linear chains. For a convex $Y \subseteq \mathbb{R}^N$ let $LC_n(Y) \subseteq C_n(Y)$ be generated by the affine simplices $[w_0, \dots, w_n]$; it is a subcomplex. For $b \in Y$ the cone $b: LC_n(Y) \rightarrow LC_{n+1}(Y)$, $[w_0, \dots, w_n] \mapsto [b, w_0, \dots, w_n]$, satisfies $\partial b(\lambda) = \lambda - b(\partial \lambda)$ for $n \geq 1$, and $\partial b(\lambda) = \lambda - \varepsilon(\lambda)[b]$ for $n = 0$, by the formula for ∂ . For an affine simplex λ let b_{λ} be its barycentre $\frac{1}{n+1} \sum_i w_i$. Define $S = \text{id}$ on LC_0 and $S(\lambda) = b_{\lambda}(S \partial \lambda)$ for $n \geq 1$; define $T = 0$ on LC_0 and $T(\lambda) = b_{\lambda}(\lambda - T \partial \lambda)$ for $n \geq 1$.

By induction on n : $\partial S \lambda = S \partial \lambda - b_{\lambda}(\partial S \partial \lambda) = S \partial \lambda - b_{\lambda}(S \partial \partial \lambda) = S \partial \lambda$ for $n \geq 2$, and for $n = 1$, $\partial S \lambda = S \partial \lambda - \varepsilon(\partial \lambda)[b_{\lambda}] = S \partial \lambda$. Similarly, using $\partial T \partial \lambda = \partial \lambda - S \partial \lambda$ by induction, $\partial T \lambda = \lambda - T \partial \lambda - b_{\lambda}(\partial \lambda - \partial T \partial \lambda) = \lambda - T \partial \lambda - b_{\lambda}(S \partial \lambda) = \lambda - T \partial \lambda - S \lambda$. Affine maps preserve barycentres and cones, so S and T commute with $g_{\#}$ for affine maps g .

General chains. Put $S\sigma = \sigma_{\#}S\iota_n$ and $T\sigma = \sigma_{\#}T\iota_n$. Naturality is clear, and since $\sigma \circ \delta^i = \sigma_{\#}\delta^i$ with δ^i affine, $\partial S\sigma = \sigma_{\#}S\partial\iota_n = \sum_i (-1)^i \sigma_{\#}\delta^i S\iota_{n-1} = S\partial\sigma$, and likewise for T . By naturality $S^m\sigma = \sigma_{\#}S^m\iota_n$.

Diameters. The diameter of an affine simplex is the largest distance between its vertices, since $|\sum_i t_i w_i - y| \leq \max_i |w_i - y|$. The simplices of $S\lambda$ are of the form $[b_{\lambda}, u_0, \dots, u_{n-1}]$ with $[u_0, \dots, u_{n-1}]$ a simplex of S applied to a face of λ . For a vertex w_i , $|b_{\lambda} - w_i| = |\sum_j (w_j - w_i)|/(n+1) \leq \frac{n}{n+1} \text{diam } \lambda$, so all points of λ are within this distance of b_{λ} ; and by induction the simplices of S of a face have diameter at most $\frac{n-1}{n} \text{diam } \lambda$. So the simplices of $S\lambda$ have diameter at most $\frac{n}{n+1} \text{diam } \lambda$. Iterate, using $\text{diam } \Delta^n = \sqrt{2}$. $\square$

Proposition 19.5.3. The inclusion $\iota: C^{\mathcal{U}}(X) \rightarrow C(X)$ is a chain homotopy equivalence. Hence it induces isomorphisms on homology.

Proof. For a singular simplex $\sigma: \Delta^n \rightarrow X$ let $m(\sigma)$ be the least $m \geq 0$ such that every subset of Δ^n of diameter at most $(n/(n+1))^m \sqrt{2}$ lies in $\sigma^{-1}(\text{int } U)$ for some $U \in \mathcal{U}$. It exists by the Lebesgue number lemma (Lemma 16.7.3), and by Lemma 19.5.2, $S^i \sigma \in C^{\mathcal{U}}(X)$ for $i \geq m(\sigma)$. The face maps are isometric embeddings and $\frac{n-1}{n} < \frac{n}{n+1}$, so $m(\sigma \circ \delta^j) \leq m(\sigma)$.

Let $D_m = \sum_{0 \leq i < m} TS^i$, so that $\partial D_m + D_m \partial = \sum_i (1 - S)S^i = 1 - S^m$. Define $D\sigma = D_{m(\sigma)}\sigma$ and $\rho = 1 - \partial D - D\partial$. Then ρ is a chain map, and

$$\rho(\sigma) = S^{m(\sigma)}\sigma + (D_{m(\sigma)} - D)\partial\sigma = S^{m(\sigma)}\sigma + \sum_j (-1)^j \sum_{m(\sigma\delta^j) \leq i < m(\sigma)} TS^i(\sigma\delta^j)$$

lies in $C^{\mathcal{U}}(X)$, since T preserves $C^{\mathcal{U}}(X)$. So $\rho: C(X) \rightarrow C^{\mathcal{U}}(X)$ satisfies $1 - \iota\rho = \partial D + D\partial$. As D maps $C^{\mathcal{U}}(X)$ into itself, the same identity shows $1 - \rho\iota = \partial D + D\partial$ on $C^{\mathcal{U}}(X)$. $\square$

Theorem 19.5.4 (Excision). *Let $Z \subseteq A \subseteq X$ with $\bar{Z} \subseteq \text{int } A$. Then the inclusion $(X \setminus Z, A \setminus Z) \rightarrow (X, A)$ induces isomorphisms on all H_n . Equivalently, if $A, B \subseteq X$ with $\text{int } A \cup \text{int } B = X$, then $H_n(B, A \cap B) \rightarrow H_n(X, A)$ is an isomorphism.*

Proof. The two statements correspond via $B = X \setminus Z$, since $\text{int } B = X \setminus \bar{Z}$. Let $\mathcal{U} = \{A, B\}$. The map $C(B)/C(A \cap B) \rightarrow C^{\mathcal{U}}(X)/C(A)$ is an isomorphism of complexes: it is surjective as every generator of $C^{\mathcal{U}}$ lies in $C(A)$ or $C(B)$, and its kernel is $C(B) \cap C(A) = C(A \cap B)$. The map $C^{\mathcal{U}}(X)/C(A) \rightarrow C(X)/C(A)$ induces isomorphisms on homology, by Proposition 19.5.3 and the five lemma applied to the long exact sequences of $0 \rightarrow C(A) \rightarrow C^{\mathcal{U}}(X) \rightarrow C^{\mathcal{U}}(X)/C(A) \rightarrow 0$ and $0 \rightarrow C(A) \rightarrow C(X) \rightarrow C(X, A) \rightarrow 0$. $\square$

Theorem 19.5.5 (Mayer–Vietoris). *If $A, B \subseteq X$ with $\text{int } A \cup \text{int } B = X$, there is a natural long exact sequence*

$$\cdots \rightarrow H_n(A \cap B) \xrightarrow{(i_*, -j_*)} H_n(A) \oplus H_n(B) \xrightarrow{k_* + l_*} H_n(X) \xrightarrow{\partial} H_{n-1}(A \cap B) \rightarrow \cdots,$$

where i, j, k, l are the inclusions, and similarly in reduced homology if $A \cap B \neq \emptyset$.

Proof. The sequence $0 \rightarrow C(A \cap B) \xrightarrow{(x \mapsto (x, -x))} C(A) \oplus C(B) \xrightarrow{((x, y) \mapsto x+y)} C^{\{A, B\}}(X) \rightarrow 0$ is exact: the kernel of the second map consists of pairs $(x, -x)$ with $x \in C(A) \cap C(B) = C(A \cap B)$. Take the long exact sequence and replace $H_n(C^{\{A, B\}}(X))$ by $H_n(X)$ using Proposition 19.5.3. For reduced homology augment each complex by $\mathbb{Z}, \mathbb{Z} \oplus \mathbb{Z}, \mathbb{Z}$. $\square$

19.6 Spheres, good pairs and degree

Theorem 19.6.1. *For $n \geq 0$, $\tilde{H}_k(S^n) \cong \mathbb{Z}$ for $k = n$ and 0 otherwise. For $n \geq 1$ the Mayer–Vietoris connecting map is an isomorphism $\tilde{H}_k(S^n) \rightarrow \tilde{H}_{k-1}(S^{n-1})$, where $S^{n-1} \subseteq S^n$ is the equator $\{x_{n+1} = 0\}$.*

Proof. For $n = 0$, S^0 is two points and $\tilde{H}_0(S^0) \cong \mathbb{Z}$, generated by the difference of the two points. For $n \geq 1$ let $A = \{x \in S^n : x_{n+1} > -1/2\}$ and $B = \{x_{n+1} < 1/2\}$. By stereographic projection from $\mp e_{n+1}$ (Example 16.9.4), A and B are homeomorphic to open balls, hence contractible, and $A \cap B$ deformation retracts onto the equator via $H(x, t) = ((1-t)x + t(x', 0))/\|(1-t)x + t(x', 0)\|$, where $x = (x', x_{n+1})$ with $x' \neq 0$. The reduced Mayer–Vietoris sequence gives the isomorphism, and induction gives the result. $\square$

Definition 19.6.2. A pair (X, A) is *good* if A is a nonempty closed subspace that is a deformation retract of some open neighbourhood V of A in X .

Proposition 19.6.3. *For a good pair, the quotient map $q: (X, A) \rightarrow (X/A, A/A)$ induces isomorphisms $H_n(X, A) \cong H_n(X/A, A/A) \cong \tilde{H}_n(X/A)$. For example, for $n \geq 1$, (D^n, S^{n-1}) is good and $H_k(D^n, S^{n-1}) \cong \tilde{H}_k(S^n) \cong \tilde{H}_{k-1}(S^{n-1})$.*

Proof. Consider the commutative diagram

$$\begin{array}{ccccc} H_n(X, A) & \longrightarrow & H_n(X, V) & \longleftarrow & H_n(X \setminus A, V \setminus A) \\ \downarrow q_* & & \downarrow q_* & & \downarrow q_* \\ H_n(X/A, A/A) & \longrightarrow & H_n(X/A, V/A) & \longleftarrow & H_n(X/A \setminus A/A, V/A \setminus A/A) \end{array}$$

The upper left map is an isomorphism by the long exact sequence of the triple (X, V, A) , since $H_n(V, A) = 0$ (Example 19.4.3). The upper right map is an isomorphism by excision, as $\bar{A} = A \subseteq V = \text{int } V$. Since V is open and saturated, the deformation retraction of V induces one of V/A onto the point A/A (Lemma 17.2.4), so the lower maps are isomorphisms for the same reasons. The right vertical map is induced by a homeomorphism of pairs, because q restricts to a bijective quotient map $X \setminus A \rightarrow X/A \setminus A/A$ (again by Lemma 17.2.4, as $X \setminus A$ is open and saturated). So the left vertical map is an isomorphism. The last isomorphism is Example 19.4.3. For (D^n, S^{n-1}) take $V = D^n \setminus \{0\}$ (Example 17.2.3); D^n/S^{n-1} is compact, and the map $D^n \rightarrow S^n$, $x \mapsto (2\sqrt{1 - \|x\|^2} x, 2\|x\|^2 - 1)$, induces a continuous bijection $D^n/S^{n-1} \rightarrow S^n$, hence a homeomorphism. The last isomorphism follows from the long exact sequence of (D^n, S^{n-1}) . $\square$

Definition 19.6.4. The *degree* of a map $f: S^n \rightarrow S^n$ is the integer $\deg f$ such that f_* is multiplication by $\deg f$ on $\tilde{H}_n(S^n) \cong \mathbb{Z}$.

Proposition 19.6.5. 1. $\deg \text{id} = 1$, $\deg(fg) = \deg f \deg g$, and homotopic maps have the same degree; a map that is not surjective has degree 0.

2. A reflection $x \mapsto (x_1, \dots, -x_i, \dots, x_{n+1})$ has degree -1 , and the antipodal map has degree $(-1)^{n+1}$.

3. A map $f: S^n \rightarrow S^n$ without fixed points is homotopic to the antipodal map, so $\deg f = (-1)^{n+1}$.

Proof. (1) Functoriality and homotopy invariance; a non-surjective map factors through $S^n \setminus \{\text{pt}\} \cong \mathbb{R}^n$, which is contractible.

(2) Let r be the reflection in the first coordinate. For $n = 0$ it swaps the two points and acts by -1 on $\tilde{H}_0(S^0)$. For $n \geq 1$, r preserves the sets A, B of the proof of Theorem 19.6.1 and commutes with the deformation retraction of $A \cap B$ onto the equator, on which it is again the reflection in the first coordinate. By naturality of the Mayer–Vietoris sequence, $\deg r$ on S^n equals $\deg r$ on S^{n-1} , hence equals -1 . A reflection in the i -th coordinate is $\pi r \pi^{-1}$ for a coordinate permutation π , so has degree -1 by (1). The antipodal map is the composite of $n + 1$ reflections.

(3) $(1 - t)f(x) - tx$ never vanishes: otherwise $f(x) = \frac{t}{1-t}x$ with $\|f(x)\| = \|x\| = 1$ forces $f(x) = x$. Normalising gives a homotopy from f to $-\text{id}$. $\square$

Corollary 19.6.6. 1. (Brouwer) There is no retraction $D^{n+1} \rightarrow S^n$, and every map $D^{n+1} \rightarrow D^{n+1}$ has a fixed point.

2. (Hairy ball theorem) S^n has a continuous nowhere vanishing tangent vector field ($v: S^n \rightarrow \mathbb{R}^{n+1}$ with $\langle v(x), x \rangle = 0$) if and only if n is odd.

3. (Invariance of dimension) If nonempty open subsets $U \subseteq \mathbb{R}^m$ and $V \subseteq \mathbb{R}^n$ are homeomorphic, then $m = n$.

Proof. (1) $\tilde{H}_n(S^n) \neq 0$ while D^{n+1} is contractible, so the identity of $\tilde{H}_n(S^n)$ cannot factor through $\tilde{H}_n(D^{n+1}) = 0$. The formula in the proof of Corollary 17.4.8 turns a fixed point free map into a retraction.

(2) If $n = 2k - 1$, $v(x) = (-x_2, x_1, \dots, -x_{2k}, x_{2k-1})$ works. Conversely, if v never vanishes, let $u = v/\|v\|$. Since $x \perp u(x)$ are unit vectors, $(1-t)x + tu(x)$ and $(1-t)u(x) - tx$ never vanish, and normalising gives homotopies $\text{id} \simeq u \simeq -\text{id}$. So $1 = (-1)^{n+1}$ and n is odd.

(3) Let $x \in U$. By excision (removing the closed set $\mathbb{R}^m \setminus U$ from $\mathbb{R}^m \setminus \{x\}$) and the long exact sequence, $H_k(U, U \setminus \{x\}) \cong H_k(\mathbb{R}^m, \mathbb{R}^m \setminus \{x\}) \cong \tilde{H}_{k-1}(\mathbb{R}^m \setminus \{x\}) \cong \tilde{H}_{k-1}(S^{m-1})$ for $m \geq 1$, which is nonzero exactly for $k = m$. A homeomorphism $U \rightarrow V$ gives isomorphisms of these local homology groups. For $m = 0$ or $n = 0$ the spaces are points, and a nonempty open subset of $\mathbb{R}^n$ is a point only if $n = 0$. $\square$

19.7 Cellular homology

Lemma 19.7.1. *Let X be a CW complex (Definition 17.7.1).*

1. $H_k(X^n, X^{n-1})$ is zero for $k \neq n$, and free abelian with basis the classes e_α of the characteristic maps $\Phi_\alpha: (D^n, S^{n-1}) \rightarrow (X^n, X^{n-1})$ applied to a fixed generator of $H_n(D^n, S^{n-1}) \cong \mathbb{Z}$, for $k = n$.
2. $H_k(X^n) = 0$ for $k > n$.
3. The inclusion $X^n \rightarrow X$ induces an isomorphism $H_k(X^n) \rightarrow H_k(X)$ for $k < n$.

Proof. (1) X^{n-1} is closed in X^n (as in the proof of Proposition 17.5.4), and it is a deformation retract of the open set obtained by removing the centres of the n -cells, via the radial deformation of $D^n \setminus \{0\}$ onto S^{n-1} , which descends to the quotient by Lemma 17.2.4. So (X^n, X^{n-1}) is a good pair, and so is $(\coprod_\alpha D_\alpha^n, \coprod_\alpha S_\alpha^{n-1})$. The characteristic maps induce a continuous bijection $\coprod_\alpha D_\alpha^n / \coprod_\alpha S_\alpha^{n-1} \rightarrow X^n / X^{n-1}$, which is a homeomorphism: a set whose preimage in $\coprod_\alpha D_\alpha^n$ is open has open preimage in $X^{n-1} \sqcup \coprod_\alpha D_\alpha^n$, since that preimage meets X^{n-1} in $\emptyset$ or X^{n-1} . By Proposition 19.6.3 and its naturality, $H_k(X^n, X^{n-1}) \cong H_k(\coprod_\alpha D_\alpha^n, \coprod_\alpha S_\alpha^{n-1}) = \bigoplus_\alpha H_k(D^n, S^{n-1})$, which is $\bigoplus_\alpha \mathbb{Z}$ for $k = n$ and 0 otherwise. (For $n = 0$, $X^{-1} = \emptyset$ and $H_k(X^0)$ is computed by Proposition 19.2.4.)

(2) By the exact sequence $H_k(X^{n-1}) \rightarrow H_k(X^n) \rightarrow H_k(X^n, X^{n-1})$ and induction on n , starting from $X^{-1} = \emptyset$.

(3) For $k < n$ the exact sequence $H_{k+1}(X^{n+1}, X^n) \rightarrow H_k(X^n) \rightarrow H_k(X^{n+1}, X^n)$ has outer terms zero, so $H_k(X^n) \cong H_k(X^m)$ for all $m \geq n$. A singular chain has compact image, which lies in a finite subcomplex (Theorem 17.7.3(2)), hence in some X^m . So every cycle of X is a cycle of some X^m , and a cycle of X^n that bounds in X bounds in some X^m ; this gives surjectivity and injectivity. $\square$

Definition 19.7.2. The *cellular chain complex* of a CW complex X has $C_n^{\text{CW}}(X) = H_n(X^n, X^{n-1})$ and differential d_n the composite $H_n(X^n, X^{n-1}) \xrightarrow{\partial} H_{n-1}(X^{n-1}) \xrightarrow{j_{n-1}} H_{n-1}(X^{n-1}, X^{n-2})$. Then $d_{n-1}d_n = 0$ because it contains two consecutive maps $\partial \circ j_{n-1}$ of the long exact sequence of (X^{n-1}, X^{n-2}) . Its homology is the *cellular homology* $H_n^{\text{CW}}(X)$.

Theorem 19.7.3. $H_n^{\text{CW}}(X) \cong H_n(X)$ for every CW complex X .

Proof. Write $j_n: H_n(X^n) \rightarrow H_n(X^n, X^{n-1})$, which is injective since $H_n(X^{n-1}) = 0$ by Lemma 19.7.1(2), and $\partial_{n+1}: H_{n+1}(X^{n+1}, X^n) \rightarrow H_n(X^n)$, so $d_{n+1} = j_n \partial_{n+1}$. Since j_n is injective, $\ker d_n = \ker(j_{n-1} \partial_n) = \ker \partial_n = \text{im } j_n$, and j_n maps $\text{im } \partial_{n+1}$ isomorphically onto $\text{im } d_{n+1}$. Hence $H_n^{\text{CW}}(X) \cong \text{im } j_n / \text{im } d_{n+1} \cong H_n(X^n) / \text{im } \partial_{n+1}$. By the exact sequence $H_{n+1}(X^{n+1}, X^n) \xrightarrow{\partial_{n+1}}$

$H_n(X^n) \rightarrow H_n(X^{n+1}) \rightarrow H_n(X^{n+1}, X^n) = 0$, this is $H_n(X^{n+1})$, which is $H_n(X)$ by Lemma 19.7.1(3). $\square$

Theorem 19.7.4 (Cellular boundary formula). *For $n \geq 2$, $d_n(e_\alpha^n) = \sum_\beta d_{\alpha\beta} e_\beta^{n-1}$, where $d_{\alpha\beta}$ is the degree of the composite $S^{n-1} \xrightarrow{\varphi_\alpha} X^{n-1} \rightarrow X^{n-1}/(X^{n-1} \setminus e_\beta^{n-1}) \cong S^{n-1}$ of the attaching map with the map collapsing the complement of the cell e_β^{n-1} (identified with S^{n-1} via Φ_β). For $n = 1$, $d_1(e_\alpha^1)$ is the difference of the endpoint and the starting point of the 1-cell.*

This is [Hat02, Section 2.2, Cellular Boundary Formula]; the proof is a diagram chase using naturality of the long exact sequences for the characteristic maps.

Example 19.7.5. 1. If all cells of X are in even dimensions, all d_n vanish and $H_n(X)$ is free with basis the n -cells. For $\mathbb{CP}^n$ (one cell in each even dimension $\leq 2n$), $H_k(\mathbb{CP}^n) \cong \mathbb{Z}$ for k even with $0 \leq k \leq 2n$, and 0 otherwise.

2. For the closed orientable surface Σ_g of genus g (Example 17.7.2), $d_1 = 0$ (one 0-cell) and $d_2 = 0$ (each 1-cell appears twice in the attaching word with opposite signs), so $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^{2g}$, $H_2 = \mathbb{Z}$.

3. For $\mathbb{RP}^n$ with one cell in each dimension, d_k is multiplication by $1 + (-1)^k$ [Hat02, Example 2.42]. So $H_k(\mathbb{RP}^n)$ is $\mathbb{Z}$ for $k = 0$, $\mathbb{Z}/2\mathbb{Z}$ for k odd with $0 < k < n$, $\mathbb{Z}$ for $k = n$ odd, and 0 otherwise.

Definition 19.7.6. The Euler characteristic of a finite CW complex with c_n cells of dimension n is $\chi(X) = \sum_n (-1)^n c_n$.

Proposition 19.7.7. $\chi(X) = \sum_n (-1)^n \dim_{\mathbb{Q}}(H_n(X) \otimes \mathbb{Q})$. In particular $\chi(X)$ depends only on the homotopy type of X .

Proof. $\mathbb{Q} = S^{-1}\mathbb{Z}$ is a flat $\mathbb{Z}$ -module (Proposition 12.3.2), so $H_n(C^{CW} \otimes \mathbb{Q}) \cong H_n(C^{CW}) \otimes \mathbb{Q} \cong H_n(X) \otimes \mathbb{Q}$. For a bounded complex V of finite-dimensional vector spaces, rank-nullity (Theorem 9.2.2) gives $\dim V_n = \dim Z_n + \dim B_{n-1}$ and $\dim H_n = \dim Z_n - \dim B_n$, so $\sum (-1)^n \dim V_n = \sum (-1)^n \dim H_n$. Apply this to $C^{CW} \otimes \mathbb{Q}$, with $\dim = c_n$. $\square$

Example 19.7.8. $\chi(S^n) = 1 + (-1)^n$, $\chi(\Sigma_g) = 2 - 2g$, $\chi(\mathbb{CP}^n) = n + 1$, $\chi(\mathbb{RP}^n) = 1$ for n even and 0 for n odd. As $\chi(\Sigma_g)$ determines g , surfaces of different genus are not homotopy equivalent.

19.8 Homology with coefficients

Definition 19.8.1. For an abelian group G , $C_n(X, A; G) = C_n(X, A) \otimes_{\mathbb{Z}} G$, and $H_n(X, A; G)$ is its homology. For a ring R these are R -modules.

Theorem 19.8.2. 1. Homotopy invariance, the long exact sequences, excision and Mayer–Vietoris hold for $H_\bullet(-; G)$, and cellular homology with coefficients, computed from $C_\bullet^{CW}(X) \otimes G$, agrees with $H_\bullet(X; G)$.

2. (Universal coefficients) There is a natural short exact sequence $0 \rightarrow H_n(X, A) \otimes G \rightarrow H_n(X, A; G) \rightarrow \text{Tor}_1^{\mathbb{Z}}(H_{n-1}(X, A), G) \rightarrow 0$. It splits, non-naturally [Hat02, Theorem 3A.3].

Proof. (1) The complexes $C(X, A)$ are free, so the short exact sequences of chain complexes used above are degreewise split and remain exact after $\otimes G$; chain homotopies remain chain homotopies. The arguments therefore apply verbatim, and the proof of Theorem 19.7.3 goes through with $H_n(X^n, X^{n-1}; G) \cong C_n^{CW}(X) \otimes G$. (2) Apply the Künneth formula (Theorem 13.9.1) to the free complex $C(X, A)$ and the complex G concentrated in degree 0. The splitting is quoted. $\square$

Example 19.8.3. With coefficients in $\mathbb{F}_2 = \mathbb{Z}/2\mathbb{Z}$ all cellular differentials of $\mathbb{RP}^n$ vanish, since $1 + (-1)^k$ is even; so $H_k(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2$ for $0 \leq k \leq n$. With coefficients in a field F of characteristic 0, $H_\bullet(X; F) \cong H_\bullet(X) \otimes F$, because F is a $\mathbb{Q}$ -vector space, hence a direct sum of copies of the flat module $\mathbb{Q}$, so $\text{Tor}_1^{\mathbb{Z}}(-, F) = 0$ (Proposition 13.8.4).

19.9 The first homology group

Theorem 19.9.1 (Hurewicz in degree one). *Let X be path-connected and $x_0 \in X$. Regarding a loop as a singular 1-simplex via $\Delta^1 \cong I$, $(t_0, t_1) \mapsto t_1$, gives a well-defined homomorphism $h: \pi_1(X, x_0) \rightarrow H_1(X)$, which induces an isomorphism $\pi_1(X, x_0)^{\text{ab}} \cong H_1(X)$ from the abelianisation (Proposition 6.9.2).*

Proof. For paths we write $f \sim g$ if $f - g \in C_1(X)$ is a boundary. (i) A constant path c satisfies $c \sim 0$: it is the boundary of the constant 2-simplex. (ii) If $f \simeq g$ relative to endpoints via $F: I^2 \rightarrow X$, divide I^2 into two triangles along the diagonal and compose with F ; the boundary of the difference of the two resulting 2-simplices is $f - g$ plus constant paths, so $f \sim g$. (iii) $f * g \sim f + g$: the 2-simplex $\sigma = (f * g) \circ \pi$, where $\pi: \Delta^2 \rightarrow I$ is the affine map with $e_0 \mapsto 0$, $e_1 \mapsto 1/2$, $e_2 \mapsto 1$, has $\partial\sigma = g - f * g + f$. (iv) Hence $f + \bar{f} \sim f * \bar{f} \sim c \sim 0$. So h is well defined and a homomorphism, and it factors through π_1^{ab} as $H_1(X)$ is abelian.

Choose for each $x \in X$ a path γ_x from x_0 to x , with γ_{x_0} constant. For a singular 1-simplex τ let $L(\tau) = [\gamma_{\tau(0)} * \tau * \bar{\gamma}_{\tau(1)}]$ in $\pi_1(X, x_0)^{\text{ab}}$, written additively, and extend L linearly to $C_1(X)$. For a singular 2-simplex σ with edges τ_0, τ_1, τ_2 (so $\partial\sigma = \tau_0 - \tau_1 + \tau_2$), $L(\partial\sigma)$ is the class of the loop $\gamma_{\sigma(e_0)} * \tau_2 * \tau_0 * \bar{\tau}_1 * \bar{\gamma}_{\sigma(e_0)}$, which is null-homotopic because the loop around the boundary of Δ^2 is null-homotopic in the contractible space Δ^2 . So L induces $\bar{L}: H_1(X) \rightarrow \pi_1(X, x_0)^{\text{ab}}$. Clearly $\bar{L}h = \text{id}$. For a cycle $z = \sum n_i \tau_i$, by (iii) and (iv), $hL(z) = \sum n_i (\gamma_{\tau_i(0)} + \tau_i - \gamma_{\tau_i(1)}) = z - \Gamma(\partial z) = z$ in $H_1(X)$, where $\Gamma: C_0(X) \rightarrow C_1(X)$ is the homomorphism $x \mapsto \gamma_x$. So h and $\bar{L}$ are inverse isomorphisms. $\square$

Remark 19.9.2. Homotopy invariance, the long exact sequence, excision, additivity over disjoint unions and the value on a point are the *Eilenberg–Steenrod axioms*. On CW complexes they determine homology, as the proof of Theorem 19.7.3 shows. Other constructions satisfying them, such as simplicial, Čech or sheaf-theoretic homology on suitable spaces, agree with singular homology there; the comparison with de Rham cohomology is Chapter 35.

19.10 Exercises

Exercise 19.10.1. Show that S^m and S^n are homotopy equivalent only if $m = n$, and that $\mathbb{R}^m \setminus \{0\}$ and $\mathbb{R}^n \setminus \{0\}$ are homeomorphic only if $m = n$.

Solution. By Theorem 19.6.1, $\tilde{H}_\bullet(S^m)$ is nonzero only in degree m . The spaces $\mathbb{R}^m \setminus \{0\}$ and S^{m-1} are homotopy equivalent (Example 17.2.3). $\square$

Exercise 19.10.2. The *suspension* SX is the quotient of $X \times [-1, 1]$ collapsing $X \times \{1\}$ and $X \times \{-1\}$ to two points. Show that $\tilde{H}_{n+1}(SX) \cong \tilde{H}_n(X)$ for all n .

Solution. Let A and B be the images of $X \times (-1/2, 1]$ and $X \times [-1, 1/2)$. They are contractible (slide to the cone points; the homotopies descend by Lemma 17.2.4), and $A \cap B \cong X \times (-1/2, 1/2) \simeq X$. The reduced Mayer–Vietoris sequence gives the isomorphism. $\square$

Exercise 19.10.3. Show that $\tilde{H}_n(\bigvee_\alpha X_\alpha) \cong \bigoplus_\alpha \tilde{H}_n(X_\alpha)$ if the basepoints x_α are such that each (X_α, x_α) is a good pair. Compute $H_\bullet$ of a wedge of k circles.

Solution. The pair $(\coprod X_\alpha, \coprod \{x_\alpha\})$ is good with quotient $\bigvee X_\alpha$, so $\tilde{H}_n(\bigvee X_\alpha) \cong H_n(\coprod X_\alpha, \coprod \{x_\alpha\}) = \bigoplus H_n(X_\alpha, x_\alpha) = \bigoplus \tilde{H}_n(X_\alpha)$. For k circles, $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}^k$, and $H_n = 0$ for $n \geq 2$; this agrees with Theorem 19.9.1, since the abelianisation of a free group of rank k is $\mathbb{Z}^k$. $\square$

Exercise 19.10.4. Using Theorem 19.9.1 and Remark 17.4.7, show that the degree of a map $S^1 \rightarrow S^1$ defined via π_1 agrees with the homological degree.

Exercise 19.10.5. Let X be a finite CW complex and $f: X \rightarrow X$ a cellular map. Define the Lefschetz number $\Lambda(f) = \sum_n (-1)^n \operatorname{tr}(f_* | H_n(X; \mathbb{Q}))$. Show that $\Lambda(f) = \sum_n (-1)^n \operatorname{tr}(f_\# | C_n^{CW}(X) \otimes \mathbb{Q})$, and deduce that $\Lambda(\operatorname{id}) = \chi(X)$.

Exercise 19.10.6. The Klein bottle K is obtained from $S^1 \vee S^1$ by attaching a 2-cell along $aba^{-1}b$. Compute $\pi_1(K)$, $H_\bullet(K)$ and $H_\bullet(K; \mathbb{F}_2)$.

Chapter 20

Singular Cohomology

20.1 Introduction

Cohomology is obtained by dualising singular chains. As groups it carries the same information as homology, by the universal coefficient theorem, but it has a natural ring structure given by the cup product, which distinguishes spaces with the same homology (Example 20.5.2). Cohomology classes are the objects of the Hodge conjecture: a Hodge class is a rational cohomology class of a special type, and the conjecture asks whether it is a combination of classes of subvarieties. The cap product relates cohomology and homology and leads to Poincaré duality (Chapter 21). We follow [Hat02, Chapter 3].

Prerequisites. Chapter 19 (Singular Homology).

Throughout, R is a commutative ring and G an abelian group, written additively.

20.2 Cochains and cohomology

Definition 20.2.1. For a pair (X, A) let $C^n(X, A; G) = \text{Hom}_{\mathbb{Z}}(C_n(X, A), G)$, the group of *singular n -cochains*, with coboundary $\delta: C^n \rightarrow C^{n+1}$, $\delta\varphi = \varphi \circ \partial$. So $C^n(X, A; G)$ is the group of functions from singular n -simplices of X to G vanishing on simplices in A , and $(\delta\varphi)(\sigma) = \sum_i (-1)^i \varphi(\sigma \circ \delta^i)$. The *singular cohomology* $H^n(X, A; G)$ is the cohomology of this complex, and $H^n(X; G) = H^n(X, \emptyset; G)$. A map of pairs f induces $f^\# = \text{Hom}(f_\#, G)$ and $f^*: H^n(Y, B; G) \rightarrow H^n(X, A; G)$, contravariantly. Reduced cohomology $\tilde{H}^n(X; G)$ is obtained by dualising the augmented chain complex. We write $H^n(X)$ for $H^n(X; \mathbb{Z})$. The *Kronecker pairing* $\langle -, - \rangle: H^n(X, A; G) \times H_n(X, A) \rightarrow G$ is $\langle [\varphi], [c] \rangle = \varphi(c)$.

Theorem 20.2.2. 1. *Homotopic maps of pairs induce the same map on cohomology.*

2. *For $B \subseteq A \subseteq X$ there is a natural long exact sequence $\cdots \rightarrow H^n(X, A; G) \rightarrow H^n(X, B; G) \rightarrow H^n(A, B; G) \xrightarrow{\delta} H^{n+1}(X, A; G) \rightarrow \cdots$, and $\langle \delta\alpha, c \rangle = \langle \alpha, \partial c \rangle$ for $\alpha \in H^n(A; G)$ and $c \in H_{n+1}(X, A)$.*

3. *(Excision) If $Z \subseteq A \subseteq X$ with $\bar{Z} \subseteq \text{int } A$, then $H^n(X, A; G) \rightarrow H^n(X \setminus Z, A \setminus Z; G)$ is an isomorphism.*

4. *(Mayer–Vietoris) If $\text{int } A \cup \text{int } B = X$, there is a long exact sequence $\cdots \rightarrow H^n(X; G) \rightarrow H^n(A; G) \oplus H^n(B; G) \rightarrow H^n(A \cap B; G) \rightarrow H^{n+1}(X; G) \rightarrow \cdots$.*

5. *$H^0(X; G)$ is the group of functions from the set of path components of X to G .*

Proof. Applying $\text{Hom}(-, G)$ turns chain homotopies into cochain homotopies, which gives (1) from Theorem 19.3.1. The sequences $0 \rightarrow C_n(A, B) \rightarrow C_n(X, B) \rightarrow C_n(X, A) \rightarrow 0$ and

$0 \rightarrow C_n(A \cap B) \rightarrow C_n(A) \oplus C_n(B) \rightarrow C_n^{\{A,B\}}(X) \rightarrow 0$ consist of free abelian groups and split degreewise, since the quotients are free on the simplices not in the subspace. So they remain exact after $\text{Hom}(-, G)$, and Theorem 13.4.3 gives the long exact sequences. The connecting map sends $[\varphi]$ to the class of $\delta\tilde{\varphi}$ for an extension $\tilde{\varphi}$ of φ to X , which gives the formula for the pairing. The chain homotopy equivalence $C^{\mathcal{U}}(X) \rightarrow C(X)$ of Proposition 19.5.3 dualises to a cochain homotopy equivalence, which gives (3) as in Theorem 19.5.4 and allows the replacement of $H^n(\text{Hom}(C^{\{A,B\}}(X), G))$ by $H^n(X; G)$ in (4). For (5), a 0-cocycle is a function φ on points with $\varphi(\gamma(1)) - \varphi(\gamma(0)) = 0$ for every path γ . $\square$

Theorem 20.2.3 (Universal coefficients). *There is a natural short exact sequence*

$$0 \rightarrow \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(X, A), G) \rightarrow H^n(X, A; G) \rightarrow \text{Hom}(H_n(X, A), G) \rightarrow 0,$$

where the second map is $\alpha \mapsto \langle \alpha, - \rangle$. It splits, non-naturally. If $G = F$ is a field and $A = \emptyset$, then $H^n(X; F) \cong \text{Hom}_F(H_n(X; F), F)$.

Proof. The first statement is Theorem 13.9.2 for the free complex $C(X, A)$. For a field, $\text{Hom}_{\mathbb{Z}}(C_n(X), F) = \text{Hom}_F(C_n(X) \otimes F, F)$, and $\text{Hom}_F(-, F)$ is exact, so it commutes with taking cohomology. $\square$

Lemma 20.2.4. $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}, G) = 0$ and $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/m\mathbb{Z}, G) \cong G/mG$; $\text{Ext}_{\mathbb{Z}}^1$ commutes with finite direct sums. Hence, if $H_{n-1}(X)$ and $H_n(X)$ are finitely generated, $H^n(X; \mathbb{Z}) \cong \mathbb{Z}^{b_n} \oplus T_{n-1}$, where b_n is the rank of $H_n(X)$ and T_{n-1} the torsion subgroup of $H_{n-1}(X)$.

Proof. Use the projective resolution $0 \rightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \rightarrow 0$ and Theorem 13.8.2: $\text{Ext}_{\mathbb{Z}}^1(\mathbb{Z}/m\mathbb{Z}, G)$ is the cokernel of $G \xrightarrow{m} G$. Additivity follows from additivity of Hom and of derived functors. For the last statement, write $H_n(X) \cong \mathbb{Z}^{b_n} \oplus T_n$ and $H_{n-1}(X) \cong \mathbb{Z}^{b_{n-1}} \oplus T_{n-1}$ with T_n, T_{n-1} finite direct sums of groups $\mathbb{Z}/m\mathbb{Z}$ (Theorem 6.11.6). By the split universal coefficient sequence, $H^n(X) \cong \text{Hom}(H_n(X), \mathbb{Z}) \oplus \text{Ext}^1(H_{n-1}(X), \mathbb{Z})$; here $\text{Hom}(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}) = 0$ gives $\text{Hom}(H_n(X), \mathbb{Z}) \cong \mathbb{Z}^{b_n}$, and $\text{Ext}^1(\mathbb{Z}/m\mathbb{Z}, \mathbb{Z}) \cong \mathbb{Z}/m\mathbb{Z}$ gives $\text{Ext}^1(H_{n-1}(X), \mathbb{Z}) \cong T_{n-1}$. $\square$

Example 20.2.5. 1. $\tilde{H}^k(S^n; G) \cong G$ for $k = n$ and 0 otherwise.

2. By Example 19.7.5 and Lemma 20.2.4, $H^k(\mathbb{R}P^n; \mathbb{Z})$ is $\mathbb{Z}$ for $k = 0$, $\mathbb{Z}/2\mathbb{Z}$ for k even with $0 < k \leq n$, $\mathbb{Z}$ for $k = n$ odd, and 0 otherwise: torsion moves up one degree compared with homology. With coefficients in $\mathbb{F}_2$, $H^k(\mathbb{R}P^n; \mathbb{F}_2) \cong \mathbb{F}_2$ for $0 \leq k \leq n$.

Theorem 20.2.6 (Cellular cohomology). *For a CW complex X , $H^n(X; G)$ is isomorphic to the cohomology of the complex $\text{Hom}(C_{\bullet}^{\text{CW}}(X), G)$ with the transposed differentials.*

Proof. By the universal coefficient theorem and Lemma 19.7.1: $H^k(X^n, X^{n-1}; G)$ is $\text{Hom}(C_n^{\text{CW}}(X), G)$ for $k = n$ and 0 otherwise; $H^k(X^n; G) = 0$ for $k > n$, because $H_k(X^n) = 0$ for $k > n$ and $H_n(X^n)$ is a subgroup of the free group $C_n^{\text{CW}}(X)$, hence free (Proposition 13.8.5). From the long exact sequence of (X, X^n) and Lemma 19.7.1(3), $H_k(X, X^n) = 0$ for $k \leq n$, hence $H^k(X, X^n; G) = 0$ for $k \leq n$ and $H^k(X; G) \rightarrow H^k(X^n; G)$ is an isomorphism for $k < n$. The proof of Theorem 19.7.3 now applies with all arrows reversed. The coboundary $H^n(X^n, X^{n-1}) \rightarrow H^n(X^n) \rightarrow H^{n+1}(X^{n+1}, X^n)$ is the transpose of d_{n+1} by the formula $\langle \delta\alpha, c \rangle = \langle \alpha, \partial c \rangle$. $\square$

Corollary 20.2.7. *Let X be a CW complex with finitely many cells in each dimension and no cells of odd dimension. Then $H^k(X; G) = 0$ for k odd, and $H^k(X; G) \cong G^{c_k}$ for k even, where c_k is the number of k -cells. The same holds for homology, and $\chi(X) = \sum_k c_k$ when X is finite.*

Proof. The cellular chain complex has $C_k^{CW}(X) = 0$ for k odd, so all its differentials vanish, being maps into or out of zero groups. By Theorem 20.2.6 the cohomology is $\text{Hom}(C_k^{CW}(X), G) = \text{Hom}(\mathbb{Z}^{c_k}, G) = G^{c_k}$ in each even degree k , and 0 in odd degrees; the homology statement is Theorem 19.7.3. The Euler characteristic is the alternating sum of the ranks. $\square$

Example 20.2.8. $\mathbb{CP}^n$ has one cell in each even dimension $0, 2, \dots, 2n$ (Example 19.7.5), so $H^{2k}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}$ for $0 \leq k \leq n$ and all other groups vanish, and $\chi(\mathbb{CP}^n) = n + 1$. The product $S^2 \times S^2$, with cells $e^0, e^2 \times e^0, e^0 \times e^2, e^2 \times e^2$, has $H^0 = H^4 = \mathbb{Z}$ and $H^2 = \mathbb{Z}^2$. For $\mathbb{RP}^n$, which has one cell in each dimension $\leq n$, the corollary does not apply, and indeed its cohomology has torsion (Example 20.2.5).

20.3 The cup product

Definition 20.3.1. For $\varphi \in C^k(X; R)$ and $\psi \in C^l(X; R)$, the *cup product* $\varphi \smile \psi \in C^{k+l}(X; R)$ is

$$(\varphi \smile \psi)(\sigma) = \varphi(\sigma \circ [e_0, \dots, e_k]) \cdot \psi(\sigma \circ [e_k, \dots, e_{k+l}]),$$

the product being taken in R : the “front k -face” times the “back l -face”. We write $\sigma[[e_0, \dots, e_k]]$ for $\sigma \circ [e_0, \dots, e_k]$.

Lemma 20.3.2. $\delta(\varphi \smile \psi) = \delta\varphi \smile \psi + (-1)^k \varphi \smile \delta\psi$. The cup product of cochains is associative, R -bilinear, has unit the 0-cochain 1 with value 1 on every point, and is natural: $f^\#(\varphi \smile \psi) = f^\#\varphi \smile f^\#\psi$.

Proof. Let $\sigma: \Delta^{k+l+1} \rightarrow X$ and $n = k + l + 1$. Then

$$\begin{aligned} (\delta\varphi \smile \psi)(\sigma) &= \sum_{i=0}^{k+1} (-1)^i \varphi(\sigma[[e_0, \dots, \widehat{e}_i, \dots, e_{k+1}]] \psi(\sigma[[e_{k+1}, \dots, e_n]]), \\ (-1)^k (\varphi \smile \delta\psi)(\sigma) &= \sum_{i=k}^n (-1)^i \varphi(\sigma[[e_0, \dots, e_k]] \psi(\sigma[[e_k, \dots, \widehat{e}_i, \dots, e_n]]). \end{aligned}$$

The term $i = k + 1$ of the first sum cancels the term $i = k$ of the second. The remaining terms are $\sum_i (-1)^i (\varphi \smile \psi)(\sigma[[e_0, \dots, \widehat{e}_i, \dots, e_n]])$, which is $(\varphi \smile \psi)(\partial\sigma) = \delta(\varphi \smile \psi)(\sigma)$. The other statements are immediate from the definition. $\square$

Definition 20.3.3. By Lemma 20.3.2, the cup product of two cocycles is a cocycle, and the product of a cocycle and a coboundary is a coboundary. It therefore induces $\smile: H^k(X; R) \times H^l(X; R) \rightarrow H^{k+l}(X; R)$, making $H^*(X; R) = \bigoplus_k H^k(X; R)$ an associative graded R -algebra with unit, the *cohomology ring*, and f^* a homomorphism of graded rings. If φ vanishes on chains in A so does $\varphi \smile \psi$, which gives a product $H^k(X, A; R) \times H^l(X; R) \rightarrow H^{k+l}(X, A; R)$.

Theorem 20.3.4 (Graded commutativity). For $\alpha \in H^k(X; R)$ and $\beta \in H^l(X; R)$, $\alpha \smile \beta = (-1)^{kl} \beta \smile \alpha$.

Sketch of proof. Let $\bar{\sigma} = \sigma \circ [e_n, \dots, e_0]$ be σ with its vertices in reverse order, and $\rho(\sigma) = \varepsilon_n \bar{\sigma}$ with $\varepsilon_n = (-1)^{n(n+1)/2}$. A direct computation shows that ρ is a chain map and that $\rho^\# \varphi \smile \rho^\# \psi = (-1)^{kl} \rho^\#(\psi \smile \varphi)$, using $\varepsilon_{k+l} = (-1)^{kl} \varepsilon_k \varepsilon_l$. The omitted verification is that ρ is chain homotopic to the identity, by the homotopy $P(\sigma) = \sum_i (-1)^i \varepsilon_{n-i} \sigma \circ \pi \circ [v_0, \dots, v_i, w_n, \dots, w_i]$, where $v_j = (e_j, 0)$ and $w_j = (e_j, 1)$ are the vertices of $\Delta^n \times I$ and $\pi: \Delta^n \times I \rightarrow \Delta^n$ is the projection; see [Hat02, Theorem 3.14]. Then $\rho^* = \text{id}$ on cohomology, and the formula follows. $\square$

Example 20.3.5. For a wedge $X \vee Y$ of CW complexes (at 0-cells), the inclusions induce an isomorphism of rings $\tilde{H}^*(X \vee Y; R) \cong \tilde{H}^*(X; R) \times \tilde{H}^*(Y; R)$, where products between the two factors vanish: the isomorphism of groups follows from Exercise 19.10.3 and the universal coefficient theorem, and it is induced by the restrictions to X and Y , which are ring homomorphisms. In particular all cup products $\tilde{H}^k(S^m) \times \tilde{H}^l(S^m) \rightarrow \tilde{H}^{k+l}(S^m)$ vanish, for degree reasons.

20.4 The Künneth formula

Definition 20.4.1. The cross product of $a \in H^k(X; R)$ and $b \in H^l(Y; R)$ is $a \times b = p_X^*(a) \smile p_Y^*(b) \in H^{k+l}(X \times Y; R)$, where p_X, p_Y are the projections. It is R -bilinear and induces $H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$, a ring homomorphism for the product $(a \otimes b)(c \otimes d) = (-1)^{|b||c|} ac \otimes bd$ on the tensor product.

Theorem 20.4.2 (Künneth formula). *Let X and Y be CW complexes such that $H^k(Y; R)$ is a finitely generated free R -module for every k (for instance R a field and Y a finite complex). Then the cross product $H^*(X; R) \otimes_R H^*(Y; R) \rightarrow H^*(X \times Y; R)$ is an isomorphism of graded rings.*

This is [Hat02, Theorem 3.15]. The ring homomorphism property follows from graded commutativity: $(a \times b)(c \times d) = p_X^* a p_Y^* b p_X^* c p_Y^* d = (-1)^{|b||c|} p_X^*(ac) p_Y^*(bd)$. The homology Künneth formula for fields, and the Eilenberg–Zilber theorem $C_*(X \times Y) \simeq C_*(X) \otimes C_*(Y)$ underlying it, are in [Hat02, Section 3.B].

Example 20.4.3. $H^*(S^1; \mathbb{Z}) = \Lambda_{\mathbb{Z}}[x]$ with $|x| = 1$ (an exterior algebra, as $x^2 \in H^2(S^1) = 0$). By induction and the Künneth formula, the cohomology ring of the torus $T^n = (S^1)^n$ is the exterior algebra $\Lambda_{\mathbb{Z}}[x_1, \dots, x_n]$ with $|x_i| = 1$, so $H^k(T^n) \cong \mathbb{Z}^{\binom{n}{k}}$. Similarly $H^*(S^2 \times S^2; \mathbb{Z}) = \mathbb{Z}[a, b]/(a^2, b^2)$ with $|a| = |b| = 2$.

20.5 Cohomology rings of projective spaces

Theorem 20.5.1. $H^*(\mathbb{RP}^n; \mathbb{F}_2) \cong \mathbb{F}_2[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 1$, and $H^*(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}[\alpha]/(\alpha^{n+1})$ with $|\alpha| = 2$. Also $H^*(\mathbb{RP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[\alpha]$ and $H^*(\mathbb{CP}^\infty; \mathbb{Z}) = \mathbb{Z}[\alpha]$.

This is [Hat02, Theorem 3.12]; the additive structure was computed above, and the content is that the generator in each degree is a power of α . A proof via Poincaré duality is given in Chapter 21.

Example 20.5.2. $\mathbb{CP}^2$ and $S^2 \vee S^4$ have the same homology and cohomology groups ($\mathbb{Z}$ in degrees 0, 2, 4), but they are not homotopy equivalent: in $H^*(\mathbb{CP}^2)$ the square of a generator of H^2 generates H^4 , whereas in $H^*(S^2 \vee S^4)$ it is zero by Example 20.3.5. Likewise $S^2 \times S^2$ and $S^2 \vee S^2 \vee S^4$ have the same groups but different rings.

Corollary 20.5.3 (Borsuk–Ulam). *There is no map $g: S^n \rightarrow S^{n-1}$ with $g(-x) = -g(x)$ for all x . Consequently every map $f: S^n \rightarrow \mathbb{R}^n$ identifies some pair of antipodal points.*

Proof. For $n = 1$ an odd map $S^1 \rightarrow S^0$ cannot exist, since its image would be a single point. Let $n \geq 2$ and suppose g exists; it induces $\bar{g}: \mathbb{RP}^n \rightarrow \mathbb{RP}^{n-1}$. For a path-connected space X , the universal coefficient theorem and Theorem 19.9.1 give $H^1(X; \mathbb{F}_2) \cong \text{Hom}(H_1(X), \mathbb{F}_2) \cong \text{Hom}(\pi_1(X), \mathbb{F}_2)$, naturally. For $m \geq 1$ let $w_m: \pi_1(\mathbb{RP}^m) \rightarrow \mathbb{F}_2$ be the monodromy of the two-sheeted covering $S^m \rightarrow \mathbb{RP}^m$ (Example 17.4.9): $w_m[\gamma] = 0$ if and only if the lifts of γ are loops. It is a homomorphism (Corollary 17.4.4(3)) and nonzero. Since $\pi_1(\mathbb{RP}^n) \cong \mathbb{Z}/2\mathbb{Z}$, w_n corresponds to the generator α of $H^1(\mathbb{RP}^n; \mathbb{F}_2)$, and $\pi_1(\mathbb{RP}^n)$ is generated by the image of a

path γ in S^n from x to $-x$. Its image under $\bar{g}$ is the image of the path $g \circ \gamma$ from $g(x)$ to $-g(x)$, whose lift is not a loop; so $w_{n-1}(\bar{g}_*[\gamma]) = 1 = w_n[\gamma]$ and $\bar{g}^*(w_{n-1}) = \alpha$. Then $\bar{g}^*(w_{n-1}^n) = \alpha^n \neq 0$ by Theorem 20.5.1, but $w_{n-1}^n \in H^n(\mathbb{RP}^{n-1}; \mathbb{F}_2) = 0$, a contradiction. If $f(x) \neq f(-x)$ for all x , then $g(x) = (f(x) - f(-x))/\|f(x) - f(-x)\|$ is an odd map $S^n \rightarrow S^{n-1}$. $\square$

20.6 Cup length and coverings by contractible sets

Lemma 20.6.1. *Let A, B be open subsets of X . The cup product of cochains induces a product $H^k(X, A; R) \times H^l(X, B; R) \rightarrow H^{k+l}(X, A \cup B; R)$, compatible with the absolute cup product under the maps $H^\bullet(X, C; R) \rightarrow H^\bullet(X; R)$ for $C = A, B, A \cup B$.*

Proof. If φ vanishes on simplices in A and ψ on simplices in B , then $\varphi \smile \psi$ vanishes on simplices in A (because of its front face) and on simplices in B (because of its back face), hence on the subcomplex $C(A) + C(B) \subseteq C(A \cup B)$. By Lemma 20.3.2, this gives a product with values in the cohomology of $\text{Hom}(C(X)/(C(A) + C(B)), R)$. It remains to identify this cohomology with $H^\bullet(X, A \cup B; R)$. The inclusion $C(A) + C(B) \rightarrow C(A \cup B)$ is a chain homotopy equivalence by Proposition 19.5.3, applied to the open cover $\{A, B\}$ of $A \cup B$, so it induces isomorphisms on cohomology with coefficients in R . The sequences $0 \rightarrow C(A) + C(B) \rightarrow C(X) \rightarrow C(X)/(C(A) + C(B)) \rightarrow 0$ and $0 \rightarrow C(A \cup B) \rightarrow C(X) \rightarrow C(X, A \cup B) \rightarrow 0$ split degreewise, the quotients being free on the simplices not in the subcomplex, so they give long exact cohomology sequences after applying $\text{Hom}(-, R)$, and the inclusions give a map between them. The five lemma (Lemma 13.4.2) shows that $H^\bullet(X, A \cup B; R) \rightarrow H^\bullet(\text{Hom}(C(X)/(C(A) + C(B)), R))$ is an isomorphism. Compatibility with the absolute product holds on cochains. $\square$

Proposition 20.6.2. *Let X be covered by open sets $U_1, \dots, U_n$ such that each inclusion $U_i \rightarrow X$ is homotopic to a constant map. Then $\alpha_1 \smile \dots \smile \alpha_n = 0$ for all $\alpha_i \in H^{k_i}(X; R)$ with $k_i \geq 1$.*

Proof. The restriction $H^k(X; R) \rightarrow H^k(U_i; R)$ is induced by the inclusion, which is homotopic to a map factoring through a point, so it is zero for $k \geq 1$, as $H^k(\text{pt}; R) = 0$ (Theorem 20.2.2(1)). By the long exact sequence of the pair (X, U_i) , each α_i lifts to $\tilde{\alpha}_i \in H^{k_i}(X, U_i; R)$. By Lemma 20.6.1, applied $n - 1$ times, $\tilde{\alpha}_1 \smile \dots \smile \tilde{\alpha}_n$ lies in $H^\bullet(X, U_1 \cup \dots \cup U_n; R) = H^\bullet(X, X; R) = 0$ and maps to $\alpha_1 \smile \dots \smile \alpha_n$. $\square$

Corollary 20.6.3. 1. *The torus T^n cannot be covered by n open sets that are contractible in T^n , but it can be covered by $n + 1$ of them.*

2. $\mathbb{CP}^n$ cannot be covered by n open sets contractible in $\mathbb{CP}^n$, and it is covered by the $n + 1$ standard affine charts, each homeomorphic to $\mathbb{C}^n$.
3. $\mathbb{RP}^n$ cannot be covered by n open sets contractible in $\mathbb{RP}^n$.

Proof. By Proposition 20.6.2 it suffices to exhibit n classes of positive degree with nonzero product: $x_1 \smile \dots \smile x_n$ generates $H^n(T^n)$ (Example 20.4.3), and $\alpha^n \neq 0$ in $H^{2n}(\mathbb{CP}^n; \mathbb{Z})$ and in $H^n(\mathbb{RP}^n; \mathbb{F}_2)$ (Theorem 20.5.1). For the covering of $T^n = \mathbb{R}^n/\mathbb{Z}^n$ by $n + 1$ sets, let U_j , $0 \leq j \leq n$, be the image of the open set of points $x \in \mathbb{R}^n$ whose coordinates avoid the interval $[\frac{j}{n+1} + \frac{1}{4(n+1)}, \frac{j}{n+1} + \frac{3}{4(n+1)}]$ modulo 1: each coordinate then ranges in an open interval of $\mathbb{R}/\mathbb{Z}$, so U_j is homeomorphic to an open cube, which is contractible. Every x lies in some U_j , because its n coordinates can meet at most n of the $n + 1$ disjoint intervals. $\square$

Remark 20.6.4. The least m such that X is covered by $m + 1$ open sets contractible in X is the Lusternik–Schnirelmann category of X . Proposition 20.6.2 bounds it from below by the cup length,

the largest number of classes of positive degree with nonzero product. By Corollary 20.6.3 the category of T^n is n . Lusternik and Schnirelmann introduced this invariant to bound from below the number of critical points of smooth functions on closed manifolds.

20.7 The cap product

Definition 20.7.1. For $\sigma: \Delta^n \rightarrow X$ and $\varphi \in C^k(X; R)$ with $k \leq n$, the *cap product* is

$$\sigma \frown \varphi = \varphi(\sigma| [e_0, \dots, e_k]) \cdot \sigma| [e_k, \dots, e_n] \in C_{n-k}(X; R),$$

extended R -linearly to $C_n(X; R) = C_n(X) \otimes R$.

Proposition 20.7.2. 1. $\partial(\sigma \frown \varphi) = (-1)^k(\partial\sigma \frown \varphi - \sigma \frown \delta\varphi)$, so the cap product induces $\frown: H_n(X; R) \times H^k(X; R) \rightarrow H_{n-k}(X; R)$, and relative versions $H_n(X, A; R) \times H^k(X; R) \rightarrow H_{n-k}(X, A; R)$ and $H_n(X, A; R) \times H^k(X, A; R) \rightarrow H_{n-k}(X; R)$.

2. (Naturality) $f_*(c) \frown \varphi = f_*(c \frown f^*\varphi)$ for $f: X \rightarrow Y$, $c \in H_n(X; R)$ and $\varphi \in H^k(Y; R)$.

3. $\psi(c \frown \varphi) = (\varphi \smile \psi)(c)$ for $\psi \in C^{n-k}(X; R)$, and $(c \frown \varphi) \frown \psi = c \frown (\varphi \smile \psi)$.

Proof. (1) We have $\partial(\sigma \frown \varphi) = \sum_{i \geq k} (-1)^{i-k} \varphi(\sigma| [e_0, \dots, e_k]) \sigma| [e_k, \dots, \widehat{e}_i, \dots, e_n]$, while

$$\partial\sigma \frown \varphi = \sum_{i \leq k} (-1)^i \varphi(\sigma| [e_0, \dots, \widehat{e}_i, \dots, e_{k+1}]) \sigma| [e_{k+1}, \dots, e_n] + \sum_{i \geq k+1} (-1)^i \varphi(\sigma| [e_0, \dots, e_k]) \sigma| [e_k, \dots, \widehat{e}_i, \dots, e_n]$$

and $\sigma \frown \delta\varphi = \sum_{i \leq k+1} (-1)^i \varphi(\sigma| [e_0, \dots, \widehat{e}_i, \dots, e_{k+1}]) \sigma| [e_{k+1}, \dots, e_n]$. Subtracting, what remains is the second sum together with the term $-(-1)^{k+1} \varphi(\sigma| [e_0, \dots, e_k]) \sigma| [e_{k+1}, \dots, e_n]$, which is the term $i = k$; this is $(-1)^k \partial(\sigma \frown \varphi)$. Hence cycle $\frown$ cocycle is a cycle, boundary $\frown$ cocycle and cycle $\frown$ coboundary are boundaries. For the relative versions, $\sigma \frown \varphi$ lies in $C(A)$ if σ does, and vanishes if φ vanishes on A and σ lies in A . (2) and (3) are immediate from the definitions: $\psi(\sigma \frown \varphi) = \varphi(\sigma| [e_0, \dots, e_k]) \psi(\sigma| [e_k, \dots, e_n]) = (\varphi \smile \psi)(\sigma)$. $\square$

Remark 20.7.3. For a closed oriented n -manifold M with fundamental class $[M] \in H_n(M; \mathbb{Z})$, the map $H^k(M; R) \rightarrow H_{n-k}(M; R)$, $\alpha \mapsto [M] \frown \alpha$, is an isomorphism: this is Poincaré duality (Chapter 21). Combined with Proposition 20.7.2(3) it shows that the cup product pairing $H^k(M; F) \times H^{n-k}(M; F) \rightarrow F$, $(\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle$, is nondegenerate over a field F .

20.8 Exercises

Exercise 20.8.1. Compute $H^k(\mathbb{RP}^n; \mathbb{Z}/4\mathbb{Z})$ from the cellular cochain complex, and check the result with the universal coefficient theorem.

Solution. The cellular cochain complex is $\mathbb{Z}/4 \rightarrow \mathbb{Z}/4 \rightarrow \dots$ with differential $\text{Hom}(d_{k+1}, \mathbb{Z}/4)$, multiplication by $1 + (-1)^{k+1}$: zero out of even degrees and multiplication by 2 out of odd degrees. So $H^0 = \mathbb{Z}/4$; for $0 < k < n$, $H^k \cong \mathbb{Z}/2$ (for k odd the kernel of 2 is $2\mathbb{Z}/4$ and the image of the previous zero map is 0; for k even the kernel is all of $\mathbb{Z}/4$ and the image is $2\mathbb{Z}/4$); and H^n is $\mathbb{Z}/4$ for n odd and $\mathbb{Z}/2$ for n even. With universal coefficients, for $0 < k < n$ odd: $\text{Hom}(\mathbb{Z}/2, \mathbb{Z}/4) \oplus \text{Ext}(0, \mathbb{Z}/4) = \mathbb{Z}/2$; for k even: $\text{Hom}(0, \mathbb{Z}/4) \oplus \text{Ext}(\mathbb{Z}/2, \mathbb{Z}/4) = \mathbb{Z}/4/2\mathbb{Z}/4 = \mathbb{Z}/2$. $\square$

Exercise 20.8.2. Let Σ_g be the closed orientable surface of genus g . Using the map $\Sigma_g \rightarrow \bigvee_{i=1}^g T^2$ collapsing $g-1$ circles, show that $H^1(\Sigma_g) \cong \mathbb{Z}^{2g}$ has a basis a_i, b_i with $a_i \smile b_i = -b_i \smile a_i$ a generator of $H^2(\Sigma_g) \cong \mathbb{Z}$, and all other products of basis elements zero.

Exercise 20.8.3. For a map $f: S^{2n-1} \rightarrow S^n$ with $n \geq 2$ let $X_f = S^n \cup_f D^{2n}$. Show that $H^n(X_f) \cong \mathbb{Z} \cdot a$ and $H^{2n}(X_f) \cong \mathbb{Z} \cdot b$ and define the Hopf invariant $h(f)$ by $a^2 = h(f)b$. Show that $h(f) = 0$ if n is odd, and that h of the Hopf map $S^3 \rightarrow S^2$ is ± 1 , using $X_f = \mathbb{CP}^2$.

Exercise 20.8.4. Show that $S^2 \vee S^4$ is not a retract of $\mathbb{CP}^2$, even up to homotopy: there are no maps $i: S^2 \vee S^4 \rightarrow \mathbb{CP}^2$ and $r: \mathbb{CP}^2 \rightarrow S^2 \vee S^4$ with $ri \simeq \text{id}$. Show also that there is no map $\mathbb{CP}^2 \rightarrow S^2$ inducing an isomorphism on H^2 .

Solution. A map $f: \mathbb{CP}^2 \rightarrow S^2$ with f^* an isomorphism on H^2 would give $f^*(x)^2 = f^*(x^2) = 0$ for a generator x of $H^2(S^2)$, while $f^*(x)$ generates $H^2(\mathbb{CP}^2)$ and its square generates $H^4(\mathbb{CP}^2)$. If $S^2 \vee S^4 \xrightarrow{i} \mathbb{CP}^2 \xrightarrow{r} S^2 \vee S^4$ composed to the identity (up to homotopy), then i^* would be surjective, and r^* injective; a generator $u \in H^2(S^2 \vee S^4)$ would satisfy $r^*(u)^2 = r^*(u^2) = 0$ with $r^*(u) = \pm$ a generator of $H^2(\mathbb{CP}^2)$ (as r^* is injective on $H^2 \cong \mathbb{Z}$ and $i^*r^* = \text{id}$ forces r^* to be split injective), contradicting $\alpha^2 \neq 0$. $\square$

Chapter 21

Topological Manifolds and Poincaré Duality

21.1 Introduction

A closed oriented n -manifold M has a fundamental class $[M] \in H_n(M; \mathbb{Z})$, and capping with it is an isomorphism $H^k(M; R) \cong H_{n-k}(M; R)$ (Theorem 21.6.2). Duality is what turns a closed oriented submanifold, or a complex subvariety, of dimension d into a cohomology class of degree $n - d$, and it makes the cup product pairing between complementary degrees nondegenerate. We follow [Hat02, Section 3.3].

Prerequisites. Chapter 20 (Singular Cohomology).

For a space X and $A \subseteq X$ we write $H_i(X | A; R) = H_i(X, X \setminus A; R)$ and $H^i(X | A; R) = H^i(X, X \setminus A; R)$, and $H_i(X | x)$ for $A = \{x\}$.

21.2 Topological manifolds

Definition 21.2.1. A *topological n -manifold* is a Hausdorff, second countable space in which every point has an open neighbourhood homeomorphic to $\mathbb{R}^n$ (Definition 18.5.1, where the role of each axiom is discussed). A *chart* is such a neighbourhood with a homeomorphism to $\mathbb{R}^n$. A manifold is *closed* if it is compact. A *manifold with boundary* is defined in the same way with neighbourhoods homeomorphic to open subsets of the half-space $\{x \in \mathbb{R}^n : x_n \geq 0\}$; its *boundary* ∂M consists of the points that correspond to points with $x_n = 0$.

Lemma 21.2.2. For a point x of an n -manifold M , $H_i(M | x; R) \cong R$ for $i = n$ and 0 otherwise. In particular the dimension of a nonempty manifold is well defined. For an open ball B of finite radius in a chart and $x \in B$, the restriction $H_i(M | B; R) \rightarrow H_i(M | x; R)$ is an isomorphism.

Proof. Let $V \ni x$ be a chart. The closure of B in V is compact, hence closed in M , so excision of $M \setminus V$ gives $H_i(M | B) \cong H_i(V | B) \cong H_i(\mathbb{R}^n | B)$, and likewise for x . Both $\mathbb{R}^n \setminus B$ and $\mathbb{R}^n \setminus \{x\}$ deformation retract onto a sphere centred at x of radius larger than that of B , by radial projection from x ; so the restriction is an isomorphism, and $H_i(\mathbb{R}^n | x) \cong \tilde{H}_{i-1}(\mathbb{R}^n \setminus \{x\}) \cong \tilde{H}_{i-1}(S^{n-1}; R)$ by the long exact sequence. The dimension is well defined by Corollary 19.6.6(3). $\square$

Example 21.2.3. $\mathbb{R}^n$, open subsets of manifolds, S^n (charts by stereographic projection), and products of manifolds (of dimension $m + n$) are manifolds; hence so is the torus T^n . The projective spaces $\mathbb{RP}^n$ and $\mathbb{CP}^n$ are closed manifolds of dimensions n and $2n$ (Chapter 41), and so are the surfaces Σ_g . The disc D^n is a manifold with boundary S^{n-1} . The line with two origins (Counterexample 16.4.11) is locally Euclidean but not Hausdorff.

21.3 Orientations

Definition 21.3.1. Let M be an n -manifold. A *coordinate ball* is an open ball of finite radius in a chart. Let M_R be the set of pairs (x, α) with $x \in M$ and $\alpha \in H_n(M | x; R)$, with $p(x, \alpha) = x$. For a coordinate ball B and $\alpha_B \in H_n(M | B; R)$ let $U(\alpha_B)$ be the set of pairs (x, α_x) with $x \in B$ and α_x the image of α_B . An R -orientation of M is a continuous section $x \mapsto \mu_x$ of p such that each μ_x generates $H_n(M | x; R) \cong R$ as an R -module. A $\mathbb{Z}$ -orientation is an *orientation*, and M is (R) -orientable if it has one. The *orientation double covering* $\tilde{M} \subseteq M_{\mathbb{Z}}$ consists of the pairs (x, α) with α a generator of $H_n(M | x; \mathbb{Z}) \cong \mathbb{Z}$.

Proposition 21.3.2. *The sets $U(\alpha_B)$ form a basis of a topology on M_R for which p is a covering map, and $\tilde{M} \rightarrow M$ is a two-sheeted covering. Let M be connected.*

1. M is orientable if and only if $\tilde{M}$ has two components; then M has exactly two orientations.
2. If $\pi_1(M)$ has no subgroup of index 2, for instance if M is simply connected, then M is orientable.
3. M is $\mathbb{F}_2$ -orientable, and an orientable manifold is R -orientable for every R .

Proof. If $(x, \alpha) \in U(\alpha_B) \cap U(\alpha_{B'})$, choose a coordinate ball $B'' \ni x$ inside $B \cap B'$. The images of α_B and $\alpha_{B'}$ in $H_n(M | B'')$ both restrict to α , so they are equal by Lemma 21.2.2, and their U is contained in both sets. By the same lemma, p maps $U(\alpha_B)$ bijectively onto B , and $p^{-1}(B)$ is the disjoint union of the sets $U(\alpha_B)$, $\alpha_B \in H_n(M | B)$. For $B' \subseteq B$, $p^{-1}(B') \cap U(\alpha_B) = U(\alpha_{B'})$ with $\alpha_{B'}$ the restriction, and $p(U(\alpha_{B'})) = B'$; so $p|_{U(\alpha_B)}$ is a homeomorphism onto B , and p is a covering map. The generators form a union of sheets, two over each coordinate ball.

(1) A section s of a covering over a connected space has open image (near $s(x)$ it lies in one sheet, by continuity) and closed image (it is the set of points e with $s(p(e)) = e$, and the covering space is Hausdorff because M is and distinct sheets are disjoint open sets). So an orientation μ gives two components $\{\mu_x\}$ and $\{-\mu_x\}$ of $\tilde{M}$. Conversely, each sheet over a coordinate ball is connected and so lies in a single component; hence the number of points of a component C of $\tilde{M}$ over x is locally constant, hence constant; if $\tilde{M}$ is not connected it is 1, and $C \rightarrow M$ is a bijective covering map, hence a homeomorphism whose inverse is an orientation. The orientations are the sections of $\tilde{M}$, which are the inverses of the two components.

(2) If M is not orientable, $\tilde{M}$ is a connected two-sheeted covering, and $p_*\pi_1(\tilde{M})$ has index 2 by Corollary 17.4.4(3).

(3) $H_n(M | x; \mathbb{F}_2)$ has a unique generator, and $x \mapsto$ this generator is continuous since it lies in $U(\alpha_B)$ for the generator α_B . The coefficient homomorphism $\mathbb{Z} \rightarrow R$ induces a continuous map $M_{\mathbb{Z}} \rightarrow M_R$ over M sending generators to generators. $\square$

21.4 The fundamental class

Lemma 21.4.1 (Relative Mayer–Vietoris). *Let Y_1, Y_2 be open subsets of a space X . There are natural long exact sequences*

$$\cdots \rightarrow H_i(X, Y_1 \cap Y_2) \rightarrow H_i(X, Y_1) \oplus H_i(X, Y_2) \rightarrow H_i(X, Y_1 \cup Y_2) \rightarrow H_{i-1}(X, Y_1 \cap Y_2) \rightarrow \cdots$$

and $\cdots \rightarrow H^i(X, Y_1 \cup Y_2) \rightarrow H^i(X, Y_1) \oplus H^i(X, Y_2) \rightarrow H^i(X, Y_1 \cap Y_2) \rightarrow H^{i+1}(X, Y_1 \cup Y_2) \rightarrow \cdots$, with any coefficients.

Proof. The sequence $0 \rightarrow C(X)/C(Y_1 \cap Y_2) \rightarrow C(X)/C(Y_1) \oplus C(X)/C(Y_2) \rightarrow C(X)/(C(Y_1) + C(Y_2)) \rightarrow 0$, with maps $x \mapsto (x, -x)$ and $(x, y) \mapsto x + y$, is exact, because $C(Y_1) \cap C(Y_2) = C(Y_1 \cap Y_2)$. All terms are free abelian groups on sets of simplices, so it is degreewise

split, and it remains exact after tensoring with R or applying $\text{Hom}(-, R)$. The inclusion $C(Y_1) + C(Y_2) \rightarrow C(Y_1 \cup Y_2)$ is a quasi-isomorphism by Proposition 19.5.3 (the Y_i are open in $Y_1 \cup Y_2$), so by the five lemma the last complex computes $H_\bullet(X, Y_1 \cup Y_2)$; dually in cohomology. $\square$

Lemma 21.4.2. *Let M be an n -manifold and $A \subseteq M$ compact.*

1. *If $x \mapsto \alpha_x$ is a continuous section of M_R over A , there is a unique $\alpha_A \in H_n(M | A; R)$ restricting to α_x for every $x \in A$.*
2. *$H_i(M | A; R) = 0$ for $i > n$.*

Proof. We omit R from the notation.

Step 1: if the lemma holds for compact A, B and $A \cap B$, it holds for $A \cup B$. Apply Lemma 21.4.1 with $Y_1 = M \setminus A$ and $Y_2 = M \setminus B$:

$$H_{i+1}(M | A \cap B) \rightarrow H_i(M | A \cup B) \xrightarrow{\Phi} H_i(M | A) \oplus H_i(M | B) \xrightarrow{\Psi} H_i(M | A \cap B),$$

where Φ consists of restrictions and Ψ is the difference of restrictions (up to the signs of the maps). For $i > n$ the outer terms vanish, giving (2). For $i = n$, Φ is injective, which gives uniqueness since restrictions to points factor through A or B . For existence, α_A and α_B restrict to elements of $H_n(M | A \cap B)$ that both restrict to α_x at every point, hence agree by uniqueness for $A \cap B$; so (α_A, α_B) lies in the kernel of Ψ and comes from some $\alpha_{A \cup B}$.

Step 2: $M = \mathbb{R}^n$ and A compact convex. Choose $x \in A$. Both $\mathbb{R}^n \setminus A$ and $\mathbb{R}^n \setminus \{x\}$ deformation retract onto a large sphere centred at x (moving points along rays from x , which leave the convex set A once), so $H_i(\mathbb{R}^n | A) \rightarrow H_i(\mathbb{R}^n | x)$ is an isomorphism, giving (2). For (1), let α_A be the element restricting to α_x . Choosing a coordinate ball $B \supseteq A$, the image of α_A at points of A is a continuous section (it comes from $H_n(\mathbb{R}^n | B) \cong H_n(\mathbb{R}^n | A)$). It agrees with the given section at x , so everywhere, since A is connected (Lemma 17.4.2).

Step 3: $M = \mathbb{R}^n$ and $A = A_1 \cup \dots \cup A_m$ with each A_j compact convex. Induction on m using Step 1, since $(A_1 \cup \dots \cup A_{m-1}) \cap A_m = \bigcup_{j < m} (A_j \cap A_m)$ is a union of $m - 1$ compact convex sets.

Step 4: $M = \mathbb{R}^n$ and A arbitrary compact. Let $\alpha \in H_i(\mathbb{R}^n | A)$ be represented by a chain z with ∂z a chain in $\mathbb{R}^n \setminus A$. The images of the simplices of ∂z form a compact set C disjoint from A . Cover $\mathbb{R}^n$ by the closed cubes of a grid of side smaller than the distance from A to C divided by $\sqrt{n}$, and let K be the union of those meeting A ; then $A \subseteq K$, $K \cap C = \emptyset$, and z defines $\alpha_K \in H_i(\mathbb{R}^n | K)$ mapping to α . If $i > n$, $\alpha_K = 0$ by Step 3, so $\alpha = 0$. If $i = n$ and α restricts to 0 at every point of A , then α_K restricts to 0 at every point of K : each cube Q meets A , and the restriction of α_K to Q is determined by its value at a point of $Q \cap A$ (Step 2). By uniqueness in Step 3, $\alpha_K = 0$, so $\alpha = 0$; this is uniqueness. For existence, $M_R \rightarrow \mathbb{R}^n$ is a trivial covering $\mathbb{R}^n \times R$ (the restrictions from a large ball are isomorphisms), so a section over A is a locally constant function $A \rightarrow R$. By compactness it takes finitely many values, on disjoint compact sets A_r , which have positive distances from each other. Choose the grid so that twice the diameter of a cube is smaller than these distances. Then two cubes meeting different sets A_r are disjoint, so the unions K_r of the cubes meeting A_r are disjoint closed sets covering K , and the section extends to the locally constant function equal to r on K_r . Step 3 gives α_K , and its image in $H_n(\mathbb{R}^n | A)$ is the required class.

Step 5: general M . A compact $A \subseteq M$ is a finite union $A_1 \cup \dots \cup A_m$ of compact sets each contained in a chart $V_j \cong \mathbb{R}^n$. By excision, $H_i(M | A_j) \cong H_i(V_j | A_j)$, compatibly with restrictions to points, so the lemma holds for sets contained in a chart by Step 4. Conclude by induction on m using Step 1, as $(A_1 \cup \dots \cup A_{m-1}) \cap A_m$ is contained in the chart V_m . $\square$

Theorem 21.4.3. *Let M be a closed connected n -manifold. Then $H_i(M; R) = 0$ for $i > n$, and the restriction $H_n(M; R) \rightarrow H_n(M | x; R) \cong R$ is injective for every x . Its image is R if M is R -orientable,*

and $\{r \in R : 2r = 0\}$ otherwise. If M is R -oriented by μ , the unique class $[M] \in H_n(M; R)$ restricting to μ_x for all x is the fundamental class. In particular $H_n(M; \mathbb{Z}) \cong \mathbb{Z}$ if M is orientable and 0 otherwise.

Proof. Apply Lemma 21.4.2 with $A = M$, noting $H_i(M | M) = H_i(M)$: classes in $H_n(M)$ correspond to sections of M_R over M . A section over the connected space M is determined by its value at x , by Lemma 17.4.2. The set M_r of pairs (y, α) with $\alpha = \pm r\mu_y$ for a local generator μ_y is a union of components of M_R (it is open and closed, as can be checked on the sets $U(\alpha_B)$). If $r = -r$, then $M_r \rightarrow M$ is a homeomorphism, and there is a section with value r . If $r \neq -r$, then $M_r \cong \tilde{M}$ (identify $(y, \pm r\mu_y)$ with $(y, \pm\mu_y)$), which has a section if and only if M is orientable. Writing a general element of R as r times a generator gives the description of the image; for R -orientability, an R -orientation takes values $u\mu_y$ with u a unit, and $u \neq -u$ unless $2 = 0$ in R . $\square$

21.5 Cohomology with compact supports

Lemma 21.5.1. *Let I be a directed set (Example 5.11.2). Colimits of diagrams $I \rightarrow \mathbf{Ab}$ exist, their underlying sets are the colimits of the underlying sets (Proposition 5.11.4), and they are exact: a colimit of exact sequences is exact. Hence they commute with homology of chain complexes. If $\phi: J \rightarrow I$ is an order-preserving map of directed sets with cofinal image (every $i \in I$ has some j with $i \leq \phi(j)$), then the induced map $\text{colim}_J A_{\phi(j)} \rightarrow \text{colim}_I A_i$ is an isomorphism.*

Proof. On the set $\text{colim}_i A_i = (\coprod_i A_i)/\sim$, define $[a_i] + [b_j] = [u(a_i) + v(b_j)]$ after mapping both to a common k ; this is well defined by Proposition 5.11.4 and makes the colimit an abelian group with the universal property in $\mathbf{Ab}$. If $A_i \xrightarrow{f_i} B_i \xrightarrow{g_i} C_i$ are exact and $[b_i]$ maps to 0, then $g_i(b_i)$ becomes 0 in some C_k ; replacing i by k we may assume $g_i(b_i) = 0$, so $b_i = f_i(a_i)$. So the colimit sequence is exact. Homology is kernel modulo image, hence preserved. For ϕ with cofinal image, every element of colim_I comes from some A_i and hence from $A_{\phi(j)}$ with $\phi(j) \geq i$. If an element of $A_{\phi(j)}$ vanishes in colim_I , it vanishes in some A_i with $i \geq \phi(j)$; choose j'' with $\phi(j'') \geq i$ and $j' \geq j, j''$; then it vanishes in $A_{\phi(j')}$, hence in colim_J . $\square$

Definition 21.5.2. The *cohomology with compact supports* of a Hausdorff space X is $H_c^k(X; R) = \text{colim}_K H^k(X | K; R)$, the colimit over the compact subsets $K \subseteq X$ directed by inclusion, along the restriction maps $H^k(X | K) \rightarrow H^k(X | L)$ for $K \subseteq L$. If X is compact, $H_c^k(X; R) = H^k(X; R)$. For an open $U \subseteq X$ and compact $K \subseteq U$, excision gives $H^k(U | K) \cong H^k(X | K)$; these isomorphisms induce a map $H_c^k(U; R) \rightarrow H_c^k(X; R)$ (extension by zero).

Example 21.5.3. The open balls centred at 0 are cofinal among compact subsets of $\mathbb{R}^n$, and for balls $B \subseteq B'$ the restriction $H^k(\mathbb{R}^n | B) \rightarrow H^k(\mathbb{R}^n | B')$ is an isomorphism, as in Lemma 21.2.2. So $H_c^k(\mathbb{R}^n; R) \cong R$ for $k = n$ and 0 otherwise, whereas $H^k(\mathbb{R}^n; R)$ is R in degree 0. Compactly supported cohomology is not a homotopy invariant.

Definition 21.5.4. Let M be an R -oriented n -manifold. For compact K let $\mu_K \in H_n(M | K; R)$ be the class of Lemma 21.4.2 restricting to the orientation. For $K \subseteq L$, μ_L restricts to μ_K , and naturality of the cap product $H_n(M | K) \times H^k(M | K) \rightarrow H_{n-k}(M)$ (Proposition 20.7.2) shows that the maps $\varphi \mapsto \mu_K \cap \varphi$ are compatible. They induce the *duality map* $D_M: H_c^k(M; R) \rightarrow H_{n-k}(M; R)$. If M is closed, $D_M(\varphi) = [M] \cap \varphi$.

21.6 Poincaré duality

Lemma 21.6.1. *Let $M = U \cup V$ with U, V open. There are long exact Mayer–Vietoris sequences*

$$\cdots \rightarrow H_c^k(U \cap V) \rightarrow H_c^k(U) \oplus H_c^k(V) \rightarrow H_c^k(M) \rightarrow H_c^{k+1}(U \cap V) \rightarrow \cdots$$

and $\cdots \rightarrow H_{n-k}(U \cap V) \rightarrow H_{n-k}(U) \oplus H_{n-k}(V) \rightarrow H_{n-k}(M) \rightarrow H_{n-k-1}(U \cap V) \rightarrow \cdots$, and the duality maps $D_{U \cap V}$, $D_U \oplus D_V$, D_M form a diagram between them that commutes up to sign.

Proof. For compact $K \subseteq U$ and $L \subseteq V$, Lemma 21.4.1 with $Y_1 = M \setminus K$, $Y_2 = M \setminus L$ and excision give an exact sequence $\cdots \rightarrow H^k(U \cap V | K \cap L) \rightarrow H^k(U | K) \oplus H^k(V | L) \rightarrow H^k(M | K \cup L) \rightarrow H^{k+1}(U \cap V | K \cap L) \rightarrow \cdots$, natural in the pair (K, L) . Take the colimit over such pairs (Lemma 21.5.1). The sets $K \cap L$ are cofinal among compact subsets of $U \cap V$ (take $K = L = C$), and the sets $K \cup L$ are cofinal among compact subsets of M : given compact $C \subseteq M$, the disjoint closed subsets $C \setminus V$ and $C \setminus U$ of the normal space C can be separated, giving an open $W \subseteq C$ with $C \setminus V \subseteq W \subseteq \bar{W} \subseteq U$, and then $K = \bar{W} \subseteq U$ and $L = C \setminus W \subseteq V$ are compact with $K \cup L = C$. This gives the first sequence; the second is Theorem 19.5.5. Commutativity of the squares not involving connecting maps is naturality of the cap product. Commutativity up to sign of the remaining square is [Hat02, Lemma 3.36]. $\square$

Theorem 21.6.2 (Poincaré duality). *For every R -oriented n -manifold M , the duality map $D_M: H_c^k(M; R) \rightarrow H_{n-k}(M; R)$ is an isomorphism for all k . In particular, if M is closed, $H^k(M; R) \rightarrow H_{n-k}(M; R)$, $\alpha \mapsto [M] \frown \alpha$, is an isomorphism.*

Proof. Open subsets of M inherit R -orientations, and the duality maps are compatible with the maps $H_c^k(U) \rightarrow H_c^k(M)$ and $H_{n-k}(U) \rightarrow H_{n-k}(M)$ by naturality of the cap product. We omit R .

(A) If D_U , D_V and $D_{U \cap V}$ are isomorphisms, so is $D_{U \cup V}$. By Lemma 21.6.1 and the five lemma (signs do not affect it).

(B) If $M = \bigcup_i U_i$ for a family of open sets directed by inclusion and all D_{U_i} are isomorphisms, so is D_M . Every compact subset of M lies in some U_i , so by cofinality $H_c^k(M) = \operatorname{colim}_i H_c^k(U_i)$. Every singular chain has compact image, so $H_{n-k}(M) = \operatorname{colim}_i H_{n-k}(U_i)$. The duality maps are compatible, so $D_M = \operatorname{colim}_i D_{U_i}$ is an isomorphism.

(C) $M = U \subseteq \mathbb{R}^n$ convex open. If $K \subseteq U$ is compact, cover it by finitely many closed balls $D_1, \dots, D_m \subseteq U$; their convex hull is the image of the compact set $\Delta^{m-1} \times D_1 \times \cdots \times D_m$ under $(t, x) \mapsto \sum_j t_j x_j$, hence a compact convex subset of U containing K . So compact convex sets C are cofinal, and for $x \in C$ the restriction $H^k(U | C) \rightarrow H^k(U | x)$ is an isomorphism (excise to $\mathbb{R}^n$ and argue as in Step 2 of the proof of Lemma 21.4.2). Hence $H_c^k(U)$ is 0 for $k \neq n$, and so is $H_{n-k}(U)$, as U is contractible. For $k = n$, $H^n(U | C) \cong \operatorname{Hom}(H_n(U | C; \mathbb{Z}), R)$ by universal coefficients, and the composite of D_U with the augmentation $H_0(U) \cong R$ sends φ to $\varphi(\mu_C)$ by Proposition 20.7.2(3) (with $\psi = 1$). Since μ_C is a unit times the image of a generator of $H_n(U | C; \mathbb{Z}) \cong \mathbb{Z}$, this is an isomorphism.

(D) $M = U \subseteq \mathbb{R}^n$ open. Finite unions of convex open sets satisfy duality by (C) and (A), by induction on the number of sets, since $(U_1 \cup \cdots \cup U_{m-1}) \cap U_m$ is a union of $m - 1$ convex open sets. The open set U is the union of countably many open balls $B_1, B_2, \dots$ (Proposition 16.3.10), hence the increasing union of the sets $B_1 \cup \cdots \cup B_m$; apply (B).

(E) General M . By Zorn's lemma and (B), there is an open $U \subseteq M$ maximal among open sets for which duality holds. If $U \neq M$, choose a chart $V \cong \mathbb{R}^n$ not contained in U . Duality holds for V and for $U \cap V$, which is an open subset of $\mathbb{R}^n$, by (D); so it holds for $U \cup V$ by (A), contradicting maximality. $\square$

Corollary 21.6.3. *Let M be a closed F -oriented n -manifold over a field F . The pairing $H^k(M; F) \times H^{n-k}(M; F) \rightarrow F$, $(\alpha, \beta) \mapsto \langle \alpha \smile \beta, [M] \rangle$, is nondegenerate: $\alpha \mapsto \langle \alpha \smile -, [M] \rangle$ is injective, and it is an isomorphism $H^k(M; F) \cong \operatorname{Hom}_F(H^{n-k}(M; F), F)$ when these spaces are finite-dimensional (which is always the case [Hat02, Appendix]).*

Proof. By Proposition 20.7.2(3), $\langle \alpha \smile \beta, [M] \rangle = \langle \beta, [M] \frown \alpha \rangle$. By the universal coefficient theorem over a field, $H^{n-k}(M; F) = \operatorname{Hom}_F(H_{n-k}(M; F), F)$, so $\langle \alpha \smile -, [M] \rangle$ is the functional

“evaluate on $[M] \frown \alpha$ ” on this dual space. Since $\alpha \mapsto [M] \frown \alpha$ is injective, and a nonzero vector is detected by the dual space, the map is injective; for finite dimensions it is bijective by counting dimensions. $\square$

Corollary 21.6.4. *Let M be a closed n -manifold whose integral homology is finitely generated, and define $\chi(M) = \sum_i (-1)^i \text{rank } H_i(M; \mathbb{Z})$. Then $\chi(M) = \sum_i (-1)^i \dim_F H_i(M; F)$ for every field F , and $\chi(M) = 0$ if n is odd.*

Proof. By the universal coefficient theorem (Theorem 19.8.2), $\dim_F H_i(M; F) = \text{rank } H_i(M) + t_i + t_{i-1}$, where t_i is the number of cyclic summands of $H_i(M)$ of order divisible by the characteristic of F (these contribute to $H_i \otimes F$ and to $\text{Tor}_1(H_i, F)$). The t terms cancel in the alternating sum. Take $F = \mathbb{F}_2$, for which M is oriented (Proposition 21.3.2). By duality and universal coefficients, $\dim H_i(M; \mathbb{F}_2) = \dim H^{n-i}(M; \mathbb{F}_2) = \dim H_{n-i}(M; \mathbb{F}_2)$, so $\chi(M) = (-1)^n \chi(M)$. $\square$

Definition 21.6.5. Let M be a closed oriented manifold of dimension $4m$. The *intersection form* is the symmetric bilinear form $Q(\alpha, \beta) = \langle \alpha \smile \beta, [M] \rangle$ on $H^{2m}(M; \mathbb{R})$; it is nondegenerate by Corollary 21.6.3, and symmetric by Theorem 20.3.4. Its *signature* $\sigma(M)$ is the number of positive minus the number of negative eigenvalues. On the free group $H^{2m}(M; \mathbb{Z})/\text{torsion}$ the integral form is unimodular (determinant ± 1) [Hat02, Proposition 3.38]. In dimension $4m + 2$ the form on H^{2m+1} is skew-symmetric and nondegenerate.

Example 21.6.6. 1. For $\mathbb{CP}^2$ with the orientation for which $\langle \alpha^2, [\mathbb{CP}^2] \rangle = 1$, $Q = (1)$ and $\sigma = 1$. For $S^2 \times S^2$, in the basis a, b of Example 20.4.3, $Q = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma = 0$. The connected sum $\mathbb{CP}^2 \# \overline{\mathbb{CP}^2}$ (with reversed orientation on the second summand) has the same Betti numbers as $S^2 \times S^2$, but its integral form is $\text{diag}(1, -1)$, which takes odd values, whereas the form of $S^2 \times S^2$ takes only even values on the diagonal; so the forms are not isomorphic over $\mathbb{Z}$ and the two manifolds are not homotopy equivalent.

2. (Cohomology of $\mathbb{CP}^n$ via duality.) The inclusion $\mathbb{CP}^{n-1} \rightarrow \mathbb{CP}^n$ is an isomorphism on H^i for $i \leq 2n - 2$ by cellular cohomology. By induction on n , α^{n-1} generates $H^{2n-2}(\mathbb{CP}^n)$. Since $\mathbb{CP}^n$ is a closed simply connected manifold, it is orientable, and by unimodularity of the pairing $H^2 \times H^{2n-2} \rightarrow H^{2n} \cong \mathbb{Z}$ between free groups of rank 1, $\alpha \smile \alpha^{n-1}$ generates H^{2n} . This proves the part of Theorem 20.5.1 about $\mathbb{CP}^n$; the argument for $\mathbb{RP}^n$ with $\mathbb{F}_2$ -coefficients is the same.

21.7 Other dualities

Theorem 21.7.1 (Lefschetz duality). *Let M be a compact R -orientable n -manifold with boundary. There is a fundamental class $[M] \in H_n(M, \partial M; R)$ restricting to local orientations, and cap product with it gives isomorphisms $H^k(M, \partial M; R) \cong H_{n-k}(M; R)$ and $H^k(M; R) \cong H_{n-k}(M, \partial M; R)$.*

This is [Hat02, Theorem 3.43]; the proof uses a collar neighbourhood of ∂M to reduce to Theorem 21.6.2 for the interior of M .

Theorem 21.7.2 (Alexander duality). *Let $K \subsetneq S^n$ be a compact, locally contractible, nonempty subspace. Then $\tilde{H}_i(S^n \setminus K; \mathbb{Z}) \cong \tilde{H}^{n-i-1}(K; \mathbb{Z})$ for all i .*

This is [Hat02, Theorem 3.44]. The idea: $\tilde{H}_i(S^n \setminus K) \cong H_c^{n-i}(S^n \setminus K)$ by duality, and $H_c^{n-i}(S^n \setminus K)$ is the colimit of $H^{n-i}(S^n, U)$ over open neighbourhoods U of K , which for locally contractible K is $H^{n-i}(S^n, K) \cong \tilde{H}^{n-i-1}(K)$.

Corollary 21.7.3 (Jordan–Brouwer separation theorem). *If $K \subseteq S^n$ is homeomorphic to S^{n-1} , then $S^n \setminus K$ has exactly two path components. If K is homeomorphic to D^k , then $S^n \setminus K$ is path-connected.*

Proof. By Theorem 21.7.2, $\tilde{H}_0(S^n \setminus K) \cong \tilde{H}^{n-1}(S^{n-1}) \cong \mathbb{Z}$, respectively $\tilde{H}^{n-1}(D^k) = 0$. Since $S^n \setminus K$ is an open subset of a manifold, its path components are its components, and $\tilde{H}_0$ is free of rank one less than their number (Proposition 19.2.4). $\square$

21.8 Borel–Moore homology and fundamental classes

Remark 21.8.1. For a locally compact space X , *Borel–Moore homology* $H_i^{BM}(X; R)$ is the homology of locally finite, possibly infinite, singular chains; for reasonable X (for instance a closed subspace of a manifold that is locally contractible, or a CW pair) it can be computed as $H_i(X^+, \{\infty\}; R)$, where X^+ is the one-point compactification. It is covariant for proper maps, contravariant for open inclusions, and a manifold M oriented over R has a fundamental class $[M] \in H_n^{BM}(M; R)$ even when it is not compact, with $H^k(M; R) \cong H_{n-k}^{BM}(M; R)$. If Z is a closed subset of an R -oriented n -manifold M , then

$$H_i^{BM}(Z; R) \cong H^{n-i}(M, M \setminus Z; R),$$

which for $Z = M$ is the duality above. See [Bre97, Chapter V], [Ive86] and [Ful98, Chapter 19].

Remark 21.8.2. These isomorphisms produce cohomology classes from subspaces. If $N \subseteq M$ is a closed oriented submanifold of dimension d (or, more generally, a closed subspace of “dimension” d with a fundamental class $[N] \in H_d^{BM}(N; R)$), its image under $H_d^{BM}(N) \cong H^{n-d}(M, M \setminus N) \rightarrow H^{n-d}(M)$ is the *cohomology class* of N . When N and M are compact it is the Poincaré dual of $i_*[N] \in H_d(M)$. A complex subvariety Z of complex dimension d of a complex manifold of dimension m has a fundamental class in $H_{2d}^{BM}(Z; \mathbb{Z})$ because its singular locus has real codimension at least 2; this gives the *cycle class* $[Z] \in H^{2m-2d}(X; \mathbb{Z})$ of Chapter 59, and the Hodge conjecture asks which rational classes are combinations of such $[Z]$. Under these identifications, cup product corresponds to intersection of cycles meeting transversally.

21.9 Exercises

Exercise 21.9.1. Let M be a closed connected n -manifold with finitely generated homology. Show that the torsion subgroup of $H_{n-1}(M; \mathbb{Z})$ is 0 if M is orientable and $\mathbb{Z}/2\mathbb{Z}$ otherwise.

Solution. By universal coefficients, $H_n(M; \mathbb{Z}/m) \cong H_n(M) \otimes \mathbb{Z}/m \oplus \text{Tor}_1(H_{n-1}(M), \mathbb{Z}/m)$, and $\text{Tor}_1(\mathbb{Z}/a, \mathbb{Z}/m) \cong \mathbb{Z}/\gcd(a, m)$. If M is orientable, $H_n(M) = \mathbb{Z}$ and $H_n(M; \mathbb{Z}/m) \cong \mathbb{Z}/m$ (Theorem 21.4.3), so all $\text{Tor}_1(H_{n-1}, \mathbb{Z}/m)$ vanish and H_{n-1} is torsion-free. Otherwise $H_n(M) = 0$ and $H_n(M; \mathbb{Z}/m) \cong \{r : 2r = 0\}$: for m an odd prime this is 0, so there is no odd torsion; for $m = 2$ it is $\mathbb{Z}/2$, so there is exactly one cyclic summand of even order; for $m = 4$ it is $\mathbb{Z}/2$, which forces that summand to be $\mathbb{Z}/2$. $\square$

Exercise 21.9.2. Show that $\mathbb{RP}^n$ is orientable if and only if n is odd.

Solution. By Example 19.7.5, $H_n(\mathbb{RP}^n; \mathbb{Z})$ is $\mathbb{Z}$ for n odd and 0 for n even; apply Theorem 21.4.3. $\square$

Exercise 21.9.3. Show that a closed simply connected 3-manifold has the homology of S^3 .

Solution. It is orientable (Proposition 21.3.2), so $H_3 \cong \mathbb{Z}$ and $H_i = 0$ for $i > 3$. $H_1 = \pi_1^{\text{ab}} = 0$. By duality $H_2 \cong H^1 \cong \text{Hom}(H_1, \mathbb{Z}) \oplus \text{Ext}^1(H_0, \mathbb{Z}) = 0$. $\square$

Exercise 21.9.4. Show that $\sigma(M) = 0$ if M admits an orientation-reversing self-homeomorphism, and compute the intersection form of T^4 .

Exercise 21.9.5. Let $K \subseteq S^3$ be homeomorphic to S^1 . Show that $H_1(S^3 \setminus K; \mathbb{Z}) \cong \mathbb{Z}$ and $H_2(S^3 \setminus K; \mathbb{Z}) = 0$.

Chapter 22

Cohomology Operations

22.1 Introduction

A cohomology operation is a natural transformation between cohomology functors, such as $x \mapsto x^2$. Because cohomology is represented by Eilenberg–MacLane spaces, operations are the same as cohomology classes of these spaces (Proposition 22.3.2). The Steenrod squares are the most important operations with $\mathbb{F}_2$ -coefficients; they refine the cup product and give obstructions that cup products cannot see, such as the theorem that a map of Hopf invariant one $S^{2n-1} \rightarrow S^n$ exists only if n is a power of 2 (Corollary 22.4.4). For the Hodge conjecture, operations show that some integral cohomology classes of smooth projective varieties cannot be classes of algebraic cycles (Remark 22.6.2). This chapter is partly a survey: the existence theorems are quoted from [Hat02], [SE62] and [MT68], and what follows from them is proved.

Prerequisites. Chapter 21 (Topological Manifolds and Poincaré Duality) and Chapter 17 (Homotopy and Covering Spaces).

22.2 Eilenberg–MacLane spaces

Definition 22.2.1. Let G be a group and $n \geq 1$, with G abelian if $n \geq 2$. An *Eilenberg–MacLane space* $K(G, n)$ is a connected CW complex with $\pi_n(K(G, n)) \cong G$ and $\pi_i(K(G, n)) = 0$ for $i \neq n$.

Theorem 22.2.2. For every $n \geq 1$ and every group G (abelian if $n \geq 2$) there is an Eilenberg–MacLane space $K(G, n)$, which can be chosen with a single 0-cell and no other cells of dimension less than n ; any two are homotopy equivalent.

For the proof, which attaches cells to a wedge of n -spheres to realise a presentation of G and then kills the higher homotopy groups, see [Hat02, Section 4.2].

Example 22.2.3. 1. S^1 is a $K(\mathbb{Z}, 1)$, since its universal covering $\mathbb{R}$ is contractible and $\pi_i(S^1) \cong \pi_i(\mathbb{R}) = 0$ for $i \geq 2$ (Proposition 17.8.3). More generally, a connected CW complex with contractible universal covering is a $K(\pi_1, 1)$; this applies to the tori T^n and to the surfaces Σ_g with $g \geq 1$.

2. $\mathbb{RP}^\infty = \bigcup_n \mathbb{RP}^n$ is a $K(\mathbb{Z}/2\mathbb{Z}, 1)$ and $\mathbb{CP}^\infty$ is a $K(\mathbb{Z}, 2)$: they are the quotients of the contractible space $S^\infty = \bigcup_n S^n$ by $\{\pm 1\}$ and by the circle group, and the long exact sequence of the fibration $S^1 \rightarrow S^\infty \rightarrow \mathbb{CP}^\infty$ gives $\pi_2(\mathbb{CP}^\infty) \cong \pi_1(S^1)$ [Hat02, Section 4.2].

Theorem 22.2.4 (Representability). *Let G be abelian and $n \geq 1$. There is a class $\iota_n \in H^n(K(G, n); G)$ such that for every CW complex X the map $\langle X, K(G, n) \rangle \rightarrow H^n(X; G)$, $[f] \mapsto f^*(\iota_n)$, is a bijection, natural in X .*

This is [Hat02, Theorem 4.57]. The class ι_n corresponds under $H^n(K(G, n); G) \cong \text{Hom}(H_n(K(G, n)), G) \cong \text{Hom}(\pi_n(K(G, n)), G) = \text{Hom}(G, G)$ (universal coefficients and the Hurewicz theorem) to the identity.

22.3 Cohomology operations

Definition 22.3.1. A cohomology operation of type $(G, m; H, n)$ is a family of maps of sets $\theta_X: H^m(X; G) \rightarrow H^n(X; H)$, one for each space X , such that $f^*\theta_Y = \theta_X f^*$ for every map $f: X \rightarrow Y$. It is *additive* if every θ_X is a homomorphism.

Proposition 22.3.2. *Restricted to pointed CW complexes, operations of type $(G, m; H, n)$ correspond bijectively to elements of $H^n(K(G, m); H)$, via $\theta \mapsto \theta(\iota_m)$.*

Proof. On the category whose objects are pointed CW complexes and whose morphisms are pointed homotopy classes of maps, $H^m(-; G)$ is represented by $(K(G, m), \iota_m)$, by Theorem 22.2.4 and Definition 5.6.6, and $H^n(-; H)$ is a functor to **Set** by homotopy invariance. Operations are natural transformations between them, and the Yoneda lemma (Theorem 5.6.1) identifies these with $H^n(K(G, m); H)$. Explicitly, $u \in H^n(K(G, m); H)$ gives $\theta(x) = f_x^*(u)$, where $f_x: X \rightarrow K(G, m)$ is the map with $f_x^*(\iota_m) = x$. $\square$

Example 22.3.3. 1. For a commutative ring R , $x \mapsto x^k$ is an operation of type $(R, m; R, km)$, corresponding to ι_m^k ; it is not additive in general.
 2. A homomorphism $G \rightarrow H$ induces an additive operation of type $(G, m; H, m)$, for example the reduction $\rho: H^m(X; \mathbb{Z}) \rightarrow H^m(X; \mathbb{Z}/p\mathbb{Z})$.
 3. Since $H^n(K(G, m); H) = 0$ for $0 < n < m$ ($K(G, m)$ may be chosen with no cells in dimensions $0 < i < m$), there are no nonzero operations lowering degree, except constant ones for $n = 0$.

Proposition 22.3.4 (Bockstein homomorphisms). *Let p be a prime. The short exact sequences $0 \rightarrow \mathbb{Z} \xrightarrow{p} \mathbb{Z} \rightarrow \mathbb{Z}/p \rightarrow 0$ and $0 \rightarrow \mathbb{Z}/p \rightarrow \mathbb{Z}/p^2 \rightarrow \mathbb{Z}/p \rightarrow 0$ induce connecting homomorphisms $\tilde{\beta}: H^n(X; \mathbb{Z}/p) \rightarrow H^{n+1}(X; \mathbb{Z})$ and $\beta: H^n(X; \mathbb{Z}/p) \rightarrow H^{n+1}(X; \mathbb{Z}/p)$, the Bockstein operations. They satisfy $\beta = \rho\tilde{\beta}$, $\beta\beta = 0$, the image of $\tilde{\beta}$ is the p -torsion subgroup $\{y : py = 0\}$ of $H^{n+1}(X; \mathbb{Z})$, and the kernel of $\tilde{\beta}$ is the image of the reduction $\rho: H^n(X; \mathbb{Z}) \rightarrow H^n(X; \mathbb{Z}/p)$.*

Proof. Since each $C_n(X)$ is free, $\text{Hom}(C_n(X), -)$ is exact, so a short exact sequence of coefficient groups gives a short exact sequence of cochain complexes and a long exact sequence (Theorem 13.4.3): $H^n(X; \mathbb{Z}) \xrightarrow{p} H^n(X; \mathbb{Z}) \xrightarrow{\rho} H^n(X; \mathbb{Z}/p) \xrightarrow{\tilde{\beta}} H^{n+1}(X; \mathbb{Z}) \xrightarrow{p} H^{n+1}(X; \mathbb{Z})$. Exactness gives the descriptions of the image and the kernel of $\tilde{\beta}$. The reduction $\mathbb{Z} \rightarrow \mathbb{Z}/p^2$ gives a morphism from the first short exact sequence to the second, and naturality of connecting homomorphisms gives $\beta = \rho\tilde{\beta}$. Finally $\beta\beta = \rho\tilde{\beta}\rho\tilde{\beta} = 0$ because $\tilde{\beta}\rho = 0$. $\square$

22.4 Steenrod squares

Theorem 22.4.1 (Steenrod). *There are unique additive operations $\text{Sq}^i: H^n(X; \mathbb{F}_2) \rightarrow H^{n+i}(X; \mathbb{F}_2)$, $i \geq 0$, defined for all spaces and pairs and natural, such that:*

1. $Sq^0 = \text{id}$;
2. $Sq^n(x) = x^2$ if $|x| = n$, and $Sq^i(x) = 0$ if $i > |x|$;
3. (Cartan formula) $Sq^i(xy) = \sum_{j+k=i} Sq^j(x)Sq^k(y)$;
4. they commute with the suspension isomorphisms $\tilde{H}^n(X) \cong \tilde{H}^{n+1}(SX)$ and with the connecting homomorphisms of pairs.

Moreover Sq^1 is the Bockstein β for $p = 2$, and for $a < 2b$ the Adem relations

$$Sq^a Sq^b = \sum_{c=0}^{\lfloor a/2 \rfloor} \binom{b-c-1}{a-2c} Sq^{a+b-c} Sq^c$$

hold, with binomial coefficients taken modulo 2.

See [SE62] and [Hat02, Section 4.L]. Writing $Sq = \sum_i Sq^i$ (a finite sum on each class), the Cartan formula says that $Sq: H^*(X; \mathbb{F}_2) \rightarrow H^*(X; \mathbb{F}_2)$ is a ring homomorphism. There are analogous *reduced power operations* $P^i: H^n(X; \mathbb{F}_p) \rightarrow H^{n+2i(p-1)}(X; \mathbb{F}_p)$ for odd primes p , with $P^i(x) = x^p$ for $|x| = 2i$.

Proposition 22.4.2. *In $H^*(\mathbb{RP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[\alpha]$, $Sq^i(\alpha^n) = \binom{n}{i} \alpha^{n+i}$. In $H^*(\mathbb{CP}^\infty; \mathbb{F}_2) = \mathbb{F}_2[x]$ with $|x| = 2$, $Sq^{2i}(x^n) = \binom{n}{i} x^{n+i}$ and $Sq^{2i+1} = 0$.*

Proof. By axioms (1) and (2), $Sq(\alpha) = \alpha + \alpha^2$. Since Sq is a ring homomorphism, $Sq(\alpha^n) = (\alpha + \alpha^2)^n = \alpha^n(1 + \alpha)^n = \sum_i \binom{n}{i} \alpha^{n+i}$; compare degrees. For $\mathbb{CP}^\infty$, $Sq^1(x) \in H^3 = 0$, so $Sq(x) = x + x^2$ and the same computation applies, all odd squares vanishing for degree reasons. The rings are those of Theorem 20.5.1. $\square$

Lemma 22.4.3. *If $i \geq 1$ is not a power of 2, then Sq^i is decomposable: it is a sum of composites $Sq^a Sq^b$ with $0 < a, b < i$.*

Proof. Choose k with $2^k < i < 2^{k+1}$ and let $a = i - 2^k$, so $0 < a < 2^k$ and $a < 2 \cdot 2^k$. In $\mathbb{F}_2[t]$ we have $(1+t)^{2^k} = 1 + t^{2^k}$ (the Frobenius map is a ring homomorphism), hence $(1+t)^{2^k-1} = (1+t^{2^k})/(1+t) = 1 + t + \cdots + t^{2^k-1}$, so $\binom{2^k-1}{a}$ is odd. The Adem relation for $Sq^a Sq^{2^k}$ has $c = 0$ term $\binom{2^k-1}{a} Sq^i = Sq^i$, so $Sq^i = Sq^a Sq^{2^k} + \sum_{c \geq 1} \binom{2^k-c-1}{a-2c} Sq^{i-c} Sq^c$, and all indices lie strictly between 0 and i . $\square$

Corollary 22.4.4 (Hopf invariant one). *Let $n \geq 2$ and $f: S^{2n-1} \rightarrow S^n$, and let $X_f = S^n \cup_f D^{2n}$, with $H^n(X_f; \mathbb{Z}) = \mathbb{Z}a$ and $H^{2n}(X_f; \mathbb{Z}) = \mathbb{Z}b$. Define the Hopf invariant $h(f)$ by $a^2 = h(f)b$. If $h(f)$ is odd, then n is a power of 2.*

Proof. X_f has cells in dimensions 0, n , $2n$, so by cellular cohomology its integral cohomology is free of rank one in these degrees and zero elsewhere, and the reductions $\bar{a}, \bar{b}$ generate the $\mathbb{F}_2$ -cohomology. Reduction is a ring homomorphism, so $Sq^n(\bar{a}) = \bar{a}^2 = h(f)\bar{b} \neq 0$. If n were not a power of 2, Lemma 22.4.3 would write $Sq^n(\bar{a})$ as a sum of terms $Sq^a Sq^b(\bar{a})$ with $0 < b < n$, and $Sq^b(\bar{a}) \in H^{n+b}(X_f; \mathbb{F}_2) = 0$. $\square$

Remark 22.4.5. The Hopf maps $S^3 \rightarrow S^2$, $S^7 \rightarrow S^4$ and $S^{15} \rightarrow S^8$ have Hopf invariant one. Adams proved, using secondary cohomology operations, that maps of Hopf invariant one exist only for $n = 1, 2, 4, 8$ [Ada60]. Consequences include the facts that $\mathbb{R}^n$ admits a bilinear multiplication without zero divisors only for $n = 1, 2, 4, 8$, and that S^{n-1} is an H -space only for $n = 1, 2, 4, 8$.

22.5 Cohomology of Eilenberg–MacLane spaces

Definition 22.5.1. A sequence $I = (i_1, \dots, i_r)$ of positive integers is *admissible* if $i_j \geq 2i_{j+1}$ for all j . Its *degree* is $\sum_j i_j$, its *excess* is $e(I) = \sum_j (i_j - 2i_{j+1}) = i_1 - i_2 - \dots - i_r$ (with $i_{r+1} = 0$), and $Sq^I = Sq^{i_1} \dots Sq^{i_r}$. The empty sequence is admissible with excess 0.

Theorem 22.5.2 (Serre). 1. The Steenrod algebra $\mathcal{A}_2$, the algebra of operations $H^*(-; \mathbb{F}_2) \rightarrow H^*(-; \mathbb{F}_2)$ generated by the Sq^i modulo the Adem relations, has the admissible monomials Sq^I as an $\mathbb{F}_2$ -basis.

2. $H^*(K(\mathbb{Z}/2, n); \mathbb{F}_2)$ is the polynomial algebra on the classes $Sq^I \iota_n$ with I admissible and $e(I) < n$.

3. $H^*(K(\mathbb{Z}, n); \mathbb{F}_2)$ is the polynomial algebra on the classes $Sq^I \iota_n$ with I admissible, $e(I) < n$ and $i_r \neq 1$.

This is [Ser53]; see also [MT68].

Example 22.5.3. For $n = 3$, the admissible sequences of excess less than 3 not ending in 1 are $\emptyset, (2), (4, 2), (8, 4, 2), \dots$. So $H^*(K(\mathbb{Z}, 3); \mathbb{F}_2) = \mathbb{F}_2[\iota_3, Sq^2 \iota_3, Sq^4 Sq^2 \iota_3, \dots]$ with generators in degrees 3, 5, 9, 17, $\dots$. In low degrees: $H^3 = \langle \iota_3 \rangle$, $H^4 = 0$, $H^5 = \langle Sq^2 \iota_3 \rangle$, $H^6 = \langle \iota_3^2 \rangle$, where $\iota_3^2 = Sq^3 \iota_3 = Sq^1 Sq^2 \iota_3$ by the Adem relation $Sq^1 Sq^2 = Sq^3$. Since $Sq^1 = \rho\tilde{\beta}$ is nonzero on $Sq^2 \iota_3$, the integral class $\tilde{\beta} Sq^2 \iota_3 \in H^6(K(\mathbb{Z}, 3); \mathbb{Z})$ is a nonzero 2-torsion class. In general $x \mapsto \tilde{\beta} Sq^2 \rho(x)$ is an integral operation raising degree by 3, defined in every degree, whose reduction modulo 2 is $\rho\tilde{\beta} Sq^2 \rho(x) = Sq^1 Sq^2 \rho(x) = Sq^3 \rho(x)$; it appears in Remark 22.6.2.

22.6 Complex cobordism and algebraic cycles

Remark 22.6.1. Complex cobordism MU^* is a generalised cohomology theory: it satisfies all Eilenberg–Steenrod axioms except the dimension axiom. Its coefficient ring is $MU^* = MU^*(pt) \cong \mathbb{Z}[x_1, x_2, \dots]$ with x_i in degree $-2i$, generated geometrically by classes of complex projective spaces and hypersurfaces [Sto68], and Quillen showed that it carries the universal one-dimensional commutative formal group law [Qui69]. For a compact complex manifold X there is a natural map $MU^*(X) \rightarrow H^*(X; \mathbb{Z})$, and a closed complex submanifold $Z \subseteq X$ of codimension c defines a class in $MU^{2c}(X)$ whose image is the cohomology class of Z (Chapter 21). See [Rav86] for the homotopy-theoretic side and [Tho54] for the origins of cobordism.

Remark 22.6.2. Atiyah and Hirzebruch showed that on a smooth projective complex variety X the integral cohomology class of every algebraic cycle lies in the image of $MU^*(X) \rightarrow H^*(X; \mathbb{Z})$, using resolution of singularities, and that cohomology operations of odd degree vanish on this image; in particular $Sq^3 \rho(x) = 0$ for the class x of a codimension-2 cycle. They constructed smooth projective varieties with torsion classes $x \in H^4(X; \mathbb{Z})$ with $Sq^3 \rho(x) \neq 0$. Such a class is a Hodge class (torsion classes are Hodge classes trivially), but it is not the class of an algebraic cycle, which disproves the Hodge conjecture with integral coefficients [AH62]. Totaro refined this by showing that the cycle class map factors as $CH^*(X) \rightarrow MU^*(X) \otimes_{MU^*} \mathbb{Z} \rightarrow H^*(X; \mathbb{Z})$ [Tot97]. These results are developed in Chapter 69.

22.7 Exercises

Exercise 22.7.1. Deduce from the Adem relations that $Sq^1 Sq^1 = 0$, $Sq^1 Sq^2 = Sq^3$ and $Sq^2 Sq^2 = Sq^3 Sq^1$.

Solution. For $(a, b) = (1, 1)$ the only term is $c = 0$ with coefficient $\binom{0}{1} = 0$. For $(1, 2)$: $\binom{1}{1}\text{Sq}^3\text{Sq}^0 = \text{Sq}^3$. For $(2, 2)$: $c = 0$ gives $\binom{1}{2} = 0$ and $c = 1$ gives $\binom{0}{0}\text{Sq}^3\text{Sq}^1$. $\square$

Exercise 22.7.2. Show that there is no space X with $H^*(X; \mathbb{F}_2) \cong \mathbb{F}_2[x]$ and $|x| = 3$.

Solution. We would have $x^2 = \text{Sq}^3x = \text{Sq}^1\text{Sq}^2x$, and $\text{Sq}^2x \in H^5(X; \mathbb{F}_2) = 0$, so $x^2 = 0$, contradicting that H^* is a polynomial ring. (More generally, the argument of Corollary 22.4.4 shows that $|x|$ must be a power of 2, and by Adams' theorem (Remark 22.4.5) only $|x| = 1, 2, 4, 8$ can occur.) $\square$

Exercise 22.7.3. Show that $\beta(\alpha^n) = n\alpha^{n+1}$ in $H^*(\mathbb{RP}^\infty; \mathbb{F}_2)$, directly from the cellular cochain complex, and compare with Proposition 22.4.2.

Exercise 22.7.4. Show that the attaching map $S^3 \rightarrow S^2$ of the 4-cell of $\mathbb{CP}^2$ is not null-homotopic, using Sq^2 or the cup product, and deduce that $\pi_3(S^2) \neq 0$.

Solution. If it were null-homotopic, $\mathbb{CP}^2$ would be homotopy equivalent to $S^2 \vee S^4$ (attaching along homotopic maps gives homotopy equivalent spaces [Hat02, Proposition 0.18]), whose cup squares in degree 2 vanish; but $\text{Sq}^2(x) = x^2 \neq 0$ in $H^4(\mathbb{CP}^2; \mathbb{F}_2)$ by Theorem 20.5.1. $\square$

Part IV

Sheaves

Chapter 23

Sheaves on Topological Spaces

23.1 Introduction

A sheaf on a space records data that are defined locally and glue uniquely: continuous, smooth or holomorphic functions, sections of a bundle, solutions of a differential equation. Sheaves are the language in which complex and algebraic geometry and Hodge theory are written: the cohomology of sheaves (Chapter 24) unifies singular, de Rham and Dolbeault cohomology, and the comparison between them is the starting point of Hodge theory. This chapter develops the basic theory: sheaves and stalks, sheafification, the abelian category of sheaves, direct and inverse images, local systems, and sheaves of modules. References are [God58], [Bre97], [KS90], [Har77, Chapter II] and [The].

Prerequisites. Chapter 16 (General Topology), Chapter 5 (Categories), Chapter 13 (Homological Algebra) and Chapter 17 (Homotopy and Covering Spaces).

Throughout, X is a topological space and $\text{Open}(X)$ is the set of open subsets of X , ordered by inclusion and viewed as a category.

23.2 Presheaves and sheaves

Definition 23.2.1. A *presheaf* $\mathcal{F}$ on X with values in a category $\mathcal{C}$ (sets, abelian groups, rings, modules) is a functor $\mathcal{F} : \text{Open}(X)^{\text{op}} \rightarrow \mathcal{C}$: an object $\mathcal{F}(U)$, the *sections* over U , for each open U , and *restriction maps* $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$, $s \mapsto s|_V$, for $V \subseteq U$, with $s|_U = s$ and $(s|_V)|_W = s|_W$. We also write $\Gamma(U, \mathcal{F}) = \mathcal{F}(U)$. A *morphism* of presheaves is a natural transformation. A presheaf of sets (or abelian groups, ...) is a *sheaf* if for every open cover $U = \bigcup_i U_i$:

1. (Locality) if $s, t \in \mathcal{F}(U)$ satisfy $s|_{U_i} = t|_{U_i}$ for all i , then $s = t$;
2. (Gluing) if $s_i \in \mathcal{F}(U_i)$ satisfy $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j , there is $s \in \mathcal{F}(U)$ with $s|_{U_i} = s_i$ for all i .

Sheaves form full subcategories $\text{Sh}(X) \subseteq \text{PSh}(X)$ and $\text{Ab}(X) \subseteq \text{PAb}(X)$. Applied to the empty cover of $\emptyset$, the axioms say that $\mathcal{F}(\emptyset)$ is a single point (the zero group).

Example 23.2.2. 1. For a space Y , $U \mapsto \{\text{continuous maps } U \rightarrow Y\}$ is a sheaf of sets, because continuity is local (pasting lemma, Lemma 16.4.4). For $Y = \mathbb{R}$ or $\mathbb{C}$ it is a sheaf of rings C_X . Likewise smooth functions on a manifold and holomorphic functions on an open subset of $\mathbb{C}$ form sheaves of rings.

2. For a continuous map $p : E \rightarrow X$, the continuous sections $s : U \rightarrow E$ with $ps = \text{id}_U$ form a sheaf of sets, the *sheaf of sections* of p .

3. Let A be a set with at least two elements. The *constant presheaf* $U \mapsto A$ (with $\emptyset \mapsto$ a point) is not a sheaf if X has two disjoint nonempty open subsets U_1, U_2 : sections $a \neq b$ on them do not glue to a section on $U_1 \sqcup U_2$. The *constant sheaf* A_X is $U \mapsto \{\text{locally constant maps } U \rightarrow A\}$.
4. For $x \in X$ and a set (group) A , the *skyscraper sheaf* x_*A has $x_*A(U) = A$ if $x \in U$ and a point otherwise.
5. The presheaf of bounded continuous functions on $\mathbb{R}$ satisfies locality but not gluing.

Definition 23.2.3. The *stalk* of a presheaf $\mathcal{F}$ at x is $\mathcal{F}_x = \text{colim}_{U \ni x} \mathcal{F}(U)$, the colimit over the open neighbourhoods of x directed by reverse inclusion (Example 5.11.2). By Proposition 5.11.4, its elements are *germs* s_x of sections $s \in \mathcal{F}(U)$, $x \in U$, where $s_x = t_x$ for $t \in \mathcal{F}(V)$ if and only if $s|_W = t|_W$ for some open $W \ni x$ in $U \cap V$. For presheaves of abelian groups the stalk is an abelian group (Lemma 21.5.1). A morphism φ induces maps $\varphi_x: \mathcal{F}_x \rightarrow \mathcal{G}_x$.

Proposition 23.2.4. Let $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ be a morphism of sheaves.

1. Two sections $s, t \in \mathcal{F}(U)$ with $s_x = t_x$ for all $x \in U$ are equal.
2. φ_U is injective for all U if and only if φ_x is injective for all x .
3. φ is an isomorphism if and only if φ_x is bijective for all x .

Proof. (1) Each $x \in U$ has an open neighbourhood $W_x \subseteq U$ with $s|_{W_x} = t|_{W_x}$; apply locality to the cover (W_x) . (2) If all φ_x are injective and $\varphi(s) = \varphi(t)$, then $\varphi(s)_x = \varphi(t)_x$ gives $s_x = t_x$ for all x , so $s = t$ by (1). Conversely, if $\varphi_x(s_x) = \varphi_x(t_x)$ with $s, t \in \mathcal{F}(U)$, then $\varphi(s)|_W = \varphi(t)|_W$ for some $W \ni x$, so $s|_W = t|_W$ and $s_x = t_x$. (3) An isomorphism induces bijections on stalks. Conversely, by (2) each φ_U is injective. Let $t \in \mathcal{G}(U)$. For each $x \in U$ choose $s^{(x)} \in \mathcal{F}(W_x)$ with $\varphi(s^{(x)})_x = t_x$; shrinking W_x , $\varphi(s^{(x)}) = t|_{W_x}$. On $W_x \cap W_y$, $\varphi(s^{(x)}) = \varphi(s^{(y)})$, so $s^{(x)} = s^{(y)}$ there by injectivity. By gluing there is s with $s|_{W_x} = s^{(x)}$, and $\varphi(s) = t$ by locality. $\square$

Proposition 23.2.5 (Gluing sheaves). Let $(U_i)_{i \in I}$ be an open cover of X , let $\mathcal{F}_i$ be a sheaf on U_i , and write $U_{ij} = U_i \cap U_j$, $U_{ijk} = U_i \cap U_j \cap U_k$. Let $\varphi_{ij}: \mathcal{F}_j|_{U_{ij}} \rightarrow \mathcal{F}_i|_{U_{ij}}$ be isomorphisms with $\varphi_{ii} = \text{id}$ and $\varphi_{ij} \circ \varphi_{jk} = \varphi_{ik}$ on U_{ijk} (the cocycle condition). Then there is a sheaf $\mathcal{F}$ on X with isomorphisms $\psi_i: \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$ such that $\psi_i = \varphi_{ij} \circ \psi_j$ on U_{ij} , and $(\mathcal{F}, (\psi_i))$ is unique up to unique isomorphism.

Proof. For U open let $\mathcal{F}(U)$ be the set of families $(s_i)_i$ with $s_i \in \mathcal{F}_i(U \cap U_i)$ and $s_i = \varphi_{ij}(s_j)$ on $U \cap U_{ij}$ for all i, j , with componentwise restriction. The sheaf axioms hold componentwise, since each $\mathcal{F}_i$ is a sheaf and the compatibility conditions can be checked locally (by locality of the $\mathcal{F}_i$). For $V \subseteq U_i$ define $\psi_i((s_j)_j) = s_i$. Its inverse sends $t \in \mathcal{F}_i(V)$ to the family $s_j = \varphi_{ji}(t|_{V \cap U_j})$; this family is compatible because $\varphi_{jk}(s_k) = \varphi_{jk}\varphi_{ki}(t) = \varphi_{ji}(t) = s_j$, and $s_i = t$ because $\varphi_{ii} = \text{id}$. Conversely a compatible family is determined by its i -th member, as $s_j = \varphi_{ji}(s_i)$; so ψ_i is an isomorphism, and $\psi_i = \varphi_{ij}\psi_j$ is the compatibility condition. If $(\mathcal{G}, (\psi'_i))$ is another solution, the isomorphisms $\psi_i^{-1}\psi'_i: \mathcal{G}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$ agree on the overlaps, since $\psi_i^{-1}\psi'_i = \psi_j^{-1}\varphi_{ij}^{-1}\varphi_{ij}\psi'_j = \psi_j^{-1}\psi'_j$; morphisms of sheaves that agree on overlaps glue uniquely, by the sheaf axioms of the target applied to the images of sections, so they define the unique isomorphism $\mathcal{G} \rightarrow \mathcal{F}$ compatible with the ψ 's. $\square$

Example 23.2.6. Vector bundles are the basic instance: on a cover trivialising a bundle, the sheaves of sections $\mathcal{F}_i \cong C^r_{U_i}$ are glued by the transition functions, and the cocycle condition of Proposition 23.2.5 is the cocycle condition for transition functions. Local systems (Section 23.5) are glued in the same way from constant sheaves, with locally constant transition functions.

23.3 Sheafification

Theorem 23.3.1. *Let $\mathcal{F}$ be a presheaf of sets (abelian groups). For U open let $\mathcal{F}^\#(U)$ be the set of functions s assigning to each $x \in U$ a germ $s(x) \in \mathcal{F}_x$ such that every $x \in U$ has an open neighbourhood $W \subseteq U$ and $t \in \mathcal{F}(W)$ with $s(y) = t_y$ for all $y \in W$. With restriction of functions, $\mathcal{F}^\#$ is a sheaf, the sheafification of $\mathcal{F}$, and $\theta: \mathcal{F} \rightarrow \mathcal{F}^\#, t \mapsto (y \mapsto t_y)$, has the following properties.*

1. $\theta_x: \mathcal{F}_x \rightarrow \mathcal{F}_x^\#$ is bijective for all x .
2. Every morphism $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ to a sheaf factors uniquely as $\varphi = \psi \circ \theta$. So sheafification is left adjoint to the inclusion of sheaves into presheaves.
3. If $\mathcal{F}$ is a sheaf, θ is an isomorphism.

Proof. The conditions defining $\mathcal{F}^\#(U)$ are local, so $\mathcal{F}^\#$ is a sheaf; for abelian groups, the pointwise operations preserve the condition.

(1) Surjectivity: an element of $\mathcal{F}_x^\#$ is the germ of some $s \in \mathcal{F}^\#(U)$, which near x equals $\theta(t)$, so the germ is $\theta_x(t_x)$. Injectivity: if $\theta(t)_x = \theta(t')_x$, then in particular $t_x = \theta(t)(x) = \theta(t')(x) = t'_x$.

(2) Let $s \in \mathcal{F}^\#(U)$, and choose an open cover (W_i) of U and $t_i \in \mathcal{F}(W_i)$ with $s|_{W_i} = \theta(t_i)$. On $W_i \cap W_j$ the germs of t_i and t_j agree, hence so do the germs of $\varphi(t_i)$ and $\varphi(t_j)$, and by Proposition 23.2.4(1) $\varphi(t_i) = \varphi(t_j)$ there. Let $\psi(s)$ be the section of $\mathcal{G}$ obtained by gluing. It does not depend on the choices (two choices agree on a common refinement), and ψ is a morphism with $\psi\theta = \varphi$. Uniqueness: any ψ' with $\psi'\theta = \varphi$ agrees with ψ on each $\theta(t_i)$, hence on s by locality.

(3) By (1) and Proposition 23.2.4(3), applied to the morphism θ of sheaves. $\square$

Proposition 23.3.2. *$\text{Ab}(X)$ is an abelian category. The kernel of $\varphi: \mathcal{F} \rightarrow \mathcal{G}$ is $U \mapsto \ker \varphi_U$; its cokernel is the sheafification of the presheaf $U \mapsto \text{coker } \varphi_U$. Taking stalks is exact: $(\ker \varphi)_x = \ker \varphi_x$ and $(\text{coker } \varphi)_x = \text{coker } \varphi_x$. A sequence of sheaves is exact if and only if it is exact on all stalks. The functor $\Gamma(U, -)$ is left exact but not exact in general.*

Proof. Sums of morphisms and finite direct sums $(\mathcal{F} \oplus \mathcal{G})(U) = \mathcal{F}(U) \oplus \mathcal{G}(U)$ make $\text{Ab}(X)$ additive. The presheaf kernel is a sheaf and is a kernel in $\text{Ab}(X)$. By Theorem 23.3.1(2), the sheafified presheaf cokernel is a cokernel. Stalks are directed colimits, which are exact (Lemma 21.5.1), and sheafification does not change stalks; this gives the formulas for stalks. If φ is a monomorphism, the canonical morphism from $\mathcal{F}$ to the kernel of its cokernel is bijective on stalks, because Ab is abelian; so it is an isomorphism by Proposition 23.2.4; similarly for epimorphisms. Images and exactness are defined by kernels and cokernels, so they are computed on stalks. Left exactness of Γ holds because kernels are computed sectionwise.

For the failure of right exactness, let $X = \mathbb{R}$, $p \neq q$ in X , and consider $\mathbb{Z}_X \rightarrow p_*\mathbb{Z} \oplus q_*\mathbb{Z}$, sending a locally constant function to its values at p and q (when defined). On stalks it is surjective: at p it is $\mathbb{Z} \rightarrow \mathbb{Z}$, the identity, and at points $x \neq p, q$ the target stalk is 0, since x has a neighbourhood containing neither p nor q . On global sections it is $\mathbb{Z} \rightarrow \mathbb{Z} \oplus \mathbb{Z}$, $n \mapsto (n, n)$, because $\mathbb{R}$ is connected, and this is not surjective. $\square$

Counterexample 23.3.3. Surjectivity on stalks is a local lifting property, and global sections can fail to lift for topological reasons. Let $T = \mathbb{R}/\mathbb{Z}$ with quotient map $q: \mathbb{R} \rightarrow T$. The covering $p: \mathbb{R} \rightarrow S^1$ of Proposition 17.4.5 induces a continuous bijection $\mathbb{R}/4\mathbb{Z} \rightarrow S^1$ from a compact space to a Hausdorff space, hence a homeomorphism (Proposition 16.6.2), so

$T \cong \mathbb{R}/4\mathbb{Z} \cong S^1$ and q is a covering map. On any space X , let $C_{\mathbb{R}}$ and C_T be the sheaves of continuous maps to $\mathbb{R}$ and to the abelian group T , and consider

$$0 \rightarrow \mathbb{Z}_X \rightarrow C_{\mathbb{R}} \xrightarrow{f \mapsto q \circ f} C_T \rightarrow 0.$$

The kernel of $f \mapsto q \circ f$ consists of the continuous maps to $\mathbb{Z}$, that is, $\mathbb{Z}_X$. On stalks the map is surjective: a continuous $g: W \rightarrow T$ near x takes a small neighbourhood of x into an evenly covered open set, over which q has a continuous inverse on each sheet. So the sequence is exact (Proposition 23.3.2). But for $X = T$, the identity $\text{id}_T \in C_T(T)$ is not of the form $q \circ f$: a continuous $f: T \rightarrow \mathbb{R}$ with $q \circ f = \text{id}_T$ would be a lift of id_T through q , which by the lifting criterion (Proposition 17.6.1) requires $\pi_1(T) \subseteq q_*\pi_1(\mathbb{R}) = 0$, whereas $\pi_1(T) \cong \pi_1(S^1) \cong \mathbb{Z}$ (Theorem 17.4.6). So $\Gamma(T, -)$ is not right exact on this sequence; the obstruction will reappear as the nonzero group $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$ (Chapter 24).

23.4 Direct and inverse images

Definition 23.4.1. Let $f: X \rightarrow Y$ be continuous. The *direct image* of a sheaf $\mathcal{F}$ on X is the sheaf $f_*\mathcal{F}(V) = \mathcal{F}(f^{-1}V)$ on Y . The *inverse image* of a sheaf $\mathcal{G}$ on Y is the sheafification $f^{-1}\mathcal{G}$ of the presheaf $f_p^{-1}\mathcal{G}(U) = \text{colim}_{V \supseteq f(U)} \mathcal{G}(V)$, the colimit over open $V \supseteq f(U)$ directed by reverse inclusion. For $i: Z \rightarrow X$ the inclusion of a subspace, $\mathcal{F}|_Z = i^{-1}\mathcal{F}$ is the *restriction*; for Z open, $\mathcal{F}|_Z(W) = \mathcal{F}(W)$.

Proposition 23.4.2. 1. $(f^{-1}\mathcal{G})_x \cong \mathcal{G}_{f(x)}$; hence f^{-1} is exact on sheaves of abelian groups. The functor f_* is left exact.
 2. There is a bijection $\text{Hom}_{\text{Sh}(X)}(f^{-1}\mathcal{G}, \mathcal{F}) \cong \text{Hom}_{\text{Sh}(Y)}(\mathcal{G}, f_*\mathcal{F})$, natural in $\mathcal{F}$ and $\mathcal{G}$: f^{-1} is left adjoint to f_* .
 3. $(g \circ f)_* = g_* \circ f_*$ and $(g \circ f)^{-1} \cong f^{-1} \circ g^{-1}$.

Proof. (1) The pairs (U, V) with $x \in U$ and $V \supseteq f(U)$ map cofinally to the open neighbourhoods V of $f(x)$ (take $U = f^{-1}V$), so $(f_p^{-1}\mathcal{G})_x = \text{colim}_{V \ni f(x)} \mathcal{G}(V)$ by Lemma 21.5.1, and sheafification preserves stalks. Exactness follows from Proposition 23.3.2. Kernels of sheaves are computed sectionwise, so f_* preserves them.

(2) By Theorem 23.3.1(2), morphisms $f^{-1}\mathcal{G} \rightarrow \mathcal{F}$ are morphisms of presheaves $f_p^{-1}\mathcal{G} \rightarrow \mathcal{F}$, that is, families of maps $\mathcal{G}(V) \rightarrow \mathcal{F}(U)$ for $V \supseteq f(U)$, compatible with restrictions in U and V . Such a family determines maps $\mathcal{G}(V) \rightarrow \mathcal{F}(f^{-1}V)$ (take $U = f^{-1}V$), compatible with restrictions, that is, a morphism $\mathcal{G} \rightarrow f_*\mathcal{F}$. Conversely a morphism $\psi: \mathcal{G} \rightarrow f_*\mathcal{F}$ gives $\mathcal{G}(V) \xrightarrow{\psi_V} \mathcal{F}(f^{-1}V) \rightarrow \mathcal{F}(U)$ by restriction, as $U \subseteq f^{-1}V$. These constructions are mutually inverse and natural.

(3) The first equality is clear; the second follows from (2) and uniqueness of adjoints (Proposition 5.10.3). $\square$

Definition 23.4.3. Let $j: U \rightarrow X$ be the inclusion of an open subset and $\mathcal{F}$ a sheaf of abelian groups on U . The *extension by zero* $j_!\mathcal{F}$ is the sheafification of the presheaf $W \mapsto \mathcal{F}(W)$ if $W \subseteq U$ and $W \mapsto 0$ otherwise.

Proposition 23.4.4. $(j_!\mathcal{F})_x = \mathcal{F}_x$ for $x \in U$ and 0 for $x \notin U$. If $Z = X \setminus U$ with inclusion i , then for every sheaf of abelian groups $\mathcal{F}$ on X there is a short exact sequence $0 \rightarrow j_!(\mathcal{F}|_U) \rightarrow \mathcal{F} \rightarrow i_*(\mathcal{F}|_Z) \rightarrow 0$.

Proof. For $x \in U$ the open neighbourhoods contained in U are cofinal; for $x \notin U$ no neighbourhood is contained in U . The first map is induced, via Theorem 23.3.1(2), by the inclusions $\mathcal{F}(W) \rightarrow \mathcal{F}(W)$ for $W \subseteq U$; the second is the unit $\mathcal{F} \rightarrow i_* i^{-1} \mathcal{F}$ of the adjunction. On stalks at $x \in U$, the sequence is $0 \rightarrow \mathcal{F}_x \rightarrow \mathcal{F}_x \rightarrow 0 \rightarrow 0$, since x has a neighbourhood missing the closed set Z . At $z \in Z$, the open subsets $W \cap Z$ are cofinal among neighbourhoods of z in Z , so $(i_* i^{-1} \mathcal{F})_z = (i^{-1} \mathcal{F})_z = \mathcal{F}_z$, and the sequence is $0 \rightarrow 0 \rightarrow \mathcal{F}_z \rightarrow \mathcal{F}_z \rightarrow 0$. $\square$

23.5 Locally constant sheaves and local systems

Definition 23.5.1. A sheaf $\mathcal{F}$ of sets is *locally constant* if every point has an open neighbourhood U with $\mathcal{F}|_U$ isomorphic to a constant sheaf A_U . A *local system* of K -vector spaces (for a field K) is a locally constant sheaf of K -vector spaces whose stalks are finite-dimensional; local systems of abelian groups or modules are defined similarly.

Proposition 23.5.2. Let $\mathcal{F}$ be a sheaf of sets on X . The étalé space $E_{\mathcal{F}} = \bigsqcup_x \mathcal{F}_x$, with the topology whose basis is the sets $\{s_x : x \in U\}$ for $s \in \mathcal{F}(U)$, has a local homeomorphism $p: E_{\mathcal{F}} \rightarrow X$, and $\mathcal{F}$ is isomorphic to the sheaf of sections of p . If $\mathcal{F}$ is locally constant and X is locally connected, p is a covering map (except that fibres may be empty), and conversely the sheaf of sections of such a covering is locally constant.

Proof. If $s_x = t_x$ for $s \in \mathcal{F}(U)$, $t \in \mathcal{F}(V)$, the two basic sets contain the basic set of $s|_W$ for some $W \ni x$; so these sets form a basis, and p maps the basic set of s homeomorphically onto U , with inverse $\hat{s}: x \mapsto s_x$. The map $s \mapsto \hat{s}$ from $\mathcal{F}(U)$ to continuous sections over U is injective by Proposition 23.2.4(1); it is surjective, because a continuous section σ is locally of the form $\hat{s}$ (near x , σ lies in the basic set of some s with $s_x = \sigma(x)$), and these local s glue. If $\mathcal{F}|_U \cong A_U$ with U connected, then $p^{-1}(U) \cong U \times A$ with A discrete, because the germs of the constant sections $a \in A$ give disjoint open sets covering $p^{-1}(U)$. Conversely, over a connected evenly covered open set, sections of a covering correspond to locally constant choices of sheets. $\square$

Theorem 23.5.3. Let X be path-connected, locally path-connected and semilocally simply connected, and $x \in X$. Taking the stalk at x , with the monodromy action, gives equivalences of categories between

1. locally constant sheaves of sets on X and sets with a right action of $\pi_1(X, x)$;
2. local systems of K -vector spaces on X and finite-dimensional right $K[\pi_1(X, x)]$ -modules, that is, representations $\rho: \pi_1(X, x)^{\text{op}} \rightarrow \text{GL}(\mathcal{F}_x)$.

Sketch of proof. By Proposition 23.5.2, locally constant sheaves correspond to covering spaces (with possibly empty fibres), and morphisms of sheaves to maps of coverings. A covering is the disjoint union of its components, which are connected coverings of X , and the fibre over x with its monodromy action (Corollary 17.4.4) is a right $\pi_1(X, x)$ -set whose orbits correspond to the components. For connected coverings, the classification theorem (Theorem 17.6.6) identifies isomorphism classes with transitive π_1 -sets $H \backslash \pi_1(X, x)$, and the lifting criterion (Proposition 17.6.1) identifies maps of coverings with equivariant maps of fibres; this gives (1). The omitted verification is that these identifications are functorial; see [Sza09, Chapter 2]. For (2), a local system is a locally constant sheaf of sets with a K -vector space structure on the stalks compatible with monodromy, and the monodromy action is then K -linear. $\square$

Example 23.5.4. 1. On a simply connected space every local system is constant.

2. On S^1 , the rank one local systems of $\mathbb{C}$ -vector spaces correspond to $\lambda \in \mathbb{C}^\times$, the monodromy around the generator of $\pi_1(S^1) \cong \mathbb{Z}$. For $\lambda \neq 1$ the local system $\mathcal{L}_\lambda$ has no nonzero global sections, since a global section is a vector fixed by monodromy. The real local system with monodromy -1 is the sheaf of locally constant sections of the Möbius band, viewed as a line bundle over S^1 .
3. For a manifold M , the *orientation sheaf* or_M , with stalks $H_n(M \mid x; \mathbb{Z})$, is the local system of rank one associated with the orientation double covering (Proposition 21.3.2); it is constant if and only if M is orientable.
4. For a family of spaces $f: E \rightarrow B$ that is locally trivial, the cohomology groups of the fibres form local systems $R^k f_* \mathbb{Z}$ on B (Chapter 24); their monodromy is a central object in the theory of variations of Hodge structure (Chapter 66).

23.6 Ringed spaces and sheaves of modules

Definition 23.6.1. A *ringed space* is a pair $(X, \mathcal{O}_X)$ of a space and a sheaf of commutative rings. It is *locally ringed* if all stalks $\mathcal{O}_{X,x}$ are local rings. A *morphism* $(f, f^\#): (X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ is a continuous map f with a morphism of sheaves of rings $f^\#: \mathcal{O}_Y \rightarrow f_* \mathcal{O}_X$; for locally ringed spaces one requires the induced maps $\mathcal{O}_{Y,f(x)} \rightarrow \mathcal{O}_{X,x}$ to be local homomorphisms. An $\mathcal{O}_X$ -*module* is a sheaf of abelian groups $\mathcal{F}$ with $\mathcal{O}_X(U)$ -module structures on $\mathcal{F}(U)$ compatible with restrictions.

Example 23.6.2. A space with its sheaf of continuous real functions, a smooth manifold with smooth functions (Chapter 31), a complex manifold with holomorphic functions (Chapter 41) and a scheme (Chapter 50) are locally ringed spaces: in each case the germs vanishing at x form the unique maximal ideal of the stalk, since a germ not vanishing at x is invertible near x .

Definition 23.6.3. Let $\mathcal{F}, \mathcal{G}$ be $\mathcal{O}_X$ -modules. The *tensor product* $\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$ is the sheafification of $U \mapsto \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$. The sheaf $\mathcal{H}om_{\mathcal{O}_X}(\mathcal{F}, \mathcal{G})$ has sections over U the morphisms $\mathcal{F}|_U \rightarrow \mathcal{G}|_U$ of $\mathcal{O}_U$ -modules, and $\mathcal{F}^\vee = \mathcal{H}om(\mathcal{F}, \mathcal{O}_X)$ is the *dual*. An $\mathcal{O}_X$ -module is *locally free of rank r* if every point has an open neighbourhood U with $\mathcal{F}|_U \cong \mathcal{O}_U^{\oplus r}$, and *invertible* if it is locally free of rank 1. For a morphism f of ringed spaces and an $\mathcal{O}_Y$ -module $\mathcal{G}$, the *pullback* is $f^* \mathcal{G} = f^{-1} \mathcal{G} \otimes_{f^{-1} \mathcal{O}_Y} \mathcal{O}_X$.

Proposition 23.6.4. 1. $\mathcal{H}om(\mathcal{F}, \mathcal{G})$ is a sheaf, and $(\mathcal{F} \otimes \mathcal{G})_x \cong \mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$.

2. If $\mathcal{E}$ and $\mathcal{E}'$ are locally free of ranks r and r' , then $\mathcal{E} \otimes \mathcal{E}'$ is locally free of rank rr' , $\mathcal{E}^\vee$ is locally free of rank r , and the evaluation map $\mathcal{E} \otimes \mathcal{E}^\vee \rightarrow \mathcal{O}_X$ is an isomorphism if $r = 1$.
3. The isomorphism classes of invertible sheaves form an abelian group $\text{Pic}(X)$ under $\otimes$, with identity $\mathcal{O}_X$ and inverse $\mathcal{L}^\vee$.

Proof. (1) Morphisms defined on the members of an open cover and agreeing on overlaps glue uniquely, because sections of $\mathcal{G}$ do. For the stalk, sheafification does not change stalks, so it suffices to compare $\text{colim}_{U \ni x} \mathcal{F}(U) \otimes_{\mathcal{O}_X(U)} \mathcal{G}(U)$ with $\mathcal{F}_x \otimes_{\mathcal{O}_{X,x}} \mathcal{G}_x$. The map $[s \otimes t] \mapsto s_x \otimes t_x$ is well defined; an inverse sends $s_x \otimes t_x$ to the class of $s|_W \otimes t|_W$ on a common neighbourhood W , which is well defined because every relation defining the tensor product over $\mathcal{O}_{X,x}$ involves finitely many germs and already holds over some neighbourhood of x [Har77, Chapter II, Section 5].

(2) All statements are local, so we may assume $\mathcal{E} = \mathcal{O}^r$ and $\mathcal{E}' = \mathcal{O}^{r'}$. Then $\mathcal{O}^r \otimes \mathcal{O}^{r'} \cong \mathcal{O}^{rr'}$ (the presheaf tensor product is already a sheaf in this case), $\mathcal{H}om(\mathcal{O}^r, \mathcal{O}) \cong \mathcal{O}^r$, and for

$r = 1$ evaluation $\mathcal{O} \otimes \mathcal{O} \rightarrow \mathcal{O}$ is multiplication, an isomorphism. Since the evaluation map is a globally defined morphism, and being an isomorphism can be checked on stalks (Proposition 23.2.4), it is an isomorphism.

(3) Tensor product is associative and commutative up to natural isomorphism, $\mathcal{O}_X \otimes \mathcal{L} \cong \mathcal{L}$, and (2) gives inverses. $\square$

Remark 23.6.5. If $E \rightarrow X$ is a (real or complex, topological, smooth or holomorphic) vector bundle of rank r , its sheaf of sections is a locally free module of rank r over the corresponding sheaf of functions, and every locally free sheaf arises in this way, uniquely up to isomorphism (Chapter 33). Under this correspondence, $\text{Pic}(X)$ is the group of line bundles; for a complex manifold it is computed by sheaf cohomology via the exponential sequence (Chapter 42).

23.7 Exercises

Exercise 23.7.1. Show that the sheafification of the constant presheaf with value A is the constant sheaf A_X of locally constant functions, and that $A_X = \pi^{-1}A$ for $\pi: X \rightarrow \text{pt}$.

Solution. The stalks of the constant presheaf are all A , and a section of the sheafification is a function $U \rightarrow A$ locally equal to a constant, that is, a locally constant function. For the second statement, $\pi_p^{-1}A(U) = A$ for $U \neq \emptyset$. $\square$

Exercise 23.7.2. Let $i: \{x\} \rightarrow X$. Show that $x_*A = i_*A$ and $\text{Hom}(\mathcal{F}, x_*A) \cong \text{Hom}(\mathcal{F}_x, A)$ for every sheaf $\mathcal{F}$ of abelian groups.

Solution. The first statement is the definition. By Proposition 23.4.2(2) and (1), $\text{Hom}(\mathcal{F}, i_*A) \cong \text{Hom}(i^{-1}\mathcal{F}, A) = \text{Hom}(\mathcal{F}_x, A)$, since a sheaf on a point is a single group. $\square$

Exercise 23.7.3. Give an example of a continuous map f and a sheaf $\mathcal{F}$ such that $(f_*\mathcal{F})_y$ is not isomorphic to the global sections of $\mathcal{F}$ on the fibre $f^{-1}(y)$.

Exercise 23.7.4. Let $X = \mathbb{C} \setminus \{0\}$ with the sheaf $\mathcal{O}$ of holomorphic functions and $\mathcal{O}^\times$ of nowhere vanishing ones. Accepting that holomorphic functions have local logarithms of nonvanishing functions but z has no global logarithm, show that $\exp: \mathcal{O} \rightarrow \mathcal{O}^\times$, $f \mapsto e^{2\pi i f}$, is an epimorphism of sheaves but not surjective on global sections, and identify its kernel.

Exercise 23.7.5. For $j: U \rightarrow X$ open, show that $j_!$ is left adjoint to j^{-1} on sheaves of abelian groups, and that j^{-1} is left adjoint to j_* .

Chapter 24

Sheaf Cohomology

24.1 Introduction

The global sections functor $\Gamma(X, -)$ is left exact but not exact (Proposition 23.3.2), and sheaf cohomology $H^i(X, \mathcal{F})$ is its right derived functor. It can be computed by several kinds of acyclic resolutions: flasque, soft or fine sheaves (Theorem 24.4.5), Čech complexes of acyclic covers (Theorem 24.5.3), and hypercohomology of complexes. On reasonable spaces the cohomology of a constant sheaf is singular cohomology (Theorem 24.6.1); on a manifold the de Rham complex is a fine resolution of the constant sheaf, which gives the de Rham theorem (Chapter 35). References: [Har77, Chapter III], [God58], [Bre97], [KS90] and [Voi02b, Chapter 4].

Prerequisites. Chapter 23 (Sheaves on Topological Spaces), Chapter 14 (Spectral Sequences), Chapter 15 (Derived Categories) and Chapter 20 (Singular Cohomology).

24.2 Cohomology as a derived functor

Lemma 24.2.1. *Let $L: \mathcal{B} \rightarrow \mathcal{A}$ be an exact functor between abelian categories with right adjoint $R: \mathcal{A} \rightarrow \mathcal{B}$. Then R preserves injective objects. In particular f_* preserves injective sheaves for every continuous f , and so does restriction j^{-1} to an open subset $j: U \rightarrow X$.*

Proof. For I injective, $\text{Hom}_{\mathcal{B}}(-, R(I)) \cong \text{Hom}_{\mathcal{A}}(L(-), I)$ is a composite of exact functors, hence exact. For f_* use Proposition 23.4.2. For j^{-1} we show that $j_!$ is an exact left adjoint: a morphism $j_! \mathcal{F} \rightarrow \mathcal{G}$ is, by Theorem 23.3.1, a compatible family of maps $\mathcal{F}(W) \rightarrow \mathcal{G}(W)$ for $W \subseteq U$, that is, a morphism $\mathcal{F} \rightarrow \mathcal{G}|_U$; and $j_!$ is exact on stalks (Proposition 23.4.4). $\square$

Theorem 24.2.2. *The category $\text{Ab}(X)$ has enough injectives, and so does the category of $\mathcal{O}_X$ -modules on a ringed space.*

Proof. For each x embed $\mathcal{F}_x$ into an injective abelian group I_x (Theorem 13.5.4). The skyscraper sheaf $x_* I_x$ is injective, since $\text{Hom}(\mathcal{G}, x_* I_x) \cong \text{Hom}(\mathcal{G}_x, I_x)$ (Exercise 23.7.2) is exact in $\mathcal{G}$. A product of injective objects is injective, and $\mathcal{F} \rightarrow \prod_x x_* I_x$ is a monomorphism, being injective on stalks. For $\mathcal{O}_X$ -modules use injective $\mathcal{O}_{X,x}$ -modules instead. $\square$

Definition 24.2.3. The *sheaf cohomology* groups are $H^i(X, \mathcal{F}) = R^i \Gamma(X, -)(\mathcal{F})$ (Definition 13.6.4). For a complex $\mathcal{K}^\bullet$ of sheaves, bounded below, the *hypercohomology* is $\mathbb{H}^i(X, \mathcal{K}^\bullet) = H^i(R\Gamma(X, \mathcal{K}^\bullet))$ (Definition 15.4.4). For a continuous $f: X \rightarrow Y$, the *higher direct images* are $R^i f_* \mathcal{F}$.

Remark 24.2.4. For an $\mathcal{O}_X$ -module $\mathcal{F}$, derived functors computed in $\mathcal{O}_X$ -modules and in $\text{Ab}(X)$ agree, because injective $\mathcal{O}_X$ -modules are flasque (Lemma 24.3.2 applies verbatim with $j_! \mathcal{O}_U \rightarrow \mathcal{O}_X$) and flasque sheaves are acyclic in both settings.

24.3 Flasque sheaves

Definition 24.3.1. A sheaf $\mathcal{F}$ is *flasque* if all restriction maps $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ are surjective.

Lemma 24.3.2. *Injective sheaves of abelian groups are flasque. Direct images and restrictions to open subsets of flasque sheaves are flasque, and so are products.*

Proof. Let I be injective and $V \subseteq U$ open, with inclusions j_V, j_U into X . The morphism $j_{V!} \mathbb{Z}_V \rightarrow j_{U!} \mathbb{Z}_U$ is injective on stalks, so $\text{Hom}(j_{U!} \mathbb{Z}_U, I) \rightarrow \text{Hom}(j_{V!} \mathbb{Z}_V, I)$ is surjective. By the adjunction in the proof of Lemma 24.2.1, $\text{Hom}(j_{U!} \mathbb{Z}_U, I) = \text{Hom}(\mathbb{Z}_U, I|_U) = I(U)$, a morphism from the constant sheaf being determined by the image of 1; so $I(U) \rightarrow I(V)$ is surjective. The other statements are immediate from the definitions. $\square$

Proposition 24.3.3. *Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \xrightarrow{\pi} \mathcal{F}'' \rightarrow 0$ be exact with $\mathcal{F}'$ flasque. Then $\mathcal{F}(U) \rightarrow \mathcal{F}''(U)$ is surjective for all U . If also $\mathcal{F}$ is flasque, so is $\mathcal{F}''$. Flasque sheaves are Γ -acyclic, so $H^i(X, \mathcal{F})$ can be computed with flasque resolutions.*

Proof. We may take $U = X$. Let $s \in \mathcal{F}''(X)$, and consider pairs (V, t) with V open, $t \in \mathcal{F}(V)$ and $\pi(t) = s|_V$, ordered by extension. A chain has an upper bound by gluing, and it is nonempty, since s lifts locally. Let (V, t) be maximal and suppose $x \notin V$. Choose $W \ni x$ and $t' \in \mathcal{F}(W)$ lifting $s|_W$. On $V \cap W$, $t - t'$ lies in $\mathcal{F}'(V \cap W)$, which extends to $u \in \mathcal{F}'(X)$ since $\mathcal{F}'$ is flasque. Then t and $t' + u|_W$ agree on $V \cap W$ and glue to a lift over $V \cup W$, contradicting maximality. If $\mathcal{F}$ is flasque and $s'' \in \mathcal{F}''(V)$, lift it to $\mathcal{F}(V)$, extend to $\mathcal{F}(X)$ and map to $\mathcal{F}''(X)$.

Let $\mathcal{F}$ be flasque and embed it into an injective I with cokernel $\mathcal{G}$, which is flasque by the above and Lemma 24.3.2. The long exact sequence gives $H^1(X, \mathcal{F}) = \text{coker}(I(X) \rightarrow \mathcal{G}(X)) = 0$ and $H^i(X, \mathcal{F}) \cong H^{i-1}(X, \mathcal{G})$ for $i \geq 2$, which vanishes by induction on i for all flasque sheaves simultaneously. The last statement is Proposition 13.6.6. $\square$

Example 24.3.4. For every sheaf $\mathcal{F}$, the sheaf $C^0(\mathcal{F}) = \prod_x x_* \mathcal{F}_x$ of possibly discontinuous sections is flasque, and $\mathcal{F} \rightarrow C^0(\mathcal{F})$ is injective. Repeating with the cokernel gives the *Godement resolution*, a canonical flasque resolution, functorial and exact in $\mathcal{F}$.

24.4 Soft and fine sheaves

In this section X is paracompact Hausdorff. For a subset $Z \subseteq X$ we write $\mathcal{F}(Z) = \Gamma(Z, \mathcal{F}|_Z)$; by Proposition 23.5.2 this is the set of continuous sections over Z of the étalé space of $\mathcal{F}$. Sections over the members of a locally finite closed cover of Z that agree on overlaps glue, by the pasting lemma applied locally.

Lemma 24.4.1. *Let $Z \subseteq X$ be closed. Every $s \in \mathcal{F}(Z)$ extends to a section over some open neighbourhood of Z .*

Proof. Each $z \in Z$ has an open neighbourhood W_z in X and $t_z \in \mathcal{F}(W_z)$ with germs $s(y)$ at all $y \in W_z \cap Z$. Take a locally finite open refinement $(U_i)_{i \in I}$ of the cover $\{W_z\} \cup \{X \setminus Z\}$, choosing for each i either a section $t_i = t_z|_{U_i}$ (first kind) or $U_i \subseteq X \setminus Z$ (second kind), and closed sets $\overline{V_i} \subseteq U_i$ with (V_i) still covering (Lemma 16.10.4, Proposition 16.10.3). For $x \in X$

let $J(x) = \{i : x \in \overline{V_i}\}$, a finite set. Let W be the set of x such that $J(x)$ contains only indices of the first kind and the germs $(t_i)_x$, $i \in J(x)$, all agree.

$Z \subseteq W$, since for $z \in Z$ all $i \in J(z)$ are of the first kind and $(t_i)_z = s(z)$. W is open: for $x \in W$ choose an open $N \ni x$ that meets only finitely many $\overline{V_i}$, avoids those with $i \notin J(x)$, lies in $\bigcap_{i \in J(x)} U_i$, and on which the sections t_i , $i \in J(x)$, agree; then $J(y) \subseteq J(x)$ for $y \in N$, so $N \subseteq W$. On such an $N = N_x$, put $t^{(x)} = t_i|_{N_x}$ for any $i \in J(x)$ (nonempty, since the V_i cover). At $y \in N_x \cap N_{x'}$ both $t^{(x)}$ and $t^{(x')}$ have the germ $(t_k)_y$ for $k \in J(y)$, so they agree, and they glue to $t \in \mathcal{F}(W)$ with $t_z = s(z)$ on Z . $\square$

Definition 24.4.2. A sheaf $\mathcal{F}$ on X is *soft* if $\mathcal{F}(X) \rightarrow \mathcal{F}(Z)$ is surjective for every closed Z . The *support* of an endomorphism η of $\mathcal{F}$ is the closure of $\{x : \eta_x \neq 0\}$. A sheaf of abelian groups is *fine* if for every locally finite open cover (U_i) there are endomorphisms η_i with $\text{supp } \eta_i \subseteq U_i$ and $\sum_i \eta_i = \text{id}$ (a *partition of unity* of $\mathcal{F}$).

Example 24.4.3. The sheaf C_X of continuous real functions is fine, by partitions of unity (Theorem 16.10.6): take η_i to be multiplication by ψ_i . More generally every sheaf of modules over C_X is fine, and on a smooth manifold every module over the sheaf of smooth functions is fine, using smooth partitions of unity (Chapter 31). The constant sheaf $\mathbb{Z}_X$ on a connected space with more than one point is not soft.

Lemma 24.4.4. 1. *Flasque sheaves and fine sheaves are soft.*

2. *If $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \xrightarrow{\pi} \mathcal{F}'' \rightarrow 0$ is exact and $\mathcal{F}'$ is soft, then $\mathcal{F}(X) \rightarrow \mathcal{F}''(X)$ is surjective.*

3. *If moreover $\mathcal{F}$ is soft, then $\mathcal{F}''$ is soft.*

Proof. (1) Let $s \in \mathcal{F}(Z)$ and extend it to an open $W \supseteq Z$ by Lemma 24.4.1. If $\mathcal{F}$ is flasque, extend further to X . If $\mathcal{F}$ is fine, take $\eta_1 + \eta_2 = \text{id}$ for the cover $\{W, X \setminus Z\}$. The section $\eta_1(s)$ of $\mathcal{F}(W)$ vanishes on $W \setminus \text{supp } \eta_1$, so it glues with 0 on $X \setminus \text{supp } \eta_1$ to a global section t ; at $z \in Z$, $t_z = s_z - (\eta_2)_z(s_z) = s_z$ since $z \notin \text{supp } \eta_2$.

(2) Choose a locally finite open cover (U_i) with lifts $s_i \in \mathcal{F}(U_i)$ of $s|_{U_i}$ for a given $s \in \mathcal{F}''(X)$, and closed $\overline{V_i} \subseteq U_i$ covering X . Consider pairs (J, t) with $J \subseteq I$ and $t \in \mathcal{F}(Z_J)$ lifting s , where $Z_J = \bigcup_{i \in J} \overline{V_i}$, which is closed (Lemma 16.10.2). Chains have upper bounds by gluing along the locally finite closed cover, so Zorn's lemma gives a maximal (J, t) . If $i \notin J$, the difference $t - s_i$ on $Z_J \cap \overline{V_i}$ is a section of $\mathcal{F}'$ (its image under π is 0), which extends to $u \in \mathcal{F}'(X)$. Then t and $s_i + u$ agree on $Z_J \cap \overline{V_i}$ and glue to a lift over $Z_{J \cup \{i\}}$, a contradiction. So $J = I$.

(3) A closed subspace Z is paracompact Hausdorff, the restricted sequence on Z is exact, and $\mathcal{F}'|_Z$ is soft. By (2) applied on Z , $\mathcal{F}(Z) \rightarrow \mathcal{F}''(Z)$ is surjective, and $\mathcal{F}(X) \rightarrow \mathcal{F}(Z)$ is surjective. $\square$

Theorem 24.4.5. *On a paracompact Hausdorff space, soft sheaves, in particular fine sheaves, are Γ -acyclic. Hence cohomology can be computed with soft or fine resolutions.*

Proof. Identical to the proof of Proposition 24.3.3, using Lemma 24.4.4 and the fact that injective sheaves are flasque, hence soft. $\square$

Corollary 24.4.6. *Let X be paracompact Hausdorff. Then $H^i(X, \mathcal{F}) = 0$ for $i > 0$ for every sheaf of C_X -modules, for instance the sheaf of sections of a real vector bundle.*

Proof. Such a sheaf is fine (Example 24.4.3) and is its own resolution. $\square$

24.5 Čech cohomology

Definition 24.5.1. Let $\mathcal{U} = (U_i)_{i \in I}$ be an open cover of X and $\mathcal{F}$ a presheaf of abelian groups. Write $U_{i_0 \dots i_p} = U_{i_0} \cap \dots \cap U_{i_p}$. An *alternating Čech p -cochain* is a family $\alpha = (\alpha_{i_0 \dots i_p})$, indexed by all $(i_0, \dots, i_p) \in I^{p+1}$, with $\alpha_{i_0 \dots i_p} \in \mathcal{F}(U_{i_0 \dots i_p})$, which changes sign when two indices are exchanged and vanishes when two indices are equal. They form a group $\check{C}^p(\mathcal{U}, \mathcal{F})$, with differential

$$(d\alpha)_{i_0 \dots i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \alpha_{i_0 \dots \widehat{i_k} \dots i_{p+1}}|_{U_{i_0 \dots i_{p+1}}}.$$

Its cohomology is the *Čech cohomology* $\check{H}^p(\mathcal{U}, \mathcal{F})$. Choosing a well-ordering of I , $\check{C}^p(\mathcal{U}, \mathcal{F}) \cong \prod_{i_0 < \dots < i_p} \mathcal{F}(U_{i_0 \dots i_p})$. For a sheaf, $\check{H}^0(\mathcal{U}, \mathcal{F}) = \mathcal{F}(X)$ by the sheaf axioms. The *sheafified Čech complex* $\check{C}^p(\mathcal{U}, \mathcal{F})$ is the sheaf $V \mapsto \check{C}^p(\mathcal{U} \cap V, \mathcal{F}|_V)$, with augmentation $\mathcal{F} \rightarrow \check{C}^0$ given by restriction.

Lemma 24.5.2. *For every sheaf $\mathcal{F}$ and open cover $\mathcal{U}$, the sequence $0 \rightarrow \mathcal{F} \rightarrow \check{C}^0(\mathcal{U}, \mathcal{F}) \rightarrow \check{C}^1(\mathcal{U}, \mathcal{F}) \rightarrow \dots$ is exact. If $\mathcal{F}$ is flasque, then $\check{H}^p(\mathcal{U}, \mathcal{F}) = 0$ for $p > 0$.*

Proof. Let $x \in X$ and choose j with $x \in U_j$. For open $V \subseteq U_j$ define $k: \check{C}^p(\mathcal{U} \cap V, \mathcal{F}) \rightarrow \check{C}^{p-1}(\mathcal{U} \cap V, \mathcal{F})$ by $(k\alpha)_{i_0 \dots i_{p-1}} = \alpha_{ji_0 \dots i_{p-1}}$, which makes sense since $V \cap U_{ji_0 \dots i_{p-1}} = V \cap U_{i_0 \dots i_{p-1}}$, and $k = \text{restriction to } V \cap U_j = V$ in degree 0. Then $(dk\alpha)_{i_0 \dots i_p} = \sum_k (-1)^k \alpha_{ji_0 \dots \widehat{i_k} \dots i_p}$ and $(kd\alpha)_{i_0 \dots i_p} = \alpha_{i_0 \dots i_p} - \sum_k (-1)^k \alpha_{ji_0 \dots \widehat{i_k} \dots i_p}$, so $dk + kd = \text{id}$, and similarly in degree 0 with the augmentation. Passing to the colimit over $V \ni x$, the complex of stalks is contractible, hence exact.

If $\mathcal{F}$ is flasque, each $\check{C}^p(\mathcal{U}, \mathcal{F})$ is a product of sheaves $(j_{i_0 \dots i_p})_*(\mathcal{F}|_{U_{i_0 \dots i_p}})$, hence flasque (Lemma 24.3.2); so the resolution is flasque, its global sections $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ compute $H^\bullet(X, \mathcal{F})$ (Proposition 24.3.3), and this vanishes in positive degrees. $\square$

Theorem 24.5.3 (Leray). *Let $\mathcal{U}$ be an open cover of X such that $H^q(U_{i_0 \dots i_p}, \mathcal{F}) = 0$ for all $q > 0$ and all finite intersections. Then $\check{H}^p(\mathcal{U}, \mathcal{F}) \cong H^p(X, \mathcal{F})$ for all p .*

Proof. Let $\mathcal{F} \rightarrow I^\bullet$ be an injective resolution and consider the first-quadrant double complex $C^{p,q} = \check{C}^p(\mathcal{U}, I^q)$. For fixed q , the row $\check{C}^\bullet(\mathcal{U}, I^q)$ is exact in positive degrees with kernel $I^q(X)$ in degree 0, by Lemmas 24.5.2 and 24.3.2. For fixed p , the column is $\prod_{i_0 < \dots < i_p} I^\bullet(U_{i_0 \dots i_p})$; since $I^\bullet|_U$ is an injective resolution of $\mathcal{F}|_U$ (Lemma 24.2.1) and products of abelian groups are exact, its cohomology is $\prod H^q(U_{i_0 \dots i_p}, \mathcal{F})$, which is 0 for $q > 0$ and $\check{C}^p(\mathcal{U}, \mathcal{F})$ for $q = 0$. By Lemma 13.7.2, applied once to rows and once to columns, both $\Gamma(X, I^\bullet)$ and $\check{C}^\bullet(\mathcal{U}, \mathcal{F})$ are quasi-isomorphic to the total complex. Taking cohomology gives the theorem. $\square$

Remark 24.5.4. Refinements of covers induce maps on Čech cohomology, well defined up to homotopy, and the colimit over all covers is the Čech cohomology $\check{H}^p(X, \mathcal{F})$. There is always a natural map $\check{H}^p(X, \mathcal{F}) \rightarrow H^p(X, \mathcal{F})$, which is an isomorphism for $p \leq 1$, and for all p if X is paracompact Hausdorff [God58, Chapter II, Section 5]. Cohomology classes in $\check{H}^1(\mathcal{U}, \mathcal{F})$ are represented by cocycles (g_{ij}) with $g_{ij} + g_{jk} = g_{ik}$; for nonabelian sheaves of groups such as $\text{GL}_n(\mathcal{O}_X)$ the same formula (written multiplicatively) classifies locally free sheaves of rank n (Chapter 33).

24.6 Comparison with singular cohomology

Theorem 24.6.1. *Let X be a paracompact Hausdorff, locally contractible space (for instance a manifold or a CW complex) and A an abelian group. Then $H^p(X, A_X) \cong H^p(X; A)$, the singular cohomology, naturally in X (and compatibly with cup products [Bre97, Chapter III]).*

Sketch of proof. Let $\mathcal{S}^p$ be the sheafification of the presheaf $U \mapsto C^p(U; A)$ of singular cochains. Since every point has arbitrarily small contractible neighbourhoods, the stalks of the cohomology presheaves $U \mapsto H^p(U; A)$ vanish for $p > 0$ and equal A for $p = 0$, so $0 \rightarrow A_X \rightarrow \mathcal{S}^\bullet$ is a resolution (sheafification preserves stalks and stalks are exact). The omitted verifications are that the sheaves $\mathcal{S}^p$ are soft on a paracompact space, and that $\Gamma(X, \mathcal{S}^\bullet) = C^\bullet(X; A)/C_0^\bullet(X; A)$, where $C_0^\bullet$ consists of cochains vanishing on sufficiently small simplices of some open cover, which is acyclic by the subdivision argument of Proposition 19.5.3; see [Bre97, Chapter III]. Then Theorem 24.4.5 gives $H^p(X, A_X) \cong H^p(C^\bullet(X; A)/C_0^\bullet) \cong H^p(X; A)$. $\square$

Example 24.6.2. $H^p(\mathbb{R}^n, \mathbb{Z}) = 0$ for $p > 0$, $H^p(S^n, \mathbb{Z})$ is $\mathbb{Z}$ for $p = 0, n$, and on a closed oriented surface of genus g , $H^1(\Sigma_g, \mathbb{Z}) \cong \mathbb{Z}^{2g}$. On spaces that are not locally contractible sheaf and singular cohomology can differ: for the topologist's sine curve T of Counterexample 16.5.8, singular cohomology sees two path components, $H^0(T; \mathbb{Z}) = \mathbb{Z}^2$, while $H^0(T, \mathbb{Z}_T) = \mathbb{Z}$ since T is connected.

Example 24.6.3. Cover S^1 by two open arcs U, V with $U \cap V = W_1 \sqcup W_2$, two disjoint arcs. Let $\mathcal{L}$ be the local system of rank one with monodromy -1 , obtained by gluing $\mathbb{Z}_U$ and $\mathbb{Z}_V$ by the identity over W_1 and by -1 over W_2 (Proposition 23.2.5 and Definition 23.5.1). The arcs and their intersections are homeomorphic to intervals, so the constant sheaf $\mathbb{Z}$ has no higher cohomology on them (Theorem 24.6.1), and Leray's theorem (Theorem 24.5.3) lets us compute with the cover $\{U, V\}$. In the trivialisation over V , the Čech complex is $\mathbb{Z}^2 \rightarrow \mathbb{Z}^2$, $(a, b) \mapsto (b - a, b + a)$, one coordinate for each W_i . Its kernel is 0 and its image is $\{(x, y) : x \equiv y \pmod{2}\}$, so

$$H^0(S^1, \mathcal{L}) = 0, \quad H^1(S^1, \mathcal{L}) \cong \mathbb{Z}/2.$$

For the constant sheaf the differential is $(a, b) \mapsto (b - a, b - a)$, which gives $H^0 = H^1 = \mathbb{Z}$, in accordance with Theorem 24.6.1. With coefficients in $\mathbb{Q}$ the twisted local system has $H^0 = H^1 = 0$, since the determinant 2 of the differential becomes invertible: a nontrivial local system can have vanishing rational cohomology on a space whose ordinary cohomology does not vanish.

24.7 Hypercohomology and the Leray spectral sequence

Proposition 24.7.1. *Let $\mathcal{K}^\bullet$ be a complex of sheaves with $\mathcal{K}^p = 0$ for $p < 0$.*

1. *A quasi-isomorphism $\mathcal{K}^\bullet \rightarrow \mathcal{L}^\bullet$ induces isomorphisms on hypercohomology.*
2. *There are convergent spectral sequences $E_1^{p,q} = H^q(X, \mathcal{K}^p) \Rightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet)$ and $E_2^{p,q} = H^p(X, \mathcal{H}^q(\mathcal{K}^\bullet)) \Rightarrow \mathbb{H}^{p+q}(X, \mathcal{K}^\bullet)$, where $\mathcal{H}^q$ are the cohomology sheaves.*
3. *If every $\mathcal{K}^p$ is Γ -acyclic, $\mathbb{H}^p(X, \mathcal{K}^\bullet) = H^p(\Gamma(X, \mathcal{K}^\bullet))$. If $\mathcal{F} \rightarrow \mathcal{K}^\bullet$ is a resolution, then $\mathbb{H}^p(X, \mathcal{K}^\bullet) = H^p(X, \mathcal{F})$.*

Proof. (1) Quasi-isomorphisms are isomorphisms in the derived category, and $R\Gamma$ is a functor on it. (2) is Proposition 15.4.8. (3) The first statement is Lemma 15.4.6, and the second follows from (1) applied to $\mathcal{F} \rightarrow \mathcal{K}^\bullet$, with $\mathcal{F}$ in degree 0. $\square$

Remark 24.7.2. On a complex manifold X the holomorphic de Rham complex $\Omega_X^\bullet$ is a resolution of $\mathbb{C}_X$ (the holomorphic Poincaré lemma), so $H^k(X; \mathbb{C}) \cong \mathbb{H}^k(X, \Omega_X^\bullet)$, and the first spectral sequence of Proposition 24.7.1 becomes the *Hodge-to-de Rham* (Frölicher) spectral sequence $E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X; \mathbb{C})$. Its degeneration for compact Kähler manifolds is a form of the Hodge decomposition (Chapter 45).

Theorem 24.7.3 (Leray spectral sequence). *Let $f: X \rightarrow Y$ be continuous and $\mathcal{F}$ a sheaf of abelian groups on X . Then $R^q f_* \mathcal{F}$ is the sheafification of the presheaf $V \mapsto H^q(f^{-1}V, \mathcal{F})$, and there is a convergent spectral sequence $E_2^{p,q} = H^p(Y, R^q f_* \mathcal{F}) \Rightarrow H^{p+q}(X, \mathcal{F})$.*

Proof. Let $\mathcal{F} \rightarrow I^\bullet$ be an injective resolution. Then $R^q f_* \mathcal{F} = \mathcal{H}^q(f_* I^\bullet)$, the sheafification of the presheaf of cohomology groups $V \mapsto H^q(I^\bullet(f^{-1}V))$, and $H^q(I^\bullet(f^{-1}V)) = H^q(f^{-1}V, \mathcal{F})$ because $I^\bullet|_{f^{-1}V}$ is an injective resolution (Lemma 24.2.1). Since $\Gamma(X, -) = \Gamma(Y, -) \circ f_*$ and f_* preserves injectives, which are $\Gamma(Y, -)$ -acyclic, the Grothendieck spectral sequence (Theorem 14.6.3) applies. $\square$

Example 24.7.4. If f is a closed embedding, f_* is exact (compute stalks as in Proposition 23.4.4), so $H^p(Y, f_* \mathcal{F}) \cong H^p(X, \mathcal{F})$. For a locally trivial fibre bundle $f: E \rightarrow B$ with fibre F over a locally contractible paracompact base, $R^q f_* \mathbb{Z}_E$ is the local system with stalks $H^q(F; \mathbb{Z})$, and the Leray spectral sequence becomes the *Serre spectral sequence* $E_2^{p,q} = H^p(B, \mathcal{H}^q(F)) \Rightarrow H^{p+q}(E; \mathbb{Z})$ with local coefficients [Bre97].

24.8 Cohomology with compact supports

Definition 24.8.1. Let X be locally compact Hausdorff. The *support* of a section $s \in \mathcal{F}(X)$ is the closed set $\{x : s_x \neq 0\}$. Let $\Gamma_c(X, \mathcal{F})$ be the group of sections with compact support, a left exact functor, and $H_c^i(X, \mathcal{F}) = R^i \Gamma_c(X, \mathcal{F})$.

Remark 24.8.2. Flasque sheaves are Γ_c -acyclic, and on locally compact Hausdorff spaces that are countable at infinity, so are soft sheaves; so H_c^* can be computed by fine resolutions. For a manifold M and an abelian group A , $H_c^i(M, A_M)$ agrees with the compactly supported singular cohomology of Definition 21.5.2, and Poincaré duality generalises to *Verdier duality* $R\Gamma_c(X, \mathcal{F})^\vee \cong R\mathrm{Hom}(\mathcal{F}, \omega_X)$ in the derived category, with $\omega_X = or_X[n]$ for an n -manifold [KS90, Chapter III], [Ive86].

24.9 Exercises

Exercise 24.9.1. Compute $\check{H}^\bullet(\mathcal{U}, \mathbb{Z}_{S^1})$ for the cover of S^1 by two open arcs whose intersection has two components, and check with Theorems 24.5.3 and 24.6.1 that $H^1(S^1, \mathbb{Z}) \cong \mathbb{Z}$.

Solution. With $I = \{0, 1\}$, $\check{C}^0 = \mathbb{Z}_{S^1}(U_0) \oplus \mathbb{Z}_{S^1}(U_1) = \mathbb{Z}^2$, $\check{C}^1 = \mathbb{Z}_{S^1}(U_0 \cap U_1) = \mathbb{Z}^2$ and $d(a, b) = (b - a, b - a)$. So $\check{H}^0 = \mathbb{Z}$ and $\check{H}^1 = \mathbb{Z}^2 / \{(c, c)\} \cong \mathbb{Z}$. The arcs and the components of their intersection are contractible, hence have no higher cohomology with constant coefficients, so Leray's theorem applies. $\square$

Exercise 24.9.2. Show that skyscraper sheaves are flasque, and that $H^i(X, x_* A) = 0$ for $i > 0$.

Solution. Restriction maps are identities or maps to 0. Flasque sheaves are acyclic (Proposition 24.3.3). $\square$

Exercise 24.9.3. Let X be an irreducible space (every nonempty open subset is dense). Show that every constant sheaf on X is flasque, and deduce $H^i(X, A_X) = 0$ for $i > 0$. (This applies to algebraic varieties with the Zariski topology, and shows that sheaf cohomology of constant sheaves in that topology does not see the classical topology.)

Solution. Nonempty open subsets of an irreducible space are connected (two disjoint nonempty open subsets would contradict density), so locally constant functions on them are constant, and $A_X(U) = A$ for $U \neq \emptyset$, with identity restriction maps. $\square$

Exercise 24.9.4. For open $U, V \subseteq X$ and a sheaf $\mathcal{F}$, construct a long exact Mayer–Vietoris sequence $\cdots \rightarrow H^i(U \cup V, \mathcal{F}) \rightarrow H^i(U, \mathcal{F}) \oplus H^i(V, \mathcal{F}) \rightarrow H^i(U \cap V, \mathcal{F}) \rightarrow H^{i+1}(U \cup V, \mathcal{F}) \rightarrow \cdots$. (Hint: for flasque I , $0 \rightarrow I(U \cup V) \rightarrow I(U) \oplus I(V) \rightarrow I(U \cap V) \rightarrow 0$ is exact.)

Part V

Analysis

Chapter 25

Real Analysis

25.1 Introduction

This chapter develops calculus in one and several real variables with complete proofs, starting slowly from limits of sequences and of functions: the sequential characterisation of limits, continuous functions and the theorems of Bolzano, Weierstrass and Heine–Cantor, the mean value theorems, de l'Hôpital's rule, Taylor's formula and convexity; then series and power series, the exponential and trigonometric functions (including a definition of π), the Riemann integral, uniform convergence of sequences of functions (with a continuous nowhere differentiable function), derivatives of maps between Euclidean spaces, the inverse and implicit function theorems, ordinary differential equations, and the approximation and compactness theorems of Stone–Weierstrass and Arzelà–Ascoli. The topological background (metric spaces, compactness, completeness, uniform convergence, the Banach fixed point theorem) is in Chapter 16. References: [Rud76], [Die60], [Spi65], [Lan93] and [Har02].

Prerequisites. Chapter 16 (General Topology), Chapter 4 (Number Systems) and Chapter 9 (Linear Algebra); Theorem 25.7.3 also uses Chapter 17 (Homotopy and Covering Spaces).

Throughout, $\mathbb{K}$ is $\mathbb{R}$ or $\mathbb{C}$, and $\mathbb{R}^n$ and $\mathbb{C}^n$ carry the Euclidean norm unless otherwise stated.

25.2 Limits of sequences

We recall that a real sequence $(a_n)_{n \in \mathbb{N}}$ *converges* to $a \in \mathbb{R}$ if for every $\varepsilon > 0$ there is N with $|a_n - a| < \varepsilon$ for all $n \geq N$ (Definition 4.7.5). This is convergence in the metric space $\mathbb{R}$ (Chapter 16). In this section we develop the theory of limits of sequences directly from this definition.

Definition 25.2.1. Let (a_n) be a real sequence.

1. If (a_n) converges to a we write $a_n \rightarrow a$ or $\lim_n a_n = a$, and call a the *limit*.
2. (a_n) *diverges to $+\infty$* , written $a_n \rightarrow +\infty$, if for every $M \in \mathbb{R}$ there is N with $a_n > M$ for all $n \geq N$; similarly $a_n \rightarrow -\infty$ if for every M eventually $a_n < M$.
3. (a_n) is *regular* if it converges or diverges to $\pm\infty$, and *oscillating* otherwise.
4. (a_n) is *bounded* if there is C with $|a_n| \leq C$ for all n ; *bounded above* or *below* are defined similarly.
5. (a_n) is *increasing* if $a_n \leq a_{n+1}$ for all n , *strictly increasing* if $a_n < a_{n+1}$, and similarly *decreasing*; it is *monotone* if it is increasing or decreasing.

6. A subsequence of (a_n) is a sequence $(a_{n_k})_{k \in \mathbb{N}}$ with $n_0 < n_1 < n_2 < \dots$.

A property holds *eventually*, or *for large n* , if there is N such that it holds for all $n \geq N$.

Proposition 25.2.2. Let (a_n) be a real sequence.

1. A sequence has at most one limit.
2. A convergent sequence is bounded.
3. (Permanence of sign) If $a_n \rightarrow a$ and $a > 0$, then $a_n > a/2 > 0$ for large n .
4. Changing, adding or removing finitely many terms does not affect convergence or the limit.
5. If $a_n \rightarrow a$, every subsequence converges to a ; the same holds for divergence to $\pm\infty$.

Proof. (1) Suppose $a_n \rightarrow a$ and $a_n \rightarrow b$ with $a \neq b$, and let $\varepsilon = |a - b|/2$. For n larger than both thresholds, $|a - b| \leq |a - a_n| + |a_n - b| < 2\varepsilon = |a - b|$, a contradiction. (2) Take $\varepsilon = 1$: for $n \geq N$, $|a_n| \leq |a_n - a| + |a| < |a| + 1$, so $|a_n| \leq \max(|a_0|, \dots, |a_{N-1}|, |a| + 1)$ for all n . (3) Take $\varepsilon = a/2$: then $a_n > a - a/2 = a/2$ for large n . (4) The definitions only involve the terms with $n \geq N$, and N may be enlarged. (5) By induction $n_k \geq k$, so $k \geq N$ implies $n_k \geq N$. $\square$

Lemma 25.2.3 (Bernoulli's inequality). For $h \geq -1$ and $n \in \mathbb{N}$, $(1 + h)^n \geq 1 + nh$.

Proof. For $n = 0$ both sides are 1. If $(1 + h)^n \geq 1 + nh$, then, as $1 + h \geq 0$, $(1 + h)^{n+1} \geq (1 + nh)(1 + h) = 1 + (n + 1)h + nh^2 \geq 1 + (n + 1)h$. $\square$

Proposition 25.2.4 (Algebra of limits). Let $a_n \rightarrow a$ and $b_n \rightarrow b$ be real sequences and $c \in \mathbb{R}$. Then $a_n + b_n \rightarrow a + b$, $ca_n \rightarrow ca$, $a_nb_n \rightarrow ab$ and $|a_n| \rightarrow |a|$. If $b \neq 0$, then $b_n \neq 0$ for large n and $a_n/b_n \rightarrow a/b$.

Proof. Given $\varepsilon > 0$, for large n we have $|a_n - a| < \varepsilon/2$ and $|b_n - b| < \varepsilon/2$, so $|(a_n + b_n) - (a + b)| < \varepsilon$. The reverse triangle inequality $||a_n| - |a|| \leq |a_n - a|$ gives $|a_n| \rightarrow |a|$, and $|ca_n - ca| = |c| |a_n - a|$ gives $ca_n \rightarrow ca$. For the product, let C bound $|a_n|$ (Proposition 25.2.2(2)); then

$$|a_nb_n - ab| \leq |a_n| |b_n - b| + |b| |a_n - a| \leq C|b_n - b| + |b| |a_n - a|,$$

which is less than ε for large n . For the quotient it suffices to show $1/b_n \rightarrow 1/b$. As $|b_n| \rightarrow |b| > 0$, permanence of sign gives $|b_n| > |b|/2$ for large n , and then $|1/b_n - 1/b| = |b - b_n|/(|b_n||b|) \leq 2|b_n - b|/|b|^2$, which tends to 0. $\square$

Proposition 25.2.5 (Comparison and squeeze). Let $(a_n), (b_n), (c_n)$ be real sequences.

1. If $a_n \leq b_n$ for large n , $a_n \rightarrow a$ and $b_n \rightarrow b$, then $a \leq b$.
2. (Squeeze theorem) If $a_n \leq c_n \leq b_n$ for large n and $a_n \rightarrow L$, $b_n \rightarrow L$, then $c_n \rightarrow L$.
3. If $a_n \leq b_n$ for large n and $a_n \rightarrow +\infty$, then $b_n \rightarrow +\infty$.
4. If $|a_n| \leq b_n$ for large n and $b_n \rightarrow 0$, then $a_n \rightarrow 0$. In particular, if (a_n) is bounded and $b_n \rightarrow 0$, then $a_nb_n \rightarrow 0$.

Proof. (1) If $a > b$, then $a_n - b_n \rightarrow a - b > 0$ by Proposition 25.2.4, so $a_n - b_n > 0$ for large n by permanence of sign, a contradiction. (2) Given $\varepsilon > 0$, for large n we have $L - \varepsilon < a_n \leq c_n \leq b_n < L + \varepsilon$. (3) Eventually $b_n \geq a_n > M$. (4) Apply (2) to $-b_n \leq a_n \leq b_n$; for the last statement, $|a_nb_n| \leq C|b_n|$. $\square$

Counterexample 25.2.6. Strict inequalities do not pass to the limit: $1/(n+1) > 0$ for all n , but the limit is 0. The convergence $|a_n| \rightarrow |a|$ does not imply $a_n \rightarrow a$ unless $a = 0$: the sequence $a_n = (-1)^n$ oscillates although $|a_n| \rightarrow 1$.

Proposition 25.2.7 (Infinite limits). *Let $a_n \rightarrow +\infty$.*

1. *If (b_n) is bounded below, then $a_n + b_n \rightarrow +\infty$.*
2. *If $b_n \geq c$ for large n , for some $c > 0$, then $a_n b_n \rightarrow +\infty$.*
3. *$a_n > 0$ for large n , and $1/a_n \rightarrow 0$. Conversely, if $b_n > 0$ and $b_n \rightarrow 0$, then $1/b_n \rightarrow +\infty$.*

Proof. (1) If $b_n \geq -C$, then for any M , eventually $a_n > M + C$, so $a_n + b_n > M$. (2) Eventually $a_n > M/c$ with $M > 0$, so $a_n b_n > M$. (3) Eventually $a_n > 1/\varepsilon > 0$, so $0 < 1/a_n < \varepsilon$. Conversely, eventually $0 < b_n < 1/M$ for $M > 0$, so $1/b_n > M$. $\square$

Counterexample 25.2.8 (Indeterminate forms). If $a_n \rightarrow +\infty$ and $b_n \rightarrow -\infty$, the limits alone do not determine the behaviour of $a_n + b_n$: with $a_n = n$, the choices $b_n = -n + c$, $b_n = -n/2$, $b_n = -2n$ and $b_n = -n + (-1)^n$ give the limit c , the limit $+\infty$, the limit $-\infty$, and an oscillating sequence. Similarly, if $a_n \rightarrow 0$ and $b_n \rightarrow +\infty$, then with $a_n = 1/(n+1)$ the choices $b_n = c(n+1)$, $(n+1)^2$ and $(-1)^n(n+1)$ show that $a_n b_n$ can converge to any c , diverge, or oscillate. The same happens for the forms ∞/∞ and $0/0$.

Proposition 25.2.9 (Fundamental limits). 1. *For $q \in \mathbb{R}$: $q^n \rightarrow 0$ if $|q| < 1$, $q^n \rightarrow 1$ if $q = 1$, $q^n \rightarrow +\infty$ if $q > 1$, and (q^n) oscillates if $q \leq -1$.*

2. *For $a > 0$, $a^{1/n} \rightarrow 1$, and $n^{1/n} \rightarrow 1$.*
3. *For $k \in \mathbb{N}$ and $a > 1$, $n^k/a^n \rightarrow 0$: exponentials beat powers.*
4. *For every $a \in \mathbb{R}$, $a^n/n! \rightarrow 0$, and $n!/n^n \rightarrow 0$.*

Proof. (1) If $q > 1$, write $q = 1 + h$ with $h > 0$; by Lemma 25.2.3, $q^n \geq 1 + nh \rightarrow +\infty$. If $0 < |q| < 1$, then $|q|^{-n} \rightarrow +\infty$, so $|q^n| \rightarrow 0$ by Proposition 25.2.7(3), and $q^n \rightarrow 0$ by Proposition 25.2.5(4). If $q = -1$ the terms alternate between ± 1 ; if $q < -1$, $|q^n| \rightarrow +\infty$ and the signs alternate, so the sequence is unbounded above and below and has no limit. (2) Let $a \geq 1$ and $a^{1/n} = 1 + h_n$ with $h_n \geq 0$ (the n -th root exists by Proposition 4.7.4). By Bernoulli, $a \geq 1 + nh_n$, so $0 \leq h_n \leq (a-1)/n \rightarrow 0$. For $a < 1$ apply this to $1/a$ and use Proposition 25.2.4. For $n^{1/n} = 1 + h_n$ with $n \geq 2$, the binomial formula gives $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)} \rightarrow 0$. (3) Write $a = 1 + h$. For $n \geq 2k+2$ the binomial formula gives $a^n \geq \binom{n}{k+1} h^{k+1} \geq \frac{(n/2)^{k+1}}{(k+1)!} h^{k+1}$, since each of the $k+1$ factors $n, n-1, \dots, n-k$ is at least $n/2$. Hence $0 \leq n^k/a^n \leq 2^{k+1}(k+1)!/(h^{k+1}n) \rightarrow 0$. (4) Choose an integer $N > 2|a|$. For $n \geq N$, $|a|^n/n! \leq \frac{|a|^N}{N!} 2^{N-n}$, which tends to 0 by (1). Finally $n!/n^n = \frac{1}{n} \cdot \frac{2}{n} \cdots \frac{n}{n} \leq \frac{1}{n}$. $\square$

Proposition 25.2.10. *Every monotone sequence is regular. An increasing sequence converges to its supremum if it is bounded above, and diverges to $+\infty$ otherwise; a decreasing sequence converges to its infimum if it is bounded below, and diverges to $-\infty$ otherwise.*

Proof. Let (a_n) be increasing. If it is bounded above, let $s = \sup_n a_n$ (Definition 4.7.1). For $\varepsilon > 0$, $s - \varepsilon$ is not an upper bound, so $a_N > s - \varepsilon$ for some N , and then $s - \varepsilon < a_N \leq a_n \leq s$ for $n \geq N$. If it is not bounded above, for every M some $a_N > M$, and then $a_n \geq a_N > M$ for $n \geq N$. The decreasing case follows by passing to $(-a_n)$. $\square$

Lemma 25.2.11. *Every real sequence has a monotone subsequence.*

Proof. Call n a *peak* if $a_n \geq a_m$ for all $m > n$. If there are infinitely many peaks $n_0 < n_1 < \dots$, then $a_{n_0} \geq a_{n_1} \geq \dots$ is a decreasing subsequence. Otherwise let n_0 be larger than all peaks. As n_0 is not a peak, there is $n_1 > n_0$ with $a_{n_1} > a_{n_0}$; as n_1 is not a peak, there is $n_2 > n_1$ with $a_{n_2} > a_{n_1}$; continuing (by recursion), we obtain a strictly increasing subsequence. $\square$

Theorem 25.2.12 (Bolzano–Weierstrass). *Every bounded real sequence has a convergent subsequence. Every real sequence has a regular subsequence: a sequence that is unbounded above has a subsequence diverging to $+\infty$.*

Proof. By Lemma 25.2.11 there is a monotone subsequence, which is regular by Proposition 25.2.10, and convergent if the sequence is bounded. If (a_n) is unbounded above, choose recursively $n_{k+1} > n_k$ with $a_{n_{k+1}} > k + 1$, which is possible because only finitely many terms have index $\leq n_k$. $\square$

Theorem 25.2.13 (Cauchy criterion). *A real sequence converges if and only if it is a Cauchy sequence.*

Proof. If $a_n \rightarrow a$, then $|a_n - a_m| \leq |a_n - a| + |a - a_m| < \varepsilon$ when both are within $\varepsilon/2$ of a . Conversely, let (a_n) be Cauchy. Taking $\varepsilon = 1$ shows that (a_n) is bounded, so by Theorem 25.2.12 a subsequence converges to some a . Given $\varepsilon > 0$, choose N with $|a_n - a_m| < \varepsilon/2$ for $m, n \geq N$, and k with $n_k \geq N$ and $|a_{n_k} - a| < \varepsilon/2$. Then $|a_n - a| \leq |a_n - a_{n_k}| + |a_{n_k} - a| < \varepsilon$ for $n \geq N$. (This is a second proof of Proposition 4.7.6 for $\mathbb{R}$.) $\square$

Counterexample 25.2.14. It is not enough that consecutive terms get close. The partial sums $H_n = \sum_{k=1}^n 1/k$ of the harmonic series satisfy $H_{n+1} - H_n = 1/(n+1) \rightarrow 0$, but $H_{2n} - H_n = \sum_{k=n+1}^{2n} 1/k \geq n \cdot \frac{1}{2n} = \frac{1}{2}$, so (H_n) is not Cauchy; being increasing, it diverges to $+\infty$. Likewise $\sqrt{n+1} - \sqrt{n} = 1/(\sqrt{n+1} + \sqrt{n}) \rightarrow 0$ while $\sqrt{n} \rightarrow +\infty$.

Proposition 25.2.15 (Subsequence criterion). 1. $a_n \rightarrow a$ if and only if every subsequence of (a_n) has a further subsequence converging to a .

2. A bounded sequence all of whose convergent subsequences have the same limit a converges to a .

Proof. (1) One direction is Proposition 25.2.2(5). Conversely, if $a_n \not\rightarrow a$, there is $\varepsilon > 0$ such that $|a_n - a| \geq \varepsilon$ for infinitely many n ; these terms form a subsequence none of whose subsequences converges to a . (2) By Theorem 25.2.12, every subsequence is bounded and has a convergent subsequence, whose limit is a by hypothesis; apply (1). $\square$

Proposition 25.2.16 (Stolz–Cesàro). *Let (b_n) be strictly increasing with $b_n \rightarrow +\infty$, and let (a_n) be a real sequence with $\frac{a_{n+1} - a_n}{b_{n+1} - b_n} \rightarrow L \in \mathbb{R}$. Then $a_n/b_n \rightarrow L$. In particular (Cesàro), if $a_n \rightarrow L$, then the averages $\frac{1}{n}(a_1 + \dots + a_n) \rightarrow L$.*

Proof. Let $\varepsilon > 0$ and choose N such that $b_N > 0$ and $|(a_{k+1} - a_k) - L(b_{k+1} - b_k)| \leq \varepsilon(b_{k+1} - b_k)$ for $k \geq N$. Summing over $N \leq k < n$ gives $|a_n - a_N - L(b_n - b_N)| \leq \varepsilon(b_n - b_N) \leq \varepsilon b_n$. Dividing by $b_n > 0$,

$$\left| \frac{a_n}{b_n} - L \right| \leq \varepsilon + \frac{|a_N - Lb_N|}{b_n},$$

and the last term is less than ε for large n , as $b_n \rightarrow +\infty$. For Cesàro, apply this to $a'_n = a_1 + \dots + a_n$ and $b_n = n$. $\square$

Counterexample 25.2.17. The converse of Cesàro's theorem fails: $a_n = (-1)^n$ does not converge, but its averages tend to 0, since the partial sums are 0 or -1 .

25.3 Limits of functions

To treat finite and infinite limits at once we use the *extended real line* $\bar{\mathbb{R}} = \mathbb{R} \cup \{-\infty, +\infty\}$, ordered so that $-\infty < x < +\infty$ for $x \in \mathbb{R}$.

Definition 25.3.1. For $a \in \bar{\mathbb{R}}$ and $\delta > 0$, the *basic neighbourhood* $U_\delta(a)$ is $(a - \delta, a + \delta)$ if $a \in \mathbb{R}$, $(1/\delta, +\infty)$ if $a = +\infty$, and $(-\infty, -1/\delta)$ if $a = -\infty$. Let $D \subseteq \mathbb{R}$. A point $a \in \bar{\mathbb{R}}$ is an *accumulation point* of D if every $U_\delta(a)$ contains a point of $D \setminus \{a\}$. A point of D that is not an accumulation point is *isolated*.

A sequence (x_n) tends to $a \in \bar{\mathbb{R}}$ if and only if for every $\delta > 0$, eventually $x_n \in U_\delta(a)$: for $a \in \mathbb{R}$ this is convergence, and for $a = \pm\infty$ divergence to $\pm\infty$. Every $a \in \bar{\mathbb{R}}$ that is an accumulation point of D is the limit of a sequence in $D \setminus \{a\}$: choose $x_n \in U_{1/(n+1)}(a) \cap D \setminus \{a\}$.

Definition 25.3.2. Let $f: D \rightarrow \mathbb{R}$, let $a \in \bar{\mathbb{R}}$ be an accumulation point of D , and $L \in \bar{\mathbb{R}}$. We write $\lim_{x \rightarrow a} f(x) = L$, or $f(x) \rightarrow L$ as $x \rightarrow a$, if for every $\varepsilon > 0$ there is $\delta > 0$ such that $f(x) \in U_\varepsilon(L)$ for all $x \in D \cap U_\delta(a)$ with $x \neq a$. For $a, L \in \mathbb{R}$ this reads: for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - L| < \varepsilon$ whenever $x \in D$ and $0 < |x - a| < \delta$. The *right limit* $\lim_{x \rightarrow a^+} f(x)$ is the limit of the restriction of f to $D \cap (a, +\infty)$, when a is an accumulation point of that set, and the *left limit* $\lim_{x \rightarrow a^-} f(x)$ is defined similarly. The same definitions apply to $\mathbb{C}$ - or $\mathbb{R}^m$ -valued functions and finite limits, with $|\cdot|$ the modulus or the Euclidean norm.

The value of f at a , if $a \in D$, plays no role in the definition.

Proposition 25.3.3. Let $f: D \rightarrow \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D .

1. The limit of f at a , if it exists, is unique.
2. If $\lim_{x \rightarrow a} f(x) = L \in \mathbb{R}$, then f is bounded on $D \cap U_\delta(a) \setminus \{a\}$ for some $\delta > 0$; if $L > 0$, then $f(x) > L/2$ for $x \in D \cap U_\delta(a) \setminus \{a\}$ for some $\delta > 0$.
3. If $a \in \mathbb{R}$ is an accumulation point of both $D \cap (a, +\infty)$ and $D \cap (-\infty, a)$, then $\lim_{x \rightarrow a} f(x) = L$ if and only if both one-sided limits exist and equal L .

Proof. (1) If $L \neq L'$ are both limits, choose ε with $U_\varepsilon(L) \cap U_\varepsilon(L') = \emptyset$, and δ working for both; since a is an accumulation point, some $x \in D \cap U_\delta(a) \setminus \{a\}$ exists, and $f(x)$ lies in both neighbourhoods. (2) Take $\varepsilon = 1$, respectively $\varepsilon = L/2$. (3) If both one-sided limits equal L , take the smaller of the two δ ; the converse is clear. $\square$

Theorem 25.3.4 (Sequential characterisation of limits). Let $f: D \rightarrow \mathbb{R}$, let $a \in \bar{\mathbb{R}}$ be an accumulation point of D , and $L \in \bar{\mathbb{R}}$. The following are equivalent:

1. $\lim_{x \rightarrow a} f(x) = L$;
2. for every sequence (x_n) in $D \setminus \{a\}$ with $x_n \rightarrow a$, we have $f(x_n) \rightarrow L$.

Proof. (1) $\Rightarrow$ (2). Let $\varepsilon > 0$ and choose δ as in Definition 25.3.2. Eventually $x_n \in U_\delta(a)$, and $x_n \in D \setminus \{a\}$, so eventually $f(x_n) \in U_\varepsilon(L)$.

(2) $\Rightarrow$ (1). Suppose (1) fails. Then there is $\varepsilon > 0$ such that for every $\delta > 0$ some $x \in D \cap U_\delta(a) \setminus \{a\}$ has $f(x) \notin U_\varepsilon(L)$. Choosing such a point x_n for $\delta = 1/(n+1)$, for each n (this uses the axiom of countable choice), we obtain a sequence in $D \setminus \{a\}$ with $x_n \rightarrow a$ and $f(x_n) \notin U_\varepsilon(L)$ for all n , so $f(x_n) \not\rightarrow L$, contradicting (2). $\square$

Theorem 25.3.4, sometimes called the *bridge theorem*, transfers every result about limits of sequences to limits of functions, and it is the standard way to show that a limit does *not* exist: exhibit two sequences tending to a along which f has different limits.

Corollary 25.3.5 (Rules for limits of functions). Let $f, g, h: D \rightarrow \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D .

1. If $f(x) \rightarrow L$ and $g(x) \rightarrow M$ with $L, M \in \mathbb{R}$, then $f + g \rightarrow L + M$, $fg \rightarrow LM$, and, if $M \neq 0$, $f/g \rightarrow L/M$ (where f/g is defined near a).
2. If $f \leq g$ near a and $f \rightarrow L$, $g \rightarrow M$, then $L \leq M$.
3. (Squeeze) If $f \leq h \leq g$ near a and $f(x) \rightarrow L$, $g(x) \rightarrow L$, then $h(x) \rightarrow L$.
4. The rules for infinite limits of Proposition 25.2.7 hold for functions.

Proof. Apply Theorem 25.3.4 and Propositions 25.2.4, 25.2.5 and 25.2.7 to $f(x_n)$, $g(x_n)$, $h(x_n)$ for sequences $x_n \rightarrow a$ in $D \setminus \{a\}$. For the quotient, $g(x) \neq 0$ near a by Proposition 25.3.3(2) applied to $|g|$. $\square$

Proposition 25.3.6 (Limits of composite functions). Let $f: D \rightarrow \mathbb{R}$ and $g: E \rightarrow \mathbb{R}$ with $f(D) \subseteq E$. Suppose $\lim_{x \rightarrow a} f(x) = b$ and $\lim_{y \rightarrow b} g(y) = L$, and that one of the following holds:

1. $f(x) \neq b$ for all $x \in D \setminus \{a\}$ near a ;
2. $b \in E$ and $g(b) = L$.

Then $\lim_{x \rightarrow a} g(f(x)) = L$.

Proof. Let $x_n \rightarrow a$ in $D \setminus \{a\}$. Then $f(x_n) \rightarrow b$ by Theorem 25.3.4. In case (1), eventually $f(x_n) \in E \setminus \{b\}$, so $g(f(x_n)) \rightarrow L$ by the same theorem. In case (2), given $\varepsilon > 0$ choose η with $g(y) \in U_\varepsilon(L)$ for $y \in E \cap U_\eta(b)$, including $y = b$; eventually $f(x_n) \in U_\eta(b)$, so $g(f(x_n)) \in U_\varepsilon(L)$. Apply the theorem again. $\square$

Counterexample 25.3.7. Neither hypothesis can be dropped. Let $f(x) = 0$ for all x , and $g(y) = 0$ for $y \neq 0$, $g(0) = 1$. Then $\lim_{x \rightarrow 0} f(x) = 0$ and $\lim_{y \rightarrow 0} g(y) = 0$, but $g(f(x)) = 1$ for all x .

Proposition 25.3.8 (Cauchy criterion for functions). Let $f: D \rightarrow \mathbb{R}$ and let $a \in \bar{\mathbb{R}}$ be an accumulation point of D . Then $\lim_{x \rightarrow a} f(x)$ exists in $\mathbb{R}$ if and only if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ for all $x, y \in D \cap U_\delta(a) \setminus \{a\}$.

Proof. If $f(x) \rightarrow L$, take δ for $\varepsilon/2$ and use the triangle inequality. Conversely, for every sequence $x_n \rightarrow a$ in $D \setminus \{a\}$, the sequence $(f(x_n))$ is Cauchy, hence converges (Theorem 25.2.13). If $x_n \rightarrow a$ and $y_n \rightarrow a$ are two such sequences, the interleaved sequence $x_0, y_0, x_1, y_1, \dots$ also tends to a , so $(f(x_n))$ and $(f(y_n))$ have the same limit, being subsequences of a convergent sequence. So all these limits equal a common L , and $f(x) \rightarrow L$ by Theorem 25.3.4. $\square$

Proposition 25.3.9 (Monotone functions). Let $f: (a, b) \rightarrow \mathbb{R}$ be increasing, with $-\infty \leq a < b \leq +\infty$. For every $c \in (a, b)$ the one-sided limits exist and

$$\lim_{x \rightarrow c^-} f(x) = \sup_{a < x < c} f(x) \leq f(c) \leq \inf_{c < x < b} f(x) = \lim_{x \rightarrow c^+} f(x),$$

and $\lim_{x \rightarrow b^-} f(x) = \sup f \in \mathbb{R} \cup \{+\infty\}$, $\lim_{x \rightarrow a^+} f(x) = \inf f \in \mathbb{R} \cup \{-\infty\}$. The analogous statements hold for decreasing functions.

Proof. The set $\{f(x) : a < x < c\}$ is bounded above by $f(c)$; let s be its supremum, so $s \leq f(c)$. Given $\varepsilon > 0$ there is $x_0 < c$ with $f(x_0) > s - \varepsilon$, and for $x_0 < x < c$, $s - \varepsilon < f(x_0) \leq f(x) \leq s$. So $\lim_{x \rightarrow c^-} f(x) = s$. The other statements are proved in the same way (for $b = +\infty$ or an unbounded supremum, use neighbourhoods of $+\infty$), and the decreasing case follows by passing to $-f$. $\square$

- Counterexample 25.3.10.** 1. The sign function $\operatorname{sgn}(x) = x/|x|$ on $\mathbb{R} \setminus \{0\}$ has one-sided limits ± 1 at 0, and no limit there.
2. $1/x$ has no limit at 0: its one-sided limits are $+\infty$ and $-\infty$.
3. The *Dirichlet function* $\mathbf{1}_{\mathbb{Q}}$, equal to 1 on $\mathbb{Q}$ and 0 on $\mathbb{R} \setminus \mathbb{Q}$, has no limit at any $a \in \mathbb{R}$. Both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$: $\mathbb{Q}$ by Proposition 4.7.3, and $\mathbb{R} \setminus \mathbb{Q}$ because $q\sqrt{2}$ is irrational for rational $q \neq 0$ (Proposition 4.6.4) and $x/\sqrt{2} < q < y/\sqrt{2}$ is possible for $x < y$. So there are rational $x_n \rightarrow a$ and irrational $y_n \rightarrow a$, all $\neq a$, and $\mathbf{1}_{\mathbb{Q}}(x_n) \rightarrow 1$, $\mathbf{1}_{\mathbb{Q}}(y_n) \rightarrow 0$.

25.4 Continuous functions of a real variable

Definition 25.4.1. Let $D \subseteq \mathbb{R}$ and $f: D \rightarrow \mathbb{R}$. Then f is *continuous at* $a \in D$ if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(a)| < \varepsilon$ for all $x \in D$ with $|x - a| < \delta$, and *continuous* if it is continuous at every point of D .

This is continuity for the metric of $\mathbb{R}$ restricted to D (Proposition 16.3.5). If $a \in D$ is an accumulation point of D , then f is continuous at a if and only if $\lim_{x \rightarrow a} f(x) = f(a)$; if a is isolated, every f is continuous at a .

Proposition 25.4.2. Let $f, g: D \rightarrow \mathbb{R}$ and $a \in D$.

1. f is continuous at a if and only if $f(x_n) \rightarrow f(a)$ for every sequence (x_n) in D with $x_n \rightarrow a$.
2. If f and g are continuous at a , so are $f + g$, fg , $|f|$, $\max(f, g)$, $\min(f, g)$, and f/g if $g(a) \neq 0$.
3. If f is continuous at a and $h: E \rightarrow \mathbb{R}$, with $f(D) \subseteq E$, is continuous at $f(a)$, then $h \circ f$ is continuous at a .
4. Polynomials are continuous on $\mathbb{R}$, and rational functions are continuous where the denominator does not vanish.

Proof. (1) The proof of Theorem 25.3.4 applies verbatim, now allowing $x_n = a$. (2) follows from (1) and Proposition 25.2.4, since $\max(f, g) = \frac{1}{2}(f + g + |f - g|)$ and $\min(f, g) = \frac{1}{2}(f + g - |f - g|)$. (3) If $x_n \rightarrow a$, then $f(x_n) \rightarrow f(a)$ and so $h(f(x_n)) \rightarrow h(f(a))$. (4) Constants and the identity are continuous; apply (2). $\square$

Definition 25.4.3. Let f be defined on an open interval containing a , and discontinuous at a . The discontinuity is *removable* if $\lim_{x \rightarrow a} f(x)$ exists in $\mathbb{R}$ (and differs from $f(a)$); it is a *jump* if both one-sided limits exist in $\mathbb{R}$ and differ; otherwise it is *essential*.

Example 25.4.4. The function equal to 1 for $x \neq 0$ and 0 at 0 has a removable discontinuity at 0. The function equal to $\operatorname{sgn}(x)$ for $x \neq 0$ and 0 at 0 has a jump. The function equal to $1/x$ for $x \neq 0$ and 0 at 0 has an essential discontinuity, and the Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ has an essential discontinuity at every point (Counterexample 25.3.10).

Theorem 25.4.5. Let $f: I \rightarrow \mathbb{R}$ be monotone on an interval I . Then every discontinuity of f at an interior point of I is a jump, and the set of points of discontinuity of f is countable.

Proof. We may assume f increasing. At an interior point c the one-sided limits $f(c^-) \leq f(c) \leq f(c^+)$ exist by Proposition 25.3.9, and f is continuous at c if and only if $f(c^-) = f(c^+)$. So every interior discontinuity is a jump, with nonempty open interval $J_c = (f(c^-), f(c^+))$. If $c < c'$ are interior discontinuity points and $c < x < c'$, then $f(c^+) \leq f(x) \leq f(c'^-)$, so

$J_c \cap J_{c'} = \emptyset$. The set $\mathbb{Q}$ is countable (see the proof of Proposition 4.8.6); fix an injection $\iota: \mathbb{Q} \rightarrow \mathbb{N}$ and send c to the rational number $q \in J_c$ with the smallest $\iota(q)$, which exists as J_c contains rational numbers. This map is injective, since the J_c are disjoint, so the interior discontinuities form a countable set. Adding the at most two endpoints of I keeps it countable. $\square$

Example 25.4.6 (Thomae's function). Let $T(x) = 1/q$ if $x = p/q$ with $p \in \mathbb{Z}$, $q \geq 1$ coprime, and $T(x) = 0$ for irrational x . Then T is continuous at every irrational number and discontinuous at every rational number.

Indeed, at a rational a , $T(a) > 0$, but there are irrational $y_n \rightarrow a$ (Counterexample 25.3.10) with $T(y_n) = 0$. At an irrational a , let $\varepsilon > 0$. The rationals p/q in $(a - 1, a + 1)$ with $q \leq 1/\varepsilon$ are finitely many, and none equals a ; let δ be smaller than 1 and than their distances to a . For $|x - a| < \delta$, either x is irrational and $T(x) = 0$, or $x = p/q$ with $q > 1/\varepsilon$, so $|T(x) - T(a)| = 1/q < \varepsilon$.

Remark 25.4.7. There is no function $\mathbb{R} \rightarrow \mathbb{R}$ that is continuous exactly at the rational numbers: the set of points of continuity of any function is a countable intersection of open sets, and $\mathbb{Q}$ is not, by the Baire category theorem (Theorem 16.7.7). By contrast, $x\mathbf{1}_{\mathbb{Q}}(x)$ is continuous exactly at 0 (Exercise 25.14.9).

Theorem 25.4.8 (Bolzano's theorem on zeros). Let $f: [a, b] \rightarrow \mathbb{R}$ be continuous with $f(a) < 0 < f(b)$. Then $f(c) = 0$ for some $c \in (a, b)$.

Proof. We construct intervals by bisection. Let $a_0 = a$, $b_0 = b$. Given $[a_n, b_n]$ with $f(a_n) < 0 \leq f(b_n)$, let $m = (a_n + b_n)/2$; if $f(m) < 0$ put $a_{n+1} = m$, $b_{n+1} = b_n$, and otherwise $a_{n+1} = a_n$, $b_{n+1} = m$. Then $f(a_{n+1}) < 0 \leq f(b_{n+1})$, (a_n) is increasing, (b_n) is decreasing, and $b_n - a_n = (b - a)/2^n \rightarrow 0$. By Proposition 25.2.10 both converge, to the same limit $c \in [a, b]$. By continuity, $f(c) = \lim f(a_n) \leq 0$ and $f(c) = \lim f(b_n) \geq 0$ (Proposition 25.2.5(1)), so $f(c) = 0$, and $c \neq a, b$. $\square$

Corollary 25.4.9. Let I be an interval and $f: I \rightarrow \mathbb{R}$ continuous.

1. (Intermediate value theorem) f takes every value between any two of its values, so $f(I)$ is an interval.
2. If f is injective, it is strictly monotone.
3. Every continuous map $f: [a, b] \rightarrow [a, b]$ has a fixed point.
4. Every real polynomial of odd degree has a real root.

Proof. (1) If $f(x) < y < f(z)$, apply Theorem 25.4.8 to $f - y$ on the interval between x and z (to $y - f$ if $x > z$). This is also the case $X = I$ of Corollary 16.5.5. (2) Let $a < b$ in I , and assume $f(a) < f(b)$ (otherwise replace f by $-f$). Let $x < y$ in I and put $h(t) = f((1-t)b + ty) - f((1-t)a + tx)$ for $t \in [0, 1]$. Since $(1-t)a + tx < (1-t)b + ty$ and f is injective, $h(t) \neq 0$ for all t . As h is continuous and $h(0) > 0$, (1) gives $h(1) = f(y) - f(x) > 0$. So f is strictly increasing. (3) $g(x) = f(x) - x$ satisfies $g(a) \geq 0 \geq g(b)$; if neither is 0, apply (1). (4) We may take $p(x) = x^{2k+1} + c_{2k}x^{2k} + \cdots + c_0$. For $|x| \geq 1 + \sum |c_i|$ we have $|c_{2k}x^{2k} + \cdots + c_0| \leq \sum |c_i||x|^{2k} < |x|^{2k+1}$, so $p(x)$ has the sign of x there. Apply (1). $\square$

Theorem 25.4.10 (Weierstrass). A continuous function $f: [a, b] \rightarrow \mathbb{R}$ is bounded and attains its maximum and its minimum.

Proof. If f were unbounded, there would be $x_n \in [a, b]$ with $|f(x_n)| > n$. By Theorem 25.2.12 a subsequence converges to some c , and $c \in [a, b]$ by Proposition 25.2.5(1). Then $f(x_{n_k}) \rightarrow f(c)$, so $(f(x_{n_k}))$ is bounded, a contradiction. Let $s = \sup f$ and choose x_n with $f(x_n) > s - 1/(n+1)$; a subsequence converges to some $c \in [a, b]$, and $f(c) = \lim f(x_{n_k}) = s$ by the squeeze theorem. The minimum is the maximum of $-f$. (This is the case $X = [a, b]$ of Corollary 16.6.5.) $\square$

Counterexample 25.4.11. Each hypothesis of Theorem 25.4.10 is needed: x on $(0, 1)$ attains neither its supremum nor its infimum; $1/x$ on $(0, 1]$ is unbounded; x on $[0, +\infty)$ is unbounded; and the function on $[0, 1]$ equal to x for $x < 1$ and to 0 at 1 has supremum 1, which is not attained.

Definition 25.4.12. A function $f: D \rightarrow \mathbb{R}$ is *uniformly continuous* if for every $\varepsilon > 0$ there is $\delta > 0$ with $|f(x) - f(y)| < \varepsilon$ for all $x, y \in D$ with $|x - y| < \delta$. It is *Lipschitz* with constant $K \geq 0$ if $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in D$.

Proposition 25.4.13. Let $f: D \rightarrow \mathbb{R}$.

1. Lipschitz functions are uniformly continuous, and uniformly continuous functions are continuous.
2. f is uniformly continuous if and only if $f(x_n) - f(y_n) \rightarrow 0$ for all sequences $(x_n), (y_n)$ in D with $x_n - y_n \rightarrow 0$.
3. (Heine–Cantor) If $D = [a, b]$ and f is continuous, then f is uniformly continuous.
4. A uniformly continuous function maps Cauchy sequences to Cauchy sequences. A uniformly continuous $f: (a, b) \rightarrow \mathbb{R}$ on a bounded interval extends to a continuous function on $[a, b]$.

Proof. (1) Take $\delta = \varepsilon/K$ (any δ if $K = 0$); the second statement is clear. (2) If f is uniformly continuous and $x_n - y_n \rightarrow 0$, eventually $|x_n - y_n| < \delta$ and so $|f(x_n) - f(y_n)| < \varepsilon$. If f is not uniformly continuous, there is $\varepsilon > 0$ and, for each n , points x_n, y_n with $|x_n - y_n| < 1/(n+1)$ and $|f(x_n) - f(y_n)| \geq \varepsilon$. (3) If not, take x_n, y_n as in the last sentence. By Theorem 25.2.12, $x_{n_k} \rightarrow c \in [a, b]$, and then $y_{n_k} \rightarrow c$. By continuity $f(x_{n_k}) - f(y_{n_k}) \rightarrow f(c) - f(c) = 0$, a contradiction. (4) If (x_n) is Cauchy and δ corresponds to ε , eventually $|x_n - x_m| < \delta$, so $|f(x_n) - f(x_m)| < \varepsilon$. For the extension to a : for any sequence $x_n \rightarrow a$ in (a, b) , $(f(x_n))$ is Cauchy, hence converges; by (2), two such sequences give the same limit (as $x_n - y_n \rightarrow 0$), say L . By Theorem 25.3.4, $\lim_{x \rightarrow a^+} f(x) = L$, and setting $f(a) = L$ gives a function continuous at a ; similarly at b . $\square$

Counterexample 25.4.14. 1. $f(x) = x^2$ on $\mathbb{R}$ is continuous but not uniformly continuous: with $x_n = n + 1$ and $y_n = n + 1 + 1/(n + 1)$, $x_n - y_n \rightarrow 0$ but $f(y_n) - f(x_n) \geq 2$.

2. $f(x) = 1/x$ on $(0, 1)$ is continuous but not uniformly continuous: with $x_n = 1/(n + 2)$ and $y_n = 1/(n + 3)$, $f(y_n) - f(x_n) = 1$; it has no continuous extension to $[0, 1)$, in accordance with Proposition 25.4.13(4).
3. $f(x) = \sqrt{x}$ on $[0, +\infty)$ is uniformly continuous, since $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$, but not Lipschitz, since $\sqrt{x}/x \rightarrow +\infty$ as $x \rightarrow 0^+$. So the implications in (1) are strict.

25.5 Upper and lower limits, and series

Definition 25.5.1. For a real sequence (a_n) let $\limsup_n a_n = \inf_n \sup_{k \geq n} a_k$ and $\liminf_n a_n = \sup_n \inf_{k \geq n} a_k$, with values in the extended real line.

Proposition 25.5.2. 1. A bounded monotone real sequence converges, to its supremum or infimum.

2. (a_n) converges to $a \in \mathbb{R}$ if and only if $\limsup a_n = \liminf a_n = a$.
3. (Bolzano–Weierstrass) Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. (1) is Proposition 25.2.10. (2) The sequences $\sup_{k \geq n} a_k$ and $\inf_{k \geq n} a_k$ are monotone and bracket a_n . If both converge to a , so does a_n ; conversely, if $|a_k - a| < \varepsilon$ for $k \geq N$, both lie within ε of a for $n \geq N$. (3) A bounded set lies in a closed cube, which is compact (Theorem 16.6.4) and hence sequentially compact (Theorem 16.7.4). $\square$

Definition 25.5.3. Let V be a normed space. A series $\sum_n v_n$ converges to s if the partial sums $s_N = \sum_{n \leq N} v_n$ converge to s , and it converges absolutely if $\sum_n \|v_n\| < \infty$.

Proposition 25.5.4. Let V be a Banach space.

1. An absolutely convergent series converges, and every rearrangement $\sum_n v_{\sigma(n)}$ (σ a bijection of $\mathbb{N}$) converges to the same sum.
2. (Comparison) If $\|v_n\| \leq c_n$ and $\sum c_n < \infty$, then $\sum v_n$ converges absolutely.
3. (Root and ratio tests) If $\limsup \|v_n\|^{1/n} < 1$, or $v_n \neq 0$ and $\limsup \|v_{n+1}\|/\|v_n\| < 1$, then $\sum v_n$ converges absolutely. If $\limsup \|v_n\|^{1/n} > 1$, it diverges.
4. (Geometric series) For $z \in \mathbb{C}$ with $|z| < 1$, $\sum_n z^n = 1/(1 - z)$.
5. (Weierstrass M-test) If $f_n: X \rightarrow V$ satisfy $\sup_x \|f_n(x)\| \leq M_n$ with $\sum M_n < \infty$, then $\sum f_n$ converges uniformly.

Proof. (1) $\|s_M - s_N\| \leq \sum_{N < n \leq M} \|v_n\|$, so the partial sums are Cauchy. For a rearrangement, given ε choose N with $\sum_{n \geq N} \|v_n\| < \varepsilon$ and then M with $\{0, \dots, N-1\} \subseteq \sigma(\{0, \dots, M\})$; for $m \geq M$, $\|\sum_{k \leq m} v_{\sigma(k)} - s_{N-1}\| \leq \sum_{n \geq N} \|v_n\| < \varepsilon$, and $\|s_{N-1} - s\| \leq \varepsilon$. (2) follows from (1). (3) If $\limsup \|v_n\|^{1/n} < q < 1$, then $\|v_n\| \leq q^n$ for large n ; compare with $\sum q^n$ using (4). If the ratio $\limsup$ is $< q < 1$, then $\|v_n\| \leq Cq^n$ for large n . If $\limsup \|v_n\|^{1/n} > 1$, then $\|v_n\| > 1$ infinitely often, so $v_n \not\rightarrow 0$ and the partial sums are not Cauchy. (4) $(1 - z) \sum_{n \leq N} z^n = 1 - z^{N+1}$, and $|z|^N \rightarrow 0$: if $0 < |z| < 1$ write $1/|z| = 1 + h$, so $|z|^{-N} \geq 1 + Nh$. (5) The partial sums are bounded and uniformly Cauchy, and the space $B(X, V)$ of bounded maps is complete (Proposition 16.7.2). $\square$

Lemma 25.5.5 (Alternating series). If $t_0 \geq t_1 \geq \dots \geq 0$ and $t_k \rightarrow 0$, then $S = \sum_k (-1)^k t_k$ converges and $S_{2m+1} \leq S \leq S_{2m}$ for all m , where S_N are the partial sums.

Proof. $S_{2m+2} - S_{2m} = -t_{2m+1} + t_{2m+2} \leq 0$ and $S_{2m+3} - S_{2m+1} = t_{2m+2} - t_{2m+3} \geq 0$, and $S_{2m+1} = S_{2m} - t_{2m+1} \leq S_{2m}$. So the even partial sums decrease, the odd ones increase, every odd one is below every even one, and their difference tends to 0. $\square$

Proposition 25.5.6 (Cauchy product). If $\sum a_n$ and $\sum b_n$ are absolutely convergent series in $\mathbb{C}$, then $c_n = \sum_{k=0}^n a_k b_{n-k}$ defines an absolutely convergent series with $\sum c_n = (\sum a_n)(\sum b_n)$.

Proof. Let $A = \sum |a_n|$ and $B = \sum |b_n|$. Then $\sum_{n \leq N} |c_n| \leq AB$, so $\sum c_n$ converges absolutely. The difference $\sum_{n \leq 2N} c_n - (\sum_{k \leq N} a_k)(\sum_{l \leq N} b_l)$ is the sum of $a_k b_l$ over $k + l \leq 2N$ with $k > N$ or $l > N$, bounded by $B \sum_{k > N} |a_k| + A \sum_{l > N} |b_l| \rightarrow 0$. $\square$

Theorem 25.5.7. Let $a_n \in \mathbb{C}$ and $R = 1/\limsup_n |a_n|^{1/n} \in [0, \infty]$, the radius of convergence. The power series $\sum_n a_n z^n$ converges absolutely for $|z| < R$, uniformly on $|z| \leq r$ for every $r < R$, and diverges for $|z| > R$. The series $\sum_n n a_n z^{n-1}$ and $\sum_n n^2 a_n z^n$ have the same radius of convergence.

Proof. For $|z| \leq r < R$ choose $r < q < R$; then $|a_n| \leq q^{-n}$ for large n , so $|a_n z^n| \leq (r/q)^n$, and the M-test applies. For $|z| > R$, $\limsup |a_n z^n|^{1/n} > 1$. Finally $n^{1/n} \rightarrow 1$: writing $n^{1/n} = 1 + h_n$, the binomial formula gives $n \geq \binom{n}{2} h_n^2$, so $h_n^2 \leq 2/(n-1)$. Hence $\limsup |n^k a_n|^{1/n} = \limsup |a_n|^{1/n}$. $\square$

25.6 Differentiation in one variable

Definition 25.6.1. Let $I \subseteq \mathbb{R}$ be an interval with more than one point and $f: I \rightarrow \mathbb{K}^m$. Then f is *differentiable* at $a \in I$ with derivative $f'(a)$ if $(f(x) - f(a))/(x - a) \rightarrow f'(a)$ as $x \rightarrow a$ in $I \setminus \{a\}$. Equivalently, there is a function $\phi: I \rightarrow \mathbb{K}^m$, continuous at a , with $f(x) - f(a) = \phi(x)(x - a)$; then $f'(a) = \phi(a)$. The same definition with I replaced by an open subset of $\mathbb{C}$ and x complex defines the *complex derivative*.

Proposition 25.6.2. *Differentiable functions are continuous. Sums, products and quotients (where the denominator does not vanish) of differentiable scalar functions are differentiable with the usual formulas, and if f is differentiable at a and g at $f(a)$, then $(g \circ f)'(a) = g'(f(a))f'(a)$.*

Proof. Write $f(x) - f(a) = \phi(x)(x - a)$ and $g(y) - g(b) = \psi(y)(y - b)$ with ϕ, ψ continuous at $a, b = f(a)$. Continuity is clear. For products, $f g(x) - f g(a) = (\phi(x)g(x) + f(a)\chi(x))(x - a)$ if $g(x) - g(a) = \chi(x)(x - a)$. For $1/f$, $1/f(x) - 1/f(a) = -\phi(x)(x - a)/(f(x)f(a))$. For composites, $g(f(x)) - g(f(a)) = \psi(f(x))\phi(x)(x - a)$ and $\psi \circ f$ is continuous at a . Evaluate the continuous factors at a . $\square$

Definition 25.6.3. Let $f: I \rightarrow \mathbb{R}$ on an interval I . Then f has a *local maximum* at $c \in I$ if there is $\delta > 0$ with $f(x) \leq f(c)$ for all $x \in I$ with $|x - c| < \delta$, a *strict local maximum* if $f(x) < f(c)$ for these $x \neq c$, and similarly for (strict) local minima. A *local extremum* is a local maximum or minimum.

Lemma 25.6.4 (Fermat). *If $f: I \rightarrow \mathbb{R}$ has a local extremum at an interior point c of I and is differentiable at c , then $f'(c) = 0$.*

Proof. Say c is a local maximum. For $x > c$ close to c , $(f(x) - f(c))/(x - c) \leq 0$, so $f'(c) \leq 0$ by Corollary 25.3.5(2); for $x < c$ the quotient is ≥ 0 , so $f'(c) \geq 0$. $\square$

Counterexample 25.6.5. The converse fails: $f(x) = x^3$ has $f'(0) = 0$ and no extremum at 0. The hypothesis that c is interior is needed: $f(x) = x$ on $[0, 1]$ has a maximum at 1 with $f'(1) = 1$. And $|x|$ has a minimum at 0, where it is not differentiable.

Theorem 25.6.6 (Mean value theorem). *Let $f, g: [a, b] \rightarrow \mathbb{R}$ be continuous and differentiable on (a, b) .*

1. (Rolle) *If $f(a) = f(b)$, then $f'(\xi) = 0$ for some $\xi \in (a, b)$.*
2. *There is $\xi \in (a, b)$ with $f(b) - f(a) = f'(\xi)(b - a)$, and (Cauchy) some η with $(f(b) - f(a))g'(\eta) = (g(b) - g(a))f'(\eta)$.*
3. *If $f' = 0$ on (a, b) , f is constant; if $f' \geq 0$ (> 0), f is increasing (strictly increasing).*

Proof. (1) By Corollary 16.6.5, f attains a maximum and a minimum; if both are at endpoints, f is constant. Otherwise one is at some $\xi \in (a, b)$, and $f'(\xi) = 0$ by Lemma 25.6.4. (2) Apply (1) to $f(x) - \frac{f(b)-f(a)}{b-a}(x - a)$, and to $(f(b) - f(a))g(x) - (g(b) - g(a))f(x)$. (3) Apply (2) on subintervals. $\square$

Proposition 25.6.7. *Let $f: I \rightarrow \mathbb{R}$ be continuous and strictly monotone. Then $J = f(I)$ is an interval, $f^{-1}: J \rightarrow I$ is continuous, and if f is differentiable at a with $f'(a) \neq 0$, then f^{-1} is differentiable at $f(a)$ with derivative $1/f'(a)$.*

Proof. J is an interval by Corollary 16.5.5. A strictly monotone bijection of intervals maps subintervals onto subintervals, so f^{-1} maps open subintervals of J (relative to J) to open subintervals of I , and is continuous. Write $f(x) - f(a) = \phi(x)(x - a)$; strict monotonicity gives $\phi(x) \neq 0$ for $x \neq a$, and $\phi(a) = f'(a) \neq 0$. So $f^{-1}(y) - a = (y - f(a))/\phi(f^{-1}(y))$ with $1/(\phi \circ f^{-1})$ continuous at $f(a)$. $\square$

Proposition 25.6.8 (Limits of derivatives). *Let $f: [a, b) \rightarrow \mathbb{R}$ be continuous, differentiable on (a, b) , and suppose $\lim_{x \rightarrow a^+} f'(x) = L \in \mathbb{R}$. Then f is differentiable at a (from the right) with $f'(a) = L$. Consequently, if f is differentiable on an open interval containing c and f' has one-sided limits at c , both equal $f'(c)$: a derivative has neither removable nor jump discontinuities.*

Proof. For $a < x < b$, the mean value theorem gives $\xi_x \in (a, x)$ with $(f(x) - f(a))/(x - a) = f'(\xi_x)$. If $x_n \rightarrow a^+$, then $\xi_{x_n} \rightarrow a^+$ by the squeeze theorem, so $f'(\xi_{x_n}) \rightarrow L$; by Theorem 25.3.4, the difference quotient tends to L . For the second statement, apply this on each side of c . $\square$

Theorem 25.6.9 (de l'Hôpital). *Let $f, g: (a, b) \rightarrow \mathbb{R}$ be differentiable, with $-\infty \leq a < b \leq +\infty$, and $g'(x) \neq 0$ for all x . Suppose that $\lim_{x \rightarrow a^+} f'(x)/g'(x) = L \in \mathbb{R}$ and that either*

1. $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^+} g(x) = 0$, or
2. $\lim_{x \rightarrow a^+} |g(x)| = +\infty$ and $L \in \mathbb{R}$.

Then $\lim_{x \rightarrow a^+} f(x)/g(x) = L$. The same holds for left limits at b , and hence for two-sided limits.

Proof. Since $g' \neq 0$, g is injective on (a, b) by Rolle's theorem, so $g(x) \neq g(y)$ for $x \neq y$, and by the Cauchy mean value theorem (Theorem 25.6.6(2)), for $x < y$ in (a, b) there is $\xi \in (x, y)$ with

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(\xi)}{g'(\xi)}. \quad (*)$$

Case (1) with $a \in \mathbb{R}$: extend f and g by $f(a) = g(a) = 0$; they become continuous on $[a, b)$, and $(*)$ holds for $x = a$ as well, with $g(y) \neq 0$ for $y > a$ by Rolle's theorem. Let $y_n \rightarrow a^+$ and choose $\xi_n \in (a, y_n)$ with $f(y_n)/g(y_n) = f'(\xi_n)/g'(\xi_n)$. Then $\xi_n \rightarrow a^+$, so $f(y_n)/g(y_n) \rightarrow L$, and Theorem 25.3.4 gives the claim. If $a = -\infty$, apply the case just proved to $F(t) = f(-1/t)$ and $G(t) = g(-1/t)$ on an interval $(0, \eta)$, noting $F'(t)/G'(t) = f'(-1/t)/g'(-1/t)$ and using Proposition 25.3.6.

Case (2). Let $\varepsilon > 0$ and choose $c \in (a, b)$ with $|f'(\xi)/g'(\xi) - L| < \varepsilon$ for $a < \xi < c$. Fix $y \in (a, c)$. For $a < x < y$, $(*)$ gives $Q(x) = (f(x) - f(y))/(g(x) - g(y))$ with $|Q(x) - L| < \varepsilon$. Since $|g(x)| \rightarrow +\infty$, $g(x) \neq 0$ for x close to a , and

$$\frac{f(x)}{g(x)} - L = (Q(x) - L) \left(1 - \frac{g(y)}{g(x)}\right) - L \frac{g(y)}{g(x)} + \frac{f(y)}{g(x)}.$$

As $x \rightarrow a^+$ with y fixed, the last two terms tend to 0 and $1 - g(y)/g(x) \rightarrow 1$, so $|f(x)/g(x) - L| < 2\varepsilon$ for x close to a . Left limits at b follow by applying the result to $x \mapsto f(-x)$, $g(-x)$. $\square$

Theorem 25.6.10 (Darboux). *Let $f: [a, b] \rightarrow \mathbb{R}$ be differentiable, with one-sided derivatives at the endpoints. Then f' takes every value between $f'(a)$ and $f'(b)$.*

Proof. Let $f'(a) < y < f'(b)$ (the other case follows by passing to $-f$) and $g(x) = f(x) - yx$. Then $g'(a) < 0 < g'(b)$. By Theorem 25.4.10, g attains a minimum at some $c \in [a, b]$. As $g'(a) < 0$, $(g(x) - g(a))/(x - a) < 0$ for $x > a$ close to a (Proposition 25.3.3(2)), so $g(x) < g(a)$ and $c \neq a$; similarly $c \neq b$. By Lemma 25.6.4, $g'(c) = 0$, that is, $f'(c) = y$. $\square$

Theorem 25.6.11 (Taylor). *Let $f: [a, x] \rightarrow \mathbb{R}$ (or $[x, a]$) have n continuous derivatives, with $f^{(n+1)}$ existing on the open interval. Then there is ξ strictly between a and x with*

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(a)}{k!} (x-a)^k + \frac{f^{(n+1)}(\xi)}{(n+1)!} (x-a)^{n+1}.$$

Proof. Let $g(t) = f(x) - \sum_{k \leq n} f^{(k)}(t)(x-t)^k/k! - M(x-t)^{n+1}/(n+1)!$, with M chosen so that $g(a) = 0$. Also $g(x) = 0$. In the derivative the sum telescopes: $g'(t) = -f^{(n+1)}(t)(x-t)^n/n! + M(x-t)^n/n!$. By Rolle's theorem $g'(\xi) = 0$ for some ξ between a and x , so $M = f^{(n+1)}(\xi)$. $\square$

Proposition 25.6.12 (Taylor's formula with Peano remainder). *Let $n \geq 1$, let $f: I \rightarrow \mathbb{R}$ be $n-1$ times differentiable on an interval I , and suppose $f^{(n)}(a)$ exists at a point $a \in I$. Let $T_n(x) = \sum_{k=0}^n f^{(k)}(a)(x-a)^k/k!$. Then*

$$\lim_{x \rightarrow a} \frac{f(x) - T_n(x)}{(x-a)^n} = 0,$$

that is, $f(x) = T_n(x) + o((x-a)^n)$ as $x \rightarrow a$.

Proof. Let $R = f - T_n$. Then R is $n-1$ times differentiable, $R^{(k)}(a) = 0$ for $0 \leq k \leq n$, and each $R^{(k)}$ with $k \leq n-1$ is continuous at a . For $n=1$ the claim is $(R(x) - R(a))/(x-a) \rightarrow R'(a) = 0$. For $n \geq 2$, apply Theorem 25.6.9(1) $n-1$ times, on each side of a , to the quotients $R^{(j)}(x)/(x-a)^{n-j}$, $0 \leq j \leq n-2$, whose numerators and denominators tend to 0 and whose denominators have nonvanishing derivatives for $x \neq a$. It remains to see that the last quotient tends to 0:

$$\frac{R^{(n-1)}(x)}{n!(x-a)} = \frac{1}{n!} \cdot \frac{R^{(n-1)}(x) - R^{(n-1)}(a)}{x-a} \rightarrow \frac{R^{(n)}(a)}{n!} = 0.$$

$\square$

Corollary 25.6.13 (Second derivative test). *Let f be differentiable on an open interval containing c , with $f'(c) = 0$, and suppose $f''(c)$ exists. If $f''(c) > 0$, then f has a strict local minimum at c ; if $f''(c) < 0$, a strict local maximum.*

Proof. By Proposition 25.6.12 with $n=2$, $(f(x) - f(c))/(x-c)^2 \rightarrow f''(c)/2 > 0$, so $f(x) - f(c) > 0$ for $x \neq c$ close to c (Proposition 25.3.3(2)). $\square$

Counterexample 25.6.14. If $f''(c) = 0$ the test says nothing: x^4 , $-x^4$ and x^3 all have vanishing first and second derivatives at 0, and have a minimum, a maximum and no extremum there.

Definition 25.6.15. A function $f: I \rightarrow \mathbb{R}$ on an interval is *convex* if $f((1-t)x + ty) \leq (1-t)f(x) + tf(y)$ for all $x, y \in I$ and $t \in [0, 1]$, and *strictly convex* if the inequality is strict whenever $x \neq y$ and $0 < t < 1$. It is (strictly) *concave* if $-f$ is (strictly) convex.

Lemma 25.6.16 (Three slopes). *A function $f: I \rightarrow \mathbb{R}$ is convex if and only if for all $x < y < z$ in I ,*

$$\frac{f(y) - f(x)}{y - x} \leq \frac{f(z) - f(x)}{z - x} \leq \frac{f(z) - f(y)}{z - y}.$$

Proof. Write $y = (1-t)x + tz$ with $t = (y-x)/(z-x) \in (0, 1)$. Convexity at this triple says $(z-x)f(y) \leq (z-y)f(x) + (y-x)f(z)$. Multiplying out, each of the two displayed inequalities is equivalent to this one. Since every $t \in (0, 1)$ and $x < z$ arise, and the cases $t \in \{0, 1\}$ or $x = y$ are trivial, the lemma follows (for $x > y$ exchange the roles). $\square$

Proposition 25.6.17. *Let $f: I \rightarrow \mathbb{R}$ on an open interval I .*

1. *If f is convex, it is continuous, it has one-sided derivatives $f'_-(x) \leq f'_+(x)$ at every point, and $f'_+(x) \leq \frac{f(y)-f(x)}{y-x} \leq f'_-(y)$ for $x < y$.*
2. *If f is differentiable, the following are equivalent: f is convex; f' is increasing; $f(y) \geq f(x) + f'(x)(y-x)$ for all $x, y \in I$.*

3. If f is twice differentiable, f is convex if and only if $f'' \geq 0$; if $f'' > 0$, f is strictly convex.
4. (Jensen) If f is convex, $x_1, \dots, x_k \in I$, $t_i \geq 0$ and $\sum t_i = 1$, then $f(\sum t_i x_i) \leq \sum t_i f(x_i)$.

Proof. (1) By Lemma 25.6.16, the slope $s(x, y) = (f(y) - f(x))/(y - x)$ is increasing in each variable. For fixed x , $s(x, y)$ decreases as $y \downarrow x$ and is bounded below by $s(w, x)$ for any $w < x$ in I (which exists, I being open); so $f'_+(x) = \lim_{y \rightarrow x^+} s(x, y)$ exists by Proposition 25.3.9, and $f'_+(x) \geq s(w, x)$. Letting $w \uparrow x$ gives $f'_-(x)$, with $f'_-(x) \leq f'_+(x)$. The inequality for $x < y$ follows from the monotonicity of slopes. The existence of one-sided derivatives implies continuity, as $f(y) - f(x) = s(x, y)(y - x) \rightarrow 0$. (2) If f is convex and differentiable, (1) gives $f'(x) = f'_+(x) \leq f'_-(y) = f'(y)$ for $x < y$. If f' is increasing and $y > x$, the mean value theorem gives $f(y) - f(x) = f'(\xi)(y - x) \geq f'(x)(y - x)$; for $y < x$, $f'(\xi) \leq f'(x)$ and $y - x < 0$ give the same inequality. Finally, if the tangent inequality holds, let $z = (1 - t)x + ty$; adding $(1 - t)$ times $f(x) \geq f(z) + f'(z)(x - z)$ and t times $f(y) \geq f(z) + f'(z)(y - z)$ gives $(1 - t)f(x) + tf(y) \geq f(z)$. (3) $f'' \geq 0$ if and only if f' is increasing (Theorem 25.6.6(3) and, conversely, $f'' = \lim$ of nonnegative quotients); apply (2). If $f'' > 0$, f' is strictly increasing and the inequalities in the proof of (2) become strict for $y \neq x$, hence so does the final one for $x \neq y$, $0 < t < 1$. (4) Induction on k : for $t_k < 1$, $\sum t_i x_i = (1 - t_k) \sum_{i < k} \frac{t_i}{1 - t_k} x_i + t_k x_k$, and the inner point lies in I because I is an interval. $\square$

Counterexample 25.6.18. A convex function need not be differentiable: $|x|$ has $f'_-(0) = -1 < 1 = f'_+(0)$. On a closed interval a convex function need not be continuous at the endpoints: the function equal to 0 on $[0, 1)$ and to 1 at 1 is convex. The converse of the last statement of (3) fails: x^4 is strictly convex, but its second derivative vanishes at 0.

Theorem 25.6.19. Let $f(z) = \sum_n a_n z^n$ have radius of convergence $R > 0$. Then f is complex differentiable on $\{|z| < R\}$ with $f'(z) = \sum_n n a_n z^{n-1}$. Hence f has complex derivatives of all orders there, and $a_n = f^{(n)}(0)/n!$. The same holds for real z in $(-R, R)$.

Proof. Let $g(z) = \sum n a_n z^{n-1}$, which has radius R by Theorem 25.5.7. Fix $|z| < \rho < R$ and $h \neq 0$ with $|z| + |h| \leq \rho$. By the binomial formula, $(z + h)^n - z^n - n z^{n-1} h = \sum_{k \geq 2} \binom{n}{k} z^{n-k} h^k$, and $\binom{n}{k} \leq \binom{n}{2} \binom{n-2}{k-2}$ for $k \geq 2$, so its absolute value is at most $\binom{n}{2} |h|^2 (|z| + |h|)^{n-2} \leq n^2 |h|^2 \rho^{n-2}$. Hence $|(f(z + h) - f(z))/h - g(z)| \leq |h| \sum_n n^2 |a_n| \rho^{n-2}$, and the last series converges. Iterating gives all derivatives, and evaluating at 0 gives a_n . $\square$

25.7 The exponential and trigonometric functions

Definition 25.7.1. For $z \in \mathbb{C}$ let $\exp(z) = e^z = \sum_{n \geq 0} z^n/n!$ (radius of convergence ∞ , by the ratio test), and for $x \in \mathbb{R}$ let $\cos x = \operatorname{Re} e^{ix}$ and $\sin x = \operatorname{Im} e^{ix}$, so that $\cos x = \sum_k (-1)^k x^{2k}/(2k)!$ and $\sin x = \sum_k (-1)^k x^{2k+1}/(2k+1)!$.

Proposition 25.7.2. 1. $e^{z+w} = e^z e^w$, $e^z \neq 0$, $\overline{e^z} = e^{\bar{z}}$, and $\exp' = \exp$.

2. $\exp$ restricts to a strictly increasing bijection $\mathbb{R} \rightarrow (0, \infty)$, whose inverse $\log$ is differentiable with $\log'(x) = 1/x$ and $\log(xy) = \log x + \log y$.
3. $\cos^2 x + \sin^2 x = 1$, $\cos' = -\sin$, $\sin' = \cos$, and $\cos(x + y) = \cos x \cos y - \sin x \sin y$, $\sin(x + y) = \sin x \cos y + \cos x \sin y$.

Proof. (1) By Proposition 25.5.6 and the binomial formula, $e^z e^w = \sum_n \sum_k \frac{z^k w^{n-k}}{k!(n-k)!} = \sum_n (z + w)^n/n!$. So $e^z e^{-z} = 1$. Conjugation is continuous and commutes with the partial sums. The derivative is Theorem 25.6.19. (2) For real x , $e^x = (e^{x/2})^2 > 0$, so $\exp$ is strictly increasing by Theorem 25.6.6. For $x \geq 0$, $e^x \geq 1 + x$, and $e^{-x} = 1/e^x$; so $\exp$ is unbounded above and takes

values arbitrarily close to 0, and by the intermediate value theorem its image is $(0, \infty)$. Apply Proposition 25.6.7. (3) $|e^{ix}|^2 = e^{ix} \overline{e^{ix}} = e^{ix} e^{-ix} = 1$; $(e^{ix})' = ie^{ix}$ by Theorem 25.6.19 applied to the real power series $\sum_n i^n x^n / n!$; the addition formulas are the real and imaginary parts of $e^{i(x+y)} = e^{ix} e^{iy}$. $\square$

Theorem 25.7.3. *The function $\cos$ has a unique zero in $(0, 2)$; call it $\pi/2$. Then $e^{i\pi/2} = i$, the map $t \mapsto e^{it}$ is a surjective continuous homomorphism $\mathbb{R} \rightarrow S^1 = \{|z| = 1\}$ with kernel $2\pi\mathbb{Z}$, and $p(t) = e^{2\pi it}$ is a covering map $\mathbb{R} \rightarrow S^1$. Every $z \neq 0$ is $re^{i\theta}$ with $r = |z|$ and θ unique modulo 2π .*

Proof. For $0 < x \leq 2$ the terms $x^{2k+1}/(2k+1)!$ decrease, since consecutive ratios are $x^2/((2k+2)(2k+3)) < 1$; by Lemma 25.5.5, $\sin x \geq x - x^3/6 > 0$. So $\cos$ is strictly decreasing on $[0, 2]$. The terms $4^k/(2k)!$ of $\cos 2$ decrease from $k = 1$ on, so $\cos 2 \leq 1 - 2 + 16/24 < 0$, while $\cos 0 = 1$; by the intermediate value theorem there is a unique zero $\pi/2 \in (0, 2)$. Then $\sin(\pi/2) = 1$, so $e^{i\pi/2} = i$ and $e^{2\pi i} = i^4 = 1$.

Kernel: if $0 < t < 2\pi$, write $e^{it/4} = u + iv$ with $u, v > 0$ (as $0 < t/4 < \pi/2$). If $e^{it} = (u + iv)^4 = 1$, then $\text{Im}(u + iv)^4 = 4uv(u^2 - v^2) = 0$ forces $u^2 = v^2 = 1/2$, and then $\text{Re}(u + iv)^4 = u^4 - 6u^2v^2 + v^4 = -1$, a contradiction. With periodicity, the kernel is $2\pi\mathbb{Z}$.

Surjectivity: for $a + ib \in S^1$ with $a, b \geq 0$, choose $t \in [0, \pi/2]$ with $\cos t = a$ (intermediate value theorem); then $\sin t = \sqrt{1 - a^2} = b$. The other quadrants are obtained by multiplying by powers of $i = e^{i\pi/2}$. For the covering property, p is injective on $[a, a + 1)$ and identifies only a and $a + 1$ in $[a, a + 1]$, so the argument of Proposition 17.4.5 applies verbatim. The polar form follows by applying this to $z/|z|$. $\square$

Remark 25.7.4. The proof of Theorem 17.4.6 uses only that $\mathbb{R}$ is simply connected and that the covering map has period 4 with fibres the translates of $4\mathbb{Z}$; it applies verbatim to $t \mapsto e^{2\pi it}$, with period 1, so $\pi_1(S^1) \cong \mathbb{Z}$ is generated by the loop $t \mapsto e^{2\pi it}$. For $x > 0$ and $s \in \mathbb{C}$ we define $x^s = e^{s \log x}$; then $\frac{d}{dx} x^s = s x^{s-1}$.

Proposition 25.7.5 (Notable limits). 1. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1$, $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$ and $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$.

2. For every $x \in \mathbb{R}$, $(1 + x/n)^n \rightarrow e^x$; in particular $(1 + 1/n)^n \rightarrow e = \exp(1)$.

3. For $k \in \mathbb{N}$ and $a > 0$: $x^k e^{-ax} \rightarrow 0$ and $(\log x)/x^a \rightarrow 0$ as $x \rightarrow +\infty$, and $x^a \log x \rightarrow 0$ as $x \rightarrow 0^+$.

Proof. (1) The first three are the derivatives of $\exp$ at 0, of $\log$ at 1 and of $\sin$ at 0 (Proposition 25.7.2), which are $e^0 = 1$, $1/1$ and $\cos 0 = 1$. For the last, apply Theorem 25.6.9(1): $\frac{1 - \cos x}{x^2}$ has the same limit as $\frac{\sin x}{2x} \rightarrow \frac{1}{2}$. (2) For $x = 0$ this is clear. Otherwise, for $n > |x|$, $1 + x/n > 0$ and $(1 + x/n)^n = \exp(n \log(1 + x/n))$, with $n \log(1 + x/n) = x \cdot \frac{\log(1+x/n)}{x/n} \rightarrow x$ by (1) and Theorem 25.3.4 (as $x/n \rightarrow 0$ and $x/n \neq 0$). By continuity of $\exp$, $(1 + x/n)^n \rightarrow e^x$. (3) For $x > 0$, $e^{ax} \geq (ax)^{k+1}/(k+1)!$, one term of the exponential series, so $0 < x^k e^{-ax} \leq (k+1)!/(a^{k+1}x) \rightarrow 0$. For $x = e^t$, $(\log x)/x^a = t e^{-at}$, which tends to 0 as $t \rightarrow +\infty$ by the case $k = 1$; since $\log x \rightarrow +\infty$ as $x \rightarrow +\infty$, Proposition 25.3.6 applies. Similarly $x^a \log x = -t e^{-at}$ for $x = e^{-t}$, and $t \rightarrow +\infty$ as $x \rightarrow 0^+$. $\square$

Counterexample 25.7.6. 1. $\sin(1/x)$ has no limit at 0: along $x_n = 1/(2\pi n + \pi/2)$ it is 1, along $y_n = 1/(\pi n)$ it is 0 ($n \geq 1$), and Theorem 25.3.4 applies. Since $x_n - y_n \rightarrow 0$, $\sin(1/x)$ is continuous but not uniformly continuous on $(0, 1)$. On the other hand $x \sin(1/x) \rightarrow 0$ by the squeeze theorem, as $|x \sin(1/x)| \leq |x|$.

2. A derivative need not be continuous. Let $f(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $f(0) = 0$. Then $f'(0) = \lim_{x \rightarrow 0} x \sin(1/x) = 0$, and $f'(x) = 2x \sin(1/x) - \cos(1/x)$ for $x \neq 0$, which has no limit at 0. In accordance with Proposition 25.6.8 and Theorem 25.6.10, the discontinuity of f' at 0 is essential.
3. The converse of de l'Hôpital's rule fails. With f as in (2) and $g(x) = x$, $f(x)/g(x) = x \sin(1/x) \rightarrow 0$, while $f'(x)/g'(x) = 2x \sin(1/x) - \cos(1/x)$ has no limit as $x \rightarrow 0$.

Corollary 25.7.7 (Inequality of arithmetic and geometric means). For $x_1, \dots, x_k > 0$ and weights $t_i \geq 0$ with $\sum t_i = 1$, $\prod_i x_i^{t_i} \leq \sum_i t_i x_i$. In particular $\sqrt[k]{x_1 \cdots x_k} \leq \frac{1}{k}(x_1 + \cdots + x_k)$, with equality if and only if all x_i are equal.

Proof. $\log'' = -1/x^2 < 0$, so $\log$ is concave (Proposition 25.6.17(3)), and by Jensen's inequality $\sum t_i \log x_i \leq \log \sum t_i x_i$. Apply $\exp$, which is increasing. For the equality statement, let $m = \frac{1}{k} \sum x_i$. As $\log'$ is strictly decreasing, the mean value theorem gives $\log x < \log m + (x - m)/m$ for every $x > 0$ with $x \neq m$ (the tangent inequality of Proposition 25.6.17(2) for $-\log$, made strict). Averaging over $x = x_1, \dots, x_k$ gives $\frac{1}{k} \sum \log x_i < \log m$ unless all $x_i = m$. $\square$

Lemma 25.7.8. The function $\beta(t) = e^{-1/t}$ for $t > 0$ and $\beta(t) = 0$ for $t \leq 0$ has derivatives of all orders on $\mathbb{R}$, all vanishing at 0. The function $\gamma(t) = \beta(t)/(\beta(t) + \beta(1-t))$ has derivatives of all orders, takes values in $[0, 1]$, and equals 0 for $t \leq 0$ and 1 for $t \geq 1$.

Proof. By induction, on $t > 0$, $\beta^{(k)}(t) = p_k(1/t)e^{-1/t}$ for a polynomial p_k . Since $e^s \geq s^m/m!$ for $s > 0$, $s^j e^{-s} \rightarrow 0$ as $s \rightarrow \infty$; so $\beta^{(k)}(t) \rightarrow 0$ as $t \rightarrow 0^+$, and the difference quotients of $\beta^{(k)}$ at 0 tend to 0, which by induction shows $\beta^{(k+1)}(0) = 0$. The denominator of γ never vanishes (one of $t, 1-t$ is positive), so γ has derivatives of all orders by Proposition 25.6.2, and the other properties are clear. $\square$

Counterexample 25.7.9. The function β of Lemma 25.7.8 has all derivatives zero at 0, so its Taylor series at 0 is zero, but $\beta(t) \neq 0$ for $t > 0$. Smooth functions need not be given by their Taylor series; complex differentiable functions are (Chapter 27).

25.8 The Riemann integral

Definition 25.8.1. Let $f: [a, b] \rightarrow \mathbb{R}$ be bounded. For a partition $P: a = t_0 < \cdots < t_k = b$ let $U(f, P) = \sum_i (t_i - t_{i-1}) \sup_{[t_{i-1}, t_i]} f$ and $L(f, P)$ the same with $\inf$. The function is *Riemann integrable* if $\sup_P L(f, P) = \inf_P U(f, P)$, and this common value is $\int_a^b f = \int_a^b f(t) dt$. A $\mathbb{C}^m$ -valued function is integrable if its real and imaginary components are, with the integral taken componentwise. We set $\int_b^a f = -\int_a^b f$.

Proposition 25.8.2. 1. If P' refines P , then $L(f, P) \leq L(f, P') \leq U(f, P') \leq U(f, P)$; and $L(f, P) \leq U(f, Q)$ for all P, Q . So f is integrable if and only if for every $\varepsilon > 0$ there is P with $U(f, P) - L(f, P) < \varepsilon$.

2. Continuous functions are integrable, and $\sum_i (t_i - t_{i-1}) f(s_i) \rightarrow \int_a^b f$ for any choice of $s_i \in [t_{i-1}, t_i]$ as $\max_i (t_i - t_{i-1}) \rightarrow 0$.
3. For continuous f, g : the integral is linear, $f \leq g$ implies $\int f \leq \int g$, $\int_a^c f = \int_a^b f + \int_b^c f$, $|\int_a^b f| \leq \int_a^b |f|$ for $\mathbb{C}^m$ -valued f with the Euclidean norm, and if $f_n \rightarrow f$ uniformly, $\int f_n \rightarrow \int f$.

Proof. (1) Adding a point to a partition can only increase infima and decrease suprema on the pieces; for P, Q compare both with $P \cup Q$. (2) By uniform continuity (Corollary 16.7.5), for $\varepsilon > 0$ there is δ such that f varies by less than ε on intervals of length $< \delta$; for partitions of mesh $< \delta$, $U - L \leq \varepsilon(b - a)$, and Riemann sums lie between L and U . (3) Linearity and additivity follow from (2), monotonicity from $L(f, P) \leq L(g, P)$, and uniform limits from $|\int (f_n - f)| \leq (b - a) \sup |f_n - f|$. For the norm inequality let $v = \int f \in \mathbb{C}^m$; then $\|v\|^2 = \operatorname{Re} \langle \int f, v \rangle = \int \operatorname{Re} \langle f(t), v \rangle dt \leq \int \|f\| \|v\|$ by Cauchy–Schwarz (Proposition 9.9.2). $\square$

Theorem 25.8.3 (Fundamental theorem of calculus). *Let $f: [a, b] \rightarrow \mathbb{C}^m$ be continuous. Then $F(x) = \int_a^x f$ is differentiable with $F' = f$. If $G: [a, b] \rightarrow \mathbb{C}^m$ is continuously differentiable, then $\int_a^b G' = G(b) - G(a)$.*

Proof. $|(F(x + h) - F(x))/h - f(x)| = |\frac{1}{h} \int_x^{x+h} (f - f(x))| \leq \sup_{|t-x| \leq |h|} |f(t) - f(x)| \rightarrow 0$. For the second statement, $\int_a^x G' - G(x)$ has derivative 0, hence is constant by Theorem 25.6.6 applied to real and imaginary components. $\square$

Corollary 25.8.4. *For continuously differentiable f, g and $\varphi: [c, d] \rightarrow [a, b]$: $\int_a^b f'g = [fg]_a^b - \int_a^b fg'$ and $\int_c^d f(\varphi(s))\varphi'(s)ds = \int_{\varphi(c)}^{\varphi(d)} f$ for continuous f . Moreover $\|G(b) - G(a)\| \leq (b - a) \sup \|G'\|$ for G continuously differentiable.*

Proof. Apply Theorem 25.8.3 to fg , to $F \circ \varphi$ with $F(x) = \int_a^x f$, and use the norm inequality of Proposition 25.8.2. $\square$

Proposition 25.8.5. *Let $f: [c, d] \times [a, b] \rightarrow \mathbb{C}$ be continuous.*

1. $x \mapsto \int_a^b f(x, t) dt$ is continuous, and if $\partial_x f$ exists and is continuous on $[c, d] \times [a, b]$, it is differentiable with derivative $\int_a^b \partial_x f(x, t) dt$.
2. (Fubini for continuous functions) $\int_c^d \int_a^b f(x, t) dt dx = \int_a^b \int_c^d f(x, t) dx dt$.

Proof. (1) Continuity follows from uniform continuity of f . For the derivative, by the mean value theorem applied to real and imaginary parts, $(f(x + h, t) - f(x, t))/h - \partial_x f(x, t)$ is bounded by the oscillation of $\partial_x f$ over $|s - x| \leq |h|$, which tends to 0 uniformly in t by uniform continuity. (2) Let $\Phi(y) = \int_c^y \int_a^b f dt dx$ and $\Psi(y) = \int_a^b \int_c^y f dx dt$. By Theorem 25.8.3 and (1) (applied to $g(y, t) = \int_c^y f(x, t) dx$, with $\partial_y g = f$ continuous), $\Phi' = \Psi' = \int_a^b f(y, t) dt$, and $\Phi(c) = \Psi(c) = 0$. $\square$

25.9 Sequences of functions

Definition 25.9.1. Let D be a set and $f_n, f: D \rightarrow \mathbb{K}^m$. Then $f_n \rightarrow f$ *pointwise* if $f_n(x) \rightarrow f(x)$ for every $x \in D$, and *uniformly* if $\|f_n - f\|_\infty = \sup_{x \in D} \|f_n(x) - f(x)\| \rightarrow 0$.

The difference is the order of quantifiers: for pointwise convergence the threshold N may depend on x , for uniform convergence it may not. Uniform convergence implies pointwise convergence.

Proposition 25.9.2. 1. (Cauchy criterion) (f_n) converges uniformly on D if and only if for every $\varepsilon > 0$ there is N with $\|f_n - f_m\|_\infty < \varepsilon$ for $m, n \geq N$.

2. If D is a metric space, the f_n are continuous and $f_n \rightarrow f$ uniformly, then f is continuous.

3. If $f_n \rightarrow f$ uniformly on $[a, b]$ and the f_n are continuous, then $\int_a^b f_n \rightarrow \int_a^b f$.

Proof. (1) If $f_n \rightarrow f$ uniformly, use the triangle inequality. Conversely, for each x the sequence $(f_n(x))$ is Cauchy, hence converges to some $f(x)$ (Theorem 25.2.13, componentwise). Letting $m \rightarrow \infty$ in $\|f_n(x) - f_m(x)\| < \varepsilon$ gives $\|f_n(x) - f(x)\| \leq \varepsilon$ for $n \geq N$ and all x . (2) is Proposition 16.3.8. (3) is Proposition 25.8.2(3). $\square$

Counterexample 25.9.3. 1. $f_n(x) = x^n$ on $[0, 1]$ converges pointwise to the function equal to 0 on $[0, 1)$ and 1 at 1, which is discontinuous; so the convergence is not uniform. On $[0, 1)$ the limit 0 is continuous, but $\sup_{[0,1)} x^n = 1$, so the convergence is still not uniform. On $[0, r]$ with $r < 1$ it is uniform, as $\sup = r^n \rightarrow 0$.

2. Pointwise convergence does not commute with integration. Let $f_n(x) = nx(1 - x^2)^n$ on $[0, 1]$. For $0 < x \leq 1$, $f_n(x) \rightarrow 0$ by Proposition 25.2.9(3) with $a = (1 - x^2)^{-1}$ (or trivially if $x = 1$), and $f_n(0) = 0$. But, substituting $u = 1 - x^2$, $\int_0^1 f_n = \frac{n}{2} \int_0^1 u^n du = \frac{n}{2(n+1)} \rightarrow \frac{1}{2} \neq 0$.

3. Uniform convergence does not commute with differentiation. $f_n(x) = x^n/n$ converges uniformly to 0 on $[0, 1]$, since $|f_n| \leq 1/n$, but $f'_n(1) = 1$ for all n .

Theorem 25.9.4. Let $f_n: [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. Suppose that (f'_n) converges uniformly to g and that $(f_n(x_0))$ converges for some $x_0 \in [a, b]$. Then (f_n) converges uniformly to a continuously differentiable f with $f' = g$.

Proof. The function g is continuous by Proposition 25.9.2(2). Let $L = \lim f_n(x_0)$ and $f(x) = L + \int_{x_0}^x g$, which is continuously differentiable with $f' = g$ by Theorem 25.8.3. By the same theorem $f_n(x) = f_n(x_0) + \int_{x_0}^x f'_n$, so $|f_n(x) - f(x)| \leq |f_n(x_0) - L| + (b - a)\|f'_n - g\|_\infty$, which tends to 0 uniformly in x . $\square$

Theorem 25.9.5 (Dini). Let K be a compact metric space and $f_n: K \rightarrow \mathbb{R}$ continuous functions with $f_1 \leq f_2 \leq \dots$, converging pointwise to a continuous function f . Then the convergence is uniform.

Proof. Let $g_n = f - f_n$, continuous, ≥ 0 , decreasing in n and tending to 0 pointwise. For $\varepsilon > 0$ the sets $K_n = \{x : g_n(x) \geq \varepsilon\}$ are closed, decreasing, and have empty intersection. By compactness some K_N is empty (the complements form an open cover with a finite subcover, and the K_n decrease), and then $0 \leq g_n < \varepsilon$ on K for $n \geq N$. $\square$

Counterexample 25.9.6. Each hypothesis of Dini's theorem is needed. On the noncompact $[0, 1)$, $f_n(x) = -x^n$ increases to the continuous function 0, not uniformly. On $[0, 1]$, $f_n(x) = -x^n$ increases to a discontinuous limit, not uniformly. And without monotonicity, the continuous functions $f_n(x) = \max(0, 1 - |nx - 2|)$, bumps of height 1 moving towards 0, converge pointwise to 0 on $[0, 1]$ but not uniformly.

Corollary 25.9.7 (Series of functions). Let $u_n: [a, b] \rightarrow \mathbb{R}$ be continuous and suppose $\sum_n \|u_n\|_\infty < \infty$. Then $\sum_n u_n$ converges uniformly to a continuous function, and $\int_a^b \sum_n u_n = \sum_n \int_a^b u_n$. If moreover the u_n are continuously differentiable and $\sum_n \|u'_n\|_\infty < \infty$, then $\sum_n u_n$ is continuously differentiable with derivative $\sum_n u'_n$.

Proof. Apply the Weierstrass M-test (Proposition 25.5.4(5)) and Propositions 25.9.2(2),(3) to the partial sums, and Theorem 25.9.4 to the partial sums and their derivatives. $\square$

Theorem 25.9.8 (A continuous nowhere differentiable function). Let $d(x)$ be the distance from $x \in \mathbb{R}$ to the nearest integer. The Takagi function $T(x) = \sum_{m \geq 0} 2^{-m} d(2^m x)$ is continuous on $\mathbb{R}$ and differentiable at no point.

Proof. Since $0 \leq d \leq \frac{1}{2}$ and d is continuous (it is 1-Lipschitz), the series converges uniformly by the M -test, and T is continuous.

We first note: if f is differentiable at x and $u_n \leq x \leq v_n$ with $u_n < v_n$ and $v_n - u_n \rightarrow 0$, then $\frac{f(v_n) - f(u_n)}{v_n - u_n} \rightarrow f'(x)$. Indeed, with $\lambda_n = (v_n - x)/(v_n - u_n) \in [0, 1]$,

$$\frac{f(v_n) - f(u_n)}{v_n - u_n} = \lambda_n \frac{f(v_n) - f(x)}{v_n - x} + (1 - \lambda_n) \frac{f(x) - f(u_n)}{x - u_n},$$

where a term with weight 0 (when $v_n = x$ or $u_n = x$) is omitted; both quotients tend to $f'(x)$, and so does their convex combination.

Fix x , and for $n \geq 1$ let $u_n = k_n/2^n \leq x < v_n = (k_n + 1)/2^n$ with $k_n \in \mathbb{Z}$. For $m \geq n$, $2^m u_n$ and $2^m v_n$ are integers, so $d(2^m u_n) = d(2^m v_n) = 0$. For $m < n$, the numbers $2^m u_n$ and $2^m v_n$ are consecutive multiples of $2^{m-n} \leq \frac{1}{2}$, so they lie in a common interval $[\frac{j}{2}, \frac{j+1}{2}]$, on which d is affine with slope ± 1 . Hence

$$q_n = \frac{T(v_n) - T(u_n)}{v_n - u_n} = \sum_{m=0}^{n-1} 2^{-m} \frac{d(2^m v_n) - d(2^m u_n)}{v_n - u_n} = \sum_{m=0}^{n-1} \varepsilon_{m,n}, \quad \varepsilon_{m,n} \in \{1, -1\}.$$

So q_n is an integer of the same parity as n , and (q_n) cannot converge: consecutive terms differ by an odd integer, so $|q_{n+1} - q_n| \geq 1$. By the preliminary remark, T is not differentiable at x . $\square$

25.10 Derivatives in several variables

Lemma 25.10.1. *All norms on a finite-dimensional real or complex vector space are equivalent (they define the same topology), and every linear map $A: V \rightarrow W$ between finite-dimensional normed spaces is continuous, with finite operator norm $\|A\| = \sup_{\|v\| \leq 1} \|Av\|$. Then $\|Av\| \leq \|A\| \|v\|$ and $\|AB\| \leq \|A\| \|B\|$.*

Proof. Identify V with $\mathbb{R}^n$ by a basis and let $\|\cdot\|$ be a norm. By the triangle inequality and Cauchy–Schwarz, $\|x\| \leq \sum |x_i| \|e_i\| \leq C \|x\|_2$, so $\|\cdot\|$ is continuous for the Euclidean topology. On the compact unit sphere of $\|\cdot\|_2$ it attains a minimum $c > 0$, so $c \|x\|_2 \leq \|x\| \leq C \|x\|_2$, and Lemma 16.3.4 applies. A linear map is Lipschitz for Euclidean norms, since $\|Ax\| \leq \sum |x_i| \|Ae_i\|$. The inequalities follow from the definition. $\square$

Definition 25.10.2. Let $U \subseteq \mathbb{R}^n$ be open and $f: U \rightarrow \mathbb{R}^m$. Then f is *differentiable* at $a \in U$ if there is a linear map $Df(a): \mathbb{R}^n \rightarrow \mathbb{R}^m$ with

$$f(a + h) = f(a) + Df(a)h + r(h), \quad \|r(h)\|/\|h\| \rightarrow 0 \text{ as } h \rightarrow 0.$$

The *partial derivatives* are $\partial_j f(a) = \frac{d}{dt} f(a + te_j)|_{t=0}$. The map f is C^1 if it is differentiable everywhere and $Df: U \rightarrow \text{Hom}(\mathbb{R}^n, \mathbb{R}^m)$ is continuous; C^k if Df is C^{k-1} ; and *smooth* (C^∞) if it is C^k for all k . A C^k -*diffeomorphism* is a bijection between open sets which is C^k with C^k inverse.

Proposition 25.10.3. 1. $Df(a)$ is unique, f is continuous at a , and the matrix of $Df(a)$ is the Jacobian matrix $(\partial_j f_i(a))$.

2. (Chain rule) If f is differentiable at a and g at $f(a)$, then $g \circ f$ is differentiable at a and $D(g \circ f)(a) = Dg(f(a)) \circ Df(a)$.

3. If all partial derivatives of f exist near a and are continuous at a , then f is differentiable at a . So f is C^1 if and only if its partial derivatives exist and are continuous, and C^k if and only if all partial derivatives of order at most k exist and are continuous.
4. (Mean value inequality) If f is C^1 and the segment $[x, y]$ lies in U , then $\|f(y) - f(x)\| \leq \sup_{z \in [x, y]} \|Df(z)\| \|y - x\|$.

Proof. (1) If A and B both work, $\|(A - B)h\|/\|h\| \rightarrow 0$, so $A = B$ by linearity. Taking $h = te_j$ gives the partial derivatives. (2) Let $b = f(a)$, $k = f(a + h) - b = Df(a)h + r_f(h)$. Then $g(f(a + h)) - g(b) - Dg(b)Df(a)h = Dg(b)r_f(h) + r_g(k)$. For small h , $\|k\| \leq (\|Df(a)\| + 1)\|h\|$, so both terms are $o(\|h\|)$. (3) We may assume $m = 1$. Write $f(a + h) - f(a) = \sum_j [f(a + h_1e_1 + \cdots + h_je_j) - f(a + h_1e_1 + \cdots + h_{j-1}e_{j-1})]$. By the mean value theorem the j -th term is $h_j \partial_j f(\xi_j)$ with ξ_j within $\|h\|$ of a , and $\partial_j f(\xi_j) \rightarrow \partial_j f(a)$. The statement about C^k follows by induction. (4) Let $g(t) = f(x + t(y - x))$; by the chain rule $g'(t) = Df(x + t(y - x))(y - x)$, and Corollary 25.8.4 applies. $\square$

Counterexample 25.10.4. Let $f(x, y) = xy/(x^2 + y^2)$ for $(x, y) \neq 0$ and $f(0) = 0$. Both partial derivatives exist everywhere (f vanishes on the axes), but $f(t, t) = 1/2$ for $t \neq 0$, so f is not continuous at 0, let alone differentiable.

Theorem 25.10.5 (Schwarz). If $f: U \rightarrow \mathbb{R}$ is C^2 , then $\partial_i \partial_j f = \partial_j \partial_i f$.

Proof. Restricting to the plane of the i -th and j -th coordinates, we may assume $n = 2$. For a rectangle $R = [x_0, x_1] \times [y_0, y_1] \subseteq U$, Theorem 25.8.3 applied twice gives $\int_{y_0}^{y_1} \int_{x_0}^{x_1} \partial_x \partial_y f \, dx \, dy = f(x_1, y_1) - f(x_0, y_1) - f(x_1, y_0) + f(x_0, y_0)$, and the same for $\int_{x_0}^{x_1} \int_{y_0}^{y_1} \partial_y \partial_x f \, dy \, dx$. By Proposition 25.8.5(2), the continuous function $g = \partial_x \partial_y f - \partial_y \partial_x f$ has integral 0 over every small rectangle. If $g(p) > 0$, then $g > g(p)/2$ on a small rectangle around p , and its integral is positive, a contradiction. $\square$

Counterexample 25.10.6. For $f(x, y) = xy(x^2 - y^2)/(x^2 + y^2)$, $f(0) = 0$, one computes $\partial_x f(0, y) = -y$ and $\partial_y f(x, 0) = x$, so $\partial_y \partial_x f(0) = -1 \neq 1 = \partial_x \partial_y f(0)$. The second partial derivatives exist but are not continuous at 0.

Theorem 25.10.7 (Taylor's theorem in several variables). Let $f: U \rightarrow \mathbb{R}$ be C^{k+1} and $[a, a + h] \subseteq U$. With multi-index notation $\partial^\alpha = \partial_1^{\alpha_1} \cdots \partial_n^{\alpha_n}$, $h^\alpha = h_1^{\alpha_1} \cdots h_n^{\alpha_n}$ and $\alpha! = \alpha_1! \cdots \alpha_n!$, there is $\xi \in (a, a + h)$ with

$$f(a + h) = \sum_{|\alpha| \leq k} \frac{\partial^\alpha f(a)}{\alpha!} h^\alpha + \sum_{|\alpha| = k+1} \frac{\partial^\alpha f(\xi)}{\alpha!} h^\alpha.$$

Proof. Apply Theorem 25.6.11 to $g(t) = f(a + th)$ on $[0, 1]$. By the chain rule and Theorem 25.10.5, $g^{(j)}(t) = \sum_{|\alpha|=j} \frac{j!}{\alpha!} \partial^\alpha f(a + th) h^\alpha$ (the multinomial formula). $\square$

Corollary 25.10.8. For $0 < r < R$ there is a smooth function $\mathbb{R}^n \rightarrow [0, 1]$ equal to 1 on the closed ball of radius r and to 0 outside the open ball of radius R .

Proof. Take $x \mapsto 1 - \gamma((\|x\|^2 - r^2)/(R^2 - r^2))$ with γ as in Lemma 25.7.8; it is smooth by the chain rule, since $\|x\|^2$ is a polynomial. $\square$

25.11 The inverse and implicit function theorems

Lemma 25.11.1. If $A \in \text{End}(\mathbb{R}^n)$ has $\|A\| < 1$, then $1 - A$ is invertible with inverse $\sum_k A^k$ and $\|(1 - A)^{-1}\| \leq 1/(1 - \|A\|)$. The group $\text{GL}_n(\mathbb{R})$ is open in $M_n(\mathbb{R})$, and inversion $\text{GL}_n(\mathbb{R}) \rightarrow \text{GL}_n(\mathbb{R})$ is smooth.

Proof. The series converges absolutely in the complete space $M_n(\mathbb{R})$ (Proposition 25.5.4), and $(1 - A) \sum_{k \leq N} A^k = 1 - A^{N+1} \rightarrow 1$. The determinant is a polynomial in the entries, hence continuous, and $\mathrm{GL}_n(\mathbb{R}) = \det^{-1}(\mathbb{R} \setminus \{0\})$. By the adjugate formula (Proposition 9.5.3), the entries of A^{-1} are polynomials in the entries of A divided by $\det A$, so inversion is smooth. $\square$

Theorem 25.11.2 (Inverse function theorem). *Let $U \subseteq \mathbb{R}^n$ be open, $f: U \rightarrow \mathbb{R}^n$ of class C^k with $k \geq 1$, and $a \in U$ with $Df(a)$ invertible. There are open sets $V \ni a$ and $W \ni f(a)$ such that $f: V \rightarrow W$ is a C^k -diffeomorphism, and $D(f^{-1})(f(x)) = Df(x)^{-1}$.*

Proof. Replacing f by $x \mapsto Df(a)^{-1}(f(a+x) - f(a))$, we may assume $a = 0$, $f(0) = 0$ and $Df(0) = 1$. Let $g(x) = x - f(x)$, so $Dg(0) = 0$; choose $r > 0$ with $\overline{B}(0, r) \subseteq U$ and $\|Dg(x)\| \leq 1/2$ there. By the mean value inequality, $\|g(x) - g(x')\| \leq \frac{1}{2}\|x - x'\|$ on $\overline{B}(0, r)$, and $\|g(x)\| \leq r/2$.

For $y \in W = B(0, r/2)$, the map $T_y(x) = y + g(x)$ sends $\overline{B}(0, r)$ into $B(0, r)$ and is a contraction with constant $1/2$. By the Banach fixed point theorem (Theorem 16.7.6) there is a unique $x \in \overline{B}(0, r)$ with $x = y + g(x)$, that is, $f(x) = y$, and $x \in B(0, r)$. Let $V = B(0, r) \cap f^{-1}(W)$, open; then $f: V \rightarrow W$ is bijective. From $x - x' = f(x) - f(x') + g(x) - g(x')$ we get $\|x - x'\| \leq 2\|f(x) - f(x')\|$, so f^{-1} is continuous. For $x \in V$, $Df(x) = 1 - Dg(x)$ is invertible with $\|Df(x)^{-1}\| \leq 2$ (Lemma 25.11.1).

Let $y = f(x)$ and $y + \kappa = f(x + h)$ with $x, x + h \in V$, so $\|h\| \leq 2\|\kappa\|$. Then $f^{-1}(y + \kappa) - f^{-1}(y) - Df(x)^{-1}\kappa = -Df(x)^{-1}(f(x + h) - f(x) - Df(x)h)$, whose norm is at most $2\|r(h)\| = o(\|\kappa\|)$. So f^{-1} is differentiable with $D(f^{-1}) = \iota \circ Df \circ f^{-1}$, where ι is inversion. If f^{-1} is C^j with $j < k$, this composite is C^j , so f^{-1} is C^{j+1} ; by induction f^{-1} is C^k (composites of C^j maps are C^j by the chain rule and induction). $\square$

Theorem 25.11.3 (Implicit function theorem). *Let $U \subseteq \mathbb{R}^n \times \mathbb{R}^m$ be open, $F: U \rightarrow \mathbb{R}^m$ of class C^k , $(a, b) \in U$ with $F(a, b) = 0$, and suppose the partial derivative $D_y F(a, b): \mathbb{R}^m \rightarrow \mathbb{R}^m$ is invertible. There are open sets $A \ni a$ and $B \ni b$ and a C^k map $g: A \rightarrow B$ such that for $(x, y) \in A \times B$: $F(x, y) = 0$ if and only if $y = g(x)$.*

Proof. The map $\Phi(x, y) = (x, F(x, y))$ has invertible derivative at (a, b) , so by Theorem 25.11.2 it restricts to a C^k -diffeomorphism of an open set, which we may shrink to $A' \times B$, onto an open set containing $(a, 0)$. Its inverse has the form $(x, z) \mapsto (x, \psi(x, z))$. Let $A = \{x : (x, 0) \in \Phi(A' \times B)\}$, open, and $g(x) = \psi(x, 0)$. $\square$

Theorem 25.11.4 (Local submersions and immersions). *Let $f: U \rightarrow \mathbb{R}^m$ be C^k , $U \subseteq \mathbb{R}^n$ open, $a \in U$.*

1. *If $Df(a)$ is surjective, there is a C^k -diffeomorphism ψ from a neighbourhood of a onto an open subset of $\mathbb{R}^n$ with $f \circ \psi^{-1}(x_1, \dots, x_n) = (x_1, \dots, x_m)$.*
2. *If $Df(a)$ is injective, there is a C^k -diffeomorphism ϕ from a neighbourhood of $f(a)$ onto an open subset of $\mathbb{R}^m$ with $\phi \circ f(x) = (x_1, \dots, x_n, 0, \dots, 0)$ near a .*

Proof. We may assume $a = 0$ and $f(0) = 0$. (1) After permuting coordinates, the first m columns of the Jacobian are linearly independent. Then $\psi(x) = (f(x), x_{m+1}, \dots, x_n)$ has invertible derivative at 0, and $f \circ \psi^{-1}$ is the projection. (2) After permuting target coordinates, the first n rows of the Jacobian are independent. Let $\Theta(x, y) = f(x) + (0, y)$ for $x \in U$, $y \in \mathbb{R}^{m-n}$; $D\Theta(0)$ is invertible, and $\phi = \Theta^{-1}$ satisfies $\phi(f(x)) = (x, 0)$. $\square$

Remark 25.11.5. More generally, if Df has constant rank r near a , there are local diffeomorphisms in source and target in which f becomes $(x_1, \dots, x_n) \mapsto (x_1, \dots, x_r, 0, \dots, 0)$ (the constant rank theorem [Lee13, Theorem 4.12]). All results of this section hold with the same proofs over $\mathbb{C}$ for complex differentiable (holomorphic) maps, once holomorphic functions are known to be C^∞ (Chapter 29).

25.12 Ordinary differential equations

Definition 25.12.1. Let $D \subseteq \mathbb{R} \times \mathbb{R}^n$ be open and $F: D \rightarrow \mathbb{R}^n$ continuous. F is *locally Lipschitz* in x if every point of D has a neighbourhood N and L with $\|F(t, x) - F(t, y)\| \leq L\|x - y\|$ for $(t, x), (t, y) \in N$; by the mean value inequality this holds if F is C^1 . A *solution* of $x' = F(t, x)$ is a differentiable map $\varphi: J \rightarrow \mathbb{R}^n$ on an interval with $(t, \varphi(t)) \in D$ and $\varphi'(t) = F(t, \varphi(t))$. By Theorem 25.8.3, this is equivalent to continuity of φ and $\varphi(t) = \varphi(t_0) + \int_{t_0}^t F(s, \varphi(s)) ds$.

Lemma 25.12.2 (Grönwall). *If $u: [t_0, t_1] \rightarrow [0, \infty)$ is continuous and $u(t) \leq C + L \int_{t_0}^t u$ with $C, L \geq 0$, then $u(t) \leq Ce^{L(t-t_0)}$.*

Proof. Let $v(t) = C + L \int_{t_0}^t u$. Then $v' = Lu \leq Lv$, so $(e^{-L(t-t_0)}v)' \leq 0$, and $u \leq v \leq v(t_0)e^{L(t-t_0)} = Ce^{L(t-t_0)}$. $\square$

Theorem 25.12.3 (Picard–Lindelöf). *Let F be continuous and locally Lipschitz in x , and $(t_0, x_0) \in D$.*

1. *There is $\delta > 0$ and a solution on $[t_0 - \delta, t_0 + \delta]$ with $\varphi(t_0) = x_0$.*
2. *Two solutions on intervals $J_1, J_2 \ni t_0$ with the same value at t_0 agree on $J_1 \cap J_2$.*
3. *There is a unique maximal solution, defined on an open interval (α, β) ; if $\beta < \infty$, then $(t, \varphi(t))$ leaves every compact subset of D as $t \rightarrow \beta$, and similarly at α .*

Proof. (1) Choose $a, b > 0$ such that $R = [t_0 - a, t_0 + a] \times \overline{B}(x_0, b) \subseteq D$ and F is L -Lipschitz in x on R ; let $M = \sup_R \|F\|$ (finite by compactness) and $\delta = \min(a, b/(M+1), 1/(2L+1))$. On the complete metric space X of continuous maps $[t_0 - \delta, t_0 + \delta] \rightarrow \overline{B}(x_0, b)$ with the supremum metric (Proposition 16.7.2), $T\varphi(t) = x_0 + \int_{t_0}^t F(s, \varphi(s)) ds$ satisfies $\|T\varphi(t) - x_0\| \leq M\delta \leq b$ and $\|T\varphi - T\psi\| \leq L\delta\|\varphi - \psi\| \leq \frac{1}{2}\|\varphi - \psi\|$. Its fixed point is a solution.

(2) The set of $t \in J_1 \cap J_2$ where the solutions agree is closed and contains t_0 . If they agree at t_1 , near $(t_1, \varphi(t_1))$ F is L -Lipschitz, and $u = \|\varphi_1 - \varphi_2\|$ satisfies $u(t) \leq L \int_{t_1}^t u$ for t near t_1 , so $u = 0$ near t_1 by Lemma 25.12.2 (with $C = 0$, on both sides). So the set is open, and equals the connected set $J_1 \cap J_2$.

(3) By (2), the solutions with $\varphi(t_0) = x_0$ glue to a solution on the union of their intervals, which is maximal; the interval is open because a solution on a closed end could be continued by (1). Suppose $\beta < \infty$ and $(t_k, \varphi(t_k)) \in K$ for a compact $K \subseteq D$ and $t_k \rightarrow \beta$. By compactness there are a, b, L, M as in (1) valid for all rectangles centred at points of K (cover K by finitely many rectangles of half the size), hence a uniform δ . For $t_k > \beta - \delta/2$, the solution through $(t_k, \varphi(t_k))$ exists up to $t_k + \delta > \beta$ and extends φ by (2), a contradiction. $\square$

Counterexample 25.12.4. 1. Without the Lipschitz condition uniqueness fails: $x' = 3x^{2/3}$, $x(0) = 0$ has the solutions $x = 0$ and $x = t^3$.

2. Solutions need not exist for all time: $x' = x^2$, $x(0) = 1$ has maximal solution $x = 1/(1-t)$ on $(-\infty, 1)$.

Proposition 25.12.5. 1. *If φ, ψ are solutions on $[t_0, t_1]$ and $\|F(t, \varphi(t)) - F(t, \psi(t))\| \leq L\|\varphi(t) - \psi(t)\|$ for all t , then $\|\varphi(t) - \psi(t)\| \leq \|\varphi(t_0) - \psi(t_0)\|e^{L(t-t_0)}$.*

2. (Linear equations) *If $A: I \rightarrow M_n(\mathbb{R})$ and $b: I \rightarrow \mathbb{R}^n$ are continuous on an open interval I , the maximal solutions of $x' = A(t)x + b(t)$ are defined on all of I .*

Proof. (1) $u = \|\varphi - \psi\|$ satisfies $u(t) \leq u(t_0) + L \int_{t_0}^t u$; apply Lemma 25.12.2. (2) On a compact subinterval $[c, d] \ni t_0$ with $\|A\| \leq L$ and $\|b\| \leq B$, $\|\varphi(t)\| \leq \|\varphi(t_0)\| + B(d - c) + L \int \|\varphi\|$, so φ is bounded there by Grönwall, its graph stays in a compact set, and by Theorem 25.12.3(3) the maximal interval contains $[c, d]$. $\square$

Theorem 25.12.6 (Smooth dependence). *If F is C^k with $k \geq 1$, the flow $\Phi(t, t_0, x_0)$, the value at t of the maximal solution with $\varphi(t_0) = x_0$, is defined on an open subset of $\mathbb{R} \times D$ and is C^k there. Its derivative in x_0 solves the linearised equation $Y' = D_x F(t, \varphi(t))Y$, $Y(t_0) = 1$.*

This is [Har02, Chapter V]; see also [Lan93]. Continuity of Φ follows from Proposition 25.12.5(1) and the uniform δ in the proof of Theorem 25.12.3.

25.13 Stone–Weierstrass and Arzelà–Ascoli

Lemma 25.13.1. *The polynomials $p_0 = 0$, $p_{k+1}(t) = p_k(t) + (t^2 - p_k(t)^2)/2$ converge uniformly to $|t|$ on $[-1, 1]$, and $p_k(0) = 0$.*

Proof. By induction $0 \leq p_k(t) \leq p_{k+1}(t) \leq |t|$: indeed $|t| - p_{k+1} = (|t| - p_k)(1 - (|t| + p_k)/2)$ with $0 \leq (|t| + p_k)/2 \leq |t| \leq 1$, and $p_{k+1} - p_k = (t^2 - p_k^2)/2 \geq 0$. Iterating, $0 \leq |t| - p_k(t) \leq |t|(1 - |t|/2)^k$, and the maximum of $s(1 - s/2)^k$ on $[0, 1]$ is at most $2/(k + 1)$ (it is attained at $s = 2/(k + 1)$ when $k \geq 1$). $\square$

Theorem 25.13.2 (Stone–Weierstrass). *Let K be a compact Hausdorff space and $\mathcal{A} \subseteq C(K, \mathbb{R})$ a subalgebra containing the constants and separating points. Then $\mathcal{A}$ is dense for the supremum norm. The same holds for $\mathcal{A} \subseteq C(K, \mathbb{C})$ if $\mathcal{A}$ is also closed under complex conjugation.*

Proof. The closure $\bar{\mathcal{A}}$ is a closed subalgebra. If $f \in \bar{\mathcal{A}}$ with $\|f\| \leq c$, then $|f| = c \lim_k p_k(f/c) \in \bar{\mathcal{A}}$ by Lemma 25.13.1; hence $\max(f, g) = (f + g + |f - g|)/2$ and $\min(f, g)$ lie in $\bar{\mathcal{A}}$. For $x \neq y$ and reals α, β there is $h \in \mathcal{A}$ with $h(x) = \alpha$, $h(y) = \beta$ (take $h = \alpha + (\beta - \alpha)(u - u(x))/(u(y) - u(x))$ with u separating x and y).

Let $f \in C(K, \mathbb{R})$ and $\varepsilon > 0$. Fix x . For each y choose $h_y \in \mathcal{A}$ with $h_y(x) = f(x)$ and $h_y(y) = f(y)$; the open sets $\{h_y < f + \varepsilon\}$ cover K , and the minimum g_x of finitely many of these h_y lies in $\bar{\mathcal{A}}$, with $g_x(x) = f(x)$ and $g_x < f + \varepsilon$. The open sets $\{g_x > f - \varepsilon\}$ cover K , and the maximum g of finitely many g_x lies in $\bar{\mathcal{A}}$ and satisfies $\|g - f\| < \varepsilon$. For the complex case, the real parts $(f + \bar{f})/2$ of elements of $\mathcal{A}$ form a real subalgebra with the same properties, which is dense in $C(K, \mathbb{R})$; so $\mathcal{A}$ is dense in $C(K, \mathbb{C})$. $\square$

Corollary 25.13.3. *Polynomials are dense in $C(K, \mathbb{R})$ for every compact $K \subseteq \mathbb{R}^n$, and trigonometric polynomials $\sum_{|k| \leq N} c_k e^{2\pi i k t}$ are dense in the space of continuous functions $\mathbb{R} \rightarrow \mathbb{C}$ of period 1.*

Proof. The first statement is immediate. For the second, continuous functions of period 1 are continuous functions on $S^1 = \mathbb{R}/\mathbb{Z}$ (Theorem 25.7.3), and the polynomials in z and $\bar{z}$ restricted to S^1 form a conjugation-closed separating algebra; on S^1 , $z^k \bar{z}^l = z^{k-l}$ corresponds to $e^{2\pi i(k-l)t}$. $\square$

Theorem 25.13.4 (Arzelà–Ascoli). *Let K be a compact Hausdorff space and $\mathcal{F} \subseteq C(K, \mathbb{R}^m)$ a family that is equicontinuous (for every x and $\varepsilon > 0$ there is a neighbourhood U of x with $\|f(x') - f(x)\| < \varepsilon$ for all $x' \in U$ and $f \in \mathcal{F}$) and pointwise bounded. Then $\mathcal{F}$ has compact closure in $C(K, \mathbb{R}^m)$ with the supremum metric.*

Proof. $C(K, \mathbb{R}^m)$ is complete, so by Theorem 16.7.4 it suffices to show that $\mathcal{F}$ is totally bounded. Let $\varepsilon > 0$. Choose finitely many points $x_1, \dots, x_k$ with neighbourhoods U_i as in the definition covering K . The set $\{(f(x_1), \dots, f(x_k)) : f \in \mathcal{F}\}$ is bounded in $\mathbb{R}^{mk}$, hence totally bounded,

so there are $f_1, \dots, f_N \in \mathcal{F}$ such that every $f \in \mathcal{F}$ satisfies $\max_i \|f(x_i) - f_j(x_i)\| < \varepsilon$ for some j . For $x \in U_i$, $\|f(x) - f_j(x)\| \leq \|f(x) - f(x_i)\| + \|f(x_i) - f_j(x_i)\| + \|f_j(x_i) - f_j(x)\| < 3\varepsilon$. $\square$

25.14 Exercises

Exercise 25.14.1. Show that $e = \exp(1)$ is irrational.

Solution. If $e = p/q$, then $x = q!(e - \sum_{k \leq q} 1/k!)$ is a positive integer, but $x = \sum_{k > q} q!/k! < \sum_{j \geq 1} (q+1)^{-j} = 1/q \leq 1$. $\square$

Exercise 25.14.2. Show that $\sum 1/n$ diverges while $\sum (-1)^n/n$ converges, and that the latter series can be rearranged to converge to any given real number.

Solution. $\sum_{2^j < n \leq 2^{j+1}} 1/n \geq 1/2$, so the partial sums are unbounded; the alternating series converges by Lemma 25.5.5. For a target c , add positive terms $1/(2k)$ until the partial sum exceeds c , then negative terms until it is below c , and so on; since both the positive and the negative parts diverge and the terms tend to 0, the partial sums converge to c . $\square$

Exercise 25.14.3. Show that $f_n(x) = \sqrt{x^2 + 1/n}$ are smooth and converge uniformly on $\mathbb{R}$ to $|x|$, which is not differentiable. Prove that if $f_n \rightarrow f$ pointwise and $f'_n \rightarrow g$ uniformly on an interval, with f_n continuously differentiable, then f is differentiable with $f' = g$.

Solution. $0 \leq \sqrt{x^2 + 1/n} - |x| = \frac{1/n}{\sqrt{x^2 + 1/n} + |x|} \leq 1/\sqrt{n}$. For the second part, $f_n(x) = f_n(a) + \int_a^x f'_n$ converges to $f(a) + \int_a^x g$, so $f(x) = f(a) + \int_a^x g$ and Theorem 25.8.3 applies. $\square$

Exercise 25.14.4. Show that $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$ is a local diffeomorphism $(0, \infty) \times \mathbb{R} \rightarrow \mathbb{R}^2 \setminus \{0\}$, but not injective, and compute its Jacobian determinant.

Exercise 25.14.5. (Peano's theorem.) Show that if F is only continuous, $x' = F(t, x)$ still has local solutions, by applying the Arzelà–Ascoli theorem to the Euler polygons φ_k with step $1/k$.

Exercise 25.14.6. Show that a compact subset of $C(K, \mathbb{R}^m)$ is equicontinuous and pointwise bounded.

Exercise 25.14.7. Compute $\lim_n \frac{n^2+1}{2n^2-n}$, $\lim_n (\sqrt{n^2+n} - n)$ and $\lim_n (2^n + 3^n)^{1/n}$.

Solution. Dividing by n^2 , the first is $\frac{1+1/n^2}{2-1/n} \rightarrow \frac{1}{2}$. For the second, $\sqrt{n^2+n} - n = \frac{n}{\sqrt{n^2+n} + n} = \frac{1}{\sqrt{1+1/n} + 1} \rightarrow \frac{1}{2}$. For the third, $3 \leq (2^n + 3^n)^{1/n} \leq (2 \cdot 3^n)^{1/n} = 3 \cdot 2^{1/n} \rightarrow 3$ (Proposition 25.2.9(2)), so the limit is 3 by the squeeze theorem. $\square$

Exercise 25.14.8. Show that $\limsup_n a_n$ is the largest element of $\bar{\mathbb{R}}$ that is a limit of a subsequence of (a_n) , and $\liminf_n a_n$ the smallest.

Solution. Let $S_n = \sup_{k \geq n} a_k$, a decreasing sequence with limit $L = \limsup_n a_n$ (Proposition 25.2.10). If $a_{n_k} \rightarrow \ell$, then $a_{n_k} \leq S_n$ for $n_k \geq n$, so $\ell \leq S_n$ for all n , and $\ell \leq L$. Conversely, if $L \in \mathbb{R}$, choose recursively $n_k > n_{k-1}$ with $S_{n_{k-1}+1} - \frac{1}{k} < a_{n_k} \leq S_{n_{k-1}+1}$; both bounds tend to L , so $a_{n_k} \rightarrow L$ by the squeeze theorem. If $L = +\infty$ the sequence is unbounded above and Theorem 25.2.12 applies; if $L = -\infty$ then $a_n \leq S_n \rightarrow -\infty$. The statement for $\liminf$ follows by passing to $(-a_n)$. $\square$

Exercise 25.14.9. Show that $f(x) = x \mathbf{1}_{\mathbb{Q}}(x)$ is continuous at 0 and discontinuous at every $a \neq 0$.

Solution. $|f(x)| \leq |x|$, so $f(x) \rightarrow 0 = f(0)$ as $x \rightarrow 0$. For $a \neq 0$, choose rational $x_n \rightarrow a$ and irrational $y_n \rightarrow a$ (Counterexample 25.3.10); then $f(x_n) = x_n \rightarrow a$ and $f(y_n) = 0$, so f has no limit at a (Theorem 25.3.4), in particular it is not continuous there. $\square$

Exercise 25.14.10. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be continuous with $f(x+y) = f(x) + f(y)$ for all x, y . Show that $f(x) = cx$ with $c = f(1)$.

Solution. $f(0) = f(0) + f(0)$ gives $f(0) = 0$, then $f(-x) = -f(x)$, and by induction $f(nx) = nf(x)$ for $n \in \mathbb{N}$, hence for $n \in \mathbb{Z}$. For $x = p/q$ with $q \geq 1$, $qf(p/q) = f(p) = pc$, so $f(x) = cx$ on $\mathbb{Q}$. For real x , choose rational $x_n \rightarrow x$ (density of $\mathbb{Q}$); by continuity $f(x) = \lim f(x_n) = \lim cx_n = cx$. $\square$

Exercise 25.14.11. Show that a continuous periodic function $f: \mathbb{R} \rightarrow \mathbb{R}$ is bounded and uniformly continuous.

Solution. Let $p > 0$ be a period. On $[0, 2p]$, f is bounded and uniformly continuous (Theorem 25.4.10 and Proposition 25.4.13(3)), and every value of f is taken on $[0, p]$. Given ε , let $\delta < p$ work on $[0, 2p]$. If $|x - y| < \delta$, translating both by the same multiple of p we may assume $0 \leq \min(x, y) < p$, and then $x, y \in [0, 2p]$, so $|f(x) - f(y)| < \varepsilon$. $\square$

Exercise 25.14.12. Show that $n(a^{1/n} - 1) \rightarrow \log a$ for every $a > 0$.

Solution. Let $t = \log a$, so $a^{1/n} = e^{t/n}$. If $t = 0$ both sides vanish. Otherwise $n(e^{t/n} - 1) = t \cdot \frac{e^{t/n} - 1}{t/n} \rightarrow t$ by Proposition 25.7.5(1) and Theorem 25.3.4, since $t/n \rightarrow 0$ and $t/n \neq 0$. $\square$

Chapter 26

Measure and Integration

26.1 Introduction

The Lebesgue integral extends the Riemann integral to a class of functions that is closed under the limits needed in analysis: the monotone and dominated convergence theorems (Theorems 26.4.2 and 26.4.4) and the completeness of the spaces L^p (Theorem 26.5.3) are the basis of the functional analysis used to prove the Hodge theorem. This chapter constructs Lebesgue measure, proves the convergence theorems, Fubini's theorem and the change of variables formula, and the Radon–Nikodym theorem. References: [Rud87] and [Fol99].

Prerequisites. Chapter 25 (Real Analysis).

We use the extended half-line $[0, \infty]$ with $a + \infty = \infty$ and $0 \cdot \infty = 0$.

26.2 Measures

Definition 26.2.1. A σ -algebra on a set X is a set $\mathcal{A}$ of subsets containing X and closed under complements and countable unions. The σ -algebra *generated* by a family of subsets is the intersection of all σ -algebras containing it; for a topological space, the *Borel* σ -algebra $\mathcal{B}(X)$ is generated by the open sets. A *measure* on $(X, \mathcal{A})$ is a map $\mu: \mathcal{A} \rightarrow [0, \infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_k A_k) = \sum_k \mu(A_k)$ for countable disjoint families. It is *finite* if $\mu(X) < \infty$, *σ -finite* if X is a countable union of sets of finite measure, and *complete* if subsets of measure zero sets are measurable. A property holds *almost everywhere* (a.e.) if it holds outside a set of measure zero.

Proposition 26.2.2. Let μ be a measure. If $A \subseteq B$, then $\mu(A) \leq \mu(B)$; $\mu(\bigcup_k A_k) \leq \sum_k \mu(A_k)$; if $A_1 \subseteq A_2 \subseteq \dots$, then $\mu(\bigcup_k A_k) = \lim_k \mu(A_k)$; and if $A_1 \supseteq A_2 \supseteq \dots$ with $\mu(A_1) < \infty$, then $\mu(\bigcap_k A_k) = \lim_k \mu(A_k)$.

Proof. $\mu(B) = \mu(A) + \mu(B \setminus A)$. Replacing A_k by $A_k \setminus \bigcup_{j < k} A_j$ gives disjoint sets with the same union and smaller measures. For increasing sets, $\bigcup_k A_k = \bigsqcup_k (A_k \setminus A_{k-1})$, and the partial sums of the measures are $\mu(A_k)$. For decreasing sets apply this to $A_1 \setminus A_k$. $\square$

Definition 26.2.3. A map $f: (X, \mathcal{A}) \rightarrow (Y, \mathcal{B})$ is *measurable* if $f^{-1}(B) \in \mathcal{A}$ for $B \in \mathcal{B}$. Functions to $[-\infty, \infty]$, $\mathbb{R}$ or $\mathbb{C}$ are measurable for the Borel σ -algebras. A *simple function* is a finite combination $\sum_i a_i \mathbf{1}_{A_i}$ of indicator functions of measurable sets.

Proposition 26.2.4. 1. $f: X \rightarrow [-\infty, \infty]$ is measurable if and only if $\{f > a\}$ is measurable for all $a \in \mathbb{R}$.

2. If $f_1, \dots, f_k: X \rightarrow \mathbb{R}$ are measurable and $\Phi: \mathbb{R}^k \rightarrow \mathbb{R}$ is continuous, $\Phi(f_1, \dots, f_k)$ is measurable; so sums and products of measurable functions are measurable, and $f: X \rightarrow \mathbb{C}$ is measurable if and only if its real and imaginary parts are.
3. Pointwise sup, inf, lim sup and lim inf of sequences of measurable functions are measurable.
4. Every measurable $f: X \rightarrow [0, \infty]$ is the pointwise limit of an increasing sequence of simple functions $0 \leq s_1 \leq s_2 \leq \dots$.

Proof. (1) The sets $(a, \infty]$ generate the Borel σ -algebra of $[-\infty, \infty]$, and the sets B with $f^{-1}(B)$ measurable form a σ -algebra. (2) Open subsets of $\mathbb{R}^k$ are countable unions of products of open intervals, and $(f_1, \dots, f_k)^{-1}$ of such a product is a finite intersection of measurable sets; so $(f_1, \dots, f_k)$ is measurable for Borel sets, and so is its composite with Φ . (3) $\{\sup_n f_n > a\} = \bigcup_n \{f_n > a\}$, $\inf_n f_n = -\sup_n (-f_n)$, and $\limsup f_n = \inf_n \sup_{k \geq n} f_k$. (4) Let $s_n = \min(n, 2^{-n} \lfloor 2^n f \rfloor)$. $\square$

26.3 Outer measures and Lebesgue measure

Definition 26.3.1. An *outer measure* on X is a map $\mu^*: \mathcal{P}(X) \rightarrow [0, \infty]$ with $\mu^*(\emptyset) = 0$, monotone and countably subadditive. A set A is μ^* -*measurable* if $\mu^*(E) = \mu^*(E \cap A) + \mu^*(E \setminus A)$ for all $E \subseteq X$.

Theorem 26.3.2 (Carathéodory). *The μ^* -measurable sets form a σ -algebra $\mathcal{M}$, and $\mu^*|_{\mathcal{M}}$ is a complete measure.*

Proof. By subadditivity, A is measurable if and only if $\mu^*(E \cap A) + \mu^*(E \setminus A) \leq \mu^*(E)$ for all E . The definition is symmetric in A and $X \setminus A$. For measurable A, B and any E ,

$$\mu^*(E \cap (A \cup B)) + \mu^*(E \setminus (A \cup B)) \leq \mu^*(E \cap A) + \mu^*((E \setminus A) \cap B) + \mu^*((E \setminus A) \setminus B) = \mu^*(E \cap A) + \mu^*(E \setminus A) = \mu^*(E),$$

so $\mathcal{M}$ is closed under finite unions. Let $A_1, A_2, \dots \in \mathcal{M}$ be disjoint, $B_n = A_1 \cup \dots \cup A_n$ and $B = \bigcup_n A_n$. Measurability of A_n gives $\mu^*(E \cap B_n) = \mu^*(E \cap A_n) + \mu^*(E \cap B_{n-1})$, so $\mu^*(E \cap B_n) = \sum_{j \leq n} \mu^*(E \cap A_j)$, and $\mu^*(E) = \mu^*(E \cap B_n) + \mu^*(E \setminus B_n) \geq \sum_{j \leq n} \mu^*(E \cap A_j) + \mu^*(E \setminus B)$. Letting $n \rightarrow \infty$ and using subadditivity, $\mu^*(E) \geq \sum_j \mu^*(E \cap A_j) + \mu^*(E \setminus B) \geq \mu^*(E \cap B) + \mu^*(E \setminus B)$. So $B \in \mathcal{M}$, and $E = B$ gives countable additivity. General countable unions reduce to disjoint ones. If $\mu^*(A) = 0$, then $\mu^*(E \cap A) + \mu^*(E \setminus A) \leq \mu^*(E)$, so A is measurable. $\square$

Definition 26.3.3. A *box* in $\mathbb{R}^n$ is a product $Q = \prod_i [a_i, b_i]$ with volume $v(Q) = \prod_i (b_i - a_i)$. The *Lebesgue outer measure* of $A \subseteq \mathbb{R}^n$ is $\lambda^*(A) = \inf\{\sum_k v(Q_k) : A \subseteq \bigcup_k Q_k\}$, the infimum over countable covers by boxes. The λ^* -measurable sets are *Lebesgue measurable*, and $\lambda = \lambda^n$ is *Lebesgue measure*.

Theorem 26.3.4. λ^* is an outer measure, and $\lambda^*(Q) = v(Q)$ for boxes. Every Borel set is Lebesgue measurable, so λ is a complete measure on a σ -algebra containing $\mathcal{B}(\mathbb{R}^n)$. It is translation invariant and regular: for measurable E , $\lambda(E) = \inf\{\lambda(U) : U \supseteq E \text{ open}\} = \sup\{\lambda(K) : K \subseteq E \text{ compact}\}$.

Proof. Countable subadditivity: given covers of A_k with total volume $\leq \lambda^*(A_k) + \varepsilon 2^{-k}$, their union covers $\bigcup_k A_k$.

Volume of a box. Clearly $\lambda^*(Q) \leq v(Q)$. Let $Q \subseteq \bigcup_k Q_k$ and $\varepsilon > 0$. Enlarge each Q_k to a box Q'_k whose interior contains Q_k , with $v(Q'_k) \leq v(Q_k) + \varepsilon 2^{-k}$. By compactness finitely many interiors cover Q , say those of $Q'_1, \dots, Q'_N$. The endpoints of Q and of the Q'_k in each coordinate define a grid dividing Q into boxes C with disjoint interiors, and $v(Q) = \sum_C v(C)$ (volume is multiplicative and length is additive). The centre of each C

lies in some Q'_k , and then $C \subseteq Q'_k$ because no grid line crosses the interior of C . The cells contained in Q'_k have disjoint interiors, so their volumes add up to at most $v(Q'_k)$. Hence $v(Q) \leq \sum_k v(Q'_k) \leq \sum_k v(Q_k) + \varepsilon$.

Borel sets. The half-spaces $H = \{x_i \leq c\}$ generate $\mathcal{B}(\mathbb{R}^n)$, since open sets are countable unions of boxes with rational vertices. Given E and a cover by boxes Q_k with $\sum v(Q_k) \leq \lambda^*(E) + \varepsilon$, the boxes $Q_k \cap H$ and $Q_k \cap \{x_i \geq c\}$ cover $E \cap H$ and $E \setminus H$ and have total volume $\sum_k v(Q_k)$. So $\lambda^*(E \cap H) + \lambda^*(E \setminus H) \leq \lambda^*(E) + \varepsilon$, and H is measurable by Theorem 26.3.2.

Invariance and regularity. Translation maps boxes to boxes of the same volume. Enlarging the covering boxes slightly into open boxes gives outer regularity. For inner regularity, first let E be bounded, contained in a closed box Q . Given ε , choose open $U \supseteq Q \setminus E$ with $\lambda(U) \leq \lambda(Q \setminus E) + \varepsilon$; then $K = Q \setminus U \subseteq E$ is compact with $\lambda(E \setminus K) \leq \lambda(U \setminus (Q \setminus E)) \leq \varepsilon$. For general E , apply this to $E \cap [-m, m]^n$ and use continuity from below. $\square$

Counterexample 26.3.5. There are subsets of $\mathbb{R}$ that are not Lebesgue measurable. Choose (by the axiom of choice) a set $V \subseteq [0, 1]$ containing exactly one element of each class of $\mathbb{R}/\mathbb{Q}$. The translates $V + q$, $q \in \mathbb{Q} \cap [-1, 1]$, are disjoint and their union lies between $[0, 1]$ and $[-1, 2]$. If V were measurable, translation invariance and countable additivity would give $1 \leq \sum_q \lambda(V) \leq 3$, which is impossible whether $\lambda(V) = 0$ or $\lambda(V) > 0$.

Lemma 26.3.6 (Dynkin). *Let $\mathcal{P}$ be a family of subsets of X closed under finite intersections, and $\mathcal{L} \supseteq \mathcal{P}$ a family containing X , closed under complements and countable disjoint unions. Then $\mathcal{L}$ contains the σ -algebra generated by $\mathcal{P}$. Consequently two measures that agree on $\mathcal{P}$ and are finite on an increasing sequence of sets of $\mathcal{P}$ exhausting X agree on $\sigma(\mathcal{P})$.*

Proof. Let $\mathcal{L}_0$ be the smallest such family containing $\mathcal{P}$. For $A \in \mathcal{P}$, the sets B with $A \cap B \in \mathcal{L}_0$ form such a family (for complements use $A \setminus B = X \setminus ((X \setminus A) \cup (A \cap B))$) containing $\mathcal{P}$, hence $\mathcal{L}_0$. Repeating with $A \in \mathcal{L}_0$ shows that $\mathcal{L}_0$ is closed under finite intersections, hence under all countable unions (disjointify), so it is a σ -algebra. For the consequence, if $\mu = \nu$ on $\mathcal{P}$ and $\mu(X_m) = \nu(X_m) < \infty$, the sets B with $\mu(B \cap X_m) = \nu(B \cap X_m)$ form such a family; let $m \rightarrow \infty$. $\square$

26.4 The integral

Definition 26.4.1. For a simple function $s = \sum_i a_i \mathbf{1}_{A_i} \geq 0$ with disjoint A_i , $\int s \, d\mu = \sum_i a_i \mu(A_i)$ (independent of the representation, by passing to a common refinement). For measurable $f: X \rightarrow [0, \infty]$, $\int f \, d\mu = \sup\{\int s \, d\mu : 0 \leq s \leq f \text{ simple}\}$, and $\int_E f = \int f \mathbf{1}_E$. A measurable $f: X \rightarrow \mathbb{C}$ is *integrable* if $\int |f| < \infty$; then $\int f = \int (\operatorname{Re} f)^+ - \int (\operatorname{Re} f)^- + i(\int (\operatorname{Im} f)^+ - \int (\operatorname{Im} f)^-)$, where $g^\pm = \max(\pm g, 0)$.

Theorem 26.4.2 (Monotone convergence). *If $0 \leq f_1 \leq f_2 \leq \dots$ are measurable and $f = \lim f_n$, then $\int f = \lim \int f_n$.*

Proof. The integrals increase and are at most $\int f$. Let $0 \leq s \leq f$ be simple and $0 < c < 1$, and $E_n = \{f_n \geq cs\}$, which increase to X . Then $\int f_n \geq c \int_{E_n} s$, and $A \mapsto \int_A s$ is a measure, so $\int_{E_n} s \rightarrow \int s$ (Proposition 26.2.2). Hence $\lim \int f_n \geq c \int s$ for all such c and s . $\square$

Corollary 26.4.3. 1. *For measurable $f, g \geq 0$ and $a \geq 0$, $\int (af + g) = a \int f + \int g$, and $\int \sum_n f_n = \sum_n \int f_n$.*

2. (Fatou) *For measurable $f_n \geq 0$, $\int \liminf f_n \leq \liminf \int f_n$.*

3. *The integral is linear on integrable functions and $|\int f| \leq \int |f|$.*

4. For measurable $f \geq 0$, $\int f = 0$ if and only if $f = 0$ a.e.; and $\mu(\{f \geq t\}) \leq \frac{1}{t} \int f$ (Chebyshev).

Proof. (1) For simple functions additivity is clear; approximate by increasing simple functions (Proposition 26.2.4(4)) and use monotone convergence. (2) Apply monotone convergence to $g_n = \inf_{k \geq n} f_k$, which satisfy $\int g_n \leq \inf_{k \geq n} \int f_k$. (3) Linearity follows from (1) by decomposing into positive and negative parts. For complex f , write $\int f = |\int f| e^{i\theta}$; then $|\int f| = \int \operatorname{Re}(e^{-i\theta} f) \leq \int |f|$. (4) $t \mathbf{1}_{\{f \geq t\}} \leq f$; and $\{f > 0\} = \bigcup_n \{f \geq 1/n\}$. $\square$

Theorem 26.4.4 (Dominated convergence). Let f_n be measurable with $f_n \rightarrow f$ a.e. and $|f_n| \leq g$ a.e. for an integrable g . Then f is integrable and $\int |f_n - f| \rightarrow 0$; in particular $\int f_n \rightarrow \int f$.

Proof. Modifying on a null set, we may assume convergence and domination everywhere. Then $|f| \leq g$, and Fatou's lemma applied to $2g - |f_n - f| \geq 0$ gives $\int 2g \leq \int 2g - \limsup \int |f_n - f|$. $\square$

Proposition 26.4.5. For continuous $f: [a, b] \rightarrow \mathbb{C}$, the Lebesgue integral $\int_{[a,b]} f d\lambda$ equals the Riemann integral.

Proof. For a partition P with mesh tending to 0, the step functions $\sum_i \min_{[t_{i-1}, t_i]} \operatorname{Re} f \cdot \mathbf{1}_{[t_{i-1}, t_i]}$ converge to $\operatorname{Re} f$ uniformly, and their Lebesgue integrals are the lower sums (Theorem 26.3.4). Apply dominated convergence, and treat the imaginary part in the same way. $\square$

Proposition 26.4.6. Every bounded Riemann integrable function $f: [a, b] \rightarrow \mathbb{R}$ (Definition 25.8.1) is Lebesgue measurable and Lebesgue integrable, with the same integral.

Proof. Choose partitions $P_1, P_2, \dots$, each refining the previous one, with $U(f, P_k) - L(f, P_k) < 1/k$ (Proposition 25.8.2(1)). Let ϕ_k and ψ_k be the step functions equal, on the interior of each interval of P_k , to the infimum and supremum of f there, and to f at the finitely many partition points. Then $\phi_k \leq f \leq \psi_k$, ϕ_k increases and ψ_k decreases (refinement), and their Lebesgue integrals are $L(f, P_k)$ and $U(f, P_k)$, since points have measure zero. The limits $\phi = \lim \phi_k$ and $\psi = \lim \psi_k$ are measurable, $\phi \leq f \leq \psi$, and by monotone convergence $\int (\psi - \phi) = \lim (U(f, P_k) - L(f, P_k)) = 0$. So $\psi = \phi$ a.e. (Corollary 26.4.3(4)), hence $f = \phi$ a.e., and f is measurable because Lebesgue measure is complete. Finally $\int f d\lambda = \lim \int \phi_k d\lambda = \lim L(f, P_k)$, which is the Riemann integral. $\square$

Counterexample 26.4.7. The converse fails. The Dirichlet function $\mathbf{1}_{\mathbb{Q}}$ on $[0, 1]$ is Lebesgue integrable with integral 0, since $\mathbb{Q}$ is countable and hence of measure zero. It is not Riemann integrable: every nondegenerate interval contains rational and irrational points, so $L(\mathbf{1}_{\mathbb{Q}}, P) = 0$ and $U(\mathbf{1}_{\mathbb{Q}}, P) = 1$ for every partition P . It is the increasing pointwise limit of the Riemann integrable functions $\mathbf{1}_{\{q_1, \dots, q_n\}}$, each of integral 0, for an enumeration of $\mathbb{Q} \cap [0, 1]$: the Riemann integral does not behave well under monotone limits, which is what the Lebesgue integral repairs (Theorem 26.4.2).

26.5 Lebesgue spaces

Definition 26.5.1. For $1 \leq p < \infty$, $L^p(\mu)$ is the space of measurable $f: X \rightarrow \mathbb{C}$ with $\|f\|_p = (\int |f|^p)^{1/p} < \infty$, modulo equality a.e. $L^\infty(\mu)$ consists of the essentially bounded functions with $\|f\|_\infty = \inf\{c : |f| \leq c \text{ a.e.}\}$, modulo equality a.e. On $L^2(\mu)$ we have the inner product $\langle f, g \rangle = \int f \bar{g}$.

Proposition 26.5.2. Let $1 \leq p, q \leq \infty$ with $1/p + 1/q = 1$. Then $\int |fg| \leq \|f\|_p \|g\|_q$ (Hölder) and $\|f + g\|_p \leq \|f\|_p + \|g\|_p$ (Minkowski). So $L^p(\mu)$ is a normed space.

Proof. The cases $p = 1$ or ∞ are clear. For $1 < p < \infty$ and $a, b > 0$, concavity of $\log$ ($\log'' < 0$) gives $\log(\frac{a^p}{p} + \frac{b^q}{q}) \geq \log a + \log b$, that is, $ab \leq a^p/p + b^q/q$. If $\|f\|_p = \|g\|_q = 1$, integrating $|fg| \leq |f|^p/p + |g|^q/q$ gives $\int |fg| \leq 1$; the general case follows by scaling. For Minkowski, $|f + g|^p \leq |f|^p + |g|^p$ and Hölder with $(p-1)q = p$ give $\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p)\|f + g\|_p^{p-1}$; also $|f + g|^p \leq 2^p(|f|^p + |g|^p)$ shows $\|f + g\|_p < \infty$. Functions with $\|f\|_p = 0$ vanish a.e. by Corollary 26.4.3(4). $\square$

Theorem 26.5.3 (Riesz–Fischer). $L^p(\mu)$ is a Banach space for $1 \leq p \leq \infty$, and every convergent sequence in L^p , $p < \infty$, has a subsequence converging a.e.

Proof. Let $p < \infty$ and (f_n) Cauchy. Choose $n_1 < n_2 < \dots$ with $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$, and let $g = \sum_k |f_{n_{k+1}} - f_{n_k}|$. By monotone convergence and Minkowski, $\|g\|_p \leq 1$, so $g < \infty$ a.e., and $f = f_{n_1} + \sum_k (f_{n_{k+1}} - f_{n_k})$ converges absolutely a.e. Since $|f - f_{n_k}| \leq \sum_{j \geq k} |f_{n_{j+1}} - f_{n_j}|$, Minkowski gives $\|f - f_{n_k}\|_p \leq 2^{1-k}$, and a Cauchy sequence with a convergent subsequence converges. For $p = \infty$, outside a null set $|f_n - f_m| \leq \|f_n - f_m\|_\infty$ for all n, m , so (f_n) converges uniformly there. $\square$

Proposition 26.5.4. For $1 \leq p < \infty$, the continuous functions with compact support are dense in $L^p(\mathbb{R}^n, \lambda)$.

Proof. By Proposition 26.2.4(4) and dominated convergence, simple functions $\sum a_i \mathbf{1}_{E_i}$ with $\lambda(E_i) < \infty$ are dense. Given E with $\lambda(E) < \infty$ and $\varepsilon > 0$, choose compact $K \subseteq E \subseteq U$ open with $\lambda(U \setminus K) < \varepsilon$ (Theorem 26.3.4) and, by Lemma 16.9.5, a continuous $\phi: \mathbb{R}^n \rightarrow [0, 1]$ with compact support in U and $\phi = 1$ on K . Then $\|\mathbf{1}_E - \phi\|_p^p \leq \lambda(U \setminus K) < \varepsilon$. $\square$

26.6 Modes of convergence

Definition 26.6.1. Let f_n, f be measurable functions on a measure space (X, μ) with values in $\mathbb{C}$. Then $f_n \rightarrow f$ *almost everywhere* if $f_n(x) \rightarrow f(x)$ for x outside a null set; *in measure* if $\mu(\{|f_n - f| > \varepsilon\}) \rightarrow 0$ for every $\varepsilon > 0$; and *in L^p* if $\|f_n - f\|_p \rightarrow 0$.

Theorem 26.6.2 (Egorov). Let $\mu(X) < \infty$ and $f_n \rightarrow f$ almost everywhere. For every $\delta > 0$ there is a measurable E with $\mu(X \setminus E) < \delta$ such that $f_n \rightarrow f$ uniformly on E .

Proof. For $k, n \geq 1$ let $E_{k,n} = \bigcup_{m \geq n} \{|f_m - f| \geq 1/k\}$. For fixed k these sets decrease in n , and their intersection consists of points where $|f_m - f| \geq 1/k$ for infinitely many m , a null set. As $\mu(X) < \infty$, $\mu(E_{k,n}) \rightarrow 0$ as $n \rightarrow \infty$ (Proposition 26.2.2). Choose n_k with $\mu(E_{k,n_k}) < \delta 2^{-k}$ and let $E = X \setminus \bigcup_k E_{k,n_k}$. Then $\mu(X \setminus E) < \delta$, and on E , $|f_m - f| < 1/k$ for all $m \geq n_k$, for every k . $\square$

Proposition 26.6.3. 1. If $f_n \rightarrow f$ in L^p for some $1 \leq p < \infty$, then $f_n \rightarrow f$ in measure.

2. If $f_n \rightarrow f$ in measure, a subsequence converges to f almost everywhere.

3. If $\mu(X) < \infty$ and $f_n \rightarrow f$ almost everywhere, then $f_n \rightarrow f$ in measure.

Proof. (1) By Chebyshev's inequality (Corollary 26.4.3(4)) applied to $|f_n - f|^p$, $\mu(\{|f_n - f| > \varepsilon\}) \leq \varepsilon^{-p} \|f_n - f\|_p^p$. (2) Choose $n_1 < n_2 < \dots$ with $\mu(A_k) < 2^{-k}$, where $A_k = \{|f_{n_k} - f| > 2^{-k}\}$. The set A of points lying in infinitely many A_k is contained in $\bigcup_{k \geq K} A_k$ for every K , so $\mu(A) \leq \sum_{k \geq K} 2^{-k} = 2^{1-K}$ for all K , and $\mu(A) = 0$. For $x \notin A$, $|f_{n_k}(x) - f(x)| \leq 2^{-k}$ for large k . (3) Given $\varepsilon, \delta > 0$, Egorov's theorem gives E with $\mu(X \setminus E) < \delta$ and $|f_n - f| < \varepsilon$ on E for large n , so $\mu(\{|f_n - f| > \varepsilon\}) < \delta$ for large n . $\square$

Counterexample 26.6.4. No other implications hold in general. Take $X = [0, 1]$ or $\mathbb{R}$ with Lebesgue measure.

1. $f_n = n\mathbf{1}_{(0,1/n)}$ on $[0, 1]$ converges to 0 everywhere, but $\int f_n = 1$: almost everywhere convergence and convergence in measure do not imply L^1 convergence, even on a finite measure space.
2. $f_n = \mathbf{1}_{[n, n+1]}$ on $\mathbb{R}$ converges to 0 everywhere but not in measure, as $\lambda(\{f_n > \frac{1}{2}\}) = 1$: part (3) of Proposition 26.6.3 and Egorov's theorem need $\mu(X) < \infty$.
3. (The typewriter sequence.) For $n = 2^k + j$ with $0 \leq j < 2^k$ let $f_n = \mathbf{1}_{[j2^{-k}, (j+1)2^{-k}]}$ on $[0, 1]$. Then $\|f_n\|_1 = 2^{-k} \rightarrow 0$, so $f_n \rightarrow 0$ in L^1 and in measure; but every $x \in [0, 1]$ lies in one of the intervals for each k and outside most of them, so $f_n(x) = 1$ and $f_n(x) = 0$ both happen infinitely often, and $(f_n(x))$ converges for no x . The subsequence in part (2) of Proposition 26.6.3 is necessary.

26.7 Product measures and Fubini's theorem

Definition 26.7.1. For measurable spaces $(X, \mathcal{A})$ and $(Y, \mathcal{B})$, the *product σ -algebra* $\mathcal{A} \otimes \mathcal{B}$ on $X \times Y$ is generated by the *rectangles* $A \times B$. For $E \subseteq X \times Y$, the sections are $E_x = \{y : (x, y) \in E\}$ and $E^y = \{x : (x, y) \in E\}$.

Theorem 26.7.2. Let μ and ν be σ -finite measures on X and Y . For $E \in \mathcal{A} \otimes \mathcal{B}$, the sections are measurable, the functions $x \mapsto \nu(E_x)$ and $y \mapsto \mu(E^y)$ are measurable, and

$$(\mu \times \nu)(E) = \int \nu(E_x) d\mu(x) = \int \mu(E^y) d\nu(y)$$

defines the unique measure on $\mathcal{A} \otimes \mathcal{B}$ with $(\mu \times \nu)(A \times B) = \mu(A)\nu(B)$.

Proof. Assume first that μ and ν are finite. The sets E for which all claims hold (including equality of the two integrals) contain the rectangles, which are closed under finite intersections, and they are closed under complements and countable disjoint unions (by additivity and monotone convergence). Measurability of sections holds for all $E \in \mathcal{A} \otimes \mathcal{B}$ since the sets with measurable sections form a σ -algebra. By Lemma 26.3.6 the claims hold on $\mathcal{A} \otimes \mathcal{B}$, and countable additivity follows from monotone convergence. For σ -finite measures write $X \times Y$ as an increasing union of $X_m \times Y_m$ with finite measures and pass to the limit. Uniqueness follows from Lemma 26.3.6. $\square$

Theorem 26.7.3 (Tonelli and Fubini). Let μ, ν be σ -finite.

1. If $f: X \times Y \rightarrow [0, \infty]$ is measurable, then $x \mapsto \int f(x, y) d\nu(y)$ is measurable and $\int_{X \times Y} f d(\mu \times \nu) = \int \left(\int f(x, y) d\nu(y) \right) d\mu(x) = \int \left(\int f(x, y) d\mu(x) \right) d\nu(y)$.
2. If $f \in L^1(\mu \times \nu)$, then $f(x, -) \in L^1(\nu)$ for a.e. x , the function $x \mapsto \int f(x, y) d\nu(y)$ (defined a.e.) is integrable, and the same equalities hold.

Proof. (1) For indicator functions this is Theorem 26.7.2; it extends to simple functions by linearity and to all $f \geq 0$ by monotone convergence. (2) Apply (1) to $|f|$ to see that $\int |f(x, y)| d\nu(y) < \infty$ for a.e. x , and then to the positive and negative parts of the real and imaginary parts. $\square$

Counterexample 26.7.4. Integrability is needed in (2). On $\mathbb{N} \times \mathbb{N}$ with counting measures let $f(n, n) = 1$, $f(n, n+1) = -1$ and $f = 0$ otherwise. Then $\sum_m \sum_n f(n, m) = 1$, since only column $m = 0$ has nonzero sum, whereas $\sum_n \sum_m f(n, m) = 0$.

Remark 26.7.5. λ^{m+n} and $\lambda^m \times \lambda^n$ agree on boxes, hence on Borel sets by Lemma 26.3.6, and λ^{m+n} is the completion of $\lambda^m \times \lambda^n$. So Fubini's theorem applies to Lebesgue measure on $\mathbb{R}^{m+n} = \mathbb{R}^m \times \mathbb{R}^n$ for Borel functions, and for Lebesgue measurable functions with "for almost every x " qualifications.

26.8 Change of variables

Lemma 26.8.1. For $T \in \text{GL}_n(\mathbb{R})$ and Borel $E \subseteq \mathbb{R}^n$, $\lambda(T(E)) = |\det T| \lambda(E)$.

Proof. Both sides are multiplicative in T , and every invertible matrix is a product of *elementary* matrices (by Gaussian elimination): permutations of coordinates, scalings of one coordinate by $c \neq 0$, and shears $x_i \mapsto x_i + cx_j$ ($i \neq j$). Permutations and scalings map boxes to boxes, multiplying volumes by $|\det T|$, so they multiply outer measure by $|\det T|$. A shear preserves each line parallel to the i -th axis and acts on it by a translation; by Theorem 26.7.3 and Remark 26.7.5, integrating the one-dimensional measures of these sections shows $\lambda(T(E)) = \lambda(E)$, and $\det T = 1$. $\square$

Theorem 26.8.2 (Change of variables). Let $U, V \subseteq \mathbb{R}^n$ be open and $\phi: U \rightarrow V$ a C^1 -diffeomorphism. For every measurable $f: V \rightarrow [0, \infty]$, and every $f \in L^1(V)$,

$$\int_V f d\lambda = \int_U f(\phi(x)) |\det D\phi(x)| d\lambda(x).$$

Proof. We use the norm $\|x\|_\infty$ and the corresponding operator norm; the mean value inequality (Proposition 25.10.3(4)) holds for it with the same proof. Cubes are balls $Q = \{x : \|x - a\|_\infty \leq h\}$.

Step 1. Let $Q \subseteq U$ be a cube with centre a and $T \in \text{GL}_n(\mathbb{R})$, and let $M = \sup_{y \in Q} \|T^{-1} D\phi(y)\|$. By the mean value inequality applied to $T^{-1}\phi$, $\phi(Q) \subseteq \phi(a) + T(\{\|z\|_\infty \leq Mh\})$, so $\lambda(\phi(Q)) \leq |\det T| M^n \lambda(Q)$ by Lemma 26.8.1.

Step 2: $\lambda(\phi(Q)) \leq \int_Q |\det D\phi|$ for cubes $Q \subseteq U$. Let $\varepsilon > 0$. Since $D\phi$ is uniformly continuous on Q and $\|D\phi^{-1}\|$ is bounded there, we can divide Q into subcubes Q_j with centres x_j such that $\|D\phi(x_j)^{-1} D\phi(y)\| \leq 1 + \varepsilon$ for $y \in Q_j$, and $|\det D\phi|$ varies by less than ε on each Q_j . Step 1 with $T = D\phi(x_j)$ gives $\lambda(\phi(Q)) \leq \sum_j \lambda(\phi(Q_j)) \leq (1 + \varepsilon)^n \sum_j |\det D\phi(x_j)| \lambda(Q_j) \leq (1 + \varepsilon)^n (\int_Q |\det D\phi| + \varepsilon \lambda(Q))$.

Step 3: $\lambda(\phi(E)) \leq \int_E |\det D\phi|$ for measurable $E \subseteq U$. An open $W \subseteq U$ is a countable union of cubes with disjoint interiors (dyadic cubes), whose boundaries are null, so the inequality holds for W . Let $E \subseteq K$ with $K \subseteq U$ compact, and $C = \sup$ of $|\det D\phi|$ on a compact neighbourhood $N \subseteq U$ of K . By regularity choose open W with $E \subseteq W \subseteq N$ and $\lambda(W \setminus E) < \delta$; then $\lambda(\phi(E)) \leq \lambda(\phi(W)) \leq \int_E |\det D\phi| + C\delta$. This shows that ϕ maps null sets to null sets, hence Lebesgue measurable sets to Lebesgue measurable sets (a measurable set is a Borel set up to a null set), and gives the inequality for E in a compact subset; exhaust U by compact sets for general E .

Step 4. By Step 3, $\int_V \mathbf{1}_{\phi(E)} \leq \int_U \mathbf{1}_{\phi(E)} \circ \phi |\det D\phi|$; by linearity and monotone convergence, $\int_V f \leq \int_U (f \circ \phi) |\det D\phi|$ for all measurable $f \geq 0$. Applying this to ϕ^{-1} and the function $(f \circ \phi) |\det D\phi|$ on U , and using $D\phi(\phi^{-1}(y)) D(\phi^{-1})(y) = 1$, gives the reverse inequality. The L^1 case follows by linearity. $\square$

Example 26.8.3. With polar coordinates $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$, a diffeomorphism $(0, \infty) \times (0, 2\pi) \rightarrow \mathbb{R}^2$ minus a closed half-line (a null set), with Jacobian determinant r , Tonelli's theorem gives $(\int_{\mathbb{R}^2} e^{-x^2} dx)^2 = \int_{\mathbb{R}^2} e^{-\|x\|^2} = \int_0^{2\pi} \int_0^\infty e^{-r^2} r dr d\theta = \pi$. So $\int_{\mathbb{R}} e^{-x^2} dx = \sqrt{\pi}$.

26.9 The Radon–Nikodym and Riesz representation theorems

Lemma 26.9.1. *Let H be a complex inner product space that is complete (a Hilbert space), for instance $L^2(\mu)$. Every continuous linear functional $\Lambda: H \rightarrow \mathbb{C}$ is of the form $\Lambda(f) = \langle f, g \rangle$ for a unique $g \in H$.*

Proof. If $\Lambda = 0$ take $g = 0$. Otherwise $N = \ker \Lambda$ is a closed proper subspace. Choose $u \notin N$ and let $d = \inf_{v \in N} \|u - v\| > 0$. For $v_k \in N$ with $\|u - v_k\| \rightarrow d$, the parallelogram law $\|v_k - v_l\|^2 = 2\|u - v_k\|^2 + 2\|u - v_l\|^2 - 4\|u - \frac{v_k + v_l}{2}\|^2$ shows that (v_k) is Cauchy; its limit v satisfies $\|u - v\| = d$. For $w \in N$ and $t \in \mathbb{C}$, $\|u - v - tw\|^2 \geq d^2$ forces $\langle u - v, w \rangle = 0$. So $z = u - v \neq 0$ is orthogonal to N . For $f \in H$, $\Lambda(f)z - \Lambda(z)f \in N$, so $\Lambda(f)\|z\|^2 = \Lambda(z)\langle f, z \rangle$, and $g = \Lambda(z)z/\|z\|^2$ works. Uniqueness: if $\langle f, g - g' \rangle = 0$ for all f , take $f = g - g'$. $\square$

Theorem 26.9.2 (Radon–Nikodym). *Let μ and ν be σ -finite measures on $(X, \mathcal{A})$ with $\nu \ll \mu$, meaning $\mu(E) = 0 \Rightarrow \nu(E) = 0$. There is a measurable $h \geq 0$, unique a.e., with $\nu(E) = \int_E h \, d\mu$ for all E .*

Proof. Assume μ and ν finite; the σ -finite case follows by decomposing X . Let $\rho = \mu + \nu$. On $L^2(\rho)$, $f \mapsto \int f \, d\nu$ is continuous, since $|\int f \, d\nu| \leq \int |f| \, d\rho \leq \rho(X)^{1/2} \|f\|_2$. By Lemma 26.9.1 there is g with $\int f \, d\nu = \int f g \, d\rho$ for all $f \in L^2(\rho)$, and g may be taken real. Taking $f = \mathbf{1}_E$ gives $\nu(E) = \int_E g \, d\rho$, so $g \geq 0$ ρ -a.e.; and $\int_E (1 - g) \, d\nu = \int_E g \, d\mu$ (for bounded f , $\int f(1 - g) \, d\nu = \int f g \, d\mu$). For $E = \{g \geq 1\}$ the left side is ≤ 0 and the right side is $\geq \mu(E)$, so $\mu(E) = 0$ and $\nu(E) = 0$. Applying the identity to $f = \mathbf{1}_E(1 + g + \cdots + g^k)$ gives $\int_E (1 - g^{k+1}) \, d\nu = \int_E (g + \cdots + g^{k+1}) \, d\mu$. As $k \rightarrow \infty$, monotone convergence gives $\nu(E) = \int_E h \, d\mu$ with $h = g/(1 - g)$. If h' also works, $\int_E (h - h') \, d\mu = 0$ for $E = \{h > h'\}$ and $\{h < h'\}$ (intersected with sets of finite measure), so $h = h'$ a.e. $\square$

Theorem 26.9.3 (Riesz–Markov). *Let X be a locally compact Hausdorff space and $\Lambda: C_c(X, \mathbb{R}) \rightarrow \mathbb{R}$ a linear map with $\Lambda(f) \geq 0$ for $f \geq 0$. There is a unique regular Borel measure μ , finite on compact sets, with $\Lambda(f) = \int f \, d\mu$ for all $f \in C_c(X, \mathbb{R})$.*

This is [Rud87, Theorem 2.14]. For $X = \mathbb{R}^n$ and Λ the Riemann integral of continuous functions with compact support, μ is Lebesgue measure restricted to Borel sets. On a smooth manifold, a volume form defines such a functional, and hence a measure (Chapter 34).

26.10 Exercises

Exercise 26.10.1. Show that the Cantor set, obtained from $[0, 1]$ by repeatedly removing open middle thirds, is compact, uncountable and of Lebesgue measure zero.

Solution. It is an intersection of closed subsets of $[0, 1]$, hence compact. After k steps it is covered by 2^k intervals of length 3^{-k} , so its measure is at most $(2/3)^k$. Its points are the numbers $\sum_k a_k 3^{-k}$ with $a_k \in \{0, 2\}$, and distinct sequences give distinct numbers, so it is in bijection with $\{0, 1\}^{\mathbb{N}}$, which is uncountable (Theorem 3.6.3). $\square$

Exercise 26.10.2. (Borel–Cantelli.) If $\sum_k \mu(A_k) < \infty$, show that almost every point lies in only finitely many A_k .

Solution. By monotone convergence $\int \sum_k \mathbf{1}_{A_k} = \sum_k \mu(A_k) < \infty$, so $\sum_k \mathbf{1}_{A_k} < \infty$ a.e. $\square$

Exercise 26.10.3. Show that $L^q(\mu) \subseteq L^p(\mu)$ for $p \leq q$ if $\mu(X) < \infty$, and that there is no inclusion in either direction for Lebesgue measure on $\mathbb{R}$.

Solution. By Hölder with exponents q/p and its conjugate, $\int |f|^p \leq (\int |f|^q)^{p/q} \mu(X)^{1-p/q}$. On $\mathbb{R}$, $|x|^{-1/2} \mathbf{1}_{[-1,1]}$ is in L^1 but not L^2 , and $(1 + |x|)^{-1}$ is in L^2 but not L^1 . $\square$

Exercise 26.10.4. For $f \in L^1(\mathbb{R}^n)$ and $g \in L^p(\mathbb{R}^n)$, show using Fubini's theorem that $(f * g)(x) = \int f(x - y)g(y) dy$ is defined a.e. and $\|f * g\|_p \leq \|f\|_1 \|g\|_p$.

Exercise 26.10.5. Show that the volume of the unit ball in $\mathbb{R}^n$ is $\omega_n = \pi^{n/2} / \Gamma(\frac{n}{2} + 1)$, where $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$, by computing $\int_{\mathbb{R}^n} e^{-\|x\|^2}$ in two ways.

Chapter 27

Complex Analysis

27.1 Introduction

A function of one complex variable that is differentiable once is analytic, and is determined by its values on any set with a limit point. This rigidity, which has no analogue in real analysis, is the source of the special features of complex geometry: on a compact complex manifold there are few holomorphic functions, and the interplay between holomorphic and topological invariants leads to Hodge theory. This chapter proves Cauchy's theorem in homotopy form (Theorem 27.3.3), the Cauchy integral formula and its consequences, the residue theorem, and the Riemann mapping theorem (Theorem 27.6.5). References: [Ahl78], [SS03], [Con78] and [Rud87, Chapters 10–16].

Prerequisites. Chapter 25 (Real Analysis) and Chapter 17 (Homotopy and Covering Spaces).

We write $D(a, r) = \{z : |z - a| < r\}$, $\overline{D}(a, r)$ for its closure, and $\mathbb{D} = D(0, 1)$. All domains (= nonempty connected open subsets of $\mathbb{C}$) are denoted U, V .

27.2 Holomorphic functions

Definition 27.2.1. Let $U \subseteq \mathbb{C}$ be open. A function $f : U \rightarrow \mathbb{C}$ is *holomorphic* if the limit $f'(a) = \lim_{h \rightarrow 0} (f(a + h) - f(a))/h$, $h \in \mathbb{C}^\times$, exists for every $a \in U$. An *entire* function is one holomorphic on $\mathbb{C}$.

Proposition 27.2.2 (Cauchy–Riemann). *Identify $\mathbb{C}$ with $\mathbb{R}^2$ and write $f = u + iv$. Then f is holomorphic at a if and only if f is differentiable at a as a map of real variables (Definition 25.10.2) and its derivative is complex linear, which holds if and only if*

$$\partial_x u = \partial_y v, \quad \partial_y u = -\partial_x v \quad \text{at } a.$$

Then $f'(a) = \partial_x u + i\partial_x v$. Sums, products, quotients and composites of holomorphic functions are holomorphic, with the usual rules.

Proof. Complex differentiability at a says $f(a + h) = f(a) + ch + o(|h|)$ with $c = f'(a) \in \mathbb{C}$, which is real differentiability with $Df(a)$ the real linear map $h \mapsto ch$, that is, with matrix $\begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$ for $c = \alpha + i\beta$. Comparing with the Jacobian matrix $\begin{pmatrix} \partial_x u & \partial_y u \\ \partial_x v & \partial_y v \end{pmatrix}$ gives the equations. The rules follow as in Proposition 25.6.2, whose proofs are valid over $\mathbb{C}$. $\square$

Example 27.2.3. Polynomials are entire; $1/z$ is holomorphic on $\mathbb{C}^\times$; power series are holomorphic in their disc of convergence, with the termwise derivative (Theorem 25.6.19); $\exp$ is entire with $\exp' = \exp$. The function $\bar{z}$ is nowhere holomorphic, since $\partial_x u = 1 \neq -1 = \partial_y v$.

Definition 27.2.4. A *path* is a piecewise continuously differentiable map $\gamma: [a, b] \rightarrow \mathbb{C}$; it is *closed* if $\gamma(a) = \gamma(b)$. Its image is γ^* and its length is $L(\gamma) = \int_a^b |\gamma'(t)| dt$. For continuous f on γ^* ,

$$\int_{\gamma} f(z) dz = \int_a^b f(\gamma(t)) \gamma'(t) dt.$$

Lemma 27.2.5. The integral is unchanged by increasing C^1 reparametrisations, changes sign under reversal, and $|\int_{\gamma} f| \leq L(\gamma) \sup_{\gamma^*} |f|$. If $f = F'$ for a holomorphic F on an open set containing γ^* , then $\int_{\gamma} f = F(\gamma(b)) - F(\gamma(a))$; in particular $\int_{\gamma} f = 0$ for closed γ . For the circle $\gamma(t) = a + re^{2\pi it}$, $t \in [0, 1]$, one has $\int_{\gamma} (z - a)^n dz = 0$ for $n \neq -1$ and $= 2\pi i$ for $n = -1$.

Proof. Reparametrisation and reversal follow from the substitution rule (Corollary 25.8.4), the estimate from Proposition 25.8.2. If $f = F'$, then $(F \circ \gamma)' = f(\gamma)\gamma'$ on each C^1 piece, so the integral telescopes by the fundamental theorem of calculus. For the circle and $n \neq -1$, $(z - a)^n$ has the primitive $(z - a)^{n+1}/(n+1)$; for $n = -1$ the integral is $\int_0^1 2\pi i dt$. $\square$

27.3 Cauchy's theorem

Theorem 27.3.1 (Goursat). Let U be open, $p \in U$, and $f: U \rightarrow \mathbb{C}$ continuous on U and holomorphic on $U \setminus \{p\}$. Then $\int_{\partial T} f = 0$ for every closed triangle $T \subseteq U$, where ∂T is its boundary traversed once.

Proof. First assume f is holomorphic on all of U . Let $I = \int_{\partial T} f$. Joining the midpoints of the sides divides T into four triangles whose boundary integrals add up to I (the inner edges cancel), so one of them, T_1 , has $|\int_{\partial T_1} f| \geq |I|/4$. Iterating gives nested triangles T_n with $\text{diam } T_n = 2^{-n} \text{diam } T$, $L(\partial T_n) = 2^{-n} L(\partial T)$ and $|\int_{\partial T_n} f| \geq 4^{-n} |I|$. Their intersection is a point z_0 . Given $\varepsilon > 0$, holomorphy at z_0 gives δ with $|f(z) - f(z_0) - f'(z_0)(z - z_0)| \leq \varepsilon |z - z_0|$ for $|z - z_0| < \delta$. The affine function $f(z_0) + f'(z_0)(z - z_0)$ has a primitive, so its integral over ∂T_n vanishes, and for large n , $|\int_{\partial T_n} f| \leq \varepsilon \text{diam}(T_n) L(\partial T_n) = \varepsilon 4^{-n} \text{diam}(T) L(\partial T)$. Hence $|I| \leq \varepsilon \text{diam}(T) L(\partial T)$ for all ε .

If f is only continuous at p : if $p \notin T$ we are done. If p is a vertex of T , write T as the union of a small triangle T' with vertex p and two triangles not containing p ; then $|\int_{\partial T} f| = |\int_{\partial T'} f| \leq L(\partial T') \sup_T |f| \rightarrow 0$ as T' shrinks. For general $p \in T$, join p to the vertices. $\square$

Corollary 27.3.2. Let U be convex, $p \in U$, and f continuous on U and holomorphic on $U \setminus \{p\}$. Then f has a primitive on U , and $\int_{\gamma} f = 0$ for every closed path γ in U .

Proof. Fix $z_0 \in U$ and set $F(z) = \int_{[z_0, z]} f$, the integral over the segment. For $z, z+h \in U$, the triangle with vertices $z_0, z, z+h$ lies in U , so $F(z+h) - F(z) = \int_{[z, z+h]} f$ by Theorem 27.3.1. Hence $|F(z+h) - F(z) - f(z)h| \leq |h| \sup_{[z, z+h]} |f - f(z)| = o(|h|)$ by continuity, so $F' = f$. Apply Lemma 27.2.5. $\square$

Theorem 27.3.3 (Cauchy's theorem, homotopy form). Let f be holomorphic on an open $U \subseteq \mathbb{C}$ and let γ_0, γ_1 be paths in U that are homotopic in U , either as closed paths or with fixed endpoints. Then $\int_{\gamma_0} f = \int_{\gamma_1} f$. In particular $\int_{\gamma} f = 0$ for every closed path that is null-homotopic in U , and every holomorphic function on a simply connected open set has a primitive.

Proof. Let $H: [0, 1]^2 \rightarrow U$ be the homotopy, $H(s, 0) = \gamma_0(s)$, $H(s, 1) = \gamma_1(s)$. Its image is compact, so there is $\rho > 0$ with $D(H(s, t), \rho) \subseteq U$ for all (s, t) , and by uniform continuity

there is N with $|H(s, t) - H(s', t')| < \rho$ whenever $|s - s'|, |t - t'| \leq 1/N$. Put $z_{jk} = H(j/N, k/N)$ and let $D_{jk} = D(z_{jk}, \rho)$, so that H maps the square $[j/N, (j+1)/N] \times [k/N, (k+1)/N]$ into D_{jk} . Let P_k be the closed polygonal path through $z_{0k}, z_{1k}, \dots, z_{Nk}$.

Each segment $[z_{jk}, z_{j+1,k}]$ and the corresponding piece of $H(-, k/N)$ lie in the convex set D_{jk} , where f has a primitive by Corollary 27.3.2; so the integrals of f over them agree, and $\int_{\gamma_0} f = \int_{P_0} f$, $\int_{\gamma_1} f = \int_{P_N} f$ (for γ_0, γ_1 piecewise C^1 ; the intermediate paths are only continuous, and the polygonal paths replace them). The closed polygon with vertices $z_{jk}, z_{j+1,k}, z_{j+1,k+1}, z_{j,k+1}$ lies in the convex set D_{jk} , so the integral of f over it vanishes. Summing over j , the contributions of the segments $[z_{jk}, z_{j,k+1}]$ cancel in pairs (for closed homotopies because $z_{0k} = z_{Nk}$; for fixed endpoints because those segments are constant), so $\int_{P_k} f = \int_{P_{k+1}} f$. Hence $\int_{P_0} f = \int_{P_N} f$. The last statement follows by fixing $z_0 \in U$ and setting $F(z) = \int_{\gamma} f$ for any path from z_0 to z , which is independent of the path since U is simply connected; then $F' = f$ as in Corollary 27.3.2. $\square$

Definition 27.3.4. For a closed path γ and $a \notin \gamma^*$, the *winding number* is $n(\gamma, a) = \frac{1}{2\pi i} \int_{\gamma} \frac{dz}{z-a}$.

Proposition 27.3.5. $n(\gamma, a)$ is an integer; it is constant on connected components of $\mathbb{C} \setminus \gamma^*$ and zero on the unbounded component. Homotopic closed paths in $\mathbb{C} \setminus \{a\}$ have the same winding number about a , and for the circle $\gamma(t) = a + re^{2\pi i t}$ one has $n(\gamma, a) = 1$.

Proof. Let $\gamma: [0, 1] \rightarrow \mathbb{C}$ and $\varphi(t) = \int_0^t \frac{\gamma'(s)}{\gamma(s)-a} ds$. Then $(e^{-\varphi}(\gamma - a))' = 0$ on each C^1 piece, so $e^{\varphi(1)} = (\gamma(1) - a)/(\gamma(0) - a) = 1$ and $\varphi(1) \in 2\pi i\mathbb{Z}$ by Theorem 25.7.3. The function $a \mapsto n(\gamma, a)$ is continuous on $\mathbb{C} \setminus \gamma^*$ and integer valued, hence locally constant, and it tends to 0 as $|a| \rightarrow \infty$. Homotopy invariance is Theorem 27.3.3 applied to $z \mapsto 1/(z-a)$ on $\mathbb{C} \setminus \{a\}$, and the circle was computed in Lemma 27.2.5. $\square$

27.4 The Cauchy integral formula

Theorem 27.4.1. Let U be convex, f holomorphic on U , γ a closed path in U and $a \in U \setminus \gamma^*$. Then

$$n(\gamma, a)f(a) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-a} dz.$$

In particular, if $\overline{D}(a, r) \subseteq U$ and γ is the circle $\partial D(a, r)$ traversed once, then $f(w) = \frac{1}{2\pi i} \int_{\gamma} \frac{f(z)}{z-w} dz$ for $w \in D(a, r)$.

Proof. Define $g(z) = (f(z) - f(a))/(z-a)$ for $z \neq a$ and $g(a) = f'(a)$. Then g is continuous on U and holomorphic on $U \setminus \{a\}$, so $\int_{\gamma} g = 0$ by Corollary 27.3.2. For the second statement, note that $n(\gamma, w) = n(\gamma, a) = 1$ for $w \in D(a, r)$ by Proposition 27.3.5, and apply the formula in the convex set $D(a, r')$ for $r < r'$ with $\overline{D}(a, r') \subseteq U$. $\square$

Theorem 27.4.2. Let f be holomorphic on an open U and $D(a, R) \subseteq U$. Then $f(w) = \sum_{n \geq 0} c_n (w-a)^n$ on $D(a, R)$, with $c_n = \frac{1}{2\pi i} \int_{\partial D(a, r)} \frac{f(z)}{(z-a)^{n+1}} dz = \frac{f^{(n)}(a)}{n!}$ for any $0 < r < R$. In particular holomorphic functions have holomorphic derivatives of all orders, and $|f^{(n)}(a)| \leq n! r^{-n} \sup_{|z-a|=r} |f|$ (Cauchy estimates).

Proof. Fix $|w-a| < r < R$. For $|z-a| = r$, $\frac{1}{z-w} = \sum_{n \geq 0} \frac{(w-a)^n}{(z-a)^{n+1}}$, converging uniformly in z since $|w-a|/r < 1$. Multiplying by $f(z)/2\pi i$ and integrating term by term (uniform convergence, Proposition 25.8.2) gives the expansion with the stated c_n , by Theorem 27.4.1. A power series is infinitely complex differentiable with $c_n = f^{(n)}(a)/n!$ (Theorem 25.6.19), and the estimate follows from Lemma 27.2.5. $\square$

- Corollary 27.4.3.** 1. (Liouville) A bounded entire function is constant; more generally, if $|f(z)| \leq C(1 + |z|)^m$ for an entire f , then f is a polynomial of degree at most m .
2. (Fundamental theorem of algebra) Every nonconstant polynomial over $\mathbb{C}$ has a root (cf. Chapter 8).
3. (Morera) A continuous function on an open set whose integral over the boundary of every closed triangle contained in the set vanishes is holomorphic.
4. (Weierstrass) If holomorphic functions f_n on U converge uniformly on compact subsets to f , then f is holomorphic and $f_n^{(k)} \rightarrow f^{(k)}$ uniformly on compact subsets.

Proof. (1) By the Cauchy estimates with $r \rightarrow \infty$, $f^{(n)}(a) = 0$ for $n > m$ and all a . (2) If p has no root, $1/p$ is entire and tends to 0 at infinity, hence is bounded and constant. (3) Locally on a disc the hypothesis gives a primitive F as in Corollary 27.3.2, and F is holomorphic, hence so is $F' = f$ by Theorem 27.4.2. (4) The uniform limit is continuous and satisfies Morera's condition by Theorem 27.3.1 and uniform convergence; convergence of derivatives follows from the Cauchy integral formula for $f^{(k)}$ on circles inside U . $\square$

Theorem 27.4.4 (Identity theorem). Let f be holomorphic on a domain U . If the zero set of f has a limit point in U , then $f \equiv 0$. Consequently two holomorphic functions agreeing on a set with a limit point in U are equal, and the zeros of a nonzero f are isolated, each of finite order m with $f(z) = (z - a)^m g(z)$, g holomorphic and $g(a) \neq 0$.

Proof. Let A be the set of $a \in U$ with $f^{(n)}(a) = 0$ for all n ; it is closed in U , and open because f vanishes identically near such a point by Theorem 27.4.2. If $z_k \rightarrow a$ with $z_k \neq a$ and $f(z_k) = 0$, and if some derivative at a were nonzero, then with m minimal, $f(z) = (z - a)^m g(z)$ with $g(a) \neq 0$ by the power series expansion, so f would have no zeros near a other than a . Hence $a \in A$, and $A = U$ by connectedness. $\square$

Theorem 27.4.5 (Maximum modulus principle). Let f be holomorphic on a domain U . If $|f|$ has a local maximum in U , then f is constant. If moreover U is bounded and f extends continuously to $\bar{U}$, then $|f|$ attains its maximum on ∂U .

Proof. Theorem 27.4.1 on circles gives the mean value property $f(a) = \int_0^1 f(a + re^{2\pi it}) dt$. If $|f| \leq |f(a)|$ on $D(a, R)$, then for $r < R$, $|f(a)| \leq \int_0^1 |f(a + re^{2\pi it})| dt \leq |f(a)|$, so $|f| = |f(a)|$ on every such circle by continuity, and $|f|$ is constant on $D(a, R)$. Then $f/\bar{f}$ is constant; if $f(a) \neq 0$ this makes $\bar{f} = c/f$ holomorphic, so both f and $\bar{f}$ satisfy the Cauchy–Riemann equations and $f' = 0$ on the disc. In either case f is constant on $D(a, R)$, hence on U by Theorem 27.4.4. The last statement follows from compactness of $\bar{U}$. $\square$

27.5 Singularities, Laurent series and residues

Theorem 27.5.1 (Laurent expansion). Let f be holomorphic on the annulus $A = \{r < |z - a| < R\}$, $0 \leq r < R \leq \infty$. There are unique $c_n \in \mathbb{C}$, $n \in \mathbb{Z}$, with $f(z) = \sum_{n \in \mathbb{Z}} c_n (z - a)^n$, converging absolutely and uniformly on compact subsets of A , and $c_n = \frac{1}{2\pi i} \int_{\partial D(a, \rho)} \frac{f(z)}{(z - a)^{n+1}} dz$ for any $r < \rho < R$.

Proof. Fix w with $r < r_1 < |w - a| < r_2 < R$. The function $g(z) = (f(z) - f(w))/(z - w)$, extended by $f'(w)$ at w , is continuous on A and holomorphic off w , hence holomorphic by Corollary 27.4.3(3) and Theorem 27.3.1. The circles $\partial D(a, r_1)$ and $\partial D(a, r_2)$ are homotopic

as closed paths in A (move radially), so $\int_{\partial D(a, r_2)} g = \int_{\partial D(a, r_1)} g$ by Theorem 27.3.3. Since $n(\partial D(a, r_2), w) = 1$ and $n(\partial D(a, r_1), w) = 0$, this gives

$$f(w) = \frac{1}{2\pi i} \int_{\partial D(a, r_2)} \frac{f(z)}{z-w} dz - \frac{1}{2\pi i} \int_{\partial D(a, r_1)} \frac{f(z)}{z-w} dz.$$

In the first integral expand $\frac{1}{z-w} = \sum_{n \geq 0} \frac{(w-a)^n}{(z-a)^{n+1}}$, in the second $\frac{-1}{z-w} = \sum_{n \geq 1} \frac{(z-a)^{n-1}}{(w-a)^n}$, both uniformly convergent on the respective circles, and integrate term by term. The resulting coefficients are independent of the radius, again by homotopy invariance. Uniqueness follows by integrating the series against $(z-a)^{-n-1}$ term by term. $\square$

Definition 27.5.2. Let f be holomorphic on a punctured disc $D(a, R) \setminus \{a\}$, with Laurent coefficients c_n . The singularity at a is *removable* if $c_n = 0$ for all $n < 0$, a *pole of order m* if $c_{-m} \neq 0$ and $c_n = 0$ for $n < -m$, and *essential* otherwise. The *residue* is $\text{Res}_a f = c_{-1}$. A function holomorphic on U except for poles at a discrete closed subset is *meromorphic*; the *order* $\text{ord}_a f$ is m at a zero of order m and $-m$ at a pole of order m .

Proposition 27.5.3. Let f be holomorphic on $D(a, R) \setminus \{a\}$.

1. (Riemann) If f is bounded near a , the singularity is removable.
2. The singularity is a pole if and only if $|f(z)| \rightarrow \infty$ as $z \rightarrow a$.
3. (Casorati–Weierstrass) If the singularity is essential, $f(D(a, \varepsilon) \setminus \{a\})$ is dense in $\mathbb{C}$ for every $\varepsilon > 0$.

Proof. (1) If $|f| \leq M$ near a , then $|c_{-n}| \leq M\rho^n$ for small ρ by Theorem 27.5.1, so $c_{-n} = 0$ for $n \geq 1$. (2) If f has a pole of order m , then $(z-a)^m f(z)$ extends holomorphically with nonzero value at a , so $|f| \rightarrow \infty$. Conversely, if $|f| \rightarrow \infty$, then $1/f$ is holomorphic and nonvanishing near a with limit 0, so by (1) it extends with a zero of some order m at a , and $f = 1/(1/f)$ has a pole of order m . (3) If the image of some punctured disc misses $D(w, \delta)$, then $g = 1/(f-w)$ is bounded there, hence extends holomorphically by (1); then $f = w + 1/g$ has a removable singularity or a pole by (2). $\square$

Example 27.5.4. The function $f(z) = e^{1/z}$ on $\mathbb{C} \setminus \{0\}$ has Laurent series $\sum_{n \geq 0} z^{-n}/n!$, so 0 is an essential singularity. More than Casorati–Weierstrass holds here: every $w \neq 0$ is taken infinitely often in every punctured disc around 0. Indeed, if $w = e^c$, the points $z_k = 1/(c + 2\pi i k)$, $k \in \mathbb{Z}$, satisfy $f(z_k) = w$ and tend to 0. The value 0 is never taken. On the real axis, $e^{1/x} \rightarrow \infty$ as $x \rightarrow 0^+$ and $\rightarrow 0$ as $x \rightarrow 0^-$, while along the imaginary axis $|e^{1/(iy)}| = 1$: the behaviour near an essential singularity depends on the direction of approach.

Counterexample 27.5.5. Real analysis does not see the radius of convergence. The function $g(x) = 1/(1+x^2)$ is real-analytic on all of $\mathbb{R}$, but its Taylor series $\sum_n (-1)^n x^{2n}$ at 0 has radius of convergence 1. The reason is complex: g extends to the meromorphic function $1/(1+z^2)$ with poles at $\pm i$, at distance 1 from 0. By Theorem 27.4.2 the series converges on $D(0, 1)$, and it cannot converge on a larger disc: its sum would be holomorphic there, equal to $1/(1+z^2)$ on $D(0, 1)$ by the identity theorem, hence bounded near i , whereas $|1/(1+z^2)| \rightarrow \infty$ as $z \rightarrow i$. Similarly $x \mapsto e^{-1/x^2}$ (extended by 0) is smooth on $\mathbb{R}$ with all derivatives zero at 0, so it is not real-analytic there; as a complex function, e^{-1/z^2} has an essential singularity at 0, and it is unbounded along the imaginary axis.

Lemma 27.5.6. Let h be holomorphic on an open U and define $g: U \times U \rightarrow \mathbb{C}$ by $g(z, w) = (h(z) - h(w))/(z - w)$ for $z \neq w$ and $g(z, z) = h'(z)$. Then g is continuous, and holomorphic in each variable separately.

Proof. Holomorphy in each variable off the diagonal is clear, and at $z = w$ the singularity is removable by Proposition 27.5.3(1), since g is bounded near it by the estimate below. For continuity at (a, a) , take $\overline{D}(a, r) \subseteq U$; for $z, w \in D(a, r)$ the segment $[w, z]$ lies in the disc and $g(z, w) = \int_0^1 h'(w + t(z - w)) dt$ (by the fundamental theorem of calculus applied to $t \mapsto h(w + t(z - w))$, also valid when $z = w$). Since h' is continuous, hence uniformly continuous on $\overline{D}(a, r/2)$, this expression is continuous in (z, w) there. $\square$

Theorem 27.5.7 (Cauchy's theorem, homology form). *Let h be holomorphic on an open $U \subseteq \mathbb{C}$ and let γ be a closed path in U with $n(\gamma, z) = 0$ for every $z \in \mathbb{C} \setminus U$. Then $\int_\gamma h = 0$, and $n(\gamma, w)h(w) = \frac{1}{2\pi i} \int_\gamma \frac{h(z)}{z-w} dz$ for every $w \in U \setminus \gamma^*$.*

Proof. (Dixon.) With g as in Lemma 27.5.6, let $\varphi(w) = \frac{1}{2\pi i} \int_\gamma g(z, w) dz$ for $w \in U$. It is continuous, since g is uniformly continuous on $\gamma^* \times K$ for compact $K \subseteq U$, and holomorphic: for a closed triangle $T \subseteq U$, Fubini's theorem for continuous functions (Proposition 25.8.5) and Theorem 27.3.1 give $\int_{\partial T} \varphi(w) dw = \frac{1}{2\pi i} \int_\gamma \left(\int_{\partial T} g(z, w) dw \right) dz = 0$, so Morera's theorem applies. Let $V = \{w \in \mathbb{C} \setminus \gamma^* : n(\gamma, w) = 0\}$, an open set, and $\psi(w) = \frac{1}{2\pi i} \int_\gamma \frac{h(z)}{z-w} dz$ on V , which is holomorphic (same argument). For $w \in U \cap V$, $\varphi(w) = \psi(w) - n(\gamma, w)h(w) = \psi(w)$. By hypothesis $U \cup V = \mathbb{C}$, so φ and ψ define an entire function F . The unbounded component of $\mathbb{C} \setminus \gamma^*$ lies in V , and $\psi(w) \rightarrow 0$ as $|w| \rightarrow \infty$; so F is bounded, hence constant equal to 0 by Liouville's theorem (Corollary 27.4.3). Thus $\varphi \equiv 0$, which is the integral formula. Applying it to $z \mapsto (z - w)h(z)$ at the point w gives $0 = \frac{1}{2\pi i} \int_\gamma h$. $\square$

Theorem 27.5.8 (Residue theorem). *Let U be simply connected and open, $S \subseteq U$ discrete and closed in U , and f holomorphic on $U \setminus S$. For every closed path γ in $U \setminus S$,*

$$\frac{1}{2\pi i} \int_\gamma f = \sum_{a \in S} n(\gamma, a) \operatorname{Res}_a f,$$

a finite sum.

Proof. Let $W = \{z \in \mathbb{C} \setminus \gamma^* : n(\gamma, z) \neq 0\}$, a union of bounded components of $\mathbb{C} \setminus \gamma^*$, so $K = \overline{W} \subseteq W \cup \gamma^*$ is compact. If $z \notin U$, then $1/(w - z)$ is holomorphic on the simply connected U , hence has a primitive, so $n(\gamma, z) = 0$; therefore $K \subseteq U$, and $S \cap K$ is finite, say $\{a_1, \dots, a_k\}$, because S is discrete and closed in U . Let $P_j(z) = \sum_{n \geq 1} c_{-n}^{(j)}(z - a_j)^{-n}$ be the principal part of f at a_j . It converges locally uniformly on $\mathbb{C} \setminus \{a_j\}$, because the Laurent series of f converges on the punctured disc and the principal part, as a power series in $1/(z - a_j)$, then has infinite radius of convergence; so P_j is holomorphic there. The function $h = f - \sum_j P_j$ has removable singularities at $a_1, \dots, a_k$, hence extends holomorphically to $U' = U \setminus (S \setminus \{a_1, \dots, a_k\})$. For $z \notin U'$ we have $n(\gamma, z) = 0$: either $z \notin U$, or $z \in S \setminus \{a_1, \dots, a_k\}$, which was excluded from K . So $\int_\gamma h = 0$ by Theorem 27.5.7. Finally $\int_\gamma P_j = 2\pi i n(\gamma, a_j) c_{-1}^{(j)}$, since the terms with $n \geq 2$ have primitives, and $n(\gamma, a) = 0$ for the remaining $a \in S$. $\square$

Example 27.5.9. $\int_{-\infty}^{\infty} \frac{dx}{1+x^2} = \pi$. Integrate $f(z) = 1/(1+z^2)$ over the boundary of the half-disc $\{|z| \leq R, \operatorname{Im} z \geq 0\}$: the only singularity inside is the simple pole at i with residue $\lim_{z \rightarrow i} (z - i)f(z) = 1/(2i)$, so the integral is π . The contribution of the semicircle is at most $\pi R/(R^2 - 1) \rightarrow 0$, and the integral over $[-R, R]$ tends to the improper integral.

Theorem 27.5.10 (Argument principle and Rouché). *Let U be simply connected, f meromorphic on U and γ a closed path in U avoiding the zeros and poles of f . Then $\frac{1}{2\pi i} \int_\gamma \frac{f'}{f} = \sum_a n(\gamma, a) \operatorname{ord}_a f$. If g is holomorphic on U and $|f - g| < |f|$ on γ^* , with f holomorphic, then f and g have the same number of zeros counted with multiplicity and weighted by $n(\gamma, -)$.*

Proof. Near a zero or pole, $f(z) = (z - a)^m h(z)$ with $h(a) \neq 0$ and $m = \text{ord}_a f$, so $f'/f = m/(z - a) + h'/h$ has residue m at a ; apply Theorem 27.5.8. For Rouché's theorem, the functions $f_t = f + t(g - f)$, $t \in [0, 1]$, have no zeros on γ^* because $|f_t| \geq |f| - |g - f| > 0$; the integer $N(t) = \frac{1}{2\pi i} \int_\gamma f'_t/f_t$ depends continuously on t (the integrand is continuous in (t, z) and γ^* is compact), hence is constant. $\square$

Theorem 27.5.11 (Open mapping theorem). *A nonconstant holomorphic function on a domain is an open map. If it is injective, its derivative has no zeros and its inverse is holomorphic.*

Proof. Let $w_0 = f(z_0)$. By Theorem 27.4.4 the zero of $f - w_0$ at z_0 has finite order $m \geq 1$ and is isolated, so there is r with $|f - w_0| \geq \delta > 0$ on $|z - z_0| = r$ and no other zero in $\overline{D}(z_0, r)$. For $|w - w_0| < \delta$, Rouché's theorem applied to $f - w_0$ and $f - w$ on the circle shows that $f - w$ has m zeros in $D(z_0, r)$; so $D(w_0, \delta) \subseteq f(D(z_0, r))$. If f is injective, then $m = 1$ at every point, that is, $f' \neq 0$; and f^{-1} is continuous because f is open, and differentiable with derivative $1/f'(f^{-1})$ by the computation of Proposition 25.6.7, valid over $\mathbb{C}$. $\square$

27.6 Conformal maps and the Riemann mapping theorem

Lemma 27.6.1 (Schwarz). *Let $f: \mathbb{D} \rightarrow \mathbb{D}$ be holomorphic with $f(0) = 0$. Then $|f(z)| \leq |z|$ and $|f'(0)| \leq 1$, with equality for some $z \neq 0$ or in the second inequality only if $f(z) = \lambda z$ with $|\lambda| = 1$.*

Proof. The function $g(z) = f(z)/z$, extended by $f'(0)$ at 0, is holomorphic on $\mathbb{D}$ (Proposition 27.5.3(1)). On $|z| = r$, $|g| \leq 1/r$, so by the maximum principle $|g| \leq 1/r$ on $D(0, r)$; letting $r \rightarrow 1$ gives $|g| \leq 1$. Equality at an interior point forces g to be a constant of modulus 1. $\square$

Proposition 27.6.2. *For $a \in \mathbb{D}$ the Möbius map $\phi_a(z) = \frac{z-a}{1-\bar{a}z}$ is a biholomorphic self-map of $\mathbb{D}$ with $\phi_a(a) = 0$ and inverse ϕ_{-a} . Every biholomorphic self-map of $\mathbb{D}$ is $z \mapsto \lambda \phi_a(z)$ with $|\lambda| = 1$.*

Proof. For $|z| < 1$ and $|a| < 1$, $|1 - \bar{a}z|^2 - |z - a|^2 = (1 - |a|^2)(1 - |z|^2) > 0$, so ϕ_a maps $\mathbb{D}$ into $\mathbb{D}$, and a computation gives $\phi_{-a}\phi_a = \text{id}$. If F is a biholomorphic self-map with $F(a) = 0$, then $g = F \circ \phi_{-a}$ fixes 0, and Lemma 27.6.1 applied to g and g^{-1} gives $|g'(0)| = 1$, hence $g(z) = \lambda z$. $\square$

Theorem 27.6.3 (Montel). *Let $\mathcal{F}$ be a family of holomorphic functions on an open U that is uniformly bounded on every compact subset. Then every sequence in $\mathcal{F}$ has a subsequence converging uniformly on compact subsets to a holomorphic function.*

Proof. Let $K \subseteq U$ be compact and 2ρ the distance from K to $\mathbb{C} \setminus U$ (or 1 if $U = \mathbb{C}$), and let K' be the set of points at distance $\leq \rho$ from K , a compact subset of U . By the Cauchy estimates, $|f'| \leq \rho^{-1} \sup_{K'} |f|$ on K for all $f \in \mathcal{F}$, so $\mathcal{F}$ is equicontinuous on K (mean value inequality). By the Arzelà–Ascoli theorem (Theorem 25.13.4) a sequence has a subsequence converging uniformly on K . Exhausting U by compact sets $K_1 \subseteq K_2 \subseteq \cdots$ and taking a diagonal subsequence gives uniform convergence on all compact subsets; the limit is holomorphic by Corollary 27.4.3(4). $\square$

Theorem 27.6.4 (Hurwitz). *Let $f_n \rightarrow f$ uniformly on compact subsets of a domain U , with f_n holomorphic and f nonconstant. If every f_n is nowhere zero, so is f ; if every f_n is injective, so is f .*

Proof. Suppose $f(z_0) = 0$. Since $f \neq 0$, there is r with $\overline{D}(z_0, r) \subseteq U$ and $f \neq 0$ on the circle $|z - z_0| = r$, so $|f| \geq \delta > 0$ there. Then $f_n \rightarrow f$ and $f'_n \rightarrow f'$ uniformly on the circle, so $\frac{1}{2\pi i} \int f'_n/f_n \rightarrow \frac{1}{2\pi i} \int f'/f \geq 1$, and by Theorem 27.5.10 f_n has a zero inside for large n . For injectivity, if $f(z_1) = f(z_2)$ with $z_1 \neq z_2$, apply the first part on a punctured neighbourhood of z_2 to the functions $f_n - f_n(z_1)$, which are nowhere zero there. $\square$

Theorem 27.6.5 (Riemann mapping theorem). *Every simply connected domain $U \subsetneq \mathbb{C}$ is biholomorphic to $\mathbb{D}$. More precisely, for $z_0 \in U$ there is a unique biholomorphic $\psi: U \rightarrow \mathbb{D}$ with $\psi(z_0) = 0$ and $\psi'(z_0) > 0$.*

Proof. Let $\mathcal{F}$ be the set of injective holomorphic maps $U \rightarrow \mathbb{D}$ sending z_0 to 0.

$\mathcal{F} \neq \emptyset$. Choose $b \notin U$. Since $z - b$ is nowhere zero on the simply connected U , the function $(z - b)'/(z - b)$ has a primitive h (Theorem 27.3.3); adjusting by a constant, $e^h = z - b$. Let $\phi = e^{h/2}$, so $\phi^2 = z - b$. Then ϕ is injective, and $\phi(z_1) = -\phi(z_2)$ implies $z_1 = z_2$ and $\phi(z_1) = 0$, which is impossible; so $\phi(U)$ and $-\phi(U)$ are disjoint. By Theorem 27.5.11, $\phi(U)$ contains a disc $D(w_0, \rho)$, hence $|\phi(z) + w_0| \geq \rho$ on U , and $z \mapsto \rho/(\phi(z) + w_0)$ maps U injectively into $\mathbb{D}$. Composing with a suitable ϕ_a (Proposition 27.6.2) gives an element of $\mathcal{F}$.

An extremal element. Injective holomorphic maps have nonvanishing derivative, so $\eta = \sup_{\psi \in \mathcal{F}} |\psi'(z_0)| > 0$. Take $\psi_n \in \mathcal{F}$ with $|\psi'_n(z_0)| \rightarrow \eta$; by Theorem 27.6.3 a subsequence converges uniformly on compact subsets to a holomorphic ψ with $|\psi'(z_0)| = \eta > 0$, so ψ is nonconstant, hence injective by Theorem 27.6.4, with $\psi(z_0) = 0$ and $|\psi| \leq 1$; by the open mapping theorem $\psi(U) \subseteq \mathbb{D}$. So $\psi \in \mathcal{F}$ and $|\psi'(z_0)| = \eta$.

Surjectivity. Suppose $\alpha \in \mathbb{D} \setminus \psi(U)$. Then $\phi_\alpha \circ \psi$ is nowhere zero on U , so as above it has a holomorphic square root g , which is injective because g^2 is. Let $F = \phi_{g(z_0)} \circ g \in \mathcal{F}$. Writing $s(w) = w^2$, we have $\psi = \phi_{-\alpha} \circ s \circ \phi_{-g(z_0)} \circ F = \Phi \circ F$ with $\Phi: \mathbb{D} \rightarrow \mathbb{D}$ holomorphic and $\Phi(0) = 0$. As s is not injective, Φ is not injective, so Φ is not a rotation and $|\Phi'(0)| < 1$ by Lemma 27.6.1. Then $\eta = |\psi'(z_0)| = |\Phi'(0)| |F'(z_0)| < |F'(z_0)| \leq \eta$, a contradiction. So ψ is a biholomorphism onto $\mathbb{D}$, and multiplying by a unimodular constant makes $\psi'(z_0) > 0$.

Uniqueness. If ψ_1, ψ_2 both work, $\psi_2 \circ \psi_1^{-1}$ is an automorphism of $\mathbb{D}$ fixing 0 with positive derivative there, hence the identity by Proposition 27.6.2. $\square$

Remark 27.6.6. A real function u on an open $U \subseteq \mathbb{C}$ is *harmonic* if it is C^2 with $\partial_x^2 u + \partial_y^2 u = 0$. Real and imaginary parts of holomorphic functions are harmonic, by the Cauchy–Riemann equations and Theorem 25.10.5. Conversely, on a simply connected U every harmonic u is the real part of a holomorphic function: the function $\partial_x u - i \partial_y u$ is holomorphic (its Cauchy–Riemann equations are the harmonicity of u and the symmetry of second derivatives), so it has a primitive f , and $\operatorname{Re} f$ differs from u by a constant. Harmonic functions therefore satisfy the mean value property and the maximum principle; this is the one-variable case of the Hodge theory of Chapter 40.

27.7 Exercises

Exercise 27.7.1. Show that $\mathbb{D}$ and $\mathbb{C}$ are not biholomorphic, although they are homeomorphic.

Solution. A biholomorphism $\mathbb{C} \rightarrow \mathbb{D}$ would be a bounded entire function, hence constant (Corollary 27.4.3). They are homeomorphic by $z \mapsto z/(1 + |z|)$ (Counterexample 16.7.9). $\square$

Exercise 27.7.2. Show that every biholomorphic map $\mathbb{C} \rightarrow \mathbb{C}$ is $z \mapsto az + b$ with $a \neq 0$.

Solution. Let f be such a map and consider $g(z) = f(1/z)$ on $\mathbb{C}^\times$. If 0 were an essential singularity of g , then by Casorati–Weierstrass the image $f(\{|z| > 1\})$ would be dense, while $f(\mathbb{D})$ is open and disjoint from it, a contradiction. So g has a pole or removable singularity at 0, which means $|f(z)| \leq C|z|^m$ for large $|z|$; by Liouville’s theorem f is a polynomial, and it is injective only if $\deg f = 1$. $\square$

Exercise 27.7.3. Use Rouché’s theorem to show that $z^5 + 3z + 1$ has exactly one root in $\mathbb{D}$ and all five roots in $D(0, 2)$.

Exercise 27.7.4. Compute $\int_0^{2\pi} \frac{dt}{a+\cos t}$ for $a > 1$ by writing $z = e^{it}$ and using the residue theorem.

Exercise 27.7.5. Give an example of a domain U and a nowhere vanishing holomorphic f on U with no holomorphic logarithm, and relate this to $\pi_1(U)$.

Chapter 28

Functional Analysis

28.1 Introduction

Hodge theory rests on an analytic theorem: on a compact Riemannian manifold every cohomology class has a unique harmonic representative. The proof solves the equation $\Delta u = f$ in a space of generalised functions and then shows that the solution is smooth. The functional analysis needed for the first step is in this chapter: Hilbert spaces and orthogonal projections, the consequences of Baire's theorem, compact operators, the spectral theorem for compact self-adjoint operators (Theorem 28.5.3) and the Fredholm alternative (Theorem 28.5.4). References: [Bre11], [Rud91], [RS80].

Prerequisites. Chapter 26 (Measure and Integration) and Chapter 9 (Linear Algebra).

All vector spaces are over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

28.2 Banach spaces and bounded operators

Definition 28.2.1. A linear map $T: X \rightarrow Y$ between normed spaces is *bounded* if $\|T\| = \sup_{\|x\| \leq 1} \|Tx\| < \infty$; equivalently (as for finite dimensions, Lemma 25.10.1) T is continuous. The bounded operators form a normed space $B(X, Y)$, and $X^* = B(X, \mathbb{K})$ is the *dual space*.

Proposition 28.2.2. 1. If Y is complete, so is $B(X, Y)$; in particular X^* is always a Banach space.

2. $C(K, \mathbb{K})$ with the supremum norm (K compact Hausdorff), $L^p(\mu)$ for $1 \leq p \leq \infty$, and the sequence spaces ℓ^p are Banach spaces.

3. (Riesz) If the closed unit ball of a normed space X is compact, then X is finite-dimensional.

Proof. (1) A Cauchy sequence (T_n) is pointwise Cauchy, and the pointwise limit T is linear with $\|T - T_n\| \leq \limsup_m \|T_m - T_n\| \rightarrow 0$. (2) Proposition 16.7.2 and Theorem 26.5.3. (3) Suppose X is infinite-dimensional. We construct unit vectors x_n with $\|x_n - x_m\| \geq 1/2$: given the finite-dimensional, hence closed (all norms on it are equivalent and it is complete), subspace Y_n spanned by $x_1, \dots, x_n$, choose $x \notin Y_n$ and $y \in Y_n$ with $\|x - y\| \leq 2 \operatorname{dist}(x, Y_n)$; then $x_{n+1} = (x - y)/\|x - y\|$ has distance at least $1/2$ from Y_n . Such a sequence has no convergent subsequence. $\square$

28.3 Hilbert spaces

Definition 28.3.1. A *Hilbert space* is a vector space with an inner product (Definition 9.9.1) that is complete for the norm $\|x\| = \langle x, x \rangle^{1/2}$. Vectors are *orthogonal* if $\langle x, y \rangle = 0$; for $S \subseteq H$, $S^\perp = \{x : \langle x, s \rangle = 0 \forall s \in S\}$, a closed subspace.

Example 28.3.2. $L^2(\mu)$ with $\langle f, g \rangle = \int f \bar{g}$ (Theorem 26.5.3); ℓ^2 ; finite-dimensional inner product spaces. The space $C([0, 1])$ with the L^2 inner product is not complete, and L^2 is its completion (Exercise 16.11.5).

Theorem 28.3.3. Let H be a Hilbert space and $C \subseteq H$ nonempty, closed and convex. Every $x \in H$ has a unique nearest point $P_C x \in C$. If M is a closed subspace, P_M is linear with $\|P_M\| \leq 1$, $x - P_M x \in M^\perp$, and $H = M \oplus M^\perp$. Moreover $(M^\perp)^\perp = M$, and $M^\perp = 0$ if and only if M is dense.

Proof. Let $d = \inf_{c \in C} \|x - c\|$ and $c_k \in C$ with $\|x - c_k\| \rightarrow d$. By the parallelogram law, $\|c_k - c_l\|^2 = 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4\|x - \frac{c_k + c_l}{2}\|^2 \leq 2\|x - c_k\|^2 + 2\|x - c_l\|^2 - 4d^2 \rightarrow 0$, using $\frac{c_k + c_l}{2} \in C$. So (c_k) is Cauchy and converges to some $c \in C$ with $\|x - c\| = d$; uniqueness follows from the same identity. For a closed subspace M and $m \in M$, $t \in \mathbb{K}$, the inequality $\|x - P_M x - tm\| \geq \|x - P_M x\|$ forces $\langle x - P_M x, m \rangle = 0$. Hence $x = P_M x + (x - P_M x) \in M + M^\perp$, and $M \cap M^\perp = 0$, which also gives linearity and $\|P_M\| \leq 1$ (Pythagoras). Finally $M \subseteq (M^\perp)^\perp$, and if $x \in (M^\perp)^\perp$ then $x - P_M x \in M^\perp \cap (M^\perp)^\perp = 0$. If M is a subspace, $\overline{M}^\perp = M^\perp$, so $M^\perp = 0$ if and only if $\overline{M} = H$. $\square$

Theorem 28.3.4 (Riesz representation). For every $\Lambda \in H^*$ there is a unique $y \in H$ with $\Lambda(x) = \langle x, y \rangle$ for all x , and $\|\Lambda\| = \|y\|$. Consequently every bounded operator $T: H \rightarrow H$ has a unique adjoint T^* with $\langle Tx, y \rangle = \langle x, T^*y \rangle$; it satisfies $\|T^*\| = \|T\|$ and $(\ker T)^\perp = \overline{\text{im } T^*}$.

Proof. Existence and uniqueness of y are Lemma 26.9.1, whose proof used only completeness; the norm equality follows from Cauchy–Schwarz and $\Lambda(y) = \|y\|^2$. For fixed y , $x \mapsto \langle Tx, y \rangle$ is a bounded functional, so it is $\langle x, T^*y \rangle$ for a unique T^*y , and T^* is linear and bounded with $\|T^*\| = \|T\|$ (from $\|T\| = \sup_{\|x\|, \|y\| \leq 1} |\langle Tx, y \rangle|$, which is symmetric in T and T^*). Finally $x \in \ker T$ if and only if $\langle x, T^*y \rangle = 0$ for all y , that is, $x \in (\text{im } T^*)^\perp$; take orthogonal complements. $\square$

Theorem 28.3.5 (Orthonormal bases). Let $(e_i)_{i \in I}$ be an orthonormal family in a Hilbert space H . The following are equivalent: its span is dense; $x = \sum_i \langle x, e_i \rangle e_i$ for all x (with at most countably many nonzero terms, in any order); $\|x\|^2 = \sum_i |\langle x, e_i \rangle|^2$ for all x (Parseval). Such a family is an orthonormal basis; every Hilbert space has one, and H is separable if and only if it has a countable one, in which case $H \cong \ell^2(I)$.

Proof. For finite $J \subseteq I$, $P_J x = \sum_{i \in J} \langle x, e_i \rangle e_i$ is the orthogonal projection onto the span of $(e_i)_{i \in J}$, so $\sum_{i \in J} |\langle x, e_i \rangle|^2 \leq \|x\|^2$ (Bessel); hence at most countably many coefficients are nonzero and, by completeness, $\sum_i \langle x, e_i \rangle e_i$ converges (its partial sums are Cauchy by Pythagoras), to the projection of x onto the closed span M . So the three conditions each say that $M = H$, using $\|x\|^2 = \|P_M x\|^2 + \|x - P_M x\|^2$. Zorn's lemma gives a maximal orthonormal family, whose closed span M satisfies $M^\perp = 0$ (a unit vector in $M^\perp$ would enlarge the family), so $M = H$. If H is separable, an orthonormal family is discrete ($\|e_i - e_j\| = \sqrt{2}$) hence countable. The map $x \mapsto (\langle x, e_i \rangle)$ is then an isometric isomorphism onto $\ell^2(I)$. $\square$

Example 28.3.6. The functions $e_n(t) = e^{2\pi i n t}$, $n \in \mathbb{Z}$, form an orthonormal basis of $L^2(S^1)$, where $S^1 = \mathbb{R}/\mathbb{Z}$ with Lebesgue measure: they are orthonormal by direct computation, their span (the trigonometric polynomials) is dense in $C(S^1)$ for the supremum norm (Corollary 25.13.3), hence in L^2 , and $C(S^1)$ is dense in $L^2(S^1)$: by Proposition 26.5.4 the continuous functions with compact support in $(0, 1)$ are dense in $L^2([0, 1])$, and they extend to continuous functions on S^1 . So every $f \in L^2(S^1)$ is the L^2 -limit of its Fourier series $\sum_n \hat{f}(n) e_n$ with $\hat{f}(n) = \int_0^1 f(t) e^{-2\pi i n t} dt$, and $\|f\|_2^2 = \sum_n |\hat{f}(n)|^2$.

28.4 The Hahn–Banach and Baire theorems

Theorem 28.4.1 (Hahn–Banach). *Let X be a vector space over $\mathbb{K}$ and $p: X \rightarrow \mathbb{R}$ with $p(x + y) \leq p(x) + p(y)$ and $p(\lambda x) = |\lambda|p(x)$ (a seminorm). Every linear functional ℓ on a subspace Y with $|\ell| \leq p$ on Y extends to a linear functional on X with $|\ell| \leq p$. Consequently, for a normed X and $x_0 \neq 0$ there is $\Lambda \in X^*$ with $\|\Lambda\| = 1$ and $\Lambda(x_0) = \|x_0\|$; in particular X^* separates points and $\|x\| = \sup_{\|\Lambda\| \leq 1} |\Lambda(x)|$.*

Proof. Real case: consider pairs (Z, m) of a subspace $Z \supseteq Y$ and an extension $m \leq p$, ordered by extension; chains have upper bounds, so Zorn's lemma gives a maximal (Z, m) . If $x \notin Z$, then for $z, z' \in Z$, $m(z) + m(z') = m(z + z') \leq p(z + x) + p(z' - x)$, so $\sup_{z'} (m(z') - p(z' - x)) \leq \inf_z (p(z + x) - m(z))$; choosing c between them extends m to $Z + \mathbb{K}x$ by $m(z + tx) = m(z) + tc$, contradicting maximality. Complex case: apply the real case to $\operatorname{Re} \ell$ and set $\ell(x) = (\operatorname{Re} \ell)(x) - i(\operatorname{Re} \ell)(ix)$. For the consequence take $Y = \mathbb{K}x_0$, $\ell(tx_0) = t\|x_0\|$ and $p = \|\cdot\|$. $\square$

Theorem 28.4.2. *Let X, Y be Banach spaces.*

1. (Uniform boundedness) *If $\mathcal{T} \subseteq B(X, Y)$ satisfies $\sup_{T \in \mathcal{T}} \|Tx\| < \infty$ for every x , then $\sup_{T \in \mathcal{T}} \|T\| < \infty$.*
2. (Open mapping) *A surjective $T \in B(X, Y)$ is open; if it is bijective, T^{-1} is bounded.*
3. (Closed graph) *A linear $T: X \rightarrow Y$ whose graph is closed in $X \times Y$ is bounded.*

Proof. (1) The sets $F_n = \{x : \|Tx\| \leq n \ \forall T \in \mathcal{T}\}$ are closed with union X , so by Baire's theorem (Theorem 16.7.7) some F_n contains a ball $\overline{B}(x_0, r)$; then $\|Tx\| \leq 2n/r$ for $\|x\| \leq 1$, using $x = ((x_0 + rx) - x_0)/r$.

(2) By Baire's theorem applied to $Y = \bigcup_n \overline{T(B(0, n))}$, the closure of $T(B(0, 1))$ contains a ball $B(y_0, \delta)$; by symmetry and convexity it contains $B(0, \delta)$. A standard iteration removes the closure: given $\|y\| < \delta$, choose x_1 with $\|x_1\| < 1$ and $\|y - Tx_1\| < \delta/2$, then x_2 with $\|x_2\| < 1/2$ and $\|y - Tx_1 - Tx_2\| < \delta/4$, and so on; the series $\sum x_k$ converges to some x with $\|x\| < 2$ and $Tx = y$. So $T(B(0, 2)) \supseteq B(0, \delta)$, which gives openness by translation and scaling.

(3) The graph G is a Banach space, and the projection $G \rightarrow X$ is bounded and bijective, so its inverse $x \mapsto (x, Tx)$ is bounded by (2). $\square$

Counterexample 28.4.3. 1. *An unbounded operator.* Differentiation $D: C^1([0, 1]) \rightarrow C([0, 1])$, both with the supremum norm, is linear but not bounded: $\|x^n\|_\infty = 1$ and $\|D(x^n)\|_\infty = n$. It is closed (if $f_k \rightarrow f$ and $f'_k \rightarrow g$ uniformly, then $f' = g$, Theorem 25.9.4), which does not contradict the closed graph theorem because $C^1([0, 1])$ is not complete for the supremum norm.

2. *An incomplete normed space.* The polynomials, with the supremum norm on $[0, 1]$: the partial sums of $\sum x^k/k!$ form a Cauchy sequence whose uniform limit e^x is not a polynomial, since all its derivatives are nonzero.
3. *Completeness is needed for uniform boundedness.* On the space c_{00} of finitely supported sequences with the supremum norm, the functionals $T_n(x) = nx_n$ satisfy $\sup_n |T_n x| < \infty$ for each x , but $\|T_n\| = n$.
4. *Discontinuous linear functionals.* Every infinite-dimensional normed space X has one. Choose linearly independent unit vectors $e_1, e_2, \dots$ and extend them, by Zorn's lemma (Theorem 3.7.2), to a maximal linearly independent set B , which is a basis of X as a vector space. The linear functional with $\ell(e_n) = n$ and $\ell = 0$ on $B \setminus \{e_1, e_2, \dots\}$ is unbounded. In finite dimensions every linear functional is continuous (Lemma 25.10.1).

28.5 Compact operators

Definition 28.5.1. $T \in B(X, Y)$ is *compact* if the closure of $T(B(0, 1))$ is compact; equivalently, every bounded sequence (x_n) has a subsequence with (Tx_{n_k}) convergent.

Proposition 28.5.2. 1. *Compact operators form a closed subspace of $B(X, Y)$, and STR is compact if T is compact and S, R are bounded.*

2. *Operators of finite rank are compact; on Hilbert spaces, T is compact if and only if it is a norm limit of finite rank operators.*

3. *If H is a Hilbert space, $T \in B(H)$ is compact if and only if T^* is.*

Proof. (1) Sums and scalar multiples are clear, and compositions with bounded operators map bounded sets to bounded sets. If $T_n \rightarrow T$ with T_n compact, then $T(B)$ is totally bounded: given ε , choose n with $\|T - T_n\| < \varepsilon$ and cover $T_n(B)$ by finitely many ε -balls; so $\overline{T(B)}$ is compact (Theorem 16.7.4). (2) A bounded set in a finite-dimensional space is relatively compact (Theorem 16.6.4). Conversely let T be compact and $\varepsilon > 0$; cover $\overline{T(B)}$ by finitely many balls $B(y_i, \varepsilon)$, let P be the orthogonal projection onto the span of the y_i (Theorem 28.3.3); then $\|Tx - PTx\| \leq \|Tx - y_i\| < \varepsilon$ for a suitable i , since P is a projection and $\|Tx - PTx\| = \text{dist}(Tx, \text{im } P)$. So $\|T - PT\| \leq \varepsilon$ with PT of finite rank. (3) follows from (2), since $(PT)^* = T^*P$ has finite rank and $\|T^* - (PT)^*\| = \|T - PT\|$. $\square$

Theorem 28.5.3 (Spectral theorem for compact self-adjoint operators). *Let $H \neq 0$ be a Hilbert space and $T \in B(H)$ compact and self-adjoint ($T = T^*$).*

1. $\|T\| = \sup_{\|x\|=1} |\langle Tx, x \rangle|$, and $\|T\|$ or $-\|T\|$ is an eigenvalue of T .
2. H has an orthonormal basis consisting of eigenvectors of T . The nonzero eigenvalues are real, form a finite or countable set with 0 as the only possible accumulation point, and each nonzero eigenvalue has finite-dimensional eigenspace.

Proof. (1) Let $s = \sup_{\|x\|=1} |\langle Tx, x \rangle| \leq \|T\|$. For x, y with $\|x\|, \|y\| \leq 1$, the identity $\langle T(x + y), x + y \rangle - \langle T(x - y), x - y \rangle = 4 \text{Re} \langle Tx, y \rangle$ (using self-adjointness) and the parallelogram law give $4 \text{Re} \langle Tx, y \rangle \leq s(\|x + y\|^2 + \|x - y\|^2) = 2s(\|x\|^2 + \|y\|^2) \leq 4s$, so $\|T\| = \sup |\langle Tx, y \rangle| \leq s$. Choose unit vectors x_n with $\langle Tx_n, x_n \rangle \rightarrow \lambda$ where $|\lambda| = \|T\|$ (note $\langle Tx, x \rangle$ is real). Then $\|Tx_n - \lambda x_n\|^2 = \|Tx_n\|^2 - 2\lambda \langle Tx_n, x_n \rangle + \lambda^2 \leq 2\lambda^2 - 2\lambda \langle Tx_n, x_n \rangle \rightarrow 0$. By compactness, after passing to a subsequence $Tx_n \rightarrow y$; if $\lambda \neq 0$ then $x_n \rightarrow y/\lambda$, so y/λ is a unit vector with $Ty/\lambda = \lambda y/\lambda$. If $\lambda = 0$ then $T = 0$ and every vector is an eigenvector.

(2) Eigenvalues are real and eigenvectors for distinct eigenvalues are orthogonal, as in finite dimensions. If $|\lambda| \geq \varepsilon > 0$, an orthonormal family in the λ -eigenspace is mapped by T to a family of vectors at mutual distance $|\lambda|\sqrt{2}$, so it must be finite by compactness; the same argument shows that only finitely many eigenvalues satisfy $|\lambda| \geq \varepsilon$. Let M be the closed span of all eigenvectors. It is T -invariant, hence so is $M^\perp$ (if $x \in M^\perp$ and $m \in M$, $\langle Tx, m \rangle = \langle x, Tm \rangle = 0$), and $T|_{M^\perp}$ is compact and self-adjoint; by (1) it has an eigenvector unless $M^\perp = 0$ or $T|_{M^\perp} = 0$. An eigenvector in $M^\perp$ is impossible, and if $T|_{M^\perp} = 0$ then $M^\perp$ consists of eigenvectors for the eigenvalue 0, so $M^\perp \subseteq M$ and $M^\perp = 0$. Hence $M = H$, and an orthonormal basis of each eigenspace, together, is an orthonormal basis of H by Theorem 28.3.5. $\square$

Theorem 28.5.4 (Fredholm alternative). *Let H be a Hilbert space, $K \in B(H)$ compact and $T = I - K$.*

1. $\ker T$ is finite-dimensional and $\text{im } T$ is closed, with $\text{im } T = (\ker T^*)^\perp$.

2. T is injective if and only if it is surjective.
3. $\dim \ker T = \dim \ker T^*$; so the equation $Tx = y$ is solvable if and only if $y \perp \ker T^*$, and the solution is unique up to $\ker T$, which has the same finite dimension as $\ker T^*$.

Proof. (1) On $\ker T$ we have $K = I$, so the unit ball of $\ker T$ is compact and $\ker T$ is finite-dimensional (Proposition 28.2.2(3)). For closedness, let $M = (\ker T)^\perp$; it suffices to show $\|Tx\| \geq c\|x\|$ on M for some $c > 0$, since then images of Cauchy sequences are Cauchy. If not, there are unit vectors $x_n \in M$ with $Tx_n \rightarrow 0$; passing to a subsequence, $Kx_n \rightarrow y$, so $x_n = Tx_n + Kx_n \rightarrow y$ with $\|y\| = 1$, $y \in M$ and $Ty = 0$, a contradiction. Then $\operatorname{im} T$ is closed and $\operatorname{im} T = \overline{\operatorname{im} T} = (\ker T^*)^\perp$ by Theorem 28.3.4 applied to T^* .

(2) Suppose T is injective but not surjective. Then $H \supsetneq TH \supsetneq T^2H \supsetneq \cdots$ is a strictly decreasing chain of closed subspaces (each $T^k H$ is closed by (1) applied to $I - K_k$ with $K_k = I - T^k$ compact, since $T^k = (I - K)^k = I - K_k$ with K_k a sum of products involving K). Choose unit vectors $u_k \in T^k H \cap (T^{k+1} H)^\perp$. For $l > k$, $Ku_k - Ku_l = u_k - (Tu_k + u_l - Tu_l)$ and $Tu_k + u_l - Tu_l \in T^{k+1} H$, so $\|Ku_k - Ku_l\| \geq 1$, contradicting compactness. Conversely, if T is surjective, then T^* is injective: indeed $\ker T^* = (\operatorname{im} T)^\perp = 0$; applying the previous argument to $T^* = I - K^*$ (with K^* compact) shows T^* is surjective, hence $\ker T = (\operatorname{im} T^*)^\perp = 0$.

(3) This is the statement that the index of $I - K$ is zero; see [Bre11, Theorem 6.6]. $\square$

Remark 28.5.5. $T \in B(X, Y)$ is Fredholm if $\ker T$ and $\operatorname{coker} T = Y/\operatorname{im} T$ are finite-dimensional; then $\operatorname{im} T$ is closed, and the index $\operatorname{ind} T = \dim \ker T - \dim \operatorname{coker} T$ is unchanged by compact perturbations and is locally constant in T [Bre11, Chapter 6]. Elliptic operators on compact manifolds are Fredholm (Chapter 39); the index is then computed by the Atiyah–Singer theorem, and its vanishing for the Laplacian is what produces harmonic representatives.

Counterexample 28.5.6. Self-adjointness is needed in the spectral theorem.

1. The Volterra operator $Vf(x) = \int_0^x f(t) dt$ on $L^2([0, 1])$ is compact but has no eigenvalues. It is compact: by the Cauchy–Schwarz inequality $|Vf(x) - Vf(y)| \leq |x - y|^{1/2} \|f\|_2$ and $|Vf(x)| \leq \|f\|_2$, so V maps the unit ball to a bounded equicontinuous set of continuous functions, which is relatively compact in $C([0, 1])$ by the Arzelà–Ascoli theorem (Theorem 25.13.4), hence in L^2 . If $Vf = \lambda f$ with $\lambda \neq 0$, then $f = \lambda^{-1} Vf$ is continuous, hence continuously differentiable, with $f' = \lambda^{-1} f$ and $f(0) = 0$, so $f = 0$; and $Vf = 0$ forces $\int_0^x f = 0$ for all x , hence $f = 0$ almost everywhere.
2. The right shift $S(x_1, x_2, \dots) = (0, x_1, x_2, \dots)$ on ℓ^2 is an isometry without eigenvalues ($Sx = \lambda x$ gives $\lambda x_1 = 0$, $\lambda x_{k+1} = x_k$, hence $x = 0$), and it is not surjective. Its adjoint, the left shift, has every λ with $|\lambda| < 1$ as an eigenvalue, with eigenvector $(1, \lambda, \lambda^2, \dots)$. So an injective bounded operator on an infinite-dimensional space need not be surjective, unlike in finite dimensions.

28.6 Weak topologies

Definition 28.6.1. The *weak topology* on a normed space X is the initial topology (Definition 16.4.1) for all $\Lambda \in X^*$; $x_n \rightarrow x$ weakly means $\Lambda(x_n) \rightarrow \Lambda(x)$ for all Λ . The *weak-* topology* on X^* is the initial topology for the evaluations at points of X .

Theorem 28.6.2 (Banach–Alaoglu). *The closed unit ball of X^* is weak-* compact. If H is a separable Hilbert space, every bounded sequence in H has a weakly convergent subsequence.*

Proof. Map $\Lambda \mapsto (\Lambda(x))_{x \in X}$ into $\prod_{x \in X} \overline{D}(0, \|x\|)$, which is compact by Tychonoff's theorem (Theorem 16.6.7). The image consists of the families satisfying the (closed) linearity conditions $\lambda_{ax+by} = a\lambda_x + b\lambda_y$, so it is a closed subset, hence compact, and the map is a homeomorphism onto its image by definition of the weak-* topology. For the second statement let (x_n) be bounded and (y_k) a dense sequence in H . By a diagonal argument, after passing to a subsequence $\langle x_n, y_k \rangle$ converges for every k , hence $\langle x_n, y \rangle$ converges for every y (uniformly approximate y by the y_k , using boundedness). The limit is a bounded conjugate-linear functional of y , so it is $\langle x, y \rangle$ for some x by Theorem 28.3.4. $\square$

Counterexample 28.6.3. Weak convergence does not imply norm convergence: an orthonormal sequence (e_n) in a Hilbert space satisfies $e_n \rightharpoonup 0$ by Bessel's inequality, while $\|e_n\| = 1$. By Proposition 28.2.2(3), the closed unit ball of an infinite-dimensional space is never norm compact, although it is weakly compact for Hilbert spaces.

28.7 Exercises

Exercise 28.7.1. Show that $(ST)^* = T^*S^*$, $T^{**} = T$ and $\|T^*T\| = \|T\|^2$ for bounded operators on a Hilbert space.

Solution. The first two follow from uniqueness of adjoints. For the last, $\|T^*T\| \leq \|T\|^2$, and $\|Tx\|^2 = \langle T^*Tx, x \rangle \leq \|T^*T\| \|x\|^2$. $\square$

Exercise 28.7.2. Let $k \in L^2([0, 1]^2)$ and $(Tf)(x) = \int_0^1 k(x, y)f(y) dy$. Show that T is a compact operator on $L^2([0, 1])$ with $\|T\| \leq \|k\|_2$, and that T^* has kernel $\overline{k(y, x)}$.

Solution. Boundedness follows from Cauchy–Schwarz and Fubini's theorem. Kernels of the form $\sum_i \phi_i(x)\psi_i(y)$ give finite rank operators and are dense in $L^2([0, 1]^2)$ (products of an orthonormal basis of $L^2([0, 1])$ form an orthonormal basis of $L^2([0, 1]^2)$), so T is a limit of finite rank operators. The formula for T^* is Fubini's theorem. $\square$

Exercise 28.7.3. Show that the functional $f \mapsto f(0)$ on $C([-1, 1])$ is not continuous for the L^2 norm, and deduce that it does not extend to $L^2([-1, 1])$.

Exercise 28.7.4. Let T be compact self-adjoint with eigenvalues λ_k and orthonormal eigenvectors e_k . Show that $Tx = \sum_k \lambda_k \langle x, e_k \rangle e_k$, that T is a norm limit of the finite rank operators obtained by truncating this sum, and that $\|T\| = \max_k |\lambda_k|$.

Chapter 29

Several Complex Variables

29.1 Introduction

In several variables holomorphic functions are still given by convergent power series, but new phenomena appear: functions extend across small holes (Theorem 29.2.9), the local ring of convergent power series is a Noetherian unique factorisation domain (Theorem 29.3.4), and the equation $\bar{\partial}u = f$ is solvable on polydiscs (Theorem 29.4.2), which is the local statement behind Dolbeault cohomology and the Hodge decomposition. References: [Hör90b], [GR65], [GH78, Chapter 0].

Prerequisites. Chapter 27 (Complex Analysis) and Chapter 12 (Commutative Algebra).

For $a \in \mathbb{C}^n$ and $r = (r_1, \dots, r_n)$ with $r_j > 0$, the *polydisc* is $\Delta(a, r) = \{z : |z_j - a_j| < r_j\}$ and its *distinguished boundary* is $\{|z_j - a_j| = r_j \ \forall j\}$. We write $\partial/\partial z_j = \frac{1}{2}(\partial_{x_j} - i\partial_{y_j})$ and $\partial/\partial \bar{z}_j = \frac{1}{2}(\partial_{x_j} + i\partial_{y_j})$, so that a C^1 function is holomorphic in z_j if and only if $\partial f/\partial \bar{z}_j = 0$ (Proposition 27.2.2).

29.2 Holomorphic functions of several variables

Definition 29.2.1. Let $U \subseteq \mathbb{C}^n$ be open. A function $f : U \rightarrow \mathbb{C}$ is *holomorphic* if it is continuous and holomorphic in each variable separately. The set of such functions is $O(U)$.

Theorem 29.2.2. Let f be holomorphic on a neighbourhood of the closure of $\Delta = \Delta(a, r)$. Then for $z \in \Delta$

$$f(z) = \frac{1}{(2\pi i)^n} \int_{|\zeta_n - a_n|=r_n} \cdots \int_{|\zeta_1 - a_1|=r_1} \frac{f(\zeta)}{(\zeta_1 - z_1) \cdots (\zeta_n - z_n)} d\zeta_1 \cdots d\zeta_n,$$

and $f(z) = \sum_{\alpha \in \mathbb{N}^n} c_\alpha (z - a)^\alpha$ with $c_\alpha = \partial^\alpha f(a)/\alpha!$, the series converging absolutely and uniformly on compact subsets of Δ . In particular holomorphic functions are C^∞ , all their partial derivatives are holomorphic, and $|c_\alpha| \leq \sup |f| \cdot r^{-\alpha}$ (Cauchy estimates).

Proof. Apply the one-variable Cauchy formula (Theorem 27.4.1) in each variable in turn, using continuity to integrate iterated integrals (Proposition 25.8.5). Expanding each factor $1/(\zeta_j - z_j)$ in a geometric series, as in Theorem 27.4.2, and multiplying out gives the power series, uniformly convergent on compact subsets; term-by-term differentiation identifies the coefficients. Smoothness and holomorphy of the derivatives follow from differentiating the integral formula under the integral sign. $\square$

Proposition 29.2.3. Let $U \subseteq \mathbb{C}^n$ be a domain.

1. (Identity theorem) If $f \in O(U)$ vanishes on a nonempty open subset, then $f = 0$.

2. (Maximum principle) If $|f|$ has a local maximum in U , then f is constant.
3. If $f_k \in \mathcal{O}(U)$ converge uniformly on compact subsets, the limit is holomorphic, and all partial derivatives converge uniformly on compact subsets.

Proof. (1) The set where all derivatives $\partial^\alpha f$ vanish is open by Theorem 29.2.2 and closed, hence is U . (2) Restrict to complex lines through the point and apply Theorem 27.4.5: f is constant on a polydisc, hence on U by (1). (3) Separate holomorphy passes to uniform limits (Corollary 27.4.3), and convergence of derivatives follows from the Cauchy estimates applied to $f_k - f_l$ on slightly smaller polydiscs. $\square$

Lemma 29.2.4. Let A be a complex $n \times n$ matrix and $A_{\mathbb{R}}$ the real-linear map of $\mathbb{R}^{2n} = \mathbb{C}^n$ it defines. Then $\det A_{\mathbb{R}} = |\det A|^2$. In particular $A_{\mathbb{R}}$ is invertible if and only if A is, and a holomorphic map with invertible complex Jacobian preserves orientation.

Proof. The map $A \mapsto A_{\mathbb{R}}$ is multiplicative, so both sides are invariant under conjugation $A \mapsto PAP^{-1}$ by an invertible complex matrix. By induction on n , choosing an eigenvector (which exists because $\mathbb{C}$ is algebraically closed, Theorem 8.7.3) and completing it to a basis, A is conjugate to an upper triangular matrix T with diagonal entries $\lambda_1, \dots, \lambda_n$. In the real basis $e_1, ie_1, \dots, e_n, ie_n$, the matrix of $T_{\mathbb{R}}$ is block upper triangular with 2×2 diagonal blocks $\begin{pmatrix} a_j & -b_j \\ b_j & a_j \end{pmatrix}$, the matrices of multiplication by $\lambda_j = a_j + ib_j$, of determinant $a_j^2 + b_j^2 = |\lambda_j|^2$. So $\det T_{\mathbb{R}} = \prod_j |\lambda_j|^2 = |\det T|^2$. $\square$

Proposition 29.2.5 (Holomorphic inverse and implicit function theorems). 1. Let $f: U \rightarrow \mathbb{C}^n$ be holomorphic on an open $U \subseteq \mathbb{C}^n$ (that is, each component is holomorphic), and let $a \in U$ with $\det(\partial f_i / \partial z_j(a)) \neq 0$. Then there are open neighbourhoods V of a and W of $f(a)$ such that $f: V \rightarrow W$ is bijective with holomorphic inverse.

2. Let $F: U \rightarrow \mathbb{C}^m$ be holomorphic on an open $U \subseteq \mathbb{C}^n \times \mathbb{C}^m$, with $F(a, b) = 0$ and $\det(\partial F_i / \partial w_j(a, b)) \neq 0$, where w are the coordinates on $\mathbb{C}^m$. Then there are neighbourhoods V of a and W of b and a holomorphic $g: V \rightarrow W$ such that $\{(z, w) \in V \times W : F(z, w) = 0\}$ is the graph of g .

Proof. (1) By Theorem 29.2.2 f is C^∞ , and by the Cauchy–Riemann equations (Proposition 27.2.2) in each variable its real derivative at z is the real-linear map underlying the complex Jacobian $J(z) = (\partial f_i / \partial z_j(z))$. By Lemma 29.2.4 it is invertible at a , so the inverse function theorem (Theorem 25.11.2) gives V, W and a C^∞ inverse $g: W \rightarrow V$. Differentiating $f \circ g = \text{id}$, the real derivative of g at w is $J(g(w))^{-1}$, the inverse of a $\mathbb{C}$ -linear map, hence $\mathbb{C}$ -linear. A C^1 map whose real derivative is everywhere $\mathbb{C}$ -linear satisfies the Cauchy–Riemann equations in each variable, so g is holomorphic (Definition 29.2.1).

(2) Apply (1) to $\Phi(z, w) = (z, F(z, w))$, whose complex Jacobian at (a, b) is block triangular with determinant $\det(\partial F_i / \partial w_j(a, b)) \neq 0$. Its local holomorphic inverse has the form $\Psi(z, u) = (z, h(z, u))$, and $g(z) = h(z, 0)$ is holomorphic, with $F(z, w) = 0$ for (z, w) near (a, b) exactly when $w = g(z)$. $\square$

Corollary 29.2.6 (Liouville). Let f be holomorphic on $\mathbb{C}^n$ with $|f(z)| \leq C(1 + \|z\|)^N$ for all z . Then f is a polynomial of degree at most N . In particular bounded holomorphic functions on $\mathbb{C}^n$ are constant.

Proof. Theorem 29.2.2, applied on the polydiscs $\Delta(0, (r, \dots, r))$ for all r , shows that the Taylor series $\sum_\alpha c_\alpha z^\alpha$ at 0 converges to f on all of $\mathbb{C}^n$, with $|c_\alpha| \leq C(1 + \sqrt{n}r)^N r^{-|\alpha|}$. For $|\alpha| > N$ the right side tends to 0 as $r \rightarrow \infty$, so $c_\alpha = 0$. $\square$

Example 29.2.7. Domains of convergence of power series in several variables need not be polydiscs. The series $\sum_{k \geq 0} (z_1 z_2)^k$ converges, to $1/(1 - z_1 z_2)$, exactly on $\{|z_1 z_2| < 1\}$, an unbounded domain containing both coordinate axes. It is invariant under $(z_1, z_2) \mapsto (e^{i\theta_1} z_1, e^{i\theta_2} z_2)$ (a *Reinhardt domain*), as is the domain of convergence of every power series centred at 0, because convergence of $\sum c_\alpha z^\alpha$ depends only on the moduli $|z_j|$.

Counterexample 29.2.8. The continuity assumption in Definition 29.2.1 is needed for real variables but not, by a theorem of Hartogs, for complex ones [Hör90b]. On $\mathbb{R}^2$ let $f(x, y) = xy/(x^2 + y^2)$ and $f(0, 0) = 0$. For each fixed y , $x \mapsto f(x, y)$ is real-analytic (it is 0 if $y = 0$, and a rational function with nonvanishing denominator otherwise), and likewise in y . But $f(t, t) = \frac{1}{2}$ for $t \neq 0$, so f is not continuous at the origin.

Theorem 29.2.9 (Hartogs). *Let $n \geq 2$, let $\Delta = \Delta(0, r) \subseteq \mathbb{C}^n$ be a polydisc and $K \subseteq \Delta$ a compact subset such that $\Delta \setminus K$ is connected and $K \subseteq \Delta(0, s)$ for some s with $s_j < r_j$. Then every $f \in \mathcal{O}(\Delta \setminus K)$ extends to a holomorphic function on Δ . In particular holomorphic functions of $n \geq 2$ variables have no isolated singularities and no isolated zeros.*

Proof. Write $z = (z', z_n)$ and choose ρ with $s_n < \rho < r_n$. For $z' \in \Delta'(0, r')$ and $|z_n| < \rho$ set

$$F(z', z_n) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} \frac{f(z', \zeta)}{\zeta - z_n} d\zeta,$$

which makes sense because $(z', \zeta) \notin K$ when $|\zeta| = \rho > s_n$. Differentiating under the integral sign, F is holomorphic in z_n , and it is continuous and holomorphic in each variable z'_j because f is. So F is holomorphic on $\Delta' \times \Delta(0, \rho)$, where $\Delta' = \Delta'(0, r')$. If $|z'_j| > s_j$ for some j , the whole disc $\{z'\} \times \Delta(0, \rho)$ avoids K , so the one-variable Cauchy formula gives $F(z', z_n) = f(z', z_n)$ there.

Now fix z_n with $s_n < |z_n| < \rho$. The functions $z' \mapsto F(z', z_n)$ and $z' \mapsto f(z', z_n)$ are both holomorphic on the connected set Δ' (the slice avoids K , since $|z_n| > s_n$) and agree on a nonempty open subset, hence agree on Δ' by Proposition 29.2.3. So $F = f$ on $\Delta' \times \{s_n < |z_n| < \rho\}$, which is exactly the overlap of $\Delta' \times \Delta(0, \rho)$ and $\Delta \cap \{|z_n| > s_n\}$. These two open sets cover Δ because $K \subseteq \{|z_n| < s_n\}$, so F and f glue to a holomorphic function G on Δ . Finally $G = f$ on all of $\Delta \setminus K$, since both are holomorphic there, $\Delta \setminus K$ is connected, and they agree on a nonempty open subset. $\square$

29.3 The Weierstrass theorems

Definition 29.3.1. Let $\mathcal{O}_n = \mathcal{O}_{\mathbb{C}^n, 0}$ be the ring of germs at 0 of holomorphic functions, that is, of convergent power series $\sum_\alpha c_\alpha z^\alpha$. A germ f is *regular of order k in z_n* if $f(0, \dots, 0, z_n) = cz_n^k + \mathcal{O}(z_n^{k+1})$ with $c \neq 0$. A *Weierstrass polynomial* of degree k is $W = z_n^k + a_{k-1}(z')z_n^{k-1} + \dots + a_0(z')$ with $a_j \in \mathcal{O}_{n-1}$ and $a_j(0) = 0$.

Lemma 29.3.2. $\mathcal{O}_n$ is an integral domain and a local ring with maximal ideal $\mathfrak{m} = \{f : f(0) = 0\}$; a germ is a unit if and only if $f(0) \neq 0$. After a linear change of coordinates, any nonzero germ becomes regular of some finite order in z_n .

Proof. Germs with $f(0) \neq 0$ are invertible, since $1/f$ is holomorphic near 0; so the nonunits form the ideal $\mathfrak{m}$ and the ring is local (Proposition 12.3.2(4)). If $fg = 0$ with $f \neq 0$, then g vanishes on a nonempty open subset of the complement of the zero set of f , which is dense, so $g = 0$ by the identity theorem. For the last statement, choose $v \in \mathbb{C}^n$ with $f(tv) \neq 0$ for small $t \neq 0$ (possible since the zero set of f is not a neighbourhood of 0), and a linear change of coordinates taking v to e_n ; then $t \mapsto f(tv)$ has a zero of finite order at 0. $\square$

Theorem 29.3.3 (Weierstrass preparation and division). *Let $f \in \mathcal{O}_n$ be regular of order k in z_n .*

1. (Preparation) $f = uW$ with $u \in \mathcal{O}_n$ a unit and W a Weierstrass polynomial of degree k , both unique.
2. (Division) Every $g \in \mathcal{O}_n$ can be written uniquely as $g = qf + \rho$ with $q \in \mathcal{O}_n$ and $\rho \in \mathcal{O}_{n-1}[z_n]$ of degree $< k$.

Proof. Choose $\varepsilon > 0$ such that $z_n \mapsto f(0, z_n)$ has $z_n = 0$ as its only zero in $|z_n| \leq \varepsilon$, and then, by continuity of f and the argument principle (Theorem 27.5.10), $\delta > 0$ such that for $|z'| < \delta$ the function $z_n \mapsto f(z', z_n)$ has no zero with $|z_n| = \varepsilon$ and exactly k zeros $w_1(z'), \dots, w_k(z')$ in $|z_n| < \varepsilon$, counted with multiplicity.

(1) Let $W(z', z_n) = \prod_{j=1}^k (z_n - w_j(z'))$, whose coefficients are, up to sign, the elementary symmetric functions $e_m(z')$ of the roots. The power sums $p_m(z') = \sum_j w_j(z')^m = \frac{1}{2\pi i} \int_{|\zeta|=\varepsilon} \zeta^m \frac{\partial_\zeta f(z', \zeta)}{f(z', \zeta)} d\zeta$ are holomorphic in z' (differentiate under the integral sign), by the residue computation in the proof of Theorem 27.5.10. From $E(t) = \prod_j (1 - w_j t) = \sum_m (-1)^m e_m t^m$ we get $E'(t)/E(t) = -\sum_{m \geq 1} p_m t^{m-1}$, so comparing coefficients expresses each e_m as a polynomial with rational coefficients in $p_1, \dots, p_m$; hence the e_m , and therefore the coefficients of W , are holomorphic. Since all roots tend to 0 as $z' \rightarrow 0$, $W(0, z_n) = z_n^k$ and W is a Weierstrass polynomial. The quotient $u = f/W$ extends holomorphically across the zeros in each slice (the zeros of f and W agree with multiplicities), and $u(z', z_n) = \frac{1}{2\pi i} \int_{|\zeta|=\varepsilon} \frac{f(z', \zeta)}{W(z', \zeta)} \frac{d\zeta}{\zeta - z_n}$ shows that u is holomorphic in both variables; it has no zeros, so it is a unit. If $f = uW = u'W'$ with W, W' Weierstrass polynomials, then W and W' have the same zeros with multiplicities in each slice, hence are equal.

(2) By (1) we may assume $f = W$ is a Weierstrass polynomial. Define

$$q(z) = \frac{1}{2\pi i} \int_{|\zeta|=\varepsilon} \frac{g(z', \zeta)}{W(z', \zeta)} \cdot \frac{d\zeta}{\zeta - z_n}, \quad \rho = g - qW,$$

so q is holomorphic. Using the one-variable Cauchy formula for g ,

$$\rho(z) = \frac{1}{2\pi i} \int_{|\zeta|=\varepsilon} g(z', \zeta) \frac{W(z', \zeta) - W(z', z_n)}{W(z', \zeta)(\zeta - z_n)} d\zeta,$$

and since W is monic of degree k in the last variable, $(W(z', \zeta) - W(z', z_n))/(\zeta - z_n)$ is a polynomial in z_n of degree $k-1$ whose coefficients are holomorphic in (z', ζ) . So $\rho \in \mathcal{O}_{n-1}[z_n]$ has degree $< k$. For uniqueness, if $q_1W + \rho_1 = q_2W + \rho_2$, then $(q_1 - q_2)W = \rho_2 - \rho_1$; for fixed small z' the left-hand side has at least k zeros in $|z_n| < \varepsilon$ with multiplicity, while the right-hand side is a polynomial of degree $< k$, so both sides vanish. For general $f = uW$ take $q = u^{-1}q_W$. $\square$

Theorem 29.3.4. $\mathcal{O}_n$ is a Noetherian local ring and a unique factorisation domain, of Krull dimension n .

Proof. Induction on n ; $\mathcal{O}_0 = \mathbb{C}$ is a field. Suppose $\mathcal{O}_{n-1}$ is Noetherian and a UFD. By Hilbert's basis theorem (Theorem 12.5.3), $\mathcal{O}_{n-1}[z_n]$ is Noetherian, and by Theorem 7.6.11 it is a UFD.

Noetherian. Let $I \subseteq \mathcal{O}_n$ be a nonzero ideal and $f \in I$ nonzero; after a linear change of coordinates f is regular of some order k in z_n (Lemma 29.3.2). By Weierstrass division, I is generated by f together with $I \cap \mathcal{O}_{n-1}[z_n]$, which is finitely generated because $\mathcal{O}_{n-1}[z_n]$ is Noetherian.

Factorial. Let $f \in \mathcal{O}_n$ be nonzero and not a unit; after a change of coordinates and by preparation, $f = uW$ with W a Weierstrass polynomial. Factor $W = W_1 \cdots W_m$ into

irreducibles in $\mathcal{O}_{n-1}[z_n]$; by Gauss's lemma (Lemma 7.6.10) and induction each factor may be taken to be a Weierstrass polynomial. Uniqueness of the factorisation in $\mathcal{O}_n$ follows from uniqueness in $\mathcal{O}_{n-1}[z_n]$ together with the uniqueness statement in Theorem 29.3.3: a factorisation in $\mathcal{O}_n$ of a Weierstrass polynomial produces, after preparation of each factor, a factorisation into Weierstrass polynomials, and irreducible Weierstrass polynomials are irreducible in $\mathcal{O}_n$ (a factorisation in $\mathcal{O}_n$ can be normalised in the same way).

Dimension. The chain of primes $0 \subsetneq (z_1) \subsetneq (z_1, z_2) \subsetneq \cdots \subsetneq (z_1, \dots, z_n) = \mathfrak{m}$ has length n , so $\dim \mathcal{O}_n \geq n$; equality holds because $\mathfrak{m}$ is generated by the n elements $z_1, \dots, z_n$ (Corollary 12.9.6). $\square$

29.4 The Dolbeault–Grothendieck lemma

Lemma 29.4.1 (Cauchy transform). *Let $g: \mathbb{C} \rightarrow \mathbb{C}$ be C^∞ with compact support and $u(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{g(z-w)}{w} dA(w)$, where dA is Lebesgue measure. Then u is C^∞ and $\partial u / \partial \bar{z} = g$. If g depends smoothly (or holomorphically) on additional parameters, so does u .*

Proof. The function $1/|w|$ is integrable near 0, so the integral converges and, by dominated convergence and the smoothness of g , we may differentiate under the integral sign: u is C^∞ and $\partial u / \partial \bar{z} = \frac{1}{\pi} \int (\partial g / \partial \bar{z})(z-w) w^{-1} dA(w)$. Write $w = \rho e^{i\theta}$, so $dA = \rho d\rho d\theta$ and $w^{-1} dA = e^{-i\theta} d\rho d\theta$, and set $F(\rho, \theta) = g(z - \rho e^{i\theta})$. From $\partial_\rho F = -(e^{i\theta} g_z + e^{-i\theta} g_{\bar{z}})$ and $\partial_\theta F = i\rho(-e^{i\theta} g_z + e^{-i\theta} g_{\bar{z}})$ we get $e^{-i\theta} g_{\bar{z}} = -\frac{1}{2} \partial_\rho F + \frac{1}{2i\rho} \partial_\theta F$. Hence

$$\frac{\partial u}{\partial \bar{z}} = \frac{1}{\pi} \int_0^\infty \int_0^{2\pi} \left(-\frac{1}{2} \partial_\rho F + \frac{1}{2i\rho} \partial_\theta F \right) d\rho d\theta = \frac{1}{\pi} \int_0^{2\pi} \frac{1}{2} g(z) d\theta = g(z),$$

since $\int_0^\infty \partial_\rho F d\rho = -g(z)$ by the fundamental theorem of calculus and compact support, and $\int_0^{2\pi} \partial_\theta F d\theta = 0$ by periodicity. Dependence on parameters follows in the same way from differentiation under the integral sign. $\square$

Theorem 29.4.2 ($\bar{\partial}$ -Poincaré lemma). *Let $\Delta' \subseteq \Delta \subseteq \mathbb{C}^n$ be polydiscs centred at 0 with $\overline{\Delta'} \subseteq \Delta$, and let $f_1, \dots, f_n$ be C^∞ functions on Δ with*

$$\frac{\partial f_j}{\partial \bar{z}_l} = \frac{\partial f_l}{\partial \bar{z}_j} \quad \text{for all } j, l.$$

Then there is a C^∞ function u on Δ' with $\partial u / \partial \bar{z}_j = f_j$ for all j . In the language of differential forms (Chapter 43), every $\bar{\partial}$ -closed $(0, 1)$ -form on Δ is $\bar{\partial}$ -exact on Δ' , and the same holds for (p, q) -forms.

Proof. We prove by descending induction on k the following statement: if $f_j = 0$ for $j > k$, then on any polydisc Δ'' with $\overline{\Delta''} \subseteq \Delta$ there is u with $\partial u / \partial \bar{z}_j = f_j$ for all j . For $k = 0$ take $u = 0$. Let $k \geq 1$ and choose a C^∞ function $\chi(z_k)$ with compact support in the k -th disc of Δ , equal to 1 on a neighbourhood of the closure of the k -th disc of Δ'' . Put $g = \chi f_k$, extended by zero, and

$$v(z) = \frac{1}{\pi} \int_{\mathbb{C}} \frac{g(z_1, \dots, z_{k-1}, z_k - w, z_{k+1}, \dots, z_n)}{w} dA(w),$$

which is C^∞ on Δ by Lemma 29.4.1 and satisfies $\partial v / \partial \bar{z}_k = g$, so $\partial v / \partial \bar{z}_k = f_k$ on Δ'' . For $j > k$ we have $\partial g / \partial \bar{z}_j = \chi \partial f_k / \partial \bar{z}_j = \chi \partial f_j / \partial \bar{z}_k = 0$ (because χ depends only on z_k and $f_j = 0$), so $\partial v / \partial \bar{z}_j = 0$ by differentiating under the integral sign.

Now set $f'_j = f_j - \partial v / \partial \bar{z}_j$. These are C^∞ on Δ , satisfy the same compatibility conditions (second derivatives commute, Theorem 25.10.5), and $f'_j = 0$ for $j \geq k$ on Δ'' . By the induction

hypothesis applied on Δ'' (with $k - 1$), there is u_0 on any polydisc compactly contained in Δ'' , in particular on Δ' if we choose Δ'' between them, with $\partial u_0 / \partial \bar{z}_j = f'_j$; then $u = u_0 + v$ works. $\square$

29.5 Analytic sets

Definition 29.5.1. A subset A of an open $U \subseteq \mathbb{C}^n$ is *analytic* if every point of U has a neighbourhood V such that $A \cap V$ is the common zero set of finitely many holomorphic functions on V . A point $a \in A$ is *regular* of dimension d if near a the set A is, after a biholomorphic change of coordinates, an open subset of $\mathbb{C}^d \times \{0\}$; otherwise it is *singular*.

Example 29.5.2. The zero set of a single nonzero $f \in \mathcal{O}(U)$ is an analytic *hypersurface*; by Theorem 29.3.3 it is, in suitable local coordinates, the zero set of a Weierstrass polynomial, hence a finite branched covering of a polydisc in $\mathbb{C}^{n-1}$. The set $\{z_1 z_2 = 0\} \subseteq \mathbb{C}^2$ is analytic with singular point the origin; $\{z_1^2 = z_2^3\}$ is analytic, irreducible, with a cusp at the origin.

Theorem 29.5.3. Let $A \subseteq U$ be analytic.

1. The regular points of A are dense in A , and the singular locus A_{sing} is a closed analytic subset of U contained in A , nowhere dense in A .
2. Locally, A has finitely many irreducible components, and an irreducible germ of dimension d admits a local projection to a polydisc in $\mathbb{C}^d$ that is a finite branched covering, branched over a proper analytic subset.
3. (Riemann extension) If $A \subseteq U$ is analytic of dimension $\leq n - 1$ and f is holomorphic on $U \setminus A$ and locally bounded near A , then f extends holomorphically to U .

These are proved in [GR65, Chapters II–III] and [GR84, Chapters 1–6]; the key tools are the Weierstrass theorems and the local parametrisation theorem. Part (3) for a hypersurface follows from the one-variable Riemann extension theorem (Proposition 27.5.3) applied on lines transverse to A , together with Theorem 29.2.2.

Remark 29.5.4. The sheaf $\mathcal{O}_X$ of holomorphic functions on a complex manifold is *coherent* (Oka's theorem), and so is the ideal sheaf of an analytic subset (Cartan); these are the foundations of the theory of coherent analytic sheaves used in Chapter 54, where they are compared with the algebraic theory [GR84].

29.6 Exercises

Exercise 29.6.1. Show that $\{z \in \mathbb{C}^2 : |z_1| < 1, |z_2| < 1\} \setminus \{0\}$ and $\mathbb{C}^2 \setminus \{0\}$ have the property that every holomorphic function extends to the full polydisc, respectively to $\mathbb{C}^2$, and deduce that these domains are not biholomorphic to any domain in $\mathbb{C}$.

Solution. Extension is Theorem 29.2.9 with $K = \{0\}$. In $\mathbb{C}$, the function $1/z$ is holomorphic on the punctured disc and does not extend; so no biholomorphism with a punctured planar domain can exist, since biholomorphisms preserve the algebra of holomorphic functions. $\square$

Exercise 29.6.2. Apply the Weierstrass preparation theorem to $f(z_1, z_2) = z_2^2 - z_1$ and to $f(z_1, z_2) = z_1 z_2$ at the origin, and explain why a linear change of coordinates is needed in the second case.

Exercise 29.6.3. Let f be C^∞ on a neighbourhood of the closed unit disc. Show that there is a C^∞ function u on the open disc with $\partial u / \partial \bar{z} = f$, and that u is unique up to adding a holomorphic function.

Solution. Multiply f by a cut-off function equal to 1 on a slightly smaller disc and apply Lemma 29.4.1; exhaust the disc. The difference of two solutions satisfies $\partial u / \partial \bar{z} = 0$, hence is holomorphic (Proposition 27.2.2). $\square$

Exercise 29.6.4. Show that a holomorphic function on a bounded domain $U \subseteq \mathbb{C}^n$, continuous on $\bar{U}$, attains the maximum of its modulus on ∂U , and give an example showing that for a polydisc the maximum is already attained on the distinguished boundary.

Chapter 30

Distributions and Sobolev Spaces

30.1 Introduction

To solve $\Delta u = f$ one first finds a weak solution in a space of distributions and then proves that it is smooth. This chapter sets up the two tools that make this work: the Fourier transform, which turns differentiation into multiplication, and the Sobolev spaces H^s , in which the elliptic estimate (Theorem 30.5.3) and Rellich's compactness theorem (Theorem 30.4.4) hold. On a compact manifold these local statements are patched together with partitions of unity to prove the Hodge theorem (Chapter 40). References: [Hör90a], [Rud91, Chapters 6–7], [Bre11], [War83, Chapter 6].

Prerequisites. Chapter 28 (Functional Analysis).

Multi-index notation is as in Theorem 25.10.7; we write $D^\alpha = \partial^\alpha$ and $\langle \xi \rangle = (1 + |\xi|^2)^{1/2}$.

30.2 Distributions

Definition 30.2.1. Let $U \subseteq \mathbb{R}^n$ be open and $\mathcal{D}(U) = C_c^\infty(U)$. A sequence $\varphi_k \rightarrow \varphi$ in $\mathcal{D}(U)$ means: all φ_k have support in a fixed compact subset of U , and $D^\alpha \varphi_k \rightarrow D^\alpha \varphi$ uniformly for every α . A *distribution* is a linear map $T: \mathcal{D}(U) \rightarrow \mathbb{C}$ such that $T(\varphi_k) \rightarrow T(\varphi)$ whenever $\varphi_k \rightarrow \varphi$; equivalently, for every compact $K \subseteq U$ there are C and N with $|T(\varphi)| \leq C \sum_{|\alpha| \leq N} \sup |D^\alpha \varphi|$ for all φ supported in K . The space of distributions is $\mathcal{D}'(U)$, and $T_k \rightarrow T$ means $T_k(\varphi) \rightarrow T(\varphi)$ for every φ .

Example 30.2.2. 1. Every $f \in L^1_{\text{loc}}(U)$ defines $T_f(\varphi) = \int f \varphi$, and $T_f = T_g$ if and only if $f = g$ a.e.; so $\mathcal{D}'(U)$ contains the locally integrable functions.

2. The *Dirac distribution* $\delta_a(\varphi) = \varphi(a)$ is not of this form: if $\delta_0 = T_f$, then $\int f \varphi = 0$ for all φ supported away from 0, so $f = 0$ a.e., while $\delta_0 \neq 0$.

3. $\text{pv } \frac{1}{x}(\varphi) = \lim_{\varepsilon \rightarrow 0} \int_{|x| > \varepsilon} \frac{\varphi(x)}{x} dx$ is a distribution on $\mathbb{R}$ of order 1.

Definition 30.2.3. For $T \in \mathcal{D}'(U)$, α a multi-index and $g \in C^\infty(U)$, set $(D^\alpha T)(\varphi) = (-1)^{|\alpha|} T(D^\alpha \varphi)$ and $(gT)(\varphi) = T(g\varphi)$. The *support* of T is the complement of the largest open set on which T vanishes.

Proposition 30.2.4. *These are distributions; $D^\alpha D^\beta T = D^{\alpha+\beta} T$; for $f \in C^k(U)$ the distributional derivatives of order $\leq k$ are the classical ones; and differentiation is continuous: $T_k \rightarrow T$ implies $D^\alpha T_k \rightarrow D^\alpha T$.*

Proof. Continuity is clear from the definition, and the formula for $D^\alpha T$ is forced by integration by parts, which for $f \in C^k$ and $\varphi \in C_c^\infty$ gives $\int (D^\alpha f) \varphi = (-1)^{|\alpha|} \int f D^\alpha \varphi$ (Fubini's theorem

and the fundamental theorem of calculus in one variable at a time, Proposition 25.8.5). The other statements are immediate. $\square$

Example 30.2.5. The Heaviside function $H = \mathbf{1}_{[0,\infty)}$ on $\mathbb{R}$ has $H' = \delta_0$, since $-\int_0^\infty \varphi' = \varphi(0)$. Hence $\delta'_0(\varphi) = -\varphi'(0)$, and the function $\log|x|$ has distributional derivative $\text{pv } \frac{1}{x}$.

Proposition 30.2.6. Let $I \subseteq \mathbb{R}$ be an open interval. A distribution T on I with $T' = 0$ is a constant function: $T = T_c$ for some $c \in \mathbb{C}$.

Proof. Fix $\psi \in \mathcal{D}(I)$ with $\int \psi = 1$ and put $c = T(\psi)$. For $\varphi \in \mathcal{D}(I)$, the function $\varphi - (\int \varphi)\psi$ has integral 0 and compact support in I , so it is the derivative of $\chi(x) = \int_{-\infty}^x (\varphi - (\int \varphi)\psi)$, which also has compact support in I (it vanishes to the left of both supports, and to the right because the total integral is 0). Hence $T(\varphi) - (\int \varphi)c = T(\chi') = -T'(\chi) = 0$, that is, $T(\varphi) = \int c\varphi$. $\square$

Proposition 30.2.7. Let $\rho \in L^1(\mathbb{R}^n)$ with $\int \rho = 1$ and $\rho_\varepsilon(x) = \varepsilon^{-n}\rho(x/\varepsilon)$. Then $T_{\rho_\varepsilon} \rightarrow \delta_0$ in $\mathcal{D}'(\mathbb{R}^n)$ as $\varepsilon \rightarrow 0$.

Proof. For $\varphi \in \mathcal{D}(\mathbb{R}^n)$, the substitution $x = \varepsilon y$ gives $\int \rho_\varepsilon \varphi - \varphi(0) = \int \rho(y)(\varphi(\varepsilon y) - \varphi(0)) dy$. The integrand tends to 0 pointwise and is bounded by $2 \sup |\varphi| |\rho|$, so the integral tends to 0 by dominated convergence (Theorem 26.4.4). $\square$

30.3 The Fourier transform

Definition 30.3.1. The Schwartz space $\mathcal{S}(\mathbb{R}^n)$ consists of the $f \in C^\infty(\mathbb{R}^n)$ with $p_{\alpha\beta}(f) = \sup_x |x^\alpha D^\beta f(x)| < \infty$ for all α, β ; convergence in $\mathcal{S}$ means convergence in all these seminorms. Its continuous dual $\mathcal{S}'(\mathbb{R}^n)$ is the space of tempered distributions. For $f \in \mathcal{S}$,

$$\hat{f}(\xi) = \mathcal{F} f(\xi) = \int_{\mathbb{R}^n} f(x) e^{-2\pi i x \cdot \xi} dx.$$

Theorem 30.3.2. $\mathcal{F}$ maps $\mathcal{S}$ continuously to $\mathcal{S}$, with $\mathcal{F}(D^\alpha f)(\xi) = (2\pi i \xi)^\alpha \hat{f}(\xi)$ and $\mathcal{F}(x^\alpha f)(\xi) = (-2\pi i)^{-|\alpha|} D^\alpha \hat{f}(\xi)$. Moreover:

1. $\mathcal{F}(e^{-\pi|x|^2}) = e^{-\pi|\xi|^2}$;
2. (Inversion) $f(x) = \int \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$, so $\mathcal{F}$ is a bijection of $\mathcal{S}$ with inverse $\mathcal{F}^{-1}g(x) = \hat{g}(-x)$;
3. (Plancherel) $\int f \bar{g} = \int \hat{f} \overline{\hat{g}}$; in particular $\|\hat{f}\|_2 = \|f\|_2$, and $\mathcal{F}$ extends to a unitary operator on $L^2(\mathbb{R}^n)$.

Proof. Differentiation under the integral sign and integration by parts give the two formulas, and they imply that $\mathcal{F} f$ is smooth and rapidly decreasing, with continuity of $\mathcal{F}$ in the seminorms.

(1) For $n = 1$ let $g(\xi) = \int e^{-\pi x^2} e^{-2\pi i x \xi} dx$. Differentiating under the integral and integrating by parts, $g'(\xi) = -2\pi \xi g(\xi)$, so $g(\xi) = g(0) e^{-\pi \xi^2}$, and $g(0) = 1$ by Example 26.8.3. The general case follows by Fubini's theorem, since $e^{-\pi|x|^2} = \prod_j e^{-\pi x_j^2}$.

(2) For $f, g \in \mathcal{S}$, Fubini's theorem gives the multiplication formula $\int \hat{f} g = \int f \hat{g}$. Apply it with $g(\xi) = e^{2\pi i x \cdot \xi} e^{-\pi \varepsilon^2 |\xi|^2}$, whose Fourier transform is $\varepsilon^{-n} e^{-\pi|x-y|^2/\varepsilon^2}$ by (1) and the scaling and translation rules. The right-hand side is a convolution of f with an approximate identity, which converges to $f(x)$ as $\varepsilon \rightarrow 0$ by uniform continuity, and the left-hand side converges to $\int \hat{f}(\xi) e^{2\pi i x \cdot \xi} d\xi$ by dominated convergence.

(3) Let $h = \bar{g}$; then $\hat{h} = \bar{g}$ by (2) and $\overline{\mathcal{F}u} = \mathcal{F}^{-1}u$. The multiplication formula with f and h gives $\int \hat{f}\bar{\hat{g}} = \int f\bar{g}$. The space $\mathcal{S}$ is dense in L^2 : it contains C_c^∞ , and every continuous function with compact support is a uniform limit of its convolutions with a smooth approximate identity (Exercise 26.10.4), which lie in C_c^∞ , while such functions are dense in L^2 by Proposition 26.5.4. Hence $\mathcal{F}$ extends uniquely to an isometry of L^2 , which is surjective because its image contains the dense subspace $\mathcal{S}$. $\square$

Definition 30.3.3. For $T \in \mathcal{S}'$, $\hat{T}(\varphi) = T(\hat{\varphi})$. By Theorem 30.3.2 this defines a linear bijection of $\mathcal{S}'$ extending the transform on $\mathcal{S}$ and on L^2 , with $\mathcal{F}(D^\alpha T) = (2\pi i\xi)^\alpha \hat{T}$.

Example 30.3.4. $\hat{\delta}_0 = 1$ and $\hat{1} = \delta_0$; the Fourier transform of a polynomial is a combination of derivatives of δ_0 . Compactly supported distributions are tempered, and so are the functions in $L^p(\mathbb{R}^n)$ for $1 \leq p \leq \infty$.

30.4 Sobolev spaces

Definition 30.4.1. For $s \in \mathbb{R}$, $H^s(\mathbb{R}^n) = \{u \in \mathcal{S}' : \hat{u} \in L^2_{\text{loc}} \text{ and } \|u\|_s^2 = \int \langle \xi \rangle^{2s} |\hat{u}(\xi)|^2 d\xi < \infty\}$, a Hilbert space with the evident inner product. On the torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, a distribution u has Fourier coefficients $\hat{u}(k) = u(e^{-2\pi i k \cdot x})$, $k \in \mathbb{Z}^n$, and $H^s(T^n) = \{u : \|u\|_s^2 = \sum_{k \in \mathbb{Z}^n} \langle k \rangle^{2s} |\hat{u}(k)|^2 < \infty\}$.

Proposition 30.4.2. $H^0 = L^2$, and for a non-negative integer m , H^m is the space of $u \in L^2$ whose distributional derivatives of order $\leq m$ lie in L^2 , with an equivalent norm $(\sum_{|\alpha| \leq m} \|D^\alpha u\|_2^2)^{1/2}$. The spaces H^s decrease in s , C^∞ functions (with compact support, respectively on the torus) are dense in each, and the pairing $(u, v) \mapsto \int u \bar{v}$ identifies H^{-s} with the dual of H^s .

Proof. $H^0 = L^2$ is Plancherel's theorem (Theorem 30.3.2), or Parseval's identity on the torus (Example 28.3.6). For integer m , $\mathcal{F}(D^\alpha u) = (2\pi i\xi)^\alpha \hat{u}$, and $\sum_{|\alpha| \leq m} |(2\pi i\xi)^\alpha|^2$ is bounded above and below by multiples of $\langle \xi \rangle^{2m}$. Density follows by truncating and smoothing the Fourier transform, and duality from the Riesz representation theorem applied to the weighted L^2 spaces. $\square$

Theorem 30.4.3 (Sobolev embedding). *If $s > k + n/2$, then every $u \in H^s$ (on $\mathbb{R}^n$ or on T^n) has a representative of class C^k , and $\|u\|_{C^k} \leq C\|u\|_s$. Hence $\bigcap_s H^s$ consists of the smooth functions.*

Proof. On the torus, for $|\alpha| \leq k$, $\sum_k |(2\pi i k)^\alpha \hat{u}(k)| \leq (\sum_k \langle k \rangle^{2s} |\hat{u}(k)|^2)^{1/2} (\sum_k \langle k \rangle^{2(|\alpha|-s)})^{1/2}$ by Cauchy-Schwarz, and the second factor is finite because $2(s - |\alpha|) > n$ (compare with $\int \langle \xi \rangle^{2(|\alpha|-s)} d\xi$, which converges by polar coordinates). So the Fourier series of $D^\alpha u$ converges absolutely and uniformly, and u is C^k with the stated bound. On $\mathbb{R}^n$ the same computation with integrals shows that $\widehat{D^\alpha u} \in L^1$, so $D^\alpha u$ is continuous and bounded by Fourier inversion. $\square$

Theorem 30.4.4 (Rellich). *For $s > t$ the inclusion $H^s(T^n) \rightarrow H^t(T^n)$ is a compact operator.*

Proof. Let P_N be the truncation $\widehat{P_N u}(k) = \hat{u}(k)$ for $|k| \leq N$ and 0 otherwise, a finite rank operator. For $u \in H^s$,

$$\|u - P_N u\|_t^2 = \sum_{|k| > N} \langle k \rangle^{2t} |\hat{u}(k)|^2 \leq \langle N \rangle^{2(t-s)} \sum_{|k| > N} \langle k \rangle^{2s} |\hat{u}(k)|^2 \leq \langle N \rangle^{2(t-s)} \|u\|_s^2,$$

so the inclusion is the norm limit of finite rank operators, hence compact (Proposition 28.5.2). $\square$

Counterexample 30.4.5. Rellich's theorem fails on $\mathbb{R}^n$: translates $u_j(x) = u(x - j)$ of a fixed nonzero $u \in C_c^\infty$ have constant H^s norms and no subsequence converging in H^t , since they are eventually supported in disjoint sets. Compactness holds for the inclusions $H_K^s \rightarrow H^t$ of the subspaces of distributions supported in a fixed compact K , which is the form used on manifolds.

30.5 Elliptic estimates

Definition 30.5.1. Let $P(D) = \sum_{|\alpha| \leq m} a_\alpha D^\alpha$ with constant coefficients $a_\alpha \in \mathbb{C}$. Its *principal symbol* is $\sigma(\xi) = \sum_{|\alpha|=m} a_\alpha (2\pi i \xi)^\alpha$, and P is *elliptic* if $\sigma(\xi) \neq 0$ for $\xi \neq 0$.

Example 30.5.2. The Laplacian $\Delta = -\sum_j \partial_j^2$ has symbol $4\pi^2|\xi|^2$ and is elliptic; the heat operator $\partial_t - \Delta$ and the wave operator are not.

Theorem 30.5.3 (Elliptic estimate and regularity). *Let $P(D)$ be elliptic of order m with constant coefficients, on the torus T^n . For every s there is C with*

$$\|u\|_{s+m} \leq C (\|P(D)u\|_s + \|u\|_s) \quad \text{for all } u \in H^{s+m}(T^n).$$

Moreover, if $u \in H^t(T^n)$ for some t and $P(D)u \in H^s(T^n)$, then $u \in H^{s+m}(T^n)$; in particular $P(D)u \in C^\infty$ implies $u \in C^\infty$.

Proof. On the Fourier side $P(D)$ is multiplication by $p(k) = \sum_{|\alpha| \leq m} a_\alpha (2\pi i k)^\alpha$. By homogeneity and compactness of the unit sphere, $|\sigma(\xi)| \geq c|\xi|^m$ for some $c > 0$, and the lower-order terms are $O(|\xi|^{m-1})$, so there are $c' > 0$ and R with $|p(k)| \geq c'\langle k \rangle^m$ for $|k| \geq R$. Hence

$$\sum_{|k| \geq R} \langle k \rangle^{2(s+m)} |\hat{u}(k)|^2 \leq c'^{-2} \sum_{|k| \geq R} \langle k \rangle^{2s} |p(k)\hat{u}(k)|^2 \leq c'^{-2} \|P(D)u\|_s^2,$$

while the finitely many terms with $|k| < R$ are bounded by $\langle R \rangle^{2m} \|u\|_s^2$. This gives the estimate. For regularity, the same computation shows that $\sum_k \langle k \rangle^{2(s+m)} |\hat{u}(k)|^2$ is finite whenever $\sum_k \langle k \rangle^{2s} |p(k)\hat{u}(k)|^2$ is; smoothness follows from Theorem 30.4.3, since then $u \in H^s$ for all s . $\square$

Remark 30.5.4. For operators with variable coefficients the estimate is proved by freezing the coefficients at a point, using the constant-coefficient case and absorbing the error terms; on a compact manifold one patches the local estimates with a partition of unity. The result is that an elliptic operator P of order m between sections of vector bundles over a compact manifold satisfies $\|u\|_{s+m} \leq C(\|Pu\|_s + \|u\|_s)$, is Fredholm as an operator $H^{s+m} \rightarrow H^s$ by Rellich's theorem, and has smooth kernel; this is proved in Chapter 39 and applied to the Laplacian in Chapter 40. The general theory is in [Hör90a, Chapters 7–10] and [War83, Chapter 6].

30.6 Exercises

Exercise 30.6.1. Show that $\varepsilon^{-n} \rho(x/\varepsilon) \rightarrow \delta_0$ in $\mathcal{D}'(\mathbb{R}^n)$ as $\varepsilon \rightarrow 0$, for any $\rho \in C_c^\infty$ with $\int \rho = 1$.

Solution. $\int \varepsilon^{-n} \rho(x/\varepsilon) \varphi(x) dx = \int \rho(y) \varphi(\varepsilon y) dy \rightarrow \varphi(0) \int \rho$ by dominated convergence. $\square$

Exercise 30.6.2. Show that $E(x) = \frac{1}{2}|x|$ satisfies $-E'' = -\delta_0$ on $\mathbb{R}$, and that on $\mathbb{R}^n$ with $n \geq 3$ the function $c_n|x|^{2-n}$ is a fundamental solution of the Laplacian for a suitable constant c_n .

Exercise 30.6.3. Show that $H^{n/2}(T^n)$ is not contained in $C^0(T^n)$, so the condition $s > n/2$ in Theorem 30.4.3 is sharp.

Exercise 30.6.4. Using the Fourier transform, solve the heat equation $\partial_t u = -\Delta u$ on T^n with initial datum $u_0 \in H^s$, and show that $u(t) \in C^\infty$ for every $t > 0$.

Solution. $\hat{u}(k, t) = e^{-4\pi^2|k|^2 t} \hat{u}_0(k)$ solves the equation coefficientwise, and for $t > 0$ the factors decay faster than any power of $\langle k \rangle$, so $u(t) \in H^r$ for all r and Theorem 30.4.3 applies. $\square$

Part VI

Differential Geometry

Chapter 31

Smooth Manifolds

31.1 Introduction

A smooth manifold is a space that looks locally like $\mathbb{R}^n$ in a way compatible with calculus. This chapter sets up manifolds, tangent spaces, submanifolds and vector fields, and proves the two facts that make global constructions possible: the regular value theorem (Theorem 31.4.3) and quotients by group actions (Theorem 31.5.1), which produce manifolds, with counterexamples marking the limits of both, and the existence of smooth partitions of unity (Theorem 31.6.1), which glues local constructions into global ones. Complex manifolds are defined in the same way with holomorphic transition maps (Chapter 41). References: [Lee13], [War83], [Hir76]. **Prerequisites.** Chapter 25 (Real Analysis), Chapter 16 (General Topology), Chapter 18 (Topological Manifolds and Group Actions) and Chapter 26 (Measure and Integration).

“Smooth” means C^∞ ; everything below works for C^k with the obvious changes.

31.2 Manifolds and smooth maps

Definition 31.2.1. Let M be a topological n -manifold (Definition 18.5.1). A *chart* is a homeomorphism $\varphi: U \rightarrow \varphi(U) \subseteq \mathbb{R}^n$ from an open $U \subseteq M$ onto an open subset. Two charts are *compatible* if the *transition map* $\psi \circ \varphi^{-1}: \varphi(U \cap V) \rightarrow \psi(U \cap V)$ is a diffeomorphism. A *smooth atlas* is a set of pairwise compatible charts covering M , and a *smooth structure* is a maximal smooth atlas; a *smooth manifold* is a topological manifold with a smooth structure. A map $f: M \rightarrow N$ of smooth manifolds is *smooth* if $\psi \circ f \circ \varphi^{-1}$ is smooth for all charts φ of M and ψ of N where the composite is defined; f is a *diffeomorphism* if it is bijective with smooth inverse. We write $C^\infty(M)$ for the smooth functions $M \rightarrow \mathbb{R}$.

Lemma 31.2.2. Every smooth atlas $\mathcal{A}$ on a topological manifold is contained in a unique maximal smooth atlas, namely the set $\mathcal{A}^{\max}$ of all charts compatible with every chart of $\mathcal{A}$.

Proof. Let $\varphi: U \rightarrow \mathbb{R}^n$ and $\psi: V \rightarrow \mathbb{R}^n$ be in $\mathcal{A}^{\max}$ and $p \in U \cap V$. Choose $\chi \in \mathcal{A}$ defined at p . Near $\varphi(p)$, $\psi \circ \varphi^{-1} = (\psi \circ \chi^{-1}) \circ (\chi \circ \varphi^{-1})$ is a composite of smooth maps, and so is its inverse; as p was arbitrary, φ and ψ are compatible. So $\mathcal{A}^{\max}$ is a smooth atlas containing $\mathcal{A}$. A chart compatible with all of $\mathcal{A}^{\max}$ is compatible with $\mathcal{A}$, hence lies in $\mathcal{A}^{\max}$, so $\mathcal{A}^{\max}$ is maximal; and a maximal atlas containing $\mathcal{A}$ consists of charts compatible with $\mathcal{A}$, so it is contained in, hence equal to, $\mathcal{A}^{\max}$. $\square$

Lemma 31.2.3 (Smooth manifold chart lemma). Let M be a set and $(U_\alpha, \varphi_\alpha)_{\alpha \in A}$ a family of subsets $U_\alpha \subseteq M$ with injections $\varphi_\alpha: U_\alpha \rightarrow \mathbb{R}^n$ such that

1. each $\varphi_\alpha(U_\alpha)$ and each $\varphi_\alpha(U_\alpha \cap U_\beta)$ is open in $\mathbb{R}^n$;

2. each $\varphi_\beta \circ \varphi_\alpha^{-1}: \varphi_\alpha(U_\alpha \cap U_\beta) \rightarrow \varphi_\beta(U_\alpha \cap U_\beta)$ is smooth;
3. countably many of the U_α cover M ;
4. for $p \neq q$ in M , either some U_α contains both, or there are disjoint $U_\alpha \ni p$ and $U_\beta \ni q$.

Then M has a unique topology and smooth structure for which the $(U_\alpha, \varphi_\alpha)$ are smooth charts.

Proof. Declare $W \subseteq M$ open if $\varphi_\alpha(W \cap U_\alpha)$ is open in $\mathbb{R}^n$ for every α . This is a topology. Each U_β is open, since $\varphi_\alpha(U_\beta \cap U_\alpha)$ is open by (1), and $\varphi_\beta: U_\beta \rightarrow \varphi_\beta(U_\beta)$ is a homeomorphism: a subset $V \subseteq U_\beta$ is open if and only if every $\varphi_\alpha(V \cap U_\alpha) = (\varphi_\alpha \circ \varphi_\beta^{-1})(\varphi_\beta(V) \cap \varphi_\beta(U_\alpha \cap U_\beta))$ is open, and by (2), with the transition maps being homeomorphisms between open sets (their inverses are also transition maps), this holds exactly when $\varphi_\beta(V)$ is open. So M is locally Euclidean. Condition (4) gives Hausdorffness, since points in a common U_α are separated inside the Hausdorff space U_α . Each U_α is second countable, so (3) gives second countability. The charts are pairwise compatible by (2), so they form a smooth atlas, which extends uniquely to a smooth structure (Lemma 31.2.2). The topology is forced, as the φ_α must be homeomorphisms from open sets. $\square$

Counterexample 31.2.4. Condition (4) is needed: the two charts of the line with two origins (Counterexample 18.5.2), each the identity on a copy of $\mathbb{R}$, satisfy (1)–(3), with transition map the identity of $\mathbb{R} \setminus \{0\}$, but the two origins lie in no common chart and in no disjoint charts, and the resulting space is not Hausdorff.

- Example 31.2.5.**
1. $\mathbb{R}^n$ and its open subsets; $GL_n(\mathbb{R})$, an open subset of $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$.
 2. S^n , with the two charts given by stereographic projection (Example 16.9.4); the transition map is $y \mapsto y/\|y\|^2$ on $\mathbb{R}^n \setminus \{0\}$, which is smooth.
 3. $\mathbb{RP}^n = (\mathbb{R}^{n+1} \setminus \{0\})/\mathbb{R}^\times$ with the charts $U_i = \{x_i \neq 0\}$, $\varphi_i([x]) = (x_0/x_i, \dots, \widehat{x_i/x_i}, \dots, x_n/x_i)$; the transition maps are rational with nonvanishing denominators. Likewise $\mathbb{CP}^n$ is a smooth $2n$ -manifold, and its transition maps are holomorphic (Chapter 41).
 4. Products $M \times N$, with products of charts. The torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$, with charts induced by the covering map.
 5. Every smooth manifold is a locally ringed space (M, C_M^∞) (Example 23.6.2), and a map is smooth if and only if it pulls back smooth functions to smooth functions.

Counterexample 31.2.6. The same topological manifold can carry non-diffeomorphic smooth structures: $\mathbb{R}^4$ has uncountably many, and S^7 has 28 up to orientation-preserving diffeomorphism (Milnor); in dimensions ≤ 3 the smooth structure is unique. The map $t \mapsto t^3$ is a homeomorphism $\mathbb{R} \rightarrow \mathbb{R}$ that is not a diffeomorphism, but the two smooth structures it defines on $\mathbb{R}$ are diffeomorphic (by $t \mapsto t^{1/3}$). See [Hir76] and [Mil97].

31.3 Tangent vectors

Definition 31.3.1. Let $p \in M$. A *derivation at p* is a linear map $v: C^\infty(M) \rightarrow \mathbb{R}$ with $v(fg) = v(f)g(p) + f(p)v(g)$. The *tangent space* $T_p M$ is the vector space of derivations at p . A smooth map $f: M \rightarrow N$ induces the *differential* $d_p f: T_p M \rightarrow T_{f(p)} N$, $(d_p f(v))(g) = v(g \circ f)$.

Proposition 31.3.2. 1. *Derivations are local: if $f = g$ near p then $v(f) = v(g)$; so $T_p M$ may be computed in any chart.*

2. If $\varphi = (x^1, \dots, x^n)$ is a chart at p , the derivations $\partial/\partial x^i|_p: f \mapsto \partial_i(f \circ \varphi^{-1})(\varphi(p))$ form a basis of $T_p M$; in particular $\dim T_p M = n$.
3. $d_p(g \circ f) = d_{f(p)}g \circ d_p f$ and $d_p \text{id} = \text{id}$; in charts, $d_p f$ is the Jacobian matrix. A diffeomorphism has invertible differentials.

Proof. (1) If f vanishes near p , choose a bump function χ (Corollary 25.10.8 transported by a chart) with $\chi(p) = 1$ and support in the set where $f = 0$; then $\chi f = 0$, so $0 = v(\chi f) = v(\chi)f(p) + \chi(p)v(f) = v(f)$. Note also $v(1) = v(1 \cdot 1) = 2v(1)$, so v kills constants.

(2) We may assume $M = \mathbb{R}^n$ (open subsets are handled by (1) and bump functions) and $p = 0$. By Taylor's theorem with integral remainder, $f(x) = f(0) + \sum_i x^i g_i(x)$ with g_i smooth and $g_i(0) = \partial_i f(0)$: indeed $f(x) - f(0) = \int_0^1 \frac{d}{dt} f(tx) dt = \sum_i x^i \int_0^1 \partial_i f(tx) dt$. Applying v and using $v(\text{const}) = 0$ gives $v(f) = \sum_i v(x^i) \partial_i f(0)$, so $v = \sum_i v(x^i) \partial/\partial x^i|_0$. The $\partial/\partial x^i|_0$ are linearly independent because they take the value δ_{ij} on the coordinate functions x^j . (3) is immediate from the definitions and (2). $\square$

Proposition 31.3.3. For a smooth curve $\gamma: (-\varepsilon, \varepsilon) \rightarrow M$, the velocity $\gamma'(0) \in T_{\gamma(0)} M$, $\gamma'(0)f = (f \circ \gamma)'(0)$, is a derivation at $\gamma(0)$. Every $v \in T_p M$ is the velocity of a smooth curve, and for smooth $f: M \rightarrow N$, $d_p f(\gamma'(0)) = (f \circ \gamma)'(0)$.

Proof. The Leibniz rule for $\gamma'(0)$ is the product rule for $(fg) \circ \gamma$. Given $v = \sum_i v^i \partial/\partial x^i|_p$ in a chart φ with $\varphi(p) = 0$, let $\gamma(t) = \varphi^{-1}(tv^1, \dots, tv^n)$ for small t ; then $\gamma'(0)f = \sum_i v^i \partial_i(f \circ \varphi^{-1})(0) = v(f)$ by Proposition 31.3.2(2). Finally $d_p f(\gamma'(0))(g) = \gamma'(0)(g \circ f) = (g \circ f \circ \gamma)'(0) = (f \circ \gamma)'(0)(g)$. $\square$

Definition 31.3.4. The *tangent bundle* is $TM = \bigsqcup_{p \in M} T_p M$ with the projection $\pi(v) = p$ for $v \in T_p M$. A chart $\varphi = (x^i)$ on U gives a bijection $\pi^{-1}(U) \rightarrow \varphi(U) \times \mathbb{R}^n$, $v \mapsto (\varphi(p), (v(x^1), \dots, v(x^n)))$; these define a topology and a smooth structure making TM a smooth $2n$ -manifold and π smooth, since the transition maps $(x, \xi) \mapsto (\tau(x), D\tau(x)\xi)$ are smooth. A *vector field* is a smooth map $X: M \rightarrow TM$ with $\pi \circ X = \text{id}$; the set of vector fields is $\mathfrak{X}(M)$. A vector field acts on functions by $(Xf)(p) = X_p(f)$, and $X \mapsto (f \mapsto Xf)$ identifies $\mathfrak{X}(M)$ with the derivations of the $\mathbb{R}$ -algebra $C^\infty(M)$.

31.4 Submanifolds

Definition 31.4.1. A smooth $f: M \rightarrow N$ is an *immersion* if every $d_p f$ is injective, a *submersion* if every $d_p f$ is surjective, and an *embedding* if it is an immersion that is a homeomorphism onto its image. A subset $S \subseteq M$ is a *submanifold* of dimension k if every $p \in S$ has a chart $\varphi: U \rightarrow \mathbb{R}^n$ of M with $\varphi(U \cap S) = \varphi(U) \cap (\mathbb{R}^k \times \{0\})$.

Proposition 31.4.2. Let $f: M \rightarrow N$ be smooth and $p \in M$.

1. If f is an immersion at p , there are charts around p and $f(p)$ in which f is $x \mapsto (x, 0)$; if f is a submersion at p , there are charts in which f is $(x, y) \mapsto x$.
2. The image of an embedding is a submanifold, and a submanifold is a smooth manifold for which the inclusion is an embedding.
3. An injective immersion of a compact manifold is an embedding.

Proof. (1) is Theorem 25.11.4 read in charts. (2) For an embedding f , the local form in (1) shows that $f(M)$ is locally $\mathbb{R}^k \times \{0\}$ in suitable charts; being a homeomorphism onto its image guarantees that the chart can be shrunk so that it meets $f(M)$ only in that slice. The converse is clear from the definition of submanifold. (3) A continuous injection from a compact space to a Hausdorff space is a homeomorphism onto its image (Proposition 16.6.2). $\square$

Theorem 31.4.3 (Regular value theorem). *Let $f: M^n \rightarrow N^m$ be smooth and $q \in N$ a regular value: $d_p f$ is surjective for every $p \in f^{-1}(q)$. Then $f^{-1}(q)$ is a submanifold of dimension $n - m$, with $T_p(f^{-1}(q)) = \ker d_p f$.*

Proof. By Proposition 31.4.2(1) there are charts in which f is the projection $(x, y) \mapsto x$, in which $f^{-1}(q)$ is $\{0\} \times \mathbb{R}^{n-m}$. Differentiating $f|_{f^{-1}(q)}$, which is constant, gives $T_p(f^{-1}(q)) \subseteq \ker d_p f$, and the dimensions agree. $\square$

Example 31.4.4. $S^n = f^{-1}(1)$ for $f(x) = \|x\|^2$, with $d_x f(v) = 2\langle x, v \rangle \neq 0$ on S^n . The orthogonal group $O(n) = F^{-1}(I)$ for $F(A) = A^T A$, viewed as a map to symmetric matrices: $d_A F(X) = A^T X + X^T A$ is surjective onto symmetric matrices for $A \in O(n)$ (take $X = AS/2$), so $O(n)$ is a submanifold of dimension $n^2 - n(n+1)/2 = n(n-1)/2$. Similarly $\mathrm{SL}_n(\mathbb{R}) = \det^{-1}(1)$ has dimension $n^2 - 1$, since $d_A \det(X) = \det(A) \mathrm{tr}(A^{-1}X)$.

Counterexample 31.4.5. At a critical value the level set can fail to be a submanifold, or not. For $f(x, y) = xy$ on $\mathbb{R}^2$, 0 is a critical value ($df = 0$ at the origin) and $f^{-1}(0)$ is the union of the coordinate axes, which is not even a topological manifold (Counterexample 18.5.5). For $f(x, y) = x^2$, 0 is also a critical value, but $f^{-1}(0)$ is the y -axis, a submanifold. So the regular value theorem gives a sufficient condition only.

Lemma 31.4.6. *Let $S \subseteq \mathbb{R}^2$ be a 1-dimensional submanifold and $p \in S$. Then near p , S is the graph of a smooth function over one of the coordinate axes: there are $i \in \{1, 2\}$, an open interval J and a smooth $g: J \rightarrow \mathbb{R}$ such that near p , $S = \{(s, g(s)) : s \in J\}$ if $i = 1$, or $S = \{(g(s), s) : s \in J\}$ if $i = 2$.*

Proof. The inclusion $\iota: S \rightarrow \mathbb{R}^2$ is an embedding (Proposition 31.4.2(2)), so $d_p \iota(T_p S)$ is a line, and for some coordinate projection pr_i the composite $\mathrm{pr}_i \circ \iota: S \rightarrow \mathbb{R}$ has nonzero differential at p . By the inverse function theorem (Theorem 25.11.2, in charts) it restricts to a diffeomorphism from a neighbourhood of p in S onto an open interval J ; composing its inverse with the other coordinate gives g . $\square$

Counterexample 31.4.7. 1. *An injective immersion that is not an embedding.* Let $\beta(t) = (\sin 2t, \sin t)$ for $t \in (-\pi, \pi)$. Then $\beta'(t) = (2 \cos 2t, \cos t)$ never vanishes, since $\cos t = 0$ forces $\cos 2t = -1$. It is injective: if $\beta(s) = \beta(t)$ with $s \neq t$ in $(-\pi, \pi)$, then $\sin s = \sin t$ gives $s = \pm\pi - t$, so $\sin 2s = -\sin 2t$, and $\sin 2s = \sin 2t$ forces $\sin 2t = 0$, that is $t \in \{0, \pm\pi/2\}$; but then $s = \pm\pi - t$ is either $\pm\pi \notin (-\pi, \pi)$ or equal to t , a contradiction. The image, a figure eight, equals $\beta([-\pi, \pi])$ because $\beta(\pm\pi) = \beta(0)$, so it is compact; since $(-\pi, \pi)$ is not compact, β is not a homeomorphism onto its image.

2. *A dense immersed line.* Let α be irrational and $\beta: \mathbb{R} \rightarrow T^2 = S^1 \times S^1$, $\beta(t) = (e^{2\pi i t}, e^{2\pi i \alpha t})$. It is an immersion, and injective: $\beta(s) = \beta(t)$ forces $s - t \in \mathbb{Z}$ and $\alpha(s - t) \in \mathbb{Z}$, so $s = t$. The closure of $\beta(\mathbb{R})$ contains $\{1\} \times S^1$, since the points $\beta(n) = (1, e^{2\pi i \alpha n})$, $n \in \mathbb{Z}$, are dense there by Lemma 18.3.4; and it is invariant under multiplication by $\beta(t)$, since β is a homomorphism and multiplication is a homeomorphism; so it contains $\{e^{2\pi i t}\} \times S^1$ for all t , and $\beta(\mathbb{R})$ is dense. For $z = e^{2\pi i t_0}$, $\beta(\mathbb{R}) \cap (\{z\} \times S^1) = \beta(t_0 + \mathbb{Z})$ is countable. If $\beta(\mathbb{R})$ were a submanifold, it would be locally closed, being locally a slice in a chart, and a dense locally closed set is open (it is open in its closure). But a nonempty open subset of T^2 meets some circle $\{z\} \times S^1$ in a nonempty open subset of that circle, which is uncountable. So $\beta(\mathbb{R})$ is not a submanifold.

3. *A smooth homeomorphism onto a non-submanifold.* The map $\gamma(t) = (t^2, t^3)$ is smooth and a homeomorphism of $\mathbb{R}$ onto the cusp $C = \{y^2 = x^3\}$, with inverse $(x, y) \mapsto y^{1/3}$, but $\gamma'(0) = 0$. The cusp is not a submanifold near 0: by Lemma 31.4.6 it would be a graph over one axis near 0, but over the x -axis it has two points $(x, \pm x^{3/2})$ for small $x > 0$, and over the y -axis it is $x = y^{2/3}$, which is not differentiable at 0.

Definition 31.4.8. A subset A of a smooth n -manifold has *measure zero* if $\varphi(A \cap U)$ has Lebesgue measure zero for every chart; by the change of variables formula (Theorem 26.8.2) it suffices to check this for the charts of one atlas.

Theorem 31.4.9 (Sard). *Let $f: M \rightarrow N$ be smooth. The set of critical values of f (the images of points where $d_p f$ is not surjective) has measure zero in N ; in particular regular values are dense. If $\dim M < \dim N$, the image of f has measure zero.*

Proof. The last statement follows from the first, since then every point is critical; alternatively, compose with the projection $\mathbb{R}^{\dim N} \rightarrow \mathbb{R}^{\dim M}$ and use that a smooth map is locally Lipschitz, so it maps sets of measure zero to sets of measure zero and $\mathbb{R}^{\dim M} \times \{0\}$ has measure zero. The general case is [Mil97, Chapter 2] or [Lee13, Chapter 6]; the proof is a careful induction on the order of vanishing of the derivatives, using Taylor's theorem and a covering argument in charts, together with the fact that a countable union of sets of measure zero has measure zero (Proposition 26.2.2) and that manifolds are covered by countably many charts. $\square$

31.5 Constructions of smooth manifolds

Open subsets, products and regular level sets of smooth manifolds are smooth manifolds (Example 31.2.5 and Theorem 31.4.3). Two further constructions are quotients by group actions and spaces of linear subspaces.

Theorem 31.5.1. *Let a discrete group G act freely and properly on a smooth manifold M by diffeomorphisms. Then M/G has a unique smooth structure for which $\pi: M \rightarrow M/G$ is a local diffeomorphism. A map $f: M/G \rightarrow N$ is smooth if and only if $f \circ \pi$ is smooth.*

Proof. By Proposition 18.6.1(3), M/G is a topological manifold and π is a covering map. Every $x \in M$ has a neighbourhood U , which we may take inside the domain of a chart φ , with $gU \cap U = \emptyset$ for $g \neq e$ (Theorem 18.4.3); $\pi|_U$ is a homeomorphism onto the open set $\pi(U)$, and we use $\varphi \circ (\pi|_U)^{-1}$ as a chart. For two such charts, from (U, φ) and (V, ψ) , and a point $\pi(u) = \pi(v)$ in $\pi(U) \cap \pi(V)$ with $u \in U, v \in V$, we have $v = gu$ for a unique g , and for u' near u , the unique point of V in the orbit of u' is gu' (as $gu' \in V$ for u' near u). So near $\varphi(u)$ the transition map is $\psi \circ g \circ \varphi^{-1}$, which is smooth, g being a diffeomorphism. These charts form a smooth atlas, and in them π is given by $\varphi \circ (\pi|_U)^{-1} \circ \pi \circ \varphi^{-1} = \text{id}$, so π is a local diffeomorphism. Uniqueness: if π is a local diffeomorphism for some smooth structure, the maps $(\pi|_U)^{-1}$ are smooth, so the charts above belong to that structure. Finally, if $f \circ \pi$ is smooth then $f = (f \circ \pi) \circ (\pi|_U)^{-1}$ is smooth on each $\pi(U)$. $\square$

Example 31.5.2. The torus $T^n = \mathbb{R}^n / \mathbb{Z}^n$, the projective space $\mathbb{RP}^n = S^n / \{\pm 1\}$, the lens spaces $S^3 / (\mathbb{Z}/p)$ and the Klein bottle (Example 18.4.5) are smooth manifolds, with smooth covering maps from $\mathbb{R}^n, S^n, S^3$ and $\mathbb{R}^2$. The smooth structure on $\mathbb{RP}^n$ agrees with the one given by the charts of Example 31.2.5(3): in the chart $x_i \neq 0$, the composite with π is $x \mapsto (x_j/x_i)_{j \neq i}$, which is smooth, with a smooth local inverse. By the same argument, a covering space of a smooth manifold has a unique smooth structure making the covering map a local diffeomorphism.

Lemma 31.5.3. *Two linear subspaces $W, W' \subseteq \mathbb{R}^n$ of the same dimension k have a common complement: a subspace Q with $W \oplus Q = W' \oplus Q = \mathbb{R}^n$.*

Proof. Induction on $n - k$. If $k = n$, take $Q = 0$. Otherwise W and W' are proper subspaces, and $\mathbb{R}^n \neq W \cup W'$: if $w \in W \setminus W'$ and $w' \in W' \setminus W$ (if one contains the other they are equal, and any vector outside W works), then $w + w'$ lies in neither. Choose $v \notin W \cup W'$. By induction $W \oplus \mathbb{R}v$ and $W' \oplus \mathbb{R}v$ have a common complement Q' , and $Q = Q' \oplus \mathbb{R}v$ is a common complement of W and W' , by counting dimensions. $\square$

Proposition 31.5.4. *The set $\text{Gr}_k(\mathbb{R}^n)$ of k -dimensional linear subspaces of $\mathbb{R}^n$ has a natural structure of smooth manifold of dimension $k(n - k)$, with charts defined as follows. For a decomposition $\mathbb{R}^n = P \oplus Q$ with $\dim P = k$, let $U_Q = \{W : W \cap Q = 0\}$; every $W \in U_Q$ is the graph $\{x + Ax : x \in P\}$ of a unique linear map $A : P \rightarrow Q$, and $\varphi_{P,Q}(W) = A \in \text{Hom}(P, Q) \cong \mathbb{R}^{k(n-k)}$. Moreover $\text{Gr}_1(\mathbb{R}^{n+1}) = \mathbb{RP}^n$.*

Proof. If $W \cap Q = 0$, then $W \oplus Q = \mathbb{R}^n$ by dimensions, so the projection $W \rightarrow P$ along Q is an isomorphism and W is the graph of $A = \text{pr}_Q \circ (\text{pr}_P|_W)^{-1}$; conversely every graph lies in U_Q . So $\varphi_{P,Q} : U_Q \rightarrow \text{Hom}(P, Q)$ is a bijection. We verify the conditions of Lemma 31.2.3. For another decomposition $P' \oplus Q'$, the graph W of A lies in $U_{Q'}$ if and only if the linear map $\text{pr}_{P'} \circ (\text{id} + A) : P \rightarrow P'$ is invertible, an open condition on A (nonvanishing of a determinant in bases); and then W is the graph of $A' = \text{pr}_{Q'} \circ (\text{id} + A) \circ (\text{pr}_{P'} \circ (\text{id} + A))^{-1}$, whose entries are rational functions of those of A with nonvanishing denominators, by Cramer's rule. This gives (1) and (2). The $\binom{n}{k}$ charts with P and Q spanned by standard basis vectors cover, since every W has a complement spanned by standard basis vectors (extend a basis of W by standard basis vectors); this gives (3). For (4), two subspaces W, W' have a common complement Q (Lemma 31.5.3), so both lie in U_Q . For $k = 1$, a line is determined by a nonzero vector up to scaling, and the charts $\varphi_{\langle e_i \rangle, Q_i}$ with Q_i spanned by the other basis vectors are the charts of $\mathbb{RP}^n$. $\square$

Remark 31.5.5. $\text{Gr}_k(\mathbb{R}^n)$ is compact: the map from the compact group $O(n)$ sending a matrix to the span of its first k columns is continuous (in the charts above it is given by rational functions) and surjective. It is the orbit space $O(n)/(O(k) \times O(n - k))$ (Proposition 18.3.2(4)). The complex Grassmannian $\text{Gr}_k(\mathbb{C}^n)$ is constructed in the same way, with holomorphic transition maps (Chapter 41).

31.6 Partitions of unity

Theorem 31.6.1. *Let M be a smooth manifold and $(U_i)_{i \in I}$ an open cover. There is a smooth partition of unity subordinate to it: smooth functions $\psi_i : M \rightarrow [0, 1]$ with $\text{supp } \psi_i \subseteq U_i$, locally finite supports, and $\sum_i \psi_i = 1$.*

Proof. M is locally compact, Hausdorff and second countable, hence paracompact with an exhaustion by compact sets (Proposition 16.10.7). Refine the cover by charts: every point lies in a chart contained in some U_i , and inside it we may take a coordinate ball. As in the proof of Proposition 16.10.7 we may choose a locally finite refinement by coordinate balls B_k , with smaller concentric balls B'_k still covering M . By Corollary 25.10.8, transported by the charts and extended by zero, there are smooth $\chi_k : M \rightarrow [0, 1]$ equal to 1 on $\overline{B'_k}$ with support in B_k . The sum $\chi = \sum_k \chi_k$ is smooth (locally finite) and ≥ 1 everywhere, so $g_k = \chi_k / \chi$ are smooth, sum to 1, and have supports in B_k . Choosing $\varphi(k)$ with $B_k \subseteq U_{\varphi(k)}$ and setting $\psi_i = \sum_{\varphi(k)=i} g_k$ gives the partition of unity, exactly as in the proof of Theorem 16.10.6. $\square$

Corollary 31.6.2. 1. *For a closed $A \subseteq M$ and an open $U \supseteq A$ there is a smooth $\chi : M \rightarrow [0, 1]$ equal to 1 on A with support in U .*

2. *Every smooth function on a closed subset (that is, a function that is locally the restriction of a smooth function) extends to a smooth function on M .*

3. *M admits a proper smooth function $M \rightarrow \mathbb{R}$.*

Proof. (1) Take the partition of unity subordinate to $\{U, M \setminus A\}$ and let χ be the function corresponding to U . (2) Patch local extensions with a partition of unity. (3) With a partition

of unity ψ_k subordinate to a locally finite cover by relatively compact sets, $f = \sum_k k\psi_k$ is smooth, and $f \geq m$ outside the union of the first m sets, so f is proper. $\square$

Remark 31.6.3. The same argument produces global objects out of local ones whenever the objects in question can be averaged: Riemannian metrics (Chapter 38), connections (Chapter 36) and Hermitian metrics on bundles all exist on any smooth manifold for this reason, while there is no such argument for complex structures or for nowhere vanishing vector fields, which are subject to topological obstructions.

31.7 Vector fields and flows

Theorem 31.7.1. *Let $X \in \mathfrak{X}(M)$. For every $p \in M$ there is a unique maximal integral curve $\gamma_p: (\alpha_p, \beta_p) \rightarrow M$ with $\gamma_p(0) = p$ and $\gamma'_p(t) = X_{\gamma_p(t)}$. The set $\mathcal{D} = \{(t, p) : \alpha_p < t < \beta_p\}$ is open in $\mathbb{R} \times M$, the flow $\Phi(t, p) = \gamma_p(t)$ is smooth, and $\Phi(t + s, p) = \Phi(t, \Phi(s, p))$ whenever both sides are defined. If M is compact, or X has compact support, then $\mathcal{D} = \mathbb{R} \times M$ and $t \mapsto \Phi(t, -)$ is a homomorphism from $\mathbb{R}$ to the diffeomorphism group.*

Proof. In a chart, $\gamma' = X(\gamma)$ is an ordinary differential equation with smooth right-hand side, so Theorem 25.12.3 gives local existence and uniqueness, and Theorem 25.12.6 gives smoothness of the flow and openness of its domain. Uniqueness lets local solutions be glued into a maximal one, and the group law follows from uniqueness, both sides being integral curves through $\Phi(s, p)$ at time 0. If X has compact support K , the curves through points outside K are constant, and those through points of K cannot leave the compact set K , so they are defined for all time by Theorem 25.12.3(3). $\square$

Definition 31.7.2. For $X, Y \in \mathfrak{X}(M)$, the *Lie bracket* is the vector field with $[X, Y]f = X(Yf) - Y(Xf)$.

Proposition 31.7.3. *$[X, Y]$ is a vector field; the bracket is bilinear, antisymmetric, satisfies the Jacobi identity and $[fX, gY] = fg[X, Y] + f(Xg)Y - g(Yf)X$, so $\mathfrak{X}(M)$ is a Lie algebra. In coordinates, $[X, Y] = \sum_{i,j} (X^j \partial_j Y^i - Y^j \partial_j X^i) \partial_i$. Moreover $[X, Y] = \left. \frac{d}{dt} \right|_{t=0} (\Phi_{-t})_* Y$, the Lie derivative $L_X Y$, and $[X, Y] = 0$ if and only if the flows of X and Y commute.*

Proof. The second-order terms cancel in $X(Yf) - Y(Xf)$, so $[X, Y]$ is a derivation of $C^\infty(M)$, hence a vector field; the algebraic identities are direct computations, as is the coordinate formula. For the Lie derivative, write in a chart $Y_{\Phi_t(p)} = D\Phi_t(p)Y_p + O(t^2)$ and expand $\Phi_t = \text{id} + tX + O(t^2)$: the derivative at $t = 0$ of $(D\Phi_{-t})Y_{\Phi_t(p)}$ is $DY(p)X(p) - DX(p)Y(p)$, which is the coordinate expression of $[X, Y]$. If the flows commute then $(\Phi_{-t})_* Y = Y$ for all t , so $[X, Y] = 0$; conversely, if $[X, Y] = 0$ then $t \mapsto (\Phi_{-t})_* Y$ has vanishing derivative (by the group law, its derivative at t_0 is $(\Phi_{-t_0})_*[X, Y]$), so $(\Phi_t)_*$ preserves Y and hence its flow, which is the statement that the flows commute. $\square$

Theorem 31.7.4 (Frobenius). *Let $D \subseteq TM$ be a smooth distribution of rank k (a subbundle, Chapter 33) that is involutive: $[X, Y]$ is a section of D whenever X and Y are. Then every point has a chart in which D is spanned by $\partial/\partial x^1, \dots, \partial/\partial x^k$; the manifold is locally a union of k -dimensional integral manifolds.*

Sketch of proof. Choose local sections $X_1, \dots, X_k$ spanning D near p ; after a linear change we may assume $X_i = \partial/\partial x^i + \sum_{j>k} a_i^j \partial/\partial x^j$, and involutivity forces $[X_i, X_j] = 0$ (the bracket lies in D and has no $\partial/\partial x^i$ components for $i \leq k$). Commuting vector fields have commuting flows (Proposition 31.7.3), and $(t_1, \dots, t_k, y) \mapsto \Phi_{t_1}^1 \circ \dots \circ \Phi_{t_k}^k(y)$ is a diffeomorphism onto a

neighbourhood of p taking the coordinate fields to the X_i . The omitted verification is that this map has invertible differential and defines a chart; see [Lee13, Chapter 19] or [War83, Chapter 1]. $\square$

Theorem 31.7.5 (Whitney embedding, compact case). *Every compact smooth n -manifold embeds in $\mathbb{R}^N$ for some N . (In fact $N = 2n + 1$ suffices, and $2n$ for $n > 0$; see [Hir76, Chapter 1].)*

Proof. Cover M by finitely many charts $\varphi_i: U_i \rightarrow \mathbb{R}^n$, $i = 1, \dots, m$, with smaller open sets V_i still covering and bump functions χ_i equal to 1 on $\overline{V_i}$ with support in U_i (Corollary 31.6.2). The map $F = (\chi_1\varphi_1, \dots, \chi_m\varphi_m, \chi_1, \dots, \chi_m): M \rightarrow \mathbb{R}^{mn+m}$ is smooth; it is an immersion, since on V_i the component $\chi_i\varphi_i = \varphi_i$ has injective differential, and it is injective: if $F(p) = F(q)$, choose i with $p \in V_i$; then $\chi_i(q) = \chi_i(p) = 1$, so $q \in U_i$ and $\varphi_i(q) = \varphi_i(p)$, whence $p = q$. By Proposition 31.4.2(3) it is an embedding. $\square$

31.8 Exercises

Exercise 31.8.1. Show that $T_x S^n = x^\perp \subseteq \mathbb{R}^{n+1}$ under the standard identification, and that TS^1 is diffeomorphic to $S^1 \times \mathbb{R}$.

Solution. The first statement is Theorem 31.4.3 applied to $\|x\|^2$. For S^1 , the vector field $X_{(a,b)} = (-b, a)$ is nowhere zero, and $(p, t) \mapsto tX_p$ is a diffeomorphism $S^1 \times \mathbb{R} \rightarrow TS^1$. $\square$

Exercise 31.8.2. Let M be a compact manifold with a nowhere vanishing vector field X . Show that the flow of X is defined for all time and gives a smooth action of $\mathbb{R}$ on M , and that M has Euler characteristic 0 (using the Poincaré–Hopf theorem, Chapter 37).

Exercise 31.8.3. Give an example of an injective immersion $\mathbb{R} \rightarrow \mathbb{R}^2$ that is not an embedding, and of an immersion of S^1 in $\mathbb{R}^2$ that is not injective.

Exercise 31.8.4. Compute $[X, Y]$ for $X = \partial_x$, $Y = x\partial_y$ on $\mathbb{R}^2$, and verify that the corresponding flows do not commute.

Solution. $[X, Y] = \partial_y$. The flow of X translates in x and that of Y shears; applying them in the two orders to the origin gives points differing by a translation in y . $\square$

Exercise 31.8.5. Show that $W \mapsto W^\perp$ is a diffeomorphism $\text{Gr}_k(\mathbb{R}^n) \rightarrow \text{Gr}_{n-k}(\mathbb{R}^n)$.

Solution. It is a bijection, its own inverse up to exchanging k and $n - k$. Use charts with P and Q spanned by complementary sets of standard basis vectors, so $Q = P^\perp$. If W is the graph of $A: P \rightarrow Q$, then $y + By$ with $y \in Q$ and $B: Q \rightarrow P$ is orthogonal to all $x + Ax$ if and only if $\langle x, By \rangle + \langle Ax, y \rangle = 0$ for all x , that is, $B = -A^T$. So $W^\perp$ is the graph of $-A^T$, and in the charts $\varphi_{P,Q}$ and $\varphi_{Q,P}$ the map is the linear map $A \mapsto -A^T$. $\square$

Exercise 31.8.6. Show that the smooth structure on $\mathbb{R}/\mathbb{Z}$ given by Theorem 31.5.1 makes $t \mapsto e^{2\pi it}$ a diffeomorphism $\mathbb{R}/\mathbb{Z} \rightarrow S^1$, where $S^1 \subseteq \mathbb{R}^2$ has the structure of a regular level set.

Solution. The induced map $\mathbb{R}/\mathbb{Z} \rightarrow S^1$ is a bijection, and smooth because its composite with π is $t \mapsto (\cos 2\pi t, \sin 2\pi t)$. Its differential is nonzero, since the velocity $2\pi(-\sin 2\pi t, \cos 2\pi t) \neq 0$ and π is a local diffeomorphism; so it is a bijective local diffeomorphism, hence a diffeomorphism. $\square$

Chapter 32

Lie Groups

32.1 Introduction

A Lie group is a group that is also a smooth manifold, with smooth group operations. Its local structure is captured by a finite-dimensional Lie algebra, and many of the spaces of Hodge theory are homogeneous spaces of Lie groups: projective spaces, Grassmannians, flag manifolds, and the period domains that classify Hodge structures (Chapter 66). This chapter proves the correspondence between one-parameter subgroups and the Lie algebra, Cartan's closed subgroup theorem (Theorem 32.4.1) and the construction of homogeneous spaces (Theorem 32.4.2). References: [Kna02], [BtD85], [War83].

Prerequisites. Chapter 31 (Smooth Manifolds) and Chapter 11 (Representations of Groups and Lie Algebras).

32.2 Lie groups and their Lie algebras

Definition 32.2.1. A *Lie group* is a group G with a smooth structure for which multiplication $G \times G \rightarrow G$ and inversion $G \rightarrow G$ are smooth. A *homomorphism of Lie groups* is a smooth group homomorphism. For $g \in G$, left and right translation are the diffeomorphisms $L_g(x) = gx$ and $R_g(x) = xg$.

Example 32.2.2. $(\mathbb{R}^n, +)$; the torus $\mathbb{R}^n/\mathbb{Z}^n$; $GL_n(\mathbb{R})$ and $GL_n(\mathbb{C})$, open subsets of matrix spaces, with polynomial multiplication and rational inversion; the *classical groups* SL_n , $O(n) = \{A : A^T A = I\}$, $SO(n)$, $U(n) = \{A : A^* A = I\}$ and $SU(n)$, which are submanifolds by Example 31.4.4 and subgroups, hence Lie groups. Products of Lie groups are Lie groups, and the identity component G^0 is an open normal subgroup.

Definition 32.2.3. A vector field X on G is *left-invariant* if $d(L_g)X = X \circ L_g$ for all g . The *Lie algebra* of G is $\mathfrak{g} = T_e G$, with the bracket transported from the left-invariant vector fields.

Proposition 32.2.4. The map $X \mapsto X_e$ is a bijection from left-invariant vector fields to $T_e G$, its inverse sending v to the field $g \mapsto d(L_g)_e v$; left-invariant fields are smooth and closed under the Lie bracket. Hence $\mathfrak{g}$ is a Lie algebra of dimension $\dim G$ (Definition 11.4.1), and a homomorphism $\varphi : G \rightarrow H$ induces a homomorphism of Lie algebras $d_e \varphi : \mathfrak{g} \rightarrow \mathfrak{h}$.

Proof. Left-invariance determines the field from its value at e , and $g \mapsto d(L_g)_e v$ is smooth because it is the image of (g, v) under the smooth map $G \times \mathfrak{g} \rightarrow TG$ induced by multiplication. If X, Y are left-invariant, so is $[X, Y]$, since $d(L_g)$ preserves brackets: for a diffeomorphism F , $[F_* X, F_* Y] = F_* [X, Y]$, as both sides act on functions by $f \mapsto (X(Y(f \circ F)) - Y(X(f \circ F))) \circ F^{-1}$. For the last statement, $\varphi \circ L_g = L_{\varphi(g)} \circ \varphi$ shows that $d_e \varphi$ intertwines the corresponding

left-invariant fields, which are therefore φ -related; and brackets of φ -related fields are φ -related. $\square$

Example 32.2.5. For $G \subseteq \mathrm{GL}_n$ a matrix group, $\mathfrak{g} \subseteq M_n$ and the bracket is the commutator $XY - YX$: indeed for $G = \mathrm{GL}_n$ the left-invariant field with value X at I is $A \mapsto AX$, and a computation in the coordinates of M_n gives $[X, Y] = XY - YX$. Thus $\mathfrak{gl}_n = M_n$, $\mathfrak{sl}_n = \{X : \mathrm{tr} X = 0\}$, $\mathfrak{o}(n) = \{X : X^T = -X\}$, $\mathfrak{u}(n) = \{X : X^* = -X\}$ and $\mathfrak{su}(n) = \mathfrak{u}(n) \cap \mathfrak{sl}_n(\mathbb{C})$, by differentiating the defining equations at I .

32.3 The exponential map

Theorem 32.3.1. For $X \in \mathfrak{g}$ the left-invariant field X is complete, and its flow through e is a homomorphism $\gamma_X : \mathbb{R} \rightarrow G$; conversely every homomorphism $\mathbb{R} \rightarrow G$ arises this way. Setting $\exp(X) = \gamma_X(1)$ defines a smooth map $\exp : \mathfrak{g} \rightarrow G$ with $d_0 \exp = \mathrm{id}$, so $\exp$ is a diffeomorphism from a neighbourhood of 0 onto a neighbourhood of e . It satisfies $\exp((s+t)X) = \exp(sX)\exp(tX)$, $\exp(-X) = \exp(X)^{-1}$, and $\varphi(\exp X) = \exp(d_e \varphi(X))$ for every homomorphism φ . For matrix groups, $\exp(X) = \sum_k X^k/k!$.

Proof. Let γ be the integral curve of X with $\gamma(0) = e$, defined on $(-\varepsilon, \varepsilon)$. For fixed s , both $t \mapsto \gamma(s)\gamma(t)$ and $t \mapsto \gamma(s+t)$ are integral curves through $\gamma(s)$ (left-invariance), so they agree where defined; this extends γ to all of $\mathbb{R}$ and gives the homomorphism property. Conversely, a homomorphism $\gamma : \mathbb{R} \rightarrow G$ is the integral curve of the left-invariant field with value $\gamma'(0)$, by the same computation. Smoothness of $(t, X) \mapsto \gamma_X(t)$ follows from smooth dependence of solutions of differential equations on parameters (Theorem 25.12.6), applied to the field on $G \times \mathfrak{g}$ whose value at (g, X) is $(d(L_g)X, 0)$. Since $\gamma_{sX}(t) = \gamma_X(st)$ we get $\exp(sX) = \gamma_X(s)$, hence $d_0 \exp(X) = \frac{d}{dt}\big|_0 \exp(tX) = X$ and the group laws; the inverse function theorem gives the local diffeomorphism. Naturality holds because $t \mapsto \varphi(\exp tX)$ is a homomorphism $\mathbb{R} \rightarrow H$ with derivative $d_e \varphi(X)$ at 0. For matrix groups, $t \mapsto \sum_k t^k X^k/k!$ converges (Proposition 25.5.4), is a homomorphism by the Cauchy product, and has derivative X at $t = 0$. $\square$

Definition 32.3.2. The adjoint representation is $\mathrm{Ad} : G \rightarrow \mathrm{GL}(\mathfrak{g})$, $\mathrm{Ad}(g) = d_e(c_g)$ where $c_g(x) = gxg^{-1}$; its differential is $\mathrm{ad} = d_e \mathrm{Ad} : \mathfrak{g} \rightarrow \mathfrak{gl}(\mathfrak{g})$.

Proposition 32.3.3. Ad is a homomorphism of Lie groups, $\mathrm{ad}(X)(Y) = [X, Y]$, and $\mathrm{Ad}(\exp X) = e^{\mathrm{ad} X}$. Moreover $\exp(X)\exp(Y) = \exp(X + Y + \frac{1}{2}[X, Y] + \cdots)$ for X, Y small, where the omitted terms are iterated brackets (the Baker–Campbell–Hausdorff formula [Kna02, Chapter 1]).

Proof. Ad is smooth because $(g, x) \mapsto gxg^{-1}$ is, and it is a homomorphism since $c_{gh} = c_g \circ c_h$. For the bracket, note that c_g is a Lie group automorphism, so $\mathrm{Ad}(g)$ preserves brackets; and $\frac{d}{dt}\big|_0 \mathrm{Ad}(\exp tX)Y$ is by definition the Lie derivative of Y along X , which is $[X, Y]$ (Proposition 31.7.3), since the flow of the left-invariant field X is right translation by $\exp(tX)$ and $\mathrm{Ad}(\exp tX)$ is the induced map on left-invariant fields. Finally $t \mapsto \mathrm{Ad}(\exp tX)$ is a homomorphism $\mathbb{R} \rightarrow \mathrm{GL}(\mathfrak{g})$ with derivative $\mathrm{ad} X$, hence equals $e^{t \mathrm{ad} X}$ by Theorem 32.3.1. $\square$

Proposition 32.3.4. Let G be a connected Lie group.

1. G is generated by $\exp(\mathfrak{g})$, indeed by $\exp(U)$ for any neighbourhood U of 0.
2. A homomorphism of Lie groups $\varphi : G \rightarrow H$ is determined by $d_e \varphi$.
3. If G is abelian, then $\exp : \mathfrak{g} \rightarrow G$ is a surjective homomorphism of groups, and $\mathfrak{g}$ is abelian.

Proof. (1) By Theorem 32.3.1, $\exp(U)$ contains a neighbourhood V of e , and we may take $V = V^{-1}$. By Proposition 18.2.3(3), V generates G . (2) $\varphi(\exp X) = \exp(d_e \varphi(X))$, so φ is determined on a generating set. (3) If G is abelian, conjugation is trivial, so $\text{Ad} = 1$ and $\text{ad} = 0$, and $\mathfrak{g}$ is abelian by Proposition 32.3.3. For commuting X, Y , the curve $t \mapsto \exp(tX) \exp(tY)$ is a homomorphism $\mathbb{R} \rightarrow G$ (as G is abelian) with derivative $X + Y$ at 0, so it equals $\exp(t(X + Y))$ by Theorem 32.3.1; hence $\exp$ is a homomorphism, whose image is a subgroup containing a neighbourhood of e , hence all of G by (1). $\square$

Lemma 32.3.5. *A discrete subgroup Γ of $\mathbb{R}^n$ is of the form $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_b$ for linearly independent vectors $v_1, \dots, v_b$.*

Proof. Induction on the dimension of the span V of Γ ; the case $V = 0$ is trivial. As Γ is discrete, it meets every bounded set in a finite set (a bounded infinite set would have an accumulation point, and differences of nearby points of Γ are small nonzero elements of Γ , while discreteness gives a ball around 0 containing no other element of Γ). Choose a basis $w_1, \dots, w_b$ of V contained in Γ , let W be the span of $w_1, \dots, w_{b-1}$, and $\Gamma' = \Gamma \cap W$, which by induction is $\mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_{b-1}$ with v_i spanning W . The elements of Γ of the form $\sum_{i < b} s_i v_i + s w_b$ with $s_i \in [0, 1]$ and $s \in (0, 1]$ form a finite nonempty set (it contains w_b); choose one, v_b , with the smallest s , say s_0 . For $\gamma \in \Gamma$, write $\gamma = \sum_{i < b} c_i v_i + c w_b$ and let $k = \lfloor c/s_0 \rfloor$. Then $\gamma - k v_b$ has w_b -coefficient in $[0, s_0)$, and subtracting a suitable element of Γ' moves its other coefficients into $[0, 1]$; by minimality of s_0 the w_b -coefficient is 0, so $\gamma - k v_b \in \Gamma'$. Hence $\Gamma = \Gamma' \oplus \mathbb{Z}v_b$. $\square$

Theorem 32.3.6. *Every connected abelian Lie group is isomorphic to $\mathbb{R}^a \times T^b$ for some a, b ; in particular every compact connected abelian Lie group is a torus.*

Proof. By Proposition 32.3.4(3), $\exp: \mathfrak{g} \cong \mathbb{R}^n \rightarrow G$ is a surjective homomorphism, and it is a local diffeomorphism at 0, hence everywhere (translate). Its kernel Γ is a discrete subgroup, since $\exp$ is injective near 0. By Lemma 32.3.5, $\Gamma = \mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_b$; completing to a basis $v_1, \dots, v_n$ of $\mathbb{R}^n$ identifies $\mathbb{R}^n/\Gamma$ with $T^b \times \mathbb{R}^{n-b}$. The induced bijective homomorphism $\mathbb{R}^n/\Gamma \rightarrow G$ is a local diffeomorphism, hence an isomorphism of Lie groups. If G is compact, $n - b = 0$. $\square$

Counterexample 32.3.7. Connectedness is needed: $O(2)$ has the abelian Lie algebra $\mathfrak{o}(2) \cong \mathbb{R}$, but it is not abelian, as a reflection does not commute with a rotation by an angle other than 0 or π .

32.4 Subgroups and homogeneous spaces

Theorem 32.4.1 (Cartan). *A closed subgroup H of a Lie group G is an embedded submanifold, hence a Lie group, with Lie algebra $\mathfrak{h} = \{X \in \mathfrak{g} : \exp(tX) \in H \text{ for all } t \in \mathbb{R}\}$.*

Proof. Let $\mathfrak{h}$ be as stated. It is closed under scalars by definition. It is closed under addition and brackets: by the product formulas $\exp(t(X + Y)) = \lim_k (\exp(tX/k) \exp(tY/k))^k$ and $\exp(t^2[X, Y]) = \lim_k (\exp(tX/k) \exp(tY/k) \exp(-tX/k) \exp(-tY/k))^{k^2}$, which follow from the Baker–Campbell–Hausdorff formula or from a direct Taylor expansion, and H is closed. So $\mathfrak{h}$ is a Lie subalgebra.

Choose a complement $\mathfrak{m}$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$. We claim that $\exp(\mathfrak{m})$ meets H only at e near 0: otherwise there are $Y_k \in \mathfrak{m} \setminus \{0\}$ with $Y_k \rightarrow 0$ and $\exp Y_k \in H$; passing to a subsequence, $Y_k/\|Y_k\| \rightarrow Y \in \mathfrak{m}$ with $\|Y\| = 1$. For $t \in \mathbb{R}$ let $m_k = \lfloor t/\|Y_k\| \rfloor$; then $m_k Y_k \rightarrow tY$ and $\exp(m_k Y_k) = (\exp Y_k)^{m_k} \in H$, so $\exp(tY) \in H$ by closedness, giving $Y \in \mathfrak{h} \cap \mathfrak{m} = 0$, a contradiction.

The map $\mathfrak{h} \times \mathfrak{m} \rightarrow G, (X, Y) \mapsto \exp X \exp Y$, has invertible differential at $(0, 0)$, hence is a diffeomorphism near 0; by the claim, in a small enough neighbourhood it carries $\mathfrak{h} \times \{0\}$ onto H . Translating by elements of H gives charts of G in which H is a slice, so H is an embedded submanifold; the group operations are smooth as restrictions, and $T_e H = \mathfrak{h}$. $\square$

Theorem 32.4.2. *Let $H \subseteq G$ be a closed subgroup. The quotient G/H has a unique smooth structure such that the projection $\pi: G \rightarrow G/H$ is a submersion; it is a manifold of dimension $\dim G - \dim H$ on which G acts smoothly and transitively. If a Lie group G acts smoothly and transitively on a manifold M and H is the stabiliser of $p \in M$, then $G/H \rightarrow M, gH \mapsto gp$, is a G -equivariant diffeomorphism.*

Proof. With $\mathfrak{m}$ as in the previous proof, the map $\mathfrak{m} \rightarrow G/H, Y \mapsto \pi(\exp Y)$, is injective near 0 and its image is open; these maps, translated by G , are the charts. The transition maps are smooth because multiplication in G is, and π becomes the projection $(X, Y) \mapsto Y$ in the corresponding charts, so it is a submersion; uniqueness follows because a submersion determines the smooth structure on its image via local sections. The action $G \times G/H \rightarrow G/H$ is smooth because it is so after composing with the submersion $\text{id} \times \pi$. For the last statement, consider the orbit map $f: G \rightarrow M, g \mapsto gp$, which is smooth and surjective. If df were nowhere surjective, every point of M would be a critical value, and M would have measure zero in itself by Sard's theorem (Theorem 31.4.9), which is false. So df is surjective at some g_0 , hence everywhere, because $f \circ L_g = g \cdot f$ and the action of g is a diffeomorphism of M . Thus f is a submersion, and so is the induced smooth bijection $G/H \rightarrow M$ (as π is a surjective submersion); a bijective submersion is a diffeomorphism, by the inverse function theorem and equality of dimensions. $\square$

- Example 32.4.3.**
1. $S^n = O(n+1)/O(n) = SO(n+1)/SO(n)$ and $S^{2n+1} = U(n+1)/U(n)$.
 2. $\mathbb{RP}^n = O(n+1)/(O(n) \times O(1))$ and $\mathbb{CP}^n = U(n+1)/(U(n) \times U(1))$; more generally the Grassmannian $\text{Gr}(k, n)$ of k -planes in $\mathbb{R}^n$ is $O(n)/(O(k) \times O(n-k))$, a compact manifold of dimension $k(n-k)$, and similarly over $\mathbb{C}$.
 3. Flag manifolds $U(n)/(U(n_1) \times \cdots \times U(n_r))$; these and their open subsets are the period domains parametrising Hodge structures (Chapter 66).
 4. $\text{GL}_n(\mathbb{R})/O(n)$ is the space of positive definite symmetric matrices, and the upper half plane is $\text{SL}_2(\mathbb{R})/\text{SO}(2)$, the period domain for Hodge structures of weight one and rank two (Chapter 62).

Proposition 32.4.4. *Let H be a closed subgroup of a Lie group G . If H and G/H are connected, so is G . Consequently $\text{SO}(n)$, $U(n)$, $\text{SU}(n)$ and $\text{GL}_n(\mathbb{C})$ are connected, while $O(n)$ and $\text{GL}_n(\mathbb{R})$ have exactly two components.*

Proof. If $G = A \sqcup B$ with A, B open, each coset gH , homeomorphic to H , is connected and so lies in A or in B . Hence A and B are unions of cosets, and $\pi(A), \pi(B)$ are disjoint open sets covering G/H (the projection is open, Proposition 18.3.2(1)); as G/H is connected, one of them is empty. For the examples, $\text{SO}(1)$ and $U(1) = S^1$ are connected, and $\text{SO}(n)/\text{SO}(n-1) \cong S^{n-1}$ and $U(n)/U(n-1) \cong S^{2n-1}$ (Example 32.4.3) are connected for $n \geq 2$, so induction applies; similarly for $\text{SU}(n)$, using $\text{SU}(1) = \{1\}$. For $\text{GL}_n(\mathbb{C})$, given A invertible, $t \mapsto \det((1-t)I + tA)$ is a nonzero polynomial with finitely many roots in $\mathbb{C}$, so a path from 0 to 1 in $\mathbb{C}$ avoiding them gives a path in $\text{GL}_n(\mathbb{C})$ from I to A . The sign of the determinant separates $O(n)$ and $\text{GL}_n(\mathbb{R})$ into two open sets, and each is connected: $O(n)^- = r \text{SO}(n)$ for a reflection r , and $\text{GL}_n^+(\mathbb{R})$ retracts onto $\text{SO}(n)$ by the Gram–Schmidt process, which is continuous. $\square$

32.5 Compact Lie groups

Theorem 32.5.1 (Haar measure). *On a compact Lie group G there is a unique regular Borel probability measure μ that is invariant under all left translations; it is also right invariant and invariant under inversion. Consequently $\int_G f(gx) d\mu(x) = \int_G f(x) d\mu(x)$ for all g and all continuous f .*

This is [BtD85, Chapter I]; the measure can be constructed by integrating a left-invariant volume form (Chapter 34), which exists because $\Lambda^{\dim G} \mathfrak{g}^*$ is one-dimensional, and normalising. Uniqueness follows by convolving two invariant measures.

Corollary 32.5.2. *Every finite-dimensional representation of a compact Lie group G on a complex vector space admits a G -invariant Hermitian inner product, and is a direct sum of irreducible subrepresentations.*

Proof. Average any Hermitian inner product: $\langle v, w \rangle_G = \int_G \langle gv, gw \rangle d\mu(g)$, which is Hermitian, positive definite and invariant. The orthogonal complement of a subrepresentation is then a subrepresentation, and induction on the dimension gives the decomposition, exactly as in Maschke's theorem (Theorem 11.2.4). $\square$

Proposition 32.5.3. *If G is compact, $\mathfrak{g}$ has an inner product for which every $\text{Ad}(g)$ is orthogonal and every $\text{ad}(X)$ is skew-symmetric. Consequently the Killing form $B(X, Y) = \text{tr}(\text{ad } X \text{ ad } Y)$ is negative semidefinite.*

Proof. Averaging an inner product over Haar measure (Theorem 32.5.1) as in Corollary 32.5.2 gives an Ad -invariant inner product; differentiating $\langle \text{Ad}(\exp tX)Y, \text{Ad}(\exp tX)Z \rangle = \langle Y, Z \rangle$ at $t = 0$ gives skew-symmetry of $\text{ad } X$ (Proposition 32.3.3). For a skew-symmetric A , $\text{tr}(A^2) = -\text{tr}(A^T A) \leq 0$. $\square$

Remark 32.5.4. Deeper structure theory, which we quote when needed: a connected Lie group has a unique simply connected covering group, and Lie algebra homomorphisms integrate to group homomorphisms from the simply connected group (Lie's second theorem); Lie subalgebras correspond to connected Lie subgroups; every compact connected Lie group is a quotient of a product of a torus and simply connected compact simple groups, which are classified by their Dynkin diagrams; and every connected compact Lie group is the maximal compact subgroup of a unique complex reductive group, its *complexification* (for instance $U(n) \subseteq \text{GL}_n(\mathbb{C})$). See [Kna02] and [BtD85]. The unitary trick, which deduces complete reducibility of representations of complex reductive groups from Corollary 32.5.2, is used in Chapter 63.

32.6 Exercises

Exercise 32.6.1. Show that $\exp: \mathfrak{sl}_2(\mathbb{R}) \rightarrow \text{SL}_2(\mathbb{R})$ is not surjective, although $\exp$ is surjective for compact connected Lie groups and for $\text{GL}_n(\mathbb{C})$.

Solution. An element of the image with distinct real eigenvalues has positive eigenvalues, and $\text{diag}(-2, -1/2) \in \text{SL}_2(\mathbb{R})$ has negative ones; it is not $\exp$ of a real matrix, since the eigenvalues of $\exp X$ are exponentials of those of X and complex eigenvalues of a real matrix come in conjugate pairs. $\square$

Exercise 32.6.2. Show that $\text{SU}(2)$ is diffeomorphic to S^3 , that its Lie algebra is $\mathfrak{su}(2) \cong \mathbb{R}^3$ with the cross product, and that the adjoint representation gives a double covering $\text{SU}(2) \rightarrow \text{SO}(3)$.

Exercise 32.6.3. Show that the image of the homomorphism $\mathbb{R} \rightarrow T^2, t \mapsto (t, \alpha t) \bmod \mathbb{Z}^2$, with α irrational, is a dense subgroup that is not closed and not an embedded submanifold; so “closed” cannot be dropped in Theorem 32.4.1.

Exercise 32.6.4. Verify that $\text{Gr}(k, n) = O(n)/(O(k) \times O(n - k))$ has dimension $k(n - k)$ and is connected, and identify $\text{Gr}(1, n)$ with $\mathbb{RP}^{n-1}$.

Chapter 33

Vector Bundles

33.1 Introduction

A vector bundle is a family of vector spaces parametrised by a manifold, locally but not globally a product. Tangent and cotangent bundles, bundles of differential forms, and the line bundles that produce the divisors and the Chern classes of algebraic geometry are all of this kind. This chapter constructs bundles from transition functions, develops the linear algebra operations on them, and classifies complex line bundles by $H^2(M; \mathbb{Z})$ (Theorem 33.4.2), which is the first Chern class. References: [Lee13], [MS74], [GH78, Chapter 0].

Prerequisites. Chapter 31 (Smooth Manifolds), Chapter 10 (Multilinear Algebra) and Chapter 23 (Sheaves on Topological Spaces).

Let $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ and let M be a smooth manifold.

33.2 Vector bundles and cocycles

Definition 33.2.1. A *vector bundle* of rank r over M is a smooth map $\pi: E \rightarrow M$ together with $\mathbb{K}$ -vector space structures on the fibres $E_p = \pi^{-1}(p)$, such that every point of M has an open neighbourhood U and a diffeomorphism $\Phi: \pi^{-1}(U) \rightarrow U \times \mathbb{K}^r$ (a *local trivialisation*) with $\text{pr}_1 \circ \Phi = \pi$, linear on each fibre. A *morphism* of vector bundles over M is a smooth $f: E \rightarrow F$ with $\pi_F \circ f = \pi_E$, linear on fibres; *line bundle* means rank 1.

Example 33.2.2. The trivial bundle $M \times \mathbb{K}^r$; the tangent bundle TM (Definition 31.3.4) and its dual, the cotangent bundle T^*M ; the Möbius bundle over S^1 , obtained from $[0, 1] \times \mathbb{R}$ by gluing $(0, t)$ to $(1, -t)$; the *tautological* line bundle $\mathcal{O}(-1)$ on $\mathbb{RP}^n$ or $\mathbb{CP}^n$, whose fibre over a line ℓ is ℓ itself; the normal bundle $N_{S/M} = (TM|_S)/TS$ of a submanifold.

Proposition 33.2.3. The tangent bundle of a Lie group G is trivial: $G \times \mathfrak{g} \rightarrow TG, (g, v) \mapsto d(L_g)_e(v)$, is an isomorphism of vector bundles. In particular $S^1, S^3 \cong \text{SU}(2)$ and the tori T^n have trivial tangent bundles.

Proof. The map is smooth, as the differential of multiplication restricted to $G \times T_e G$, it covers the identity of G , and it is a linear isomorphism on each fibre since L_g is a diffeomorphism. A bijective morphism of vector bundles over M is an isomorphism, because in local trivialisations it is given by a smooth family of invertible matrices, whose inverses are smooth by Cramer's rule. For S^3 note that $\text{SU}(2) = \left\{ \begin{pmatrix} a & -\bar{b} \\ b & \bar{a} \end{pmatrix} : |a|^2 + |b|^2 = 1 \right\}$ is diffeomorphic to $S^3 \subseteq \mathbb{C}^2$. $\square$

Counterexample 33.2.4. The Möbius bundle E over S^1 (Example 33.2.2) is not trivial. A nowhere vanishing section would correspond to a continuous $s: [0, 1] \rightarrow \mathbb{R} \setminus \{0\}$ with

$s(1) = -s(0)$, and by the intermediate value theorem s would vanish somewhere; and a real line bundle with a nowhere vanishing section s is trivial, by $(p, t) \mapsto ts(p)$. The tautological line bundle of $\mathbb{RP}^1$ is isomorphic to E : parametrising $\mathbb{RP}^1$ by $\theta \in [0, \pi]$, with θ corresponding to the line through $u(\theta) = (\cos \theta, \sin \theta)$, the section u trivialises the bundle over $[0, \pi]$ and $u(\pi) = -u(0)$, which is the gluing of E . Hence the tautological line bundle of $\mathbb{RP}^n$, which restricts to it on $\mathbb{RP}^1 \subseteq \mathbb{RP}^n$, is nontrivial as well.

Proposition 33.2.5. *Let (U_i) be an open cover of M .*

1. *A vector bundle with trivialisations Φ_i over U_i determines transition functions $g_{ij}: U_i \cap U_j \rightarrow \mathrm{GL}_r(\mathbb{K})$, smooth, with $\Phi_i \Phi_j^{-1}(p, v) = (p, g_{ij}(p)v)$; they satisfy the cocycle condition $g_{ij}g_{jk} = g_{ik}$ on triple intersections, and $g_{ii} = 1$.*
2. *Conversely, every such family (g_{ij}) arises from a vector bundle, unique up to isomorphism, namely $E = (\bigsqcup_i U_i \times \mathbb{K}^r) / \sim$ with $(p, v)_j \sim (p, g_{ij}(p)v)_i$.*
3. *Two cocycles $(g_{ij}), (g'_{ij})$ define isomorphic bundles if and only if there are smooth $h_i: U_i \rightarrow \mathrm{GL}_r(\mathbb{K})$ with $g'_{ij} = h_i g_{ij} h_j^{-1}$. Hence isomorphism classes of rank r bundles correspond to $\check{H}^1(M, \mathrm{GL}_r)$, Čech cohomology with values in the sheaf of smooth $\mathrm{GL}_r(\mathbb{K})$ -valued functions (Remark 24.5.4).*

Proof. (1) $\Phi_i \Phi_j^{-1}$ is a diffeomorphism of $(U_i \cap U_j) \times \mathbb{K}^r$ commuting with the projection and linear on fibres, hence of the stated form with g_{ij} smooth (its entries are obtained by evaluating on constant sections). The cocycle condition is immediate. (2) The relation is an equivalence relation precisely because of the cocycle condition; the quotient is a smooth manifold with charts induced from $U_i \times \mathbb{K}^r$, and the projection and fibrewise linear structures are well defined. (3) An isomorphism $E \rightarrow E'$ is given in the trivialisations by smooth maps h_i with values in GL_r , and compatibility on overlaps is exactly $g'_{ij} = h_i g_{ij} h_j^{-1}$; this is the statement that the two cocycles define the same Čech cohomology class. $\square$

Definition 33.2.6. A section of E over $U \subseteq M$ open is a smooth $s: U \rightarrow E$ with $\pi \circ s = \mathrm{id}$; they form a $C^\infty(U)$ -module $\Gamma(U, E)$, and $U \mapsto \Gamma(U, E)$ is a sheaf $\mathcal{E}$ of C_M^∞ -modules. A *frame* over U is a tuple of sections that is a basis of each fibre.

Proposition 33.2.7. $\mathcal{E}$ is locally free of rank r (Definition 23.6.3): a trivialisation over U is the same as a frame over U , and gives $\mathcal{E}|_U \cong (C_U^\infty)^r$. Conversely every locally free C_M^∞ -module of rank r is the sheaf of sections of a vector bundle, unique up to isomorphism, and this correspondence is an equivalence between vector bundles and locally free sheaves.

Proof. A trivialisation Φ gives the frame $s_i(p) = \Phi^{-1}(p, e_i)$, and conversely a frame defines $\Phi(v) = (p, (v^1, \dots, v^r))$ where $v = \sum v^i s_i(p)$; smoothness in both directions follows because the coefficients are smooth functions (they are the components in a trivialisation). Given a locally free sheaf $\mathcal{F}$, choose isomorphisms $\mathcal{F}|_{U_i} \cong (C_{U_i}^\infty)^r$; on overlaps they differ by automorphisms of $(C^\infty)^r$, that is, by matrices g_{ij} of smooth functions with smooth inverse, which form a cocycle. The bundle of Proposition 33.2.5(2) has $\mathcal{F}$ as its sheaf of sections. Morphisms correspond in the same way, since a morphism of bundles is determined by its action on sections. $\square$

33.3 Operations on bundles

Proposition 33.3.1. Let E, F be vector bundles over M of ranks r, s , with cocycles $(g_{ij}), (h_{ij})$, and let $f: N \rightarrow M$ be smooth.

1. There are vector bundles $E \oplus F$, $E \otimes F$, $E^\vee$, $\text{Hom}(E, F)$, $\Lambda^k E$ and $\text{Sym}^k E$ whose fibres are the corresponding constructions on the fibres, with cocycles $g_{ij} \oplus h_{ij}$, $g_{ij} \otimes h_{ij}$, $(g_{ij}^T)^{-1}$, and so on. Their sheaves of sections are the corresponding operations on locally free sheaves.
2. The pullback f^*E , with fibre $(f^*E)_q = E_{f(q)}$ and cocycle $g_{ij} \circ f$ for the cover $(f^{-1}U_i)$, is a vector bundle on N ; f^* is functorial and $(g \circ f)^* \cong f^*g^*$.
3. $\det E = \Lambda^r E$ is a line bundle with cocycle $\det g_{ij}$.

Proof. All the constructions are functorial in the fibres and depend smoothly on the transition matrices (their entries are polynomial, or rational with nonvanishing denominator, in the entries of g_{ij}), so the resulting families of matrices are smooth cocycles, and Proposition 33.2.5(2) applies. That the fibres are as stated, and the identification of the sheaves of sections, follow because both are computed locally in trivialisations. $\square$

Proposition 33.3.2. *Every vector bundle over a manifold admits a metric: a smooth family of inner products (Hermitian inner products in the complex case) on the fibres. Consequently:*

1. every subbundle has a complement, and every short exact sequence $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$ of vector bundles splits;
2. $E \cong E^\vee$ for real bundles, and $E^\vee \cong \bar{E}$ for complex bundles with Hermitian metric.

Proof. Choose trivialisations over a cover (U_i) and a subordinate partition of unity (ψ_i) (Theorem 31.6.1); the sum $\sum_i \psi_i \langle -, - \rangle_i$ of the pulled-back standard inner products is a metric, since a convex combination of inner products is an inner product. Given a subbundle $E' \subseteq E$, the orthogonal complement $E'^\perp$ is a subbundle (in a local frame adapted to E' , apply Gram–Schmidt, Theorem 9.9.3, which depends smoothly on the frame) and $E = E' \oplus E'^\perp$; a surjection $E \rightarrow E''$ then splits by inverting $E'^\perp \rightarrow E''$. Statement (2) is the fibrewise isomorphism $v \mapsto \langle -, v \rangle$, which is smooth. $\square$

Proposition 33.3.3. 1. *A vector bundle E of rank r is trivial if and only if it has r sections $s_1, \dots, s_r$ that are linearly independent at every point (a global frame).*

2. *For a real line bundle L , $L \otimes L$ is trivial. For a complex line bundle L , $L \otimes \bar{L}$ is trivial, so $\bar{L} \cong L^\vee$.*

In particular the isomorphism classes of real line bundles form a group under $\otimes$ in which every element has order at most 2.

Proof. (1) A frame gives the morphism $M \times \mathbb{K}^r \rightarrow E$, $(p, a) \mapsto \sum_i a_i s_i(p)$, which is bijective on each fibre, hence an isomorphism (as in the proof of Proposition 33.2.3); conversely the standard basis vectors are a frame of the trivial bundle. (2) Choose a metric (Proposition 33.3.2). In the real case the map $L \otimes L \rightarrow M \times \mathbb{R}$, $v \otimes w \mapsto \langle v, w \rangle$, is fibrewise a nonzero linear map between lines, hence an isomorphism; smoothness is checked in a local frame. In the complex case use $v \otimes \bar{w} \mapsto \langle v, w \rangle$ on $L \otimes \bar{L}$. The group structure on line bundles is Proposition 33.3.1 and the uniqueness of tensor products up to isomorphism, with the trivial bundle as unit and $L^\vee$ as inverse, since $L \otimes L^\vee \cong \text{End}(L)$ has the nowhere vanishing section id . $\square$

Example 33.3.4. For the Möbius bundle E over S^1 , $E \otimes E$ is trivial by Proposition 33.3.3, although E is not (Counterexample 33.2.4). Directly: $E \otimes E$ is glued from $[0, 1] \times \mathbb{R}$ by $(0, t) \sim (1, (-1)^2 t) = (1, t)$, which is the trivial bundle. Likewise $E \oplus E$ is trivial: it is glued by $-\text{id}_{\mathbb{R}^2}$, the rotation by π , and the rotations by πs , $0 \leq s \leq 1$, connect it to the identity. The same argument shows that a real bundle glued from $[0, 1] \times \mathbb{R}^r$ by a matrix A is trivial when A can be joined to the identity by a path in $\text{GL}_r(\mathbb{R})$: if A_t is such a path, constant near its ends, with $A_0 = \text{id}$ and $A_1 = A$, the sections $t \mapsto A_t e_i$ satisfy the gluing condition and form a global frame.

Proposition 33.3.5. *Let $f: E \rightarrow F$ be a morphism of vector bundles over M whose rank on the fibres is constant. Then $\ker f \subseteq E$ and $\operatorname{im} f \subseteq F$ are subbundles, and $\operatorname{coker} f = F/\operatorname{im} f$ is a vector bundle. For a submanifold $S \subseteq M$ the normal bundle $N_{S/M} = (TM|_S)/TS$ is a vector bundle, and if $S = \phi^{-1}(q)$ for a regular value q of a smooth $\phi: M \rightarrow N$, then $d\phi$ induces an isomorphism $N_{S/M} \cong S \times T_q N$, so $N_{S/M}$ is trivial.*

Proof. The question is local, so let $E = U \times \mathbb{K}^r$, $F = U \times \mathbb{K}^s$ and f a smooth family of matrices $A(p)$ of constant rank k . Near p_0 some $k \times k$ minor of A is nonzero; after permuting bases, $A = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}$ with A_{11} invertible near p_0 . The first k columns of A are linearly independent and, as the rank is k , span the image; so they form a frame of $\operatorname{im} f$, which is therefore a subbundle. The vectors $\begin{pmatrix} -A_{11}^{-1}A_{12}w \\ w \end{pmatrix}$, $w \in \mathbb{K}^{r-k}$, lie in the kernel (the last $s - k$ rows are combinations of the first k), and they span it by dimension count; they form a frame of $\ker f$ depending smoothly on p . Completing a frame of $\operatorname{im} f$ by constant vectors to a frame of F gives a frame of $\operatorname{coker} f$. For the normal bundle apply this to the inclusion $TS \rightarrow TM|_S$, of constant rank $\dim S$. If $S = \phi^{-1}(q)$ with q regular, $d\phi: TM|_S \rightarrow S \times T_q N$ is surjective with kernel TS (Theorem 31.4.3), so it induces the stated isomorphism. $\square$

Example 33.3.6. TS^n is stably trivial: $TS^n \oplus (S^n \times \mathbb{R}) \cong S^n \times \mathbb{R}^{n+1}$, because the normal bundle of $S^n = f^{-1}(1)$, $f(x) = |x|^2$, is trivial by Proposition 33.3.5, and $TS^n \oplus N_{S^n/\mathbb{R}^{n+1}} \cong T\mathbb{R}^{n+1}|_{S^n}$ by Proposition 33.3.2(1). Nevertheless TS^2 is not trivial, by the hairy ball theorem (S^2 has no nowhere vanishing vector field, as $\chi(S^2) \neq 0$, Theorem 37.5.1); so a bundle is not determined by its stable class.

Example 33.3.7. On $\mathbb{CP}^n$ there is an exact sequence of holomorphic vector bundles $0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}(1)^{\oplus(n+1)} \rightarrow T\mathbb{CP}^n \rightarrow 0$ (the Euler sequence), where $\mathcal{O}(1) = \mathcal{O}(-1)^\vee$ (Chapter 42). Over the smooth category it splits by Proposition 33.3.2, but it does not split holomorphically: holomorphic splittings are much rarer, which is why holomorphic bundles carry more information than smooth ones.

33.4 Line bundles and the first Chern class

Definition 33.4.1. The Picard group $\operatorname{Pic}^\infty(M)$ is the set of isomorphism classes of smooth complex line bundles with the tensor product; by Proposition 23.6.4 it is a group, with inverse the dual.

Theorem 33.4.2. *Let $\mathbb{C}_M^\times$ be the sheaf of smooth nowhere vanishing complex functions. There are natural isomorphisms*

$$\operatorname{Pic}^\infty(M) \cong H^1(M, \mathbb{C}_M^\times) \xrightarrow{\sim} H^2(M; \mathbb{Z}),$$

*the second induced by the connecting homomorphism of the exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathbb{C}_M^\times \xrightarrow{\exp(2\pi i \cdot)} \mathbb{C}_M^\times \rightarrow 0$. The resulting class $c_1(L) \in H^2(M; \mathbb{Z})$ is the first Chern class; it satisfies $c_1(L \otimes L') = c_1(L) + c_1(L')$, $c_1(f^*L) = f^*c_1(L)$, and L is trivial if and only if $c_1(L) = 0$.*

Proof. Line bundles are classified by $\check{H}^1$ of the sheaf of smooth nonvanishing functions by Proposition 33.2.5(3), and Čech and sheaf cohomology agree in degree one (Remark 24.5.4); the group structures agree because the cocycle of a tensor product is the product of the cocycles. The exponential sequence is exact as a sequence of sheaves: surjectivity holds because a nonvanishing function has a smooth logarithm on any set where it is defined and

contractible, for instance a small ball, and the kernel of $\exp(2\pi i \cdot)$ is the constant sheaf $\mathbb{Z}$ (Theorem 25.7.3). Its long exact sequence reads

$$H^1(M, \underline{\mathbb{C}}_M) \rightarrow H^1(M, \underline{\mathbb{C}}_M^\times) \rightarrow H^2(M, \mathbb{Z}_M) \rightarrow H^2(M, \underline{\mathbb{C}}_M),$$

and the outer terms vanish because $\underline{\mathbb{C}}_M$ is a sheaf of modules over C_M , hence fine, hence acyclic (Corollary 24.4.6); manifolds are paracompact (Proposition 16.10.7). Finally $H^2(M, \mathbb{Z}_M) \cong H^2(M; \mathbb{Z})$ by Theorem 24.6.1, since manifolds are locally contractible. Additivity and naturality follow from the corresponding properties of the cocycles and of the connecting homomorphism, and triviality of L means exactly that its class in $H^1(M, \underline{\mathbb{C}}_M^\times)$ vanishes. $\square$

Example 33.4.3. $H^2(S^2; \mathbb{Z}) \cong \mathbb{Z}$, so the complex line bundles on $S^2 = \mathbb{CP}^1$ are classified by an integer, the degree; the tautological bundle $\mathcal{O}(-1)$ has $c_1 = -1$ with the standard orientation. On $\mathbb{CP}^n$, $H^2 \cong \mathbb{Z}$ is generated by $c_1(\mathcal{O}(1))$. Every complex line bundle on a surface of genus g is classified by its degree in $H^2 \cong \mathbb{Z}$, while on a simply connected manifold with $H^2 = 0$, such as S^3 or S^4 , every complex line bundle is trivial. Real line bundles are classified in the same way by $H^1(M; \mathbb{Z}/2\mathbb{Z})$, using the sheaf of nonvanishing real functions and the sequence $0 \rightarrow \mathbb{Z}/2 \rightarrow \mathbb{R}^\times \rightarrow \mathbb{R}_{>0}^\times \rightarrow 0$; the resulting class is the first Stiefel–Whitney class $w_1(L)$, which vanishes if and only if L is orientable.

Remark 33.4.4. For bundles of higher rank the classification is not by a single class: isomorphism classes of rank r complex bundles on M correspond to homotopy classes of maps $M \rightarrow \text{BU}(r)$, and the Chern classes $c_k(E) \in H^{2k}(M; \mathbb{Z})$ are pulled back from universal classes on the classifying space. They are constructed and computed in Chapter 37; for line bundles c_1 agrees with the class above.

33.5 Exercises

Exercise 33.5.1. Show that the Möbius bundle $L \rightarrow S^1$ is nontrivial, that $L \otimes L$ is trivial, and that L has no nowhere vanishing section.

Solution. A nowhere vanishing section would give a trivialisation; on the other hand a section is a continuous function $f: [0, 1] \rightarrow \mathbb{R}$ with $f(1) = -f(0)$, which must vanish somewhere by the intermediate value theorem. The transition function of $L \otimes L$ is the square of that of L , namely 1. $\square$

Exercise 33.5.2. Show that TS^2 is not trivial, using that a trivialisation would give a nowhere vanishing vector field (Corollary 19.6.6).

Exercise 33.5.3. Show that the tautological line bundle on $\mathbb{CP}^n$ is $\mathcal{O}(-1) = \{(\ell, v) : v \in \ell\} \subseteq \mathbb{CP}^n \times \mathbb{C}^{n+1}$, that it is a submanifold and a line bundle, and compute its transition functions for the standard cover.

Solution. On $U_i = \{x_i \neq 0\}$ the map $\ell \mapsto (\ell, (x_0/x_i, \dots, 1, \dots, x_n/x_i))$ is a nowhere vanishing section s_i , giving a trivialisation; then $s_j = (x_i/x_j)s_i$, so $g_{ij} = x_i/x_j$ as functions on $U_i \cap U_j$. $\square$

Exercise 33.5.4. Let $S \subseteq M$ be a submanifold with normal bundle $N = (TM|_S)/TS$. Show that if $S = f^{-1}(q)$ for a submersion f , then $N \cong S \times \mathbb{R}^m$ is trivial, and that the normal bundle of $\mathbb{RP}^{n-1} \subseteq \mathbb{RP}^n$ is the tautological bundle.

Chapter 34

Differential Forms and Integration

34.1 Introduction

Differential forms are the objects that can be integrated over oriented submanifolds, and the exterior derivative is the operation that makes Stokes' theorem (Theorem 34.4.4) true: the integral of $d\omega$ over a manifold with boundary is the integral of ω over the boundary. Everything in Hodge theory is written in this language: de Rham cohomology (Chapter 35), harmonic forms (Chapter 40), and the decomposition of forms on a complex manifold into types (Chapter 43). References: [Lee13], [War83], [Spi65].

Prerequisites. Chapter 33 (Vector Bundles) and Chapter 26 (Measure and Integration).

34.2 Differential forms

Definition 34.2.1. A *differential k -form* on a smooth manifold M is a smooth section of $\Lambda^k T^*M$; the space of k -forms is $\Omega^k(M)$, and $\Omega^\bullet(M) = \bigoplus_k \Omega^k(M)$ with the wedge product (Proposition 10.3.3) is a graded-commutative algebra over $C^\infty(M)$. Equivalently, a k -form assigns to each p an alternating k -linear map $(T_p M)^k \rightarrow \mathbb{R}$, smoothly in p . In a chart $(x^1, \dots, x^n)$ every k -form is uniquely $\omega = \sum_{i_1 < \dots < i_k} \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}$ with smooth coefficients. For $f: N \rightarrow M$ smooth, the *pullback* $f^*\omega$ is defined by $(f^*\omega)_p(v_1, \dots, v_k) = \omega_{f(p)}(d_p f v_1, \dots, d_p f v_k)$; it is an algebra homomorphism and $(g \circ f)^* = f^* g^*$.

Theorem 34.2.2. There is a unique family of linear maps $d: \Omega^k(M) \rightarrow \Omega^{k+1}(M)$, for all M and k , such that: df is the differential of f for $f \in \Omega^0(M) = C^\infty(M)$; $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{|\alpha|} \alpha \wedge d\beta$; and $d \circ d = 0$. It is local, given in a chart by $d(\sum_I \omega_I dx^I) = \sum_I d\omega_I \wedge dx^I$, and natural: $f^*(d\omega) = d(f^*\omega)$. Moreover, for vector fields,

$$d\omega(X_0, \dots, X_k) = \sum_i (-1)^i X_i \left(\omega(X_0, \dots, \widehat{X}_i, \dots, X_k) \right) + \sum_{i < j} (-1)^{i+j} \omega([X_i, X_j], X_0, \dots, \widehat{X}_i, \dots, \widehat{X}_j, \dots, X_k).$$

Proof. Uniqueness and locality. If ω vanishes on an open set U and $p \in U$, choose a bump function χ supported in U with $\chi(p) = 1$ (Corollary 31.6.2); then $\chi\omega = 0$, so $0 = d(\chi\omega) = d\chi \wedge \omega + \chi d\omega$, and evaluating at p gives $d\omega_p = 0$. Hence d is local and may be computed in charts, where the three rules force the coordinate formula, using $d(dx^I) = 0$.

Existence. Define d in each chart by the coordinate formula. It satisfies the three rules there: the Leibniz rule is a computation with the product rule, and $dd = 0$ follows from the symmetry of second partial derivatives (Theorem 25.10.5) together with the antisymmetry of $dx^i \wedge dx^j$. By uniqueness, the definitions in two charts agree on the overlap, so they define a global operator.

Naturality holds for 0-forms by the chain rule, hence for df and for wedge products of such, hence in general by the coordinate formula. The invariant formula is checked by expanding both sides in a chart: both are C^∞ -multilinear and alternating in the X_i (the bracket terms cancel the derivatives of the coefficients), so it suffices to verify it on coordinate fields, where all brackets vanish and it reduces to the coordinate formula. $\square$

Definition 34.2.3. For $X \in \mathfrak{X}(M)$ the *interior product* $\iota_X: \Omega^k \rightarrow \Omega^{k-1}$ is $(\iota_X \omega)(X_1, \dots, X_{k-1}) = \omega(X, X_1, \dots, X_{k-1})$ (Definition 10.4.4), and the *Lie derivative* is $L_X \omega = \frac{d}{dt} \Big|_{t=0} \Phi_t^* \omega$, where Φ is the flow of X .

Proposition 34.2.4 (Cartan's formulas). $L_X = d\iota_X + \iota_X d$ on forms; L_X commutes with d ; $\frac{d}{dt} \Phi_t^* \omega = \Phi_t^* L_X \omega$; and $L_{[X,Y]} = L_X L_Y - L_Y L_X$.

Proof. Both L_X and $d\iota_X + \iota_X d$ are derivations of degree 0 of $\Omega^\bullet(M)$ commuting with d (for the second, $d(d\iota_X + \iota_X d) = d\iota_X d = (d\iota_X + \iota_X d)d$), and both are local. On functions, $L_X f = Xf = \iota_X df$, and on df both give $d(Xf)$. Since forms are locally sums of products $f dg_1 \wedge \dots \wedge dg_k$, the two derivations agree. The formula $\frac{d}{dt} \Phi_t^* \omega = \Phi_t^* L_X \omega$ follows from the group law $\Phi_{t+s} = \Phi_t \Phi_s$, and the last identity from the corresponding one for vector fields (Proposition 31.7.3) by evaluating on forms. $\square$

Example 34.2.5. On $\mathbb{R}^3$ the Euclidean metric and orientation identify 1-forms and 2-forms with vector fields, by $F = (F_1, F_2, F_3) \mapsto F_1 dx + F_2 dy + F_3 dz$ and $F \mapsto F_1 dy \wedge dz + F_2 dz \wedge dx + F_3 dx \wedge dy$, and 3-forms with functions. Under these identifications d on 0-, 1- and 2-forms is the gradient, the curl and the divergence, and $d \circ d = 0$ becomes $\text{curl grad} = 0$ and $\text{div curl} = 0$.

Definition 34.2.6. A form ω is *closed* if $d\omega = 0$ and *exact* if $\omega = d\eta$ for some form η . Exact forms are closed, since $d \circ d = 0$.

Definition 34.2.7. For a 1-form α on M and a piecewise smooth curve $\gamma: [a, b] \rightarrow M$, the *line integral* is $\int_\gamma \alpha = \int_a^b \alpha_{\gamma(t)}(\dot{\gamma}(t)) dt$, the sum of the integrals of $\gamma^* \alpha$ over the smooth pieces.

Proposition 34.2.8. 1. The line integral is unchanged under orientation-preserving reparametrisation $\gamma \circ \phi$, $\phi: [c, d] \rightarrow [a, b]$ increasing and piecewise smooth with $\phi(c) = a$, $\phi(d) = b$, and changes sign under reversal.

2. (Fundamental theorem for line integrals) $\int_\gamma df = f(\gamma(b)) - f(\gamma(a))$.

3. On a connected manifold, a 1-form α is exact if and only if $\int_\gamma \alpha = 0$ for every closed piecewise smooth curve γ .

Proof. (1) is the substitution rule (Corollary 25.8.4) on each piece, as $(\gamma \circ \phi)^* \alpha = \phi^* \gamma^* \alpha$. (2) $\gamma^* df = d(f \circ \gamma) = (f \circ \gamma)' dt$, and Theorem 25.8.3 applies on each piece; the intermediate values cancel. (3) If $\alpha = df$, closed curves give 0 by (2). Conversely, fix p_0 and define $f(p) = \int_{\gamma} \alpha$ for any piecewise smooth curve γ from p_0 to p ; such curves exist since M is connected and locally path-connected by piecewise smooth curves (use straight segments in charts), and the value does not depend on γ , because for two curves γ, γ' the concatenation of γ with the reverse of γ' is closed. In a chart around p , with p at the origin, computing f along γ followed by the segment from p to $p + se_i$ gives $f(p + se_i) - f(p) = \int_0^s \alpha_i(p + ue_i) du$, so $\partial_i f = \alpha_i$ by Theorem 25.8.3; hence f is smooth and $df = \alpha$. $\square$

Counterexample 34.2.9. Closed forms need not be exact. On $M = \mathbb{R}^2 \setminus \{0\}$, the *angle form* $\alpha = (x dy - y dx)/(x^2 + y^2)$ is closed, as a direct computation shows, but along the unit circle

$\gamma(t) = (\cos t, \sin t)$, $\gamma^*\alpha = (\cos^2 t + \sin^2 t) dt = dt$, so $\int_\gamma \alpha = 2\pi \neq 0$, and α is not exact by Proposition 34.2.8(3). On the half-plane $\{x > 0\}$ it is exact, equal to $d \arctan(y/x)$: exactness is a global question, measured by de Rham cohomology (Chapter 35).

34.3 Orientations

Definition 34.3.1. An *orientation* of a smooth n -manifold M is an equivalence class of nowhere vanishing n -forms, two being equivalent if they differ by a positive function; M is *orientable* if such a form exists, and *oriented* once one is chosen. A chart is *positively oriented* if $dx^1 \wedge \cdots \wedge dx^n$ represents the orientation.

Proposition 34.3.2. M is orientable if and only if it has an atlas whose transition maps have positive Jacobian determinant, if and only if the line bundle $\det T^*M$ is trivial. For connected M there are then exactly two orientations. This agrees with the topological notion of Definition 21.3.1.

Proof. A nowhere vanishing n -form μ gives such an atlas: take charts with $\mu = f dx^1 \wedge \cdots \wedge dx^n$, $f > 0$ (compose with a reflection if necessary), and on overlaps the transition Jacobian is the ratio of two positive functions. Conversely, given such an atlas, patch the forms $dx^1 \wedge \cdots \wedge dx^n$ with a partition of unity: the sum is nowhere vanishing because in each chart it is a positive combination of positive multiples of $dx^1 \wedge \cdots \wedge dx^n$. A nowhere vanishing section of the line bundle $\det T^*M = \Lambda^n T^*M$ is the same as a trivialisation. For the last statement, in a positively oriented chart a generator of $H_n(M | p; \mathbb{Z})$ is determined by the orientation of $\mathbb{R}^n$, and this choice is locally constant, giving a section of the orientation covering; conversely such a section determines a compatible atlas. $\square$

Counterexample 34.3.3. The open Möbius band is not orientable. Let $\mathbb{Z}$ act on $\mathbb{R} \times (-1, 1)$ by $g^k(x, t) = (x+k, (-1)^k t)$, freely and properly, and let B be the smooth quotient (Theorem 31.5.1). A nowhere vanishing 2-form μ on B would pull back to a nowhere vanishing form $f dx \wedge dt$ on $\mathbb{R} \times (-1, 1)$ invariant under g . As $g^*(f dx \wedge dt) = (f \circ g) dx \wedge d(-t) = -(f \circ g) dx \wedge dt$, invariance means $f \circ g = -f$. But f is continuous and nowhere zero on the connected set $\mathbb{R} \times (-1, 1)$, hence of constant sign (Corollary 16.5.5), which contradicts $f(g(x, t)) = -f(x, t)$.

34.4 Integration and Stokes' theorem

Definition 34.4.1. Let M be an oriented n -manifold and ω an n -form with compact support contained in a positively oriented chart $\varphi: U \rightarrow \mathbb{R}^n$. Write $(\varphi^{-1})^*\omega = f dx^1 \wedge \cdots \wedge dx^n$ and set $\int_M \omega = \int_{\varphi(U)} f d\lambda$. For general compactly supported ω , choose a partition of unity (ψ_i) subordinate to a cover by positively oriented charts and set $\int_M \omega = \sum_i \int_M \psi_i \omega$.

Proposition 34.4.2. The integral is well defined, linear, and satisfies $\int_M \omega = \int_N f^* \omega$ for an orientation-preserving diffeomorphism $f: N \rightarrow M$; it changes sign if the orientation is reversed.

Proof. For ω supported in two charts, the two definitions agree by the change of variables formula (Theorem 26.8.2): if $\tau = \varphi \circ \psi^{-1}$ is the transition map, then the coefficient functions are related by $g = (f \circ \tau) \det D\tau$ with $\det D\tau > 0$, so $\int g = \int (f \circ \tau) |\det D\tau| = \int f$. The sum over a partition of unity is finite (compact support meets finitely many supports) and independent of the choices: for two partitions (ψ_i) , (χ_j) , expand $\sum_i \int \psi_i \omega = \sum_{i,j} \int \chi_j \psi_i \omega = \sum_j \int \chi_j \omega$. Linearity and the behaviour under diffeomorphisms follow, again by change of variables. $\square$

Definition 34.4.3. A *smooth n -manifold with boundary* is defined as in Definition 31.2.1 with charts taking values in open subsets of the half-space $\mathbb{H}^n = \{x \in \mathbb{R}^n : x^n \geq 0\}$; the *boundary*

∂M is the set of points corresponding to $x^n = 0$, a smooth $(n-1)$ -manifold. If M is oriented, ∂M carries the *induced orientation*, defined by declaring a chart of ∂M positively oriented when the n -form $dx^1 \wedge \cdots \wedge dx^n$ of a positively oriented chart of M is $(-1)^n v \wedge (dx^1 \wedge \cdots \wedge dx^{n-1})$ for an outward-pointing covector v ; concretely, $(x^1, \dots, x^{n-1})$ is positively oriented on ∂M when $(-1)^n(x^1, \dots, x^n)$ is on M .

Theorem 34.4.4 (Stokes). *Let M be an oriented n -manifold with boundary and ω an $(n-1)$ -form with compact support. Then*

$$\int_M d\omega = \int_{\partial M} \omega,$$

where ∂M has the induced orientation. If $\partial M = \emptyset$, the integral of an exact compactly supported form vanishes.

Proof. By a partition of unity and linearity we may assume that ω is supported in a single positively oriented chart, so that M is $\mathbb{R}^n$ or $\mathbb{H}^n$ and ω has compact support. Write $\omega = \sum_i (-1)^{i-1} f_i dx^1 \wedge \cdots \wedge \widehat{dx^i} \wedge \cdots \wedge dx^n$, so that $d\omega = (\sum_i \partial_i f_i) dx^1 \wedge \cdots \wedge dx^n$. By Fubini's theorem (Theorem 26.7.3) and the fundamental theorem of calculus, $\int \partial_i f_i d\lambda = 0$ for $i < n$ and also for $i = n$ in the case of $\mathbb{R}^n$, since f_i has compact support. In the case of $\mathbb{H}^n$, the only surviving term is

$$\int_{\mathbb{H}^n} \partial_n f_n d\lambda = \int_{\mathbb{R}^{n-1}} \left(\int_0^\infty \partial_n f_n dx^n \right) dx^1 \cdots dx^{n-1} = - \int_{\mathbb{R}^{n-1}} f_n(x', 0) dx',$$

while $\int_{\partial M} \omega = \int_{\mathbb{R}^{n-1}} (-1)^{n-1} f_n(x', 0) dx'$ computed with the induced orientation, which introduces the sign $(-1)^n$; the two agree. $\square$

Counterexample 34.4.5. Stokes' theorem needs compact support. On $M = \mathbb{R}$, without boundary, the 0-form $\arctan$ has $\int_{\mathbb{R}} d\arctan = \int_{\mathbb{R}} \frac{dx}{1+x^2} = \pi \neq 0$. On $M = [0, +\infty)$, with boundary the point 0 carrying the induced orientation -1 , the constant function 1 has $\int_M d1 = 0$ while $\int_{\partial M} 1 = -1$.

Example 34.4.6. For $M = [a, b]$ with its standard orientation, $\partial M = \{b\} - \{a\}$ with the induced orientation, and Stokes' theorem for a function f is the fundamental theorem of calculus, $\int_a^b f' = f(b) - f(a)$. For a compact region $M \subseteq \mathbb{R}^2$ with smooth boundary and $\omega = P dx + Q dy$, it is Green's formula $\int_M (\partial_x Q - \partial_y P) dx dy = \int_{\partial M} P dx + Q dy$, the boundary being traversed with M on the left.

Example 34.4.7. For $M \subseteq \mathbb{R}^3$ a compact region with smooth boundary, Stokes' theorem specialises to the divergence theorem $\int_M \operatorname{div} F dV = \int_{\partial M} \langle F, \nu \rangle dA$ and to the classical Stokes and Green formulas, after identifying vector fields with 1- or 2-forms using the Euclidean metric and orientation (Theorem 10.5.3). On a compact oriented manifold without boundary, $\int_M d\omega = 0$, so integration descends to de Rham cohomology (Chapter 35).

Corollary 34.4.8. *On a compact oriented n -manifold M without boundary, $\int_M : \Omega^n(M) \rightarrow \mathbb{R}$ is surjective, and $\int_M \mu > 0$ for any μ representing the orientation. Hence M carries a volume form, and any two differ by a positive function.*

Proof. In a positively oriented chart the integral of a nonnegative bump n -form is positive; a form representing the orientation is positive in every positively oriented chart, so its integral is positive by the definition via a partition of unity. $\square$

34.5 Exercises

Exercise 34.5.1. Compute the pullback of $dx \wedge dy$ under polar coordinates $(r, \theta) \mapsto (r \cos \theta, r \sin \theta)$, and the exterior derivative of $\alpha = (x dy - y dx)/(x^2 + y^2)$ on $\mathbb{R}^2 \setminus \{0\}$. Show that α is closed but not exact, by integrating over the unit circle.

Solution. The pullback is $r dr \wedge d\theta$. A computation gives $d\alpha = 0$, while $\int_{S^1} \alpha = 2\pi \neq 0$; an exact form would integrate to 0 over the closed curve S^1 by Stokes' theorem. $\square$

Exercise 34.5.2. Show that $\mathbb{RP}^n$ is orientable if and only if n is odd, by computing the effect of the antipodal map on a volume form of S^n (compare Exercise 21.9.2).

Exercise 34.5.3. Let μ be a volume form and X a vector field. Show that $L_X \mu = (\operatorname{div} X)\mu$ defines the divergence, that $\operatorname{div} X = \sum_i \partial_i X^i$ in coordinates with $\mu = dx^1 \wedge \cdots \wedge dx^n$, and that the flow of X preserves μ if and only if $\operatorname{div} X = 0$.

Solution. $L_X \mu$ is an n -form, hence a function times μ . By Cartan's formula $L_X \mu = d\iota_X \mu$, and computing $\iota_X \mu$ in coordinates gives $\sum_i (-1)^{i-1} X^i dx^1 \wedge \cdots \widehat{dx^i} \cdots \wedge dx^n$, whose differential is $(\sum_i \partial_i X^i)\mu$. The flow preserves μ if and only if $\frac{d}{dt} \Phi_t^* \mu = \Phi_t^* L_X \mu$ vanishes. $\square$

Exercise 34.5.4. Let $f: M \rightarrow N$ be a smooth map between compact oriented n -manifolds without boundary. Show that $\int_M f^* \omega = \deg(f) \int_N \omega$ for a suitable integer $\deg(f)$ independent of ω , using Sard's theorem and the local form of f near regular values.

Chapter 35

de Rham Cohomology

35.1 Introduction

The de Rham complex of a smooth manifold computes its real cohomology: closed forms modulo exact forms are singular cohomology classes with real coefficients (Theorem 35.3.1). This is the bridge between analysis and topology on which Hodge theory is built: the Hodge theorem selects a canonical harmonic representative in each de Rham class (Chapter 40), and on a complex manifold the type decomposition of harmonic forms gives the Hodge decomposition (Chapter 45). References: [BT82], [War83], [GH78, Chapter 0].

Prerequisites. Chapter 34 (Differential Forms and Integration), Chapter 24 (Sheaf Cohomology), Chapter 21 (Topological Manifolds and Poincaré Duality) and Chapter 30 (Distributions and Sobolev Spaces).

35.2 The de Rham complex

Definition 35.2.1. The *de Rham complex* of a smooth manifold M is $0 \rightarrow \Omega^0(M) \xrightarrow{d} \Omega^1(M) \xrightarrow{d} \cdots$, a complex by Theorem 34.2.2. Its cohomology $H_{\text{dR}}^k(M) = \ker(d: \Omega^k \rightarrow \Omega^{k+1})/\text{im}(d: \Omega^{k-1} \rightarrow \Omega^k)$ is the *de Rham cohomology*; its elements are classes $[\omega]$ of *closed* forms modulo *exact* ones. The wedge product makes $H_{\text{dR}}^\bullet(M)$ a graded-commutative $\mathbb{R}$ -algebra, and a smooth map $f: N \rightarrow M$ induces an algebra homomorphism $f^*: H_{\text{dR}}^\bullet(M) \rightarrow H_{\text{dR}}^\bullet(N)$. Replacing $\Omega^\bullet(M)$ by the compactly supported forms $\Omega_c^\bullet(M)$ gives $H_{\text{dR},c}^\bullet(M)$, functorial for open inclusions (extension by zero) and for proper maps.

Proposition 35.2.2. $H_{\text{dR}}^0(M) \cong \mathbb{R}^{\pi_0(M)}$, the locally constant functions. The wedge product is well defined on classes, and f^* commutes with d , hence descends to cohomology.

Proof. A function with $df = 0$ has vanishing differential, hence is constant on connected components (Theorem 25.6.6 in charts, and connectedness). If α, β are closed then $d(\alpha \wedge \beta) = 0$, and if $\alpha = d\gamma$ then $\alpha \wedge \beta = d(\gamma \wedge \beta)$, because $d(\gamma \wedge \beta) = d\gamma \wedge \beta \pm \gamma \wedge d\beta$ and $d\beta = 0$; naturality of d is Theorem 34.2.2. $\square$

Theorem 35.2.3 (Homotopy invariance and the Poincaré lemma). *Let $i_0, i_1: M \rightarrow M \times \mathbb{R}$ be $p \mapsto (p, 0)$ and $p \mapsto (p, 1)$. There is a linear map $K: \Omega^k(M \times \mathbb{R}) \rightarrow \Omega^{k-1}(M)$ with*

$$i_1^* \omega - i_0^* \omega = d(K\omega) + K(d\omega).$$

Consequently smoothly homotopic maps induce the same map on de Rham cohomology, homotopy equivalent manifolds have isomorphic de Rham cohomology, and $H_{\text{dR}}^k(U) = 0$ for $k > 0$ and $H_{\text{dR}}^0(U) = \mathbb{R}$ for every convex open $U \subseteq \mathbb{R}^n$ (the Poincaré lemma).

Proof. Write a form on $M \times \mathbb{R}$ uniquely as $\omega = \alpha_t + dt \wedge \beta_t$, where α_t, β_t are families of forms on M depending smoothly on t and containing no dt . Set $K\omega = \int_0^1 \beta_t dt$. Writing $d = d_M + dt \wedge \partial_t$ in this decomposition, a computation gives

$$d(K\omega) + K(d\omega) = \int_0^1 d_M \beta_t dt + \int_0^1 (\partial_t \alpha_t - d_M \beta_t) dt = \alpha_1 - \alpha_0 = i_1^* \omega - i_0^* \omega,$$

using the fundamental theorem of calculus and differentiation under the integral sign (Proposition 25.8.5).

If $F: M \times \mathbb{R} \rightarrow N$ is smooth with $F \circ i_0 = f$ and $F \circ i_1 = g$, then for closed ω on N , $g^* \omega - f^* \omega = d(KF^* \omega)$, so $f^* = g^*$ on cohomology. Continuous homotopies may be replaced by smooth ones: continuous maps between manifolds are homotopic to smooth maps, and continuously homotopic smooth maps are smoothly homotopic [Hir76, Chapter 2]. A convex U is smoothly contractible, by the straight-line homotopy, so its de Rham cohomology is that of a point. $\square$

Theorem 35.2.4 (Mayer–Vietoris). *Let $M = U \cup V$ with U, V open. There is a long exact sequence*

$$\cdots \rightarrow H_{\text{dR}}^k(M) \rightarrow H_{\text{dR}}^k(U) \oplus H_{\text{dR}}^k(V) \rightarrow H_{\text{dR}}^k(U \cap V) \xrightarrow{\delta} H_{\text{dR}}^{k+1}(M) \rightarrow \cdots$$

Proof. The sequence of complexes $0 \rightarrow \Omega^\bullet(M) \rightarrow \Omega^\bullet(U) \oplus \Omega^\bullet(V) \rightarrow \Omega^\bullet(U \cap V) \rightarrow 0$, with maps $\omega \mapsto (\omega|_U, \omega|_V)$ and $(\alpha, \beta) \mapsto \alpha|_{U \cap V} - \beta|_{U \cap V}$, is exact: injectivity and exactness in the middle are clear, and surjectivity holds because for a partition of unity $\{\psi_U, \psi_V\}$ subordinate to $\{U, V\}$ (Theorem 31.6.1) and $\gamma \in \Omega^\bullet(U \cap V)$, the forms $\psi_V \gamma$ on U and $-\psi_U \gamma$ on V , both extended by zero, have difference γ . Apply Theorem 13.4.3. $\square$

Example 35.2.5. $H_{\text{dR}}^k(S^n) \cong \mathbb{R}$ for $k = 0, n$ and 0 otherwise ($n \geq 1$): cover S^n by two contractible open sets whose intersection is homotopy equivalent to S^{n-1} (Theorem 19.6.1) and induct with Mayer–Vietoris. For the circle, the generator of H^1 is the form $d\theta$, which is closed but not exact; its integral over S^1 is 2π (Exercise 34.5.1). In general $\int_M: H_{\text{dR}}^n(M) \rightarrow \mathbb{R}$ is well defined for compact oriented M by Stokes' theorem (Theorem 34.4.4).

35.3 The de Rham theorem

Theorem 35.3.1 (de Rham). *For every smooth manifold M there are natural isomorphisms*

$$H_{\text{dR}}^k(M) \cong H^k(M, \mathbb{R}_M) \cong H^k(M; \mathbb{R}),$$

the first to sheaf cohomology of the constant sheaf, the second to singular cohomology with real coefficients. Under these isomorphisms the wedge product corresponds to the cup product, and the isomorphism is induced by integration: the class of a closed k -form ω corresponds to the singular cochain $\sigma \mapsto \int_{\Delta^k} \sigma^ \omega$ on smooth simplices.*

Proof. Let $\underline{\Omega}^k$ be the sheaf $U \mapsto \Omega^k(U)$. The sequence of sheaves

$$0 \rightarrow \mathbb{R}_M \rightarrow \underline{\Omega}^0 \xrightarrow{d} \underline{\Omega}^1 \xrightarrow{d} \cdots$$

is exact: exactness at $\underline{\Omega}^0$ is Proposition 35.2.2, and exactness at $\underline{\Omega}^k$ for $k \geq 1$ holds on stalks, because every point has a basis of convex coordinate neighbourhoods, on which closed forms are exact by the Poincaré lemma (Theorem 35.2.3). Each $\underline{\Omega}^k$ is a module over the sheaf of smooth functions, hence fine, hence acyclic on the paracompact space M (Example 24.4.3

and Theorem 24.4.5). So this is an acyclic resolution of $\mathbb{R}_M$ and Proposition 24.7.1(3) gives $H^k(M, \mathbb{R}_M) \cong H^k(\Gamma(M, \underline{\Omega}^\bullet)) = H_{\text{dR}}^k(M)$. The second isomorphism is Theorem 24.6.1, since manifolds are locally contractible and paracompact, and it is compatible with products. The description by integration over smooth simplices is the standard comparison of the two resolutions; see [War83, Chapter 5] or [Bre97, Chapter III]. $\square$

Corollary 35.3.2. *For a compact manifold M the spaces $H_{\text{dR}}^k(M)$ are finite-dimensional, they vanish for $k > \dim M$, and $\dim H_{\text{dR}}^k(M) = b_k(M)$, the k -th Betti number. For a compact oriented n -manifold, $H_{\text{dR}}^n(M) \cong \mathbb{R}$ for each connected component, with the isomorphism given by integration.*

Proof. Compact manifolds have finitely generated singular homology [Hat02, Appendix], so Theorem 35.3.1 and the universal coefficient theorem give finite dimension and the identification with Betti numbers; the vanishing above the dimension holds because there are no nonzero forms of degree $> n$. For the top degree, $H^n(M; \mathbb{R}) \cong H_0(M; \mathbb{R}) \cong \mathbb{R}$ per component by Poincaré duality (Theorem 21.6.2), and integration is nonzero on the class of a volume form (Corollary 34.4.8). $\square$

Proposition 35.3.3. *Let M be a connected manifold and $x_0 \in M$. There is an isomorphism*

$$H_{\text{dR}}^1(M) \xrightarrow{\sim} \text{Hom}(\pi_1(M, x_0), \mathbb{R}),$$

sending $[\omega]$ to the homomorphism whose value on the class of a piecewise smooth loop γ is the period $\int_\gamma \omega$. In particular $H_{\text{dR}}^1(M) = 0$ if M is simply connected, and a closed 1-form is exact if and only if all its periods vanish.

Proof. Compose the isomorphism $H_{\text{dR}}^1(M) \cong H^1(M; \mathbb{R})$ of Theorem 35.3.1, given by integration over smooth simplices, with $H^1(M; \mathbb{R}) \cong \text{Hom}_{\mathbb{R}}(H_1(M; \mathbb{R}), \mathbb{R}) = \text{Hom}(H_1(M), \mathbb{R})$ (Theorem 20.2.3 over the field $\mathbb{R}$) and with $\text{Hom}(H_1(M), \mathbb{R}) \cong \text{Hom}(\pi_1(M, x_0), \mathbb{R})$, which holds by the Hurewicz theorem (Theorem 19.9.1) because homomorphisms to the abelian group $\mathbb{R}$ factor through the abelianisation. The composite is the period map.

The last statement can also be seen directly. If $\omega = df$, then $\int_\gamma \omega = f(\gamma(1)) - f(\gamma(0)) = 0$ for loops. Conversely, if all periods vanish, $f(x) = \int_{x_0}^x \omega$ along any piecewise smooth path is well defined, since two paths differ by a loop, and in a chart around x that is a ball, integrating along segments shows $df = \omega$, by the Poincaré lemma on the ball (Theorem 35.2.3): there $\omega = dg$, and $f - g$ is constant. $\square$

Example 35.3.4. On $M = \mathbb{R}^2 \setminus \{0\}$ the angle form $\omega = \frac{x dy - y dx}{x^2 + y^2}$ is closed, and its period on the unit circle is 2π , so it is not exact. Since $\pi_1(M) \cong \mathbb{Z}$, Proposition 35.3.3 gives $H_{\text{dR}}^1(M) = \mathbb{R}[\omega]$: a closed 1-form η on M is exact if and only if $\int_{S^1} \eta = 0$, and in general $\eta - \frac{1}{2\pi}(\int_{S^1} \eta)\omega$ is exact. On the simply connected $\mathbb{R}^3 \setminus \{0\}$, by contrast, every closed 1-form is exact.

Counterexample 35.3.5. De Rham cohomology of a noncompact manifold can be infinite-dimensional. On $M = \mathbb{C} \setminus \mathbb{Z}$ let α_n be the angle form centred at n , the pullback of the angle form of Counterexample 34.2.9 by $z \mapsto z - n$. It is closed, and its integral over the circle γ_m of radius $\frac{1}{2}$ around m is 2π if $m = n$ and 0 otherwise (for $m \neq n$ the circle lies in a half-plane, where α_n is exact). If $\sum_n c_n \alpha_n = df$ for finitely many nonzero c_n , integrating over γ_m gives $2\pi c_m = 0$ (Proposition 34.2.8). So the classes $[\alpha_n]$, $n \in \mathbb{Z}$, are linearly independent in $H_{\text{dR}}^1(M)$.

Proposition 35.3.6. *Let a finite group G act freely on M by diffeomorphisms, and $\pi: M \rightarrow M/G$ the quotient (Theorem 31.5.1). Then π^* induces an isomorphism $H_{\text{dR}}^k(M/G) \cong H_{\text{dR}}^k(M)^G$ onto the classes fixed by G .*

Proof. As π is a local diffeomorphism and $\pi \circ g = \pi$, π^* identifies $\Omega^\bullet(M/G)$ with the complex $\Omega^\bullet(M)^G$ of G -invariant forms: an invariant form descends, because locally π has smooth inverses that differ by elements of G . The averaging operator $A = \frac{1}{|G|} \sum_{g \in G} g^*$ commutes with d and is a projection of $\Omega^\bullet(M)$ onto $\Omega^\bullet(M)^G$. The map $H^k(\Omega^\bullet(M)^G) \rightarrow H_{\text{dR}}^k(M)^G$ is injective: if an invariant ω equals $d\eta$, then $\omega = A\omega = d(A\eta)$ with $A\eta$ invariant. It is surjective: if $[\omega]$ is G -fixed, then $A\omega$ is invariant, closed, and cohomologous to ω , since each $g^*\omega$ is. $\square$

Example 35.3.7. $H_{\text{dR}}^k(\mathbb{R}P^n) \cong \mathbb{R}$ for $k = 0$, and for $k = n$ if n is odd; it vanishes otherwise. By Proposition 35.3.6 and Example 35.2.5, only the action of the antipodal map a on $H_{\text{dR}}^n(S^n) \cong \mathbb{R}$ matters. The volume form of S^n is $\iota_X \Omega$ restricted to S^n , where $\Omega = dx^1 \wedge \cdots \wedge dx^{n+1}$ and $X = \sum x^i \partial_i$; as a is the restriction of $-\text{id}$, $a^* \Omega = (-1)^{n+1} \Omega$ and a preserves X , so a^* acts on H^n by $(-1)^{n+1}$, and the invariant part is H^n for n odd and 0 for n even. In particular $\mathbb{R}P^n$ is not orientable for n even, by Corollary 35.3.2.

Theorem 35.3.8 (Künneth formula). *If M has a finite good cover, the map $\bigoplus_{p+q=k} H_{\text{dR}}^p(M) \otimes H_{\text{dR}}^q(N) \rightarrow H_{\text{dR}}^k(M \times N)$, $[\alpha] \otimes [\beta] \mapsto [\text{pr}_M^* \alpha \wedge \text{pr}_N^* \beta]$, is an isomorphism for every manifold N .*

Proof. Write ψ_M for the map. It is natural in M for restrictions to open subsets. If $M \cong \mathbb{R}^m$, then $\text{pr}_N: M \times N \rightarrow N$ is a homotopy equivalence and $H_{\text{dR}}^\bullet(M) = \mathbb{R}$ in degree 0, so ψ_M is pr_N^* , an isomorphism by Theorem 35.2.3. For open $U, V \subseteq M$, tensoring the Mayer–Vietoris sequence of U, V with the vector space $H_{\text{dR}}^q(N)$ and summing over q gives an exact sequence, which ψ maps to the Mayer–Vietoris sequence of $U \times N, V \times N$. The squares involving restrictions commute, and so does the one with the connecting maps: with a partition of unity $\{\rho_U, \rho_V\}$ on $U \cup V$, the connecting maps are $[\omega] \mapsto [d(\rho_V \omega)]$ on both sides, and for closed β , $d(\text{pr}_M^*(\rho_V \omega) \wedge \text{pr}_N^* \beta) = \text{pr}_M^* d(\rho_V \omega) \wedge \text{pr}_N^* \beta$. So if ψ is an isomorphism for U, V and $U \cap V$, it is one for $U \cup V$ by the five lemma (Lemma 13.4.2). Now let $U_1, \dots, U_k$ be a finite good cover of M ; we show by induction on k that ψ is an isomorphism for every open set covered by k open sets forming a good cover. For $k = 1$ this is the case $\mathbb{R}^m$. For the induction step apply the five lemma to $W = U_1 \cup \cdots \cup U_{k-1}$ and U_k , noting that $W \cap U_k$ is covered by the $k - 1$ sets $U_i \cap U_k$, which form a good cover. $\square$

Corollary 35.3.9. $H_{\text{dR}}^\bullet(T^n)$ is the exterior algebra on the classes of $dx^1, \dots, dx^n$, so $b_k(T^n) = \binom{n}{k}$; and $H_{\text{dR}}^\bullet(S^1 \times S^2)$ has dimensions 1, 1, 1, 1.

Proof. By Example 35.2.5, $H^\bullet(S^1)$ is spanned by 1 and $[d\theta]$, and $T^n = (S^1)^n$ has a finite good cover; Theorem 35.3.8 and induction give the tensor product of n copies of $\mathbb{R} \oplus \mathbb{R}[d\theta]$, which is the exterior algebra because $d\theta \wedge d\theta = 0$ and the wedge product is graded commutative. The second statement is the same computation with $H^\bullet(S^2)$. $\square$

Theorem 35.3.10 (Poincaré duality in de Rham cohomology). *Let M be an oriented n -manifold. The pairing*

$$H_{\text{dR}}^k(M) \times H_{\text{dR},c}^{n-k}(M) \rightarrow \mathbb{R}, \quad ([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta,$$

is well defined and nondegenerate if M has a finite good cover (for instance if M is compact); in that case $H_{\text{dR}}^k(M) \cong H_{\text{dR},c}^{n-k}(M)^$. For compact oriented M this is the statement that $\int_M \alpha \wedge \beta$ is a perfect pairing between $H_{\text{dR}}^k(M)$ and $H_{\text{dR}}^{n-k}(M)$.*

Proof. The pairing is well defined by Stokes' theorem: if α is closed and $\beta = d\gamma$ has compact support, then $\alpha \wedge \beta = \pm d(\alpha \wedge \gamma)$ has compact support and integrates to zero. Nondegeneracy is proved by induction on the number of sets in a finite good cover, one in which all nonempty intersections are diffeomorphic to $\mathbb{R}^n$ (such covers exist on any manifold, and finite ones on

compact manifolds), comparing the Mayer–Vietoris sequences of $H_{\text{dR}}^\bullet$ and $H_{\text{dR},c}^\bullet$ with the five lemma; see [BT82, Chapter I, Section 5]. Alternatively it follows from Theorem 35.3.1 and topological Poincaré duality (Corollary 21.6.3), since the wedge product corresponds to the cup product. $\square$

Theorem 35.3.11 (Cartan–Chevalley–Eilenberg). *Let G be a compact connected Lie group. Every de Rham class of G has a unique representative that is invariant under left and right translations, and the bi-invariant forms are closed. Consequently $H_{\text{dR}}^\bullet(G) \cong (\Lambda^\bullet \mathfrak{g}^*)^G$, the space of $\text{Ad}(G)$ -invariant alternating forms on the Lie algebra.*

Proof. $G \times G$ acts on G by $(a, b) \cdot x = axb^{-1}$; let μ be the Haar probability measure on the compact connected group $K = G \times G$ (Theorem 32.5.1) and, for a form ω , let $A\omega = \int_K k^* \omega \, d\mu(k)$, defined pointwise; $A\omega$ is smooth and bi-invariant, A commutes with d (differentiation under the integral sign), and $A\omega = \omega$ for bi-invariant ω .

Averaging does not change classes. Let ω be closed. Each $k \in K$ is connected to the identity by a smooth path, so $k^* \omega$ is cohomologous to ω (Theorem 35.2.3). For a closed form η of complementary degree, Fubini’s theorem gives $\int_G A\omega \wedge \eta = \int_K (\int_G k^* \omega \wedge \eta) \, d\mu(k) = \int_G \omega \wedge \eta$, since the inner integral depends only on the class of $k^* \omega$ (Stokes). By Poincaré duality (Theorem 35.3.10), $[A\omega] = [\omega]$.

Bi-invariant forms are closed. The inversion $\iota(x) = x^{-1}$ satisfies $\iota \circ L_a = R_{a^{-1}} \circ \iota$, so it maps bi-invariant forms to bi-invariant forms, and $d_e \iota = -\text{id}$ (differentiate $\exp(tX)^{-1} = \exp(-tX)$). A bi-invariant form is determined by its value at e , so $\iota^* \omega = (-1)^k \omega$ for a bi-invariant k -form. As $d\omega$ is bi-invariant of degree $k+1$, $(-1)^{k+1} d\omega = \iota^* d\omega = d\iota^* \omega = (-1)^k d\omega$, so $d\omega = 0$.

Uniqueness. If a bi-invariant ω is exact, $\omega = d\eta$, then $\omega = A\omega = d(A\eta) = 0$, because $A\eta$ is bi-invariant, hence closed. So bi-invariant forms map isomorphically onto $H_{\text{dR}}^\bullet(G)$. A left-invariant form is determined by its value $\omega_e \in \Lambda^\bullet \mathfrak{g}^*$, and it is also right-invariant if and only if ω_e is invariant under $\text{Ad}(g) = d_e(L_g \circ R_{g^{-1}})$. $\square$

Example 35.3.12. For $G = \text{SU}(2)$, $\mathfrak{g} = \mathfrak{su}(2) \cong \mathbb{R}^3$ and Ad maps $\text{SU}(2)$ onto $\text{SO}(3)$ (Exercise 32.6.2). The $\text{SO}(3)$ -invariants in $\Lambda^\bullet \mathbb{R}^3$ are the constants and the multiples of the determinant $e_1^* \wedge e_2^* \wedge e_3^*$, since Λ^1 and $\Lambda^2 \cong \mathbb{R}^3$ are irreducible nontrivial representations. So $H^\bullet(\text{SU}(2))$ is $\mathbb{R}$ in degrees 0 and 3, in accordance with $\text{SU}(2) \cong S^3$.

35.4 Currents

Definition 35.4.1. Let M be an oriented n -manifold. A k -current is a linear functional T on $\Omega_c^{n-k}(M)$ that is continuous in the sense of distributions: $T(\varphi_j) \rightarrow 0$ whenever the coefficients of φ_j , in any chart, have supports in a fixed compact set and converge to 0 together with all derivatives. Currents form a space $\mathcal{D}'_k(M)$, and dT is defined by $(dT)(\varphi) = (-1)^{k+1} T(d\varphi)$, so that $d \circ d = 0$.

Example 35.4.2. 1. A locally integrable k -form η defines a current $T_\eta(\varphi) = \int_M \eta \wedge \varphi$, and $dT_\eta = T_{d\eta}$ for smooth η by Stokes’ theorem; so currents generalise forms in the same way as distributions generalise functions (Chapter 30).

2. A compact oriented k -dimensional submanifold $Z \subseteq M$ defines the *current of integration* $[Z](\varphi) = \int_Z \varphi$ for $\varphi \in \Omega_c^k(M)$. Stokes’ theorem gives $d[Z] = 0$ when $\partial Z = \emptyset$, so $[Z]$ has a cohomology class.

Theorem 35.4.3. *On an oriented manifold the complex of currents computes de Rham cohomology: the inclusion of smooth forms into currents is a quasi-isomorphism. Under this identification and*

Poincaré duality, the class of $[Z]$ for a compact oriented k -dimensional submanifold Z of a compact oriented n -manifold M is the Poincaré dual of Z : it is the unique class $\eta_Z \in H_{\text{dR}}^{n-k}(M)$ with

$$\int_Z \varphi = \int_M \eta_Z \wedge \varphi \quad \text{for all closed } \varphi \in \Omega^k(M).$$

This is proved in [dR84, Chapter IV] and [GH78, Chapter 0, Section 4]; the first statement is a sheaf-theoretic argument like the proof of Theorem 35.3.1, applied to the fine resolution of $\mathbb{R}_M$ by sheaves of currents, and the second follows by comparing both sides with the topological Poincaré dual (Remark 21.8.2).

Remark 35.4.4. Currents are the natural home for the cycle classes of Hodge theory: an analytic subvariety Z of a compact complex manifold, possibly singular, defines a current of integration because its singular locus has measure zero and $\int_{Z_{\text{reg}}} \varphi$ converges (Chapter 59); its class in $H^{2n-2k}(X; \mathbb{Z})$ is the cycle class whose Hodge-theoretic characterisation the Hodge conjecture concerns.

35.5 Exercises

Exercise 35.5.1. Compute $H_{\text{dR}}^\bullet(T^n)$ and show that it is the exterior algebra on H^1 , generated by the forms $d\theta^1, \dots, d\theta^n$.

Solution. By Theorem 35.3.1 and Example 20.4.3, $H_{\text{dR}}^\bullet(T^n) \cong \Lambda^\bullet \mathbb{R}^n$. The classes of the closed forms $d\theta^i$, which are not exact since $\int_{S^1_i} d\theta^i = 2\pi$, generate: their products are the $\binom{n}{k}$ linearly independent classes of $d\theta^{i_1} \wedge \dots \wedge d\theta^{i_k}$, which can be detected by integration over the corresponding subtori. $\square$

Exercise 35.5.2. Show that $H_{\text{dR},c}^0(\mathbb{R}) = 0$ and $H_{\text{dR},c}^1(\mathbb{R}) \cong \mathbb{R}$, with the isomorphism given by integration; so compactly supported de Rham cohomology is not homotopy invariant.

Solution. A compactly supported function with $df = 0$ is constant, hence zero. The map $\int_{\mathbb{R}} : \Omega_c^1(\mathbb{R}) \rightarrow \mathbb{R}$ is surjective, and $\int f dx$ integrates to zero if and only if $F(x) = \int_{-\infty}^x f$ has compact support, in which case $f dx = dF$ is exact. $\square$

Exercise 35.5.3. Let $f : M \rightarrow N$ be a smooth map of compact connected oriented n -manifolds. Show that $f^* : H_{\text{dR}}^n(N) \rightarrow H_{\text{dR}}^n(M)$ is multiplication by the degree of f , and that a map of nonzero degree is surjective.

Exercise 35.5.4. Prove the Künneth formula $H_{\text{dR}}^\bullet(M \times N) \cong H_{\text{dR}}^\bullet(M) \otimes H_{\text{dR}}^\bullet(N)$ for manifolds with finite good covers, by induction with Mayer–Vietoris.

Chapter 36

Connections and Curvature

36.1 Introduction

A connection is a way of differentiating sections of a vector bundle; its curvature measures the failure of second derivatives to commute, and is the source of characteristic classes (Chapter 37). Flat connections are the same as local systems (Theorem 36.3.5), which is the differential-geometric form of the monodromy correspondence of Chapter 23 and the starting point of the theory of variations of Hodge structure (Chapter 66), where the Gauss–Manin connection appears. References: [KN63], [MS74], [GH78, Chapter 0].

Prerequisites. Chapter 33 (Vector Bundles), Chapter 34 (Differential Forms and Integration) and Chapter 32 (Lie Groups).

For a vector bundle $E \rightarrow M$ we write $\Omega^k(M, E) = \Gamma(M, \Lambda^k T^*M \otimes E)$ for the E -valued k -forms, so that $\Omega^0(M, E) = \Gamma(M, E)$.

36.2 Connections

Definition 36.2.1. A *connection* on a vector bundle $E \rightarrow M$ is a map $\nabla: \Gamma(M, E) \rightarrow \Omega^1(M, E)$ that is $\mathbb{R}$ -linear and satisfies the Leibniz rule

$$\nabla(fs) = df \otimes s + f\nabla s, \quad f \in C^\infty(M), s \in \Gamma(M, E).$$

For a vector field X we write $\nabla_X s = (\nabla s)(X)$, so $\nabla_{fX} s = f\nabla_X s$ and $\nabla_X(fs) = (Xf)s + f\nabla_X s$.

Proposition 36.2.2. *Connections are local operators, and in a local frame $(e_1, \dots, e_r)$ over U a connection is given by $\nabla e_j = \sum_i \theta_j^i \otimes e_i$ for a matrix $\theta = (\theta_j^i)$ of 1-forms, the connection form; then $\nabla(\sum_j s^j e_j) = \sum_i (ds^i + \sum_j \theta_j^i s^j) e_i$. Under a change of frame $e' = eg$ with $g: U \rightarrow \text{GL}_r$, the connection form transforms as $\theta' = g^{-1}\theta g + g^{-1}dg$. Every vector bundle over a manifold admits a connection, and the set of connections is an affine space over $\Omega^1(M, \text{End } E)$.*

Proof. Locality follows from the Leibniz rule with a bump function, as in Theorem 34.2.2. The formulas in a frame are immediate from the Leibniz rule, and the transformation rule follows by expanding $\nabla(e_k g_j^k)$. The trivial bundle $U \times \mathbb{R}^r$ has the connection d ; patching the pullbacks of such connections with a partition of unity (Theorem 31.6.1) gives a connection, because the Leibniz rule is preserved by convex combinations: if $\nabla = \sum_i \psi_i \nabla^i$ with $\sum \psi_i = 1$, then $\nabla(fs) = \sum_i \psi_i (df \otimes s + f\nabla^i s) = df \otimes s + f\nabla s$. The difference of two connections is C^∞ -linear, hence a bundle map $E \rightarrow T^*M \otimes E$. $\square$

Definition 36.2.3. The *exterior covariant derivative* $d_\nabla: \Omega^k(M, E) \rightarrow \Omega^{k+1}(M, E)$ is determined by $d_\nabla(\alpha \otimes s) = d\alpha \otimes s + (-1)^k \alpha \wedge \nabla s$, and the *curvature* is $F_\nabla = d_\nabla \circ \nabla: \Gamma(M, E) \rightarrow \Omega^2(M, E)$.

Proposition 36.2.4. F_∇ is $C^\infty(M)$ -linear, hence an element of $\Omega^2(M, \text{End } E)$, and

$$F_\nabla(X, Y)s = \nabla_X \nabla_Y s - \nabla_Y \nabla_X s - \nabla_{[X, Y]} s.$$

In a frame with connection form θ one has $F = d\theta + \theta \wedge \theta$, and under a change of frame $F' = g^{-1} F g$. Moreover $d_\nabla F = 0$ in the sense that $dF + \theta \wedge F - F \wedge \theta = 0$ (the Bianchi identity), and $d_\nabla \circ d_\nabla = F_\nabla \wedge -$ on $\Omega^\bullet(M, E)$.

Proof. For $f \in C^\infty(M)$, $F_\nabla(fs) = d_\nabla(df \otimes s + f \nabla s) = -df \wedge \nabla s + df \wedge \nabla s + f d_\nabla \nabla s = f F_\nabla(s)$, so F_∇ is tensorial. The formula in terms of vector fields follows by evaluating on X, Y and using $d\alpha(X, Y) = X\alpha(Y) - Y\alpha(X) - \alpha([X, Y])$ (Theorem 34.2.2). In a frame, $F(e_j) = d_\nabla(\sum_i \theta_j^i \otimes e_i) = \sum_i (d\theta_j^i) e_i - \sum_i \theta_j^i \wedge \nabla e_i = \sum_i (d\theta_j^i + \sum_k \theta_k^i \wedge \theta_j^k) e_i$. The transformation rule follows from that of θ by a computation, and differentiating $F = d\theta + \theta \wedge \theta$ gives $dF = d\theta \wedge \theta - \theta \wedge d\theta = F \wedge \theta - \theta \wedge F$. The last identity is checked on $\alpha \otimes s$. $\square$

Example 36.2.5. On the trivial bundle, $\nabla = d + \theta$ for a matrix of 1-forms θ , with curvature $d\theta + \theta \wedge \theta$; for a line bundle θ is a single 1-form and $F = d\theta$ is closed. A connection on E induces connections on $E^\vee$, $E \otimes F$, $\Lambda^k E$ and $\text{End } E$ by the Leibniz rule, with the corresponding curvatures; for instance $\nabla^{\text{End}}(A) = \nabla \circ A - A \circ \nabla$.

Proposition 36.2.6. Let ∇ be a connection on $E \rightarrow M$ with curvature F , and ∇' a connection on $E' \rightarrow M$ with curvature F' .

1. For smooth $f: N \rightarrow M$ there is a unique connection $f^*\nabla$ on f^*E with $(f^*\nabla)(f^*s) = f^*(\nabla s)$ for sections s of E ; its curvature is f^*F .
2. The connection $\nabla \otimes 1 + 1 \otimes \nabla'$ on $E \otimes E'$ has curvature $F \otimes 1 + 1 \otimes F'$, and the dual connection on $E^\vee$, $(\nabla^\vee \phi)(s) = d(\phi(s)) - \phi(\nabla s)$, has curvature $-F^T$.
3. For a line bundle, ∇ and $\nabla + \alpha$, with α a 1-form, have curvatures differing by $d\alpha$; so the curvature of a line bundle is a closed 2-form whose class does not depend on the connection.

Proof. (1) In a local frame (e_j) of E with connection form θ , the pullback frame (f^*e_j) of f^*E and the form $f^*\theta$ define a connection, and the transformation rule of Proposition 36.2.2 is preserved by pullback, so these local connections glue. The condition determines $f^*\nabla$ on the pulled back frames, hence everywhere by the Leibniz rule. The curvature is $d(f^*\theta) + f^*\theta \wedge f^*\theta = f^*(d\theta + \theta \wedge \theta)$ (Proposition 36.2.4). (2) In frames $(e_j), (e'_k)$ with connection forms θ, θ' , the frame $(e_j \otimes e'_k)$ has connection form $\theta \otimes 1 + 1 \otimes \theta'$, and as the two summands commute (they act on different factors), d of it plus its square is $(d\theta + \theta \wedge \theta) \otimes 1 + 1 \otimes (d\theta' + \theta' \wedge \theta')$, the cross terms cancelling by graded commutativity. The dual frame has connection form $-\theta^T$, and $d(-\theta^T) + \theta^T \wedge \theta^T = -(d\theta + \theta \wedge \theta)^T$, as $(\theta \wedge \theta)^T = -\theta^T \wedge \theta^T$ for matrices of 1-forms. (3) For rank one, θ is a 1-form and $\theta \wedge \theta = 0$, so $F = d\theta$, and replacing θ by $\theta + \alpha$ adds $d\alpha$. Closedness is the Bianchi identity. $\square$

36.3 Parallel transport, holonomy and flatness

Definition 36.3.1. A section s along a smooth curve $\gamma: [0, 1] \rightarrow M$ is *parallel* if $\nabla_{\dot{\gamma}} s = 0$. *Parallel transport* along γ is the map $P_\gamma: E_{\gamma(0)} \rightarrow E_{\gamma(1)}$ sending v to $s(1)$, where s is the parallel section with $s(0) = v$.

Proposition 36.3.2. Parallel sections along γ exist and are unique for each initial value, so P_γ is a well-defined linear isomorphism, depending smoothly on the data and satisfying $P_{\gamma \circ \delta} = P_\delta \circ P_\gamma$ and $P_{\bar{\gamma}} = P_\gamma^{-1}$.

Proof. In a local frame the equation $\nabla_{\dot{\gamma}} s = 0$ reads $\dot{s}^i + \sum_j \theta_j^i(\dot{\gamma}) s^j = 0$, a linear system of ordinary differential equations with smooth coefficients; by Proposition 25.12.5(2) its solutions exist on the whole interval, are unique, and depend linearly on the initial value, smoothly on parameters (Theorem 25.12.6). Linearity gives invertibility, with inverse the transport along the reversed curve, and the composition rule follows from uniqueness. $\square$

Proposition 36.3.3 (Homotopy invariance of pullbacks). *Let $E \rightarrow N$ be a vector bundle and $F: M \times \mathbb{R} \rightarrow N$ a smooth map, and put $f = F(-, 0)$ and $g = F(-, 1)$. Then $f^*E \cong g^*E$ as vector bundles on M . In particular every vector bundle on a star-shaped open subset of $\mathbb{R}^n$, for instance on $\mathbb{R}^n$ or on a ball, is trivial.*

Proof. Choose a connection on $E' = F^*E$ (Proposition 36.2.2). For $x \in M$ let $\gamma_x(t) = (x, t)$, $0 \leq t \leq 1$, and let $P_x: E'_{(x,0)} \rightarrow E'_{(x,1)}$ be parallel transport along γ_x . Since $E'_{(x,0)} = E_{f(x)} = (f^*E)_x$ and $E'_{(x,1)} = (g^*E)_x$, the P_x are linear isomorphisms $(f^*E)_x \rightarrow (g^*E)_x$, and they depend smoothly on x (Proposition 36.3.2), so they form an isomorphism of vector bundles $f^*E \rightarrow g^*E$. For a star-shaped U , with centre p , apply this to $F(x, t) = p + t(x - p)$ and $N = U$: then $g = \text{id}$ and f is constant, so $E = \text{id}^*E \cong f^*E = U \times E_p$ is trivial. $\square$

Definition 36.3.4. The *holonomy group* of ∇ at p is $\text{Hol}_p(\nabla) = \{P_\gamma : \gamma \text{ a loop at } p\} \subseteq \text{GL}(E_p)$, a subgroup. A connection is *flat* if $F_\nabla = 0$.

Theorem 36.3.5. *For a connection ∇ on $E \rightarrow M$ the following are equivalent:*

1. $F_\nabla = 0$;
2. every point has a neighbourhood with a frame of parallel sections (a flat frame), in which $\nabla = d$;
3. parallel transport along a path depends only on its homotopy class with fixed endpoints.

In that case the sheaf $\mathcal{E}^\nabla$ of parallel sections is a local system of rank r with $\mathcal{E}^\nabla \otimes_{\mathbb{R}_M} C_M^\infty \cong \mathcal{E}$, and on a connected M this gives an equivalence between flat bundles and representations of $\pi_1(M, p)$, the representation being the holonomy

$$\pi_1(M, p) \rightarrow \text{GL}(E_p), \quad [\gamma] \mapsto P_\gamma^{-1}.$$

Proof. (2) $\Rightarrow$ (1) is clear, since d has zero curvature, and curvature is local.

(1) $\Rightarrow$ (2). Work over a coordinate ball U with a frame, so that $E|_U = U \times \mathbb{R}^r$ and $\nabla = d + \theta$ with $\theta = \sum_i \theta_i dx^i$, a matrix of 1-forms. On $U \times \mathbb{R}^r$ consider the rank n distribution spanned by the vector fields $X_i = \partial/\partial x^i - \sum_{j,k} (\theta_i)_j^k v^j \partial/\partial v^k$. A computation gives $[X_i, X_j] = -\sum_{k,l} (F_{ij})_l^k v^l \partial/\partial v^k$, where $F = \sum_{i < j} F_{ij} dx^i \wedge dx^j$; so the distribution is involutive if and only if $F = 0$. If $F = 0$, the Frobenius theorem (Theorem 31.7.4) gives integral manifolds; each is transverse to the fibres $\{x\} \times \mathbb{R}^r$, hence locally the graph of a map $s: U' \rightarrow \mathbb{R}^r$, and the condition that the graph is an integral manifold is exactly $\partial_i s + \theta_i s = 0$, that is, $\nabla s = 0$. Taking the integral manifolds through $(p, v_1), \dots, (p, v_r)$ for a basis (v_i) gives r parallel sections near p , which form a frame because they are linearly independent at p , hence nearby.

(2) $\Rightarrow$ (3). Let $H: [0, 1]^2 \rightarrow M$ be a homotopy with fixed endpoints from γ_0 to γ_1 . Cover the square by finitely many small squares each mapped by H into a neighbourhood carrying a flat frame, as in the proof of Theorem 27.3.3. In a flat frame parallel transport along any path is the identity matrix, so the transport along the two edge paths of each small square agree; moving across the squares one at a time shows that the transports along γ_0 and γ_1 coincide.

(3) $\Rightarrow$ (2). On a simply connected neighbourhood U of a point p , define $s_i(q) = P_\gamma(v_i)$ for a path γ from p to q in U and a basis (v_i) of E_p ; this is well defined by (3), smooth by smooth dependence of solutions of differential equations, and gives a frame of parallel sections, in which the connection form vanishes.

For the last statement, parallel sections form a sheaf that is locally constant by (2), hence a local system, and a local frame of parallel sections is a frame of E , which gives the isomorphism with $\mathcal{E}$. Conversely a local system $\mathcal{L}$ determines the bundle $E = \mathcal{L} \otimes_{\mathbb{R}_M} C_M^\infty$ with the connection d in any flat frame; these constructions are inverse to each other. The correspondence with representations of π_1 is Theorem 23.5.3, and the monodromy of the local system of parallel sections is given by parallel transport; the inverse appears because monodromy was defined as a right action. $\square$

Example 36.3.6. Flat connections on the trivial line bundle over $S^1 = \mathbb{R}/2\pi\mathbb{Z}$: every connection is $d + a(\theta) d\theta$, it is flat, and parallel transport around the circle is multiplication by $\exp(-\int_0^{2\pi} a)$. Two such connections are isomorphic (by a gauge transformation $g: S^1 \rightarrow \mathbb{C}^\times$, which changes $a d\theta$ by $g^{-1}dg$) if and only if their holonomies agree, because the 1-forms $g^{-1}dg$ are exactly those with $\int_{S^1} g^{-1}dg \in 2\pi i\mathbb{Z}$ (a nonvanishing function has a continuous logarithm along the circle whose increase is $2\pi i$ times its winding number). So flat line bundles with connection on S^1 are classified by the holonomy in $\mathbb{C}^\times$, in accordance with Theorem 36.3.5, since $\pi_1(S^1) = \mathbb{Z}$.

36.4 Metric connections

Definition 36.4.1. Let E carry a Euclidean metric $\langle -, - \rangle$ (or a Hermitian metric in the complex case). A connection is *metric* if $d\langle s, t \rangle = \langle \nabla s, t \rangle + \langle s, \nabla t \rangle$.

Proposition 36.4.2. *Metric connections exist, and ∇ is metric if and only if parallel transport is an isometry, if and only if in every orthonormal local frame the connection form is skew-symmetric (skew-Hermitian in the complex case); then the curvature takes values in the skew-symmetric (skew-Hermitian) endomorphisms, that is, in the Lie algebra of the structure group.*

Proof. Existence: patch the standard connections of orthonormal local frames with a partition of unity, as in Proposition 36.2.2; the metric condition is preserved by convex combinations. If ∇ is metric and s, t are parallel along γ , then $\frac{d}{dt}\langle s, t \rangle = 0$, so transport is an isometry; conversely differentiate. In an orthonormal frame the metric condition reads $0 = \theta_j^i + \theta_i^j$ (respectively $\theta_j^i + \overline{\theta_i^j}$), and the statement about curvature follows from $F = d\theta + \theta \wedge \theta$. $\square$

Example 36.4.3. On a complex vector bundle with a Hermitian metric over a complex manifold, there is a unique connection that is metric and compatible with the holomorphic structure ($\nabla^{0,1} = \bar{\partial}$): the *Chern connection* (Chapter 42). Its curvature is a $(1, 1)$ -form, and for line bundles $\frac{i}{2\pi}F$ represents the first Chern class in de Rham cohomology, which is the differential-geometric form of Theorem 33.4.2.

36.5 Principal bundles

Definition 36.5.1. Let G be a Lie group. A *principal G -bundle* is a smooth surjection $\pi: P \rightarrow M$ with a free right action of G preserving the fibres and acting simply transitively on them, which is locally trivial: every point of M has a neighbourhood U with a G -equivariant diffeomorphism $\pi^{-1}(U) \cong U \times G$ over U . As for vector bundles, principal bundles are classified by cocycles $g_{ij}: U_i \cap U_j \rightarrow G$, that is, by $\check{H}^1(M, \underline{G})$.

Example 36.5.2. The frame bundle $\text{Fr}(E)$ of a rank r real vector bundle, whose fibre over p is the set of bases of E_p , is a principal $\text{GL}_r(\mathbb{R})$ -bundle, and E is recovered as the associated bundle $\text{Fr}(E) \times_{\text{GL}_r} \mathbb{R}^r$; a metric reduces the structure group to $O(r)$, an orientation to GL_r^+ , and a complex structure to $\text{GL}_{r/2}(\mathbb{C})$. Covering spaces are principal bundles with discrete structure group.

Remark 36.5.3. A connection on a principal bundle is a G -equivariant splitting of $0 \rightarrow \ker d\pi \rightarrow TP \rightarrow \pi^*TM \rightarrow 0$, equivalently a $\mathfrak{g}$ -valued 1-form ω on P restricting to the Maurer–Cartan form on fibres and satisfying $R_g^*\omega = \text{Ad}(g^{-1})\omega$. Its curvature is $\Omega = d\omega + \frac{1}{2}[\omega, \omega]$, and connections on P correspond to connections on all associated vector bundles, compatible with the structure group. This is the framework of gauge theory and of Chern–Weil theory, which we use in Chapter 37; see [KN63, Chapters I–II].

36.6 Exercises

Exercise 36.6.1. Compute the connection and curvature forms of the Levi-Civita connection of the round sphere S^2 in a local orthonormal frame, and verify that the curvature is $F = \text{vol}$ times a skew-symmetric matrix, of total integral 4π (compare the Gauss–Bonnet theorem, Chapter 37).

Exercise 36.6.2. Give an example of a flat bundle that is not trivial, and of a bundle admitting no flat connection.

Solution. The Möbius bundle over S^1 is flat (its transition function is locally constant) but not trivial (Exercise 33.5.1). The tautological bundle $O(-1)$ on $\mathbb{CP}^1$ admits no flat connection: by Theorem 36.3.5 a flat bundle on a simply connected manifold is determined by a representation of the trivial group, hence is trivial, while $O(-1)$ is not trivial because $c_1(O(-1)) \neq 0$ (Example 33.4.3). $\square$

Exercise 36.6.3. Show that for a line bundle with connection, the holonomy is given by $\exp(-\int_\Sigma F)$ for a loop bounding a surface Σ , and deduce that flat line bundles on a simply connected manifold are trivial.

Exercise 36.6.4. Verify that a connection on E induces connections on $E^\vee$ and $E \otimes F$, and that the curvature of $E \otimes F$ is $F_E \otimes 1 + 1 \otimes F_F$.

Chapter 37

Characteristic Classes

37.1 Introduction

Characteristic classes measure how far a vector bundle is from being trivial. For complex bundles they are the Chern classes $c_k(E) \in H^{2k}(M; \mathbb{Z})$, which can be constructed either topologically or, in de Rham cohomology, from the curvature of any connection by Chern–Weil theory (Theorem 37.2.2). They are central to the Hodge conjecture: Chern classes of algebraic vector bundles are Hodge classes, the Chern character appears in the Riemann–Roch theorem, and the class of a divisor is the first Chern class of a line bundle (Chapter 53). References: [MS74], [BT82], [Hir66], [GH78, Chapter 1].

Prerequisites. Chapter 36 (Connections and Curvature), Chapter 35 (de Rham Cohomology) and Chapter 22 (Cohomology Operations).

37.2 Chern–Weil theory

Definition 37.2.1. An *invariant polynomial* of degree k is a symmetric k -linear form $\tilde{P}$ on $\mathfrak{gl}_r(\mathbb{C}) = M_r(\mathbb{C})$ with $\tilde{P}(gA_1g^{-1}, \dots, gA_kg^{-1}) = \tilde{P}(A_1, \dots, A_k)$ for $g \in \mathrm{GL}_r(\mathbb{C})$; we write $P(A) = \tilde{P}(A, \dots, A)$. Differentiating the invariance along $g = \exp(tB)$ gives

$$\sum_{i=1}^k \tilde{P}(A_1, \dots, [B, A_i], \dots, A_k) = 0.$$

Multilinear forms are extended to matrices of differential forms by multiplying the entries with the wedge product; then $\tilde{P}(\alpha_1, \dots, \alpha_k)$ is a form of degree $\sum \deg \alpha_i$, and it is symmetric in the graded sense.

Theorem 37.2.2 (Chern–Weil). Let $E \rightarrow M$ be a complex vector bundle of rank r , ∇ a connection with curvature F , and P an invariant polynomial of degree k . Then:

1. $P(F)$ is a well-defined global $2k$ -form (computed in any local frame) and $dP(F) = 0$;
2. the class $[P(F)] \in H_{\mathrm{dR}}^{2k}(M)$ does not depend on ∇ ;
3. the resulting map from invariant polynomials to $H_{\mathrm{dR}}^{\bullet}(M)$ is natural: $[P(F_{f^*\nabla})] = f^*[P(F_{\nabla})]$ for smooth $f: N \rightarrow M$.

Proof. (1) Under a change of frame the curvature matrix transforms by conjugation (Proposition 36.2.4), so $P(F)$ is unchanged and defines a global form. For closedness, note that for matrices of forms and the covariant differential $d_{\nabla}\alpha = d\alpha + \theta \wedge \alpha - (-1)^{\deg \alpha} \alpha \wedge \theta$ one has

$d\tilde{P}(\alpha_1, \dots, \alpha_k) = \sum_i (-1)^{\deg \alpha_1 + \dots + \deg \alpha_{i-1}} \tilde{P}(\alpha_1, \dots, d_{\nabla} \alpha_i, \dots, \alpha_k)$: the terms involving θ cancel by the infinitesimal invariance of Definition 37.2.1. Applying this with all $\alpha_i = F$ and using the Bianchi identity $d_{\nabla} F = 0$ gives $dP(F) = 0$.

(2) Let ∇_0, ∇_1 be connections and $\alpha = \nabla_1 - \nabla_0 \in \Omega^1(M, \text{End } E)$. The family $\nabla_t = \nabla_0 + t\alpha$ has curvature $F_t = F_0 + t d_{\nabla_0} \alpha + t^2 \alpha \wedge \alpha$, so $\frac{d}{dt} F_t = d_{\nabla_t} \alpha$. Hence

$$\frac{d}{dt} P(F_t) = k \tilde{P}(d_{\nabla_t} \alpha, F_t, \dots, F_t) = k d \tilde{P}(\alpha, F_t, \dots, F_t),$$

using the formula of (1) and $d_{\nabla_t} F_t = 0$. Integrating in t from 0 to 1 gives $P(F_1) - P(F_0) = d \left(k \int_0^1 \tilde{P}(\alpha, F_t, \dots, F_t) dt \right)$, an exact form.

(3) The pullback connection has curvature f^*F , and P commutes with pullback. $\square$

Definition 37.2.3. The *Chern forms* of (E, ∇) are defined by

$$\det \left(I + \frac{i}{2\pi} F \right) = 1 + c_1(E, \nabla) + c_2(E, \nabla) + \dots, \quad c_k(E, \nabla) \in \Omega^{2k}(M),$$

and the *Chern classes* in de Rham cohomology are $c_k(E) = [c_k(E, \nabla)] \in H_{\text{dR}}^{2k}(M)$, independent of ∇ by Theorem 37.2.2. The *total Chern class* is $c(E) = \sum_k c_k(E)$.

Proposition 37.2.4. 1. $c_0 = 1$ and $c_k(E) = 0$ for $k > r$; $c_k(f^*E) = f^*c_k(E)$.

2. $c(E \oplus E') = c(E)c(E')$ (the Whitney formula).

3. For a line bundle L with Hermitian metric and Chern connection, $c_1(L) = [\frac{i}{2\pi} F]$, which is the image in $H^2(M; \mathbb{R})$ of the integral class of Theorem 33.4.2.

4. $c_k(\bar{E}) = (-1)^k c_k(E)$, and for the dual $c_k(E^\vee) = (-1)^k c_k(E)$.

Proof. (1) The coefficients of $\det(I + tA)$ are the elementary symmetric functions of the eigenvalues, so they vanish above degree r ; naturality is Theorem 37.2.2(3). (2) Choose connections ∇, ∇' ; their direct sum has block diagonal curvature, and the determinant of a block diagonal matrix is the product of the determinants. (4) A conjugate-linear isomorphism $E \cong \bar{E}$ carries a connection to the conjugate one with curvature $\bar{F}$, and $\det(I + \frac{i}{2\pi} \bar{F}) = \det(I - \frac{i}{2\pi} F)$; comparing degrees gives the sign, and $E^\vee \cong \bar{E}$ for a Hermitian metric (Proposition 33.3.2).

(3) For a line bundle the curvature is a 2-form F and $c_1(L, \nabla) = \frac{i}{2\pi} F$. To identify the class, represent L by a cocycle (g_{ij}) and choose a partition of unity; the Čech–de Rham comparison ([BT82, Chapter II]) identifies the image of the integral class under $H^2(M; \mathbb{Z}) \rightarrow H_{\text{dR}}^2(M)$ with the class of the 2-form obtained from $\frac{1}{2\pi i} d \log g_{ij}$, which is exactly $\frac{i}{2\pi} F$ for a connection built from the same trivialisations; see [GH78, Chapter 1]. $\square$

Example 37.2.5. For the tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^1 = S^2$ with the metric induced from $\mathbb{C}^2$, the Chern connection has curvature $F = -\frac{dz \wedge d\bar{z}}{(1+|z|^2)^2}$ in the chart z , and $\int_{\mathbb{CP}^1} \frac{i}{2\pi} F = -1$; so $c_1(\mathcal{O}(-1))$ is -1 times the positive generator of $H^2(\mathbb{CP}^1; \mathbb{Z}) \cong \mathbb{Z}$. Consequently $c_1(\mathcal{O}(1)) = h$ is the positive generator, and $c(T\mathbb{CP}^n) = (1+h)^{n+1}$ by the Euler sequence (Example 33.3.7) and the Whitney formula; for instance $c_1(T\mathbb{CP}^n) = (n+1)h$.

37.3 Chern classes: axioms and the splitting principle

Theorem 37.3.1. *There is a unique way to assign to every complex vector bundle E over a paracompact space classes $c_k(E) \in H^{2k}(M; \mathbb{Z})$ such that: $c_0 = 1$; $c_k(f^*E) = f^*c_k(E)$; $c(E \oplus E') = c(E)c(E')$; and $c_1(\mathcal{O}(-1))$ is the negative generator of $H^2(\mathbb{CP}^1; \mathbb{Z})$. Their images in de Rham cohomology are the Chern–Weil classes of Definition 37.2.3.*

This is [MS74, Chapter 14], where the classes are constructed from the Thom isomorphism or from the cohomology of projective bundles; the comparison with Chern–Weil theory is [MS74, Appendix C] or [GH78, Chapter 1].

Theorem 37.3.2 (Splitting principle). *For every complex vector bundle $E \rightarrow M$ of rank r there is a space $F(E)$ (the flag bundle) and a map $\pi: F(E) \rightarrow M$ such that π^* is injective on cohomology and π^*E is a direct sum of line bundles $L_1 \oplus \cdots \oplus L_r$. Writing $x_i = c_1(L_i)$, the Chern roots, one has $c(E) = \prod_i (1 + x_i)$, and every symmetric polynomial identity in the x_i gives an identity between characteristic classes.*

This is [BT82, Chapter 21] or [MS74, Chapter 14]: one takes the bundle of full flags in the fibres, whose cohomology is a free module over $H^*(M)$ by the Leray–Hirsch theorem, and the pullback of E acquires a filtration with line bundle quotients, which splits after a further pullback or, over $\mathbb{R}$, by choosing a metric.

Definition 37.3.3. With Chern roots as above, the *Chern character* and the *Todd class* are

$$\text{ch}(E) = \sum_i e^{x_i} = r + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \cdots, \quad \text{td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}} = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \cdots,$$

elements of $H^*(M; \mathbb{Q})$, well defined because they are symmetric power series in the roots, hence polynomials in the Chern classes.

Proposition 37.3.4. $\text{ch}(E \oplus E') = \text{ch}(E) + \text{ch}(E')$ and $\text{ch}(E \otimes E') = \text{ch}(E)\text{ch}(E')$; the Todd class is multiplicative in short exact sequences. Hence ch induces a ring homomorphism from the K-theory of M to $H^{\text{even}}(M; \mathbb{Q})$.

Proof. By the splitting principle we may assume $E = \bigoplus L_i$ and $E' = \bigoplus L'_j$; then the Chern roots of $E \oplus E'$ are the x_i and y_j , and those of $E \otimes E'$ are the $x_i + y_j$, because $c_1(L \otimes L') = c_1(L) + c_1(L')$ (Theorem 33.4.2). The statements follow from $e^{x+y} = e^x e^y$ and from multiplicativity of products over the roots. Short exact sequences of bundles split topologically (Proposition 33.3.2), which gives the last statements. $\square$

37.4 The Thom isomorphism in de Rham cohomology

In this section $\pi: E \rightarrow M$ is a smooth real vector bundle of rank n over a manifold M of dimension m . It is *oriented* if it has an atlas of local trivialisations $E|_U \cong U \times \mathbb{R}^n$ whose transition functions take values in $\text{GL}_n^+(\mathbb{R})$, the matrices of positive determinant; such trivialisations are called oriented.

Definition 37.4.1. A form ω on E has *compact vertical support* if $\text{supp } \omega \cap \pi^{-1}(K)$ is compact for every compact $K \subseteq M$. These forms form a subcomplex $\Omega_{\text{cv}}^\bullet(E)$ of $\Omega^\bullet(E)$, whose cohomology is written $H_{\text{cv}}^\bullet(E)$. Restriction to $E|_U$ for $U \subseteq M$ open preserves compact vertical support.

Let $t = (t_1, \dots, t_n)$ be the fibre coordinates of an oriented trivialisation $E|_U \cong U \times \mathbb{R}^n$. Every $\omega \in \Omega_{\text{cv}}^k(E|_U)$ is uniquely a sum $\omega = \omega' + \omega''$, where ω'' is the part containing $dt_1 \wedge \cdots \wedge dt_n$ and ω' contains fewer than n of the dt_j . Locally on U , in coordinates x of M , $\omega'' = \sum_I f_I(x, t) dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$.

Definition 37.4.2. For an oriented bundle, *integration along the fibres* $\pi_*: \Omega_{\text{cv}}^k(E) \rightarrow \Omega^{k-n}(M)$ is defined in an oriented trivialisation by $\pi_*\omega' = 0$ and

$$\pi_*\left(\sum_I f_I(x, t) dx^I \wedge dt_1 \wedge \cdots \wedge dt_n\right) = \sum_I \left(\int_{\mathbb{R}^n} f_I(x, t) dt\right) dx^I.$$

Proposition 37.4.3. *Integration along the fibres is well defined, commutes with restriction to open subsets of M and with d , $d\pi_* = \pi_*d$, and satisfies the projection formula $\pi_*(\pi^*\alpha \wedge \omega) = \alpha \wedge \pi_*\omega$ for $\alpha \in \Omega^\bullet(M)$ and $\omega \in \Omega_{\text{cv}}^\bullet(E)$. Reversing the orientation of E changes the sign of π_* .*

Proof. The integrals converge, since $f_I(x, -)$ has compact support, locally uniformly in x . A change of oriented trivialisation $t' = g(x)t$ with $\det g > 0$ gives $dt'_1 \wedge \cdots \wedge dt'_n = \det g(x) dt_1 \wedge \cdots \wedge dt_n$ plus terms containing some dx^i and fewer dt_j ; so the parts ω' and ω'' are the same in both trivialisations, and the integrals agree by the change of variables formula (Theorem 26.8.2), with Jacobian $\det g(x) = |\det g(x)|$. A change of coordinates in M does not affect the fibre integrals. So π_* is well defined, and it clearly commutes with restrictions. For the projection formula, $\pi^*\alpha \wedge \omega'$ has fewer than n factors dt_j , and $\pi^*(dx^K) \wedge f dx^I \wedge dt_1 \wedge \cdots \wedge dt_n = f dx^K \wedge dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$; by linearity in α over $C^\infty(M)$ this gives the formula. For d : if $\omega = f dx^I \wedge dt_J$ with $|J| < n$, the part of $d\omega$ with n factors dt is a sum of terms $\pm \partial_{t_j} f dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, whose fibre integrals vanish because $\int_{\mathbb{R}} \partial_{t_j} f dt_j = 0$ for compactly supported f (Theorem 25.8.3); so $\pi_*d\omega = 0 = d\pi_*\omega$. If $\omega = f dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, then $d\omega = \sum_i \partial_{x_i} f dx^i \wedge dx^I \wedge dt_1 \wedge \cdots \wedge dt_n$, and $\pi_*d\omega = \sum_i (\int \partial_{x_i} f dt) dx^i \wedge dx^I = d\pi_*\omega$ by differentiation under the integral sign (Proposition 25.8.5). Reversing the orientation replaces $dt_1 \wedge \cdots \wedge dt_n$ by $-dt_1 \wedge \cdots \wedge dt_n$ in an oriented trivialisation. $\square$

Lemma 37.4.4. *For every manifold N and $n \geq 0$, integration along the fibres of the trivial bundle $N \times \mathbb{R}^n \rightarrow N$ induces isomorphisms $H_{\text{cv}}^k(N \times \mathbb{R}^n) \rightarrow H^{k-n}(N)$.*

Proof. We treat $n = 1$ in a form that allows induction. Let $N' = N \times \mathbb{R}^{n-1}$ and consider the projection $p: N' \times \mathbb{R} \rightarrow N'$ with fibre coordinate t , and the subcomplexes $C^\bullet \subseteq \Omega^\bullet(N' \times \mathbb{R})$ and $C'^\bullet \subseteq \Omega^\bullet(N')$ of forms whose supports meet the preimage of every compact subset of N in a compact set. Every $\omega \in C^k$ is uniquely $\omega = \alpha_t + \beta_t \wedge dt$, with α_t, β_t families of forms on N' depending smoothly on t and without dt , and $p_*\omega = \int_{\mathbb{R}} \beta_t dt$, which lies in C'^{k-1} . Choose $e \in C_c^\infty(\mathbb{R})$ with $\int e = 1$, let $E(t) = \int_{-\infty}^t e$, and define

$$e_*(\beta) = p^*\beta \wedge e(t) dt, \quad K\omega = \int_{-\infty}^t \beta_u du - E(t) \int_{-\infty}^\infty \beta_u du.$$

Then $e_*: C'^{k-1} \rightarrow C^k$ commutes with d (as $d(e dt) = 0$), $p_*e_* = \text{id}$, and $K\omega$ lies in C^{k-1} : it vanishes for t below the support of ω , and also above it, where $\int_{-\infty}^t \beta_u du = \int_{-\infty}^\infty \beta_u du$ and $E(t) = 1$ (using that the t -supports are uniformly bounded over compact subsets of N). We claim that on C^k ,

$$dK\omega - Kd\omega = (-1)^{k-1}(\omega - e_*p_*\omega).$$

If $\omega = \alpha_t$ has no dt , then $K\omega = 0$, $e_*p_*\omega = 0$, and the dt -part of $d\omega$ is $(-1)^k \partial_t \alpha_t \wedge dt$, so $Kd\omega = (-1)^k \int_{-\infty}^t \partial_u \alpha_u du = (-1)^k \alpha_t$ (the second term vanishes as $\int_{\mathbb{R}} \partial_u \alpha_u du = 0$), which gives the claim. If $\omega = \beta_t \wedge dt$, write $d_{N'}$ for the exterior derivative along N' and $g_t = K\omega$. Then $d\omega = d_{N'}\beta_t \wedge dt$, so $Kd\omega = d_{N'}g_t$ by differentiation under the integral sign, while $dK\omega = d_{N'}g_t + (-1)^{k-1} \partial_t g_t \wedge dt$ with $\partial_t g_t = \beta_t - e(t) \int_{\mathbb{R}} \beta_u du$. Hence $dK\omega - Kd\omega = (-1)^{k-1}(\beta_t \wedge dt - e(t)(\int_{\mathbb{R}} \beta_u du) \wedge dt) = (-1)^{k-1}(\omega - e_*p_*\omega)$, proving the claim. So p_* and e_* are inverse isomorphisms on cohomology.

For $n = 1$ this is the lemma. For general n , integration over $\mathbb{R}^n$ is the composite of integration over the last coordinate, which maps the complex of forms on $N \times \mathbb{R}^n$ with compact vertical support over N to the corresponding complex on $N \times \mathbb{R}^{n-1}$, and integration over $\mathbb{R}^{n-1}$; by Fubini's theorem (in the form of Proposition 25.8.5) and the orientation convention $dt_1 \wedge \cdots \wedge dt_n = (dt_1 \wedge \cdots \wedge dt_{n-1}) \wedge dt_n$, and induction on n , the lemma follows. $\square$

Theorem 37.4.5 (Thom isomorphism). *Let $\pi: E \rightarrow M$ be an oriented vector bundle of rank n over a manifold with a finite good cover (for instance a compact manifold). Then $\pi_*: H_{\text{cv}}^k(E) \rightarrow H^{k-n}(M)$ is an isomorphism for every k . There is a unique class $\Phi \in H_{\text{cv}}^n(E)$ with $\pi_*\Phi = 1$, the Thom class, and the inverse isomorphism is $\alpha \mapsto \pi^*\alpha \wedge \Phi$.*

Proof. For open $U, V \subseteq M$ and a partition of unity $\{\rho_U, \rho_V\}$ subordinate to $\{U, V\}$ on $U \cup V$, the sequence $0 \rightarrow \Omega_{\text{cv}}(E|_{U \cup V}) \rightarrow \Omega_{\text{cv}}(E|_U) \oplus \Omega_{\text{cv}}(E|_V) \rightarrow \Omega_{\text{cv}}(E|_{U \cap V}) \rightarrow 0$ is exact, exactly as in Theorem 35.2.4, using the functions $\pi^*\rho_U, \pi^*\rho_V$: multiplying a form on $E|_{U \cap V}$ with compact vertical support by $\pi^*\rho_V$ and extending by zero gives a form on $E|_U$ with compact vertical support, because a compact $K \subseteq U$ meets $\text{supp } \rho_V$ in a compact subset of $U \cap V$. By Proposition 37.4.3, π_* maps this sequence to the Mayer–Vietoris sequence of U, V in de Rham cohomology, commuting with restrictions, and it commutes with the connecting maps, which are given on both sides by $\omega \mapsto d(\pi^*\rho_V \omega)$ and $\beta \mapsto d(\rho_V \beta)$, since $\pi_*d(\pi^*\rho_V \omega) = d(\rho_V \pi_*\omega)$ by the projection formula. So the long exact sequences form a commutative ladder.

Choose a finite good cover $\{U_1, \dots, U_N\}$ of M subordinate to a cover by trivialising open sets (possible by [BT82, Chapter I, Section 5]): all nonempty finite intersections of the U_i are diffeomorphic to $\mathbb{R}^m$ and E is trivial over each. We show by induction on k that π_* is an isomorphism over any union of k open sets forming such a good cover. For $k = 1$ this is Lemma 37.4.4. For the induction step write the union as $W \cup U$ with W a union of $k - 1$ of the sets; then $W \cap U$ is the union of the $k - 1$ sets $U_i \cap U$, which again form such a cover. By induction π_* is an isomorphism over W, U and $W \cap U$, and by the five lemma (Lemma 13.4.2) over $W \cup U$.

The class $\Phi = \pi_*^{-1}(1)$ is unique. By the projection formula $\pi_*(\pi^*\alpha \wedge \Phi) = \alpha \wedge \pi_*\Phi = \alpha$, so $\alpha \mapsto \pi^*\alpha \wedge \Phi$ is the inverse of π_* . $\square$

Definition 37.4.6. For E and M as in Theorem 37.4.5, the *Euler class* of E in de Rham cohomology is $e(E) = s^*\Phi \in H^n(M)$, where $s: M \rightarrow E$ is the zero section and Φ is viewed as a class in $H^n(E)$.

Proposition 37.4.7. *Let M be compact and E, F oriented bundles over M of ranks n, n' .*

1. $e(f^*E) = f^*e(E)$ for a smooth map $f: N \rightarrow M$ of compact manifolds, and reversing the orientation of E changes the sign of $e(E)$.
2. If n is odd, $e(E) = 0$.
3. If E has a nowhere vanishing section, $e(E) = 0$.
4. $e(E \oplus F) = e(E) \wedge e(F)$, with $E \oplus F$ oriented by an oriented frame of E followed by one of F .

Proof. We use that a closed form $\phi \in \Omega_{\text{cv}}^n(E)$ represents Φ if and only if $\pi_*\phi = 1$: $\pi_*\phi$ is a closed 0-form, that is, locally constant, and it represents $\pi_*[\phi]$. (1) The bundle map $\tilde{f}: f^*E \rightarrow E$ is an orientation-preserving isomorphism on fibres, so $\tilde{f}^*\phi$ has compact vertical support and, computing in pulled back trivialisations, fibre integral $f^*\pi_*\phi = 1$; hence it represents the Thom class of f^*E , and $s_N^*\tilde{f}^*\phi = f^*s^*\phi$. The orientation statement follows from the last statement of Proposition 37.4.3. (2) Let $a(v) = -v$. In oriented trivialisations the change of variables $t \mapsto -t$ has Jacobian $(-1)^n = -1$, so $\pi_*a^*\phi = -\pi_*\phi = -1$, and $a^*\phi$ represents $-\Phi$. As $a \circ s = s$, $e(E) = [s^*\phi] = [s^*a^*\phi] = -e(E)$. (3) Choose a metric on E and let

σ be a section with $|\sigma| \geq c > 0$ (by compactness). For $\lambda > 0$ let $m_\lambda(v) = \lambda v$; then $m_\lambda^* \phi$ has fibre integral 1 (Jacobian $\lambda^n > 0$), so it represents Φ , and its support is $\lambda^{-1} \text{supp } \phi$. As $\text{supp } \phi$ lies in some $\{|v| \leq R\}$, for $\lambda > R/c$ the form $m_\lambda^* \phi$ vanishes along $\sigma(M)$, so $\sigma^* m_\lambda^* \phi = 0$. The sections s and σ are homotopic through $t\sigma$, so $e(E) = s^*[m_\lambda^* \phi] = \sigma^*[m_\lambda^* \phi] = 0$ by homotopy invariance (Theorem 35.2.3). (4) Let $p_1: E \oplus F \rightarrow E$ and $p_2: E \oplus F \rightarrow F$ be the projections and ϕ, ψ Thom forms. Then $p_1^* \phi \wedge p_2^* \psi$ has compact vertical support, and in product trivialisations with fibre coordinates (t, u) its component containing $dt_1 \wedge \cdots \wedge dt_n \wedge du_1 \wedge \cdots \wedge du_n$ is the product of the components of ϕ and ψ of top vertical degree, so by Fubini its fibre integral is $\pi_* \phi \cdot \pi_* \psi = 1$. Hence it represents the Thom class of $E \oplus F$, and pulling back by the zero section, which is (s_E, s_F) , gives $e(E \oplus F) = e(E) \wedge e(F)$. $\square$

Remark 37.4.8. The Thom class links Poincaré duality and characteristic classes. If $S \subseteq M$ is a compact oriented submanifold of a compact oriented manifold, with normal bundle ν identified with a tubular neighbourhood of S , the Thom form of ν , extended by zero, represents the Poincaré dual of S in M ; restricted to S it gives $e(\nu)$, the *self-intersection* of S . Applied to the diagonal of $M \times M$, whose normal bundle is TM , this yields $\int_M e(TM) = \chi(M)$, part (3) of Theorem 37.5.1 [BT82].

37.5 Euler, Pontryagin and Stiefel–Whitney classes

Theorem 37.5.1. *Let $E \rightarrow M$ be an oriented real vector bundle of rank n over a compact manifold.*

1. (Thom isomorphism) $H^k(M; \mathbb{Z}) \cong H^{k+n}(E, E \setminus 0; \mathbb{Z})$, given by cup product with a Thom class $u \in H^n(E, E \setminus 0; \mathbb{Z})$ restricting to a generator on each fibre.
2. The Euler class $e(E) \in H^n(M; \mathbb{Z})$ is the restriction of u to the zero section. It is natural, satisfies $e(E \oplus E') = e(E)e(E')$, vanishes if E has a nowhere vanishing section, and for a complex bundle of rank r one has $e(E) = c_r(E)$.
3. (Gauss–Bonnet–Chern and Poincaré–Hopf) For a compact oriented n -manifold, $\int_M e(TM) = \chi(M)$, and the sum of the indices of the zeros of a vector field with isolated zeros equals $\chi(M)$.

These are [MS74, Chapters 9–11] and [BT82, Chapter 6]. With real coefficients, (1) is Theorem 37.4.5 (whose proof works over any manifold with a finite good cover), and the de Rham image of the Euler class is the class of Definition 37.4.6, whose properties were proved in Proposition 37.4.7. In the Chern–Weil picture, $e(TM)$ is represented by the Pfaffian of the curvature of a metric connection divided by $(2\pi)^{n/2}$, which is Chern’s proof of the Gauss–Bonnet theorem.

Definition 37.5.2. For a real vector bundle E , the *Pontryagin classes* are $p_k(E) = (-1)^k c_{2k}(E \otimes \mathbb{C}) \in H^{4k}(M; \mathbb{Z})$, and the *Stiefel–Whitney classes* $w_k(E) \in H^k(M; \mathbb{Z}/2\mathbb{Z})$ are the mod 2 characteristic classes determined by the same axioms with $\mathbb{Z}/2$ coefficients; $w_1(E) = 0$ if and only if E is orientable.

Remark 37.5.3. Hirzebruch’s signature theorem expresses the signature of a compact oriented $4m$ -manifold (Definition 21.6.5) as $\int_M L(p_1, \dots, p_m)$ for a universal polynomial in the Pontryagin classes, and the Riemann–Roch–Hirzebruch theorem computes the holomorphic Euler characteristic of a holomorphic bundle on a compact complex manifold as $\int_X \text{ch}(E) \text{td}(TX)$ [Hir66]. Both are special cases of the Atiyah–Singer index theorem (Chapter *efindex*-theorems), and both are used in Chapter 47 and Chapter 58.

Remark 37.5.4. Isomorphism classes of rank r complex bundles over a paracompact space M correspond to homotopy classes of maps $M \rightarrow \text{BU}(r)$, where $\text{BU}(r)$ is the infinite

Grassmannian $\text{Gr}_r(\mathbb{C}^\infty)$, and $H^\bullet(\text{BU}(r); \mathbb{Z}) = \mathbb{Z}[c_1, \dots, c_r]$ is a polynomial ring on the universal Chern classes; every characteristic class is a pullback of a universal one [MS74, Chapter 14]. For line bundles, $\text{BU}(1) = \mathbb{CP}^\infty = K(\mathbb{Z}, 2)$, which recovers Theorem 33.4.2 and Theorem 22.2.4.

Example 37.5.5. On a smooth projective complex variety, the Chern classes of an algebraic vector bundle are algebraic cycle classes, hence Hodge classes; for a line bundle, $c_1(L)$ is the class of a divisor (Chapter 53), and the Lefschetz theorem on $(1, 1)$ -classes (Chapter 65) says conversely that every integral $(1, 1)$ -class is of this form. This is the case $p = 1$ of the Hodge conjecture, and it is the only case known in general.

37.6 Exercises

Exercise 37.6.1. Using the Euler sequence, compute $c(T\mathbb{CP}^2)$ and $\int_{\mathbb{CP}^2} c_2(T\mathbb{CP}^2)$, and check that the latter is $\chi(\mathbb{CP}^2) = 3$.

Solution. $c(T\mathbb{CP}^2) = (1 + h)^3 = 1 + 3h + 3h^2$, so $c_2 = 3h^2$ and $\int_{\mathbb{CP}^2} c_2 = 3$ since $\int h^2 = 1$. By Theorem 37.5.1, $c_2(T\mathbb{CP}^2) = e(T\mathbb{CP}^2)$ integrates to $\chi(\mathbb{CP}^2) = 3$ (Example 19.7.8). $\square$

Exercise 37.6.2. Show that a bundle with a flat connection has vanishing Chern classes in de Rham cohomology, and give an example where the integral Chern classes are nonzero.

Solution. Chern–Weil forms are computed from the curvature, which is zero. A flat line bundle on a space with torsion in $H^2(M; \mathbb{Z})$, for instance a nontrivial real line bundle on $\mathbb{RP}^2$ complexified, has torsion $c_1 \neq 0$ mapping to zero in de Rham cohomology. $\square$

Exercise 37.6.3. Show that $\text{ch}(L) = e^{c_1(L)}$ for a line bundle, and compute ch and td of $T\mathbb{CP}^1$.

Exercise 37.6.4. Show that S^4 admits no almost complex structure, using that c_1 would lie in $H^2(S^4) = 0$ and $c_2 = e(TS^4)$ would have to satisfy $\int_{S^4} c_2 = \chi(S^4) = 2$ while also $c_1^2 = p_1 + 2c_2$ with $p_1(TS^4) = 0$.

Chapter 38

Riemannian Geometry

38.1 Introduction

A Riemannian metric turns a manifold into a metric space, produces a canonical connection (Theorem 38.2.6) and, together with an orientation, defines the Hodge star, the codifferential and the Laplacian (Definition 38.7.2). Harmonic forms are the kernel of that Laplacian, and the Hodge theorem (Chapter 40) says that on a compact manifold they represent de Rham cohomology. Along the way we develop the local and global theory of geodesics: covariant derivatives along curves, the Gauss lemma, the fact that geodesics are locally the shortest curves, and the Hopf–Rinow theorem, all with complete proofs; parallel transport and holonomy, with the holonomy principle, and the theorems of Ambrose–Singer, de Rham and Berger; and Jacobi fields, conjugate points and the first comparison theorems. References: [dC92], [Lee18], [KN63], [Joy00], [War83], [GH78, Chapter 0].

Prerequisites. Chapter 36 (Connections and Curvature).

Conventions. (M, g) is a Riemannian manifold of dimension n , and we write $\langle v, w \rangle = g(v, w)$ and $|v| = \langle v, v \rangle^{1/2}$. Curves are piecewise smooth unless stated otherwise. In local coordinates we use the summation convention: an index repeated once up and once down is summed over.

38.2 Metrics and the Levi-Civita connection

Definition 38.2.1. A *Riemannian metric* on a smooth manifold M is a smooth section g of $\text{Sym}^2 T^*M$ that is an inner product on each tangent space; (M, g) is a *Riemannian manifold*. The *length* of a piecewise smooth curve is $L(\gamma) = \int_a^b |\dot{\gamma}(t)| dt$ with $|v| = g(v, v)^{1/2}$, and $d(p, q)$ is the infimum of the lengths of curves from p to q . An *isometry* is a diffeomorphism preserving g .

Proposition 38.2.2. Every smooth manifold admits a Riemannian metric, and d is a metric inducing the topology of M .

Proof. Patch the Euclidean metrics of charts with a partition of unity (Theorem 31.6.1); a convex combination of inner products is an inner product. Symmetry and the triangle inequality for d are clear, and $d(p, q) > 0$ for $p \neq q$ because in a chart around p the metric is comparable to the Euclidean one on a compact neighbourhood, so curves leaving a small ball have length bounded below. The same comparison shows that the d -balls form a basis of the topology. $\square$

Example 38.2.3. The Euclidean metric on $\mathbb{R}^n$; the induced metric on a submanifold, for instance the round metric on $S^n \subseteq \mathbb{R}^{n+1}$; the hyperbolic metric $(dx^2 + dy^2)/y^2$ on the upper

half plane; a left-invariant metric on a Lie group, obtained from any inner product on $\mathfrak{g}$, which is bi-invariant when G is compact (average over Haar measure, Theorem 32.5.1); and the Fubini–Study metric on $\mathbb{CP}^n$ (Chapter 44).

Proposition 38.2.4. *Let a discrete group G act freely and properly on M by isometries of g . Then M/G carries a unique Riemannian metric for which $\pi: M \rightarrow M/G$ is a local isometry.*

Proof. By Theorem 31.5.1, π is a local diffeomorphism. For $x \in M$ define the metric on $T_{\pi(x)}(M/G)$ by transporting g_x through $d_x\pi$. This does not depend on the choice of x in its orbit: $\pi \circ h = \pi$ for $h \in G$, so $d_{hx}\pi \circ d_xh = d_x\pi$, and d_xh is an isometry $T_xM \rightarrow T_{hx}M$. On an open set $\pi(U)$ over which π has the smooth inverse $(\pi|_U)^{-1}$ the metric is the pullback of g by this inverse, hence smooth. Uniqueness is clear. $\square$

Example 38.2.5. The flat torus $\mathbb{R}^n/\mathbb{Z}^n$ and the flat Klein bottle $\mathbb{R}^2/G$ (Example 18.4.5), since translations and the maps $(x, y) \mapsto (x + \frac{1}{2}, -y)$ are Euclidean isometries; and the round metric on $\mathbb{RP}^n = S^n/\{\pm 1\}$, as $-\text{id}$ is an isometry of S^n .

Theorem 38.2.6 (Fundamental theorem of Riemannian geometry). *On a Riemannian manifold there is a unique connection ∇ on TM that is metric ($Xg(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z)$) and torsion-free ($\nabla_X Y - \nabla_Y X = [X, Y]$): the Levi-Civita connection. It is determined by the Koszul formula*

$$2g(\nabla_X Y, Z) = Xg(Y, Z) + Yg(X, Z) - Zg(X, Y) + g([X, Y], Z) - g([X, Z], Y) - g([Y, Z], X).$$

In coordinates, $\nabla_{\partial_i} \partial_j = \sum_k \Gamma_{ij}^k \partial_k$ with $\Gamma_{ij}^k = \frac{1}{2} \sum_l g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij})$.

Proof. If ∇ is metric and torsion-free, expanding the three terms $Xg(Y, Z)$, $Yg(X, Z)$, $-Zg(X, Y)$ with the metric property and using torsion-freeness to replace $\nabla_X Y - \nabla_Y X$ by $[X, Y]$ gives the Koszul formula; since g is nondegenerate, the formula determines $\nabla_X Y$, which proves uniqueness. Conversely, define $\nabla_X Y$ by the Koszul formula. The right-hand side is $\mathbb{R}$ -bilinear, C^∞ -linear in X and Z and satisfies the Leibniz rule in Y (the terms with derivatives of f in $Y \mapsto fY$ combine to $2(Xf)g(Y, Z)$), so ∇ is a connection; exchanging X and Y shows it is torsion-free, and adding the formulas for $g(\nabla_X Y, Z)$ and $g(Y, \nabla_X Z)$ gives $Xg(Y, Z)$. The coordinate formula follows by taking X, Y, Z coordinate fields, whose brackets vanish. $\square$

In coordinates, torsion-freeness says $\Gamma_{ij}^k = \Gamma_{ji}^k$, and the metric property for coordinate fields says

$$\partial_i g_{jk} = \Gamma_{ij}^l g_{lk} + \Gamma_{ik}^l g_{jl}. \quad (*)$$

Definition 38.2.7. A *geodesic* is a curve with $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$; in coordinates, $\ddot{\gamma}^k + \sum_{i,j} \Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j = 0$. By Theorem 25.12.3 there is a unique maximal geodesic with given initial position and velocity, depending smoothly on them; the *exponential map* $\exp_p(v) = \gamma_v(1)$ is defined on a neighbourhood of $0 \in T_p M$.

38.3 Covariant derivatives along curves

A *vector field along* a smooth curve $\gamma: I \rightarrow M$ is a smooth map $V: I \rightarrow TM$ with $V(t) \in T_{\gamma(t)}M$; for instance the velocity $\dot{\gamma}$, or $Y \circ \gamma$ for a vector field Y on M .

Proposition 38.3.1. *There is a unique operator D_t sending vector fields along γ to vector fields along γ such that D_t is $\mathbb{R}$ -linear, $D_t(fV) = \dot{f}V + fD_tV$ for smooth $f: I \rightarrow \mathbb{R}$, and $D_t(Y \circ \gamma) = \nabla_{\dot{\gamma}}Y$ for every vector field Y defined near $\gamma(I)$. In coordinates, for $V = V^k \partial_k$,*

$$D_t V = (\dot{V}^k + \Gamma_{ij}^k(\gamma) \dot{\gamma}^i V^j) \partial_k.$$

Moreover $\frac{d}{dt} \langle V, W \rangle = \langle D_t V, W \rangle + \langle V, D_t W \rangle$. A vector field along γ is parallel in the sense of Definition 36.3.1 if and only if $D_t V = 0$.

Proof. In a chart, $V = V^k(\partial_k \circ \gamma)$, and the axioms force $D_t V = \dot{V}^k \partial_k + V^k \nabla_{\dot{\gamma}} \partial_k = (\dot{V}^k + \dot{\gamma}^i V^j \Gamma_{ij}^k) \partial_k$, since $\nabla_{\dot{\gamma}} \partial_j = \dot{\gamma}^i \Gamma_{ij}^k \partial_k$. This proves uniqueness. Conversely, the formula satisfies the axioms in each chart (for $Y = Y^k \partial_k$, $\frac{d}{dt} Y^k(\gamma) = \dot{\gamma}^i \partial_i Y^k$ matches the coordinate expression of $\nabla_{\dot{\gamma}} Y$), and by uniqueness the local definitions agree on overlaps. For the metric property, differentiate $\langle V, W \rangle = g_{jk}(\gamma) V^j W^k$: the derivative of $g_{jk}(\gamma)$ is $\dot{\gamma}^i \partial_i g_{jk}$, and by (*) the result equals $\langle D_t V, W \rangle + \langle V, D_t W \rangle$. The last statement is the local expression of $\nabla_{\dot{\gamma}} s = 0$ in the coordinate frame. $\square$

Lemma 38.3.2 (Symmetry lemma). *Let $f: (-\varepsilon, \varepsilon) \times (a, b) \rightarrow M$ be smooth, and write D_s, D_t for the covariant derivatives along the curves $s \mapsto f(s, t)$ and $t \mapsto f(s, t)$. Then $D_s \partial_t f = D_t \partial_s f$.*

Proof. In coordinates both sides equal $(\partial_s \partial_t f^k + \Gamma_{ij}^k \partial_s f^i \partial_t f^j) \partial_k$, by the symmetry $\Gamma_{ij}^k = \Gamma_{ji}^k$. $\square$

Proposition 38.3.3. *A curve γ is a geodesic (Definition 38.2.7) if and only if $D_t \dot{\gamma} = 0$. Geodesics have constant speed, and parallel transport along any curve is a linear isometry.*

Proof. The coordinate expression of $D_t \dot{\gamma}$ is the geodesic equation. If $D_t \dot{\gamma} = 0$ then $\frac{d}{dt} |\dot{\gamma}|^2 = 2 \langle D_t \dot{\gamma}, \dot{\gamma} \rangle = 0$. If V and W are parallel, $\langle V, W \rangle$ is constant by Proposition 38.3.1. $\square$

38.4 The exponential map and minimising geodesics

Geodesics are the projections to M of the integral curves of a vector field on TM , the *geodesic spray*, which in the coordinates (x, v) of TM is $v^k \partial_{x^k} - \Gamma_{ij}^k(x) v^i v^j \partial_{v^k}$. By Theorem 31.7.1, the set $\mathcal{E} \subseteq TM$ of vectors v for which γ_v is defined on $[0, 1]$ is open, and $\exp: \mathcal{E} \rightarrow M$ is smooth. Since $\gamma_{cv}(t) = \gamma_v(ct)$, each $\mathcal{E}_p = \mathcal{E} \cap T_p M$ is star-shaped about 0, and $\exp_p(tv) = \gamma_v(t)$.

Proposition 38.4.1. $d_0 \exp_p = \text{id}$, so $\exp_p$ is a diffeomorphism from a neighbourhood of 0 onto a neighbourhood of p , giving normal coordinates in which $g_{ij}(p) = \delta_{ij}$ and $\Gamma_{ij}^k(p) = 0$. Geodesics through p are the images of lines through 0.

Proof. Since $\exp_p(tv) = \gamma_v(t)$, the differential of $\exp_p$ at 0 sends v to $\dot{\gamma}_v(0) = v$, and the inverse function theorem applies. In normal coordinates for an orthonormal basis of $T_p M$, $g_{ij}(p) = \delta_{ij}$, and the lines $t \mapsto tv$ are geodesics, so the geodesic equation at $t = 0$ gives $\Gamma_{ij}^k(p) v^i v^j = 0$ for all v ; as Γ_{ij}^k is symmetric in i, j , it vanishes at p . $\square$

Theorem 38.4.2 (Gauss lemma). *Let $v \in \mathcal{E}_p$ and $w \in T_p M$, and identify $T_v(T_p M)$ with $T_p M$. Then $\langle d_v \exp_p(v), d_v \exp_p(w) \rangle = \langle v, w \rangle$.*

Proof. Since $\mathcal{E}_p$ is open and star-shaped, $f(s, t) = \exp_p(t(v + sw))$ is defined and smooth for $|s|$ small and $t \in [0, 1]$. Let $T = \partial_t f$ and $S = \partial_s f$. For fixed $s, t \mapsto f(s, t)$ is the geodesic with initial velocity $v + sw$, so $D_t T = 0$ and $|T|^2 = |v + sw|^2$ for all t . Using Proposition 38.3.1 and Lemma 38.3.2,

$$\partial_t \langle S, T \rangle = \langle D_t S, T \rangle + \langle S, D_t T \rangle = \langle D_s T, T \rangle = \frac{1}{2} \partial_s |T|^2 = \frac{1}{2} \partial_s |v + sw|^2 = \langle v + sw, w \rangle.$$

At $s = 0$ this is $\langle v, w \rangle$. Also $f(s, 0) = p$, so $S(s, 0) = 0$. Integrating, $\langle S, T \rangle(0, 1) = \langle v, w \rangle$. Finally $T(0, 1) = \frac{d}{dt} \exp_p(tv)|_{t=1} = d_v \exp_p(v)$ and $S(0, 1) = \frac{d}{ds} \exp_p(v + sw)|_{s=0} = d_v \exp_p(w)$. $\square$

Theorem 38.4.3 (Geodesics are locally minimising). *Let $\varepsilon > 0$ be such that $\exp_p$ is a diffeomorphism from $B(0, \varepsilon) \subseteq T_p M$ onto an open set U (a normal ball).*

1. *For $|v| < \varepsilon$, every curve from p to $\exp_p(v)$ has length $\geq |v|$, with equality if and only if it is a monotone reparametrisation of the radial geodesic $t \mapsto \exp_p(tv)$, $t \in [0, 1]$.*
2. *$d(p, \exp_p(v)) = |v|$ for $|v| < \varepsilon$, and $\exp_p(B(0, r))$ is the metric ball $\{x : d(p, x) < r\}$ for $0 < r \leq \varepsilon$.*

Proof. For $x \in U$ let $\rho(x) = |\exp_p^{-1}(x)|$, a continuous function, smooth on $U \setminus \{p\}$. Let $c: [a, b] \rightarrow U \setminus \{p\}$ be a piecewise smooth curve and write $c(t) = \exp_p(\rho(t)u(t))$ with $\rho(t) = \rho(c(t)) > 0$ and $|u(t)| = 1$. Then $\dot{c} = d \exp_p(\dot{\rho}u + \rho\dot{u})$ at the point ρu , with $\langle u, \dot{u} \rangle = 0$. By the Gauss lemma, $d \exp_p(u)$ has length 1 and is orthogonal to $d \exp_p(\rho\dot{u})$ (both at ρu), so

$$|\dot{c}|^2 = \dot{\rho}^2 + |d \exp_p(\rho\dot{u})|^2 \geq \dot{\rho}^2, \quad \text{hence} \quad L(c) \geq \int_a^b |\dot{\rho}| dt \geq \rho(b) - \rho(a). \quad (**)$$

Now let $c: [a, b] \rightarrow M$ go from p to $q = \exp_p(v)$, $0 < |v| = r < \varepsilon$. Suppose first that c stays in U . Let a' be the last time with $c(a') = p$; applying $(**)$ on $[a', b]$ for $a'' \downarrow a'$ gives $L(c) \geq r$. If $L(c) = r$, then c is constant on $[a, a']$, and on $(a', b]$ equality holds in $(**)$: $\dot{\rho} \geq 0$ and $d \exp_p(\rho\dot{u}) = 0$, so $\dot{u} = 0$ ($d \exp_p$ being injective on $B(0, \varepsilon)$), u is constant and c is a monotone reparametrisation of the radial geodesic. Conversely such reparametrisations have length r . If c leaves U , choose $r < r' < \varepsilon$. The set $K = \exp_p(\bar{B}(0, r'))$ is compact, hence closed, and contains p ; so c leaves K , and by continuity $\rho(c(t))$ takes the value r' at some first time t_2 with $c([a, t_2]) \subseteq K$. The first case applied to $c|_{[a, t_2]}$ gives $L(c) \geq r' > r$. This proves (1).

For (2), the radial geodesic has length $|v|$, so $d(p, \exp_p(v)) = |v|$ by (1). The ball $\exp_p(B(0, r))$ lies in the metric ball of radius r . Conversely, if $d(p, x) < r$, choose a curve c from p to x with $L(c) < r' < r$; if $x \notin \exp_p(B(0, r'))$, the argument above shows $L(c) \geq r'$, a contradiction. $\square$

Remark 38.4.4. Every point also has a *geodesically convex* neighbourhood, in which any two points are joined by a unique minimising geodesic, contained in the neighbourhood [dC92, Chapter 3].

Lemma 38.4.5 (Uniformly normal neighbourhoods). *Every $q \in M$ has a neighbourhood W and a $\delta > 0$ such that for every $x \in W$, $\exp_x$ is defined on $B(0, \delta) \subseteq T_x M$, is a diffeomorphism from it onto an open set, and $\exp_x(B(0, \delta)) \supseteq W$.*

Proof. Consider $E: \mathcal{E} \rightarrow M \times M$, $E(v) = (\pi(v), \exp(v))$. In coordinates (x, v) near $0_q \in TM$, its differential at 0_q is $\begin{pmatrix} I & 0 \\ I & I \end{pmatrix}$, since $\exp(0_x) = x$ and $d_0 \exp_x = \text{id}$; so by the inverse function theorem E is a diffeomorphism from a neighbourhood O of 0_q onto a neighbourhood of (q, q) . Shrinking, we may take $O = \{v \in T_x M : x \in W', |v| < \delta\}$ for a neighbourhood W' of q and some $\delta > 0$ (the norm is continuous on TM), and choose a neighbourhood $W \subseteq W'$ of q with $W \times W \subseteq E(O)$. For $x \in W$, $\exp_x$ is injective on $B(0, \delta)$ with invertible differential, hence a diffeomorphism onto an open set; and for $y \in W$, $(x, y) = E(v)$ with $v \in O$, that is $y = \exp_x(v)$ with $|v| < \delta$. $\square$

Lemma 38.4.6. *Let $c: [a, b] \rightarrow M$ be a piecewise smooth curve of constant speed with $L(c) = d(c(a), c(b))$. Then c is a geodesic; in particular it is smooth.*

Proof. Every subarc of c is also minimising, since otherwise replacing it by a shorter curve would shorten c . Let $t_0 \in [a, b]$, and take W and δ for $q = c(t_0)$ as in Lemma 38.4.5. For $t_1 < t_0 < t_2$ close to t_0 (or $t_1 = t_0$ at the endpoints), $c([t_1, t_2]) \subseteq W$ and $L(c|_{[t_1, t_2]}) < \delta$. Then $c|_{[t_1, t_2]}$ is a minimising curve from $x = c(t_1)$ to a point of the normal ball $\exp_x(B(0, \delta))$, so by Theorem 38.4.3(1) it is a monotone reparametrisation of a radial geodesic from x , and having constant speed, it is that geodesic, affinely reparametrised. So c is a geodesic near every t_0 . $\square$

Theorem 38.4.7 (Hopf–Rinow). *For a connected Riemannian manifold (M, g) the following are equivalent: (M, d) is a complete metric space; the geodesics are defined for all time; closed bounded subsets are compact. In that case any two points are joined by a minimising geodesic. In particular compact Riemannian manifolds are geodesically complete.*

Proof. We show more: the three conditions are also equivalent to (d) $\exp_p$ is defined on all of $T_p M$ for some p ; and (d) implies that every q is joined to p by a minimising geodesic.

Step 1: (d) gives minimising geodesics from p . Let $q \neq p$, $r = d(p, q)$, and choose $0 < \delta < r$ with $\exp_p$ a diffeomorphism on $B(0, \delta')$ for some $\delta' > \delta$. The geodesic sphere $S = \exp_p(\{|w| = \delta\})$ is compact; let $x_0 = \exp_p(\delta u) \in S$, $|u| = 1$, minimise $d(\cdot, q)$ on S . Every curve from p to q meets S (by the argument in the proof of Theorem 38.4.3), and points of S have distance δ from p , so $r = \delta + d(x_0, q)$. Let $\gamma(t) = \exp_p(tu)$, defined for all t by (d), and let $T = \{t \in [\delta, r] : d(\gamma(t), q) = r - t\}$, a closed set containing δ . Let $t_0 = \max T$ and suppose $t_0 < r$. Apply the same construction at $\gamma(t_0)$: for small $\eta > 0$ there is x_1 on the geodesic sphere of radius η about $\gamma(t_0)$ with $d(\gamma(t_0), q) = \eta + d(x_1, q)$, so $d(x_1, q) = r - t_0 - \eta$ and $d(p, x_1) \geq d(p, q) - d(x_1, q) = t_0 + \eta$. The curve going along γ from p to $\gamma(t_0)$ and then along the radial geodesic to x_1 has length $t_0 + \eta$, so it is minimising, and by Lemma 38.4.6 (after parametrising by arc length) it is a geodesic, hence equal to γ on $[0, t_0 + \eta]$. So $x_1 = \gamma(t_0 + \eta)$ and $t_0 + \eta \in T$, a contradiction. Hence $t_0 = r$, $\gamma(r) = q$ and $L(\gamma|_{[0, r]}) = r$.

Step 2: the equivalences. If closed bounded sets are compact, a Cauchy sequence, being bounded, has a convergent subsequence, hence converges; so M is complete. If M is complete, let $\gamma: [0, b) \rightarrow M$ be a maximal unit-speed geodesic with $b < \infty$. For $t_k \uparrow b$, $d(\gamma(t_k), \gamma(t_l)) \leq |t_k - t_l|$, so $(\gamma(t_k))$ is Cauchy and converges to some q . Take W, δ for q as in Lemma 38.4.5; then geodesics with initial point in W and speed 1 are defined on $(-\delta, \delta)$. Choose $t_k > b - \delta$ with $\gamma(t_k) \in W$; the geodesic with initial velocity $\dot{\gamma}(t_k)$ is defined up to time $t_k + \delta > b$ and extends γ by uniqueness, a contradiction. So geodesics are defined for all time, which implies (d). Finally, assume (d) and let K be closed and bounded, say $K \subseteq \{x : d(p, x) \leq R\}$. By Step 1, this ball is contained in $\exp_p(\bar{B}(0, R))$, which is compact; so K is compact. The last statements follow from Step 1 applied at every point, and from the compactness of closed bounded subsets of a compact space. $\square$

Counterexample 38.4.8. Completeness is needed. In $M = \mathbb{R}^2 \setminus \{0\}$ with the Euclidean metric, the geodesic $t \mapsto (1 - t, 0)$ is not defined at $t = 1$, the Cauchy sequence $(1/k, 0)$ has no limit, and the set $\{x \in M : |x| \leq 1\}$ is closed in M and bounded but not compact. The points $(1, 0)$ and $(-1, 0)$ have distance 2, the infimum of the lengths of curves avoiding the origin, but no curve of length 2 joins them in M , since in $\mathbb{R}^2$ the only such curve is the segment through the origin.

Remark 38.4.9. Minimising geodesics need not be unique: on the round sphere every half great circle joins two antipodal points, and has length π . And geodesics stop minimising: a great circle arc longer than π is a geodesic but not the shortest curve between its endpoints. Both phenomena are explained by conjugate points (Section 38.6).

38.5 Parallel transport and holonomy

The *holonomy group* $\text{Hol}_p(g) \subseteq \text{GL}(T_p M)$ of (M, g) at p is the holonomy group of the Levi-Civita connection (Definition 36.3.4), for piecewise smooth loops at p . The *restricted holonomy group* $\text{Hol}_p^0(g)$ is the subgroup of transports along null-homotopic loops.

Proposition 38.5.1. *Let (M, g) be connected and $p, q \in M$.*

1. $\text{Hol}_p(g) \subseteq \text{O}(T_p M, g_p)$.
2. For a curve σ from p to q , $\text{Hol}_q(g) = P_\sigma \text{Hol}_p(g) P_\sigma^{-1}$; so the holonomy groups at different points are conjugate.
3. $\text{Hol}_p^0(g)$ is a normal subgroup of $\text{Hol}_p(g)$, and $[\gamma] \mapsto P_\gamma^{-1} \text{Hol}_p^0(g)$ is a surjective homomorphism $\pi_1(M, p) \rightarrow \text{Hol}_p(g) / \text{Hol}_p^0(g)$. In particular $\text{Hol}_p = \text{Hol}_p^0$ if M is simply connected.
4. M is orientable if and only if $\text{Hol}_p(g) \subseteq \text{SO}(T_p M)$.
5. $\text{Hol}_p^0(g)$ is trivial if and only if g is flat, $R = 0$.

Proof. (1) is Proposition 38.3.3. (2) Loops γ at p correspond bijectively to loops $\bar{\sigma} * \gamma * \sigma$ at q , with transport $P_\sigma P_\gamma P_\sigma^{-1}$ (Proposition 36.3.2). (3) If β is null-homotopic and α is a loop at p , then $\bar{\alpha} * \beta * \alpha$ is null-homotopic, with transport $P_\alpha P_\beta P_\alpha^{-1}$; so Hol_p^0 is normal. If $\gamma \simeq \gamma'$, then $\bar{\gamma} * \gamma'$ is null-homotopic and $P_{\gamma'} P_\gamma^{-1} \in \text{Hol}_p^0$, so the map is well defined on homotopy classes, and it is multiplicative since $P_{\gamma * \delta}^{-1} = P_\gamma^{-1} P_\delta^{-1}$. Surjectivity is clear. (4) If M is oriented, a parallel frame along a curve is continuous, so the set of times at which it is positively oriented is open and closed; hence parallel transport preserves orientations. Conversely, if $\text{Hol}_p \subseteq \text{SO}(T_p M)$, fix an orientation of $T_p M$ and transport it along curves from p ; by (2) the result does not depend on the curve, and near each point the transported frames along radial geodesics form a smooth local frame, so the orientation is continuous. (5) By Theorem 36.3.5, $R = 0$ if and only if parallel transport depends only on the homotopy class of the path with fixed endpoints, which is equivalent to the triviality of transport around null-homotopic loops. $\square$

Theorem 38.5.2 (Holonomy principle). *Let (M, g) be connected and let E be a tensor bundle $T^{\otimes a} \otimes (T^*)^{\otimes b}$ of M with the connection induced by ∇ . A tensor $\tau_p \in E_p$ is the value at p of a parallel section of E if and only if it is invariant under $\text{Hol}_p(g)$, acting on E_p through its action on $T_p M$; the parallel section is then unique. Similarly, a subspace $V \subseteq T_p M$ is the fibre at p of a parallel subbundle of TM if and only if it is $\text{Hol}_p(g)$ -invariant.*

Proof. Parallel transport on E is induced by that on TM . If τ is parallel, then along every loop γ at p the section $\tau \circ \gamma$ is parallel, so $P_\gamma \tau_p = \tau_p$. Conversely, if τ_p is invariant, define $\tau_q = P_\sigma \tau_p$ for any curve σ from p to q . If σ' is another such curve, $P_{\sigma'}^{-1} P_\sigma$ is the transport along the loop $\sigma * \bar{\sigma}'$ at p , which lies in Hol_p and fixes τ_p ; so τ_q is well defined. Near a point q_0 , τ_q is the transport of τ_{q_0} along the radial geodesic from q_0 to q in a normal ball, which depends smoothly on q by smooth dependence of solutions of differential equations; so τ is smooth. For every curve c , $\tau \circ c$ is the transport of $\tau_{c(0)}$ along c , hence parallel; so $\nabla \tau = 0$. Uniqueness holds because a parallel section is determined by its value at one point of a connected manifold. The statement for subbundles is proved in the same way. $\square$

Example 38.5.3. 1. On $\mathbb{R}^n$ the Christoffel symbols vanish in the standard coordinates, so parallel transport is the identity and $\text{Hol} = \{1\}$; the same holds for the flat torus, since parallel transport in the quotient is induced from $\mathbb{R}^n$ and translations have trivial differential.

2. For the flat Klein bottle $\mathbb{R}^2/G$ (Example 38.2.5), a loop at $\pi(\tilde{x})$ lifts to a curve from $\tilde{x}$ to $h\tilde{x}$ for some $h \in G$; transport in $\mathbb{R}^2$ is the identity, and $d\pi_{h\tilde{x}} = d\pi_{\tilde{x}} \circ (dh)^{-1}$, so the holonomy is $(dh)^{-1}$, the inverse of the linear part of h . Hence $\text{Hol} = \{1, \text{diag}(1, -1)\}$: the Klein bottle is flat, $\text{Hol}^0 = \{1\}$, but $\text{Hol} \neq \{1\}$, and $\text{Hol} \not\subseteq \text{SO}(2)$, in accordance with non-orientability.
3. For the round sphere S^2 , $\text{Hol}_p = \text{Hol}_p^0$ is a nontrivial subgroup of $\text{SO}(2)$ by Proposition 38.5.1(3),(4),(5), as S^2 is simply connected, orientable and not flat. Since Hol_p^0 is a connected Lie subgroup (Theorem 38.5.4 below), it is $\text{SO}(2)$. Concretely, transport around the boundary of the octant $\{x, y, z \geq 0\}$ rotates tangent vectors by $\pi/2$, the area enclosed.

Theorem 38.5.4 (Ambrose–Singer). *$\text{Hol}_p(g)$ is a Lie subgroup of $\text{O}(T_p M)$ with identity component $\text{Hol}_p^0(g)$, and the Lie algebra of $\text{Hol}_p(g)$ is spanned by the endomorphisms $P_\sigma^{-1} \circ R(P_\sigma u, P_\sigma v) \circ P_\sigma$ of $T_p M$, for all curves σ starting at p and all $u, v \in T_p M$.*

This is [KN63, Chapter II]; see also [Joy00]. The theorem makes Proposition 38.5.1(5) quantitative: curvature measures infinitesimal holonomy.

Theorem 38.5.5 (de Rham decomposition). *If the holonomy representation of a connected Riemannian manifold on $T_p M$ is reducible, $T_p M = V_1 \oplus V_2$ with both summands Hol_p -invariant and nonzero, then M is locally isometric to a Riemannian product. If moreover M is complete and simply connected, it is globally isometric to a product $M_1 \times M_2$ of Riemannian manifolds with $T_p M_i = V_i$.*

Sketch of proof. By Theorem 38.5.2, V_1 and V_2 extend to parallel, mutually orthogonal distributions D_1 and D_2 . A parallel distribution is involutive, as $[X, Y] = \nabla_X Y - \nabla_Y X$ lies in D_i when X, Y do, so by the Frobenius theorem (Theorem 31.7.4) there are local coordinates (x, y) with D_1 spanned by the ∂_x and D_2 by the ∂_y ; the omitted verification is that in these coordinates the metric has the product form $g_1(x) + g_2(y)$, which follows from the parallelism of D_1 and D_2 . The global statement is [KN63, Chapter IV]. $\square$

Theorem 38.5.6 (Berger). *Let (M, g) be a simply connected Riemannian manifold of dimension n that is not locally a product (its holonomy representation is irreducible) and not locally symmetric ($\nabla R \neq 0$). Then $\text{Hol}(g)$ is one of: $\text{SO}(n)$; $\text{U}(m)$ with $n = 2m \geq 4$ (Kähler metrics); $\text{SU}(m)$ with $n = 2m \geq 4$ (Ricci-flat Kähler, Calabi–Yau); $\text{Sp}(m)$ with $n = 4m \geq 8$ (hyper-Kähler); $\text{Sp}(m)\text{Sp}(1)$ with $n = 4m \geq 8$ (quaternionic Kähler); G_2 with $n = 7$; $\text{Spin}(7)$ with $n = 8$.*

This is [Ber55]; Berger’s list also contained $\text{Spin}(9)$ in dimension 16, which was later shown to occur only for symmetric metrics. See [Joy00] for the modern treatment and for the existence of metrics with holonomy G_2 and $\text{Spin}(7)$.

Remark 38.5.7. By the holonomy principle, the reduction of the holonomy group to a subgroup of $\text{O}(n)$ is the same as the existence of a parallel tensor with that stabiliser. For instance $\text{Hol} \subseteq \text{U}(m)$ if and only if there is a parallel orthogonal almost complex structure J , that is, (M, g, J) is Kähler (Chapter 44); and $\text{Hol} \subseteq \text{SU}(m)$ if and only if there is moreover a parallel complex volume form, which for a simply connected Kähler manifold is equivalent to Ricci-flatness (Chapter 48). This is the dictionary between Riemannian holonomy and complex geometry that underlies Part VII.

38.6 Curvature

Definition 38.6.1. The *Riemann curvature tensor* is the curvature of the Levi-Civita connection (Proposition 36.2.4), $R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z$, viewed as the 4-tensor

$R(X, Y, Z, W) = g(R(X, Y)Z, W)$. For a plane $\sigma \subseteq T_p M$ with orthonormal basis u, v , the sectional curvature is $K(\sigma) = R(u, v, v, u)$; the Ricci tensor is $\text{Ric}(X, Y) = \sum_i R(e_i, X, Y, e_i)$ for an orthonormal frame, and the scalar curvature is $s = \sum_i \text{Ric}(e_i, e_i)$.

Proposition 38.6.2. *R is tensorial and satisfies $R(X, Y, Z, W) = -R(Y, X, Z, W) = -R(X, Y, W, Z) = R(Z, W, X, Y)$ and the first Bianchi identity $R(X, Y)Z + R(Y, Z)X + R(Z, X)Y = 0$. The sectional curvatures determine R .*

Proof. Tensoriality and the first antisymmetry are general facts about curvature. The second follows by polarising $R(X, Y, Z, Z) = 0$, which holds because the connection is metric: $0 = XYg(Z, Z)/2 - \dots$ expands to $g(R(X, Y)Z, Z)$. The first Bianchi identity follows from torsion-freeness and the Jacobi identity, and the symmetry in the pairs is a formal consequence of the other three, obtained by adding the Bianchi identities for the four cyclic permutations. The last statement is the standard polarisation argument [dC92, Chapter 4]. $\square$

Example 38.6.3. Euclidean space has $R = 0$; the round sphere of radius 1 has constant sectional curvature 1 and $\text{Ric} = (n - 1)g$; hyperbolic space has constant curvature -1 . A Lie group with a bi-invariant metric has $R(X, Y)Z = \frac{1}{4}[[X, Y], Z]$ for left-invariant fields, hence non-negative sectional curvature. A metric has constant sectional curvature κ if and only if $R(X, Y)Z = \kappa(\langle Y, Z \rangle X - \langle X, Z \rangle Y)$, since the right-hand side has the symmetries of Proposition 38.6.2 and sectional curvature κ .

Lemma 38.6.4. *For f as in Lemma 38.3.2 and a vector field V along f , $D_s D_t V - D_t D_s V = R(\partial_s f, \partial_t f)V$.*

Proof. Write $V = V^k(\partial_k \circ f)$. In $D_s D_t V$ the terms containing derivatives of the V^k are symmetric in s and t , so the left side is $V^k(D_s D_t - D_t D_s)(\partial_k \circ f)$. Now $D_t(\partial_k \circ f) = \partial_t f^i(\nabla_{\partial_i} \partial_k) \circ f$, and $D_s D_t(\partial_k \circ f) = \partial_s \partial_t f^i(\nabla_{\partial_i} \partial_k) \circ f + \partial_t f^i \partial_s f^j(\nabla_{\partial_j} \nabla_{\partial_i} \partial_k) \circ f$. Antisymmetrising and using $[\partial_i, \partial_j] = 0$ gives $\partial_s f^j \partial_t f^i R(\partial_j, \partial_i) \partial_k = R(\partial_s f, \partial_t f) \partial_k$. $\square$

Definition 38.6.5. A Jacobi field along a geodesic γ is a vector field J along γ with $D_t^2 J + R(J, \dot{\gamma})\dot{\gamma} = 0$. A point $\gamma(b)$ is conjugate to $\gamma(a)$ along γ if there is a nonzero Jacobi field vanishing at a and b .

The Jacobi equation is a linear second order equation, so a Jacobi field is determined by $J(a)$ and $D_t J(a)$, and the Jacobi fields along γ form a vector space of dimension $2n$.

Proposition 38.6.6. 1. *If $f(s, t)$ is a family of geodesics $t \mapsto f(s, t)$, then $J = \partial_s f(0, -)$ is a Jacobi field along $\gamma = f(0, -)$.*

2. *For $v \in \mathcal{E}_p$ and $w \in T_p M$, $J(t) = d_{tv} \exp_p(tw)$ is the Jacobi field along $\gamma(t) = \exp_p(tv)$ with $J(0) = 0$ and $D_t J(0) = w$.*

3. *$d_v \exp_p$ is singular if and only if $\exp_p(v)$ is conjugate to p along $t \mapsto \exp_p(tv)$, $t \in [0, 1]$.*

Proof. (1) With $T = \partial_t f$ and $S = \partial_s f$, Lemmas 38.3.2 and 38.6.4 give $D_t D_t S = D_t D_s T = D_s D_t T + R(T, S)T = R(T, S)T = -R(S, T)T$, since $D_t T = 0$. (2) Apply (1) to $f(s, t) = \exp_p(t(v + sw))$: $\partial_s f(0, t) = d_{tv} \exp_p(tw)$. It vanishes at $t = 0$, and $D_t S(0, 0) = D_s T(0, 0) = \frac{d}{ds}(v + sw) = w$, the covariant derivative along the constant curve $s \mapsto p$ being the ordinary derivative in $T_p M$. (3) By (2) and uniqueness, the Jacobi fields vanishing at 0 are exactly the $d_{tv} \exp_p(tw)$, and at $t = 1$ this is $d_v \exp_p(w)$. $\square$

Example 38.6.7. If g has constant sectional curvature κ and γ has unit speed, then for a parallel unit field $E \perp \dot{\gamma}$, $R(E, \dot{\gamma})\dot{\gamma} = \kappa E$ by Example 38.6.3, and the Jacobi fields normal to γ with $J(0) = 0$ are $J(t) = s_\kappa(t)E(t)$ with $s_\kappa(t) = t$, $\sin(\sqrt{\kappa}t)/\sqrt{\kappa}$ or $\sinh(\sqrt{-\kappa}t)/\sqrt{-\kappa}$ for $\kappa = 0$, > 0 , < 0 . On the unit sphere they vanish again at $t = \pi$: antipodal points are conjugate, which explains Remark 38.4.9. For $\kappa \leq 0$ there are no conjugate points.

Theorem 38.6.8 (Cartan–Hadamard). *Let (M, g) be complete and connected with sectional curvature $K \leq 0$. Then for every p , $\exp_p: T_p M \rightarrow M$ is a covering map; if M is simply connected, it is a diffeomorphism, so M is diffeomorphic to $\mathbb{R}^n$.*

Sketch of proof. Let J be a Jacobi field along a geodesic γ with $J(0) = 0$ and $D_t J(0) \neq 0$, and $\phi = |J|^2$. Then $\phi'' = 2|D_t J|^2 - 2\langle R(J, \dot{\gamma})\dot{\gamma}, J \rangle = 2|D_t J|^2 - 2K(J, \dot{\gamma})(|J|^2|\dot{\gamma}|^2 - \langle J, \dot{\gamma} \rangle^2) \geq 0$, where $K(J, \dot{\gamma})$ is the sectional curvature of the plane they span (if they are independent). So ϕ is convex with $\phi(0) = \phi'(0) = 0$ and $\phi''(0) > 0$, hence positive for $t > 0$: there are no conjugate points, and $\exp_p$ is a local diffeomorphism on $T_p M$ (defined everywhere by Theorem 38.4.7) by Proposition 38.6.6(3). The omitted verification is that a local diffeomorphism from a complete Riemannian manifold (here $T_p M$ with the pulled back metric, whose radial lines are geodesics) that is a local isometry is a covering map [dC92, Chapter 7]. $\square$

Theorem 38.6.9 (Bonnet–Myers). *Let (M, g) be complete and connected with $\text{Ric} \geq (n-1)\kappa g$ for some $\kappa > 0$. Then M has diameter at most $\pi/\sqrt{\kappa}$; hence M is compact, and $\pi_1(M)$ is finite.*

This is [dC92, Chapter 9]. The proof computes the second variation of length along a minimising geodesic of length $L > \pi/\sqrt{\kappa}$ in the directions $\sin(\pi t/L)E_i(t)$, for a parallel orthonormal frame $E_i \perp \dot{\gamma}$, and finds that their sum is negative, contradicting minimality. Finiteness of π_1 follows by applying the diameter bound to the universal cover with the pulled back metric, which is compact, so the fibres of the covering are finite.

Remark 38.6.10. Theorems 38.6.8 and 38.6.9 are the simplest *comparison theorems*, which compare a manifold with a space of constant curvature through its Jacobi fields. The Rauch comparison theorem [dC92, Chapter 10] and the Bishop–Gromov volume comparison [Lee18] are their refinements, and they lead to the sphere theorems and to Gromov’s compactness theorem.

Theorem 38.6.11 (Gauss–Bonnet). *Let M be a compact oriented Riemannian surface, with Gaussian curvature K (the sectional curvature of its tangent planes) and area form dA . Then*

$$\int_M K dA = 2\pi\chi(M).$$

Proof. Let J be the rotation by $+\pi/2$ in each tangent plane, for the metric and the orientation: $Je_1 = e_2$ for every oriented orthonormal frame (e_1, e_2) . It is a smooth endomorphism with $J^2 = -1$, so TM becomes a complex line bundle, with i acting as J , whose underlying oriented real bundle is TM . Near a point choose an oriented orthonormal frame (e_1, e_2) and define the 1-form ω by $\nabla_X e_1 = \omega(X)e_2$; since ∇ is metric, $\langle \nabla_X e_1, e_1 \rangle = 0$ and $\nabla_X e_2 = -\omega(X)e_1$. Hence $\nabla J = J\nabla$, so ∇ is a connection on the complex line bundle, and in the complex frame e_1 it reads $\nabla e_1 = \omega \otimes Je_1 = i\omega \otimes e_1$: its connection form is $\theta = i\omega$ and its curvature is $F = d\theta = i d\omega$ (Proposition 36.2.4).

We compare $d\omega$ with K . From the definition of the curvature tensor,

$$R(X, Y)e_1 = \nabla_X(\omega(Y)e_2) - \nabla_Y(\omega(X)e_2) - \omega([X, Y])e_2 = (X\omega(Y) - Y\omega(X) - \omega([X, Y]))e_2 = d\omega(X, Y)e_2,$$

the terms $\mp\omega(X)\omega(Y)e_1$ cancelling. By Definition 38.6.1 and Proposition 38.6.2, $K = R(e_1, e_2, e_2, e_1) = -R(e_1, e_2, e_1, e_2) = -\langle R(e_1, e_2)e_1, e_2 \rangle = -d\omega(e_1, e_2)$. So $d\omega = -K e^1 \wedge e^2 = -K dA$ and $F = -iK dA$.

By Chern–Weil theory (Theorem 37.2.2 and Definition 37.2.3), $c_1(TM)$ is represented by $\frac{i}{2\pi}F = \frac{1}{2\pi}K dA$ for this connection. For a complex line bundle the Euler class of the underlying oriented real bundle is c_1 (Theorem 37.5.1(2)), and $\int_M e(TM) = \chi(M)$ by Theorem 37.5.1(3). Hence $\frac{1}{2\pi} \int_M K dA = \chi(M)$. $\square$

Corollary 38.6.12. *Let Σ_g be the closed orientable surface of genus g , so that $\chi(\Sigma_g) = 2 - 2g$ (Example 19.7.5).*

1. *The integral of the Gaussian curvature does not depend on the metric: for the round unit sphere, $K = 1$ and the area is $4\pi = 2\pi \cdot 2$.*
2. *If Σ_g carries a metric with $K > 0$ everywhere, then $g = 0$; if it carries a metric with $K \geq 0$ everywhere, then $g \leq 1$; if it carries one with $K < 0$ everywhere, then $g \geq 2$.*
3. *The torus admits flat metrics, as $\mathbb{R}^2/\mathbb{Z}^2$, but no metric with $K > 0$ everywhere or $K < 0$ everywhere.*

Proof. (1) is Theorem 38.6.11. (2) The sign of $\int_M K dA = 2\pi(2 - 2g)$ is positive, non-negative or negative accordingly, and $2 - 2g > 0$ forces $g = 0$, $2 - 2g \geq 0$ forces $g \leq 1$, and $2 - 2g < 0$ forces $g \geq 2$. (3) The torus has $\chi = 0$, so K cannot have constant nonzero sign, while the flat metric induced from $\mathbb{R}^2$ has $K = 0$ (Example 38.6.3). $\square$

38.7 The Hodge star and the Laplacian

Definition 38.7.1. Let (M, g) be an oriented Riemannian n -manifold. The metric induces inner products on $\Lambda^k T_p^*M$ (Proposition 10.5.2), a volume form vol_g , and the *Hodge star* $*$: $\Omega^k(M) \rightarrow \Omega^{n-k}(M)$, characterised pointwise by $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \text{vol}_g$ (Theorem 10.5.3); it satisfies $** = (-1)^{k(n-k)}$ on k -forms. In coordinates $\text{vol}_g = \sqrt{\det(g_{ij})} dx^1 \wedge \cdots \wedge dx^n$ in a positively oriented chart.

Definition 38.7.2. On a compact oriented Riemannian manifold define the L^2 inner product on $\Omega^k(M)$ by

$$(\alpha, \beta) = \int_M \alpha \wedge *\beta = \int_M \langle \alpha, \beta \rangle \text{vol}_g,$$

the *codifferential* $\delta = (-1)^{n(k+1)+1} * d* : \Omega^k(M) \rightarrow \Omega^{k-1}(M)$, and the *Hodge Laplacian* $\Delta = d\delta + \delta d$. A form is *harmonic* if $\Delta\alpha = 0$; the space of harmonic k -forms is $\mathcal{H}^k(M)$.

Proposition 38.7.3. *On a compact oriented Riemannian manifold:*

1. $(d\alpha, \beta) = (\alpha, \delta\beta)$ for $\alpha \in \Omega^{k-1}$, $\beta \in \Omega^k$: the codifferential is the formal adjoint of d ;
2. Δ is self-adjoint and non-negative: $(\Delta\alpha, \alpha) = \|d\alpha\|^2 + \|\delta\alpha\|^2$;
3. $\Delta\alpha = 0$ if and only if $d\alpha = 0$ and $\delta\alpha = 0$;
4. $*\Delta = \Delta*$, so $*$ maps harmonic k -forms isomorphically to harmonic $(n - k)$ -forms;
5. Δ on functions is $-\sum_i \partial_i^2$ in normal coordinates at a point, and in general $\Delta f = -\frac{1}{\sqrt{\det g}} \sum_{i,j} \partial_i(\sqrt{\det g} g^{ij} \partial_j f)$.

Proof. (1) By Stokes' theorem (Theorem 34.4.4), $0 = \int_M d(\alpha \wedge *\beta) = \int_M d\alpha \wedge *\beta + (-1)^{k-1} \int_M \alpha \wedge d*\beta$. Using $** = (-1)^{k(n-k)}$ one rewrites $d*\beta = \pm **d*\beta$, and the sign in the definition of δ is chosen exactly so that the identity becomes $(d\alpha, \beta) = (\alpha, \delta\beta)$.

(2) $(\Delta\alpha, \beta) = (d\delta\alpha, \beta) + (\delta d\alpha, \beta) = (\delta\alpha, \delta\beta) + (d\alpha, d\beta)$, which is symmetric; taking $\beta = \alpha$ gives the formula. (3) If $\Delta\alpha = 0$ then $\|d\alpha\|^2 + \|\delta\alpha\|^2 = 0$, so both vanish; the converse is clear. (4) follows from the definition of δ and $** = \pm 1$: $*d\delta = \delta d*$ and $*\delta d = \delta d*$ up to the signs, which cancel. (5) is a computation with the coordinate formula for $*$ and d , using $\text{vol}_g = \sqrt{\det g} dx^1 \wedge \cdots \wedge dx^n$; in normal coordinates at p the first derivatives of g vanish, so $\Delta f(p) = -\sum_i \partial_i^2 f(p)$. $\square$

Remark 38.7.4. The Levi-Civita connection gives a second Laplacian, the *connection Laplacian* $\nabla^*\nabla$, and the two differ by curvature: the *Weitzenböck formula* states $\Delta = \nabla^*\nabla + \mathcal{R}$, where $\mathcal{R}$ is an endomorphism built from the curvature tensor; on 1-forms $\mathcal{R} = \text{Ric}$. Since $(\nabla^*\nabla\alpha, \alpha) = \|\nabla\alpha\|^2 \geq 0$, this gives vanishing theorems: if $\text{Ric} > 0$ on a compact oriented manifold, every harmonic 1-form vanishes, hence $b_1(M) = 0$ by the Hodge theorem (Bochner's theorem, Chapter 40). For proofs see [War83, Chapter 6] and [Lee18, Chapter 4].

Example 38.7.5. On a compact connected oriented manifold the harmonic 0-forms are the constants, since $\Delta f = 0$ implies $df = 0$; by (4) the harmonic n -forms are the multiples of vol_g . On the flat torus $\mathbb{R}^n/\mathbb{Z}^n$, the harmonic forms are exactly the forms with constant coefficients in the coordinates dx^i , a space of dimension $\binom{n}{k}$, which matches $b_k(T^n)$ (Exercise 35.5.1).

38.8 Exercises

Exercise 38.8.1. Verify that $\delta = -*d*$ on k -forms when n is even, and compute δ on 1-forms in $\mathbb{R}^3$, identifying it with minus the divergence.

Exercise 38.8.2. Show that in dimension $n = 2k$ the Hodge star on k -forms depends only on the conformal class of the metric, and that $*^2 = (-1)^k$ on Ω^k .

Solution. Under $g \mapsto \lambda g$ with $\lambda > 0$, the inner product on k -forms scales by λ^{-k} and the volume form by $\lambda^{n/2} = \lambda^k$, so $*$ is unchanged by the characterisation $\alpha \wedge *\beta = \langle \alpha, \beta \rangle \text{vol}$. The sign is $(-1)^{k(n-k)} = (-1)^{k^2} = (-1)^k$. $\square$

Exercise 38.8.3. Compute the Laplacian on functions on the round sphere S^2 in spherical coordinates, and verify that the restrictions of linear functions of $\mathbb{R}^3$ are eigenfunctions with eigenvalue 2.

Exercise 38.8.4. Let G be a compact Lie group with a bi-invariant metric. Show that the geodesics through e are the one-parameter subgroups, and that $\nabla_X Y = \frac{1}{2}[X, Y]$ for left-invariant fields.

Exercise 38.8.5. Show that the geodesics of the round sphere $S^n \subseteq \mathbb{R}^{n+1}$ are the great circles $\gamma(t) = \cos(t)p + \sin(t)v$ with $|p| = |v| = 1$ and $v \perp p$ (up to affine reparametrisation).

Solution. The reflection ρ of $\mathbb{R}^{n+1}$ fixing the plane spanned by p and v pointwise preserves S^n and its metric, so it is an isometry, and it maps geodesics to geodesics. The geodesic γ_v with $\gamma_v(0) = p$, $\dot{\gamma}_v(0) = v$ is mapped to a geodesic with the same initial data, hence to itself by uniqueness; so γ_v lies in the fixed set of ρ on S^n , the great circle through p and v . Having constant speed 1 (Proposition 38.3.3) and starting at p with velocity v , it is $\cos(t)p + \sin(t)v$. $\square$

Exercise 38.8.6. Show that the holonomy group of a Riemannian product $M_1 \times M_2$ at (p_1, p_2) is $\text{Hol}_{p_1}(M_1) \times \text{Hol}_{p_2}(M_2)$, acting on $T_{p_1}M_1 \oplus T_{p_2}M_2$.

Solution. The Levi-Civita connection of the product metric is the direct sum of those of the factors, since the direct sum connection is metric and torsion-free. So a vector (v_1, v_2) is transported along a curve (c_1, c_2) to $(P_{c_1}v_1, P_{c_2}v_2)$. Loops at (p_1, p_2) are exactly the pairs of loops at p_1 and p_2 , which gives the claim. $\square$

Exercise 38.8.7. Let (M, g) be connected with $\text{Hol}_p(g) = \{1\}$. Show that M is flat and that TM is trivial.

Solution. By Theorem 38.5.2, each vector of an orthonormal basis of $T_p M$ extends to a parallel vector field E_i ; the E_i are orthonormal everywhere (parallel transport is an isometry), so they trivialise TM . For parallel fields $R(X, Y)E_i = \nabla_X \nabla_Y E_i - \nabla_Y \nabla_X E_i - \nabla_{[X, Y]} E_i = 0$, so $R = 0$. $\square$

Chapter 39

Elliptic Operators on Compact Manifolds

39.1 Introduction

This chapter contains the analysis behind the Hodge theorem. An elliptic differential operator on a compact manifold is Fredholm between Sobolev spaces (Theorem 39.5.2), its kernel consists of smooth sections, and a self-adjoint elliptic operator has a discrete spectrum with smooth eigensections (Theorem 39.5.4). Applied to the Hodge Laplacian these statements give the decomposition of the space of forms and the existence of harmonic representatives (Chapter 40). References: [War83, Chapter 6], [Wel08, Chapter 4], [Hör90a].

Prerequisites. Chapter 38 (Riemannian Geometry) and Chapter 30 (Distributions and Sobolev Spaces).

Throughout, M is a compact smooth manifold, and E, F are complex vector bundles on M with Hermitian metrics; M carries a fixed Riemannian metric and, when needed, an orientation, so that sections can be integrated.

39.2 Differential operators and symbols

Definition 39.2.1. A linear map $P: \Gamma(M, E) \rightarrow \Gamma(M, F)$ is a *differential operator of order at most m* if in local coordinates and local frames it is given by $Pu = \sum_{|\alpha| \leq m} A_\alpha(x) \partial^\alpha u$ with smooth matrix-valued coefficients A_α . Its *principal symbol* is

$$\sigma_m(P)(x, \xi) = \sum_{|\alpha|=m} A_\alpha(x) (i\xi)^\alpha \in \text{Hom}(E_x, F_x), \quad \xi \in T_x^*M,$$

and P is *elliptic* if $\sigma_m(P)(x, \xi)$ is invertible for all x and all $\xi \neq 0$.

Proposition 39.2.2. *The principal symbol is well defined: it does not depend on the coordinates and frames, and it is characterised by*

$$\sigma_m(P)(x, d_x f) u(x) = \frac{i^m}{m!} P((f - f(x))^m u)(x)$$

for $f \in C^\infty(M)$ and $u \in \Gamma(M, E)$. Symbols are multiplicative, $\sigma_{m+m'}(P'P) = \sigma_{m'}(P')\sigma_m(P)$, so a composite of elliptic operators is elliptic, and the symbol of the formal adjoint is the adjoint of the symbol.

Proof. Expanding $P((f - f(x))^m u)$ by the Leibniz rule, all terms in which fewer than m derivatives fall on $(f - f(x))^m$ vanish at x , and the remaining term is $\sum_{|\alpha|=m} A_\alpha(x) \partial^\alpha ((f -$

$f(x))^m)(x)u(x) = m! \sum_{|\alpha|=m} A_\alpha(x)(d_x f)^\alpha u(x)$, which gives the formula. The right-hand side is defined without reference to coordinates, so the symbol is invariantly defined, and multiplicativity follows from the same formula applied twice. $\square$

- Example 39.2.3.** 1. The Hodge Laplacian $\Delta = d\delta + \delta d$ on $\Omega^k(M)$ has principal symbol $|\xi|^2 \text{id}$ (compute in normal coordinates, where the first-order terms of the metric vanish); it is elliptic.
2. The operator $d + \delta: \Omega^{\text{even}} \rightarrow \Omega^{\text{odd}}$ has symbol $i(\xi \wedge -\iota_\xi)$, whose square is $-|\xi|^2$, so it is invertible for $\xi \neq 0$ and $d + \delta$ is elliptic. Its index is the Euler characteristic (Chapter 40).
3. On a complex manifold, $\bar{\partial} + \bar{\partial}^*$ is elliptic (Chapter 43); the Cauchy–Riemann operator $\bar{\partial}$ itself is elliptic in one variable.
4. Not every operator is elliptic: the wave operator $\partial_t^2 - \Delta$ and the heat operator $\partial_t - \Delta$ are not.

Proposition 39.2.4 (Fundamental solution of the Laplacian). *On $\mathbb{R}^3$, the locally integrable function $E(x) = -\frac{1}{4\pi|x|}$ satisfies $\sum_i \partial_i^2 E = \delta_0$ in $\mathcal{D}'(\mathbb{R}^3)$. Consequently, for $f \in \mathcal{D}(\mathbb{R}^3)$ the convolution $u = E * f$ solves $\sum_i \partial_i^2 u = f$.*

Proof. E is locally integrable, as $\int_{|x| \leq 1} |x|^{-1} dx = 4\pi \int_0^1 r dr$ in spherical coordinates (Theorem 26.8.2). Away from 0, a direct computation gives $\Delta E = 0$, where $\Delta = \sum_i \partial_i^2$ here. Let $\varphi \in \mathcal{D}(\mathbb{R}^3)$ and $A_\varepsilon = \{\varepsilon \leq |x| \leq R\}$ with R large. By the divergence theorem (Example 34.4.7) applied to $E \nabla \varphi - \varphi \nabla E$ on A_ε (Green's identity), and since φ vanishes near $|x| = R$,

$$\int_{A_\varepsilon} E \Delta \varphi = \int_{A_\varepsilon} (E \Delta \varphi - \varphi \Delta E) = - \int_{|x|=\varepsilon} (E \partial_r \varphi - \varphi \partial_r E) dA,$$

where ∂_r is the radial derivative, the sign coming from the inner boundary sphere having outward normal $-x/|x|$. On $|x| = \varepsilon$, $E = -1/(4\pi\varepsilon)$ and $\partial_r E = 1/(4\pi\varepsilon^2)$; the area is $4\pi\varepsilon^2$. So the first term is at most $\varepsilon \sup |\nabla \varphi| \rightarrow 0$, and the second is $\frac{1}{4\pi\varepsilon^2} \int_{|x|=\varepsilon} \varphi dA \rightarrow \varphi(0)$ by continuity. Letting $\varepsilon \rightarrow 0$ (dominated convergence on the left), $(\Delta E)(\varphi) = \int E \Delta \varphi = \varphi(0)$. For the last statement, $\Delta(E * f) = E * \Delta f$ by differentiation under the integral sign, and $(E * \Delta f)(x) = \int E(y) \Delta_y f(x - y) dy = f(x)$ by the first part applied to $\varphi(y) = f(x - y)$. $\square$

39.3 Sobolev spaces of sections

Definition 39.3.1. Fix a finite atlas of M with trivialisations of E and a subordinate partition of unity (ψ_i) . For $s \in \mathbb{R}$ and $u \in \Gamma(M, E)$ set $\|u\|_s^2 = \sum_i \|\psi_i u\|_{H^s(\mathbb{R}^n)}^2$, using the coordinates and frames to view $\psi_i u$ as a compactly supported $\mathbb{C}^r$ -valued function. The completion is the Sobolev space $H^s(E)$ of sections.

Proposition 39.3.2. *The space $H^s(E)$ and the equivalence class of its norm do not depend on the choices. Moreover $H^0(E) = L^2(E)$ with the inner product $\int_M \langle u, v \rangle \text{vol}$; the inclusions $H^s(E) \rightarrow H^t(E)$ for $s > t$ are compact; $\bigcap_s H^s(E) = \Gamma(M, E)$; and a differential operator of order m extends to a bounded map $H^{s+m}(E) \rightarrow H^s(F)$ for every s .*

Proof. Changing charts and frames changes $\psi_i u$ by composition with a diffeomorphism and multiplication by a smooth matrix, which are bounded operations on H^s of compactly supported functions [War83, Chapter 6]; hence the norms are equivalent. For $s = 0$ the statement is Plancherel's theorem plus the fact that the two ways of integrating differ by

a bounded density. Compactness follows from the Rellich theorem (Theorem 30.4.4, in the form of Counterexample 30.4.5 for compactly supported distributions) applied to each $\psi_i u$. The identification of $\bigcap_s H^s$ with smooth sections is the Sobolev embedding theorem (Theorem 30.4.3), and boundedness of differential operators follows because differentiation lowers s by the order and multiplication by smooth functions is bounded. $\square$

39.4 Elliptic estimates and regularity

Theorem 39.4.1. *Let P be an elliptic differential operator of order m from E to F over the compact manifold M . For every s there is C_s with*

$$\|u\|_{s+m} \leq C_s (\|Pu\|_s + \|u\|_s) \quad \text{for all } u \in H^{s+m}(E),$$

and (elliptic regularity) if $u \in H^t(E)$ for some t and $Pu \in H^s(F)$, then $u \in H^{s+m}(E)$; in particular Pu smooth implies u smooth.

Sketch of proof. Locally one freezes the coefficients: near a point x_0 , P differs from the constant-coefficient elliptic operator $P_0 = \sum_{|\alpha|=m} A_\alpha(x_0) \partial^\alpha$ by an operator whose top-order coefficients are small on a small neighbourhood and whose remaining terms have order $< m$. The constant-coefficient estimate (Theorem 30.5.3, transported to $\mathbb{R}^n$ by a cut-off and to the torus) then gives $\|u\|_{s+m} \leq C(\|Pu\|_s + \|u\|_{s+m-1})$ for u supported in that neighbourhood, and the error term is absorbed using the interpolation inequality $\|u\|_{s+m-1} \leq \varepsilon \|u\|_{s+m} + C_\varepsilon \|u\|_s$, which follows from $\langle \xi \rangle^{s+m-1} \leq \varepsilon \langle \xi \rangle^{s+m} + C_\varepsilon \langle \xi \rangle^s$. Summing over a finite cover with a partition of unity gives the global estimate. Regularity is proved by the same estimates applied to difference quotients, or by constructing a *parametrix*: an operator Q with $QP = I + S$ and $PQ = I + S'$ where S, S' are smoothing operators; then $u = Q(Pu) - Su$ is as regular as Pu allows. The omitted verifications are the construction of the parametrix and the coordinate invariance of the estimates; see [War83, Chapter 6], [Wel08, Chapter 4] or [Hör90a, Chapters 17–18]. $\square$

Counterexample 39.4.2. Regularity fails for non-elliptic operators. On $\mathbb{R}^2$ with coordinates (t, x) let $P = \partial_t^2 - \partial_x^2$, whose symbol $-\tau^2 + \xi^2$ vanishes on the lines $\tau = \pm \xi$. For every locally integrable g on $\mathbb{R}$, the function $u(t, x) = g(x - t)$ satisfies $Pu = 0$ in $\mathcal{D}'(\mathbb{R}^2)$. Indeed, in the coordinates $s = x - t$, $r = x + t$ one has $\partial_t = \partial_r - \partial_s$, $\partial_x = \partial_r + \partial_s$, so $\partial_t^2 - \partial_x^2 = -4\partial_r \partial_s$ and $dt dx = \frac{1}{2} ds dr$. For a test function φ , written $\tilde{\varphi}(s, r)$ in the new coordinates,

$$\langle Pu, \varphi \rangle = \int u (\partial_t^2 - \partial_x^2) \varphi dt dx = -2 \int g(s) \left(\int \partial_r (\partial_s \tilde{\varphi}) dr \right) ds = 0,$$

since $\partial_s \tilde{\varphi}$ has compact support in r . Taking g the indicator function of $[0, \infty)$ gives a discontinuous solution of $Pu = 0$, although 0 is smooth. For elliptic operators this cannot happen: a distribution u with $\sum_j \partial_j^2 u$ smooth is smooth, by the local form of Theorem 39.4.1.

39.5 Fredholm theory and the spectral decomposition

Definition 39.5.1. The *formal adjoint* P^* of a differential operator P is the differential operator of the same order characterised by $\int_M \langle Pu, v \rangle = \int_M \langle u, P^*v \rangle$ for smooth sections; it exists, is unique, and has symbol $\sigma_m(P^*)(x, \xi) = (-1)^m \sigma_m(P)(x, \xi)^*$, so P^* is elliptic if P is. The operator P is *formally self-adjoint* if $P = P^*$; the Hodge Laplacian is (Proposition 38.7.3).

Theorem 39.5.2. *Let P be elliptic of order m on a compact manifold. Then:*

1. $\ker P = \{u \in \Gamma(M, E) : Pu = 0\}$ is finite-dimensional and consists of smooth sections, and the same holds for $\ker P^*$;
2. $P : H^{s+m}(E) \rightarrow H^s(F)$ is Fredholm for every s , with kernel and cokernel independent of s ;
3. there is an orthogonal decomposition $L^2(F) = \operatorname{im}(P) \oplus \ker P^*$, and $Pu = f$ is solvable with u smooth if and only if f is smooth and orthogonal to $\ker P^*$.

Proof. (1) Elements of the kernel are smooth by elliptic regularity. On $\ker P$ the elliptic estimate gives $\|u\|_{s+m} \leq C_s \|u\|_s$, so the H^s and H^{s+m} norms are equivalent there; since the inclusion $H^{s+m} \rightarrow H^s$ is compact (Proposition 39.3.2), the unit ball of $\ker P$ is compact, and $\ker P$ is finite-dimensional (Proposition 28.2.2(3)).

(2) Closed range: let $K \subseteq H^{s+m}$ be the orthogonal complement of $\ker P$. We claim $\|u\|_{s+m} \leq C \|Pu\|_s$ on K . If not, there are $u_j \in K$ with $\|u_j\|_{s+m} = 1$ and $Pu_j \rightarrow 0$; by compactness of the inclusion, after passing to a subsequence $u_j \rightarrow u$ in H^s , and the elliptic estimate gives $\|u_j - u_l\|_{s+m} \leq C(\|P(u_j - u_l)\|_s + \|u_j - u_l\|_s) \rightarrow 0$, so $u_j \rightarrow u$ in H^{s+m} with $\|u\|_{s+m} = 1$, $Pu = 0$ and $u \in K$, a contradiction. Hence P has closed range, and with (1) applied to P and P^* it is Fredholm; the kernel is independent of s by regularity, and so is the cokernel by (3).

(3) The operator P with domain $H^m(E)$ and its formal adjoint satisfy: $f \in L^2(F)$ is orthogonal to $\operatorname{im}(P)$ if and only if $\int \langle Pu, f \rangle = 0$ for all smooth u , that is, $P^*f = 0$ in the sense of distributions, which by elliptic regularity means $f \in \ker P^*$ and f smooth. Since the range is closed, $L^2 = \overline{\operatorname{im} P} \oplus \ker P^* = \operatorname{im} P \oplus \ker P^*$, where $\operatorname{im} P$ is taken on H^m ; and if f is smooth and orthogonal to $\ker P^*$, a solution $u \in H^m$ exists and is smooth by regularity. $\square$

Example 39.5.3. On the circle $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ let $P = d/d\theta$ on complex functions. Its symbol $i\xi$ is invertible for $\xi \neq 0$, so P is elliptic of order 1, and integration by parts gives $P^* = -d/d\theta$. Both kernels consist of the constants. Theorem 39.5.2 predicts that $u' = f$ has a smooth periodic solution exactly when f is orthogonal to $\ker P^*$, that is, when $\int_0^{2\pi} f d\theta = 0$; this is visible directly, since $u(\theta) = \int_0^\theta f$ is periodic exactly in that case. The index is $\dim \ker P - \dim \ker P^* = 0$. For the perturbation $P_c = d/d\theta + c$ with a constant $c \in \mathbb{C}$, the kernel is spanned by $e^{-c\theta}$ when this function is 2π -periodic, that is, when $c \in i\mathbb{Z}$, and is 0 otherwise; so the dimension of the kernel jumps. The adjoint $P_c^* = -d/d\theta + \bar{c}$ has kernel spanned by $e^{\bar{c}\theta}$ for exactly the same values of c , so the index stays 0, as predicted by Remark 39.5.5, since the principal symbol does not depend on c .

Theorem 39.5.4. Let P be a formally self-adjoint elliptic operator of order $m > 0$ on a compact manifold, with $P \geq 0$ in the sense that $\int \langle Pu, u \rangle \geq 0$. Then $L^2(E)$ has a complete orthonormal system of smooth eigensections of P , the eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$ are real, have finite multiplicity, and tend to $+\infty$. Every $u \in L^2(E)$ is the L^2 -limit of its eigenfunction expansion, which converges in $\Gamma(M, E)$ when u is smooth.

Proof. By Theorem 39.5.2 applied to $P + I$, which is elliptic and injective on smooth sections (as $\int \langle (P + I)u, u \rangle \geq \|u\|_{L^2}^2$), the map $P + I : H^m(E) \rightarrow L^2(E)$ is bijective, with bounded inverse G by the open mapping theorem (Theorem 28.4.2). Composing with the compact inclusion $H^m \rightarrow L^2$ shows that $G : L^2 \rightarrow L^2$ is compact, and it is self-adjoint and positive because $P + I$ is. By the spectral theorem for compact self-adjoint operators (Theorem 28.5.3) there is an orthonormal basis of $L^2(E)$ of eigenvectors of G , with eigenvalues $\mu_k > 0$ tending to 0 (they are nonzero since G is injective). Then $Pu_k = (\mu_k^{-1} - 1)u_k$, so the u_k are eigensections of P with eigenvalues $\lambda_k = \mu_k^{-1} - 1 \rightarrow \infty$, smooth by elliptic regularity, and $\lambda_k \geq 0$ by positivity. Finite multiplicity is finite-dimensionality of the eigenspaces, which are kernels of the elliptic operators $P - \lambda_k$. For smooth u , the coefficients $\langle u, u_k \rangle$ decay faster than any

power of λ_k , because $\lambda_k^N \langle u, u_k \rangle = \langle P^N u, u_k \rangle$ is bounded; with the Sobolev embedding this gives convergence in every H^s , hence in $\Gamma(M, E)$. $\square$

Remark 39.5.5. The index $\text{ind } P = \dim \ker P - \dim \ker P^*$ depends only on the homotopy class of the principal symbol, and the Atiyah–Singer index theorem computes it as a characteristic number of the symbol; for $d + \delta$ this gives the Gauss–Bonnet theorem, and for the Dolbeault operator the Riemann–Roch–Hirzebruch theorem (Chapter 47). See [LM89].

39.6 Exercises

Exercise 39.6.1. Compute the principal symbol of d , of δ and of Δ on forms, and verify that Δ is elliptic while d is not (for $n \geq 2$).

Solution. $\sigma_1(d)(x, \xi)\alpha = i\xi \wedge \alpha$ and $\sigma_1(\delta)(x, \xi)\alpha = -i\iota_\xi \alpha$, so $\sigma_2(\Delta)(x, \xi) = \xi \wedge \iota_\xi + \iota_\xi(\xi \wedge -) = |\xi|^2 \text{id}$ by the algebraic identity of Proposition 10.4.5. The symbol of d is not injective on Λ^k for $k \geq 1$, since $\xi \wedge \xi = 0$. $\square$

Exercise 39.6.2. Compute the spectrum of the Laplacian on functions on the flat torus $\mathbb{R}^n / \mathbb{Z}^n$ and verify Theorem 39.5.4 directly.

Solution. The functions $e^{2\pi i k \cdot x}$, $k \in \mathbb{Z}^n$, are eigenfunctions with eigenvalues $4\pi^2 |k|^2$, and they form an orthonormal basis of L^2 (Example 28.3.6); multiplicities are the numbers of lattice points on spheres, and the eigenvalues tend to infinity. $\square$

Exercise 39.6.3. Show that a formally self-adjoint elliptic operator on a compact manifold has index zero, and that the index of an elliptic operator changes sign when P is replaced by P^* .

Exercise 39.6.4. Give an example of a compact manifold and a smooth function f for which $\Delta u = f$ has no solution, and characterise the f for which it does.

Solution. On a compact connected oriented M , $\ker \Delta$ on functions consists of the constants, so by Theorem 39.5.2 the equation is solvable exactly when $\int_M f \text{ vol} = 0$; any f with nonzero integral, for instance $f = 1$, gives no solution. $\square$

Chapter 40

Harmonic Forms and the Hodge Theorem

40.1 Introduction

On a compact oriented Riemannian manifold every de Rham cohomology class contains exactly one harmonic form (Theorem 40.2.2). This is the theorem that makes Hodge theory possible: it turns a topological object, the cohomology of a manifold, into the solution space of an elliptic equation, on which the metric and, in the complex case, the complex structure act. The consequences drawn here are the Hodge decomposition of the space of forms, Poincaré duality via the Hodge star, and Bochner's vanishing theorem; the complex analogue is in Chapter 45. References: [War83, Chapter 6], [Wel08, Chapter 4], [GH78, Chapter 0], [Voi02b, Chapter 5].

Prerequisites. Chapter 39 (Elliptic Operators on Compact Manifolds) and Chapter 35 (de Rham Cohomology).

Throughout, (M, g) is a compact oriented Riemannian n -manifold, with the L^2 inner product, the codifferential δ and the Laplacian $\Delta = d\delta + \delta d$ of Definition 38.7.2, and $\mathcal{H}^k = \mathcal{H}^k(M, g) = \{\alpha \in \Omega^k(M) : \Delta\alpha = 0\}$.

40.2 The Hodge decomposition

Theorem 40.2.1. *For each k the space $\mathcal{H}^k$ is finite-dimensional and consists of the forms with $d\alpha = 0$ and $\delta\alpha = 0$, and there is an orthogonal direct sum decomposition*

$$\Omega^k(M) = \mathcal{H}^k \oplus d\Omega^{k-1}(M) \oplus \delta\Omega^{k+1}(M),$$

with $\ker d = \mathcal{H}^k \oplus d\Omega^{k-1}$ and $\ker \delta = \mathcal{H}^k \oplus \delta\Omega^{k+1}$. Moreover there is a unique operator $G: \Omega^k(M) \rightarrow \Omega^k(M)$, the Green operator, that vanishes on $\mathcal{H}^k$, maps $(\mathcal{H}^k)^\perp$ into itself, and satisfies $\alpha = H\alpha + \Delta G\alpha$, where H is the orthogonal projection onto $\mathcal{H}^k$; it commutes with d , δ and Δ .

Proof. Δ is elliptic, formally self-adjoint and non-negative (Example 39.2.3 and Proposition 38.7.3), so by Theorem 39.5.2 its kernel $\mathcal{H}^k$ is finite-dimensional and consists of smooth forms, and

$$\Omega^k(M) = \mathcal{H}^k \oplus \Delta\Omega^k(M)$$

orthogonally: indeed $L^2 = \text{im } \Delta \oplus \ker \Delta$ by Theorem 39.5.2(3), and a smooth form orthogonal to $\mathcal{H}^k$ is $\Delta\beta$ for a smooth β by elliptic regularity. The description of $\mathcal{H}^k$ as $\ker d \cap \ker \delta$ is Proposition 38.7.3(3).

Since $\Delta\beta = d(\delta\beta) + \delta(d\beta)$, the second summand is contained in $d\Omega^{k-1} + \delta\Omega^{k+1}$, so the three spaces span Ω^k . They are pairwise orthogonal: harmonic forms are orthogonal to $d\Omega^{k-1}$ because $(\alpha, d\beta) = (\delta\alpha, \beta) = 0$, and similarly to $\delta\Omega^{k+1}$; and $(d\beta, \delta\gamma) = (dd\beta, \gamma) = 0$. Hence the sum is direct and orthogonal. If $d\alpha = 0$ and $\alpha = \eta + d\beta + \delta\gamma$, then $0 = (d\alpha, \gamma) = (d\delta\gamma, \gamma) = \|\delta\gamma\|^2$, so $\alpha \in \mathcal{H}^k \oplus d\Omega^{k-1}$; the statement for $\ker \delta$ is symmetric.

For the Green operator, define $G\alpha = 0$ on $\mathcal{H}^k$ and, on the orthogonal complement, $G\alpha = \beta$ where β is the unique solution of $\Delta\beta = \alpha$ orthogonal to $\mathcal{H}^k$ (Theorem 39.5.2(3) gives existence and smoothness, and uniqueness holds because two solutions differ by a harmonic form). Then $\alpha = H\alpha + \Delta G\alpha$ by construction. Since Δ commutes with d , δ and Δ , and these operators preserve $\mathcal{H}^\bullet$ and its orthogonal complement (for instance $d\mathcal{H}^k = 0$ and $(d\alpha, \eta) = (\alpha, \delta\eta) = 0$ for η harmonic), the uniqueness of G forces $Gd = dG$, $G\delta = \delta G$ and $G\Delta = \Delta G$. $\square$

Theorem 40.2.2 (Hodge). *The map $\mathcal{H}^k \rightarrow H_{\text{dR}}^k(M)$, $\alpha \mapsto [\alpha]$, is an isomorphism: every de Rham class contains a unique harmonic representative, namely the one of smallest L^2 norm. In particular $H_{\text{dR}}^k(M)$ is finite-dimensional, and $\dim \mathcal{H}^k = b_k(M)$.*

Proof. Harmonic forms are closed, so the map is defined. It is injective: if $\alpha \in \mathcal{H}^k$ is exact, $\alpha = d\beta$, then $\|\alpha\|^2 = (\alpha, d\beta) = (\delta\alpha, \beta) = 0$. It is surjective: a closed α decomposes as $\alpha = \eta + d\beta + \delta\gamma$ with η harmonic, and $\delta\gamma = 0$ as in the proof of Theorem 40.2.1, so $\alpha = \eta + d\beta$ is cohomologous to η . If $\alpha = \eta + d\beta$ is any representative, then $\|\alpha\|^2 = \|\eta\|^2 + \|d\beta\|^2 \geq \|\eta\|^2$ with equality only for $\alpha = \eta$, which is the statement about the norm. Finite-dimensionality and the identification with the Betti numbers follow from Theorem 35.3.1. $\square$

Corollary 40.2.3. *The Hodge star induces an isomorphism $*$: $\mathcal{H}^k \rightarrow \mathcal{H}^{n-k}$, hence $b_k(M) = b_{n-k}(M)$, and the pairing*

$$H_{\text{dR}}^k(M) \times H_{\text{dR}}^{n-k}(M) \rightarrow \mathbb{R}, \quad ([\alpha], [\beta]) \mapsto \int_M \alpha \wedge \beta,$$

is nondegenerate; for $[\beta] = [\alpha]$ with α harmonic and nonzero it takes the value $\|\alpha\|^2 > 0$.*

Proof. $*$ commutes with Δ (Proposition 38.7.3(4)) and is an isomorphism of Ω^k onto Ω^{n-k} , hence restricts to an isomorphism on harmonic forms; combine with Theorem 40.2.2. For a nonzero harmonic α , $\int_M \alpha \wedge * \alpha = \|\alpha\|^2 > 0$, so the pairing is nondegenerate on the harmonic representatives, which represent all classes. $\square$

Remark 40.2.4. The space $\mathcal{H}^k$ depends on the metric, but its dimension does not, and the isomorphism of Theorem 40.2.2 identifies all of them with $H_{\text{dR}}^k(M)$. This is the mechanism that produces the Hodge structures of Chapter 45: on a compact Kähler manifold the harmonic forms decompose by type, and the resulting decomposition of cohomology is independent of the metric, although the harmonic representatives are not.

Proposition 40.2.5. *Let G be a compact connected Lie group with a bi-invariant Riemannian metric (Example 38.2.3) and an orientation. Then the harmonic forms on G are exactly the bi-invariant forms.*

Proof. Left and right translations are isometries, and they preserve the orientation, being connected to the identity through diffeomorphisms; so they commute with the Hodge star, and $*$ maps bi-invariant forms to bi-invariant forms. A bi-invariant form ω is closed (Theorem 35.3.11), and so is $*\omega$; hence $\delta\omega = \pm *d*\omega = 0$ and ω is harmonic (Proposition 38.7.3(3)). The bi-invariant k -forms and the harmonic k -forms both map isomorphically onto $H_{\text{dR}}^k(G)$ (Theorems 35.3.11 and 40.2.2), so the inclusion of the first space into the second is an equality. $\square$

Counterexample 40.2.6. Compactness is essential in the Hodge theorem. On $\mathbb{R}$ with the Euclidean metric, the harmonic functions are the affine functions $ax + b$, a two-dimensional space, while $H_{\text{dR}}^0(\mathbb{R}) = \mathbb{R}$; and the harmonic 1-forms $(ax + b) dx$ form a two-dimensional space although $H_{\text{dR}}^1(\mathbb{R}) = 0$. Without compactness harmonic forms need not be closed or coclosed: the function x is harmonic with $dx \neq 0$, and $x dx$ is harmonic with $\delta(x dx) = -1 \neq 0$, since $\delta = -*d*$ on 1-forms on $\mathbb{R}$. The integration by parts behind Proposition 38.7.3(3) fails for forms without compact support.

40.3 Applications

Corollary 40.3.1. For a compact oriented manifold, $\sum_k (-1)^k \dim \mathcal{H}^k = \chi(M)$, which is the index of the elliptic operator $d + \delta: \Omega^{\text{even}}(M) \rightarrow \Omega^{\text{odd}}(M)$.

Proof. The first equality is Theorem 40.2.2 together with the definition of χ as the alternating sum of Betti numbers (Corollary 21.6.4). For the second, $(d + \delta)^* = d + \delta$ exchanges even and odd forms, its kernel on even forms is $\bigoplus_{k \text{ even}} \mathcal{H}^k$ and on odd forms $\bigoplus_{k \text{ odd}} \mathcal{H}^k$, because $(d + \delta)\alpha = 0$ implies $\Delta\alpha = (d + \delta)^2\alpha = 0$; so the index is $\sum_k (-1)^k \dim \mathcal{H}^k$. $\square$

Corollary 40.3.2 (Poincaré inequality). Let M be compact and connected, and let $\lambda_1 > 0$ be the smallest nonzero eigenvalue of the Laplacian $\Delta = \delta d$ on functions. Then for every smooth function f with $\int_M f \text{ vol} = 0$,

$$\int_M f^2 \text{ vol} \leq \frac{1}{\lambda_1} \int_M |df|^2 \text{ vol},$$

with equality exactly for the eigenfunctions of eigenvalue λ_1 .

Proof. A function u with $\Delta u = 0$ satisfies $\|du\|^2 = (\delta du, u) = 0$, so it is locally constant, hence constant as M is connected: $\mathcal{H}^0 = \mathbb{R}$. By Theorem 39.5.4 there is an orthonormal basis $u_0, u_1, \dots$ of $L^2(M)$ of smooth eigenfunctions with eigenvalues $0 = \lambda^{(0)} < \lambda_1 = \lambda^{(1)} \leq \lambda^{(2)} \leq \dots$, where u_0 is constant. Write $f = \sum_k c_k u_k$; then $c_0 = (f, u_0) = 0$ by hypothesis. Since $\Delta f = \sum_k \lambda^{(k)} c_k u_k$ in L^2 ,

$$\|df\|^2 = (\Delta f, f) = \sum_{k \geq 1} \lambda^{(k)} c_k^2 \geq \lambda_1 \sum_{k \geq 1} c_k^2 = \lambda_1 \|f\|^2,$$

with equality if and only if $c_k = 0$ whenever $\lambda^{(k)} > \lambda_1$, that is, f lies in the λ_1 -eigenspace. $\square$

Example 40.3.3. On the circle $M = \mathbb{R}/2\pi\mathbb{Z}$ with its standard metric, $\Delta f = -f''$. The real solutions of $f'' = -\lambda f$ with $\lambda > 0$ are $a \cos(\sqrt{\lambda} \theta) + b \sin(\sqrt{\lambda} \theta)$, which are 2π -periodic only for $\sqrt{\lambda} \in \mathbb{Z}$; so the eigenvalues are the k^2 , with eigenfunctions $\cos k\theta, \sin k\theta$ (and 1 for $k = 0$). Hence $\lambda_1 = 1$, and Corollary 40.3.2 is Wirtinger's inequality: $\int_0^{2\pi} f^2 \leq \int_0^{2\pi} f'^2$ for 2π -periodic smooth f with $\int_0^{2\pi} f = 0$, with equality exactly for $f = a \cos \theta + b \sin \theta$. Theorem 39.5.4 also shows that the functions $1, \cos k\theta, \sin k\theta$ span a dense subspace of L^2 , which is the completeness of Fourier series.

Theorem 40.3.4 (Bochner). Let (M, g) be a compact oriented Riemannian manifold with $\text{Ric} \geq 0$. Then every harmonic 1-form is parallel, and $b_1(M) \leq n$. If moreover $\text{Ric} > 0$ at some point, then $b_1(M) = 0$.

Proof. By the Weitzenböck formula (Remark 38.7.4), on 1-forms $\Delta = \nabla^* \nabla + \text{Ric}$. For α harmonic,

$$0 = (\Delta \alpha, \alpha) = \|\nabla \alpha\|^2 + \int_M \text{Ric}(\alpha^\sharp, \alpha^\sharp) \text{ vol},$$

where $\alpha^\sharp$ is the vector field dual to α . Both terms are non-negative, hence both vanish: $\nabla\alpha = 0$, so α is parallel and determined by its value at one point, giving $\dim \mathcal{H}^1 \leq n$; and $\text{Ric}(\alpha^\sharp, \alpha^\sharp) = 0$ everywhere. If $\text{Ric} > 0$ at p and $\alpha \neq 0$, then α is nowhere zero (being parallel and nonzero somewhere), so $\text{Ric}(\alpha^\sharp, \alpha^\sharp)(p) > 0$, a contradiction; hence $\mathcal{H}^1 = 0$ and $b_1(M) = 0$ by Theorem 40.2.2. $\square$

Example 40.3.5. 1. On the round sphere S^n with $n \geq 2$, $\text{Ric} > 0$, so $b_1 = 0$, which agrees with Example 35.2.5.

2. On a flat torus the harmonic forms are those with constant coefficients (Example 38.7.5), so Theorem 40.2.2 recovers $b_k(T^n) = \binom{n}{k}$.

3. A compact oriented manifold with a metric of positive Ricci curvature has $b_1 = 0$; by Bochner's argument applied to the other Laplacians, positivity assumptions on curvature produce vanishing theorems in all degrees. In the complex setting the same mechanism gives the Kodaira vanishing theorem (Chapter 47).

Remark 40.3.6. The proof of Theorem 40.2.1 used only that $(\Omega^\bullet, d)$ is a complex of sections of vector bundles whose Laplacian is elliptic: an *elliptic complex*. The same argument applies verbatim to any elliptic complex, in particular to the Dolbeault complex $(\Omega^{p,\bullet}, \bar{\partial})$ on a compact complex manifold, giving the isomorphism between Dolbeault cohomology and $\bar{\partial}$ -harmonic forms (Chapter 43), which is how the Hodge decomposition of Chapter 45 is proved. There is also a proof of the Hodge theorem by the heat equation, which constructs the harmonic projection as $\lim_{t \rightarrow \infty} e^{-t\Delta}$ and yields the local index formula [LM89].

40.4 Exercises

Exercise 40.4.1. Show that on a compact connected oriented Riemannian n -manifold, $\mathcal{H}^0 = \mathbb{R}$ and $\mathcal{H}^n = \mathbb{R} \text{vol}_g$, and deduce $b_0 = b_n = 1$.

Solution. A harmonic function has $df = 0$, hence is constant; apply $*$ for the top degree (Corollary 40.2.3). $\square$

Exercise 40.4.2. Show that the Green operator is self-adjoint, that it is compact as an operator on L^2 , and that $\alpha \mapsto H\alpha$ is the orthogonal projection onto $\mathcal{H}^k$ with $H = I - \Delta G$.

Exercise 40.4.3. Give an example of a compact manifold and two metrics for which the harmonic representatives of a fixed cohomology class differ.

Solution. On a torus $\mathbb{R}^2/\mathbb{Z}^2$ with the flat metric, the harmonic representative of the class of dx is dx ; after a conformal change $g' = e^{2f}g$ with non-constant f , the codifferential changes and the harmonic representative of the same class is a different 1-form, since δ' differs from δ . $\square$

Exercise 40.4.4. Let M be a compact orientable surface of genus g . Using Theorem 40.2.2 and Corollary 40.2.3, show that $\dim \mathcal{H}^1 = 2g$ and that the intersection pairing on $\mathcal{H}^1$ is a nondegenerate alternating form.

Part VII

Complex Geometry

Chapter 41

Complex Manifolds

41.1 Introduction

A complex manifold is a manifold with holomorphic transition functions. The rigidity of holomorphic functions makes these spaces very different from smooth manifolds: compact complex manifolds have only constant holomorphic functions, and their geometry is recorded instead by the cohomology of holomorphic bundles and by the decomposition of differential forms into types, which is the subject of Hodge theory. This chapter sets up complex manifolds, the operators ∂ and $\bar{\partial}$, analytic subvarieties and blow-ups. References: [GH78, Chapter 0], [Huy05, Chapter 2], [Voi02b, Chapter 3].

Prerequisites. Chapter 31 (Smooth Manifolds), Chapter 29 (Several Complex Variables), Chapter 10 (Multilinear Algebra) and Chapter 33 (Vector Bundles).

41.2 Complex manifolds

Definition 41.2.1. A *complex manifold* of dimension n is a topological $2n$ -manifold with an atlas of charts $\varphi_i: U_i \rightarrow \mathbb{C}^n$ whose transition maps are holomorphic; a map of complex manifolds is *holomorphic* if it is so in charts, and a *biholomorphism* is a bijective holomorphic map with holomorphic inverse. The sheaf of holomorphic functions is $\mathcal{O}_X$, making $(X, \mathcal{O}_X)$ a locally ringed space. Every complex manifold is in particular a smooth $2n$ -manifold, canonically oriented by the orientation of $\mathbb{C}^n$ (transition maps have Jacobian determinant $|\det_{\mathbb{C}}|^2 > 0$).

- Example 41.2.2.**
1. Open subsets of $\mathbb{C}^n$; $\mathbb{CP}^n$ with the charts of Example 31.2.5, whose transition maps are rational and nowhere singular.
 2. A *complex torus* $X = \mathbb{C}^n / \Lambda$ for a lattice $\Lambda \cong \mathbb{Z}^{2n}$; these are compact, and for $n = 1$ they are the elliptic curves. Two complex tori are rarely biholomorphic even when they are diffeomorphic, which is the first appearance of moduli (Chapter 62).
 3. Grassmannians $\mathrm{Gr}(k, n)$ of k -planes in $\mathbb{C}^n$, with charts given by graphs of linear maps; they are compact complex manifolds of dimension $k(n - k)$.
 4. A smooth analytic hypersurface $\{f = 0\} \subseteq \mathbb{CP}^n$, for f a homogeneous polynomial with $df \neq 0$ on the zero set, is a compact complex manifold of dimension $n - 1$, by the implicit function theorem (Theorem 25.11.3, valid for holomorphic maps by Remark 25.11.5).
 5. Not every even-dimensional smooth manifold admits a complex structure: S^4 does not (Exercise 37.6.4), and whether S^6 does is open.

Proposition 41.2.3. *A holomorphic function on a compact connected complex manifold is constant. Consequently a compact complex manifold has no nonconstant holomorphic map to $\mathbb{C}^n$, and $\mathcal{O}_X(X) = \mathbb{C}$.*

Proof. $|f|$ attains a maximum at some p ; in a chart around p , the maximum principle (Proposition 29.2.3) makes f constant near p , and the set where f equals $f(p)$ is open and closed by the identity theorem. $\square$

Corollary 41.2.4. *Every compact complex submanifold of $\mathbb{C}^n$ is a finite set of points. In particular no compact complex manifold of positive dimension, such as $\mathbb{CP}^m$ or a complex torus, embeds holomorphically in $\mathbb{C}^n$, in contrast with the Whitney embedding theorem for smooth manifolds (Theorem 31.7.5).*

Proof. The coordinate functions restrict to holomorphic functions on each connected component, which are constant by Proposition 41.2.3; so each component is a point, and a compact discrete set is finite. $\square$

Theorem 41.2.5. *Let $X = \mathbb{C}^n/\Lambda$ and $Y = \mathbb{C}^m/\Lambda'$ be complex tori. Every holomorphic map $f: X \rightarrow Y$ is induced by an affine map $\mathbb{C}^n \rightarrow \mathbb{C}^m$, $z \mapsto Az + b$, with A a complex $m \times n$ matrix such that $A\Lambda \subseteq \Lambda'$. In particular f is a group homomorphism followed by a translation, and $X \cong Y$ as complex manifolds if and only if $n = m$ and $\Lambda' = A\Lambda$ for some $A \in \mathrm{GL}_n(\mathbb{C})$.*

Proof. The projection $\mathbb{C}^n \rightarrow X$ is a universal covering, as $\mathbb{C}^n$ is simply connected and Λ acts freely and properly (Theorem 18.4.2). By the lifting criterion (Proposition 17.6.1), $f \circ \pi$ lifts to a continuous map $F: \mathbb{C}^n \rightarrow \mathbb{C}^m$, holomorphic because π' is a local biholomorphism. For $\lambda \in \Lambda$, $F(z + \lambda) - F(z) \in \Lambda'$ for all z , and by continuity and discreteness of Λ' it is a constant $c(\lambda)$. Hence each partial derivative $\partial F_j / \partial z_k$ is Λ -periodic, so it is a holomorphic function on the compact manifold X , hence constant (Proposition 41.2.3). So $F(z) = Az + b$, and $A\lambda = c(\lambda) \in \Lambda'$. If f is a biholomorphism, apply this to f and f^{-1} : their lifts compose to a lift of the identity, which is a translation by an element of Λ , so A is invertible with $A\Lambda = \Lambda'$ and $n = m$; the converse is clear. $\square$

Corollary 41.2.6. *Every one-dimensional complex torus is biholomorphic to $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\mathrm{Im} \tau > 0$, and $E_\tau \cong E_{\tau'}$ if and only if $\tau' = \frac{a\tau+b}{c\tau+d}$ for some $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$.*

Proof. A lattice $\mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ is carried by $z \mapsto z/\omega_1$ to $\mathbb{Z} + \mathbb{Z}\tau$ with $\tau = \omega_2/\omega_1 \notin \mathbb{R}$, and replacing ω_2 by $-\omega_2$ if necessary, $\mathrm{Im} \tau > 0$. By Theorem 41.2.5, $E_\tau \cong E_{\tau'}$ if and only if $\alpha(\mathbb{Z} + \mathbb{Z}\tau) = \mathbb{Z} + \mathbb{Z}\tau'$ for some $\alpha \in \mathbb{C}^\times$. Then $\alpha\tau = a\tau' + b$ and $\alpha = c\tau' + d$ for an integer matrix, invertible over $\mathbb{Z}$ because the inverse change of basis is also integral; so $\tau = \frac{a\tau'+b}{c\tau'+d}$, and the determinant is +1 because both τ and τ' lie in the upper half plane, as $\mathrm{Im} \frac{a\tau'+b}{c\tau'+d} = \frac{(ad-bc)\mathrm{Im} \tau'}{|c\tau'+d|^2}$. Conversely such a matrix gives the lattice isomorphism. The relation is symmetric, so τ' is also of this form in terms of τ . $\square$

Counterexample 41.2.7. Diffeomorphic complex manifolds need not be biholomorphic: all E_τ are diffeomorphic to $S^1 \times S^1$, but by Corollary 41.2.6 $E_i \not\cong E_{2i}$, since $\frac{ai+b}{ci+d} = 2i$ with integers of determinant 1 would give, comparing imaginary parts, $|ci+d|^2 = \frac{1}{2}$, impossible for integers c, d .

Proposition 41.2.8. *Let $f: X \rightarrow Y$ be a nonconstant holomorphic map of connected Riemann surfaces (complex manifolds of dimension 1), and $p \in X$. There are holomorphic coordinates z centred at p and w centred at $f(p)$ in which f is $z \mapsto z^k$ for a unique integer $k = e_p \geq 1$, the ramification index. Consequently f is open, its fibres are discrete, and if X is compact and Y connected, then f is surjective and has finite fibres.*

Proof. Choose coordinates z at p and w at $f(p)$ with $z(p) = 0$, $w(f(p)) = 0$. In them f is a holomorphic function of one variable with $f(0) = 0$, not identically zero near 0 by the identity theorem (Theorem 27.4.4) and connectedness of X . So $f(z) = z^k g(z)$ with $k \geq 1$ and $g(0) \neq 0$. On a small disc where g has no zeros, g'/g has a primitive L (Corollary 27.3.2), and $(ge^{-L})' = 0$, so $g = ce^L$ with $c \neq 0$; writing $c = e^{c'}$, the function $h = e^{(L+c')/k}$ satisfies $h^k = g$. The new coordinate $\tilde{z} = zh(z)$ has derivative $h(0) \neq 0$ at 0, so it is a local biholomorphism (Theorem 27.5.11, or Proposition 29.2.5), and $f = \tilde{z}^k$. The integer k is determined by f , as the number of preimages near p of points $w \neq 0$ near 0. The map $z \mapsto z^k$ is open with discrete fibres, so f is. If X is compact, $f(X)$ is compact, hence closed, and open, so $f(X) = Y$ when Y is connected; the fibres are discrete and closed in a compact space, hence finite. $\square$

Proposition 41.2.9. *Let $f: X \rightarrow Y$ be a nonconstant holomorphic map of compact connected Riemann surfaces. The number $\deg f = \sum_{p \in f^{-1}(q)} e_p$ does not depend on $q \in Y$. It equals the number of preimages of every point q outside the finite set $f(\{p : e_p > 1\})$ of branch points.*

Proof. The ramification points, where f' vanishes in coordinates, are the zeros of a holomorphic function locally, hence discrete, hence finitely many in the compact X . Fix q_0 with preimages $p_1, \dots, p_r$ (Proposition 41.2.8), and choose disjoint coordinate discs D_i around p_i on which f is $z \mapsto z^{e_i}$ onto a disc D around q_0 . The set $f(X \setminus \bigcup_i D_i)$ is compact and does not contain q_0 , so there is a neighbourhood $D' \subseteq D$ of q_0 avoiding it. For $q \in D' \setminus \{q_0\}$, all preimages of q lie in $\bigcup_i D_i$, and in D_i the equation $z^{e_i} = w(q) \neq 0$ has exactly e_i solutions, each with ramification index 1. So the weighted count equals $\sum_i e_i$ on D' . Hence $q \mapsto \deg_q f$ is locally constant, hence constant on the connected Y . $\square$

Corollary 41.2.10. *Every holomorphic map from a simply connected compact connected complex manifold, such as $\mathbb{CP}^1$, to a complex torus is constant. In particular $\mathbb{CP}^1$ is not biholomorphic, and admits no nonconstant holomorphic map, to an elliptic curve.*

Proof. Let $f: X \rightarrow T = \mathbb{C}^m/\Lambda$. The projection $\mathbb{C}^m \rightarrow T$ is a covering (Theorem 18.4.2), so by the lifting criterion (Proposition 17.6.1) f lifts to a continuous $F: X \rightarrow \mathbb{C}^m$, which is holomorphic because the projection is a local biholomorphism. By Proposition 41.2.3 each component of F is constant, so f is constant. The projective line is covered by the two charts U_0, U_1 , each homeomorphic to $\mathbb{C}$, with path-connected intersection $\mathbb{C}^\times$; by the Seifert–van Kampen theorem (Theorem 17.5.1) $\pi_1(\mathbb{CP}^1)$ is a quotient of the free product of two trivial groups, hence trivial. $\square$

41.3 Almost complex structures and forms of type (p, q)

Definition 41.3.1. An *almost complex structure* on a smooth manifold M is an endomorphism J of TM with $J^2 = -\text{id}$. It makes each $T_p M$ a complex vector space, and gives the decomposition $T_p M \otimes \mathbb{C} = T_p^{1,0} \oplus T_p^{0,1}$ into the $\pm i$ eigenspaces, with $T^{0,1} = \overline{T^{1,0}}$, and dually $\Lambda^k T_p^* M \otimes \mathbb{C} = \bigoplus_{p+q=k} \Lambda^{p,q}$ (Proposition 10.6.2). A complex manifold has a canonical almost complex structure, given in holomorphic coordinates $z^j = x^j + iy^j$ by $J\partial_{x^j} = \partial_{y^j}$, $J\partial_{y^j} = -\partial_{x^j}$; then $T^{1,0}$ has frame $\partial/\partial z^j$ and $\Lambda^{p,q}$ has frame $dz^I \wedge d\bar{z}^J$ with $|I| = p$, $|J| = q$. An almost complex structure is *integrable* if it arises from a complex structure.

Definition 41.3.2. On a complex manifold, $\Omega^{p,q}(X)$ denotes the smooth (p, q) -forms, so $\Omega^k(X, \mathbb{C}) = \bigoplus_{p+q=k} \Omega^{p,q}(X)$. Writing $d = \partial + \bar{\partial}$ with $\partial: \Omega^{p,q} \rightarrow \Omega^{p+1,q}$ and $\bar{\partial}: \Omega^{p,q} \rightarrow \Omega^{p,q+1}$ the components of d , one has in coordinates

$$\partial \left(\sum f_{IJ} dz^I \wedge d\bar{z}^J \right) = \sum \frac{\partial f_{IJ}}{\partial z^k} dz^k \wedge dz^I \wedge d\bar{z}^J, \quad \bar{\partial} \left(\sum f_{IJ} dz^I \wedge d\bar{z}^J \right) = \sum \frac{\partial f_{IJ}}{\partial \bar{z}^k} d\bar{z}^k \wedge dz^I \wedge d\bar{z}^J.$$

Proposition 41.3.3. *On a complex manifold $d = \partial + \bar{\partial}$, and $\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0$. A function is holomorphic if and only if $\bar{\partial}f = 0$, and a $(p, 0)$ -form α is holomorphic (has holomorphic coefficients) if and only if $\bar{\partial}\alpha = 0$; the sheaf of holomorphic p -forms is denoted Ω_X^p , with $\Omega_X^0 = \mathcal{O}_X$ and $\Omega_X^n = K_X$ the canonical bundle.*

Proof. In holomorphic coordinates d of a (p, q) -form has components of types $(p+1, q)$ and $(p, q+1)$ only, by the coordinate formula for d ; these are ∂ and $\bar{\partial}$. Then $0 = d^2$ decomposes into its components of types $(p+2, q)$, $(p+1, q+1)$ and $(p, q+2)$, which are ∂^2 , $\partial\bar{\partial} + \bar{\partial}\partial$ and $\bar{\partial}^2$. The statement about functions is the Cauchy–Riemann equations (Definition 29.2.1), and for forms it is the same condition on the coefficients. $\square$

Theorem 41.3.4 (Newlander–Nirenberg). *An almost complex structure J on a smooth manifold is integrable if and only if the Nijenhuis tensor vanishes,*

$$N_J(X, Y) = [JX, JY] - J[JX, Y] - J[X, JY] - [X, Y] = 0,$$

equivalently if and only if $T^{0,1}$ is closed under the Lie bracket, equivalently if and only if $d = \partial + \bar{\partial}$ on 1-forms with respect to the induced type decomposition.

The implication “integrable $\Rightarrow N_J = 0$ ” is a computation in holomorphic coordinates, where the brackets of the coordinate fields vanish; the converse is a hard analytic theorem, proved in [Huy05, Chapter 2] and, in the real-analytic case, by the Frobenius theorem (Theorem 31.7.4) applied to $T^{0,1}$ after complexification.

Proposition 41.3.5. *Every almost complex structure on a manifold of real dimension 2 has vanishing Nijenhuis tensor. Consequently, by Theorem 41.3.4, every oriented surface with a Riemannian metric is a Riemann surface, with J the rotation by $\pi/2$ in each tangent plane.*

Proof. N_J is antisymmetric and C^∞ -bilinear, so it suffices to evaluate it on X and JX for a local nowhere vanishing field X , which together with JX form a frame. Using $J^2 = -1$, $N_J(X, JX) = [JX, -X] - J[JX, JX] - J[X, -X] - [X, JX] = [X, JX] - [X, JX] = 0$. On an oriented surface with a metric, rotation by $\pi/2$ in the positive sense is a smooth endomorphism with $J^2 = -1$. $\square$

41.4 Analytic subvarieties and blow-ups

Definition 41.4.1. An *analytic subvariety* of a complex manifold X is a closed subset that is locally the common zero set of finitely many holomorphic functions (Definition 29.5.1); it is *smooth*, that is, a complex submanifold, at its regular points. By Theorem 29.5.3 the singular locus is a proper analytic subvariety, the regular locus is dense, and Z has a well-defined dimension on each irreducible component.

Proposition 41.4.2. *Let $Z \subseteq X$ be an analytic subvariety of pure complex dimension k in a compact complex manifold of dimension n . Then integration over the regular locus, $\varphi \mapsto \int_{Z_{\text{reg}}} \varphi$, converges for every smooth $2k$ -form φ and defines a closed current of degree $2n - 2k$; its class in $H^{2n-2k}(X; \mathbb{R})$ is the cycle class $[Z]$. Moreover $\int_Z \varphi \geq 0$ when φ is a positive (k, k) -form.*

Sketch of proof. Convergence follows from the local structure of analytic sets: near a regular point Z is a submanifold, and near a singular point it is a finite branched cover of a polydisc (Theorem 29.5.3), so the volume of $Z \cap K$ is finite for compact K (in a chart, Wirtinger’s inequality bounds the volume by the integral of a power of the standard form $\frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j$ of $\mathbb{C}^n$). Closedness, that is, Stokes’ theorem across the singular locus, is proved by cutting out a neighbourhood of the singular set, whose boundary has small area because the singular locus has real codimension at least 2. The positivity statement is local linear algebra. Details are in [GH78, Chapter 0, Section 2] and [Voi02b, Chapter 11]. $\square$

Remark 41.4.3. If X carries a Kähler form ω (Chapter 44) and $Z \neq \emptyset$, then $[Z] \neq 0$: pairing with the closed form ω^k gives $\int_Z \omega^k > 0$, the volume of Z up to a constant. On non-Kähler manifolds cycle classes can vanish; for instance on the Hopf surface of Exercise 41.5.3 the elliptic curves $\{z_1 = 0\}/\langle 2 \rangle$ have zero class, since H^2 of the Hopf surface vanishes.

Definition 41.4.4. The *blow-up* of $\mathbb{C}^n$ at the origin is

$$\mathrm{Bl}_0 \mathbb{C}^n = \{(z, \ell) \in \mathbb{C}^n \times \mathbb{CP}^{n-1} : z \in \ell\},$$

with the projection π to $\mathbb{C}^n$. It is a complex manifold, π is a biholomorphism over $\mathbb{C}^n \setminus \{0\}$, and $E = \pi^{-1}(0) \cong \mathbb{CP}^{n-1}$ is the *exceptional divisor*. For a point p of a complex manifold X one defines $\mathrm{Bl}_p X$ by gluing $\mathrm{Bl}_0 \mathbb{C}^n$ in a chart, and similarly along a complex submanifold $Z \subseteq X$, with $E = \mathbb{P}(N_{Z/X})$.

Proposition 41.4.5. $\mathrm{Bl}_p X$ is a compact complex manifold if X is, π is proper and holomorphic, and

$$H^k(\mathrm{Bl}_p X; \mathbb{Z}) \cong H^k(X; \mathbb{Z}) \oplus \bigoplus_{0 < j < n} H^{k-2j}(\mathrm{pt}; \mathbb{Z}),$$

so blowing up a point adds one generator in each even degree $2, 4, \dots, 2n - 2$. The exceptional divisor has normal bundle $\mathcal{O}(-1)$ in $\mathrm{Bl}_0 \mathbb{C}^n$.

Proof. In the chart $\ell = [1 : w_2 : \dots : w_n]$ the map $(u, w_2, \dots, w_n) \mapsto (u, uw_2, \dots, uw_n)$ identifies $\mathbb{C}^n$ with an open subset of $\mathrm{Bl}_0 \mathbb{C}^n$, which shows that the blow-up is a complex manifold and that π is holomorphic and biholomorphic away from E ; properness holds because $\mathbb{CP}^{n-1}$ is compact. In these coordinates E is $\{u = 0\}$, and the transition functions of its normal bundle are those of the tautological bundle, so $N_E \cong \mathcal{O}(-1)$. For the cohomology, use the Mayer-Vietoris sequence for $\mathrm{Bl}_p X = (\mathrm{Bl}_p X \setminus E) \cup \pi^{-1}(B)$ with B a small ball, together with $\pi^{-1}(B) \simeq \mathbb{CP}^{n-1}$ and $\mathrm{Bl}_p X \setminus E \cong X \setminus \{p\}$, and Example 20.2.5. $\square$

Remark 41.4.6. Blow-ups are the basic birational modifications: Hironaka's resolution of singularities states that every analytic (or algebraic, in characteristic zero) variety becomes smooth after finitely many blow-ups along smooth centres, and the weak factorisation theorem says that any two birational smooth projective varieties are connected by blow-ups and blow-downs. Both are used to reduce statements about the Hodge conjecture to the smooth projective case (Chapter 70).

41.5 Exercises

Exercise 41.5.1. Show that every complex torus of dimension 1 is biholomorphic to $\mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\mathrm{Im} \tau > 0$, and that two such are biholomorphic if and only if the parameters are related by an element of $\mathrm{SL}_2(\mathbb{Z})$ acting by Möbius transformations.

Exercise 41.5.2. Verify that the transition maps of $\mathbb{CP}^n$ are holomorphic and that $\mathbb{CP}^1$ is biholomorphic to the Riemann sphere $\mathbb{C} \cup \{\infty\}$, with S^2 as underlying smooth manifold.

Exercise 41.5.3. The *Hopf surface* is $X = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$. Show that it is a compact complex manifold diffeomorphic to $S^1 \times S^3$, and deduce that $b_1(X) = 1$; conclude that X carries no Kähler metric (Chapter 45), since odd Betti numbers of compact Kähler manifolds are even.

Solution. The action is free and properly discontinuous, so the quotient is a complex manifold, and it is compact because the annulus $1 \leq |z| \leq 2$ maps onto it. The diffeomorphism with $S^1 \times S^3$ comes from the polar decomposition $\mathbb{C}^2 \setminus \{0\} \cong S^3 \times \mathbb{R}_{>0}$, with the action becoming translation on the second factor. Then $b_1(S^1 \times S^3) = 1$ by the Künneth formula. $\square$

Exercise 41.5.4. Show that S^2 and S^6 admit almost complex structures (for S^6 use the octonionic cross product), and that S^{2n} is integrable.

Chapter 42

Holomorphic Vector Bundles

42.1 Introduction

Holomorphic line bundles are the objects that encode divisors, projective embeddings and, through their first Chern classes, the $(1, 1)$ -part of cohomology. The exponential sequence (Theorem 42.2.4) computes the Picard group in terms of cohomology and already contains the case $p = 1$ of the Hodge conjecture: the classes of divisors are exactly the integral classes of type $(1, 1)$ (Chapter 65). References: [GH78, Chapter 1], [Huy05, Chapters 2–4], [Voi02b, Chapter 4].

Prerequisites. Chapter 41 (Complex Manifolds), Chapter 24 (Sheaf Cohomology) and Chapter 37 (Characteristic Classes).

X denotes a complex manifold of dimension n .

42.2 Holomorphic bundles and sheaves

Definition 42.2.1. A *holomorphic vector bundle* of rank r on X is a complex manifold E with a holomorphic map $\pi: E \rightarrow X$ and $\mathbb{C}$ -vector space structures on the fibres, locally biholomorphic to $U \times \mathbb{C}^r$ over U and linear on fibres; equivalently, it is given by holomorphic transition functions $g_{ij}: U_i \cap U_j \rightarrow \mathrm{GL}_r(\mathbb{C})$ satisfying the cocycle condition (Proposition 33.2.5). Its sheaf of holomorphic sections is a locally free $\mathcal{O}_X$ -module of rank r , and this is an equivalence between holomorphic bundles and locally free sheaves, as in Proposition 33.2.7.

Example 42.2.2. The holomorphic tangent bundle $T^{1,0}X$, with transition functions the holomorphic Jacobians; the bundles $\Omega_X^p = \Lambda^p(T^{1,0}X)^\vee$ of holomorphic p -forms and the canonical bundle $K_X = \Omega_X^n$; the tautological bundle $\mathcal{O}(-1)$ on $\mathbb{CP}^n$ (Exercise 33.5.3), its dual $\mathcal{O}(1)$ and the powers $\mathcal{O}(k) = \mathcal{O}(1)^{\otimes k}$. The holomorphic sections of $\mathcal{O}(k)$ on $\mathbb{CP}^n$ are the homogeneous polynomials of degree k for $k \geq 0$, and 0 for $k < 0$ (Exercise 42.6.1).

Proposition 42.2.3. The isomorphism classes of holomorphic line bundles on X form a group $\mathrm{Pic}(X)$ under $\otimes$, and

$$\mathrm{Pic}(X) \cong H^1(X, \mathcal{O}_X^*),$$

where $\mathcal{O}_X^*$ is the sheaf of nowhere vanishing holomorphic functions.

Proof. As in Proposition 33.2.5 and Theorem 33.4.2, line bundles correspond to Čech cocycles with values in $\mathcal{O}_X^*$ modulo coboundaries, and $\check{H}^1 = H^1$ (Remark 24.5.4); tensor product corresponds to multiplication of cocycles. $\square$

Theorem 42.2.4 (Exponential sequence). *The sequence of sheaves $0 \rightarrow \mathbb{Z}_X \rightarrow \mathcal{O}_X \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}_X^* \rightarrow 0$ is exact, and its long exact sequence reads*

$$\cdots \rightarrow H^1(X, \mathbb{Z}) \rightarrow H^1(X, \mathcal{O}_X) \rightarrow \text{Pic}(X) \xrightarrow{c_1} H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X) \rightarrow \cdots$$

The map c_1 is the first Chern class of the underlying smooth line bundle (Theorem 33.4.2). In particular, for compact X :

1. c_1 is surjective onto the kernel of $H^2(X, \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X)$;
2. its kernel $\text{Pic}^0(X)$ is $H^1(X, \mathcal{O}_X)/\text{im } H^1(X, \mathbb{Z})$;
3. if $H^1(X, \mathcal{O}_X) = H^2(X, \mathcal{O}_X) = 0$, then $c_1: \text{Pic}(X) \rightarrow H^2(X, \mathbb{Z})$ is an isomorphism.

Proof. Exactness: the kernel of $\exp(2\pi i \cdot)$ is the constant sheaf $\mathbb{Z}$, and surjectivity holds on stalks because a nonvanishing holomorphic function on a small ball has a holomorphic logarithm (integrate df/f , which is possible on a simply connected domain by Theorem 27.3.3 applied in one variable at a time, or note that the ball is contractible and use the same argument as in Theorem 33.4.2). The long exact sequence is Theorem 13.4.3, using $\text{Pic}(X) \cong H^1(X, \mathcal{O}_X^*)$. That the connecting map is the topological first Chern class follows by comparing with the corresponding sequence for smooth functions, into which this one maps, and using that the sheaf of smooth functions is acyclic. The three consequences are exactness of the long sequence. $\square$

Remark 42.2.5. When X is compact Kähler, the Hodge decomposition (Chapter 45) shows that $H^1(X, \mathbb{Z})$ maps onto a lattice in the complex vector space $H^1(X, \mathcal{O}_X)$ of dimension $b_1/2$, so $\text{Pic}^0(X)$ is a compact complex torus, the *Picard variety*, and the image of c_1 , the *Néron–Severi group*, is the group of integral classes of type $(1, 1)$ (Chapter 65).

Lemma 42.2.6. *Let L be a holomorphic line bundle on X .*

1. L is trivial if and only if it has a nowhere vanishing holomorphic section.
2. If X is compact and connected and both L and its dual $L^\vee$ have nonzero holomorphic sections, then L is trivial. In particular, if L is nontrivial and $H^0(X, L) \neq 0$, then $H^0(X, L^\vee) = 0$.

Proof. (1) A nowhere vanishing section s gives the map $\mathcal{O}_X \rightarrow L$, $f \mapsto fs$, which is an isomorphism on every fibre; conversely the trivial bundle has the section 1.

(2) First, if U is a connected open set and a, b are holomorphic functions on U , neither identically zero, then ab is not identically zero: otherwise $U \setminus a^{-1}(0)$ is open, nonempty by the identity theorem (Proposition 29.2.3), and b vanishes on it, so $b = 0$ by the identity theorem again. Next, a nonzero section s of L does not vanish identically on any nonempty open set: the set of points near which s vanishes identically is open, and it is closed by the identity theorem applied in local frames, so it is empty because X is connected and $s \neq 0$.

Now let $s \in H^0(X, L)$ and $t \in H^0(X, L^\vee)$ be nonzero. The pairing $t(s)$ is a holomorphic function on X , hence a constant c (Proposition 41.2.3). In a connected coordinate neighbourhood U with a holomorphic frame e and dual frame $e^\vee$, write $s = ae$ and $t = be^\vee$; then $t(s) = ab$ with a, b not identically zero on U , so $c \neq 0$. Therefore $a(x)b(x) = c \neq 0$ everywhere, s vanishes nowhere, and L is trivial by (1). $\square$

Definition 42.2.7. Let L be a smooth complex line bundle on X , and let $\mathcal{A}^{0,q}(L)$ denote the smooth L -valued $(0, q)$ -forms. A $\bar{\partial}$ -operator on L is a $\mathbb{C}$ -linear map $D: \mathcal{A}^0(L) \rightarrow \mathcal{A}^{0,1}(L)$ with $D(fs) = \bar{\partial}f \otimes s + fDs$ for smooth functions f . In a local smooth frame e it has the form $D(fe) = (\bar{\partial}f + fa_e)e$ for a $(0, 1)$ -form $a_e = De/e$. The operator is *integrable* if $\bar{\partial}a_e = 0$ for every local frame e .

Proposition 42.2.8. 1. If L is a holomorphic line bundle, the operator $\bar{\partial}_L(fe) = \bar{\partial}f \otimes e$, for e a holomorphic frame, is a well-defined integrable $\bar{\partial}$ -operator, and a smooth section s is holomorphic if and only if $\bar{\partial}_L s = 0$.

2. Integrability does not depend on the frame: if $e' = ge$, then $a_{e'} = a_e + g^{-1}\bar{\partial}g$ and $\bar{\partial}a_{e'} = \bar{\partial}a_e$.

3. Conversely, for every integrable $\bar{\partial}$ -operator D on a smooth complex line bundle L there is a unique structure of holomorphic line bundle on L with $D = \bar{\partial}_L$.

In particular, on a Riemann surface, where there are no $(0, 2)$ -forms, every $\bar{\partial}$ -operator is integrable.

Proof. (2) $D(e') = D(ge) = (\bar{\partial}g + ga_e)e = (g^{-1}\bar{\partial}g + a_e)e'$. Moreover $\bar{\partial}(g^{-1}\bar{\partial}g) = -g^{-2}\bar{\partial}g \wedge \bar{\partial}g + g^{-1}\bar{\partial}^2g = 0$, since a 1-form wedged with itself vanishes and $\bar{\partial}^2 = 0$ (Proposition 41.3.3).

(1) Two holomorphic frames differ by a holomorphic g , with $\bar{\partial}g = 0$, so by the computation in (2) the operators defined by the two frames agree; the Leibniz rule is clear, and $a_e = 0$ for a holomorphic frame, so $\bar{\partial}_L$ is integrable. A section fe is holomorphic if and only if f is, that is, $\bar{\partial}f = 0$.

(3) Let $x \in X$, choose a coordinate polydisc Δ around x on which L has a smooth frame e , and write $a_e = \sum_j f_j d\bar{z}_j$. The condition $\bar{\partial}a_e = 0$ says $\partial f_j / \partial \bar{z}_l = \partial f_l / \partial \bar{z}_j$, so by the $\bar{\partial}$ -Poincaré lemma (Theorem 29.4.2) there is a smooth u on a smaller polydisc Δ' with $\bar{\partial}u = a_e$. The frame $e' = e^{-u}e$ satisfies $De' = (-\bar{\partial}u + a_e)e' = 0$. Cover X by open sets U_i with frames e_i such that $De_i = 0$. On overlaps $e_i = g_{ij}e_j$ with g_{ij} smooth and nowhere zero, and $0 = De_i = \bar{\partial}g_{ij} \otimes e_j$, so the g_{ij} are holomorphic and define a holomorphic structure on L for which the e_i are holomorphic frames. For a section $s = fe_i$ we get $Ds = \bar{\partial}f \otimes e_i = \bar{\partial}_L s$, so $D = \bar{\partial}_L$. For uniqueness, the holomorphic frames of any structure with $\bar{\partial}_L = D$ are the local frames killed by D , so the holomorphic atlas is determined by D . $\square$

Example 42.2.9. On the smooth trivial line bundle $X \times \mathbb{C}$, the $\bar{\partial}$ -operators are the $D_a = \bar{\partial} + a$ with a a $(0, 1)$ -form, integrable exactly when $\bar{\partial}a = 0$. A holomorphic isomorphism between the structures defined by D_a and D_b is multiplication by a smooth nowhere vanishing function g taking D_a -holomorphic sections to D_b -holomorphic ones; since $D_b(gs) = g(D_a s + (g^{-1}\bar{\partial}g + b - a)s)$, this happens if and only if $a - b = g^{-1}\bar{\partial}g$. In particular, if $a - b = \bar{\partial}u$ for a smooth function u , then $g = e^u$ is an isomorphism. So $a \mapsto (X \times \mathbb{C}, D_a)$ defines a map from the $\bar{\partial}$ -closed $(0, 1)$ -forms modulo $\bar{\partial}$ of functions to $\text{Pic}(X)$, whose image consists of line bundles that are trivial as smooth bundles; compare the map $H^1(X, \mathcal{O}_X) \rightarrow \text{Pic}(X)$ of Theorem 42.2.4, and Chapter 43, where the quotient above is the Dolbeault group $H_{\bar{\partial}}^{0,1}(X)$.

42.3 Divisors

Definition 42.3.1. A *divisor* on X is a locally finite formal sum $D = \sum_i a_i Z_i$ with $a_i \in \mathbb{Z}$ and Z_i irreducible analytic hypersurfaces; they form the group $\text{Div}(X)$. A *meromorphic function* is a section of the sheaf $\mathcal{M}_X$ of quotients f/g of holomorphic functions with $g \neq 0$; the order $\text{ord}_Z(f) \in \mathbb{Z}$ of a meromorphic function along an irreducible hypersurface Z is defined by writing $f = u h^m$ near a regular point of Z , where h is a local defining function of Z and u is a unit, using that $\mathcal{O}_{X,x}$ is a unique factorisation domain (Theorem 29.3.4). The divisor of f is $(f) = \sum_Z \text{ord}_Z(f) Z$, and divisors of this form are *principal*.

Proposition 42.3.2. There is a homomorphism $\text{Div}(X) \rightarrow \text{Pic}(X)$, $D \mapsto \mathcal{O}(D)$, defined by taking local defining functions f_i for D on a cover and the cocycle $g_{ij} = f_i/f_j \in \mathcal{O}^*$. Its kernel is the group of principal divisors, and $\mathcal{O}(D)$ has a canonical meromorphic section with divisor D ; for D effective (all $a_i \geq 0$) this section is holomorphic and vanishes exactly on D . Conversely a nonzero meromorphic section of a line bundle L has a divisor D with $L \cong \mathcal{O}(D)$.

Proof. The functions f_i/f_j are nowhere vanishing and holomorphic on overlaps, because two local defining functions of the same divisor differ by a unit (Theorem 29.3.4), and they clearly form a cocycle; changing the f_i changes the cocycle by a coboundary. The local functions f_i glue to a meromorphic section of $\mathcal{O}(D)$, whose divisor is D . If $\mathcal{O}(D)$ is trivial, the cocycle is a coboundary $f_i/f_j = u_i/u_j$, so the f_i/u_i glue to a global meromorphic function with divisor D ; conversely, principal divisors give trivial cocycles. The last statement follows by taking the divisor of the section and comparing cocycles. $\square$

Example 42.3.3. On $\mathbb{CP}^n$ every irreducible hypersurface is the zero set of an irreducible homogeneous polynomial, and $\mathcal{O}(D) \cong \mathcal{O}(\deg D)$; the map $\deg: \text{Div}(\mathbb{CP}^n) \rightarrow \mathbb{Z}$ induces the isomorphism $\text{Pic}(\mathbb{CP}^n) \cong \mathbb{Z}$, which also follows from Theorem 42.2.4 since $H^1(\mathbb{CP}^n, \mathcal{O}) = H^2(\mathbb{CP}^n, \mathcal{O}) = 0$ (Chapter 43). On a complex torus $X = \mathbb{C}^n/\Lambda$, by contrast, $H^1(X, \mathcal{O}_X) \neq 0$ and $\text{Pic}^0(X)$ is a nontrivial complex torus, the dual torus.

Counterexample 42.3.4. Holomorphic line bundles are not determined by their topology. Let $X = \mathbb{C}/\Lambda$ with $\Lambda = \mathbb{Z}\omega_1 + \mathbb{Z}\omega_2$ a lattice, numbered so that $\text{Im}(\omega_2/\omega_1) > 0$, $\pi: \mathbb{C} \rightarrow X$ the projection, and $p \neq q$ points of X . The line bundle $L = \mathcal{O}(p - q) = \mathcal{O}(p) \otimes \mathcal{O}(q)^\vee$ is trivial as a smooth bundle but not as a holomorphic bundle.

Smooth triviality. Lift $q - p$ to $c \in \mathbb{C}$ and let $\tau(x) = x + c$, a biholomorphism of X with $\tau(p) = q$. A local defining function of q composed with τ is a local defining function of p , so $\tau^*\mathcal{O}(q) \cong \mathcal{O}(p)$. The translations $x \mapsto x + tc$, $0 \leq t \leq 1$, form a homotopy from the identity to τ , so by Theorem 33.4.2 and homotopy invariance (Theorem 20.2.2), $c_1(\mathcal{O}(p)) = \tau^*c_1(\mathcal{O}(q)) = c_1(\mathcal{O}(q))$. Hence $c_1(L) = 0$ and L is smoothly trivial, by the same theorem.

Holomorphic nontriviality. If L were holomorphically trivial, the divisor $p - q$ would be principal (Proposition 42.3.2): there would be a meromorphic function f on X whose only pole is a simple pole at q . Then $F = f \circ \pi$ is meromorphic on $\mathbb{C}$, invariant under translation by Λ , with simple poles exactly at the points of $\pi^{-1}(q)$. Choose a lift $\tilde{q}$ of q and let P be the parallelogram with vertices $a, a + \omega_1, a + \omega_1 + \omega_2, a + \omega_2$, where $a = \tilde{q} - \frac{1}{2}(\omega_1 + \omega_2)$. A boundary point $\tilde{q} + (s - \frac{1}{2})\omega_1 + (t - \frac{1}{2})\omega_2$ has s or t in $\{0, 1\}$, so it is not in $\tilde{q} + \Lambda$; and $\tilde{q}$ is the only point of $\tilde{q} + \Lambda$ inside P . The boundary ∂P , traversed through the vertices in the order listed, is positively oriented, with winding number 1 about $\tilde{q}$ and 0 about the points outside P . By the residue theorem (Theorem 27.5.8), $\int_{\partial P} F dz = 2\pi i \text{Res}_{\tilde{q}} F \neq 0$, the residue at a simple pole being nonzero. On the other hand the integrals over opposite sides of P cancel: the side from $a + \omega_2$ to $a + \omega_1 + \omega_2$ is the translate by ω_2 of the side from a to $a + \omega_1$, traversed in the opposite direction, and $F(z + \omega_2) = F(z)$; similarly for the other pair. So $\int_{\partial P} F dz = 0$, a contradiction.

Thus L is a nontrivial element of $\text{Pic}^0(X)$, the kernel of c_1 in Theorem 42.2.4. The same argument shows that no meromorphic function on X has a single pole which is simple.

42.4 Hermitian metrics and the Chern connection

Theorem 42.4.1. Let E be a holomorphic vector bundle with a Hermitian metric h . There is a unique connection ∇ on E , the Chern connection, that is metric and satisfies $\nabla^{0,1} = \bar{\partial}$. In a holomorphic frame with $H_{ij} = h(e_i, e_j)$ one has $\theta = \bar{H}^{-1} \partial \bar{H}$ and curvature $F = \bar{\partial} \theta$, a $(1, 1)$ -form with values in $\text{End } E$. For a line bundle with local weight $h = e^{-\varphi}$ in a holomorphic frame, $F = \partial \bar{\partial} \varphi$ and

$$c_1(L) = \left[\frac{i}{2\pi} F \right] = \left[\frac{i}{2\pi} \partial \bar{\partial} \varphi \right] \in H^2(X; \mathbb{R}),$$

a class of type $(1, 1)$.

Proof. Write $\nabla = \nabla^{1,0} + \nabla^{0,1}$. The condition $\nabla^{0,1} = \bar{\partial}$ fixes the $(0,1)$ -part, and the metric condition $d h(s, t) = h(\nabla s, t) + h(s, \nabla t)$ splits into its $(1,0)$ - and $(0,1)$ -parts, the first of which reads $\partial h(s, t) = h(\nabla^{1,0} s, t) + h(s, \bar{\partial} t)$ and determines $\nabla^{1,0}$; this proves uniqueness, and the resulting formula in a holomorphic frame, $\theta = \bar{H}^{-1} \partial \bar{H}$, defines a connection with the two properties, which proves existence. Then $F = d\theta + \theta \wedge \theta$ has no $(0,2)$ -part because $\bar{\partial}^2 = 0$ and $\nabla^{0,1} = \bar{\partial}$, and no $(2,0)$ -part by the metric condition; the formula $F = \bar{\partial} \theta$ follows. For a line bundle $H = e^{-\varphi}$ is real, so $\theta = H^{-1} \partial H = \partial \log H = -\partial \varphi$ and $F = \bar{\partial} \theta = -\bar{\partial} \partial \varphi = \partial \bar{\partial} \varphi$; the Chern class is computed by Chern–Weil theory (Proposition 37.2.4). $\square$

Definition 42.4.2. A Hermitian line bundle (L, h) is *positive* if the real $(1,1)$ -form $\frac{i}{2\pi} F$ is positive definite, that is, if it is the fundamental form of a Hermitian metric on X (Chapter 44); L is *positive* if it admits such a metric. A line bundle is *ample* if a power of it gives an embedding of X in projective space (Chapter 47).

Example 42.4.3. On $\mathbb{CP}^n$, give $\mathcal{O}(-1)$ the metric induced from the standard Hermitian metric of $\mathbb{C}^{n+1}$. On $U_0 = \{x_0 \neq 0\}$, with coordinates $w_j = x_j/x_0$, the section $\sigma(w) = (1, w_1, \dots, w_n)$ is a holomorphic frame with $h(\sigma, \sigma) = 1 + |w|^2 = e^{-\varphi}$, so by Theorem 42.4.1 $\frac{i}{2\pi} F = -\frac{i}{2\pi} \partial \bar{\partial} \log(1 + |w|^2) = -\omega_{FS}$, where ω_{FS} , the *Fubini–Study form*, is the global $(1,1)$ -form given on U_0 by $\frac{i}{2\pi} \partial \bar{\partial} \log(1 + |w|^2)$ (on U_j one uses $\log(|x|^2/|x_j|^2)$ in place of $\log(1 + |w|^2) = \log(|x|^2/|x_0|^2)$; on overlaps these differ by $\log|x_k/x_j|^2$, the logarithm of the squared modulus of a nowhere vanishing holomorphic function, which $\partial \bar{\partial}$ kills); hence $\mathcal{O}(1)$ is positive, with $\frac{i}{2\pi} F = \omega_{FS}$ and $c_1(\mathcal{O}(1)) = [\omega_{FS}]$, the positive generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$ (Example 37.2.5).

42.5 The canonical bundle and adjunction

Proposition 42.5.1 (Adjunction). *Let $Z \subseteq X$ be a smooth hypersurface with associated line bundle $\mathcal{O}(Z)$. Then the normal bundle is $N_{Z/X} \cong \mathcal{O}(Z)|_Z$, and*

$$K_Z \cong (K_X \otimes \mathcal{O}(Z))|_Z.$$

For a smooth hypersurface of degree d in $\mathbb{CP}^n$ one gets $K_Z \cong \mathcal{O}(d - n - 1)|_Z$.

Proof. A local defining function f_i of Z gives, by df_i , an isomorphism of the conormal bundle $N_{Z/X}^\vee$ with $\mathcal{O}(-Z)|_Z$: the transition functions of the conormal bundle are the f_i/f_j restricted to Z , computed from $df_i = (f_i/f_j) df_j$ modulo T^*Z . Taking determinants in the exact sequence $0 \rightarrow N_{Z/X}^\vee \rightarrow \Omega_X^1|_Z \rightarrow \Omega_Z^1 \rightarrow 0$ gives $K_X|_Z \cong K_Z \otimes N_{Z/X}^\vee \cong K_Z \otimes \mathcal{O}(-Z)|_Z$, which is the formula. For $\mathbb{CP}^n$ use $K_{\mathbb{CP}^n} \cong \mathcal{O}(-n-1)$, which follows from the Euler sequence (Example 33.3.7) by taking determinants, and $\mathcal{O}(Z) = \mathcal{O}(d)$. $\square$

Example 42.5.2. A smooth plane curve of degree d in $\mathbb{CP}^2$ has $K_Z \cong \mathcal{O}(d-3)|_Z$, whence genus $g = (d-1)(d-2)/2$ by Riemann–Roch (Chapter 55); a smooth quadric surface in $\mathbb{CP}^3$ has $K \cong \mathcal{O}(-2)$ and is $\mathbb{CP}^1 \times \mathbb{CP}^1$; a smooth quartic surface in $\mathbb{CP}^3$ has trivial canonical bundle and is a K3 surface, one of the basic examples in Hodge theory.

42.6 Exercises

Exercise 42.6.1. Show that the holomorphic sections of $\mathcal{O}(k)$ on $\mathbb{CP}^n$ are the homogeneous polynomials of degree k in the homogeneous coordinates, and that there are none for $k < 0$.

Solution. A section is given by functions s_i on U_i with $s_i = (x_i/x_j)^{-k} s_j$; the function $F(x) = x_i^k s_i(x/x_i)$ is independent of i , holomorphic on $\mathbb{C}^{n+1} \setminus \{0\}$ and homogeneous of degree k . By Hartogs' theorem (Theorem 29.2.9) it extends to $\mathbb{C}^{n+1}$, and a homogeneous holomorphic function of degree k is a polynomial for $k \geq 0$ and zero for $k < 0$, by looking at its power series. $\square$

Exercise 42.6.2. Let X be a compact Riemann surface. Show that $\deg L = \int_X c_1(L)$ defines a homomorphism $\text{Pic}(X) \rightarrow \mathbb{Z}$, that $\deg \mathcal{O}(D) = \sum_p a_p$ for $D = \sum a_p p$, and that a line bundle of negative degree has no nonzero holomorphic sections.

Exercise 42.6.3. Compute the Chern connection and its curvature for the trivial line bundle with metric $e^{-|z|^2}$ on $\mathbb{C}$, and verify that the curvature is a positive $(1, 1)$ -form.

Exercise 42.6.4. For a complex torus $X = \mathbb{C}/\Lambda$ of dimension one, show that $H^1(X, \mathcal{O}_X) \cong \mathbb{C}$ and deduce that $\text{Pic}^0(X)$ is a complex torus of dimension one.

Chapter 43

Dolbeault Cohomology

43.1 Introduction

Dolbeault cohomology is the cohomology of the $\bar{\partial}$ -operator on forms of type (p, q) . By the Dolbeault theorem (Theorem 43.2.3) it computes the sheaf cohomology of the holomorphic p -forms, and on a compact complex manifold harmonic theory for $\bar{\partial}$ makes it finite-dimensional and gives Serre duality (Theorem 43.3.4). The numbers $h^{p,q} = \dim H^q(X, \Omega_X^p)$ are the *Hodge numbers*; for compact Kähler manifolds they are the dimensions of the pieces of the Hodge decomposition (Chapter 45). References: [GH78, Chapter 0], [Voi02b, Chapter 4], [Huy05, Chapters 2–4], [Wel08].

Prerequisites. Chapter 42 (Holomorphic Vector Bundles) and Chapter 39 (Elliptic Operators on Compact Manifolds).

X is a complex manifold of dimension n and E a holomorphic vector bundle on it.

43.2 The Dolbeault theorem

Definition 43.2.1. Let $\mathcal{A}^{p,q}(E)$ be the sheaf of smooth E -valued (p, q) -forms and $\mathcal{A}^{p,q}(X, E)$ its global sections. In a holomorphic frame of E set $\bar{\partial}_E(\sum_j \alpha_j e_j) = \sum_j \bar{\partial} \alpha_j e_j$; this is independent of the frame because the transition functions are holomorphic, and $\bar{\partial}_E^2 = 0$. The *Dolbeault cohomology* is $H_{\bar{\partial}}^{p,q}(X, E) = H^q(\mathcal{A}^{p,\bullet}(X, E), \bar{\partial}_E)$; for E trivial we write $H_{\bar{\partial}}^{p,q}(X)$.

Lemma 43.2.2. Let $\Delta' \subseteq \Delta$ be polydiscs with $\overline{\Delta'} \subseteq \Delta$ and α a smooth (p, q) -form on Δ with $q \geq 1$ and $\bar{\partial} \alpha = 0$. Then $\alpha = \bar{\partial} \beta$ on Δ' for some smooth $(p, q-1)$ -form β .

Proof. Writing $\alpha = \sum_I dz^I \wedge \alpha_I$ with α_I of type $(0, q)$, the condition $\bar{\partial} \alpha = 0$ holds componentwise, since $\bar{\partial}(dz^I) = 0$; so we may assume $p = 0$. For $q = 1$ this is Theorem 29.4.2. For general q the same induction applies: if α involves only $d\bar{z}^1, \dots, d\bar{z}^k$, write $\alpha = d\bar{z}^k \wedge \alpha_1 + \alpha_2$ with α_1, α_2 not involving $d\bar{z}^k, \dots, d\bar{z}^n$; the equation $\bar{\partial} \alpha = 0$ implies that the coefficients of α_1 are holomorphic in $z_{k+1}, \dots, z_n$. Solving $\partial g / \partial \bar{z}_k = (\text{coefficient of } \alpha_1)$ with the Cauchy transform of Lemma 29.4.1, after a cut-off in z_k alone, gives a form γ with $\alpha - \bar{\partial} \gamma$ involving only $d\bar{z}^1, \dots, d\bar{z}^{k-1}$ on a slightly smaller polydisc; by induction on k we conclude. The details are exactly as in the proof of Theorem 29.4.2; see [GH78, Chapter 0, Section 2]. $\square$

Theorem 43.2.3 (Dolbeault). The sequence of sheaves $0 \rightarrow \Omega_X^p(E) \rightarrow \mathcal{A}^{p,0}(E) \xrightarrow{\bar{\partial}_E} \mathcal{A}^{p,1}(E) \xrightarrow{\bar{\partial}_E} \dots$ is a fine resolution of the sheaf $\Omega_X^p(E) = \Omega_X^p \otimes E$ of holomorphic E -valued p -forms. Consequently

$$H_{\bar{\partial}}^{p,q}(X, E) \cong H^q(X, \Omega_X^p \otimes E).$$

Proof. Exactness at $\mathcal{A}^{p,0}$ says that the E -valued $(p, 0)$ -forms killed by $\bar{\partial}_E$ are the holomorphic ones (Proposition 41.3.3), and exactness at $\mathcal{A}^{p,q}$ for $q \geq 1$ holds on stalks by Lemma 43.2.2, applied in a holomorphic frame of E on small polydiscs. The sheaves $\mathcal{A}^{p,q}(E)$ are modules over the sheaf of smooth functions, hence fine and acyclic (Theorem 24.4.5); apply Proposition 24.7.1(3). $\square$

Theorem 43.2.4 (Holomorphic Poincaré lemma). *The holomorphic de Rham complex $0 \rightarrow \mathbb{C}_X \rightarrow \Omega_X^0 \xrightarrow{\partial} \Omega_X^1 \xrightarrow{\partial} \cdots \rightarrow \Omega_X^n \rightarrow 0$ is exact. Hence $H^k(X; \mathbb{C}) \cong \mathbb{H}^k(X, \Omega_X^\bullet)$, and there is a spectral sequence (the Frölicher or Hodge-to-de Rham spectral sequence)*

$$E_1^{p,q} = H^q(X, \Omega_X^p) \Rightarrow H^{p+q}(X; \mathbb{C}).$$

Proof. On holomorphic forms $d = \partial$. Consider the double complex of sheaves $\mathcal{A}^{p,q}$ with differentials ∂ and $\bar{\partial}$, whose total complex is the complexified de Rham complex. By Theorem 43.2.3 each column $(\mathcal{A}^{p,\bullet}, \bar{\partial})$ is a resolution of Ω_X^p ; by Lemma 13.7.2, applied to stalks, the inclusion $\Omega_X^\bullet \rightarrow \text{Tot}(\mathcal{A}^{\bullet,\bullet}) = \mathcal{A}_\mathbb{C}^\bullet$ is a quasi-isomorphism. The de Rham complex is a resolution of $\mathbb{C}_X$ (Theorem 35.3.1), so $\Omega_X^\bullet$ has cohomology sheaf $\mathbb{C}_X$ in degree 0 and none elsewhere, which is the exactness. The isomorphism with $H^k(X; \mathbb{C})$ follows from Proposition 24.7.1(1) and Theorem 35.3.1, and the spectral sequence is the first one of Proposition 24.7.1(2). $\square$

Proposition 43.2.5. *Let $f: X \rightarrow Y$ be a holomorphic map of complex manifolds. Pullback of forms maps $\mathcal{A}^{p,q}(Y)$ into $\mathcal{A}^{p,q}(X)$ and commutes with ∂ and $\bar{\partial}$, so it induces maps $f^*: H_{\bar{\partial}}^{p,q}(Y) \rightarrow H_{\bar{\partial}}^{p,q}(X)$, with $(g \circ f)^* = f^* \circ g^*$. The wedge product induces products $H_{\bar{\partial}}^{p,q}(X) \times H_{\bar{\partial}}^{p',q'}(X) \rightarrow H_{\bar{\partial}}^{p+p',q+q'}(X)$, making $\bigoplus_{p,q} H_{\bar{\partial}}^{p,q}(X)$ a graded-commutative algebra (for the total degree $p+q$), and f^* is a homomorphism of algebras.*

Proof. In local holomorphic coordinates w_j on Y , $f^*dw_j = d(w_j \circ f) = \partial(w_j \circ f)$ because $w_j \circ f$ is holomorphic, so f^*dw_j has type $(1, 0)$, and by conjugation $f^*d\bar{w}_j$ has type $(0, 1)$; hence f^* preserves types. Since $f^*d = df^*$ and $d = \partial + \bar{\partial}$ with ∂ of type $(1, 0)$ and $\bar{\partial}$ of type $(0, 1)$ (Proposition 41.3.3), comparing the components of type $(p+1, q)$ and $(p, q+1)$ of $f^*d\alpha = df^*\alpha$ for α of type (p, q) gives $f^*\partial\alpha = \partial f^*\alpha$ and $f^*\bar{\partial}\alpha = \bar{\partial} f^*\alpha$. So f^* maps $\bar{\partial}$ -closed forms to $\bar{\partial}$ -closed forms and $\bar{\partial}$ -exact forms to $\bar{\partial}$ -exact forms. Functoriality holds already for forms.

Similarly, taking the component of type $(p+p', q+q'+1)$ of the Leibniz rule $d(\alpha \wedge \beta) = d\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge d\beta$ gives $\bar{\partial}(\alpha \wedge \beta) = \bar{\partial}\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge \bar{\partial}\beta$. Hence the wedge product of $\bar{\partial}$ -closed forms is $\bar{\partial}$ -closed, and if $\bar{\partial}\beta = 0$ then $\bar{\partial}(\gamma \wedge \beta) = \bar{\partial}(\gamma \wedge \beta)$ is exact; so the product is well defined on classes. Graded commutativity and compatibility with f^* hold for forms. $\square$

Definition 43.2.6. The Bott–Chern and Aeppli cohomology groups of X are

$$H_{BC}^{p,q}(X) = \frac{\{\alpha \in \mathcal{A}^{p,q}(X) : \partial\alpha = \bar{\partial}\alpha = 0\}}{\partial\bar{\partial}\mathcal{A}^{p-1,q-1}(X)}, \quad H_A^{p,q}(X) = \frac{\{\alpha \in \mathcal{A}^{p,q}(X) : \partial\bar{\partial}\alpha = 0\}}{\partial\mathcal{A}^{p-1,q}(X) + \bar{\partial}\mathcal{A}^{p,q-1}(X)}.$$

Lemma 43.2.7. *The identity on forms induces natural linear maps*

$$H_{BC}^{p,q}(X) \rightarrow H_{\bar{\partial}}^{p,q}(X), \quad H_{BC}^{p,q}(X) \rightarrow H_{dR}^{p+q}(X; \mathbb{C}), \quad H_{\bar{\partial}}^{p,q}(X) \rightarrow H_A^{p,q}(X).$$

Complex conjugation induces isomorphisms $H_{BC}^{p,q}(X) \cong H_{BC}^{q,p}(X)$ of real vector spaces, conjugate-linear over $\mathbb{C}$.

Proof. The quotients in Definition 43.2.6 make sense because $\partial\bar{\partial}\beta$ is ∂ - and $\bar{\partial}$ -closed and $\partial\alpha = 0, \bar{\partial}\alpha = 0$ imply $\partial\bar{\partial}\alpha = 0$ (using $\partial^2 = \bar{\partial}^2 = 0$ and $\partial\bar{\partial} = -\bar{\partial}\partial$). If $\partial\alpha = \bar{\partial}\alpha = 0$, then α is $\bar{\partial}$ -closed and $d\alpha = 0$; and $\partial\bar{\partial}\beta = -\bar{\partial}(\partial\beta)$ is $\bar{\partial}$ -exact, while $\partial\bar{\partial}\beta = d(\bar{\partial}\beta)$ is d -exact because $\bar{\partial}^2\beta = 0$. This gives the first two maps. For the third, a $\bar{\partial}$ -closed form is $\partial\bar{\partial}$ -closed, and $\bar{\partial}$ -exact forms lie in the denominator of H_A . Finally $\overline{\partial\alpha} = \bar{\partial}\bar{\alpha}$, so conjugation preserves the conditions defining H_{BC} and maps $\partial\bar{\partial}\mathcal{A}^{p-1,q-1}$ onto $\partial\bar{\partial}\mathcal{A}^{q-1,p-1} = \partial\bar{\partial}\mathcal{A}^{q-1,p-1}$. $\square$

Remark 43.2.8. Unlike $H_{\bar{\partial}}^{p,q}$, the Bott–Chern groups have a definition symmetric in p and q , and they map to de Rham cohomology. On a compact Kähler manifold the $\partial\bar{\partial}$ -lemma (Chapter 45) says that a d -closed form of pure type which is d -exact or $\bar{\partial}$ -exact is of the form $\partial\bar{\partial}\beta$; this is exactly the injectivity of the first two maps of Lemma 43.2.7. On general compact complex manifolds these maps need not be injective, and Bott–Chern cohomology is a finer invariant than Dolbeault or de Rham cohomology.

43.3 Harmonic theory for $\bar{\partial}$

Definition 43.3.1. Let X be compact with a Hermitian metric and E a Hermitian holomorphic bundle. With the L^2 inner products on $\mathcal{A}^{p,q}(X, E)$, let $\bar{\partial}_E^*$ be the formal adjoint of $\bar{\partial}_E$ (Definition 39.5.1) and $\square_E = \bar{\partial}_E\bar{\partial}_E^* + \bar{\partial}_E^*\bar{\partial}_E$, the $\bar{\partial}$ -Laplacian. The space of $\bar{\partial}$ -harmonic forms is $\mathcal{H}^{p,q}(X, E) = \ker \square_E$.

Lemma 43.3.2. In the situation of Definition 43.3.1, a form $\alpha \in \mathcal{A}^{p,q}(X, E)$ is $\square_E$ -harmonic if and only if $\bar{\partial}_E\alpha = 0$ and $\bar{\partial}_E^*\alpha = 0$. In particular, for $q = 0$ the harmonic forms are exactly the holomorphic sections of $\Omega_X^p \otimes E$, for every choice of the metrics, and

$$\mathcal{H}^{p,0}(X, E) = H_{\bar{\partial}}^{p,0}(X, E) = H^0(X, \Omega_X^p \otimes E).$$

Proof. Since X is compact, the defining property of formal adjoints gives $\langle \square_E\alpha, \alpha \rangle = \|\bar{\partial}_E\alpha\|^2 + \|\bar{\partial}_E^*\alpha\|^2$ in L^2 , so $\square_E\alpha = 0$ forces $\bar{\partial}_E\alpha = \bar{\partial}_E^*\alpha = 0$; the converse is clear from the definition of $\square_E$. For $q = 0$, $\bar{\partial}_E^*$ maps $\mathcal{A}^{p,0}$ to $\mathcal{A}^{p,-1} = 0$, so the harmonic $(p, 0)$ -forms are the $\bar{\partial}_E$ -closed ones, which are the holomorphic sections of $\Omega_X^p \otimes E$ (Proposition 41.3.3, in a holomorphic frame of E); and $H_{\bar{\partial}}^{p,0}(X, E)$ is the kernel of $\bar{\partial}_E$ on $\mathcal{A}^{p,0}(X, E)$, as there are no forms of type $(p, -1)$. $\square$

Theorem 43.3.3. $\square_E$ is elliptic, with principal symbol $\frac{1}{2}|\xi|^2$. Consequently $\mathcal{H}^{p,q}(X, E)$ is finite-dimensional, there is an orthogonal decomposition $\mathcal{A}^{p,q}(X, E) = \mathcal{H}^{p,q}(X, E) \oplus \bar{\partial}_E\mathcal{A}^{p,q-1}(X, E) \oplus \bar{\partial}_E^*\mathcal{A}^{p,q+1}(X, E)$, and every Dolbeault class has a unique harmonic representative:

$$H^q(X, \Omega_X^p \otimes E) \cong H_{\bar{\partial}}^{p,q}(X, E) \cong \mathcal{H}^{p,q}(X, E).$$

In particular the Hodge numbers $h^{p,q}(X) = \dim H^q(X, \Omega_X^p)$ of a compact complex manifold are finite.

Proof. The symbol of $\bar{\partial}$ is $i\xi^{0,1}\wedge-$ and that of $\bar{\partial}^*$ is $-i\iota_{\xi^{0,1}}$, so the symbol of $\square_E$ is $\xi^{0,1}\wedge\iota + \iota\xi^{0,1}\wedge = |\xi^{0,1}|^2 = \frac{1}{2}|\xi|^2$, by the same computation as in Exercise 39.6.1. The rest is the proof of Theorems 40.2.1 and 40.2.2 with d, δ replaced by $\bar{\partial}_E, \bar{\partial}_E^*$ (Remark 40.3.6): the only inputs were ellipticity, self-adjointness and $\bar{\partial}_E^2 = 0$. Combine with Theorem 43.2.3. $\square$

Theorem 43.3.4 (Serre duality). Let X be a compact complex manifold of dimension n and E a holomorphic vector bundle. The pairing

$$H^q(X, E) \times H^{n-q}(X, K_X \otimes E^\vee) \rightarrow H^n(X, K_X) \xrightarrow{\int} \mathbb{C}, \quad ([\alpha], [\beta]) \mapsto \int_X \alpha \wedge \beta,$$

computed with Dolbeault representatives of types $(0, q)$ and $(n, n - q)$ and the evaluation $E \otimes E^\vee \rightarrow \mathbb{C}$, is perfect. In particular $h^{p,q} = h^{n-p, n-q}$ for the Hodge numbers.

Proof. The pairing is well defined by Stokes' theorem, since $\int_X \bar{\partial}(\alpha \wedge \gamma) = \int_X d(\alpha \wedge \gamma) = 0$ for forms of total type $(n, n - 1)$. Choose Hermitian metrics and define the conjugate-linear operator $\bar{*}_E: \mathcal{A}^{0,q}(E) \rightarrow \mathcal{A}^{n, n-q}(E^\vee)$ by $\alpha \wedge \bar{*}_E \beta = \langle \alpha, \beta \rangle \text{vol}$, using the metric to identify $\bar{E}$ with $E^\vee$. As for the real Hodge star, $\bar{\partial}^* = -\bar{*} \bar{\partial} \bar{*}$, so $\bar{*}_E$ maps $\square_E$ -harmonic forms bijectively onto $\square_{K \otimes E^\vee}$ -harmonic forms. For a nonzero harmonic α , $\int_X \alpha \wedge \bar{*}_E \alpha = \|\alpha\|^2 > 0$, so the pairing is nondegenerate on harmonic representatives, which by Theorem 43.3.3 represent all classes. For the Hodge numbers take $E = \Omega_X^p$, so $K_X \otimes E^\vee \cong \Omega_X^{n-p}$ by the perfect pairing $\Omega^p \otimes \Omega^{n-p} \rightarrow K_X$. $\square$

43.4 Examples and the Frölicher inequality

Proposition 43.4.1. *For a compact complex manifold, $b_k(X) \leq \sum_{p+q=k} h^{p,q}(X)$, and $\chi(X) = \sum_{p,q} (-1)^{p+q} h^{p,q}(X)$. Equality holds for all k if and only if the Frölicher spectral sequence degenerates at E_1 .*

Proof. In a spectral sequence of finite-dimensional vector spaces, $\dim E_{r+1}^{p,q} \leq \dim E_r^{p,q}$ and $\sum_{p+q=k} \dim E_\infty^{p,q} = b_k$ (Theorem 14.3.4), so $b_k \leq \sum_{p+q=k} h^{p,q}$, with equality for all k if and only if all differentials d_r , $r \geq 1$, vanish. The Euler characteristic is unchanged when passing to cohomology of a differential of total degree one, so it is the same on every page. $\square$

Example 43.4.2. 1. For a complex torus $X = \mathbb{C}^n / \Lambda$, the forms $dz^I \wedge d\bar{z}^J$ with constant coefficients are harmonic for the flat metric and span each $\mathcal{H}^{p,q}$, so $h^{p,q} = \binom{n}{p} \binom{n}{q}$, and $\sum_{p+q=k} h^{p,q} = \binom{2n}{k} = b_k$: the Frölicher spectral sequence degenerates.

2. For $\mathbb{CP}^n$, $h^{p,q} = 1$ if $p = q \leq n$ and 0 otherwise. Since $b_{2k} = 1$ and $b_{2k+1} = 0$, this follows from the Hodge decomposition (Chapter 45) once one knows $h^{k,k} \geq 1$, which is witnessed by the class of ω_{FS}^k ; see [GH78, Chapter 0, Section 7].

3. For the Hopf surface of Exercise 41.5.3, $b_1 = 1$, $h^{0,1} = 1$ and $h^{1,0} = 0$. So $b_1 = h^{1,0} + h^{0,1}$ (the Frölicher spectral sequence of a compact complex surface always degenerates), but $h^{1,0} \neq h^{0,1}$: the Hodge symmetry $h^{p,q} = h^{q,p}$ of compact Kähler manifolds fails [BHPVdV04, Chapter IV].

Example 43.4.3. The Hodge numbers of $\mathbb{CP}^1$ can be computed directly. First $h^{0,0} = 1$ by Proposition 41.2.3. A holomorphic 1-form is $f(z) dz$ on the chart $\mathbb{C}$, with f entire; in the chart $w = 1/z$ at infinity, $dz = -w^{-2} dw$, so $-f(1/w)w^{-2}$ must extend holomorphically to $w = 0$. Hence $|f(z)| \leq C|z|^{-2}$ for $|z| \geq 1$, so f is bounded, hence constant by Liouville's theorem (Corollary 27.4.3), and the constant is 0. So $h^{1,0} = 0$ by Lemma 43.3.2. Serre duality (Theorem 43.3.4, with $n = 1$ and $E = \mathcal{O}$ or $E = \Omega^1$) gives $h^{0,1} = h^{1,0} = 0$ and $h^{1,1} = h^{0,0} = 1$. Since $b_0 = b_2 = 1$ and $b_1 = 0$, equality holds in Proposition 43.4.1. Moreover $H^1(\mathbb{CP}^1, \mathcal{O}) = 0$, and $H^2(\mathbb{CP}^1, \mathcal{O}) = 0$ because there are no forms of type $(0, 2)$ on a curve; by Theorem 42.2.4, $c_1: \text{Pic}(\mathbb{CP}^1) \rightarrow H^2(\mathbb{CP}^1; \mathbb{Z}) \cong \mathbb{Z}$ is an isomorphism.

Counterexample 43.4.4. Dolbeault cohomology is not a homotopy invariant, unlike de Rham cohomology. The plane $\mathbb{C}$ is contractible, so $H_{\text{dR}}^k(\mathbb{C}) = 0$ for $k > 0$ and $H_{\text{dR}}^0(\mathbb{C}) = \mathbb{C}$. But $H_{\bar{\partial}}^{0,0}(\mathbb{C})$ and $H_{\bar{\partial}}^{1,0}(\mathbb{C})$ are the spaces of entire functions and of holomorphic 1-forms $f(z) dz$ (there is nothing to divide by in type $(p, 0)$), both infinite-dimensional: they contain the z^k and the $z^k dz$. The two computations are reconciled by $\partial: H_{\bar{\partial}}^{0,0}(\mathbb{C}) \rightarrow H_{\bar{\partial}}^{1,0}(\mathbb{C})$, $f \mapsto f' dz$,

which is surjective, because every entire function has an entire primitive (Corollary 27.3.2), and has kernel the constants. In the Frölicher spectral sequence of Theorem 43.2.4 the first differential is induced by ∂ , and here it cancels the two infinite-dimensional groups against each other, leaving $H_{\text{dR}}^0(\mathbb{C}) = \mathbb{C}$.

Remark 43.4.5. Cohomology of line bundles on projective space is computed by Bott's formula: $H^q(\mathbb{CP}^n, \mathcal{O}(k))$ vanishes for $0 < q < n$ and all k , H^0 = homogeneous polynomials of degree k , and $H^n(\mathbb{CP}^n, \mathcal{O}(k)) \cong H^0(\mathbb{CP}^n, \mathcal{O}(-k - n - 1))^\vee$ by Serre duality, since $K_{\mathbb{CP}^n} = \mathcal{O}(-n - 1)$ (Proposition 42.5.1). Vanishing theorems for positive bundles are the subject of Chapter 47; the algebraic computation is in Chapter 52.

43.5 Exercises

Exercise 43.5.1. Let X be a compact Riemann surface of genus g . Using Serre duality and the Frölicher inequality, show that $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1}$, and that $h^{1,0} + h^{0,1} \geq 2g$ with equality if and only if the Frölicher spectral sequence degenerates (it always does in dimension one, by Chapter 45).

Solution. $h^{0,0} = 1$ since holomorphic functions are constant (Proposition 41.2.3); Serre duality gives $h^{1,1} = h^{0,0}$ and $h^{1,0} = h^{0,1}$. The inequality is Proposition 43.4.1 for $k = 1$, with $b_1 = 2g$. $\square$

Exercise 43.5.2. Show that $H_{\bar{\partial}}^{0,1}(\mathbb{C}) = 0$, but that $H_{\bar{\partial}}^{0,1}(\mathbb{C}^2 \setminus \{0\})$ is infinite-dimensional.

Exercise 43.5.3. For a compact Riemann surface X and a line bundle L , show that $H^1(X, L) \cong H^0(X, K_X \otimes L^\vee)^\vee$, and deduce that $H^1(X, L) = 0$ if $\deg L > 2g - 2$.

Exercise 43.5.4. Show that the holomorphic Euler characteristic $\chi(X, E) = \sum_q (-1)^q \dim H^q(X, E)$ of a compact complex manifold is the index of the elliptic operator $\bar{\partial}_E + \bar{\partial}_E^* : \mathcal{A}^{0,\text{even}}(E) \rightarrow \mathcal{A}^{0,\text{odd}}(E)$.

Chapter 44

Kähler Manifolds

44.1 Introduction

A Kähler metric is a Hermitian metric whose fundamental 2-form is closed. This single condition makes the Riemannian and the complex structure compatible to second order (Proposition 44.3.2), it holds on every projective manifold (Corollary 44.4.2), and it is exactly what the Hodge decomposition of Chapter 45 needs. This chapter develops Kähler metrics and the pointwise linear algebra of the Lefschetz operator (Theorem 44.5.2), which becomes the hard Lefschetz theorem in Chapter 46. References: [GH78, Chapter 0], [Huy05, Chapters 1 and 3], [Voi02b, Chapters 3 and 6].

Prerequisites. Chapter 43 (Dolbeault Cohomology), Chapter 38 (Riemannian Geometry) and Chapter 11 (Representations of Groups and Lie Algebras).

Throughout, X is a complex manifold of complex dimension n and J is its almost complex structure (multiplication by i on tangent vectors). Forms are complex valued unless said otherwise, and $\Lambda^{p,q}$ denotes forms of type (p, q) .

44.2 Hermitian metrics and the fundamental form

Definition 44.2.1. A Hermitian metric on X is a Riemannian metric g with $g(JU, JV) = g(U, V)$ for all tangent vectors U, V . Its *fundamental form* is the real 2-form $\omega(U, V) = g(JU, V)$.

Proposition 44.2.2. Let g be a Hermitian metric and extend it complex bilinearly to complexified tangent vectors. In holomorphic coordinates $z^j = x^j + iy^j$ put $h_{j\bar{k}} = 2g(\partial/\partial z^j, \partial/\partial \bar{z}^k)$. Then $(h_{j\bar{k}})$ is a positive definite Hermitian matrix at every point and

$$\omega = \frac{i}{2} \sum_{j,k} h_{j\bar{k}} dz^j \wedge d\bar{z}^k,$$

a real form of type $(1, 1)$. Conversely, every real $(1, 1)$ -form $\frac{i}{2} \sum h_{j\bar{k}} dz^j \wedge d\bar{z}^k$ whose matrix $(h_{j\bar{k}})$ is positive definite (such a form is called **positive**) is the fundamental form of exactly one Hermitian metric. For the standard metric of $\mathbb{C}^n$ one gets $\omega_0 = \frac{i}{2} \sum_j dz^j \wedge d\bar{z}^j = \sum_j dx^j \wedge dy^j$.

Proof. The J -invariance of g says that g pairs the $+i$ - and $-i$ -eigenspaces of J with each other and each of them with itself trivially: $g(\partial_j, \partial_k) = g(J\partial_j, J\partial_k) = i^2 g(\partial_j, \partial_k)$ forces $g(\partial_j, \partial_k) = 0$, where $\partial_j = \partial/\partial z^j$, and similarly for $\bar{\partial}_j = \partial/\partial \bar{z}^j$. So $g = \frac{1}{2} \sum h_{j\bar{k}} (dz^j \otimes d\bar{z}^k + d\bar{z}^k \otimes dz^j)$. Since g is real, $\overline{h_{j\bar{k}}} = h_{k\bar{j}}$, and for $U = \sum u^j \partial_j + \overline{u^j} \bar{\partial}_j$ real, $g(U, U) = \sum h_{j\bar{k}} u^j \overline{u^k}$, so $(h_{j\bar{k}})$ is positive definite. Since $J\partial_j = i\partial_j$ and $J\bar{\partial}_k = -i\bar{\partial}_k$, $\omega(\partial_j, \bar{\partial}_k) = g(i\partial_j, \bar{\partial}_k) = \frac{i}{2} h_{j\bar{k}}$, $\omega(\bar{\partial}_k, \partial_j) = -\frac{i}{2} h_{j\bar{k}}$ and ω vanishes on pairs of the same type; this is the displayed formula. Conversely, a

positive $(1, 1)$ -form determines $h_{j\bar{k}}$ and hence g by the same formulas, and g is then a J -invariant Riemannian metric. For $\mathbb{C}^n$, $g(\partial_j, \bar{\partial}_k) = \frac{1}{4}(g(\partial_{x^j}, \partial_{x^k}) + g(\partial_{y^j}, \partial_{y^k})) = \frac{1}{2}\delta_{jk}$, and $\frac{i}{2}dz \wedge d\bar{z} = dx \wedge dy$. $\square$

Proposition 44.2.3. *For a Hermitian metric, $\omega^n/n! = \text{vol}_g$ is the Riemannian volume form for the canonical orientation. The restriction of ω to a complex submanifold Y is the fundamental form of the restricted metric. Hence for a nonempty compact complex submanifold, or analytic subvariety, Z of pure dimension k ,*

$$\int_Z \frac{\omega^k}{k!} = \text{vol}(Z_{\text{reg}}) > 0.$$

Proof. At a point choose, by Gram–Schmidt for the Hermitian form $(h_{j\bar{k}})$, a linear change of coordinates making $h_{j\bar{k}} = \delta_{jk}$ there. Then $\omega = \sum_j dx^j \wedge dy^j$ at that point, so $\omega^n = n! dx^1 \wedge dy^1 \wedge \cdots \wedge dx^n \wedge dy^n$, which is the volume form of the orthonormal basis $\partial_{x^1}, \partial_{y^1}, \dots$ in the canonical orientation. The tangent spaces of Y are J -stable, so the restricted metric is Hermitian, and $\omega(U, V) = g(JU, V)$ for U, V tangent to Y is its fundamental form. The formula for Z follows by applying the first statement to Z_{reg} , which is a nonempty open complex manifold of dimension k ; the integral converges by Proposition 41.4.2. $\square$

44.3 The Kähler condition

Definition 44.3.1. A Hermitian metric is *Kähler* if its fundamental form is closed, $d\omega = 0$; then ω is called a *Kähler form*. A complex manifold is *Kähler* if it admits a Kähler metric, and the class $[\omega] \in H^2(X; \mathbb{R})$ is then a *Kähler class*.

Proposition 44.3.2. *For a Hermitian metric on X the following are equivalent:*

1. $d\omega = 0$;
2. around every point p there are holomorphic coordinates centred at p with $h_{j\bar{k}} = \delta_{jk} + O(|z|^2)$ (the metric osculates to order two to the flat metric);
3. locally $\omega = i\partial\bar{\partial}\varphi$ for a smooth real function φ , a local Kähler potential.

Proof. (2) $\Rightarrow$ (1): in the coordinates of (2), the coefficients of $d\omega$ at p are first derivatives of the $h_{j\bar{k}}$ at 0, which vanish; since p is arbitrary, $d\omega = 0$. (3) $\Rightarrow$ (1): $d(i\partial\bar{\partial}\varphi) = i(\partial + \bar{\partial})\partial\bar{\partial}\varphi = 0$ by Proposition 41.3.3.

(1) $\Rightarrow$ (2): after a linear change of coordinates, $h_{j\bar{k}}(0) = \delta_{jk}$. By Taylor's theorem

$$h_{j\bar{k}} = \delta_{jk} + \sum_l (a_{jkl}z^l + b_{jkl}\bar{z}^l) + O(|z|^2),$$

and $\overline{h_{j\bar{k}}} = h_{k\bar{j}}$ gives $b_{jkl} = \overline{a_{kjl}}$. The $(2, 1)$ -component of $d\omega$ is $\partial\omega = \frac{i}{2} \sum \partial_l h_{j\bar{k}} dz^l \wedge dz^j \wedge d\bar{z}^k$, and its vanishing at 0 says that $a_{jkl} = a_{lkj}$. Define $w^k = z^k + \frac{1}{2} \sum_{j,l} a_{jkl} z^j \bar{z}^l$. Since $dw^k = dz^k$ at 0, the map $z \mapsto w$ is a diffeomorphism near 0 (Theorem 25.11.2), and its inverse is holomorphic because its derivative, the inverse of a complex linear map, is complex linear; so the w^k are holomorphic coordinates near 0. By the symmetry $a_{jkl} = a_{lkj}$, $dw^k = dz^k + \sum_{j,l} a_{jkl} z^j \bar{z}^l dz^l$. Hence

$$\sum_k dw^k \wedge d\bar{w}^k = \sum_{j,k} \left(\delta_{jk} + \sum_l a_{jkl} z^l + \sum_l \overline{a_{kjl}} \bar{z}^l \right) dz^j \wedge d\bar{z}^k + O(|z|^2),$$

so $\omega - \frac{i}{2} \sum_k dw^k \wedge d\bar{w}^k$ has coefficients $O(|z|^2)$ in the basis $dz^j \wedge d\bar{z}^k$, hence also in the basis $dw^j \wedge d\bar{w}^k$, which differs from it by a smooth change of basis, and $|z|$ and $|w|$ are comparable near 0. This is (2).

(1) $\Rightarrow$ (3): on a polydisc Δ around a point, $\omega = d\eta$ for a real 1-form η by the Poincaré lemma (Theorem 35.2.3). Write $\eta = \eta^{1,0} + \eta^{0,1}$ with $\eta^{1,0} = \overline{\eta^{0,1}}$. The $(0,2)$ -component of $d\eta = \omega$ is $\bar{\partial}\eta^{0,1} = 0$, so on a smaller polydisc $\eta^{0,1} = \bar{\partial}f$ for a smooth complex function f by Lemma 43.2.2, and then $\eta^{1,0} = \partial\bar{f}$. The $(1,1)$ -component gives

$$\omega = \partial\eta^{0,1} + \bar{\partial}\eta^{1,0} = \partial\bar{\partial}f + \bar{\partial}\partial\bar{f} = \partial\bar{\partial}(f - \bar{f}),$$

so $\varphi = -i(f - \bar{f}) = 2\operatorname{Im} f$ is real and $i\partial\bar{\partial}\varphi = \omega$. $\square$

Remark 44.3.3. The Kähler condition has several other equivalent forms: J is parallel for the Levi-Civita connection; the Levi-Civita connection of g , transported to $T^{1,0}X$, is the Chern connection of the Hermitian holomorphic bundle $T^{1,0}X$ (Theorem 42.4.1); the holonomy group of g is contained in $U(n)$ [Huy05, Chapter 4]. Condition (2) is the useful one in practice: any first-order identity between operators built from g and J that holds on $\mathbb{C}^n$ with the flat metric holds on every Kähler manifold, since it can be checked at the centre of such coordinates. The Kähler identities of Chapter 45 are proved this way.

44.4 Examples

Proposition 44.4.1 (Fubini–Study metric). *Let σ be a holomorphic section of the projection $\mathbb{C}^{n+1} \setminus \{0\} \rightarrow \mathbb{CP}^n$ over an open set U . The form*

$$\omega_{FS} = \frac{i}{2\pi} \partial\bar{\partial} \log |\sigma|^2$$

does not depend on σ , and so defines a real $(1,1)$ -form on $\mathbb{CP}^n$. It is a Kähler form, invariant under the action of $U(n+1)$, and $[\omega_{FS}] = c_1(\mathcal{O}(1))$, the positive generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$.

Proof. Another section is $\lambda\sigma$ with λ holomorphic and nowhere zero on U (shrinking U so that $\log \lambda$ exists), and $\log |\lambda\sigma|^2 = \log |\sigma|^2 + \log \lambda + \overline{\log \lambda}$. Since $\bar{\partial} \log \lambda = 0$ and $\partial \log \bar{\lambda} = 0$, $\partial\bar{\partial}$ kills the last two terms, so ω_{FS} is well defined; it is real because $i\partial\bar{\partial}u = -i\bar{\partial}\partial u = i\partial\bar{\partial}u$ for real u , and closed by Proposition 44.3.2. For $A \in U(n+1)$, $A \circ \sigma$ is a section over $A(U)$ with $|A\sigma| = |\sigma|$, so ω_{FS} is $U(n+1)$ -invariant. As $U(n+1)$ acts transitively on $\mathbb{CP}^n$, positivity need only be checked at $[1 : 0 : \cdots : 0]$. With the section $\sigma(w) = (1, w_1, \dots, w_n)$ in affine coordinates, $\log |\sigma|^2 = \log(1 + |w|^2) = |w|^2 + O(|w|^4)$, so at $w = 0$

$$\omega_{FS} = \frac{i}{2\pi} \sum_j dw^j \wedge d\bar{w}^j = \frac{1}{\pi} \sum_j du^j \wedge dv^j \quad (w^j = u^j + iv^j),$$

which is positive. Finally, σ is a nowhere-vanishing section of $\mathcal{O}(-1) \subseteq \mathbb{CP}^n \times \mathbb{C}^{n+1}$, and the dual frame σ^* of $\mathcal{O}(1)$ has norm $|\sigma^*|^2 = |\sigma|^{-2} = e^{-\varphi}$ for the dual of the standard metric, with $\varphi = \log |\sigma|^2$. By Theorem 42.4.1, $c_1(\mathcal{O}(1)) = [\frac{i}{2\pi} \partial\bar{\partial}\varphi] = [\omega_{FS}]$. This is the positive generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$ by Example 42.4.3. $\square$

Corollary 44.4.2. *A complex submanifold of a Kähler manifold is Kähler with the restricted metric. In particular every projective manifold, that is, every closed complex submanifold $X \subseteq \mathbb{CP}^N$, is Kähler, and its Kähler class $[\omega_{FS}|_X]$ is the restriction of the hyperplane class $c_1(\mathcal{O}(1))$.*

Proof. By Proposition 44.2.3, the fundamental form of the restricted metric is the restriction of ω , and restriction of forms commutes with d . $\square$

- Example 44.4.3.** 1. $\mathbb{C}^n$ with ω_0 , and complex tori $\mathbb{C}^n/\Lambda$ with the form induced by ω_0 , which is invariant under translations. Products of Kähler manifolds are Kähler, with $\omega = \text{pr}_1^* \omega_1 + \text{pr}_2^* \omega_2$.
2. Every Hermitian metric on a Riemann surface is Kähler, since $d\omega$ is a 3-form on a real surface.
3. Every smooth projective variety is Kähler (Corollary 44.4.2). The blow-up of a compact Kähler manifold along a compact complex submanifold is again Kähler [Voi02b, Chapter 3].
4. Not every compact Kähler manifold is projective: a complex torus $\mathbb{C}^n/\Lambda$ with $n \geq 2$ and Λ general has no nonconstant meromorphic functions, hence no embedding into projective space [BL04]; see also Chapter 57. Voisin even showed that there are compact Kähler manifolds that are not homotopy equivalent to any projective manifold [Voi04].

Proposition 44.4.4. *Let X be a compact Kähler manifold with Kähler form ω . Then $[\omega]^k \neq 0$ in $H^{2k}(X; \mathbb{R})$ for $0 \leq k \leq n$; in particular $b_{2k}(X) \geq 1$. The cycle class (Proposition 41.4.2) of a nonempty analytic subvariety Z of pure dimension k is nonzero in $H^{2n-2k}(X; \mathbb{R})$.*

Proof. By Proposition 44.2.3, $\int_X \omega^k \wedge \omega^{n-k} = n! \text{vol}(X) > 0$. If $\omega^k = d\eta$ were exact, then $\omega^n = d(\eta \wedge \omega^{n-k})$ would be exact and its integral would vanish by Stokes' theorem. For Z , the class $[Z]$ is represented by the current of integration over Z_{reg} , whose value on the closed form ω^k is $\int_Z \omega^k = k! \text{vol}(Z_{\text{reg}}) > 0$; since a closed current that is exact vanishes on closed forms, $[Z] \neq 0$ (Theorem 35.4.3). $\square$

Lemma 44.4.5 (Wirtinger inequality for 2-planes). *Let g be a Hermitian metric with fundamental form ω , and let u, v be orthonormal tangent vectors at a point. Then $|\omega(u, v)| \leq 1$, with equality if and only if $v = \pm Ju$, that is, if and only if the real 2-plane spanned by u and v is a complex line.*

Proof. $\omega(u, v) = g(Ju, v)$, and $|Ju| = |u| = 1$ because g is J -invariant. By the Cauchy–Schwarz inequality $|g(Ju, v)| \leq |Ju||v| = 1$, with equality if and only if $v = \pm Ju$. A real plane containing u and Ju is J -stable, hence a complex line. $\square$

Proposition 44.4.6. *Let (X, ω) be a Kähler manifold and $C \subseteq X$ a compact complex curve (a compact complex submanifold of dimension 1). If Σ is a compact oriented surface and $f: \Sigma \rightarrow X$ an immersion with $f_*[\Sigma] = [C]$ in $H_2(X; \mathbb{R})$, then $\text{area}(f) \geq \text{area}(C) = \int_C \omega$, with equality if $f(\Sigma)$ is a complex curve with its complex orientation. So holomorphic curves minimise area in their homology class.*

Proof. At a point of Σ choose an oriented orthonormal basis u, v of $T\Sigma$ for the pulled-back metric; the area form of f^*g takes the value 1 on (u, v) , while $f^*\omega$ takes the value $\omega(df u, df v) \leq 1$ by Lemma 44.4.5. So $f^*\omega \leq dA$ pointwise, and

$$\text{area}(f) = \int_{\Sigma} dA \geq \int_{\Sigma} f^*\omega = \int_C \omega = \text{area}(C),$$

where the middle equality holds because ω is closed and $f_*[\Sigma] = [C]$ (Theorem 35.3.1), and the last is Proposition 44.2.3 with $k = 1$. If $f(\Sigma)$ is a complex curve with the complex orientation, equality holds pointwise by the equality case of the lemma. $\square$

Remark 44.4.7. The same argument, with the Wirtinger inequality $\frac{\omega^k}{k!} \leq \text{vol}$ on real $2k$ -planes, shows that compact complex submanifolds of any dimension minimise volume in their homology class [GH78, Chapter 0, Section 2]. A closed form that is bounded by the volume on every plane, with equality on a class of submanifolds, is called a *calibration*; the Kähler form is the first example, and the holomorphic volume form of a Calabi–Yau manifold calibrates the *special Lagrangian* submanifolds [Joy00].

Counterexample 44.4.8. The Hopf surface $X = (\mathbb{C}^2 \setminus \{0\})/\langle z \mapsto 2z \rangle$ is a compact complex manifold diffeomorphic to $S^1 \times S^3$ (Exercise 41.5.3), and it is not Kähler: $H^2(X; \mathbb{R}) = 0$, whereas a compact Kähler manifold of positive dimension has $H^2 \neq 0$ by Proposition 44.4.4. Another obstruction, that $b_1(X) = 1$ is odd, is explained in Chapter 45.

44.5 Linear algebra of the Lefschetz operator

Definition 44.5.1. Let V be a real vector space of dimension $2n$ with a complex structure J and a J -invariant inner product g , with fundamental form ω . Let $\Lambda^\bullet = \Lambda^\bullet V^* \otimes_{\mathbb{R}} \mathbb{C}$ with the Hermitian inner product induced by g , for which the wedge products of an orthonormal basis of V^* form an orthonormal basis. Define:

- the *Lefschetz operator* $L(\alpha) = \omega \wedge \alpha$, of degree $+2$;
- its adjoint $\Lambda = L^*$, of degree -2 ;
- the *counting operator* $H = \sum_k (k - n) \pi_k$, where π_k is the projection onto Λ^k .

On a Hermitian manifold these operators act pointwise on forms. A form α with $\Lambda\alpha = 0$ is called *primitive*, and P^k denotes the primitive forms of degree k .

Theorem 44.5.2. *The operators satisfy*

$$[L, \Lambda] = H, \quad [H, L] = 2L, \quad [H, \Lambda] = -2\Lambda,$$

so $e = L$, $f = \Lambda$, $h = H$ define a representation of $\mathfrak{sl}_2$ on $\Lambda^\bullet$ in which Λ^k is the weight space of weight $k - n$. Consequently:

1. for $0 \leq k \leq n$, $L^{n-k}: \Lambda^k \rightarrow \Lambda^{2n-k}$ is an isomorphism;
2. $P^k = 0$ for $k > n$, and every $\alpha \in \Lambda^k$ can be written uniquely as $\alpha = \sum_{j \geq 0} L^j \alpha_j$ with $\alpha_j \in P^{k-2j}$ (the Lefschetz or primitive decomposition);
3. for $k \leq n$, a form $\alpha \in \Lambda^k$ is primitive if and only if $L^{n-k+1} \alpha = 0$;
4. L, Λ map $\Lambda^{p,q}$ to $\Lambda^{p+1,q+1}$ and $\Lambda^{p-1,q-1}$, and if α has type (p, q) then each α_j has type $(p - j, q - j)$.

Proof. The last two commutators hold because L raises the degree by 2 and Λ lowers it by 2: for $\alpha \in \Lambda^k$, $HL\alpha - LH\alpha = (k + 2 - n)L\alpha - (k - n)L\alpha = 2L\alpha$, and similarly for Λ .

For $[L, \Lambda] = H$, choose an orthonormal basis $e_1, Je_1, \dots, e_n, Je_n$ of V (possible by induction: $e_1^\perp$ is J -stable when e_1 is a unit vector and $e_1 \perp Je_1$, as $g(e_1, Je_1) = g(Je_1, J^2 e_1) = -g(Je_1, e_1)$). Let $x^1, y^1, \dots, x^n, y^n$ be the dual basis, so that $\omega = \sum_j \omega_j$ with $\omega_j = x^j \wedge y^j$. The algebra $\Lambda^\bullet$ is the graded tensor product of the four-dimensional algebras $A_j = \Lambda^\bullet \langle x^j, y^j \rangle$, with the product inner product; $L = \sum_j L_j$ where L_j is multiplication by ω_j in the j -th factor, and so $\Lambda = \sum_j \Lambda_j$ with Λ_j the adjoint of L_j acting in the j -th factor. Since L_j and Λ_j have even degree, operators in different factors commute, and

$$[L, \Lambda] = \sum_{j,k} [L_j, \Lambda_k] = \sum_j [L_j, \Lambda_j].$$

In A_j , with orthonormal basis $1, x^j, y^j, x^j \wedge y^j$, the operator L_j maps $1 \mapsto x^j \wedge y^j$ and kills the other basis vectors, and Λ_j maps $x^j \wedge y^j \mapsto 1$ and kills the others. So $[L_j, \Lambda_j]$ is -1 on 1 , 0 on x^j, y^j and $+1$ on $x^j \wedge y^j$: it is multiplication by $(\deg_j - 1)$, where $\deg_j$ is the degree in

the j -th factor. On a product of basis elements of total degree $k = \sum_j \deg_j$, $[L, \Lambda]$ is therefore multiplication by $\sum_j (\deg_j - 1) = k - n$, which is H .

The consequences are statements about finite-dimensional $\mathfrak{sl}_2$ -representations. The map $(e, f, h) \mapsto (f, e, -h)$ is an automorphism of $\mathfrak{sl}_2$, so $e' = \Lambda$, $f' = L$, $h' = -H$ is again a representation, in which Λ^k has weight $n - k$. Corollary 11.5.4 for (e', f', h') says: $f'^j = L^j$ is an isomorphism from weight j to weight $-j$, which for $j = n - k$ is (1); the primitive vectors $\ker e' = \ker \Lambda$ occur only in weights $j \geq 0$, that is, in degrees $k \leq n$; and

$$\Lambda^\bullet = \bigoplus_{j \geq 0} \bigoplus_{i=0}^j L^i P^{n-j},$$

which, restricted to degree k , is (2). For (3), write $\alpha = \sum_i L^i \alpha_i$ with $\alpha_i \in P^{k-2i}$. A primitive vector of degree $m \leq n$ generates an irreducible representation with basis $\alpha, L\alpha, \dots, L^{n-m}\alpha$ and $L^{n-m+1}\alpha = 0$ (Theorem 11.5.2). So $L^{n-k+1}L^i\alpha_i = 0$ for $i = 0$, while for $i \geq 1$ the exponent $n - k + 1 + i \leq n - (k - 2i)$ and $L^{n-k+1+i}\alpha_i \neq 0$ if $\alpha_i \neq 0$. These terms lie in different summands of the direct sum above, so $L^{n-k+1}\alpha = 0$ if and only if $\alpha_i = 0$ for all $i \geq 1$, that is, if and only if $\alpha = \alpha_0$ is primitive.

For (4), L has type $(1, 1)$ because ω does (Proposition 44.2.2); since the spaces $\Lambda^{p,q}$ are mutually orthogonal (the basis $dz^I \wedge d\bar{z}^K$ of an orthonormal complex frame is orthogonal), the adjoint Λ has type $(-1, -1)$. For $\theta \in \mathbb{R}$ let T_θ act on $\Lambda^{p,q}$ by $e^{i(p-q)\theta}$. It commutes with L and Λ , hence maps $L^j P^{k-2j}$ to itself, and by uniqueness in (2) $(T_\theta \alpha)_j = T_\theta(\alpha_j)$. If α has type (p, q) then $T_\theta \alpha = e^{i(p-q)\theta} \alpha$, so $T_\theta \alpha_j = e^{i(p-q)\theta} \alpha_j$ for all θ , and α_j has only components of types (p', q') with $p' - q' = p - q$ and $p' + q' = k - 2j$, that is, $(p', q') = (p - j, q - j)$. $\square$

Remark 44.5.3. For a primitive form $\alpha \in P^k$ of type (p, q) , the Hodge star operator is given by *Weil's formula*

$$*\alpha = (-1)^{k(k+1)/2} i^{p-q} \frac{L^{n-k}\alpha}{(n-k)!},$$

and it follows that $i^{p-q}(-1)^{k(k-1)/2} \alpha \wedge \bar{\alpha} \wedge \omega^{n-k}$ is a positive multiple of the volume form when $\alpha \neq 0$ [Huy05, Section 1.2]. For instance, when $n = 1$ and $\alpha = dz$, one has $*dz = -i dz$ and $i dz \wedge d\bar{z} = 2 dx \wedge dy$. Integrated over a compact Kähler manifold, this positivity becomes the Hodge–Riemann bilinear relations of Chapter 46, the source of the polarisations of the Hodge structures of Part IX.

44.6 Exercises

Exercise 44.6.1. Compute $\int_{\mathbb{CP}^1} \omega_{FS}$ directly in the affine coordinate w , and deduce that $\int_{\mathbb{CP}^n} \omega_{FS}^n = 1$.

Solution. On $\mathbb{CP}^1$, $\omega_{FS} = \frac{i}{2\pi} \partial \bar{\partial} \log(1 + |w|^2) = \frac{i}{2\pi} \frac{dw \wedge d\bar{w}}{(1 + |w|^2)^2}$, and the point at infinity has measure zero. In polar coordinates $\frac{i}{2} dw \wedge d\bar{w} = r dr d\theta$, so

$$\int_{\mathbb{CP}^1} \omega_{FS} = \frac{1}{\pi} \int_0^{2\pi} \int_0^\infty \frac{r dr}{(1 + r^2)^2} d\theta = \frac{1}{\pi} \cdot 2\pi \cdot \frac{1}{2} = 1.$$

A line generates $H_2(\mathbb{CP}^n; \mathbb{Z})$ (it is the 2-skeleton of the standard cell structure), so $[\omega_{FS}]$ is the generator α of $H^2(\mathbb{CP}^n; \mathbb{Z})$ that is positive on lines, and α^n generates $H^{2n}(\mathbb{CP}^n; \mathbb{Z})$ (Theorem 20.5.1); it is positive on the fundamental class since ω_{FS}^n is a positive multiple of the volume form (Proposition 44.2.3). So $\int_{\mathbb{CP}^n} \omega_{FS}^n = 1$. $\square$

Exercise 44.6.2. Show that $\dim P^k = \binom{2n}{k} - \binom{2n}{k-2}$ for $k \leq n$, and list the dimensions of P^0, P^1, P^2 when $n = 2$.

Solution. By Theorem 44.5.2(2), $\Lambda^k = P^k \oplus L\Lambda^{k-2}$, and $L: \Lambda^{k-2} \rightarrow \Lambda^k$ is injective for $k \leq n$ by (1) (as L^{n-k+2} is injective on Λ^{k-2}). For $n = 2$ the dimensions are 1, 4 and $6 - 1 = 5$. $\square$

Exercise 44.6.3. Show that the Kähler form of a compact Kähler manifold of positive dimension is never of the form $i\partial\bar{\partial}\varphi$ for a smooth function φ on all of X , while on $\mathbb{C}^n$ one has $\omega_0 = i\partial\bar{\partial}(\frac{1}{2}|z|^2)$.

Solution. $i\partial\bar{\partial}\varphi = d(i\bar{\partial}\varphi)$, because $d = \partial + \bar{\partial}$ and $\bar{\partial}\bar{\partial}\varphi = 0$. So $[\omega] = 0$, contradicting Proposition 44.4.4 for $k = 1$. On $\mathbb{C}^n$, $i\partial\bar{\partial}\frac{1}{2}\sum_j z^j\bar{z}^j = \frac{i}{2}\sum_j dz^j \wedge d\bar{z}^j = \omega_0$. $\square$

Exercise 44.6.4. Let $C \subseteq X$ be the image of the line $\{z_1 = 0\}$ in the Hopf surface of Counterexample 44.4.8. Show that C is an elliptic curve and give a second proof that X is not Kähler using $\int_C \omega$.

Solution. $C = (\mathbb{C}^\times)/\langle 2 \rangle \cong \mathbb{C}/(2\pi i\mathbb{Z} + \log 2\mathbb{Z})$ via the exponential, a complex torus of dimension one. If ω were a Kähler form, then $\omega = d\eta$ because $H^2(X; \mathbb{R}) = 0$, so $\int_C \omega = \int_C d\eta = 0$ by Stokes' theorem; but $\int_C \omega = \text{vol}(C) > 0$ by Proposition 44.2.3. $\square$

Exercise 44.6.5. Show that the set of Kähler classes of a compact Kähler manifold is an open convex cone in the real vector space $H^{1,1}(X) \cap H^2(X; \mathbb{R})$ of classes represented by closed real $(1, 1)$ -forms.

Solution. If ω_1, ω_2 are Kähler forms and $s, t > 0$, then $s\omega_1 + t\omega_2$ is a closed positive real $(1, 1)$ -form, hence Kähler; this gives convexity and the cone property. For openness, choose closed real $(1, 1)$ -forms $\beta_1, \dots, \beta_m$ whose classes form a basis of the space. Positivity is an open condition on the coefficient matrix and X is compact, so $\omega + \sum_i c_i \beta_i$ is positive for all small $|c_i|$. $\square$

Exercise 44.6.6. Let $V = V_1 \oplus V_2$ be an orthogonal sum of Hermitian vector spaces of complex dimensions n_1, n_2 . Show that $L = L_1 \otimes 1 + 1 \otimes L_2$, $\Lambda = \Lambda_1 \otimes 1 + 1 \otimes \Lambda_2$ on $\Lambda^\bullet V^* = \Lambda^\bullet V_1^* \otimes \Lambda^\bullet V_2^*$, and that the $\mathfrak{sl}_2$ -representation of Theorem 44.5.2 is the tensor product of those for V_1 and V_2 . Deduce the decomposition of $\Lambda^\bullet(\mathbb{C}^n)^*$ into irreducibles using the Clebsch–Gordan formula.

Solution. $\omega = \omega_1 + \omega_2$, and the tensor product inner product is the induced one, which gives the formulas for L and Λ (the operators have even degree, so there are no signs). The counting operators satisfy $H = H_1 \otimes 1 + 1 \otimes H_2$ since $n = n_1 + n_2$; this is the tensor product representation. For $n = 1$ the representation on $\Lambda^\bullet(\mathbb{C})^* \cong \mathbb{C}^4$ is $V(1) \oplus 2V(0)$ (from the proof of Theorem 44.5.2: $1, x \wedge y$ span $V(1)$, and x, y are primitive of weight 0). Hence $\Lambda^\bullet(\mathbb{C}^n)^* \cong (V(1) \oplus 2V(0))^{\otimes n}$, which decomposes by the Clebsch–Gordan formula $V(a) \otimes V(b) \cong V(a+b) \oplus V(a+b-2) \oplus \dots \oplus V(|a-b|)$ (Exercise 11.6.2). $\square$

Chapter 45

The Hodge Decomposition

45.1 Introduction

On a compact Kähler manifold the complex cohomology splits as

$$H^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X), \quad \overline{H^{p,q}(X)} = H^{q,p}(X), \quad H^{p,q}(X) \cong H^q(X, \Omega_X^p)$$

(Theorem 45.5.2). This *Hodge decomposition* is the basic structure of the whole subject: Part IX axiomatises it as a Hodge structure, and the Hodge conjecture asks which rational classes of type (p, p) come from algebraic cycles. The proof combines the Hodge theorem of Chapter 40 with the *Kähler identities* (Theorem 45.3.3), which give $\Delta_d = 2\Delta_{\bar{d}}$, so that the Laplacian preserves types. We also prove the $\partial\bar{\partial}$ -lemma, derive the topological restrictions on Kähler manifolds, and compute examples. References: [GH78, Chapter 0, Sections 6–7], [Huy05, Chapter 3], [Voi02b, Chapter 6], [Wel08, Chapter V].

Prerequisites. Chapter 44 (Kähler Manifolds) and Chapter 40 (Harmonic Forms and the Hodge Theorem).

45.2 Conventions

Notation 45.2.1. Let X be a complex manifold of complex dimension n with a Hermitian metric g and fundamental form ω . Complex forms carry the Hermitian inner product $\langle \alpha, \beta \rangle$ extending the real one of Definition 38.7.1 (linear in α , conjugate linear in β). The Hodge star $*$ is extended complex linearly, so that $\alpha \wedge *\bar{\beta} = \langle \alpha, \beta \rangle \text{vol}_g$, and when X is compact the L^2 product is $(\alpha, \beta) = \int_X \alpha \wedge *\bar{\beta}$. The operators d , δ and $\Delta = \Delta_d = d\delta + \delta d$ are extended complex linearly; since they are real, δ is still the formal adjoint of d . We write ∂^* , $\bar{\partial}^*$ for the formal adjoints of ∂ , $\bar{\partial}$, and

$$\Delta_{\partial} = \partial\partial^* + \partial^*\partial, \quad \Delta_{\bar{\partial}} = \bar{\partial}\bar{\partial}^* + \bar{\partial}^*\bar{\partial},$$

so that $\Delta_{\bar{\partial}}$ is the operator $\square$ of Definition 43.3.1. The operators L , Λ , H are those of Definition 44.5.1, applied pointwise, and $\Pi^{p,q}$ is the projection onto forms of type (p, q) . Finally $c(\alpha) = \bar{\alpha}$ is complex conjugation of forms. It is conjugate linear and satisfies $(c\alpha, c\beta) = \overline{(\alpha, \beta)}$, $c\partial c = \bar{\partial}$, and $cLc = L$, $c\Lambda c = \Lambda$, since ω is real.

Lemma 45.2.2. *The complex linear Hodge star maps $\Lambda^{p,q}$ to $\Lambda^{n-q, n-p}$, forms of different types are orthogonal, and*

$$\partial^* = - * \bar{\partial}^*, \quad \bar{\partial}^* = - * \partial^*, \quad \delta = \partial^* + \bar{\partial}^*,$$

where ∂^* has type $(-1, 0)$ and $\bar{\partial}^*$ has type $(0, -1)$. Moreover $c\partial^*c = \bar{\partial}^*$.

Proof. At a point choose holomorphic coordinates with $h_{j\bar{k}} = \delta_{jk}$ there (as in the proof of Proposition 44.2.3). Then $dz^j = dx^j + i dy^j$ with dx^j, dy^j orthonormal, so $\langle dz^j, dz^k \rangle = 2\delta_{jk}$ and $\langle dz^j, d\bar{z}^k \rangle = \langle dx^j, dx^k \rangle - \langle dy^j, dy^k \rangle = 0$ (the term $i \cdot (-i)$ is -1). Hence the products $dz^I \wedge d\bar{z}^J$ are pairwise orthogonal, and forms of different types are orthogonal.

If $\beta \in \Lambda^{p,q}$, then $\alpha \wedge * \beta = \langle \alpha, \bar{\beta} \rangle \text{vol} = 0$ for every α of degree $p + q$ and type other than (q, p) , since $\bar{\beta}$ has type (q, p) . The wedge pairing $\Lambda^{r,s} \times \Lambda^{n-r,n-s} \rightarrow \Lambda^{n,n}$ is perfect and pairs $\Lambda^{r,s}$ trivially with all other types of complementary degree, so the component of $*\beta$ of type $(n - r, n - s)$ vanishes for every $(r, s) \neq (q, p)$. Thus $*\beta$ has type $(n - q, n - p)$.

The real dimension of X is even, so $\delta = -*d*$ by Definition 38.7.2. On a form of type (p, q) , the operator $-*\partial*$ has type $(0, -1)$, since $(p, q) \mapsto (n - q, n - p) \mapsto (n - q + 1, n - p) \mapsto (p, q - 1)$, and likewise $-*\bar{\partial}*$ has type $(-1, 0)$. So $\delta = D' + D''$ with $D' = -*\partial*$ of type $(-1, 0)$ and $D'' = -*\bar{\partial}*$ of type $(0, -1)$. For α of type $(p - 1, q)$ and β of type (p, q) , orthogonality of types gives $(\partial\alpha, \beta) = (d\alpha, \beta) = (\alpha, \delta\beta) = (\alpha, D'\beta)$, so $D' = \partial^*$; similarly $D'' = \bar{\partial}^*$. Finally $(c\partial^*c\alpha, \beta) = (\partial^*c\alpha, c\beta) = (c\alpha, \partial c\beta) = (\alpha, c\partial c\beta) = (\alpha, \bar{\partial}\beta)$. $\square$

45.3 The Kähler identities

Lemma 45.3.1. *Let P be a differential operator on forms, defined on every Hermitian manifold of dimension n , whose coefficients at each point, in any holomorphic coordinates, are given by universal expressions in the values at that point of the $h_{j\bar{k}}$ and their first derivatives. If $P = 0$ for the flat metric on $\mathbb{C}^n$, then $P = 0$ on every Kähler manifold of dimension n .*

Proof. Let $p \in X$ and choose holomorphic coordinates centred at p with $h_{j\bar{k}} = \delta_{jk} + O(|z|^2)$ (Proposition 44.3.2). In this chart, the metric of X and the flat metric $h_{j\bar{k}} = \delta_{jk}$ have the same values and the same first derivatives at p . So the coefficients of P at p are the same for both metrics, and those of the flat metric vanish. $\square$

Lemma 45.3.2. *Let W be a finite-dimensional complex vector space with a Hermitian inner product, and extend the inner product to $\Lambda^\bullet W$ as above. For $\theta \in W$ let $\varepsilon(\theta) = \theta \wedge -$ and let $\varepsilon(\theta)^*$ be its adjoint. Then*

$$\varepsilon(\theta)\varepsilon(\theta') = -\varepsilon(\theta')\varepsilon(\theta), \quad \varepsilon(\varphi)^*\varepsilon(\theta) + \varepsilon(\theta)\varepsilon(\varphi)^* = \langle \theta, \varphi \rangle \text{id}.$$

Proof. The first identity holds because $\theta \wedge \theta' = -\theta' \wedge \theta$. Both sides of the second are linear in θ and conjugate linear in φ , so it suffices to take $\theta = e_a, \varphi = e_b$ from an orthonormal basis $e_1, \dots, e_m$ and to evaluate on $e_I = e_{i_1} \wedge \dots \wedge e_{i_r}, i_1 < \dots < i_r$. The adjoint $\varepsilon(e_b)^*$ sends e_I to $(-1)^{s-1} e_{I \setminus b}$ if $b = i_s \in I$, and to 0 if $b \notin I$. If $a = b$, exactly one of the two terms is nonzero on e_I , depending on whether $a \in I$, and it equals e_I . If $a \neq b$, both terms vanish unless $b \in I$ and $a \notin I$. In that case moving e_b into place in $e_a \wedge e_I$ costs one extra sign compared with e_I , so the two terms cancel. $\square$

Theorem 45.3.3 (Kähler identities). *On a Kähler manifold,*

$$[\Lambda, \bar{\partial}] = -i\partial^*, \quad [\Lambda, \partial] = i\bar{\partial}^*, \quad [\bar{\partial}^*, L] = i\partial, \quad [\partial^*, L] = -i\bar{\partial},$$

and $[L, d] = 0, [\Lambda, \delta] = 0$.

Proof. The operator $P = [\Lambda, \bar{\partial}] + i\partial^*$ satisfies the hypothesis of Lemma 45.3.1. Indeed Λ and $*$ are algebraic operators whose coefficients are universal functions of the $h_{j\bar{k}}$, and P is obtained from them and from $\bar{\partial}$ by composition, using $\partial^* = -*\bar{\partial}*$ (Lemma 45.2.2). The derivatives in $\bar{\partial}$ fall at most once on these coefficients. So it suffices to prove $P = 0$ on $\mathbb{C}^n$ with the flat metric.

There $\omega = \frac{i}{2} \sum_k dz^k \wedge d\bar{z}^k$. Write $e_k = \varepsilon(dz^k)$ and $\bar{e}_k = \varepsilon(d\bar{z}^k)$, which are constant operators. Then

$$L = \frac{i}{2} \sum_k e_k \bar{e}_k, \quad \Lambda = L^* = -\frac{i}{2} \sum_k \bar{e}_k^* e_k^*, \quad \bar{\partial} = \sum_l \bar{e}_l \partial_{\bar{z}^l}, \quad \partial = \sum_k e_k \partial_{z^k},$$

where $\partial_{z^k}, \partial_{\bar{z}^l}$ act on the coefficients and commute with the constant operators. By Lemma 45.3.2 and the inner products $\langle d\bar{z}^l, d\bar{z}^k \rangle = 2\delta_{kl}$, $\langle d\bar{z}^l, dz^k \rangle = 0$ computed in Lemma 45.2.2, we have $e_k^* \bar{e}_l = -\bar{e}_l e_k^*$ and $\bar{e}_k^* \bar{e}_l = 2\delta_{kl} - \bar{e}_l \bar{e}_k^*$. Hence

$$\bar{e}_k^* e_k^* \bar{e}_l = -\bar{e}_k^* \bar{e}_l e_k^* = -2\delta_{kl} e_k^* + \bar{e}_l \bar{e}_k^* e_k^*, \quad \text{that is,} \quad [\bar{e}_k^* e_k^*, \bar{e}_l] = -2\delta_{kl} e_k^*,$$

and therefore $[\Lambda, \bar{\partial}] = -\frac{i}{2} \sum_k (-2) e_k^* \partial_{\bar{z}^k} = i \sum_k e_k^* \partial_{\bar{z}^k}$. On the other hand, integration by parts against compactly supported forms shows that the formal adjoint of ∂_{z^k} acting on coefficients is $-\partial_{\bar{z}^k}$, because $\int \partial_z f \cdot \bar{u} = -\int f \overline{\partial_z u}$. So $\partial^* = \sum_k (\partial_{z^k})^* e_k^* = -\sum_k e_k^* \partial_{\bar{z}^k}$, and $[\Lambda, \bar{\partial}] = -i\partial^*$.

Conjugating with c and using $c\Lambda c = \Lambda$, $c\bar{\partial}c = \partial$, $c\partial^*c = \bar{\partial}^*$ and $c(-i)c = i$ gives $[\Lambda, \partial] = i\bar{\partial}^*$. Taking formal adjoints of the first two identities gives $[\bar{\partial}^*, L] = (-i\partial^*)^* = i\partial$ and $[\partial^*, L] = (i\bar{\partial}^*)^* = -i\bar{\partial}$. Finally, $d(\omega \wedge \alpha) = \omega \wedge d\alpha$ because $d\omega = 0$ and ω has even degree, so $[L, d] = 0$, and its adjoint is $[\Lambda, \delta] = 0$. $\square$

Remark 45.3.4. The signs in the Kähler identities depend on conventions: the normalisation $\omega = \frac{i}{2} \sum h_{j\bar{k}} dz^j \wedge d\bar{z}^k$ used here, the sign of δ , and whether inner products are linear in the first or in the second variable. They agree with [Huy05, Proposition 3.1.12]. The proof via Lemma 45.3.1 is the typical use of the osculation property of Kähler metrics. The identities fail for non-Kähler Hermitian metrics, where torsion terms involving $d\omega$ appear.

45.4 Comparison of Laplacians

Theorem 45.4.1. *On a Kähler manifold, $\partial\bar{\partial}^* + \bar{\partial}^*\partial = 0$, $\bar{\partial}\partial^* + \partial^*\bar{\partial} = 0$, and*

$$\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_{\partial}.$$

Consequently Δ_d preserves types: it commutes with every $\Pi^{p,q}$, with $\partial, \bar{\partial}, \partial^, \bar{\partial}^*$, and with c . Moreover Δ_d commutes with $*$, L and Λ .*

Proof. By Theorem 45.3.3, $i\bar{\partial}^* = [\Lambda, \partial]$, so

$$i(\partial\bar{\partial}^* + \bar{\partial}^*\partial) = \partial\Lambda\partial - \partial\partial\Lambda + \Lambda\partial\partial - \partial\Lambda\partial = 0,$$

using $\partial^2 = 0$. The second identity is its conjugate. Expanding,

$$\Delta_d = (\partial + \bar{\partial})(\partial^* + \bar{\partial}^*) + (\partial^* + \bar{\partial}^*)(\partial + \bar{\partial}) = \Delta_{\partial} + \Delta_{\bar{\partial}} + (\partial\bar{\partial}^* + \bar{\partial}^*\partial) + (\bar{\partial}\partial^* + \partial^*\bar{\partial}) = \Delta_{\partial} + \Delta_{\bar{\partial}}.$$

Using $\partial^* = i[\Lambda, \bar{\partial}]$ and $\bar{\partial}^* = -i[\Lambda, \partial]$,

$$\Delta_{\partial} = i(\partial\Lambda\bar{\partial} - \partial\bar{\partial}\Lambda + \Lambda\bar{\partial}\partial - \bar{\partial}\Lambda\partial), \quad \Delta_{\bar{\partial}} = -i(\bar{\partial}\Lambda\partial - \bar{\partial}\partial\Lambda + \Lambda\partial\bar{\partial} - \partial\Lambda\bar{\partial}),$$

so $\Delta_{\partial} - \Delta_{\bar{\partial}} = i(-(\partial\bar{\partial} + \bar{\partial}\partial)\Lambda + \Lambda(\bar{\partial}\partial + \partial\bar{\partial})) = 0$ by Proposition 41.3.3. Hence $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_{\partial}$.

$\Delta_{\bar{\partial}}$ has type $(0,0)$, so Δ_d commutes with the $\Pi^{p,q}$. Δ_d commutes with d and δ , hence with their components of each type, which are $\partial, \bar{\partial}, \partial^*, \bar{\partial}^*$. It is real, so it commutes with c , and it commutes with $*$ by Proposition 38.7.3. For L : by the Kähler identities $[L, \delta] = -[\partial^*, L] - [\bar{\partial}^*, L] = i(\bar{\partial} - \partial)$ and $[L, d] = 0$, so

$$[L, \Delta_d] = d[L, \delta] + [L, \delta]d = i((\partial + \bar{\partial})(\bar{\partial} - \partial) + (\bar{\partial} - \partial)(\partial + \bar{\partial})) = 0,$$

again because $\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0$. Taking adjoints, $[\Lambda, \Delta_d] = 0$. $\square$

45.5 The Hodge decomposition

Definition 45.5.1. For a complex manifold X , let $H^{p,q}(X) \subseteq H^{p+q}(X; \mathbb{C})$ be the subspace of classes that can be represented by a closed form of type (p, q) . For a Hermitian metric, let $\mathcal{H}_{\mathbb{C}}^k$ be the complex Δ_d -harmonic k -forms and $\mathcal{H}^{p,q} = \mathcal{H}_{\mathbb{C}}^{p+q} \cap \mathcal{A}^{p,q}(X)$.

Theorem 45.5.2 (Hodge decomposition). *Let X be a compact Kähler manifold. Then*

1. $\mathcal{H}_{\mathbb{C}}^k = \bigoplus_{p+q=k} \mathcal{H}^{p,q}$, and $\mathcal{H}^{p,q}$ is also the space of $\Delta_{\bar{\partial}}$ -harmonic, and of Δ_{∂} -harmonic, forms of type (p, q) ;
2. $H^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}(X)$, and $\mathcal{H}^{p,q} \rightarrow H^{p,q}(X)$, $\alpha \mapsto [\alpha]$, is an isomorphism;
3. $\overline{H^{p,q}(X)} = H^{q,p}(X)$ (Hodge symmetry);
4. $H^{p,q}(X) \cong H_{\bar{\partial}}^{p,q}(X) \cong H^q(X, \Omega_X^p)$.

The subspaces $H^{p,q}(X)$ depend only on the complex structure of X , not on the Kähler metric.

Proof. (1) If $\Delta_d \alpha = 0$ then $\Delta_d \Pi^{p,q} \alpha = \Pi^{p,q} \Delta_d \alpha = 0$ by Theorem 45.4.1, so the components of a harmonic form are harmonic. The second statement holds because $\Delta_d = 2\Delta_{\bar{\partial}} = 2\Delta_{\partial}$.

(2) Applying Theorem 40.2.2 to real and imaginary parts, and Theorem 35.3.1, the map $\mathcal{H}_{\mathbb{C}}^k \rightarrow H^k(X; \mathbb{C})$ is an isomorphism. By (1), $H^k(X; \mathbb{C})$ is the direct sum of the images of the $\mathcal{H}^{p,q}$, and each image lies in $H^{p,q}(X)$ because harmonic forms are closed. Conversely, let α be a closed form of type (p, q) and let h be its harmonic projection. By Theorem 40.2.1 (applied to real and imaginary parts), $\alpha = h + d\beta$ for some β , and h is the orthogonal projection of α onto $\mathcal{H}_{\mathbb{C}}^k$. Since forms of different types are orthogonal (Lemma 45.2.2), α is orthogonal to $\mathcal{H}^{r,s}$ for $(r, s) \neq (p, q)$, so $h \in \mathcal{H}^{p,q}$ by (1). So $H^{p,q}(X)$ is exactly the image of $\mathcal{H}^{p,q}$, and (2) follows.

(3) Δ_d is a real operator, so c maps $\mathcal{H}^{p,q}$ onto $\mathcal{H}^{q,p}$. Alternatively, the conjugate of a closed (p, q) -form is a closed (q, p) -form.

(4) By (1), $\mathcal{H}^{p,q}$ is the space of $\Delta_{\bar{\partial}}$ -harmonic (p, q) -forms, which is isomorphic to $H_{\bar{\partial}}^{p,q}(X) \cong H^q(X, \Omega_X^p)$ by Theorem 43.3.3.

The definition of $H^{p,q}(X)$ does not involve a metric. □

Corollary 45.5.3. *Let X be a compact Kähler manifold of dimension n , with Hodge numbers $h^{p,q} = \dim H^q(X, \Omega_X^p)$. Then:*

1. $b_k = \sum_{p+q=k} h^{p,q}$, so the Frölicher spectral sequence degenerates at E_1 ;
2. $h^{p,q} = h^{q,p} = h^{n-p, n-q} = h^{n-q, n-p}$;
3. b_{2k+1} is even for every k ;
4. $h^{p,p} \geq 1$ for $0 \leq p \leq n$.

Proof. (1) is Theorem 45.5.2(2) and (4) together with Proposition 43.4.1. In (2), the first equality is Hodge symmetry and the second is Serre duality (Theorem 43.3.4). For (3), $b_{2k+1} = \sum_{p+q=2k+1} h^{p,q} = 2 \sum_{p < q, p+q=2k+1} h^{p,q}$, since $p \neq q$ when $p + q$ is odd. For (4), ω^p is a closed form of type (p, p) and $[\omega]^p \neq 0$ by Proposition 44.4.4. □

Remark 45.5.4. The Hodge numbers are displayed in the *Hodge diamond*, with $h^{n,n}$ at the top, $h^{0,0}$ at the bottom, and $h^{p,q}$ in row $p + q$. Row k adds up to b_k , and the diamond is symmetric

under the reflections $h^{p,q} \leftrightarrow h^{q,p}$ (Hodge symmetry) and $h^{p,q} \leftrightarrow h^{n-p,n-q}$ (Serre duality). For a compact Kähler surface it reads

$$\begin{array}{ccccc}
 & & h^{2,2} & & \\
 & h^{2,1} & & h^{1,2} & \\
 h^{2,0} & & h^{1,1} & & h^{0,2} \\
 & h^{1,0} & & h^{0,1} & \\
 & & h^{0,0} & &
 \end{array}
 =
 \begin{array}{ccccc}
 & & 1 & & \\
 & q & & q & \\
 p_g & & h^{1,1} & & p_g \\
 & q & & q & \\
 & & 1 & &
 \end{array}$$

where $q = h^{0,1}$ is the irregularity and $p_g = h^{2,0}$ the geometric genus.

Remark 45.5.5. The Hodge filtration $F^p H^k(X; \mathbb{C}) = \bigoplus_{r \geq p} H^{r,k-r}(X)$ determines the decomposition, since $H^{p,q} = F^p \cap \overline{F}^q$ when $p + q = k$. Because the Frölicher spectral sequence degenerates, F^p is also the image of the hypercohomology of the truncated complex $\Omega_X^{\geq p}$ in $H^k(X; \mathbb{C})$ [Voi02b, Chapter 8]. The filtration varies holomorphically in families, whereas the decomposition does not. This is why the theory of Hodge structures in Part IX is phrased with filtrations.

Proposition 45.5.6. *On a compact Kähler manifold every holomorphic p -form is closed and harmonic, and $H^0(X, \Omega_X^p) = \mathcal{H}^{p,0} \cong H^{p,0}(X) \subseteq H^p(X; \mathbb{C})$. In particular a holomorphic form that is exact, or merely cohomologous to zero, vanishes, and $b_1(X) = 2 \dim H^0(X, \Omega_X^1)$.*

Proof. If α is a holomorphic p -form, then $\bar{\partial}\alpha = 0$ and $\bar{\partial}^* \alpha = 0$, the latter because $\bar{\partial}^* \alpha$ would have type $(p, -1)$. So $\Delta_{\bar{\partial}} \alpha = 0$, hence $\Delta_d \alpha = 0$ and α is closed and harmonic. Conversely, if $\alpha \in \mathcal{H}^{p,0}$ then $0 = (\Delta_{\bar{\partial}} \alpha, \alpha) = \|\bar{\partial}\alpha\|^2 + \|\bar{\partial}^* \alpha\|^2$, so $\bar{\partial}\alpha = 0$ and α is holomorphic. The harmonic representative of a class is unique, so a harmonic form cohomologous to zero vanishes. Finally $b_1 = h^{1,0} + h^{0,1} = 2h^{1,0}$. $\square$

Proposition 45.5.7. *Let $f: X \rightarrow Y$ be a holomorphic map of compact Kähler manifolds. Then $f^* H^{p,q}(Y) \subseteq H^{p,q}(X)$. The cup product maps $H^{p,q} \times H^{r,s}$ into $H^{p+r,q+s}$. If $\dim X = n$, the Poincaré duality pairing $H^k(X; \mathbb{C}) \times H^{2n-k}(X; \mathbb{C}) \rightarrow \mathbb{C}$ restricts to perfect pairings $H^{p,q}(X) \times H^{n-p,n-q}(X) \rightarrow \mathbb{C}$, and it is zero on $H^{p,q} \times H^{r,s}$ for all other pairs of types.*

Proof. Pullback by a holomorphic map and wedge product preserve types of forms, and the wedge product corresponds to the cup product (Theorem 35.3.1). For $(r, s) \neq (n-p, n-q)$ with $r + s = 2n - p - q$, the wedge product of forms of types (p, q) and (r, s) has type $(p+r, q+s) \neq (n, n)$ and so is zero, being a $2n$ -form. The pairing on $H^k \times H^{2n-k}$ is perfect (Theorem 35.3.10), and by Theorem 45.5.2 it is block diagonal with blocks $H^{p,q} \times H^{n-p,n-q}$. Each block is therefore perfect. $\square$

Proposition 45.5.8. *Let X be a compact Kähler manifold of dimension n and $Z \subseteq X$ an analytic subvariety of pure codimension p . Then its cycle class $[Z] \in H^{2p}(X; \mathbb{C})$ (Proposition 41.4.2) lies in $H^{p,p}(X)$.*

Proof. Let $k = n - p = \dim Z$. The restriction of a form of type (r, s) to the complex manifold Z_{reg} of dimension k has type (r, s) there, so it vanishes if $r + s = 2k$ and $(r, s) \neq (k, k)$. Hence $\int_Z \beta = 0$ for every closed form β of type $(r, s) \neq (k, k)$ with $r + s = 2k$. By definition of the cycle class, $\int_X \eta \wedge \beta = \int_Z \beta$ for a closed form η representing $[Z]$ (Theorem 35.4.3). Write $[Z] = \sum_{a+b=2p} z^{a,b}$ with $z^{a,b} \in H^{a,b}(X)$. By Proposition 45.5.7, $z^{a,b}$ pairs only with $H^{n-a,n-b}$, and it pairs to zero with it unless $(n-a, n-b) = (k, k)$. So $z^{a,b}$ pairs to zero with everything unless $(a, b) = (p, p)$, and by perfectness $z^{a,b} = 0$ for $(a, b) \neq (p, p)$. $\square$

Remark 45.5.9. The cycle class of Z is the image of an integral class in $H^{2p}(X; \mathbb{Z})$ (Chapter 59). So algebraic cycles give *Hodge classes*, the elements of $H^{2p}(X; \mathbb{Q}) \cap H^{p,p}(X)$. The Hodge conjecture (Chapter 64) predicts that on a projective manifold every Hodge class is a rational linear combination of cycle classes.

45.6 The $\partial\bar{\partial}$ -lemma

Lemma 45.6.1. *Let X be compact Kähler. Then $\mathcal{A}^{p,q}(X) = \mathcal{H}^{p,q} \oplus \partial\mathcal{A}^{p-1,q}(X) \oplus \bar{\partial}^*\mathcal{A}^{p+1,q}(X)$ orthogonally, and $\ker \partial = \mathcal{H}^{p,q} \oplus \partial\mathcal{A}^{p-1,q}(X)$. The same holds with $\bar{\partial}, \bar{\partial}^*$ in place of ∂, ∂^* .*

Proof. For $\bar{\partial}$ the decomposition is Theorem 43.3.3, with $\mathcal{H}^{p,q}$ the $\Delta_{\bar{\partial}}$ -harmonic forms (Theorem 45.5.2(1)). Applying the conjugate linear isometry c to the decomposition of $\mathcal{A}^{q,p}(X)$, and using $c\bar{\partial}c = \partial$, $c\bar{\partial}^*c = \partial^*$ and $c\mathcal{H}^{q,p} = \mathcal{H}^{p,q}$, gives the decomposition for ∂ . If $\partial\alpha = 0$ and $\alpha = h + \partial\beta + \bar{\partial}^*\gamma$, then $0 = (\partial\alpha, \gamma) = (\partial\bar{\partial}^*\gamma, \gamma) = \|\partial^*\gamma\|^2$, because $\partial h = 0$ and $\partial^2 = 0$. So $\alpha \in \mathcal{H}^{p,q} \oplus \text{im } \partial$; the argument for $\bar{\partial}$ is the same. $\square$

Theorem 45.6.2 ($\partial\bar{\partial}$ -lemma). *Let X be a compact Kähler manifold and α a d -closed form of type (p, q) . The following are equivalent:*

1. α is d -exact;
2. α is ∂ -exact;
3. α is $\bar{\partial}$ -exact;
4. $\alpha = \partial\bar{\partial}\gamma$ for a form γ of type $(p-1, q-1)$.

Proof. (4) implies the others: $\partial\bar{\partial}\gamma = d(\bar{\partial}\gamma) = \partial(\bar{\partial}\gamma) = \bar{\partial}(-\partial\gamma)$.

Assume (1), (2) or (3). Then α is orthogonal to $\mathcal{H}^{p,q}$: for h harmonic, $(d\beta, h) = (\beta, \delta h) = 0$, and likewise with $\partial^*h = 0$ or $\bar{\partial}^*h = 0$ (harmonic forms are killed by all of these, by Theorem 45.5.2(1)). Since $\partial\alpha = 0$ (the $(p+1, q)$ -part of $d\alpha = 0$), Lemma 45.6.1 gives $\alpha = \partial\gamma$ with γ of type $(p-1, q)$. Decompose $\gamma = h' + \bar{\partial}\mu + \bar{\partial}^*\nu$ by the same lemma. Then $\partial h' = 0$, and $\partial\bar{\partial}^*\nu = -\bar{\partial}^*\partial\nu$ by Theorem 45.4.1, so

$$\alpha = \partial\bar{\partial}\mu - \bar{\partial}^*\partial\nu.$$

Now $\bar{\partial}\alpha = 0$ and $\bar{\partial}\partial\bar{\partial}\mu = -\partial\bar{\partial}\bar{\partial}\mu = 0$, hence $\bar{\partial}\bar{\partial}^*\partial\nu = 0$ and $\|\bar{\partial}^*\partial\nu\|^2 = (\bar{\partial}\bar{\partial}^*\partial\nu, \partial\nu) = 0$. Therefore $\alpha = \partial\bar{\partial}\mu$, with μ of type $(p-1, q-1)$. $\square$

Corollary 45.6.3. *Let X be compact Kähler. Every Dolbeault class in $H_{\bar{\partial}}^{p,q}(X)$ has a d -closed representative, and $[\alpha] \mapsto [\alpha]_{\bar{\partial}}$, for α a closed (p, q) -form, is a well-defined isomorphism $H^{p,q}(X) \rightarrow H_{\bar{\partial}}^{p,q}(X)$ that does not depend on any metric.*

Proof. If α and α' are closed (p, q) -forms in the same de Rham class, then $\alpha - \alpha'$ is a d -exact closed form of type (p, q) , hence $\bar{\partial}$ -exact by Theorem 45.6.2. So the map is well defined. For a Kähler metric, restricting it to harmonic representatives gives the isomorphism $\mathcal{H}^{p,q} \cong H_{\bar{\partial}}^{p,q}(X)$ of Theorem 43.3.3. The harmonic representative of a Dolbeault class is d -closed. $\square$

Corollary 45.6.4. *Let X be compact Kähler and let α, α' be closed real $(1, 1)$ -forms with $[\alpha] = [\alpha']$ in $H^2(X; \mathbb{R})$. Then $\alpha' = \alpha + i\partial\bar{\partial}\varphi$ for a smooth real function φ . In particular any two Kähler forms with the same Kähler class differ by $i\partial\bar{\partial}\varphi$.*

Proof. By Theorem 45.6.2, $\alpha' - \alpha = \partial\bar{\partial}\mu$ for a complex function μ . Since $\alpha' - \alpha$ is real, $\partial\bar{\partial}\mu = \partial\bar{\partial}\bar{\mu} = \bar{\partial}\partial\bar{\mu} = -\partial\bar{\partial}\bar{\mu}$. So $\alpha' - \alpha = \frac{1}{2}\partial\bar{\partial}(\mu - \bar{\mu}) = i\partial\bar{\partial}\operatorname{Im}\mu$, and $\varphi = \operatorname{Im}\mu$ works. $\square$

Remark 45.6.5. Deligne, Griffiths, Morgan and Sullivan deduced from the $\partial\bar{\partial}$ -lemma that the real homotopy type of a compact Kähler manifold is determined by its real cohomology ring (the manifold is *formal*). In particular all Massey products vanish, which gives obstructions to Kählerness beyond the Hodge numbers [DGMS75]. The $\partial\bar{\partial}$ -lemma, and hence the Hodge decomposition, also hold for compact complex manifolds bimeromorphic to Kähler ones [DGMS75].

45.7 Examples

Example 45.7.1. Let X be a compact Riemann surface. It is Kähler (Example 44.4.3), so b_1 is even, and its Hodge diamond has $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1} = g$, where $g = b_1/2$. So the genus g , a topological invariant, equals both the dimension of the space of holomorphic 1-forms $H^0(X, K_X)$ and $\dim H^1(X, \mathcal{O}_X)$. For the Riemann sphere $g = 0$; for an elliptic curve $\mathbb{C}/\Lambda$, $g = 1$ and $H^0(X, K_X)$ is spanned by dz .

Example 45.7.2. For $\mathbb{CP}^n$, $b_{2k} = 1$ for $0 \leq k \leq n$ and all other Betti numbers vanish (Theorem 20.5.1). By Corollary 45.5.3(4), $h^{k,k} \geq 1$, so $H^{2k}(\mathbb{CP}^n; \mathbb{C}) = H^{k,k}$ is spanned by $[\omega_{FS}]^k$, and $h^{p,q} = 0$ for $p \neq q$. In particular $\mathbb{CP}^n$ has no nonzero holomorphic forms of positive degree, and $H^q(\mathbb{CP}^n, \mathcal{O}) = 0$ for $q > 0$.

Example 45.7.3. Let $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ with $\operatorname{Im}\tau > 0$, and let a, b be the classes in $H_1(E_\tau; \mathbb{Z})$ of the images of the segments $[0, 1]$ and $[0, \tau]$. Integration identifies $H^1(E_\tau; \mathbb{C})$ with $\mathbb{C}^2$ through $[\alpha] \mapsto (\int_a \alpha, \int_b \alpha)$. Then $H^{1,0}$ is spanned by $[dz] \mapsto (1, \tau)$ and $H^{0,1}$ by $[d\bar{z}] \mapsto (1, \bar{\tau})$. The lattice $H^1(E_\tau; \mathbb{Z})$ and the vector space $H^1(E_\tau; \mathbb{C})$ do not depend on τ , but the line $H^{1,0} = [1 : \tau] \in \mathbb{P}^1$ does, and it recovers τ . So the Hodge decomposition captures the complex structure, which is not a topological invariant. This is the first example of a *period map* (Chapter 66).

Example 45.7.4. For a smooth quartic surface $X \subseteq \mathbb{CP}^3$ (a K3 surface) the Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 0 & & 0 & \\ 1 & & 20 & & 1 \\ & 0 & & 0 & \\ & & 1 & & \end{array}$$

Here $b_1 = 0$ by the Lefschetz hyperplane theorem (Chapter 47), $K_X \cong \mathcal{O}_X$ by adjunction (Proposition 42.5.1) so that $p_g = 1$, and $b_2 = 22$ comes from the Euler characteristic $\chi = c_2(X) = 24$ [BHPVdV04, Chapter VIII]. The subspace $H^{1,1} \cap H^2(X; \mathbb{Q})$ is where the Hodge conjecture for divisors, the Lefschetz (1, 1)-theorem, lives.

Example 45.7.5. Let C and C' be compact Riemann surfaces of genera g and g' , so $h^{0,0} = h^{1,1} = 1$ and $h^{1,0} = h^{0,1} = g$ for C (Example 45.7.1). By Exercise 45.8.1, the surface $S = C \times C'$ has

$$h^{1,0}(S) = g + g', \quad h^{2,0}(S) = gg', \quad h^{1,1}(S) = 2 + 2gg',$$

since $H^{1,1}(S)$ receives $H^{1,1}(C) \otimes H^{0,0}(C')$, $H^{0,0}(C) \otimes H^{1,1}(C')$, $H^{1,0}(C) \otimes H^{0,1}(C')$ and $H^{0,1}(C) \otimes H^{1,0}(C')$. As a check, $b_2(S) = 2h^{2,0} + h^{1,1} = 2 + 4gg' = 1 + (2g)(2g') + 1$ by the Künneth

formula. For two elliptic curves, $g = g' = 1$, the Hodge diamond is

$$\begin{array}{ccccc} & & 1 & & \\ & 2 & & 2 & \\ 1 & & 4 & & 1 \\ & 2 & & 2 & \\ & & 1 & & \end{array}$$

which is also that of every two-dimensional complex torus (Example 43.4.2). The $H^{1,1}$ of S always contains the classes of the two fibres, and the remaining $2gg'$ dimensions come from $H^1(C) \otimes H^1(C')$; which rational classes lie there is governed by the homomorphisms between the Jacobians of C and C' (for elliptic curves this is worked out in Chapter 64).

Counterexample 45.7.6. The Hopf surface (Exercise 41.5.3) has $b_1 = 1$, which is odd, so by Corollary 45.5.3(3) it is not Kähler, which gives a second proof of Counterexample 44.4.8. It also shows how the theorem fails: $h^{1,0} = 0 \neq 1 = h^{0,1}$ (Example 43.4.2).

Counterexample 45.7.7. Let G be the complex Heisenberg group of matrices $\begin{pmatrix} 1 & z_1 & z_3 \\ 0 & 1 & z_2 \\ 0 & 0 & 1 \end{pmatrix}$, $z \in \mathbb{C}^3$, and $\Gamma \subseteq G$ the subgroup with entries in $\mathbb{Z}[i]$. The Iwasawa manifold $X = \Gamma \backslash G$ is a compact complex manifold of dimension 3 carrying a holomorphic 1-form that is not closed. So X is not Kähler, by Proposition 45.5.6.

Proof. Left multiplication by an element with entries (a_1, a_2, a_3) is $(z_1, z_2, z_3) \mapsto (z_1 + a_1, z_2 + a_2, z_3 + a_1z_2 + a_3)$. This map is holomorphic, and Γ acts freely and properly discontinuously, being a discrete subgroup acting by translations of the group. So X is a complex manifold. It is compact: using $a_1, a_2 \in \mathbb{Z}[i]$ one moves z_1, z_2 into the unit square, and then using a_3 one moves z_3 there, so the image of a compact set surjects onto X . The form $\theta = dz_3 - z_1 dz_2$ is invariant under left multiplication: $d(z_3 + a_1z_2 + a_3) - (z_1 + a_1)d(z_2 + a_2) = dz_3 - z_1 dz_2$. So it descends to a holomorphic 1-form on X , and $d\theta = -dz_1 \wedge dz_2 \neq 0$. $\square$

45.8 Exercises

Exercise 45.8.1. Let X and Y be compact Kähler manifolds. Show that $h^{p,q}(X \times Y) = \sum h^{p_1,q_1}(X) h^{p_2,q_2}(Y)$, the sum over $p_1 + p_2 = p$, $q_1 + q_2 = q$.

Solution. $X \times Y$ is Kähler (Example 44.4.3). The Künneth isomorphism $H^\bullet(X; \mathbb{C}) \otimes H^\bullet(Y; \mathbb{C}) \rightarrow H^\bullet(X \times Y; \mathbb{C})$, $a \otimes b \mapsto \text{pr}_X^* a \cup \text{pr}_Y^* b$ (Theorem 20.4.2), maps $H^{p_1,q_1}(X) \otimes H^{p_2,q_2}(Y)$ into $H^{p_1+p_2,q_1+q_2}(X \times Y)$ by Proposition 45.5.7. The source is the direct sum of these pieces and the target is the direct sum of its $H^{p,q}$. So the isomorphism maps $\bigoplus_{p_1+p_2=p, q_1+q_2=q} H^{p_1,q_1} \otimes H^{p_2,q_2}$ onto $H^{p,q}(X \times Y)$. $\square$

Exercise 45.8.2. Calabi and Eckmann constructed complex structures on $S^{2a+1} \times S^{2b+1}$ for all $a, b \geq 0$ [CE53]. Show that none of them is Kähler, except on $S^1 \times S^1$.

Solution. The complex dimension is $a + b + 1 \geq 1$. By the Künneth formula, the cohomology of $S^{2a+1} \times S^{2b+1}$ lives in degrees 0, $2a+1$, $2b+1$ and $2a+2b+2$. If $a, b \geq 1$, then $H^2 = 0$, contradicting Proposition 44.4.4. If $a = 0 < b$, then $b_1 = 1$ is odd, contradicting Corollary 45.5.3(3), and the case $b = 0 < a$ is symmetric. The remaining case $a = b = 0$ is a torus of real dimension 2, and every complex structure on it is Kähler (Example 44.4.3). $\square$

Exercise 45.8.3. Show that ω_{FS} is the only $U(n+1)$ -invariant Kähler form on $\mathbb{CP}^n$ in its cohomology class.

Solution. Let ω' be another one. By Corollary 45.6.4, $\omega' = \omega_{FS} + i\partial\bar{\partial}\varphi$ with φ real. For $g \in U(n+1)$, $g^*\omega' = \omega'$ and $g^*\omega_{FS} = \omega_{FS}$, so $i\partial\bar{\partial}(\varphi \circ g) = i\partial\bar{\partial}\varphi$, since g is holomorphic. Averaging over $U(n+1)$ with its normalised Haar measure gives an invariant function $\tilde{\varphi} = \int \varphi \circ g \, dg$ with $i\partial\bar{\partial}\tilde{\varphi} = i\partial\bar{\partial}\varphi$ (differentiate under the integral). As $U(n+1)$ acts transitively, $\tilde{\varphi}$ is constant. Hence $i\partial\bar{\partial}\varphi = 0$ and $\omega' = \omega_{FS}$. $\square$

Exercise 45.8.4. Let X be compact Kähler. Show that the map $H^1(X; \mathbb{R}) \rightarrow H^1(X, \mathcal{O}_X)$ induced by the inclusion $\mathbb{R}_X \rightarrow \mathcal{O}_X$ is an isomorphism of real vector spaces. (This is the starting point for the Picard variety $H^1(X, \mathcal{O}_X)/H^1(X; \mathbb{Z})$.)

Solution. Projecting forms to their $(0, q)$ -components defines a morphism from the de Rham resolution of $\mathbb{C}_X$ to the Dolbeault resolution of $\mathcal{O}_X$ over the inclusion $\mathbb{C}_X \rightarrow \mathcal{O}_X$, because the $(0, q+1)$ -part of $d\alpha$ is $\bar{\partial}$ of the $(0, q)$ -part of α . So the map sends $[\alpha]$ to $[\alpha^{0,1}]_{\bar{\partial}}$. Both sides have real dimension $b_1 = 2h^{0,1}$. If α is a real harmonic 1-form with $\alpha^{0,1}$ $\bar{\partial}$ -exact, then $\alpha^{0,1}$ is harmonic (by Theorem 45.5.2(1)) and $\bar{\partial}$ -exact, hence zero. Then $\alpha^{1,0} = \overline{\alpha^{0,1}} = 0$, so $\alpha = 0$ and the map is injective. $\square$

Exercise 45.8.5. Let X be compact Kähler. Show that L and Λ map harmonic forms to harmonic forms, and that the primitive components (Theorem 44.5.2) of a harmonic form are harmonic.

Solution. L and Λ commute with Δ_d (Theorem 45.4.1), so they preserve $\ker \Delta_d$. Let $\alpha = \sum_j L^j \alpha_j$ be the primitive decomposition of a harmonic form. Then $\Lambda \Delta_d \alpha_j = \Delta_d \Lambda \alpha_j = 0$, so $\Delta_d \alpha_j$ is primitive, and $0 = \Delta_d \alpha = \sum_j L^j (\Delta_d \alpha_j)$ is a primitive decomposition of 0. By uniqueness, every $\Delta_d \alpha_j$ vanishes. $\square$

Exercise 45.8.6. Show that on a compact complex surface every holomorphic 1-form is closed, even without assuming the surface is Kähler. (Hint: for a holomorphic 1-form θ , compute $\int_X d\theta \wedge \overline{d\theta}$ in two ways.)

Solution. $d\theta = \partial\theta$ is a holomorphic 2-form, so $d\theta \wedge \overline{d\theta}$ is a form of type $(2, 2)$. In local coordinates, $d\theta = f \, dz_1 \wedge dz_2$, and $d\theta \wedge \overline{d\theta} = |f|^2 \, dz_1 \wedge dz_2 \wedge d\bar{z}_1 \wedge d\bar{z}_2$, which is $4|f|^2 \, dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$, a non-negative multiple of the volume form. On the other hand $d\theta \wedge \overline{d\theta} = d(\theta \wedge \overline{d\theta})$, because $\overline{d\theta}$ is closed. So its integral vanishes by Stokes' theorem, which forces $f = 0$. The Iwasawa manifold (Counterexample 45.7.7) shows that this fails in dimension 3. $\square$

Chapter 46

Hard Lefschetz and the Hodge–Riemann Relations

46.1 Introduction

Let X be a compact Kähler manifold of dimension n with Kähler form ω . Cup product with the Kähler class is an operator L on $H^\bullet(X; \mathbb{R})$ of degree 2. The *hard Lefschetz theorem* (Theorem 46.2.2) says that

$$L^{n-k}: H^k(X; \mathbb{R}) \rightarrow H^{2n-k}(X; \mathbb{R})$$

is an isomorphism for $k \leq n$. The proof takes the pointwise $\mathfrak{sl}_2$ -action of Theorem 44.5.2 and passes it to harmonic forms, using that L and Λ commute with the Laplacian. The result is a canonical decomposition of cohomology into primitive pieces. On them the intersection form, twisted by powers of ω , has definite signs: these are the *Hodge–Riemann bilinear relations* (Theorem 46.4.2). Their consequences include the *Hodge index theorem* (Theorem 46.5.1) and, in Part IX, the polarisations of the Hodge structures of projective manifolds. References: [GH78, Chapter 0, Section 7], [Huy05, Section 3.3], [Voi02b, Chapter 6].

Prerequisites. Chapter 45 (The Hodge Decomposition).

Throughout, X is a compact Kähler manifold of dimension n with Kähler form ω , and we use the notation of Notation 45.2.1. We identify $H^k(X; \mathbb{C})$ with the space $\mathcal{H}_{\mathbb{C}}^k$ of complex harmonic forms and $H^k(X; \mathbb{R})$ with the real harmonic forms (Theorem 40.2.2).

46.2 The $\mathfrak{sl}_2$ -action on cohomology

Definition 46.2.1. The *Lefschetz operator* on cohomology is $L: H^k(X; \mathbb{R}) \rightarrow H^{k+2}(X; \mathbb{R})$, $L(a) = [\omega] \cup a$, which is represented on forms by $\alpha \mapsto \omega \wedge \alpha$. A class $a \in H^k(X; \mathbb{R})$ with $k \leq n$ is *primitive* if $L^{n-k+1}a = 0$. We write $P^k(X) = P^k(X; \mathbb{R})$ for the primitive classes of degree k , $P^k(X; \mathbb{C})$ for their complexification, and $P^{p,q}(X) = P^{p+q}(X; \mathbb{C}) \cap H^{p,q}(X)$.

Theorem 46.2.2 (Hard Lefschetz). *Let X be a compact Kähler manifold of dimension n with Kähler class $[\omega]$. Then:*

1. *the pointwise operators L , Λ and H preserve the space $\mathcal{H}^\bullet$ of harmonic forms, which becomes a finite-dimensional representation of $\mathfrak{sl}_2$ in which $\mathcal{H}^k$ has weight $k - n$;*
2. *for $0 \leq k \leq n$, $L^{n-k}: H^k(X; \mathbb{R}) \rightarrow H^{2n-k}(X; \mathbb{R})$ is an isomorphism;*

3. (Lefschetz decomposition) for every k ,

$$H^k(X; \mathbb{R}) = \bigoplus_{j \geq 0} L^j P^{k-2j}(X),$$

and L^j is injective on $P^{k-2j}(X)$;

4. a class of degree $k \leq n$ is primitive if and only if its harmonic representative α satisfies $\Lambda\alpha = 0$, that is, α is primitive at every point;

5. the decompositions are compatible with the Hodge decomposition: $P^k(X; \mathbb{C}) = \bigoplus_{p+q=k} P^{p,q}(X)$ and $H^{p,q}(X) = \bigoplus_j L^j P^{p-j,q-j}(X)$.

Proof. (1) By Theorem 45.4.1, L and Λ commute with Δ_d , so they preserve $\mathcal{H}^\bullet = \ker \Delta_d$; so does H , which acts on $\mathcal{H}^k$ as multiplication by $k - n$. The commutation relations $[L, \Lambda] = H$, $[H, L] = 2L$, $[H, \Lambda] = -2\Lambda$ hold pointwise by Theorem 44.5.2, hence on $\mathcal{H}^\bullet$. The space $\mathcal{H}^\bullet$ is finite-dimensional by Theorem 40.2.1, and it is real since all three operators are real.

(2) and (3) now follow exactly as statements (1) and (2) of Theorem 44.5.2 were derived from Corollary 11.5.4: that argument used only a finite-dimensional $\mathfrak{sl}_2$ -representation in which the degree k part has weight $k - n$. Under the Hodge isomorphism, L on harmonic forms is the cup product with $[\omega]$, since $\omega \wedge \alpha$ is harmonic and represents $[\omega] \cup [\alpha]$ (Theorem 35.3.1). This gives (2), and it gives (3) with the primitive subspaces defined as $\ker \Lambda \cap \mathcal{H}^{k-2j}$.

(4) Part (3) of Theorem 44.5.2 was also derived from the representation theory alone. Applied to $\mathcal{H}^\bullet$, it says that for $\alpha \in \mathcal{H}^k$ with $k \leq n$, $\Lambda\alpha = 0$ if and only if $L^{n-k+1}\alpha = 0$. So the subspaces $\ker \Lambda \cap \mathcal{H}^k$ used in (3) are the $P^k(X)$ of Definition 46.2.1.

(5) L maps $H^{p,q}$ to $H^{p+1,q+1}$. So the kernel of L^{n-k+1} on $H^k(X; \mathbb{C}) = \bigoplus_{p+q=k} H^{p,q}$ is the direct sum of its kernels on the summands, which gives the first statement. The second follows by comparing types in (3). $\square$

Remark 46.2.3. Primitivity is defined by L alone, so $P^k(X)$ and the Lefschetz decomposition depend only on the Kähler class $[\omega]$, not on the metric. The operator Λ on cohomology was defined using harmonic forms, but it too depends only on $[\omega]$, by Proposition 46.2.4.

Proposition 46.2.4. *Let V be a finite-dimensional vector space over a field K of characteristic zero, graded by degree. Let e and h be endomorphisms such that h acts on the degree k part by $k - n$ and $[h, e] = 2e$. Then there is at most one endomorphism f with $[e, f] = h$ and $[h, f] = -2f$. If $K \subseteq K'$ is a field extension and such an f exists over K' , then it is defined over K . In particular, if X is projective and $[\omega]$ is the restriction of the hyperplane class, then L and Λ preserve $H^\bullet(X; \mathbb{Q})$, and the Lefschetz decomposition is defined over $\mathbb{Q}$.*

Proof. It suffices to treat $K = \mathbb{C}$ for uniqueness, after extending scalars. If f and f' both work, then $g = f - f'$ satisfies $[e, g] = 0$ and $[h, g] = -2g$. The Lie algebra $\mathfrak{sl}_2$ acts on $\text{End}(V)$ through the triple (e, f, h) by $x \cdot \phi = [x, \phi]$. In this representation g is killed by e and has weight -2 . By Theorem 11.5.2, a nonzero vector killed by e has a non-negative weight, so $g = 0$. The conditions on f are linear equations with coefficients in K . A system of linear equations over K that has a solution over K' has one over K , since the rank of a matrix does not change under field extension, and by uniqueness the two solutions coincide. For the last statement, $[\omega] = c_1(\mathcal{O}(1))|_X$ lies in the image of $H^2(X; \mathbb{Z})$ (Proposition 44.4.1), so L preserves $H^\bullet(X; \mathbb{Q})$, and the rest follows. $\square$

Corollary 46.2.5. *Let X be a compact Kähler manifold of dimension n . Then $L: H^k \rightarrow H^{k+2}$ is injective for $k < n$ and surjective for $k \geq n - 1$. Consequently*

$$b_k \leq b_{k+2} \quad (k < n), \quad h^{p,q} \leq h^{p+1,q+1} \quad (p+q < n), \quad \dim P^k = b_k - b_{k-2} \quad (k \leq n).$$

Proof. For $k < n$, L^{n-k} is injective on H^k and equals $L^{n-k-1} \circ L$, so L is injective on H^k . For $k \geq n-1$, put $j = 2n - k - 2 \leq n-1$. Then $L^{n-j} = L \circ L^{n-j-1}$ maps H^j onto $H^{2n-j} = H^{k+2}$, and L^{n-j-1} lands in H^k , so L is onto H^{k+2} . The inequality for Hodge numbers follows since L has type $(1, 1)$, and the last formula from $H^k = P^k \oplus LH^{k-2}$ with L injective on H^{k-2} . $\square$

Example 46.2.6. Let $X = \mathbb{C}^n/\Lambda$ be a complex torus with a flat Kähler metric, so that ω has constant coefficients. The harmonic forms are the forms with constant coefficients (Example 43.4.2), so $H^\bullet(X; \mathbb{R}) \cong \Lambda^\bullet V^*$ with $V = \mathbb{R}^{2n}$, and L is the wedge product with the constant form ω . In this case Theorem 46.2.2 is exactly the linear algebra of Theorem 44.5.2. For instance, for $n = 2$, $H^2(X; \mathbb{R})$ has dimension 6 and is the sum of $\mathbb{R}[\omega]$ and the 5-dimensional primitive part (Exercise 44.6.2).

Example 46.2.7. The manifold $S^2 \times S^3 \times S^3$ has Poincaré polynomial $(1+t^2)(1+t^3)^2 = 1 + t^2 + 2t^3 + 2t^5 + t^6 + t^8$. Its real dimension is 8, so a Kähler structure would have $n = 4$, and Corollary 46.2.5 would give $b_2 \leq b_4$. But $b_2 = 1 > 0 = b_4$, so no complex structure on this manifold is Kähler.

46.3 Pointwise Hodge–Riemann relations

Proposition 46.3.1. *Let V be a complex vector space of dimension n with a Hermitian inner product and fundamental form ω , as in Definition 44.5.1. Let α be a primitive form of type (p, q) and degree $k = p + q \leq n$. Then*

$$i^{p-q}(-1)^{k(k-1)/2} \alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)! |\alpha|^2 \text{vol}.$$

More generally, if α and β are primitive of type (p, q) , then $i^{p-q}(-1)^{k(k-1)/2} \alpha \wedge \bar{\beta} \wedge \omega^{n-k} = (n-k)! \langle \alpha, \beta \rangle \text{vol}$.

Proof. The second statement follows from the first by polarisation, since both sides are Hermitian forms in (α, β) . Choose a basis with $h_{j\bar{k}} = \delta_{jk}$, so that $dz_j = dx_j + i dy_j$ with dx_j, dy_j orthonormal, and put $v_j = dx_j \wedge dy_j = \frac{i}{2} dz_j \wedge d\bar{z}_j$. Then $\omega = \sum_j v_j$, and $\text{vol} = v_1 \wedge \cdots \wedge v_n$. For a set R of indices write $v_R = \bigwedge_{j \in R} v_j$, and for disjoint A, B write $D = A \cup B$ and $dz_A \wedge d\bar{z}_B$ for the wedge products in increasing order.

Step 1: splitting. Every monomial $dz_I \wedge d\bar{z}_J$ equals $\pm(2/i)^{|I \cap J|} dz_A \wedge d\bar{z}_B \wedge v_C$ with $C = I \cap J$, $A = I \setminus C$, $B = J \setminus C$. So $\Lambda^{p,q}$ is the orthogonal direct sum of the subspaces $dz_A \wedge d\bar{z}_B \wedge S_D$, where S_D is spanned by the v_R with $R \cap D = \emptyset$. Write $\alpha = \sum_{A,B} dz_A \wedge d\bar{z}_B \wedge f_{AB}$ with $f_{AB} \in S_D$ homogeneous of degree $r = p - |A| = q - |B|$ in the v_j . In the proof of Theorem 44.5.2, $\Lambda = \sum_j \Lambda_j$, where Λ_j acts on the j -th factor, kills 1, dz_j and $d\bar{z}_j$, and sends v_j to 1. Hence $\Lambda(dz_A \wedge d\bar{z}_B \wedge f) = dz_A \wedge d\bar{z}_B \wedge \Lambda f$, where $\Lambda v_R = \sum_{j \in R} v_{R \setminus j}$, and α is primitive if and only if every f_{AB} is.

Step 2: the summands are orthogonal for the left-hand side. Let α and β lie in the summands of (A, B) and (A', B') . For $j \in A$, the form α contains dz_j but not $d\bar{z}_j$. A top-degree monomial needs exactly one $d\bar{z}_j$, and it cannot come from a power of ω or from a v_j in β , both of which would repeat dz_j . So it must come from $\bar{\beta} = d\bar{z}_{A'} \wedge dz_{B'} \wedge \bar{f}'$, which forces $j \in A'$. Thus $\alpha \wedge \bar{\beta} \wedge \omega^{n-k} = 0$ unless $A \subseteq A'$, and by symmetry $A = A'$ and $B = B'$. The summands are also orthogonal for $\langle \cdot, \cdot \rangle$, so it suffices to treat $\alpha = dz_A \wedge d\bar{z}_B \wedge f$ with $|A| = a$, $|B| = b$, $f = \sum_{|R|=r} f_R v_R \in S_D$ primitive, and $m = n - a - b$ indices outside D . Then $n - k = m - 2r$.

Step 3: the v -part. Only the v_j with $j \notin D$ survive in ω^{n-k} after wedging with $dz_A \wedge d\bar{z}_B \wedge d\bar{z}_A \wedge dz_B$. Moreover $(\sum_{j \notin D} v_j)^{m-2r} = (m-2r)! \sum_{|T|=m-2r} v_T$, and $v_R v_{R'} v_T$ is the product of

all $v_j, j \notin D$, exactly when R, R', T partition the complement of D . So

$$f \wedge \bar{f} \wedge \left(\sum_{j \notin D} v_j \right)^{m-2r} = (m-2r)! \sum_{R \cap R' = \emptyset} f_R \overline{f_{R'}} v_{\{1, \dots, n\} \setminus D}.$$

We claim that $\sum_{R \cap R' = \emptyset} f_R \overline{f_{R'}} = (-1)^r \sum_R |f_R|^2$ for primitive f . Primitivity says that $\sum_{R' \supseteq Q} f_{R'} = 0$ for every set Q of size $r-1$. For $|S| = s < r$, each $R' \supseteq S$ of size r contains exactly $r-s$ sets Q of size $r-1$ with $S \subseteq Q$, so $\sum_{R' \supseteq S} f_{R'} = \frac{1}{r-s} \sum_Q \sum_{R' \supseteq Q} f_{R'} = 0$. By inclusion–exclusion, $[R \cap R' = \emptyset] = \sum_{S \subseteq R \cap R'} (-1)^{|S|}$. Hence

$$\sum_{R'} \overline{f_{R'}} [R \cap R' = \emptyset] = \sum_{S \subseteq R} (-1)^{|S|} \sum_{R' \supseteq S} \overline{f_{R'}} = (-1)^r \overline{f_R},$$

because only $S = R$ survives. Multiplying by f_R and summing over R proves the claim.

Step 4: the dz -part and the signs. Reordering gives

$$dz_A \wedge d\bar{z}_B \wedge d\bar{z}_A \wedge dz_B = (-1)^{ab+b} (dz_A \wedge d\bar{z}_A) \wedge (dz_B \wedge d\bar{z}_B) = (-1)^{ab+b+a(a-1)/2+b(b-1)/2} (-2i)^{a+b} v_D,$$

using $dz_A \wedge d\bar{z}_A = (-1)^{a(a-1)/2} \bigwedge_{j \in A} (dz_j \wedge d\bar{z}_j)$ and $dz_j \wedge d\bar{z}_j = -2i v_j$. The forms f and $\bar{f}$ have even degree, so they can be moved freely. Therefore $i^{p-q} (-1)^{k(k-1)/2} \alpha \wedge \bar{\alpha} \wedge \omega^{n-k}$ equals $(m-2r)! \sum_R |f_R|^2 \text{ vol}$ times

$$i^{a-b} (-2i)^{a+b} (-1)^{k(k-1)/2+ab+b+a(a-1)/2+b(b-1)/2+r}.$$

Now $i^{a-b} (-i)^{a+b} = (-1)^{a+b} i^{2a} = (-1)^b$. With $s = a+b$ and $k = s+2r$, we have $k(k-1)/2 \equiv s(s-1)/2 + r \equiv a(a-1)/2 + b(b-1)/2 + ab + r \pmod{2}$. So the total sign is $+1$ and the factor is 2^{a+b} . Finally $|\alpha|^2 = |dz_A \wedge d\bar{z}_B|^2 |f|^2 = 2^{a+b} \sum_R |f_R|^2$, since the $dz_j, d\bar{z}_j$ are orthogonal of squared length 2 (Lemma 45.2.2) and the v_R are orthonormal. As $m-2r = n-k$, this is the assertion. $\square$

Remark 46.3.2. Proposition 46.3.1 also follows from Weil’s formula $*\alpha = (-1)^{k(k+1)/2} i^{p-q} L^{n-k} \alpha / (n-k)!$ for primitive α of type (p, q) (Remark 44.5.3), using $\beta \wedge *\bar{\alpha} = \langle \beta, \alpha \rangle \text{ vol}$. The direct computation above avoids proving Weil’s formula in general.

46.4 The Hodge–Riemann bilinear relations

Definition 46.4.1. For $k \leq n$ define the bilinear form Q on $H^k(X; \mathbb{R})$, and its complex bilinear extension to $H^k(X; \mathbb{C})$, by

$$Q(a, b) = (-1)^{k(k-1)/2} \int_X a \cup b \cup [\omega]^{n-k}.$$

Theorem 46.4.2 (Hodge–Riemann bilinear relations). *Let X be a compact Kähler manifold of dimension n and $k \leq n$. Then:*

1. $Q(b, a) = (-1)^k Q(a, b)$, and Q is nondegenerate on $H^k(X; \mathbb{R})$;
2. $Q(H^{p,q}, H^{p',q'}) = 0$ unless $(p', q') = (q, p)$;
3. the Lefschetz decomposition is Q -orthogonal: $Q(L^j P^{k-2j}, L^{j'} P^{k-2j'}) = 0$ for $j \neq j'$;
4. for $0 \neq a \in P^{p,q}(X)$, $p+q=k$, we have $i^{p-q} Q(a, \bar{a}) > 0$.

Proof. (1) The symmetry is graded commutativity of the cup product. For nondegeneracy, if $Q(a, b) = 0$ for all b , then $a \cup [\omega]^{n-k}$ pairs trivially with H^k , so $L^{n-k}a = 0$ by Poincaré duality (Theorem 35.3.10), and $a = 0$ by Theorem 46.2.2.

(2) is Proposition 45.5.7, as L^{n-k} has type $(n-k, n-k)$.

(3) Let $j < j'$, $a \in P^{k-2j}$ and $b \in P^{k-2j'}$. Then $Q(L^j a, L^{j'} b) = \pm \int_X b \cup L^{n-k+j+j'} a$, and $n-k+j+j' \geq n-(k-2j)+1$, so $L^{n-k+j+j'} a = 0$ by primitivity. The case $j > j'$ follows by (1).

(4) Let α be the harmonic representative of a . It has type (p, q) (Theorem 45.5.2) and is primitive at every point (Theorem 46.2.2(4)), and $\bar{\alpha}$ represents $\bar{a}$. By Proposition 46.3.1,

$$i^{p-q} Q(a, \bar{a}) = \int_X i^{p-q} (-1)^{k(k-1)/2} \alpha \wedge \bar{\alpha} \wedge \omega^{n-k} = (n-k)! \|\alpha\|^2 > 0.$$

□

Remark 46.4.3. In the language of Chapter 62: if $[\omega]$ is integral, then $P^k(X; \mathbb{Q})$ with the decomposition $P^k(X; \mathbb{C}) = \bigoplus P^{p,q}$ and the form Q is a *polarised Hodge structure* of weight k . By Theorem 46.4.2(3), $H^k(X; \mathbb{Q})$ is an orthogonal direct sum of such structures, with Q twisted by signs on the summands. This is why the Hodge structures of projective manifolds are polarisable, a fact that underlies the semisimplicity arguments of Part IX.

Example 46.4.4. For a compact Riemann surface ($n = 1$) and $k = 1$, every class is primitive, and part (4) says that $i \int_X \alpha \wedge \bar{\alpha} > 0$ for every nonzero holomorphic 1-form α . These are *Riemann's bilinear relations*. For $E_\tau = \mathbb{C}/(\mathbb{Z} + \tau\mathbb{Z})$ and $\alpha = dz$,

$$i \int_{E_\tau} dz \wedge d\bar{z} = i \cdot (-2i) \text{area}(E_\tau) = 2 \text{Im } \tau > 0,$$

so the positivity condition $\text{Im } \tau > 0$ on the period τ is an instance of the Hodge–Riemann relations. It is the condition cutting out the period domain, here the upper half plane (Chapter 66).

46.5 The Hodge index theorem

Theorem 46.5.1 (Hodge index theorem). *Let X be a compact Kähler manifold of even dimension n . The signature of the intersection form $(a, b) \mapsto \int_X a \cup b$ on $H^n(X; \mathbb{R})$ is*

$$\sigma(X) = \sum_{p,q=0}^n (-1)^p h^{p,q}(X).$$

Proof. Let $I(a, b) = \int_X a \cup b$ on H^n . By Theorem 46.4.2(3) and the Hodge decomposition, $H^n(X; \mathbb{R})$ is the I -orthogonal direct sum of the real subspaces $W_{j,p,q} = L^j(P^{p,q} + P^{q,p}) \cap H^n(X; \mathbb{R})$ over $j \geq 0$ and $p \leq q$ with $p+q = n-2j =: k$. Here $P^{p,q}$ is paired by I only with $P^{q,p}$ (Theorem 46.4.2(2)). We compute the sign of I on each piece. Note that k is even. For $a \in P^{p,q}$ we have $I(L^j a, L^j \bar{a}) = \int a \cup \bar{a} \cup [\omega]^{2j} = (-1)^{k(k-1)/2} Q(a, \bar{a})$, since $n-k = 2j$. So $I(L^j a, L^j \bar{a}) = (-1)^{k(k-1)/2} i^{q-p} \cdot i^{p-q} Q(a, \bar{a})$ has the sign $\varepsilon_{p,q} = (-1)^{k(k-1)/2 + (q-p)/2}$. Writing $k = 2m$, $k(k-1)/2 \equiv m$ and $(q-p)/2 = m-p$, so $\varepsilon_{p,q} = (-1)^p$.

- If $p < q$, the real points of $L^j(P^{p,q} \oplus P^{q,p})$ are the $L^j(a + \bar{a})$, $a \in P^{p,q}$, and $I(L^j(a + \bar{a}), L^j(a + \bar{a})) = 2I(L^j a, L^j \bar{a})$, because $I(L^j a, L^j a) = 0$ by type. So I is definite of sign $(-1)^p$ on a real space of dimension $2 \dim P^{p,q}$.
- If $p = q$, then I is definite of sign $(-1)^p$ on the real points of $L^j P^{p,p}$, which has real dimension $\dim P^{p,p}$.

Hence $\sigma(X) = \sum_{p+q \leq n, p+q \text{ even}} (-1)^p \dim P^{p,q}$. By Theorem 46.2.2(5) and Corollary 46.2.5, $\dim P^{p,q} = h^{p,q} - h^{p-1,q-1}$ for $p+q \leq n$. Substituting $p' = p-1, q' = q-1$ in the second sum gives

$$\sigma(X) = \sum_{p+q=n} (-1)^p h^{p,q} + 2 \sum_{p+q < n, p+q \text{ even}} (-1)^p h^{p,q}.$$

On the other hand, in $\sum_{p,q} (-1)^p h^{p,q}$ the terms with $p+q$ odd cancel in pairs by Hodge symmetry $h^{p,q} = h^{q,p}$, since then $(-1)^p = -(-1)^q$. The terms with $p+q > n$ even equal those with $p+q < n$ by Serre duality $h^{p,q} = h^{n-p,n-q}$, since n is even. So the two expressions agree. $\square$

Corollary 46.5.2. *Let X be a compact Kähler surface with irregularity $q = h^{0,1}$ and geometric genus $p_g = h^{2,0}$. The intersection form on $H^2(X; \mathbb{R})$ has signature $(2p_g + 1, h^{1,1} - 1)$, and its restriction to $H_{\mathbb{R}}^{1,1} = H^{1,1}(X) \cap H^2(X; \mathbb{R})$ has signature $(1, h^{1,1} - 1)$. So $b^+(X) = 2p_g + 1$, and a real $(1, 1)$ -class a with $\int_X a \cup [\omega] = 0$ satisfies $\int_X a^2 \leq 0$, with equality only for $a = 0$.*

Proof. For $n = 2$ the pieces in the proof of Theorem 46.5.1 are: $L(P^{0,0}) = \mathbb{R}[\omega]$ with sign $+$; $P^{0,2} + P^{2,0} = H^{0,2} \oplus H^{2,0}$ with sign $+$ and real dimension $2p_g$ (every class of type $(2, 0)$ is primitive since $H^{3,1} = 0$); and $P^{1,1}$ with sign $-$ and dimension $h^{1,1} - 1$. The real $(1, 1)$ -classes are $\mathbb{R}[\omega] \oplus P_{\mathbb{R}}^{1,1}$. A class $a \in H_{\mathbb{R}}^{1,1}$ with $\int a \cup [\omega] = 0$ is orthogonal to $[\omega]$, so it lies in $P_{\mathbb{R}}^{1,1}$, where the form is negative definite. $\square$

Remark 46.5.3. For a projective surface, applied to classes of divisors, Corollary 46.5.2 is the classical Hodge index theorem: the intersection form on the Néron–Severi group has exactly one positive eigenvalue (Chapter 58). Grothendieck’s standard conjecture of Hodge type asks for the analogue of Theorem 46.4.2(4) for algebraic cycles in arbitrary characteristic (Chapter 73).

46.6 Further remarks and counterexamples

Remark 46.6.1. Hard Lefschetz is a special property of Kähler manifolds. Thurston gave a compact symplectic 4-manifold, a nilmanifold with $b_1 = 3$, that admits no Kähler structure [Thu76]. Its first Betti number is odd, so Corollary 45.5.3 already rules out a Kähler structure. Moreover, for every symplectic form ω the Lefschetz map $[\omega] \cup -: H^1 \rightarrow H^3$ fails to be an isomorphism: otherwise $(a, b) \mapsto \int a \cup b \cup [\omega]$ would be a nondegenerate skew-symmetric form on the odd-dimensional space H^1 . For singular projective varieties, hard Lefschetz fails for ordinary cohomology but holds for intersection cohomology [BBD82]. In characteristic p , Deligne proved hard Lefschetz for ℓ -adic cohomology as part of his second proof of the Weil conjectures [Del80].

Counterexample 46.6.2. The sign in Theorem 46.4.2(4) genuinely requires primitivity. On $X = \mathbb{CP}^1 \times \mathbb{CP}^1$ with $\omega = \text{pr}_1^* \omega_{FS} + \text{pr}_2^* \omega_{FS}$, let h_1, h_2 be the pullbacks of the generators, so $[\omega] = h_1 + h_2$, $h_1^2 = h_2^2 = 0$ and $\int h_1 h_2 = 1$. For $k = 2$, $Q(a, a) = -\int a^2$. Then $Q(h_1 - h_2, h_1 - h_2) = 2 > 0$ for the primitive class $h_1 - h_2$, as predicted for type $(1, 1)$, while $Q([\omega], [\omega]) = -2 < 0$ for the non-primitive class $[\omega]$.

46.7 Exercises

Exercise 46.7.1. Compute the signature of $\mathbb{CP}^2$, of $\mathbb{CP}^1 \times \mathbb{CP}^1$, and of a smooth quartic surface in $\mathbb{CP}^3$ from Theorem 46.5.1, using the Hodge numbers of Chapter 45.

Solution. For a surface $\sigma = 2 + 2p_g - h^{1,1}$ (the q -terms cancel). For $\mathbb{CP}^2$: $p_g = 0$, $h^{1,1} = 1$, so $\sigma = 1$. For $\mathbb{CP}^1 \times \mathbb{CP}^1$: $p_g = 0$ and $h^{1,1} = 2$ by Exercise 45.8.1, so $\sigma = 0$, matching the hyperbolic form of Counterexample 46.6.2. For the quartic (Example 45.7.4): $p_g = 1$ and $h^{1,1} = 20$, so $\sigma = -16$. $\square$

Exercise 46.7.2. Let X be compact Kähler of dimension $n \geq 1$ and $0 \neq a \in H^1(X; \mathbb{R})$. Show that $a \cup [\omega]^{n-1} \neq 0$, and that $(a, b) \mapsto \int_X a \cup b \cup [\omega]^{n-1}$ is a symplectic form on $H^1(X; \mathbb{R})$.

Solution. The first statement is hard Lefschetz for $k = 1$. The form is Q for $k = 1$, so it is skew-symmetric and nondegenerate by Theorem 46.4.2(1). This gives another proof that b_1 is even. $\square$

Exercise 46.7.3. Let X be a compact Kähler surface with $b_2(X) = 1$. Show that $p_g = 0$ and $h^{1,1} = 1$, and that a generator e of $H^2(X; \mathbb{Z})/\text{torsion} \cong \mathbb{Z}$ satisfies $\int_X e^2 = 1$.

Solution. By Theorem 45.5.2, $b_2 = 2p_g + h^{1,1}$, and $h^{1,1} \geq 1$ (Corollary 45.5.3), so $p_g = 0$ and $h^{1,1} = 1$. By Corollary 46.5.2 the intersection form has signature $(1, 0)$, so $\int e^2 > 0$. By Poincaré duality the cup product pairing on $H^2(X; \mathbb{Z})/\text{torsion}$ is unimodular (Corollary 21.6.3), so $\int e^2 = \pm 1$, hence 1. $\square$

Exercise 46.7.4. Determine the Lefschetz decomposition of $H^\bullet(\mathbb{CP}^1 \times \mathbb{CP}^1; \mathbb{R})$ for $[\omega] = h_1 + h_2$, and for $[\omega] = h_1 + 2h_2$.

Solution. In both cases $P^0 = \mathbb{R}$ and $H^4 = L^2 P^0$. For $[\omega] = h_1 + h_2$, P^2 is the kernel of $L: H^2 \rightarrow H^4$, $a_1 h_1 + a_2 h_2 \mapsto (a_1 + a_2) h_1 h_2$, which is spanned by $h_1 - h_2$, and $H^2 = L P^0 \oplus P^2$. For $[\omega] = h_1 + 2h_2$, L sends $a_1 h_1 + a_2 h_2$ to $(2a_1 + a_2) h_1 h_2$, so P^2 is spanned by $h_1 - 2h_2$. The primitive subspace depends on the Kähler class. $\square$

Exercise 46.7.5. Show that the character of the $\mathfrak{sl}_2$ -representation $H^\bullet(X; \mathbb{R})$ determines the Betti numbers and vice versa, and write $H^\bullet(\mathbb{CP}^n; \mathbb{R})$ and $H^\bullet$ of a compact Riemann surface of genus g as sums of irreducibles $V(m)$.

Solution. The weight $k - n$ has multiplicity b_k , so the character is $\sum_k b_k t^{k-n}$. For $\mathbb{CP}^n$, $H^\bullet = V(n)$, spanned by $1, [\omega], \dots, [\omega]^n$. For a Riemann surface, $H^\bullet = V(1) \oplus 2g V(0)$: the classes 1 and $[\omega]$ span $V(1)$, and H^1 consists of primitive classes of weight 0. $\square$

Chapter 47

Vanishing and Embedding Theorems

47.1 Introduction

This chapter answers the question: which compact Kähler manifolds are projective? The answer is Kodaira's embedding theorem (Theorem 47.6.5). A compact complex manifold embeds in projective space if and only if it carries a positive line bundle, if and only if it has a Kähler form with integral cohomology class. The analytic input is the Kodaira–Nakano vanishing theorem (Theorem 47.4.1). It comes from the Bochner–Kodaira–Nakano identity, a version of the Kähler identities with coefficients in a Hermitian bundle, in which the curvature appears. Vanishing also gives a Hodge-theoretic proof of the Lefschetz hyperplane theorem (Theorem 47.5.1). Together these results form the bridge from Kähler geometry to algebraic geometry: by Chow's theorem (Chapter 54) a projective manifold is algebraic, so the Hodge theory of the previous chapters applies to smooth projective varieties. References: [GH78, Chapter 1, Sections 2–4], [Huy05, Chapter 5], [Voi02b, Chapter 7], [Kod54].

Prerequisites. Chapter 46 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 42 (Holomorphic Vector Bundles).

Throughout, X is a compact complex manifold of dimension n . For a line bundle L with a Hermitian metric h , we write F_h for the curvature of its Chern connection and $\omega_h = \frac{i}{2\pi}F_h$ for its *curvature form*, a closed real $(1, 1)$ -form representing $c_1(L)$ (Theorem 42.4.1). We write $L^k = L^{\otimes k}$ and $L(-D) = L \otimes \mathcal{O}(-D)$.

47.2 Positive line bundles

Proposition 47.2.1. *Positivity (Definition 42.4.2) has the following properties.*

1. $\mathcal{O}(1)$ on $\mathbb{CP}^N$ is positive, and if $f: Y \rightarrow X$ is a holomorphic immersion and L is positive, then f^*L is positive.
2. Metrics multiply under tensor products and curvature forms add: $\omega_{h_1 \otimes h_2} = \omega_{h_1} + \omega_{h_2}$ and $\omega_{h^{-1}} = -\omega_h$ for the dual metric. Hence tensor products of positive bundles are positive.
3. If L is positive and M is any line bundle, then $L^k \otimes M$ is positive for all $k \geq k_0$.
4. If (L, h) is positive, then ω_h is a Kähler form. So X is Kähler and $c_1(L)_{\mathbb{R}}$ is a Kähler class.
5. Conversely, if a Kähler form ω represents $c_1(L)_{\mathbb{R}}$, then L has a Hermitian metric with $\omega_h = \omega$. In particular L is positive.

Proof. (1) The first statement is Example 42.4.3. The Chern connection of f^*h is the pullback, so $\omega_{f^*h} = f^*\omega_h$, and the pullback of a positive $(1, 1)$ -form by a holomorphic immersion

is positive, since the restriction of a positive definite Hermitian form to a subspace is positive definite. (2) In local holomorphic frames the weights add: $h_1 \otimes h_2 = e^{-(\varphi_1 + \varphi_2)}$, and $h^{-1} = e^\varphi$. Apply the formula $F = \partial\bar{\partial}\varphi$. (3) Choose metrics. Since X is compact and ω_{h_L} is positive, there is C with $\omega_{h_M} \geq -C\omega_{h_L}$. Then $\omega_{h_L^k \otimes h_M} = k\omega_{h_L} + \omega_{h_M} \geq (k - C)\omega_{h_L}$, which is positive for $k > C$. (4) ω_h is a closed positive real $(1, 1)$ -form, hence a Kähler form (Proposition 44.2.2). (5) X is compact Kähler with Kähler form ω . Let h_0 be any metric on L . Then ω_{h_0} and ω are cohomologous closed real $(1, 1)$ -forms, so $\omega = \omega_{h_0} + i\partial\bar{\partial}\psi$ for a smooth real function ψ (Corollary 45.6.4). The metric $h = h_0 e^{-2\pi\psi}$ has local weights $\varphi = \varphi_0 + 2\pi\psi$, so $\omega_h = \omega_{h_0} + i\partial\bar{\partial}\psi = \omega$. $\square$

Lemma 47.2.2. *Let X be a compact Kähler manifold. The map $H^2(X; \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X)$ induced by $\mathbb{Z}_X \subseteq \mathcal{O}_X$ is the composite of $H^2(X; \mathbb{Z}) \rightarrow H^2(X; \mathbb{C})$ with the map $[\alpha] \mapsto [\alpha^{0,2}]_{\bar{\partial}} \in H_{\bar{\partial}}^{0,2}(X) \cong H^2(X, \mathcal{O}_X)$. Consequently, every class $a \in H^2(X; \mathbb{Z})$ whose image in $H^2(X; \mathbb{C})$ lies in $H^{1,1}(X)$ is $c_1(L)$ for a holomorphic line bundle L .*

Proof. The inclusion $\mathbb{Z}_X \rightarrow \mathcal{O}_X$ factors through $\mathbb{C}_X$. The de Rham complex $\mathcal{A}_{\mathbb{C}}^\bullet$ resolves $\mathbb{C}_X$ and the Dolbeault complex $\mathcal{A}^{0,\bullet}$ resolves $\mathcal{O}_X$ (Theorems 35.3.1 and 43.2.3), by fine, hence acyclic, sheaves. Projection onto the $(0, q)$ -component commutes with the differentials: the $(0, q+1)$ -part of $d\alpha$ is $\bar{\partial}$ applied to the $(0, q)$ -part of α , because ∂ raises the first degree and $\bar{\partial}$ of a form of type (r, s) with $r \geq 1$ has type $(r, s+1)$. So it is a morphism of resolutions over $\mathbb{C}_X \rightarrow \mathcal{O}_X$, and it induces the map on cohomology (Proposition 13.6.6, which is natural). If the image of a lies in $H^{1,1}(X)$, it is represented by a closed form of type $(1, 1)$, whose $(0, 2)$ -part is zero. So a maps to 0 in $H^2(X, \mathcal{O}_X)$, and $a = c_1(L)$ for some L by Theorem 42.2.4(1). $\square$

Definition 47.2.3. A *Hodge manifold* is a compact complex manifold with a Kähler form ω whose class $[\omega] \in H^2(X; \mathbb{R})$ lies in the image of $H^2(X; \mathbb{Z})$; such an ω is a *Hodge metric*, and $[\omega]$ an *integral Kähler class*.

Corollary 47.2.4. *A compact complex manifold carries a positive line bundle if and only if it is a Hodge manifold.*

Proof. If L is positive, then ω_h is a Kähler form representing the integral class $c_1(L)$ (Proposition 47.2.1(4)). Conversely, let ω be a Hodge metric and $a \in H^2(X; \mathbb{Z})$ a class mapping to $[\omega]$. As ω has type $(1, 1)$, Lemma 47.2.2 gives L with $c_1(L) = a$, so $c_1(L)_{\mathbb{R}} = [\omega]$, and L is positive by Proposition 47.2.1(5). $\square$

Proposition 47.2.5. *On a compact Riemann surface X , a line bundle L is positive if and only if $\deg L = \int_X c_1(L) > 0$. Moreover $\deg K_X = 2g - 2$, where $g = b_1(X)/2$ is the genus.*

Proof. If L is positive, $\deg L = \int_X \omega_h > 0$. Conversely, if $\deg L > 0$, choose any Hermitian metric on X and rescale its fundamental form to a Kähler form ω with $\int_X \omega = \deg L$ (every Hermitian metric on a curve is Kähler, Example 44.4.3). Since $H^2(X; \mathbb{R}) \cong \mathbb{R}$ via integration, $[\omega] = c_1(L)_{\mathbb{R}}$, and Proposition 47.2.1(5) applies. For the last statement, $K_X = T^*X$, so $\deg K_X = -\int_X c_1(TX) = -\int_X e(TX) = -\chi(X) = -(2 - 2g)$ by Theorem 37.5.1 and Proposition 37.2.4(4). $\square$

47.3 The Bochner–Kodaira–Nakano identity

Notation 47.3.1. Let X be a Hermitian manifold and (E, h) a Hermitian holomorphic vector bundle, with Chern connection $D = D' + \bar{\partial}_E$, where D' has type $(1, 0)$. On $\mathcal{A}^{p,q}(X, E)$ we use

the pointwise inner product induced by g and h and, for X compact, the L^2 product. The operators L and Λ act on the form part. Let D'^* and $\bar{\partial}_E^*$ be the formal adjoints, and put

$$\Delta' = D'D'^* + D'^*D', \quad \Delta'' = \bar{\partial}_E\bar{\partial}_E^* + \bar{\partial}_E^*\bar{\partial}_E.$$

The curvature $F = D^2$ is a $(1, 1)$ -form with values in $\text{End } E$, and as operators $F = D'\bar{\partial}_E + \bar{\partial}_ED'$, since $D'^2 = 0$ and $\bar{\partial}_E^2 = 0$ (Theorem 42.4.1).

Lemma 47.3.2. *Let (E, h) be a Hermitian holomorphic vector bundle and $p \in X$. There is a holomorphic frame $e_1, \dots, e_r$ of E near p with $h(e_a, e_b) = \delta_{ab} + O(|z|^2)$ in holomorphic coordinates z centred at p . In such a frame the connection form of D vanishes at p .*

Proof. Start with a holomorphic frame that is orthonormal at p , and write $H_{ab} = h(e_a, e_b) = \delta_{ab} + \sum_l (\partial_l H_{ab}(0)z^l + \bar{\partial}_l H_{ab}(0)\bar{z}^l) + O(|z|^2)$. Put $e'_a = e_a - \sum_{b,l} \partial_l H_{ab}(0)z^l e_b$, again a holomorphic frame near p . To first order,

$$h(e'_a, e'_b) = H_{ab} - \sum_l \partial_l H_{ab}(0)z^l - \sum_l \overline{\partial_l H_{ba}(0)}\bar{z}^l + O(|z|^2),$$

and $\overline{\partial_l H_{ba}} = \bar{\partial}_l \overline{H_{ba}} = \bar{\partial}_l H_{ab}$ because H is Hermitian. So all first-order terms cancel. The connection form $\bar{H}^{-1}\partial\bar{H}$ of Theorem 42.4.1 involves only first derivatives of H , so it vanishes at p . $\square$

Theorem 47.3.3 (Kähler identities with coefficients). *Let X be Kähler and (E, h) a Hermitian holomorphic vector bundle. Then*

$$[\Lambda, \bar{\partial}_E] = -iD'^*, \quad [\Lambda, D'] = i\bar{\partial}_E^*.$$

Proof. Both identities are between first-order differential operators. In holomorphic coordinates and a holomorphic frame of E , the coefficients of these operators at a point are universal expressions in the metric coefficients $h_{j\bar{k}}$ of X , the matrix H of h , and their first derivatives at that point. Indeed $\bar{\partial}_E$ acts componentwise, $D' = \partial + \theta$ where the connection form θ involves first derivatives of H , and the formal adjoints are obtained by integration by parts, which differentiates the pointwise inner products and the volume form once. Given p , choose coordinates osculating to order two (Proposition 44.3.2) and a frame as in Lemma 47.3.2. The data then agree to first order at p with the flat metric on $\mathbb{C}^n$ and the trivial bundle $\mathbb{C}^r$ with the standard metric. For these, an E -valued form is an r -tuple of scalar forms, $\bar{\partial}_E = \bar{\partial}$ and $D' = \partial$ act componentwise, and the identities are Theorem 45.3.3. So both sides agree at p , as in Lemma 45.3.1. $\square$

Theorem 47.3.4 (Bochner–Kodaira–Nakano identity). *Let X be Kähler and (E, h) a Hermitian holomorphic vector bundle with curvature F . Then on $\mathcal{A}^{p,q}(X, E)$*

$$\Delta'' = \Delta' + [iF, \Lambda],$$

where iF acts by wedge product combined with the action of $\text{End } E$.

Proof. By Theorem 47.3.3, $\bar{\partial}_E^* = -i[\Lambda, D']$ and $D'^* = i[\Lambda, \bar{\partial}_E]$. Writing $\bar{\partial}$ for $\bar{\partial}_E$,

$$\begin{aligned} \Delta'' &= -i(\bar{\partial}\Lambda D' - \bar{\partial}D'\Lambda + \Lambda D'\bar{\partial} - D'\Lambda\bar{\partial}), \\ \Delta' &= i(D'\Lambda\bar{\partial} - D'\bar{\partial}\Lambda + \Lambda\bar{\partial}D' - \bar{\partial}\Lambda D'). \end{aligned}$$

Subtracting, the terms $\bar{\partial}\Lambda D'$ and $D'\Lambda\bar{\partial}$ cancel, and

$$\Delta'' - \Delta' = i(\bar{\partial}D' + D'\bar{\partial})\Lambda - i\Lambda(D'\bar{\partial} + \bar{\partial}D') = iF\Lambda - \Lambda iF = [iF, \Lambda].$$

$\square$

47.4 The Kodaira–Nakano vanishing theorem

Theorem 47.4.1 (Kodaira–Nakano vanishing). *Let X be a compact complex manifold of dimension n and L a positive line bundle. Then*

$$H^q(X, \Omega_X^p \otimes L) = 0 \quad \text{for } p + q > n, \quad H^q(X, \Omega_X^p \otimes L^{-1}) = 0 \quad \text{for } p + q < n.$$

Proof. Choose a metric h on L with $\omega = \omega_h$ positive, and give X the Kähler metric with fundamental form ω . Then $iF_h = 2\pi\omega$, so $[iF_h, \Lambda] = 2\pi[L, \Lambda] = 2\pi H$, which is multiplication by $2\pi(p+q-n)$ on $\mathcal{A}^{p,q}(X, L)$ (Theorem 44.5.2). By Theorem 43.3.3, every class in $H^q(X, \Omega_X^p \otimes L)$ is represented by some α with $\Delta''\alpha = 0$. By Theorem 47.3.4,

$$0 = (\Delta''\alpha, \alpha) = \|D'\alpha\|^2 + \|D'^*\alpha\|^2 + 2\pi(p+q-n)\|\alpha\|^2.$$

If $p+q > n$, all three terms are non-negative, so $\alpha = 0$. The second statement follows by Serre duality (Theorem 43.3.4): $H^q(X, \Omega_X^p \otimes L^{-1})$ is dual to $H^{n-q}(X, K_X \otimes (\Omega_X^p)^\vee \otimes L)$, and $K_X \otimes (\Omega_X^p)^\vee \cong \Omega_X^{n-p}$ by the perfect pairing $\Omega_X^p \otimes \Omega_X^{n-p} \rightarrow K_X$. Here $(n-p) + (n-q) > n$. $\square$

Corollary 47.4.2 (Kodaira vanishing). *If L is positive, then $H^q(X, K_X \otimes L) = 0$ and $H^{n-q}(X, L^{-1}) = 0$ for $q > 0$. More generally, if $N \otimes K_X^{-1}$ is positive, then $H^q(X, N) = 0$ for $q > 0$.*

Proof. Take $p = n$ in Theorem 47.4.1, and $p = 0$ in the second statement. For the last claim write $N = K_X \otimes (N \otimes K_X^{-1})$. $\square$

Example 47.4.3. On $\mathbb{CP}^n$, $K = \mathcal{O}(-n-1)$ (Proposition 42.5.1). Since $\mathcal{O}(k) \otimes K^{-1} = \mathcal{O}(k+n+1)$ is positive for $k > -n-1$, we get $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $q > 0$ and $k > -n-1$. By the second statement of Theorem 47.4.1 with $p = 0$, also $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $q < n$ and $k < 0$. Together, $H^q(\mathbb{CP}^n, \mathcal{O}(k)) = 0$ for $0 < q < n$ and all k , which is part of Bott's formula (Remark 43.4.5).

Corollary 47.4.4. *Let X be a compact Riemann surface of genus g and L a line bundle with $\deg L > 2g - 2$. Then $H^1(X, L) = 0$.*

Proof. $\deg(L \otimes K_X^{-1}) = \deg L - (2g - 2) > 0$, so $L \otimes K_X^{-1}$ is positive (Proposition 47.2.5), and Corollary 47.4.2 applies. $\square$

Counterexample 47.4.5. Positivity cannot be dropped. For $L = \mathcal{O}_X$, $H^n(X, K_X) \cong H^0(X, \mathcal{O}_X)^\vee \cong \mathbb{C}$ by Serre duality. On an elliptic curve E , where $K_E \cong \mathcal{O}_E$ (because dz is a nowhere vanishing holomorphic 1-form), every line bundle of degree 0 has degree $\leq 2g - 2 = 0$, and $H^1(E, \mathcal{O}_E) \cong \mathbb{C}$ is nonzero, so Corollary 47.4.4 is sharp.

Remark 47.4.6. The Kodaira vanishing theorem has been generalised to nef and big line bundles (Kawamata–Viehweg) [Laz04, Chapter 4], and it has an algebraic proof by reduction to positive characteristic [DI87]. However, it fails in characteristic p in general [Ray78]. Its dual form $H^q(X, L^{-1}) = 0$ for $q < n$ will be used for coherent cohomology in Chapter 52.

47.5 The Lefschetz hyperplane theorem

Theorem 47.5.1 (Lefschetz hyperplane theorem). *Let X be a compact Kähler manifold of dimension n and $Y \subseteq X$ a smooth hypersurface such that $\mathcal{O}(Y)$ is positive, for instance a smooth hyperplane section of a projective manifold. Then the restriction map*

$$H^k(X; \mathbb{Q}) \rightarrow H^k(Y; \mathbb{Q})$$

is an isomorphism for $k < n - 1$ and injective for $k = n - 1$. The same holds for $H^q(X, \Omega_X^p) \rightarrow H^q(Y, \Omega_Y^p)$ with $k = p + q$.

Proof. Write $\mathcal{O}(-Y)$ for the ideal sheaf of Y . It is locally generated by a defining function, since in local coordinates with $Y = \{z_1 = 0\}$ a holomorphic function g vanishing on Y satisfies $g(z) = z_1 \int_0^1 \partial_1 g(tz_1, z_2, \dots, z_n) dt$. Holomorphic functions on Y extend locally (constantly in z_1), so $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$ is onto. Tensoring $0 \rightarrow \mathcal{O}_X(-Y) \rightarrow \mathcal{O}_X \rightarrow i_*\mathcal{O}_Y \rightarrow 0$ with Ω_X^p gives

$$0 \rightarrow \Omega_X^p \otimes \mathcal{O}(-Y) \rightarrow \Omega_X^p \rightarrow i_*(\Omega_X^p|_Y) \rightarrow 0.$$

By Theorem 47.4.1, $H^q(X, \Omega_X^p \otimes \mathcal{O}(-Y)) = 0$ for $p + q < n$. Since $H^q(X, i_*\mathcal{G}) = H^q(Y, \mathcal{G})$ (Example 24.7.4), the long exact sequence shows that $H^q(X, \Omega_X^p) \rightarrow H^q(Y, \Omega_X^p|_Y)$ is an isomorphism for $p + q < n - 1$ and injective for $p + q = n - 1$.

Next, the conormal sequence $0 \rightarrow \mathcal{O}(-Y)|_Y \rightarrow \Omega_X^1|_Y \rightarrow \Omega_Y^1 \rightarrow 0$ (Proposition 42.5.1) gives, by taking exterior powers of a sequence of bundles whose first term is a line bundle, the exact sequence

$$0 \rightarrow \Omega_Y^{p-1} \otimes \mathcal{O}(-Y)|_Y \rightarrow \Omega_X^p|_Y \rightarrow \Omega_Y^p \rightarrow 0.$$

Locally this is the decomposition $\Lambda^p(A \oplus C) = \Lambda^p C \oplus (A \otimes \Lambda^{p-1} C)$ for a line A . Now $\mathcal{O}(Y)|_Y$ is positive on Y (Proposition 47.2.1(1)) and $\dim Y = n - 1$. So $H^q(Y, \Omega_Y^{p-1} \otimes \mathcal{O}(-Y)|_Y) = 0$ for $(p - 1) + q < n - 1$, and hence $H^q(Y, \Omega_X^p|_Y) \rightarrow H^q(Y, \Omega_Y^p)$ is an isomorphism for $p + q < n - 1$ and injective for $p + q = n - 1$. Composing proves the statement for $H^q(\Omega^p)$.

The composite $H^q(X, \Omega_X^p) \rightarrow H^q(Y, \Omega_Y^p)$ is induced by restriction of forms. Under the canonical isomorphisms $H^q(\Omega^p) \cong H^{p,q}$ of Corollary 45.6.3 on X and on Y (which is Kähler by Corollary 44.4.2), it corresponds to the restriction $H^{p,q}(X) \rightarrow H^{p,q}(Y)$, since restriction of forms preserves types and commutes with d and $\bar{\partial}$. By the Hodge decomposition, $H^k(X; \mathbb{C}) \rightarrow H^k(Y; \mathbb{C})$ is therefore an isomorphism for $k < n - 1$ and injective for $k = n - 1$. This map is the complexification of the restriction on rational cohomology, and rank is unchanged by extension of scalars. $\square$

Remark 47.5.2. Lefschetz's theorem holds with integer coefficients and for homotopy groups: $\pi_k(X, Y) = 0$ for $k \leq n - 1$, so that $H_k(Y; \mathbb{Z}) \rightarrow H_k(X; \mathbb{Z})$ is an isomorphism for $k < n - 1$ and surjective for $k = n - 1$. The proof uses Morse theory on the affine variety $X \setminus Y$, which has the homotopy type of a CW complex of dimension $\leq n$ (Andreotti–Frankel) [Mil63, Section 7].

Example 47.5.3. Let $Y \subseteq \mathbb{CP}^{n+1}$ be a smooth hypersurface of degree d , so $\dim Y = n$ and $\mathcal{O}(Y) = \mathcal{O}(d)$ is positive. Then $H^k(Y; \mathbb{Q}) \cong H^k(\mathbb{CP}^{n+1}; \mathbb{Q})$ for $k < n$, so $b_k(Y) = b_k(\mathbb{CP}^n)$ for $k < n$, and by Poincaré duality also for $k > n$. Only the middle cohomology $H^n(Y)$ is interesting. For a surface in $\mathbb{CP}^3$, $b_1 = 0$, which was used for the quartic surface in Example 45.7.4. Also $h^{0,q}(Y) = 0$ for $0 < q < n$, that is, $H^q(Y, \mathcal{O}_Y) = 0$ in that range.

47.6 The Kodaira embedding theorem

Definition 47.6.1. Let L be a holomorphic line bundle on X . A point x is a *base point* of L if every $s \in H^0(X, L)$ vanishes at x . If L has no base points, the map $\varphi_L: X \rightarrow \mathbb{P}(H^0(X, L)^\vee)$ sends x to the hyperplane of sections vanishing at x . After choosing a basis $s_0, \dots, s_N$ of $H^0(X, L)$, it is $x \mapsto [s_0(x) : \dots : s_N(x)]$, computed in any local frame, so it is holomorphic. L is *very ample* if φ_L is an embedding, and *ample* if some power L^k is very ample. A compact complex manifold is *projective* if it admits a holomorphic embedding into some $\mathbb{CP}^N$.

Lemma 47.6.2. Let L have no base points. Then φ_L is an embedding if and only if

1. for all $x \neq y$ there is $s \in H^0(X, L)$ with $s(x) = 0 \neq s(y)$ (sections separate points), and

2. for all x , the differentials at x of the sections vanishing at x span $L_x \otimes T_x^* X$ (sections separate tangents), where $T_x^* X$ is the holomorphic cotangent space.

Proof. $\varphi_L(x) = \varphi_L(y)$ exactly when the hyperplanes of sections vanishing at x and at y coincide, so (1) is injectivity. Near x , choose the basis with $s_0(x) \neq 0$. Then φ_L is given in affine coordinates by the functions s_j/s_0 . A section s vanishing at x satisfies $d_x s = d_x(s/s_0) \otimes s_0(x)$. Every s is congruent modulo multiples of s_0 to a section vanishing at x , and $d_x(s_0/s_0) = 0$, so the differentials $d_x(s/s_0)$ of sections vanishing at x span the same space as the $d_x(s_j/s_0)$. Thus (2) says that the differentials of the affine coordinates of φ_L at x span $T_x^* X$, that is, φ_L is an immersion at x . An injective immersion of a compact manifold is an embedding (Proposition 31.4.2), and the image of a holomorphic embedding is a complex submanifold by the holomorphic implicit function theorem. $\square$

Lemma 47.6.3. Let $n \geq 2$, let $x_1, \dots, x_s \in X$ be distinct points, and let $\sigma: \tilde{X} \rightarrow X$ be the blow-up at these points, with exceptional divisors $E_i = \sigma^{-1}(x_i) \cong \mathbb{P}(T_{x_i} X)$ and $E = \sum_i E_i$ (Definition 41.4.4). Then:

1. $\mathcal{O}(E_i)|_{E_i} \cong \mathcal{O}_{E_i}(-1)$;
2. $K_{\tilde{X}} \cong \sigma^* K_X \otimes \mathcal{O}((n-1)E)$;
3. for every line bundle M on X , $\sigma^*: H^0(X, M) \rightarrow H^0(\tilde{X}, \sigma^* M)$ is bijective, and it maps the sections vanishing at all x_i onto $H^0(\tilde{X}, \sigma^* M(-E))$;
4. if $s \in H^0(X, M)$ vanishes at x_i , the restriction of $\sigma^* s \in H^0(\tilde{X}, \sigma^* M(-E))$ to E_i is the section of $M_{x_i} \otimes \mathcal{O}_{E_i}(1)$ given by the linear form $d_{x_i} s \in M_{x_i} \otimes T_{x_i}^* X$ on $T_{x_i} X$;
5. over a coordinate ball B around x_i , $\sigma^{-1}(B) = \{(z, \ell) \in B \times \mathbb{CP}^{n-1} : z \in \ell\}$ (with x_i at the origin), $\mathcal{O}(E_i)$ restricts to the tautological bundle $\text{pr}_2^* \mathcal{O}(-1)$, and the canonical section s_{E_i} of $\mathcal{O}(E_i)$ corresponds to the tautological section $(z, \ell) \mapsto z$.

Proof. (5) The tautological section vanishes exactly on E_i , to first order: in the chart $(u, w_2, \dots, w_n) \mapsto (u, uw_2, \dots, uw_n)$ it is u times the frame $(1, w_2, \dots, w_n)$. So it is a section with divisor E_i , and the bundle is $\mathcal{O}(E_i)$ by Proposition 42.3.2. (1) Restrict (5) to $E_i = \{0\} \times \mathbb{CP}^{n-1}$; alternatively combine Proposition 41.4.5 with adjunction. (2) In the chart above, $\sigma^*(dz_1 \wedge \dots \wedge dz_n) = u^{n-1} du \wedge dw_2 \wedge \dots \wedge dw_n$. So the pullback of a local frame of K_X is a section of $K_{\tilde{X}}$ vanishing to order $n-1$ along E , which gives $\sigma^* K_X \cong K_{\tilde{X}} \otimes \mathcal{O}(-(n-1)E)$. (3) σ^* is injective since σ is surjective. Given $\tilde{s} \in H^0(\tilde{X}, \sigma^* M)$, its restriction to $\tilde{X} \setminus E \cong X \setminus \{x_i\}$ is a section of M there. In a local frame near x_i it is a holomorphic function on a punctured ball, which extends by Hartogs' theorem (Theorem 29.2.9), since $n \geq 2$. The extension s satisfies $\sigma^* s = \tilde{s}$ off E , hence everywhere. A section $\sigma^* s$ vanishes on E if and only if $s(x_i) = 0$ for all i , and a section vanishing on the smooth hypersurface E is divisible by s_E . (4) Write $s = f e$ in a local frame with $f(0) = 0$, $f = \sum_j a_j z_j + O(|z|^2)$. In the chart, $f \circ \sigma = u(a_1 + \sum_{j \geq 2} a_j w_j) + O(u^2)$, and dividing by the local equation u of E and setting $u = 0$ gives $a_1 + \sum_{j \geq 2} a_j w_j$. This is the linear form $\sum_j a_j z_j$ written in the trivialisation of $\mathcal{O}_{\mathbb{CP}^{n-1}}(1)$ over $\{z_1 \neq 0\}$ by z_1 , and $\sum a_j z_j = d_0 f$. $\square$

Lemma 47.6.4. Let L be positive, M any line bundle and $l \geq 1$ an integer. Let $s \in \{1, 2\}$ and $(x_1, \dots, x_s)$ a tuple of distinct points. There are a neighbourhood W of $(x_1, \dots, x_s)$ in X^s and k_0 such that, for every tuple $(x'_1, \dots, x'_s) \in W$ of distinct points and every $k \geq k_0$, the line bundle $\sigma^*(L^k \otimes M) \otimes \mathcal{O}(-lE)$ on the blow-up $\tilde{X}$ at $x'_1, \dots, x'_s$ is positive. For $n = 1$ the same holds with $\tilde{X} = X$ and $E = \sum_i x'_i$.

Proof. Choose disjoint coordinate charts around the x_i containing closed coordinate balls $\bar{B}(x_i, 3\varepsilon)$, and let W be the set of tuples with $x'_i \in B(x_i, \varepsilon)$. Let ω_0 be the Euclidean Kähler form of the chart. Fix metrics on L and M . By compactness there are constants $c, C > 0$ with $\omega_{h_M} \geq -C\omega_{h_L}$ on X and $\omega_{h_L} \geq c\omega_0$ on each $\bar{B}(x_i, 3\varepsilon)$.

Fix x'_i and write $w = z - z(x'_i)$. Put a metric on $\mathcal{O}(E_i)$ as follows. On $\sigma^{-1}(B(x'_i, \varepsilon))$ it is the metric h_1 induced by (5) of Lemma 47.6.3 from the tautological bundle, so that $|s_{E_i}|_{h_1} = |w|$. On $\tilde{X} \setminus E_i$ it is the metric h_2 with $|s_{E_i}|_{h_2} = 1$. Glue them by $h = \rho(|w|)h_1 + (1 - \rho(|w|))h_2$, where ρ is a smooth cut-off equal to 1 on $[0, \varepsilon/2]$ and to 0 on $[3\varepsilon/4, \infty)$. Let θ_i be the curvature form of the dual metric on $\mathcal{O}(-E_i)$. Then:

- on $\sigma^{-1}(B(x'_i, \varepsilon/2))$, $\theta_i = \text{pr}_2^* \omega_{FS} \geq 0$, since $\mathcal{O}(-E_i) \cong \text{pr}_2^* \mathcal{O}(1)$ with the Fubini–Study metric (Proposition 44.4.1);
- outside $\sigma^{-1}(B(x'_i, 3\varepsilon/4))$, $\theta_i = 0$, since s_{E_i} is a holomorphic frame of constant norm;
- on the region $\varepsilon/2 \leq |w| \leq 3\varepsilon/4$, $\theta_i = \frac{i}{2\pi} \partial \bar{\partial} \log(\rho(|w|)|w|^2 + 1 - \rho(|w|))$, a smooth form whose coefficients depend continuously on (z, x'_i) . So $\theta_i \geq -C_1 \omega_0$ there, with C_1 independent of $x'_i \in B(x_i, \varepsilon)$.

The curvature form of $N_k = \sigma^*(L^k \otimes M) \otimes \mathcal{O}(-lE)$ with the product metric is $\Theta_k = \sigma^*(k\omega_{h_L} + \omega_{h_M}) + l \sum_i \theta_i \geq (k - C)\sigma^* \omega_{h_L} + l \sum_i \theta_i$. Let $k > C + lC_1/c$. Away from the balls $B(x'_i, 3\varepsilon/4)$, σ is biholomorphic and $\Theta_k \geq (k - C)\sigma^* \omega_{h_L} > 0$. On the annuli, $\Theta_k \geq ((k - C)c - lC_1)\sigma^* \omega_0 > 0$. On $\sigma^{-1}(B(x'_i, \varepsilon/2))$,

$$\Theta_k \geq (k - C)c \sigma^* \omega_0 + l \text{pr}_2^* \omega_{FS} \geq \min((k - C)c, l) (\text{pr}_1^* \omega_0 + \text{pr}_2^* \omega_{FS})|_{\sigma^{-1}(B)},$$

which is positive, because $\sigma^{-1}(B)$ is a complex submanifold of $B \times \mathbb{CP}^{n-1}$, on which $\text{pr}_1^* \omega_0 + \text{pr}_2^* \omega_{FS}$ is a Kähler form (Corollary 44.4.2). For $n = 1$ the same construction works with $\mathbb{CP}^0$ a point: h_1 is the flat metric, $\text{pr}_2^* \omega_{FS} = 0$, and $\sigma^* \omega_0 = \omega_0$ is positive. So $k_0 = \lfloor C + lC_1/c \rfloor + 1$ works for all tuples in W . $\square$

Theorem 47.6.5 (Kodaira embedding theorem). *Let X be a compact complex manifold and L a positive line bundle. Then there is k_0 such that L^k is very ample for all $k \geq k_0$. In particular X is projective.*

Proof. Step 1: vanishing on blow-ups. Let $\sigma: \tilde{X} \rightarrow X$ be the blow-up at distinct points $x'_1, \dots, x'_s$ ($s \in \{1, 2\}$) and $m \in \{1, 2\}$. By Lemma 47.6.3(2),

$$\sigma^* L^k \otimes \mathcal{O}(-mE) = K_{\tilde{X}} \otimes N_k, \quad N_k = \sigma^*(L^k \otimes K_X^{-1}) \otimes \mathcal{O}(-(m+n-1)E),$$

also for $n = 1$. By Lemma 47.6.4 with $M = K_X^{-1}$ and $l = m + n - 1 \geq 1$, every tuple has a neighbourhood W and a k_0 such that N_k is positive for $k \geq k_0$ and tuples in W . Then $H^1(\tilde{X}, \sigma^* L^k \otimes \mathcal{O}(-mE)) = 0$ by Corollary 47.4.2, applied on the compact complex manifold $\tilde{X}$.

Step 2: consequences. For $s = 1, m = 1$, the exact sequence $0 \rightarrow \sigma^* L^k(-E) \rightarrow \sigma^* L^k \rightarrow \sigma^* L^k|_E \rightarrow 0$ and Step 1 show that $H^0(\tilde{X}, \sigma^* L^k) \rightarrow H^0(E, \sigma^* L^k|_E)$ is onto. The bundle $\sigma^* L^k|_E = E \times L_{x'}^k$ is trivial and E is compact and connected, so the target is $L_{x'}^k$. By Lemma 47.6.3(3), $H^0(X, L^k) \rightarrow L_{x'}^k$ is onto, so x' is not a base point. For $s = 1, m = 2$, the sequence $0 \rightarrow \sigma^* L^k(-2E) \rightarrow \sigma^* L^k(-E) \rightarrow \sigma^* L^k(-E)|_E \rightarrow 0$ shows that $H^0(\tilde{X}, \sigma^* L^k(-E))$ maps onto $H^0(E, L_{x'}^k \otimes \mathcal{O}_E(1))$, using Lemma 47.6.3(1). This space contains every $v \otimes \lambda$ with λ a linear form on $T_{x'}^* X$. By parts (3) and (4) of that lemma, the differentials of the sections of L^k vanishing at x' span $L_{x'}^k \otimes T_{x'}^* X$. For $s = 2, m = 1$, similarly $H^0(X, L^k) \rightarrow L_{x'_1}^k \oplus L_{x'_2}^k$ is onto, so some section vanishes at x'_1 but not at x'_2 . For $n = 1$ the same arguments work with E a sum of points and $\mathcal{O}(-x')|_{x'} \cong T_{x'}^* X$.

Step 3: uniformity. Cover the compact space X by finitely many neighbourhoods W from Step 1 for $s = 1, m \in \{1, 2\}$. We get k_1 such that for all $k \geq k_1$, L^k has no base points and its sections separate tangents at every point. So φ_{L^k} is an immersion for all $k \geq k_1$. An immersion is locally injective (Proposition 31.4.2), so every x has a neighbourhood V_x on which $\varphi_{L^{k_1}}$ is injective. Then $U = \bigcup_x V_x \times V_x$ is a neighbourhood of the diagonal in $X \times X$ such that $\varphi_{L^{k_1}}(x) \neq \varphi_{L^{k_1}}(y)$ for $(x, y) \in U$ with $x \neq y$. The complement $K = (X \times X) \setminus U$ is compact and consists of pairs of distinct points, so finitely many neighbourhoods from Step 1 with $s = 2$ cover it. This gives k_2 such that sections of L^k separate the points of every pair in K for $k \geq k_2$.

Step 4: conclusion. Let $k \geq \max(k_2, 2k_1)$. For $(x, y) \in U$ with $x \neq y$, pick $s \in H^0(X, L^{k_1})$ with $s(x) = 0 \neq s(y)$ and $t \in H^0(X, L^{k-k_1})$ with $t(y) \neq 0$, which exists because $k - k_1 \geq k_1$. Then $st \in H^0(X, L^k)$ vanishes at x but not at y . Together with Step 3, sections of L^k separate all pairs of points and all tangents, so L^k is very ample by Lemma 47.6.2. $\square$

Corollary 47.6.6. *For a compact complex manifold X the following are equivalent:*

1. X is projective;
2. X carries a positive line bundle;
3. X is a Hodge manifold;
4. X is Kähler and some Kähler class lies in the image of $H^2(X; \mathbb{Q}) \rightarrow H^2(X; \mathbb{R})$.

Proof. (1) $\Rightarrow$ (2): restrict $\mathcal{O}(1)$ (Proposition 47.2.1(1)). (2) $\Leftrightarrow$ (3) is Corollary 47.2.4, and (2) $\Rightarrow$ (1) is Theorem 47.6.5. (3) $\Rightarrow$ (4) is clear. For (4) $\Rightarrow$ (3), if $[\omega]$ is rational, then $N[\omega]$ is integral for a suitable integer $N > 0$, and $N\omega$ is a Hodge metric. $\square$

Corollary 47.6.7. *A compact Kähler manifold with $H^0(X, \Omega_X^2) = 0$ is projective. In particular every compact Riemann surface is projective, and so is every compact Kähler surface with $p_g = 0$.*

Proof. By the Hodge decomposition, $H^{2,0} = 0 = H^{0,2}$, so $H^2(X; \mathbb{R}) = H^{1,1}(X) \cap H^2(X; \mathbb{R})$ consists of classes of closed real $(1, 1)$ -forms. The Kähler classes form an open subset of this space: if ω is Kähler and $\beta_1, \dots, \beta_m$ are closed real $(1, 1)$ -forms whose classes form a basis, then $\omega + \sum_i c_i \beta_i$ is positive for all small $|c_i|$, by compactness. The image of $H^2(X; \mathbb{Q})$ is dense in $H^2(X; \mathbb{R}) = H^2(X; \mathbb{Q}) \otimes \mathbb{R}$. So there is a rational Kähler class, and Corollary 47.6.6 applies. For a Riemann surface $\Omega^2 = 0$. $\square$

Remark 47.6.8. By Chow's theorem (Chapter 54), a closed complex submanifold of $\mathbb{CP}^N$ is the zero set of homogeneous polynomials, so projective manifolds are smooth projective varieties. Not every compact Kähler manifold is projective: a complex torus $\mathbb{C}^n / \Lambda$ is projective if and only if Λ satisfies the Riemann bilinear relations, that is, some integral class of type $(1, 1)$ is positive, which fails for a general lattice when $n \geq 2$ (Chapter 57, [BL04]).

47.7 Exercises

Exercise 47.7.1. Let X be a compact Riemann surface of genus g and L a line bundle of degree $\geq 2g + 1$. Show directly, without blowing up, that L is very ample.

Solution. For points x, y , possibly equal, $\deg L(-x - y) = \deg L - 2 > 2g - 2$, so $H^1(X, L(-x - y)) = 0$ by Corollary 47.4.4. The exact sequence $0 \rightarrow L(-x - y) \rightarrow L \rightarrow L \otimes (\mathcal{O}_X / \mathcal{O}_{-x-y}) \rightarrow 0$ shows that $H^0(X, L)$ maps onto the two-dimensional space $L_x \oplus L_y$ if $x \neq y$, and onto the 1-jets $L_x \otimes \mathcal{O}_{X,x} / \mathfrak{m}_x^2$ if $x = y$. This is separation of points and tangents. Base-point freeness follows similarly from $H^1(X, L(-x)) = 0$. $\square$

Exercise 47.7.2. Let L be a positive line bundle on X and E a holomorphic vector bundle of the form $M_1 \oplus \cdots \oplus M_r$ with M_i line bundles. Show that $H^q(X, E \otimes L^k) = 0$ for all $q > 0$ and $k \gg 0$. (The statement holds for all holomorphic vector bundles, and even for coherent sheaves; this is Serre's vanishing theorem, see Chapter 52.)

Solution. Cohomology commutes with finite direct sums. For each i , $M_i \otimes L^k \otimes K_X^{-1}$ is positive for $k \gg 0$ by Proposition 47.2.1(3), so $H^q(X, M_i \otimes L^k) = 0$ for $q > 0$ by Corollary 47.4.2. $\square$

Exercise 47.7.3. Let $Y \subseteq \mathbb{CP}^3$ be a smooth surface of degree d . Show that $q(Y) = 0$ and $p_g(Y) = h^0(\mathbb{CP}^3, \mathcal{O}(d-4))$, using Theorem 47.5.1, adjunction and Kodaira vanishing.

Solution. $q = h^{0,1}(Y) = h^{0,1}(\mathbb{CP}^3) = 0$ by the Lefschetz theorem for $H^1(\mathcal{O})$ ($p + q = 1 < 2$). By adjunction $K_Y = \mathcal{O}(d-4)|_Y$, so $p_g = h^0(Y, \mathcal{O}_Y(d-4))$. The sequence $0 \rightarrow \mathcal{O}(-4) \rightarrow \mathcal{O}(d-4) \rightarrow \mathcal{O}_Y(d-4) \rightarrow 0$ on $\mathbb{CP}^3$ and $H^1(\mathbb{CP}^3, \mathcal{O}(-4)) = 0$ (Example 47.4.3) give $H^0(\mathbb{CP}^3, \mathcal{O}(d-4)) \rightarrow H^0(Y, \mathcal{O}_Y(d-4))$ onto. It is injective because $H^0(\mathbb{CP}^3, \mathcal{O}(-4)) = 0$. For $d = 4$ this gives $p_g = 1$. $\square$

Exercise 47.7.4. Show that a compact complex manifold with $H^2(X; \mathbb{Q}) = 0$ and $\dim X \geq 1$ is never projective, and deduce again that the Hopf surface is not projective.

Solution. A projective manifold has an integral Kähler class $[\omega]$, and $[\omega] \neq 0$ (Proposition 44.4.4). The Hopf surface has $H^2 = 0$ (Counterexample 44.4.8). $\square$

Exercise 47.7.5. Show that products of projective manifolds and blow-ups of projective manifolds at finitely many points are projective.

Solution. If ω_1, ω_2 are Hodge metrics on X_1, X_2 , then $\text{pr}_1^* \omega_1 + \text{pr}_2^* \omega_2$ is a Hodge metric on $X_1 \times X_2$, so Corollary 47.6.6 applies. For a blow-up at one or two points, Lemma 47.6.4 with $M = \mathcal{O}_X$ and $l = 1$ shows that $\sigma^* L^k \otimes \mathcal{O}(-E)$ is positive for $k \gg 0$ when L is positive. The same proof works for any finite number of points, with pairwise disjoint balls. $\square$

Exercise 47.7.6. Let X be a compact Riemann surface and L a line bundle with $\deg L < 0$. Deduce $H^0(X, L) = 0$ from Theorem 47.4.1, and show directly from Theorem 47.3.4 that a holomorphic section s of a Hermitian line bundle whose curvature satisfies $iF < 0$ everywhere vanishes.

Solution. L^{-1} is positive (Proposition 47.2.5), and the second statement of Theorem 47.4.1 with $p = q = 0 < 1 = n$ gives $H^0(X, L) = 0$. Directly: on sections ($p = q = 0$), $\Delta s = 0$, so $[iF, \Delta]s = -\Delta(iF s) = -f s$, where $iF = f \omega$ with $f < 0$ (for $n = 1$, $\Delta \omega = 1$). Since $\Delta'' s = \bar{\partial}^* \bar{\partial} s = 0$ for holomorphic s , $0 = \|D' s\|^2 + \|D'' s\|^2 - \int_X f |s|^2 \omega$, and all terms are non-negative, so $s = 0$. $\square$

Chapter 48

Calabi–Yau Manifolds

48.1 Introduction

A compact Kähler manifold with vanishing first Chern class carries, in every Kähler class, a unique Ricci-flat Kähler metric. This is Yau’s solution of the Calabi conjecture (Theorem 48.3.1). Such manifolds, and in particular those with trivial canonical bundle, are called Calabi–Yau manifolds. They are among the richest examples in Hodge theory. Their Hodge numbers are constrained but far from trivial, their deformations are unobstructed and governed by $H^{n-1,1}$, and they are the setting of mirror symmetry. Elliptic curves, K3 surfaces and the quintic threefold are the basic examples. We compute the Hodge numbers of the quintic threefold completely (Proposition 48.4.4). References: [GHJ03], [Joy00, Chapter 6], [Yau78], [Bea83].

Prerequisites. Chapter 47 (Vanishing and Embedding Theorems), Chapter 46 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 37 (Characteristic Classes).

Throughout, X is a compact Kähler manifold of dimension n with Kähler form $\omega = \frac{i}{2} \sum h_{j\bar{k}} dz^j \wedge d\bar{z}^k$ in local holomorphic coordinates (Proposition 44.2.2). We write $c_1(X) = c_1(T^{1,0}X) = -c_1(K_X)$.

48.2 The Ricci form

Definition 48.2.1. The *Ricci form* of the Kähler metric ω is the real $(1, 1)$ -form given in local holomorphic coordinates by

$$\text{Ric}(\omega) = -i\partial\bar{\partial} \log \det(h_{j\bar{k}}).$$

The metric is *Ricci-flat* if $\text{Ric}(\omega) = 0$ and *Kähler–Einstein* if $\text{Ric}(\omega) = \lambda\omega$ for a constant λ .

Proposition 48.2.2. $\text{Ric}(\omega)$ is a well-defined closed real $(1, 1)$ -form, equal to 2π times the curvature form $\frac{i}{2\pi}F$ of the anticanonical bundle $K_X^{-1} = \Lambda^n T^{1,0}X$ with the metric induced by ω . Hence

$$\left[\frac{1}{2\pi} \text{Ric}(\omega) \right] = c_1(X) \in H^2(X; \mathbb{R}).$$

Proof. In a holomorphic chart, $e = \partial_1 \wedge \cdots \wedge \partial_n$ is a holomorphic frame of K_X^{-1} whose norm for the induced metric is $|e|^2 = \det(g(\partial_j, \bar{\partial}_k)) = 2^{-n} \det(h_{j\bar{k}})$. So the local weight is $\varphi = -\log \det(h_{j\bar{k}}) + n \log 2$, and by Theorem 42.4.1 the curvature is $F = \partial\bar{\partial}\varphi = -\partial\bar{\partial} \log \det(h_{j\bar{k}})$, with $c_1(K_X^{-1}) = [\frac{i}{2\pi}F] = [\frac{1}{2\pi} \text{Ric}(\omega)]$. This also shows that $\text{Ric}(\omega)$ is independent of the chart: under a change of coordinates, $\det(h_{j\bar{k}})$ is multiplied by $|J|^2$ with J the holomorphic Jacobian determinant, and $\partial\bar{\partial} \log |J|^2 = 0$. It is real and closed as a curvature form of a line bundle. $\square$

Remark 48.2.3. $\text{Ric}(\omega)(U, V) = \text{Ric}_g(JU, V)$, where Ric_g is the Ricci tensor of the underlying Riemannian metric [Huy05, Chapter 4]. So Ricci-flat Kähler metrics are Ricci-flat Riemannian metrics, and Proposition 48.2.2 shows that $c_1(X)_{\mathbb{R}} = 0$ is a necessary condition for a Kähler metric to be Ricci-flat.

48.3 The Calabi conjecture

Theorem 48.3.1 (Calabi–Yau). *Let X be a compact Kähler manifold with Kähler form ω , and let ρ be a closed real $(1, 1)$ -form with $[\rho] = 2\pi c_1(X)$. Then there is a unique Kähler form ω' with $[\omega'] = [\omega]$ and $\text{Ric}(\omega') = \rho$. In particular, if $c_1(X)_{\mathbb{R}} = 0$, every Kähler class contains a unique Ricci-flat Kähler form.*

Uniqueness is due to Calabi and existence to Yau [Yau78]. Proposition 48.3.2 reduces the theorem to a nonlinear elliptic equation and proves uniqueness; existence is proved by the continuity method with a priori estimates up to order 2 and 3 [Joy00, Chapter 5].

Proposition 48.3.2. *In the situation of Theorem 48.3.1, there is a smooth real function F with $\rho = \text{Ric}(\omega) - i\partial\bar{\partial}F$ and $\int_X (e^F - 1)\omega^n = 0$. A Kähler form $\omega' = \omega + i\partial\bar{\partial}\varphi$ in the class of ω satisfies $\text{Ric}(\omega') = \rho$ if and only if*

$$(\omega + i\partial\bar{\partial}\varphi)^n = e^F \omega^n$$

(the complex Monge–Ampère equation). Solutions of this equation are unique up to additive constants, so ω' is unique.

Proof. ρ and $\text{Ric}(\omega)$ are cohomologous closed real $(1, 1)$ -forms, so $\rho = \text{Ric}(\omega) - i\partial\bar{\partial}F$ by Corollary 45.6.4, and adding a constant to F achieves the normalisation. Every Kähler form in $[\omega]$ is $\omega + i\partial\bar{\partial}\varphi$ for a real φ , by the same corollary. The ratio of volume forms $\omega'^n/\omega^n = \det(h'_{j\bar{k}})/\det(h_{j\bar{k}})$ is a positive function, so $\text{Ric}(\omega') - \text{Ric}(\omega) = -i\partial\bar{\partial}\log(\omega'^n/\omega^n)$. Hence $\text{Ric}(\omega') = \rho$ if and only if $i\partial\bar{\partial}(\log(\omega'^n/\omega^n) - F) = 0$, if and only if $\log(\omega'^n/\omega^n) = F + c$ with c constant (a real function with $\partial\bar{\partial}u = 0$ on a compact manifold is pluriharmonic, hence harmonic for the Kähler Laplacian, hence constant). Integrating, $\int \omega'^n = \int \omega^n$ because the forms are cohomologous, and the normalisation of F forces $c = 0$.

For uniqueness, let $\omega_1 = \omega + i\partial\bar{\partial}\varphi_1$ and $\omega_2 = \omega + i\partial\bar{\partial}\varphi_2$ with $\omega_1^n = \omega_2^n$, and let $\psi = \varphi_1 - \varphi_2$. Then

$$0 = \omega_1^n - \omega_2^n = i\partial\bar{\partial}\psi \wedge T, \quad T = \sum_{k=0}^{n-1} \omega_1^k \wedge \omega_2^{n-1-k}.$$

Multiply by ψ and integrate. Since T is closed, Stokes' theorem gives $0 = \int_X \psi i\partial\bar{\partial}\psi \wedge T = -\int_X i\partial\psi \wedge \bar{\partial}\psi \wedge T$. At each point, ω_1 and ω_2 are positive, so T is a sum of products of positive $(1, 1)$ -forms, and $i\partial\psi \wedge \bar{\partial}\psi \wedge T$ is a non-negative multiple of the volume form, positive where $\partial\psi \neq 0$ (diagonalise ω_1 and ω_2 simultaneously at the point). Hence $\partial\psi = 0$, so $d\psi = 0$ as ψ is real, and ψ is constant. $\square$

48.4 Calabi–Yau manifolds and their Hodge numbers

Definition 48.4.1. A Calabi–Yau manifold is a compact Kähler manifold X with trivial canonical bundle $K_X \cong \mathcal{O}_X$. Equivalently, there is a nowhere vanishing holomorphic n -form Ω , the holomorphic volume form. It is strict if moreover $h^{p,0}(X) = 0$ for $0 < p < n$.

Proposition 48.4.2. *Let X be a Calabi–Yau manifold of dimension n .*

1. $c_1(X) = 0$, so X carries Ricci-flat Kähler metrics (Theorem 48.3.1). Moreover $h^{0,q} = h^{n,q}$ for all q ; in particular $h^{n,0} = h^{0,n} = 1$.
2. Contraction with Ω is an isomorphism of holomorphic bundles $T^{1,0}X \rightarrow \Omega_X^{n-1}$, $v \mapsto \iota_v \Omega$, and more generally $\Lambda^p T^{1,0}X \cong \Omega_X^{n-p}$. Hence $H^1(X, T_X) \cong H^{n-1,1}(X)$.

Proof. (1) $c_1(X) = -c_1(K_X) = 0$, and $h^{n,q} = h^q(X, K_X) = h^q(X, \mathcal{O}_X) = h^{0,q}$ (Theorem 45.5.2(4)). For $q = 0$ this is 1, and $h^{0,n} = h^{n,0}$ by Hodge symmetry. (2) At a point choose coordinates with $\Omega = f dz^1 \wedge \cdots \wedge dz^n$, $f \neq 0$. Then $\iota_{\partial_j} \Omega = (-1)^{j-1} f dz^1 \wedge \cdots \widehat{dz^j} \cdots \wedge dz^n$, and these form a basis of Ω^{n-1} , so contraction is a holomorphic bundle isomorphism; the same computation with $\iota_{\partial_{j_1}} \cdots \iota_{\partial_{j_p}}$ gives $\Lambda^p T^{1,0}X \cong \Omega_X^{n-p}$. By Dolbeault's theorem, $H^1(X, T_X) \cong H^1(X, \Omega^{n-1}) \cong H^{n-1,1}(X)$. $\square$

Example 48.4.3. 1. Complex tori $\mathbb{C}^n/\Lambda$, with $\Omega = dz^1 \wedge \cdots \wedge dz^n$ and the flat metric, which is Ricci-flat. They are not strict for $n \geq 2$.

2. Elliptic curves are the Calabi–Yau manifolds of dimension 1, since K is trivial exactly in genus 1.
3. A K3 surface is a strict Calabi–Yau surface, that is, $K_X \cong \mathcal{O}$ and $h^{1,0} = 0$. Smooth quartic surfaces in $\mathbb{CP}^3$ are K3 surfaces (Example 45.7.4).
4. A smooth hypersurface of degree $n + 2$ in $\mathbb{CP}^{n+1}$ has trivial canonical bundle by adjunction, $K = \mathcal{O}(n + 2 - n - 2) = \mathcal{O}$ (Proposition 42.5.1), and for $n \geq 2$ it is strict by the Lefschetz hyperplane theorem (Theorem 47.5.1), since $h^{p,0}(X) = h^{p,0}(\mathbb{CP}^{n+1}) = 0$ for $0 < p < n$.
5. Products of Calabi–Yau manifolds are Calabi–Yau, with $\Omega = \text{pr}_1^* \Omega_1 \wedge \text{pr}_2^* \Omega_2$; they are not strict.

Proposition 48.4.4 (The quintic threefold). *Let $X \subseteq \mathbb{CP}^4$ be a smooth quintic hypersurface. Then X is a strict Calabi–Yau threefold with Hodge diamond determined by*

$$h^{0,0} = h^{3,0} = 1, \quad h^{1,0} = h^{2,0} = 0, \quad h^{1,1} = 1, \quad h^{2,1} = 101,$$

and Euler characteristic $\chi(X) = -200$.

Proof. K_X is trivial and $h^{1,0} = h^{2,0} = 0$ by Example 48.4.3(4), with $n = 3$. By the Lefschetz hyperplane theorem, $b_1(X) = 0$ and $b_2(X) = b_2(\mathbb{CP}^4) = 1$, so $h^{1,1} = b_2 - 2h^{2,0} = 1$. By Poincaré duality $b_4 = 1$ and $b_5 = 0$, and by the Hodge decomposition $b_3 = h^{3,0} + h^{2,1} + h^{1,2} + h^{0,3} = 2 + 2h^{2,1}$. So $\chi(X) = 1 + 1 - b_3 + 1 + 1 = 2 - 2h^{2,1}$.

We compute $\chi(X) = \int_X c_3(TX)$ (Theorem 37.5.1). Let h be the hyperplane class. The tangent bundle of $\mathbb{CP}^4$ has total Chern class $(1 + h)^5$ (Example 37.2.5). The normal bundle of X is $\mathcal{O}(X)|_X = \mathcal{O}(5)|_X$ (Proposition 42.5.1), with $c = 1 + 5h$. From $0 \rightarrow TX \rightarrow T\mathbb{CP}^4|_X \rightarrow N \rightarrow 0$ and the Whitney formula,

$$c(TX) = \frac{(1 + h)^5}{1 + 5h} = (1 + 5h + 10h^2 + 10h^3)(1 - 5h + 25h^2 - 125h^3) = 1 + 10h^2 - 40h^3$$

restricted to X , modulo h^4 . So $c_1(X) = 0$, as it must be, and $\chi(X) = -40 \int_X h^3$. Finally $\int_X h^3 = 5$: the class of X is Poincaré dual to $c_1(\mathcal{O}(X)) = 5h$ [GH78, Chapter 1, Section 1], and $\int_{\mathbb{CP}^4} h^4 = 1$. Hence $\chi(X) = -200$ and $h^{2,1} = 101$. $\square$

Remark 48.4.5. The number 101 is the number of moduli of quintic threefolds: the quintic polynomials in five variables form a space of dimension $\binom{9}{4} = 126$, and removing scalars and the 24-dimensional group PGL_5 leaves $126 - 1 - 24 = 101$ parameters. This matches $h^1(X, T_X) = h^{2,1}$ (Proposition 48.4.2), which is the dimension of the space of first-order deformations of the complex structure. By the Bogomolov–Tian–Todorov theorem the deformations of a Calabi–Yau manifold are unobstructed, so its Kuranishi space is smooth of dimension $h^{n-1,1}$ [Tia87], [Tod89].

Proposition 48.4.6 (Complete intersection Calabi–Yau threefolds). *Let $X = V(f_1, \dots, f_r) \subseteq \mathbb{CP}^{3+r}$ with f_i homogeneous of degree $d_i \geq 2$ and $\sum_i d_i = 4 + r$, and assume that every partial intersection $V(f_1, \dots, f_j)$, $1 \leq j \leq r$, is smooth of codimension j . Then X is a strict Calabi–Yau threefold with $h^{1,1} = 1$, and*

$(d_1, \dots, d_r)$	(5)	(2, 4)	(3, 3)	(2, 2, 3)	(2, 2, 2, 2)
$c_2(X)$	$10h^2$	$7h^2$	$6h^2$	$5h^2$	$4h^2$
$c_3(X)$	$-40h^3$	$-22h^3$	$-16h^3$	$-12h^3$	$-8h^3$
$\int_X h^3 = \prod_i d_i$	5	8	9	12	16
$\chi(X)$	-200	-176	-144	-144	-128
$h^{2,1}(X)$	101	89	73	73	65

These five are all the possibilities with every $d_i \geq 2$.

Proof. Write $X_j = V(f_1, \dots, f_j)$, so that $X_j \subseteq X_{j-1}$ is a smooth hypersurface cut out by a section of $\mathcal{O}(d_j)$. By adjunction (Proposition 42.5.1) applied r times, $K_X = \mathcal{O}(\sum_i d_i - 4 - r)|_X = \mathcal{O}_X$. By the Lefschetz hyperplane theorem (Theorem 47.5.1) applied to each $X_j \subseteq X_{j-1}$, $H^k(X) \cong H^k(\mathbb{CP}^{3+r})$ for $k < 3$ (with $\mathbb{Q}$ -coefficients); so $b_1 = 0$ and $b_2 = 1 = 2h^{2,0} + h^{1,1}$, and since $h^{1,1} \geq 1$ (the Kähler class is nonzero, Proposition 44.4.4), $h^{1,0} = h^{2,0} = 0$ and $h^{1,1} = 1$, which makes X strict. As in the proof of Proposition 48.4.4, $b_4 = 1$, $b_3 = 2 + 2h^{2,1}$ and $\chi(X) = 2 - 2h^{2,1}$.

The normal bundle of X is $\bigoplus_i \mathcal{O}(d_i)|_X$ (adjunction at each step), so by the Whitney formula $c(TX) = (1 + h)^{4+r} \prod_i (1 + d_i h)^{-1}$ restricted to X , modulo h^4 . For instance for (2, 4),

$$(1 + 6h + 15h^2 + 20h^3)(1 - 6h + 28h^2 - 120h^3) = 1 + 7h^2 - 22h^3,$$

using $(1 + 2h)^{-1}(1 + 4h)^{-1} = (1 - 2h + 4h^2 - 8h^3)(1 - 4h + 16h^2 - 64h^3)$ modulo h^4 ; the other columns are the same computation. The degree $\int_X h^3 = \prod_i d_i$ holds because the class of X is $\prod_i d_i h^r$, each hypersurface having class $d_i h$ (as in the proof of Proposition 48.4.4). Then $\chi(X) = \int_X c_3(TX)$ (Theorem 37.5.1) gives the fourth row, and $h^{2,1} = 1 - \chi/2$ the last. Finally, $\sum_i (d_i - 1) = 4$ with all $d_i \geq 2$ has exactly the five solutions listed, up to order. $\square$

48.5 Structure and symmetries

Theorem 48.5.1 (Beauville–Bogomolov decomposition). *Let X be a compact Kähler manifold with $c_1(X)_{\mathbb{R}} = 0$. Then there is a finite étale cover $\tilde{X} \rightarrow X$ with*

$$\tilde{X} \cong T \times \prod_i Y_i \times \prod_j Z_j,$$

where T is a complex torus, the Y_i are simply connected strict Calabi–Yau manifolds with holonomy $\mathrm{SU}(\dim Y_i)$, and the Z_j are simply connected hyperkähler manifolds, with holonomy $\mathrm{Sp}(\dim Z_j/2)$ and $H^0(Z_j, \Omega^2)$ spanned by a symplectic form.

This is [Bea83]. Its proof uses the Ricci-flat metric of Theorem 48.3.1 and the de Rham and Berger classifications of holonomy groups. In dimension 2 the only simply connected factors are K3 surfaces, which are both strict Calabi–Yau and hyperkähler.

Remark 48.5.2. The first step is Bochner’s argument. For a Ricci-flat metric, harmonic 1-forms are parallel (Theorem 40.3.4), and more generally every holomorphic p -form is parallel. So the holomorphic forms are determined by the holonomy representation, and $h^{p,0}$ is the dimension of the space of holonomy-invariant forms of type $(p, 0)$ at a point [GHJ03]. For holonomy exactly $SU(n)$ the only invariants are 1 and the volume form, which gives $h^{p,0} = 0$ for $0 < p < n$.

Remark 48.5.3. *Mirror symmetry* predicts that strict Calabi–Yau threefolds come in pairs $(X, X^\vee)$ with $h^{1,1}(X) = h^{2,1}(X^\vee)$ and $h^{2,1}(X) = h^{1,1}(X^\vee)$, the complex geometry of X being equivalent to the symplectic geometry of $X^\vee$. The mirror of the quintic has $h^{1,1} = 101$ and $h^{2,1} = 1$, and the resulting prediction of the numbers of rational curves on the quintic [CdLOGP91] was a starting point of enumerative mirror symmetry.

Remark 48.5.4. For projective Calabi–Yau manifolds the Hodge conjecture is known in some degrees. On any projective threefold it holds in all degrees, by the Lefschetz $(1, 1)$ -theorem in degree 2 and hard Lefschetz in degree 4 (Chapter 65); the interesting group H^3 carries no Hodge classes, being of odd degree. For K3 surfaces it holds in all degrees (Chapter 68). For higher-dimensional hyperkähler varieties it is largely open.

48.6 Exercises

Exercise 48.6.1. Show that a smooth quartic surface $X \subseteq \mathbb{CP}^3$ has $\chi(X) = 24$ by computing $c_2(TX)$ as in Proposition 48.4.4, and deduce $h^{1,1} = 20$.

Solution. $c(TX) = (1 + h)^4 / (1 + 4h) = (1 + 4h + 6h^2)(1 - 4h + 16h^2) = 1 + 6h^2$ modulo h^3 , so $c_1 = 0$ and $\chi(X) = 6 \int_X h^2 = 6 \cdot 4 = 24$. With $b_1 = b_3 = 0$ and $b_0 = b_4 = 1$ we get $b_2 = 22$, and $h^{1,1} = b_2 - 2h^{2,0} = 22 - 2 = 20$. $\square$

Exercise 48.6.2. Show that the Fubini–Study metric on $\mathbb{CP}^n$ is Kähler–Einstein with $\text{Ric}(\omega_{FS}) = 2\pi(n + 1)\omega_{FS}$.

Solution. By $U(n + 1)$ -invariance (Proposition 44.4.1), $\text{Ric}(\omega_{FS})$ is an invariant closed real $(1, 1)$ -form. At the point $w = 0$ of the affine chart, $\omega_{FS} = \frac{i}{2\pi} \partial \bar{\partial} \log(1 + |w|^2)$ has $h_{j\bar{k}} = \frac{1}{\pi} \frac{(1 + |w|^2) \delta_{jk} - \bar{w}_j w_k}{(1 + |w|^2)^2}$, whose determinant is $\pi^{-n} (1 + |w|^2)^{-n-1}$. So $\text{Ric}(\omega_{FS}) = (n + 1) i \partial \bar{\partial} \log(1 + |w|^2) = 2\pi(n + 1)\omega_{FS}$ on the chart, hence everywhere. $\square$

Exercise 48.6.3. Show that a smooth hypersurface $X \subseteq \mathbb{CP}^{n+1}$ of degree d has $c_1(X) = (n + 2 - d)h|_X$, so that it is Fano ($c_1 > 0$), Calabi–Yau or of general type ($c_1 < 0$) according as $d < n + 2$, $d = n + 2$ or $d > n + 2$.

Solution. $c(TX) = (1 + h)^{n+2} / (1 + dh)$ has degree-one part $(n + 2)h - dh$. Alternatively, $K_X = \mathcal{O}(d - n - 2)|_X$ by adjunction. $\square$

Part VIII

Algebraic Geometry

Chapter 49

Algebraic Varieties

49.1 Introduction

This chapter develops classical algebraic geometry over an algebraically closed field: affine and projective varieties, their regular functions and morphisms, dimension, tangent spaces and smoothness, rational maps, and blow-ups. The Nullstellensatz of Chapter 12 translates geometry into commutative algebra. Closed subsets correspond to radical ideals, varieties to prime ideals, points to maximal ideals, and affine varieties to finitely generated domains. Chapter 50 generalises everything to schemes, which are needed for non-closed fields, for families, and for nilpotents. The classical language is enough for most statements about complex projective manifolds, and it is the one in which the Hodge conjecture is usually stated. References: [Har77, Chapter I], [GH78, Chapter 1].

Prerequisites. Chapter 12 (Commutative Algebra), Chapter 16 (General Topology) and Chapter 23 (Sheaves on Topological Spaces).

Throughout, k is an algebraically closed field, of any characteristic. Rings are commutative with 1, and $k[x_1, \dots, x_n]$ is the polynomial ring, with its usual addition and multiplication.

49.2 Affine algebraic sets and the Zariski topology

Definition 49.2.1. *Affine n -space* is $\mathbb{A}^n = k^n$. For a subset $T \subseteq A = k[x_1, \dots, x_n]$ put $Z(T) = \{a \in \mathbb{A}^n : f(a) = 0 \text{ for all } f \in T\}$, and for $Y \subseteq \mathbb{A}^n$ put $I(Y) = \{f \in A : f(a) = 0 \text{ for all } a \in Y\}$. The sets $Z(T)$ are the *algebraic sets*.

Proposition 49.2.2. 1. $Z(0) = \mathbb{A}^n$, $Z(1) = \emptyset$, $Z(T) = Z((T))$, and $Z(I) \cup Z(J) = Z(IJ) = Z(I \cap J)$ and $\bigcap_{\alpha} Z(I_{\alpha}) = Z(\sum_{\alpha} I_{\alpha})$ for ideals. So the algebraic sets are the closed sets of a topology on $\mathbb{A}^n$, the Zariski topology.

2. $I(Y)$ is a radical ideal, $Z(I(Y)) = \overline{Y}$, and $I(Z(J)) = \sqrt{J}$ for every ideal J .

3. $Y \mapsto I(Y)$ and $J \mapsto Z(J)$ are mutually inverse inclusion-reversing bijections between closed subsets of $\mathbb{A}^n$ and radical ideals of A . Under them, points correspond to maximal ideals.

Proof. (1) If $a \notin Z(I) \cup Z(J)$, pick $f \in I$, $g \in J$ with $f(a) \neq 0 \neq g(a)$; then $fg \in IJ$ does not vanish at a . So $Z(IJ) \subseteq Z(I) \cup Z(J) \subseteq Z(I \cap J) \subseteq Z(IJ)$. The rest is immediate. (2) If f^m vanishes on Y , so does f , so $I(Y)$ is radical. $Z(I(Y))$ is a closed set containing Y , and if $Z(J) \supseteq Y$ then $J \subseteq I(Y)$, so $Z(J) \supseteq Z(I(Y))$; hence $Z(I(Y)) = \overline{Y}$. The equality $I(Z(J)) = \sqrt{J}$ is the Nullstellensatz (Theorem 12.7.2(3)), the inclusion $\supseteq$ being clear. (3) follows from (2), and points correspond to maximal ideals by Theorem 12.7.2(1). $\square$

Definition 49.2.3. A topological space Y is *irreducible* if it is nonempty and not the union of two proper closed subsets; equivalently, every nonempty open subset is dense. It is *Noetherian* if every descending chain of closed subsets is eventually constant.

Proposition 49.2.4. 1. A closed set $Y \subseteq \mathbb{A}^n$ is irreducible if and only if $I(Y)$ is prime.

2. $\mathbb{A}^n$, and every subspace of it, is Noetherian.

3. Every Noetherian space Y is a finite union $Y = Y_1 \cup \cdots \cup Y_r$ of irreducible closed subsets, unique up to order if $Y_i \not\subseteq Y_j$ for $i \neq j$; the Y_i are the irreducible components.

Proof. (1) If $Y = Y_1 \cup Y_2$ with Y_i proper closed, pick $f_i \in I(Y_i) \setminus I(Y)$ by Proposition 49.2.2(3). Then $f_1 f_2 \in I(Y)$, so $I(Y)$ is not prime. Conversely, if $f g \in I(Y)$, then $Y = (Y \cap Z(f)) \cup (Y \cap Z(g))$, so $Y \subseteq Z(f)$ or $Y \subseteq Z(g)$ by irreducibility. (Also $Y \neq \emptyset$ if and only if $I(Y) \neq A$.) (2) A descending chain of closed sets gives an ascending chain of ideals, which stabilises because A is Noetherian (Theorem 12.5.3). For a subspace, its closed sets are traces of closed sets of $\mathbb{A}^n$. (3) If some closed subset is not such a finite union, choose a minimal one Y' , which exists by the Noetherian property. It is not irreducible, so $Y' = Y'_1 \cup Y'_2$ with smaller closed sets, which are finite unions, a contradiction. For uniqueness, if $Y_1 \cup \cdots \cup Y_r = Y'_1 \cup \cdots \cup Y'_s$ are irredundant, then $Y'_1 = \bigcup_i (Y'_1 \cap Y_i)$ forces $Y'_1 \subseteq Y_i$ for some i , and similarly $Y_i \subseteq Y'_j$, so $j = 1$ and $Y'_1 = Y_i$; induct. $\square$

Definition 49.2.5. An *affine variety* is an irreducible closed subset $Y \subseteq \mathbb{A}^n$ with the induced topology. Its *coordinate ring* is $A(Y) = k[x_1, \dots, x_n]/I(Y)$, a finitely generated k -algebra that is a domain. A *quasi-affine variety* is a nonempty open subset of an affine variety.

Example 49.2.6. 1. $\mathbb{A}^n$ is an affine variety, since $I(\mathbb{A}^n) = 0$ is prime (k is infinite, so a polynomial vanishing everywhere is 0).

2. If f is irreducible, then (f) is prime because $k[x_1, \dots, x_n]$ is a unique factorisation domain (Theorem 7.6.11), so the *hypersurface* $Z(f)$ is an affine variety with $I(Z(f)) = (f)$.

3. The closed subsets of $\mathbb{A}^1$ are $\mathbb{A}^1$ and the finite sets, since $k[x]$ is a principal ideal domain. So the Zariski topology is not Hausdorff: any two nonempty open sets meet.

4. The Zariski topology on $\mathbb{A}^2$ is not the product topology on $\mathbb{A}^1 \times \mathbb{A}^1$. The closed sets of the product topology are the finite unions of points, horizontal and vertical lines and $\mathbb{A}^2$, whereas the parabola $Z(y - x^2)$ is also Zariski closed.

5. The *twisted cubic* $Y = \{(t, t^2, t^3)\} = Z(y - x^2, z - x^3) \subseteq \mathbb{A}^3$ is an affine variety with $A(Y) \cong k[x, y, z]/(y - x^2, z - x^3) \cong k[t]$, a domain.

49.3 Regular functions and morphisms

Definition 49.3.1. Let Y be a quasi-affine variety in $\mathbb{A}^n$. A function $f: Y \rightarrow k$ is *regular at* $P \in Y$ if there are an open neighbourhood $U \ni P$ and polynomials g, h with h nowhere zero on U and $f = g/h$ on U . It is *regular* on an open set if it is regular at each of its points. The regular functions on open subsets form a sheaf of k -algebras $\mathcal{O}_Y$. The *local ring* at P is the stalk $\mathcal{O}_{Y,P}$, and the *function field* $K(Y)$ is the set of regular functions on nonempty open subsets modulo agreement on a nonempty open subset.

Lemma 49.3.2. A regular function $f: Y \rightarrow k = \mathbb{A}^1$ is continuous for the Zariski topologies. If two regular functions on an irreducible Y agree on a nonempty open subset, they are equal.

Proof. The closed subsets of $\mathbb{A}^1$ are finite sets, so it suffices that $f^{-1}(a)$ is closed. Locally $f = g/h$, and $f^{-1}(a) \cap U = Z(g - ah) \cap U$ is closed in U ; being closed is a local property. For the second statement, $\{f = f'\} = (f - f')^{-1}(0)$ is closed and contains a dense open subset. $\square$

Theorem 49.3.3. *Let Y be an affine variety with coordinate ring $A = A(Y)$. For $P \in Y$ let $\mathfrak{m}_P \subseteq A$ be the maximal ideal of functions vanishing at P . Then:*

1. $\mathcal{O}_{Y,P} \cong A_{\mathfrak{m}_P}$;
2. for $f \in A$, the basic open set $D(f) = Y \setminus Z(f)$ satisfies $\mathcal{O}(D(f)) \cong A_f$; in particular $\mathcal{O}(Y) \cong A$;
3. $K(Y) \cong \text{Frac}(A)$, and the basic open sets form a basis of the topology.

Proof. The basic open sets form a basis because $Y \setminus Z(J) = \bigcup_{f \in J} D(f)$. A polynomial vanishing on Y lies in $I(Y)$, so A is a ring of functions on Y . (1) Send $g/h \in A_{\mathfrak{m}_P}$, $h(P) \neq 0$, to the germ of the function g/h on $D(h)$. This is surjective by definition. It is injective: if g/h vanishes near P , then g vanishes on a nonempty open subset of Y , hence on Y by Lemma 49.3.2, so $g = 0$. So all local rings embed compatibly in $\text{Frac}(A)$, and a regular function on an open set U is determined by its germs, so $\mathcal{O}(U) \subseteq \bigcap_{P \in U} A_{\mathfrak{m}_P} \subseteq \text{Frac}(A)$, with equality by the definition of regularity.

(2) The points of $D(f)$ correspond to the maximal ideals of A not containing f , which are the maximal ideals of A_f (Proposition 12.3.2). So we must show $\bigcap_{\mathfrak{m} \not\ni f} A_{\mathfrak{m}} = A_f$ inside $\text{Frac}(A)$. The inclusion $\supseteq$ is clear. Let φ lie in the intersection, and let $J = \{a \in A_f : a\varphi \in A_f\}$, an ideal of A_f . For each maximal ideal $\mathfrak{n}$ of A_f we have $\varphi = g/h$ with $h \notin \mathfrak{n}$, so $h \in J \setminus \mathfrak{n}$. Thus J lies in no maximal ideal, $1 \in J$, and $\varphi \in A_f$. Taking $f = 1$ gives $\mathcal{O}(Y) = A$.

(3) Every nonempty open set contains a nonempty basic open set, and $\mathcal{O}(D(f)) = A_f$ with restriction maps the inclusions of subrings of $\text{Frac}(A)$. So $K(Y) = \bigcup_{f \neq 0} A_f = \text{Frac}(A)$. $\square$

Definition 49.3.4. Let X, Y be varieties, meaning for now quasi-affine varieties (Definition 49.4.4 below extends this to quasi-projective ones). A *morphism* $\varphi: X \rightarrow Y$ is a continuous map such that for every open $V \subseteq Y$ and every regular f on V , the function $f \circ \varphi$ is regular on $\varphi^{-1}(V)$. Varieties and morphisms form a category, and an isomorphism is a morphism with an inverse morphism. Equivalently, a morphism is a morphism of ringed spaces $(X, \mathcal{O}_X) \rightarrow (Y, \mathcal{O}_Y)$ whose map of sheaves is composition of functions (Definition 23.6.1).

Proposition 49.3.5. *Let X be a variety and $Y \subseteq \mathbb{A}^n$ an affine variety. Then $\varphi \mapsto \varphi^*$, $\varphi^*(f) = f \circ \varphi$, is a bijection*

$$\text{Hom}(X, Y) \rightarrow \text{Hom}_{k\text{-alg}}(A(Y), \mathcal{O}(X)).$$

Consequently $Y \mapsto A(Y)$ is an anti-equivalence between the category of affine varieties and that of finitely generated k -algebras that are domains.

Proof. Given a k -algebra map $\theta: A(Y) \rightarrow \mathcal{O}(X)$, let $\xi_i = \theta(\bar{x}_i)$ and define $\psi(P) = (\xi_1(P), \dots, \xi_n(P))$. For $f \in I(Y)$, $f(\psi(P)) = \theta(\bar{f})(P) = 0$, so ψ maps into Y , and $f \circ \psi = \theta(\bar{f})$ for all polynomials f . Then $\psi^{-1}(Y \cap Z(f)) = \theta(\bar{f})^{-1}(0)$ is closed (Lemma 49.3.2), so ψ is continuous. A function that is locally g/h pulls back to $(g \circ \psi)/(h \circ \psi)$, a quotient of regular functions, hence regular, so ψ is a morphism with $\psi^* = \theta$. Conversely, a morphism φ is determined by the functions $x_i \circ \varphi = \varphi^*(\bar{x}_i)$. For the anti-equivalence, use $\mathcal{O}(X) = A(X)$ for affine X (Theorem 49.3.3). Every finitely generated domain $B = k[x_1, \dots, x_n]/\mathfrak{p}$ is $A(Z(\mathfrak{p}))$, since $I(Z(\mathfrak{p})) = \sqrt{\mathfrak{p}} = \mathfrak{p}$. $\square$

Example 49.3.6. 1. $\mathbb{A}^1 \setminus \{0\}$ is isomorphic to the affine variety $Z(xy - 1) \subseteq \mathbb{A}^2$, with coordinate ring $k[x, x^{-1}]$. More generally $D(f) \subseteq Y$ is isomorphic to the affine variety with coordinate ring $A(Y)_f \cong A(Y)[t]/(tf - 1)$.

2. $\mathbb{A}^2 \setminus \{0\}$ is not isomorphic to an affine variety. Its regular functions are $\mathcal{O}(D(x)) \cap \mathcal{O}(D(y)) = k[x, y]_x \cap k[x, y]_y = k[x, y]$ by Theorem 49.3.3. So if it were affine, the inclusion into $\mathbb{A}^2$ would induce an isomorphism of coordinate rings and would therefore be an isomorphism by Proposition 49.3.5, which it is not.
3. The morphism $\mathbb{A}^1 \rightarrow Z(y^2 - x^3)$, $t \mapsto (t^2, t^3)$, is a bijective morphism that is not an isomorphism: it corresponds to the inclusion $k[t^2, t^3] \subseteq k[t]$, which is not surjective.

49.4 Projective varieties

Definition 49.4.1. *Projective n -space* is $\mathbb{P}^n = (k^{n+1} \setminus \{0\})/k^\times$, with points written $[a_0 : \cdots : a_n]$. Let $S = k[x_0, \dots, x_n]$ with its grading by degree, and $S_+ = (x_0, \dots, x_n)$. For a set T of homogeneous polynomials, $Z(T) \subseteq \mathbb{P}^n$ is the set of points at which all $f \in T$ vanish (this is well defined since $f(\lambda a) = \lambda^{\deg f} f(a)$). For $Y \subseteq \mathbb{P}^n$, $I(Y)$ is the ideal generated by the homogeneous polynomials vanishing on Y , a homogeneous ideal. The sets $Z(\mathfrak{a})$, $\mathfrak{a}$ a homogeneous ideal, are the closed sets of the Zariski topology on $\mathbb{P}^n$, as in Proposition 49.2.2(1).

Proposition 49.4.2 (Projective Nullstellensatz). *Let $\mathfrak{a} \subseteq S$ be a homogeneous ideal. Then $Z(\mathfrak{a}) = \emptyset$ if and only if $\sqrt{\mathfrak{a}} \supseteq S_+$, and otherwise $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$. Hence closed subsets of $\mathbb{P}^n$ correspond bijectively to homogeneous radical ideals not containing S_+ , and irreducible closed subsets to homogeneous primes not containing S_+ .*

Proof. Let $C \subseteq \mathbb{A}^{n+1}$ be the affine zero set of $\mathfrak{a}$. Since $\mathfrak{a}$ is homogeneous, C is a union of lines through 0 (or empty), and $Z(\mathfrak{a}) = (C \setminus \{0\})/k^\times$. So $Z(\mathfrak{a}) = \emptyset$ if and only if $C \subseteq \{0\}$, if and only if $\sqrt{\mathfrak{a}} \supseteq I(\{0\}) = S_+$ by the affine Nullstellensatz. Otherwise $C \neq \{0\}$ contains 0 if $\mathfrak{a} \neq S$, and a homogeneous f of positive degree vanishes on $Z(\mathfrak{a})$ if and only if it vanishes on C , if and only if $f \in \sqrt{\mathfrak{a}}$. The radical of a homogeneous ideal is homogeneous: if $f^m \in \mathfrak{a}$ and f_d is the top component of f , then f_d^m is the top component of f^m , so it lies in $\mathfrak{a}$ (Lemma 7.10.4), and $f_d \in \sqrt{\mathfrak{a}}$; now induct on the number of components of $f - f_d$. Hence $I(Z(\mathfrak{a})) = \sqrt{\mathfrak{a}}$.

For irreducibility, argue as in Proposition 49.2.4(1) with homogeneous f, g . This suffices because a homogeneous ideal $\mathfrak{p}$ such that $fg \in \mathfrak{p}$ implies $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$ for homogeneous f, g is prime. Indeed, let $fg \in \mathfrak{p}$ with $f, g \notin \mathfrak{p}$. Subtracting from f and g their homogeneous components of high degree that lie in $\mathfrak{p}$ does not change the hypotheses, so we may assume that the top components f_d and g_e are not in $\mathfrak{p}$. But $f_d g_e$ is the top component of fg , so it lies in $\mathfrak{p}$, a contradiction. $\square$

Proposition 49.4.3. *Let $U_i = \mathbb{P}^n \setminus Z(x_i)$. The map $\varphi_i: U_i \rightarrow \mathbb{A}^n$, $[a] \mapsto (a_j/a_i)_{j \neq i}$, is a homeomorphism for the Zariski topologies. Under it, closed subsets of U_i correspond to closed subsets of $\mathbb{A}^n$ through dehomogenisation $f \mapsto f(x_0, \dots, 1, \dots, x_n)$ and homogenisation.*

Proof. φ_i is bijective with inverse $(b_j) \mapsto [b_0 : \cdots : 1 : \cdots : b_n]$. For homogeneous f of degree d , $\varphi_i(Z(f) \cap U_i)$ is the zero set of the dehomogenisation $f(x_0, \dots, 1, \dots, x_n)$. Conversely, for $g \in k[y_j : j \neq i]$ of degree d , the homogenisation $x_i^d g(x_0/x_i, \dots)$ is a homogeneous polynomial whose zero set in U_i is $\varphi_i^{-1}(Z(g))$. So φ_i and its inverse are closed maps. $\square$

Definition 49.4.4. *A projective variety* is an irreducible closed subset $Y \subseteq \mathbb{P}^n$, and its *homogeneous coordinate ring* is $S(Y) = S/I(Y)$, a graded domain. A *quasi-projective variety* is a nonempty open subset of a projective variety. A function $f: Y \rightarrow k$ on a quasi-projective variety is *regular at P* if near P it is g/h with g, h homogeneous of the same degree and h nowhere zero. This gives the sheaf $\mathcal{O}_Y$, the local rings $\mathcal{O}_{Y,P}$ and the function field $K(Y)$ as before. From now on a *variety* is a quasi-projective variety, and morphisms are defined as in Definition 49.3.4.

Proposition 49.4.5. *Let $Y \subseteq \mathbb{P}^n$ be a quasi-projective variety. Then φ_i restricts to an isomorphism of $Y \cap U_i$ (if nonempty) onto a quasi-affine variety, and onto an affine variety if Y is projective, with coordinate ring the degree-zero part $S(Y)_{(x_i)}$ of $S(Y)_{x_i}$. In particular, every quasi-affine variety is quasi-projective (via $\mathbb{A}^n \cong U_0$), every variety has an open cover by affine varieties, and $K(Y) \cong K(Y \cap U_i)$.*

Proof. By Proposition 49.4.3, φ_i is a homeomorphism onto its image, which is locally closed, and irreducible since $Y \cap U_i$ is a nonempty open subset of an irreducible space. A quotient g/h of homogeneous polynomials of degree d equals the quotient of the dehomogenisations of g and h on U_i , and conversely a quotient of polynomials in the affine coordinates can be homogenised to a quotient of homogeneous polynomials of the same degree. So regular functions correspond. If Y is projective, $\varphi_i(Y \cap U_i)$ is closed, and dehomogenisation identifies its ideal with the degree-zero part of the localised ideal. A quasi-affine variety is covered by basic open sets $D(f)$ of its closure, which are affine (Example 49.3.6(1)). Finally, every nonempty open subset of Y meets U_i in a dense open subset, so the function fields agree. $\square$

Theorem 49.4.6. *If Y is a projective variety, then $\mathcal{O}(Y) = k$. Consequently every morphism from a projective variety to an affine variety is constant.*

Proof. Let $f \in \mathcal{O}(Y)$ and $L = \text{Frac}(S(Y))$. We view $K(Y)$ as the degree-zero part of L , by $g/h \mapsto g/h$. Let Σ be the set of i with $x_i \notin I(Y)$; the other x_i are zero in $S(Y)$. For $i \in \Sigma$, f restricted to $Y \cap U_i$ lies in $S(Y)_{(x_i)}$ by Proposition 49.4.5 and Theorem 49.3.3, so $x_i^{N_i} f \in S(Y)_{N_i}$ for some N_i , as elements of L . Let $N = \sum_{i \in \Sigma} N_i$. The space $S(Y)_N$ is spanned by the monomials of degree N in the x_i , $i \in \Sigma$, and each of them is divisible by $x_i^{N_i}$ for some $i \in \Sigma$. So $S(Y)_N \cdot f \subseteq S(Y)_N$, and by induction $S(Y)_N \cdot f^q \subseteq S(Y)_N$ for all $q \geq 1$. Choose $i \in \Sigma$. Then $x_i^N f^q \in S(Y)$ for all q , so $S(Y)[f] \subseteq x_i^{-N} S(Y)$, a finitely generated $S(Y)$ -module. As $S(Y)$ is Noetherian, $S(Y)[f]$ is a finitely generated $S(Y)$ -module, and f is integral over $S(Y)$ (Proposition 12.6.2(1)). Taking the degree-zero parts of the coefficients of an integral equation shows that f is integral over $S(Y)_0 = k$. As k is algebraically closed and f lies in a field, $f \in k$. For the second statement, compose with the coordinate functions of the target. $\square$

Example 49.4.7. 1. A projective hypersurface is $Z(f) \subseteq \mathbb{P}^n$ with f irreducible and homogeneous. Plane curves are the case $n = 2$.

2. The d -uple (Veronese) embedding $v_d: \mathbb{P}^n \rightarrow \mathbb{P}^N$, $N = \binom{n+d}{n} - 1$, sends a to the vector of all monomials of degree d in a . Its image is closed, cut out by the quadratic relations $z_M z_{M'} = z_{M''} z_{M'''}$ whenever $MM' = M''M'''$, and v_d is an isomorphism onto it [Har77, Chapter I, Exercises 2.12 and 3.4]. On the image, the inverse is given on $\{z_{x_i^d} \neq 0\}$ by $z \mapsto [z_{x_i^{d-1}x_0} : \cdots : z_{x_i^{d-1}x_n}]$. For $n = 1$, $d = 3$ the image is the twisted cubic in $\mathbb{P}^3$, the closure of the affine twisted cubic.
3. The Segre embedding $\mathbb{P}^m \times \mathbb{P}^n \rightarrow \mathbb{P}^{mn+m+n}$, $(a, b) \mapsto (a_i b_j)$, has closed image $Z(z_{ij} z_{kl} - z_{il} z_{kj})$. This image is used to give $\mathbb{P}^m \times \mathbb{P}^n$ the structure of a projective variety. For $m = n = 1$ it is the smooth quadric surface $Z(z_{00} z_{11} - z_{01} z_{10})$ in $\mathbb{P}^3$.

Proposition 49.4.8. *Let $X \subseteq \mathbb{A}^m$ and $Y \subseteq \mathbb{A}^n$ be affine varieties. Then $X \times Y \subseteq \mathbb{A}^{m+n}$ is an affine variety with coordinate ring $A(X) \otimes_k A(Y)$, and it is the product of X and Y in the category of varieties.*

Proof. $X \times Y = Z(I(X) + I(Y))$ is closed. It is irreducible: if $X \times Y = Z_1 \cup Z_2$ with Z_i closed, each fibre $\{x\} \times Y \cong Y$ is irreducible, so it lies in Z_1 or in Z_2 . The set $X_i = \{x : \{x\} \times Y \subseteq Z_i\} = \bigcap_{y \in Y} \{x : (x, y) \in Z_i\}$ is closed, and $X = X_1 \cup X_2$, so $X = X_1$ or $X = X_2$. The natural map $A(X) \otimes_k A(Y) \rightarrow \{\text{functions on } X \times Y\}$ is injective: write an element as $\sum_i a_i \otimes b_i$ with the b_i linearly independent over k ; if it vanishes, then for each x the function $\sum_i a_i(x) b_i$ vanishes

on Y , so all $a_i(x) = 0$, so all $a_i = 0$. Its image is the ring of polynomial functions on $X \times Y$, which is $A(X \times Y)$. The universal property follows from Proposition 49.3.5 and the universal property of $\otimes_k$ for k -algebras (Proposition 7.10.2). $\square$

49.5 Dimension

Definition 49.5.1. The *dimension* of a topological space is the supremum of the lengths r of chains $Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_r$ of irreducible closed subsets. For an irreducible closed subset $Z \subseteq Y$, the *codimension* of Z in Y is the supremum of the lengths of such chains with $Z_0 = Z$. Varieties of dimension 1 and 2 are *curves* and *surfaces*.

Lemma 49.5.2. Let A be a finitely generated k -algebra that is a domain, of dimension d . Then every maximal chain of primes of A has length d . Consequently $\text{ht } \mathfrak{p} + \dim A/\mathfrak{p} = d$ for every prime $\mathfrak{p}$, and every maximal ideal has height d .

Proof. Induction on $d = \text{trdeg}_k \text{Frac}(A)$ (Theorem 12.9.3). If $d = 0$, A is a domain algebraic over k , hence a field (Proposition 12.6.2(3)), and the only chain is 0. Let $d > 0$ and let $0 = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m$ be a maximal chain. Then $m \geq 1$, and no prime lies strictly between 0 and $\mathfrak{p}_1$. By Noether normalisation (Theorem 12.6.5), A is finite over a polynomial ring $B = k[y_1, \dots, y_d]$. Let $\mathfrak{q} = \mathfrak{p}_1 \cap B$.

The prime $\mathfrak{q}$ is nonzero. Indeed, take $a \in \mathfrak{p}_1 \setminus 0$ and an equation $a^r + b_1 a^{r-1} + \cdots + b_r = 0$ over B with r minimal. Then $b_r \neq 0$, since otherwise we could divide by a , and $b_r \in aA \cap B \subseteq \mathfrak{q}$. If there were a prime $0 \subsetneq \mathfrak{q}' \subsetneq \mathfrak{q}$, then going down for the integral extension $B \subseteq A$ of domains with B normal (Remark 12.6.4; B is a unique factorisation domain, hence normal by Proposition 12.6.2(4)) would give a prime $\mathfrak{p}' \subsetneq \mathfrak{p}_1$ lying over $\mathfrak{q}'$, so $\mathfrak{p}' \neq 0$, which is a contradiction. So $\mathfrak{q}$ is a minimal nonzero prime of the unique factorisation domain B . It contains an irreducible factor f of any of its nonzero elements, and (f) is a nonzero prime, so $\mathfrak{q} = (f)$. If f involves y_i , the images of the other y_j in $B/(f)$ are algebraically independent, since a relation $g(y_j : j \neq i) \in (f)$ is impossible for $g \neq 0$, and y_i is algebraic over them. So $\text{trdeg} \text{Frac}(B/\mathfrak{q}) = d - 1$. As $A/\mathfrak{p}_1$ is integral over $B/\mathfrak{q}$, it also has transcendence degree $d - 1$, and by induction the maximal chain $\mathfrak{p}_1/\mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m/\mathfrak{p}_1$ of $A/\mathfrak{p}_1$ has length $d - 1$. Hence $m = d$.

For a prime $\mathfrak{p}$, a chain of length $\text{ht } \mathfrak{p}$ ending at $\mathfrak{p}$ and a chain of length $\dim A/\mathfrak{p}$ starting at $\mathfrak{p}$ cannot be refined, since they have maximal length. Their concatenation is therefore a maximal chain of A , of length d . $\square$

Theorem 49.5.3. Let Y be a variety.

1. $\dim Y = \text{trdeg}_k K(Y)$; if Y is affine, $\dim Y = \dim A(Y)$.
2. $\dim U = \dim Y$ for every nonempty open $U \subseteq Y$, and $\dim \mathcal{O}_{Y,P} = \dim Y$ for every $P \in Y$.
3. If $Z \subsetneq Y$ is closed and irreducible, then $\dim Z < \dim Y$, and $\dim Z + \text{codim}(Z, Y) = \dim Y$.

Proof. First let Y be affine with $A = A(Y)$. Irreducible closed subsets of Y correspond to primes of A , reversing inclusions (Propositions 49.2.2 and 49.2.4), so $\dim Y = \dim A = \text{trdeg} \text{Frac}(A) = \text{trdeg} K(Y)$ by Theorem 12.9.3 and Theorem 49.3.3. Also $\dim \mathcal{O}_{Y,P} = \dim A_{\mathfrak{m}_P} = \text{ht } \mathfrak{m}_P = \dim Y$ by Lemma 49.5.2. For U open, $Z \mapsto \bar{Z}$ maps chains of irreducible closed subsets of U injectively to chains in Y , so $\dim U \leq \dim Y$. Conversely, let $P \in U$ and take a chain of primes of length $\dim Y$ contained in $\mathfrak{m}_P$. It gives a chain of irreducible closed subsets of Y all containing P , and their intersections with U form a chain of the same length in U , since a nonempty open subset of an irreducible space is dense.

Statement (3) for affine Y is Lemma 49.5.2, since $\text{codim}(Z, Y) = \text{ht } I(Z)/I(Y)$ and $\text{ht} \geq 1$ for a nonzero prime.

In general, cover Y by affine open subsets Y_i (Proposition 49.4.5); they have $K(Y_i) = K(Y)$, so $\dim Y_i = \text{trdeg } K(Y)$. Given a chain $Z_0 \subsetneq \cdots \subsetneq Z_r$ in Y , choose Y_i meeting Z_0 ; the sets $Z_j \cap Y_i$ form a chain in Y_i (their closures are the Z_j). So $\dim Y = \max_i \dim Y_i = \text{trdeg } K(Y)$, and the other statements reduce to the affine case in the same way: a proper closed irreducible Z meets some Y_i in a proper subset, since otherwise Z would contain the dense set Y_i . $\square$

Theorem 49.5.4 (Krull's theorem for varieties). *Let Y be an affine variety of dimension r and $f \in A(Y)$ neither zero nor a unit. Then every irreducible component of $Y \cap Z(f)$ has dimension $r - 1$. A variety $Y \subseteq \mathbb{A}^n$ has dimension $n - 1$ if and only if $Y = Z(f)$ for an irreducible polynomial f .*

Proof. The components of $Z(f)$ correspond to the minimal primes over (f) in $A(Y)$. They have height ≤ 1 by Krull's principal ideal theorem (Theorem 12.9.5), hence height 1 as they are nonzero, hence dimension $r - 1$ by Lemma 49.5.2. For the second statement, $Z(f)$ has dimension $n - 1$ by the first. Conversely, if $\dim Y = n - 1$, then $I(Y)$ has height 1. It contains an irreducible factor f of any of its nonzero elements, and $0 \subsetneq (f) \subseteq I(Y)$ with (f) prime, so $I(Y) = (f)$. $\square$

Theorem 49.5.5 (Projective dimension theorem). *Let $Y \subseteq \mathbb{P}^n$ be a projective variety of dimension $r \geq 1$ and f a homogeneous polynomial of positive degree with $f \notin I(Y)$. Then $Y \cap Z(f)$ is nonempty and all its irreducible components have dimension $r - 1$.*

Proof. Let $C(Y) \subseteq \mathbb{A}^{n+1}$ be the affine cone, the affine variety with coordinate ring $S(Y)$. The fraction field of $S(Y)$ is $K(Y)(x_i)$ for any $x_i \notin I(Y)$, with x_i transcendental over $K(Y)$ (elements of $K(Y)$ have degree zero), so $\dim C(Y) = r + 1$ by Theorem 49.5.3. The same holds for every projective variety, in particular for the components of $Y \cap Z(f)$, whose cones are the components of $C(Y) \cap Z(f)$ other than $\{0\}$. The set $C(Y) \cap Z(f)$ contains 0, and f is not a unit in $S(Y)$, so by Theorem 49.5.4 all its components have dimension $r \geq 1$. In particular they are not $\{0\}$, so they are cones over components of $Y \cap Z(f)$ of dimension $r - 1$. $\square$

Corollary 49.5.6. *Any $r \leq \dim Y$ hypersurfaces in $\mathbb{P}^n$ meet a projective variety Y . In particular two curves in $\mathbb{P}^2$ always meet, and $\mathbb{P}^2$ is not isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, which contains disjoint curves.*

Proof. Apply Theorem 49.5.5 repeatedly to a component of dimension $\geq \dim Y - j$ after j steps. (If some f vanishes on the current component, the dimension does not drop.) The curves $\mathbb{P}^1 \times \{0\}$ and $\mathbb{P}^1 \times \{\infty\}$ are disjoint. $\square$

49.6 Tangent spaces and smoothness

Definition 49.6.1. Let P be a point of a variety Y , with local ring $(\mathcal{O}_{Y,P}, \mathfrak{m}_P)$ and residue field k . The Zariski tangent space is $T_P Y = (\mathfrak{m}_P / \mathfrak{m}_P^2)^\vee$. The variety is *nonsingular* (or *smooth*) at P if $\mathcal{O}_{Y,P}$ is a regular local ring (Definition 12.11.1), that is, if $\dim_k T_P Y = \dim Y$, and *singular* at P otherwise. Y is nonsingular if it is nonsingular at every point.

Lemma 49.6.2. *For every Noetherian local ring $(R, \mathfrak{m}, k)$, $\dim R \leq \dim_k \mathfrak{m} / \mathfrak{m}^2$. Hence $\dim T_P Y \geq \dim Y$.*

Proof. By Nakayama's lemma (Corollary 12.4.3), $\mathfrak{m}$ is generated by $e = \dim \mathfrak{m} / \mathfrak{m}^2$ elements, and $\mathfrak{m}$ is minimal over them, so $\dim R = \text{ht } \mathfrak{m} \leq e$ by Theorem 12.9.5. The second statement uses $\dim \mathcal{O}_{Y,P} = \dim Y$. $\square$

Theorem 49.6.3 (Jacobian criterion). *Let $Y \subseteq \mathbb{A}^n$ be an affine variety of dimension r with $I(Y) = (f_1, \dots, f_t)$, and $P \in Y$. Then $\dim T_P Y = n - \text{rank } J(P)$, where $J = (\partial f_i / \partial x_j)$ is the Jacobian matrix. So Y is nonsingular at P if and only if $\text{rank } J(P) = n - r$.*

Proof. Let $M = (x_1 - P_1, \dots, x_n - P_n) \subseteq k[x]$. The linear map $\theta: M \rightarrow k^n$, $g \mapsto (\partial g / \partial x_j(P))_j$, is onto with kernel M^2 (expand g in powers of $x - P$), so $M/M^2 \cong k^n$. The maximal ideal $\mathfrak{m}$ of $A(Y)$ at P satisfies $\mathfrak{m}/\mathfrak{m}^2 \cong M/(M^2 + I(Y))$, and localising at $\mathfrak{m}$ does not change $\mathfrak{m}/\mathfrak{m}^2$. The image of $I(Y)$ in $M/M^2 \cong k^n$ is spanned by the $\theta(f_i)$, the rows of $J(P)$, because $\theta(g f_i) = g(P)\theta(f_i)$ as $f_i(P) = 0$. So $\dim \mathfrak{m}/\mathfrak{m}^2 = n - \text{rank } J(P)$. $\square$

Example 49.6.4. 1. The cusp $Z(y^2 - x^3)$ and the node $Z(y^2 - x^2(x + 1))$ (characteristic $\neq 2$) are singular exactly at the origin, where the Jacobian $(\partial f / \partial x, \partial f / \partial y)$ vanishes.

2. A hypersurface $Z(f) \subseteq \mathbb{A}^n$ is singular exactly at the common zeros of f and all $\partial f / \partial x_j$.

3. $\mathbb{A}^n, \mathbb{P}^n$ and the smooth quadric $Z(xw - yz) \subseteq \mathbb{P}^3$ are nonsingular.

49.7 Rational maps and birational equivalence

Lemma 49.7.1. *Two morphisms $\varphi, \psi: X \rightarrow Y$ of varieties that agree on a nonempty open subset are equal.*

Proof. Embed Y in some $\mathbb{P}^n$. The set $E = \{x : \varphi(x) = \psi(x)\}$ is closed. Indeed, near a point of X , φ and ψ map into standard open sets U_a and U_b , so on a neighbourhood V they are given by regular functions $\varphi = [f_0 : \dots : f_n]$ with $f_a = 1$ and $\psi = [g_0 : \dots : g_n]$ with $g_b = 1$. On V , E is the common zero set of the regular functions $f_i g_j - f_j g_i$. The closed set E contains a dense open subset of the irreducible X , so $E = X$. $\square$

Definition 49.7.2. A rational map $\varphi: X \dashrightarrow Y$ is an equivalence class of pairs (U, φ_U) , where $U \subseteq X$ is nonempty open and $\varphi_U: U \rightarrow Y$ is a morphism, two pairs being equivalent if the morphisms agree on the intersection (which is possible by Lemma 49.7.1). It is *dominant* if $\varphi_U(U)$ is dense. A *birational map* is a dominant rational map with a dominant rational inverse, and X, Y are *birational* if there is one. A variety birational to $\mathbb{P}^n$ is *rational*.

Theorem 49.7.3. *The functor $X \mapsto K(X)$ is an anti-equivalence between the category of varieties with dominant rational maps and the category of finitely generated field extensions of k . In particular X and Y are birational if and only if $K(X) \cong K(Y)$ over k , if and only if they have isomorphic nonempty open subsets.*

Proof. A dominant φ defined on U pulls back regular functions on nonempty open subsets of Y to regular functions on nonempty open subsets of U , since preimages of dense open sets under a dominant map are nonempty. This gives $\varphi^*: K(Y) \rightarrow K(X)$. Conversely, let $\theta: K(Y) \rightarrow K(X)$ be a k -embedding. Choose an affine open $V \subseteq Y$ with $A(V) = k[y_1, \dots, y_n]$. The elements $\theta(y_i)$ are regular on some nonempty open $U \subseteq X$, and θ restricts to $A(V) \rightarrow \mathcal{O}(U)$. By Proposition 49.3.5 this is φ^* for a morphism $\varphi: U \rightarrow V$. It is dominant: if its image lay in a proper closed subset $Z(g)$ with $0 \neq g \in A(V)$, then $\theta(g) = 0$, contradicting injectivity. These constructions are inverse to each other by Lemma 49.7.1 and Theorem 49.3.3. Every finitely generated extension $K = k(y_1, \dots, y_n)$ is $K(V)$ for the affine variety with coordinate ring $k[y_1, \dots, y_n] \subseteq K$ (Proposition 49.3.5).

If $\varphi: X \dashrightarrow Y$ and $\psi: Y \dashrightarrow X$ are inverse birational maps, defined on U and V , then $\psi \circ \varphi = \text{id}$ on $\varphi^{-1}(V)$ and $\varphi \circ \psi = \text{id}$ on $\psi^{-1}(U)$. So φ restricts to an isomorphism $\varphi^{-1}(\psi^{-1}(U)) \rightarrow \psi^{-1}(U)$. The converse is clear. $\square$

Proposition 49.7.4. *Every variety of dimension r is birational to a hypersurface in $\mathbb{A}^{r+1}$.*

Proof. The finitely generated extension $K(Y)/k$ of transcendence degree r is separably generated: there are $x_1, \dots, x_r$, algebraically independent, such that $K(Y)$ is a finite separable extension of $k(x_1, \dots, x_r)$. In characteristic 0 this holds for every transcendence basis. In characteristic p it holds because k is perfect [Mat86, Section 26]. By the primitive element theorem (Theorem 8.6.3), $K(Y) = k(x_1, \dots, x_r, y)$ with y separable algebraic over $k(x)$. Clearing denominators in the minimal polynomial of y gives an irreducible $f \in k[x_1, \dots, x_r, t]$ with $f(x, y) = 0$, and $K(Z(f)) \cong K(Y)$. Apply Theorem 49.7.3. $\square$

Theorem 49.7.5. *The singular points of a variety Y form a proper closed subset.*

Proof. Closedness is local, so let $Y \subseteq \mathbb{A}^n$ be affine of dimension r with $I(Y) = (f_1, \dots, f_t)$. By Lemma 49.6.2 and Theorem 49.6.3, $\text{rank } J(P) \leq n - r$ everywhere, and the singular set is the common zero set of the $(n - r)$ -minors of J , which is closed. It is proper: by Proposition 49.7.4 and Theorem 49.7.3, Y has an open subset isomorphic to an open subset of a hypersurface $Z(f) \subseteq \mathbb{A}^{r+1}$ with f irreducible, so it suffices to find a nonsingular point on $Z(f)$. The singular points are $Z(f, \partial f / \partial x_1, \dots)$ (Example 49.6.4). If they were all of $Z(f)$, then each $\partial f / \partial x_j$ would lie in $I(Z(f)) = (f)$, hence vanish, by degree. In characteristic 0 this makes f constant. In characteristic p it makes f a polynomial in $x_1^p, \dots, x_{r+1}^p$, hence a p -th power since k is perfect, contradicting irreducibility. So the nonsingular points of $Z(f)$ form a nonempty open set, which meets the open subset isomorphic to an open subset of Y , because $Z(f)$ is irreducible. Isomorphic open subsets have isomorphic local rings. $\square$

Example 49.7.6. 1. $\mathbb{P}^n$ and $\mathbb{A}^n$ are birational, and so are $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$: all three have function field $k(x_1, x_2)$ for $n = 2$. By Corollary 49.5.6, $\mathbb{P}^1 \times \mathbb{P}^1$ and $\mathbb{P}^2$ are not isomorphic.

2. The cuspidal cubic $Z(y^2 - x^3)$ is rational: $t = y/x$ gives $K = k(t)$. It is singular, so it is not isomorphic to $\mathbb{P}^1$ or $\mathbb{A}^1$.

3. The Cremona transformation $\mathbb{P}^2 \dashrightarrow \mathbb{P}^2$, $[x : y : z] \mapsto [yz : xz : xy]$, is birational and equal to its own inverse, but it is not defined at the three coordinate points.

4. A nonsingular plane cubic is not rational. This follows from the genus (Chapter 55): rational nonsingular projective curves are isomorphic to $\mathbb{P}^1$, which has genus 0, while plane cubics have genus 1.

49.8 Blow-ups

Definition 49.8.1. The *blow-up* of $\mathbb{A}^n$ at 0 is the closed subset $\text{Bl}_0 \mathbb{A}^n = \{(x, y) \in \mathbb{A}^n \times \mathbb{P}^{n-1} : x_i y_j = x_j y_i \text{ for all } i, j\}$ with the projection $\pi : \text{Bl}_0 \mathbb{A}^n \rightarrow \mathbb{A}^n$. For a closed subvariety $Y \subseteq \mathbb{A}^n$ through 0, the *blow-up* (or *strict transform*) $\tilde{Y}$ is the closure of $\pi^{-1}(Y \setminus \{0\})$ in $\text{Bl}_0 \mathbb{A}^n$.

Proposition 49.8.2. $\text{Bl}_0 \mathbb{A}^n$ is a nonsingular variety of dimension n . The map π is an isomorphism over $\mathbb{A}^n \setminus \{0\}$, and $E = \pi^{-1}(0) = \{0\} \times \mathbb{P}^{n-1}$ parametrises the lines through 0. For Y as above, $\tilde{Y} \rightarrow Y$ is birational, and $\tilde{Y} \cap E$ consists of limits of secant directions of Y at 0.

Proof. On $\{y_i \neq 0\}$ put $y_i = 1$. The equations become $x_j = x_i y_j$, so this open set is the graph of a morphism $\mathbb{A}^n \rightarrow \mathbb{A}^{n-1}$, which is isomorphic to $\mathbb{A}^n$ with coordinates $(x_i, y_j)_{j \neq i}$. These charts are irreducible of dimension n and nonsingular, and pairwise they meet in nonempty open sets, so $\text{Bl}_0 \mathbb{A}^n$ is irreducible, of dimension n and nonsingular. Over $x \neq 0$, the equations force $y = [x]$, so $x \mapsto (x, [x])$ is an inverse of π there. Over 0 the equations are empty, so $E = \{0\} \times \mathbb{P}^{n-1}$. For Y , the map $\tilde{Y} \rightarrow Y$ is an isomorphism over $Y \setminus \{0\}$, which is dense in $\tilde{Y}$ by definition, and a point $(0, [v]) \in \tilde{Y}$ is a limit of points $(x, [x])$ with $x \in Y \setminus \{0\}$. $\square$

Example 49.8.3. Let $\text{char } k \neq 2$. For the node $Y = Z(y^2 - x^2(x + 1))$, in the chart $y = xt$ the preimage is $Z(x^2(t^2 - x - 1))$, the union of $E = Z(x)$ (with multiplicity 2) and the strict transform $\tilde{Y} = Z(t^2 - x - 1)$. The latter is nonsingular and meets E at the two points $t = \pm 1$, the two tangent directions of the branches: blowing up separates the branches. For the cusp $Z(y^2 - x^3)$, the strict transform is $Z(t^2 - x)$, nonsingular and meeting E at one point. In both cases one blow-up *resolves* the singularity.

Remark 49.8.4. Blow-ups at points glue to blow-ups of any variety at a nonsingular point, and more generally along closed subvarieties. For $k = \mathbb{C}$ this is the construction of Definition 41.4.4. Hironaka proved that in characteristic 0 every variety is birational to a nonsingular projective variety, and in fact that singularities can be resolved by a sequence of blow-ups along nonsingular centres [Hir64]. In positive characteristic this is known only in low dimension.

49.9 Exercises

Exercise 49.9.1. Show that $\mathbb{A}^1$ minus finitely many points is affine, and determine its coordinate ring.

Solution. If the points are $a_1, \dots, a_m$, the set is $D(f)$ for $f = \prod_i (x - a_i)$, which is affine with coordinate ring $k[x]_f$ by Example 49.3.6(1). $\square$

Exercise 49.9.2. Let $\text{char } k \neq 2$. Show that every irreducible conic $Z(q) \subseteq \mathbb{P}^2$ (q a quadratic form) is isomorphic to $\mathbb{P}^1$.

Solution. Diagonalise q by a linear change of coordinates. It has rank 3, since a quadratic form of rank ≤ 2 over an algebraically closed field factors into linear forms. Over k it is then equivalent to $xz - y^2$. The morphism $\mathbb{P}^1 \rightarrow Z(xz - y^2)$, $[s : t] \mapsto [s^2 : st : t^2]$, is the Veronese map v_2 , an isomorphism onto its image (Example 49.4.7). Its inverse is given by $[x : y]$ on $\{x \neq 0\}$ and by $[y : z]$ on $\{z \neq 0\}$. $\square$

Exercise 49.9.3. Let $\text{char } k \neq 2$. Show that the cone $Z(x^2 + y^2 - z^2) \subseteq \mathbb{A}^3$ is singular only at the origin, and compute its tangent space there.

Solution. The Jacobian is $(2x, 2y, -2z)$, which vanishes only at 0, and $\dim = 2$, so the cone is nonsingular away from 0 (Theorem 49.6.3). At 0 the Jacobian is zero, so $T_0Y = k^3$ has dimension $3 > 2$. $\square$

Exercise 49.9.4. Show that n hypersurfaces in $\mathbb{P}^n$ have a common point, and that the Segre quadric $Z(xw - yz) \subseteq \mathbb{P}^3$ contains two families of lines such that two lines of the same family are disjoint and two lines of different families meet in one point.

Solution. The first statement is Corollary 49.5.6 with $Y = \mathbb{P}^n$. The quadric is the image of the Segre map $\mathbb{P}^1 \times \mathbb{P}^1 \rightarrow \mathbb{P}^3$, $([a_0 : a_1], [b_0 : b_1]) \mapsto [a_0b_0 : a_0b_1 : a_1b_0 : a_1b_1]$, which is a bijection onto it. The images of $\{a\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{b\}$ are lines. Two lines of the same family are images of disjoint sets, and $\{a\} \times \mathbb{P}^1$ meets $\mathbb{P}^1 \times \{b\}$ exactly in (a, b) . $\square$

Exercise 49.9.5. Show that the projective twisted cubic $C = v_3(\mathbb{P}^1) \subseteq \mathbb{P}^3$ is cut out by the three quadrics $xz - y^2$, $yw - z^2$, $xw - yz$, but not by any two of them.

Solution. The three quadrics say that the matrix $\begin{pmatrix} x & y & z \\ y & z & w \end{pmatrix}$ has rank ≤ 1 . Such a point with $x \neq 0$ is $[1 : t : t^2 : t^3]$ with $t = y/x$, and with $x = 0$ it is $[0 : 0 : 0 : 1]$, so the common zeros are exactly C . The quadrics $xz - y^2$ and $yw - z^2$ also vanish on the line $y = z = 0$, which is not contained in C . For the other pairs, $\{xz = y^2, xw = yz\}$ contains the line $x = y = 0$, and $\{yw = z^2, xw = yz\}$ contains the line $z = w = 0$. $\square$

Chapter 50

Schemes

50.1 Introduction

A scheme is a locally ringed space that is locally the spectrum of a ring. Schemes extend the varieties of Chapter 49 in three directions that Hodge theory needs. They work over arbitrary base rings, so that a variety defined over $\mathbb{Q}$, or a family over a base, is a single object; this is how the conjugates X^σ of the theory of absolute Hodge classes are defined (Chapter 71). They allow nilpotent functions, which carry infinitesimal information such as tangent vectors and deformations. And they have non-closed points, which make the generic behaviour of families visible. References: [Har77, Chapter II], [The].

Prerequisites. Chapter 49 (Algebraic Varieties) and Chapter 12 (Commutative Algebra).

Rings are commutative with 1. For a ring A we use the notation $V(I)$, $D(f)$ of Definition 12.2.1. For a prime $\mathfrak{p}$, $k(\mathfrak{p}) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ is its residue field.

50.2 The spectrum of a ring

Proposition 50.2.1. *Let A be a ring and $X = \text{Spec } A$.*

1. *The sets $V(I)$ are the closed sets of a topology, the Zariski topology: $V(IJ) = V(I \cap J) = V(I) \cup V(J)$ and $V(\sum I_\alpha) = \bigcap V(I_\alpha)$. Moreover $V(I) \subseteq V(J)$ if and only if $\sqrt{I} \subseteq \sqrt{J}$.*
2. *The sets $D(f)$ form a basis of the topology, $D(f) \cap D(g) = D(fg)$, and $D(f) \subseteq D(g)$ if and only if $f \in \sqrt{(g)}$.*
3. *The closure of $\{\mathfrak{p}\}$ is $V(\mathfrak{p})$. So the closed points are the maximal ideals, and $\mathfrak{p} \mapsto V(\mathfrak{p})$ is a bijection from X to the set of irreducible closed subsets; $\mathfrak{p}$ is the unique generic point of $V(\mathfrak{p})$.*
4. *X and every $D(f)$ are quasi-compact.*
5. *A ring map $\varphi: A \rightarrow B$ induces a continuous map $\text{Spec } B \rightarrow \text{Spec } A$, $\mathfrak{q} \mapsto \varphi^{-1}(\mathfrak{q})$, and the preimage of $D(f)$ is $D(\varphi(f))$.*

Proof. (1) A prime contains IJ if and only if it contains I or J , and it contains $\sum I_\alpha$ if and only if it contains every I_α . The last statement holds because $\sqrt{I}$ is the intersection of the primes containing I (Proposition 12.2.2). (2) $X \setminus V(I) = \bigcup_{f \in I} D(f)$, and $\mathfrak{p}$ omits fg if and only if it omits both. $D(f) \subseteq D(g)$ means $V(g) \subseteq V(f)$, that is, $f \in \sqrt{(g)}$ by (1). (3) The smallest $V(I)$ containing $\mathfrak{p}$ is $V(\mathfrak{p})$. If $V(I)$ is irreducible, then $\sqrt{I}$ is prime: if $fg \in \sqrt{I}$, then $V(I) = (V(I) \cap V(f)) \cup (V(I) \cap V(g))$, so $V(I) \subseteq V(f)$, say, and $f \in \sqrt{I}$. Then $V(I) = V(\sqrt{I})$ is the closure of the point $\sqrt{I}$, which is unique because $V(\mathfrak{p}) = V(\mathfrak{q})$ forces $\mathfrak{p} = \mathfrak{q}$. (4) If $X = \bigcup_\alpha D(f_\alpha)$,

then no prime contains all f_α , so they generate the unit ideal, and $1 = \sum_{i=1}^r a_i f_{\alpha_i}$ involves finitely many. For $D(f) \cong \operatorname{Spec} A_f$ (by Proposition 12.3.2(3)) the same argument applies. (5) $\varphi^{-1}(\mathfrak{q})$ omits f exactly when $\mathfrak{q}$ omits $\varphi(f)$. $\square$

Definition 50.2.2. The *structure sheaf* $\mathcal{O} = \mathcal{O}_{\operatorname{Spec} A}$ assigns to an open $U \subseteq \operatorname{Spec} A$ the ring of functions $s: U \rightarrow \bigsqcup_{\mathfrak{p} \in U} A_{\mathfrak{p}}$ with $s(\mathfrak{p}) \in A_{\mathfrak{p}}$ that are *locally quotients*: every point of U has a neighbourhood $V \subseteq U$ and $a, f \in A$ with $f \notin \mathfrak{p}$ and $s(\mathfrak{p}) = a/f$ in $A_{\mathfrak{p}}$ for all $\mathfrak{p} \in V$. The locally ringed space $(\operatorname{Spec} A, \mathcal{O})$ is the *affine scheme* of A .

Theorem 50.2.3. Let A be a ring.

1. For every $\mathfrak{p}$, the stalk $\mathcal{O}_{\mathfrak{p}}$ is isomorphic to $A_{\mathfrak{p}}$; in particular $\operatorname{Spec} A$ is a locally ringed space.
2. For every $f \in A$, $\mathcal{O}(D(f)) \cong A_f$; in particular $\Gamma(\operatorname{Spec} A, \mathcal{O}) \cong A$.

Proof. (1) Evaluation at $\mathfrak{p}$ gives $\mathcal{O}_{\mathfrak{p}} \rightarrow A_{\mathfrak{p}}$. It is surjective, since a/f with $f \notin \mathfrak{p}$ defines a section on $D(f)$. It is injective: if s, t agree at $\mathfrak{p}$ and are $a/f, b/g$ near $\mathfrak{p}$, then $h(ag - bf) = 0$ for some $h \notin \mathfrak{p}$, so $a/f = b/g$ in every $A_{\mathfrak{q}}$ with $\mathfrak{q} \in D(hgh)$, a neighbourhood of $\mathfrak{p}$.

(2) Define $\psi: A_f \rightarrow \mathcal{O}(D(f))$ by $a/f^n \mapsto (\mathfrak{p} \mapsto a/f^n)$. *Injective*: if $a/f^n = 0$ in every $A_{\mathfrak{p}}$, $\mathfrak{p} \in D(f)$, then the annihilator of a is contained in no prime of $D(f)$. So $V(\operatorname{Ann}(a)) \subseteq V(f)$, and $f \in \sqrt{\operatorname{Ann}(a)}$, that is, $f^m a = 0$ and $a/f^n = 0$ in A_f . *Surjective*: let $s \in \mathcal{O}(D(f))$. Cover $D(f)$ by open sets on which $s = a_i/g_i$. Shrinking them to basic open sets $D(h_i) \subseteq D(g_i)$, and writing $h_i^m = c g_i$, so that $a_i/g_i = c a_i/h_i^m$, we may assume $s = a_i/h_i$ on $D(h_i)$. By (4) of Proposition 50.2.1, finitely many $D(h_1), \dots, D(h_r)$ suffice. On $D(h_i h_j)$ both a_i/h_i and a_j/h_j represent s . By the injectivity just proved, applied to $D(h_i h_j)$, we get $(h_i h_j)^N (a_i h_j - a_j h_i) = 0$, and we may take one N for all pairs. Replacing a_i by $a_i h_i^N$ and h_i by h_i^{N+1} we get $a_i h_j = a_j h_i$. Since $D(f) \subseteq \bigcup D(h_i)$, $f^n = \sum_i b_i h_i$ for some n and b_i . Let $a = \sum_i b_i a_i$. Then $h_j a = \sum_i b_i a_i h_j = \sum_i b_i h_i a_j = f^n a_j$, so $a/f^n = a_j/h_j$ on $D(h_j)$ for every j , and $s = \psi(a/f^n)$. $\square$

Theorem 50.2.4. For every locally ringed space X and ring A , the map

$$\operatorname{Hom}_{\operatorname{LRS}}(X, \operatorname{Spec} A) \rightarrow \operatorname{Hom}_{\operatorname{Ring}}(A, \Gamma(X, \mathcal{O}_X)), \quad (f, f^\#) \mapsto f^\#_{\operatorname{Spec} A},$$

is bijective. In particular $A \mapsto \operatorname{Spec} A$ is an anti-equivalence between rings and affine schemes, and a ring map φ induces the morphism $\operatorname{Spec} \varphi$ whose map on stalks is the localisation $A_{\varphi^{-1}(\mathfrak{q})} \rightarrow B_{\mathfrak{q}}$.

Proof. Given $\psi: A \rightarrow \Gamma(X, \mathcal{O}_X)$, define $f(x) = \psi_x^{-1}(\mathfrak{m}_x)$, where $\psi_x: A \rightarrow \mathcal{O}_{X,x}$. This is a prime. Then $f^{-1}(D(a)) = X_{\psi(a)} = \{x : \psi(a)_x \notin \mathfrak{m}_x\}$, which is open, since a germ that is a unit is a unit nearby. So f is continuous. On $X_{\psi(a)}$ the section $\psi(a)$ is invertible, so ψ induces $A_a \rightarrow \mathcal{O}_X(X_{\psi(a)})$. By Theorem 50.2.3 these maps define a morphism of sheaves $\mathcal{O}_{\operatorname{Spec} A} \rightarrow f_* \mathcal{O}_X$ on the basis of basic open sets, hence a morphism of sheaves. On the stalk at x it is the map $A_{f(x)} \rightarrow \mathcal{O}_{X,x}$ induced by ψ_x , which is local by the definition of $f(x)$. Conversely, let $(f, f^\#)$ be a morphism with global map ψ . Since $f_x^\#: A_{f(x)} \rightarrow \mathcal{O}_{X,x}$ is local and compatible with ψ_x , we get $f(x) A_{f(x)} = (f_x^\#)^{-1}(\mathfrak{m}_x)$, so $f(x) = \psi_x^{-1}(\mathfrak{m}_x)$. On $D(a)$ the map $f^\#$ is determined by ψ through the universal property of A_a . So the two constructions are inverse. For $X = \operatorname{Spec} B$, $\Gamma(X, \mathcal{O}_X) = B$ by Theorem 50.2.3. $\square$

50.3 Schemes and their properties

Definition 50.3.1. A *scheme* is a locally ringed space $(X, \mathcal{O}_X)$ in which every point has an open neighbourhood U such that $(U, \mathcal{O}_X|_U)$ is isomorphic to an affine scheme. A *morphism* of schemes is a morphism of locally ringed spaces. For a scheme S , a scheme *over* S is a

morphism $X \rightarrow S$, and morphisms over S commute with the structure maps. We write $X(T) = \text{Hom}_S(T, X)$ for the T -valued points, and $X(R) = X(\text{Spec } R)$. The residue field at x is $k(x) = \mathcal{O}_{X,x}/\mathfrak{m}_x$.

Proposition 50.3.2. *Let X_i be schemes, $U_{ij} \subseteq X_i$ open subschemes with $U_{ii} = X_i$, and $\varphi_{ij}: U_{ij} \rightarrow U_{ji}$ isomorphisms such that $\varphi_{ij}(U_{ij} \cap U_{ik}) = U_{ji} \cap U_{jk}$ and $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$ on $U_{ij} \cap U_{ik}$. Then there is a scheme X , unique up to unique isomorphism, with open subschemes X_i glued along the φ_{ij} . Morphisms defined on the X_i that agree on overlaps glue to a morphism $X \rightarrow Y$.*

Proof. Glue the topological spaces along the homeomorphisms φ_{ij} , then glue the sheaves, which is possible by the cocycle condition. The result is locally ringed and locally affine because each X_i is. Morphisms of locally ringed spaces glue because continuous maps and sheaf maps do (Proposition 23.6.4). $\square$

Example 50.3.3. 1. $\text{Spec } k$ for a field k is a point. $\text{Spec } \mathbb{Z}$ has closed points (p) with residue fields $\mathbb{F}_p$, and a generic point (0) with residue field $\mathbb{Q}$, dense in $\text{Spec } \mathbb{Z}$.

2. *Affine space* over A is $\mathbb{A}_A^n = \text{Spec } A[x_1, \dots, x_n]$. For $A = k$ algebraically closed, its closed points are the points of k^n (Theorem 12.7.2), and it has one further point for each irreducible closed subset of dimension > 0 , namely its generic point.

3. *Projective space* $\mathbb{P}_A^n$ is glued from $U_i = \text{Spec } A[x_0/x_i, \dots, x_n/x_i]$ along $U_i \supseteq D(x_j/x_i) \cong D(x_i/x_j) \subseteq U_j$. It is also $\text{Proj } A[x_0, \dots, x_n]$ (Section 50.6).

4. The *affine line with doubled origin* is glued from two copies of $\mathbb{A}_k^1$ along the identity of $\mathbb{A}_k^1 \setminus \{0\}$. It is a scheme that is not isomorphic to any variety (Example 50.5.3).

5. $\text{Spec } k[\varepsilon]/(\varepsilon^2)$ is a single point with a non-reduced local ring. A morphism $\text{Spec } k[\varepsilon]/(\varepsilon^2) \rightarrow X$ over k is a k -point x of X together with a tangent vector in $(\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$ (Exercise 50.10.2).

Definition 50.3.4. A scheme X is *reduced* if all local rings are reduced, *irreducible* if its space is, *integral* if every $\mathcal{O}_X(U)$, $U \neq \emptyset$, is a domain, *locally Noetherian* if it has an affine open cover by spectra of Noetherian rings, and *Noetherian* if moreover it is quasi-compact. A morphism $f: X \rightarrow Y$ is *locally of finite type* if there are affine open covers $Y = \bigcup \text{Spec } B_i$ and $f^{-1}(\text{Spec } B_i) = \bigcup_j \text{Spec } A_{ij}$ with each A_{ij} a finitely generated B_i -algebra, *of finite type* if moreover finitely many A_{ij} suffice for each i , and *finite* if $f^{-1}(\text{Spec } B_i) = \text{Spec } A_i$ with A_i a finite B_i -module. An *open immersion* is an isomorphism onto an open subscheme. A *closed immersion* is a morphism that is a homeomorphism onto a closed subset and such that $f^\#: \mathcal{O}_Y \rightarrow f_*\mathcal{O}_X$ is surjective. A *closed subscheme* is an equivalence class of closed immersions.

Proposition 50.3.5. *A scheme is integral if and only if it is reduced and irreducible. An integral scheme X has a unique generic point η , and its function field is $K(X) = \mathcal{O}_{X,\eta}$, which is the fraction field of A for every nonempty affine open $\text{Spec } A \subseteq X$.*

Proof. If X is integral, it is reduced. If it were reducible, there would be disjoint nonempty open sets U_1, U_2 , and $\mathcal{O}(U_1 \sqcup U_2) = \mathcal{O}(U_1) \times \mathcal{O}(U_2)$ would not be a domain. Conversely, let X be reduced and irreducible, and let $U = \text{Spec } A$ be a nonempty affine open. Then A is reduced with irreducible spectrum, so its nilradical 0 is prime (Proposition 50.2.1(3)), and A is a domain. For general U and $fg = 0$ in $\mathcal{O}(U)$, U is the union of the closed sets where the germs of f and of g lie in the maximal ideal. On each affine open these are $V(f)$ and $V(g)$. By irreducibility one of them, say $V(f)$, is all of U , so f is nilpotent on each affine open, hence 0 . The generic point of X is the generic point of each nonempty open subset, which for $\text{Spec } A$ is the prime (0) , with local ring $A_{(0)} = \text{Frac}(A)$. $\square$

Proposition 50.3.6. *Let X be a scheme and K a field. Morphisms $\text{Spec } K \rightarrow X$ correspond bijectively to pairs of a point $x \in X$ and a field homomorphism $k(x) \rightarrow K$. If X is of finite type over an algebraically closed field k , then the closed points of X are exactly the points with $k(x) = k$, and $X(k)$ is the set of closed points.*

Proof. A morphism from the one-point space $\text{Spec } K$ has image a point x , and the stalk map $\mathcal{O}_{X,x} \rightarrow K$ is local, so it kills $\mathfrak{m}_x$ and factors through $k(x)$. Conversely such data define a morphism, by Theorem 50.2.4 applied on an affine neighbourhood. For X of finite type over k , a point x in an affine open $\text{Spec } A$ is closed in $\text{Spec } A$ if and only if its prime is maximal, if and only if $k(x) = k$, by Zariski's lemma (Theorem 12.7.1) and algebraic closedness. If $k(x) = k$, then x is closed in every affine open containing it, and its closure does not meet any open set not containing x , since open sets are stable under generisation. So x is closed in X . Conversely, a closed point of X is closed in an affine open containing it. $\square$

50.4 Fibre products

Theorem 50.4.1. *Fibre products exist in the category of schemes. For affine schemes, $\text{Spec } A \times_{\text{Spec } R} \text{Spec } B = \text{Spec}(A \otimes_R B)$.*

Sketch of proof. For affine schemes this follows from Theorem 50.2.4 and the universal property of $A \otimes_R B$ (Proposition 7.10.2): a morphism $T \rightarrow \text{Spec}(A \otimes_R B)$ is a ring map $A \otimes_R B \rightarrow \Gamma(T, \mathcal{O}_T)$, that is, a compatible pair of ring maps from A and B . In general one glues. If $X \times_S Y$ exists and $U \subseteq X$ is open, then the preimage of U in $X \times_S Y$ is $U \times_S Y$. Hence, covering S , then X and Y by affine opens, the affine fibre products glue along these canonical open subschemes by Proposition 50.3.2. The omitted verifications are the cocycle conditions, which follow from the uniqueness of fibre products, and the universal property of the glued scheme, which is local [Har77, Theorem II.3.3]. $\square$

Definition 50.4.2. For a morphism $f: X \rightarrow Y$ and $y \in Y$, the *fibre* is $X_y = X \times_Y \text{Spec } k(y)$, a scheme over the residue field $k(y)$ whose underlying space is homeomorphic to $f^{-1}(y)$ [Har77, Exercise II.3.10]. For $Y' \rightarrow Y$, the *base change* of X is $X_{Y'} = X \times_Y Y'$. A property of morphisms is *stable under base change* if $X_{Y'} \rightarrow Y'$ has it whenever $X \rightarrow Y$ does.

Example 50.4.3. 1. $\mathbb{A}_R^m \times_R \mathbb{A}_R^n = \mathbb{A}_R^{m+n}$, since $R[x] \otimes_R R[y] = R[x, y]$.

2. $\text{Spec } \mathbb{C} \times_{\text{Spec } \mathbb{R}} \text{Spec } \mathbb{C} = \text{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}) = \text{Spec}(\mathbb{C} \times \mathbb{C})$, two points. So a fibre product of one-point schemes can have several points, and the underlying set of a fibre product is not the fibre product of the underlying sets.
3. For $f: \text{Spec } \mathbb{Z}[i] \rightarrow \text{Spec } \mathbb{Z}$, the fibre over (p) is $\text{Spec } \mathbb{F}_p[x]/(x^2 + 1)$. It is two reduced points if $p \equiv 1 \pmod{4}$, one point with residue field $\mathbb{F}_{p^2}$ if $p \equiv 3 \pmod{4}$, and the non-reduced point $\text{Spec } \mathbb{F}_2[x]/(x+1)^2$ if $p = 2$. The generic fibre is $\text{Spec } \mathbb{Q}(i)$.

50.5 Separated and proper morphisms

Definition 50.5.1. A morphism $f: X \rightarrow Y$ is *separated* if the diagonal $\Delta: X \rightarrow X \times_Y X$ is a closed immersion. It is *proper* if it is separated, of finite type, and *universally closed*: every base change $X_{Y'} \rightarrow Y'$ is a closed map. A scheme over a field k is *complete* if it is proper over $\text{Spec } k$.

Proposition 50.5.2. *Every morphism of affine schemes is separated. Open and closed immersions are separated, and so are compositions and base changes of separated morphisms. A morphism $X \rightarrow Y$ is separated if and only if the diagonal has closed image.*

Proof. For $\text{Spec } A \rightarrow \text{Spec } R$ the diagonal corresponds to the multiplication $A \otimes_R A \rightarrow A$, which is surjective, so Δ is a closed immersion (a surjection of rings $B \rightarrow B/I$ gives a homeomorphism onto $V(I)$ and a surjection of sheaves, by Theorem 50.2.3). For the remaining statements, see [Har77, Corollary II.4.2 and Corollary II.4.6]. The last one holds because Δ is always a locally closed immersion: over an affine open of Y and affine opens U of X , the diagonal maps U by a closed immersion into the open subscheme $U \times_Y U$. $\square$

Example 50.5.3. The affine line with doubled origin X is not separated over k . Let $i_1, i_2: \mathbb{A}_k^1 \rightarrow X$ be the two open immersions. The morphism $(i_1, i_2): \mathbb{A}_k^1 \rightarrow X \times_k X$ maps $t \neq 0$ into the diagonal but maps 0 to the pair of the two origins, which is not in the diagonal. So the preimage of $\Delta(X)$ is $\mathbb{A}_k^1 \setminus \{0\}$, which is not closed. Hence $\Delta(X)$ is not closed.

Theorem 50.5.4 (Valuative criteria). *Let $f: X \rightarrow Y$ be a morphism of finite type with X Noetherian. For every valuation ring R with fraction field K and every commutative square formed by $\text{Spec } K \rightarrow X$ and $\text{Spec } R \rightarrow Y$, consider lifts $\text{Spec } R \rightarrow X$ making both triangles commute. Then f is separated if and only if there is at most one lift for every such square, and proper if and only if there is exactly one. It suffices to consider discrete valuation rings.*

This is [Har77, Theorems II.4.3 and II.4.7 and Exercise II.4.11]. Geometrically: a punctured disc $\text{Spec } K$ in X whose image extends over the centre of $\text{Spec } R$ in Y extends in at most one way (separated: limits are unique) or in exactly one way (proper: limits exist).

Theorem 50.5.5. *For every ring A , $\mathbb{P}_A^n \rightarrow \text{Spec } A$ is proper. Consequently every projective morphism, that is, a closed immersion into some $\mathbb{P}_Y^n$ followed by the projection, is proper, and every projective variety is complete.*

Proof. Since $\mathbb{P}_A^n = \mathbb{P}_{\mathbb{Z}}^n \times \text{Spec } A$ and properness is stable under base change [Har77, Corollary II.4.8], we may take $A = \mathbb{Z}$. The morphism is of finite type, being covered by the $n + 1$ affine spaces U_i . It is separated: on $U_i \times U_j$ the diagonal is $U_i \cap U_j \rightarrow U_i \times U_j$, given by the ring map $\mathbb{Z}[x/x_i] \otimes \mathbb{Z}[x/x_j] \rightarrow \mathbb{Z}[x/x_i]_{x_j/x_i}$, which is surjective because x_i/x_j is the image of an element of the second factor. By Proposition 50.5.2 this suffices. For properness, use Theorem 50.5.4. A morphism $\text{Spec } K \rightarrow \mathbb{P}_{\mathbb{Z}}^n$ lands in some U_i , so by Theorem 50.2.4 it is given by elements $a_j = x_j/x_i \in K$ with $a_i = 1$. Let v be the valuation of R (with $v(0) = \infty$) and choose m with $v(a_m)$ minimal; then $a_m \neq 0$. All $b_j = a_j/a_m$ lie in R , and the ring map $\mathbb{Z}[x_0/x_m, \dots, x_n/x_m] \rightarrow R, x_j/x_m \mapsto b_j$, defines $\text{Spec } R \rightarrow U_m$. On $\text{Spec } K$ it agrees with the given morphism, because on $U_i \cap U_m$ the coordinates $x_j/x_i = (x_j/x_m)(x_m/x_i)$ are sent to $b_j/b_i = a_j$. Uniqueness follows from separatedness. Closed immersions are proper, and compositions of proper morphisms are proper [Har77, Corollary II.4.8]. $\square$

50.6 Projective schemes

Definition 50.6.1. Let $S = \bigoplus_{d \geq 0} S_d$ be a graded ring and $S_+ = \bigoplus_{d > 0} S_d$. The set $\text{Proj } S$ consists of the homogeneous primes not containing S_+ , with the topology whose closed sets are the $V_+(a)$, a homogeneous. For homogeneous $f \in S_+$ let $D_+(f) = \{\mathfrak{p} : f \notin \mathfrak{p}\}$, and let $S_{(f)}$ be the degree-zero part of S_f , and $S_{(\mathfrak{p})}$ the degree-zero part of the localisation at the homogeneous elements not in $\mathfrak{p}$. The structure sheaf consists of the functions $s(\mathfrak{p}) \in S_{(\mathfrak{p})}$ that are locally quotients a/f with a, f homogeneous of the same degree.

Proposition 50.6.2. *$\text{Proj } S$ is a scheme. The stalk at $\mathfrak{p}$ is $S_{(\mathfrak{p})}$, the sets $D_+(f)$ form an open cover, and $D_+(f) \cong \text{Spec } S_{(f)}$. There is a morphism $\text{Proj } S \rightarrow \text{Spec } S_0$. If $S = A[x_0, \dots, x_n]$, then $\text{Proj } S = \mathbb{P}_A^n$, and if $S = A[x_0, \dots, x_n]/I$ with I homogeneous, then $\text{Proj } S$ is a closed subscheme of $\mathbb{P}_A^n$.*

Proof. The stalks are computed as in Theorem 50.2.3(1). Every $\mathfrak{p} \in \text{Proj } S$ omits some homogeneous element of S_+ , so the $D_+(f)$ cover. Let f have degree $e > 0$. The map $\mathfrak{p} \mapsto (\mathfrak{p}S_f) \cap S_{(f)}$ is a homeomorphism $D_+(f) \rightarrow \text{Spec } S_{(f)}$. Its inverse sends $\mathfrak{q}$ to the homogeneous ideal whose degree d part is $\{a \in S_d : a^e / f^d \in \mathfrak{q}\}$; checking that this is a homogeneous prime and that the two maps are inverse homeomorphisms is routine [Har77, Proposition II.2.5]. Under it, $S_{(\mathfrak{p})}$ corresponds to the localisation of $S_{(f)}$ at $\mathfrak{q}$, and the locally-quotient descriptions of the two structure sheaves match, so the ringed spaces are isomorphic. The inclusions $S_0 \rightarrow S_{(f)}$ glue to the morphism to $\text{Spec } S_0$. For $S = A[x_0, \dots, x_n]$, $S_{(x_i)} = A[x_0/x_i, \dots, x_n/x_i]$, and the gluing agrees with Example 50.3.3(3). A surjection $S \rightarrow S/I$ induces surjections $S_{(x_i)} \rightarrow (S/I)_{(x_i)}$, hence closed immersions $\text{Spec}(S/I)_{(x_i)} \rightarrow \text{Spec } S_{(x_i)}$, which glue to a closed immersion $\text{Proj } S/I \rightarrow \mathbb{P}_A^n$. $\square$

50.7 Varieties as schemes

Theorem 50.7.1. *Let k be algebraically closed. There is a fully faithful functor t from the category of varieties over k (Chapter 49) to the category of schemes over k . The space $t(V)$ is the set of irreducible closed subsets of V , and V is homeomorphic to the subspace of closed points of $t(V)$. The image consists of the integral separated schemes of finite type over k that admit an immersion into some $\mathbb{P}_k^n$, and $t(V)$ is proper over k if and only if V is projective.*

Sketch of proof. For an affine variety V with coordinate ring A , $t(V) = \text{Spec } A$. Irreducible closed subsets of V correspond to primes of A (Proposition 49.2.4), and points to maximal ideals. By Theorem 49.3.3 and Theorem 50.2.3, the regular functions on an open subset $U \subseteq V$ are the sections of $\mathcal{O}_{\text{Spec } A}$ over the corresponding open subset. In general $t(V)$ is glued from affine pieces. Full faithfulness reduces, by gluing, to the affine case, where both categories are anti-equivalent to the category of finitely generated domains (Proposition 49.3.5 and Theorem 50.2.4). The omitted verifications are that the construction is compatible with gluing and that quasi-projective varieties give separated schemes, which is Proposition 50.5.2 and Theorem 50.5.5; see [Har77, Proposition II.2.6 and Proposition II.4.10]. $\square$

Remark 50.7.2. From now on, a *variety over a field k* is an integral separated scheme of finite type over k . For k algebraically closed and quasi-projective varieties this agrees with Chapter 49 by Theorem 50.7.1. There are complete varieties that are not projective (Nagata, Hironaka), but every complete curve is projective, and so is every smooth complete surface; see [Har77, Appendix B].

50.8 Flat and smooth morphisms

Definition 50.8.1. A morphism $f: X \rightarrow Y$ is *flat* at x if $\mathcal{O}_{X,x}$ is a flat $\mathcal{O}_{Y,f(x)}$ -module (Definition 7.9.7), and *flat* if it is flat at every point. It is *faithfully flat* if moreover it is surjective.

Proposition 50.8.2. 1. *Open immersions are flat, and $\mathbb{A}_Y^n \rightarrow Y$ and $\mathbb{P}_Y^n \rightarrow Y$ are flat. Compositions and base changes of flat morphisms are flat.*

2. *Let R be a principal ideal domain, for instance a discrete valuation ring. An R -module is flat if and only if it is torsion-free. Hence $X \rightarrow \text{Spec } R$ is flat if and only if every local ring $\mathcal{O}_{X,x}$ is torsion-free over R . If X is integral, this holds if and only if X dominates $\text{Spec } R$.*

Proof. (1) Localisations are flat (Proposition 12.3.2(2)), and so are free modules such as $A[x_1, \dots, x_n]$ over A ; local rings of $\mathbb{A}_A^n$ are localisations of $A[x]$. The statements on composition and base change are [Har77, Proposition III.9.2]. (2) If M is flat and $0 \neq t \in R$, tensoring

the injection $t: R \rightarrow R$ with M shows that t is injective on M . Conversely, a torsion-free M is the union of its finitely generated submodules, which are torsion-free, hence free (Theorem 7.11.3). A union of flat submodules is flat, since an element of the kernel of $N' \otimes M \rightarrow N \otimes M$ comes from some finitely generated submodule of M . For integral X , torsion-freeness fails exactly when some nonzero $t \in R$ vanishes on X , that is, when X maps into a closed point. $\square$

Example 50.8.3. 1. The family $X = \operatorname{Spec} k[t, x, y]/(xy - t) \rightarrow \mathbb{A}_k^1 = \operatorname{Spec} k[t]$ is flat, since X is integral and dominates $\mathbb{A}^1$. Its fibres are the smooth conics $xy = t$ for $t \neq 0$, which degenerate to the pair of lines $xy = 0$.

2. The blow-up $\operatorname{Bl}_0 \mathbb{A}^2 \rightarrow \mathbb{A}^2$ is not flat. Its fibres have dimension 0 except over the origin, where the fibre is $\mathbb{P}^1$, while flat morphisms of finite type between varieties have equidimensional fibres [Har77, Proposition III.9.5].

3. A closed immersion $\operatorname{Spec} A/I \rightarrow \operatorname{Spec} A$ with $0 \neq I$ and A a domain is not flat, since A/I is torsion.

Definition 50.8.4. A morphism $f: X \rightarrow Y$ is *smooth of relative dimension n* if every point of X has affine open neighbourhoods $\operatorname{Spec} B$ of x and $\operatorname{Spec} A$ of $f(x)$, with $f(\operatorname{Spec} B) \subseteq \operatorname{Spec} A$, such that

$$B \cong A[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r) \quad \text{with } \det(\partial g_i / \partial x_{n+j})_{1 \leq i, j \leq r} \text{ invertible in } B.$$

It is *étale* if it is smooth of relative dimension 0.

Proposition 50.8.5. Let k be algebraically closed and X a variety over k of dimension n . Then $X \rightarrow \operatorname{Spec} k$ is smooth (of relative dimension n) if and only if X is nonsingular in the sense of Definition 49.6.1.

Proof. Both conditions are local, so let X be affine. Suppose $B = k[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r)$ with the minor invertible, and let P be a closed point. Every irreducible component of $Z(g_1, \dots, g_r) \subseteq \mathbb{A}^{n+r}$ has dimension $\geq n$ by Theorem 49.5.4 applied r times, so $\dim \mathcal{O}_{X,P} \geq n$. The Jacobian of (g_i) has rank r at P , so $\dim T_P X \leq n$ by the computation in Theorem 49.6.3, which only used generators of the ideal defining B . By Lemma 49.6.2, $\mathcal{O}_{X,P}$ is regular of dimension n . Conversely, let $X \subseteq \mathbb{A}^N$ be nonsingular at P with $I(X) = (f_1, \dots, f_t)$. The Jacobian has rank $r = N - n$ at P , so after renumbering, $\det(\partial f_i / \partial x_{n+j})_{i,j \leq r}$ is nonzero at P , hence invertible on a neighbourhood $D(h)$. The scheme $X' = Z(f_1, \dots, f_r) \cap D(h)$ is smooth of relative dimension n by construction. So by the first part its local ring at P is regular of dimension n , in particular a domain (Remark 12.11.3). Hence X' has a unique irreducible component through P , of dimension n , which contains $X \cap D(h)$ and so equals it near P . As X' is reduced near P , $X = X'$ near P . $\square$

Counterexample 50.8.6. Over a non-perfect field, regular does not imply smooth. Let p be an odd prime, $k = \mathbb{F}_p(t)$ and $X = \operatorname{Spec} k[x, y]/(y^2 - x^p + t)$. At the closed point $\mathfrak{m} = (y, x^p - t)$, whose residue field is $k(t^{1/p})$, the relation $x^p - t = y^2 \in \mathfrak{m}^2$ shows that $\mathfrak{m}/\mathfrak{m}^2$ is spanned by y . So X is regular there, of dimension 1. But X is not smooth over k . Smoothness is stable under base change, since a presentation as in Definition 50.8.4 base changes to one. After base change to $\bar{k}$, with $s = t^{1/p}$, the curve becomes $y^2 = (x-s)^p$, whose Jacobian $(-p(x-s)^{p-1}, 2y) = (0, 2y)$ vanishes at the point $(s, 0)$. That point is singular by Theorem 49.6.3, so $X_{\bar{k}}$ is not smooth by Proposition 50.8.5.

50.9 Base change and varieties over non-closed fields

Definition 50.9.1. Let X be a scheme over a field k and $K \supseteq k$ a field extension. The *base change* is $X_K = X \times_{\operatorname{Spec} k} \operatorname{Spec} K$. The scheme X is *geometrically reduced, irreducible or integral* if $X_{\bar{k}}$ is reduced, irreducible or integral for an algebraic closure $\bar{k}$. A *geometric point* is a morphism $\operatorname{Spec} \Omega \rightarrow X$ with Ω algebraically closed.

Example 50.9.2. 1. $X = \operatorname{Spec} \mathbb{R}[x]/(x^2 + 1) = \operatorname{Spec} \mathbb{C}$ is integral over $\mathbb{R}$, but $X_{\mathbb{C}} = \operatorname{Spec} \mathbb{C} \sqcup \operatorname{Spec} \mathbb{C}$ is not irreducible.

2. $X = \operatorname{Spec} \mathbb{F}_p(t)[x]/(x^p - t)$ is a point with residue field $\mathbb{F}_p(t^{1/p})$, integral, but $X_{\bar{k}}$ is non-reduced, since $x^p - t = (x - t^{1/p})^p$.

3. The conic $X = \operatorname{Spec} \mathbb{R}[x, y]/(x^2 + y^2 + 1)$ has no $\mathbb{R}$ -points, but it is a nonempty integral scheme whose closed points have residue field $\mathbb{C}$. $X(\mathbb{C})$ is an affine conic with an action of complex conjugation without fixed points.

4. For a variety X over k and a Galois extension K/k , $\operatorname{Gal}(K/k)$ acts on $X(K) = \operatorname{Hom}_k(\operatorname{Spec} K, X)$ through its action on $\operatorname{Spec} K$, and $X(k) \subseteq X(K)^{\operatorname{Gal}(K/k)}$. For a variety over $\mathbb{Q}$, the set $X(\bar{\mathbb{Q}})$ with its Galois action is the starting point of the ℓ -adic cohomology of Chapter 61.

Remark 50.9.3. A variety X over $\mathbb{C}$ is defined by finitely many polynomials, so it is defined over a finitely generated subfield $k_0 \subseteq \mathbb{C}$: $X = X_0 \times_{k_0} \mathbb{C}$. Moreover, X_0 extends to a scheme of finite type over a finitely generated $\mathbb{Z}$ -algebra, a technique called *spreading out*. For an automorphism σ of $\mathbb{C}$, the *conjugate variety* is $X^\sigma = X \times_{\mathbb{C}, \sigma} \mathbb{C}$. It is a scheme isomorphic to X as an abstract scheme but not over $\mathbb{C}$, and $X^\sigma(\mathbb{C})$ is in general not homeomorphic to $X(\mathbb{C})$ [Ser64]. These constructions are used in the theory of absolute Hodge classes (Chapter 71).

50.10 Exercises

Exercise 50.10.1. Describe the points of $\operatorname{Spec} \mathbb{Z}[x]$: show that they are (0) , the primes (f) with f irreducible, and the maximal ideals (p, f) with p prime and f irreducible modulo p .

Solution. Let $\mathfrak{p}$ be a prime. If $\mathfrak{p} \cap \mathbb{Z} = (p)$ with p prime, then $\mathfrak{p}$ corresponds to a prime of $\mathbb{F}_p[x]$, a principal ideal domain, namely 0 or $(\bar{f})$ with $\bar{f}$ irreducible. This gives (p) and (p, f) , and the latter is maximal. If $\mathfrak{p} \cap \mathbb{Z} = 0$, then $\mathfrak{p}$ corresponds to a prime of $\mathbb{Q}[x]$ (localisation at $\mathbb{Z} \setminus 0$), which is 0 or (g) with g irreducible. Then $\mathfrak{p} = g\mathbb{Q}[x] \cap \mathbb{Z}[x] = (f)$, where f is g made primitive (Gauss's lemma, Lemma 7.6.10). The primes (p) are the primes (f) with $f = p$ constant. $\square$

Exercise 50.10.2. Let X be a scheme over an algebraically closed field k and $D = \operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. Show that $X(D)$ is in bijection with pairs of a closed point $x \in X(k)$ and a vector of $(\mathfrak{m}_x/\mathfrak{m}_x^2)^\vee$.

Solution. The image of D is a point x , and the morphism is a local k -algebra map $\varphi: \mathcal{O}_{X,x} \rightarrow k[\varepsilon]$. Composing with $k[\varepsilon] \rightarrow k$ gives $k(x) = k$. Write $\varphi(a) = a(x) + \delta(a)\varepsilon$. Then δ is a k -linear derivation at x that kills k and $\mathfrak{m}_x^2$, since $\varepsilon^2 = 0$. So δ is a linear form on $\mathfrak{m}_x/\mathfrak{m}_x^2$, and conversely every such form defines a local homomorphism. $\square$

Exercise 50.10.3. Let R be a discrete valuation ring with fraction field K and residue field κ . Describe $\operatorname{Spec} R$ and the fibres of $X \rightarrow \operatorname{Spec} R$ for $X = \operatorname{Spec} R[x, y]/(xy - \pi)$, where π is a uniformiser.

Solution. $\text{Spec } R$ has two points: the closed point $\mathfrak{m}$ and the generic point (0) , which is open. The generic fibre is the smooth conic $xy = \pi$ over K , isomorphic to $\mathbb{A}_K^1 \setminus \{0\}$. The special fibre is $xy = 0$ over κ , two lines meeting in a point. X is a domain dominating $\text{Spec } R$, so the morphism is flat (Proposition 50.8.2). $\square$

Exercise 50.10.4. Show that $\text{Proj } k[x, y, z]/(xz - y^2) \cong \mathbb{P}_k^1$ for every field k .

Solution. The ring $S = k[x, y, z]/(xz - y^2)$ is isomorphic to the subring $k[s^2, st, t^2] \subseteq k[s, t]$ of polynomials of even degree, with degrees halved. For a graded ring T and $T^{(2)} = \bigoplus_d T_{2d}$ one has $\text{Proj } T^{(2)} \cong \text{Proj } T$, since $D_+(f^2)$ and $D_+(f)$ have the same ring $T_{(f)} = T_{(f^2)}^{(2)}$ [Har77, Exercise II.5.13]. Hence $\text{Proj } S \cong \text{Proj } k[s, t] = \mathbb{P}_k^1$. $\square$

Chapter 51

Quasi-coherent and Coherent Sheaves

51.1 Introduction

Quasi-coherent sheaves are to schemes what modules are to rings. On an affine scheme $\text{Spec } A$ they are exactly the sheaves $\widetilde{M}$ attached to A -modules, and in general they are glued from such. Coherent sheaves, the finitely generated ones, include vector bundles, ideal sheaves of subvarieties, structure sheaves of closed subschemes and sheaves of differentials. Their cohomology is the subject of Chapter 52. This chapter develops the basic theory: the equivalence between modules and quasi-coherent sheaves on affine schemes, closed subschemes and ideal sheaves, the twisting sheaves $\mathcal{O}(n)$ and Serre's theorem that $\mathcal{F}(n)$ is globally generated for $n \gg 0$, and sheaves of differentials, with the Euler sequence and the canonical sheaf of $\mathbb{P}^n$. References: [Har77, Sections II.5 and II.8], [The].

Prerequisites. Chapter 50 (Schemes).

For a ringed space $(X, \mathcal{O}_X)$, $\mathcal{O}_X$ -modules, their tensor products, $\mathcal{H}om$, direct and inverse images are as in Definition 23.6.3. For $f \in \Gamma(X, \mathcal{O}_X)$ we write $X_f = \{x \in X : f_x \notin \mathfrak{m}_x\}$, an open set on which f is invertible.

51.2 The sheaf associated to a module

Definition 51.2.1. Let A be a ring and M an A -module. The sheaf $\widetilde{M}$ on $\text{Spec } A$ assigns to an open U the set of functions $s: U \rightarrow \bigsqcup_{\mathfrak{p} \in U} M_{\mathfrak{p}}$ with $s(\mathfrak{p}) \in M_{\mathfrak{p}}$ that are locally of the form m/f with $m \in M$ and $f \in A$ nonvanishing on the neighbourhood. It is an $\mathcal{O}_{\text{Spec } A}$ -module.

Proposition 51.2.2. Let A be a ring, M, N modules and $X = \text{Spec } A$.

1. $(\widetilde{M})_{\mathfrak{p}} \cong M_{\mathfrak{p}}$ and $\widetilde{M}(D(f)) \cong M_f$; in particular $\Gamma(X, \widetilde{M}) = M$.
2. $M \mapsto \widetilde{M}$ is an exact, fully faithful functor from A -modules to $\mathcal{O}_X$ -modules, commuting with direct sums, and $(M \otimes_A N)^{\sim} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$.
3. For a ring map $A \rightarrow B$ and $f: \text{Spec } B \rightarrow \text{Spec } A$: $f_*(\widetilde{N}) \cong ({}_A N)^{\sim}$ for a B -module N , and $f^*(\widetilde{M}) \cong (B \otimes_A M)^{\sim}$.

Proof. (1) The proof of Theorem 50.2.3 applies word for word with A replaced by M in the numerators. (2) Exactness can be checked on stalks, where it is exactness of localisation (Proposition 12.3.2(2)). Full faithfulness: a morphism $\widetilde{M} \rightarrow \widetilde{N}$ is determined by its effect on global sections, by (1) on basic open sets, and every A -linear map induces one. Direct sums and tensor products commute with localisation, and the sheaf tensor product has stalks $M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}$; the natural map is therefore an isomorphism on stalks. (3) For $g \in A$ with

image $\bar{g} \in B$, $f_*(\tilde{N})(D(g)) = \tilde{N}(D(\bar{g})) = N_{\bar{g}} = (N_A)_g$, compatibly with restrictions. For the pullback, both sides represent the functor $\mathcal{G} \mapsto \text{Hom}_{\mathcal{O}_{\text{Spec } A}}(\tilde{M}, f_*\mathcal{G})$ on quasi-coherent $\mathcal{G}$, by adjunction and (2); see [Har77, Proposition II.5.2]. $\square$

51.3 Quasi-coherent sheaves

Definition 51.3.1. An $\mathcal{O}_X$ -module $\mathcal{F}$ on a scheme X is *quasi-coherent* if X has an affine open cover $X = \bigcup \text{Spec } A_i$ with $\mathcal{F}|_{\text{Spec } A_i} \cong \tilde{M}_i$ for A_i -modules M_i . If X is Noetherian, $\mathcal{F}$ is *coherent* if moreover the M_i can be taken finitely generated.

Lemma 51.3.2. Let X be a scheme that is Noetherian, or quasi-compact and separated. Let $\mathcal{L}$ be an invertible sheaf, that is, locally isomorphic to $\mathcal{O}_X$, $f \in \Gamma(X, \mathcal{L})$, and $X_f = \{x : f_x \notin \mathfrak{m}_x \mathcal{L}_x\}$. Let $\mathcal{F}$ be quasi-coherent.

1. If $s \in \Gamma(X, \mathcal{F})$ restricts to 0 on X_f , then $s \otimes f^n = 0$ in $\Gamma(X, \mathcal{F} \otimes \mathcal{L}^n)$ for some n .
2. If $t \in \mathcal{F}(X_f)$, then $t \otimes f^n$ extends to a global section of $\mathcal{F} \otimes \mathcal{L}^n$ for some n .

Proof. Cover X by finitely many affine opens $U_i = \text{Spec } A_i$ with $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$ and $\mathcal{F}|_{U_i} = \tilde{M}_i$. Every $U_i \cap U_j$ is quasi-compact: every open subset of a Noetherian scheme is, and in the separated case $U_i \cap U_j$ is affine, being the preimage of the closed subscheme $\Delta(X)$ under $U_i \times U_j \rightarrow X \times X$. Under the trivialisation f becomes $f_i \in A_i$ and $X_f \cap U_i = D(f_i)$. (1) $s|_{U_i} \in M_i$ vanishes in $(M_i)_{f_i}$ by Proposition 51.2.2(1), so $f_i^n s|_{U_i} = 0$ for large n , which we take uniform over the finitely many i . (2) $t|_{D(f_i)} \in (M_i)_{f_i}$, so $f_i^n t$ extends to a section t_i over U_i . On $U_i \cap U_j$, which is quasi-compact and so covered by finitely many affine opens, $t_i - t_j$ vanishes on X_f . By (1) applied to $U_i \cap U_j$, $f^m(t_i - t_j) = 0$ there. So the sections $f^m t_i$ glue to a global section of $\mathcal{F} \otimes \mathcal{L}^{n+m}$ extending $t \otimes f^{n+m}$. $\square$

Theorem 51.3.3. Let $X = \text{Spec } A$. An $\mathcal{O}_X$ -module $\mathcal{F}$ is quasi-coherent if and only if the natural map $\Gamma(X, \mathcal{F})^\sim \rightarrow \mathcal{F}$ is an isomorphism. Hence $M \mapsto \tilde{M}$ is an equivalence between A -modules and quasi-coherent $\mathcal{O}_X$ -modules, with inverse $\Gamma(X, -)$. A sequence of quasi-coherent sheaves on X is exact if and only if the sequence of global sections is exact.

Proof. The natural map is defined on $D(g)$ by $M_g \rightarrow \mathcal{F}(D(g))$, $m/g^n \mapsto g^{-n}m|_{D(g)}$, where $M = \Gamma(X, \mathcal{F})$. If $\mathcal{F}$ is quasi-coherent, apply Lemma 51.3.2 with $\mathcal{L} = \mathcal{O}_X$ and $f = g$ to the affine, hence quasi-compact and separated, scheme X : part (2) gives surjectivity and part (1) injectivity of $M_g \rightarrow \mathcal{F}(D(g))$. So the map is an isomorphism on a basis. The converse is clear. The equivalence follows with Proposition 51.2.2(2). For exactness, kernels and cokernels of maps $\tilde{M} \rightarrow \tilde{N}$ are the sheaves associated to the kernel and cokernel, since $\sim$ is exact. So a sequence of quasi-coherent sheaves is isomorphic to $\sim$ applied to its sequence of global sections, and $\sim$ is exact and faithful: $\tilde{M} = 0$ implies $M = 0$. $\square$

Corollary 51.3.4. On any scheme, kernels, cokernels, images, direct sums and tensor products of quasi-coherent sheaves are quasi-coherent, and pullbacks of quasi-coherent sheaves are quasi-coherent. $\mathcal{F}$ is quasi-coherent if and only if $\mathcal{F}|_U \cong \mathcal{F}(U)^\sim$ for every affine open U . On a Noetherian scheme the same statements hold for coherent sheaves (with pullbacks along morphisms of Noetherian schemes), and on $\text{Spec } A$ with A Noetherian, $\tilde{M}$ is coherent if and only if M is finitely generated.

Proof. All statements are local, and for affine schemes they follow from Theorem 51.3.3 and Proposition 51.2.2. For the characterisation of quasi-coherence on every affine open, restrict a cover with $\mathcal{F}|_{U_i} = \tilde{M}_i$ to the affine open U : then $\mathcal{F}|_U$ is quasi-coherent in the sense of the definition on U , and Theorem 51.3.3 applies. For coherence, submodules and quotients of

finitely generated modules over a Noetherian ring are finitely generated (Proposition 12.5.2). If $\tilde{M}$ is coherent on $\text{Spec } A$, then M is locally, on a finite cover by $D(f_i)$, generated by finitely many elements m_{ij}/f_i^n ; the m_{ij} generate M , since a module is zero when all its localisations at the f_i vanish. $\square$

Proposition 51.3.5. *Let $f: X \rightarrow Y$ be a morphism with X Noetherian, or with X quasi-compact and separated. If $\mathcal{F}$ is quasi-coherent on X , then $f_*\mathcal{F}$ is quasi-coherent on Y .*

Proof. The question is local on Y , so let Y be affine. Cover X by finitely many affine opens U_i . The intersections $U_{ij} = U_i \cap U_j$ are covered by finitely many affine opens U_{ijk} (in the separated case U_{ij} itself is affine). By the sheaf axiom,

$$0 \rightarrow f_*\mathcal{F} \rightarrow \bigoplus_i f_*(\mathcal{F}|_{U_i}) \rightarrow \bigoplus_{i,j,k} f_*(\mathcal{F}|_{U_{ijk}})$$

is exact. Each term on the right is the direct image of a quasi-coherent sheaf under a morphism of affine schemes, which is quasi-coherent by Proposition 51.2.2(3). So $f_*\mathcal{F}$ is a kernel of quasi-coherent sheaves. $\square$

Proposition 51.3.6. *Let X be a scheme. For a closed immersion $i: Y \rightarrow X$ the ideal sheaf $\mathcal{I}_Y = \ker(\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y)$ is quasi-coherent, and $Y \mapsto \mathcal{I}_Y$ is a bijection from closed subschemes of X to quasi-coherent ideal sheaves of $\mathcal{O}_X$. For $X = \text{Spec } A$, every closed subscheme is $\text{Spec } A/I$ for a unique ideal I . If X is Noetherian, $\mathcal{I}_Y$ is coherent.*

Sketch of proof. The question is local, so let $X = \text{Spec } A$. The underlying space of Y is closed in X , so Y is quasi-compact. It is separated, since a closed immersion is separated (Proposition 50.5.2). By Proposition 51.3.5, $i_*\mathcal{O}_Y$ is quasi-coherent, so the kernel $\mathcal{I}_Y = \tilde{I}$ is quasi-coherent with $I = \Gamma(X, \mathcal{I}_Y)$. As $\mathcal{O}_X \rightarrow i_*\mathcal{O}_Y$ is onto, $i_*\mathcal{O}_Y \cong (A/I)^\sim$, and its support $V(I)$ is the image of Y . Then $Y \rightarrow \text{Spec } A/I$ is a homeomorphism with an isomorphism of structure sheaves, so $Y \cong \text{Spec } A/I$. Conversely, a quasi-coherent ideal $\tilde{I}$ defines $\text{Spec } A/I$, and on a general X these glue. The omitted verifications are the compatibilities of these constructions with restriction to smaller affine opens [Har77, Corollary II.5.10]. $\square$

Definition 51.3.7. The *support* of a sheaf $\mathcal{F}$ is $\{x : \mathcal{F}_x \neq 0\}$. For a coherent sheaf it is closed: on $\text{Spec } A$ with M finitely generated, $\text{Supp } \tilde{M} = V(\text{Ann } M)$. The *fibre* of $\mathcal{F}$ at x is $\mathcal{F}(x) = \mathcal{F}_x \otimes k(x)$.

Proposition 51.3.8. *Let $\mathcal{F}$ be coherent on a Noetherian scheme X .*

1. (Nakayama) *If sections $s_1, \dots, s_r$ of $\mathcal{F}$ near x span $\mathcal{F}(x)$, they generate $\mathcal{F}$ on a neighbourhood of x .*
2. *$x \mapsto \dim_{k(x)} \mathcal{F}(x)$ is upper semicontinuous.*
3. *If X is reduced, $\mathcal{F}$ is locally free (locally isomorphic to $\mathcal{O}_X^r$) if and only if this function is locally constant.*

Proof. (1) By Nakayama's lemma (Corollary 12.4.3) the s_i generate the stalk $\mathcal{F}_x$. The cokernel of $\mathcal{O}^r \rightarrow \mathcal{F}$ is coherent with x outside its closed support. (2) follows from (1): generators at x generate nearby, so the fibre dimension cannot jump up near x . (3) If $X = \text{Spec } A$ is reduced and the fibre dimension is r near x , choose $s_1, \dots, s_r$ as in (1) and let $\mathcal{K}$ be the kernel of the resulting surjection $\mathcal{O}^r \rightarrow \mathcal{F}$. For each y nearby, $\mathcal{O}^r(y) \rightarrow \mathcal{F}(y)$ is a surjection of r -dimensional spaces, hence injective. So an element $(a_1, \dots, a_r)$ of $\Gamma(\mathcal{K})$ has all a_i vanishing at every point, hence nilpotent, hence 0. So $\mathcal{K} = 0$. $\square$

51.4 Locally free sheaves and the Picard group

Definition 51.4.1. An $\mathcal{O}_X$ -module $\mathcal{E}$ is *locally free of rank r* if it is locally isomorphic to $\mathcal{O}_X^r$, and *invertible* if $r = 1$. Its dual is $\mathcal{E}^\vee = \mathcal{H}om(\mathcal{E}, \mathcal{O}_X)$, its determinant $\det \mathcal{E} = \Lambda^r \mathcal{E}$. The *Picard group* $\text{Pic}(X)$ is the group of isomorphism classes of invertible sheaves under $\otimes$, with inverse $\mathcal{L}^{-1} = \mathcal{L}^\vee$.

Proposition 51.4.2. $\text{Pic}(X) \cong H^1(X, \mathcal{O}_X^*)$. *Locally free sheaves of rank r on X correspond to isomorphism classes of algebraic vector bundles: schemes $E \rightarrow X$ locally isomorphic to $\mathbb{A}_U^r \rightarrow U$ with linear transition functions in $\text{GL}_r(\mathcal{O}_X(U \cap V))$. The correspondence sends E to its sheaf of sections.*

Proof. An invertible sheaf trivialised on an open cover (U_i) by generators e_i gives units $g_{ij} \in \mathcal{O}^*(U_i \cap U_j)$ with $e_j = g_{ij}e_i$. They form a Čech cocycle, changing the e_i changes it by a coboundary, and tensor product multiplies cocycles. Conversely a cocycle glues copies of $\mathcal{O}_{U_i}$. So invertible sheaves trivialised on (U_i) correspond to $\check{H}^1((U_i), \mathcal{O}^*)$, and passing to the limit over covers gives $\check{H}^1(X, \mathcal{O}^*) = H^1(X, \mathcal{O}^*)$ (Remark 24.5.4). The same cocycles with values in GL_r glue the $\mathbb{A}_{U_i}^r$ to a vector bundle, as for manifolds (Proposition 33.2.5). $\square$

51.5 Twisting sheaves on projective schemes

Definition 51.5.1. Let S be a graded ring and M a graded S -module. The sheaf $\widetilde{M}$ on $\text{Proj } S$ has sections over $D_+(f)$ equal to $M_{(f)}$, the degree-zero part of M_f , and stalks $M_{(\mathfrak{p})}$. For $n \in \mathbb{Z}$, $M(n)$ is M with degrees shifted, $M(n)_d = M_{n+d}$, and $\mathcal{O}(n) = S(n)^\sim$. For an $\mathcal{O}$ -module $\mathcal{F}$ put $\mathcal{F}(n) = \mathcal{F} \otimes \mathcal{O}(n)$ and $\Gamma_*(\mathcal{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathcal{F}(n))$. For a scheme X with a closed immersion into $\mathbb{P}_A^r$, $\mathcal{O}_X(1)$ is the restriction of $\mathcal{O}(1)$.

Proposition 51.5.2. *Let S be generated by S_1 as an S_0 -algebra and $X = \text{Proj } S$. Then $\mathcal{O}(n)$ is invertible, $\mathcal{O}(n) \otimes \mathcal{O}(m) \cong \mathcal{O}(n+m)$, and $\widetilde{M}(n) \cong (M(n))^\sim$. For $S = A[x_0, \dots, x_r]$ and $X = \mathbb{P}_A^r$,*

$$\Gamma(X, \mathcal{O}(n)) \cong S_n \quad \text{for all } n \in \mathbb{Z},$$

the homogeneous polynomials of degree n (zero for $n < 0$); so $\Gamma_(\mathcal{O}_X) = S$.*

Proof. Since S is generated by S_1 , the $D_+(f)$ with $f \in S_1$ cover X , and on $D_+(f)$ the module $S(n)_{(f)}$ is free over $S_{(f)}$ with basis f^n . This proves invertibility, and the other isomorphisms follow similarly from $M(n)_{(f)} = M_{(f)} \cdot f^n$. For $\mathbb{P}_A^r$, a global section of $\mathcal{O}(n)$ is a compatible family of degree- n elements $t_i \in S_{x_i}$. The x_i are non-zero-divisors, so all S_{x_i} embed in $S' = S_{x_0 \cdots x_r}$ and compatibility means that the t_i are one element t of $\bigcap_i S_{x_i}$, whose homogeneous components lie in the intersection. An element of S' is a sum of Laurent monomials x^a , $a \in \mathbb{Z}^{r+1}$, with coefficients in A . It lies in S_{x_i} exactly when only monomials with $a_j \geq 0$ for $j \neq i$ occur. Hence $\bigcap_i S_{x_i} = S$, and the degree- n part is S_n . $\square$

Theorem 51.5.3 (Serre). *Let X be a closed subscheme of $\mathbb{P}_A^r$ with A Noetherian, and $\mathcal{F}$ a coherent sheaf on X . Then there is n_0 such that for all $n \geq n_0$ the sheaf $\mathcal{F}(n)$ is generated by finitely many global sections. Consequently $\mathcal{F}$ is a quotient of a finite direct sum of sheaves $\mathcal{O}_X(-n)$.*

Proof. Let $i: X \rightarrow \mathbb{P}_A^r$ be the inclusion. Then $i_*\mathcal{F}$ is coherent (Proposition 51.3.5, with finiteness checked on affines), and $i_*(\mathcal{F}(n)) = (i_*\mathcal{F})(n)$ since $\mathcal{O}_X(1) = i^*\mathcal{O}(1)$. So we may assume $X = \mathbb{P}_A^r$. Cover X by $U_i = D_+(x_i) = X_{x_i}$, where $x_i \in \Gamma(X, \mathcal{O}(1))$, and write $\mathcal{F}|_{U_i} = \widetilde{M_i}$ with M_i generated by finitely many sections s_{ij} . By Lemma 51.3.2(2) with $\mathcal{L} = \mathcal{O}(1)$ and $f = x_i$, for n large each $s_{ij} \otimes x_i^n$ extends to a global section of $\mathcal{F}(n)$, and we can take one n for

the finitely many sections. On U_i , x_i^n generates $\mathcal{O}(n)$, so the sections $s_{ij} \otimes x_i^n$ generate $\mathcal{F}(n)|_{U_i}$. The same holds for all larger n (multiply by x_i). Finally, N global sections generating $\mathcal{F}(n)$ give a surjection $\mathcal{O}^N \rightarrow \mathcal{F}(n)$, that is, $\mathcal{O}(-n)^N \rightarrow \mathcal{F}$. $\square$

Definition 51.5.4. An invertible sheaf $\mathcal{L}$ on a scheme X of finite type over A is *very ample* (relative to A) if $\mathcal{L} \cong i^*\mathcal{O}(1)$ for an immersion $i: X \rightarrow \mathbb{P}_A^r$, and *ample* if for every coherent $\mathcal{F}$ the sheaf $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for $n \gg 0$. By Theorem 51.5.3, very ample sheaves on projective schemes over Noetherian rings are ample. The converse relation, that some power of an ample sheaf is very ample, is in Chapter 53.

51.6 Sheaves of differentials

Definition 51.6.1. Let $f: X \rightarrow Y$ be a morphism of schemes. The *sheaf of relative differentials* $\Omega_{X/Y}$ is the quasi-coherent $\mathcal{O}_X$ -module obtained by gluing the sheaves $(\Omega_{B/A})^\sim$ on affine opens $\text{Spec } B \subseteq f^{-1}(\text{Spec } A)$, together with the derivation $d: \mathcal{O}_X \rightarrow \Omega_{X/Y}$. The gluing is possible because Ω commutes with localisation (Proposition 12.12.2(5)). For X smooth of relative dimension n over Y , the *canonical sheaf* is $\omega_{X/Y} = \Lambda^n \Omega_{X/Y}$, and the *tangent sheaf* is $\mathcal{T}_{X/Y} = \Omega_{X/Y}^\vee$.

- Proposition 51.6.2.**
1. For morphisms $X \rightarrow Y \rightarrow Z$ there is an exact sequence $f^*\Omega_{Y/Z} \rightarrow \Omega_{X/Z} \rightarrow \Omega_{X/Y} \rightarrow 0$.
 2. For a closed subscheme $Y \subseteq X$ over Z with ideal sheaf $\mathcal{I}$ there is an exact sequence $\mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_{X/Z}|_Y \rightarrow \Omega_{Y/Z} \rightarrow 0$ (the conormal sequence).
 3. Ω commutes with base change: $\Omega_{X_{Y'}/Y'} \cong g'^*\Omega_{X/Y}$.
 4. If $X \rightarrow Y$ is smooth of relative dimension n , then $\Omega_{X/Y}$ is locally free of rank n . In particular, a nonsingular variety of dimension n over an algebraically closed field has $\Omega_{X/k}$ locally free of rank n .

Proof. (1), (2) are Proposition 12.12.2(3), (4) on affine opens. (3) holds because $\Omega_{(B \otimes_A A')/A'} \cong \Omega_{B/A} \otimes_A A'$, which follows from the universal property. (4) Locally $B = A[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r)$ with $\det(\partial g_i / \partial x_{n+j})$ invertible. By Proposition 12.12.2(2), (4), $\Omega_{B/A}$ is the quotient of the free module on $dx_1, \dots, dx_{n+r}$ by the relations $dg_i = \sum_j (\partial g_i / \partial x_j) dx_j = 0$. Since the $r \times r$ block for $dx_{n+1}, \dots, dx_{n+r}$ is invertible, these relations express $dx_{n+1}, \dots, dx_{n+r}$ uniquely in terms of $dx_1, \dots, dx_n$, and $\Omega_{B/A}$ is free on $dx_1, \dots, dx_n$. The last statement uses Proposition 50.8.5. $\square$

Theorem 51.6.3 (Euler sequence). On $X = \mathbb{P}_A^r$ there is an exact sequence of locally free sheaves

$$0 \rightarrow \Omega_{X/A} \rightarrow \mathcal{O}(-1)^{r+1} \rightarrow \mathcal{O}_X \rightarrow 0,$$

where the second map sends the basis section e_j to x_j . Consequently $\omega_{\mathbb{P}_A^r/A} \cong \mathcal{O}(-r-1)$.

Proof. Write $\mathcal{O}(-1)^{r+1} = (S(-1)^{r+1})^\sim$ with basis $e_0, \dots, e_r$ of degree 1, and let π be the map $e_j \mapsto x_j$. It is surjective: on U_i , the section e_i/x_i maps to 1. On $U_i = \text{Spec } A[x_0/x_i, \dots, x_r/x_i]$, Ω is free on the $d(x_j/x_i)$, $j \neq i$. Define

$$\varphi_i: \Omega|_{U_i} \rightarrow \mathcal{O}(-1)^{r+1}|_{U_i}, \quad d(x_j/x_i) \mapsto \frac{1}{x_i^2} (x_i e_j - x_j e_i).$$

The composite with π is 0. The kernel of π on U_i consists of the $\sum_j a_j e_j$ (a_j of degree -1) with $\sum_j a_j x_j = 0$, that is, $a_i = -\sum_{j \neq i} a_j x_j/x_i$, and then $\sum_j a_j e_j = \sum_{j \neq i} (a_j x_i) \varphi_i(d(x_j/x_i))$ with $a_j x_i$

of degree 0. The images $\varphi_i(d(x_j/x_i))$ are independent, since the coefficient of e_j is $1/x_i$. So φ_i is an isomorphism onto $\ker \pi|_{U_i}$. On $U_i \cap U_k$, writing $x_j/x_i = (x_j/x_k)(x_k/x_i)$ and $d(x_k/x_i) = -(x_k/x_i)^2 d(x_i/x_k)$ shows $\varphi_k(d(x_j/x_i)) = \varphi_i(d(x_j/x_i))$ by a direct computation. So the φ_i glue. Taking determinants in the exact sequence gives $\omega \otimes \mathcal{O} \cong \det(\mathcal{O}(-1)^{r+1}) = \mathcal{O}(-r-1)$. $\square$

Proposition 51.6.4 (Adjunction). *Let $Y \subseteq X$ be nonsingular varieties over an algebraically closed field, of dimensions $n-1$ and n , with ideal sheaf $\mathcal{I}$. Then $\mathcal{I}$ is invertible, $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{I}|_Y$ is the conormal sheaf, the conormal sequence $0 \rightarrow \mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_X|_Y \rightarrow \Omega_Y \rightarrow 0$ is exact on the left, and*

$$\omega_Y \cong \omega_X \otimes \mathcal{I}^{-1}|_Y.$$

For a nonsingular hypersurface Y of degree d in $\mathbb{P}^n$, $\omega_Y \cong \mathcal{O}_Y(d-n-1)$.

Proof. Near a point of Y , $\mathcal{O}_X$ is a regular local ring, hence a unique factorisation domain (Remark 12.11.3), and the height-one prime of Y is principal, generated by g . So $\mathcal{I}$ is invertible and $\mathcal{I}/\mathcal{I}^2 = \mathcal{I} \otimes \mathcal{O}_Y$. The map $\mathcal{I}/\mathcal{I}^2 \rightarrow \Omega_X|_Y$ sends g to dg . Both sides of the right-exact conormal sequence are locally free, of ranks 1, n and $n-1$ (Proposition 51.6.2(4)). A surjection of locally free sheaves of ranks n and $n-1$ has locally free kernel of rank 1, which is the image of $\mathcal{I}/\mathcal{I}^2$. A surjection between invertible sheaves is an isomorphism, since locally it is multiplication by a function that generates the unit ideal. So the sequence is exact. Taking determinants gives $\omega_X|_Y \cong \omega_Y \otimes \mathcal{I}/\mathcal{I}^2$. For a hypersurface of degree d , $\mathcal{I} = \mathcal{O}(-d)$ and $\omega_{\mathbb{P}^n} = \mathcal{O}(-n-1)$. $\square$

51.7 Exercises

Exercise 51.7.1. Let $X = \operatorname{Spec} R$ for a discrete valuation ring R with fraction field K , and let $\mathcal{F}$ be the sheaf with $\mathcal{F}(X) = 0$ and $\mathcal{F}(\{\eta\}) = K$ on the open generic point. Show that $\mathcal{F}$ is an $\mathcal{O}_X$ -module that is not quasi-coherent, while j_*K for $j: \{\eta\} \rightarrow X$ is quasi-coherent.

Solution. If $\mathcal{F}$ were quasi-coherent, it would be $\widetilde{\Gamma(X, \mathcal{F})} = 0$ (Theorem 51.3.3), but $\mathcal{F}_\eta = K \neq 0$. On the other hand $j_*K = \widetilde{K}$, since $\widetilde{K}(\{\eta\}) = K_{(0)} = K$ and $\widetilde{K}(X) = K$. $\square$

Exercise 51.7.2. Show that $\mathcal{O}(-1)$ on $\mathbb{P}_k^r$ is the sheaf of sections of the tautological line bundle $\{(x, \ell) \in \mathbb{A}^{r+1} \times \mathbb{P}^r : x \in \ell\}$, and that $\mathcal{O}(1)$ has no nonzero section vanishing on a hyperplane twice.

Solution. On U_i the tautological bundle is trivialised by the section $\ell \mapsto (x_0/x_i, \dots, x_r/x_i)$, and on $U_i \cap U_j$ these trivialisations differ by x_i/x_j . This is the cocycle of $\mathcal{O}(-1)$, whose generator on U_i is $1/x_i$: $1/x_j = (x_i/x_j)(1/x_i)$. A section of $\mathcal{O}(1)$ is a linear form by Proposition 51.5.2, and a nonzero linear form vanishes on its hyperplane to order exactly 1. $\square$

Exercise 51.7.3. Using Proposition 51.6.4, show that a nonsingular plane cubic has trivial canonical sheaf and that a nonsingular plane quartic has $\omega \cong \mathcal{O}(1)$, so that $\Gamma(Y, \omega_Y)$ contains the three linear forms. (In fact it is spanned by them, and the genus is 3; see Chapter 55.)

Solution. $d - n - 1 = 3 - 3 = 0$ and $4 - 3 = 1$. The restriction $\Gamma(\mathbb{P}^2, \mathcal{O}(1)) \rightarrow \Gamma(Y, \mathcal{O}_Y(1))$ is injective because a linear form vanishing on Y would be divisible by the quartic equation. So $\Gamma(Y, \omega_Y)$ contains a 3-dimensional space. $\square$

Exercise 51.7.4. Compute $\Omega_{X/k}$ for the cusp $X = \operatorname{Spec} k[x, y]/(y^2 - x^3)$ and show that it is not locally free.

Solution. By Example 12.12.3, $\Omega = (B dx \oplus B dy)/(2y dy - 3x^2 dx)$. Its fibre at the origin has dimension 2, while at other points it has dimension 1 (where $x \neq 0$ or $y \neq 0$, the relation is nonzero modulo the maximal ideal, in characteristic $\neq 2, 3$). As X is reduced, Proposition 51.3.8(3) shows that Ω is not locally free. $\square$

Chapter 52

Cohomology of Coherent Sheaves

52.1 Introduction

The sheaf cohomology of Chapter 24 applied to quasi-coherent sheaves on schemes is computable. On affine schemes it vanishes in positive degrees (Theorem 52.2.3), so on separated schemes it is computed by Čech complexes of affine covers. On projective space everything can be computed explicitly (Theorem 52.3.1). For a projective scheme over a field the cohomology groups of coherent sheaves are finite-dimensional, those of $\mathcal{F}(n)$ vanish for $n \gg 0$, and the Euler characteristic $\chi(\mathcal{F}(n))$ is a polynomial in n , the Hilbert polynomial. Serre duality relates H^i and H^{n-i} . For complex projective manifolds these groups coincide with the Dolbeault cohomology of Chapter 43 by GAGA (Chapter 54), and so compute the Hodge numbers. References: [Har77, Chapter III], [The].

Prerequisites. Chapter 51 (Quasi-coherent and Coherent Sheaves) and Chapter 24 (Sheaf Cohomology).

For an A -module M and $f_1, \dots, f_r \in A$, the Čech complex $\check{C}^\bullet(f; M)$ is $\prod_{i_0} M_{f_{i_0}} \rightarrow \prod_{i_0 < i_1} M_{f_{i_0} f_{i_1}} \rightarrow \dots$ with the alternating differential of Definition 24.5.1. It is the Čech complex of M for the cover of $\bigcup_i D(f_i)$ by the $D(f_i)$.

52.2 Vanishing on affine schemes

Lemma 52.2.1. *If $f_1, \dots, f_r$ generate the unit ideal of A , then the augmented Čech complex $0 \rightarrow M \rightarrow \check{C}^\bullet(f; M)$ is exact for every A -module M .*

Proof. A module that becomes zero after inverting each f_j is zero, because every prime omits some f_j (Proposition 12.3.3). Localisation is exact, so it suffices to show that the complex becomes exact after inverting f_j . After inverting f_j it is the augmented Čech complex of M_{f_j} for the elements $f_1, \dots, f_r$ of A_{f_j} , one of which is a unit. The operator $(k\alpha)_{i_0 \dots i_{p-1}} = \alpha_{ji_0 \dots i_{p-1}}$ is then defined, since $M_{f_j f_{i_0} \dots} = M_{f_{i_0} \dots}$, and $dk + kd = \text{id}$ as in the proof of Lemma 24.5.2. So the localised complex is contractible. $\square$

Lemma 52.2.2 (Cartan). *Let X be a topological space and $\mathcal{B}$ a basis of quasi-compact open sets closed under finite intersections. Let $\mathcal{F}$ be a sheaf of abelian groups such that $\check{H}^p(\mathcal{V}, \mathcal{F}) = 0$ for all $p > 0$, all $U \in \mathcal{B}$ and all finite covers $\mathcal{V}$ of U by members of $\mathcal{B}$. Then $H^p(U, \mathcal{F}) = 0$ for all $U \in \mathcal{B}$ and $p > 0$.*

Proof. Embed $\mathcal{F}$ into an injective sheaf $\mathcal{I}$ (Theorem 24.2.2) and let $\mathcal{Q} = \mathcal{I}/\mathcal{F}$.

Step 1: $\mathcal{I}(U) \rightarrow \mathcal{Q}(U)$ is onto for $U \in \mathcal{B}$. A section $s \in \mathcal{Q}(U)$ lifts locally, and by quasi-compactness we find a finite cover of U by $U_1, \dots, U_m \in \mathcal{B}$ and lifts $t_i \in \mathcal{I}(U_i)$. Then

$t_i - t_j \in \mathcal{F}(U_i \cap U_j)$ is a Čech 1-cocycle, so by hypothesis $t_i - t_j = c_j - c_i$ with $c_i \in \mathcal{F}(U_i)$. The sections $t_i + c_i$ agree on overlaps and glue to a lift of s .

Step 2: $\mathcal{Q}$ satisfies the hypothesis. For a finite cover $\mathcal{V}$ of $U \in \mathcal{B}$ by members of $\mathcal{B}$, all intersections lie in $\mathcal{B}$, so by Step 1 the sequence $0 \rightarrow \check{C}^\bullet(\mathcal{V}, \mathcal{F}) \rightarrow \check{C}^\bullet(\mathcal{V}, \mathcal{I}) \rightarrow \check{C}^\bullet(\mathcal{V}, \mathcal{Q}) \rightarrow 0$ is exact. Injective sheaves are flasque (Lemma 24.3.2), so $\check{H}^p(\mathcal{V}, \mathcal{I}) = 0$ for $p > 0$ (Lemma 24.5.2). The long exact sequence gives $\check{H}^p(\mathcal{V}, \mathcal{Q}) \cong \check{H}^{p+1}(\mathcal{V}, \mathcal{F}) = 0$.

Step 3. By the long exact sequence of $0 \rightarrow \mathcal{F} \rightarrow \mathcal{I} \rightarrow \mathcal{Q} \rightarrow 0$ on U , $H^1(U, \mathcal{F}) = \text{coker}(\mathcal{I}(U) \rightarrow \mathcal{Q}(U)) = 0$ by Step 1, and $H^{p+1}(U, \mathcal{F}) \cong H^p(U, \mathcal{Q})$ for $p \geq 1$. By Step 2 and induction on p , applied to all sheaves satisfying the hypothesis at once, $H^p(U, \mathcal{F}) = 0$ for all $p \geq 1$. $\square$

Theorem 52.2.3 (Serre). *Let $X = \text{Spec } A$ and $\mathcal{F}$ a quasi-coherent sheaf on X . Then $H^p(X, \mathcal{F}) = 0$ for all $p > 0$.*

Proof. Take $\mathcal{B}$ to be the basic open sets $D(g)$. They are quasi-compact (Proposition 50.2.1) and $D(g) \cap D(h) = D(gh)$. A finite cover of $D(g) \cong \text{Spec } A_g$ by basic open sets $D(f_i)$ with $D(f_i) \subseteq D(g)$ is a cover of $\text{Spec } A_g$ by the $D(f_i)$, and the f_i generate the unit ideal of A_g . By Theorem 51.3.3, $\mathcal{F} = \tilde{M}$ and the Čech complex of the cover is $\check{C}^\bullet(f; M_g)$, which is exact in positive degrees by Lemma 52.2.1. Apply Lemma 52.2.2. $\square$

Counterexample 52.2.4. Theorem 52.2.3 needs quasi-coherence. Let $X = \mathbb{A}_k^1 = \text{Spec } k[t]$ with its Zariski topology, and let $0, 1$ be the corresponding closed points. Consider the morphism of sheaves of abelian groups

$$\mathbb{Z}_X \rightarrow (i_0)_*\mathbb{Z} \oplus (i_1)_*\mathbb{Z}$$

to the skyscraper sheaves at 0 and 1 , taking the values of a locally constant function at the two points, and let $\mathcal{K}$ be its kernel. It is surjective on stalks: the stalk of $(i_0)_*\mathbb{Z}$ at a point y is $\mathbb{Z}$ if every open neighbourhood of y contains 0 , that is, if $y = 0$, since $\{0\}$ is closed, and 0 otherwise; and at $y = 0$ the map on stalks is the identity of $\mathbb{Z}$. So $0 \rightarrow \mathcal{K} \rightarrow \mathbb{Z}_X \rightarrow (i_0)_*\mathbb{Z} \oplus (i_1)_*\mathbb{Z} \rightarrow 0$ is exact. Since X is irreducible, every nonempty open subset is irreducible, hence connected, so $\mathbb{Z}_X(U) = \mathbb{Z}$ for all nonempty U , with the identity as restriction maps; thus $\mathbb{Z}_X$ is flasque and $H^1(X, \mathbb{Z}_X) = 0$ (Proposition 24.3.3), and $\Gamma(X, \mathbb{Z}_X) = \mathbb{Z}$ maps to $\mathbb{Z} \oplus \mathbb{Z}$ diagonally. The long exact sequence gives

$$H^1(X, \mathcal{K}) \cong (\mathbb{Z} \oplus \mathbb{Z})/\mathbb{Z} \cong \mathbb{Z} \neq 0,$$

although X is affine. The sheaf $\mathcal{K}$, the extension by zero of $\mathbb{Z}$ from $X \setminus \{0, 1\}$ (Definition 23.4.3), is not a sheaf of $\mathcal{O}_X$ -modules.

Corollary 52.2.5. *Let X be a quasi-compact separated scheme, $\mathcal{U}$ a finite cover by affine open subsets and $\mathcal{F}$ quasi-coherent. Then $H^p(X, \mathcal{F}) \cong \check{H}^p(\mathcal{U}, \mathcal{F})$. In particular $H^p(X, \mathcal{F}) = 0$ for $p \geq |\mathcal{U}|$.*

Proof. Finite intersections of affine opens in a separated scheme are affine (as in the proof of Lemma 51.3.2), so Theorem 52.2.3 and Leray's theorem (Theorem 24.5.3) apply. The alternating Čech complex of a cover with m members vanishes in degrees $\geq m$. $\square$

Remark 52.2.6. Conversely, Serre's criterion says that a Noetherian scheme on which H^1 of every quasi-coherent (even every coherent) ideal sheaf vanishes is affine [Har77, Theorem III.3.7]. Grothendieck's vanishing theorem says that $H^p(X, \mathcal{F}) = 0$ for $p > \dim X$ for every sheaf of abelian groups on a Noetherian space X [Har77, Theorem III.2.7].

52.3 The cohomology of projective space

Theorem 52.3.1. *Let A be a Noetherian ring, $S = A[x_0, \dots, x_r]$ and $X = \mathbb{P}_A^r$ with $r \geq 1$.*

1. $\bigoplus_n H^0(X, \mathcal{O}(n)) \cong S$.
2. $H^i(X, \mathcal{O}(n)) = 0$ for $0 < i < r$ and all n .
3. $H^r(X, \mathcal{O}(n))$ is the free A -module with basis the Laurent monomials $x_0^{a_0} \cdots x_r^{a_r}$ with all $a_i \leq -1$ and $\sum a_i = n$. In particular $H^r(X, \mathcal{O}(-r-1)) \cong A$, spanned by $(x_0 \cdots x_r)^{-1}$, and $H^r(X, \mathcal{O}(n)) = 0$ for $n > -r-1$.
4. The multiplication pairing $H^0(X, \mathcal{O}(n)) \times H^r(X, \mathcal{O}(-n-r-1)) \rightarrow H^r(X, \mathcal{O}(-r-1)) \cong A$ is perfect.

Proof. Let $\mathcal{F} = \bigoplus_n \mathcal{O}(n)$, a quasi-coherent sheaf. By Corollary 52.2.5 with the standard cover $U_i = D_+(x_i)$, $H^i(X, \mathcal{O}(n))$ is the degree- n part of the cohomology of the graded complex

$$\prod_{i_0} S_{x_{i_0}} \rightarrow \prod_{i_0 < i_1} S_{x_{i_0} x_{i_1}} \rightarrow \cdots \rightarrow S_{x_0 \cdots x_r},$$

which is the Čech complex of $\mathcal{F}$. (1) is Proposition 51.5.2. (3) H^r is the cokernel of $\prod_k S_{x_0 \cdots \widehat{x_k} \cdots x_r} \rightarrow S_{x_0 \cdots x_r}$. The target is free on all Laurent monomials, and the image is spanned by those in which some exponent a_k is ≥ 0 . The monomials with all $a_i \leq -1$ give a basis of the cokernel. (4) In the basis of (3), $x^b \cdot x^a$ with $b_i \geq 0$, $\sum b_i = n$, and $a_i \leq -1$, $\sum a_i = -n-r-1$ is zero in H^r unless $b_i + a_i = -1$ for all i . So the monomial bases $\{x^b\}$ and $\{x^{-b-1}\}$ are dual.

(2) By induction on r . The case $r = 1$ is empty. Let $r \geq 2$ and $M^i = \bigoplus_n H^i(X, \mathcal{O}(n))$. Every element of M^i , $i > 0$, is killed by a power of x_r . Indeed, localising the Čech complex at x_r gives the Čech complex of S_{x_r} for the cover of U_r by the affine sets $U_i \cap U_r$. Its cohomology is $H^i(U_r, \mathcal{F}|_{U_r}) = 0$ for $i > 0$, as U_r is affine (Theorem 52.2.3), and localisation is exact. Multiplication by x_r is injective on M^i for $0 < i < r$. Let $H = V_+(x_r) \cong \mathbb{P}_A^{r-1}$. The exact sequence $0 \rightarrow S(-1) \xrightarrow{x_r} S \rightarrow S/(x_r) \rightarrow 0$ gives $0 \rightarrow \mathcal{O}(n-1) \rightarrow \mathcal{O}(n) \rightarrow \mathcal{O}_H(n) \rightarrow 0$, with $H^i(X, \mathcal{O}_H(n)) = H^i(H, \mathcal{O}_H(n))$ (Example 24.7.4). In the long exact sequence

$$H^{i-1}(H, \mathcal{O}_H(n)) \rightarrow H^i(X, \mathcal{O}(n-1)) \xrightarrow{x_r} H^i(X, \mathcal{O}(n)),$$

the left term vanishes for $2 \leq i \leq r-1$ by induction ($0 < i-1 < r-1$). For $i = 1$, the map $H^0(X, \mathcal{O}(n)) = S_n \rightarrow H^0(H, \mathcal{O}_H(n)) = (S/x_r)_n$ is onto, so again x_r is injective on H^1 . An element killed by a power of x_r , on which x_r is injective, is zero. So $M^i = 0$ for $0 < i < r$. $\square$

Corollary 52.3.2. *On $\mathbb{P}_k^1$, $h^0(\mathcal{O}(n)) = \max(n+1, 0)$ and $h^1(\mathcal{O}(n)) = \max(-n-1, 0)$. So $\chi(\mathcal{O}(n)) = n+1$ for all n , and $H^1(\mathbb{P}^1, \mathcal{O}(-2)) \cong k$.*

Proof. Count the monomials $x_0^{a_0} x_1^{a_1}$ with $a_0 + a_1 = n$ and $a_i \geq 0$, respectively $a_i \leq -1$. $\square$

52.4 Finiteness and Serre vanishing

Theorem 52.4.1 (Serre). *Let X be a projective scheme over a Noetherian ring A with a very ample sheaf $\mathcal{O}_X(1)$, and $\mathcal{F}$ a coherent sheaf on X . Then:*

1. $H^i(X, \mathcal{F})$ is a finitely generated A -module for every i ;
2. there is n_0 with $H^i(X, \mathcal{F}(n)) = 0$ for all $i > 0$ and $n \geq n_0$.

Proof. Pushing forward along the closed immersion $X \rightarrow \mathbb{P}_A^r$ changes neither the cohomology nor coherence (Example 24.7.4), so let $X = \mathbb{P}_A^r$. Both statements hold for $i > r$ by Corollary 52.2.5. We use descending induction on i , for all coherent sheaves simultaneously. By Theorem 51.5.3 there is an exact sequence $0 \rightarrow \mathcal{R} \rightarrow \mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$ with $\mathcal{E}$ a finite direct sum of sheaves $\mathcal{O}(q_j)$ and $\mathcal{R}$ coherent (Corollary 51.3.4). By Theorem 52.3.1, $H^i(\mathcal{E})$ is finitely generated and $H^i(\mathcal{E}(n)) = 0$ for $i > 0$ and n large. The exact sequence $H^i(\mathcal{E}(n)) \rightarrow H^i(\mathcal{F}(n)) \rightarrow H^{i+1}(\mathcal{R}(n))$ and the induction hypothesis for $\mathcal{R}$ give both statements for $\mathcal{F}$ in degree i . For (1), A is Noetherian, so an extension of a submodule of a finitely generated module by a quotient of a finitely generated module is finitely generated. $\square$

Corollary 52.4.2. *For a projective scheme X over a field k and $\mathcal{F}$ coherent, all $H^i(X, \mathcal{F})$ are finite-dimensional, and they vanish for $i > \dim X$. In particular $\Gamma(X, \mathcal{O}_X)$ is a finite-dimensional k -algebra, equal to k if X is geometrically integral.*

Proof. Finiteness is Theorem 52.4.1(1), and vanishing above the dimension is Grothendieck's theorem (Remark 52.2.6). If X is integral, $\Gamma(X, \mathcal{O}_X)$ is a domain finite over k , hence a field finite over k . If X is geometrically integral, its base change to $\bar{k}$ is $\Gamma(X_{\bar{k}}, \mathcal{O}) = \bar{k}$ by Theorem 49.4.6 (cohomology commutes with flat base change of fields, by the Čech description), so $\Gamma(X, \mathcal{O}_X) = k$. $\square$

52.5 Euler characteristic and Hilbert polynomial

Definition 52.5.1. For a coherent sheaf $\mathcal{F}$ on a projective scheme X over a field k , the *Euler characteristic* is $\chi(\mathcal{F}) = \sum_i (-1)^i h^i(\mathcal{F})$, where $h^i = \dim_k H^i$. It is additive in short exact sequences, by the long exact sequence.

Theorem 52.5.2. *Let $X \subseteq \mathbb{P}_k^r$ be a projective scheme over an infinite field k and $\mathcal{F}$ a coherent sheaf on X . There is a polynomial $P_{\mathcal{F}} \in \mathbb{Q}[t]$, the Hilbert polynomial, with $\chi(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for all $n \in \mathbb{Z}$. Its degree is $\dim \operatorname{Supp} \mathcal{F}$ ($-\infty$ if $\mathcal{F} = 0$), and $h^0(\mathcal{F}(n)) = P_{\mathcal{F}}(n)$ for $n \gg 0$.*

Proof. Induction on $d = \dim \operatorname{Supp} \mathcal{F}$. If $d \leq 0$, the support is finite, $\mathcal{O}(1)$ is trivial on a neighbourhood of it, so $\mathcal{F}(n) \cong \mathcal{F}$ and χ is constant, equal to $h^0(\mathcal{F})$, which is nonzero if $\mathcal{F} \neq 0$ (a sheaf with finite support is a sum of skyscrapers). Let $d \geq 1$. The *associated points* of $\mathcal{F}$, the points x such that $\mathfrak{m}_x$ is an associated prime of $\mathcal{F}_x$, are finite in number (Proposition 12.8.2, on a finite affine cover). Since k is infinite, there is a linear form h vanishing at none of them: S_1 is not a finite union of proper subspaces. Then multiplication by h is injective on $\mathcal{F}(-1) \rightarrow \mathcal{F}$, because its kernel would have an associated point among those of $\mathcal{F}$. So there is an exact sequence

$$0 \rightarrow \mathcal{F}(n-1) \xrightarrow{h} \mathcal{F}(n) \rightarrow \mathcal{F}_H(n) \rightarrow 0, \quad \mathcal{F}_H = \mathcal{F}/h\mathcal{F}.$$

$\operatorname{Supp} \mathcal{F}_H = \operatorname{Supp} \mathcal{F} \cap V_+(h)$ has dimension $d-1$ by Theorem 49.5.5, since h vanishes at no generic point of a component of the support (these are associated points). By induction $\chi(\mathcal{F}(n)) - \chi(\mathcal{F}(n-1)) = P_{\mathcal{F}_H}(n)$ is a polynomial of degree $d-1$, so $\chi(\mathcal{F}(n))$ is a polynomial of degree d . The last statement follows from Theorem 52.4.1(2). $\square$

Definition 52.5.3. For a closed subscheme $X \subseteq \mathbb{P}_k^r$ of dimension d with Hilbert polynomial $P_X = P_{\mathcal{O}_X}$, the *degree* of X is $d!$ times the leading coefficient of P_X , and the *arithmetic genus* is $p_a(X) = (-1)^d (P_X(0) - 1)$.

Example 52.5.4. 1. $P_{\mathbb{P}^r}(n) = \binom{n+r}{r}$ for all n , by Theorem 52.3.1. So $\mathbb{P}^r$ has degree 1 and $p_a = 0$.

2. For a hypersurface $X = V_+(f)$ of degree d , the sequence $0 \rightarrow \mathcal{O}(n-d) \rightarrow \mathcal{O}(n) \rightarrow \mathcal{O}_X(n) \rightarrow 0$ gives $P_X(n) = \binom{n+r}{r} - \binom{n-d+r}{r}$, of degree $r-1$ with leading coefficient $d/(r-1)!$. So X has degree d .
3. For a plane curve of degree d , $P_X(n) = dn + 1 - \frac{(d-1)(d-2)}{2}$, so $p_a = \frac{(d-1)(d-2)}{2}$: 0 for lines and conics, 1 for cubics, 3 for quartics.
4. The twisted cubic $C = v_3(\mathbb{P}^1) \subseteq \mathbb{P}^3$ has $\mathcal{O}_C(n) = \mathcal{O}_{\mathbb{P}^1}(3n)$, so $P_C(n) = 3n + 1$: degree 3, genus 0.

52.6 Serre duality

Theorem 52.6.1 (Serre duality on projective space). *Let $X = \mathbb{P}_k^r$ and $\omega = \mathcal{O}(-r-1)$, with the isomorphism $H^r(X, \omega) \cong k$ of Theorem 52.3.1. For every coherent $\mathcal{F}$ the pairing $\text{Hom}(\mathcal{F}, \omega) \times H^r(X, \mathcal{F}) \rightarrow H^r(X, \omega) \cong k$ is perfect.*

Proof. Both $\mathcal{F} \mapsto \text{Hom}(\mathcal{F}, \omega)$ and $\mathcal{F} \mapsto H^r(X, \mathcal{F})^\vee$ are contravariant and left exact: the second because $H^{r+1} = 0$, so H^r is right exact. The pairing gives a natural transformation between them, which is an isomorphism for $\mathcal{F} = \mathcal{O}(q)$ by Theorem 52.3.1(4), since $\text{Hom}(\mathcal{O}(q), \omega) = H^0(\mathcal{O}(-q-r-1))$, and hence for finite sums of such. Every coherent $\mathcal{F}$ has a presentation $\mathcal{E}_1 \rightarrow \mathcal{E}_0 \rightarrow \mathcal{F} \rightarrow 0$ by finite sums of twisting sheaves (Theorem 51.5.3, applied twice). By the five lemma the transformation is an isomorphism for $\mathcal{F}$. $\square$

Theorem 52.6.2 (Serre duality). *Let X be a nonsingular projective variety of dimension n over an algebraically closed field k , with canonical sheaf $\omega_X = \Lambda^n \Omega_{X/k}$. There is an isomorphism $H^n(X, \omega_X) \cong k$, and for every locally free sheaf $\mathcal{E}$ and every i the pairing*

$$H^i(X, \mathcal{E}) \times H^{n-i}(X, \mathcal{E}^\vee \otimes \omega_X) \rightarrow H^n(X, \omega_X) \cong k$$

is perfect. More generally $\text{Ext}^i(\mathcal{F}, \omega_X) \cong H^{n-i}(X, \mathcal{F})^\vee$ for coherent $\mathcal{F}$.

This is [Har77, Theorem III.7.6 and Corollary III.7.7], where it is deduced from Theorem 52.6.1 and a computation of Ext sheaves for the embedding $X \subseteq \mathbb{P}^r$, and [Har77, Corollary III.7.12] for the identification of the dualising sheaf with ω_X . For $k = \mathbb{C}$, via GAGA (Chapter 54), it agrees with the analytic Serre duality of Theorem 43.3.4.

Corollary 52.6.3. *For a nonsingular projective variety over k put $h^{p,q} = \dim H^q(X, \Omega_X^p)$. Then $h^{p,q} = h^{n-p, n-q}$, and $\chi(\omega_X) = (-1)^n \chi(\mathcal{O}_X)$.*

Proof. Ω^p is locally free, and $(\Omega^p)^\vee \otimes \omega \cong \Omega^{n-p}$ by the perfect pairing $\Omega^p \otimes \Omega^{n-p} \rightarrow \omega$. Apply Theorem 52.6.2. For the second, take $\mathcal{E} = \mathcal{O}_X$. $\square$

52.7 Semicontinuity and base change

Theorem 52.7.1 (Semicontinuity). *Let $f: X \rightarrow Y$ be a projective morphism of Noetherian schemes and $\mathcal{F}$ a coherent sheaf on X , flat over Y . For $y \in Y$ let $\mathcal{F}_y$ be the restriction to the fibre X_y . Then:*

1. $y \mapsto h^i(X_y, \mathcal{F}_y)$ is upper semicontinuous for every i ;
2. $y \mapsto \chi(\mathcal{F}_y)$ is locally constant, and so is the Hilbert polynomial of $\mathcal{F}_y$;
3. (cohomology and base change) if Y is integral and $h^i(X_y, \mathcal{F}_y)$ is constant, then $R^i f_* \mathcal{F}$ is locally free and $R^i f_* \mathcal{F} \otimes k(y) \cong H^i(X_y, \mathcal{F}_y)$ for all y .

This is [Har77, Theorem III.12.8, Corollary III.12.9 and Theorem III.9.9]. The proof represents the cohomology of all fibres by one bounded complex of finitely generated locally free sheaves on Y . The theorem implies that Hodge numbers are upper semicontinuous in algebraic families; in the Kähler case they are in fact constant, by the Hodge decomposition and the constancy of Betti numbers in smooth families (Chapter 66).

Counterexample 52.7.2. Individual h^i can jump, although χ cannot. For the family of line bundles $\mathcal{L}_t$ of degree 0 on a fixed elliptic curve E , parametrised by $t \in E$ (the Poincaré bundle on $E \times E$), one has $h^0(\mathcal{L}_t) = 1$ if $\mathcal{L}_t$ is trivial and 0 otherwise, while $\chi(\mathcal{L}_t) = 0$ is constant (Chapter 55).

52.8 Exercises

Exercise 52.8.1. Compute $H^1(\mathbb{A}_k^2 \setminus \{0\}, \mathcal{O})$ with the Čech complex of the cover $D(x) \cup D(y)$, and deduce again that $\mathbb{A}_k^2 \setminus \{0\}$ is not affine.

Solution. The complex is $k[x, y]_x \times k[x, y]_y \rightarrow k[x, y]_{xy}$, $(a, b) \mapsto b - a$. Its cokernel has basis the monomials $x^{-i}y^{-j}$ with $i, j \geq 1$, so H^1 is infinite-dimensional. The scheme is quasi-compact and separated, so the Čech complex computes cohomology (Corollary 52.2.5), and it is not affine by Theorem 52.2.3. $\square$

Exercise 52.8.2. Let $C \subseteq \mathbb{P}_k^2$ be a plane curve of degree $d \geq 1$. Show that $h^0(\mathcal{O}_C) = 1$ and $h^1(\mathcal{O}_C) = \frac{(d-1)(d-2)}{2}$.

Solution. The sequence $0 \rightarrow \mathcal{O}(-d) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_C \rightarrow 0$ and Theorem 52.3.1 give $0 \rightarrow k \rightarrow H^0(\mathcal{O}_C) \rightarrow H^1(\mathcal{O}(-d)) = 0 \rightarrow 0$ and $0 \rightarrow H^1(\mathcal{O}_C) \rightarrow H^2(\mathcal{O}(-d)) \rightarrow H^2(\mathcal{O}) = 0$. The space $H^2(\mathbb{P}^2, \mathcal{O}(-d))$ has basis the monomials $x^a y^b z^c$ with $a, b, c \leq -1$ and $a + b + c = -d$, and there are $\binom{d-1}{2}$ of them. $\square$

Exercise 52.8.3. Show that $\chi(\mathbb{P}^r, \mathcal{O}(n)) = \binom{n+r}{r}$ for all $n \in \mathbb{Z}$, interpreting the binomial coefficient as the polynomial $\frac{(n+1)\cdots(n+r)}{r!}$, and check the value at $n = -r - 1$ against Serre duality.

Solution. For $n \geq 0$ only H^0 contributes and has dimension $\binom{n+r}{r}$. For $-r - 1 < n < 0$ all cohomology vanishes and the polynomial vanishes too. For $n \leq -r - 1$, only H^r contributes, with dimension equal to the number of monomials with exponents ≤ -1 summing to n , which is $\binom{-n-1}{r}$. This equals $(-1)^r \binom{n+r}{r}$ as polynomials in n . At $n = -r - 1$ we get $(-1)^r$, matching $h^r(\omega) = h^0(\mathcal{O}) = 1$. $\square$

Exercise 52.8.4. Let $X \subseteq \mathbb{P}_k^r$ be a projective variety of dimension $d \geq 1$ over an infinite field, and H a hyperplane not containing X . Show that $X \cap H$ has the same degree as X .

Solution. From $0 \rightarrow \mathcal{O}_X(n-1) \rightarrow \mathcal{O}_X(n) \rightarrow \mathcal{O}_{X \cap H}(n) \rightarrow 0$ we get $P_{X \cap H}(n) = P_X(n) - P_X(n-1)$. If P_X has leading term $e n^d / d!$, the difference has leading term $e n^{d-1} / (d-1)!$. So both have degree e . $\square$

Chapter 53

Divisors and Line Bundles

53.1 Introduction

Divisors are the codimension-one cycles. They are the first case of the algebraic cycles whose cohomology classes the Hodge conjecture is about, and the case in which the conjecture is a theorem (the Lefschetz $(1, 1)$ -theorem, Chapter 65). This chapter treats Weil divisors and the class group, Cartier divisors and their identification with line bundles, linear systems and morphisms to projective space, ample line bundles, intersection numbers on surfaces, and the Néron–Severi group. References: [Har77, Sections II.6, II.7 and V.1], [Laz04, Chapter 1], [Bea96, Chapter I].

Prerequisites. Chapter 52 (Cohomology of Coherent Sheaves).

Throughout, schemes are Noetherian and separated. $K(X)$ is the function field of an integral scheme and $\mathcal{K}_X$ the constant sheaf $K(X)$.

53.2 Weil divisors

Definition 53.2.1. Let X be an integral scheme that is *regular in codimension one*: every local ring $\mathcal{O}_{X,x}$ of dimension one is regular, hence a discrete valuation ring (Theorem 12.11.2). A *prime divisor* is a closed integral subscheme Y of codimension one; its generic point η_Y has local ring a DVR with valuation v_Y on $K(X)$. A *Weil divisor* is an element $D = \sum n_Y Y$ of the free abelian group $\text{Div}(X)$ on the prime divisors; it is *effective*, $D \geq 0$, if all $n_Y \geq 0$. For $f \in K(X)^*$ put $(f) = \sum_Y v_Y(f) Y$, a *principal divisor*. Divisors D, D' are *linearly equivalent*, $D \sim D'$, if $D - D'$ is principal, and the *class group* is $\text{Cl}(X) = \text{Div}(X)/\{\text{principal divisors}\}$.

Lemma 53.2.2. For $f \in K(X)^*$, $v_Y(f) = 0$ for all but finitely many Y , so (f) is a divisor, and $f \mapsto (f)$ is a homomorphism $K(X)^* \rightarrow \text{Div}(X)$.

Proof. f is a regular function with regular inverse on some nonempty open U . The complement $X \setminus U$ is a proper closed subset of the Noetherian space X , so it contains only finitely many prime divisors (they are among its finitely many irreducible components). For Y meeting U , f is a unit at η_Y , so $v_Y(f) = 0$. Additivity is $v_Y(fg) = v_Y(f) + v_Y(g)$. $\square$

Proposition 53.2.3. 1. Let A be a Noetherian domain. If A is a unique factorisation domain, then $\text{Cl}(\text{Spec } A) = 0$. In particular $\text{Cl}(\mathbb{A}_k^n) = 0$.

2. If $Z \subseteq X$ is closed and $U = X \setminus Z$, there is an exact sequence $\bigoplus \mathbb{Z} Z_i \rightarrow \text{Cl}(X) \rightarrow \text{Cl}(U) \rightarrow 0$, the sum running over the irreducible components Z_i of Z of codimension one.
3. For $X = \mathbb{P}_k^n$, the degree $\deg \sum n_Y Y = \sum n_Y \deg Y$, where $\deg Y$ is the degree of a homogeneous generator of the ideal of Y , induces $\text{Cl}(\mathbb{P}_k^n) \cong \mathbb{Z}$, generated by a hyperplane H .

Proof. (1) A prime divisor of $\text{Spec } A$ corresponds to a prime $\mathfrak{p}$ of height one. It contains an irreducible factor f of any of its nonzero elements, and (f) is a nonzero prime in a UFD, so $\mathfrak{p} = (f)$. Then $Y = (f)$ is principal, because $v_{(f)}(f) = 1$ and $v_{(g)}(f) = 0$ for irreducibles g not associate to f . (2) $Y \mapsto Y \cap U$ gives a surjection $\text{Div}(X) \rightarrow \text{Div}(U)$, since a prime divisor of U is the trace of its closure. Its kernel is generated by the Z_i , and principal divisors map to principal divisors, $(f)|_U = (f|_U)$. (3) A prime divisor of $\mathbb{P}^n$ is $V_+(g)$ with g irreducible homogeneous, by Theorem 49.5.4 applied to the cone. For $f = g/h$ with g, h homogeneous of the same degree, (f) has degree $\deg g - \deg h = 0$, so $\deg$ is defined on Cl . It is onto, as $\deg H = 1$. If $D = \sum n_i Y_i$ with $Y_i = V_+(g_i)$ has degree 0, then $f = \prod_i g_i^{n_i}$ is a quotient of homogeneous polynomials of the same degree, hence an element of $K(\mathbb{P}^n)^*$, and $(f) = D$ because $v_{Y_i}(g_i) = \delta_{ij}$. So $\deg$ is injective. $\square$

Example 53.2.4. Let k be a field of characteristic $\neq 2$ and $X = \text{Spec } k[x, y, z]/(xy - z^2)$, the quadric cone. It is normal, and the line $Y = \{y = z = 0\}$ is a prime divisor with $2Y = (y)$, since y vanishes to order 2 along Y (on Y , x is a unit near the generic point and $y = z^2/x$). One shows $\text{Cl}(X) \cong \mathbb{Z}/2\mathbb{Z}$, generated by Y [Har77, Example II.6.5.2]. So Y is not locally principal at the vertex, although $2Y$ is.

53.3 Cartier divisors and invertible sheaves

Definition 53.3.1. Let X be an integral scheme. A *Cartier divisor* is a global section of $\mathcal{K}_X^*/\mathcal{O}_X^*$. It is given by an open cover (U_i) and $f_i \in K(X)^*$ with $f_i/f_j \in \mathcal{O}^*(U_i \cap U_j)$; the f_i are *local equations*. It is *principal* if it is the image of some $f \in K(X)^*$, and *effective* if all $f_i \in \mathcal{O}(U_i)$. The group of Cartier divisors modulo principal ones is $\text{CaCl}(X)$. For a Cartier divisor D , $\mathcal{O}(D) \subseteq \mathcal{K}_X$ is the subsheaf generated by f_i^{-1} on U_i .

Proposition 53.3.2. Let X be an integral scheme.

1. $\mathcal{O}(D)$ is invertible, $\mathcal{O}(D - D') \cong \mathcal{O}(D) \otimes \mathcal{O}(D')^{-1}$, and $D \mapsto \mathcal{O}(D)$ induces an isomorphism $\text{CaCl}(X) \cong \text{Pic}(X)$.
2. If X is moreover regular in codimension one and locally factorial, meaning that all local rings are unique factorisation domains, for instance if X is regular (Remark 12.11.3), then Weil and Cartier divisors coincide, and $\text{Cl}(X) \cong \text{CaCl}(X) \cong \text{Pic}(X)$.

Proof. (1) On U_i , $\mathcal{O}(D)$ is free with generator f_i^{-1} , so it is invertible, and multiplication of local equations corresponds to tensor product of sub- $\mathcal{O}_X$ -modules of $\mathcal{K}_X$. $\mathcal{O}(D) \cong \mathcal{O}_X$ if and only if there is a global generator f^{-1} , if and only if D is principal, so the map is injective on classes. It is onto: an invertible sheaf $\mathcal{L}$ on an integral scheme embeds in $\mathcal{L} \otimes \mathcal{K}_X \cong \mathcal{K}_X$ (a constant sheaf, as X is irreducible), and a local generator e_i of $\mathcal{L}$ becomes some $f_i^{-1} \in K(X)^*$, so $\mathcal{L} \cong \mathcal{O}(D)$ with local equations f_i . (2) A Cartier divisor gives the Weil divisor $\sum_Y v_Y(f_Y)Y$, where f_Y is a local equation near η_Y . Conversely a Weil divisor is locally principal: near x , each prime divisor through x corresponds to a height-one prime of the UFD $\mathcal{O}_{X,x}$, which is principal, generated by some g . So D has the local equation $\prod g^{n_Y}$ on a neighbourhood of x . These constructions are inverse, and principal divisors correspond. $\square$

Example 53.3.3. $\text{Pic}(\mathbb{P}_k^n) \cong \mathbb{Z}$, generated by $\mathcal{O}(1) = \mathcal{O}(H)$, by Propositions 53.2.3(3) and 53.3.2. For the quadric cone of Example 53.2.4, $\text{Pic}(X) = 0$ while $\text{Cl}(X) = \mathbb{Z}/2$: the line Y is a Weil divisor that is not Cartier.

53.4 Linear systems and morphisms to projective space

Proposition 53.4.1. *Let X be a scheme over a ring A . Morphisms $\varphi: X \rightarrow \mathbb{P}_A^n$ over A correspond bijectively to invertible sheaves $\mathcal{L}$ with sections $s_0, \dots, s_n \in \Gamma(X, \mathcal{L})$ generating $\mathcal{L}$, up to isomorphism of such data. The correspondence sends φ to $\varphi^*\mathcal{O}(1)$ and $s_i = \varphi^*x_i$.*

Proof. Given $(\mathcal{L}, s_i)$, the open sets X_{s_i} cover X because the s_i generate. On X_{s_i} the quotients s_j/s_i are regular functions, and the ring map $A[x_0/x_i, \dots, x_n/x_i] \rightarrow \Gamma(X_{s_i}, \mathcal{O})$, $x_j/x_i \mapsto s_j/s_i$, defines $X_{s_i} \rightarrow U_i$ (Theorem 50.2.4). These glue, since the quotients satisfy the relations of the gluing of $\mathbb{P}^n$. Then $\varphi^*\mathcal{O}(1) \cong \mathcal{L}$, the generator x_i of $\mathcal{O}(1)$ on U_i being sent to s_i . Conversely φ^*x_i generate $\varphi^*\mathcal{O}(1)$, and φ is recovered on each $\varphi^{-1}(U_i) = X_{\varphi^*x_i}$ from the quotients. $\square$

Definition 53.4.2. Let X be a projective variety over an algebraically closed field k and D a Cartier divisor. The *complete linear system* $|D|$ is the set of effective divisors linearly equivalent to D . A *linear system* is a subset corresponding to a linear subspace of $\Gamma(X, \mathcal{O}(D))$ as below, and its *base points* are the common points of its members.

Proposition 53.4.3. *In the situation of Definition 53.4.2, sending a nonzero $s \in \Gamma(X, \mathcal{O}(D))$ to its divisor of zeros $(s)_0$ gives a bijection $(\Gamma(X, \mathcal{O}(D)) \setminus 0)/k^* \cong |D|$. So $|D|$ is a projective space of dimension $h^0(\mathcal{O}(D)) - 1$, finite by Corollary 52.4.2.*

Proof. A section s of $\mathcal{O}(D) \subseteq \mathcal{K}$ is an element $f \in K(X)$ with $f_i f$ regular on each U_i , and $(s)_0$ is the effective divisor with local equations $f_i f$, which equals $D + (f)$. Conversely, if $D' = D + (f) \geq 0$, then $f \in \Gamma(X, \mathcal{O}(D))$. Two sections give the same divisor if and only if their quotient is a unit on X , that is, an element of $\Gamma(X, \mathcal{O}_X)^* = k^*$ (Corollary 52.4.2). $\square$

Proposition 53.4.4. *Let X be a projective scheme over an algebraically closed field k and $\varphi: X \rightarrow \mathbb{P}_k^n$ the morphism given by sections $s_0, \dots, s_n$ generating $\mathcal{L}$, spanning $V \subseteq \Gamma(X, \mathcal{L})$. Then φ is a closed immersion if and only if V separates points (for closed $P \neq Q$ there is $s \in V$ vanishing at P but not at Q) and separates tangent vectors (for every closed P , the sections vanishing at P span $\mathfrak{m}_P \mathcal{L}_P / \mathfrak{m}_P^2 \mathcal{L}_P$).*

This is [Har77, Proposition II.7.3]; it is the algebraic version of Lemma 47.6.2. The proof uses that a proper morphism that is injective on closed points and on tangent spaces is a closed immersion, by Nakayama's lemma applied to $\varphi_*\mathcal{O}_X$.

53.5 Ample line bundles

Theorem 53.5.1. *Let X be of finite type over a Noetherian ring A and $\mathcal{L}$ invertible. Then $\mathcal{L}$ is ample (Definition 51.5.4) if and only if $\mathcal{L}^m$ is very ample over A for some $m > 0$.*

Proof. If $\mathcal{L}^m$ is very ample, X is an open subscheme of a projective scheme $\bar{X}$ with $\mathcal{L}^m = \mathcal{O}(1)|_X$. A coherent $\mathcal{F}$ on X extends to a coherent sheaf on $\bar{X}$ [Har77, Exercise II.5.15]. Applying Theorem 51.5.3 to the extensions of $\mathcal{F}, \mathcal{F} \otimes \mathcal{L}, \dots, \mathcal{F} \otimes \mathcal{L}^{m-1}$ shows that $\mathcal{F} \otimes \mathcal{L}^n$ is globally generated for $n \gg 0$.

Conversely, let $\mathcal{L}$ be ample. *Step 1: affine cover by sets X_{s_i} .* Let $P \in X$, U an affine neighbourhood on which $\mathcal{L}$ is trivial, and $\mathcal{I}$ the ideal sheaf of the reduced closed subscheme $X \setminus U$. For some $n > 0$, $\mathcal{I} \otimes \mathcal{L}^n$ is globally generated, so there is $s \in \Gamma(X, \mathcal{I} \otimes \mathcal{L}^n) \subseteq \Gamma(X, \mathcal{L}^n)$ with $s_P \notin \mathfrak{m}_P \mathcal{L}_P^n$. Then $P \in X_s \subseteq U$, and X_s is a basic open subset of the affine U (where s becomes a function), hence affine. By quasi-compactness finitely many $X_{s_1}, \dots, X_{s_k}$ cover X , and replacing the s_i by powers we may assume they are all sections of the same $\mathcal{L}^n$.

Step 2: coordinates. Let $B_i = \Gamma(X_{s_i}, \mathcal{O})$, a finitely generated A -algebra with generators b_{ij} . By Lemma 51.3.2(2) there is N such that $s_i^N b_{ij}$ extends to a global section c_{ij} of $\mathcal{L}^{nN}$ for all i, j . The sections s_i^N and c_{ij} of $\mathcal{L}^{nN}$ generate it, because the s_i^N have no common

zero. Let $\varphi: X \rightarrow \mathbb{P}_A^M$ be the morphism they define (Proposition 53.4.1), with homogeneous coordinates y_i, y_{ij} corresponding to s_i^N, c_{ij} .

Step 3: φ is an immersion. We have $\varphi^{-1}(D_+(y_i)) = X_{s_i}$, and the map $X_{s_i} \rightarrow D_+(y_i)$ corresponds to the ring map $A[y/y_i] \rightarrow B_i$ sending y_{ij}/y_i to $c_{ij}/s_i^N = b_{ij}$. It is surjective, so $X_{s_i} \rightarrow D_+(y_i)$ is a closed immersion. Hence φ is a closed immersion of X into the open subset $\bigcup_i D_+(y_i)$ of $\mathbb{P}_A^M$, that is, an immersion, and $\mathcal{L}^{nN} = \varphi^*\mathcal{O}(1)$ is very ample. $\square$

Theorem 53.5.2 (Cohomological criterion). *Let X be proper over a Noetherian ring A and $\mathcal{L}$ invertible. Then $\mathcal{L}$ is ample if and only if for every coherent $\mathcal{F}$ there is n_0 with $H^i(X, \mathcal{F} \otimes \mathcal{L}^n) = 0$ for all $i > 0$ and $n \geq n_0$.*

Proof. If $\mathcal{L}$ is ample, some $\mathcal{L}^m$ is very ample (Theorem 53.5.1). As X is proper, the immersion is closed, so X is projective. Apply Theorem 52.4.1(2) to $\mathcal{F} \otimes \mathcal{L}^r, 0 \leq r < m$.

Conversely, let P be a closed point with ideal sheaf $\mathcal{I}_P$. From $0 \rightarrow \mathcal{I}_P \mathcal{F} \rightarrow \mathcal{F} \rightarrow \mathcal{F} \otimes k(P) \rightarrow 0$ and $H^1(\mathcal{I}_P \mathcal{F} \otimes \mathcal{L}^n) = 0$ for n large, $\Gamma(\mathcal{F} \otimes \mathcal{L}^n) \rightarrow \mathcal{F} \otimes \mathcal{L}^n \otimes k(P)$ is onto. By Nakayama's lemma (Proposition 51.3.8(1)), $\mathcal{F} \otimes \mathcal{L}^n$ is then generated by global sections on a neighbourhood of P , a neighbourhood that depends on n . Apply this to $\mathcal{F} = \mathcal{O}_X$ to find n_0 and a neighbourhood V of P on which $\mathcal{L}^{n_0}$ is globally generated. Apply it to $\mathcal{F} \otimes \mathcal{L}^r, 0 \leq r < n_0$, to find n_1 and neighbourhoods V_r on which $\mathcal{F} \otimes \mathcal{L}^{r+n_1}$ is globally generated. On $W_P = V \cap \bigcap_r V_r$, every $\mathcal{F} \otimes \mathcal{L}^n$ with $n \geq n_1$ is globally generated, being a tensor product $(\mathcal{F} \otimes \mathcal{L}^{r+n_1}) \otimes (\mathcal{L}^{n_0})^q$. The W_P for closed points P cover the quasi-compact X , since every nonempty closed subset contains a closed point, so finitely many suffice. $\square$

Example 53.5.3. 1. $\mathcal{O}(d)$ on $\mathbb{P}_k^n, n \geq 1$, is ample if and only if $d > 0$: for $d > 0$ it is very ample (Example 49.4.7(2)), and for $d \leq 0$ and $\mathcal{F} = \mathcal{O}(-1)$ the sheaf $\mathcal{F} \otimes \mathcal{O}(d)^m = \mathcal{O}(dm - 1)$ is nonzero without nonzero global sections (Proposition 51.5.2).

2. The restriction of an ample sheaf to a closed subscheme is ample, and a tensor product of ample sheaves is ample.
3. For a smooth projective variety over $\mathbb{C}$, a line bundle is ample if and only if the associated holomorphic line bundle is positive (Definition 42.4.2). This combines Theorem 47.6.5 with GAGA (Chapter 54).

53.6 Intersection numbers on surfaces

Definition 53.6.1. Let X be a nonsingular projective surface over an algebraically closed field k . For invertible sheaves $\mathcal{L}, \mathcal{M}$ put

$$(\mathcal{L} \cdot \mathcal{M}) = \chi(\mathcal{O}_X) - \chi(\mathcal{L}^{-1}) - \chi(\mathcal{M}^{-1}) + \chi(\mathcal{L}^{-1} \otimes \mathcal{M}^{-1}),$$

and for divisors $D \cdot E = (\mathcal{O}(D) \cdot \mathcal{O}(E)), D^2 = D \cdot D$.

Proposition 53.6.2. *The pairing $(D, E) \mapsto D \cdot E$ is symmetric, and for an effective divisor C (for instance a curve) and any divisor D ,*

$$D \cdot C = \chi(\mathcal{O}_C) - \chi(\mathcal{O}_X(-D)|_C).$$

Proof. Symmetry is clear. Tensor $0 \rightarrow \mathcal{O}(-C) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_C \rightarrow 0$ with $\mathcal{O}$ and with $\mathcal{O}(-D)$ and use additivity of χ : $\chi(\mathcal{O}) - \chi(\mathcal{O}(-C)) = \chi(\mathcal{O}_C)$ and $\chi(\mathcal{O}(-D)) - \chi(\mathcal{O}(-D-C)) = \chi(\mathcal{O}(-D)|_C)$. Subtracting the second equation from the first gives the formula. $\square$

Theorem 53.6.3. *The pairing $(D, E) \mapsto D \cdot E$ is bilinear on $\text{Pic}(X)$. For an irreducible curve C , $D \cdot C = \deg(\mathcal{O}(D)|_{\tilde{C}})$, the degree of the pullback to the normalisation. If C and D are curves meeting transversally in finitely many points, then $C \cdot D = \#(C \cap D)$, and if C, D are distinct irreducible curves, $C \cdot D \geq 0$.*

This is [Har77, Theorem V.1.1, Proposition V.1.4 and Exercise V.1.1]; see also [Bea96, Chapter I], where Definition 53.6.1 is taken as the definition. By Proposition 53.6.2 and the Riemann–Roch theorem on curves (Chapter 55), $D \cdot C$ is the degree of $\mathcal{O}(D)|_C$, which is additive in D . Linearity in the other variable follows by writing any divisor as a difference of very ample ones, which are linearly equivalent to nonsingular curves by Bertini’s theorem.

Example 53.6.4. 1. On $\mathbb{P}^2$, $H \cdot H = \chi(\mathcal{O}) - 2\chi(\mathcal{O}(-1)) + \chi(\mathcal{O}(-2)) = 1 - 0 + 0 = 1$ by Theorem 52.3.1. By bilinearity, curves of degrees d and e satisfy $C \cdot D = de$ (Bézout’s theorem).

2. On $\mathbb{P}^1 \times \mathbb{P}^1$, with fibres $F_1 = \{pt\} \times \mathbb{P}^1$ and $F_2 = \mathbb{P}^1 \times \{pt\}$, $\chi(\mathcal{O}(a, b)) = (a+1)(b+1)$ by the Künneth formula for the Čech complexes, and the definition gives $F_1^2 = F_2^2 = 0$ and $F_1 \cdot F_2 = 1$.

3. If $\pi: \tilde{X} \rightarrow X$ is the blow-up of a point with exceptional curve $E \cong \mathbb{P}^1$, then $E^2 = -1$, since $\mathcal{O}(E)|_E \cong \mathcal{O}_{\mathbb{P}^1}(-1)$ (Lemma 47.6.3(1) in the complex case; the algebraic proof is the same).

Theorem 53.6.5 (Hodge index theorem). *Let H be an ample divisor on a nonsingular projective surface X . If $D \cdot H = 0$ and D is not numerically trivial (Definition 53.7.1), then $D^2 < 0$. Equivalently, the intersection form on $\text{Num}(X) \otimes \mathbb{R}$ has signature $(1, \rho - 1)$.*

This is [Har77, Theorem V.1.9], proved there with Riemann–Roch on surfaces. Over $\mathbb{C}$ it is the part of Corollary 46.5.2 concerning classes of divisors, once intersection numbers are identified with cup products of cycle classes (Chapter 59).

Theorem 53.6.6 (Nakai–Moishezon criterion). *A divisor D on a nonsingular projective surface is ample if and only if $D^2 > 0$ and $D \cdot C > 0$ for every irreducible curve C .*

This is [Har77, Theorem V.1.10]. In higher dimensions the criterion reads $D^{\dim V} \cdot V > 0$ for every closed subvariety V [Laz04, Theorem 1.2.23].

53.7 The Néron–Severi group

Definition 53.7.1. Let X be a nonsingular projective variety over an algebraically closed field. A divisor D is *numerically trivial*, $D \equiv 0$, if $\deg(\mathcal{O}(D)|_C) = 0$ for every irreducible curve $C \subseteq X$. $\text{Num}(X) = \text{Pic}(X)/\equiv$. Two divisors are *algebraically equivalent* if they are members of a family of divisors parametrised by a connected base, and $\text{NS}(X) = \text{Pic}(X)/\{\text{algebraic equivalence}\}$ is the *Néron–Severi group*. The rank of $\text{Num}(X)$ is the *Picard number* $\rho(X)$.

Theorem 53.7.2 (Theorem of the base). *$\text{NS}(X)$ is a finitely generated abelian group, algebraic equivalence implies numerical equivalence, and $\text{Num}(X)$ is a free abelian group of finite rank $\rho(X)$, the quotient of $\text{NS}(X)$ by its torsion.*

This is due to Severi and Néron; see [Laz04, Section 1.1]. Over $\mathbb{C}$ it follows from Hodge theory. The kernel $\text{Pic}^0(X)$ of the first Chern class is the Picard torus (Remark 42.2.5), and $\text{NS}(X)$ embeds into $H^2(X; \mathbb{Z})$ with image the integral classes of type $(1, 1)$ (Chapter 65). So $\rho(X) \leq h^{1,1}(X)$.

- Example 53.7.3.** 1. $\rho(\mathbb{P}^n) = 1$ and $\rho(\mathbb{P}^1 \times \mathbb{P}^1) = 2$, since $\text{Pic}(\mathbb{P}^1 \times \mathbb{P}^1) \cong \mathbb{Z}^2$, generated by the two rulings.
2. For a quartic surface $X \subseteq \mathbb{P}^3$, $1 \leq \rho(X) \leq h^{1,1} = 20$ (Example 45.7.4). The very general quartic has $\rho = 1$ (Noether–Lefschetz), and the Fermat quartic $x^4 + y^4 + z^4 + w^4 = 0$ has $\rho = 20$.
3. For $E \times E'$ a product of elliptic curves, $\rho = 2 + \text{rank Hom}(E, E')$, which is 2, 3 or 4 (Chapter 57).

53.8 Exercises

Exercise 53.8.1. Let $Y \subseteq \mathbb{P}_k^n$ be an irreducible hypersurface of degree d . Show that $\text{Cl}(\mathbb{P}^n \setminus Y) \cong \mathbb{Z}/d\mathbb{Z}$.

Solution. By Proposition 53.2.3(2), $\text{Cl}(\mathbb{P}^n \setminus Y) = \text{Cl}(\mathbb{P}^n)/\mathbb{Z}[Y] = \mathbb{Z}/d\mathbb{Z}$, using (3) and $\deg Y = d$. $\square$

Exercise 53.8.2. Show that the conics in $\mathbb{P}^2$ form a linear system $|2H|$ of dimension 5 without base points, and that the morphism it defines is the Veronese embedding $v_2: \mathbb{P}^2 \rightarrow \mathbb{P}^5$.

Solution. $\Gamma(\mathbb{P}^2, \mathcal{O}(2))$ is the 6-dimensional space of quadratic forms (Proposition 51.5.2), so $\dim |2H| = 5$ by Proposition 53.4.3. The monomials x_i^2 have no common zero, so there are no base points, and the morphism of Proposition 53.4.1 given by the six quadratic monomials is v_2 . $\square$

Exercise 53.8.3. Let C be an irreducible curve on a nonsingular projective surface with $C^2 < 0$. Show that C is the only member of $|C|$, that is, $h^0(\mathcal{O}(C)) = 1$.

Solution. Let $D \in |C|$ and write $D = aC + D'$ with $a \geq 0$ and D' effective not containing C . If $a \geq 1$, then $(a-1)C + D'$ is effective and linearly equivalent to 0. By Proposition 53.4.3 with $h^0(\mathcal{O}) = 1$, the only effective divisor in $|0|$ is 0, so $a = 1$, $D' = 0$ and $D = C$. If $a = 0$, then $C^2 = C \cdot D' \geq 0$ by Theorem 53.6.3, since D' is a sum of curves different from C . This contradicts $C^2 < 0$. $\square$

Exercise 53.8.4. Show that if $\mathcal{L}$ is ample on X and $Y \subseteq X$ is a closed subscheme, then $\mathcal{L}|_Y$ is ample, using Theorem 53.5.2 (for X proper over a Noetherian ring).

Solution. For coherent $\mathcal{G}$ on Y , $i_*\mathcal{G}$ is coherent on X and $H^j(Y, \mathcal{G} \otimes \mathcal{L}|_Y^n) = H^j(X, i_*\mathcal{G} \otimes \mathcal{L}^n)$ by the projection formula and Example 24.7.4. These vanish for $j > 0$ and $n \gg 0$. $\square$

Chapter 54

Analytic Spaces and GAGA

54.1 Introduction

A complex algebraic variety X has an underlying complex analytic space X^{an} : the set $X(\mathbb{C})$ with the Euclidean topology and the sheaf of holomorphic functions. For projective varieties the two worlds agree. Chow's theorem says that closed analytic subsets of $\mathbb{P}^n$ are algebraic, and Serre's GAGA theorems say that coherent algebraic and analytic sheaves, and their cohomology, correspond. Consequently the Hodge theory of compact Kähler manifolds (Part VII) applies to smooth projective varieties, and the Hodge filtration can be defined algebraically through the algebraic de Rham complex, whose hypercohomology computes $H^\bullet(X^{\text{an}}; \mathbb{C})$ by Grothendieck's comparison theorem. This algebraicity is the starting point of the theory of absolute Hodge classes (Chapter 71). References: [Ser56], [Har77, Appendix B], [GH78, Chapter 1], [Gro66].

Prerequisites. Chapter 53 (Divisors and Line Bundles), Chapter 47 (Vanishing and Embedding Theorems) and Chapter 41 (Complex Manifolds).

54.2 Complex analytic spaces

Definition 54.2.1. A *local model* is a closed subset $V = \{f_1 = \cdots = f_k = 0\}$ of an open set $U \subseteq \mathbb{C}^n$, with f_i holomorphic on U , together with the sheaf of rings $\mathcal{O}_V = (\mathcal{O}_U / (f_1, \dots, f_k))|_V$. A *complex analytic space* is a Hausdorff locally ringed space over $\mathbb{C}$ that is locally isomorphic to a local model. Morphisms are morphisms of locally ringed spaces over $\mathbb{C}$. A sheaf of $\mathcal{O}$ -modules is *coherent* if it is locally the cokernel of a map $\mathcal{O}^p \rightarrow \mathcal{O}^q$.

Theorem 54.2.2 (Oka and Cartan). *The structure sheaf of a complex analytic space is coherent, and the ideal sheaf of holomorphic functions vanishing on a closed analytic subset is coherent. Kernels, cokernels and images of morphisms of coherent sheaves are coherent.*

This is [GR84, Chapters 2 and 4]. The proof is based on the Weierstrass division theorem (Theorem 29.3.3). Complex manifolds are complex analytic spaces, and conversely an analytic space whose local rings are regular is a complex manifold.

54.3 The analytification functor

Proposition 54.3.1. *Let X be a scheme locally of finite type over $\mathbb{C}$. There is a complex analytic space X^{an} with a morphism of locally ringed spaces $\varphi: X^{\text{an}} \rightarrow X$, such that every morphism from a complex analytic space to X (as locally ringed spaces over $\mathbb{C}$) factors uniquely through φ . The*

underlying set of X^{an} is $X(\mathbb{C})$, the set of closed points. $X \mapsto X^{\text{an}}$ is a functor, and for X affine, $X = \text{Spec } \mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_k)$, X^{an} is the local model $\{f = 0\} \subseteq \mathbb{C}^n$.

Sketch of proof. For affine X take the local model. Its universal property follows from that of the polynomial ring: a morphism from an analytic space Z to $\mathbb{A}_{\mathbb{C}}^n$ is an n -tuple of holomorphic functions on Z (Theorem 50.2.4), and it factors through X exactly when the f_i vanish on them. By the universal property the affine constructions glue (Proposition 50.3.2). The underlying set is $X(\mathbb{C})$ because points of an analytic space have residue field $\mathbb{C}$, and these are the closed points (Proposition 50.3.6). The omitted verification is the gluing, which is formal; see [Har77, Appendix B] and [Gro71, Exposé XII]. $\square$

Proposition 54.3.2. *Let X be of finite type over $\mathbb{C}$.*

1. *X is smooth over $\mathbb{C}$ if and only if X^{an} is a complex manifold, and then $\dim_{\mathbb{C}} X^{\text{an}} = \dim X$ at each point.*
2. *X is separated if and only if X^{an} is Hausdorff, and proper if and only if X^{an} is compact.*
3. *X is connected if and only if X^{an} is connected.*
4. *For a coherent sheaf $\mathcal{F}$ on X , $\mathcal{F}^{\text{an}} = \varphi^* \mathcal{F}$ is coherent, and $\mathcal{F} \mapsto \mathcal{F}^{\text{an}}$ is exact, because $\mathcal{O}_{X^{\text{an}},x}$ is flat over $\mathcal{O}_{X,x}$. Moreover $(\Omega_{X/\mathbb{C}})^{\text{an}} = \Omega_{X^{\text{an}}}$, the sheaf of holomorphic 1-forms.*

Proof. (1) If X is smooth, locally $X = \text{Spec } \mathbb{C}[x_1, \dots, x_{n+r}]/(g_1, \dots, g_r)$ with the Jacobian minor with respect to $x_{n+1}, \dots, x_{n+r}$ invertible (Definition 50.8.4). By the holomorphic implicit function theorem (a consequence of Theorem 25.11.2 and the Cauchy–Riemann equations), X^{an} is locally the graph of a holomorphic map from an open set of $\mathbb{C}^n$, so it is a complex manifold of dimension n . Conversely, if X^{an} is a manifold of dimension n at x , the completions of $\mathcal{O}_{X,x}$ and $\mathcal{O}_{X^{\text{an}},x}$ agree (both are $\mathbb{C}[[x_1, \dots, x_N]]/(f)$), so $\mathcal{O}_{X,x}$ is regular, and X is smooth at x by Proposition 50.8.5. Statements (2)–(4) are in [Gro71, Exposé XII] and [Ser56, Section 2]. $\square$

54.4 Chow’s theorem and GAGA

Theorem 54.4.1 (Serre’s GAGA). *Let X be a projective scheme over $\mathbb{C}$.*

1. *For every coherent sheaf $\mathcal{F}$ on X the natural maps $H^i(X, \mathcal{F}) \rightarrow H^i(X^{\text{an}}, \mathcal{F}^{\text{an}})$ are isomorphisms.*
2. *$\mathcal{F} \mapsto \mathcal{F}^{\text{an}}$ is an equivalence between coherent sheaves on X and coherent analytic sheaves on X^{an} .*

This is [Ser56, Théorèmes 1–3]. The proof reduces to $X = \mathbb{P}^n$ by pushing forward, verifies (1) for the sheaves $\mathcal{O}(q)$ by comparing the two computations of their cohomology, and deduces (1) in general by induction on n and exact sequences. For (2), the analytic version of Serre’s generation theorem (Theorem 51.5.3) presents every coherent analytic sheaf as a cokernel of a map between sums of sheaves $\mathcal{O}(q)^{\text{an}}$, and (1) applied to $\mathcal{H}om$ sheaves shows that such maps are algebraic.

Corollary 54.4.2 (Chow’s theorem). *Every closed analytic subset of $\mathbb{P}_{\mathbb{C}}^n$ is the zero set of finitely many homogeneous polynomials.*

Proof. Let $Z \subseteq \mathbb{P}^{n,\text{an}}$ be closed analytic. Its ideal sheaf $\mathcal{I} \subseteq \mathcal{O}^{\text{an}}$ is coherent (Theorem 54.2.2). By Theorem 54.4.1(2), the inclusion $\mathcal{I} \rightarrow \mathcal{O}^{\text{an}}$ is the analytification of a morphism $\mathcal{J} \rightarrow \mathcal{O}_{\mathbb{P}^n}$ of coherent sheaves, which is injective because analytification is exact and faithful. So $\mathcal{J}$ is an ideal sheaf, defining a closed subscheme Y (Proposition 51.3.6), and Y^{an} has ideal sheaf $\mathcal{J}^{\text{an}} = \mathcal{I}$, so its underlying set is Z . The closed subscheme Y is cut out by homogeneous polynomials (Proposition 50.6.2 and Theorem 51.5.3). $\square$

Corollary 54.4.3. *Let X and Y be projective varieties over $\mathbb{C}$.*

1. *Every holomorphic map $X^{\text{an}} \rightarrow Y^{\text{an}}$ is the analytification of a unique morphism $X \rightarrow Y$.*
2. *$\text{Pic}(X) \rightarrow \text{Pic}(X^{\text{an}})$ is an isomorphism, and every meromorphic function on X^{an} is rational.*
3. *A compact complex manifold that admits a holomorphic embedding into some $\mathbb{CP}^N$ is X^{an} for a nonsingular projective variety X , unique up to isomorphism.*

Proof. (1) The graph of f is a closed analytic subset of $(X \times Y)^{\text{an}} \subseteq \mathbb{P}^{N,\text{an}}$ (Segre), hence algebraic by Chow's theorem, and it is a closed subvariety Γ whose projection to X is bijective and an analytic isomorphism. By Theorem 54.4.1(2) applied to $\mathcal{O}_X \rightarrow p_*\mathcal{O}_\Gamma$, whose analytification is an isomorphism, the projection is an algebraic isomorphism, and f is the composite with the second projection. Uniqueness holds because $X(\mathbb{C})$ is dense in X . (2) Invertible sheaves correspond under Theorem 54.4.1(2), since the equivalence is compatible with $\otimes$. A meromorphic function f with polar divisor D is the quotient $f s_D / s_D$ of two holomorphic sections of the analytic line bundle $\mathcal{O}(D)$. This bundle and its sections are algebraic by Theorem 54.4.1, so f is rational. (3) The image is a closed analytic subset, hence algebraic by Chow's theorem, and it is nonsingular by Proposition 54.3.2(1). Uniqueness is (1). $\square$

Corollary 54.4.4. *Every holomorphic map $f: \mathbb{CP}^n \rightarrow \mathbb{CP}^m$ is of the form $[x] \mapsto [F_0(x) : \cdots : F_m(x)]$ for homogeneous polynomials $F_0, \dots, F_m$ of a common degree $d \geq 0$ with no common zero in $\mathbb{C}^{n+1} \setminus \{0\}$. If $m < n$, every such map is constant.*

Proof. By Corollary 54.4.3(1), f is a morphism $\mathbb{P}^n \rightarrow \mathbb{P}^m$. By Proposition 53.4.1 it is given by the invertible sheaf $\mathcal{L} = f^*\mathcal{O}(1)$ and the generating sections $s_i = f^*x_i$. By Example 53.3.3, $\mathcal{L} \cong \mathcal{O}(d)$ for some $d \in \mathbb{Z}$, and by Theorem 52.3.1 its global sections are the homogeneous polynomials of degree d if $d \geq 0$ and 0 if $d < 0$; since the s_i generate, $d \geq 0$, and generating means having no common zero. If $d = 0$ the F_i are constants and f is constant. Suppose $d \geq 1$ and $m < n$. Put $Y_{-1} = \mathbb{P}^n$ and $Y_j = Y_{j-1} \cap Z(F_j)$. By induction every irreducible component of Y_j has dimension at least $n - j - 1$ and $Y_j \neq \emptyset$ for $j \leq m$: a component Y of Y_{j-1} has dimension $\geq n - j \geq n - m \geq 1$, and $Y \cap Z(F_j)$ is either Y (if $F_j \in I(Y)$) or nonempty with components of dimension $\dim Y - 1$ (Theorem 49.5.5). So $Y_m \neq \emptyset$, and the F_i have a common zero, a contradiction. $\square$

Counterexample 54.4.5. GAGA fails without properness. The exponential is a holomorphic function on $\mathbb{A}^{1,\text{an}} = \mathbb{C}$ that is not a polynomial, so $\Gamma(\mathbb{A}^1, \mathcal{O}) \neq \Gamma(\mathbb{C}, \mathcal{O}^{\text{an}})$. The holomorphic map $\exp: \mathbb{C} \rightarrow \mathbb{C}^*$ is not algebraic, since an algebraic map $\mathbb{A}^1 \rightarrow \mathbb{A}^1 \setminus 0$ is constant: it corresponds to a unit of $\mathbb{C}[t]$. Chow's theorem also fails in $\mathbb{C}^n$: the graph of $\exp$ is a closed analytic subset of $\mathbb{C}^2$ that is not algebraic, since it meets the line $y = 1$ in infinitely many points.

Counterexample 54.4.6. Not every compact complex manifold is algebraic, even when it is Kähler: a general complex torus of dimension ≥ 2 is not of the form X^{an} for any variety (Example 44.4.3), and the Hopf surface is not even Kähler. There are also nonsingular complete varieties that are not projective, such as Hironaka's example [Har77, Appendix B,

Example 3.4.1]; their analytifications are compact complex manifolds that are algebraic but not projective.

54.5 Algebraic de Rham cohomology

Definition 54.5.1. Let X be a smooth scheme over a field k . The *algebraic de Rham complex* is $\Omega_{X/k}^\bullet = (O_X \xrightarrow{d} \Omega_{X/k}^1 \xrightarrow{d} \Omega_{X/k}^2 \rightarrow \cdots)$, with $\Omega^p = \Lambda^p \Omega^1$ and d the k -linear exterior derivative. The *algebraic de Rham cohomology* is the hypercohomology $H_{\text{dR}}^n(X/k) = \mathbb{H}^n(X, \Omega_{X/k}^\bullet)$ in the Zariski topology, and the *Hodge filtration* is $F^p H_{\text{dR}}^n(X/k) = \text{im} \left(\mathbb{H}^n(X, \Omega_{X/k}^{\geq p}) \rightarrow H_{\text{dR}}^n(X/k) \right)$. The spectral sequence of the filtration by the truncations $\Omega^{\geq p}$ is the *Hodge-to-de Rham spectral sequence* $E_1^{p,q} = H^q(X, \Omega_{X/k}^p) \Rightarrow H_{\text{dR}}^{p+q}(X/k)$ (Proposition 24.7.1).

Theorem 54.5.2 (Grothendieck). *For a smooth scheme X of finite type over $\mathbb{C}$ there are natural isomorphisms $H_{\text{dR}}^n(X/\mathbb{C}) \cong H^n(X^{\text{an}}; \mathbb{C})$.*

Proof. We prove it for X projective; the general case is in [Gro66], proved with resolution of singularities. The morphism $\varphi: X^{\text{an}} \rightarrow X$ gives maps $\Omega_{X/\mathbb{C}}^p \rightarrow \varphi_* \Omega_{X^{\text{an}}}^p$ compatible with d , hence a map of hypercohomologies $\mathbb{H}^n(X, \Omega_{X/\mathbb{C}}^\bullet) \rightarrow \mathbb{H}^n(X^{\text{an}}, \Omega_{X^{\text{an}}}^\bullet)$ compatible with the filtrations by the truncations. On E_1 it is $H^q(X, \Omega_{X/\mathbb{C}}^p) \rightarrow H^q(X^{\text{an}}, \Omega_{X^{\text{an}}}^p)$, an isomorphism by Theorem 54.4.1(1) and Proposition 54.3.2(4). A morphism of first-quadrant spectral sequences that is an isomorphism on E_1 is an isomorphism on all pages. As the filtrations are finite, the five lemma gives an isomorphism on the abutments. Finally $\mathbb{H}^n(X^{\text{an}}, \Omega_{X^{\text{an}}}^\bullet) \cong H^n(X^{\text{an}}; \mathbb{C})$ by the holomorphic Poincaré lemma (Theorem 43.2.4). $\square$

Corollary 54.5.3. *Let X be a smooth projective variety over $\mathbb{C}$. Then X^{an} is a compact Kähler manifold, the Hodge-to-de Rham spectral sequence of X degenerates at E_1 , and under Theorem 54.5.2*

$$F^p H^n(X^{\text{an}}; \mathbb{C}) = \bigoplus_{r \geq p} H^{r, n-r}(X^{\text{an}}) \quad \text{corresponds to} \quad F^p H_{\text{dR}}^n(X/\mathbb{C}), \quad H^{p,q}(X^{\text{an}}) \cong H^q(X, \Omega_{X/\mathbb{C}}^p).$$

If $X = X_0 \times_k \mathbb{C}$ for a subfield $k \subseteq \mathbb{C}$ and a smooth projective X_0 over k , then $H_{\text{dR}}^n(X/\mathbb{C}) = H_{\text{dR}}^n(X_0/k) \otimes_k \mathbb{C}$, compatibly with the Hodge filtrations. So the Hodge filtration is defined over k .

Proof. X^{an} is a compact complex submanifold of $\mathbb{CP}^N$ (Proposition 54.3.2), hence Kähler (Corollary 44.4.2). The E_1 terms have total dimension $\sum_{p+q=n} h^{p,q} = b_n$ by Corollary 45.5.3, so all differentials vanish. The statement on filtrations is Remark 45.5.5 transported by the comparison, which respects the filtrations by construction. For base change, $\Omega_{X/\mathbb{C}}^p = \Omega_{X_0/k}^p \otimes_k \mathbb{C}$ (Proposition 51.6.2(3)), and hypercohomology can be computed by Čech complexes of a finite affine cover (Corollary 52.2.5), which commute with the flat base change $-\otimes_k \mathbb{C}$. $\square$

Corollary 54.5.4. *Let X be a smooth projective variety over $\mathbb{C}$ and $\sigma \in \text{Aut}(\mathbb{C})$. The conjugate variety X^σ (Remark 50.9.3) has the same Betti numbers and Hodge numbers as X . The map σ induces a σ -linear isomorphism $H_{\text{dR}}^n(X/\mathbb{C}) \rightarrow H_{\text{dR}}^n(X^\sigma/\mathbb{C})$ that preserves the Hodge filtrations.*

Proof. The scheme X^σ is obtained from X by the base change $\sigma: \text{Spec } \mathbb{C} \rightarrow \text{Spec } \mathbb{C}$. It is again smooth and projective, and $\Omega_{X^\sigma}^p = \sigma^* \Omega_X^p$. So $H^q(X^\sigma, \Omega^p) = H^q(X, \Omega^p) \otimes_{\mathbb{C}, \sigma} \mathbb{C}$ by flat base change of Čech complexes, and the same holds for de Rham cohomology and its Hodge filtration. Now apply Theorem 54.5.2 and Corollary 54.5.3. $\square$

Remark 54.5.5. The spaces X^{an} and $(X^\sigma)^{\text{an}}$ need not be homeomorphic, and can have different fundamental groups [Ser64]. The isomorphism of Corollary 54.5.4 does not preserve the rational structures $H^n(X^{\text{an}}; \mathbb{Q})$. Whether it maps Hodge classes to Hodge classes is the question of *absolute Hodge classes* (Chapter 71); Deligne proved it for abelian varieties [Del82]. Deligne and Illusie gave an algebraic proof of the degeneration of the Hodge-to-de Rham spectral sequence in characteristic 0 [DI87].

54.6 Smooth projective varieties as Kähler manifolds

Theorem 54.6.1. *The functor $X \mapsto X^{\text{an}}$ is an equivalence between the category of nonsingular projective varieties over $\mathbb{C}$ and the category of compact complex manifolds that admit a positive line bundle (equivalently, a Hodge metric, Corollary 47.2.4), with holomorphic maps. A line bundle $\mathcal{L}$ on a nonsingular projective variety is ample if and only if $\mathcal{L}^{\text{an}}$ is positive.*

Proof. Full faithfulness is Corollary 54.4.3(1). A compact complex manifold with a positive line bundle embeds into some $\mathbb{CP}^N$ (Theorem 47.6.5), so it is X^{an} by Corollary 54.4.3(3). Conversely $\mathcal{O}(1)$ restricted to $X^{\text{an}} \subseteq \mathbb{CP}^N$ is positive (Proposition 47.2.1(1)). If $\mathcal{L}$ is ample, $\mathcal{L}^m$ is very ample (Theorem 53.5.1), so $\mathcal{L}^{m,\text{an}}$ is positive, and so is $\mathcal{L}^{\text{an}}$, by dividing the curvature by m . If $\mathcal{L}^{\text{an}}$ is positive, then $\mathcal{L}^{k,\text{an}}$ is very ample for large k by Theorem 47.6.5. The sections involved are algebraic by Theorem 54.4.1(1), and the embedding is algebraic by Corollary 54.4.3(1). So $\mathcal{L}^k$ is very ample and $\mathcal{L}$ is ample. $\square$

Remark 54.6.2. Consequently the Hodge numbers $h^{p,q} = \dim H^q(X, \Omega^p)$ of a smooth projective variety are algebraic invariants, computable by coherent cohomology. The cohomology classes of algebraic cycles (Chapter 59) are Hodge classes (Proposition 45.5.8), and by Chow's theorem the analytic subvarieties of X^{an} are exactly the algebraic ones. So on a projective manifold the Hodge conjecture may equivalently be stated with analytic cycles. For compact Kähler manifolds the analogous statement with analytic cycles is false (Chapter 69).

54.7 Exercises

Exercise 54.7.1. Show that every compact Riemann surface is C^{an} for a unique nonsingular projective curve C , and that its field of meromorphic functions is a finitely generated extension of $\mathbb{C}$ of transcendence degree 1.

Solution. A compact Riemann surface is Kähler with $H^{2,0} = 0$, hence projective by Corollary 47.6.7, that is, it embeds into some $\mathbb{CP}^N$. Apply Corollary 54.4.3(3). Meromorphic functions are rational by Corollary 54.4.3(2), so the field is $K(C)$, of transcendence degree $\dim C = 1$ (Theorem 49.5.3). $\square$

Exercise 54.7.2. Let $C \subseteq \mathbb{P}_{\mathbb{C}}^2$ be a nonsingular curve of degree d . Show that C^{an} is a Riemann surface of topological genus $(d-1)(d-2)/2$.

Solution. By Exercise 52.8.2, $h^1(C, \mathcal{O}_C) = (d-1)(d-2)/2$. By GAGA this is $h^{0,1}(C^{\text{an}})$, and by the Hodge decomposition $b_1 = 2h^{0,1}$ (Example 45.7.1), so the genus $b_1/2$ equals $(d-1)(d-2)/2$. $\square$

Exercise 54.7.3. Let $X \subseteq \mathbb{P}_{\mathbb{C}}^3$ be a nonsingular quartic surface. Compute $h^{0,1}$, $h^{0,2}$ and $h^{2,0}$ algebraically.

Solution. $h^{0,q} = h^q(X, \mathcal{O}_X)$ by Corollary 54.5.3. The sequence $0 \rightarrow \mathcal{O}(-4) \rightarrow \mathcal{O} \rightarrow \mathcal{O}_X \rightarrow 0$ on $\mathbb{P}^3$ and Theorem 52.3.1 give $h^0(\mathcal{O}_X) = 1$, $h^1(\mathcal{O}_X) = 0$ and $h^2(\mathcal{O}_X) = h^3(\mathbb{P}^3, \mathcal{O}(-4)) = 1$. By Serre duality, or directly from $\omega_X \cong \mathcal{O}_X$ (Proposition 51.6.4), $h^{2,0} = h^0(\omega_X) = 1$. $\square$

Chapter 55

Algebraic Curves and Riemann Surfaces

55.1 Introduction

Curves are the one-dimensional case, in which almost everything can be computed. Over $\mathbb{C}$, nonsingular projective curves are the same as compact Riemann surfaces (Theorem 55.2.5). The Riemann–Roch theorem (Theorem 55.3.2) computes dimensions of spaces of sections, and from it follow the classification of curves of genus 0 and 1, the Riemann–Hurwitz formula, and the theory of the canonical embedding. The Jacobian of a curve is the first nontrivial example of a Hodge structure that is not of Tate type, and its theory (Abel–Jacobi) leads to the abelian varieties of Chapter 57 and the intermediate Jacobians of Chapter 67. References: [Har77, Chapter IV], [GH78, Chapter 2].

Prerequisites. Chapter 54 (Analytic Spaces and GAGA).

Throughout, k is algebraically closed, and a *curve* is a nonsingular projective variety of dimension one over k , unless stated otherwise. Points are closed points. For a curve X , divisors are finite formal sums $D = \sum n_P P$ (Weil and Cartier divisors agree by Proposition 53.3.2), $\deg D = \sum n_P$, $\mathcal{O}(D)$ is the associated invertible sheaf, $l(D) = h^0(\mathcal{O}(D))$, and K is a *canonical divisor*, $\mathcal{O}(K) \cong \omega_X = \Omega_{X/k}$.

55.2 Curves and their function fields

Proposition 55.2.1. *Let X be a nonsingular curve (not necessarily projective). Every local ring $\mathcal{O}_{X,P}$ is a discrete valuation ring of $K(X)$ with residue field k . A rational map from X to a projective variety Y extends to a morphism $X \rightarrow Y$.*

Proof. $\mathcal{O}_{X,P}$ is a regular local ring of dimension one, hence a DVR (Theorem 12.11.2), with residue field k by the Nullstellensatz. For the extension, let $Y \subseteq \mathbb{P}^n$ and let the map be defined near P except at P , given by rational functions $[f_0 : \cdots : f_n]$. Let v be the valuation at P and choose m with $v(f_m)$ minimal among the nonzero f_i . Then $f_i/f_m \in \mathcal{O}_{X,P}$ for all i , and $[f_0/f_m : \cdots : f_n/f_m]$ is a morphism on a neighbourhood of P that agrees with the map where both are defined. It lands in Y because Y is closed. (This is the valuative argument of Theorem 50.5.5.) $\square$

Theorem 55.2.2. *The functor $X \mapsto K(X)$ is an anti-equivalence between the category of curves with nonconstant morphisms and the category of function fields of one variable over k (finitely generated extensions of transcendence degree 1). Every nonconstant morphism of curves $f : X \rightarrow Y$ is finite*

and surjective, and its degree $\deg f = [K(X) : K(Y)]$ satisfies

$$\sum_{P \mapsto Q} e_P = \deg f \quad \text{for every } Q \in Y,$$

where the ramification index $e_P = v_P(t)$ for a uniformiser t at Q .

Proof. By Theorem 49.7.3, nonconstant morphisms $X \rightarrow Y$ give k -embeddings $K(Y) \rightarrow K(X)$, and every such embedding comes from a dominant rational map $X \dashrightarrow Y$, which is a morphism by Proposition 55.2.1. This gives full faithfulness. That every function field K is $K(X)$ for a curve X is [Har77, Theorem I.6.9]: the points of X are the discrete valuation rings of K containing k , and X is covered by the affine curves $\text{Spec } A'$, where A' is the integral closure of $k[t]$ in K for a nonconstant $t \in K$.

Let $f: X \rightarrow Y$ be nonconstant. Its image is closed (Theorem 50.5.5) and irreducible, not a point, hence Y . For an affine $V = \text{Spec } B \subseteq Y$, $f^{-1}(V)$ consists of the points whose valuation rings contain B . These are the points of $\text{Spec } A'$, where A' is the integral closure of B in $K(X)$, because a DVR is integrally closed and every local ring of A' at a maximal ideal is a DVR. By the finiteness of integral closure [Har77, Theorem I.3.9A], A' is a finite B -module, so f is finite. Localising at Q , $A'_Q = A' \otimes_B \mathcal{O}_{Y,Q}$ is a finitely generated torsion-free module over the DVR $\mathcal{O}_{Y,Q}$, hence free (Theorem 7.11.3), of rank $[K(X) : K(Y)]$ since $A'_Q \otimes K(Y) = K(X)$. So $\dim_k A'_Q / t A'_Q = \deg f$. The semilocal ring A'_Q has maximal ideals the points P over Q , and by the Chinese remainder theorem $A'_Q / t A'_Q \cong \prod_{P \mapsto Q} \mathcal{O}_{X,P} / t \mathcal{O}_{X,P}$, where the factor at P has dimension $v_P(t) = e_P$. $\square$

Corollary 55.2.3. *For nonconstant $f \in K(X)$, $\deg(f)_0 = \deg(f)_\infty = [K(X) : k(f)]$, where $(f)_0$ and $(f)_\infty$ are the divisors of zeros and poles. So principal divisors have degree 0, and $\deg$ induces a surjection $\text{Pic}(X) \rightarrow \mathbb{Z}$ with kernel $\text{Pic}^0(X)$. For a nonconstant morphism $f: X \rightarrow Y$ and a divisor D on Y , $\deg f^*D = \deg f \cdot \deg D$.*

Proof. f defines a morphism $X \rightarrow \mathbb{P}^1$ (Proposition 55.2.1) of degree $[K(X) : k(f)]$, and $(f)_0, (f)_\infty$ are the pullbacks of the points 0 and ∞ , with multiplicities e_P . Apply Theorem 55.2.2. The last statement is the same formula applied to each point of D . $\square$

Lemma 55.2.4. *If $\deg D < 0$ then $l(D) = 0$. If $\deg D = 0$, then $l(D) \leq 1$, with equality if and only if $D \sim 0$.*

Proof. A nonzero section of $\mathcal{O}(D)$ gives an effective divisor $D' \sim D$ (Proposition 53.4.3), and $\deg D' \geq 0$, with equality only for $D' = 0$, by Corollary 55.2.3. $\square$

Theorem 55.2.5. *The functor $X \mapsto X^{\text{an}}$ is an equivalence between complex curves and compact Riemann surfaces. The topological genus of X^{an} equals $g(X) = h^1(X, \mathcal{O}_X) = h^0(X, \omega_X)$.*

Proof. A compact Riemann surface is Kähler with $H^{2,0} = 0$, hence projective (Corollary 47.6.7), hence of the form X^{an} (Corollary 54.4.3); full faithfulness is also part of that corollary. By GAGA and the Hodge decomposition (Example 45.7.1), $b_1(X^{\text{an}})/2 = h^{0,1} = h^1(X, \mathcal{O}_X) = h^{1,0} = h^0(X, \omega_X)$. $\square$

55.3 The Riemann–Roch theorem

Definition 55.3.1. The *genus* of a curve is $g = h^1(X, \mathcal{O}_X)$. By Serre duality (Theorem 52.6.2) also $g = h^0(X, \omega_X) = l(K)$.

Theorem 55.3.2 (Riemann–Roch). *For every divisor D on a curve X of genus g ,*

$$l(D) - l(K - D) = \deg D + 1 - g.$$

Proof. First, $\chi(\mathcal{O}(D)) = \deg D + 1 - g$. This holds for $D = 0$, since $h^0(\mathcal{O}_X) = 1$ (Corollary 52.4.2) and $h^1 = g$. For a point P , the sequence $0 \rightarrow \mathcal{O}(D) \rightarrow \mathcal{O}(D + P) \rightarrow k(P) \rightarrow 0$, where $k(P)$ is the skyscraper sheaf at P , gives $\chi(\mathcal{O}(D + P)) = \chi(\mathcal{O}(D)) + 1$. Both sides of the formula change by 1 when a point is added or removed, and every divisor is reached from 0 this way. Second, $h^1(\mathcal{O}(D)) = h^0(\omega \otimes \mathcal{O}(-D)) = l(K - D)$ by Serre duality. $\square$

Corollary 55.3.3. *Let X be a curve of genus g .*

1. $\deg K = 2g - 2$.
2. *If $\deg D > 2g - 2$, then $l(D) = \deg D + 1 - g$.*
3. *If $\deg D \geq 2g$, then $|D|$ has no base points; if $\deg D \geq 2g + 1$, then D is very ample.*
4. *$X \cong \mathbb{P}^1$ if and only if $g = 0$, if and only if there are points $P \neq Q$ with $P \sim Q$.*

Proof. (1) Put $D = K$: $g - 1 = \deg K + 1 - g$. (2) $\deg(K - D) < 0$, so $l(K - D) = 0$ by Lemma 55.2.4. (3) By (2), $l(D - P) = l(D) - 1$ if $\deg D - 1 > 2g - 2$, so P is not a base point. Similarly $l(D - P - Q) = l(D) - 2$ for all P, Q (possibly equal) if $\deg D - 2 > 2g - 2$, which means that $|D|$ separates points and tangent vectors, and D is very ample by Proposition 53.4.4. (4) If $g = 0$, then $l(P) = 2$ by (2), and a nonconstant $f \in \Gamma(\mathcal{O}(P))$ has a single simple pole, so $[K(X) : k(f)] = 1$ by Corollary 55.2.3, and $X \cong \mathbb{P}^1$ by Theorem 55.2.2. On $\mathbb{P}^1$ all points are linearly equivalent. Conversely, if $P \sim Q$ with $P \neq Q$, then $l(P) \geq 2$, and as before $X \cong \mathbb{P}^1$, which has $g = h^1(\mathcal{O}_{\mathbb{P}^1}) = 0$ (Corollary 52.3.2). $\square$

Example 55.3.4. Let $g = 1$ and fix a point O . Then $K \sim 0$, since $\deg K = 0$ and $l(K) = 1$ (Lemma 55.2.4). The divisor $3O$ is very ample with $l(3O) = 3$, so it embeds X as a plane cubic. Choosing coordinates $1, x, y$ with poles of order 0, 2, 3 at O ($l(2O) = 2$, $l(3O) = 3$), the seven functions $1, x, y, x^2, xy, x^3, y^2$ in the 6-dimensional space $\Gamma(\mathcal{O}(6O))$ are linearly dependent, which gives a Weierstrass equation $y^2 + a_1xy + a_3y = x^3 + a_2x^2 + a_4x + a_6$. The map $P \mapsto \mathcal{O}(P - O)$ is a bijection $X \rightarrow \text{Pic}^0(X)$ (Exercise 55.7.1), which gives X the structure of an abelian group with identity O : the *elliptic curve*.

55.4 The Riemann–Hurwitz formula

Theorem 55.4.1 (Riemann–Hurwitz). *Let $f: X \rightarrow Y$ be a nonconstant morphism of curves of degree n such that $K(X)/K(Y)$ is separable and every ramification index e_P is prime to the characteristic (always true in characteristic 0). Then*

$$2g(X) - 2 = n(2g(Y) - 2) + \sum_{P \in X} (e_P - 1).$$

Proof. Let ω be a nonzero rational differential on Y , so that $\deg(\omega) = 2g(Y) - 2$ by Corollary 55.3.3(1). Its pullback $f^*\omega$ is a nonzero rational differential on X , by separability. At $P \mapsto Q$ let t be a uniformiser at Q and s at P , so $t = us^e$ with u a unit and $e = e_P$. Then $dt = (eus^{e-1} + s^e du/ds) ds$, which has order exactly $e - 1$ at P because $e \neq 0$ in k . Writing $\omega = h dt$ near Q gives $v_P(f^*\omega) = e_P v_Q(\omega) + e_P - 1$. Summing over P and using $\sum_{P \mapsto Q} e_P = n$ (Theorem 55.2.2) gives $\deg(f^*\omega) = n \deg(\omega) + \sum_P (e_P - 1)$. $\square$

- Example 55.4.2.** 1. A hyperelliptic curve $y^2 = h(x)$, with h a polynomial of degree $2g + 1$ or $2g + 2$ without repeated roots (characteristic $\neq 2$), has a degree-2 map to $\mathbb{P}^1$ branched at the $2g + 2$ roots of h (and ∞ if $\deg h$ is odd). Riemann–Hurwitz gives $2g(X) - 2 = 2(-2) + 2g + 2$, so $g(X) = g$.
2. For the Fermat curve $x^d + y^d = z^d$ and the projection $[x : y : z] \mapsto [x : z]$ of degree d , the ramification points are the d points with $y = 0$, each with $e = d$. So $2g - 2 = -2d + d(d - 1)$, and $g = (d - 1)(d - 2)/2$, as for every plane curve of degree d (Example 52.5.4).
3. Over $\mathbb{C}$, Riemann–Hurwitz is the Euler characteristic computation $\chi(X) = n\chi(Y) - \sum(e_p - 1)$ for a branched covering of surfaces.

Corollary 55.4.3 (Lüroth). *Let k be algebraically closed of characteristic 0. Every field L with $k \subsetneq L \subseteq k(t)$ is of the form $L = k(s)$ for some $s \in L$. Equivalently, a curve that receives a nonconstant morphism from $\mathbb{P}^1$ is isomorphic to $\mathbb{P}^1$.*

Proof. Choose $u \in L \setminus k$. It is transcendental over k , since k is algebraically closed, so $k(t)$ is algebraic and finitely generated, hence finite, over $k(u)$. As a $k(u)$ -subspace of $k(t)$, L is finite over $k(u)$, so it is a function field of one variable over k . By Theorem 55.2.2, $L = K(Y)$ for a curve Y , and the inclusion $L \subseteq k(t) = K(\mathbb{P}^1)$ comes from a nonconstant morphism $f: \mathbb{P}^1 \rightarrow Y$, separable because the characteristic is 0. By Theorem 55.4.1, $-2 = \deg f (2g(Y) - 2) + \sum_p (e_p - 1) \geq \deg f (2g(Y) - 2)$, so $g(Y) = 0$ and $Y \cong \mathbb{P}^1$ by Corollary 55.3.3(4). Hence $L \cong K(\mathbb{P}^1)$ is generated over k by one element. The second statement is the same argument. $\square$

Counterexample 55.4.4. The separability hypothesis in Theorem 55.4.1 cannot be dropped. Let $\text{char } k = p > 0$ and $F: \mathbb{P}^1 \rightarrow \mathbb{P}^1$, $[x : y] \mapsto [x^p : y^p]$. It is bijective on points, since every element of the algebraically closed field k has exactly one p -th root ($x^p - a^p = (x - a)^p$). On function fields it is the inclusion $k(t^p) \subseteq k(t)$, of degree p and purely inseparable, since t is the only root of $X^p - t^p = (X - t)^p$. At a point $a \in k$, the uniformiser $x - a^p$ at $F(a) = a^p$ pulls back to $x^p - a^p = (x - a)^p$, so $e_a = p$: every point is ramified, in accordance with $\sum_{p \mapsto Q} e_p = \deg F = p$ and a single preimage. The Riemann–Hurwitz formula would give $-2 = -2p + \sum_p (p - 1)$, a sum with infinitely many nonzero terms. The proof breaks down because pullbacks of differentials vanish: $d(x^p) = px^{p-1} dx = 0$.

55.5 Hyperelliptic curves and the canonical embedding

Definition 55.5.1. A curve of genus $g \geq 2$ is *hyperelliptic* if it has a morphism of degree 2 to $\mathbb{P}^1$, equivalently a divisor D with $\deg D = 2$ and $l(D) = 2$ (a g_2^1).

Theorem 55.5.2. *Let X be a curve of genus $g \geq 1$. Then $|K|$ has no base points. If $g \geq 2$, then K is very ample if and only if X is not hyperelliptic. Every curve of genus 2 is hyperelliptic.*

Proof. By Riemann–Roch, $l(K - P) = g - 2 + l(P)$, and $l(P) = 1$ since $X \not\cong \mathbb{P}^1$ (Corollary 55.3.3(4)). So $l(K - P) = g - 1 = l(K) - 1$ and P is not a base point. Similarly $l(K - P - Q) = g - 3 + l(P + Q)$, so $l(K - P - Q) = l(K) - 2$ for all P, Q if and only if $l(P + Q) = 1$ for all P, Q , that is, if and only if there is no g_2^1 among the divisors $P + Q$. Every g_2^1 is of this form, since it contains an effective divisor of degree 2. By Proposition 53.4.4, K is very ample exactly when X is not hyperelliptic. For $g = 2$, $\deg K = 2$ and $l(K) = 2$, so K is a g_2^1 . $\square$

Example 55.5.3. A nonsingular plane quartic has genus 3 and $\omega \cong \mathcal{O}(1)$ (Exercise 51.7.3), so the plane embedding is the canonical embedding, and plane quartics are not hyperelliptic. Conversely every non-hyperelliptic curve of genus 3 is canonically a plane quartic, since its canonical image in $\mathbb{P}^2$ is a curve of degree $\deg K = 4$.

55.6 The Jacobian and the Abel–Jacobi theorem

Definition 55.6.1. Let X be a compact Riemann surface of genus g . The *Jacobian* is the complex torus $J(X) = H^0(X, \Omega^1)^\vee / H_1(X; \mathbb{Z})$, where $\gamma \in H_1(X; \mathbb{Z})$ acts by $\omega \mapsto \int_\gamma \omega$. Fixing a base point P_0 , the *Abel–Jacobi map* is $u: X \rightarrow J(X)$, $u(P) = (\omega \mapsto \int_{P_0}^P \omega)$ modulo periods; it extends additively to divisors, and on degree-zero divisors it does not depend on P_0 .

Proposition 55.6.2. $H_1(X; \mathbb{Z}) \rightarrow H^0(X, \Omega^1)^\vee$ is injective with image a lattice, so $J(X)$ is a compact complex torus of dimension g . Moreover $\text{Pic}^0(X^{\text{an}}) \cong H^1(X, \mathcal{O})/H^1(X; \mathbb{Z}) \cong J(X)$ as complex tori.

Proof. Holomorphic 1-forms are closed (Proposition 45.5.6), so the integrals depend only on homology classes. By the Hodge decomposition, $H^1(X; \mathbb{R}) \rightarrow H^{0,1} \cong H^1(X, \mathcal{O})$ is an isomorphism of real vector spaces (Exercise 45.8.4), so the image of $H^1(X; \mathbb{Z})$ is a lattice in the g -dimensional complex space $H^1(X, \mathcal{O})$, and by Theorem 42.2.4(2), $\text{Pic}^0(X^{\text{an}}) \cong H^1(X, \mathcal{O})/H^1(X; \mathbb{Z})$. Serre duality identifies $H^1(X, \mathcal{O})$ with $H^0(X, \Omega^1)^\vee$ via $(\alpha, \omega) \mapsto \int_X \alpha \wedge \omega$ for a Dolbeault representative α , and Poincaré duality identifies $H^1(X; \mathbb{Z})$ with $H_1(X; \mathbb{Z})$. Under these identifications, the class of a closed real 1-form η Poincaré dual to γ is sent to $\omega \mapsto \int_X \eta^{0,1} \wedge \omega = \int_X \eta \wedge \omega = \int_\gamma \omega$, since $\eta^{1,0} \wedge \omega = 0$ for type reasons. So the lattices correspond. $\square$

Theorem 55.6.3 (Abel and Jacobi). *Let X be a compact Riemann surface of genus g .*

1. (Abel) *A divisor D of degree 0 is principal if and only if $u(D) = 0$ in $J(X)$. Equivalently, under Proposition 55.6.2 the class of $\mathcal{O}(D)$ in $\text{Pic}^0(X)$ corresponds to $u(D)$.*
2. (Jacobi inversion) *Every divisor class of degree g contains an effective divisor. So the map $X^{(g)} = X^g/S_g \rightarrow \text{Pic}^g(X)$, and hence $u: X^{(g)} \rightarrow J(X)$, is surjective. It is injective over a dense open subset.*

Sketch of proof. (2) By Riemann–Roch, $l(D) \geq \deg D + 1 - g = 1$ for $\deg D = g$, so D is linearly equivalent to an effective divisor. Choosing points $P_1, \dots, P_g$ successively, each outside the base locus of $|K - P_1 - \dots - P_{i-1}|$, gives $l(K - P_1 - \dots - P_g) = 0$, hence $l(P_1 + \dots + P_g) = 1$ by Riemann–Roch; for such divisors the fibre of $X^{(g)} \rightarrow \text{Pic}^g(X)$ is the single point $|D|$. (1) One direction: if $D = (f)$, let $D_t = f^*(t)$ for $t \in \mathbb{P}^1$. Then $t \mapsto u(D_t - D_\infty)$ is a holomorphic map $\mathbb{P}^1 \rightarrow J(X)$, which lifts to the universal cover $\mathbb{C}^g$ because $\mathbb{P}^1$ is simply connected, and is therefore constant. It vanishes at $t = \infty$, so $u(D) = u(D_0 - D_\infty) = 0$. The converse, which we omit, constructs a meromorphic function with divisor D as the exponential of an abelian integral of the third kind whose periods are integral [GH78, Chapter 2, Section 2]. $\square$

Remark 55.6.4. The Jacobian is determined by the weight-one Hodge structure $H^1(X; \mathbb{Z})$ with $H^1(X; \mathbb{C}) = H^{1,0} \oplus H^{0,1}$, and it is polarised by the intersection form, which satisfies the Riemann bilinear relations (Example 46.4.4). So $J(X)$ is an abelian variety (Chapter 57). By Torelli’s theorem the polarised Jacobian determines X [GH78, Chapter 2, Section 7]. Griffiths’ intermediate Jacobians (Chapter 67) generalise this construction to higher-dimensional varieties.

55.7 Exercises

Exercise 55.7.1. Let X be a curve of genus 1 with a point O . Show that $P \mapsto \mathcal{O}(P - O)$ is a bijection $X \rightarrow \text{Pic}^0(X)$.

Solution. Injective: if $P - O \sim Q - O$ with $P \neq Q$, then $P \sim Q$ and $X \cong \mathbb{P}^1$ by Corollary 55.3.3(4), contradicting $g = 1$. Surjective: for D of degree 0, $l(D + O) = 1$ by Riemann–Roch ($\deg(D + O) = 1 > 0 = 2g - 2$), so $D + O \sim P$ for an effective divisor P of degree 1, that is, a point. $\square$

Exercise 55.7.2. Show that $g(X) \geq g(Y)$ for every nonconstant morphism of curves $X \rightarrow Y$ in characteristic 0. Deduce that there is no nonconstant morphism $\mathbb{P}^1 \rightarrow Y$ if $g(Y) \geq 1$ (Lüroth’s theorem for curves).

Solution. Riemann–Hurwitz gives $2g(X) - 2 \geq n(2g(Y) - 2)$. If $g(Y) \geq 1$ the right-hand side is $\geq 2g(Y) - 2$, so $g(X) \geq g(Y)$, and if $g(Y) = 0$ there is nothing to prove. $\square$

Exercise 55.7.3. Show that a curve of genus 2 over a field of characteristic $\neq 2$ is of the form $y^2 = h(x)$ with h of degree 5 or 6 without repeated roots.

Solution. By Theorem 55.5.2, $|K|$ gives a degree-2 morphism $X \rightarrow \mathbb{P}^1$. So $K(X)$ is a quadratic extension of $k(x)$, of the form $k(x)(\sqrt{h})$ with $h \in k[x]$ squarefree (characteristic $\neq 2$). By Riemann–Hurwitz there are $2g + 2 = 6$ branch points, which are the roots of h together with ∞ if $\deg h$ is odd. So $\deg h \in \{5, 6\}$. $\square$

Exercise 55.7.4. Let $C \subseteq \mathbb{P}_{\mathbb{C}}^2$ be a nonsingular curve of degree d and $p \notin C$ a general point. By projecting from p and applying Riemann–Hurwitz, recover $g(C) = (d - 1)(d - 2)/2$.

Solution. Projection from p is a morphism $C \rightarrow \mathbb{P}^1$ of degree d . Its ramification points are the points where the line through p is tangent to C . For general p they are simple ($e = 2$), and their number is the number of tangent lines to C through p , which is the degree $d(d - 1)$ of the polar curve $\sum_i p_i \partial F / \partial x_i = 0$ meeting C (Bézout, Example 53.6.4). So $2g - 2 = -2d + d(d - 1)$ and $g = (d - 1)(d - 2)/2$. $\square$

Chapter 56

Linear Algebraic Groups

56.1 Introduction

A Hodge structure on a real vector space V is the same as an action of the real algebraic group $\mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$, the Deligne torus (Proposition 56.4.5). The symmetry group of a rational Hodge structure, its Mumford–Tate group (Chapter 63), is the smallest algebraic subgroup of $\text{GL}(V)$ defined over $\mathbb{Q}$ through which this action factors. This chapter provides the language: affine group schemes and their Hopf algebras, the embedding of every algebraic group into some GL_n , representations, tori and characters, Lie algebras, reductive groups and Weil restriction. The ground field is a field k of characteristic 0 unless stated otherwise. References: [Wat79], [Mil17], [Bor91], [DM82].

Prerequisites. Chapter 50 (Schemes), Chapter 32 (Lie Groups) and Chapter 11 (Representations of Groups and Lie Algebras).

56.2 Affine group schemes

Definition 56.2.1. An *affine group scheme* over k is an affine k -scheme $G = \text{Spec } A$ whose functor of points $R \mapsto G(R) = \text{Hom}_{k\text{-alg}}(A, R)$ takes values in groups, functorially in the k -algebra R . Equivalently (Yoneda), $A = \mathcal{O}(G)$ is a *commutative Hopf algebra*: it has k -algebra maps $\Delta: A \rightarrow A \otimes A$ (comultiplication), $\varepsilon: A \rightarrow k$ (counit) and $S: A \rightarrow A$ (antipode), dual to multiplication, identity and inverse, satisfying the dual group axioms. A *linear algebraic group* is an affine group scheme of finite type. Homomorphisms are morphisms of schemes compatible with the group laws.

- Example 56.2.2.**
1. The *additive group* $\mathbb{G}_a = \text{Spec } k[t]$, $\Delta(t) = t \otimes 1 + 1 \otimes t$, with $\mathbb{G}_a(R) = (R, +)$.
 2. The *multiplicative group* $\mathbb{G}_m = \text{Spec } k[t, t^{-1}]$, $\Delta(t) = t \otimes t$, with $\mathbb{G}_m(R) = (R^*, \cdot)$. Its subgroup $\mu_n = \text{Spec } k[t]/(t^n - 1)$ of n -th roots of unity.
 3. $\text{GL}_n = \text{Spec } k[x_{ij}, 1/\det]$ with $\Delta(x_{ij}) = \sum_l x_{il} \otimes x_{lj}$, and its closed subgroups SL_n ($\det = 1$), the orthogonal group $\text{O}(q) = \{g : q(gv) = q(v)\}$ of a nondegenerate quadratic form, the symplectic group Sp_{2g} of a nondegenerate alternating form, and the diagonal torus $\mathbb{G}_m^n$.
 4. For a finite-dimensional vector space V , $\text{GL}(V)(R) = \text{Aut}_R(V \otimes_k R)$.

Definition 56.2.3. A *representation* of G on a k -vector space V is a natural action of $G(R)$ on $V \otimes_k R$ by R -linear maps. Equivalently it is a *comodule*: a linear map $\rho: V \rightarrow V \otimes A$ with

$(\rho \otimes \text{id})\rho = (\text{id} \otimes \Delta)\rho$ and $(\text{id} \otimes \varepsilon)\rho = \text{id}$. The element $g \in G(R)$, that is $g: A \rightarrow R$, acts by $(\text{id} \otimes g) \circ \rho$ extended R -linearly. The *regular representation* is A itself with $\rho = \Delta$.

Lemma 56.2.4. *Every comodule is the union of its finite-dimensional subcomodules.*

Proof. Let $v \in V$ and choose a basis (a_i) of A . Write $\rho(v) = \sum_i v_i \otimes a_i$ (finite sum) and $\Delta(a_i) = \sum_{j,l} c_{ijl} a_j \otimes a_l$. Applying the two sides of the coassociativity identity to v gives $\sum_i \rho(v_i) \otimes a_i = \sum_{i,j,l} c_{ijl} v_i \otimes a_j \otimes a_l$, and comparing the coefficients of a_l in the last factor gives $\rho(v_l) = \sum_{i,j} c_{ijl} v_i \otimes a_j$. So $W = \text{span}(v_i)$ is a subcomodule, and $v = \sum_i \varepsilon(a_i) v_i \in W$ by the counit identity. $\square$

Theorem 56.2.5. *Every linear algebraic group G is isomorphic to a closed subgroup of some GL_n .*

Proof. Let $A = \mathcal{O}(G)$ with algebra generators $f_1, \dots, f_m$. By Lemma 56.2.4 for the regular representation, there is a finite-dimensional subcomodule $V \subseteq A$ containing all f_i . Choose a basis $v_1, \dots, v_n$ of V and write $\Delta(v_j) = \sum_i v_i \otimes a_{ij}$ with $a_{ij} \in A$. Coassociativity shows that $x_{ij} \mapsto a_{ij}$ defines a Hopf algebra map $k[x_{ij}, 1/\det] \rightarrow A$, that is, a homomorphism $\rho: G \rightarrow \text{GL}_n$ (the determinant maps to a unit, whose inverse is the image of the determinant of $S(a_{ij})$). It is surjective on rings, because $v_j = (\varepsilon \otimes \text{id})\Delta(v_j) = \sum_i \varepsilon(v_i) a_{ij}$ lies in the image, and the v_j include the generators f_i . So ρ is a closed immersion. $\square$

Remark 56.2.6. In characteristic 0 every linear algebraic group is smooth (Cartier's theorem) [Mil17], so its reduced structure causes no trouble and $\dim G = \dim \text{Lie}(G)$. In characteristic p , $\mu_p = \text{Spec } k[t]/(t^p - 1) = \text{Spec } k[t]/(t - 1)^p$ is a non-reduced group scheme with one point.

56.3 Representations and Tannakian categories

Definition 56.3.1. The finite-dimensional representations of G form an abelian category $\text{Rep}_k(G)$ with tensor products, duals and the unit object k (trivial representation), and a *fibre functor* $\omega: \text{Rep}_k(G) \rightarrow \text{Vec}_k$ forgetting the action. For a representation V , the tensor spaces are $T^{r,s}(V) = V^{\otimes r} \otimes (V^\vee)^{\otimes s}$, and a *tensor* $t \in T^{r,s}(V)$ is *fixed* by G if $g(t \otimes 1) = t \otimes 1$ for all $g \in G(R)$ and all R .

Theorem 56.3.2. 1. (Tannaka duality) G is recovered from $(\text{Rep}_k(G), \omega)$: $G(R)$ is the group of families (λ_V) of R -linear automorphisms of $V \otimes R$, natural in V and compatible with tensor products and the unit.

2. (Chevalley) Every closed subgroup $H \subseteq G$ is the stabiliser of a line L in some finite-dimensional representation W of G .

3. If $H \subseteq \text{GL}(V)$ is reductive (Definition 56.6.1), then H is the subgroup of $\text{GL}(V)$ fixing all tensors in $\bigoplus_{r,s} T^{r,s}(V)$ that are fixed by H .

Statement (1) is [DM82, Theorem 2.11], (2) is Chevalley's theorem [Mil17], and (3) is [Del82, Proposition 3.1(c)]. Statement (3) is the reason why Mumford–Tate groups, defined as stabilisers of Hodge tensors, capture all the Hodge classes in tensor spaces (Chapter 63). Without reductivity (3) fails. Every tensor fixed by the group B of invertible upper triangular 2×2 matrices is fixed by GL_2 : a B -fixed vector in a representation of GL_2 is a highest weight vector of weight 0, so it spans a trivial subrepresentation. But $B \neq \text{GL}_2$.

Proposition 56.3.3. Let G be a linear algebraic group and V a representation. For $t \in T^{r,s}(V)$ the stabiliser $G_t = \{g : gt = t\}$ is a closed subgroup of G , defined over the field of definition of t . For $v \in V$, the stabiliser of the line kv is a closed subgroup.

Proof. Write $\rho(t) = \sum_i t_i \otimes a_i$ with the t_i a basis of $T^{r,s}(V)$ containing $t = t_1$. Then $g \in G(R)$ fixes t if and only if $g(a_1) = 1$ and $g(a_i) = 0$ for $i \geq 2$, which are polynomial equations. For the line, g stabilises kv if and only if $gv \wedge v = 0$ in $\Lambda^2 V$, again polynomial equations. $\square$

56.4 Tori, characters and the Deligne torus

Definition 56.4.1. A *character* of G is a homomorphism $G \rightarrow \mathbb{G}_m$; they form an abelian group $X^*(G)$ under pointwise multiplication (written additively). A *cocharacter* is a homomorphism $\mathbb{G}_m \rightarrow G$. A *split torus* is a group isomorphic to $\mathbb{G}_m^r$, and a *torus* is a group T such that $T_{\bar{k}}$ is a split torus. T is *split* over a field extension L if T_L is.

Proposition 56.4.2. Let $T = \mathbb{G}_m^r$. Then $X^*(T) \cong \mathbb{Z}^r$, the characters being the monomials $t^m = t_1^{m_1} \cdots t_r^{m_r}$, and $O(T) = k[X^*(T)]$ is the group algebra. A representation of T is the same as a $X^*(T)$ -graded vector space $V = \bigoplus_{\chi} V_{\chi}$, where T acts on V_{χ} through χ . In particular every representation of a split torus is a direct sum of one-dimensional ones.

Proof. $O(T) = k[t_i^{\pm 1}]$ has basis the monomials t^m , which are group-like: $\Delta(t^m) = t^m \otimes t^m$. A character is a group-like unit, and the group-like elements of a group algebra are the group elements (a group-like element $\sum c_m t^m$ satisfies $\sum c_m t^m \otimes t^m = \sum c_m c_{m'} t^m \otimes t^{m'}$, forcing a single term with $c_m = 1$). For a comodule V put $V_m = \{v : \rho(v) = v \otimes t^m\}$. Writing $\rho(v) = \sum_m v_m \otimes t^m$, coassociativity gives $\rho(v_m) = v_m \otimes t^m$, so $v_m \in V_m$, and the counit gives $v = \sum_m v_m$. The sum of the V_m is direct because the t^m are linearly independent. $\square$

Remark 56.4.3. For a torus T over k , split over a finite Galois extension L , $\text{Gal}(L/k)$ acts on $X^*(T_L) \cong \mathbb{Z}^r$, and $T \mapsto X^*(T_L)$ is an anti-equivalence between tori over k split by L and free abelian groups of finite rank with a $\text{Gal}(L/k)$ -action [Mil17]. Representations of T are $X^*(T_L)$ -graded L -vector spaces V_L with a semilinear Galois action compatible with the grading.

Definition 56.4.4. The *Deligne torus* is $\mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ (Section 56.7), the real algebraic group with $\mathbb{S}(R) = (R \otimes_{\mathbb{R}} \mathbb{C})^*$. Concretely $\mathbb{S} = \text{Spec } \mathbb{R}[a, b, 1/(a^2 + b^2)]$, $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ via $z = a + ib$, and $\mathbb{S}(\mathbb{C}) = (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})^* \cong \mathbb{C}^* \times \mathbb{C}^*$, where $z \in \mathbb{C}^* = \mathbb{S}(\mathbb{R})$ maps to $(z, \bar{z})$. The characters of $\mathbb{S}_{\mathbb{C}}$ are $z^p \bar{z}^q$, $(p, q) \in \mathbb{Z}^2$. The *weight cocharacter* $w: \mathbb{G}_m \rightarrow \mathbb{S}$ is the inclusion $\mathbb{R}^* \subseteq \mathbb{C}^*$, and the *norm* $\text{Nm}: \mathbb{S} \rightarrow \mathbb{G}_m$ is $z \mapsto z\bar{z}$.

Proposition 56.4.5. Let V be a finite-dimensional real vector space. Representations $h: \mathbb{S} \rightarrow \text{GL}(V)$ correspond bijectively to decompositions $V_{\mathbb{C}} = \bigoplus_{p,q} V^{p,q}$ with $\overline{V^{p,q}} = V^{q,p}$, via

$$V^{p,q} = \{v \in V_{\mathbb{C}} : h(z)v = z^{-p} \bar{z}^{-q} v \text{ for all } z \in \mathbb{C}^*\}.$$

Under this correspondence $w(r)$ acts on $V^{p,q}$ by $r^{-(p+q)}$, so $V = \bigoplus_n V_n$ with $V_{n,\mathbb{C}} = \bigoplus_{p+q=n} V^{p,q}$ is the weight decomposition.

Proof. $\mathbb{S}_{\mathbb{C}}$ is a split torus with character group $\mathbb{Z}^2$ generated by z and $\bar{z}$. By Proposition 56.4.2, a representation of $\mathbb{S}_{\mathbb{C}}$ on $V_{\mathbb{C}}$ is a $\mathbb{Z}^2$ -grading, which we index by $(p, q) \mapsto$ the character $z^{-p} \bar{z}^{-q}$ (Deligne's sign convention). The subgroup $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$ is Zariski dense in $\mathbb{S}_{\mathbb{C}}$, since a Laurent polynomial in $z, \bar{z}$ vanishing on $\{(z, \bar{z})\}$ is zero, so the grading is determined by the action of $\mathbb{C}^*$ as displayed. The representation is defined over $\mathbb{R}$ if and only if $V_{\mathbb{C}}$ is stable under the action of $\mathbb{C}^*$ and $h(z)$ commutes with complex conjugation on $V_{\mathbb{C}}$. If $v \in V^{p,q}$, then $h(z)\bar{v} = \overline{h(z)v} = \overline{z^{-p} \bar{z}^{-q} v} = z^{-q} \bar{z}^{-p} \bar{v}$, so $\bar{v} \in V^{q,p}$. Conversely, a grading with $\overline{V^{p,q}} = V^{q,p}$ defines $h(z)$ by the formula, and $h(z)$ commutes with conjugation by the same computation, hence is real. The statement about w is the case $z = r \in \mathbb{R}^*$. $\square$

56.5 The Lie algebra

Definition 56.5.1. Let $k[\varepsilon] = k[t]/(t^2)$. The *Lie algebra* of G is $\mathrm{Lie}(G) = \ker(G(k[\varepsilon]) \rightarrow G(k))$, a k -vector space identified with the tangent space $(\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$ at the identity (Exercise 50.10.2). The bracket is induced by the commutator in $G(k[\varepsilon_1, \varepsilon_2])$, and a representation ρ of G differentiates to a representation $d\rho$ of $\mathrm{Lie}(G)$.

Example 56.5.2. $\mathrm{Lie}(\mathrm{GL}_n) = \{1 + \varepsilon X\} \cong \mathfrak{gl}_n$, with the commutator bracket. Since $\det(1 + \varepsilon X) = 1 + \varepsilon \operatorname{tr} X$, $\mathrm{Lie}(\mathrm{SL}_n) = \{X : \operatorname{tr} X = 0\}$. Similarly $\mathrm{Lie}(\mathrm{O}(q)) = \{X : B(Xv, w) + B(v, Xw) = 0\}$, where B is the bilinear form of q , and $\mathrm{Lie}(\mathbb{G}_m^r) = k^r$ with zero bracket. For $k = \mathbb{R}$ or $\mathbb{C}$, $\mathrm{Lie}(G)$ is the Lie algebra of the Lie group $G(k)$ (Definition 32.2.3).

Proposition 56.5.3. In characteristic 0, for connected G , $H \mapsto \mathrm{Lie}(H)$ is an injective, inclusion-preserving map from connected closed subgroups of G to Lie subalgebras of $\mathrm{Lie}(G)$. A subspace $W \subseteq V$ of a representation is G -stable if and only if it is $\mathrm{Lie}(G)$ -stable, and a vector is fixed by G if and only if it is killed by $\mathrm{Lie}(G)$.

See [Bor91] and [Mil17]. It fails in characteristic p : μ_p and $\{1\}$ have different Lie algebras over a field of characteristic p but the same points, and $\mathbb{G}_a$ acting on k^2 by $t \mapsto \begin{pmatrix} 1 & t^p \\ 0 & 1 \end{pmatrix}$ has Lie acting trivially without fixing everything.

56.6 Reductive groups

Definition 56.6.1. A linear algebraic group G over k is *unipotent* if it is isomorphic to a closed subgroup of the upper triangular unipotent matrices. The *unipotent radical* $R_u(G)$ is the largest connected normal unipotent subgroup of $G_{\bar{k}}$. A connected G is *reductive* if $R_u(G) = 1$, and *semisimple* if moreover its centre is finite.

Theorem 56.6.2. Let k have characteristic 0 and G be connected.

1. G is reductive if and only if every finite-dimensional representation of G is semisimple (a direct sum of irreducible representations).
2. If G is reductive, then $G = Z(G)^\circ \cdot G^{\mathrm{der}}$, where the connected centre $Z(G)^\circ$ is a torus and the derived group G^{der} is semisimple, and $Z(G)^\circ \cap G^{\mathrm{der}}$ is finite.
3. Tori, GL_n , SL_n , Sp_{2g} and $\mathrm{SO}(q)$ are reductive; $\mathbb{G}_a$ and the group of invertible upper triangular matrices are not.

See [Mil17] and [Bor91]. For example, the representation of $\mathbb{G}_a$ on k^2 by $t \mapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ has the invariant line ke_1 without an invariant complement, so $\mathbb{G}_a$ is not reductive.

Proposition 56.6.3 (Compact real forms). Let G be a connected linear algebraic group over $\mathbb{R}$ and θ an involution of $G_{\mathbb{C}}$ such that $G^{(\theta)} = \{g \in G(\mathbb{C}) : \theta(\bar{g}) = g\}$ is compact. Then G is reductive. In particular, if $G \subseteq \mathrm{GL}(V)$ and there is a positive definite inner product on V such that G is stable under $g \mapsto g^T$ (the adjoint), then G is reductive.

Proof. The group $K = G^{(\theta)}$ is a compact real form of $G(\mathbb{C})$: it is the fixed group of the antiholomorphic involution $g \mapsto \theta(\bar{g})$, so its Lie algebra $\mathfrak{k}$ is a real form of $\mathrm{Lie}(G_{\mathbb{C}})$, that is, $\mathrm{Lie}(G_{\mathbb{C}}) = \mathfrak{k} \oplus i\mathfrak{k}$. Let W be a complex representation of $G_{\mathbb{C}}$ and $W' \subseteq W$ a subrepresentation. By Corollary 32.5.2, W has a K -invariant Hermitian inner product, and the orthogonal complement W'' of W' is K -stable, hence $\mathfrak{k}$ -stable. As $\mathfrak{k}$ spans $\mathrm{Lie}(G_{\mathbb{C}})$ over $\mathbb{C}$, W'' is $\mathrm{Lie}(G_{\mathbb{C}})$ -stable, hence $G_{\mathbb{C}}$ -stable by Proposition 56.5.3. So every representation of $G_{\mathbb{C}}$, and therefore of G , is semisimple, and G is reductive by Theorem 56.6.2(1). For the second statement, extend

the inner product to a Hermitian one on $V_{\mathbb{C}}$ and take $\theta(g) = (g^*)^{-1}$, where g^* is the Hermitian adjoint of $g \in G(\mathbb{C})$. Then $\theta(\bar{g}) = (g^T)^{-1}$ in a real orthonormal basis, so $G^{(\theta)} = G(\mathbb{C}) \cap U(V_{\mathbb{C}})$, which is compact. $\square$

Remark 56.6.4. A variant of this argument, with the Weil operator of a polarisation in place of the transpose, shows that Mumford–Tate groups of polarisable Hodge structures are reductive (Chapter 63).

56.7 Weil restriction of scalars

Definition 56.7.1. Let L/k be a finite field extension and Y an affine scheme over L . The *Weil restriction* $\text{Res}_{L/k} Y$ is the functor on k -algebras $R \mapsto Y(R \otimes_k L)$.

Proposition 56.7.2. $\text{Res}_{L/k} Y$ is represented by an affine k -scheme. If Y is a linear algebraic group of dimension d , then $\text{Res}_{L/k} Y$ is a linear algebraic group of dimension $d[L:k]$. If L/k is Galois, then $(\text{Res}_{L/k} Y)_L \cong \prod_{\sigma \in \text{Gal}(L/k)} Y^{\sigma}$.

Proof. Choose a basis $e_1, \dots, e_m$ of L over k . For $Y = \mathbb{A}_L^n$, a point of $Y(R \otimes_k L)$ is an n -tuple $y_j = \sum_i x_{ij} e_i$ with $x_{ij} \in R$, so $\text{Res}_{L/k} \mathbb{A}_L^n = \mathbb{A}_k^{nm}$. If $Y = V(f_1, \dots, f_s) \subseteq \mathbb{A}_L^n$, substitute $y_j = \sum_i x_{ij} e_i$ into f_l and expand in the basis: $f_l = \sum_i F_{li}(x) e_i$ with $F_{li} \in k[x]$. Then $\text{Res}_{L/k} Y = V(F_{li}) \subseteq \mathbb{A}_k^{nm}$. The group structure is inherited functorially. For the dimension and the Galois statement, use $(R \otimes_k L) \otimes_k L \cong \prod_{\sigma} R \otimes_k L$ for R an L -algebra, which gives $\text{Res}_{L/k} Y(R) = \prod_{\sigma} Y^{\sigma}(R)$ for L -algebras R . $\square$

Example 56.7.3. 1. $\mathbb{S} = \text{Res}_{\mathbb{C}/\mathbb{R}} \mathbb{G}_m$ is a two-dimensional torus over $\mathbb{R}$ that is not split, with $\mathbb{S}_{\mathbb{C}} \cong \mathbb{G}_m^2$; complex conjugation swaps the two characters z and $\bar{z}$.

2. For a number field E , $T_E = \text{Res}_{E/\mathbb{Q}} \mathbb{G}_m$ has $T_E(\mathbb{Q}) = E^*$. If E is a CM field, the Mumford–Tate group of an abelian variety with complex multiplication by E is a subtorus of T_E (Chapter 57).

3. The kernel $\mathbb{U}(1) = \ker(\text{Nm}: \mathbb{S} \rightarrow \mathbb{G}_m)$ is the circle group, with $\mathbb{U}(1)(\mathbb{R}) = S^1$, a torus that is compact and not split over $\mathbb{R}$.

56.8 Exercises

Exercise 56.8.1. Show that $X^*(\text{GL}_n) = \mathbb{Z}$, generated by $\det$, and that $X^*(\text{SL}_n) = 0$.

Solution. A character is a unit $\chi \in k[x_{ij}, 1/\det]$ with $\Delta(\chi) = \chi \otimes \chi$. The units of this ring are $c \det^m$, since $\det$ is irreducible, and group-likeness forces $c = 1$. For SL_n , the group $\text{SL}_n(\bar{k})$ is generated by the elementary matrices $1 + tE_{ij}$, $i \neq j$, and each family $t \mapsto 1 + tE_{ij}$ is a homomorphism $\mathbb{G}_a \rightarrow \text{SL}_n$. A character of $\mathbb{G}_a$ is a group-like unit of $k[t]$, hence 1. So every character of SL_n is trivial on a generating set of $\text{SL}_n(\bar{k})$, and as SL_n is reduced, it is trivial. $\square$

Exercise 56.8.2. Describe the representations of $\mathbb{S}$ on $\mathbb{R}^2$ given by $h(a + ib) = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}$ and by $h(z) = z\bar{z} \cdot \text{id}$, in terms of the decomposition of Proposition 56.4.5.

Solution. The first is multiplication by z on $\mathbb{C} = \mathbb{R}^2$. On $V_{\mathbb{C}} = \mathbb{C}^2$ its eigenvectors are $(1, -i)$ with eigenvalue z and $(1, i)$ with eigenvalue $\bar{z}$. With $h(z)v = z^{-p}\bar{z}^{-q}v$, these are $V^{-1,0}$ and $V^{0,-1}$, a Hodge structure of weight -1 (the Hodge structure of H_1 of an elliptic curve). The second acts by $z\bar{z} = z^1\bar{z}^1$ on all of $V_{\mathbb{C}}$, so $V = V^{-1,-1}$, of weight -2 . $\square$

Exercise 56.8.3. Show directly that the group B of invertible upper triangular 2×2 matrices has a representation that is not semisimple, and compute its unipotent radical.

Solution. The standard representation on k^2 has the invariant line ke_1 , and an invariant complement would be spanned by a common eigenvector outside ke_1 , which does not exist because $\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ has only the eigenline ke_1 . The unipotent radical is the normal subgroup U of unipotent upper triangular matrices, and $B/U \cong \mathbb{G}_m^2$ is a torus. $\square$

Chapter 57

Abelian Varieties

57.1 Introduction

A complex abelian variety is a complex torus $\mathbb{C}^g/\Lambda$ that embeds in projective space. Tori are the simplest compact Kähler manifolds, and their cohomology is completely explicit: exterior powers of H^1 . For this reason abelian varieties are the main testing ground of the Hodge conjecture. They include the known cases (divisors, Lefschetz classes), the first instances of exceptional Hodge classes (Weil classes), and Deligne's theorem that Hodge classes on abelian varieties are absolutely Hodge (Part IX). This chapter proves Riemann's criterion for a torus to be algebraic, describes line bundles, polarisations, the dual and the Jacobian, and discusses endomorphisms and complex multiplication. References: [BL04], [Mum70], [GH78, Chapter 2, Section 6].

Prerequisites. Chapter 55 (Algebraic Curves and Riemann Surfaces), Chapter 53 (Divisors and Line Bundles), Chapter 45 (The Hodge Decomposition) and Chapter 56 (Linear Algebraic Groups).

Throughout, V is a complex vector space of dimension g , $\Lambda \subseteq V$ a lattice (a discrete subgroup of rank $2g$ spanning V over $\mathbb{R}$), and $T = V/\Lambda$ the complex torus, with its structure of compact complex Lie group (group law written additively). $\bar{V}^\vee$ denotes the space of conjugate-linear forms $V \rightarrow \mathbb{C}$.

57.2 Complex tori and their cohomology

Proposition 57.2.1. 1. $H_1(T; \mathbb{Z}) = \Lambda$, and cup product gives ring isomorphisms $H^k(T; \mathbb{Z}) \cong \Lambda^k \text{Hom}(\Lambda, \mathbb{Z})$, the alternating k -forms on Λ .

2. The Hodge decomposition is $H^1(T; \mathbb{C}) = \text{Hom}_{\mathbb{R}}(\Lambda \otimes \mathbb{R}, \mathbb{C}) = V^\vee \oplus \bar{V}^\vee$, with $H^{1,0} = V^\vee$ (spanned by the dz_j) and $H^{0,1} = \bar{V}^\vee$ (spanned by the $d\bar{z}_j$), and $H^{p,q}(T) = \Lambda^p V^\vee \otimes \Lambda^q \bar{V}^\vee$.

Proof. (1) T is diffeomorphic to $(S^1)^{2g}$ via a $\mathbb{Z}$ -basis of Λ , and $\mathbb{R}^{2g} \rightarrow T$ is the universal covering, so $\pi_1(T) = H_1(T; \mathbb{Z}) = \Lambda$. The ring structure follows from the Künneth formula (Theorem 20.4.2) and $H^\bullet(S^1) = \Lambda^\bullet \mathbb{Z}$. (2) With a flat metric the harmonic forms are the constant-coefficient forms (Example 43.4.2); a constant 1-form is a real-linear form on V , and it splits uniquely into complex-linear and conjugate-linear parts. $\square$

Definition 57.2.2. A complex torus is an *abelian variety* if it is projective. A homomorphism of tori is a holomorphic group homomorphism, and an *isogeny* is a surjective homomorphism with finite kernel. More generally, over any field an *abelian variety* is a complete connected group variety; such a group is automatically commutative and projective [Mum70, Chapter II].

Lemma 57.2.3. *Every holomorphic map $f: T \rightarrow T'$ of complex tori with $f(0) = 0$ is a homomorphism, induced by a complex-linear map $F: V \rightarrow V'$ with $F(\Lambda) \subseteq \Lambda'$.*

Proof. Lift f to $F: V \rightarrow V'$ with $F(0) = 0$. For $\lambda \in \Lambda$, $F(v+\lambda) - F(v) \in \Lambda'$ depends continuously on v , hence is constant. So the partial derivatives of F are Λ -periodic holomorphic functions, so they descend to holomorphic functions on the compact connected manifold T , which are constant by the maximum principle (Proposition 29.2.3). So F is linear, and $F(\Lambda) \subseteq \Lambda'$ because $F(\lambda) = F(0 + \lambda) - F(0) \in \Lambda'$. $\square$

Proposition 57.2.4. *Let $T = V/\Lambda$ be a complex torus of dimension g and $n \geq 1$.*

1. *Multiplication by n , $[n]: T \rightarrow T$, is a surjective homomorphism with kernel $T[n] = \frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$. The torsion subgroup of T is $(\Lambda \otimes \mathbb{Q})/\Lambda \cong (\mathbb{Q}/\mathbb{Z})^{2g}$, and it is dense in T .*
2. *A homomorphism $f: T \rightarrow T'$ of tori of the same dimension is an isogeny if and only if its linear lift $F: V \rightarrow V'$ is an isomorphism; then $|\ker f| = [\Lambda' : F(\Lambda)]$.*
3. *If $f: T \rightarrow T'$ is an isogeny with $|\ker f| = d$, there is an isogeny $h: T' \rightarrow T$ with $h \circ f = [d]_T$ and $f \circ h = [d]_{T'}$. In particular being isogenous is an equivalence relation.*

Proof. (1) $[n]$ is induced by $v \mapsto nv$, which is bijective on V ; so $[n]$ is surjective, and $x = v + \Lambda$ lies in the kernel if and only if $nv \in \Lambda$. A $\mathbb{Z}$ -basis of Λ identifies $\frac{1}{n}\Lambda/\Lambda$ with $(\frac{1}{n}\mathbb{Z}/\mathbb{Z})^{2g} \cong (\mathbb{Z}/n\mathbb{Z})^{2g}$, and the union over n is $(\Lambda \otimes \mathbb{Q})/\Lambda$. The rational points $\Lambda \otimes \mathbb{Q}$ are dense in $\Lambda \otimes \mathbb{R} = V$, so their image is dense in T .

(2) By Lemma 57.2.3, f is induced by a $\mathbb{C}$ -linear F with $F(\Lambda) \subseteq \Lambda'$. If F is an isomorphism, $F(\Lambda)$ is a lattice in V' contained in Λ' , hence of finite index, f is surjective, and $\ker f = F^{-1}(\Lambda')/\Lambda \cong \Lambda'/F(\Lambda)$ is finite. Conversely, if f has finite kernel, then $\ker F$ is a complex subspace whose image in T is contained in the finite group $\ker f$; a connected subspace mapping into a finite set maps to a point, so $\ker F \subseteq \Lambda$, which forces $\ker F = 0$; equal dimensions make F an isomorphism.

(3) The group $\Lambda'/F(\Lambda)$ has order d , so $d\Lambda' \subseteq F(\Lambda)$. Hence $H = dF^{-1}: V' \rightarrow V$ maps Λ' into Λ and induces a homomorphism $h: T' \rightarrow T$ with $HF = d$ and $FH = d$, which is an isogeny by (2). Reflexivity and transitivity of isogeny are clear, and (3) gives symmetry. $\square$

57.3 Line bundles and the Riemann conditions

Proposition 57.3.1. *Identify $H^2(T; \mathbb{Z})$ with the alternating forms $E: \Lambda \times \Lambda \rightarrow \mathbb{Z}$ (Proposition 57.2.1). A class E lies in $H^{1,1}(T)$ if and only if $E(iu, iv) = E(u, v)$ for all $u, v \in V$ (extending E $\mathbb{R}$ -bilinearly), if and only if $E = \operatorname{Im} H$ for a (unique) Hermitian form H on V , namely $H(u, v) = E(iu, v) + iE(u, v)$. Every such E is $c_1(L)$ for a holomorphic line bundle L , so*

$$\operatorname{NS}(T) := \operatorname{im}(c_1) = \{H \text{ Hermitian on } V : \operatorname{Im} H(\Lambda, \Lambda) \subseteq \mathbb{Z}\}.$$

Proof. A real alternating form E on V is a constant real 2-form. Its $(2, 0)$ and $(0, 2)$ components vanish exactly when E is of type $(1, 1)$ for the complex structure i , that is, $E(iu, iv) = E(u, v)$: the $(2, 0) + (0, 2)$ part is $\frac{1}{2}(E - E(i \cdot, i \cdot))$. Given this, $H(u, v) = E(iu, v) + iE(u, v)$ is Hermitian: $H(iu, v) = E(-u, v) + iE(iu, v) = i(E(iu, v) + iE(u, v)) = iH(u, v)$, and $\overline{H(v, u)} = E(iv, u) - iE(v, u) = E(iu, v) + iE(u, v)$, using $E(iv, u) = -E(u, iv) = -E(iu, i^2v) = E(iu, v)$. Conversely Im of a Hermitian form satisfies the condition. The last statement is Lemma 47.2.2. $\square$

Theorem 57.3.2 (Riemann). *A complex torus $T = V/\Lambda$ is an abelian variety if and only if there is a positive definite Hermitian form H on V with $\operatorname{Im} H(\Lambda, \Lambda) \subseteq \mathbb{Z}$. Such an H (or its alternating form $E = \operatorname{Im} H$) is a polarisation; it is the first Chern class of an ample line bundle.*

Proof. For a constant real 2-form ω put $E_\omega = -\omega$ and $H_\omega(u, v) = E_\omega(iu, v) + iE_\omega(u, v)$. If ω is of type $(1, 1)$, then H_ω is Hermitian (Proposition 57.3.1) with $\operatorname{Re} H_\omega(u, u) = -\omega(iu, u) = \omega(u, iu)$. So ω is a positive $(1, 1)$ -form, that is, the fundamental form of a Hermitian metric (Proposition 44.2.2), if and only if H_ω is positive definite. Conversely every Hermitian H is H_ω for $\omega = -\operatorname{Im} H$.

Suppose H is positive definite with $\operatorname{Im} H$ integral on Λ . Then $\omega = -\operatorname{Im} H$ is a positive constant $(1, 1)$ -form, closed because its coefficients are constant, and its class is integral. So T is a Hodge manifold (Definition 47.2.3) and projective by Corollary 47.6.6.

Conversely, let T be projective, with an ample line bundle L and a metric of positive curvature form ω_h (Theorem 54.6.1). Average over translations: $\bar{\omega} = \int_T t_x^* \omega_h dx$ for the normalised Haar measure (Theorem 32.5.1). Each $t_x^* \omega_h$ is positive and cohomologous to ω_h , because translations are homotopic to the identity, so $\bar{\omega}$ is a positive translation-invariant closed $(1, 1)$ -form in the integral class $c_1(L)$. A translation-invariant form has constant coefficients. So $H_{\bar{\omega}}$ is positive definite, and $\operatorname{Im} H_{\bar{\omega}} = -\bar{\omega}$ is integral on Λ , being (up to sign) the integral class $c_1(L)$ in $H^2(T; \mathbb{Z}) = \Lambda^2 \operatorname{Hom}(\Lambda, \mathbb{Z})$. $\square$

Remark 57.3.3. The proof shows that ample line bundles correspond to positive definite H once a constant form ω is matched with the alternating form $-\omega$; this is the convention of [BL04], where $c_1(L) = \operatorname{Im} H$. With the opposite matching one gets negative definite forms instead; Riemann's condition is insensitive to this, since $-H$ is positive definite if H is negative definite.

Proposition 57.3.4. *Let E be a polarisation. There is a basis $\lambda_1, \dots, \lambda_g, \mu_1, \dots, \mu_g$ of Λ with $E(\lambda_j, \mu_k) = d_j \delta_{jk}$ and $E(\lambda_j, \lambda_k) = E(\mu_j, \mu_k) = 0$, where $d_1 \mid d_2 \mid \dots \mid d_g$ are positive integers, the type of the polarisation. It is principal if all $d_j = 1$. Then $\frac{1}{d_1} \mu_1, \dots, \frac{1}{d_g} \mu_g$ is a $\mathbb{C}$ -basis of V , and in it $\Lambda = \mathbb{Z}\mathbb{Z}^g + D\mathbb{Z}^g$ with $D = \operatorname{diag}(d_j)$ and Z a symmetric complex $g \times g$ matrix with $\operatorname{Im} Z$ positive definite. Conversely every such Z in the Siegel upper half space $\mathfrak{H}_g$ defines a polarised abelian variety of type D .*

Proof. The symplectic basis is the elementary divisor theorem for alternating forms over $\mathbb{Z}$ (Frobenius), proved like the Smith normal form (Theorem 7.11.2); see [BL04]. Since H is positive definite, the real span of the μ_j is E -isotropic, and H is real and positive on it, so the μ_j are $\mathbb{C}$ -linearly independent: a complex relation $\sum c_j \mu_j = 0$ with $c = a + ib$ gives $\sum a_j \mu_j = -i \sum b_j \mu_j$, and pairing with H forces $a = b = 0$. Writing $\lambda_j = \sum_k Z_{kj} \mu_k / d_k$, the conditions that the λ 's span an isotropic subspace and that H is positive definite translate into $Z = Z^T$ and $\operatorname{Im} Z > 0$ by a direct computation [BL04]. The converse is the same computation read backwards together with Theorem 57.3.2. $\square$

Example 57.3.5. For $g = 1$, $\mathfrak{H}_1$ is the upper half plane, and $\Lambda = \mathbb{Z}\tau + \mathbb{Z}$ with $\operatorname{Im} \tau > 0$. Every one-dimensional torus is an abelian variety, as Riemann's condition only asks $\operatorname{Im} \tau > 0$ (compare Example 46.4.4). The alternating form $E(\tau, 1) = 1$ gives $H(u, v) = u\bar{v} / \operatorname{Im} \tau$.

Counterexample 57.3.6. For $g \geq 2$ the general complex torus is not algebraic. Fix a real basis identifying V with $\mathbb{R}^{2g}$ and Λ with $\mathbb{Z}^{2g}$, so that complex tori correspond to complex structures J on $\mathbb{R}^{2g}$. For each of the countably many nonzero integral alternating forms E , the condition $E(Ju, Jv) = E(u, v)$ defines a proper closed real-analytic subset of the space of complex structures when $g \geq 2$ [BL04]. Outside the countable union of these subsets, $\operatorname{NS}(T) = 0$ by Proposition 57.3.1, so there is no ample line bundle and T is not projective. For such tori the Hodge conjecture in degree 2 has no analytic cycles to work with; more generally the analytic version of the conjecture fails for some complex tori (Chapter 69).

Theorem 57.3.7 (Appell–Humbert). *Holomorphic line bundles on T correspond bijectively to pairs (H, α) with H Hermitian, $E = \operatorname{Im} H$ integral on Λ , and $\alpha: \Lambda \rightarrow U(1)$ a semicharacter:*

$\alpha(\lambda + \mu) = \alpha(\lambda)\alpha(\mu)e^{i\pi E(\lambda, \mu)}$. The bundle $L(H, \alpha)$ is the quotient of $V \times \mathbb{C}$ by $\lambda \cdot (v, t) = (v + \lambda, \alpha(\lambda)e^{\pi H(v, \lambda) + \frac{\pi}{2}H(\lambda, \lambda)}t)$, and $c_1(L(H, \alpha)) = E$. Its sections are the theta functions, and if H is positive definite of type $(d_1, \dots, d_g)$, then $h^0(L) = d_1 \cdots d_g$. Moreover L^2 is globally generated and L^3 is very ample (Lefschetz).

This is [BL04]; see also [Mum70].

57.4 The dual abelian variety

Proposition 57.4.1. $\text{Pic}^0(T) \cong \hat{T} = \bar{V}^\vee / \hat{\Lambda}$, where $\hat{\Lambda} = \{l \in \bar{V}^\vee : \text{Im } l(\Lambda) \subseteq \mathbb{Z}\}$, a complex torus of dimension g , the dual torus. A polarisation H defines a homomorphism $\varphi_H: T \rightarrow \hat{T}$, $v \mapsto H(v, \cdot)$, which is an isogeny of degree $(d_1 \cdots d_g)^2$, and an isomorphism if the polarisation is principal. If T is an abelian variety, so is $\hat{T}$.

Proof. By the exponential sequence, $\text{Pic}^0(T) = H^1(T, \mathcal{O})/H^1(T; \mathbb{Z})$ (Theorem 42.2.4), and $H^1(T, \mathcal{O}) = H^{0,1} = \bar{V}^\vee$ (Proposition 57.2.1). The map $H^1(T; \mathbb{R}) = \text{Hom}_{\mathbb{R}}(V, \mathbb{R}) \rightarrow \bar{V}^\vee$ sends a real linear form a to its conjugate-linear part l_a , and $a = 2 \text{Re } l_a$. So the image of $H^1(T; \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z})$ is $\{l : 2 \text{Re } l(\Lambda) \subseteq \mathbb{Z}\}$, which the $\mathbb{C}$ -linear automorphism $l \mapsto 2il$ of $\bar{V}^\vee$ carries onto $\hat{\Lambda}$, since $\text{Im}(2il) = 2 \text{Re } l$. The map $v \mapsto H(v, \cdot)$ is complex-linear from V to $\bar{V}^\vee$ and maps Λ into $\hat{\Lambda}$ because $\text{Im } H(\Lambda, \Lambda) \subseteq \mathbb{Z}$. It is injective since H is nondegenerate, so its kernel on T is $\hat{\Lambda}'/\Lambda$ with $\hat{\Lambda}' = \{v : \text{Im } H(v, \Lambda) \subseteq \mathbb{Z}\}$, of order $\det E = (d_1 \cdots d_g)^2$ in a symplectic basis. The inverse of H gives a polarisation on $\hat{T}$ [BL04]. $\square$

57.5 Jacobians

Theorem 57.5.1. Let C be a compact Riemann surface of genus g . The Jacobian $J(C) = H^0(C, \Omega^1)^\vee / H_1(C; \mathbb{Z})$ (Definition 55.6.1) is a principally polarised abelian variety, the polarisation being the intersection form on $H_1(C; \mathbb{Z})$.

Proof. $J(C)$ is a complex torus of dimension g (Proposition 55.6.2). The intersection form E on $H_1(C; \mathbb{Z})$ is unimodular (Poincaré duality, Corollary 21.6.3), so a symplectic basis $a_1, \dots, a_g, b_1, \dots, b_g$ exists with $E(a_j, b_k) = \delta_{jk}$. Extend E $\mathbb{R}$ -linearly to $V = H^0(\Omega^1)^\vee \cong H_1(C; \mathbb{R})$. Under Poincaré duality $H_1(C; \mathbb{R}) \cong H^1(C; \mathbb{R})$ and the real isomorphism $H^1(C; \mathbb{R}) \rightarrow H^{1,0}$, $\alpha \mapsto \alpha^{1,0}$, the form E becomes $\pm \int_C \alpha \wedge \beta = \pm 2 \text{Re} \int_C \alpha^{1,0} \wedge \overline{\beta^{1,0}}$, and the complex structure of V becomes that of $H^{1,0}$ (up to conjugation). So $E(iu, iv) = E(u, v)$, and the Hermitian form $H(u, v) = E(iu, v) + iE(u, v)$ is, up to sign and a positive factor, $i \int_C \alpha^{1,0} \wedge \overline{\beta^{1,0}}$, which is definite by the Hodge–Riemann relations (Theorem 46.4.2, Example 46.4.4). Replacing E by $-E$ if necessary, H is positive definite, and $J(C)$ is an abelian variety by Theorem 57.3.2. The polarisation is principal as E is unimodular. $\square$

Remark 57.5.2. By Torelli's theorem the principally polarised Jacobian determines C . The Jacobians form a subvariety of dimension $3g - 3$ of the moduli space $\mathfrak{S}_g / \text{Sp}_{2g}(\mathbb{Z})$ of principally polarised abelian varieties, which has dimension $g(g+1)/2$. For $g \leq 3$ the dimensions agree and the general principally polarised abelian variety is a Jacobian; for $g \geq 4$ it is not, and characterising Jacobians is the *Schottky problem* [BL04].

57.6 Endomorphisms and complex multiplication

Proposition 57.6.1. Let $E = \mathbb{C}/(\mathbb{Z}\tau + \mathbb{Z})$. Then $\text{End}(E)$ is $\mathbb{Z}$, or an order in the imaginary quadratic field $\mathbb{Q}(\tau)$; the latter happens if and only if τ is quadratic over $\mathbb{Q}$.

Proof. By Lemma 57.2.3 an endomorphism is multiplication by some $\alpha \in \mathbb{C}$ with $\alpha\Lambda \subseteq \Lambda$. Then $\alpha = a + b\tau$ and $\alpha\tau = c + d\tau$ with integers a, b, c, d . If $b \neq 0$, eliminating α gives $b\tau^2 + (a - d)\tau - c = 0$, so τ is quadratic, imaginary as $\text{Im } \tau > 0$, and $\alpha \in \mathbb{Q}(\tau)$ is integral (it is an eigenvalue of an integer matrix). If $b = 0$ then $\alpha = a \in \mathbb{Z}$. Conversely, if $a\tau^2 + b\tau + c = 0$ with integers $a \neq 0, b, c$, then $\alpha = a\tau$ satisfies $\alpha \cdot 1 = a\tau \in \Lambda$ and $\alpha\tau = -b\tau - c \in \Lambda$, so multiplication by $\alpha \notin \mathbb{Z}$ is an endomorphism. $\square$

Definition 57.6.2. An abelian variety A of dimension g has *complex multiplication* (CM) if $\text{End}(A) \otimes \mathbb{Q}$ contains a commutative semisimple subalgebra of dimension $2g$ over $\mathbb{Q}$, for instance a CM field F (a totally imaginary quadratic extension of a totally real field) with $[F : \mathbb{Q}] = 2g$.

Theorem 57.6.3. Let A be an abelian variety over $\mathbb{C}$.

1. (Poincaré reducibility) A is isogenous to a product $A_1^{n_1} \times \cdots \times A_r^{n_r}$ with the A_i simple and pairwise non-isogenous, and $\text{End}(A) \otimes \mathbb{Q} \cong \prod_i M_{n_i}(D_i)$ with $D_i = \text{End}(A_i) \otimes \mathbb{Q}$ division algebras.
2. A polarisation defines the Rosati involution $f \mapsto f^\dagger$ on $\text{End}(A) \otimes \mathbb{Q}$, and $\text{tr}(f f^\dagger) > 0$ for $f \neq 0$.
3. (Shimura–Taniyama) If A has CM by a CM field F of degree $2g$, then $A \cong \mathbb{C}^g / \Phi(\mathfrak{a})$ for a fractional ideal $\mathfrak{a}$ of F and a CM type $\Phi = \{\phi_1, \dots, \phi_g\}$, a set of embeddings $F \rightarrow \mathbb{C}$ containing exactly one of each pair of complex conjugate embeddings.

This is [BL04] and [Mum70]. The endomorphism algebras D_i are classified by Albert into four types. For the Hodge conjecture, CM abelian varieties are the case where the Mumford–Tate group is a torus (Chapter 63), and where exceptional Hodge classes first appear (Chapter 68).

- Example 57.6.4.**
1. $E_i = \mathbb{C}/\mathbb{Z}[i]$ has $\text{End}(E_i) = \mathbb{Z}[i]$ and $E_\rho = \mathbb{C}/\mathbb{Z}[\rho]$, $\rho = e^{2\pi i/3}$, has $\text{End}(E_\rho) = \mathbb{Z}[\rho]$.
 2. For E without CM, $\text{End}(E \times E) \otimes \mathbb{Q} = M_2(\mathbb{Q})$, and $\text{NS}(E \times E) \otimes \mathbb{Q}$ has rank 3; for E with CM it has rank 4. This is Example 53.7.3(3).
 3. The Jacobian of the curve $y^2 = x^5 - 1$ has CM by $\mathbb{Q}(\zeta_5)$, a CM field of degree $4 = 2g$.

57.7 Exercises

Exercise 57.7.1. Let E_1, E_2 be elliptic curves. Show that $\text{NS}(E_1 \times E_2) \cong \mathbb{Z}^2 \oplus \text{Hom}(E_1, E_2)$, and deduce that its rank is 2, 3 or 4.

Solution. Write $\Lambda = \Lambda_1 \oplus \Lambda_2$. An integral alternating form of type $(1, 1)$ on Λ restricts to such forms on Λ_1 and Λ_2 , each a multiple of the generator $(\mathbb{Z}^2)$, and its mixed part $E(\lambda_1, \lambda_2)$ is a pairing of type $(1, 1)$ between Λ_1 and Λ_2 , which via E_1 's principal polarisation is the same as a $\mathbb{Z}$ -linear map $\Lambda_1 \rightarrow \Lambda_2$ commuting with i after identification, that is, a homomorphism $E_1 \rightarrow E_2$ (Lemma 57.2.3). By Proposition 57.6.1 and its analogue for homomorphisms, $\text{Hom}(E_1, E_2)$ has rank 0, 1 or 2. $\square$

Exercise 57.7.2. Show that the Hodge classes in $H^2(T; \mathbb{Q})$ of a complex torus form the space $\text{NS}(T) \otimes \mathbb{Q}$, and that for $T = E^g$, E without CM, this space has dimension $g(g+1)/2$.

Solution. The first statement is Proposition 57.3.1. For E^g , $\text{NS} \otimes \mathbb{Q}$ is the space of $g \times g$ Hermitian matrices with entries in $\text{End}(E) \otimes \mathbb{Q} = \mathbb{Q}$ that are Hermitian for the Rosati involution, that is, symmetric rational matrices, of dimension $g(g+1)/2$ [BL04]. $\square$

Exercise 57.7.3. Show that multiplication by n on T is an isogeny of degree n^{2g} with kernel $\frac{1}{n}\Lambda/\Lambda \cong (\mathbb{Z}/n)^{2g}$.

Solution. The map is induced by $v \mapsto nv$ on V , surjective, with kernel $\{v : nv \in \Lambda\}/\Lambda = \frac{1}{n}\Lambda/\Lambda$, which has n^{2g} elements since $\Lambda \cong \mathbb{Z}^{2g}$. $\square$

Chapter 58

Algebraic Cycles and Chow Groups

58.1 Introduction

An algebraic cycle is a formal integer combination of subvarieties. The Hodge conjecture asks which cohomology classes come from cycles, so cycles and the relations between them are the objects on the algebraic side of the conjecture. This chapter introduces the Chow groups, cycles modulo rational equivalence, with their functoriality (proper pushforward, flat pullback) and the intersection product on smooth varieties, and compares the adequate equivalence relations (rational, algebraic, homological, numerical) that appear in the standard conjectures. We follow Fulton [Ful98]. Many constructions are quoted from there, but the computations for affine and projective spaces and the elementary exact sequences are proved. Chapter 59 attaches cohomology classes to cycles.

Prerequisites. Chapter 53 (Divisors and Line Bundles) and Chapter 50 (Schemes).

Throughout, schemes are separated and of finite type over a field k , a *variety* is an integral such scheme, and *subvariety* means closed subvariety.

58.2 Cycles and rational equivalence

Definition 58.2.1. The group $Z_k(X)$ of k -cycles is the free abelian group on the k -dimensional subvarieties of X ; write $Z_\bullet(X) = \bigoplus_k Z_k(X)$. For a closed subscheme $Y \subseteq X$ with irreducible components Y_i of dimension k and generic points η_i , its *fundamental cycle* is $[Y] = \sum_i \ell(\mathcal{O}_{Y, \eta_i})[Y_i]$, where ℓ is the length of the Artinian local ring. For a $(k+1)$ -dimensional subvariety W and $f \in K(W)^*$, the divisor $\text{div}(f) = \sum_V \text{ord}_V(f)[V] \in Z_k(X)$, the sum over codimension-one subvarieties V of W , is defined with the order function $\text{ord}_V(f) = \ell(\mathcal{O}_{W,V}/(a)) - \ell(\mathcal{O}_{W,V}/(b))$ for $f = a/b$ [Ful98, Section 1.2]. The subgroup of *cycles rationally equivalent to zero* is generated by these divisors, and the *Chow group* is

$$\text{CH}_k(X) = Z_k(X) / \{\text{cycles rationally equivalent to } 0\}.$$

For X equidimensional of dimension n put $\text{CH}^p(X) = \text{CH}_{n-p}(X)$.

- Example 58.2.2.**
1. $\text{CH}_n(X) = Z_n(X)$ is free on the n -dimensional irreducible components.
 2. If X is a nonsingular variety, ord_V is the valuation of the DVR $\mathcal{O}_{X,V}$ (Definition 53.2.1), so $\text{CH}^1(X) = \text{Cl}(X) \cong \text{Pic}(X)$ (Proposition 53.3.2).
 3. For a nonsingular projective curve C , $\text{CH}_0(C) = \text{CH}^1(C) = \text{Pic}(C)$, and the degree gives $\text{CH}_0(C) \rightarrow \mathbb{Z}$ with kernel $\text{Pic}^0(C)$ (Corollary 55.2.3); over $\mathbb{C}$ this kernel is the Jacobian (Theorem 55.6.3).

Proposition 58.2.3 (Localisation sequence). *Let $Y \subseteq X$ be closed and $U = X \setminus Y$. Then the sequence*

$$\mathrm{CH}_k(Y) \xrightarrow{i_*} \mathrm{CH}_k(X) \xrightarrow{j^*} \mathrm{CH}_k(U) \rightarrow 0$$

is exact, where i_ is induced by the inclusion of cycles and j^* by restriction $[V] \mapsto [V \cap U]$.*

Proof. On cycles, $0 \rightarrow Z_k(Y) \rightarrow Z_k(X) \rightarrow Z_k(U) \rightarrow 0$ is exact: $V \mapsto V \cap U$ is a bijection from k -dimensional subvarieties of X not contained in Y to those of U (with inverse the closure), and those contained in Y are the subvarieties of Y . Both maps preserve rational equivalence, since the divisor of $f \in K(W)^*$ restricts to the divisor of f on $W \cap U$, and a divisor on $W \subseteq Y$ is a divisor on W in X . So j^* is surjective and $j^* \circ i_* = 0$. If $\alpha \in Z_k(X)$ restricts to a cycle rationally equivalent to zero on U , say $\alpha|_U = \sum \mathrm{div}(f_i)$ with $f_i \in K(W_i)^*$, $W_i \subseteq U$, then $\alpha - \sum \mathrm{div}(f_i)$, computed on the closures $\overline{W_i}$ (which have the same function fields), restricts to 0 on U , so it is supported on Y . $\square$

Proposition 58.2.4 (Homotopy invariance). *Let $p: X \times \mathbb{A}^1 \rightarrow X$ be the projection. Then $p^*: \mathrm{CH}_k(X) \rightarrow \mathrm{CH}_{k+1}(X \times \mathbb{A}^1)$, $[V] \mapsto [V \times \mathbb{A}^1]$, is surjective. More generally flat pullback along the projection of a vector bundle of rank r is surjective (in fact an isomorphism) $\mathrm{CH}_k(X) \rightarrow \mathrm{CH}_{k+r}(E)$.*

Sketch of proof. By the localisation sequence and Noetherian induction on X , it suffices to treat X affine and irreducible, and to show that every $(k+1)$ -dimensional subvariety V of $X \times \mathbb{A}^1$ is, modulo cycles pulled back from X , rationally equivalent to cycles supported over a proper closed subset of X . If V dominates a subvariety W of dimension $k+1$ of X , then V is a divisor in $W \times \mathbb{A}^1$, defined over the generic point of W by a polynomial $f \in K(W)[t]$. The divisor of f on $W \times \mathbb{A}^1$ is $[V]$ plus cycles lying over a proper closed subset of W , which are handled by induction. If V dominates a subvariety of dimension k , then $V = W \times \mathbb{A}^1$ is a pullback. The omitted verification is the bookkeeping in this induction [Ful98, Proposition 1.9 and Theorem 3.3]. $\square$

Theorem 58.2.5. $\mathrm{CH}_k(\mathbb{A}^n) = 0$ for $k < n$ and $\mathrm{CH}_n(\mathbb{A}^n) = \mathbb{Z}$. For $\mathbb{P}^n$, $\mathrm{CH}_k(\mathbb{P}^n) \cong \mathbb{Z}$ for $0 \leq k \leq n$, generated by the class of a linear subspace L_k of dimension k .

Proof. For $\mathbb{A}^n$ use induction on n and Proposition 58.2.4: $\mathrm{CH}_k(\mathbb{A}^n) = \mathrm{CH}_k(\mathbb{A}^{n-1} \times \mathbb{A}^1)$ is a quotient of $\mathrm{CH}_{k-1}(\mathbb{A}^{n-1})$, which vanishes for $k-1 < n-1$; and $\mathrm{CH}_0(\mathbb{A}^0) = \mathbb{Z}$, $\mathrm{CH}_k(\mathbb{A}^0) = 0$ for $k > 0$. For $\mathbb{P}^n$, the localisation sequence for $\mathbb{P}^{n-1} \subseteq \mathbb{P}^n$ with complement $\mathbb{A}^n$ gives $\mathrm{CH}_k(\mathbb{P}^{n-1}) \rightarrow \mathrm{CH}_k(\mathbb{P}^n) \rightarrow \mathrm{CH}_k(\mathbb{A}^n) \rightarrow 0$. For $k < n$ the right term vanishes, so $\mathrm{CH}_k(\mathbb{P}^n)$ is generated by $[L_k]$ by induction; for $k = n$ it is $\mathbb{Z}[\mathbb{P}^n]$. To see that $[L_k]$ has infinite order, use the degree: a k -cycle α and a general linear subspace M of dimension $n-k$ meet in finitely many points, and $\alpha \mapsto \#(\alpha \cap M)$ (with multiplicities) is invariant under rational equivalence and equals 1 on $[L_k]$ [Ful98, Chapter 1]. $\square$

Example 58.2.6. Let $Y = V(f) \subseteq \mathbb{P}^n$ be an irreducible hypersurface of degree d and $U = \mathbb{P}^n \setminus Y$. Then $\mathrm{CH}^1(U) \cong \mathbb{Z}/d$. Indeed $\mathrm{div}(f/x_0^d) = [Y] - d[H_0]$ on $\mathbb{P}^n$, with $H_0 = V(x_0)$, so $[Y] = d[H]$ in $\mathrm{CH}^1(\mathbb{P}^n) = \mathbb{Z}[H]$ (Theorem 58.2.5). The localisation sequence (Proposition 58.2.3) reads $\mathrm{CH}_{n-1}(Y) = \mathbb{Z}[Y] \rightarrow \mathrm{CH}_{n-1}(\mathbb{P}^n) = \mathbb{Z}[H] \rightarrow \mathrm{CH}_{n-1}(U) \rightarrow 0$, with image $d\mathbb{Z}[H]$. For instance the complement of a smooth conic in $\mathbb{P}^2$ has $\mathrm{Pic}(U) = \mathrm{CH}^1(U) \cong \mathbb{Z}/2$ (Example 58.2.2(2)), a nonzero torsion Picard group on a smooth affine surface: the restriction of a line L is not principal on U , but $2L$ restricts to the principal divisor of ℓ^2/q , with ℓ and q equations of L and of the conic.

Counterexample 58.2.7. The degree of Theorem 58.3.1(3) needs completeness. On $\mathbb{A}^1 = \mathrm{Spec} k[t]$ every point a is the divisor of the rational function $t - a$, so $[a] = 0$ in $\mathrm{CH}_0(\mathbb{A}^1)$, and $\mathrm{CH}_0(\mathbb{A}^1) = 0$, in accordance with Theorem 58.2.5. On $\mathbb{P}^1$ the same function has divisor

$[a] - [\infty]$: the pole at infinity, invisible on $\mathbb{A}^1$, is what makes the degree of a principal divisor zero rather than one.

58.3 Functoriality

- Theorem 58.3.1.** 1. (Proper pushforward) For a proper morphism $f: X \rightarrow Y$, the map $f_*[V] = \deg(V/f(V))[f(V)]$ if $\dim f(V) = \dim V$ and 0 otherwise, where $\deg(V/f(V)) = [K(V) : K(f(V))]$, induces homomorphisms $f_*: \text{CH}_k(X) \rightarrow \text{CH}_k(Y)$, functorial in f .
2. (Flat pullback) For a flat morphism $f: X \rightarrow Y$ of relative dimension r , $f^*[V] = [f^{-1}(V)]$ induces $f^*: \text{CH}_k(Y) \rightarrow \text{CH}_{k+r}(X)$.
3. For X complete, the degree $\deg: \text{CH}_0(X) \rightarrow \mathbb{Z}$, $\sum n_P[P] \mapsto \sum n_P[k(P) : k]$, is well defined; it is the pushforward to $\text{CH}_0(\text{Spec } k) = \mathbb{Z}$.

This is [Ful98, Theorem 1.4 and Theorem 1.7]. For (1), the key case is a proper surjective morphism of curves, where the pushforward of a principal divisor is principal by the norm: $f_*\text{div}(g) = \text{div}(N_{K(X)/K(Y)}(g))$. For curves this is Corollary 55.2.3 applied to g and its norm.

58.4 Chern classes and the intersection product

Definition 58.4.1. For a line bundle L on X and a k -dimensional subvariety V , write $L|_V \cong \mathcal{O}_V(D)$ for a Cartier divisor D on V and put $c_1(L) \cap [V] = [D] \in \text{CH}_{k-1}(X)$. This is well defined and extends to $c_1(L) \cap -: \text{CH}_k(X) \rightarrow \text{CH}_{k-1}(X)$ [Ful98, Chapter 2]. For a vector bundle E of rank r , with projective bundle $p: \mathbb{P}(E) \rightarrow X$ and tautological line bundle $\mathcal{O}(1)$, the Chern classes $c_i(E) \cap -$ are the unique operations with $\sum_{i=0}^r p^*c_i(E) \cap c_1(\mathcal{O}(1))^{r-i} \cap p^*\alpha = 0$ and $c_0 = 1$ (Grothendieck's definition) [Ful98, Chapter 3].

Theorem 58.4.2. The Chern classes of vector bundles on X , acting on $\text{CH}_\bullet(X)$, commute with each other and satisfy naturality under flat pullback and proper pushforward (projection formula), the Whitney formula $c(E) = c(E')c(E'')$ for $0 \rightarrow E' \rightarrow E \rightarrow E'' \rightarrow 0$, and the splitting principle. For X smooth over $\mathbb{C}$ they map to the topological Chern classes of Chapter 37 under the cycle class map.

This is [Ful98, Theorem 3.2]; the comparison with topology is in [Ful98, Chapter 19].

Theorem 58.4.3. Let X be a nonsingular variety of dimension n . There is a product $\text{CH}^p(X) \otimes \text{CH}^q(X) \rightarrow \text{CH}^{p+q}(X)$, $\alpha \otimes \beta \mapsto \alpha \cdot \beta$, making $\text{CH}^\bullet(X) = \bigoplus_p \text{CH}^p(X)$ a commutative, associative graded ring with unit $[X]$, such that:

- if subvarieties V, W intersect properly, that is, every component Z of $V \cap W$ has codimension $\text{codim } V + \text{codim } W$, then $[V] \cdot [W] = \sum_Z i(Z; V \cdot W)[Z]$ with positive intersection multiplicities, given by Serre's formula $i(Z; V \cdot W) = \sum_j (-1)^j \ell_{\mathcal{O}_{X,Z}} \text{Tor}_j^{\mathcal{O}_{X,Z}}(\mathcal{O}_V, \mathcal{O}_W)$; if the intersection is also transverse at the generic point of Z , then $i(Z; V \cdot W) = 1$;
- for a morphism $f: X \rightarrow Y$ of nonsingular varieties there is a ring homomorphism $f^*: \text{CH}^\bullet(Y) \rightarrow \text{CH}^\bullet(X)$, and for f proper the projection formula $f_*(f^*\alpha \cdot \beta) = \alpha \cdot f_*\beta$ holds;
- on $\text{CH}^1(X) = \text{Pic}(X)$ the product with CH^p is $c_1(L) \cap -$, and for $n = 2$ the product $\text{CH}^1 \times \text{CH}^1 \rightarrow \text{CH}^2 \xrightarrow{\deg} \mathbb{Z}$ is the intersection number of Definition 53.6.1.

This is [Ful98, Chapters 6–8]; the construction uses deformation to the normal cone. Serre's formula is discussed in [Ful98, Chapter 7]. Chow's moving lemma gives an older construction for quasi-projective X : cycles can be moved within their rational equivalence class to intersect properly.

Example 58.4.4. $\mathrm{CH}^\bullet(\mathbb{P}^n) = \mathbb{Z}[h]/(h^{n+1})$, with $h = [H]$ the class of a hyperplane: $h^k = [L_{n-k}]$, since k general hyperplanes meet transversally in a linear subspace. So a subvariety of codimension p and degree d has class dh^p , and Bézout's theorem says that properly intersecting subvarieties of degrees d and e intersect in a cycle of degree de .

Example 58.4.5. CH_0 can be enormous. For a complex projective surface S with $h^{2,0}(S) \neq 0$, Mumford showed that the kernel of $\deg: \mathrm{CH}_0(S) \rightarrow \mathbb{Z}$ is not “finite-dimensional”: it is not parametrised by any variety [Mum68]. By contrast, Bloch's conjecture predicts that $\mathrm{CH}_0(S) = \mathbb{Z}$ when $p_g(S) = q(S) = 0$; it is known for surfaces not of general type. These phenomena are the starting point of the Bloch–Beilinson conjectures relating Chow groups to Hodge theory.

58.5 Correspondences

Definition 58.5.1. Let X, Y be nonsingular projective varieties, X of pure dimension d . A *correspondence of degree r* from X to Y is an element of $\mathrm{CH}^{d+r}(X \times Y)$. It acts on Chow groups by $\Gamma_*(\alpha) = \mathrm{pr}_{2*}(\Gamma \cdot \mathrm{pr}_1^* \alpha)$, and correspondences compose by $\Gamma' \circ \Gamma = \mathrm{pr}_{13*}(\mathrm{pr}_{12}^* \Gamma \cdot \mathrm{pr}_{23}^* \Gamma')$.

Proposition 58.5.2. *Composition of correspondences is associative, $(\Gamma' \circ \Gamma)_* = \Gamma'_* \circ \Gamma_*$, the diagonal Δ_X acts as the identity, and the graph of a morphism $f: X \rightarrow Y$ acts as f_* , its transpose as f^* .*

This is [Ful98, Proposition 16.1.1]; it follows from the projection formula and the compatibility of pullback and pushforward in fibre squares. Correspondences are the morphisms in the category of Chow motives, and the idempotent correspondences, projectors, cut motives out of varieties (Chapter 73).

58.6 Adequate equivalence relations

Definition 58.6.1. Let X be a nonsingular projective variety and $\alpha \in Z^p(X)$.

1. α is *algebraically equivalent to zero* if there are a nonsingular projective curve C , points $c_0, c_1 \in C$ and a cycle Γ on $C \times X$ with $\alpha = \Gamma(c_1) - \Gamma(c_0)$, where $\Gamma(c) = \mathrm{pr}_{2*}(\Gamma \cdot [\{c\} \times X])$.
2. For a Weil cohomology theory $H^\bullet$ with a cycle class map $\mathrm{cl}: Z^p(X) \rightarrow H^{2p}(X)$ compatible with pullbacks, proper pushforwards, products and degrees, for instance singular cohomology when $k = \mathbb{C}$ (Chapter 59), α is *homologically equivalent to zero* if $\mathrm{cl}(\alpha) = 0$.
3. α is *numerically equivalent to zero* if $\deg(\alpha \cdot \beta) = 0$ for all $\beta \in Z^{n-p}(X)$.

We write $\mathrm{CH}^p(X) \rightarrow \mathrm{Alg}^p(X) \rightarrow \mathrm{Hom}^p(X) \rightarrow \mathrm{Num}^p(X)$ for the quotients of $Z^p(X)$ by these relations.

Proposition 58.6.2. *Rational equivalence implies algebraic equivalence, which implies homological equivalence (for any cohomology theory as in Definition 58.6.1(2)), which implies numerical equivalence.*

Proof. Rational equivalence is the special case $C = \mathbb{P}^1$ [Ful98, Proposition 1.6]: the divisor of f on W is $\Gamma(0) - \Gamma(\infty)$ for the closure Γ of the graph of $f: W \dashrightarrow \mathbb{P}^1$. If $\alpha = \Gamma(c_1) - \Gamma(c_0)$, then $\mathrm{cl}(\alpha) = \Gamma_*(\mathrm{cl}(c_1) - \mathrm{cl}(c_0))$, and $\mathrm{cl}(c_1) = \mathrm{cl}(c_0)$ in $H^2(C)$ since both are the point class, which is determined by the degree. Homological equivalence implies numerical equivalence because the cycle class map is assumed compatible with products and degrees: $\deg(\alpha \cdot \beta) = \int \mathrm{cl}(\alpha) \cup \mathrm{cl}(\beta)$. The previous step used the compatibility with pullback and pushforward. $\square$

Remark 58.6.3. All three inclusions can be strict. Rational and algebraic equivalence differ already for points on an elliptic curve, since $\text{Pic}^0(E) \neq 0$. Algebraic and homological equivalence differ in codimension ≥ 2 : the Griffiths group $\text{Hom}^p(X)/\text{Alg}^p(X)$ can be nonzero, and even not finitely generated, for 1-cycles on a general quintic threefold (Griffiths, Clemens; Chapter 67). Whether homological and numerical equivalence coincide is Grothendieck's standard conjecture D (Chapter 73); it is known for divisors (Matsusaka) and in characteristic 0 it follows from the Hodge conjecture.

58.7 Riemann–Roch

Definition 58.7.1. For a vector bundle E with Chern roots $x_1, \dots, x_r$ (splitting principle), the Chern character and Todd class are

$$\text{ch}(E) = \sum_i e^{x_i} = r + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \cdots, \quad \text{td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}} = 1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2) + \cdots$$

in $\text{CH}^\bullet(X) \otimes \mathbb{Q}$. For a coherent sheaf $\mathcal{F}$ on a nonsingular variety, $\text{ch}(\mathcal{F})$ is defined through a finite locally free resolution.

Theorem 58.7.2 (Grothendieck–Riemann–Roch). *Let $f: X \rightarrow Y$ be a proper morphism of nonsingular quasi-projective varieties and $\mathcal{F}$ a coherent sheaf on X . Then in $\text{CH}^\bullet(Y) \otimes \mathbb{Q}$*

$$\text{ch}\left(\sum_i (-1)^i R^i f_* \mathcal{F}\right) \cdot \text{td}(T_Y) = f_*(\text{ch}(\mathcal{F}) \cdot \text{td}(T_X)).$$

In particular, for X projective and $Y = \text{Spec } k$ (Hirzebruch–Riemann–Roch),

$$\chi(X, \mathcal{F}) = \int_X \text{ch}(\mathcal{F}) \text{td}(T_X).$$

This is [Ful98, Theorem 15.2 and Corollary 15.2.1]. Hirzebruch proved the second formula over $\mathbb{C}$ [Hir66]. Over $\mathbb{C}$ it is also a special case of the Atiyah–Singer index theorem for the Dolbeault complex (Chapter 39).

Example 58.7.3. 1. For a curve of genus g and a line bundle L : $\text{td}(T_C) = 1 + \frac{1}{2}c_1(T_C)$ and $\int c_1(T_C) = 2 - 2g$, so HRR gives $\chi(L) = \deg L + 1 - g$, which is Theorem 55.3.2.

2. For a surface S with canonical class K and $c_2 = c_2(T_S)$: $\text{td}(T_S) = 1 - \frac{1}{2}K + \frac{1}{12}(K^2 + c_2)$. For $\mathcal{F} = \mathcal{O}_S$ this gives Noether's formula $\chi(\mathcal{O}_S) = \frac{1}{12}(K^2 + c_2)$, and for a divisor D , $\chi(\mathcal{O}(D)) = \chi(\mathcal{O}_S) + \frac{1}{2}D \cdot (D - K)$.

3. For $S = \mathbb{P}^2$: $K^2 = 9$ and $c_2 = 3$, so $\chi(\mathcal{O}) = 1$. For a K3 surface: $K = 0$, $c_2 = 24$ (Exercise 48.6.1), so $\chi(\mathcal{O}) = 2$, matching $h^0 - h^1 + h^2 = 1 - 0 + 1$.

Example 58.7.4. Theorem 58.7.2 has content beyond the case $Y = \text{Spec } k$; here is a case where both sides can be computed. Let X be a nonsingular projective variety, $i: D \rightarrow X$ the inclusion of a nonsingular prime divisor, $d = [D] \in \text{CH}^1(X)$, and $\mathcal{F} = \mathcal{O}_D$. Since i is a closed immersion, $R^q i_* \mathcal{O}_D = 0$ for $q > 0$, and the exact sequence $0 \rightarrow \mathcal{O}_X(-D) \rightarrow \mathcal{O}_X \rightarrow i_* \mathcal{O}_D \rightarrow 0$ gives $\text{ch}(i_* \mathcal{O}_D) = 1 - e^{-d}$ (the Chern character is additive in exact sequences, by the Whitney formula of Theorem 58.4.2 and the splitting principle). So the left side is $(1 - e^{-d}) \text{td}(T_X)$. On the right, the exact sequence $0 \rightarrow T_D \rightarrow i^* T_X \rightarrow N_{D/X} \rightarrow 0$ with $N_{D/X} = i^* \mathcal{O}_X(D)$ and the multiplicativity of the Todd class give

$$\text{td}(T_D) = i^* \left(\text{td}(T_X) \cdot \frac{1 - e^{-d}}{d} \right),$$

where $(1 - e^{-d})/d = 1 - \frac{1}{2}d + \frac{1}{6}d^2 - \dots$ is the inverse of $\text{td}(\mathcal{O}_X(D))$. By the projection formula (Theorem 58.4.3(2)), $i_* i^* \beta = \beta \cdot i_* [D] = \beta \cdot d$, so the right side is $\text{td}(T_X)(1 - e^{-d})$ as well. The denominators of the Todd class cancel against the normal bundle: this is the mechanism of the general proof, which factors a projective morphism into a closed immersion and a projection $\mathbb{P}_Y^n \rightarrow Y$ and treats the two cases separately [Ful98, Chapter 15].

Example 58.7.5. Let $f: X \rightarrow Y$ be a finite étale morphism of degree n between nonsingular projective varieties (Definition 50.8.4). Then $f^* \Omega_Y \cong \Omega_X$, so $T_X \cong f^* T_Y$, and $R^q f_* \mathcal{O}_X = 0$ for $q > 0$ because f is finite, hence affine. Theorem 58.7.2 and the projection formula give

$$\text{ch}(f_* \mathcal{O}_X) \text{td}(T_Y) = f_*(f^* \text{td}(T_Y)) = \text{td}(T_Y) \cdot f_*[X] = n \text{td}(T_Y),$$

and since $\text{td}(T_Y)$ is invertible, $\text{ch}(f_* \mathcal{O}_X) = n$: the rank- n vector bundle $f_* \mathcal{O}_X$ has all its Chern classes zero in $\text{CH}^\bullet(Y) \otimes \mathbb{Q}$. Taking degrees, $\chi(X, \mathcal{O}_X) = \chi(Y, f_* \mathcal{O}_X) = n \chi(Y, \mathcal{O}_Y)$. For curves this is $1 - g(X) = n(1 - g(Y))$, the Riemann–Hurwitz formula for unramified coverings (Theorem 55.4.1).

58.8 Exercises

Exercise 58.8.1. Compute $\text{CH}^\bullet(\mathbb{P}^1 \times \mathbb{P}^1)$ using the localisation sequence, and show that it is $\mathbb{Z}[a, b]/(a^2, b^2)$ with a, b the classes of the two rulings.

Solution. Remove the two lines $\{\infty\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{\infty\}$; the complement is $\mathbb{A}^2$. Localisation and Theorem 58.2.5 show that CH_1 is generated by the two lines and CH_0 by a point, and $\text{CH}_2 = \mathbb{Z}$. The two line classes are independent, as their intersection numbers with the lines are $a \cdot a = 0$, $a \cdot b = 1$, $b \cdot b = 0$ (Example 53.6.4). The ring structure follows: $a^2 = b^2 = 0$ and $ab = [\text{pt}]$. $\square$

Exercise 58.8.2. Use Example 58.4.4 to show that two plane conics without common components meet in at most 4 points, and in exactly 4 counted with multiplicity.

Solution. $[C] \cdot [C'] = (2h)(2h) = 4h^2 = 4[\text{pt}]$ in $\text{CH}^2(\mathbb{P}^2)$, and since the intersection is proper, it equals $\sum_P i(P)[P]$ with $i(P) \geq 1$. Taking degrees, $\sum i(P) = 4$. $\square$

Exercise 58.8.3. Check Noether's formula for a smooth quartic surface and for $\mathbb{P}^1 \times \mathbb{P}^1$.

Solution. For the quartic, $K = 0$, $c_2 = 24$, and $\chi(\mathcal{O}) = 2$ as computed in Example 58.7.3. For $\mathbb{P}^1 \times \mathbb{P}^1$, $K = -2a - 2b$ has $K^2 = 8$, $c_2 = \chi_{\text{top}} = 4$, and $\frac{8+4}{12} = 1 = \chi(\mathcal{O})$. $\square$

Chapter 59

The Cycle Class Map

59.1 Introduction

A subvariety Z of codimension p in a complex projective manifold X has a cohomology class $[Z] \in H^{2p}(X; \mathbb{Z})$. Topologically it is the Poincaré dual of the fundamental class of Z , which exists although Z may be singular. Analytically it is represented by the current of integration over Z . The resulting *cycle class map* $\text{cl}: \text{CH}^p(X) \rightarrow H^{2p}(X; \mathbb{Z})$ is a ring homomorphism compatible with pullbacks, pushforwards and Chern classes. Its image consists of Hodge classes (Corollary 59.4.2), and the Hodge conjecture asks whether, after tensoring with $\mathbb{Q}$, every Hodge class is in the image. This chapter constructs the map, states its properties, and shows that the conjecture can equivalently be phrased with Chern classes of vector bundles. References: [Ful98, Chapter 19], [Voi02b, Chapter 11], [GH78, Chapter 0, Section 4].

Prerequisites. Chapter 58 (Algebraic Cycles and Chow Groups), Chapter 54 (Analytic Spaces and GAGA), Chapter 35 (de Rham Cohomology), Chapter 21 (Topological Manifolds and Poincaré Duality) and Chapter 45 (The Hodge Decomposition).

Throughout, X is a nonsingular complex variety of dimension n , identified with the complex manifold X^{an} (Chapter 54), and cohomology is singular cohomology of X^{an} .

59.2 Fundamental classes

Lemma 59.2.1. *Let S be a closed complex analytic subset of complex dimension $\leq d$ of a complex manifold. Then $H_i^{\text{BM}}(S; \mathbb{Z}) = 0$ for $i > 2d$, and $H_{2d}^{\text{BM}}(S; \mathbb{Z})$ is free on the irreducible components of dimension d .*

Sketch of proof. Analytic sets admit triangulations compatible with a stratification by complex submanifolds, the strata of S having real dimension $\leq 2d$ [Ful98, Chapter 19]. So the locally finite chain complex of S vanishes above degree $2d$, which gives the first statement. For the second, let $\Sigma = \text{Sing}(S) \cup S_{<d}$, where $S_{<d}$ is the union of the components of dimension $< d$. It has complex dimension $\leq d - 1$, and the long exact sequence $H_{2d}^{\text{BM}}(\Sigma) \rightarrow H_{2d}^{\text{BM}}(S) \rightarrow H_{2d}^{\text{BM}}(S \setminus \Sigma) \rightarrow H_{2d-1}^{\text{BM}}(\Sigma)$ has outer terms zero by the first statement. The complex manifold $S \setminus \Sigma$ is canonically oriented, of real dimension $2d$, with one connected component for each d -dimensional irreducible component of S (the regular locus of an irreducible analytic set is connected), and H_{2d}^{BM} of an oriented $2d$ -manifold is free on its connected components (Remark 21.8.1). $\square$

Definition 59.2.2. Let $Z \subseteq X$ be a subvariety of codimension p . Its *fundamental class* $[Z] \in H_{2n-2p}^{\text{BM}}(Z; \mathbb{Z})$ is the class corresponding to Z in Lemma 59.2.1. Its *cycle class* $\text{cl}(Z) \in H^{2p}(X; \mathbb{Z})$ is the image of $[Z]$ under $H_{2n-2p}^{\text{BM}}(Z) \cong H^{2p}(X, X \setminus Z) \rightarrow H^{2p}(X)$ (Remark 21.8.2).

It extends linearly to $\text{cl}: Z^p(X) \rightarrow H^{2p}(X; \mathbb{Z})$. For X compact, $\text{cl}(Z)$ is the Poincaré dual of $i_*[Z] \in H_{2n-2p}(X; \mathbb{Z})$.

Proposition 59.2.3. *Let X be compact. The image of $\text{cl}(Z)$ in $H^{2p}(X; \mathbb{R}) \cong H_{\text{dR}}^{2p}(X)$ is the class of the current of integration over Z_{reg} of Proposition 41.4.2: for every closed $(2n - 2p)$ -form φ ,*

$$\int_X \text{cl}(Z) \wedge \varphi = \int_{Z_{\text{reg}}} \varphi.$$

Sketch of proof. For Z nonsingular this is Theorem 35.4.3. In general both sides are determined by their restriction to the complement of $\Sigma = \text{Sing}(Z)$. The topological class is determined there by Lemma 59.2.1, and the current by the fact that integration over Z_{reg} is a closed current whose restriction to $X \setminus \Sigma$ is the integration current of a closed submanifold. Two closed currents of degree $2p$ that agree off an analytic set of real codimension $> 2p$ in X are equal (the difference would be a closed current supported on Σ of degree below its codimension, which vanishes by the support theorem) [GH78, Chapter 3], [Voi02b, Section 11.1]. $\square$

59.3 The cycle class map and its properties

Theorem 59.3.1. *Let X be a nonsingular complex variety.*

1. cl factors through rational equivalence and gives homomorphisms $\text{cl}: \text{CH}^p(X) \rightarrow H^{2p}(X; \mathbb{Z})$, which form a ring homomorphism $\text{CH}^\bullet(X) \rightarrow H^{2\bullet}(X; \mathbb{Z})$ for the intersection product and the cup product.
2. It is compatible with pullback along morphisms of nonsingular varieties and with proper pushforward (Gysin maps in cohomology), and with exterior products.
3. $\text{cl}(c_i(E) \cap [X]) = c_i(E)$, the topological Chern class of the underlying complex vector bundle.
4. For X projective, $\deg \alpha = \int_X \text{cl}(\alpha)$ for $\alpha \in \text{CH}^n(X)$.

This is [Ful98, Chapter 19]. For (1), rational equivalence is generated by the fibres $\Gamma(0) - \Gamma(\infty)$ of a family $\Gamma \subseteq \mathbb{P}^1 \times X$ (Proposition 58.6.2), and $\text{cl}(\Gamma(0)) = \text{cl}(\Gamma(\infty))$ because the two points of $\mathbb{P}^1$ have the same class in $H^2(\mathbb{P}^1; \mathbb{Z})$ and $\text{cl}(\Gamma(t)) = \text{pr}_{2*}(\text{cl}(\Gamma) \cup \text{pr}_1^* \text{cl}(t))$. Multiplicativity reduces, by the moving lemma or deformation to the normal cone, to transverse intersections, where it is the statement that cup product is Poincaré dual to transverse intersection (Remark 21.8.2).

Example 59.3.2. 1. On $\mathbb{P}^n$, $\text{cl}(h^k)$ is the k -th power of the positive generator of $H^2(\mathbb{CP}^n; \mathbb{Z})$, and $\text{cl}: \text{CH}^\bullet(\mathbb{P}^n) \rightarrow H^{2\bullet}(\mathbb{CP}^n; \mathbb{Z})$ is an isomorphism of rings (Example 58.4.4 and Theorem 20.5.1).

2. For a divisor D , $\text{cl}(D) = c_1(\mathcal{O}(D))$, by (3) applied to $E = \mathcal{O}(D)$ and $c_1(\mathcal{O}(D)) \cap [X] = [D]$. So on $\text{CH}^1(X) = \text{Pic}(X)$ the cycle class map is the first Chern class of Theorem 42.2.4.
3. For a point x of a projective X , $\text{cl}(x)$ is the generator of $H^{2n}(X; \mathbb{Z}) \cong \mathbb{Z}$ dual to the fundamental class. So cl on CH_0 is the degree, and it loses all the information in the huge kernel of Example 58.4.5.

Example 59.3.3. 1. Let C be a nonsingular projective curve of genus g . On $\text{CH}^1(C) = \text{Pic}(C)$ the cycle class map is the degree $\text{CH}^1(C) \rightarrow H^2(C; \mathbb{Z}) \cong \mathbb{Z}$ (Example 59.3.2(2) and (3)), which is surjective, and its kernel $\text{CH}^1(C)_{\text{hom}}$ is the group of divisor classes of

degree 0, isomorphic to the Jacobian $J(C)$ by the Abel–Jacobi theorem (Theorem 55.6.3). For $g \geq 1$ this kernel is an uncountable group; for an elliptic curve and points $p \neq q$ the class of $p - q$ is nonzero (Counterexample 42.3.4 and GAGA). For $g = 0$, cl is an isomorphism.

2. Let $Z = V(f_1, \dots, f_p) \subseteq \mathbb{P}^n$ be a complete intersection of hypersurfaces of degrees $d_1, \dots, d_p$, meeting properly and generically transversally. Then $[Z] = [V(f_1)] \cdots [V(f_p)]$ in $\text{CH}^p(\mathbb{P}^n)$, so by Theorem 59.3.1(1) and Exercise 59.6.1, $\text{cl}(Z) = d_1 \cdots d_p h^p$, and $\int \text{cl}(Z) \cup h^{n-p} = d_1 \cdots d_p$ is the degree of Z (Bézout).

59.4 Cycle classes are Hodge classes

Definition 59.4.1. Let X be a compact Kähler manifold. The group of *rational Hodge classes* of degree $2p$ is $\text{Hdg}^p(X) = H^{2p}(X; \mathbb{Q}) \cap H^{p,p}(X)$, the intersection taken inside $H^{2p}(X; \mathbb{C})$. The group of *integral Hodge classes* $\text{Hdg}^p(X; \mathbb{Z})$ is the preimage of $H^{p,p}(X)$ under $H^{2p}(X; \mathbb{Z}) \rightarrow H^{2p}(X; \mathbb{C})$; it contains the torsion subgroup.

Corollary 59.4.2. For a nonsingular projective complex variety X , $\text{cl}(\text{CH}^p(X)) \subseteq \text{Hdg}^p(X; \mathbb{Z})$, and $\text{cl}(\text{CH}^p(X)) \otimes \mathbb{Q} \subseteq \text{Hdg}^p(X)$.

Proof. By Proposition 59.2.3, the image of $\text{cl}(Z)$ in $H^{2p}(X; \mathbb{C})$ is the class of the integration current over Z , which lies in $H^{p,p}(X)$ by Proposition 45.5.8. Extend linearly. $\square$

Corollary 59.4.3. Let $X \subseteq \mathbb{P}^N$ be a nonsingular projective variety of dimension n , let $h \in H^2(X; \mathbb{Z})$ be the class of a hyperplane section, and let $Z = \sum_i a_i Z_i \in Z^p(X)$ be a nonzero effective cycle ($a_i \geq 0$). Then $\int_X \text{cl}(Z) \cup h^{n-p} > 0$. In particular $\text{cl}(Z)$ is not a torsion class, and a nonzero effective cycle is not homologically, hence not rationally, equivalent to zero.

Proof. By Example 59.3.2(2), $h = c_1(\mathcal{O}(1)|_X)$, which is represented by the restriction ω of the Fubini–Study form (Example 42.4.3 and naturality of Chern–Weil forms), a Kähler form on X . By Proposition 59.2.3 and Proposition 44.2.3,

$$\int_X \text{cl}(Z_i) \cup h^{n-p} = \int_{Z_i, \text{reg}} \omega^{n-p} = (n-p)! \text{vol}(Z_{i, \text{reg}}) > 0,$$

and summing with the coefficients $a_i \geq 0$, not all zero, gives a positive number. A torsion class, or the class 0, would give 0. $\square$

Counterexample 59.4.4. Corollary 59.4.3 is a constraint on which Hodge classes can be classes of *effective* cycles, and the Hodge conjecture cannot be strengthened to effective cycles. On $X = \mathbb{P}^1 \times \mathbb{P}^1$ with the classes a, b of the two rulings, $\int (a - b) \cup (a + b) = 0$ for the hyperplane class $a + b$ of the Segre embedding (as $a^2 = b^2 = 0$), so $a - b$ is an integral Hodge class, the class of the divisor $\{p\} \times \mathbb{P}^1 - \mathbb{P}^1 \times \{q\}$, but not the class of any effective cycle. In general the classes of effective cycles of codimension p form a monoid, closed under addition, which by the corollary meets the hyperplane $\{\alpha : \int_X \alpha \cup h^{n-p} = 0\}$ only in 0.

Remark 59.4.5. The *Hodge conjecture* (Chapter 64) asserts that $\text{cl}(\text{CH}^p(X)) \otimes \mathbb{Q} = \text{Hdg}^p(X)$ for every nonsingular projective complex variety. The integral version, $\text{cl}(\text{CH}^p(X)) = \text{Hdg}^p(X; \mathbb{Z})$, is false: Atiyah and Hirzebruch found torsion classes that are not algebraic [AH62], and Kollár found non-torsion counterexamples (Chapter 69). For $p = 1$ the integral statement is true: this is the Lefschetz $(1, 1)$ -theorem (Chapter 65), which for projective X follows from Lemma 47.2.2 together with Example 59.3.2(2) and GAGA.

59.5 Cycles and coherent sheaves

Proposition 59.5.1. *Let X be a nonsingular projective variety and $Z \subseteq X$ a subvariety of codimension p . Then $\text{ch}(\mathcal{O}_Z) = [Z] + (\text{terms in } \text{CH}^{>p}(X) \otimes \mathbb{Q})$, where ch is computed with a finite locally free resolution of $\mathcal{O}_Z$. Consequently the $\mathbb{Q}$ -subspace of $H^{2p}(X; \mathbb{Q})$ spanned by the classes of algebraic cycles equals the $\mathbb{Q}$ -subalgebra generated by the Chern classes of algebraic vector bundles on X .*

Proof. The first statement is in [Ful98, Chapter 15]: it follows from Grothendieck–Riemann–Roch (Theorem 58.7.2) applied to a resolution of singularities of Z , or directly from the fact that the lowest nonzero term of ch of a sheaf supported in codimension p is its fundamental cycle. On a nonsingular projective variety every coherent sheaf has a finite resolution by locally free sheaves [Har77, Exercises III.6.8 and III.6.9], so $\text{ch}(\mathcal{O}_Z)$ is a polynomial with rational coefficients in Chern classes of vector bundles. By descending induction on p , every $[Z]$ lies in the $\mathbb{Q}$ -algebra generated by Chern classes of vector bundles. Conversely, Chern classes of vector bundles are classes of algebraic cycles, $c_i(E) \cap [X] \in \text{CH}^i(X)$ (Theorem 59.3.1(3)), and products of cycle classes are cycle classes (Theorem 59.3.1(1)). $\square$

Remark 59.5.2. Proposition 59.5.1 suggests versions of the Hodge conjecture for compact Kähler manifolds: with analytic cycles, with Chern classes of holomorphic vector bundles, or with Chern classes of coherent analytic sheaves. All of these fail for suitable complex tori: Voisin constructed a four-dimensional torus with a nonzero Hodge class in H^4 that is not a combination of Chern classes of coherent sheaves [Voi02a]. So projectivity is essential (Chapter 69).

Remark 59.5.3. The cycle class map refines to the *Deligne cycle class* with values in Deligne cohomology $H_D^{2p}(X; \mathbb{Z}(p))$. Its restriction to homologically trivial cycles is the Abel–Jacobi map to the intermediate Jacobian (Chapter 67), which sees information about $\text{CH}^p(X)$ invisible to cohomology, such as Pic^0 for $p = 1$.

59.6 Exercises

Exercise 59.6.1. Let $X \subseteq \mathbb{P}^n$ be a hypersurface of degree d . Show that $\text{cl}(X) = dh$ in $H^2(\mathbb{CP}^n; \mathbb{Z})$.

Solution. $[X] = \text{div}(f/x_0^d) + d[H_0]$ in $Z^1(\mathbb{P}^n)$ for a defining polynomial f , so $[X] = dh$ in $\text{CH}^1(\mathbb{P}^n)$, and $\text{cl}(h)$ is the generator. Alternatively, $\text{cl}(X) = c_1(\mathcal{O}(X)) = c_1(\mathcal{O}(d)) = dh$ by Example 59.3.2(2). $\square$

Exercise 59.6.2. Let X be a nonsingular projective variety of dimension n and $\Delta \subseteq X \times X$ the diagonal. Show that $\deg(\Delta \cdot \Delta) = \chi_{\text{top}}(X)$, and more generally the *Lefschetz fixed point formula*: for a morphism $f: X \rightarrow X$ with graph Γ_f , $\deg(\Gamma_f \cdot \Delta) = \sum_i (-1)^i \text{tr}(f^* | H^i(X; \mathbb{Q}))$.

Solution. Choose a basis (e_a) of $H^\bullet(X; \mathbb{Q})$ with dual basis (e^a) for the Poincaré pairing, $\int_X e_a \cup e^b = \delta_a^b$. By the Künneth formula, $\text{cl}(\Delta) = \sum_a (-1)^{\deg e_a} e_a \otimes e^a$ (the sign makes $\int_{X \times X} \text{cl}(\Delta) \cup (\alpha \otimes \beta) = \int_X \alpha \cup \beta$). Then $\text{cl}(\Gamma_f) = (1 \times f)^* \text{cl}(\Delta)$ up to the ordering of factors, and $\int \text{cl}(\Gamma_f) \cup \text{cl}(\Delta) = \sum_a (-1)^{\deg e_a} \int_X f^* e_a \cup e^a$, which is the alternating trace. For $f = \text{id}$ the traces are the Betti numbers. Degrees of intersections are integrals of cup products by Theorem 59.3.1(4). $\square$

Exercise 59.6.3. Show that every torsion class in $H^{2p}(X; \mathbb{Z})$ is an integral Hodge class, and that for X with $H^{2p}(X; \mathbb{Z})$ torsion-free and $H^{2p}(X; \mathbb{C}) = H^{p,p}(X)$ every integral class is a Hodge class.

Solution. A torsion class maps to $0 \in H^{2p}(X; \mathbb{C})$, which lies in $H^{p,p}$. The second statement is immediate from the definition. $\square$

Chapter 60

Index Theorems and Riemann–Roch

60.1 Introduction

An elliptic operator on a compact manifold has a finite-dimensional kernel and cokernel (Theorem 39.5.2), and the difference of their dimensions, the *index*, is remarkably stable: it does not change under deformations, and the Atiyah–Singer index theorem computes it from characteristic classes. For the three classical elliptic complexes the index is a familiar invariant: the Euler characteristic, the signature, and the holomorphic Euler characteristic $\chi(X, E) = \sum_q (-1)^q \dim H^q(X, E)$. The corresponding cases of the index theorem are the Gauss–Bonnet–Chern theorem, Hirzebruch’s signature theorem and the Hirzebruch–Riemann–Roch theorem. This chapter identifies the three indices with complete proofs, states the index theorem and its three cases, proves Hirzebruch–Riemann–Roch directly for line bundles on curves and for all line bundles on projective spaces, and draws the integrality consequences (Noether’s formula, divisibility of Chern numbers) that make these theorems useful in algebraic geometry. References: [Hir66], [AS68a], [AS68b], [LM89], [GH78].

Prerequisites. Chapter 40 (Harmonic Forms and the Hodge Theorem), Chapter 43 (Dolbeault Cohomology), Chapter 37 (Characteristic Classes), Chapter 55 (Algebraic Curves), Chapter 52 (Cohomology of Coherent Sheaves) and Chapter 59 (The Cycle Class Map).

Conventions. Manifolds are compact, and characteristic classes are taken in de Rham cohomology (or rational cohomology), identified by the de Rham theorem. For a real vector bundle V we write $V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}$. For a class a in the cohomology of a compact oriented manifold, $\int_M a$ is the integral of its top-degree component.

60.2 The index of an elliptic operator

Definition 60.2.1. The *index* of an elliptic differential operator $P: \Gamma(M, E) \rightarrow \Gamma(M, F)$ on a compact manifold is $\text{ind } P = \dim \ker P - \dim \ker P^*$, where P^* is the formal adjoint (Definition 39.5.1).

By Theorem 39.5.2, $\ker P^*$ is isomorphic to the cokernel of P on smooth sections, so $\text{ind } P = \dim \ker P - \dim \text{coker } P$, and it is the Fredholm index of $P: H^{s+m}(E) \rightarrow H^s(F)$ for every s . As the Fredholm index is locally constant on the space of Fredholm operators (Chapter 28), $\text{ind } P$ is unchanged when P is deformed through a continuous family of elliptic operators of the same order; in particular it depends only on the principal symbol up to homotopy through invertible symbols.

Proposition 60.2.2 (Euler characteristic operator). *For a compact oriented Riemannian manifold, $d + \delta: \Omega^{\text{even}}(M) \rightarrow \Omega^{\text{odd}}(M)$ is elliptic with index $\chi(M)$.*

Proof. This is Corollary 40.3.1; ellipticity holds because $(d + \delta)^2 = \Delta$ has principal symbol $|\xi|^2$, so the symbol of $d + \delta$ is injective for $\xi \neq 0$, and it maps between bundles of the same rank (2^{n-1}) . $\square$

Proposition 60.2.3 (Dolbeault operator). *Let X be a compact complex manifold with a Hermitian metric and E a holomorphic vector bundle with a Hermitian metric. The operator $\bar{\partial}_E + \bar{\partial}_E^* : \mathcal{A}^{0,\text{even}}(X, E) \rightarrow \mathcal{A}^{0,\text{odd}}(X, E)$ is elliptic, and its index is $\chi(X, E) = \sum_q (-1)^q \dim H^q(X, E)$.*

Proof. Let $D = \bar{\partial}_E + \bar{\partial}_E^*$ on $\mathcal{A}^{0,\bullet}(X, E)$. As $\bar{\partial}_E^2 = 0$, $D^2 = \bar{\partial}_E \bar{\partial}_E^* + \bar{\partial}_E^* \bar{\partial}_E = \square_E$, which is elliptic with principal symbol $\frac{1}{2}|\xi|^2$ (Theorem 43.3.3); so D is elliptic, and it is formally self-adjoint, so the formal adjoint of its even part is its odd part. If $D\alpha = 0$, then $\square_E \alpha = 0$; conversely $(\square_E \alpha, \alpha) = \|D\alpha\|^2$, so $\ker D = \ker \square_E = \bigoplus_q \mathcal{H}^{0,q}(X, E)$, split into even and odd q . By Theorem 43.3.3, $\mathcal{H}^{0,q}(X, E) \cong H^q(X, E)$, and the index is $\sum_q (-1)^q \dim H^q(X, E)$. $\square$

Proposition 60.2.4 (Signature operator). *Let M be a compact oriented Riemannian manifold of dimension $4m$, and τ the operator on $\Omega^\bullet(M) \otimes \mathbb{C}$ equal to $i^{k(k-1)+2m}$ on k -forms. Then $\tau^2 = 1$, τ anticommutes with $d + \delta$, and $d + \delta : \Omega^+ \rightarrow \Omega^-$, between the ± 1 -eigenspaces of τ , is elliptic with index the signature $\sigma(M)$ (Definition 21.6.5).*

Sketch of proof. We omit the verification of $\tau^2 = 1$ and $\tau(d + \delta) = -(d + \delta)\tau$, which follow from $** = (-1)^{k(n-k)}$ and $\delta = -*d*$ in even dimension (Proposition 38.7.3) by tracking signs [LM89, Chapter 3]. Granting these, $d + \delta$ maps Ω^+ to Ω^- , the kernel of $d + \delta$ on $\Omega^\pm$ is the space $\mathcal{H}^\pm$ of harmonic forms in $\Omega^\pm$, and the index is $\dim \mathcal{H}^+ - \dim \mathcal{H}^-$, exactly as in Proposition 60.2.2. As τ commutes with Δ and maps harmonic k -forms to harmonic $(4m - k)$ -forms, for $k < 2m$ the space $\mathcal{H}^k \oplus \mathcal{H}^{4m-k}$ is τ -stable, and $\alpha \mapsto \alpha \pm \tau\alpha$ identifies $\mathcal{H}^k$ with its ± 1 -eigenspaces, which therefore have equal dimensions and cancel in the index. On $\mathcal{H}^{2m}$, $\tau = i^{2m(2m-1)+2m} = *$, and for a real harmonic $2m$ -form α with $*\alpha = \pm\alpha$, $\int_M \alpha \wedge \alpha = \pm \int_M \alpha \wedge *\alpha = \pm \|\alpha\|^2$. So the intersection form is positive definite on the self-dual and negative definite on the anti-self-dual harmonic forms, which are orthogonal for it, and by the Hodge theorem the index $\dim \mathcal{H}_+^{2m} - \dim \mathcal{H}_-^{2m}$ is the signature. $\square$

60.3 The Atiyah–Singer index theorem

Definition 60.3.1. For a complex vector bundle with Chern roots $x_1, \dots, x_r$ and a real vector bundle V whose complexification has Chern roots $\pm y_1, \dots, \pm y_s$ (and 0 if the rank is odd), define

$$\text{td}(E) = \prod_i \frac{x_i}{1 - e^{-x_i}}, \quad L(V) = \prod_j \frac{y_j}{\tanh y_j}, \quad \hat{A}(V) = \prod_j \frac{y_j/2}{\sinh(y_j/2)}.$$

L and $\hat{A}$ are even in each y_j , hence polynomials in the Pontryagin classes: $L = 1 + \frac{1}{3}p_1 + \frac{1}{45}(7p_2 - p_1^2) + \dots$ and $\hat{A} = 1 - \frac{1}{24}p_1 + \dots$. For a compact oriented $4m$ -manifold, the L -genus is $\int_M L(TM)$; for a compact complex manifold, the Todd genus is $\int_X \text{td}(T_X)$.

Theorem 60.3.2 (Atiyah–Singer). *The index of an elliptic operator on a compact manifold depends only on the class of its principal symbol in the compactly supported K-theory of the tangent bundle, and it is given by a universal cohomological formula in the Chern character of that class and the Todd class of $TM_{\mathbb{C}}$. For the three classical operators the formula gives:*

1. (Gauss–Bonnet–Chern) $\chi(M) = \int_M e(TM)$ for a compact oriented manifold;
2. (Hirzebruch signature theorem) $\sigma(M) = \int_M L(TM)$ for a compact oriented $4m$ -manifold;

3. (Hirzebruch–Riemann–Roch) $\chi(X, E) = \int_X \text{ch}(E) \text{td}(T_X)$ for a holomorphic vector bundle E on a compact complex manifold X , where T_X is the holomorphic tangent bundle.

This is [AS68a] and [AS68b]; for the formulation and the heat equation proof see [LM89]. The signature theorem and Riemann–Roch for projective manifolds were proved earlier by Hirzebruch, using Thom’s cobordism theory [Hir66]. Case (1) is also Theorem 37.5.1(3). For the Dirac operator of a spin manifold the index is the $\hat{A}$ -genus, which is therefore an integer on spin manifolds.

Remark 60.3.3. For a projective manifold X and an algebraic vector bundle E , case (3) also follows from the Grothendieck–Riemann–Roch theorem (Theorem 58.7.2): it gives $\chi(X, E) = \deg(\text{ch}(E) \text{td}(T_X))$ in the Chow ring, and the cycle class map is a ring homomorphism compatible with Chern classes and degrees (Theorem 59.3.1), so the right-hand side is $\int_X \text{ch}(E) \text{td}(T_X)$; by GAGA the algebraic and analytic Euler characteristics agree (Chapter 54).

60.4 Riemann–Roch: direct verifications

Proposition 60.4.1. *Hirzebruch–Riemann–Roch holds for line bundles on a nonsingular projective curve X of genus g : $\chi(X, \mathcal{O}(D)) = \int_X \text{ch}(\mathcal{O}(D)) \text{td}(T_X) = \deg D + 1 - g$.*

Proof. The left side is $l(D) - h^1(\mathcal{O}(D)) = l(D) - l(K - D)$ by Serre duality (Theorem 52.6.2), which is $\deg D + 1 - g$ by the Riemann–Roch theorem (Theorem 55.3.2). On the right, $\text{ch}(\mathcal{O}(D)) = 1 + c_1(\mathcal{O}(D))$ and $\text{td}(T_X) = 1 + \frac{1}{2}c_1(T_X)$ in $H^\bullet(X)$, as $H^{>2}(X) = 0$; so the integrand of degree 2 is $c_1(\mathcal{O}(D)) + \frac{1}{2}c_1(T_X)$. Now $\int_X c_1(\mathcal{O}(D)) = \deg D$, because $c_1(\mathcal{O}(D))$ is the cycle class of D and the cycle class map is compatible with degrees (Theorem 59.3.1); and $c_1(T_X) = -c_1(K_X)$ with $\deg K = 2g - 2$ (Corollary 55.3.3). So the right side is $\deg D - (g - 1)$. $\square$

Corollary 60.4.2. *Let E be a holomorphic vector bundle on a compact connected Riemann surface C of genus g , and $\deg E = \int_C c_1(E)$. Then*

$$\chi(C, E) = \deg E + \text{rk}(E)(1 - g).$$

For a line bundle this reads $\chi(C, L) = \int_C (c_1(L) + \frac{1}{2}c_1(C)) = \deg L + \frac{1}{2} \deg K_C^\vee$, where $c_1(C) = c_1(T_C)$ and $T_C = K_C^\vee$, the dual of the canonical bundle.

Proof. On a curve only classes of degree 0 and 2 survive, so $\text{ch}(E) = \text{rk}(E) + c_1(E)$ and $\text{td}(T_C) = 1 + \frac{1}{2}c_1(T_C)$ (Definition 37.3.3), and the degree-2 part of their product is $c_1(E) + \frac{1}{2}\text{rk}(E)c_1(T_C)$. Since $T_C = K_C^\vee$ (both have the local frame $\partial/\partial z$, dual to dz), $\int_C c_1(T_C) = -\deg K_C = 2 - 2g$ (Proposition 37.2.4(4) and Proposition 47.2.5). Theorem 60.3.2(3) gives $\chi(C, E) = \deg E + \text{rk}(E)(1 - g)$. $\square$

Remark 60.4.3. The same theorem appears in several guises, and it helps to translate between them. For a divisor D on a curve, $\chi(C, \mathcal{O}(D)) = h^0 - h^1$, and Serre duality $h^1(\mathcal{O}(D)) = h^0(K_C - D)$ turns Corollary 60.4.2 into the classical form $l(D) - l(K - D) = \deg D + 1 - g$ of Theorem 55.3.2. The Hirzebruch form $\int \text{ch}(E) \text{td}(X)$ writes the correction term through the Todd class of the tangent bundle, $c_1(X) := c_1(T_X) = -c_1(K_X)$; this is why $\frac{1}{2}c_1(C)$, $\frac{1}{2}\deg K_C^\vee = 1 - g$ and $-\frac{1}{2}K$ appear, for instance in the surface formula $\chi(S, L) = \chi(S, \mathcal{O}_S) + \frac{1}{2}L \cdot (L - K_S)$ of Corollary 60.4.6. In every form, the Euler characteristic depends only on topological data (degrees and Chern classes), while h^0 alone does not; vanishing theorems (Chapter 47) are what turn the formula into a computation of h^0 .

Lemma 60.4.4. *On $\mathbb{P}^n$ with hyperplane class h , $\text{td}(T_{\mathbb{P}^n}) = (h/(1 - e^{-h}))^{n+1}$ and $\text{ch}(\mathcal{O}(k)) = e^{kh}$.*

Proof. The Euler sequence $0 \rightarrow \mathcal{O} \rightarrow \mathcal{O}(1)^{\oplus(n+1)} \rightarrow T_{\mathbb{P}^n} \rightarrow 0$ (Example 33.3.7) and the multiplicativity of the Todd class in exact sequences (Proposition 37.3.4) give $\mathrm{td}(T) \mathrm{td}(\mathcal{O}) = \mathrm{td}(\mathcal{O}(1))^{n+1}$, with $\mathrm{td}(\mathcal{O}) = 1$ and $\mathrm{td}(\mathcal{O}(1)) = h/(1 - e^{-h})$, as $c_1(\mathcal{O}(1)) = h$ (Example 37.2.5). Also $\mathrm{ch}(\mathcal{O}(k)) = e^{c_1(\mathcal{O}(k))} = e^{kh}$. $\square$

Proposition 60.4.5. *For all $n \geq 1$ and $k \in \mathbb{Z}$,*

$$\chi(\mathbb{P}^n, \mathcal{O}(k)) = \int_{\mathbb{P}^n} \mathrm{ch}(\mathcal{O}(k)) \mathrm{td}(T_{\mathbb{P}^n}) = \binom{n+k}{n} = \frac{(k+1)(k+2) \cdots (k+n)}{n!}.$$

Proof. The left side. By Theorem 52.3.1, $H^q(\mathbb{P}^n, \mathcal{O}(k)) = 0$ for $0 < q < n$; $h^0(\mathcal{O}(k))$ is the number of monomials of degree k in $n+1$ variables, which is $\binom{n+k}{n}$ for $k \geq 0$ and 0 for $k < 0$; and $h^n(\mathcal{O}(k))$ is the number of Laurent monomials $x_0^{a_0} \cdots x_n^{a_n}$ with all $a_i \leq -1$ and $\sum a_i = k$, which is the number of monomials of degree $-k-n-1$, that is $\binom{-k-1}{n}$ for $k \leq -n-1$ and 0 otherwise. For $k \geq 0$ the polynomial $P(k) = (k+1) \cdots (k+n)/n!$ equals $\binom{n+k}{n}$; for $-n \leq k \leq -1$ it vanishes, as does χ ; and for $k \leq -n-1$, $P(k) = (-1)^n(-k-1)(-k-2) \cdots (-k-n)/n! = (-1)^n \binom{-k-1}{n} = (-1)^n h^n(\mathcal{O}(k))$. So $\chi(\mathbb{P}^n, \mathcal{O}(k)) = P(k)$ in all cases.

The right side. By Lemma 60.4.4 and $\int_{\mathbb{P}^n} h^n = 1$, the integral is the coefficient of h^n in $e^{kh}(h/(1 - e^{-h}))^{n+1}$, a power series convergent near 0, that is, the residue at 0 of the meromorphic function $e^{kh}(1 - e^{-h})^{-n-1}$:

$$\frac{1}{2\pi i} \oint_{|h|=\varepsilon} \frac{e^{kh} dh}{(1 - e^{-h})^{n+1}}.$$

The substitution $u = 1 - e^{-h}$ is biholomorphic near 0 with $u(0) = 0$, so it maps the small circle to a loop winding once around 0; since $e^{-h} = 1 - u$, $e^{kh} = (1 - u)^{-k}$ and $dh = du/(1 - u)$, the integral becomes $\frac{1}{2\pi i} \oint (1 - u)^{-k-1} u^{-n-1} du$, the coefficient of u^n in $(1 - u)^{-k-1} = \sum_j \binom{k+j}{j} u^j$ (Chapter 27). This coefficient is $\binom{k+n}{n} = P(k)$, read as a polynomial in k . $\square$

Corollary 60.4.6 (Noether's formula). *For a compact complex surface S , $\chi(S, \mathcal{O}_S) = \frac{1}{12} \int_S (c_1^2 + c_2)$, where $c_i = c_i(T_S)$. Consequently $\int_S (c_1^2 + c_2)$ is divisible by 12, and for a line bundle L with $l = c_1(L)$,*

$$\chi(S, L) = \chi(S, \mathcal{O}_S) + \frac{1}{2} \int_S l(l + c_1) = \chi(S, \mathcal{O}_S) + \frac{1}{2} L \cdot (L - K_S),$$

where $L \cdot M = \int_S c_1(L)c_1(M)$ is the intersection number and K_S is the canonical bundle.

Proof. By Definition 60.3.1, the degree-4 part of the Todd class is $\frac{1}{12}(c_1^2 + c_2)$ (Definition 37.3.3), and Theorem 60.3.2(3) with $E = \mathcal{O}_S$ gives the formula; the left side is an integer. For L , the degree-4 part of $(1 + l + \frac{1}{2}l^2)(1 + \frac{1}{2}c_1 + \frac{1}{12}(c_1^2 + c_2))$ is $\frac{1}{2}l^2 + \frac{1}{2}lc_1 + \frac{1}{12}(c_1^2 + c_2)$. Finally $K_S = \Lambda^2 T_S^\vee$ and, with Chern roots x_1, x_2 of T_S (Theorem 37.3.2), $T_S^\vee$ has roots $-x_1, -x_2$ (Proposition 37.2.4(4)) and $\Lambda^2 T_S^\vee$ has the single root $-x_1 - x_2$. So $c_1(K_S) = -c_1$, and $\int_S l(l + c_1) = L \cdot L - L \cdot K_S$. $\square$

Lemma 60.4.7. *For a complex vector bundle E , $(E_{\mathbb{R}})_{\mathbb{C}} \cong E \oplus \bar{E}$, and hence $p_1(E_{\mathbb{R}}) = c_1(E)^2 - 2c_2(E)$.*

Proof. The complex structure J of E extends $\mathbb{C}$ -linearly to $(E_{\mathbb{R}})_{\mathbb{C}}$, where it has eigenvalues $\pm i$; the $+i$ -eigenbundle $\{v - iJv\}$ is isomorphic to E by $v \mapsto \frac{1}{2}(v - iJv)$, and the $-i$ -eigenbundle to $\bar{E}$. By Definition 37.5.2 and the Whitney formula, $p_1(E_{\mathbb{R}}) = -c_2(E \oplus \bar{E}) = -(c_2(E) + c_1(E)c_1(\bar{E}) + c_2(\bar{E})) = c_1^2 - 2c_2$, using $c_1(\bar{E}) = -c_1(E)$ and $c_2(\bar{E}) = c_2(E)$ (Proposition 37.2.4(4)). $\square$

Corollary 60.4.8 (Gauss–Bonnet for complex manifolds). *Let X be a compact complex manifold of dimension n and $c_i(X) = c_i(T_X)$ the Chern classes of its holomorphic tangent bundle. Then*

$$\chi(X) = \int_X c_n(X) = \sum_{p,q} (-1)^{p+q} h^{p,q}(X) = \sum_p (-1)^p \chi(X, \Omega_X^p).$$

For a curve of genus g this is $\int_C c_1(C) = 2 - 2g$, and for a surface $\chi(S) = \int_S c_2(S)$, so Noether's formula reads $\chi(S, \mathcal{O}_S) = \frac{1}{12}(K_S^2 + \chi(S))$.

Proof. The real tangent bundle TX with the complex structure J is isomorphic, as a complex vector bundle, to $T_X = T^{1,0}X$, by $v \mapsto \frac{1}{2}(v - iJv)$ (as in Lemma 60.4.7), and the orientation of X is the one induced by J . So $e(TX) = c_n(T_X)$ and $\int_X c_n(X) = \chi(X)$ by Theorem 37.5.1(2),(3). The Frölicher spectral sequence gives $\chi(X) = \sum_{p,q} (-1)^{p+q} h^{p,q}(X)$ (Proposition 43.4.1), and grouping the terms with the same p gives $\sum_p (-1)^p \chi(X, \Omega_X^p)$.

The equality $\sum_p (-1)^p \chi(X, \Omega_X^p) = \int_X c_n(X)$ also follows from Hirzebruch–Riemann–Roch (Theorem 60.3.2(3)), which is how it is often presented. If $x_1, \dots, x_n$ are the Chern roots of T_X , those of $\Omega_X^1 = T_X^\vee$ are $-x_1, \dots, -x_n$, so $\text{ch}(\Omega_X^p) = \text{ch}(\Lambda^p \Omega_X^1)$ is the p -th elementary symmetric function of the e^{-x_i} , and

$$\sum_p (-1)^p \text{ch}(\Omega_X^p) \text{td}(T_X) = \prod_{i=1}^n (1 - e^{-x_i}) \cdot \prod_{i=1}^n \frac{x_i}{1 - e^{-x_i}} = x_1 \cdots x_n = c_n(X).$$

Integrating and summing Theorem 60.3.2(3) over p gives the claim. For a curve, $c_1(C) = c_1(T_C)$ has degree $2 - 2g$ (Corollary 60.4.2), and for a surface $c_1^2 = K_S^2$ since $c_1(S) = -c_1(K_S)$ (Corollary 60.4.6). $\square$

- Example 60.4.9.** 1. $\mathbb{P}^2$: $c(T) = (1 + h)^3$, so $c_1^2 = 9$, $c_2 = 3$ and Noether's formula gives $\chi(\mathcal{O}) = 12/12 = 1$, as it must. By Lemma 60.4.7, $p_1 = 9 - 6 = 3$, and the signature theorem $\sigma = \frac{1}{3} \int p_1 = 1$ agrees with the intersection form (1) on $H^2(\mathbb{P}^2) = \mathbb{Z}h$.
2. $\mathbb{P}^1 \times \mathbb{P}^1$: $c_1 = 2a + 2b$ and $c_2 = 4ab$, where a, b are the pullbacks of the point classes, with $a^2 = b^2 = 0$ and $\int ab = 1$. So $c_1^2 = 8$, $c_2 = 4$, $\chi(\mathcal{O}) = 1$, and $p_1 = 8 - 8 = 0$, in agreement with the hyperbolic intersection form on H^2 , of signature 0.
3. A K3 surface has $c_1 = 0$ and $\chi(\mathcal{O}) = h^0 - h^1 + h^2 = 1 - 0 + 1 = 2$ (Chapter 48), so Noether's formula gives $\int c_2 = 24$, the topological Euler characteristic; then $p_1 = -48$ and $\sigma = -16$. As $b_2 = 22$, the intersection form has 3 positive and 19 negative eigenvalues.

Counterexample 60.4.10. The characteristic numbers appearing in the index theorems are a priori rational, and their integrality is a real constraint. There is no compact complex surface with $\int c_1^2 = \int c_2 = 1$, since $12 \nmid 2$ would contradict Noether's formula. By the signature theorem, $\int_M p_1(TM) = 3\sigma(M)$ is divisible by 3 for every compact oriented 4-manifold M ; so no such manifold has first Pontryagin number 1, although nothing in the formal properties of characteristic classes excludes it.

Proposition 60.4.11 (Riemann–Roch on complex tori). *Let $T = \mathbb{C}^g / \Lambda$ be a complex torus and L a holomorphic line bundle on T . Then*

$$\chi(T, L) = \frac{1}{g!} \int_T c_1(L)^g.$$

If L is positive, then $H^q(T, L) = 0$ for $q > 0$, and $h^0(T, L) = \frac{1}{g!} \int_T c_1(L)^g > 0$.

Proof. The vector fields $\partial/\partial z_1, \dots, \partial/\partial z_g$ descend to T and form a global holomorphic frame of T_T , which is therefore trivial, so all its Chern classes vanish and all its Chern roots may be taken to be 0; hence $\text{td}(T_T) = 1$. With one Chern root, $\text{ch}(L) = e^{c_1(L)}$ (Definition 37.3.3). By Theorem 60.3.2(3), $\chi(T, L) = \int_T e^{c_1(L)}$, and only the term of degree $2g$ contributes. If L is positive, then since $K_T = \Omega_T^g$ is trivial (it has the frame $dz_1 \wedge \dots \wedge dz_g$), $H^q(T, L) = H^q(T, \Omega_T^g \otimes L) = 0$ for $q > 0$ by Kodaira–Nakano vanishing (Theorem 47.4.1), so $h^0 = \chi$. Finally $c_1(L)$ is represented by a positive $(1, 1)$ -form ω_L , the fundamental form of a Hermitian metric (Definition 42.4.2), and $\int_T \omega_L^g = g! \text{vol}(T) > 0$ (Proposition 44.2.3). $\square$

Example 60.4.12. 1. For $g = 1$ the proposition says $\chi(E, L) = \deg L$, in accordance with Proposition 60.4.1 for genus 1.

2. On $T = E \times E'$ with $L = \text{pr}_1^* \mathcal{O}(p) \otimes \text{pr}_2^* \mathcal{O}(q)$, $c_1(L) = a + b$ with a, b the classes of the fibres $\{p\} \times E'$ and $E \times \{q\}$, so $c_1(L)^2 = 2ab$. The bundles $\mathcal{O}(p)$ and $\mathcal{O}(q)$ have degree 1, hence are positive (Proposition 47.2.5), and the product of positive metrics has curvature form $\text{pr}_1^* \omega_1 + \text{pr}_2^* \omega_2 > 0$, so L is positive and $h^0(T, L) = \frac{1}{2} \cdot 2 = 1$: the only sections are the multiples of the product of the canonical sections. A line bundle with $h^0 = 1$ of this kind is a *principal polarisation*; its section is, in the analytic description of Chapter 57, a theta function.

60.5 Exercises

Exercise 60.5.1. Using Hirzebruch–Riemann–Roch on $\mathbb{P}^1 \times \mathbb{P}^1$, show that $\chi(\mathcal{O}(p, q)) = (p+1)(q+1)$ for the line bundle of bidegree (p, q) , and check it for $p, q \geq 0$ by counting bihomogeneous polynomials.

Solution. With $l = pa + qb$ and $c_1 = 2a + 2b$, Corollary 60.4.6 gives $\chi = 1 + \frac{1}{2} \int (pa + qb)((p+2)a + (q+2)b) = 1 + \frac{1}{2}(p(q+2) + q(p+2)) = pq + p + q + 1$. For $p, q \geq 0$ the higher cohomology vanishes (by the Künneth formula and Theorem 52.3.1) and the sections are the polynomials of bidegree (p, q) , of which there are $(p+1)(q+1)$. $\square$

Exercise 60.5.2. Show that the Todd genus of $\mathbb{P}^n$ is 1, and that the Todd genus of a compact complex manifold equals $\chi(X, \mathcal{O}_X) = \sum_q (-1)^q h^{0,q}(X)$.

Solution. The second statement is Theorem 60.3.2(3) with $E = \mathcal{O}_X$, since $\text{ch}(\mathcal{O}_X) = 1$. For $\mathbb{P}^n$, Proposition 60.4.5 with $k = 0$ gives $\binom{n}{n} = 1$. $\square$

Exercise 60.5.3. Let $S \subseteq \mathbb{P}^3$ be a smooth surface of degree d . Using the adjunction formula $c_1(T_S) = (4-d)h$ and Noether's formula, show that $\int_S c_2(T_S) = d^3 - 4d^2 + 6d$, and check that this gives 24 for a quartic.

Solution. From the exact sequence $0 \rightarrow \mathcal{O}_{\mathbb{P}^3}(-d) \rightarrow \mathcal{O}_{\mathbb{P}^3} \rightarrow \mathcal{O}_S \rightarrow 0$, additivity of χ and Proposition 60.4.5, $\chi(\mathcal{O}_S) = 1 - \frac{(1-d)(2-d)(3-d)}{6} = 1 + \frac{(d-1)(d-2)(d-3)}{6}$. As $\int_S h^2 = d$, $\int c_1^2 = (4-d)^2 d$. Noether's formula gives $\int c_2 = 12\chi(\mathcal{O}_S) - (4-d)^2 d = 12 + 2(d-1)(d-2)(d-3) - d(4-d)^2 = d^3 - 4d^2 + 6d$. For $d = 4$: $64 - 64 + 24 = 24$. $\square$

Chapter 61

Étale Cohomology

61.1 Introduction

Singular cohomology needs the complex topology. For varieties over other fields, Grothendieck replaced open subsets by étale maps and built étale cohomology. With coefficients $\mathbb{Z}/\ell^n\mathbb{Z}$, and in the limit $\mathbb{Z}_\ell$ and $\mathbb{Q}_\ell$, for a prime ℓ different from the characteristic, it has all the formal properties of singular cohomology. It agrees with singular cohomology over $\mathbb{C}$ (Artin's comparison theorem, Theorem 61.4.3) and carries an action of the Galois group of the field of definition. This Galois action is the arithmetic counterpart of the Hodge structure: the Tate conjecture (Chapter 72) is the analogue of the Hodge conjecture with Galois invariants in place of Hodge classes. Most of the foundational results are quoted from SGA 4 and Milne's book. References: [Mil80], [AGV72], [AGV73], [Del77].

Prerequisites. Chapter 50 (Schemes), Chapter 24 (Sheaf Cohomology) and Chapter 8 (Fields and Galois Theory).

Throughout, ℓ is a prime and schemes are Noetherian. For a field k , k^s is a separable closure and $G_k = \text{Gal}(k^s/k)$ the absolute Galois group, a profinite group (Remark 8.10.7).

61.2 Étale morphisms and the étale site

Proposition 61.2.1. *Let $f: X \rightarrow Y$ be a morphism locally of finite type. The following are equivalent: (a) f is étale (Definition 50.8.4); (b) f is flat and unramified, meaning $\Omega_{X/Y} = 0$; (c) f is flat, and every fibre X_y is a disjoint union of spectra of finite separable field extensions of $k(y)$. Étale morphisms are stable under composition and base change, and a morphism between étale Y -schemes is étale. For varieties over $\mathbb{C}$, f is étale if and only if f^{an} is a local biholomorphism.*

Proof. For the equivalences see [Mil80, Chapter I]. If f is étale, locally $X = \text{Spec } A[x_1, \dots, x_r]/(g_1, \dots, g_r)$ with invertible Jacobian determinant. Over $\mathbb{C}$ the holomorphic inverse function theorem then shows that f^{an} is a local biholomorphism, as in Proposition 54.3.2(1). Conversely, if f^{an} is a local biholomorphism, the completed local rings of X and Y at corresponding points agree. So f is flat and unramified at closed points, hence étale near them, and étaleness is an open condition. $\square$

Definition 61.2.2. The *small étale site* X_{et} has objects the étale morphisms $U \rightarrow X$, and a family $(U_i \rightarrow U)$ is a *covering* if the union of the images is U . An *étale sheaf* is a presheaf $\mathcal{F}$ of abelian groups on X_{et} such that for every covering the sequence $0 \rightarrow \mathcal{F}(U) \rightarrow \prod_i \mathcal{F}(U_i) \rightarrow \prod_{i,j} \mathcal{F}(U_i \times_U U_j)$ is exact. Étale sheaves form an abelian category with enough injectives, and *étale cohomology* $H^i(X_{\text{et}}, \mathcal{F})$ is the right derived functor of $\mathcal{F} \mapsto \mathcal{F}(X)$.

- Example 61.2.3.** 1. Every quasi-coherent sheaf $\mathcal{F}$ defines an étale sheaf $U \mapsto \Gamma(U, f^*\mathcal{F})$ (faithfully flat descent), and $H^i(X_{\text{ét}}, \mathcal{F}) = H^i(X, \mathcal{F})$ [Mil80, Chapter III].
2. $\mathbb{G}_m: U \mapsto \Gamma(U, \mathcal{O}_U)^*$ and $\mu_n: U \mapsto \{f \in \Gamma(U, \mathcal{O}_U) : f^n = 1\}$ are étale sheaves.
3. For an abelian group A , the constant sheaf A_X assigns to U the locally constant functions $U \rightarrow A$.

Proposition 61.2.4. For a field k , the functor $\mathcal{F} \mapsto \varinjlim_L \mathcal{F}(\text{Spec } L)$, the colimit over finite separable extensions $L \subseteq k^s$, is an equivalence between étale sheaves on $\text{Spec } k$ and discrete G_k -modules (abelian groups with a continuous action). Under it, $H^i((\text{Spec } k)_{\text{ét}}, \mathcal{F}) \cong H^i(G_k, M)$, the continuous Galois cohomology.

Sketch of proof. An étale k -scheme is a disjoint union of spectra of finite separable extensions (Proposition 61.2.1), so a sheaf is determined by its values on the $\text{Spec } L$, and the sheaf condition for the Galois covering $\text{Spec } L' \rightarrow \text{Spec } L$ (L'/L Galois) says $\mathcal{F}(L) = \mathcal{F}(L')^{\text{Gal}(L'/L)}$. This identifies sheaves with discrete G_k -modules M , with $\mathcal{F}(L) = M^{G_L}$. Global sections become M^{G_k} , and taking derived functors gives Galois cohomology [Mil80, Chapters II and III]. $\square$

Counterexample 61.2.5. The Zariski topology is too coarse to see topology. Let X be an irreducible variety and A an abelian group. Every nonempty open subset of X is irreducible, hence connected, so the constant sheaf A_X has $A_X(U) = A$ for every nonempty open U , with the identity maps as restrictions. Thus A_X is flasque, and $H^i(X_{\text{Zar}}, A_X) = 0$ for $i > 0$ (Proposition 24.3.3). For an elliptic curve E over $\mathbb{C}$ this gives $H^1(E_{\text{Zar}}, \mathbb{Z}/n) = 0$, whereas $H^1(E^{\text{an}}; \mathbb{Z}/n) \cong (\mathbb{Z}/n)^2$. The étale topology repairs this for torsion coefficients (Example 61.3.2 and Theorem 61.4.3), but not for $\mathbb{Z}$: for a normal connected variety, $H^1(X_{\text{ét}}, \mathbb{Z})$ is the group of continuous homomorphisms from the profinite étale fundamental group to $\mathbb{Z}$, which is 0 [Mil80, Chapter III]. This is why ℓ -adic cohomology is defined as an inverse limit of cohomology with finite coefficients (Definition 61.4.1).

61.3 The Kummer sequence

Proposition 61.3.1 (Kummer sequence). Let n be invertible on X . Then $0 \rightarrow \mu_n \rightarrow \mathbb{G}_m \xrightarrow{f \mapsto f^n} \mathbb{G}_m \rightarrow 0$ is an exact sequence of étale sheaves. If X is proper and connected over an algebraically closed field k with n invertible in k , there is an exact sequence

$$0 \rightarrow \text{Pic}(X)/n \rightarrow H^2(X_{\text{ét}}, \mu_n) \rightarrow \text{Br}(X)[n] \rightarrow 0, \quad H^1(X_{\text{ét}}, \mu_n) \cong \text{Pic}(X)[n],$$

where $\text{Br}(X) = H^2(X_{\text{ét}}, \mathbb{G}_m)$ is the cohomological Brauer group and $[n]$ denotes n -torsion.

Proof. Exactness on the left is clear. For surjectivity, let $f \in \Gamma(U, \mathcal{O}_U)^*$. The morphism $U' = \text{Spec}_U \mathcal{O}_U[t]/(t^n - f) \rightarrow U$ is finite, flat and surjective, and étale because $\partial(t^n - f)/\partial t = nt^{n-1}$ is invertible on U' (n and t are units). On U' , $f = t^n$ is an n -th power. So f is locally an n -th power in the étale topology. The long exact cohomology sequence uses $H^0(X, \mathbb{G}_m) = \Gamma(X, \mathcal{O}_X)^* = k^*$, which is n -divisible as k is algebraically closed, and $H^1(X_{\text{ét}}, \mathbb{G}_m) = \text{Pic}(X)$ (Hilbert's theorem 90 in Grothendieck's form, [Mil80, Chapter III]). It reads $0 \rightarrow H^1(\mu_n) \rightarrow \text{Pic}(X) \xrightarrow{n} \text{Pic}(X) \rightarrow H^2(\mu_n) \rightarrow \text{Br}(X) \xrightarrow{n} \text{Br}(X)$, which gives both statements. $\square$

Example 61.3.2. For an elliptic curve E over an algebraically closed field of characteristic prime to n , $H^1(E_{\text{ét}}, \mu_n) \cong \text{Pic}^0(E)[n] \cong E[n] \cong (\mathbb{Z}/n)^2$, using $\text{Pic}^0(E) \cong E$ (Exercise 55.7.1) and the structure of n -torsion [Mum70]. So H^1 has rank $2 = b_1$, as over $\mathbb{C}$.

61.4 ℓ -adic cohomology

Definition 61.4.1. Let X be a variety over a field k of characteristic $\neq \ell$, and $\bar{X} = X \times_k \bar{k}$. The ℓ -adic cohomology is

$$H^i(\bar{X}, \mathbb{Z}_\ell) = \varprojlim_m H^i(\bar{X}_{\text{et}}, \mathbb{Z}/\ell^m \mathbb{Z}), \quad H^i(\bar{X}, \mathbb{Q}_\ell) = H^i(\bar{X}, \mathbb{Z}_\ell) \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell.$$

The Tate twist is $\mathbb{Z}_\ell(1) = \varprojlim_m \mu_{\ell^m}(\bar{k})$, a free $\mathbb{Z}_\ell$ -module of rank one on which G_k acts through the cyclotomic character, and $\mathbb{Z}_\ell(r) = \mathbb{Z}_\ell(1)^{\otimes r}$, $H^i(\bar{X}, \mathbb{Q}_\ell(r)) = H^i(\bar{X}, \mathbb{Q}_\ell) \otimes \mathbb{Q}_\ell(r)$. The group G_k acts on $H^i(\bar{X}, \mathbb{Q}_\ell(r))$ through its action on $\bar{X}$.

Theorem 61.4.2. Let X be a nonsingular projective variety of dimension n over k , and $\ell \neq \text{char } k$.

1. $H^i(\bar{X}, \mathbb{Q}_\ell)$ is finite-dimensional, and zero outside $0 \leq i \leq 2n$.
2. (Poincaré duality) $H^{2n}(\bar{X}, \mathbb{Q}_\ell(n)) \cong \mathbb{Q}_\ell$ via a trace map, and cup product $H^i(\bar{X}, \mathbb{Q}_\ell) \times H^{2n-i}(\bar{X}, \mathbb{Q}_\ell(n)) \rightarrow \mathbb{Q}_\ell$ is perfect and G_k -equivariant.
3. (Künneth) $H^\bullet(\bar{X} \times \bar{Y}, \mathbb{Q}_\ell) \cong H^\bullet(\bar{X}, \mathbb{Q}_\ell) \otimes H^\bullet(\bar{Y}, \mathbb{Q}_\ell)$.
4. (Cycle classes) There is a cycle class map $\text{cl}_\ell: \text{CH}^p(\bar{X}) \rightarrow H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$, a ring homomorphism compatible with pullback, pushforward and degrees, and G_k -equivariant; the class of a cycle defined over a finite extension k' is fixed by $G_{k'}$.
5. (Lefschetz trace formula) For an endomorphism f of $\bar{X}$, $\deg(\Gamma_f \cdot \Delta) = \sum_i (-1)^i \text{tr}(f^* | H^i(\bar{X}, \mathbb{Q}_\ell))$.
6. (Hard Lefschetz) Cup product with the class of a hyperplane section induces isomorphisms $H^{n-i}(\bar{X}, \mathbb{Q}_\ell) \rightarrow H^{n+i}(\bar{X}, \mathbb{Q}_\ell(i))$.

Statements (1)–(5) are [Mil80, Chapter VI] and [Del77]; (6) is due to Deligne [Del80].

Theorem 61.4.3 (Artin's comparison theorem). Let X be a variety over $\mathbb{C}$ and A a finite abelian group. Then $H^i(X_{\text{et}}, A) \cong H^i(X^{\text{an}}; A)$, compatibly with cup products. Hence $H^i(X, \mathbb{Z}_\ell) \cong H^i(X^{\text{an}}; \mathbb{Z}) \otimes \mathbb{Z}_\ell$ for X nonsingular and projective (or more generally when the singular cohomology groups are finitely generated), and cl_ℓ corresponds to $\text{cl} \otimes \mathbb{Z}_\ell$ of Chapter 59.

This is [AGV73, Exposé XI, Théorème 4.4] and [Mil80, Chapter III]. The proof compares both theories on a basis of neighbourhoods given by “elementary fibrations”, which are $K(\pi, 1)$ spaces in both topologies. The isomorphism with $\mathbb{Z}_\ell$ -coefficients follows by the universal coefficient theorem and passing to the limit.

Definition 61.4.4. A Weil cohomology theory for nonsingular projective varieties over k , with coefficients in a field K of characteristic 0, is a contravariant functor $X \mapsto H^\bullet(X)$ to graded-commutative finite-dimensional K -algebras, satisfying Poincaré duality, the Künneth formula, and having a cycle class map compatible with pullback, pushforward and products, with the class of a point generating H^{2n} and satisfying the weak and hard Lefschetz theorems. Examples: singular cohomology with $\mathbb{Q}$ -coefficients for $k \subseteq \mathbb{C}$; ℓ -adic cohomology for $\ell \neq \text{char } k$ (Theorem 61.4.2); algebraic de Rham cohomology in characteristic 0 (Chapter 54); crystalline cohomology in characteristic p .

Remark 61.4.5. For X over a finite field $\mathbb{F}_q$, the Frobenius F acts on $\bar{X}$, its fixed points are $X(\mathbb{F}_{q^m})$ for F^m , and the trace formula (Theorem 61.4.2(5)) gives $\#X(\mathbb{F}_{q^m}) = \sum_i (-1)^i \text{tr}(F^{m*} | H^i)$, so the zeta function is rational. Deligne proved that the eigenvalues of Frobenius on H^i are algebraic integers of absolute value $q^{i/2}$ [Del74a], the analogue of the Hodge decomposition's weight: this is the Riemann hypothesis part of the Weil conjectures.

61.5 Galois actions and the Tate conjecture

Example 61.5.1. Let E be an elliptic curve over k and $T_\ell(E) = \varprojlim E[\ell^m](\bar{k})$ its Tate module, free of rank 2 over $\mathbb{Z}_\ell$ with an action of G_k . Then $H^1(\bar{E}, \mathbb{Z}_\ell) \cong \text{Hom}(T_\ell(E), \mathbb{Z}_\ell)$ equivariantly, by Example 61.3.2 and duality. For k a number field and E without complex multiplication, the image of G_k in $\text{GL}_2(\mathbb{Z}_\ell)$ is open (Serre's open image theorem), the ℓ -adic analogue of the fact that the Mumford–Tate group of E is GL_2 (Chapter 63).

Remark 61.5.2. By Theorem 61.4.2(4), the classes of cycles defined over k lie in the invariants $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$, and those of cycles defined over finite extensions lie in the classes fixed by an open subgroup. The Tate conjecture asserts that for k finitely generated over its prime field the $\mathbb{Q}_\ell$ -span of the cycle classes is exactly $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$ (Chapter 72).

61.6 Exercises

Exercise 61.6.1. Let k be a field. Show that $H^1((\text{Spec } k)_{\text{et}}, \mathbb{Z}/n) \cong \text{Hom}_{\text{cont}}(G_k, \mathbb{Z}/n)$, and describe it for $k = \mathbb{F}_q$.

Solution. By Proposition 61.2.4, this is $H^1(G_k, \mathbb{Z}/n)$ with trivial action, which is the group of continuous homomorphisms. For $\mathbb{F}_q$, $G_k = \hat{\mathbb{Z}}$ generated by Frobenius, so the group is $\mathbb{Z}/n$, corresponding to the unique cyclic extension of degree d for each $d \mid n$. $\square$

Exercise 61.6.2. Let k be a field and n invertible in k . Show that $H^1(G_k, \mu_n) \cong k^*/k^{*n}$ (Kummer theory).

Solution. Apply the Kummer sequence on $\text{Spec } k$ and Proposition 61.2.4: $k^* \xrightarrow{n} k^* \rightarrow H^1(G_k, \mu_n) \rightarrow H^1(G_k, (k^s)^*)$, and the last group is 0 by Hilbert's theorem 90. $\square$

Exercise 61.6.3. Show that $H^2(\mathbb{P}_k^1, \mu_n) \cong \mathbb{Z}/n$ for n prime to the characteristic, and that $H^2(\mathbb{P}_{\bar{k}}^1, \mathbb{Z}_\ell(1)) \cong \mathbb{Z}_\ell$.

Solution. In Proposition 61.3.1, $\text{Pic}(\mathbb{P}^1) = \mathbb{Z}$ and the Brauer group of a curve over an algebraically closed field vanishes (Tsen's theorem) [Mil80, Chapter III], so $H^2(\mu_n) = \mathbb{Z}/n$. Passing to the limit over $n = \ell^m$ gives $\mathbb{Z}_\ell$. $\square$

Part IX

Hodge Theory

Chapter 62

Hodge Structures

62.1 Introduction

The cohomology $H^k(X; \mathbb{Z})$ of a compact Kähler manifold comes with the Hodge decomposition of $H^k(X; \mathbb{C})$. The abstract structure that remembers exactly this, a lattice together with a decomposition of its complexification compatible with complex conjugation, is a *Hodge structure*. Hodge structures form an abelian category with tensor products (Theorem 62.3.3). Polarizable ones form a semisimple category (Theorem 62.5.3), and the cohomology of projective manifolds is polarizable. In this language the Hodge conjecture is a statement about the Hodge classes of the Hodge structures $H^{2p}(X; \mathbb{Q})$ (Chapter 64). References: [Voi02b, Chapter 7], [Del71], [PS08, Chapter 2].

Prerequisites. Chapter 46 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 56 (Linear Algebraic Groups).

Throughout, A denotes one of the rings $\mathbb{Z}$, $\mathbb{Q}$ or $\mathbb{R}$, and for an A -module H_A we write $H_{\mathbb{C}} = H_A \otimes_A \mathbb{C}$, with complex conjugation acting on the second factor.

62.2 Pure Hodge structures

Definition 62.2.1. An A -Hodge structure of weight $k \in \mathbb{Z}$ is a finitely generated A -module H_A with a decomposition

$$H_{\mathbb{C}} = \bigoplus_{p+q=k} H^{p,q}, \quad \overline{H^{p,q}} = H^{q,p}.$$

The *Hodge numbers* are $h^{p,q} = \dim_{\mathbb{C}} H^{p,q}$, and the *Hodge filtration* is the decreasing filtration $F^p H_{\mathbb{C}} = \bigoplus_{r \geq p} H^{r,k-r}$. For $A = \mathbb{Z}$ only the torsion-free quotient matters for the decomposition, since $H_{\mathbb{C}} = H_{\mathbb{Z}} \otimes \mathbb{C}$ kills torsion.

Proposition 62.2.2. *Decompositions as in Definition 62.2.1 correspond bijectively to finite decreasing filtrations $F^{\bullet}$ of $H_{\mathbb{C}}$ such that $H_{\mathbb{C}} = F^p \oplus \overline{F^{k-p+1}}$ for all p . The inverse bijection is $H^{p,q} = F^p \cap \overline{F^q}$.*

Proof. Given the decomposition, $\overline{F^{k-p+1}} = \bigoplus_{r \geq k-p+1} H^{k-r,r} = \bigoplus_{s \leq p-1} H^{s,k-s}$, which is a complement of F^p ; and $F^p \cap \overline{F^q}$ with $p+q=k$ is $\bigoplus_{r \geq p} H^{r,k-r} \cap \bigoplus_{s \leq p} H^{s,k-s} = H^{p,q}$. Conversely, given the filtration, put $H^{p,q} = F^p \cap \overline{F^q}$. Then $\overline{H^{p,q}} = H^{q,p}$. The sum of the $H^{p,q}$ is direct: $H^{p,q} \subseteq F^p$ and $H^{p',q'} \subseteq \overline{F^{q'}} \subseteq \overline{F^{k-p+1}}$ for $p' < p$. To see that it is all of $H_{\mathbb{C}}$, show $F^p = H^{p,k-p} \oplus F^{p+1}$ by descending induction. Intersecting $H_{\mathbb{C}} = F^{p+1} \oplus \overline{F^{k-p}}$ with $F^p \supseteq F^{p+1}$ gives $F^p = F^{p+1} \oplus (F^p \cap \overline{F^{k-p}}) = F^{p+1} \oplus H^{p,k-p}$. $\square$

- Example 62.2.3.** 1. The Tate structure $\mathbb{Z}(n)$ is the rank-one lattice $(2\pi i)^n \mathbb{Z} \subseteq \mathbb{C}$, of weight $-2n$ and type $(-n, -n)$. We write $H(n) = H \otimes \mathbb{Z}(n)$ for the Tate twist, of weight $k - 2n$.
2. For a compact Kähler manifold X , $H^k(X; \mathbb{Z})$ is a $\mathbb{Z}$ -Hodge structure of weight k by Theorem 45.5.2.
3. For a compact Riemann surface of genus g , H^1 has $h^{1,0} = h^{0,1} = g$. For a K3 surface, H^2 has Hodge numbers $(1, 20, 1)$.
4. A Hodge structure of weight k with $H^{p,q} = 0$ unless $p = q$ is of Tate type; for instance $H^{2p}(\mathbb{CP}^n; \mathbb{Z}) \cong \mathbb{Z}(-p)$.

Remark 62.2.4. The factor $(2\pi i)^n$ in $\mathbb{Z}(n)$ is not decoration. Integration of a closed 1-form over the circle $S^1 \subseteq \mathbb{C}^*$, or the connecting map of the exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O} \xrightarrow{\exp(2\pi i \cdot)} \mathcal{O}^* \rightarrow 0$, produce periods in $2\pi i \mathbb{Z}$, and the cycle class of a codimension- p cycle naturally lies in $H^{2p}(X; \mathbb{Z}(p))$. Over $\mathbb{Q}$ the twist only changes the weight, but it matters for the comparison with de Rham and ℓ -adic cohomology (Chapter 71).

62.3 Morphisms and the category of Hodge structures

Definition 62.3.1. A morphism of A -Hodge structures $H \rightarrow H'$ of weights k and $k + 2r$ (a morphism of type (r, r)) is an A -linear map φ with $\varphi_{\mathbb{C}}(H^{p,q}) \subseteq H'^{p+r, q+r}$. Morphisms of type $(0, 0)$ between structures of the same weight are simply called morphisms.

Proposition 62.3.2. A morphism $\varphi: H \rightarrow H'$ of Hodge structures of the same weight is strict for the Hodge filtrations: $\varphi(F^p H_{\mathbb{C}}) = F^p H'_{\mathbb{C}} \cap \varphi(H_{\mathbb{C}})$.

Proof. φ preserves the decompositions. So for $x \in H_{\mathbb{C}}$ with $\varphi(x) \in F^p$, decompose $x = \sum_r x^{r, k-r}$. Then $\varphi(x) = \sum_r \varphi(x^{r, k-r})$ with $\varphi(x^{r, k-r}) \in H'^{r, k-r}$, and $\varphi(x) \in F^p$ forces $\varphi(x^{r, k-r}) = 0$ for $r < p$. So $\varphi(x) = \varphi(\sum_{r \geq p} x^{r, k-r}) \in \varphi(F^p)$. $\square$

Theorem 62.3.3. For $A = \mathbb{Q}$ or $\mathbb{R}$, the Hodge structures of weight k form an abelian category: kernels, cokernels and images of morphisms carry induced Hodge structures of weight k . The category of all A -Hodge structures (direct sums of pure ones of various weights) has tensor products, duals and internal Hom:

$$(H \otimes H')^{p,q} = \bigoplus_{a+a'=p, b+b'=q} H^{a,b} \otimes H'^{a',b'}, \quad (H^{\vee})^{p,q} = (H^{-p,-q})^{\vee}, \quad \text{Hom}(H, H') = H^{\vee} \otimes H',$$

of weights $k + k'$, $-k$ and $k' - k$. Morphisms of type $(0, 0)$ between H and H' of the same weight are exactly the elements of $\text{Hom}(H, H')_A \cap \text{Hom}(H, H')^{0,0}$.

Proof. Let $\varphi: H \rightarrow H'$ be a morphism. Then $\ker \varphi_{\mathbb{C}} = \bigoplus_{p,q} (\ker \varphi_{\mathbb{C}} \cap H^{p,q})$, because φ preserves the decomposition, and this decomposition is stable under conjugation since φ is defined over A . So $\ker \varphi_A$ is a Hodge structure. Similarly $\varphi_{\mathbb{C}}(H_{\mathbb{C}}) = \bigoplus \varphi(H^{p,q})$ and $H'_{\mathbb{C}} / \varphi(H_{\mathbb{C}}) = \bigoplus H'^{p,q} / \varphi(H^{p,q})$. The coimage and image coincide because the underlying category of vector spaces is abelian and the Hodge structures on both are induced from $H^{p,q}$. The formulas for tensor products and duals define Hodge structures because conjugation acts factorwise, and a linear map is of type $(0, 0)$ exactly when it maps each $H^{p,q}$ into $H'^{p,q}$, that is, when it lies in $\text{Hom}(H, H')^{0,0}$. $\square$

Counterexample 62.3.4. Strictness is special. The category of filtered vector spaces is not abelian: the identity of a vector space V , from the trivial filtration $F^0 = V, F^1 = 0$ to the filtration $F^0 = F^1 = V, F^2 = 0$, is a bijective morphism of filtered spaces that is not an isomorphism, since its kernel and cokernel vanish but it is not strict. Proposition 62.3.2 excludes this for Hodge structures, because the complex conjugate filtration is also respected.

62.4 Hodge structures as representations

Proposition 62.4.1. *A real Hodge structure on $H_{\mathbb{R}}$, that is, a direct sum of real Hodge structures of various weights, is the same as a representation $h: \mathbb{S} \rightarrow \mathrm{GL}(H_{\mathbb{R}})$ of the Deligne torus, via $h(z)v = z^{-p}\bar{z}^{-q}v$ for $v \in H^{p,q}$. It is pure of weight k if and only if $h \circ w: \mathbb{G}_m \rightarrow \mathrm{GL}(H_{\mathbb{R}})$ is $r \mapsto r^{-k}$. A $\mathbb{Q}$ -structure $H_{\mathbb{Q}}$ gives a $\mathbb{Q}$ -Hodge structure if and only if the weight cocharacter $h \circ w$ is defined over $\mathbb{Q}$. The Weil operator $C = h(i)^{-1}$ acts on $H^{p,q}$ as i^{p-q} ; it is a real operator with $C^2 = (-1)^k$ on a structure of weight k .*

Proof. The first statement is Proposition 56.4.5, which also gives $h(w(r)) = r^{-(p+q)}$. The weight decomposition $H_{\mathbb{R}} = \bigoplus_k H_k$ is the eigenspace decomposition of $h \circ w$, and it is defined over $\mathbb{Q}$ exactly when $h \circ w$ is. Finally $h(i)$ acts on $H^{p,q}$ by $i^{-p}\bar{i}^{-q} = i^{-p}(-i)^{-q} = i^{q-p}$, so $C = h(i)^{-1}$ acts as i^{p-q} , and $C^2 = (-1)^{p-q} = (-1)^{p+q}$. $\square$

62.5 Polarisation

Definition 62.5.1. A polarisation of a $\mathbb{Q}$ -Hodge structure H of weight k is a bilinear form $Q: H_{\mathbb{Q}} \times H_{\mathbb{Q}} \rightarrow \mathbb{Q}$ such that

1. $Q(y, x) = (-1)^k Q(x, y)$;
2. $Q(H^{p',q'}, H^{p'',q''}) = 0$ unless $p' = q$ and $q' = p$ (for the $\mathbb{C}$ -bilinear extension);
3. $i^{p-q} Q(x, \bar{x}) > 0$ for every nonzero $x \in H^{p,q}$; equivalently $Q(Cx, \bar{x}) > 0$ for all nonzero $x \in H_{\mathbb{C}}$.

Equivalently, $Q: H \otimes H \rightarrow \mathbb{Q}(-k)$ is a morphism of Hodge structures satisfying (3). H is polarisable if it admits a polarisation.

Theorem 62.5.2. *Let X be a compact Kähler manifold of dimension n with a Kähler class $[\omega]$ in the image of $H^2(X; \mathbb{Q})$ (for instance X projective). Then for $k \leq n$ the primitive cohomology $P^k(X; \mathbb{Q})$ is a $\mathbb{Q}$ -Hodge structure polarised by $Q_k(a, b) = (-1)^{k(k-1)/2} \int_X a \cup b \cup [\omega]^{n-k}$. The whole $H^k(X; \mathbb{Q})$ is polarisable: in terms of the Lefschetz decomposition, a polarisation is*

$$\tilde{Q} \left(\sum_j L^j a_j, \sum_j L^j b_j \right) = \sum_j Q_{k-2j}(a_j, b_j), \quad a_j, b_j \in P^{k-2j}(X; \mathbb{Q}).$$

Proof. P^k is a sub-Hodge structure, as the kernel of the morphism L^{n-k+1} of type $(n-k+1, n-k+1)$ (Theorem 62.3.3), and it is defined over $\mathbb{Q}$ because $[\omega]$ is rational. Conditions (1)–(3) for Q_k on P^k are Theorem 46.4.2. For H^k , the summands $L^j P^{k-2j}$ of the Lefschetz decomposition (Theorem 46.2.2) are sub-Hodge structures defined over $\mathbb{Q}$, since L^j is a morphism of type (j, j) , and $L^j: P^{k-2j} \rightarrow H^k$ is injective. So $\tilde{Q}$ is well defined, $(-1)^k$ -symmetric, and pairs $H^{p,q}$ only with $H^{q,p}$. For $a \in P^{p',q'}$ with $p' + q' = k - 2j$, the element $x = L^j a$ lies in $H^{p'+j, q'+j}$, and $i^{(p'+j)-(q'+j)} \tilde{Q}(x, \bar{x}) = i^{p'-q'} Q_{k-2j}(a, \bar{a}) > 0$ by the primitive case. As the summands are $\tilde{Q}$ -orthogonal and the $H^{p,q}$ -components of different summands pair to zero, (3) holds on all of H^k . For comparison, $Q_k(L^j a, L^j b) = (-1)^j Q_{k-2j}(a, b)$, because both are $\pm \int a \cup b \cup [\omega]^{n-k+2j}$ and $\frac{k(k-1)}{2} - \frac{(k-2j)(k-2j-1)}{2} = 2jk - 2j^2 - j \equiv j \pmod{2}$. So $\tilde{Q}$ is Q_k with the sign changed on the summands with j odd; Q_k itself is not a polarisation of H^k in general (Counterexample 46.6.2). $\square$

Theorem 62.5.3. *The category of polarisable $\mathbb{Q}$ -Hodge structures of weight k is semisimple: every sub-Hodge structure of a polarisable Hodge structure has a complement, and every polarisable Hodge structure is a finite direct sum of irreducible ones. Sub-Hodge structures and quotients of polarisable Hodge structures are polarisable.*

Proof. Let Q polarise H and let $W \subseteq H$ be a sub-Hodge structure. Then $W_{\mathbb{C}} = \bigoplus (W_{\mathbb{C}} \cap H^{p,q})$, and the restriction of Q to W satisfies (1)–(3), since (3) is a positivity condition that restricts to subspaces. So $Q|_W$ is a polarisation, in particular nondegenerate: if $Q(x, W) = 0$ for $x \in W^{p,q}$, then $Q(Cx, \bar{x}) = 0$ forces $x = 0$. The orthogonal $W^{\perp} = \{y \in H_{\mathbb{C}} : Q(y, W) = 0\}$ is a sub-Hodge structure: if $y \in W_{\mathbb{C}}^{\perp}$ has components $y^{p',q'}$, then for $w \in W^{p,q}$, $Q(y^{p',q'}, w)$ is the only possibly nonzero term of $Q(y, w) = 0$ when $(p, q) = (q', p')$, so each $y^{p',q'} \in W_{\mathbb{C}}^{\perp}$. As $W \cap W^{\perp} = 0$ by nondegeneracy and dimensions add up, $H = W \oplus W^{\perp}$. The quotient $H/W \cong W^{\perp}$ is polarised by $Q|_{W^{\perp}}$. Semisimplicity follows by induction on the dimension. $\square$

Counterexample 62.5.4. Not every Hodge structure is polarisable. In dimension 2 every weight-one $\mathbb{Q}$ -Hodge structure is polarisable: $V_{\mathbb{C}} = V^{1,0} \oplus V^{0,1}$ with $V^{1,0}$ spanned by $(1, \tau)$, $\tau \notin \mathbb{R}$, is polarised by $\pm \det$, and it is the Hodge structure of an elliptic curve. But a weight-one Hodge structure of dimension 4 whose $H^{1,0}$ is a general 2-dimensional subspace of $\mathbb{C}^4$ with $H^{1,0} \cap \overline{H^{1,0}} = 0$ admits no polarisation: it is the Hodge structure of a non-algebraic complex torus (Counterexample 57.3.6 and Theorem 62.7.1).

Proposition 62.5.5. *Let H be a torsion-free $\mathbb{Z}$ -Hodge structure of weight k and Q a polarisation of $H_{\mathbb{C}}$. The group of automorphisms of H (automorphisms of $H_{\mathbb{Z}}$ whose complexification preserves every $H^{p,q}$) that preserve Q is finite.*

Proof. Put $h(x, y) = Q(Cx, \bar{y})$ on $H_{\mathbb{C}}$, with C the Weil operator (Proposition 62.4.1). By conditions (1)–(3) of Definition 62.5.1, h is a positive definite Hermitian form: condition (3) is $h(x, x) > 0$ for $x \neq 0$, and Hermitian symmetry follows from (1) and $Q(Cx, Cy) = Q(x, y)$, which holds by (2) because C acts on $H^{p,q} \times H^{q,p}$ by $i^{p-q}i^{q-p} = 1$. An automorphism g preserving the decomposition commutes with C , which is a scalar on each $H^{p,q}$; if it also preserves Q it preserves h . So the matrices of such g in a $\mathbb{Z}$ -basis of $H_{\mathbb{Z}}$ lie in the unitary group of h , which is a bounded subset of the real matrices (all g in it have operator norm 1 for the norm of h), and they have integer entries. A bounded set of integer matrices is finite. $\square$

62.6 Hodge classes

Definition 62.6.1. A Hodge class in a $\mathbb{Q}$ -Hodge structure H of even weight $2p$ is an element of $\text{Hdg}(H) = H_{\mathbb{Q}} \cap H^{p,p}$. Equivalently, $\text{Hdg}(H) = \text{Hom}_{\text{HS}}(\mathbb{Q}(-p), H)$.

Proposition 62.6.2. *Morphisms of Hodge structures $H \rightarrow H'$ of the same weight are exactly the Hodge classes in $\text{Hom}(H, H')$, and $\text{End}_{\text{HS}}(H) = \text{Hdg}(\text{End } H)$. Morphisms of Hodge structures map Hodge classes to Hodge classes. For a compact Kähler manifold X , $\text{Hdg}^p(X) = \text{Hdg}(H^{2p}(X; \mathbb{Q}))$ (Definition 59.4.1) is a $\mathbb{Q}$ -subalgebra of $\bigoplus_p H^{2p}(X; \mathbb{Q})$ under cup product.*

Proof. The first statement is the last part of Theorem 62.3.3. A morphism of type (r, r) maps $H_{\mathbb{Q}} \cap H^{p,p}$ into $H'_{\mathbb{Q}} \cap H^{p+r, p+r}$. Cup product $H^{2p} \otimes H^{2q} \rightarrow H^{2p+2q}$ is a morphism of Hodge structures, as the wedge product of forms of types (p, p) and (q, q) has type $(p+q, p+q)$, and it preserves rational classes. $\square$

Example 62.6.3. 1. In $H^2(X; \mathbb{Q})$ of a compact Kähler manifold, the Hodge classes are the rational classes of type $(1, 1)$; for a complex torus they form $\text{NS}(T) \otimes \mathbb{Q}$ (Exercise 57.7.2).
 2. Hodge classes can be scarce even when $H^{p,p}$ is large: for a very general projective K3 surface of given degree, the rational classes of type $(1, 1)$ in H^2 are only the multiples of the polarisation, although $h^{1,1} = 20$.
 3. The polarisation Q of a Hodge structure H of weight k is a Hodge class in $(H^{\vee})^{\otimes 2}(-k)$, a Hodge structure of weight 0.

62.7 Weight one and complex tori

Theorem 62.7.1. *The functor $T = V/\Lambda \mapsto H_1(T; \mathbb{Z}) = \Lambda$, with the decomposition $\Lambda_{\mathbb{C}} = V \oplus \bar{V}$ into the $(-1, 0)$ - and $(0, -1)$ -parts, is an equivalence between the category of complex tori and the category of torsion-free $\mathbb{Z}$ -Hodge structures of type $\{(-1, 0), (0, -1)\}$. Its inverse is $H \mapsto H_{\mathbb{C}}/(F^0 H_{\mathbb{C}} + H_{\mathbb{Z}})$. Under it, abelian varieties correspond exactly to the polarisable Hodge structures, and a polarisation of the Hodge structure is (up to sign) a Riemann form.*

Proof. $\Lambda \otimes_{\mathbb{Z}} \mathbb{R} = V$ as real vector spaces, so $\Lambda_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C} = V^{1,0} \oplus V^{0,1}$, the $\pm i$ -eigenspaces of the complex structure, and the projection to $V^{1,0} \cong V$ restricts to the identity of V on $\Lambda_{\mathbb{R}} = V$. So $V \cong \Lambda_{\mathbb{C}}/V^{0,1} = \Lambda_{\mathbb{C}}/F^0$, and $T = \Lambda_{\mathbb{C}}/(F^0 + \Lambda)$. Conversely, for a Hodge structure H of this type, $H_{\mathbb{R}} \rightarrow H_{\mathbb{C}}/F^0 = H^{-1,0}$ is an isomorphism of real vector spaces, since $H_{\mathbb{R}} \cap F^0 = H_{\mathbb{R}} \cap H^{0,-1} = 0$ (a real vector in $H^{0,-1}$ lies in $H^{-1,0} \cap H^{0,-1} = 0$). So $H_{\mathbb{Z}}$ is a lattice in the complex vector space $H^{-1,0}$. Homomorphisms of tori are $\mathbb{C}$ -linear maps preserving lattices (Lemma 57.2.3), which are exactly the morphisms of Hodge structures. A polarisation Q of weight -1 is an alternating integral (after scaling) form on Λ with $Q(V, V) = 0$ and $iQ(x, \bar{x}) > 0$ on $V^{1,0}$, up to the sign conventions; this is Riemann's condition in the form of Theorem 57.3.2. $\square$

Remark 62.7.2. Dually, $H^1(T; \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z})$ is of weight one with $H^{1,0} = V^{\vee}$. For a curve C , $H^1(C; \mathbb{Z})$ is polarised by the cup product, and the corresponding torus is the Jacobian (Theorem 57.5.1). In higher weight there is no such dictionary: a Hodge structure of weight 3 does not in general come from a complex torus in a way compatible with polarisations, which is the origin of the difference between Griffiths' intermediate Jacobians and Weil's (Chapter 67).

Corollary 62.7.3. *A complex torus with a polarisation has only finitely many group automorphisms preserving the polarisation.*

Proof. By Theorem 62.7.1, group automorphisms of the torus are the automorphisms of the Hodge structure H_1 , and a polarisation of the torus is a polarisation of this Hodge structure. Apply Proposition 62.5.5. $\square$

Counterexample 62.7.4. The polarisation is needed in Proposition 62.5.5. Let $E = \mathbb{C}/\Lambda$ be an elliptic curve and $T = E \times E = \mathbb{C}^2/\Lambda^2$. Every matrix in $\text{GL}_2(\mathbb{Z})$ acts $\mathbb{C}$ -linearly on $\mathbb{C}^2$ and preserves Λ^2 , so it is an automorphism of the Hodge structure $H_1(T; \mathbb{Z})$ (Theorem 62.7.1), and $\text{GL}_2(\mathbb{Z})$ is infinite. Consequently no single polarisation of $H_1(T)$ is preserved by all of these automorphisms; each polarisation is preserved only by a finite subgroup.

Example 62.7.5. Let $E_{\tau} = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$ with $\text{Im } \tau > 0$, and $H = H_1(E_{\tau}; \mathbb{Q}) = \mathbb{Q} + \mathbb{Q}\tau \subseteq \mathbb{C}$, of type $\{(-1, 0), (0, -1)\}$. An $\mathbb{R}$ -linear endomorphism of $H_{\mathbb{R}} = \mathbb{C}$ preserves the decomposition $H_{\mathbb{C}} = H^{-1,0} \oplus H^{0,-1}$ exactly when it commutes with the complex structure, that is, when it is multiplication by some $\alpha \in \mathbb{C}$ (as in the proof of Theorem 62.7.1). So $\text{End}_{\text{HS}}(H) = \{\alpha \in \mathbb{C} : \alpha(\mathbb{Q} + \mathbb{Q}\tau) \subseteq \mathbb{Q} + \mathbb{Q}\tau\}$, which is $\text{End}(E_{\tau}) \otimes \mathbb{Q}$: clearing denominators, α preserves $\mathbb{Q} + \mathbb{Q}\tau$ if and only if $n\alpha$ preserves $\mathbb{Z} + \mathbb{Z}\tau$ for some integer $n \geq 1$. By Proposition 57.6.1,

$$\text{End}_{\text{HS}}(H) = \begin{cases} \mathbb{Q}(\tau) & \text{if } [\mathbb{Q}(\tau) : \mathbb{Q}] = 2 \quad (E_{\tau} \text{ has complex multiplication}), \\ \mathbb{Q} & \text{otherwise.} \end{cases}$$

For instance $\tau = i$ gives $\mathbb{Q}(i)$. Elliptic curves with complex multiplication are thus exactly those whose Hodge structure has extra symmetry; they are countably many up to isomorphism.

62.8 Exercises

Exercise 62.8.1. Show that $\mathbb{Z}(m) \otimes \mathbb{Z}(n) \cong \mathbb{Z}(m+n)$ and $\mathbb{Z}(n)^\vee \cong \mathbb{Z}(-n)$, and that a rank-one $\mathbb{Q}$ -Hodge structure is isomorphic to $\mathbb{Q}(n)$ for a unique n .

Solution. The lattices multiply, $(2\pi i)^m \mathbb{Z} \otimes (2\pi i)^n \mathbb{Z} \rightarrow (2\pi i)^{m+n} \mathbb{Z}$, and the types add. A rank-one structure H has $H_{\mathbb{C}}$ one-dimensional, so it is a single $H^{p,q}$ with $\overline{H^{p,q}} = H^{q,p}$ forcing $p = q$; the weight is $2p$, and any nonzero rational vector gives an isomorphism with $\mathbb{Q}(-p)$. $\square$

Exercise 62.8.2. Compute the Hodge numbers of $\Lambda^2 H$ for the weight-one Hodge structure $H = H^1(E)$ of an elliptic curve, and of $H^1(E)^{\otimes 2}$.

Solution. $H^{1,0}$ and $H^{0,1}$ are one-dimensional. $\Lambda^2 H$ is one-dimensional of type $(1,1)$, isomorphic to $\mathbb{Q}(-1)$ (it is $H^2(E)$). $H^{\otimes 2}$ has Hodge numbers $h^{2,0} = 1$, $h^{1,1} = 2$, $h^{0,2} = 1$, and $\text{Sym}^2 H$ has $(1,1,1)$. $\square$

Exercise 62.8.3. Let $H = H^1(E_\tau; \mathbb{Q})$ with basis dual to $1, \tau$ and Q the cup product pairing. Verify condition (3) of Definition 62.5.1 directly.

Solution. $H^{1,0}$ is spanned by $[dz]$, and $iQ([dz], \overline{[dz]}) = i \int_{E_\tau} dz \wedge d\bar{z} = 2\text{Im } \tau > 0$, as in Example 46.4.4. $\square$

Chapter 63

Mumford–Tate Groups

63.1 Introduction

The Mumford–Tate group of a rational Hodge structure H is the smallest algebraic group over $\mathbb{Q}$ whose real points contain the image of the Deligne torus. It is the symmetry group of H : a rational subspace of any tensor space built from H is a sub-Hodge structure exactly when it is stable under the Mumford–Tate group, and Hodge classes are exactly the invariants of the special Mumford–Tate group (Theorem 63.3.2). So determining all Hodge classes on a variety, the first step in any attack on the Hodge conjecture, amounts to computing a group and its invariants. For abelian varieties this is a problem in the representation theory of reductive groups, and it is how the known cases, and the exceptional Weil classes, are found (Chapter 68). References: [Del82], [BL04], [GGK12].

Prerequisites. Chapter 62 (Hodge Structures) and Chapter 57 (Abelian Varieties).

Throughout, H is a $\mathbb{Q}$ -Hodge structure with representation $h: \mathbb{S} \rightarrow \mathrm{GL}(H_{\mathbb{R}})$ (Proposition 62.4.1), and $T^{m,n}(H) = H^{\otimes m} \otimes (H^{\vee})^{\otimes n}$ are the tensor spaces, which are Hodge structures (Theorem 62.3.3).

63.2 Definition and first properties

Definition 63.2.1. The *Mumford–Tate group* $\mathrm{MT}(H)$ is the smallest algebraic subgroup $G \subseteq \mathrm{GL}(H_{\mathbb{Q}})$, defined over $\mathbb{Q}$, such that $h(\mathbb{S}(\mathbb{R})) \subseteq G(\mathbb{R})$. The *Hodge group*, or special Mumford–Tate group, $\mathrm{Hg}(H)$ is the smallest algebraic subgroup defined over $\mathbb{Q}$ with $h(\mathbb{U}(1)(\mathbb{R})) \subseteq \mathrm{Hg}(H)(\mathbb{R})$, where $\mathbb{U}(1) \subseteq \mathbb{S}$ is the circle group (Example 56.7.3).

Proposition 63.2.2. 1. $\mathrm{MT}(H)$ and $\mathrm{Hg}(H)$ exist and are connected, and $\mathrm{Hg}(H) \subseteq \mathrm{MT}(H)$.

2. If H is pure of weight $k \neq 0$, then $\mathrm{MT}(H) = \mathbb{G}_m \cdot \mathrm{Hg}(H)$, where $\mathbb{G}_m$ is the group of scalars; if $k = 0$, $\mathrm{MT}(H) = \mathrm{Hg}(H)$.

3. $\mathrm{MT}(H)$ acts on each $T^{m,n}(H)$, and $\mathrm{MT}(H \oplus H') \subseteq \mathrm{MT}(H) \times \mathrm{MT}(H')$ projects onto each factor.

Proof. (1) Intersections of algebraic subgroups defined over $\mathbb{Q}$ are algebraic subgroups defined over $\mathbb{Q}$, and the Noetherian property of $\mathrm{GL}(H_{\mathbb{Q}})$ makes the intersection of all subgroups with the property a finite intersection, which still has it. The identity component of $\mathrm{MT}(H)$ is defined over $\mathbb{Q}$ and contains the image of the connected group $\mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, so it equals $\mathrm{MT}(H)$; likewise for $\mathrm{Hg}(H)$. (2) $\mathbb{S}(\mathbb{R}) = \mathbb{R}_{>0} \cdot \mathbb{U}(1)(\mathbb{R})$, and $h(r) = r^{-k}$ for $r \in \mathbb{R}_{>0}$ by the weight condition. So $h(\mathbb{S}(\mathbb{R})) \subseteq (\mathbb{G}_m \cdot \mathrm{Hg}(H))(\mathbb{R})$, which gives $\subseteq$. Conversely $\mathrm{MT}(H)$ contains the scalars $h(\mathbb{R}_{>0})$ if $k \neq 0$, hence the $\mathbb{Q}$ -group $\mathbb{G}_m$, and it contains $\mathrm{Hg}(H)$. If $k = 0$, h is trivial on $\mathbb{R}_{>0}$. (3) The action on tensors is functorial. For the direct sum, the image of $\mathrm{MT}(H \oplus H')$

in $\mathrm{GL}(H_{\mathbb{Q}})$ is a $\mathbb{Q}$ -subgroup containing $h(\mathbb{S})$, so it contains $\mathrm{MT}(H)$, and it is contained in it because $\mathrm{MT}(H) \times \mathrm{MT}(H')$ contains the image of $h \oplus h'$. $\square$

Theorem 63.2.3. *A $\mathbb{Q}$ -subspace $W \subseteq T^{m,n}(H)$ is a sub-Hodge structure if and only if it is stable under $\mathrm{MT}(H)$. A vector $t \in T^{m,n}(H)_{\mathbb{Q}}$ spans a sub-Hodge structure if and only if $\mathrm{MT}(H)$ stabilises the line $\mathbb{Q}t$.*

Proof. By Proposition 62.4.1, $W_{\mathbb{C}} = \bigoplus (W_{\mathbb{C}} \cap T^{p,q})$ if and only if $W_{\mathbb{R}}$ is stable under $h(\mathbb{S}(\mathbb{R}))$, since the $T^{p,q}$ are the eigenspaces of the torus. The stabiliser of W in $\mathrm{GL}(T^{m,n}(H)_{\mathbb{Q}})$, pulled back to $\mathrm{GL}(H_{\mathbb{Q}})$, is an algebraic subgroup defined over $\mathbb{Q}$ (Proposition 56.3.3). It contains $h(\mathbb{S}(\mathbb{R}))$ if and only if it contains $\mathrm{MT}(H)$, by minimality. The second statement is the case $W = \mathbb{Q}t$. $\square$

Corollary 63.2.4. *$\mathrm{End}_{\mathrm{HS}}(H) = \mathrm{End}(H_{\mathbb{Q}})^{\mathrm{MT}(H)}$, the endomorphisms commuting with the Mumford–Tate group. In particular H is irreducible as a Hodge structure if and only if $H_{\mathbb{Q}}$ is irreducible as a representation of $\mathrm{MT}(H)$.*

Proof. An endomorphism is a morphism of Hodge structures if and only if its graph, a subspace of $H \oplus H$, is a sub-Hodge structure. By Theorem 63.2.3 applied to $H \oplus H$ (whose Mumford–Tate group is the diagonal copy of $\mathrm{MT}(H)$), this means that the graph is $\mathrm{MT}(H)$ -stable, that is, the endomorphism commutes with $\mathrm{MT}(H)$. Sub-Hodge structures are the $\mathrm{MT}(H)$ -stable subspaces. $\square$

63.3 Reductivity and Hodge classes

Theorem 63.3.1. *If H is polarisable, then $\mathrm{MT}(H)$ and $\mathrm{Hg}(H)$ are reductive.*

Proof. By Theorem 63.2.3, the $\mathrm{MT}(H)$ -stable subspaces of $H_{\mathbb{Q}}$ are the sub-Hodge structures. Since H is polarisable, every sub-Hodge structure has a complement (Theorem 62.5.3), so $H_{\mathbb{Q}}$ is a semisimple representation of $\mathrm{MT}(H)$. It remains semisimple after extension of scalars to $\bar{\mathbb{Q}}$, as we are in characteristic 0 [Mil17]. Let $U = R_u(\mathrm{MT}(H)_{\bar{\mathbb{Q}}})$. On each irreducible summand W of $H_{\bar{\mathbb{Q}}}$, the unipotent normal subgroup U has a nonzero fixed vector (Kolchin’s theorem [Bor91]), and the fixed space W^U is stable under $\mathrm{MT}(H)$ because U is normal, so $W^U = W$. Hence U acts trivially on the faithful representation $H_{\bar{\mathbb{Q}}}$, so $U = 1$ and $\mathrm{MT}(H)$ is reductive. The same argument applies to $\mathrm{Hg}(H)$, a normal subgroup of the reductive group $\mathrm{MT}(H)$ (it is normalised by $\mathrm{MT}(H)$ because $h(\mathbb{S})$ is commutative), whose representation $H_{\bar{\mathbb{Q}}}$ remains semisimple. $\square$

Theorem 63.3.2. *Let H be a $\mathbb{Q}$ -Hodge structure, pure of weight k .*

1. *A rational tensor $t \in T^{m,n}(H)_{\mathbb{Q}}$ is a Hodge class, that is, of type (p, p) with $2p = k(m - n)$, if and only if it is fixed by $\mathrm{Hg}(H)$.*
2. *If H is polarisable, $\mathrm{Hg}(H)$ is the subgroup of $\mathrm{GL}(H_{\mathbb{Q}})$ fixing all Hodge classes in all tensor spaces $T^{m,n}(H)$.*

Proof. (1) The circle $z \in \mathbb{U}(1)(\mathbb{R})$, $|z| = 1$, acts on $T^{p,q}$ by $z^{-p}\bar{z}^{-q} = z^{q-p}$. So a rational t is fixed by $h(\mathbb{U}(1)(\mathbb{R}))$ if and only if all its components have $p = q$, that is, t is a Hodge class. The stabiliser of t is an algebraic subgroup defined over $\mathbb{Q}$ (Proposition 56.3.3), and it contains $h(\mathbb{U}(1)(\mathbb{R}))$ if and only if it contains $\mathrm{Hg}(H)$. (2) By (1), $\mathrm{Hg}(H)$ is contained in the subgroup fixing all Hodge classes. By Theorem 63.3.1 $\mathrm{Hg}(H)$ is reductive, so by Theorem 56.3.2(3) it is the subgroup fixing all tensors fixed by $\mathrm{Hg}(H)$, which by (1) are the Hodge classes. $\square$

Corollary 63.3.3. *Let A be a complex abelian variety and $H = H^1(A; \mathbb{Q})$. Then $H^\bullet(A; \mathbb{Q}) \cong \Lambda^\bullet H$ as Hodge structures, and the Hodge classes on A are $\bigoplus_p \text{Hdg}^p(A) = (\Lambda^\bullet H)^{\text{Hg}(H)}$. The same holds for all powers A^r , with $H^1(A^r) = H^{\oplus r}$ and the same Hodge group.*

Proof. $H^\bullet(A; \mathbb{Q}) = \Lambda^\bullet H^1(A; \mathbb{Q})$ by Proposition 57.2.1, compatibly with Hodge decompositions (cup product is a morphism of Hodge structures), and $\Lambda^\bullet H$ is a direct summand of the tensor algebra. Apply Theorem 63.3.2(1). For A^r , the Hodge group of $H^{\oplus r}$ is the diagonal image of $\text{Hg}(H)$ by Proposition 63.2.2(3). $\square$

Proposition 63.3.4. *Let H be pure of weight k with a polarisation Q . Then $\text{Hg}(H)$ is contained in the automorphism group $\text{Aut}(H_{\mathbb{Q}}, Q)$ of the form, which is the symplectic group $\text{Sp}(H_{\mathbb{Q}}, Q)$ for k odd and the orthogonal group $\text{O}(H_{\mathbb{Q}}, Q)$ for k even. Consequently $\text{MT}(H)$ is contained in the group of similitudes $\{g : Q(gx, gy) = \lambda(g)Q(x, y)\}$.*

Proof. Viewed as an element of $T^{0,2}(H) = (H^\vee)^{\otimes 2}$, a Hodge structure of weight $-2k$, the form Q is rational and lies in $\bigoplus_{p+q=k} (H^{p,q})^\vee \otimes (H^{q,p})^\vee$ by condition (2) of Definition 62.5.1. By Theorem 62.3.3 the dual $(H^{p,q})^\vee$ has type $(-p, -q)$, so Q has type $(-k, -k)$ and is a Hodge class. By Theorem 63.3.2(1) it is fixed by $\text{Hg}(H)$, that is, $Q(gx, gy) = Q(x, y)$ for $g \in \text{Hg}(H)$. The form is alternating for k odd and symmetric for k even. The last statement follows from Proposition 63.2.2(2), since scalars are similitudes. $\square$

Proposition 63.3.5. *For a $\mathbb{Q}$ -Hodge structure H pure of weight k , the following are equivalent: $\text{Hg}(H) = 1$; every element of $H_{\mathbb{Q}}$ is a Hodge class; H is of Tate type, that is, $k = 2p$ is even and $H \cong \mathbb{Q}(-p)^{\oplus r}$. In that case $\text{MT}(H) = \mathbb{G}_m$ (scalars) if $p \neq 0$, and $\text{MT}(H) = 1$ if $p = 0$.*

Proof. If $\text{Hg}(H) = 1$, every rational vector is fixed, hence a Hodge class by Theorem 63.3.2(1). If every rational vector is a Hodge class, then $H_{\mathbb{C}}$, spanned by $H_{\mathbb{Q}}$, equals $H^{p,p}$, and a basis of $H_{\mathbb{Q}}$ gives $H \cong \mathbb{Q}(-p)^{\oplus r}$. If H is of Tate type, $\text{U}(1)(\mathbb{R})$ acts on $H^{p,p}$ by $z^{-p}\bar{z}^{-p} = |z|^{-2p} = 1$, so the smallest $\mathbb{Q}$ -group containing its image is trivial. The last statement follows from Proposition 63.2.2(2): $\text{MT}(H) = \mathbb{G}_m \cdot \text{Hg}(H) = \mathbb{G}_m$ if the weight $2p$ is nonzero, and $\text{MT}(H) = \text{Hg}(H) = 1$ if $p = 0$. $\square$

Example 63.3.6. Let S be a nonsingular projective surface and $H = H^2(S; \mathbb{Q})$. By Proposition 63.3.5, $\text{Hg}(H)$ is trivial exactly when $h^{2,0}(S) = 0$, and then every class in $H^2(S; \mathbb{Q})$ is a Hodge class; by the Lefschetz (1, 1)-theorem (Chapter 65) all of them are then classes of divisors. When $h^{2,0} \neq 0$, the Hodge group fixes every Hodge class, in particular the class of a hyperplane section and all of $\text{NS}(S) \otimes \mathbb{Q}$, and it preserves the polarisation of Theorem 62.5.2 (Proposition 63.3.4); so it acts on the orthogonal complement of $\text{NS}(S) \otimes \mathbb{Q}$ for the intersection form, the *transcendental part*, preserving the intersection form, and it is nontrivial there.

63.4 Examples

Proposition 63.4.1. *Let E be a complex elliptic curve and $H = H^1(E; \mathbb{Q})$. If E has complex multiplication by the imaginary quadratic field K , then $\text{MT}(H) = \text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$, a torus of dimension 2, and $\text{Hg}(H)$ is the norm-one subtorus. If E has no complex multiplication, then $\text{MT}(H) = \text{GL}_2$ and $\text{Hg}(H) = \text{SL}_2$.*

Proof. By Corollary 63.2.4 and Theorem 62.7.1, the commutant of $\text{MT}(H)$ in $\text{End}(H_{\mathbb{Q}})$ is $\text{End}(E) \otimes \mathbb{Q}$. In the CM case, K acts on the 2-dimensional space $H_{\mathbb{Q}}$, making it a 1-dimensional K -vector space, so the commutant of K in $\text{GL}(H_{\mathbb{Q}})$ is the torus $T_K = \text{Res}_{K/\mathbb{Q}} \mathbb{G}_m$, and $\text{MT}(H) \subseteq T_K$. It contains the scalars (weight 1 $\neq 0$) and the Weil operator $C = h(i)^{-1}$, which acts as i on $H^{1,0}$ and $-i$ on $H^{0,1}$ and is not a scalar. So $\dim \text{MT}(H) = 2$ and $\text{MT}(H) = T_K$. Without CM, the

commutant is $\mathbb{Q}$. The group $\text{MT}(H)$ is connected and reductive (Theorem 63.3.1). A connected reductive subgroup of GL_2 is a torus or contains SL_2 . A torus of GL_2 containing a non-scalar element has as commutant a 2-dimensional commutative algebra, not $\mathbb{Q}$. So $\text{MT}(H) \supseteq \text{SL}_2$, and with the scalars, $\text{MT}(H) = \text{GL}_2$. The Hodge groups follow from Proposition 63.2.2(2) and the fact that Hg preserves the polarisation (it fixes the Hodge class Q), so lies in $\text{SL}_2 = \text{Sp}_2$. $\square$

Proposition 63.4.2. *If A is an abelian variety with complex multiplication by a CM field F of degree $2g$, then $\text{MT}(H^1(A; \mathbb{Q}))$ is a torus, contained in $\text{Res}_{F/\mathbb{Q}} \mathbb{G}_m$. Conversely, a polarisable weight-one Hodge structure whose Mumford–Tate group is a torus is of CM type.*

Proof. F acts on $H_{\mathbb{Q}}$ by Hodge endomorphisms, and $\dim_{\mathbb{Q}} H_{\mathbb{Q}} = 2g = [F : \mathbb{Q}]$, so $H_{\mathbb{Q}}$ is a 1-dimensional F -vector space. Its commutant in $\text{GL}(H_{\mathbb{Q}})$ is $\text{Res}_{F/\mathbb{Q}} \mathbb{G}_m$, a torus, which contains MT by Corollary 63.2.4; subgroups of tori that are connected are tori. Conversely, if MT is a torus T , its commutant contains the $\mathbb{Q}$ -algebra generated by $T(\mathbb{Q})$, which is commutative semisimple; one shows that it contains a maximal commutative subalgebra of dimension $2g$ [BL04]. $\square$

Theorem 63.4.3. *For a very general principally polarised abelian variety A of dimension g (outside a countable union of proper analytic subsets of the moduli space), $\text{MT}(H^1(A)) = \text{GSp}_{2g}$ and $\text{Hg}(H^1(A)) = \text{Sp}_{2g}$. Consequently $\text{Hdg}^{\bullet}(A^r)$ is generated by divisor classes for every r , and the Hodge conjecture holds for all powers of A .*

Sketch of proof. For each of the countably many $\mathbb{Q}$ -subgroups $G \subsetneq \text{GSp}_{2g}$, the set of points of the Siegel space $\mathfrak{H}_g$ (Proposition 57.3.4) where h factors through $G(\mathbb{R})$ is a proper analytic subset, since $\text{GSp}_{2g}(\mathbb{R})$ acts transitively on $\mathfrak{H}_g$ and the h of a general point generates. Outside these subsets $\text{MT} = \text{GSp}_{2g}$. By classical invariant theory for Sp_{2g} , the invariants in $\Lambda^{\bullet}(H^{\oplus r})$ are generated by the symplectic forms pairing the various copies, which are the classes of divisors [BL04]. By Corollary 63.3.3 these are all the Hodge classes. Rational Hodge classes of degree 2 are classes of divisors: a multiple lifts to an integral class of type $(1, 1)$, which is c_1 of a line bundle (Lemma 47.2.2), algebraic by GAGA (Theorem 54.4.1), whose class is that of a divisor (Example 59.3.2(2)). $\square$

Remark 63.4.4. The Mumford–Tate group is not determined by the endomorphism algebra. Mumford constructed abelian fourfolds with $\text{End}(A) = \mathbb{Z}$ whose Mumford–Tate group is much smaller than GSp_8 : a form of $\mathbb{G}_m \cdot (\text{SL}_2)^3$ acting through the tensor product of three standard representations [Mum69]. For such A the ring of Hodge classes is not detected by endomorphisms.

Remark 63.4.5. For A defined over a number field k , the Mumford–Tate conjecture predicts that the identity component of the Zariski closure of the image of G_k in $\text{GL}(H^1(\bar{A}, \mathbb{Q}_{\ell}))$ equals $\text{MT}(H^1(A; \mathbb{Q})) \otimes \mathbb{Q}_{\ell}$, under Artin’s comparison isomorphism (Theorem 61.4.3). It relates the Hodge conjecture to the Tate conjecture (Chapter 72), and it is known for elliptic curves (Serre) and in many cases for abelian varieties.

63.5 Exercises

Exercise 63.5.1. Show that $\text{MT}(\mathbb{Q}(n)) = \mathbb{G}_m$ for $n \neq 0$ and $\text{MT}(\mathbb{Q}(0)) = 1$.

Solution. $h(z)$ acts on $\mathbb{Q}(n)$, of type $(-n, -n)$, by $z^n \bar{z}^n = |z|^{2n}$. For $n \neq 0$ the image of $\mathbb{S}(\mathbb{R})$ is $\mathbb{R}_{>0}$, whose $\mathbb{Q}$ -Zariski closure in GL_1 is $\mathbb{G}_m$. For $n = 0$ the action is trivial. $\square$

Exercise 63.5.2. Let E_1, E_2 be non-isogenous elliptic curves without complex multiplication. Show that $\mathrm{Hg}(H^1(E_1) \oplus H^1(E_2)) = \mathrm{SL}_2 \times \mathrm{SL}_2$, and deduce that $\mathrm{Hdg}^1(E_1 \times E_2)$ has dimension 2.

Solution. By Proposition 63.2.2(3) the Hodge group is a connected reductive subgroup of $\mathrm{SL}_2 \times \mathrm{SL}_2$ projecting onto both factors. By Goursat's lemma for the simple groups SL_2 , it is either the product or the graph of an isomorphism $\mathrm{SL}_2 \rightarrow \mathrm{SL}_2$. In the second case the two representations would be isomorphic as Hodge group representations, giving a nonzero Hodge class in $\mathrm{Hom}(H^1(E_1), H^1(E_2))$, hence an isogeny (Theorem 62.7.1), which is excluded. The invariants of $\mathrm{SL}_2 \times \mathrm{SL}_2$ in $\Lambda^2(H_1 \oplus H_2) = \Lambda^2 H_1 \oplus (H_1 \otimes H_2) \oplus \Lambda^2 H_2$ are $\Lambda^2 H_1 \oplus \Lambda^2 H_2$, of dimension 2. $\square$

Chapter 64

The Hodge Conjecture

64.1 Introduction

On a nonsingular complex projective variety there are two ways of producing rational cohomology classes of type (p, p) . The algebro-geometric way takes the class of a subvariety of codimension p (Chapter 59). The Hodge-theoretic way takes all rational classes whose type is (p, p) . The first set is contained in the second (Proposition 64.2.2). The Hodge conjecture asserts that, after taking $\mathbb{Q}$ -linear combinations, they are equal. The inclusion is easy; the conjecture is a *construction problem*: from a class satisfying two linear conditions one must produce subvarieties, and there is no general method for producing subvarieties of codimension ≥ 2 . This chapter states the conjecture and its equivalent forms, proves the elementary reductions, tells the history of its two corrections, and records the reasons for believing it and the reasons it is hard. We follow the exposition of the source *The Hodge Conjecture* (2026) and Deligne's official problem description [Del06], checking each claim against the literature.

Prerequisites. Chapter 62 (Hodge Structures) and Chapter 59 (The Cycle Class Map).

Throughout, X is a nonsingular complex projective variety of dimension n , identified with the compact Kähler manifold X^{an} (Theorem 54.6.1), and $0 \leq p \leq n$.

64.2 Algebraic classes and Hodge classes

Definition 64.2.1. The group of *rational Hodge classes* of codimension p is $\text{Hdg}^p(X) = H^{2p}(X; \mathbb{Q}) \cap H^{p,p}(X)$ (Definition 59.4.1). The space of *algebraic classes* is the image of the cycle class map,

$$Z_{\text{alg}}^p(X) = \text{cl}(\text{CH}^p(X)) \otimes \mathbb{Q} = \left\{ \sum_i a_i \text{cl}(Z_i) : a_i \in \mathbb{Q}, Z_i \subseteq X \text{ subvarieties of codimension } p \right\} \subseteq H^{2p}(X; \mathbb{Q}).$$

Proposition 64.2.2 (The easy inclusion). $Z_{\text{alg}}^p(X) \subseteq \text{Hdg}^p(X)$.

Proof. This is Corollary 59.4.2. The content is that a form of type (a, b) with $a > \dim Z$ or $b > \dim Z$ restricts to zero on Z_{reg} (Proposition 45.5.8). $\square$

Conjecture 64.2.3 (Hodge conjecture). *For every nonsingular complex projective variety X and every p ,*

$$Z_{\text{alg}}^p(X) = \text{Hdg}^p(X).$$

That is, every rational cohomology class of type (p, p) on X is a rational linear combination of classes of algebraic subvarieties of codimension p . We write $\text{HC}(X, p)$ for the statement for given X and p , and $\text{HC}(X)$ for all p .

Remark 64.2.4. Every hypothesis is necessary. The coefficients must be rational: the integral version is false, both for torsion classes (Atiyah–Hirzebruch) and for non-torsion classes (Kollár). X must be projective: the analogous statement for compact Kähler manifolds is false (Voisin). Hodge’s original, more general conjecture on the whole cohomology was false as stated (Grothendieck). These failures are the subject of Chapter 69 and Section 64.4.

Proposition 64.2.5 (Equivalent forms). *For nonsingular projective X the following are equivalent to $\mathrm{HC}(X, p)$:*

1. (homological form) every class in $H_{2n-2p}(X; \mathbb{Q})$ whose Poincaré dual is of type (p, p) is a rational combination of fundamental classes of subvarieties of dimension $n - p$;
2. (Hodge-theoretic form) the map $\mathrm{CH}^p(X)_{\mathbb{Q}} \rightarrow \mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(0), H^{2p}(X; \mathbb{Q})(p))$ induced by cl is surjective;
3. (vector bundle form) $\mathrm{Hdg}^p(X)$ is contained in the $\mathbb{Q}$ -algebra generated by the Chern classes of algebraic vector bundles on X .

Proof. (1) is Poincaré duality, which maps fundamental classes to cycle classes (Definition 59.2.2) and Hodge classes to the corresponding classes in homology. (2) holds because $\mathrm{Hom}_{\mathrm{HS}}(\mathbb{Q}(0), H^{2p}(X; \mathbb{Q})(p))$ is the space of rational classes of type $(0, 0)$ in $H^{2p}(X; \mathbb{Q})$, which is $\mathrm{Hdg}^p(X)$ twisted (Definition 62.6.1). (3) follows from Proposition 59.5.1: the $\mathbb{Q}$ -span of algebraic classes of all codimensions equals the $\mathbb{Q}$ -algebra generated by Chern classes of vector bundles, and its degree- $2p$ part is $Z_{\mathrm{alg}}^p(X)$. $\square$

Remark 64.2.6. Applied to a product $X \times Y$, the conjecture predicts that every morphism of Hodge structures $H^k(X; \mathbb{Q}) \rightarrow H^{k+2r-2n}(Y; \mathbb{Q})(r-n)$ that is a Hodge class in $H^{2r}(X \times Y; \mathbb{Q})$ (via Künneth and Poincaré duality) is induced by an algebraic correspondence (Definition 58.5.1). This is the form in which the conjecture is used: for instance, it would make the Künneth projectors and the Lefschetz inverse algebraic (Chapter 73).

64.3 First reductions and elementary cases

Proposition 64.3.1. *$\mathrm{HC}(X, 0)$ and $\mathrm{HC}(X, n)$ hold: $\mathrm{Hdg}^0(X) = H^0(X; \mathbb{Q})$ is spanned by the classes of the connected components, and $\mathrm{Hdg}^n(X) = H^{2n}(X; \mathbb{Q})$ is spanned by the classes of points, one in each connected component.*

Proof. H^0 and H^{2n} are entirely of type $(0, 0)$ and (n, n) . The class of a connected component is its fundamental class, and the class of a point generates H^{2n} of its component (Example 59.3.2(3)). $\square$

Proposition 64.3.2 (Hard Lefschetz reduction). *Let $2p \leq n$. Then $\mathrm{HC}(X, p)$ implies $\mathrm{HC}(X, n-p)$.*

Proof. Let $h = \mathrm{cl}(H)$ be the class of a hyperplane section, an algebraic class. By Theorem 46.2.2, $L^{n-2p}: H^{2p}(X; \mathbb{Q}) \rightarrow H^{2n-2p}(X; \mathbb{Q})$, $\alpha \mapsto h^{n-2p} \cup \alpha$, is an isomorphism; it is defined over $\mathbb{Q}$ and of type $(n-2p, n-2p)$, so it maps $\mathrm{Hdg}^p(X)$ isomorphically onto $\mathrm{Hdg}^{n-p}(X)$ (its inverse is also of type $(-(n-2p), -(n-2p))$ and defined over $\mathbb{Q}$). If $\beta \in \mathrm{Hdg}^{n-p}(X)$, then $\beta = h^{n-2p} \alpha$ with $\alpha \in \mathrm{Hdg}^p(X)$, which is algebraic by hypothesis, and β is a product of algebraic classes, hence algebraic (Theorem 59.3.1(1)). $\square$

Remark 64.3.3. The case $p = 1$ is the Lefschetz $(1, 1)$ -theorem (Chapter 65). With Proposition 64.3.2 it gives $\mathrm{HC}(X, n-1)$, and therefore the Hodge conjecture for all varieties of

dimension at most 3. The first open case is $p = 2$ on fourfolds. The converse of Proposition 64.3.2, that $\mathrm{HC}(X, n - p)$ implies $\mathrm{HC}(X, p)$, is not known in general: it would require the inverse of L^{n-2p} to be induced by an algebraic correspondence, which is the Lefschetz standard conjecture (Chapter 73).

Proposition 64.3.4. *Let $f: X \rightarrow Y$ be a surjective morphism of nonsingular projective varieties of the same dimension. If $\mathrm{HC}(X, p)$ holds, then $\mathrm{HC}(Y, p)$ holds.*

Proof. f is generically finite of some degree $d \geq 1$, so $f_*[X] = d[Y]$ in $\mathrm{CH}^0(Y)$ (Theorem 58.3.1). The pullback f^* and the Gysin pushforward f_* in cohomology are morphisms of Hodge structures (of type $(0, 0)$ between the relevant degrees, as $\dim X = \dim Y$), and both map algebraic classes to algebraic classes (Theorem 59.3.1(2)). By the projection formula, $f_*f^*\beta = f_*(1) \cup \beta = d\beta$. So for $\beta \in \mathrm{Hdg}^p(Y)$, $f^*\beta \in \mathrm{Hdg}^p(X)$ is algebraic, and $\beta = \frac{1}{d}f_*(f^*\beta)$ is algebraic. $\square$

Example 64.3.5. If X has a filtration by closed subvarieties $X = X_0 \supseteq X_1 \supseteq \cdots \supseteq X_m = \emptyset$ with each $X_i \setminus X_{i+1}$ a disjoint union of affine spaces, then $H^{\mathrm{odd}}(X; \mathbb{Z}) = 0$ and $H^{2p}(X; \mathbb{Z})$ is free on the classes of the closures of the cells of codimension p [Ful98, Example 19.1.11]. So $\mathrm{HC}(X)$ holds, even integrally. This covers projective spaces, Grassmannians, flag varieties and smooth projective toric varieties; here the Hodge structures are of Tate type and there is no room for a non-algebraic Hodge class. For $\mathbb{P}^n$ it is Example 59.3.2(1).

Example 64.3.6. Let E, E' be complex elliptic curves. By Künneth, $H^2(E \times E'; \mathbb{Q}) = H^2(E) \oplus (H^1(E) \otimes H^1(E')) \oplus H^2(E')$, and the outer summands are spanned by the classes of $\{\mathrm{pt}\} \times E'$ and $E \times \{\mathrm{pt}\}$. The Hodge classes in the middle summand correspond to morphisms of Hodge structures $H_1(E) \rightarrow H_1(E')$ (after Poincaré duality on E), that is, by Theorem 62.7.1, to elements of $\mathrm{Hom}(E, E') \otimes \mathbb{Q}$. The Hodge classes are nonzero exactly when E and E' are isogenous. Each homomorphism φ has a graph $\Gamma_\varphi \subseteq E \times E'$, whose class has Künneth component in $H^1 \otimes H^1$ equal to the class corresponding to φ , so $\mathrm{HC}(E \times E', 1)$ holds. (This is also a consequence of the Lefschetz $(1, 1)$ -theorem.)

Remark 64.3.7. $\mathrm{HC}(X)$ and $\mathrm{HC}(Y)$ do not formally imply $\mathrm{HC}(X \times Y)$. The Künneth summands $H^i(X) \otimes H^j(Y)$ with i and j odd contain Hodge classes that are not products of Hodge classes on the factors. Example 64.3.6 is the simplest case, and the Weil classes on abelian fourfolds (Chapter 68) are the serious ones. For blow-ups, on the other hand, the blow-up formula $H^\bullet(\tilde{X}) = H^\bullet(X) \oplus \bigoplus_{j=1}^{c-1} H^{\bullet-2j}(W)(-j)$ for the blow-up along a nonsingular centre W of codimension c is induced by algebraic correspondences. So HC for X and W in the relevant degrees gives it for $\tilde{X}$ [Voi02b, Theorem 7.31].

64.4 History and two corrections

Remark 64.4.1 (Lefschetz). In 1924 Lefschetz proved that on a nonsingular projective surface an integral class of type $(1, 1)$ in H^2 is the class of a divisor, using normal functions. The modern proof via the exponential sequence is due to Kodaira and Spencer [KS53]. Lefschetz also proved the hyperplane theorem and, with a gap, the hard Lefschetz theorem, whose first complete proof is Hodge's, by harmonic forms. The framework of the conjecture (hyperplane sections, primitive cohomology, divisors versus cohomology) is Lefschetz's.

Remark 64.4.2 (Hodge). At the International Congress of 1950 Hodge proposed a generalisation of Lefschetz's theorem to all codimensions [Hod52]. He stated it with integral coefficients, and he also proposed a stronger conjecture about the whole cohomology. Both versions were false, for different reasons.

Remark 64.4.3 (First correction: integral to rational). Atiyah and Hirzebruch showed in 1962 that there are nonsingular projective varieties with torsion classes in $H^{2p}(X; \mathbb{Z})$ (automatically Hodge classes) that are not algebraic [AH62]. The obstruction is topological: algebraic classes come from complex vector bundles, so the differentials of the Atiyah–Hirzebruch spectral sequence, starting with $d_3 = \text{Sq}_{\mathbb{Z}}^3$, must vanish on them. Kollár later gave non-torsion counterexamples: on a very general hypersurface of suitable degree in $\mathbb{P}^4$ the generator of $H^4(X; \mathbb{Z}) \cong \mathbb{Z}$ is not algebraic, although a multiple of it is (Chapter 69). Rational coefficients remove both phenomena.

Definition 64.4.4. The *coniveau filtration* on $H^k(X; \mathbb{Q})$ is

$$N^c H^k(X; \mathbb{Q}) = \sum_Z \ker (H^k(X; \mathbb{Q}) \rightarrow H^k(X \setminus Z; \mathbb{Q})),$$

the sum over closed algebraic subsets $Z \subseteq X$ of codimension $\geq c$: the classes *supported in codimension c* .

Proposition 64.4.5. $N^c H^k(X; \mathbb{Q})$ is a sub-Hodge structure of $H^k(X; \mathbb{Q})$ contained in $F^c H^k(X; \mathbb{C})$. Moreover $N^p H^{2p}(X; \mathbb{Q}) = Z_{\text{alg}}^p(X)$.

Sketch of proof. A class supported on Z lies in the image of the Gysin map $H^{k-2c'}(\tilde{Z}; \mathbb{Q})(-c') \rightarrow H^k(X; \mathbb{Q})$ from a resolution of singularities $\tilde{Z}$ of a component of Z of codimension $c' \geq c$, by Deligne’s theory of weights; the Gysin map is a morphism of Hodge structures of type (c', c') , so its image is a sub-Hodge structure contained in $F^{c'} \subseteq F^c$ [Del71]. For $k = 2p$ and $c = p$, the Gysin image of $H^0(\tilde{Z})$ is spanned by $\text{cl}(Z)$. $\square$

Remark 64.4.6 (Second correction: Grothendieck’s amendment). Hodge’s general conjecture asserted $N^c H^k(X; \mathbb{Q}) = H^k(X; \mathbb{Q}) \cap F^c H^k(X; \mathbb{C})$. In 1969 Grothendieck observed that this is false for trivial reasons [Gro69a]: the left side is a sub-Hodge structure, while the right side, a rational subspace contained in $H^{k-c,c} \oplus \dots \oplus H^{c,k-c}$, need not be one, and he gave an abelian variety where they differ. He proposed the *generalised Hodge conjecture*: $N^c H^k(X; \mathbb{Q})$ is the largest sub-Hodge structure of $H^k(X; \mathbb{Q})$ contained in $F^c H^k(X; \mathbb{C})$. For $k = 2p$ and $c = p$ the largest sub-Hodge structure contained in $F^p \cap \overline{F^p} = H^{p,p}$ is spanned by $\text{Hdg}^p(X)$, so the generalised conjecture contains Conjecture 64.2.3.

Remark 64.4.7 (The Clay problem). In 2000 the Hodge conjecture became one of the seven Millennium Prize Problems of the Clay Mathematics Institute, with an official description by Deligne [Del06], which emphasises the reformulation through absolute Hodge classes (Chapter 71).

64.5 Why one might believe it, and why it is hard

Remark 64.5.1 (Evidence). 1. *It holds where it can be checked*: for $p \in \{0, 1, n-1, n\}$, hence in dimension ≤ 3 ; for cellular varieties; for cubic fourfolds [Zuc77]; for many abelian varieties (Chapter 68). No verified case has failed.

2. *Hodge loci are algebraic*. In a family of projective manifolds, the locus where a given integral class stays of type (p, p) is a countable union of algebraic subvarieties, by the theorem of Cattani, Deligne and Kaplan [CDK95]. This is what the conjecture predicts, since loci where cycles exist are algebraic (Hilbert schemes), and the theorem is proved without it (Chapter 66).

3. *Hodge classes on abelian varieties are absolute* [Del82]: they are compatible with all automorphisms of $\mathbb{C}$, as algebraic classes are and a random Hodge class would not be (Chapter 71).
4. *Coherence*: the conjecture follows from, and is required by, the expected properties of pure motives and the standard conjectures (Chapter 73).

Remark 64.5.2 (Difficulty). 1. *No construction machine in codimension ≥ 2 .* For divisors, an integral class of type $(1, 1)$ lifts to a line bundle by the exponential sequence, and a high twist of a line bundle has sections whose zero loci are divisors (Chapter 65). This uses that $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$ is the cohomology of a sheaf of abelian groups. In codimension $p \geq 2$, Bloch's formula $\text{CH}^p(X) \cong H^p(X, \mathcal{K}_p)$ expresses the Chow group through a sheaf of K -groups, but this transfers the difficulty rather than removing it.

2. *Deformation theory obstructs.* Hilbert schemes of subvarieties of codimension ≥ 2 are typically singular of excess dimension, and deformations of a cycle do not see the target Hodge class.

3. *The transcendental input is weak.* Hodge theory says nothing about where in X a cycle should lie. Griffiths' normal functions (Chapter 67) convert the problem into the existence of singularities of certain sections, which is as hard in general.

4. *The conjecture is close to false.* Its integral, Kähler and general versions all fail (Chapter 69). A proof must distinguish it from these neighbours.

64.6 Exercises

Exercise 64.6.1. Show that a $\mathbb{Q}$ -Hodge structure of odd weight has no Hodge classes in any Tate twist, so that Hodge classes live only in even degrees. Explain why the Künneth summands $H^{\text{odd}}(X) \otimes H^{\text{odd}}(Y)$ of a product are where new Hodge classes appear.

Solution. $V(m)$ has weight $k-2m$, odd, so it has no component of type $(0, 0)$, and $\text{Hom}_{\text{HS}}(\mathbb{Q}(0), V(m)) = 0$. The tensor product of two odd-weight structures has even weight, and it can contain Hodge classes not coming from Hodge classes on the factors, which have none in odd degree (Example 64.3.6). $\square$

Exercise 64.6.2. Verify $\text{HC}(\mathbb{P}^m \times \mathbb{P}^k)$ directly.

Solution. By Künneth, $H^\bullet = \mathbb{Q}[a, b]/(a^{m+1}, b^{k+1})$ with a, b the pullbacks of hyperplane classes, all classes are of Tate type and spanned by the monomials $a^i b^j$, which are the classes of the products of linear subspaces $L \times L'$ (Example 59.3.2(1) and Theorem 59.3.1(2)). $\square$

Exercise 64.6.3. Let a finite group G act on X with nonsingular projective quotient $Y = X/G$. Show that $\text{HC}(X, p)$ implies $\text{HC}(Y, p)$.

Solution. The quotient map $X \rightarrow Y$ is surjective and finite, of degree $|G|$, so Proposition 64.3.4 applies. $\square$

Chapter 65

The Lefschetz (1,1) Theorem

65.1 Introduction

The Lefschetz (1,1)-theorem is the one case of the Hodge conjecture that is completely understood, and it holds even with integral coefficients: on a nonsingular projective variety every integral class of type (1, 1) in H^2 is the class of a divisor (Theorem 65.3.1). Its proof uses the exponential sequence, which exists because the Picard group is the cohomology of a sheaf of abelian groups. With hard Lefschetz it gives the Hodge conjecture in codimension $n - 1$, hence in dimension at most three. This chapter proves the theorem, draws the consequences, and explains precisely which step has no analogue in higher codimension. References: [GH78, Chapter 1, Section 1], [Voi02b, Section 11.3], [KS53].

Prerequisites. Chapter 64 (The Hodge Conjecture) and Chapter 54 (Analytic Spaces and GAGA).

Throughout, X is a compact Kähler manifold of dimension n ; from Section 65.3 on it is projective.

65.2 The theorem for compact Kähler manifolds

Theorem 65.2.1 (Lefschetz (1, 1), Kähler form). *Let X be a compact Kähler manifold. The image of the first Chern class $c_1: \text{Pic}(X) \rightarrow H^2(X; \mathbb{Z})$ is exactly the group of integral Hodge classes $\text{Hdg}^1(X; \mathbb{Z})$, the classes whose image in $H^2(X; \mathbb{C})$ is of type (1, 1).*

Proof. The Chern class of a holomorphic line bundle is represented by $\frac{i}{2\pi}F$, a closed real (1,1)-form (Theorem 42.4.1), so $c_1(\text{Pic}(X)) \subseteq \text{Hdg}^1(X; \mathbb{Z})$. Conversely, let $a \in H^2(X; \mathbb{Z})$ have image of type (1, 1). By the *key lemma* (Lemma 47.2.2), the map $H^2(X; \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X)$ induced by $\mathbb{Z}_X \subseteq \mathcal{O}_X$ is the composite of $H^2(X; \mathbb{Z}) \rightarrow H^2(X; \mathbb{C})$ with the projection to $H^{0,2}(X) \cong H^2(X, \mathcal{O}_X)$. So a maps to 0 in $H^2(X, \mathcal{O}_X)$. In the long exact sequence of the exponential sequence $0 \rightarrow \mathbb{Z} \rightarrow \mathcal{O}_X \rightarrow \mathcal{O}_X^* \rightarrow 0$ (Theorem 42.2.4),

$$H^1(X, \mathcal{O}_X^*) = \text{Pic}(X) \xrightarrow{c_1} H^2(X; \mathbb{Z}) \rightarrow H^2(X, \mathcal{O}_X),$$

so a is in the image of c_1 . □

Corollary 65.2.2. $\text{NS}(X) := c_1(\text{Pic}(X)) = \text{Hdg}^1(X; \mathbb{Z})$, and $\text{Pic}(X)$ is an extension $0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \rightarrow \text{NS}(X) \rightarrow 0$ of a finitely generated abelian group by the complex torus $\text{Pic}^0(X) = H^1(X, \mathcal{O}_X)/H^1(X; \mathbb{Z})$ of dimension $h^{0,1} = b_1/2$. The Picard number $\rho(X) = \text{rank NS}(X)$ satisfies $\rho(X) \leq h^{1,1}(X)$.

Proof. The first statement is Theorem 65.2.1, and the extension is the exponential sequence. $\text{NS}(X) \subseteq H^2(X; \mathbb{Z})$ is finitely generated, and $\text{Pic}^0(X)$ is a torus by Remark 42.2.5 and Exercise 45.8.4. Finally $\text{NS}(X) \otimes \mathbb{R}$ injects into the real classes of type (1, 1), which form a real vector space of dimension $h^{1,1}$ (a real structure on $H^{1,1}$, since $\overline{H^{1,1}} = H^{1,1}$). $\square$

Proposition 65.2.3. *Let X be a compact Kähler manifold, and let $U(1)_X$ be the sheaf of locally constant functions with values in $U(1) = \{z \in \mathbb{C} : |z| = 1\}$. The inclusion $U(1)_X \subseteq \mathcal{O}_X^*$ induces a homomorphism $b: H^1(X, U(1)_X) \rightarrow \text{Pic}(X)$, whose image consists of the flat unitary line bundles, those given by locally constant transition functions of modulus one.*

1. *b maps the kernel of the connecting map $\delta: H^1(X, U(1)_X) \rightarrow H^2(X; \mathbb{Z})$ isomorphically onto $\text{Pic}^0(X)$, and this kernel is $H^1(X; \mathbb{R})/H^1(X; \mathbb{Z})$, a real torus of dimension $b_1(X)$.*
2. *A holomorphic line bundle is flat unitary if and only if its first Chern class is a torsion element of $H^2(X; \mathbb{Z})$. In particular every torsion class in $H^2(X; \mathbb{Z})$ lies in $\text{NS}(X)$.*

Proof. The map $t \mapsto e^{2\pi i t}$ gives a short exact sequence of sheaves $0 \rightarrow \mathbb{Z}_X \rightarrow \mathbb{R}_X \rightarrow U(1)_X \rightarrow 0$: on a connected neighbourhood a locally constant function is constant, and $\mathbb{R} \rightarrow U(1)$ is surjective with kernel $\mathbb{Z}$. The inclusions $\mathbb{R}_X \subseteq \mathcal{O}_X$ and $U(1)_X \subseteq \mathcal{O}_X^*$ map it to the exponential sequence of Theorem 42.2.4, and by naturality of the long exact sequence (Theorem 13.4.3) we get a commutative diagram with exact rows

$$\begin{array}{ccccccccc} H^1(X; \mathbb{Z}) & \longrightarrow & H^1(X; \mathbb{R}) & \xrightarrow{\beta} & H^1(X, U(1)_X) & \xrightarrow{\delta} & H^2(X; \mathbb{Z}) & \longrightarrow & H^2(X; \mathbb{R}) \\ \parallel & & \downarrow a & & \downarrow b & & \parallel & & \\ H^1(X; \mathbb{Z}) & \longrightarrow & H^1(X, \mathcal{O}_X) & \xrightarrow{\beta'} & \text{Pic}(X) & \xrightarrow{c_1} & H^2(X; \mathbb{Z}) & & \end{array}$$

where a is induced by $\mathbb{R}_X \subseteq \mathcal{O}_X$. By Exercise 45.8.4, a is an isomorphism of real vector spaces. By exactness, $\ker \delta = \beta(H^1(X; \mathbb{R})) \cong H^1(X; \mathbb{R})/H^1(X; \mathbb{Z})$ and $\text{Pic}^0(X) = \ker c_1 = \beta'(H^1(X, \mathcal{O}_X)) \cong H^1(X, \mathcal{O}_X)/H^1(X; \mathbb{Z})$, and a induces an isomorphism between these quotients, since the left square commutes. As $b\beta = \beta'a$, this isomorphism is the restriction of b , proving (1); the image of $H^1(X; \mathbb{Z})$ in $H^1(X; \mathbb{R})$ is a lattice by the universal coefficient theorem (Theorem 20.2.3).

(2) If $L = b(y)$, then $c_1(L) = \delta(y)$ maps to 0 in $H^2(X; \mathbb{R})$ by exactness of the top row, so it is torsion, because the kernel of $H^2(X; \mathbb{Z}) \rightarrow H^2(X; \mathbb{R})$ is the torsion subgroup (again by the universal coefficient theorem). Conversely, let $c_1(L)$ be torsion. Then it maps to 0 in $H^2(X; \mathbb{R})$, so $c_1(L) = \delta(y)$ for some y , and $L \otimes b(y)^{-1}$ has $c_1 = 0$. By (1), $L \otimes b(y)^{-1} = b(y')$ is flat unitary, and so is $L = b(yy')$. The same argument, starting from a torsion class $t = \delta(y)$, shows $t = c_1(b(y)) \in \text{NS}(X)$. $\square$

Example 65.2.4. Let $T = \mathbb{C}^2/\Lambda$ be a complex torus of dimension 2 and $\sigma = dz_1 \wedge dz_2$. By Example 43.4.2, $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 4$, with $H^{2,0}(T)$ spanned by $[\sigma]$ and $H^{0,2}(T)$ by $[\bar{\sigma}]$; and $H^2(T; \mathbb{Z}) \cong \mathbb{Z}^6$ (Example 20.4.3). A real class $a \in H^2(T; \mathbb{R})$ is of type (1, 1) if and only if $\int_T a \wedge \sigma = 0$. Indeed, write $a = a^{2,0} + a^{1,1} + a^{0,2}$ (Theorem 45.5.2), with $a^{2,0} = \overline{a^{0,2}}$ because a is real, and $a^{0,2} = c[\bar{\sigma}]$. For type reasons only $a^{0,2}$ contributes to $\int_T a \wedge \sigma$, so the integral is $c \int_T \bar{\sigma} \wedge \sigma$, and $\bar{\sigma} \wedge \sigma$ is a nonzero constant multiple of $dx_1 \wedge dy_1 \wedge dx_2 \wedge dy_2$ ($z_j = x_j + iy_j$); hence it vanishes if and only if $c = 0$, that is, $a^{0,2} = a^{2,0} = 0$.

By Theorem 65.2.1, $\text{NS}(T)$ is therefore the kernel of $\varphi: H^2(T; \mathbb{Z}) \rightarrow \mathbb{C}$, $\varphi(a) = \int_T a \wedge \sigma$. If $e_1, \dots, e_6$ is a $\mathbb{Z}$ -basis of $H^2(T; \mathbb{Z})$ with periods $\pi_k = \varphi(e_k)$, then

$$\rho(T) = 6 - \dim_{\mathbb{Q}}(\mathbb{Q}\pi_1 + \dots + \mathbb{Q}\pi_6),$$

since the kernel of $\mathbb{Z}^6 \rightarrow \mathbb{C}, m \mapsto \sum_k m_k \pi_k$, has the same rank as the kernel of its extension to $\mathbb{Q}^6$. The real extension $H^2(T; \mathbb{R}) \rightarrow \mathbb{C}$ has as kernel the real classes of type $(1, 1)$, of real dimension $h^{1,1} = 4$, so it is surjective and the π_k span $\mathbb{C}$ over $\mathbb{R}$. Hence the $\mathbb{Q}$ -dimension above is at least 2 and $\rho(T) \leq 4$, as predicted by Corollary 65.2.2. The Picard number of T is thus governed by the linear relations over $\mathbb{Q}$ among six complex numbers: $\rho(T) = 4$ when the periods span only a two-dimensional $\mathbb{Q}$ -vector space, as for $E \times E$ with $E = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}i)$ (Example 65.3.4(2), since $\text{Hom}(E, E) \supseteq \mathbb{Z}[i]$ has rank 2), and $\rho(T) = 0$ when they are linearly independent over $\mathbb{Q}$, which is the situation of Exercise 65.5.1.

65.3 The projective case: classes of divisors

Theorem 65.3.1 (Lefschetz $(1, 1)$). *Let X be a nonsingular complex projective variety. Every class in $\text{Hdg}^1(X; \mathbb{Z})$ is the class of a divisor: the cycle class map $\text{CH}^1(X) \rightarrow \text{Hdg}^1(X; \mathbb{Z})$ is surjective. In particular $\text{HC}(X, 1)$ holds, even integrally.*

Proof. By Theorem 65.2.1, a class $a \in \text{Hdg}^1(X; \mathbb{Z})$ is $c_1(L)$ for a holomorphic line bundle L on X^{an} . By GAGA (Corollary 54.4.3(2)), L is the analytification of an algebraic line bundle, still denoted L . Let H be a hyperplane section. By Serre's theorem (Theorem 51.5.3), $L(m) = L \otimes \mathcal{O}(mH)$ is globally generated for $m \gg 0$, so it has a nonzero section, whose divisor D_1 is effective with $\mathcal{O}(D_1) \cong L(m)$ (Proposition 53.3.2). Likewise $\mathcal{O}(mH)$ has the divisor $D_2 = mH$. So $L \cong \mathcal{O}(D_1 - D_2)$, and $a = c_1(L) = \text{cl}(D_1) - \text{cl}(D_2)$ by Example 59.3.2(2). $\square$

Corollary 65.3.2. *For a nonsingular projective variety X of dimension n , $\text{HC}(X, n - 1)$ holds. Consequently the Hodge conjecture holds for all nonsingular projective varieties of dimension at most 3.*

Proof. $\text{HC}(X, 1)$ holds by Theorem 65.3.1, and implies $\text{HC}(X, n - 1)$ for $n \geq 2$ by Proposition 64.3.2. Together with the trivial cases $p = 0$ and $p = n$ (Proposition 64.3.1), this covers all p when $n \leq 3$. $\square$

Remark 65.3.3. For curves on a threefold, the integral version fails in general: Proposition 64.3.2 only gives that some multiple of an integral Hodge class in H^4 is algebraic, since L^{n-2} need not be an isomorphism over $\mathbb{Z}$. Kollár's counterexamples to the integral Hodge conjecture are integral classes in H^4 of threefolds (Chapter 69). On the other hand, Voisin proved the integral Hodge conjecture for curve classes on uniruled threefolds and on Calabi–Yau threefolds [Voi06].

Example 65.3.4. 1. For $X \subseteq \mathbb{P}^3$ a nonsingular quartic surface, $\text{NS}(X)$ is a sublattice of $H^2(X; \mathbb{Z}) \cong \mathbb{Z}^{22}$ of rank $\rho \in [1, 20]$, and $\rho = 20$ is attained (for instance by the Fermat quartic).

2. For $E \times E'$, $\text{NS} \cong \mathbb{Z}^2 \oplus \text{Hom}(E, E')$ (Exercise 57.7.1), which by Theorem 65.3.1 is also the lattice of integral classes of type $(1, 1)$, a statement about Hodge structures (Example 64.3.6).

Example 65.3.5. Let $X = \mathbb{P}^1 \times \mathbb{P}^1$, which is isomorphic to a nonsingular quadric surface in $\mathbb{P}^3$ (Example 42.5.2), with projections pr_1, pr_2 , and let h be the positive generator of $H^2(\mathbb{P}^1; \mathbb{Z})$. By the Künneth formula (Theorem 20.4.2), $H^2(X; \mathbb{Z}) = \mathbb{Z}a \oplus \mathbb{Z}b$ with $a = \text{pr}_1^* h$ and $b = \text{pr}_2^* h$. By naturality of c_1 (Theorem 33.4.2), $a = c_1(\text{pr}_1^* \mathcal{O}(1))$ and $b = c_1(\text{pr}_2^* \mathcal{O}(1))$; these are the classes of the divisors $\{p\} \times \mathbb{P}^1$ and $\mathbb{P}^1 \times \{p\}$, the two rulings of the quadric. So every integral class is algebraic and $\rho(X) = b_2(X) = 2$, without any Hodge theory. Conversely this determines the Hodge numbers: $\rho(X) \leq h^{1,1}(X)$ (Corollary 65.2.2) and $h^{2,0} + h^{1,1} + h^{0,2} = b_2 = 2$ with

$h^{2,0} = h^{0,2}$ force $h^{1,1} = 2$ and $h^{2,0} = 0$. The intersection numbers are $a^2 = b^2 = 0$ and $ab = 1$: two lines of the same ruling are disjoint, and two lines of different rulings meet transversally in one point.

Theorem 65.3.6 (Noether–Lefschetz). *For $d \geq 4$, a very general nonsingular surface of degree d in $\mathbb{P}^3$ has $\text{Pic}(X) = \mathbb{Z} \cdot \mathcal{O}_X(1)$, so $\rho(X) = 1$.*

This is proved in [Voi03] and [GH85]. The proof shows, with the infinitesimal variation of Hodge structure (Chapter 66), that the locus of surfaces on which a given integral class stays of type $(1, 1)$ is a proper subvariety of the parameter space, unless the class is a multiple of the hyperplane class; a very general surface lies outside all these countably many loci. For $d \leq 3$ the statement is false: quadrics contain two families of lines, and cubic surfaces contain 27 lines.

65.4 Why the proof does not generalise

Remark 65.4.1. The proof of Theorem 65.3.1 has three steps, and none generalises to codimension $p \geq 2$.

1. *A cohomological object parametrising cycles.* Divisors modulo linear equivalence are line bundles, $\text{Pic}(X) = H^1(X, \mathcal{O}_X^*)$, the cohomology of a sheaf of abelian groups, and the exponential sequence relates it to $H^2(X; \mathbb{Z})$. For codimension p the Chow group $\text{CH}^p(X) \cong H^p(X, \mathcal{K}_p)$ (Bloch’s formula) involves a sheaf of Quillen K -groups, which has no comparison with singular cohomology that is exact in the needed way.
2. *The key lemma.* The obstruction to lifting an integral class is its image in $H^2(X, \mathcal{O}_X) = H^{0,2}$, a coherent cohomology group, and it vanishes exactly for type $(1, 1)$. For $p \geq 2$, the analogous vanishing of the components in $H^{p-1,p+1}, \dots, H^{0,2p}$ has no sheaf-theoretic interpretation producing an object like a line bundle. Deligne cohomology $H_{\mathcal{D}}^{2p}(X; \mathbb{Z}(p))$ fits into an exact sequence $0 \rightarrow J^p(X) \rightarrow H_{\mathcal{D}}^{2p}(X; \mathbb{Z}(p)) \rightarrow \text{Hdg}^p(X; \mathbb{Z}) \rightarrow 0$, so every integral Hodge class lifts to Deligne cohomology (Chapter 67); the missing step is to lift further to cycles.
3. *Sections of line bundles.* A line bundle twisted by a high multiple of an ample bundle has sections, whose zero loci are divisors. In codimension ≥ 2 there is no analogue: a vector bundle of rank p with enough sections gives cycles as zero loci of sections, but the Chern classes of such bundles need not span the Hodge classes, and producing a bundle with a prescribed Chern class is the vector bundle form of the conjecture itself (Proposition 64.2.5(3)).

65.5 Exercises

Exercise 65.5.1. Let T be a complex torus of dimension 2 with $\rho(T) = 0$. Show that T carries no nonconstant meromorphic functions and no curves.

Solution. A curve C would give a line bundle $\mathcal{O}(C)$ with $c_1(\mathcal{O}(C)) = \text{cl}(C) \neq 0$, since $\int_T \text{cl}(C) \wedge \omega = \int_C \omega > 0$ for a Kähler form ω (Proposition 44.4.4). So $\text{NS}(T) \neq 0$, a contradiction. A nonconstant meromorphic function f has a nonzero polar divisor, which is a union of curves. $\square$

Exercise 65.5.2. Let S be a nonsingular projective surface with $p_g(S) = h^{2,0}(S) = 0$. Show that every class in $H^2(S; \mathbb{Z})$ is the class of a divisor, and that $\rho(S) = b_2(S)$.

Solution. If $h^{2,0} = h^{0,2} = 0$, then $H^2(S; \mathbb{C}) = H^{1,1}$, so every integral class is an integral Hodge class, hence algebraic by Theorem 65.3.1. So $\text{NS}(S) = H^2(S; \mathbb{Z})$ and $\rho = b_2$. $\square$

Exercise 65.5.3. Show that on a nonsingular cubic threefold $X \subseteq \mathbb{P}^4$ the Hodge conjecture holds, and compute Hdg^1 and Hdg^2 .

Solution. By Corollary 65.3.2 the conjecture holds in dimension 3. By the Lefschetz hyperplane theorem and Poincaré duality, $H^2 = \mathbb{Q}h$ and $H^4 = \mathbb{Q}h^2/3$ (the class of a line, since $\int_X h^3 = 3$), both algebraic, so $\text{Hdg}^1 = H^2(X; \mathbb{Q})$ and $\text{Hdg}^2 = H^4(X; \mathbb{Q})$. The interesting group H^3 , of dimension 10 with $h^{2,1} = 5$, has no Hodge classes, being of odd degree. $\square$

Chapter 66

Variations of Hodge Structure

66.1 Introduction

In a smooth projective family $f: X \rightarrow S$, the fibres are diffeomorphic (Ehresmann), so their cohomology groups form a local system with a flat connection, the Gauss–Manin connection. The Hodge filtrations of the fibres vary holomorphically, and their derivatives move each F^p at most into F^{p-1} (Griffiths transversality). The resulting object, a *variation of Hodge structure*, is recorded by the *period map* to a period domain. Variations are the natural setting for asking how Hodge classes behave in families. The locus where a class stays of type (p, p) is an analytic subvariety, and by the theorem of Cattani, Deligne and Kaplan it is algebraic, as the Hodge conjecture predicts. References: [Voi02b, Chapters 9 and 10], [Voi03], [CMSP03], [Gri68].

Prerequisites. Chapter 62 (Hodge Structures), Chapter 36 (Connections and Curvature) and Chapter 54 (Analytic Spaces and GAGA).

Throughout, $f: X \rightarrow S$ is a *smooth projective family*: a proper holomorphic submersion of complex manifolds with connected base S , whose fibres $X_t = f^{-1}(t)$ are projective (for instance the analytification of a smooth projective morphism of varieties).

66.2 Ehresmann’s theorem and the Gauss–Manin connection

Theorem 66.2.1 (Ehresmann). *A proper surjective submersion $f: M \rightarrow B$ of smooth manifolds is a locally trivial fibre bundle: every $b \in B$ has a neighbourhood U with a diffeomorphism $f^{-1}(U) \cong U \times f^{-1}(b)$ over U .*

Proof. Take U a coordinate ball centred at b with coordinates $x^1, \dots, x^m$. Since df is surjective, the coordinate vector fields $\partial/\partial x^i$ lift locally to vector fields on $f^{-1}(U)$, and a partition of unity (Theorem 31.6.1) glues local lifts to vector fields v_i on $f^{-1}(U)$ with $df(v_i) = \partial/\partial x^i$, since the condition is affine. As f is proper, the flows of the v_i exist as long as the flows of $\partial/\partial x^i$ stay in U (Theorem 31.7.1: an integral curve cannot leave a compact set in finite time, and f^{-1} of a compact subset of U is compact). The map $U \times f^{-1}(b) \rightarrow f^{-1}(U)$, $(x, y) \mapsto \phi_{x^1}^1 \circ \dots \circ \phi_{x^m}^m(y)$, composing the flows for the times x^i , is then a diffeomorphism over U , with inverse obtained by flowing back. $\square$

Definition 66.2.2. By Ehresmann’s theorem, $\mathcal{H}_{\mathbb{Z}}^k = R^k f_* \mathbb{Z}$ is a local system on S with fibres $H^k(X_t; \mathbb{Z})$. The holomorphic vector bundle $\mathcal{H}^k = \mathcal{H}_{\mathbb{Z}}^k \otimes_{\mathbb{Z}} \mathcal{O}_S$ carries the flat connection ∇ whose flat sections are the local sections of $\mathcal{H}_{\mathbb{C}}^k$ (Theorem 36.3.5), the *Gauss–Manin connection*. Transport along a loop gives the *monodromy representation* $\pi_1(S, t) \rightarrow \mathrm{GL}(H^k(X_t; \mathbb{Z}))$.

Theorem 66.2.3. *In a smooth projective family:*

1. *the Hodge numbers $h^{p,q}(X_t)$ are independent of t ;*
2. *the Hodge filtrations $F^p H^k(X_t; \mathbb{C})$ form holomorphic subbundles $\mathcal{F}^p \subseteq \mathcal{H}^k$ (the Hodge bundles);*
3. *(Griffiths transversality) $\nabla \mathcal{F}^p \subseteq \mathcal{F}^{p-1} \otimes \Omega_S^1$.*

Sketch of proof. (1) The numbers $h^{p,q}(X_t) = \dim H^q(X_t, \Omega_{X_t}^p)$ are upper semicontinuous in t : this is Theorem 52.7.1 for an algebraic family, and Kodaira–Spencer’s semicontinuity theorem in the analytic case [Voi02b, Chapter 9]. Their sum over $p + q = k$ is the Betti number b_k , constant by Ehresmann’s theorem. A sum of upper semicontinuous integer functions that is constant forces each to be locally constant, hence constant as S is connected. (2) With constant Hodge numbers, the Hodge filtration of X_t can be computed from harmonic forms for a family of Kähler metrics, and depends smoothly on t . Holomorphicity follows from (3) applied in the $(0, 1)$ -direction: differentiating a family of classes of type $\geq p$ in an antiholomorphic direction does not leave F^p [Voi02b, Chapter 10]. (3) Trivialise the family smoothly (Theorem 66.2.1) and represent a local section of $\mathcal{F}^p$ by closed forms α_t on the fixed manifold X_0 that are of type $\geq p$ for the complex structure J_t . Differentiating in a holomorphic direction v at $t = 0$, $\nabla_v[\alpha] = [\partial_v \alpha_t|_{t=0}]$. A form of type $\geq p$ for J_t is, in terms of J_0 , a form of type $\geq p$ plus terms of order t of type $\geq p - 1$, because the $(1, 0)$ -forms for J_t differ from those for J_0 by first-order terms of type $(0, 1)$. So $\partial_v \alpha_t|_{t=0}$ has type $\geq p - 1$. $\square$

Definition 66.2.4. *A polarised variation of Hodge structure of weight k on a complex manifold S is a local system $\mathcal{H}_{\mathbb{Z}}$ of finitely generated abelian groups, a flat bilinear form Q , and a filtration of $\mathcal{H} = \mathcal{H}_{\mathbb{Z}} \otimes \mathcal{O}_S$ by holomorphic subbundles $\mathcal{F}^p$ such that each fibre is a Hodge structure of weight k polarised by Q , and $\nabla \mathcal{F}^p \subseteq \mathcal{F}^{p-1} \otimes \Omega_S^1$. By Theorem 66.2.3 and Theorem 62.5.2, the primitive cohomology of a smooth projective family, with the polarisation given by a relatively ample class, is one.*

66.3 Period domains and period maps

Definition 66.3.1. Fix a lattice $H_{\mathbb{Z}}$, a bilinear form Q with $Q(y, x) = (-1)^k Q(x, y)$, and Hodge numbers $h^{p,q}$. The *period domain* D is the set of filtrations $F^\bullet$ of $H_{\mathbb{C}}$ with $\dim F^p = \sum_{r \geq p} h^{r, k-r}$ that define Hodge structures of weight k polarised by Q . It is an open subset, in the analytic topology, of the *compact dual* $\check{D} = \{F^\bullet : Q(F^p, F^{k-p+1}) = 0\}$, a closed subvariety of a flag variety. The real group $G = \text{Aut}(H_{\mathbb{R}}, Q)$ acts transitively on D with compact isotropy groups.

Example 66.3.2. 1. Weight 1, $h^{1,0} = h^{0,1} = g$, Q symplectic: D is the Siegel upper half space $\mathfrak{H}_g$ (Proposition 57.3.4), with $G = \text{Sp}_{2g}(\mathbb{R})$, a Hermitian symmetric domain. For $g = 1$ it is the upper half plane.

2. Weight 2 with $h^{2,0} = 1$, the case of K3 surfaces ($h^{1,1} = 20$): a Hodge structure is determined by the line $F^2 = H^{2,0} = \mathbb{C}\omega$, and $D = \{[\omega] \in \mathbb{P}(H_{\mathbb{C}}) : Q(\omega, \omega) = 0, Q(\omega, \bar{\omega}) > 0\}$, an open subset of a quadric, with $G = \text{SO}(3, 19)$. This is a Hermitian symmetric domain of type IV.

3. For weight ≥ 2 with $h^{2,0} \geq 2$, or weight 3 with $h^{3,0} \neq 0$ and $h^{2,1} \neq 0$ (the quintic threefold), D is not Hermitian symmetric, and Griffiths transversality is a genuine restriction: period maps are tangent to a distribution that is not integrable.

Example 66.3.3. Over the upper half plane $\mathfrak{H}_1 = \{\tau : \operatorname{Im} \tau > 0\}$ consider the family of elliptic curves $E_\tau = \mathbb{C}/(\mathbb{Z} + \mathbb{Z}\tau)$, the quotient of $\mathbb{C} \times \mathfrak{H}_1$ by the action of $\mathbb{Z}^2$ given by $(m, n) \cdot (z, \tau) = (z + m + n\tau, \tau)$. The loops a, b given by the segments from 0 to 1 and from 0 to τ form a basis of $H_1(E_\tau; \mathbb{Z})$ that is flat, since the family is trivialised smoothly by $(x + y\tau, \tau) \mapsto (x, y)$, $x, y \in \mathbb{R}/\mathbb{Z}$. The holomorphic form dz has periods $\int_a dz = 1$ and $\int_b dz = \tau$. So in the flat coordinates given by the dual basis of H^1 , the line $F^1 = H^{1,0}(E_\tau)$ is spanned by $(1, \tau)$, and the period map $\mathfrak{H}_1 \rightarrow \mathfrak{H}_1$ of Example 66.3.2(1) is the identity. This is the *tautological* variation: every polarised variation of weight one and rank two is locally the pullback of it by its period map.

Definition 66.3.4. Given a polarised variation on S and a base point, the *period map* $\mathcal{P}: \tilde{S} \rightarrow D$ from the universal cover sends $\tilde{t}$ to the Hodge filtration of the fibre, transported to $H_{\mathbb{C}} = \mathcal{H}_{t_0}$ by the flat connection. It descends to $S \rightarrow \Gamma \backslash D$, where Γ is the image of the monodromy representation.

Proposition 66.3.5. *The period map is holomorphic and horizontal: its differential maps $T_t S$ into $\bigoplus_p \operatorname{Hom}(F^p/F^{p+1}, F^{p-1}/F^p) \subseteq T_{\mathcal{P}(t)} \check{D}$. For a family of projective manifolds its differential at t is the composite of the Kodaira–Spencer map $T_t S \rightarrow H^1(X_t, T_{X_t})$ with the contraction and cup product $H^1(X_t, T) \rightarrow \bigoplus_p \operatorname{Hom}(H^q(\Omega^p), H^{q+1}(\Omega^{p-1}))$.*

This is Theorem 66.2.3(2), (3) restated, together with Griffiths' computation of the differential [Voi02b, Chapter 10]. The second description is the *infinitesimal variation of Hodge structure*, the main tool in the proof of the Noether–Lefschetz theorem (Theorem 65.3.6).

Example 66.3.6. The *Legendre family* $E_\lambda: y^2 = x(x-1)(x-\lambda)$, $\lambda \in \mathbb{C} \setminus \{0, 1\}$, is a family of elliptic curves. With the holomorphic form $\omega_\lambda = dx/y$ and a symplectic basis a, b of $H_1(E_\lambda; \mathbb{Z})$ transported along paths, the period map is $\lambda \mapsto \tau(\lambda) = \int_b \omega_\lambda / \int_a \omega_\lambda \in \mathfrak{H}_1$. The periods $\int_\gamma \omega_\lambda$ satisfy the Picard–Fuchs equation $\lambda(1-\lambda)\omega'' + (1-2\lambda)\omega' - \frac{1}{4}\omega = 0$, a differential equation expressing the Gauss–Manin connection. The monodromy is a subgroup of finite index of $\operatorname{SL}_2(\mathbb{Z})$, the congruence subgroup $\Gamma(2)$.

Proposition 66.3.7. *Let $H_{\mathbb{Z}}, Q$ be as in Definition 66.3.1, of weight 2 with $h^{2,0} = 1$. Then for a polarised Hodge structure the filtration is determined by F^2 , namely $F^1 = (F^2)^\perp$, and every holomorphic family of such polarised Hodge structures over a complex manifold, with $\mathcal{F}^p$ holomorphic subbundles of a flat bundle $\mathcal{H}$ preserving Q and $H_{\mathbb{Z}}$, satisfies Griffiths transversality automatically.*

Proof. By condition (2) of Definition 62.5.1, $Q(H^{2,0}, H^{2,0} \oplus H^{1,1}) = 0$, so $F^1 = H^{2,0} \oplus H^{1,1} \subseteq (F^2)^\perp$; both have dimension $\dim H - 1$ (Q is nondegenerate), so they are equal. Transversality asks $\nabla \mathcal{F}^2 \subseteq \mathcal{F}^1 \otimes \Omega^1$ and $\nabla \mathcal{F}^1 \subseteq \mathcal{F}^0 \otimes \Omega^1$; the second is empty since $\mathcal{F}^0 = \mathcal{H}$. For the first, let ω be a local holomorphic generator of $\mathcal{F}^2$. Then $Q(\omega, \omega) = 0$, and since Q is flat, differentiating along a vector field v gives $2Q(\nabla_v \omega, \omega) = 0$, so $\nabla_v \omega \in \omega^\perp = \mathcal{F}^1$. $\square$

Counterexample 66.3.8. Without such a special shape, transversality is a genuine condition, and holomorphic families of Hodge structures need not be variations of Hodge structure. Let $H_{\mathbb{Z}} = \mathbb{Z}^2$ and consider Hodge structures of weight 2 with $h^{2,0} = h^{0,2} = 1$ and $h^{1,1} = 0$ (not polarised). Such a structure is a line $F^2 = H^{2,0} = \mathbb{C}\omega \subseteq \mathbb{C}^2$ with ω and $\bar{\omega}$ independent, so the Hodge structures form $D' = \mathbb{P}^1(\mathbb{C}) \setminus \mathbb{P}^1(\mathbb{R})$, two copies of the upper half plane. The tautological family over D' , with $\mathcal{F}^2 = \mathcal{F}^1$ the tautological line in the trivial flat bundle $D' \times \mathbb{C}^2$, is a holomorphic family of Hodge structures. Transversality would require $\nabla \mathcal{F}^2 \subseteq \mathcal{F}^1 = \mathcal{F}^2$, so that $\mathcal{F}^2$ would be a flat, hence locally constant, subbundle; but the tautological line varies. So this family satisfies all the axioms except polarisation and transversality, and by Theorem 66.2.3(3) it is not the variation of Hodge structure of any smooth projective family.

Corollary 66.3.9. *If the period map of a polarised variation of Hodge structure is constant, its monodromy group Γ is finite. For instance, in the Legendre family (Example 66.3.6) the monodromy is infinite, so the period map is not constant and the curves E_λ are not all isomorphic.*

Proof. Transport by the flat connection identifies the fibres over the universal cover with a fixed $(H_{\mathbb{Z}}, Q)$, and the period map is equivariant: for $\gamma \in \pi_1(S, t)$ with monodromy $\rho(\gamma)$, $\mathcal{P}(\gamma\tilde{t}) = \rho(\gamma)\mathcal{P}(\tilde{t})$ (Definition 66.3.4). If $\mathcal{P}$ is constant with value $F^\bullet$, then every $\rho(\gamma)$ preserves $F^\bullet$, $H_{\mathbb{Z}}$ and Q , so Γ lies in the finite group of Proposition 62.5.5. For the Legendre family $\Gamma = \Gamma(2)$ is infinite (it contains $\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$, of infinite order). If all E_λ were isomorphic, the $\tau(\lambda)$ would all lie in one orbit of $\mathrm{SL}_2(\mathbb{Z})$ (Chapter 57), a countable set; a continuous map from the connected space $\tilde{S}$ to $\mathfrak{H}_1$ with countable image is constant. $\square$

66.4 Global invariants and the theorem of the fixed part

Theorem 66.4.1 (Theorem of the fixed part). *Let $f: X \rightarrow S$ be a smooth projective morphism of complex varieties, $\tilde{X}$ a smooth compactification of X , and $t \in S$. Then $H^0(S, R^k f_* \mathbb{Q}) = H^k(X_t; \mathbb{Q})^{\pi_1(S, t)}$ is a sub-Hodge structure of $H^k(X_t; \mathbb{Q})$, and it is the image of the restriction map $H^k(\tilde{X}; \mathbb{Q}) \rightarrow H^k(X_t; \mathbb{Q})$ (the global invariant cycle theorem).*

This is [Del71, Théorème 4.1.1 and Corollaire 4.1.2]; the proof uses mixed Hodge theory (Chapter 70). A consequence: a Hodge class on one fibre that is invariant under monodromy is the restriction of a Hodge class on the total space, and so remains a Hodge class on every fibre. Deligne also proved that the monodromy representation on $H^k(X_t; \mathbb{Q})$ is semisimple.

66.5 Hodge loci

Definition 66.5.1. Let $\mathcal{H}$ be a variation of Hodge structure of weight $2p$ on S . For a flat section λ of $\mathcal{H}_{\mathbb{Z}}$ over a simply connected open $U \subseteq S$, the *Hodge locus* of λ is $\{t \in U : \lambda_t \in F^p \mathcal{H}_t\}$, the set where λ is a Hodge class. The *Hodge locus* of the variation is the union, over all local flat sections, of these sets, a subset of S .

Proposition 66.5.2. *The Hodge locus of a flat integral section λ over U is a closed analytic subset of U , locally defined by $\mathrm{rank} \mathcal{H} - \mathrm{rank} \mathcal{F}^p$ holomorphic equations, so each component has codimension at most $\dim \mathcal{H} / \mathcal{F}^p = h^{p-1, p+1} + h^{p-2, p+2} + \dots + h^{0, 2p}$. For a real class of weight $2p$, lying in F^p is the same as being of type (p, p) .*

Proof. The composite $U \rightarrow \mathcal{H}|_U \rightarrow \mathcal{H} / \mathcal{F}^p$, $t \mapsto$ the image of λ_t , is a holomorphic section of the holomorphic bundle $\mathcal{H} / \mathcal{F}^p$ of rank $\sum_{r < p} h^{r, 2p-r} = \sum_{j \geq 1} h^{p-j, p+j}$. Its zero set is the Hodge locus, cut out locally by as many holomorphic functions as the rank. For real λ , $\lambda \in F^p$ implies $\lambda = \bar{\lambda} \in \overline{F^p}$, so $\lambda \in F^p \cap \overline{F^p} = H^{p, p}$. $\square$

Theorem 66.5.3 (Cattani–Deligne–Kaplan). *Let $\mathcal{H}$ be a polarised variation of Hodge structure of weight $2p$ over a nonsingular complex algebraic variety S (for instance from a smooth projective family). Then the Hodge locus of $\mathcal{H}$, restricted to integral classes λ with $Q(\lambda, \lambda)$ bounded, is a finite union of algebraic subvarieties of S , and the whole Hodge locus is a countable union of algebraic subvarieties.*

This is [CDK95]. The Hodge conjecture would imply it: on the Hodge locus of λ , the class is algebraic on each fibre, and the locus where a given family of cycles exists is constructible by the theory of Hilbert schemes, with countably many Hilbert schemes involved. The

theorem is proved unconditionally with the several-variable SL_2 -orbit theorem describing the asymptotics of the period map near the boundary of S . Recent work of Bakker, Klingler and Tsimerman uses o-minimal geometry to reprove and extend it [BKT20].

Remark 66.5.4. Grothendieck's *variational Hodge conjecture* predicts that if a Hodge class λ stays of type (p, p) on all fibres of a family, and is algebraic on one fibre X_0 , then it is algebraic on all fibres. By Theorem 66.4.1 it would follow from the Hodge conjecture applied to the total space. It is known for $p = 1$ and in some other cases, and it remains open in general (Chapter 69).

66.6 Exercises

Exercise 66.6.1. Show that in a smooth projective family of curves the genus is constant, and in a smooth family of K3 surfaces the Hodge numbers of H^2 are $(1, 20, 1)$ on every fibre.

Solution. Both follow from Theorem 66.2.3(1): the genus is $h^{1,0}$. For K3 surfaces it suffices that one fibre is a K3 surface with these numbers, for instance a quartic (Exercise 48.6.1). $\square$

Exercise 66.6.2. Show that Griffiths transversality is automatic for variations of weight one: every holomorphic family of Hodge filtrations $F^1 \subseteq H_{\mathbb{C}}$ is horizontal.

Solution. For weight one the filtration is $0 = F^2 \subseteq F^1 \subseteq F^0 = H_{\mathbb{C}}$, and transversality asks $\nabla F^1 \subseteq F^0 \otimes \Omega^1$, which holds trivially. $\square$

Exercise 66.6.3. In a family of K3 surfaces with a nonconstant period map, show that the Hodge locus of an integral class $\lambda \in H^2$ that is not generically of type $(1, 1)$ has codimension at most 1.

Solution. By Proposition 66.5.2, the codimension is at most $\dim \mathcal{H}/\mathcal{F}^1 = h^{0,2} = 1$. Concretely, the locus is where $Q(\lambda, \omega_t) = 0$ for a local holomorphic generator ω_t of F^2 , one holomorphic equation. $\square$

Chapter 67

Intermediate Jacobians and Normal Functions

67.1 Introduction

Cycle classes forget much of the structure of Chow groups: a cycle whose class vanishes can still be very far from zero. Griffiths' intermediate Jacobian $J^p(X)$, a complex torus built from the odd-degree Hodge structure $H^{2p-1}(X)$, receives the *Abel–Jacobi map* from homologically trivial cycles of codimension p . For $p = 1$ this is the Picard variety and the classical Abel–Jacobi map. Deligne cohomology packages the cycle class and the Abel–Jacobi invariant into one group. Families of intermediate Jacobians and their *normal functions* give the only serious analytic method for constructing cycles; it reproves the Lefschetz $(1, 1)$ -theorem and proves the Hodge conjecture for cubic fourfolds. References: [Voi02b, Chapter 12], [Voi03], [Gri68], [CMSP03].

Prerequisites. Chapter 66 (Variations of Hodge Structure), Chapter 57 (Abelian Varieties) and Chapter 59 (The Cycle Class Map).

Throughout, X is a compact Kähler manifold of dimension n , usually projective, and $1 \leq p \leq n$.

67.2 Intermediate Jacobians

Definition 67.2.1. The p -th intermediate Jacobian of X is

$$J^p(X) = \frac{H^{2p-1}(X; \mathbb{C})}{F^p H^{2p-1}(X; \mathbb{C}) + H^{2p-1}(X; \mathbb{Z})},$$

where $H^{2p-1}(X; \mathbb{Z})$ means its image in $H^{2p-1}(X; \mathbb{C})$.

Proposition 67.2.2. $J^p(X)$ is a compact complex torus of dimension $\frac{1}{2}b_{2p-1}(X)$. Via Poincaré duality,

$$J^p(X) \cong \frac{F^{n-p+1} H^{2n-2p+1}(X; \mathbb{C})^\vee}{H_{2n-2p+1}(X; \mathbb{Z})},$$

where a homology class acts by integration. $J^1(X) = \text{Pic}^0(X)$, and $J^n(X)$ is the Albanese torus $\text{Alb}(X) = H^0(X, \Omega^1)^\vee / H_1(X; \mathbb{Z})$.

Proof. H^{2p-1} has odd weight $2p - 1$, so $H_{\mathbb{C}} = F^p \oplus \overline{F^{(2p-1)-p+1}} = F^p \oplus \overline{F^p}$ (Proposition 62.2.2). Hence the real subspace $H^{2p-1}(X; \mathbb{R})$ meets F^p trivially and maps isomorphically onto $H_{\mathbb{C}}/F^p$ (dimensions: $b_{2p-1} = 2 \dim_{\mathbb{C}} H_{\mathbb{C}}/F^p$). So the image of $H^{2p-1}(X; \mathbb{Z})$ modulo torsion is a lattice

in the complex vector space $H_{\mathbb{C}}/F^p$, of complex dimension $\frac{1}{2}b_{2p-1}$. For the second description, the Poincaré pairing $H^{2p-1} \times H^{2n-2p+1} \rightarrow \mathbb{C}$ pairs F^p with F^{n-p+1} to zero (types add up to more than n in one index) and is perfect, so $H_{\mathbb{C}}/F^p \cong (F^{n-p+1}H^{2n-2p+1})^{\vee}$, and $H^{2p-1}(X; \mathbb{Z}) \cong H_{2n-2p+1}(X; \mathbb{Z})$ modulo torsion. For $p = 1$, $H^1(X; \mathbb{C})/F^1 = H^{0,1} = H^1(X, \mathcal{O})$, giving $\text{Pic}^0(X)$ (Corollary 65.2.2); for $p = n$ the second description gives $H^0(X, \Omega^1)^{\vee}/H_1(X; \mathbb{Z})$. $\square$

Proposition 67.2.3. *A holomorphic map $f: X \rightarrow Y$ of compact Kähler manifolds induces holomorphic group homomorphisms $f^*: J^p(Y) \rightarrow J^p(X)$, with $(g \circ f)^* = f^* \circ g^*$ and $\text{id}^* = \text{id}$. If $H^{2p-1}(X; \mathbb{Q}) = 0$, then $J^p(X) = 0$.*

Proof. The pullback $f^*: H^{2p-1}(Y; \mathbb{C}) \rightarrow H^{2p-1}(X; \mathbb{C})$ maps $H^{r,s}(Y)$ into $H^{r,s}(X)$ (Proposition 45.5.7), hence $F^p H^{2p-1}(Y)$ into $F^p H^{2p-1}(X)$, and it maps the image of $H^{2p-1}(Y; \mathbb{Z})$ into the image of $H^{2p-1}(X; \mathbb{Z})$, being induced by the pullback of integral cohomology. So it induces a $\mathbb{C}$ -linear map $H^{2p-1}(Y; \mathbb{C})/F^p \rightarrow H^{2p-1}(X; \mathbb{C})/F^p$ carrying lattice into lattice, hence a holomorphic homomorphism of the quotient tori; functoriality holds already on cohomology. The last statement is the dimension formula of Proposition 67.2.2, since $b_{2p-1}(X) = \dim_{\mathbb{Q}} H^{2p-1}(X; \mathbb{Q})$. $\square$

Example 67.2.4. Let $T = \mathbb{C}^g/\Lambda$ be a complex torus and $x_0 = 0$. The holomorphic 1-forms on T are the constant forms $\sum_j a_j dz_j$ (Example 43.4.2 and Lemma 43.3.2), so $H^0(T, \Omega^1)^{\vee} \cong \mathbb{C}^g$, the functional $\omega \mapsto \int \omega$ being determined by its values on $dz_1, \dots, dz_g$. For a path γ in T from 0 to x , lifted to a path in $\mathbb{C}^g$ from 0 to $\tilde{x}$, $\int_{\gamma} dz_j = \tilde{x}_j$; for a closed loop the lift ends at a point of Λ , and every $\lambda \in \Lambda$ arises from the loop $t \mapsto t\lambda$. So the period lattice is Λ , $\text{Alb}(T) = \mathbb{C}^g/\Lambda = T$, and the Albanese map $x \mapsto \text{AJ}(x - 0)$ of Exercise 67.7.2 is the identity of T . Likewise $J^1(T) = \text{Pic}^0(T) = H^{0,1}(T)/H^1(T; \mathbb{Z})$, where $H^{0,1}(T)$ is spanned by the forms $d\bar{z}_j$ and $H^1(T; \mathbb{Z}) = \text{Hom}(\Lambda, \mathbb{Z})$ embeds by sending an $\mathbb{R}$ -linear form on $\mathbb{C}^g$ with integer values on Λ to its $(0, 1)$ -part; this is the *dual torus* of T . For $g = 1$ it is again isomorphic to T (Chapter 55).

Remark 67.2.5. The complex structure of $J^p(X)$ is the one for which F^p is killed; it varies holomorphically with X by Griffiths transversality. But $J^p(X)$ is in general not an abelian variety: the Hodge–Riemann form on H^{2p-1} is definite on each $H^{r, 2p-1-r}$ with alternating signs, so it gives a positive definite form on $H_{\mathbb{C}}/F^p$ only when the Hodge structure has *level one*, $H^{2p-1} = H^{p, p-1} \oplus H^{p-1, p}$. Weil’s intermediate Jacobian uses a different complex structure that makes it polarised, but it does not vary holomorphically. When H^{2p-1} has level one, both agree and $J^p(X)$ is a principally polarised abelian variety, for instance for curves, and for $p = 2$ on the cubic threefold (Example 67.6.4).

67.3 The Abel–Jacobi map

Definition 67.3.1. Let X be projective and $Z \in Z^p(X)$ a cycle with $\text{cl}(Z) = 0$ in $H^{2p}(X; \mathbb{Z})$. Then $Z = \partial\Gamma$ for a real $(2n - 2p + 1)$ -chain Γ (the support of Z has real dimension $2n - 2p$, and its fundamental class vanishes in $H_{2n-2p}(X; \mathbb{Z})$). The *Abel–Jacobi invariant* is

$$\text{AJ}(Z) = \left(\omega \mapsto \int_{\Gamma} \omega \right) \in \frac{F^{n-p+1}H^{2n-2p+1}(X; \mathbb{C})^{\vee}}{H_{2n-2p+1}(X; \mathbb{Z})} = J^p(X),$$

for ω closed forms of type $\geq n - p + 1$ representing classes in F^{n-p+1} .

Proposition 67.3.2. *$\text{AJ}(Z)$ is well defined, and $\text{AJ}: Z^p(X)_{\text{hom}} \rightarrow J^p(X)$ is a homomorphism. It factors through rational equivalence, giving $\text{AJ}: \text{CH}^p(X)_{\text{hom}} \rightarrow J^p(X)$. For $p = 1$, and for curves, it is the Abel–Jacobi map of Theorem 55.6.3.*

Sketch of proof. If Γ and Γ' both bound Z , then $\Gamma - \Gamma'$ is a cycle and $\int_{\Gamma - \Gamma'} \omega$ is a period, an element of the image of $H_{2n-2p+1}(X; \mathbb{Z})$. The integral does not depend on the representative ω of the class: if $\omega = d\eta$ with ω of type $\geq n - p + 1$, one can choose η of type $\geq n - p + 1$, because the degeneration of the Hodge-to-de Rham spectral sequence (Corollary 45.5.3) makes d strict for the filtration by type [Voi02b, Chapter 8]; and then $\int_{\Gamma} d\eta = \int_Z \eta = 0$ because Z has complex dimension $n - p$ and η has more than $n - p$ holomorphic indices. For rational equivalence one shows that $t \mapsto \text{AJ}(Z_t - Z_0)$ is a holomorphic map from $\mathbb{P}^1$ to the torus $J^p(X)$, hence constant, as in the proof of Theorem 55.6.3 [Voi02b, Chapter 12]. $\square$

Remark 67.3.3. For curves, the Abel–Jacobi map $\text{CH}^1(C)_{\text{hom}} \rightarrow J^1(C)$ is an isomorphism (Theorem 55.6.3). In higher dimension its kernel can be enormous. For 0-cycles on a surface S with $h^{2,0}(S) > 0$, Mumford proved that the maps $\text{Sym}^m S \times \text{Sym}^m S \rightarrow \text{CH}_0(S)_{\text{hom}}$, $(a, b) \mapsto a - b$, are not surjective for any m [Mum68]: the group $\text{CH}_0(S)_{\text{hom}}$ is “infinite-dimensional”, while its image under the Albanese map $\text{CH}_0(S)_{\text{hom}} \rightarrow \text{Alb}(S)$ is a torus of finite dimension. On the other side, Griffiths’ theorem (Section 67.5) shows that AJ can detect cycles which are homologically, but not algebraically, trivial.

67.4 Deligne cohomology

Definition 67.4.1. The *Deligne complex* is $\mathbb{Z}(p)_{\mathcal{D}} = (\mathbb{Z}(p) \rightarrow \mathcal{O}_X \rightarrow \Omega_X^1 \rightarrow \cdots \rightarrow \Omega_X^{p-1})$, with $\mathbb{Z}(p) = (2\pi i)^p \mathbb{Z}$ in degree 0, and *Deligne cohomology* is its hypercohomology $H_{\mathcal{D}}^k(X; \mathbb{Z}(p))$. For $p = 1$, $\mathbb{Z}(1)_{\mathcal{D}} \simeq \mathcal{O}_X^*[-1]$ by the exponential sequence, so $H_{\mathcal{D}}^2(X; \mathbb{Z}(1)) = \text{Pic}(X)$.

Proposition 67.4.2. For X compact Kähler there is an exact sequence

$$0 \rightarrow J^p(X) \rightarrow H_{\mathcal{D}}^{2p}(X; \mathbb{Z}(p)) \rightarrow \text{Hdg}^p(X; \mathbb{Z}(p)) \rightarrow 0.$$

For $p = 1$ it is $0 \rightarrow \text{Pic}^0(X) \rightarrow \text{Pic}(X) \rightarrow \text{NS}(X) \rightarrow 0$.

Proof. The Deligne complex is the cone, shifted, of $\mathbb{Z}(p) \rightarrow \Omega_X^{<p}$. So there is a long exact sequence

$$H^{2p-1}(X; \mathbb{Z}(p)) \rightarrow \mathbb{H}^{2p-1}(X, \Omega^{<p}) \rightarrow H_{\mathcal{D}}^{2p}(X; \mathbb{Z}(p)) \rightarrow H^{2p}(X; \mathbb{Z}(p)) \rightarrow \mathbb{H}^{2p}(X, \Omega^{<p}).$$

By the holomorphic Poincaré lemma and the degeneration of the Hodge-to-de Rham spectral sequence (Corollary 45.5.3), $\mathbb{H}^k(X, \Omega^{<p}) = H^k(X; \mathbb{C})/F^p H^k(X; \mathbb{C})$. So the kernel of the last map consists of the integral classes whose image in $H^{2p}(X; \mathbb{C})$ lies in F^p ; a real class in F^p of weight $2p$ lies in $F^p \cap \overline{F^p} = H^{p,p}$, so these are the integral Hodge classes. The cokernel of the first map is $J^p(X)$ (the twist by $(2\pi i)^p$ only rescales the lattice). $\square$

Example 67.4.3. 1. For $p = 0$ the Deligne complex is $\mathbb{Z}$ in degree 0, so $H_{\mathcal{D}}^k(X; \mathbb{Z}(0)) = H^k(X; \mathbb{Z})$.

2. For $p = 1$, $\mathbb{Z}(1)_{\mathcal{D}} \simeq \mathcal{O}_X^*[-1]$ gives $H_{\mathcal{D}}^k(X; \mathbb{Z}(1)) = H^{k-1}(X, \mathcal{O}_X^*)$: for compact connected X , $H_{\mathcal{D}}^1(X; \mathbb{Z}(1)) = \mathbb{C}^*$ (the nonzero constants, by Proposition 41.2.3) and $H_{\mathcal{D}}^2(X; \mathbb{Z}(1)) = \text{Pic}(X)$.

3. For X a point and $p \geq 1$, the complex is $\mathbb{Z}(p) \rightarrow \mathbb{C}$ in degrees 0, 1, so $H_{\mathcal{D}}^1(\text{pt}; \mathbb{Z}(p)) = \mathbb{C}/(2\pi i)^p \mathbb{Z}$ and the other groups vanish. For $p = 1$ this is $\mathbb{C}/2\pi i \mathbb{Z} \cong \mathbb{C}^*$ via $\exp$, in accordance with (2).

Deligne cohomology is thus a mixture of integral cohomology and of the holomorphic data $\mathbb{C}/F^p$; it is not a topological invariant, and for $p \geq 1$ its groups are in general neither finitely generated nor vector spaces.

Remark 67.4.4. There is a *Deligne cycle class* $\text{cl}_{\mathcal{D}}: \text{CH}^p(X) \rightarrow H_{\mathcal{D}}^{2p}(X; \mathbb{Z}(p))$ lifting cl and restricting to AJ on homologically trivial cycles [Voi02b, Chapter 12]. In these terms the integral Hodge conjecture asks whether $\text{cl}_{\mathcal{D}}$ followed by the projection to $\text{Hdg}^p(X; \mathbb{Z}(p))$ is surjective. For $p = 1$ both maps are isomorphisms, $\text{CH}^1(X) = \text{Pic}(X) = H_{\mathcal{D}}^2(X; \mathbb{Z}(1))$; for $p \geq 2$, $\text{cl}_{\mathcal{D}}$ is neither injective nor surjective in general.

67.5 Griffiths' theorem

Theorem 67.5.1 (Griffiths). *Let $X \subseteq \mathbb{P}^4$ be a general quintic threefold. There are homologically trivial 1-cycles on X , differences of lines, that are not algebraically equivalent to zero, even after multiplication by any nonzero integer. So the Griffiths group $\text{Griff}^2(X) = Z^2(X)_{\text{hom}}/Z^2(X)_{\text{alg}}$ is not torsion.*

This is [Gri69]. The argument: algebraically trivial cycles map under AJ to an abelian subvariety of $J^2(X)$ whose tangent space lies in $H^{1,2}$, which for a quintic threefold is forced to be zero by the variation of Hodge structure. But the Abel–Jacobi images of differences of lines, deformed in the family of quintics, have a nonzero infinitesimal invariant, so they are not torsion. Clemens later showed that $\text{Griff}^2(X) \otimes \mathbb{Q}$ is infinite-dimensional for the general quintic [Cle83].

67.6 Normal functions

Definition 67.6.1. Let $f: X \rightarrow S$ be a smooth projective family. The intermediate Jacobians of the fibres form a family of complex tori $\mathcal{J}^p \rightarrow S$, the quotient of the holomorphic bundle $\mathcal{H}^{2p-1}/\mathcal{F}^p$ by the local system $R^{2p-1}f_*\mathbb{Z}$. A *normal function* is a holomorphic section ν of $\mathcal{J}^p$ that is *horizontal*: its local lifts $\tilde{\nu}$ to $\mathcal{H}^{2p-1}$ satisfy $\nabla \tilde{\nu} \in \mathcal{F}^{p-1} \otimes \Omega_S^1$. A family of homologically trivial cycles Z_t gives the normal function $t \mapsto \text{AJ}(Z_t)$.

Remark 67.6.2 (Poincaré and Lefschetz). Lefschetz's original proof of the $(1, 1)$ -theorem for a surface S used normal functions. Take a Lefschetz pencil of curves $C_t \subseteq S$. An integral class λ of type $(1, 1)$ restricts to classes on the C_t that, after subtracting a multiple of the hyperplane class, have degree 0, and a topological construction produces from λ a normal function ν with $\nu(t) \in J(C_t) = \text{Pic}^0(C_t)$. By Jacobi inversion (Theorem 55.6.3), $\nu(t)$ is the class of a divisor of degree 0 on C_t , and these divisors sweep out a divisor on S whose class is λ up to multiples of the hyperplane class (*Poincaré's existence theorem*). The same strategy for $p = 2$ fails because the intermediate Jacobians of the hyperplane sections are not generated by cycles. For cubic fourfolds, where the relevant J^2 of the cubic threefold hyperplane sections is covered by Abel–Jacobi images of lines, Zucker carried it through [Zuc77].

Remark 67.6.3. Green and Griffiths showed that the Hodge conjecture for a primitive Hodge class λ of degree $2p$ on X is equivalent to the existence of singularities, over the discriminant locus of a sufficiently ample linear system, of the normal function attached to λ [GG07]. This converts the conjecture into a problem about the boundary behaviour of normal functions, studied by Brosnan, Pearlstein, Schnell and others.

Example 67.6.4. For a nonsingular cubic threefold $X \subseteq \mathbb{P}^4$, $H^3(X)$ has Hodge numbers $h^{3,0} = 0$, $h^{2,1} = 5$, so it has level one and $J^2(X)$ is a principally polarised abelian variety of dimension 5. The Abel–Jacobi map from the Fano surface of lines on X to $J^2(X)$ identifies $J^2(X)$ with the Albanese variety of the Fano surface. Clemens and Griffiths showed that $J^2(X)$ is not a product of Jacobians of curves, and deduced that X is not rational, although it is unirational [CG72].

67.7 Exercises

Exercise 67.7.1. Compute $\dim J^p(X)$ for $X = \mathbb{P}^n$, for an abelian variety of dimension g , and for a quintic threefold with $p = 2$.

Solution. $\mathbb{P}^n$ has no odd cohomology, so all $J^p = 0$. For an abelian variety, $b_{2p-1} = \binom{2g}{2p-1}$, so $\dim J^p = \frac{1}{2} \binom{2g}{2p-1}$. For the quintic, $b_3 = 204$ (Proposition 48.4.4), so $\dim J^2 = 102$. $\square$

Exercise 67.7.2. Show that for $p = n$ the Abel–Jacobi map on 0-cycles of degree zero is $\sum n_i x_i \mapsto (\omega \mapsto \sum_i n_i \int_{x_0}^{x_i} \omega)$, and that it gives the *Albanese map* $X \rightarrow \text{Alb}(X)$, $x \mapsto \text{AJ}(x - x_0)$.

Solution. For $Z = \sum n_i (x_i - x_0)$ with $\sum n_i = 0$, a chain Γ with boundary Z is a sum of paths from x_0 to x_i with multiplicities n_i , and $F^1 H^1 = H^{1,0}$ is spanned by holomorphic 1-forms (Proposition 45.5.6). The formula follows from Definition 67.3.1 with $p = n$. $\square$

Chapter 68

Known Cases and Evidence

68.1 Introduction

This chapter collects the cases in which the Hodge conjecture is known, and explains the mechanism in each. They fall into three groups. First, cases where every Hodge class is forced to be a combination of known algebraic classes: small dimension, cellular varieties, varieties whose Hodge classes are generated by divisors, and hypersurfaces outside the middle degree. Second, abelian varieties, where the Mumford–Tate group (Chapter 63) makes the Hodge classes computable; there the difficulty concentrates in the *Weil classes*. Third, geometric methods: the decomposition of the diagonal for varieties with small Chow groups, normal functions for cubic fourfolds, and the Kuga–Satake construction for K3 surfaces. We state precisely what is proved, with proofs of the elementary parts and references for the rest. References: the source *The Hodge Conjecture* (Chapter 6), [Lew99], [BL04], [Voi03].

Prerequisites. Chapter 65 (The Lefschetz (1,1) Theorem), Chapter 63 (Mumford–Tate Groups), Chapter 66 (Variations of Hodge Structure) and Chapter 67 (Intermediate Jacobians and Normal Functions).

Throughout, X is a nonsingular complex projective variety of dimension n .

68.2 Cases from general theory

Definition 68.2.1. The *Lefschetz ring* of X is the $\mathbb{Q}$ -subalgebra $\text{Lef}^\bullet(X) \subseteq \bigoplus_p H^{2p}(X; \mathbb{Q})$ generated by $\text{Hdg}^1(X)$, the rational classes of divisors. Its elements are *Lefschetz classes*.

Proposition 68.2.2. *Lefschetz classes are algebraic. Hence $\text{HC}(X)$ holds whenever $\text{Hdg}^\bullet(X) = \text{Lef}^\bullet(X)$.*

Proof. Divisor classes are algebraic by the Lefschetz (1, 1)-theorem (Theorem 65.3.1), and products of algebraic classes are algebraic (Theorem 59.3.1(1)). $\square$

Theorem 68.2.3. *The Hodge conjecture holds in the following cases:*

1. $p \in \{0, 1, n-1, n\}$, and hence for $\dim X \leq 3$ (Corollary 65.3.2);
2. cellular varieties, such as flag varieties and toric varieties (Example 64.3.5);
3. very general abelian varieties and all their powers (Theorem 63.4.3);
4. smooth complete intersections, in all degrees except the middle one (Proposition 68.4.1);
5. uniruled fourfolds [CM78] and, more generally, varieties whose group of 0-cycles is supported on a subvariety of dimension ≤ 3 , in codimension 2 [BS83];

6. cubic fourfolds [Zuc77];

7. Fermat hypersurfaces of many degrees and dimensions [Shi79].

Items (1)–(4) are proved in this book. For (5)–(7) see the references and Sections 68.5 and 68.4.

Proposition 68.2.4. *Let X and Y be nonsingular projective varieties, and suppose that $H^\bullet(Y; \mathbb{Q})$ is spanned by classes of algebraic cycles, for instance Y cellular (Example 64.3.5). Then $\mathrm{HC}(X \times Y)$ holds if and only if $\mathrm{HC}(X)$ holds.*

Proof. Choose a basis $y_j = \mathrm{cl}(W_j) \in H^{2k_j}(Y; \mathbb{Q})$ of algebraic classes; they are of type (k_j, k_j) (Corollary 59.4.2), and $H^\bullet(Y; \mathbb{Q})$ has no odd part. By the Künneth formula (Theorem 20.4.2), every $\beta \in H^{2p}(X \times Y; \mathbb{Q})$ is uniquely $\beta = \sum_j \mathrm{pr}_1^* \alpha_j \cup \mathrm{pr}_2^* y_j$ with $\alpha_j \in H^{2p-2k_j}(X; \mathbb{Q})$, and the same holds over $\mathbb{C}$. Since pullbacks and cup products preserve types (Proposition 45.5.7), the component of type (r, s) of α_j contributes a class of type $(r + k_j, s + k_j)$. If β is of type (p, p) , comparing components of type (p, p) and using the uniqueness of the decomposition shows that each α_j is of type $(p - k_j, p - k_j)$, hence a Hodge class. If $\mathrm{HC}(X)$ holds, $\alpha_j = \mathrm{cl}(Z_j)$ for a rational cycle Z_j , and $\beta = \sum_j \mathrm{cl}(Z_j \times W_j)$ is algebraic (Theorem 59.3.1(1) and (2)).

Conversely, let $\alpha \in \mathrm{Hdg}^p(X)$ and let $y_0 = \mathrm{cl}(\mathrm{pt}) \in H^{2\dim Y}(Y; \mathbb{Q})$. The class $\mathrm{pr}_1^* \alpha \cup \mathrm{pr}_2^* y_0$ is a Hodge class on $X \times Y$; it is the pullback of α to $X \times \{\mathrm{pt}\}$ pushed forward to $X \times Y$, and if it is algebraic, so is its pushforward $\mathrm{pr}_{1*}(\mathrm{pr}_1^* \alpha \cup \mathrm{pr}_2^* y_0) = \alpha \cup \mathrm{pr}_{1*} \mathrm{pr}_2^* y_0 = \alpha$, by the projection formula and $\mathrm{pr}_{1*} \mathrm{pr}_2^* y_0 = 1$. $\square$

Example 68.2.5. The Hodge conjecture holds for $S \times \mathbb{P}^m$, $S \times \mathrm{Gr}(k, n)$ and $S \times F$ with F a flag variety, for every nonsingular projective surface or threefold S , although these products can have arbitrary dimension: by Proposition 68.2.4 and Theorem 68.2.3(1),(2). The same argument shows that the Hodge classes of $X \times Y$ are all products of Hodge classes of X with classes of Y in this situation; the interesting Hodge classes on products come from factors with odd cohomology, as for $E \times E'$ (Example 64.3.6).

68.3 Abelian varieties

Remark 68.3.1. For an abelian variety A , the Hodge classes on all powers A^r are the invariants of the Hodge group $\mathrm{Hg} = \mathrm{Hg}(H^1(A))$ (Corollary 63.3.3), and the divisor classes on A^r are the invariants of the *Lefschetz group* $\mathrm{Lef}(A)$, the connected centraliser of $\mathrm{End}(A) \otimes \mathbb{Q}$ in the symplectic group of a polarisation [Mur84]. Always $\mathrm{Hg} \subseteq \mathrm{Lef}(A)$. If equality holds, the Hodge ring of every power of A is generated by divisors, and the Hodge conjecture holds for all powers of A .

Theorem 68.3.2. $\mathrm{Hg}(A) = \mathrm{Lef}(A)$, and hence the Hodge conjecture holds for all powers of A , in each of the following cases: A is a product of elliptic curves [Ima76], [Mur84]; A is simple of prime dimension (Tankeev) [Tan82]; A is of dimension at most 5 and has no factor of Weil type (Moonen–Zarhin) [MZ99]. More generally the Hodge conjecture holds for A itself (not necessarily its powers) whenever $\mathrm{Hdg}^\bullet(A) = \mathrm{Lef}^\bullet(A)$, which holds for A of dimension ≤ 3 by Theorem 68.2.3(1).

Proposition 68.3.3 (Weil classes). *Let A be a complex abelian fourfold and $K \subseteq \mathrm{End}(A) \otimes \mathbb{Q}$ an imaginary quadratic field such that, for the two embeddings $\sigma, \bar{\sigma}: K \rightarrow \mathbb{C}$, the eigenspaces of K on $H^{1,0}(A)$ both have dimension 2 (Weil type). Then $H^1(A; \mathbb{Q})$ is a 4-dimensional K -vector space. Let $V_\sigma, V_{\bar{\sigma}} \subseteq H^1(A; \mathbb{C})$ be the eigenspaces of K . There is a 2-dimensional $\mathbb{Q}$ -subspace $W_K \subseteq \Lambda^4 H^1(A; \mathbb{Q}) = H^4(A; \mathbb{Q})$ with $W_K \otimes \mathbb{C} = \Lambda^4 V_\sigma \oplus \Lambda^4 V_{\bar{\sigma}}$, and it consists of Hodge classes, the Weil classes. For a general abelian fourfold of Weil type, $\mathrm{Hdg}^2(A) = \mathrm{Lef}^2(A) \oplus W_K$ with $W_K \cap \mathrm{Lef}^2(A) = 0$, so the Weil classes are not combinations of intersections of divisors.*

Proof. K acts on $H_{\mathbb{C}} = H^1(A; \mathbb{C})$ with eigenspaces $V_{\sigma}, V_{\bar{\sigma}}$, each of dimension 4, and K acts by Hodge endomorphisms, so $V_{\sigma} = V_{\sigma}^{1,0} \oplus V_{\sigma}^{0,1}$. By hypothesis $\dim V_{\sigma}^{1,0} = 2$, and then $\dim V_{\sigma}^{0,1} = 2$. The space $\Lambda_K^4 H^1(A; \mathbb{Q}) \otimes_{\mathbb{Q}} \mathbb{C}$ decomposes as $\Lambda^4 V_{\sigma} \oplus \Lambda^4 V_{\bar{\sigma}}$, and $\Lambda^4 V_{\sigma} = \Lambda^2 V_{\sigma}^{1,0} \otimes \Lambda^2 V_{\sigma}^{0,1}$ is of type $(2, 2)$, and likewise for $\bar{\sigma}$. The subspace $\Lambda^4 V_{\sigma} \oplus \Lambda^4 V_{\bar{\sigma}}$ of $\Lambda^4 H^1(A; \mathbb{C})$ is defined over K , since the eigenspace decomposition is, and it is stable under the nontrivial automorphism of K , which exchanges V_{σ} and $V_{\bar{\sigma}}$. By Galois descent it is the complexification of a $\mathbb{Q}$ -subspace W_K , of dimension 2, whose elements are rational classes of type $(2, 2)$. That $W_K \cap \text{Lef}^2 = 0$ for general A of Weil type, where $\text{End}(A) \otimes \mathbb{Q} = K$ and the Lefschetz group is a unitary group $\text{U}(2, 2)$ while $\text{Hg} = \text{SU}(2, 2)$, is [Wei79]; see also [BL04]. $\square$

Remark 68.3.4. Weil classes are the essential difficulty for abelian varieties: they are the first Hodge classes not generated by divisors, and Moonen and Zarhin showed that in dimension ≤ 5 all exceptional Hodge classes come from Weil classes. The Hodge conjecture for Weil classes was proved by Schoen for $K = \mathbb{Q}(\sqrt{-3})$ and $K = \mathbb{Q}(i)$ in special cases [Sch88], by van Geemen in further cases, and by Markman for all abelian fourfolds of Weil type of discriminant 1 [Mar23]. Weil classes are absolute Hodge classes by Deligne's theorem (Chapter 71).

68.4 Hypersurfaces and complete intersections

Proposition 68.4.1. *Let $X \subseteq \mathbb{P}^{n+1}$ be a nonsingular hypersurface, or more generally a nonsingular complete intersection of dimension n in some $\mathbb{P}^N$, and h the hyperplane class. Then for $2p \neq n$, $H^{2p}(X; \mathbb{Q}) = \mathbb{Q}h^p$. Hence $\text{HC}(X, p)$ holds for $2p \neq n$.*

Proof. By the Lefschetz hyperplane theorem (Theorem 47.5.1), applied repeatedly for complete intersections, $H^k(X; \mathbb{Q}) \cong H^k(\mathbb{P}^N; \mathbb{Q})$ for $k < n$, which is $\mathbb{Q}h^{k/2}$ for k even. For $k > n$, Poincaré duality gives $b_k = b_{2n-k}$, which is 1 for k even, and $h^{k/2} \neq 0$ by Proposition 44.4.4, so $H^k = \mathbb{Q}h^{k/2}$. Powers of h are algebraic, being classes of complete intersections of hyperplane sections. $\square$

Remark 68.4.2. In the middle degree $n = 2p$, a very general hypersurface of degree d has $\text{Hdg}^p(X) = \mathbb{Q}h^p$ as soon as H^n is not of Tate type up to h^p , for instance for surfaces with $d \geq 4$ (Theorem 65.3.6) and for fourfolds with $d \geq 3$, so the conjecture holds there trivially. Special hypersurfaces are the interesting cases. For cubic fourfolds, Zucker proved the conjecture with normal functions (Remark 67.6.2), and Voisin proved the integral version [Voi06]. For Fermat hypersurfaces $x_0^d + \cdots + x_{n+1}^d = 0$, the Hodge classes are computable via the action of $(\mu_d)^{n+2}$ and characters (Jacobi sums), and Shioda proved the conjecture for many (d, n) , including d prime and $d = 4$ [Shi79].

68.5 Uniruled varieties and the decomposition of the diagonal

Theorem 68.5.1 (Bloch–Srinivas). *Suppose that $\text{CH}_0(X)_{\mathbb{Q}}$ is supported on a closed subvariety of dimension ≤ 3 , that is, there is $W \subseteq X$ of dimension ≤ 3 with $\text{CH}_0(W)_{\mathbb{Q}} \rightarrow \text{CH}_0(X)_{\mathbb{Q}}$ surjective. Then $\text{HC}(X, 2)$ holds. This applies to rationally connected varieties, for which $\text{CH}_0(X)_{\mathbb{Q}} = \mathbb{Q}$.*

This is [BS83]. The method is the *decomposition of the diagonal*: the hypothesis gives an equality $N\Delta_X = Z_1 + Z_2$ in $\text{CH}^n(X \times X)$ with Z_1 supported on $D \times X$ for a divisor D and Z_2 supported on $X \times W$. Letting both sides act on $H^4(X; \mathbb{Q})$ as correspondences, the identity (times N) factors through the cohomology of a resolution of D , where Hodge classes of degree 2 on a variety of lower dimension are handled by induction and the $(1, 1)$ -theorem,

plus a contribution through W , of dimension ≤ 3 , where the conjecture is known. The same argument gives the conjecture for uniruled fourfolds, first proved by Conte and Murre [CM78].

68.6 K3 surfaces, hyper-Kähler varieties and Kuga–Satake

Remark 68.6.1. For a single K3 surface the conjecture holds, by dimension. The interesting questions are on products. *Mukai’s conjecture*, proved by Buskin [Bus19], says that every rational Hodge isometry $H^2(S; \mathbb{Q}) \rightarrow H^2(S'; \mathbb{Q})$ between K3 surfaces, a Hodge class in $H^4(S \times S'; \mathbb{Q})$, is algebraic. Markman extended this to hyper-Kähler varieties of K3^[n]-type.

Remark 68.6.2 (Kuga–Satake). Kuga and Satake attached to a polarised K3 surface S an abelian variety $\mathrm{KS}(S)$, of dimension 2^{19} , built from the even Clifford algebra of the lattice of primitive classes, together with an embedding of Hodge structures $H_{\mathrm{prim}}^2(S; \mathbb{Q}) \hookrightarrow H^1(\mathrm{KS}(S)) \otimes H^1(\mathrm{KS}(S))$ [KS67]. The Hodge conjecture predicts that this embedding is induced by an algebraic correspondence between S and $\mathrm{KS}(S)^2$ (the *Kuga–Satake Hodge conjecture*). It is known for some families, for instance by Paranjape for double planes branched along six lines [Par88]. Deligne used the Kuga–Satake construction, which works in families and in characteristic p , to prove the Weil conjectures for K3 surfaces [Del72], and to show that Hodge classes on K3 surfaces and their products with abelian varieties behave like absolute Hodge classes.

68.7 The evidence assessed

Remark 68.7.1. The known cases share a feature: the Hodge classes are either generated by divisors, or produced by an explicit geometric construction adapted to the variety (normal functions for cubic fourfolds, the decomposition of the diagonal, families of cycles on special abelian fourfolds, derived equivalences of K3 surfaces). There is no general mechanism. At the same time no case has failed, the Hodge loci behave as the conjecture predicts (Theorem 66.5.3), and Hodge classes on abelian varieties satisfy the arithmetic consequence of the conjecture (Deligne, Chapter 71). The first genuinely open cases include Weil classes on general abelian fourfolds of Weil type of arbitrary discriminant, abelian varieties of dimension ≥ 6 with exceptional classes, and products of two K3 surfaces in degree 4 away from Hodge isometries.

68.8 Exercises

Exercise 68.8.1. For a nonsingular cubic fourfold $X \subseteq \mathbb{P}^5$, show that $H^4(X; \mathbb{Q})$ has Hodge numbers $(0, 1, 21, 1, 0)$ and that $\mathrm{HC}(X, p)$ holds for $p \neq 2$ by Proposition 68.4.1.

Solution. As in Proposition 48.4.4, $c(TX) = (1 + h)^6 / (1 + 3h)$, whose degree-4 part is $(15 - 3 \cdot 20 + 9 \cdot 15 - 27 \cdot 6 + 81)h^4 = 9h^4$, and $\int_X h^4 = 3$, so $\chi(X) = 27$. The odd Betti numbers vanish and $b_0 = b_2 = b_6 = b_8 = 1$ (Proposition 68.4.1), so $b_4 = 27 - 4 = 23$. The Hodge numbers $h^{4,0} = 0$, $h^{3,1} = 1$, $h^{2,2} = 21$ follow from Griffiths’ description of the primitive cohomology of hypersurfaces by residues [Voi03]. The second statement is Proposition 68.4.1 with $n = 4$. $\square$

Exercise 68.8.2. Let E be an elliptic curve without complex multiplication. Show that $\mathrm{Hdg}^\bullet(E^r) = \mathrm{Lef}^\bullet(E^r)$ for all r .

Solution. By Proposition 63.4.1, $\mathrm{Hg}(E) = \mathrm{SL}_2$, and the invariants of SL_2 acting diagonally on $\Lambda^\bullet(H^{\oplus r})$ are generated, by classical invariant theory, by the invariants in $\Lambda^2(H^{\oplus r})$, which are the divisor classes (Exercise 57.7.2). Apply Corollary 63.3.3. $\square$

Chapter 69

Variants That Fail

69.1 Introduction

The Hodge conjecture is surrounded by statements that are equally natural and demonstrably false. Each failure identifies a mechanism that Hodge theory does not control, and so a constraint on any proof. We treat four variants: the integral Hodge conjecture, false for torsion classes (Atiyah–Hirzebruch) and for non-torsion classes (Kollár); the Kähler version, false even with Chern classes of coherent sheaves (Voisin); Hodge’s general conjecture, false for formal reasons and repaired by Grothendieck; and the variational Hodge conjecture, which is open but belongs here as the statement one would need to propagate algebraicity in families. The chapter ends with a diagnostic list. References: the source *The Hodge Conjecture* (Chapter 7), [AH62], [Kol92], [Voi02a], [Gro69a].

Prerequisites. Chapter 68 (Known Cases and Evidence).

Throughout, X is a nonsingular complex projective variety of dimension n unless stated otherwise.

69.2 The integral Hodge conjecture

Definition 69.2.1. The *integral Hodge conjecture* $\text{IHC}(X, p)$ asserts that the group $\text{Hdg}^p(X; \mathbb{Z})$ of integral Hodge classes (Definition 59.4.1) is generated by classes of subvarieties. Its *defect* is the group $Z^{2p}(X) = \text{Hdg}^p(X; \mathbb{Z}) / \text{cl}(\text{CH}^p(X))$.

Proposition 69.2.2. $Z^{2p}(X)$ is finitely generated, and it is finite if and only if $\text{HC}(X, p)$ holds. $Z^2(X) = 0$, and $Z^{2n}(X) = 0$.

Proof. $\text{Hdg}^p(X; \mathbb{Z})$ is a subgroup of the finitely generated group $H^{2p}(X; \mathbb{Z})$. The group $Z^{2p}(X) \otimes \mathbb{Q}$ is $\text{Hdg}^p(X) / Z_{\text{alg}}^p(X)$, which vanishes exactly when $\text{HC}(X, p)$ holds, and a finitely generated group is finite if and only if its rationalisation vanishes. $Z^2 = 0$ is the Lefschetz $(1, 1)$ -theorem (Theorem 65.3.1), and $H^{2n}(X; \mathbb{Z})$ is generated by the classes of points, one for each component. $\square$

Theorem 69.2.3 (Atiyah–Hirzebruch). *There are nonsingular projective varieties X with torsion classes in $H^4(X; \mathbb{Z})$, automatically integral Hodge classes, that are not algebraic. In particular IHC fails.*

This is [AH62]. The obstruction is topological. A subvariety Z of codimension p defines an element $[O_Z]$ of topological K -theory $K^0(X)$, through a finite resolution by vector bundles, and its image in the graded piece of the topological filtration is $\text{cl}(Z)$ up to the factor $(p-1)!$, since $c_p(O_Z) = (-1)^{p-1}(p-1)! \text{cl}(Z)$ while $\text{ch}_p(O_Z) = \text{cl}(Z)$ (Proposition 59.5.1). So algebraic

classes survive to E_∞ in the Atiyah–Hirzebruch spectral sequence $H^s(X; K^t(\text{pt})) \Rightarrow K^{s+t}(X)$, whose first possibly nonzero differential is $d_3 = \text{Sq}_{\mathbb{Z}}^3 = \beta \circ \text{Sq}^2 \circ \rho$, the integral lift of the Steenrod square (Chapter 22). Hence an algebraic class α satisfies $\text{Sq}_{\mathbb{Z}}^3(\alpha) = 0$. Atiyah and Hirzebruch then produce, by the Godeaux–Serre construction of a quotient $X = Y/G$ of a complete intersection by a free action of a finite group G such as $(\mathbb{Z}/p)^3$, whose cohomology agrees with that of the classifying space BG in low degrees, a torsion class with $\text{Sq}_{\mathbb{Z}}^3 \neq 0$.

Remark 69.2.4. Totaro observed that the cycle class map factors through complex cobordism, $\text{CH}^p(X) \rightarrow MU^{2p}(X) \otimes_{MU^*} \mathbb{Z} \rightarrow H^{2p}(X; \mathbb{Z})$, which gives a finer obstruction [Tot97]. Soulé and Voisin produced further torsion counterexamples by other methods [SV05]. All these classes vanish in $H^{2p}(X; \mathbb{Q})$, so they do not affect the rational conjecture: Hodge theory is blind to torsion.

Theorem 69.2.5 (Kollár). *Let $\ell \geq 5$ be a prime and d a multiple of ℓ^3 . For a very general hypersurface $X \subseteq \mathbb{P}^4$ of degree d , every curve $C \subseteq X$ has degree divisible by ℓ . Consequently the generator α of $H^4(X; \mathbb{Z}) \cong \mathbb{Z}$ is an integral Hodge class that is not algebraic, and $Z^4(X) \neq 0$ contains an element of order ℓ .*

Proof of the consequence. The divisibility statement is [Kol92]. Granting it: $\dim X = 3$, and by the Lefschetz hyperplane theorem (Theorem 47.5.1) and Poincaré duality, $H^2(X; \mathbb{Z}) \cong \mathbb{Z}h$ and $H^4(X; \mathbb{Z}) \cong \mathbb{Z}$, generated by a class α with $\int_X \alpha \cup h = 1$ (the pairing $H^2 \times H^4 \rightarrow \mathbb{Z}$ is perfect on the torsion-free quotients; integral Lefschetz for H^2 uses the integral version of the hyperplane theorem, Remark 47.5.2). Since $H^4(X; \mathbb{C}) = \mathbb{C}h^2$ is of type $(2, 2)$, α is an integral Hodge class. A 1-cycle $\sum a_i C_i$ has class $(\sum a_i \deg C_i)\alpha$, because $\int \text{cl}(C) \cup h = \deg C$ (Theorem 59.3.1(4)). If all degrees are divisible by ℓ , the algebraic classes lie in $\ell\mathbb{Z}\alpha$, so α is not algebraic and $\mathbb{Z}/\ell$ is a quotient of $Z^4(X)$. $\square$

Corollary 69.2.6. *In the situation of Theorem 69.2.5, the defect $Z^4(X)$ is a finite cyclic group whose order is divisible by ℓ and divides d . In particular $\text{HC}(X, 2)$ holds although $\text{IHC}(X, 2)$ fails.*

Proof. With α the generator of $H^4(X; \mathbb{Z}) = \mathbb{Z}\alpha$ normalised by $\int_X \alpha \cup h = 1$, the class h^2 satisfies $\int_X h^2 \cup h = \deg X = d$, so $h^2 = d\alpha$. It is algebraic, the class of the intersection of X with two hyperplanes, a curve of degree d . So $d\mathbb{Z}\alpha \subseteq \text{cl}(\text{CH}^2(X)) \subseteq \ell\mathbb{Z}\alpha$, the second inclusion by Theorem 69.2.5. Hence $\text{cl}(\text{CH}^2(X)) = m\mathbb{Z}\alpha$ with $\ell \mid m \mid d$, and $Z^4(X) = \text{Hdg}^2(X; \mathbb{Z})/\text{cl}(\text{CH}^2(X)) = \mathbb{Z}\alpha/m\mathbb{Z}\alpha \cong \mathbb{Z}/m$. Rationally $\alpha = \frac{1}{d}h^2$ is algebraic. $\square$

Remark 69.2.7. The divisibility is proved by specialisation and countability. The relative Hilbert scheme of curves in the universal family of hypersurfaces has countably many components, each proper over the parameter space. For a very general member X , every curve on X therefore lies in a component that dominates the parameter space, so it specialises to a 1-cycle of the same degree on every member, in particular on a special singular member X_0 . Kollár builds X_0 from a cyclic covering of degree ℓ so that every 1-cycle on it has degree divisible by ℓ ; this is where $\ell^3 \mid d$ is used. (Kollár’s statement allows any integer ℓ prime to 6.) The Hodge structure here is as simple as possible; the obstruction is purely about degrees of curves, on which Hodge theory has nothing to say.

Theorem 69.2.8. 1. (Voisin) $\text{IHC}(X, 2)$ holds for uniruled threefolds and for Calabi–Yau threefolds [Voi06].

2. (Colliot-Thélène–Voisin) The group $Z^4(X)$ is a birational invariant of nonsingular projective varieties. Its torsion subgroup is a quotient of the unramified cohomology group $H_{\text{nr}}^3(X, \mathbb{Q}/\mathbb{Z}(2))$, and is isomorphic to it when $\text{CH}_0(X)$ is supported on a surface, for instance when X is rationally connected. Since unramified cohomology vanishes for rational varieties, a nonzero torsion element of $Z^4(X)$ obstructs rationality [CTV12].

69.3 The Kähler counterexample

Theorem 69.3.1 (Voisin). *There is a complex torus T of dimension 4 with a nonzero class in $\mathrm{Hdg}^2(T)$ that is not in the $\mathbb{Q}$ -span of the classes $\mathrm{ch}_2(\mathcal{F})$ of coherent analytic sheaves on T , nor of classes of analytic subvarieties. Hence the Hodge conjecture fails for compact Kähler manifolds in both the cycle form and the sheaf form.*

This is [Voi02a]. The torus is of Weil type (Proposition 68.3.3): the construction of Weil classes uses only Hodge theory, not projectivity. In outline, Voisin chooses the lattice so that $\mathrm{NS}(T) = 0$ and the Hodge classes in H^4 are primitive for a Kähler class ω , that is, $\int_T \alpha \wedge \omega^2 = 0$. For a coherent sheaf $\mathcal{F}$ on T , $c_1(\mathcal{F}) = 0$ because $\mathrm{NS}(T) = 0$, and a positivity property of c_2 coming from Hermite–Einstein metrics (a Bogomolov-type inequality) gives $\int_T c_2(\mathcal{F}) \wedge \omega^2 \geq 0$ with equality only if $c_2(\mathcal{F}) = 0$. Since $c_2(\mathcal{F})$ is a Hodge class, it is primitive, so $c_2(\mathcal{F}) = 0$. Analytic subvarieties are covered because their structure sheaves are coherent sheaves.

Remark 69.3.2. All the Hodge-theoretic structure (Hodge decomposition, hard Lefschetz, Hodge–Riemann relations, the $\partial\bar{\partial}$ -lemma) holds on Voisin’s torus. So no proof of the Hodge conjecture can use only the Kähler package: it must use projectivity, through sections of ample line bundles, algebraic Hilbert schemes, GAGA, or arithmetic models. For $p = 1$ the Kähler version holds in the form of Theorem 65.2.1: integral $(1, 1)$ -classes are Chern classes of holomorphic line bundles.

69.4 Hodge’s general conjecture

Proposition 69.4.1. *Hodge’s original general conjecture $N^c H^k(X; \mathbb{Q}) = H^k(X; \mathbb{Q}) \cap F^c H^k(X; \mathbb{C})$ fails in general: the left side is a sub-Hodge structure (Proposition 64.4.5), while the right side need not be.*

Proof. Let $V = H^k(X; \mathbb{Q})$ and $W = V \cap F^c$. Since V is stable under conjugation, $W \subseteq F^c \cap \overline{F^c}$, which is $\bigoplus_{c \leq p \leq k-c} V^{p, k-p}$. A rational subspace of this sum need not be the rational form of a sum of Hodge summands, that is, need not be a sub-Hodge structure. Grothendieck exhibits an abelian variety whose H^3 contains a rational line in F^1 whose smallest containing sub-Hodge structure is not contained in F^1 [Gro69a]. For such a line, the right side contains it, and the left side, a sub-Hodge structure contained in F^1 (Proposition 64.4.5), cannot. $\square$

Remark 69.4.2. The lesson is formal: the objects of Hodge theory are sub-Hodge structures, not rational subspaces satisfying a filtration condition. Grothendieck’s repaired *generalised Hodge conjecture* (Remark 64.4.6) is open in essentially every case where the Hodge conjecture is, and in many where it is known.

69.5 The variational Hodge conjecture

Conjecture 69.5.1 (Variational Hodge conjecture). *Let $f : X \rightarrow S$ be a smooth projective morphism over a connected nonsingular base, and λ a flat section of $R^{2p} f_* \mathbb{Q}$ with $\lambda_s \in \mathrm{Hdg}^p(X_s)$ for all s . If λ_{s_0} is algebraic for one s_0 , then λ_s is algebraic for all s .*

Remark 69.5.2. The variational conjecture follows from the Hodge conjecture applied to a compactification of the total space, by the theorem of the fixed part (Theorem 66.4.1). Bloch’s semiregularity theorem proves the infinitesimal version: a local complete intersection cycle whose semiregularity map is injective deforms together with its cohomology class [Blo72].

The Cattani–Deligne–Kaplan theorem (Theorem 66.5.3) says that the locus where λ stays Hodge is algebraic; the variational conjecture says that on this locus λ stays algebraic. The first is a theorem and the second is open, which summarises the state of the subject.

69.6 A diagnostic list

Remark 69.6.1. A proof of the Hodge conjecture must be able to distinguish it from its false neighbours:

1. *It must lose denominators.* Since the integral version is false (Kollár), cycles can only be produced after clearing denominators. In known cases the denominators enter through push-pull, $f_*f^* = \deg f$, or through factorials such as $c_p(\mathcal{O}_Z) = (-1)^{p-1}(p-1)!\operatorname{cl}(Z)$.
2. *It must use projectivity*, not just the Kähler package (Voisin).
3. *It must be insensitive to torsion* (Atiyah–Hirzebruch): it cannot produce integral cycles from integral Hodge classes.
4. *It must respect sub-Hodge structures* (Grothendieck): it must work with Hodge classes, not with rational classes in a filtration step.

69.7 Exercises

Exercise 69.7.1. Show that every torsion class in $H^{2p}(X; \mathbb{Z})$ is an integral Hodge class, and that $Z^{2p}(X)$ is torsion if $\operatorname{HC}(X, p)$ holds.

Solution. A torsion class maps to $0 \in H^{2p}(X; \mathbb{C})$, which lies in $H^{p,p}$. The second statement is Proposition 69.2.2. $\square$

Exercise 69.7.2. Why does Theorem 69.2.5 not contradict the fact that a quintic threefold contains lines, which have degree 1?

Solution. The theorem needs d divisible by $\ell^3 \geq 125$, and the hypersurface very general. A quintic has $d = 5$, which is not divisible by 125. $\square$

Chapter 70

Mixed Hodge Structures

70.1 Introduction

The cohomology of a smooth but non-compact variety, or of a singular projective one, is not a pure Hodge structure. For example, $H^1(\mathbb{C}^*; \mathbb{Q})$ is one-dimensional, so it cannot carry a Hodge structure of odd weight. Deligne showed that the cohomology of every complex algebraic variety carries a *mixed Hodge structure*: an increasing *weight filtration* whose graded pieces are pure Hodge structures of various weights. Morphisms of mixed Hodge structures are strict for both filtrations (Theorem 70.3.2), and this rigidity has consequences for compact varieties too. Three of them are proved in Section 70.6: the global invariant cycle theorem, used in Chapter 66; the fact that algebraicity of a Hodge class can be tested on a Zariski open subset, modulo lower dimensions; and the birational invariance of the Hodge conjecture, modulo lower dimensions. References: [Del71], [Del74b], [PS08], [Voi02b, Chapter 8].

Prerequisites. Chapter 62 (Hodge Structures), Chapter 15 (Derived Categories) and Chapter 54 (Analytic Spaces and GAGA).

Conventions. Filtrations are finite and exhaustive. $W_\bullet$ always denotes an increasing filtration ($W_{k-1} \subseteq W_k$) and $F^\bullet$ a decreasing one ($F^{p+1} \subseteq F^p$). For an increasing filtration, $\mathrm{Gr}_k^W = W_k/W_{k-1}$. The Tate structure $\mathbb{Q}(m)$ is $(2\pi i)^m \mathbb{Q} \subseteq \mathbb{C}$, pure of weight $-2m$ and type $(-m, -m)$ (Example 62.2.3), and $H(m) = H \otimes \mathbb{Q}(m)$. Varieties are complex algebraic varieties, and their cohomology is the singular cohomology of the associated analytic space.

70.2 Mixed Hodge structures

Definition 70.2.1. A *mixed Hodge structure* (H, W, F) consists of

1. a finite-dimensional $\mathbb{Q}$ -vector space H ;
2. a finite increasing filtration $W_\bullet$ of H by $\mathbb{Q}$ -subspaces, the *weight filtration*;
3. a finite decreasing filtration $F^\bullet$ of $H_{\mathbb{C}} = H \otimes_{\mathbb{Q}} \mathbb{C}$ by $\mathbb{C}$ -subspaces, the *Hodge filtration*;

such that for every k the filtration induced by F on $\mathrm{Gr}_k^W H_{\mathbb{C}} = W_k H_{\mathbb{C}} / W_{k-1} H_{\mathbb{C}}$, namely

$$F^p \mathrm{Gr}_k^W H_{\mathbb{C}} = (F^p \cap W_k H_{\mathbb{C}} + W_{k-1} H_{\mathbb{C}}) / W_{k-1} H_{\mathbb{C}},$$

is the Hodge filtration of a pure $\mathbb{Q}$ -Hodge structure of weight k on $\mathrm{Gr}_k^W H$ (Proposition 62.2.2). A *morphism* $f: (H, W, F) \rightarrow (H', W', F')$ is a $\mathbb{Q}$ -linear map with $f(W_k) \subseteq W'_k$ and $f_{\mathbb{C}}(F^p) \subseteq F'^p$ for all k, p . The *Hodge numbers* are $h^{p,q}(H) = \dim_{\mathbb{C}}(\mathrm{Gr}_{p+q}^W H_{\mathbb{C}})^{p,q}$, and the *weights* of H are the k with $\mathrm{Gr}_k^W H \neq 0$.

- Example 70.2.2.** 1. A pure Hodge structure of weight k is a mixed Hodge structure with $W_{k-1} = 0$ and $W_k = H$, and morphisms of pure Hodge structures of the same weight are the same as morphisms of the associated mixed Hodge structures. A direct sum of pure Hodge structures of different weights is a mixed Hodge structure, with W_k the sum of the summands of weight $\leq k$.
2. $H^1(\mathbb{C}^*; \mathbb{Q})$ is one-dimensional, spanned by the class of $\frac{1}{2\pi i} \frac{dz}{z}$, which integrates to 1 over the unit circle. With $W_1 = 0$, $W_2 = H^1$, $F^1 = H^1(\mathbb{C}^*; \mathbb{C})$ and $F^2 = 0$, it is the pure Tate structure $\mathbb{Q}(-1)$ of weight 2 and type $(1, 1)$. This is the structure given by Theorem 70.4.3 below, with $\bar{X} = \mathbb{P}^1$ and $D = \{0, \infty\}$.
3. Let C be a nodal plane cubic curve, obtained from $\mathbb{P}^1$ by identifying two points. Then $H^1(C; \mathbb{Q}) \cong \mathbb{Q}$ is pure of weight 0 and type $(0, 0)$ (Example 70.5.2 below): the loop through the node is not detected by holomorphic 1-forms.
4. *Kummer structures.* Let $q \in \mathbb{C}^*$, choose $\log q$, and put $\lambda = \log q / 2\pi i$. On $H = \mathbb{Q}e_0 \oplus \mathbb{Q}e_1$ let $W_{-3} = 0$, $W_{-2} = W_{-1} = \mathbb{Q}e_1$, $W_0 = H$, and $F^{-1} = H_{\mathbb{C}}$, $F^0 = \mathbb{C}(e_0 - \lambda e_1)$, $F^1 = 0$. This is a mixed Hodge structure K_q , an extension of $\mathbb{Q}(0)$ by $\mathbb{Q}(1)$ (Exercise 70.7.2). Another choice of $\log q$ changes λ by an integer n , and $e_0 \mapsto e_0 + ne_1$ is an isomorphism between the two structures, so K_q depends only on q .

Counterexample 70.2.3. A mixed Hodge structure need not be the direct sum of its graded pieces, not even after extending scalars to $\mathbb{R}$. In K_q , a complement of W_{-2} that is a sub-mixed Hodge structure is a line $L = \mathbb{Q}(e_0 + te_1)$ with $t \in \mathbb{Q}$ whose induced Hodge filtration is that of $\mathbb{Q}(0)$, that is, with $L_{\mathbb{C}} \subseteq F^0 = \mathbb{C}(e_0 - \lambda e_1)$. This forces $t = -\lambda$. So K_q splits if and only if $\lambda \in \mathbb{Q}$, that is, q is a root of unity; and the same argument with $t \in \mathbb{R}$ shows that $K_q \otimes \mathbb{R}$ does not split when $|q| \neq 1$, since then $\lambda \notin \mathbb{R}$. Carlson showed that the analogous extensions of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ over $\mathbb{Z}$ are classified exactly by $q \in \mathbb{C}/2\pi i\mathbb{Z} \cong \mathbb{C}^*$ [Car80]: Kummer structures are the Hodge-theoretic incarnation of the exponential map. Contrast this with pure polarisable Hodge structures, which form a semisimple category (Theorem 62.5.3).

70.3 The Deligne splitting and strictness

Theorem 70.3.1 (Deligne). *Let (H, W, F) be a mixed Hodge structure. For $p, q \in \mathbb{Z}$ put*

$$I^{p,q} = F^p \cap W_{p+q, \mathbb{C}} \cap \left(\overline{F^q} \cap W_{p+q, \mathbb{C}} + \sum_{j \geq 1} \overline{F^{q-j}} \cap W_{p+q-j-1, \mathbb{C}} \right).$$

Then $H_{\mathbb{C}} = \bigoplus_{p,q} I^{p,q}$, and

$$W_{k, \mathbb{C}} = \bigoplus_{p+q \leq k} I^{p,q}, \quad F^p = \bigoplus_{p' \geq p} I^{p',q}, \quad \overline{I^{p,q}} \subseteq I^{q,p} \oplus \bigoplus_{p' < q, q' < p} I^{p',q'}.$$

The projection $W_{p+q, \mathbb{C}} \rightarrow \text{Gr}_{p+q}^W H_{\mathbb{C}}$ maps $I^{p,q}$ isomorphically onto $(\text{Gr}_{p+q}^W H_{\mathbb{C}})^{p,q}$. Conversely, if W is a filtration by $\mathbb{Q}$ -subspaces and F a filtration of $H_{\mathbb{C}}$ admitting a bigrading $H_{\mathbb{C}} = \bigoplus I^{p,q}$ with these three properties, then (H, W, F) is a mixed Hodge structure.

The first part is [Del71]; see [PS08, Chapter 3] for the explicit formula. The converse is immediate: on Gr_k^W the images of the $I^{p,q}$ with $p+q = k$ give a decomposition with $\overline{H^{p,q}} = H^{q,p}$, since the error terms have $p' + q' < k$, and the filtration induced by F is $\bigoplus_{p' \geq p} H^{p',k-p'}$. The subspaces $I^{p,q}$ form the *Deligne splitting*. It splits W and F simultaneously, but it is not compatible with complex conjugation in general (Counterexample 70.2.3). Every morphism

f of mixed Hodge structures maps $I^{p,q}(H)$ into $I^{p,q}(H')$, as is clear from the formula, since f preserves W, F and, being defined over $\mathbb{Q}$, also $\bar{F}$.

Theorem 70.3.2 (Deligne). 1. Every morphism $f: H \rightarrow H'$ of mixed Hodge structures is strict for W and for F : $f(W_k) = W'_k \cap f(H)$ and $f_{\mathbb{C}}(F^p) = F'^p \cap f_{\mathbb{C}}(H_{\mathbb{C}})$.

2. Mixed Hodge structures form an abelian category, in which kernels, images and cokernels carry the induced filtrations.

3. For every k , the functor $H \mapsto \text{Gr}_k^W H$ from mixed Hodge structures to pure Hodge structures of weight k is exact.

4. $H \otimes H', H^\vee$ and $\text{Hom}(H, H') = H^\vee \otimes H'$ carry mixed Hodge structures, with $W_k(H \otimes H') = \sum_{a+b=k} W_a H \otimes W_b H', F^p(H \otimes H') = \sum_{a+b=p} F^a H \otimes F^b H', W_k(H^\vee) = (W_{-k-1} H)^\perp$ and $F^p(H^\vee) = (F^{1-p} H)^\perp$.

Proof. (1) By Theorem 70.3.1, $f_{\mathbb{C}}$ maps $I^{p,q}(H)$ into $I^{p,q}(H')$. Let $y = f(x) \in W'_{k,\mathbb{C}}$ and decompose $x = \sum x^{p,q}$ with $x^{p,q} \in I^{p,q}(H)$. Then $y = \sum f(x^{p,q})$ is the decomposition of y , so $f(x^{p,q}) = 0$ whenever $p+q > k$, and $y = f(\sum_{p+q \leq k} x^{p,q}) \in f(W_{k,\mathbb{C}})$. The same argument, with $p' \geq p$ in place of $p+q \leq k$, proves strictness for F . Strictness for W over $\mathbb{Q}$ follows from strictness over $\mathbb{C}$, because $f(W_k)$ and $W'_k \cap f(H)$ are $\mathbb{Q}$ -subspaces with the same complexification.

(2) Since $f_{\mathbb{C}}$ preserves the bigradings, its kernel $K_{\mathbb{C}}$, image $f(H_{\mathbb{C}})$ and cokernel $C_{\mathbb{C}}$ are bigraded: $K_{\mathbb{C}} = \bigoplus (K_{\mathbb{C}} \cap I^{p,q}(H))$, $f(H_{\mathbb{C}}) = \bigoplus f(I^{p,q}(H))$ and $C_{\mathbb{C}} = \bigoplus I^{p,q}(H')/f(I^{p,q}(H))$. Give $K, f(H)$ and C the filtrations induced as subspaces or quotients. In each case W_k and F^p are the sums of the pieces with $p+q \leq k$, respectively with first index $\geq p$. For $f(H)$ the filtrations induced from H (quotient) and from H' (subspace) agree, which is the strictness proved in (1). The bigradings satisfy the conjugation condition of Theorem 70.3.1, because $K, f(H)$ and C are defined over $\mathbb{Q}$ and the components of a vector in a bigraded subspace lie in it. By the converse part of Theorem 70.3.1, $K, f(H)$ and C are mixed Hodge structures, and the canonical map $H/K \rightarrow f(H)$ is an isomorphism of mixed Hodge structures.

(3) If $0 \rightarrow H' \rightarrow H \rightarrow H'' \rightarrow 0$ is an exact sequence of mixed Hodge structures, then its complexification is exact and each map preserves the bigradings, so $0 \rightarrow I^{p,q}(H') \rightarrow I^{p,q}(H) \rightarrow I^{p,q}(H'') \rightarrow 0$ is exact for each (p, q) . Since $\text{Gr}_k^W H_{\mathbb{C}} \cong \bigoplus_{p+q=k} I^{p,q}(H)$, the sequence of graded pieces is exact.

(4) Put $I^{p,q}(H \otimes H') = \bigoplus I^{a,b}(H) \otimes I^{a',b'}(H')$, the sum over $a+a'=p, b+b'=q$. This bigrading splits the filtrations W and F of the statement, and it satisfies the conjugation condition because the error terms of the factors multiply to error terms. As $W_k(H \otimes H')$ is defined over $\mathbb{Q}$, the converse part of Theorem 70.3.1 applies. For the dual, use $I^{p,q}(H^\vee) = (\bigoplus_{(p',q') \neq (-p,-q)} I^{p',q'}(H))^\perp$ in the same way. $\square$

Lemma 70.3.3 (Principle of weights). Let $f: H \rightarrow H'$ be a morphism of mixed Hodge structures.

1. If the weights of H are all $< m$ and those of H' are all $\geq m$, or the other way round, then $f = 0$.
2. If H' is pure of weight k , then $f(H) = f(W_k H)$ and $f(W_{k-1} H) = 0$.

Proof. (1) If the weights of H are $< m$ and those of H' are $\geq m$, then $H = W_{m-1} H$ and $f(H) \subseteq W'_{m-1} H' = 0$. In the other case $W_{m-1} H = 0$ and $H' = W'_{m-1} H'$, so strictness (Theorem 70.3.2(1)) gives $f(H) = f(H) \cap W'_{m-1} H' = f(W_{m-1} H) = 0$.

(2) $W'_{k-1} H' = 0$ and $W'_k H' = H'$, so $f(W_{k-1} H) \subseteq W'_{k-1} H' = 0$, and strictness gives $f(W_k H) = W'_k H' \cap f(H) = f(H)$. $\square$

Remark 70.3.4. Lemma 70.3.3 is used over and over below: a map compatible with mixed Hodge structures cannot move classes of one weight to classes of another. Its power comes from the fact that the maps it is applied to (restrictions, Gysin maps, differentials of spectral sequences) are defined topologically, while the weights are defined algebro-geometrically. It has no analogue for filtered vector spaces, where strictness fails (Counterexample 62.3.4).

70.4 The logarithmic de Rham complex

Definition 70.4.1. Let $\bar{X}$ be a complex manifold of dimension n and $D \subseteq \bar{X}$ a divisor with *simple normal crossings*: $D = D_1 \cup \cdots \cup D_N$ with each D_i a nonsingular hypersurface, and near every point there are holomorphic coordinates $z_1, \dots, z_n$ with $D = \{z_1 \cdots z_r = 0\}$. Put $X = \bar{X} \setminus D$ and let $j: X \rightarrow \bar{X}$ be the inclusion.

1. The sheaf $\Omega_{\bar{X}}^p(\log D)$ of *logarithmic p -forms* is the $\mathcal{O}_{\bar{X}}$ -submodule of $j_*\Omega_X^p$ generated locally by the wedge products of p of the forms $dz_1/z_1, \dots, dz_r/z_r, dz_{r+1}, \dots, dz_n$. It is locally free, and stable under d , which gives the *logarithmic de Rham complex* $\Omega_{\bar{X}}^\bullet(\log D)$.
2. The *weight filtration* $W_m\Omega_{\bar{X}}^p(\log D)$ is the subsheaf spanned by the forms $\alpha \wedge dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m}$ with α holomorphic: forms with at most m logarithmic factors.
3. $D^{(m)}$ is the disjoint union of the m -fold intersections $D_{i_1} \cap \cdots \cap D_{i_m}$, $i_1 < \cdots < i_m$, with $D^{(0)} = \bar{X}$. It is a complex manifold of dimension $n - m$, with the map $a_m: D^{(m)} \rightarrow \bar{X}$.

Lemma 70.4.2. The residue map $\alpha \wedge dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m} \mapsto \alpha|_{D_{i_1} \cap \cdots \cap D_{i_m}}$ induces an isomorphism of complexes

$$\text{Res}_m: \text{Gr}_m^W \Omega_{\bar{X}}^\bullet(\log D) \longrightarrow (a_m)_* \Omega_{D^{(m)}}^{\bullet-m}.$$

Sketch of proof. It is easiest to construct the inverse. For $I = \{i_1 < \cdots < i_m\}$ write $D_I = D_{i_1} \cap \cdots \cap D_{i_m}$ and $dz_I/z_I = dz_{i_1}/z_{i_1} \wedge \cdots \wedge dz_{i_m}/z_{i_m}$. Given a holomorphic form β on D_I , extend it locally to a holomorphic form $\tilde{\beta}$ on $\bar{X}$ and send it to the class of $\tilde{\beta} \wedge dz_I/z_I$ in Gr_m^W . The class does not depend on the extension: two extensions differ by a combination of forms $z_i\gamma$ and $dz_i \wedge \gamma$ with $i \in I$, and $z_i\gamma \wedge dz_I/z_I$ and $dz_i \wedge \gamma \wedge dz_I/z_I$ have at most $m - 1$ logarithmic factors, since $z_i \cdot dz_i/z_i = dz_i$ and $dz_i \wedge dz_i/z_i = 0$. It does not depend on the coordinates: replacing z_i by uz_i with u a unit changes dz_i/z_i by the holomorphic form du/u , which lowers the weight. The map is surjective by the definition of W_m , it commutes with d because $d(dz_i/z_i) = 0$, and Res_m is its inverse. That it is injective, that is, that a sum $\sum_I \tilde{\beta}_I \wedge dz_I/z_I$ lies in W_{m-1} only if every β_I vanishes, is a computation in the local basis $dz_I/z_I \wedge dz_J$, $I \subseteq \{1, \dots, r\}$, $J \subseteq \{r+1, \dots, n\}$, which we omit [Del71]. $\square$

The residue depends on the ordering $i_1 < \cdots < i_m$ through a sign, and in the comparison with rational cohomology it comes with the factor $(2\pi i)^{-m}$, because $\frac{1}{2\pi i} \frac{dz}{z}$ is the rational class of Example 70.2.2(2). This is why the Tate twist $(-m)$ appears below.

Theorem 70.4.3 (Deligne). Let X be a nonsingular variety, and $\bar{X}$ a nonsingular compactification with $D = \bar{X} \setminus X$ a simple normal crossings divisor; such a compactification exists by Nagata's compactification theorem and Hironaka's resolution of singularities [Hir64].

1. The inclusion $\Omega_{\bar{X}}^\bullet(\log D) \rightarrow j_*\Omega_X^\bullet$ is a quasi-isomorphism, and $j_*\Omega_X^\bullet$ computes $Rj_*\mathbb{C}_X$; hence $\mathbb{H}^k(\bar{X}, \Omega_{\bar{X}}^\bullet(\log D)) \cong H^k(X; \mathbb{C})$.
2. Let $W_m H^k(X; \mathbb{C})$ be the image of $\mathbb{H}^k(\bar{X}, W_{m-k}\Omega_{\bar{X}}^\bullet(\log D))$ and $F^p H^k(X; \mathbb{C})$ the image of $\mathbb{H}^k(\bar{X}, \Omega_{\bar{X}}^{\geq p}(\log D))$. Then W is defined over $\mathbb{Q}$, and $(H^k(X; \mathbb{Q}), W, F)$ is a mixed Hodge structure, independent of the choice of $\bar{X}$ and functorial for morphisms of nonsingular varieties.

3. The weights of $H^k(X; \mathbb{Q})$ lie in $[k, 2k]$, and $W_k H^k(X; \mathbb{Q}) = \text{im} (H^k(\tilde{X}; \mathbb{Q}) \rightarrow H^k(X; \mathbb{Q}))$.
4. There is a spectral sequence, the weight spectral sequence,

$$E_1^{-m, k+m} = H^{k-m}(D^{(m)}; \mathbb{Q})(-m) \implies H^k(X; \mathbb{Q}),$$

whose terms are pure Hodge structures of weight $k + m$, whose differential d_1 is an alternating sum of Gysin maps, and which degenerates at E_2 , with $E_2^{-m, k+m} = \text{Gr}_{k+m}^W H^k(X; \mathbb{Q})$. The Hodge-to-de Rham spectral sequence $E_1^{p,q} = H^q(\tilde{X}, \Omega_{\tilde{X}}^p(\log D)) \implies H^{p+q}(X; \mathbb{C})$ degenerates at E_1 .

This is [Del71, Section 3]; see also [PS08, Chapter 4]. The weight spectral sequence is the spectral sequence of the filtered complex $(\Omega_{\tilde{X}}^\bullet(\log D), W)$, with E_1 computed by Lemma 70.4.2. Once one knows that its differentials are morphisms of Hodge structures, degeneration at E_2 is the principle of weights: for $r \geq 2$, $d_r: E_r^{-m, k+m} \rightarrow E_r^{-m+r, k+m-r+1}$ goes between subquotients of pure Hodge structures of weights $k + m$ and $k + m + 1 - r \neq k + m$, so it vanishes by Lemma 70.3.3(1).

Example 70.4.4. Let $\bar{C}$ be a compact Riemann surface of genus g , $s \geq 1$, and $C = \bar{C} \setminus \{p_1, \dots, p_s\}$. Then $D^{(1)} = \{p_1, \dots, p_s\}$ and $D^{(2)} = \emptyset$, and the weight spectral sequence reduces to the exact sequence of mixed Hodge structures

$$0 \rightarrow H^1(\bar{C}; \mathbb{Q}) \rightarrow H^1(C; \mathbb{Q}) \xrightarrow{\text{Res}} \mathbb{Q}(-1)^s \xrightarrow{\Sigma} \mathbb{Q}(-1) \rightarrow 0,$$

where Res takes the residues at the punctures and Σ , the Gysin map $H^0(D^{(1)})(-1) \rightarrow H^2(\bar{C})$, is the sum. So $W_1 H^1(C) = H^1(\bar{C})$ is pure of weight 1 and $\text{Gr}_2^W H^1(C) \cong \mathbb{Q}(-1)^{s-1}$; exactness at $\mathbb{Q}(-1)^s$ is the residue theorem. For $\mathbb{P}^1$ minus the three points $0, 1, \infty$, $H^1 \cong \mathbb{Q}(-1)^2$ is spanned by the classes of $\frac{1}{2\pi i} \frac{dz}{z}$ and $\frac{1}{2\pi i} \frac{dz}{z-1}$.

Counterexample 70.4.5. The weights of $H^k(X)$ for X nonsingular need not include k or $2k$, and cohomology of even degree can be pure of odd weight. Let $E \subseteq \mathbb{P}^2$ be a smooth cubic curve and $X = \mathbb{P}^2 \setminus E$, an affine surface. The weight spectral sequence ($D^{(2)} = \emptyset$) gives the Gysin sequence

$$H^0(E)(-1) \rightarrow H^2(\mathbb{P}^2) \rightarrow H^2(X) \rightarrow H^1(E)(-1) \rightarrow H^3(\mathbb{P}^2) = 0.$$

The first map sends 1 to $\text{cl}(E) = 3h \neq 0$, so it is an isomorphism over $\mathbb{Q}$, and $H^2(X; \mathbb{Q}) \cong H^1(E; \mathbb{Q})(-1)$ is pure of weight 3.

70.5 Singular varieties

Theorem 70.5.1 (Deligne). *For every variety X and every k , $H^k(X; \mathbb{Q})$ and $H_c^k(X; \mathbb{Q})$ carry functorial mixed Hodge structures, extending those of Theorem 70.4.3. The long exact sequence of a pair (X, Z) with Z closed, and the sequence $\dots \rightarrow H_c^k(X \setminus Z) \rightarrow H_c^k(X) \rightarrow H_c^k(Z) \rightarrow \dots$, are exact sequences of mixed Hodge structures, and cup products and Künneth isomorphisms are morphisms of mixed Hodge structures. The weights of $H^k(X)$ lie in $[0, 2k]$; if X is proper they lie in $[0, k]$, and if X is nonsingular in $[k, 2k]$. If X is proper and $\pi: \tilde{X} \rightarrow X$ is a resolution of singularities, then $W_{k-1} H^k(X; \mathbb{Q}) = \ker(\pi^*: H^k(X; \mathbb{Q}) \rightarrow H^k(\tilde{X}; \mathbb{Q}))$.*

This is [Del74b], via simplicial resolutions of X by nonsingular varieties; see also [PS08, Chapter 5]. For proper X with a resolution $\pi: \tilde{X} \rightarrow X$ that is an isomorphism outside a

closed subset $Z \subseteq X$ with preimage $E = \pi^{-1}(Z)$, there is an exact sequence of mixed Hodge structures

$$\cdots \rightarrow H^{k-1}(E) \rightarrow H^k(X) \rightarrow H^k(\tilde{X}) \oplus H^k(Z) \rightarrow H^k(E) \rightarrow H^{k+1}(X) \rightarrow \cdots,$$

the Mayer–Vietoris sequence of the proper modification, which computes the mixed Hodge structure of X inductively on the dimension.

Example 70.5.2. Let C be the nodal cubic, $\pi: \mathbb{P}^1 \rightarrow C$ its normalisation, Z the node and $E = \pi^{-1}(Z) = \{0, \infty\}$. The sequence above reads

$$0 \rightarrow H^0(C) \rightarrow H^0(\mathbb{P}^1) \oplus H^0(Z) \rightarrow H^0(E) \rightarrow H^1(C) \rightarrow H^1(\mathbb{P}^1) \oplus H^1(Z) = 0.$$

The map $\mathbb{Q}^2 \rightarrow \mathbb{Q}^2$, $(a, b) \mapsto (a - b, a - b)$, has a one-dimensional cokernel, so $H^1(C) \cong \mathbb{Q}(0)$, a quotient of the pure structure $H^0(E)$ of weight 0. Accordingly $W_0 H^1(C) = \ker(H^1(C) \rightarrow H^1(\mathbb{P}^1)) = H^1(C)$, as Theorem 70.5.1 predicts, and $H^1(C)$ has no Hodge decomposition of weight 1.

70.6 Applications of weights

Corollary 70.6.1. Let $\tilde{X}$ be a nonsingular projective variety of dimension n , $U \subseteq \tilde{X}$ a nonempty open subset, and $Z = \tilde{X} \setminus U$ with irreducible components $Z_1, \dots, Z_N$ of codimensions $c_1, \dots, c_N$. Let $g_i: \tilde{Z}_i \rightarrow \tilde{X}$ be resolutions of singularities of the Z_i composed with the inclusions.

1. The image of the restriction $H^k(\tilde{X}; \mathbb{Q}) \rightarrow H^k(U; \mathbb{Q})$ is $W_k H^k(U; \mathbb{Q})$.
2. The kernel of the restriction is $\sum_i g_{i*} H^{k-2c_i}(\tilde{Z}_i; \mathbb{Q})(-c_i)$, the sum of the images of the Gysin maps.

Sketch of proof. (1) By resolution of singularities there is a proper birational morphism $\pi: \tilde{X} \rightarrow \tilde{X}$ from a nonsingular projective variety that is an isomorphism over U and such that $\tilde{X} \setminus \pi^{-1}(U)$ is a simple normal crossings divisor. By Theorem 70.4.3(3), the image of $H^k(\tilde{X}) \rightarrow H^k(U)$ is $W_k H^k(U)$. The image of $H^k(\tilde{X})$ is contained in it, because restriction from $\tilde{X}$ factors through π^* . Conversely, for $\alpha \in H^k(\tilde{X})$ the class $\pi_* \alpha \in H^k(\tilde{X})$, defined by Poincaré duality, restricts on U to $\alpha|_U$, since π is an isomorphism over U . So the two images agree.

(2) The Gysin images restrict to zero on U , since the classes are supported on Z . Conversely, the kernel is the image of $H_Z^k(\tilde{X}; \mathbb{Q}) \cong H_{2n-k}^{BM}(Z; \mathbb{Q})(-n)$ (Poincaré–Alexander duality), by a morphism of mixed Hodge structures into the pure structure $H^k(\tilde{X})$ of weight k . By Lemma 70.3.3(2) the kernel is the image of the part of weight $\leq k$, and Deligne shows that the lowest weight part of the Borel–Moore homology of Z is the image of the homology of the resolutions $\tilde{Z}_i$ [Del74b]. This last identification is the verification we omit. $\square$

Theorem 70.6.2 (Global invariant cycle theorem). Let $f: \mathcal{X} \rightarrow S$ be a smooth projective morphism of nonsingular varieties with S connected, $\tilde{\mathcal{X}}$ a nonsingular projective compactification of $\mathcal{X}$, and $t \in S$ with fibre X_t . Then

$$\mathrm{im}(H^k(\tilde{\mathcal{X}}; \mathbb{Q}) \rightarrow H^k(X_t; \mathbb{Q})) = \mathrm{im}(H^k(\mathcal{X}; \mathbb{Q}) \rightarrow H^k(X_t; \mathbb{Q})) = H^k(X_t; \mathbb{Q})^{\pi_1(S, t)},$$

and this subspace is a sub-Hodge structure of $H^k(X_t; \mathbb{Q})$.

Proof. The Leray spectral sequence $E_2^{a,b} = H^a(S, R^b f_* \mathbb{Q}) \Rightarrow H^{a+b}(\mathcal{X}; \mathbb{Q})$ of a smooth projective morphism degenerates at E_2 [Del68]. Its edge map $H^k(\mathcal{X}) \rightarrow E_\infty^{0,k} \subseteq E_2^{0,k} = H^0(S, R^k f_* \mathbb{Q}) \subseteq H^k(X_t)$ is the restriction to the fibre, and by degeneration $E_\infty^{0,k} = E_2^{0,k}$, so the restriction

is surjective onto $H^0(S, R^k f_* \mathbb{Q})$. Since $R^k f_* \mathbb{Q}$ is a local system (Theorem 66.2.1) and S is connected, $H^0(S, R^k f_* \mathbb{Q})$ is the space of $\pi_1(S, t)$ -invariants in the stalk $H^k(X_t; \mathbb{Q})$. This proves the second equality.

The restriction $r: H^k(\mathcal{X}; \mathbb{Q}) \rightarrow H^k(X_t; \mathbb{Q})$ is induced by the inclusion of nonsingular varieties $X_t \rightarrow \mathcal{X}$, so it is a morphism of mixed Hodge structures (Theorem 70.4.3(2)), and its target is pure of weight k because X_t is nonsingular and projective. By Lemma 70.3.3(2), $r(H^k(\mathcal{X})) = r(W_k H^k(\mathcal{X}))$, and $W_k H^k(\mathcal{X})$ is the image of $H^k(\bar{X})$ by Corollary 70.6.1(1). Since restriction from $\bar{X}$ to X_t factors through $\mathcal{X}$, this proves the first equality. Finally, the image of $H^k(\bar{X}) \rightarrow H^k(X_t)$ is the image of a morphism of pure Hodge structures of weight k , hence a sub-Hodge structure (Theorem 62.3.3). $\square$

Remark 70.6.3. Theorem 70.6.2 is the proof of Theorem 66.4.1 promised in Chapter 66. The only input not proved in this book is the degeneration of the Leray spectral sequence, which Deligne deduces from the hard Lefschetz theorem (Chapter 46) applied to the fibres [Del68]. The coniveau filtration is handled in the same way: $N^c H^k(X)$ is a sum of kernels of restrictions $H^k(X) \rightarrow H^k(X \setminus Z)$, which by Corollary 70.6.1(2) are images of Gysin maps and so sub-Hodge structures (Proposition 64.4.5).

Proposition 70.6.4. *Keep the notation of Corollary 70.6.1, and let $\alpha \in \text{Hdg}^p(\bar{X})$. Suppose that*

1. *there is an algebraic cycle V on $\bar{X}$ with $\mathbb{Q}$ -coefficients such that $\alpha|_U = \text{cl}(V)|_U$ in $H^{2p}(U; \mathbb{Q})$, and*
2. *for each i with $c_i \leq p$, the Hodge conjecture holds for $\tilde{Z}_i$ in codimension $p - c_i$.*

Then α is algebraic.

Proof. Put $\beta = \alpha - \text{cl}(V)$. It is a Hodge class, by Corollary 59.4.2, and $\beta|_U = 0$. By Corollary 70.6.1(2), β lies in the image of

$$g_* = \sum_i g_{i*}: A = \bigoplus_i H^{2p-2c_i}(\tilde{Z}_i; \mathbb{Q})(-c_i) \longrightarrow H^{2p}(\bar{X}; \mathbb{Q}),$$

where the summands with $c_i > p$ are zero. Each Gysin map g_{i*} is a morphism of Hodge structures of weight $2p$: it is Poincaré dual to pushforward in homology, and it maps forms of type (a, b) to currents of type $(a + c_i, b + c_i)$. The source A is a direct sum of polarisable Hodge structures of weight $2p$ (Theorem 62.5.2). By Theorem 62.5.3, $\ker g_*$ has a complement A' in A that is a sub-Hodge structure, and $g_{*|A'}: A' \rightarrow g_*(A)$ is an isomorphism of Hodge structures onto the sub-Hodge structure $g_*(A)$. The preimage $\gamma = (\gamma_i) \in A'$ of β is then a Hodge class, since isomorphisms of Hodge structures preserve Hodge classes (Proposition 62.6.2); that is, $\gamma_i \in \text{Hdg}^{p-c_i}(\tilde{Z}_i)$. By hypothesis (2), $\gamma_i = \text{cl}(V_i)$ for cycles V_i on $\tilde{Z}_i$ with $\mathbb{Q}$ -coefficients. By compatibility of the cycle class with proper pushforward (Theorem 59.3.1(2)), $\beta = \sum_i g_{i*} \text{cl}(V_i) = \text{cl}(\sum_i g_{i*} V_i)$. Hence $\alpha = \text{cl}(V + \sum_i g_{i*} V_i)$ is algebraic. $\square$

Corollary 70.6.5. *Assume that the Hodge conjecture holds for all nonsingular projective varieties of dimension $< n$, and let $\bar{X}$ be an irreducible nonsingular projective variety of dimension n and $U \subseteq \bar{X}$ a nonempty open subset. Then a Hodge class $\alpha \in \text{Hdg}^p(\bar{X})$ is algebraic if and only if $\alpha|_U$ is the restriction of the class of an algebraic cycle on $\bar{X}$.*

Proof. One direction is trivial. For the other, since $\bar{X}$ is irreducible and U is nonempty, every component of $\bar{X} \setminus U$ has codimension $c_i \geq 1$, so $\dim \tilde{Z}_i < n$ and hypothesis (2) of Proposition 70.6.4 holds. $\square$

Corollary 70.6.6. *Assume that the Hodge conjecture holds for all nonsingular projective varieties of dimension $< n$, and let $\bar{X}$ and $\bar{X}'$ be birational irreducible nonsingular projective varieties of dimension n . Then the Hodge conjecture holds for $\bar{X}$ if and only if it holds for $\bar{X}'$.*

Proof. Choose nonempty open subsets $U \subseteq \bar{X}$ and $U' \subseteq \bar{X}'$ and an isomorphism $\phi: U' \rightarrow U$; we identify U and U' through ϕ . Suppose the Hodge conjecture holds for $\bar{X}'$, and let $\alpha \in \text{Hdg}^p(\bar{X})$. The restriction $\alpha|_U$ lies in $W_{2p}H^{2p}(U; \mathbb{Q})$ by Corollary 70.6.1(1), which is a pure Hodge structure of weight $2p$, since the weights of $H^{2p}(U)$ are $\geq 2p$ (Theorem 70.4.3(3)). As the restriction is a morphism of mixed Hodge structures, $\alpha|_U$ is rational and lies in F^p , so it is a Hodge class of this pure structure (Proposition 62.2.2: in weight $2p$, $F^p \cap \bar{F}^p$ is the (p, p) -part). The restriction $r': H^{2p}(\bar{X}'; \mathbb{Q}) \rightarrow W_{2p}H^{2p}(U; \mathbb{Q})$ is a surjective morphism of pure Hodge structures of weight $2p$, by Corollary 70.6.1(1) applied to $\bar{X}'$. As in the proof of Proposition 70.6.4, semisimplicity (Theorem 62.5.3) gives a Hodge class $\alpha' \in \text{Hdg}^p(\bar{X}')$ with $r'(\alpha') = \alpha|_U$. By hypothesis $\alpha' = \text{cl}(V')$ for a cycle V' on $\bar{X}'$ with $\mathbb{Q}$ -coefficients. Let V be the closure in $\bar{X}$ of the cycle $V'|_U$. Since the cycle class commutes with restriction to open subsets (Theorem 59.3.1(2)), $\text{cl}(V)|_U = \text{cl}(V'|_U) = \alpha'|_U = \alpha|_U$. By Corollary 70.6.5, α is algebraic. The converse follows by symmetry. $\square$

Remark 70.6.7. The arguments of Proposition 70.6.4 and Corollary 70.6.6 use semisimplicity to lift Hodge classes along surjections, and this is a statement about $\mathbb{Q}$ -structures. Over $\mathbb{Z}$ an integral Hodge class in the image of a morphism need not be the image of an integral Hodge class, and the integral analogues must be proved by other means: for classes of degree 4, the birational invariance of the defect $Z^4(X)$ is a theorem of Colliot-Thélène and Voisin (Theorem 69.2.8).

70.7 Exercises

Exercise 70.7.1. Compute the mixed Hodge structure on $H^\bullet((\mathbb{C}^*)^n; \mathbb{Q})$.

Solution. By the Künneth formula, which is compatible with mixed Hodge structures (Theorem 70.5.1), and Example 70.2.2(2), $H^k((\mathbb{C}^*)^n) \cong \Lambda^k(\mathbb{Q}(-1)^n) \cong \mathbb{Q}(-k)^{\binom{n}{k}}$, pure of weight $2k$ and type (k, k) , spanned by the products of the classes $\frac{1}{2\pi i} \frac{dz_j}{z_j}$. The weights are the largest allowed by Theorem 70.4.3(3). $\square$

Exercise 70.7.2. Show that the Kummer structure K_q of Example 70.2.2(4) is a mixed Hodge structure with $\text{Gr}_{-2}^W K_q \cong \mathbb{Q}(1)$ and $\text{Gr}_0^W K_q \cong \mathbb{Q}(0)$, and that $K_q \cong \mathbb{Q}(1) \oplus \mathbb{Q}(0)$ if and only if q is a root of unity.

Solution. The only nonzero graded pieces are $\text{Gr}_{-2}^W = \mathbb{Q}e_1$ and $\text{Gr}_0^W = H/\mathbb{Q}e_1$. On Gr_{-2}^W the induced filtration is $F^{-1} = \mathbb{C}e_1$ and $F^0 = F^0 \cap \mathbb{C}e_1 = 0$, since $e_0 - \lambda e_1 \notin \mathbb{C}e_1$; this is the Hodge filtration of type $(-1, -1)$, that of $\mathbb{Q}(1)$ (identify e_1 with $2\pi i$). On Gr_0^W the image of F^0 is everything and $F^1 = 0$, which is type $(0, 0)$. So K_q is a mixed Hodge structure. An isomorphism $K_q \cong \mathbb{Q}(1) \oplus \mathbb{Q}(0)$ maps $W_{-2}K_q$ onto $\mathbb{Q}(1)$, so the preimage of $\mathbb{Q}(0)$ is a complement of W_{-2} that is a sub-mixed Hodge structure; by Counterexample 70.2.3 one exists if and only if $\lambda = \log q/2\pi i \in \mathbb{Q}$. Writing $\lambda = a/b$ with $b \geq 1$, this says $q^b = e^{2\pi i a} = 1$; conversely if $q^b = 1$ then $\log q \in 2\pi i \frac{1}{b}\mathbb{Z}$. $\square$

Exercise 70.7.3. Let X be a nonsingular variety. Show that $h^{p,q}(H^k(X)) = 0$ unless $0 \leq p, q \leq k$ and $k \leq p + q \leq 2k$.

Solution. The condition on $p + q$ is Theorem 70.4.3(3). The complex $\Omega_{\bar{X}}^{\geq k+1}(\log D)$ is zero in degrees $\leq k$, so its hypercohomology vanishes in degree k , and $F^{k+1}H^k(X; \mathbb{C}) = 0$. Hence F^{k+1} vanishes on every Gr_m^W and $h^{p,q} = 0$ for $p > k$; by the symmetry $h^{p,q} = h^{q,p}$ of pure Hodge structures, also for $q > k$. Finally $F^0 = H_{\mathbb{C}}$ because $\Omega^{\geq 0}(\log D)$ is the whole complex, so $h^{p,q} = 0$ for $p < 0$, and for $q < 0$ by symmetry. $\square$

Exercise 70.7.4. Let E be an elliptic curve with origin 0 and $a \in E, a \neq 0$. Show that $H^1(E \setminus \{0\})$ is pure of weight 1, that $H^1(E \setminus \{0, a\})$ has $\mathrm{Gr}_1^W \cong H^1(E)$ and $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)$, and compute the Hodge numbers of both.

Solution. By Example 70.4.4 with $g = 1$: for $s = 1$, $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)^0 = 0$, so $H^1(E \setminus \{0\}) = H^1(E)$, with $h^{1,0} = h^{0,1} = 1$. For $s = 2$, $\mathrm{Gr}_1^W = H^1(E)$ and $\mathrm{Gr}_2^W \cong \mathbb{Q}(-1)$, so $h^{1,0} = h^{0,1} = h^{1,1} = 1$ and $\dim H^1 = 3$. (Whether this extension splits depends on a : it splits if and only if a is a torsion point of E , by Carlson's description of extensions [Car80]; we do not prove this.) $\square$

Chapter 71

Absolute Hodge Classes

71.1 Introduction

The cohomology of a nonsingular projective variety over $\mathbb{C}$ comes in several flavours: Betti cohomology with its rational structure, algebraic de Rham cohomology with its Hodge filtration, and ℓ -adic cohomology with, for varieties defined over smaller fields, a Galois action. The Hodge conjecture concerns classes that are special in the first two at once. An automorphism σ of the field $\mathbb{C}$, in general wildly discontinuous, transforms a variety X into a *conjugate variety* X^σ and acts on de Rham and ℓ -adic cohomology, but not on Betti cohomology. Classes of algebraic cycles are carried to classes of algebraic cycles, so they remain rational and of type (p, p) after conjugation. Deligne called a Hodge class *absolute* if it shares this property. Absolute Hodge classes sit between algebraic classes and Hodge classes, and, unlike algebraic classes, one can prove things about them: Deligne's theorem that all Hodge classes on abelian varieties are absolute (Theorem 71.7.1) is the strongest general result in the direction of the Hodge conjecture. References: [Del82], [CS14], [Voi07], and the source *The Hodge Conjecture* (Chapter 8).

Prerequisites. Chapter 64 (The Hodge Conjecture), Chapter 61 (Étale Cohomology), Chapter 54 (Analytic Spaces and GAGA), Chapter 63 (Mumford–Tate Groups), Chapter 68 (Known Cases and Evidence) and Chapter 70 (Mixed Hodge Structures).

Conventions. In this chapter X is a nonsingular projective variety over $\mathbb{C}$ of dimension n , and ℓ runs over all primes. We write $\mathbb{Q}(p) = (2\pi i)^p \mathbb{Q} \subseteq \mathbb{C}$ and $H_B^k(X)(p) = H^k(X^{\text{an}}; \mathbb{Q}) \otimes \mathbb{Q}(p) \subseteq H^k(X^{\text{an}}; \mathbb{C})$ for twisted Betti cohomology, $H_{\text{dR}}^k(X) = H_{\text{dR}}^k(X/\mathbb{C})$ (Definition 54.5.1), and $H_\ell^k(X)(p) = H_{\text{ét}}^k(X, \mathbb{Q}_\ell(p))$ (Definition 61.4.1; here X is already over an algebraically closed field).

71.2 Conjugate varieties and comparison isomorphisms

Definition 71.2.1. Let $\sigma \in \text{Aut}(\mathbb{C})$. The *conjugate variety* is $X^\sigma = X \times_{\text{Spec } \mathbb{C}, \sigma} \text{Spec } \mathbb{C}$, the base change of X along the morphism $\text{Spec } \mathbb{C} \rightarrow \text{Spec } \mathbb{C}$ induced by σ . The projection $\sigma_X: X^\sigma \rightarrow X$ is an isomorphism of schemes, but not of $\mathbb{C}$ -schemes. If $X \subseteq \mathbb{P}^N$ is cut out by homogeneous polynomials f_j , then X^σ is cut out by the polynomials f_j^σ obtained by applying σ to the coefficients, and on closed points σ_X^{-1} is $(x_0 : \cdots : x_N) \mapsto (\sigma(x_0) : \cdots : \sigma(x_N))$.

Example 71.2.2. Let E be the elliptic curve $y^2 = x^3 + ax + b$. Then E^σ is $y^2 = x^3 + \sigma(a)x + \sigma(b)$, so $j(E^\sigma) = \sigma(j(E))$, and $E^\sigma \cong E$ if and only if σ fixes $j(E)$. If $j(E)$ is transcendental, then $\sigma(j(E))$ can be any transcendental number, and E has uncountably many pairwise non-isomorphic conjugates. If E is defined over $\mathbb{Q}$, then $E^\sigma = E$ for every σ , but σ_E is a discontinuous bijection of $E(\mathbb{C})$ for $\sigma \neq \text{id}$, complex conjugation.

Definition 71.2.3. For $\sigma \in \text{Aut}(\mathbb{C})$ define

1. $\sigma_*: H_{\text{dR}}^k(X) \rightarrow H_{\text{dR}}^k(X^\sigma)$, the σ -linear isomorphism induced by $\sigma_X^* \Omega_{X/\mathbb{C}}^\bullet = \Omega_{X^\sigma/\mathbb{C}}^\bullet$ (Corollary 54.5.4); it maps $F^p H_{\text{dR}}^k(X)$ onto $F^p H_{\text{dR}}^k(X^\sigma)$;
2. $\sigma_* = \sigma_X^*: H_\ell^k(X)(p) \rightarrow H_\ell^k(X^\sigma)(p)$, the pullback in étale cohomology along the isomorphism of schemes σ_X , which identifies the sheaves μ_{ℓ^m} ; it is $\mathbb{Q}_\ell$ -linear.

Remark 71.2.4. There is no map $H_B^k(X) \rightarrow H_B^k(X^\sigma)$ for general σ : σ_X is not continuous for the classical topology unless σ is the identity or complex conjugation, and X^{an} and $(X^\sigma)^{\text{an}}$ need not even be homeomorphic (Remark 54.5.5). The *comparison isomorphisms*

$$c_{\text{dR}}: H_B^k(X)(p) \otimes_{\mathbb{Q}} \mathbb{C} \xrightarrow{\sim} H_{\text{dR}}^k(X), \quad c_\ell: H_B^k(X)(p) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \xrightarrow{\sim} H_\ell^k(X)(p)$$

of Theorem 54.5.2 and Theorem 61.4.3 (the second uses $\mathbb{Z}(1) \otimes \mathbb{Z}/\ell^m \cong \mu_{\ell^m}$, $2\pi i/\ell^m \mapsto e^{2\pi i/\ell^m}$) give two further structures on $H_{\text{dR}}^k(X)$ and $H_\ell^k(X)$: a $\mathbb{Q}$ -structure and the complex conjugation it defines. The map σ_* preserves the algebraic structure (the Hodge filtration, cup products, algebraic cycle classes) but has no reason to preserve these transcendental ones. In particular it preserves F^p but not $\overline{F^q}$, so it need not map $H^{p,q}(X)$ to $H^{p,q}(X^\sigma)$.

Proposition 71.2.5. *The Betti numbers, the Hodge numbers, the dimension of $\text{Hdg}^1(X)$ and the Picard number of X and X^σ agree.*

Proof. Betti and Hodge numbers: Corollary 54.5.4. For divisors, σ_X^* gives a bijection between divisors on X and divisors on X^σ that preserves linear and algebraic equivalence, since these are defined algebraically; so the Néron–Severi groups are isomorphic and the Picard numbers agree. By the Lefschetz (1, 1)-theorem (Theorem 65.3.1), $\dim \text{Hdg}^1(X)$ is the Picard number. $\square$

71.3 Absolute Hodge classes

Definition 71.3.1 (Deligne). Write $H_{\mathbb{A}}^{2p}(X)(p) = H_{\text{dR}}^{2p}(X) \times \prod_\ell H_\ell^{2p}(X)(p)$ and $t = (t_{\text{dR}}, (t_\ell)_\ell)$ for its elements.

1. t is a *Hodge cycle* on X if there is $t_B \in H_B^{2p}(X)(p)$ with $c_{\text{dR}}(t_B) = t_{\text{dR}}$ and $c_\ell(t_B) = t_\ell$ for all ℓ , and $t_{\text{dR}} \in F^p H_{\text{dR}}^{2p}(X)$.
2. t is an *absolute Hodge cycle* if $\sigma_* t = (\sigma_* t_{\text{dR}}, (\sigma_* t_\ell)_\ell)$ is a Hodge cycle on X^σ for every $\sigma \in \text{Aut}(\mathbb{C})$.

The absolute Hodge cycles form a $\mathbb{Q}$ -vector space $C_{\text{AH}}^p(X)$: σ_* is additive and fixes rational scalars.

Lemma 71.3.2. *The map $t \mapsto (2\pi i)^{-p} t_B$ is an isomorphism from the $\mathbb{Q}$ -vector space of Hodge cycles on X onto $\text{Hdg}^p(X)$. Its inverse sends α to $(c_{\text{dR}}((2\pi i)^p \alpha), (c_\ell((2\pi i)^p \alpha))_\ell)$.*

Proof. t_B is determined by t , because c_ℓ is injective. Put $\alpha = (2\pi i)^{-p} t_B \in H^{2p}(X^{\text{an}}; \mathbb{Q})$. Since F^p is a complex subspace, $t_{\text{dR}} \in F^p$ if and only if $\alpha \in F^p$. A rational class is fixed by complex conjugation, so $\alpha \in F^p$ if and only if $\alpha \in F^p \cap \overline{F^p} = H^{p,p}$ (Proposition 62.2.2), that is, $\alpha \in \text{Hdg}^p(X)$. The inverse map is clear. $\square$

We write $\text{Hdg}_{\text{AH}}^p(X) \subseteq \text{Hdg}^p(X)$ for the image of $C_{\text{AH}}^p(X)$, and call its elements *absolute Hodge classes*.

Proposition 71.3.3. *Classes of algebraic cycles are absolute Hodge: $Z_{\text{alg}}^p(X) \subseteq \text{Hdg}_{\text{AH}}^p(X) \subseteq \text{Hdg}^p(X)$.*

Proof. Let Z be a cycle of codimension p with $\mathbb{Q}$ -coefficients. There are cycle classes $\text{cl}_{\text{dR}}(Z) \in F^p H_{\text{dR}}^{2p}(X)$ and $\text{cl}_{\ell}(Z) \in H_{\ell}^{2p}(X)(p)$ (Theorem 61.4.2(4)), both defined algebraically, and with suitable sign conventions they are the images of $(2\pi i)^p \text{cl}(Z)$ under c_{dR} and c_{ℓ} : for ℓ -adic cohomology this is Theorem 61.4.3, and for de Rham cohomology it is [Del82]. So $\text{cl}_{\mathbb{A}}(Z) = (\text{cl}_{\text{dR}}(Z), (\text{cl}_{\ell}(Z))_{\ell})$ is a Hodge cycle. Since the cycle classes are defined algebraically, they are compatible with transport of structure along the isomorphism of schemes σ_X : $\sigma_* \text{cl}_{\mathbb{A}}(Z) = \text{cl}_{\mathbb{A}}(Z^{\sigma})$, where $Z^{\sigma} = \sigma_X^{-1}(Z)$ is a cycle on X^{σ} . By the same argument on X^{σ} , this is a Hodge cycle. The second inclusion is the case $\sigma = \text{id}$. $\square$

Remark 71.3.4. The factor $(2\pi i)^p$ is forced: already for a point on $\mathbb{P}^1$, the de Rham class is $\pm 2\pi i$ times the Betti class (Exercise 71.9.2). This is why the rational structure relevant for cycle classes of codimension p is $H_B^{2p}(X)(p)$ and not $H^{2p}(X^{\text{an}}; \mathbb{Q})$.

Conjecture 71.3.5 (Absolute Hodge conjecture). *Every Hodge class is absolute Hodge: $\text{Hdg}_{\text{AH}}^p(X) = \text{Hdg}^p(X)$.*

By Proposition 71.3.3, the Hodge conjecture for X and X^{σ} in all codimensions implies Conjecture 71.3.5. The converse is not expected to be provable directly: being absolute Hodge is a condition on cohomology and its comparison isomorphisms, and it never produces a subvariety.

Remark 71.3.6. Several variants are in use. Omitting the ℓ -adic components gives *de Rham–Betti absolute* classes, which is the version in the source *The Hodge Conjecture*; Deligne’s theorem below holds for the full definition, and all known results hold for both. André’s *motivated cycles* [And96b] are a smaller class, generated by algebraic cycles and the inverse Lefschetz operators (Corollary 71.4.3); they are absolute Hodge.

71.4 Operations on absolute Hodge classes

Proposition 71.4.1. *Let $f: X \rightarrow Y$ be a morphism of nonsingular projective varieties, of relative dimension $d = \dim X - \dim Y$. Then absolute Hodge classes are preserved by*

1. *pullback $f^*: \text{Hdg}^p(Y) \rightarrow \text{Hdg}^p(X)$ and pushforward $f_*: \text{Hdg}^p(X) \rightarrow \text{Hdg}^{p-d}(Y)$;*
2. *cup product, so $\bigoplus_p \text{Hdg}_{\text{AH}}^p(X)$ is a $\mathbb{Q}$ -subalgebra of $\bigoplus_p \text{Hdg}^p(X)$;*
3. *the Künneth projections $H^{2p}(X \times Y) \rightarrow H^i(X) \otimes H^{2p-i}(Y)$: the Künneth components of an absolute Hodge class on $X \times Y$ are absolute Hodge;*
4. *correspondences: if $\gamma \in \text{Hdg}_{\text{AH}}^q(X \times Y)$ then $\gamma_* = \text{pr}_{2*}(\text{pr}_1^*(-) \cup \gamma)$ maps absolute Hodge classes on X to absolute Hodge classes on Y .*

Proof. Each operation is defined in Betti, de Rham and ℓ -adic cohomology. In de Rham and ℓ -adic cohomology it is defined algebraically, from pullbacks, cup products and trace maps, so it commutes with σ_* : for instance $\sigma_*(f^*t) = (f^{\sigma})^*(\sigma_*t)$, where $f^{\sigma}: X^{\sigma} \rightarrow Y^{\sigma}$ is the conjugate morphism. The operations are compatible with the comparison isomorphisms c_{dR} and c_{ℓ} (with the Tate twists indicated: f_* goes from $H^{2p}(X)(p)$ to $H^{2p-2d}(Y)(p-d)$), and on Betti cohomology they are morphisms of Hodge structures, so they map Hodge cycles to Hodge cycles [Del82]. Given an absolute Hodge cycle t on Y and $\sigma \in \text{Aut}(\mathbb{C})$, the cycle $\sigma_*(f^*t) = (f^{\sigma})^*(\sigma_*t)$ is the image of the Hodge cycle σ_*t on Y^{σ} under an operation that

preserves Hodge cycles, hence a Hodge cycle. The other cases are identical: for (3), the Künneth decomposition is built from pr_1^* , pr_2^* and cup product, so the projections commute with σ_* and preserve Hodge cycles, because the Künneth decomposition of $H^{2p}(X \times Y)$ is a decomposition into sub-Hodge structures; and (4) is a composite of (1) and (2). $\square$

Corollary 71.4.2. *The Künneth components $\pi_i \in H^{2n-i}(X) \otimes H^i(X) \subseteq H^{2n}(X \times X)$ of the class of the diagonal are absolute Hodge classes.*

Proof. The class of the diagonal is algebraic, hence absolute Hodge by Proposition 71.3.3; apply Proposition 71.4.1(3). $\square$

Corollary 71.4.3. *Let $h \in H^2(X)$ be the class of an ample divisor and Λ the inverse Lefschetz operator of $L = h \cup -$ (Proposition 46.2.4). The class $\lambda \in H^{2n-2}(X \times X)(n-1)$ of Λ , obtained through the Künneth decomposition and Poincaré duality, is an absolute Hodge class.*

Proof. Let $\sigma \in \text{Aut}(\mathbb{C})$ and let h^σ be the class of the conjugate divisor on X^σ , with operators L^σ and Λ^σ . In de Rham cohomology, σ_* intertwines L with L^σ (it preserves cup products and cycle classes) and preserves degrees. Hence $\sigma_* \Lambda \sigma_*^{-1}$ satisfies the equations $[L^\sigma, f] = h_{\text{deg}}^\sigma$, $[h_{\text{deg}}^\sigma, f] = -2f$, where h_{deg}^σ acts by $k - n$ in degree k , and by the uniqueness in Proposition 46.2.4 it equals $\Lambda_{\text{dR}}^\sigma$. The same argument applies in ℓ -adic cohomology, using hard Lefschetz there (Theorem 61.4.2(6)). On the other hand, the Betti operator Λ_B^σ is defined over $\mathbb{Q}$ (Proposition 46.2.4), and its images under c_{dR} and c_ℓ satisfy the defining equations, so they are $\Lambda_{\text{dR}}^\sigma$ and Λ_ℓ^σ by uniqueness again. Finally, Λ_B^σ is a morphism of Hodge structures of type $(-1, -1)$. Indeed, let $h(z)$, $z \in \mathbb{S}(\mathbb{R}) = \mathbb{C}^*$, be the action of the Deligne torus on $H^\bullet(X^\sigma; \mathbb{R})$ (Proposition 62.4.1). Since L^σ maps $H^{p,q}$ to $H^{p+1,q+1}$, $h(z)L^\sigma h(z)^{-1} = |z|^{-2}L^\sigma$, and $h(z)$ commutes with the grading operator. So $|z|^{-2}h(z)\Lambda_B^\sigma h(z)^{-1}$ solves the defining equations for L^σ , and by uniqueness $h(z)\Lambda_B^\sigma h(z)^{-1} = |z|^2\Lambda_B^\sigma$ for all z , which says exactly that Λ_B^σ maps $H^{p,q}$ to $H^{p-1,q-1}$. So the class of Λ^σ is a Hodge class on $X^\sigma \times X^\sigma$ (Proposition 62.6.2), and $\sigma_*\lambda$ is a Hodge cycle. $\square$

Remark 71.4.4. Grothendieck's Lefschetz standard conjecture asks for λ to be algebraic (Chapter 73). Corollary 71.4.3 proves the weaker statement that λ is absolute Hodge, and the Hodge conjecture would bridge the gap. This is typical: statements whose algebraic form is open are often theorems in the absolute Hodge setting.

71.5 Galois action and fields of definition

Every X is defined over a subfield of $\mathbb{C}$ finitely generated over $\mathbb{Q}$, since it is cut out by finitely many polynomials: $X = X_0 \times_k \mathbb{C}$ with $k \subseteq \mathbb{C}$ finitely generated over $\mathbb{Q}$ and X_0 a nonsingular projective k -variety. Let $\bar{k}$ be the algebraic closure of k in $\mathbb{C}$ and $G_k = \text{Gal}(\bar{k}/k)$. For $\tau \in \text{Aut}(\mathbb{C}/k)$ we have $X^\tau = X$ canonically, so τ_* acts on $H_{\mathbb{A}}^{2p}(X)(p)$. On $H_\ell^{2p}(X)(p) \cong H_{\text{et}}^{2p}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$, where the isomorphism is invariance of étale cohomology of proper varieties under extension of algebraically closed fields [Mil80, Chapter VI], the action of τ is the Galois action of $\tau|_{\bar{k}} \in G_k$.

Proposition 71.5.1. *In this situation, $\text{Aut}(\mathbb{C}/k)$ maps $C_{\text{AH}}^p(X)$ to itself, and it acts on it through a finite quotient of G_k . Consequently there is a finite extension $k' \subseteq \bar{k}$ of k such that for every absolute Hodge cycle t , the ℓ -adic components $t_\ell \in H_{\text{et}}^{2p}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$ are fixed by $G_{k'}$.*

Proof. If t is absolute Hodge and $\tau \in \text{Aut}(\mathbb{C}/k)$, then τ_*t is a Hodge cycle on $X^\tau = X$, and $\sigma_*(\tau_*t) = (\sigma\tau)_*t$ is a Hodge cycle on $X^{\sigma\tau}$ for every σ ; so τ_*t is absolute Hodge. The space

$V = C_{\text{AH}}^p(X)$ is finite-dimensional over $\mathbb{Q}$, since $t \mapsto t_B$ embeds it in $H_B^{2p}(X)(p)$. Fix a prime ℓ . The map $V \otimes \mathbb{Q}_\ell \rightarrow H_\ell^{2p}(X)(p)$, $t \otimes a \mapsto at_\ell$, is injective, because it is the restriction of the comparison isomorphism c_ℓ to $V_B \otimes \mathbb{Q}_\ell$, where $V_B = \{t_B : t \in V\}$. It is equivariant, and the action of τ on the right depends only on $\tau|_{\bar{k}}$. So the action on V is given by a homomorphism $\rho: G_k \rightarrow \text{GL}(V)$, and ρ is continuous for the ℓ -adic topology on $\text{GL}(V \otimes \mathbb{Q}_\ell)$, because the Galois action on ℓ -adic cohomology is continuous. The image $\rho(G_k)$ is therefore a compact subgroup of $\text{GL}(V \otimes \mathbb{Q}_\ell)$, and it is contained in the countable group $\text{GL}(V)$. A countable compact Hausdorff group is finite: by the Baire category theorem one of its points is not nowhere dense, hence is open, so by homogeneity every point is open, and a discrete compact space is finite. Hence $\ker \rho$ is an open subgroup of G_k , of the form $G_{k'}$ with k'/k finite, and it fixes every t , hence every t_ℓ . $\square$

Corollary 71.5.2. *If $X = X_0 \times_k \mathbb{C}$ as above, the ℓ -adic components of absolute Hodge classes are Tate classes: classes in $H_{\text{et}}^{2p}(X_0 \times_k \bar{k}, \mathbb{Q}_\ell(p))$ fixed by an open subgroup of G_k . If the absolute Hodge conjecture holds for X , then every Hodge class of X gives, through c_ℓ , a Tate class.*

Proof. The first statement is Proposition 71.5.1. For the second, a Hodge class α determines the Hodge cycle of Lemma 71.3.2, which is absolute Hodge by hypothesis, and its ℓ -adic component is $c_\ell((2\pi i)^p \alpha)$. $\square$

Remark 71.5.3. Corollary 71.5.2 is the bridge from Hodge theory to arithmetic. Algebraic classes of cycles defined over k' are Tate classes for $G_{k'}$ (Theorem 61.4.2(4)), and the Tate conjecture predicts the converse (Chapter 72). Absolute Hodge classes give Tate classes without producing cycles.

71.6 Principle B

Theorem 71.6.1 (Deligne's Principle B). *Let $f: X \rightarrow S$ be a smooth projective morphism of nonsingular complex varieties with S connected, and let t be a global section of the local system $R^{2p}f_*\mathbb{Q}(p)$ such that $t_s \in H_B^{2p}(X_s)(p)$ is (the Betti component of) a Hodge cycle for every $s \in S$. If t_{s_0} is absolute Hodge for one point $s_0 \in S$, then t_s is absolute Hodge for every $s \in S$.*

Sketch of proof. Fix $\sigma \in \text{Aut}(\mathbb{C})$. The relative de Rham cohomology $\mathcal{H}_{\text{dR}}^{2p}(X/S)$ is an algebraic vector bundle on S with an algebraic flat connection, the Gauss–Manin connection (Definition 66.2.2), which is defined algebraically by Katz and Oda [KO68]; hence conjugation by σ carries it to the Gauss–Manin connection of $f^\sigma: X^\sigma \rightarrow S^\sigma$. We use three facts. First, the de Rham components $t_{\text{dR},s}$ form a flat section of $\mathcal{H}_{\text{dR}}^{2p}$, and flat sections of the Gauss–Manin connection are algebraic, by its regularity; so $\sigma_* t_{\text{dR}}$ is a flat algebraic section over S^σ . Second, the ℓ -adic components form a global section of the étale local system $R^{2p}f_*\mathbb{Q}_\ell(p)$, and σ_* carries it to a global section over S^σ . We omit the proofs of these two facts [Del82]. Third, by hypothesis, $u = \sigma_* t$ is a Hodge cycle at the point $\sigma(s_0) \in S^\sigma$, with Betti component u_B .

Since u_ℓ is a global section, it is invariant under the monodromy of S^σ ; as c_ℓ is compatible with monodromy, so is $u_B \otimes 1$, and hence u_B is the value at $\sigma(s_0)$ of a global section of $R^{2p}f_*^\sigma\mathbb{Q}(p)$. Its de Rham and ℓ -adic incarnations are flat and agree with u_{dR} and u_ℓ at $\sigma(s_0)$, hence everywhere, because flat sections of a local system on a connected space are determined by one value. So u is rational at every point of S^σ . By the global invariant cycle theorem (Theorem 70.6.2), the global sections of $R^{2p}f_*^\sigma\mathbb{Q}(p)$ form a Hodge structure V that maps to each fibre by an injective morphism of Hodge structures: V is the quotient of H^{2p} of a nonsingular projective compactification of X^σ by the kernel of the restriction map to global sections, which does not depend on the fibre. The element $u_B \in V$ is of type $(0, 0)$ in the fibre

at $\sigma(s_0)$, hence in V , hence in every fibre. Thus $u = \sigma_* t$ is a Hodge cycle at every point of S^σ . As σ was arbitrary, t_s is absolute Hodge for every s . $\square$

Remark 71.6.2. Principle B is the absolute Hodge analogue of the variational Hodge conjecture (Conjecture 69.5.1), which is open for algebraic classes. The difference is that absolute Hodge cycles are defined by conditions on cohomology, which propagate along flat sections, while algebraic cycles are geometric objects whose existence on one fibre does not obviously imply existence on the others.

71.7 Deligne's theorem for abelian varieties

Theorem 71.7.1 (Deligne). *Let A be a complex abelian variety. Every Hodge class on A is absolute Hodge: $\mathrm{Hdg}_{\mathrm{AH}}^p(A) = \mathrm{Hdg}^p(A)$ for all p . The same holds for Hodge classes in all tensor spaces $T^{m,m'}(H^1(A; \mathbb{Q}))(r)$ of weight 0, defined analogously.*

Outline of proof. The proof is [Del82]; see also [CS14]. It combines four ingredients.

1. *Principle B* (Theorem 71.6.1).
2. *Principle A.* If $t_1, \dots, t_r$ are absolute Hodge tensors on A , and $G \subseteq \mathrm{GL}(H^1(A; \mathbb{Q}))$ is the subgroup fixing them, then every tensor fixed by G is absolute Hodge. Applied when G is the Hodge group, this reduces the theorem to finding absolute Hodge tensors whose stabiliser is $\mathrm{Hg}(A)$ (Theorem 63.3.2).
3. *Weil classes are absolute Hodge.* Let K be an imaginary quadratic field and consider abelian varieties of Weil type for K (Proposition 68.3.3 treats dimension 4; the construction is the same in every even dimension), with a fixed type of polarisation. They form a connected family over which the Weil classes form a flat family of Hodge classes. For the *split* type, the family contains abelian varieties of the form $B \otimes_{\mathbb{Q}} K$, on which the Weil classes are algebraic. By Principle B the Weil classes are absolute Hodge on every member.
4. *Reduction to CM type, and the CM case.* A Hodge class t on A is fixed by $\mathrm{Hg}(A)$, hence remains Hodge on every abelian variety in the connected family parametrised by the Mumford–Tate domain of A , and this family contains abelian varieties of CM type. By Principle B it suffices to treat A of CM type. For those, the Hodge group is a torus (Proposition 63.4.2), and one shows that it is the stabiliser of the pullbacks of split Weil classes along homomorphisms from A to abelian varieties of Weil type, together with divisor classes. Principle A and (3) conclude.

$\square$

Remark 71.7.2. Theorem 71.7.1 says that a Hodge class on an abelian variety, however transcendently it was produced, behaves under all automorphisms of $\mathbb{C}$ as if it were algebraic. Its limit is equally clear: the Weil classes on a general abelian fourfold of Weil type are absolute Hodge but, apart from the cases listed in Chapter 68, not known to be algebraic. Deligne used the theorem, with Milne, to build an unconditional Tannakian category of motives for absolute Hodge cycles containing the motives of abelian varieties [DM82]. A deformation argument of the same kind, combined with the Kuga–Satake construction, had earlier given Deligne's proof of the Weil conjectures for K3 surfaces [Del72].

71.8 Fields of definition of Hodge loci

Lemma 71.8.1. *Let $Z \subseteq \mathbb{P}_{\mathbb{C}}^N$ be a closed subvariety. If the set $\{\sigma(Z) : \sigma \in \text{Aut}(\mathbb{C}/\bar{\mathbb{Q}})\}$ is countable, then Z is defined over $\bar{\mathbb{Q}}$.*

Proof. Fix a monomial order. The homogeneous ideal of Z has a unique reduced Gröbner basis, and the field $k_0 \subseteq \mathbb{C}$ generated by its coefficients is the smallest field of definition of Z ; for $\sigma \in \text{Aut}(\mathbb{C})$ the reduced Gröbner basis of $\sigma(Z)$ is obtained by applying σ to the coefficients, so $\sigma(Z) = \tau(Z)$ if and only if $\sigma|_{k_0} = \tau|_{k_0}$. If $k_0 \not\subseteq \bar{\mathbb{Q}}$, it contains a transcendental element c , and for each transcendental $c' \in \mathbb{C}$ there is $\sigma \in \text{Aut}(\mathbb{C}/\bar{\mathbb{Q}})$ with $\sigma(c) = c'$ (extend $c \mapsto c'$ from $\bar{\mathbb{Q}}(c)$ to $\mathbb{C}$, using transcendence bases and algebraic closures). The values $\sigma(c)$, and hence the restrictions $\sigma|_{k_0}$ and the conjugates $\sigma(Z)$, are then uncountable, a contradiction. So $k_0 \subseteq \bar{\mathbb{Q}}$. $\square$

Proposition 71.8.2 (Voisin). *Let $f: X \rightarrow S$ be a smooth projective morphism of nonsingular quasi-projective varieties defined over $\bar{\mathbb{Q}}$. Let $Z \subseteq S$ be an irreducible closed subvariety, $s \in Z$ and $\alpha \in \text{Hdg}^p(X_s)$ such that the parallel transport of α along every path in Z is a Hodge class, and suppose Z is maximal with this property, that is, an irreducible component of the Hodge locus (Definition 66.5.1). If α is absolute Hodge, then Z is defined over $\bar{\mathbb{Q}}$, and its conjugates are components of Hodge loci.*

Sketch of proof. Z is algebraic by the Cattani–Deligne–Kaplan theorem (Theorem 66.5.3). By Principle B, applied over the nonsingular locus of Z , the class is absolute Hodge at every point of Z . Let $\sigma \in \text{Aut}(\mathbb{C}/\bar{\mathbb{Q}})$; then $X^\sigma = X$ and $S^\sigma = S$. As in the proof of Theorem 71.6.1, σ transports the flat family of absolute Hodge cycles over Z to a flat family of Hodge cycles over $\sigma(Z)$. So $\sigma(Z)$ is contained in a component Z' of the Hodge locus of a class, and applying the same argument to σ^{-1} and Z' shows $\sigma^{-1}(Z') \subseteq Z$, so $Z' = \sigma(Z)$. The Hodge loci of all rational classes have only countably many components in total, since there are countably many rational classes at a base point and each Hodge locus is an analytic subset with countably many components. So Z has countably many conjugates, and Lemma 71.8.1 (after embedding S in a projective space over $\bar{\mathbb{Q}}$ and taking closures) shows that Z is defined over $\bar{\mathbb{Q}}$ [Voi07]. $\square$

Corollary 71.8.3. *In a family of abelian varieties defined over $\bar{\mathbb{Q}}$, every component of the Hodge locus is defined over $\bar{\mathbb{Q}}$.*

Proof. Combine Theorem 71.7.1 with Proposition 71.8.2. $\square$

Remark 71.8.4. Voisin also showed a partial converse: if a component of a Hodge locus is defined over $\bar{\mathbb{Q}}$ and its conjugates are components of Hodge loci, then the Hodge classes over it are absolute Hodge, under a mild condition on the variation of Hodge structure [Voi07]. So the absolute Hodge conjecture is, to a large extent, a statement about the arithmetic of Hodge loci. For families of abelian varieties Corollary 71.8.3 reflects the fact that special subvarieties of Shimura varieties are defined over $\bar{\mathbb{Q}}$.

71.9 Exercises

Exercise 71.9.1. Let X be a nonsingular projective surface and $\sigma \in \text{Aut}(\mathbb{C})$. Show that σ_* maps $\text{Hdg}^1(X)$, viewed through Lemma 71.3.2, isomorphically onto $\text{Hdg}^1(X^\sigma)$, and that $\text{Hdg}^2(X) = H^4(X; \mathbb{Q})$ also consists of absolute Hodge classes. Conclude that the absolute Hodge conjecture holds for surfaces.

Solution. By the Lefschetz $(1, 1)$ -theorem, $\text{Hdg}^1(X)$ is spanned by divisor classes, which are absolute Hodge (Proposition 71.3.3), and $\sigma_* \text{cl}(D) = \text{cl}(D^\sigma)$. The same applies to σ^{-1} on X^σ , so the map is bijective. $H^4(X; \mathbb{Q})$ is spanned by the class of a point, which is algebraic. $\text{Hdg}^0 = H^0$ is spanned by the class of X . So all Hodge classes on X are algebraic, hence absolute Hodge. $\square$

Exercise 71.9.2. On $\mathbb{P}^1$ with coordinate z , the line bundle $\mathcal{O}(1)$ has transition function z on $U_0 \cap U_1$, and its de Rham Chern class is represented in Čech–de Rham cohomology by the closed form dz/z on $U_0 \cap U_1$. Using a partition of unity, show that the corresponding smooth 2-form η satisfies $\int_{\mathbb{P}^1} \eta = \pm 2\pi i$. Deduce that the de Rham class of a point is $\pm 2\pi i$ times the Betti class, so that it lies in $H_B^2(\mathbb{P}^1)(1)$.

Solution. Let ρ_0 be a smooth function on $\mathbb{P}^1$ equal to 1 for $|z| \leq 1$ and to 0 for $|z| \geq 2$, and $\rho_1 = 1 - \rho_0$. The Čech cocycle dz/z is the difference of the 1-forms $\rho_0 dz/z$ on U_1 and $-\rho_1 dz/z$ on U_0 (each extended by zero), and the corresponding global 2-form is $\eta = d(\rho_0 dz/z)$, which equals $-d(\rho_1 dz/z)$ where both are defined, since $d(dz/z) = 0$. It vanishes for $|z| \leq 1$ and for $|z| \geq 2$. By Stokes' theorem on the annulus $\frac{1}{2} \leq |z| \leq 3$, $\int \eta = \int_{|z|=3} \rho_0 dz/z - \int_{|z|=1/2} \rho_0 dz/z = 0 - 2\pi i$. The sign depends on the conventions for the Čech–de Rham isomorphism. As $\int_{\mathbb{P}^1}$ of the Betti class of a point is 1, the de Rham class is $\pm 2\pi i$ times the Betti class. $\square$

Exercise 71.9.3. Let X and Y be nonsingular projective varieties and $\gamma \in \text{Hdg}^q(X \times Y)$ an absolute Hodge class such that $\gamma_*: H^{2p}(X; \mathbb{Q}) \rightarrow H^{2p+2q-2\dim X}(Y; \mathbb{Q})$ is surjective. Show that if the absolute Hodge conjecture holds for X in codimension p , then it holds for Y in codimension $p + q - \dim X$.

Solution. γ_* is a surjective morphism of polarisable Hodge structures (up to a Tate twist), so by semisimplicity (Theorem 62.5.3) every Hodge class on Y in that degree is γ_* of a Hodge class on X , as in the proof of Proposition 70.6.4. By hypothesis that class is absolute Hodge, and γ_* preserves absolute Hodge classes (Proposition 71.4.1(4)). $\square$

Exercise 71.9.4. Let C be a nonsingular projective curve and J its Jacobian. Using the Abel–Jacobi map $C^g \rightarrow J$ and Theorem 71.7.1, show that every Hodge class on C^m , for every $m \geq 1$, is absolute Hodge.

Solution. Fix a base point and let $u: C \rightarrow J$ be the Abel–Jacobi map; $u^*: H^1(J; \mathbb{Q}) \rightarrow H^1(C; \mathbb{Q})$ is an isomorphism of Hodge structures. By the Künneth formula, $H^\bullet(C^m)$ is the direct sum of the sub-Hodge structures $H^{a_1}(C) \otimes \cdots \otimes H^{a_m}(C)$, $a_i \in \{0, 1, 2\}$, and a Hodge class on C^m is a sum of Hodge classes in these summands, which are absolute Hodge if the summands are, by Proposition 71.4.1(3). Fix a summand, let $S = \{i : a_i = 1\}$ and $T = \{i : a_i = 2\}$. Since $H^0(C)$ is spanned by 1 and $H^2(C)$ by the class of a point, a Hodge class in the summand has the form $\text{pr}_S^*(\gamma) \cup \prod_{i \in T} \text{pr}_i^*[\text{pt}]$ with γ a Hodge class in the summand $H^1(C)^{\otimes |S|}$ of $H^{|S|}(C^S)$. The pullback $(u^S)^*$ maps the corresponding Künneth summand $H^1(J)^{\otimes |S|}$ of $H^{|S|}(J^S)$ isomorphically onto $H^1(C)^{\otimes |S|}$, as a morphism of Hodge structures, so $\gamma = (u^S)^* \delta$ with δ a Hodge class on the abelian variety J^S . By Theorem 71.7.1 δ is absolute Hodge, and so is γ by Proposition 71.4.1(1). The classes $\text{pr}_i^*[\text{pt}]$ are algebraic, and cup products and pullbacks preserve absolute Hodge classes. $\square$

Chapter 72

The Tate Conjecture

72.1 Introduction

The Tate conjecture is the arithmetic analogue of the Hodge conjecture. The Hodge decomposition is replaced by the action of a Galois group on ℓ -adic cohomology, and Hodge classes by *Tate classes*, the classes fixed by an open subgroup of the Galois group. Classes of algebraic cycles defined over the base field are Tate classes, and Tate conjectured the converse for varieties over finitely generated fields [Tat65]. Over finite fields there is no Hodge theory and the Tate conjecture is the only statement available; over $\mathbb{C}$ there is no interesting Galois group and the Hodge conjecture is the only statement available; over number fields both apply, and the Mumford–Tate conjecture connects them. The best-known evidence is the case of divisors on abelian varieties, which is Tate’s isogeny theorem over finite fields [Tat66a] and Faltings’ theorem over number fields [Fal83]. References: [Tat65], [Tot17], [Mil80], and the source *The Hodge Conjecture* (Chapter 8).

Prerequisites. Chapter 61 (Étale Cohomology), Chapter 57 (Abelian Varieties) and Chapter 71 (Absolute Hodge Classes).

Conventions. In this chapter k is a field finitely generated over its prime field, $\bar{k}$ an algebraic closure, $G_k = \text{Aut}(\bar{k}/k)$, which is the Galois group of the separable closure of k in $\bar{k}$, and ℓ a prime different from the characteristic of k . X is a nonsingular projective geometrically irreducible variety over k , $\bar{X} = X \times_k \bar{k}$, and $H^i(\bar{X}, \mathbb{Q}_\ell(j))$ is ℓ -adic cohomology with the continuous action of G_k (Definition 61.4.1). For a finite extension k'/k inside $\bar{k}$ we write $G_{k'} \subseteq G_k$ for the corresponding open subgroup.

72.2 The statement

Definition 72.2.1. The space of *Tate classes of codimension p over k* is $T_\ell^p(X/k) = H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$. The space of *geometric Tate classes* is $T_\ell^p(\bar{X}) = \bigcup_{k'} T_\ell^p(X_{k'}/k')$, the union over finite extensions k'/k , that is, the classes fixed by some open subgroup of G_k .

Lemma 72.2.2. Let Z be an algebraic cycle of codimension p on $\bar{X}$ with $\mathbb{Q}$ -coefficients. Then Z is defined over a finite extension k' of k , and $\text{cl}_\ell(Z) \in T_\ell^p(X_{k'}/k')$. If Z is defined over k , then $\text{cl}_\ell(Z) \in T_\ell^p(X/k)$.

Proof. Each component of Z is cut out by finitely many polynomials with coefficients in $\bar{k}$, which lie in a finite extension k' . For $g \in G_{k'}$, g fixes Z , and the cycle class is G_k -equivariant (Theorem 61.4.2(4)), so $g \text{cl}_\ell(Z) = \text{cl}_\ell(gZ) = \text{cl}_\ell(Z)$. The same argument with G_k gives the last statement. $\square$

Conjecture 72.2.3 (Tate). *The cycle class map induces a surjection*

$$\mathrm{CH}^p(X) \otimes_{\mathbb{Z}} \mathbb{Q}_\ell \longrightarrow H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}.$$

We write $T^p(X/k)$ for this statement, and $T^p(\bar{X})$ for the statement that the geometric Tate classes are spanned over $\mathbb{Q}_\ell$ by classes of cycles on $\bar{X}$.

Lemma 72.2.4. *If k'/k is a finite extension and $T^p(X_{k'}/k')$ holds, then $T^p(X/k)$ holds. In particular $T^p(\bar{X})$ is equivalent to $T^p(X_{k'}/k')$ for all finite k'/k .*

Proof. Let k_s be the separable closure of k in k' , so that k'/k_s is purely inseparable and $G_{k'} = G_{k_s}$. First we show $T^p(X_{k_s}/k_s)$. The projection $\pi: X_{k'} \rightarrow X_{k_s}$ is finite and a homeomorphism, so for every subvariety $W \subseteq X_{k'}$ the preimage of $\pi(W)$ is W as a set, and $\pi^*[\pi(W)] = m[W]$ for some integer $m \geq 1$. As $\bar{X}$ is the same for k' and k_s , this gives $\mathrm{cl}_\ell(W) = \frac{1}{m}\mathrm{cl}_\ell(\pi(W))$ in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$. So classes of cycles on $X_{k'}$ are combinations of classes of cycles on X_{k_s} , and $T^p(X_{k_s}/k_s)$ holds.

Now let $v \in H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))^{G_k}$. It is G_{k_s} -invariant, so $v = \sum_i a_i \mathrm{cl}_\ell(Z_i)$ with $a_i \in \mathbb{Q}_\ell$ and Z_i cycles on X_{k_s} . Let $g_1, \dots, g_d$ be representatives of the cosets G_k/G_{k_s} , where $d = [k_s : k]$. On G_{k_s} -invariant vectors the operator $\pi_\Gamma = \frac{1}{d} \sum_j g_j$ does not depend on the representatives, it fixes v , and $\pi_\Gamma \mathrm{cl}_\ell(Z_i) = \frac{1}{d} \mathrm{cl}_\ell(\sum_j g_j Z_i)$. The cycle $\sum_j g_j Z_i$ on $\bar{X}$ is the pullback of the pushforward of Z_i along the finite separable morphism $X_{k_s} \rightarrow X$, hence the class of a cycle defined over k . So $v = \sum_i a_i \pi_\Gamma \mathrm{cl}_\ell(Z_i)$ is a combination of classes of cycles defined over k .

For the last statement, a geometric Tate class is fixed by some $G_{k'}$, so $T^p(X_{k'}/k')$ for all k' implies $T^p(\bar{X})$. Conversely, if $T^p(\bar{X})$ holds and v is a Tate class over k' , then $v = \sum a_i \mathrm{cl}_\ell(Z_i)$ with cycles Z_i on $\bar{X}$, all defined over some finite extension k'' of k' (Lemma 72.2.2). The argument above, with k'/k replaced by k''/k' , used only that v is invariant under the smaller Galois group and is a combination of classes of cycles defined over the larger field; so it shows that v is a combination of classes of cycles defined over k' . $\square$

Counterexample 72.2.5. The hypothesis that k is finitely generated cannot be dropped. Let X be a nonsingular projective surface over $\mathbb{C}$ with $h^{2,0}(X) > 0$, for instance a quartic surface in $\mathbb{P}^3$. For $k = \mathbb{C}$ the Galois group is trivial, so every class in $H^2(X, \mathbb{Q}_\ell(1))$ is invariant. But the $\mathbb{Q}_\ell$ -span of divisor classes has dimension $\rho(X) \leq h^{1,1} < b_2(X)$, by Artin's comparison theorem (Theorem 61.4.3) and the Lefschetz (1,1)-theorem. So $T^1(X/\mathbb{C})$ fails. The same happens over $\bar{\mathbb{Q}}$, and over $\bar{\mathbb{F}}_p$ for a surface with $\rho(X) < b_2(X)$.

Remark 72.2.6. Over a finite field $k = \mathbb{F}_q$, Tate formulated a companion conjecture: the rank of the group of codimension- p cycles on X modulo numerical equivalence equals the order of the pole of the zeta function $Z(X, q^{-s})$ at $s = p$ [Tat65]. Since the eigenvalues of Frobenius on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$ have absolute value q^p [Del74a], that order is the multiplicity of the eigenvalue q^p of the geometric Frobenius on H^{2p} . This refinement is tied to $T^p(X/k)$, to the equality of homological and numerical equivalence (Chapter 73), and to the semisimplicity of the Frobenius action; see [Tot17].

72.3 Divisors on abelian varieties

For an abelian variety A over k , the *Tate module* is $T_\ell A = \varprojlim_m A[\ell^m](\bar{k})$, a free $\mathbb{Z}_\ell$ -module of rank $2 \dim A$ with a continuous action of G_k , and $V_\ell A = T_\ell A \otimes_{\mathbb{Z}_\ell} \mathbb{Q}_\ell$. As in Example 61.5.1, $H^1(\bar{A}, \mathbb{Q}_\ell) \cong \mathrm{Hom}(V_\ell A, \mathbb{Q}_\ell)$ and $H^i(\bar{A}, \mathbb{Q}_\ell) \cong \wedge^i H^1(\bar{A}, \mathbb{Q}_\ell)$, compatibly with G_k [Mil80].

Theorem 72.3.1 (Tate, Zarhin, Faltings). *Let A and B be abelian varieties over k , where k is finite or of characteristic $\neq 2$. Then:*

1. the natural map $\mathrm{Hom}_k(A, B) \otimes_{\mathbb{Z}} \mathbb{Z}_{\ell} \rightarrow \mathrm{Hom}_{G_k}(T_{\ell}A, T_{\ell}B)$ is an isomorphism;
2. the representation of G_k on $V_{\ell}A$ is semisimple.

This is due to Tate for finite fields [Tat66a], Zarhin for finitely generated fields of characteristic $p > 2$ [Zar75], and Faltings for finitely generated fields of characteristic 0, where it is the heart of his proof of the Mordell conjecture [Fal83]. Injectivity in (1), and the fact that the cokernel is torsion-free, hold over any field [Mum70]; the content is surjectivity. Over a finite field the key finiteness input is that there are only finitely many isomorphism classes of polarised abelian varieties of given dimension and degree over k , and over number fields Faltings proves this finiteness with his height.

Corollary 72.3.2. *For an abelian variety A over k (as in Theorem 72.3.1), $T^1(A/k)$ holds: the Tate classes in $H^2(\bar{A}, \mathbb{Q}_{\ell}(1))$ are $\mathbb{Q}_{\ell}$ -linear combinations of classes of divisors defined over k .*

Sketch of proof. We use two facts from the theory of abelian varieties [Mum70]. First, $H^2(\bar{A}, \mathbb{Q}_{\ell}(1)) = \Lambda^2 H^1(\bar{A}, \mathbb{Q}_{\ell})(1)$ is the space of alternating $\mathbb{Q}_{\ell}$ -bilinear forms $V_{\ell}A \times V_{\ell}A \rightarrow \mathbb{Q}_{\ell}(1)$, and through the Weil pairing $e_{\ell}: V_{\ell}A \times V_{\ell}A^{\vee} \rightarrow \mathbb{Q}_{\ell}(1)$ such a form is $E_{\phi}(x, y) = e_{\ell}(x, \phi y)$ for a unique $\phi \in \mathrm{Hom}(V_{\ell}A, V_{\ell}A^{\vee})$ that is symmetric, $\phi^{\vee} = \phi$ (with the sign conventions of [Mum70]); a form is G_k -invariant if and only if ϕ is G_k -equivariant, because e_{ℓ} is. Second, the class $c_1(L) \in H^2(\bar{A}, \mathbb{Q}_{\ell}(1))$ of a line bundle corresponds to $\phi = \lambda_L$, the homomorphism $\lambda_L: A \rightarrow A^{\vee}$, $a \mapsto t_a^* L \otimes L^{-1}$, and $L \mapsto \lambda_L$ identifies $\mathrm{NS}(A) \otimes \mathbb{Q}$ with the symmetric elements of $\mathrm{Hom}_k(A, A^{\vee}) \otimes \mathbb{Q}$.

Now let E be a G_k -invariant class. Then $E = E_{\phi}$ with ϕ symmetric and equivariant, so by Theorem 72.3.1(1), applied to $B = A^{\vee}$, $\phi = \sum_j a_j \psi_j$ with $a_j \in \mathbb{Q}_{\ell}$ and $\psi_j \in \mathrm{Hom}_k(A, A^{\vee})$. Replacing ψ_j by $\frac{1}{2}(\psi_j + \psi_j^{\vee})$ we may take the ψ_j symmetric, since ϕ is. Each symmetric ψ_j is λ_{L_j} for some $L_j \in \mathrm{NS}(A) \otimes \mathbb{Q}$, with L_j defined over k . Hence $E = \sum_j a_j c_1(L_j)$. $\square$

Example 72.3.3. Let C_1 and C_2 be nonsingular projective curves over k with Jacobians J_1, J_2 . By the Künneth formula, $H^2(\bar{C}_1 \times \bar{C}_2) = H^2(\bar{C}_1) \oplus (H^1(\bar{C}_1) \otimes H^1(\bar{C}_2)) \oplus H^2(\bar{C}_2)$, and the Tate classes in the middle summand, twisted by (1), are the G_k -equivariant maps $V_{\ell}J_1 \rightarrow V_{\ell}J_2$. By Theorem 72.3.1 these are combinations of homomorphisms $J_1 \rightarrow J_2$, which are induced by divisors on $C_1 \times C_2$ (divisorial correspondences). So $T^1(C_1 \times C_2/k)$ holds. This was Tate's motivating application [Tat66a].

72.4 K3 surfaces and other known cases

Theorem 72.4.1. *The Tate conjecture T^1 holds in the following cases.*

1. Abelian varieties over finite fields and over finitely generated fields of characteristic $\neq 2$ (Corollary 72.3.2), and more generally varieties whose H^2 is controlled by an abelian variety, such as products of curves (Example 72.3.3).
2. K3 surfaces over finitely generated fields of characteristic 0 [And96a].
3. K3 surfaces over finite fields of characteristic $p \geq 3$: for ordinary K3 surfaces by Nygaard [Nyg83], for those of finite height by Nygaard and Ogus [NO85], and in general by work of Charles [Cha13], Maulik [Mau14] and Madapusi Pera [MP15].

The proofs for K3 surfaces use the Kuga–Satake construction (Remark 68.6.2), which embeds H^2 of a K3 surface into H^2 of an abelian variety, and transport Theorem 72.3.1 through it. The difficulty is arithmetic: the Kuga–Satake abelian variety must be defined over a finitely generated field, and in positive characteristic one needs integral models of orthogonal Shimura varieties.

Remark 72.4.2. For a surface X over a finite field, Tate showed that $T^1(X/k)$ is equivalent to the finiteness of the ℓ -primary part of the Brauer group $\text{Br}(X)$, and, with Artin, conjectured a formula for the special value of the zeta function in terms of $|\text{Br}(X)|$ and the discriminant of the Néron–Severi lattice, an analogue of the Birch and Swinnerton-Dyer conjecture [Tat66b]. The Tate conjecture for divisors thus has a cohomological content (surjectivity of a cycle map) and a finiteness content (Brauer groups) that are equivalent.

72.5 The Mumford–Tate conjecture

Let $k \subseteq \mathbb{C}$ be a number field and X nonsingular projective over k . For each i , Artin’s comparison theorem identifies $H^i(\tilde{X}, \mathbb{Q}_\ell)$ with $H^i(X_{\mathbb{C}}; \mathbb{Q}) \otimes \mathbb{Q}_\ell$ (Theorem 61.4.3, using invariance of étale cohomology under extension of algebraically closed fields [Mil80]). So G_k acts on $H \otimes \mathbb{Q}_\ell$, where $H = H^i(X_{\mathbb{C}}; \mathbb{Q})$ is a Hodge structure with Mumford–Tate group $\text{MT}(H)$ (Definition 63.2.1).

Definition 72.5.1. Let V be a finite-dimensional $\mathbb{Q}_\ell$ -vector space with a continuous representation $\rho: G_k \rightarrow \text{GL}(V)$. Its ℓ -adic algebraic monodromy group $G_\ell(V)$ is the Zariski closure of $\rho(G_k)$ in $\text{GL}(V)$, an algebraic group over $\mathbb{Q}_\ell$, and $G_\ell^0(V)$ is its identity component. For $V = H \otimes \mathbb{Q}_\ell$ with $H = H^i(X_{\mathbb{C}}; \mathbb{Q})$ we write G_ℓ and G_ℓ^0 .

Lemma 72.5.2. In the situation of Definition 72.5.1, $G_\ell^0(V)$ does not change if k is replaced by a finite extension, and the vectors of V fixed by some open subgroup of G_k are exactly the invariants of $G_\ell^0(V)$.

Proof. If $\Gamma_0 \subseteq \Gamma$ are subgroups of $\text{GL}(V)(\mathbb{Q}_\ell)$ with $\Gamma = \bigcup_{j=1}^r g_j \Gamma_0$, then the Zariski closures satisfy $\bar{\Gamma} = \bigcup_j g_j \bar{\Gamma}_0$, so $\bar{\Gamma}_0$ is a closed subgroup of finite index in $\bar{\Gamma}$; it is then a union of connected components of $\bar{\Gamma}$ and contains $\bar{\Gamma}^0$. The subgroup $\rho^{-1}(G_\ell^0(V)(\mathbb{Q}_\ell))$ has finite index in G_k and is closed, hence open, say G_{k_0} ; the Zariski closure of $\rho(G_{k_0})$ lies in $G_\ell^0(V)$ and has finite index in $G_\ell(V)$, so it equals $G_\ell^0(V)$. For any finite k'/k , the Zariski closure of $\rho(G_{k'})$ has finite index in $G_\ell(V)$, so its identity component is $G_\ell^0(V)$. This proves the first statement. For the second, the stabiliser of a vector is Zariski closed, so a vector fixed by $\rho(G_{k'})$ is fixed by its Zariski closure, which contains $G_\ell^0(V)$; conversely a $G_\ell^0(V)$ -invariant vector is fixed by $\rho(G_{k_0})$. $\square$

Conjecture 72.5.3 (Mumford–Tate). $G_\ell^0 = \text{MT}(H) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$ inside $\text{GL}(H \otimes \mathbb{Q}_\ell)$.

Theorem 72.5.4 (Deligne). If $X = A$ is an abelian variety and $H = H^1(A_{\mathbb{C}}; \mathbb{Q})$, then $G_\ell^0 \subseteq \text{MT}(H) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell$.

Sketch of proof. Let $\text{MT}' \subseteq \text{GL}(H) \times \mathbb{G}_m$ be the subgroup fixing every Hodge class in every tensor space $T^{m,m'}(H)(r)$ of weight 0, where $\mathbb{G}_m$ acts on $\mathbb{Q}(1)$ through the identity character. Its image under the first projection is $\text{MT}(H)$; this is a reformulation of Theorem 63.3.2 which we do not prove [Del82]. By Noetherianity, MT' is the stabiliser of finitely many Hodge classes $t_1, \dots, t_N$. The spaces $T^{m,m'}(H)(r)$ are Tate twists of Künneth summands of $H^\bullet(A_{\mathbb{C}}^M)$ for suitable M (using a polarisation to identify $H^\vee$ with $H(1)$), so by Theorem 71.7.1 the t_j are absolute Hodge, and by Proposition 71.5.1 their ℓ -adic incarnations are fixed by an open subgroup $G_{k'}$. The group G_k acts on $T^{m,m'}(H \otimes \mathbb{Q}_\ell)(r)$ through $(\rho_\ell, \chi_\ell): G_k \rightarrow \text{GL}(H \otimes \mathbb{Q}_\ell) \times \mathbb{G}_m(\mathbb{Q}_\ell)$, χ_ℓ the cyclotomic character. Hence $(\rho_\ell, \chi_\ell)(G_{k'}) \subseteq \text{MT}'(\mathbb{Q}_\ell)$, so $\rho_\ell(G_{k'}) \subseteq \text{MT}(H)(\mathbb{Q}_\ell)$, and taking Zariski closures, $G_\ell^0 \subseteq \text{MT}(H)_{\mathbb{Q}_\ell}$. $\square$

Proposition 72.5.5. Let A be an abelian variety over a number field $k \subseteq \mathbb{C}$ such that all endomorphisms of $A_{\mathbb{C}}$ are defined over k . Then G_ℓ and $\text{MT}(H)_{\mathbb{Q}_\ell}$ have the same commutant in $\text{End}(H \otimes \mathbb{Q}_\ell)$, namely $\text{End}(A) \otimes \mathbb{Q}_\ell$.

Proof. By Theorem 72.3.1(1) with $B = A$, the commutant of $\rho_\ell(G_k)$, which equals the commutant of its Zariski closure G_ℓ , is $\text{End}_k(A) \otimes \mathbb{Q}_\ell$, acting on $V_\ell A$ and dually on $H \otimes \mathbb{Q}_\ell$. By Corollary 63.2.4, the commutant of $\text{MT}(H)$ in $\text{End}(H)$ is $\text{End}_{\text{HS}}(H) = \text{End}(A_{\mathbb{C}}) \otimes \mathbb{Q}$, and taking commutants commutes with the flat base change $\mathbb{Q} \rightarrow \mathbb{Q}_\ell$ (it is the solution space of linear equations). As $\text{End}(A_{\mathbb{C}}) = \text{End}_k(A)$ by hypothesis, the two commutants agree. $\square$

Remark 72.5.6. Proposition 72.5.5 and Theorem 72.5.4 show that $G_\ell^0 \subseteq \text{MT}_{\mathbb{Q}_\ell}$ are reductive groups with the same commutant (G_ℓ^0 is reductive because $V_\ell A$ is semisimple, Theorem 72.3.1(2)). This is consistent with the Mumford–Tate conjecture but does not imply it: reductive subgroups with the same commutant, such as $\text{Sp}_{2g} \subseteq \text{SL}_{2g}$ acting on $\mathbb{Q}_\ell^{2g}$, can differ. The conjecture is known for elliptic curves: for E without complex multiplication, Serre’s open image theorem [Ser72] gives $G_\ell = \text{GL}_2$, which is $\text{MT}(H)_{\mathbb{Q}_\ell}$ by Proposition 63.4.1. It is known for abelian varieties of CM type, and for many abelian varieties with $\text{End}(A_{\mathbb{C}}) = \mathbb{Z}$, depending on the dimension [Pin98]; it is open in general.

Proposition 72.5.7. *Let X be a nonsingular projective variety over a number field $k \subseteq \mathbb{C}$, and $p \geq 0$. Suppose that the geometric Tate classes in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p))$ are exactly the images under c_ℓ of the classes $(2\pi i)^p \alpha \otimes a$, $\alpha \in \text{Hdg}^p(X_{\mathbb{C}})$, $a \in \mathbb{Q}_\ell$; this is what the Mumford–Tate conjecture for $H^{2p}(X_{\mathbb{C}})$ predicts. Then the Hodge conjecture for $X_{\mathbb{C}}$ in codimension p is equivalent to $T^p(\bar{X})$.*

Proof. We use that every algebraic class on $X_{\mathbb{C}}$ is the class of a cycle defined over $\bar{k}$: a cycle on $X_{\mathbb{C}}$ is defined over a finitely generated extension of $\bar{k}$, so it is a member of a family of cycles parametrised by a variety over $\bar{k}$, and specialising to a $\bar{k}$ -point does not change its cohomology class, which is locally constant in the family. So the $\mathbb{Q}$ -space Z_{alg}^p of algebraic classes is the same for $X_{\mathbb{C}}$ and $\bar{X}$, and c_ℓ identifies $Z_{\text{alg}}^p \otimes \mathbb{Q}_\ell$ with the $\mathbb{Q}_\ell$ -span of the ℓ -adic cycle classes, by Theorem 61.4.3. By hypothesis c_ℓ identifies $\text{Hdg}^p(X_{\mathbb{C}}) \otimes \mathbb{Q}_\ell$ with $T_\ell^p(\bar{X})$. Hence $T^p(\bar{X})$ says $Z_{\text{alg}}^p \otimes \mathbb{Q}_\ell = \text{Hdg}^p \otimes \mathbb{Q}_\ell$, which, since $Z_{\text{alg}}^p \subseteq \text{Hdg}^p$ are $\mathbb{Q}$ -subspaces of the same finite-dimensional space, is equivalent to $Z_{\text{alg}}^p = \text{Hdg}^p$. $\square$

Remark 72.5.8. Neither the Hodge nor the Tate conjecture is known to imply the other. The Hodge conjecture says nothing over finite fields, and the Tate conjecture over number fields is linked to it only through the Mumford–Tate conjecture, which is itself open. The two known inclusions, algebraic $\subseteq$ Hodge and algebraic $\subseteq$ Tate, are joined by Deligne’s theorem, which gives Hodge $\subseteq$ Tate for abelian varieties (Corollary 71.5.2 and Theorem 71.7.1), and Theorem 72.5.4 is the group-theoretic form of the same statement. For abelian varieties Faltings’ theorem proves the Tate conjecture for divisors, and the Lefschetz (1, 1)-theorem the Hodge conjecture for divisors; in codimension ≥ 2 both are open.

72.6 Exercises

Exercise 72.6.1. Let $k = \mathbb{F}_q$ and let F be the geometric Frobenius, acting on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$. Assuming that F acts semisimply, show that $\dim T_\ell^p(\bar{X})$ is the number of eigenvalues of F on $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$, counted with multiplicity, of the form ζq^p with ζ a root of unity.

Solution. $G_{\mathbb{F}_{q^m}}$ is topologically generated by F^m , and the geometric Frobenius acts on $\mathbb{Q}_\ell(1)$ by q^{-1} . So a class in $H^{2p}(\bar{X}, \mathbb{Q}_\ell(p)) = H^{2p}(\bar{X}, \mathbb{Q}_\ell) \otimes \mathbb{Q}_\ell(p)$ is fixed by $G_{\mathbb{F}_{q^m}}$ if and only if, as a class in $H^{2p}(\bar{X}, \mathbb{Q}_\ell)$, it satisfies $F^m v = q^{pm} v$. Since F is semisimple, the union over m of the eigenspaces of F^m for q^{pm} is the sum of the eigenspaces of F for the eigenvalues λ with $\lambda^m = q^{pm}$ for some m , that is, $\lambda = \zeta q^p$. The union is an increasing union of subspaces, so it is the largest of them. $\square$

Exercise 72.6.2. Let E be an elliptic curve over a number field $k \subseteq \mathbb{C}$ with complex multiplication by an imaginary quadratic field K , all of whose endomorphisms are defined over k . Show that G_ℓ^0 is contained in the torus $(\text{Res}_{K/\mathbb{Q}} \mathbb{G}_m)_{\mathbb{Q}_\ell}$, and deduce that $G_\ell^0 \subseteq \text{MT}(H^1(E_{\mathbb{C}}))_{\mathbb{Q}_\ell}$ directly, without Theorem 72.5.4.

Solution. The action of $K = \text{End}(E) \otimes \mathbb{Q}$ on $V_\ell E$ commutes with G_k , since the endomorphisms are defined over k . $V_\ell E$ is free of rank one over $K \otimes \mathbb{Q}_\ell$, so the commutant of $K \otimes \mathbb{Q}_\ell$ in $\text{End}(V_\ell E)$ is $K \otimes \mathbb{Q}_\ell$ itself. Hence $\rho_\ell(G_k) \subseteq (K \otimes \mathbb{Q}_\ell)^\times = (\text{Res}_{K/\mathbb{Q}} \mathbb{G}_m)(\mathbb{Q}_\ell)$, and the same holds for the Zariski closure. By Proposition 63.4.1, this torus is $\text{MT}(H^1(E_{\mathbb{C}}))$ (the same argument applies to the dual H^1). $\square$

Exercise 72.6.3. Let E be a supersingular elliptic curve over $\mathbb{F}_p$, and $k = \mathbb{F}_{p^2}$ a field over which all endomorphisms of E are defined, so that $\text{End}(E_k) \otimes \mathbb{Q}$ is a quaternion algebra of rank 4. Using Theorem 72.3.1, show that $\rho(E \times E) = 6$ over k , and that all of $H^2(\overline{E \times E}, \mathbb{Q}_\ell(1))$ consists of Tate classes over k . Why is this impossible over $\mathbb{C}$?

Solution. By Example 72.3.3, the Néron–Severi group of $E \times E$ has rank $2 + \text{rank Hom}(E, E) = 2 + 4 = 6 = b_2(E \times E)$. By Corollary 72.3.2 the Tate classes are spanned by divisor classes, and conversely divisor classes over k are Tate; since the divisor classes span a 6-dimensional space, every class in H^2 is a Tate class. Over $\mathbb{C}$ the rank of $\text{End}(E)$ is at most 2 (Proposition 57.6.1), so $\rho(E \times E) \leq 4 < 6$; equivalently $h^{2,0}(E \times E) = 1$ prevents all of H^2 from being of type $(1, 1)$. $\square$

Chapter 73

The Standard Conjectures and Motives

73.1 Introduction

In the 1960s Grothendieck proposed a small set of conjectures about algebraic cycles, the *standard conjectures*, designed to prove the Weil conjectures and to build a category of *motives*: a universal cohomology theory through which every Weil cohomology factors. Deligne proved the Weil conjectures by other means, and the standard conjectures remain open. They are much weaker than the Hodge conjecture, which implies them in characteristic 0 (Theorem 73.3.9), yet even the Lefschetz standard conjecture, which asks for one explicit class to be algebraic, is open in general. This chapter states the conjectures, proves the implications among them, constructs pure motives, proves Jannsen's semisimplicity theorem, reformulates the Hodge conjecture motivically, and describes the conjectures of Bloch and Beilinson on the kernel of the cycle class map, the other half of the theory of algebraic cycles. References: [Gro69b], [Kle68], [Kle94], [Sch94], [And04], [Jan94], and the source *The Hodge Conjecture* (Chapters 8 and 9).

Prerequisites. Chapter 58 (Algebraic Cycles and Chow Groups), Chapter 46 (Hard Lefschetz and the Hodge–Riemann Relations) and Chapter 72 (The Tate Conjecture).

Conventions. k is a field, and varieties are nonsingular projective over k ; X has pure dimension n unless stated otherwise. We fix a Weil cohomology theory $H^\bullet$ with coefficients in a field K of characteristic 0 (Definition 61.4.4): ℓ -adic cohomology of $X \times_k \bar{k}$ in general, or Betti cohomology of $X_{\mathbb{C}}$ with $K = \mathbb{Q}$ when $k \subseteq \mathbb{C}$. Tate twists are suppressed from the notation. For $\sim$ one of rational, homological (with respect to $H^\bullet$) or numerical equivalence (Definition 58.6.1) we write

$$A_{\sim}^p(X) = (Z^p(X) \otimes_{\mathbb{Z}} \mathbb{Q}) / \sim, \quad A_{\sim}^\bullet(X) = \bigoplus_p A_{\sim}^p(X),$$

so that $A_{\text{rat}}^p(X) = \text{CH}^p(X)_{\mathbb{Q}}$ and $A_{\text{hom}}^p(X)$ is the $\mathbb{Q}$ -span of the image of the cycle class map in $H^{2p}(X)$, whose elements we call *algebraic classes*. We fix an ample divisor on X with class $h \in A_{\text{hom}}^1(X)$ and write $L = h \cup - : H^i(X) \rightarrow H^{i+2}(X)$.

73.2 Correspondences in cohomology

Proposition 73.2.1. *Let X, Y, W be varieties, X of dimension n .*

1. *A class $\gamma \in H^{2n+2r}(X \times Y)$ acts by $\gamma_*(x) = \text{pr}_{2*}(\text{pr}_1^*x \cup \gamma)$, which maps $H^i(X)$ to $H^{i+2r}(Y)$, and the Künneth formula together with Poincaré duality give an isomorphism $H^{2n+2r}(X \times Y) \cong \bigoplus_i \text{Hom}_K(H^i(X), H^{i+2r}(Y))$, $\gamma \mapsto (\gamma_*|_{H^i(X)})_i$.*

2. For $\gamma \in H^\bullet(X \times Y)$ and $\delta \in H^\bullet(Y \times W)$, the class $\delta \circ \gamma = \text{pr}_{13*}(\text{pr}_{12}^* \gamma \cup \text{pr}_{23}^* \delta)$ satisfies $(\delta \circ \gamma)_* = \delta_* \circ \gamma_*$, the class of the diagonal acts as the identity, and $\text{cl}(\Gamma' \circ \Gamma) = \text{cl}(\Gamma') \circ \text{cl}(\Gamma)$ for algebraic correspondences (Definition 58.5.1).
3. (Lefschetz trace formula) For $\alpha, \beta \in H^{2n}(X \times X)$, $\int_{X \times X} \alpha \cup {}^t \beta = \sum_i (-1)^i \text{tr}((\beta \circ \alpha)_* | H^i(X))$, where ${}^t \beta$ is the pullback of β under the exchange of the factors. The right side is the supertrace $\text{str}((\beta \circ \alpha)_*)$.

These are formal consequences of the axioms of a Weil cohomology [Kle68]; (2) is the cohomological version of Proposition 58.5.2, and (3) for $\beta = \Delta_X$ is the Lefschetz trace formula of Theorem 61.4.2(5). By (2), an algebraic correspondence maps algebraic classes to algebraic classes: $\text{cl}(\Gamma)_* \text{cl}(Z) = \text{cl}(\Gamma_* Z)$. We call a K -linear endomorphism of $H^\bullet(X)$ *algebraic* if it is γ_* for γ in the $\mathbb{Q}$ -span of classes of algebraic cycles on $X \times X$. Sums and composites of algebraic endomorphisms are algebraic, and $L = (\Delta_* h)_*$ is algebraic.

Lemma 73.2.2. $A_{\text{num}}^p(X)$ is a finite-dimensional $\mathbb{Q}$ -vector space of dimension at most $\dim_K H^{2p}(X)$, and the pairing $A_{\text{num}}^p(X) \times A_{\text{num}}^{n-p}(X) \rightarrow \mathbb{Q}$, $(\alpha, \beta) \mapsto \deg(\alpha \cdot \beta)$, is perfect. If $K = \mathbb{Q}$, then $A_{\text{hom}}^p(X)$ is finite-dimensional too.

Proof. Homological equivalence implies numerical equivalence (Proposition 58.6.2), and $\deg(\alpha \cdot \beta) = \int \text{cl}(\alpha) \cup \text{cl}(\beta)$. Let $\alpha_1, \dots, \alpha_N \in A_{\text{num}}^p(X)$ be linearly independent. The map $A_{\text{num}}^{n-p}(X) \rightarrow \mathbb{Q}^N$, $\beta \mapsto (\deg(\alpha_i \cdot \beta))_i$, is surjective: otherwise a nonzero linear form on $\mathbb{Q}^N$ vanishes on its image, which gives a nontrivial combination of the α_i that pairs to zero with every cycle, that is, is zero in A_{num}^p . Choose β_j with $\deg(\alpha_i \cdot \beta_j) = \delta_{ij}$. If $\sum_i c_i \text{cl}(\alpha_i) = 0$ with $c_i \in K$, cupping with $\text{cl}(\beta_j)$ and integrating gives $c_j = 0$. So the $\text{cl}(\alpha_i)$ are K -linearly independent in $H^{2p}(X)$ and $N \leq \dim_K H^{2p}(X)$. The pairing is nondegenerate by definition, hence perfect between finite-dimensional spaces. If $K = \mathbb{Q}$, $A_{\text{hom}}^p(X)$ is a subspace of $H^{2p}(X)$. $\square$

Remark 73.2.3. For ℓ -adic cohomology, $A_{\text{hom}}^p(X)$ is a $\mathbb{Q}$ -subspace of a $\mathbb{Q}_\ell$ -vector space, and it is not known in general that it is finite-dimensional over $\mathbb{Q}$, or that $A_{\text{hom}}^p(X) \otimes_{\mathbb{Q}} \mathbb{Q}_\ell \rightarrow H^{2p}(X)$ is injective. Both follow from the equality of homological and numerical equivalence (Conjecture 73.3.3 below), by the proof of Lemma 73.2.2.

73.3 The standard conjectures

By hard Lefschetz (an axiom of Weil cohomologies; for ℓ -adic cohomology a theorem of Deligne, Theorem 61.4.2(6)), $L^{n-i}: H^i(X) \rightarrow H^{2n-i}(X)$ is an isomorphism for $i \leq n$. As in Chapter 46, this gives the Lefschetz decomposition $H^i = \bigoplus_j L^j P^{i-2j}$, with $P^j = \ker(L^{n-j+1}: H^j \rightarrow H^{2n-j+2})$ for $j \leq n$, and a unique operator Λ with $[L, \Lambda] = h_{\deg}$ and $[h_{\deg}, \Lambda] = -2\Lambda$, where $h_{\deg}$ acts on H^i by $i - n$ (Proposition 46.2.4); explicitly $\Lambda(L^r x) = r(n - j - r + 1)L^{r-1}x$ for $x \in P^j$. So $H^\bullet(X)$ is a representation of $\mathfrak{sl}_2$ with $e = L$, $f = \Lambda$, $h = h_{\deg}$, in which H^i is the weight space of weight $i - n$ and P^j consists of lowest weight vectors of weight $j - n$.

Conjecture 73.3.1 (Lefschetz standard conjecture $B(X)$). *The operator Λ is algebraic.*

Conjecture 73.3.2 (Künneth standard conjecture $C(X)$). *The Künneth components $\pi_i \in H^{2n-i}(X) \otimes H^i(X) \subseteq H^{2n}(X \times X)$ of the class of the diagonal are algebraic; equivalently, the projectors of $H^\bullet(X)$ onto the summands $H^i(X)$ are algebraic.*

Conjecture 73.3.3 (Standard conjecture $D(X)$). *Homological and numerical equivalence coincide on $Z^\bullet(X) \otimes \mathbb{Q}$: $A_{\text{hom}}^\bullet(X) = A_{\text{num}}^\bullet(X)$.*

Conjecture 73.3.4 (Hodge standard conjecture $\text{Hdg}(X)$). For $p \leq n/2$, the symmetric bilinear form $(x, y) \mapsto (-1)^p \int_X L^{n-2p}(x \cup y)$ is positive definite on the $\mathbb{Q}$ -space $A_{\text{hom}}^p(X) \cap P^{2p}(X)$ of primitive algebraic classes.

The equivalence in $C(X)$ is Proposition 73.2.1(1): $(\pi_i)_*$ is the projector onto $H^i(X)$.

Lemma 73.3.5. Let V be a finite-dimensional representation of $\mathfrak{sl}_2$ over a field of characteristic 0, with weight spaces V_w for h .

1. The projector of V onto V_w , along the other weight spaces, is a polynomial in h .
2. The projector onto the isotypic component of $V(m)$ is a polynomial in the Casimir element $c = ef + fe + \frac{1}{2}h^2$, which acts on $V(m)$ by $\frac{1}{2}m(m+2)$.
3. For $j \geq 0$, $e^j: V_{-j} \rightarrow V_j$ is bijective, and its inverse is the restriction to V_j of $P(f^j e^j) f^j$ for a polynomial P .
4. If v is a lowest weight vector of weight $-m$ and $0 \leq j \leq m$, then $f^j e^j v = c_{m,j} v$ with $c_{m,j}$ a nonzero integer.

Proof. We use the basis $v_0, \dots, v_m$ of $V(m)$ from Theorem 11.5.2, with $h v_i = (m - 2i)v_i$, $f v_i = (i + 1)v_{i+1}$, $e v_i = (m - i + 1)v_{i-1}$, and the decomposition of V into irreducibles. (1) h is diagonalisable with finitely many eigenvalues $w_1, \dots, w_s$, and the projector onto V_{w_a} is $\prod_{b \neq a} (h - w_b)/(w_a - w_b)$. (2) On v_0 , $e v_0 = 0$ gives $ef v_0 = [e, f]v_0 = h v_0 = m v_0$ and $f e v_0 = 0$, so $c v_0 = (m + \frac{1}{2}m^2)v_0$; as c commutes with e, f, h , it acts on $V(m)$ by this scalar. The scalars $\frac{1}{2}m(m+2)$ are distinct for distinct $m \geq 0$, so the projectors are Lagrange polynomials in c . (4) The lowest weight vector of $V(m)$ is v_m . Applying e^j multiplies it by $\prod_{s=0}^{j-1} (s+1) = j!$ and gives v_{m-j} ; applying f^j multiplies by $\prod_{t=0}^{j-1} (m - j + t + 1)$ and returns to v_m . All factors are positive. (3) The weight $-j$ vector of $V(m)$, when it exists ($m \geq j$, $m \equiv j \pmod{2}$), is v_i with $i = (m + j)/2$; applying e^j and then f^j multiplies it by $\prod_{s=0}^{j-1} (m - i + s + 1) \prod_{t=0}^{j-1} (i - j + t + 1)$, a product of integers $\geq (m - j)/2 + 1 \geq 1$. So $f^j e^j$ acts on V_{-j} diagonalisably with finitely many nonzero eigenvalues $c_1, \dots, c_r$, and $P(f^j e^j)$ inverts it for the Lagrange polynomial with $P(c_a) = c_a^{-1}$. Then $P(f^j e^j) f^j \circ e^j$ is the identity of V_{-j} , so e^j is injective there; as $\dim V_{-j} = \dim V_j$, it is bijective, with inverse $P(f^j e^j) f^j|_{V_j}$. $\square$

Proposition 73.3.6. Assume $B(X)$. Then:

1. $C(X)$ holds;
2. for $p \leq n/2$, $L^{n-2p}: A_{\text{hom}}^p(X) \rightarrow A_{\text{hom}}^{n-p}(X)$ is bijective (Grothendieck's conjecture $A(X, L)$);
3. if $x = \sum_j L^j x_j$ is the Lefschetz decomposition of an algebraic class, with x_j primitive, then every x_j is algebraic.

Proof. By $B(X)$, L and Λ are algebraic, hence so is every noncommutative polynomial in them, in particular $h_{\text{deg}} = [L, \Lambda]$ and the Casimir element. (1) By Lemma 73.3.5(1), the projector onto H^i is a polynomial in h_{deg} . (2) L^{n-2p} maps algebraic classes to algebraic classes and is injective on H^{2p} by hard Lefschetz. By Lemma 73.3.5(3), its inverse $H^{2n-2p} \rightarrow H^{2p}$ is the restriction of an algebraic endomorphism, so it maps algebraic classes to algebraic classes. (3) The summand $L^j P^{i-2j}$ of H^i is the intersection of H^i with the isotypic component of $V(n - i + 2j)$, since a primitive class in P^{i-2j} is a lowest weight vector of weight $i - 2j - n$. By Lemma 73.3.5(1),(2) the projector onto $L^j P^{i-2j}$ is algebraic, so $L^j x_j$ is algebraic. By Lemma 73.3.5(4), $x_j = c^{-1} \Lambda^j L^j x_j$ for a nonzero integer c , so x_j is algebraic. $\square$

Proposition 73.3.7. *If $k \subseteq \mathbb{C}$ and $H^\bullet$ is Betti cohomology, then $\text{Hdg}(X)$ holds.*

Proof. Algebraic classes in $H^{2p}(X_{\mathbb{C}}; \mathbb{Q})$ are rational classes of type (p, p) (Corollary 59.4.2). The class h is represented by a Kähler form ω , the restriction of a multiple of the Fubini–Study form. For a nonzero primitive rational class x of type (p, p) , the Hodge–Riemann relation (Theorem 46.4.2(4)) with $k = 2p$ and $\bar{x} = x$ reads $Q(x, x) > 0$, where $Q(x, x) = (-1)^{2p(2p-1)/2} \int x \cup x \cup [\omega]^{n-2p} = (-1)^p \int L^{n-2p}(x \cup x)$, since $p(2p-1) \equiv p \pmod{2}$. $\square$

Proposition 73.3.8. *$B(X)$ and $\text{Hdg}(X)$ together imply $D(X)$.*

Proof. First let $p \leq n/2$, and put $\langle x, y \rangle = \int L^{n-2p}(x \cup y)$ on H^{2p} . The Lefschetz decomposition is orthogonal for $\langle \cdot, \cdot \rangle$: for $a \in P^{2p-2j}$, $b \in P^{2p-2j'}$ and $j < j'$, $\langle L^j a, L^{j'} b \rangle = \int b \cup L^{n-2p+j+j'} a$ vanishes because $n - 2p + j + j' \geq n - (2p - 2j) + 1$, so $L^{n-2p+j+j'} a = 0$ by primitivity. Let $x \in A_{\text{hom}}^p(X)$ with $\langle x, y \rangle = 0$ for every $y \in A_{\text{hom}}^p(X)$, and write $x = \sum_j L^j x_j$. By Proposition 73.3.6(3) the x_j are algebraic, so $y = L^j x_j$ is allowed, and $0 = \langle x, L^j x_j \rangle = \langle L^j x_j, L^j x_j \rangle = \int L^{n-2(p-j)}(x_j \cup x_j)$. By $\text{Hdg}(X)$ in codimension $p - j$, $x_j = 0$. Hence $\langle \cdot, \cdot \rangle$ is nondegenerate on $A_{\text{hom}}^p(X)$. Now if $x \in A_{\text{hom}}^p(X)$ is numerically trivial, then $\langle x, y \rangle = \deg(x \cdot h^{n-2p} \cdot y) = 0$ for all algebraic y , so $x = 0$.

Now let $p > n/2$ and let $x \in A_{\text{hom}}^p(X)$ be numerically trivial. By Proposition 73.3.6(2) applied to $n - p < n/2$, $L^{2p-n}: A_{\text{hom}}^{n-p} \rightarrow A_{\text{hom}}^p$ is bijective, so $x = L^{2p-n} y$ with y algebraic, and every algebraic class of codimension p is $L^{2p-n} u$ with $u \in A_{\text{hom}}^{n-p}$. Then $\deg(y \cdot L^{2p-n} u) = \deg(x \cdot u) = 0$ for all such u , so y is numerically trivial, hence $y = 0$ by the first case, and $x = 0$. $\square$

Theorem 73.3.9. *Let $k \subseteq \mathbb{C}$ and let $H^\bullet$ be Betti cohomology. If the Hodge conjecture holds for $X \times X$, then $B(X)$, $C(X)$ and $D(X)$ hold.*

Proof. The operator Λ is defined over $\mathbb{Q}$ and is a morphism of Hodge structures of type $(-1, -1)$ (see the proof of Corollary 71.4.3), so its class in $H^{2n-2}(X \times X; \mathbb{Q})$ is a Hodge class (Proposition 62.6.2). By the Hodge conjecture it is algebraic, which is $B(X)$. Then $C(X)$ follows from Proposition 73.3.6, and $D(X)$ from Propositions 73.3.7 and 73.3.8. $\square$

Remark 73.3.10. Grothendieck showed that the standard conjectures for X and $X \times X$ over a finite field imply the Riemann hypothesis part of the Weil conjectures, by an argument modelled on Serre’s Kähler analogue: the graph of Frobenius is an algebraic correspondence, and the positivity of $\text{Hdg}(X \times X)$ bounds its eigenvalues [Gro69b], [Kle68]. Deligne’s proof [Del74a] avoided the standard conjectures. Katz and Messing then deduced $C(X)$ over finite fields from Deligne’s theorem [KM74]. In positive characteristic, $\text{Hdg}(X)$ is known in general only for surfaces, where it is the Hodge index theorem [Har77, Theorem V.1.9].

Theorem 73.3.11 (Known cases). 1. *$B(X)$ holds if $\dim X \leq 2$, if X is an abelian variety [Lie68], for products of varieties for which it holds [Kle68], for varieties whose cohomology is spanned by algebraic classes (projective spaces, Grassmannians, flag varieties), for smooth hypersurfaces and complete intersections in projective space (Exercise 73.6.3), and for hyper-Kähler varieties deformation equivalent to Hilbert schemes of points on K3 surfaces [CM13].*

2. *$C(X)$ holds over finite fields [KM74], and whenever $B(X)$ holds.*

3. *$D(X)$ holds for divisors in any characteristic, and in characteristic 0 for abelian varieties [Lie68].*

4. *$\text{Hdg}(X)$ holds in characteristic 0 (Proposition 73.3.7) and for surfaces in any characteristic.*

See [Kle94] for a survey and for the classical cases. For a surface, the components of Λ on $H^2 \rightarrow H^0$ and $H^4 \rightarrow H^2$ are expressed through h and the class of a point, and the component $H^3 \rightarrow H^1$, the inverse of $L: H^1 \rightarrow H^3$, is induced by a divisorial correspondence built from the Picard and Albanese varieties. Milne showed that Hdg for abelian varieties in positive characteristic would follow from the Hodge conjecture for complex abelian varieties of CM type [Mil02].

Remark 73.3.12. $B(X)$ asks for the algebraicity of one specific Hodge class on $X \times X$, while the Hodge conjecture asks for all of them, and still $B(X)$ is open for most varieties of dimension ≥ 3 . For a threefold, $B(X)$ essentially amounts to the algebraicity of the inverse of $L: H^2(X) \rightarrow H^4(X)$, a class in $H^2(X) \otimes H^2(X)$ whose transcendental part is not accessible by divisors. The class of Λ is absolute Hodge (Corollary 71.4.3), but no algebraic cycle representing it is known in general.

73.4 Pure motives

Definition 73.4.1. Let $\sim$ be rational, homological or numerical equivalence. The category $\mathcal{M}_\sim(k)$ of *pure motives* over k (with $\mathbb{Q}$ -coefficients) has as objects triples (X, e, m) with X a nonsingular projective variety of pure dimension, $e \in A_\sim^{\dim X}(X \times X)$ an idempotent, $e \circ e = e$, and $m \in \mathbb{Z}$, together with their finite formal direct sums. Morphisms between triples are

$$\text{Hom}((X, e, m), (Y, f, m')) = f \circ A_\sim^{\dim X + m' - m}(X \times Y) \circ e,$$

with composition of correspondences, and are extended to direct sums by matrices. The *motive* of X is $h(X) = (X, \Delta_X, 0)$, and a morphism $\phi: X \rightarrow Y$ gives $h(\phi) = {}^t\Gamma_\phi: h(Y) \rightarrow h(X)$. The *unit motive* is $\mathbf{1} = h(\text{Spec } k)$, the *Lefschetz motive* is $\mathbb{L} = (\text{Spec } k, \text{id}, -1)$, and the *Tate twist* is $(X, e, m)(r) = (X, e, m + r)$.

Composition is well defined because $\sim$ is compatible with pushforward, pullback and intersection product (Proposition 58.5.2). The category $\mathcal{M}_\sim(k)$ is $\mathbb{Q}$ -linear and additive, and *pseudo-abelian*: an idempotent endomorphism $\phi = f \circ \phi \circ f$ of (X, f, m) has image (X, ϕ, m) . The product $(X, e, m) \otimes (Y, f, m') = (X \times Y, e \times f, m + m')$ makes it a tensor category, with $(X, e, m) \otimes \mathbb{L} = (X, e, m - 1)$ [Sch94].

Example 73.4.2. In $\mathcal{M}_{\text{rat}}(k)$, $h(\mathbb{P}^1) \cong \mathbf{1} \oplus \mathbb{L}$. Let x be a k -point of $\mathbb{P}^1$, and put $e_0 = [x \times \mathbb{P}^1]$ and $e_2 = [\mathbb{P}^1 \times x]$ in $\text{CH}^1(\mathbb{P}^1 \times \mathbb{P}^1)$. The diagonal has bidegree $(1, 1)$ and $\text{CH}^1(\mathbb{P}^1 \times \mathbb{P}^1)$ is free on e_0 and e_2 , so $\Delta = e_0 + e_2$. By the composition formula of Definition 58.5.1, on $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{P}^1$:

1. $e_0 \circ e_0 = \text{pr}_{13*}([x \times \mathbb{P}^1 \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times x \times \mathbb{P}^1]) = \text{pr}_{13*}[x \times x \times \mathbb{P}^1] = e_0$;
2. $e_2 \circ e_2 = \text{pr}_{13*}([\mathbb{P}^1 \times x \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times \mathbb{P}^1 \times x]) = \text{pr}_{13*}[\mathbb{P}^1 \times x \times x] = e_2$;
3. $e_2 \circ e_0 = \text{pr}_{13*}([x \times \mathbb{P}^1 \times \mathbb{P}^1] \cdot [\mathbb{P}^1 \times \mathbb{P}^1 \times x]) = \text{pr}_{13*}[x \times \mathbb{P}^1 \times x] = 0$, since pr_{13} contracts the middle factor;
4. $e_0 \circ e_2 = \text{pr}_{13*}([\mathbb{P}^1 \times x \times \mathbb{P}^1]^2) = 0$, since $[x]^2 = 0$ in $\text{CH}^\bullet(\mathbb{P}^1)$.

So e_0, e_2 are orthogonal idempotents with sum Δ , and $h(\mathbb{P}^1) = (\mathbb{P}^1, e_0, 0) \oplus (\mathbb{P}^1, e_2, 0)$. The correspondences $a = [\mathbb{P}^1] \in \text{CH}^0(\text{Spec } k \times \mathbb{P}^1)$ and $b = [x] \in \text{CH}^1(\mathbb{P}^1 \times \text{Spec } k)$ satisfy $b \circ a = \text{id}_1$ and $a \circ b = e_0$, so $(\mathbb{P}^1, e_0, 0) \cong \mathbf{1}$. Similarly $a' = [x] \in \text{CH}^1(\text{Spec } k \times \mathbb{P}^1)$, a morphism $\mathbb{L} \rightarrow h(\mathbb{P}^1)$, and $b' = [\mathbb{P}^1] \in \text{CH}^0(\mathbb{P}^1 \times \text{Spec } k)$, a morphism $h(\mathbb{P}^1) \rightarrow \mathbb{L}$, satisfy $b' \circ a' = \text{id}_\mathbb{L}$ and $a' \circ b' = e_2$, so $(\mathbb{P}^1, e_2, 0) \cong \mathbb{L}$. In cohomology, e_0 and e_2 are the Künneth components π_0 and π_2 .

Theorem 73.4.3 (Jannsen). $\mathcal{M}_{\text{num}}(k)$ is a semisimple abelian category. Conversely, if $\sim$ is an adequate equivalence relation such that $\mathcal{M}_\sim(k)$ is semisimple abelian, then $\sim$ is numerical equivalence.

Proof of the first statement. Step 1: for every X of dimension n , the algebra $E = A_{\text{num}}^n(X \times X)$, with composition of correspondences, is semisimple. It is finite-dimensional over $\mathbb{Q}$ by Lemma 73.2.2, so its Jacobson radical R is a nilpotent two-sided ideal. Let $\alpha \in Z^n(X \times X) \otimes \mathbb{Q}$ have image in R , and let β be any cycle of dimension n on $X \times X$. Put $\gamma = \beta \circ \alpha$. Its image lies in R , so the image of $\gamma^{\circ s}$ in E is 0 for some $s \geq 1$, and then so is the image of $\gamma^{\circ t}$ for all $t \geq s$. By Proposition 73.2.1(3) with ${}^t\Delta = \Delta$,

$$\text{str}((\gamma_*)^t) = \int \text{cl}(\gamma^{\circ t}) \cup \text{cl}(\Delta) = \deg(\gamma^{\circ t} \cdot \Delta) = 0 \quad (t \geq s),$$

because $\gamma^{\circ t}$ is numerically trivial. Let $\lambda_1, \dots, \lambda_a$ and $\mu_1, \dots, \mu_b$ be the eigenvalues, with multiplicity, of γ_* on H^{even} and on H^{odd} , in an algebraic closure of K . Then $\sum_i \lambda_i^t - \sum_j \mu_j^t = 0$ for $t \geq s$, so the rational function

$$R(z) = \sum_i \frac{\lambda_i z}{1 - \lambda_i z} - \sum_j \frac{\mu_j z}{1 - \mu_j z} = \sum_{t \geq 1} \left(\sum_i \lambda_i^t - \sum_j \mu_j^t \right) z^t$$

is a polynomial. Its poles at the points $1/\lambda$, for nonzero eigenvalues λ , must therefore cancel, which means that every nonzero value occurs equally often among the λ_i and the μ_j . Hence $\sum_i \lambda_i^t = \sum_j \mu_j^t$ for all $t \geq 1$, and in particular $\text{str}(\gamma_*) = 0$. By the trace formula again, $\deg(\alpha \cdot {}^t\beta) = \text{str}((\beta \circ \alpha)_*) = 0$. As β was arbitrary and every cycle of dimension n on $X \times X$ is of the form ${}^t\beta$, α is numerically trivial. So $R = 0$, and E is semisimple.

Step 2: every object is a direct summand of some $h(Y)(m)$. The triple (X, e, m) is the image of the idempotent e of $h(X)(m)$. Given finitely many triples (X_i, e_i, m_i) , choose $m \geq \max m_i$. Then $(X_i, e_i, m_i) = (X_i, e_i, m) \otimes \mathbb{L}^{\otimes(m-m_i)}$, and $\mathbb{L}^{\otimes r}$ is a summand of $h(\mathbb{P}^1)^{\otimes r} = h((\mathbb{P}^1)^r)$ by Example 73.4.2 (which holds modulo numerical equivalence as well). So each (X_i, e_i, m_i) is a summand of $h(X_i \times (\mathbb{P}^1)^{m-m_i})(m)$, and their direct sum is a summand of $h(Y)(m)$ with Y the disjoint union of the $X_i \times (\mathbb{P}^1)^{m-m_i}$.

Step 3: conclusion. By Step 2, the endomorphism algebra of an object M is $\varepsilon E \varepsilon$ for an idempotent ε of $E = \text{End}(h(Y)(m)) = A_{\text{num}}^{\dim Y}(Y \times Y)$ (for disconnected Y , a product of such algebras over pairs of components, with the same argument). A corner $\varepsilon E \varepsilon$ of a finite-dimensional semisimple algebra is semisimple. Finally, a $\mathbb{Q}$ -linear additive pseudo-abelian category in which every object has a finite-dimensional semisimple endomorphism algebra is abelian and semisimple [Jan92]. $\square$

The converse statement is also in [Jan92]. Jannsen's theorem was a surprise: Grothendieck expected semisimplicity to require the standard conjectures.

Remark 73.4.4. What the standard conjectures are still needed for is the *Tannakian* structure. The natural commutativity constraint on $\mathcal{M}_{\text{num}}(k)$ gives $h(X)$ the wrong dimension (the supertrace $\sum (-1)^i b_i$, which can be negative); to correct it one changes the sign of the constraint on the odd part, which requires the Künneth projectors to be algebraic modulo numerical equivalence. With C and D , $\mathcal{M}_{\text{num}}(k) = \mathcal{M}_{\text{hom}}(k)$ is a semisimple Tannakian category, the realisation $H^\bullet$ is a fibre functor, and $\mathcal{M}_{\text{num}}(k)$ is the category of representations of a pro-reductive group, the *motivic Galois group* [Jan92], [And04], [DM82]. Unconditional substitutes exist: Deligne's motives for absolute Hodge cycles [DM82] and André's motives for motivated cycles [And96b] are Tannakian and contain the motives of abelian varieties.

Proposition 73.4.5 (The Hodge conjecture, motivically). *Let $k = \mathbb{C}$ and let $\sim$ be homological equivalence for Betti cohomology. The Hodge realisation, which sends (X, e, m) to the graded polarisable $\mathbb{Q}$ -Hodge structure $e_* H^\bullet(X; \mathbb{Q})(m)$, with $H^i(X)(m)$ placed in degree $i - 2m$, is a faithful functor from $\mathcal{M}_{\text{hom}}(\mathbb{C})$ to graded polarisable $\mathbb{Q}$ -Hodge structures. It is full if and only if the Hodge conjecture holds for all nonsingular complex projective varieties.*

Proof. A morphism $h(X)(m) \rightarrow h(Y)(m')$ is an element $\gamma \in A_{\text{hom}}^{n+r}(X \times Y)$ with $n = \dim X$, $r = m' - m$, and it acts by $\gamma_*: H^i(X)(m) \rightarrow H^{i+2r}(Y)(m')$, which preserves the grading. Faithfulness is Proposition 73.2.1(1): an element of A_{hom} is determined by its class, which is determined by its action. Under the isomorphism of Proposition 73.2.1(1), which is compatible with Hodge structures up to the Tate twist, the grading-preserving morphisms of Hodge structures $\bigoplus_i H^i(X)(m) \rightarrow \bigoplus_i H^{i+2r}(Y)(m')$ correspond to the Hodge classes in $H^{2n+2r}(X \times Y; \mathbb{Q})$ (Proposition 62.6.2), and the morphisms of motives to the algebraic classes. So the realisation is full on the objects $h(X)(m)$ if and only if $\text{Hdg}^{n+r}(X \times Y) = Z_{\text{alg}}^{n+r}(X \times Y)$ for all X, Y, r . Taking $X = \text{Spec } \mathbb{C}$ gives the Hodge conjecture for Y ; conversely the Hodge conjecture for all varieties includes $X \times Y$. Fullness for general objects follows, since they are summands of the $h(X)(m)$ and morphisms between summands are cut out by the idempotents. $\square$

Remark 73.4.6. Proposition 73.4.5 is the cleanest statement of what the Hodge conjecture is for: the Hodge structure of a variety would determine its motive, and every Hodge-theoretic relation between varieties would be realised by correspondences. Under the Hodge conjecture, the standard conjectures hold (Theorem 73.3.9), $\mathcal{M}_{\text{hom}}(\mathbb{C})$ is Tannakian, and the motivic Galois group of the Tannakian subcategory generated by $h(X)$ is the Mumford–Tate group of $H^\bullet(X; \mathbb{Q})$ (Chapter 63), an identification of a geometric object with a Hodge-theoretic one. If the Hodge conjecture were false, the motives for absolute Hodge cycles (Remark 73.4.4) would survive, but they would no longer be built out of subvarieties.

73.5 The Bloch–Beilinson conjectures

The Hodge conjecture concerns the image of the cycle class map. Its kernel, $\text{CH}^p(X)_{\text{hom}} = \ker(\text{cl})$, is governed by a separate set of conjectures.

Theorem 73.5.1 (Mumford). *Let S be a nonsingular complex projective surface with $h^{2,0}(S) > 0$. Then the group $\text{CH}_0(S)_{\deg 0}$ of 0-cycles of degree 0 modulo rational equivalence is infinite-dimensional: for no variety T and cycle Z on $T \times S$ is the map $T \rightarrow \text{CH}_0(S)_{\deg 0}$, $t \mapsto Z_t - Z_{t_0}$, surjective.*

This is [Mum68]; compare Example 58.4.5. The proof shows that a surjective family would force every holomorphic 2-form to vanish, by pulling it back along the family. Chow groups thus see Hodge-theoretic information that cohomology classes of cycles do not.

Conjecture 73.5.2 (Bloch). *If S is a nonsingular complex projective surface with $h^{2,0}(S) = 0$, then the Albanese map $\text{CH}_0(S)_{\deg 0} \rightarrow \text{Alb}(S)$ is an isomorphism.*

Bloch’s conjecture is known for surfaces not of general type [BKL76] and in scattered cases of general type, and open in general [Blo10]. By the Bloch–Srinivas method (Theorem 68.5.1), its conclusion implies a decomposition of the diagonal, and conversely the conclusion of Mumford’s theorem is the Hodge-theoretic obstruction to one.

Conjecture 73.5.3 (Bloch–Beilinson filtration). *For every nonsingular projective variety X over a field of characteristic 0 and every p , there is a finite decreasing filtration $F^\bullet$ on $\text{CH}^p(X)_{\mathbb{Q}}$ such that:*

1. $F^0 = \text{CH}^p(X)_{\mathbb{Q}}$, $F^1 = \text{CH}^p(X)_{\text{hom}, \mathbb{Q}}$ and $F^{p+1} = 0$;
2. the filtrations are compatible with intersection products, $F^i \cdot F^j \subseteq F^{i+j}$, and with the action of correspondences;
3. a correspondence $\Gamma \in \text{CH}^{\dim X+r}(X \times Y)_{\mathbb{Q}}$ acts on $\text{Gr}_F^v \text{CH}^p(X)_{\mathbb{Q}}$ through its cohomology class, and it acts by 0 if $\Gamma_*: H^{2p-v}(X) \rightarrow H^{2p-v+2r}(Y)$ vanishes.

See [Jan94] and [Blo10]; the formulation is due to Beilinson and Bloch, and Murre gave an equivalent one in terms of a decomposition of the diagonal into orthogonal idempotents lifting the Künneth components to Chow groups [Mur93]. On Gr_F^1 the filtration is expected to be detected by the Abel–Jacobi map (Chapter 67), and Gr_F^ν is expected to be an Ext^ν in a category of mixed motives. Beilinson and Bloch conjectured moreover that for X defined over a number field, $F^2 \mathrm{CH}^p(X_{\bar{\mathbb{Q}}})_{\mathbb{Q}} = 0$, so that the Abel–Jacobi map is injective on $\mathrm{CH}^p(X_{\bar{\mathbb{Q}}})_{\mathrm{hom}, \mathbb{Q}}$ [Bei87].

Remark 73.5.4. Conjecture 73.5.3 gives Bloch’s conjecture up to torsion. Let S be a surface with $h^{2,0}(S) = 0$. By the Lefschetz $(1, 1)$ -theorem, $H^2(S; \mathbb{Q})$ is spanned by classes of divisors, so there is a correspondence $\Gamma = \sum a_{ij} D_i \times D_j$, with divisors D_i and $a_{ij} \in \mathbb{Q}$, that acts on $H^2(S)$ as the identity. On 0-cycles $\Gamma_* = 0$, since $(D_i \times D_j) \cdot (z \times S) = 0$ for a 0-cycle z in general position. The diagonal also acts on H^2 as the identity, so by (3) with $p = \nu = 2$, the identity of $\mathrm{Gr}_F^2 \mathrm{CH}_0(S)_{\mathbb{Q}}$ equals $\Gamma_* = 0$, and $F^2 \mathrm{CH}_0(S)_{\mathbb{Q}} = 0$. The Albanese kernel is expected to lie in F^2 , which is the case when the Künneth component π_3 is realised, as Murre showed it can be, by a correspondence acting through the Albanese variety [Mur93]; then the Albanese kernel is torsion. Kimura’s notion of *finite-dimensional* motives gives another route: motives of curves and abelian varieties are finite-dimensional, and finite-dimensionality of the motive of S would imply Bloch’s conjecture for S [Kim05].

Remark 73.5.5. The Hodge conjecture describes Gr_F^0 : the image of the cycle class map is as large as Hodge theory permits. The Bloch–Beilinson conjectures describe F^1 : the kernel is as structured as motivic expectations require. Neither is known to imply the other, but a complete theory of algebraic cycles needs both, and both are expected to come from a category of mixed motives. The absolute Hodge classes of Chapter 71, the motivated cycles of André and the unconditional results of this chapter all live on the cohomological side; on the side of cycles, the Lefschetz $(1, 1)$ -theorem (Chapter 65) remains the only general method for producing an algebraic cycle from a cohomology class.

73.6 Exercises

Exercise 73.6.1. Let X be a nonsingular complex projective variety. Show that the Künneth components π_i of the class of the diagonal are Hodge classes on $X \times X$, and deduce $C(X)$ from the Hodge conjecture for $X \times X$ without using $B(X)$.

Solution. The class of the diagonal is algebraic, hence a Hodge class. The Künneth decomposition $H^{2n}(X \times X) = \bigoplus_i H^{2n-i}(X) \otimes H^i(X)$ is a decomposition into sub-Hodge structures, and the components of a Hodge class along such a decomposition are Hodge classes (the components of a rational class of type (n, n) are rational and of type (n, n)). The Hodge conjecture for $X \times X$ makes each π_i algebraic. $\square$

Exercise 73.6.2. Show that $h(\mathbb{P}^n) \cong \bigoplus_{i=0}^n \mathbb{L}^{\otimes i}$ in $\mathcal{M}_{\mathrm{rat}}(k)$, and that $B(\mathbb{P}^n)$ holds, by writing Δ explicitly as a correspondence.

Solution. Let h_1, h_2 be the pullbacks of the hyperplane class to $\mathbb{P}^n \times \mathbb{P}^n$ and $p_i = h_1^{n-i} h_2^i \in \mathrm{CH}^n(\mathbb{P}^n \times \mathbb{P}^n)$. Then $(p_i)_*(h^j) = \mathrm{pr}_{2*}(h_1^{j+n-i} h_2^i) = \deg(h^{j+n-i}) h^i = \delta_{ij} h^i$, and $p_a \circ p_b = \mathrm{pr}_{13*}(h_1^{n-b} h_2^b h_3^{n-a} h_3^a) = \deg(h^{n+b-a}) h_1^{n-b} h_3^a = \delta_{ab} p_a$. The group $\mathrm{CH}^n(\mathbb{P}^n \times \mathbb{P}^n)$ is free on the classes $h_1^{n-i} h_2^i$, and $\gamma = \sum_i a_i h_1^{n-i} h_2^i$ acts by $\gamma_*(h^j) = a_j h^j$; as Δ acts as the identity, $\Delta = \sum_i p_i$. The correspondences $a = h^i \in \mathrm{CH}^i(\mathrm{Spec} k \times \mathbb{P}^n)$, a morphism $\mathbb{L}^{\otimes i} \rightarrow h(\mathbb{P}^n)$, and $b = h^{n-i} \in \mathrm{CH}^{n-i}(\mathbb{P}^n \times \mathrm{Spec} k)$, a morphism $h(\mathbb{P}^n) \rightarrow \mathbb{L}^{\otimes i}$, satisfy $b \circ a = \deg(h^n) = 1$ and $a \circ b = h_1^{n-i} h_2^i = p_i$, so $(\mathbb{P}^n, p_i, 0) \cong \mathbb{L}^{\otimes i}$. For B : the only primitive class is $1 \in P^0$,

so $\Lambda(h^j) = j(n - j + 1)h^{j-1}$. The correspondence $\lambda = \sum_{i=1}^n i(n - i + 1)h_1^{n-i}h_2^{i-1}$ satisfies $\lambda_*(h^j) = \sum_i i(n - i + 1) \deg(h^{j+n-i})h^{i-1} = j(n - j + 1)h^{j-1}$, so $\lambda_* = \Lambda$ and Λ is algebraic. $\square$

Exercise 73.6.3. Let $X \subseteq \mathbb{P}^{n+1}$ be a nonsingular hypersurface over $\mathbb{C}$. Show that $B(X)$ holds.

Solution. By the Lefschetz hyperplane theorem and Poincaré duality, $H^i(X; \mathbb{Q})$ is $\mathbb{Q}h^{i/2}$ for $i \neq n$ even, 0 for $i \neq n$ odd, and $H^n = \mathbb{Q}h^{n/2} \oplus H_{\text{prim}}^n$ (with the first summand only for n even), where $H_{\text{prim}}^n = P^n$. The operator Λ kills P^n and acts on the powers of h by $\Lambda(h^i) = i(n - i + 1)h^{i-1}$. Let $d = \deg X = \int h^n$, and consider the algebraic class $\lambda = \frac{1}{d} \sum_{j=1}^n j(n - j + 1)h_1^{n-j}h_2^{j-1} \in H^{2n-2}(X \times X)$, where h_1, h_2 are the pullbacks of h . It acts by $\lambda_*(x) = \frac{1}{d} \sum_j j(n - j + 1) \left(\int_X x \cup h^{n-j} \right) h^{j-1}$, and for $x \in H^i$ only the term with $i = 2j$ can contribute. On h^j it gives $j(n - j + 1)h^{j-1} = \Lambda(h^j)$, because $\int h^n = d$. For $x \in P^n$, the only possible term has $j = n/2$ (so n is even), and $\int x \cup h^{n/2} = \int L^{n/2}x = 0$ because $Lx = 0$ by primitivity (if $n \geq 2$; the case $n = 0$ is trivial). So λ_* and Λ agree on a basis, and Λ is algebraic. The same argument applies to smooth complete intersections, whose cohomology has the same shape. $\square$

Exercise 73.6.4. Explain why Jannsen's theorem does not imply the standard conjecture D , although both concern numerical equivalence. Show, using the converse part of Theorem 73.4.3, that $D(X)$ for all X is equivalent to the semisimplicity of $\mathcal{M}_{\text{hom}}(k)$.

Solution. Jannsen's theorem is a statement about the quotient A_{num} of A_{hom} ; D says that the kernel of $A_{\text{hom}} \rightarrow A_{\text{num}}$ vanishes. A quotient algebra can be semisimple whatever the kernel is, so the first says nothing about the second. If D holds for all varieties, then $\mathcal{M}_{\text{hom}}(k) = \mathcal{M}_{\text{num}}(k)$, which is semisimple abelian by Jannsen. Conversely, if $\mathcal{M}_{\text{hom}}(k)$ is semisimple abelian, the converse part of the theorem says that homological equivalence is numerical equivalence, which is D for all varieties. $\square$

Exercise 73.6.5. Show that $\text{CH}_0(\mathbb{P}^2) = \mathbb{Z}$, via the degree, and $\text{CH}_0(\mathbb{P}^1 \times \mathbb{P}^1) = \mathbb{Z}$, in accordance with Bloch's conjecture.

Solution. Two points of $\mathbb{P}^2$ lie on a line $\ell \cong \mathbb{P}^1$, and any two points of $\mathbb{P}^1$ are rationally equivalent, being the zero and the pole of a rational function; pushing forward along $\ell \rightarrow \mathbb{P}^2$, the two points are rationally equivalent on $\mathbb{P}^2$. So $\text{CH}_0(\mathbb{P}^2)$ is generated by the class of one point, and the degree shows it is $\mathbb{Z}$. On $\mathbb{P}^1 \times \mathbb{P}^1$, the points (a, b) and (a', b') are joined by the chain $(a, b), (a', b), (a', b')$, whose consecutive points lie on a fibre $\cong \mathbb{P}^1$. Both surfaces have $h^{2,0} = 0$ and trivial Albanese, so this is Bloch's conjecture for them. $\square$

Part X

Miscellany

Bibliography

- [Ada60] J. F. Adams, *On the non-existence of elements of Hopf invariant one*, *Annals of Mathematics. Second Series* **72** (1960), 20–104.
- [AGV72] M. Artin, A. Grothendieck, and J.-L. Verdier, *Théorie des topos et cohomologie étale des schémas. Tome 1: Théorie des topos (SGA 4)*, *Lecture Notes in Mathematics*, vol. 269, Springer-Verlag, Berlin, 1972.
- [AGV73] ———, *Théorie des topos et cohomologie étale des schémas. Tome 3 (SGA 4)*, *Lecture Notes in Mathematics*, vol. 305, Springer-Verlag, Berlin, 1973.
- [AH62] M. F. Atiyah and F. Hirzebruch, *Analytic cycles on complex manifolds*, *Topology* **1** (1962), 25–45.
- [Ahl78] Lars V. Ahlfors, *Complex analysis*, third ed., McGraw-Hill Book Co., New York, 1978.
- [AM69] M. F. Atiyah and I. G. Macdonald, *Introduction to commutative algebra*, Addison-Wesley, Reading, MA, 1969.
- [And96a] Yves André, *On the Shafarevich and Tate conjectures for hyper-Kähler varieties*, *Mathematische Annalen* **305** (1996), no. 2, 205–248.
- [And96b] ———, *Pour une théorie inconditionnelle des motifs*, *Publications Mathématiques de l’IHÉS* **83** (1996), 5–49.
- [And04] ———, *Une introduction aux motifs (motifs purs, motifs mixtes, périodes)*, *Panoramas et Synthèses*, vol. 17, Société Mathématique de France, Paris, 2004.
- [Art42] Emil Artin, *Galois theory*, *Notre Dame Mathematical Lectures*, vol. 2, University of Notre Dame, Notre Dame, IN, 1942.
- [AS68a] Michael F. Atiyah and Isadore M. Singer, *The index of elliptic operators. I*, *Annals of Mathematics. Second Series* **87** (1968), 484–530.
- [AS68b] ———, *The index of elliptic operators. III*, *Annals of Mathematics. Second Series* **87** (1968), 546–604.
- [Axl15] Sheldon Axler, *Linear algebra done right*, third ed., *Undergraduate Texts in Mathematics*, Springer, Cham, 2015.
- [BBD82] A. A. Beilinson, J. Bernstein, and P. Deligne, *Faisceaux pervers*, *Astérisque* **100** (1982), 5–171.
- [Bea83] Arnaud Beauville, *Variétés Kähleriennes dont la première classe de Chern est nulle*, *Journal of Differential Geometry* **18** (1983), no. 4, 755–782.

- [Bea96] ———, *Complex algebraic surfaces*, second ed., London Mathematical Society Student Texts, vol. 34, Cambridge University Press, Cambridge, 1996.
- [Bei87] Alexander A. Beilinson, *Height pairing between algebraic cycles*, K-theory, Arithmetic and Geometry (Moscow, 1984–1986), Lecture Notes in Mathematics, vol. 1289, Springer-Verlag, Berlin, 1987, pp. 1–25.
- [Ber55] Marcel Berger, *Sur les groupes d'holonomie homogène des variétés à connexion affine et des variétés riemanniennes*, Bulletin de la Société Mathématique de France **83** (1955), 279–330.
- [BHPVdV04] Wolf P. Barth, Klaus Hulek, Chris A. M. Peters, and Antonius Van de Ven, *Compact complex surfaces*, second ed., Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 4, Springer-Verlag, Berlin, 2004.
- [BKL76] Spencer Bloch, Arnold Kas, and David Lieberman, *Zero cycles on surfaces with $p_g = 0$* , Compositio Mathematica **33** (1976), no. 2, 135–145.
- [BKT20] Benjamin Bakker, Bruno Klingler, and Jacob Tsimerman, *Tame topology of arithmetic quotients and algebraicity of Hodge loci*, Journal of the American Mathematical Society **33** (2020), no. 4, 917–939.
- [BL04] Christina Birkenhake and Herbert Lange, *Complex abelian varieties*, second ed., Grundlehren der Mathematischen Wissenschaften, vol. 302, Springer-Verlag, Berlin, 2004.
- [Blo72] Spencer Bloch, *Semi-regularity and de Rham cohomology*, Inventiones Mathematicae **17** (1972), 51–66.
- [Blo10] ———, *Lectures on algebraic cycles*, second ed., New Mathematical Monographs, vol. 16, Cambridge University Press, Cambridge, 2010.
- [Bor91] Armand Borel, *Linear algebraic groups*, second ed., Graduate Texts in Mathematics, vol. 126, Springer-Verlag, New York, 1991.
- [Bre97] Glen E. Bredon, *Sheaf theory*, second ed., Graduate Texts in Mathematics, vol. 170, Springer-Verlag, New York, 1997.
- [Bre11] Haim Brezis, *Functional analysis, sobolev spaces and partial differential equations*, Universitext, Springer, New York, 2011.
- [BS83] Spencer Bloch and V. Srinivas, *Remarks on correspondences and algebraic cycles*, American Journal of Mathematics **105** (1983), no. 5, 1235–1253.
- [BT82] Raoul Bott and Loring W. Tu, *Differential forms in algebraic topology*, Graduate Texts in Mathematics, vol. 82, Springer-Verlag, New York-Berlin, 1982.
- [BtD85] Theodor Bröcker and Tammo tom Dieck, *Representations of compact lie groups*, Graduate Texts in Mathematics, vol. 98, Springer-Verlag, New York, 1985.
- [Bus19] Nikolay Buskin, *Every rational Hodge isometry between two K3 surfaces is algebraic*, Journal für die reine und angewandte Mathematik **755** (2019), 127–150.
- [Car80] James A. Carlson, *Extensions of mixed Hodge structures*, Journées de Géométrie Algébrique d'Angers, Juillet 1979, Sijthoff & Noordhoff, Alphen aan den Rijn, 1980.

- [CDK95] Eduardo Cattani, Pierre Deligne, and Aroldo Kaplan, *On the locus of Hodge classes*, Journal of the American Mathematical Society **8** (1995), no. 2, 483–506.
- [CdLOGP91] Philip Candelas, Xenia C. de la Ossa, Paul S. Green, and Linda Parkes, *A pair of Calabi–Yau manifolds as an exactly soluble superconformal theory*, Nuclear Physics B **359** (1991), no. 1, 21–74.
- [CE53] Eugenio Calabi and Beno Eckmann, *A class of compact, complex manifolds which are not algebraic*, Annals of Mathematics. Second Series **58** (1953), 494–500.
- [CE56] Henri Cartan and Samuel Eilenberg, *Homological algebra*, Princeton University Press, Princeton, NJ, 1956.
- [CG72] C. Herbert Clemens and Phillip A. Griffiths, *The intermediate Jacobian of the cubic threefold*, Annals of Mathematics. Second Series **95** (1972), 281–356.
- [Cha13] François Charles, *The Tate conjecture for K3 surfaces over finite fields*, Inventiones Mathematicae **194** (2013), no. 1, 119–145.
- [CK90] C. C. Chang and H. J. Keisler, *Model theory*, third ed., Studies in Logic and the Foundations of Mathematics, vol. 73, North-Holland, Amsterdam, 1990.
- [Cle83] Herbert Clemens, *Homological equivalence, modulo algebraic equivalence, is not finitely generated*, Publications Mathématiques de l’IHÉS **58** (1983), 19–38.
- [CM78] Alberto Conte and Jacob P. Murre, *The Hodge conjecture for fourfolds admitting a covering by rational curves*, Mathematische Annalen **238** (1978), no. 1, 79–88.
- [CM13] François Charles and Eyal Markman, *The standard conjectures for holomorphic symplectic varieties deformation equivalent to Hilbert schemes of K3 surfaces*, Compositio Mathematica **149** (2013), no. 3, 481–494.
- [CMSP03] James Carlson, Stefan Müller-Stach, and Chris Peters, *Period mappings and period domains*, Cambridge Studies in Advanced Mathematics, vol. 85, Cambridge University Press, Cambridge, 2003.
- [Coh63] Paul J. Cohen, *The independence of the continuum hypothesis*, Proceedings of the National Academy of Sciences of the U.S.A. **50** (1963), 1143–1148.
- [Con78] John B. Conway, *Functions of one complex variable I*, second ed., Graduate Texts in Mathematics, vol. 11, Springer-Verlag, New York, 1978.
- [CS14] François Charles and Christian Schnell, *Notes on absolute Hodge classes*, Hodge Theory, Mathematical Notes, vol. 49, Princeton University Press, Princeton, NJ, 2014.
- [CTV12] Jean-Louis Colliot-Thélène and Claire Voisin, *Cohomologie non ramifiée et conjecture de Hodge entière*, Duke Mathematical Journal **161** (2012), no. 5, 735–801.
- [dC92] Manfredo Perdigão do Carmo, *Riemannian geometry*, Mathematics: Theory & Applications, Birkhäuser Boston, Boston, MA, 1992.
- [Ded72] Richard Dedekind, *Stetigkeit und irrationale Zahlen*, Vieweg, Braunschweig, 1872.
- [Del68] Pierre Deligne, *Théorème de Lefschetz et critères de dégénérescence de suites spectrales*, Publications Mathématiques de l’IHÉS **35** (1968).

- [Del71] ———, *Théorie de Hodge. II*, Institut des Hautes Études Scientifiques. Publications Mathématiques **40** (1971), 5–57.
- [Del72] ———, *La conjecture de Weil pour les surfaces K3*, *Inventiones Mathematicae* **15** (1972), 206–226.
- [Del74a] ———, *La conjecture de Weil. I*, *Publications Mathématiques de l’IHÉS* **43** (1974), 273–307.
- [Del74b] ———, *Théorie de Hodge. III*, Institut des Hautes Études Scientifiques. Publications Mathématiques **44** (1974), 5–77.
- [Del77] ———, *Cohomologie étale (SGA 4 $\frac{1}{2}$)*, *Lecture Notes in Mathematics*, vol. 569, Springer-Verlag, Berlin, 1977.
- [Del80] ———, *La conjecture de Weil. II*, *Publications Mathématiques de l’IHÉS* **52** (1980), 137–252.
- [Del82] ———, *Hodge cycles on abelian varieties (notes by J. S. Milne)*, *Hodge Cycles, Motives, and Shimura Varieties*, *Lecture Notes in Mathematics*, vol. 900, Springer-Verlag, Berlin, 1982, pp. 9–100.
- [Del06] ———, *The Hodge conjecture*, *The Millennium Prize Problems*, Clay Math. Inst., Cambridge, MA, 2006, pp. 45–53.
- [DF04] David S. Dummit and Richard M. Foote, *Abstract algebra*, third ed., John Wiley & Sons, Hoboken, NJ, 2004.
- [DGMS75] Pierre Deligne, Phillip Griffiths, John Morgan, and Dennis Sullivan, *Real homotopy theory of Kähler manifolds*, *Inventiones Mathematicae* **29** (1975), no. 3, 245–274.
- [DI87] Pierre Deligne and Luc Illusie, *Relèvements modulo p^2 et décomposition du complexe de de Rham*, *Inventiones Mathematicae* **89** (1987), no. 2, 247–270.
- [Die60] J. Dieudonné, *Foundations of modern analysis*, *Pure and Applied Mathematics*, vol. 10, Academic Press, New York-London, 1960.
- [DM82] Pierre Deligne and James S. Milne, *Tannakian categories*, *Hodge Cycles, Motives, and Shimura Varieties*, *Lecture Notes in Mathematics*, vol. 900, Springer-Verlag, Berlin, 1982, pp. 101–228.
- [dR84] Georges de Rham, *Differentiable manifolds*, *Grundlehren der mathematischen Wissenschaften*, vol. 266, Springer-Verlag, Berlin, 1984.
- [Eis95] David Eisenbud, *Commutative algebra with a view toward algebraic geometry*, *Graduate Texts in Mathematics*, vol. 150, Springer-Verlag, New York, 1995.
- [EML45] Samuel Eilenberg and Saunders Mac Lane, *General theory of natural equivalences*, *Transactions of the American Mathematical Society* **58** (1945), 231–294.
- [End01] Herbert B. Enderton, *A mathematical introduction to logic*, second ed., Harcourt/Academic Press, Burlington, MA, 2001.
- [Eng89] Ryszard Engelking, *General topology*, revised and completed ed., *Sigma Series in Pure Mathematics*, vol. 6, Heldermann Verlag, Berlin, 1989.

- [Fal83] Gerd Faltings, *Endlichkeitssätze für abelsche Varietäten über Zahlkörpern*, *Inventiones Mathematicae* **73** (1983), no. 3, 349–366.
- [FH91] William Fulton and Joe Harris, *Representation theory. A first course*, Graduate Texts in Mathematics, vol. 129, Springer-Verlag, New York, 1991.
- [Fol99] Gerald B. Folland, *Real analysis: Modern techniques and their applications*, second ed., Pure and Applied Mathematics, John Wiley & Sons, New York, 1999.
- [Ful98] William Fulton, *Intersection theory*, second ed., *Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge*, vol. 2, Springer-Verlag, Berlin, 1998.
- [G30] Kurt Gödel, *Die Vollständigkeit der Axiome des logischen Funktionenkalküls*, *Monatshefte für Mathematik und Physik* **37** (1930), 349–360.
- [G31] ———, *Über formal unentscheidbare Sätze der Principia Mathematica und verwandter Systeme I*, *Monatshefte für Mathematik und Physik* **38** (1931), 173–198.
- [G38] ———, *The consistency of the axiom of choice and of the generalized continuum-hypothesis*, *Proceedings of the National Academy of Sciences of the U.S.A.* **24** (1938), 556–557.
- [GG07] Mark Green and Phillip Griffiths, *Algebraic cycles and singularities of normal functions*, *Algebraic Cycles and Motives. Vol. 1*, London Mathematical Society Lecture Note Series, vol. 343, Cambridge University Press, Cambridge, 2007, pp. 206–263.
- [GGK12] Mark Green, Phillip Griffiths, and Matt Kerr, *Mumford–Tate groups and domains: Their geometry and arithmetic*, *Annals of Mathematics Studies*, vol. 183, Princeton University Press, Princeton, NJ, 2012.
- [GH78] Phillip Griffiths and Joseph Harris, *Principles of algebraic geometry*, Wiley-Interscience, New York, 1978.
- [GH85] ———, *On the Noether–Lefschetz theorem and some remarks on codimension-two cycles*, *Mathematische Annalen* **271** (1985), no. 1, 31–51.
- [GHJ03] Mark Gross, Daniel Huybrechts, and Dominic Joyce, *Calabi–Yau manifolds and related geometries*, Universitext, Springer-Verlag, Berlin, 2003.
- [GM03] Sergei I. Gelfand and Yuri I. Manin, *Methods of homological algebra*, second ed., Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2003.
- [God58] Roger Godement, *Topologie algébrique et théorie des faisceaux*, Hermann, Paris, 1958.
- [GR65] Robert C. Gunning and Hugo Rossi, *Analytic functions of several complex variables*, Prentice-Hall, Englewood Cliffs, NJ, 1965.
- [GR84] Hans Grauert and Reinhold Remmert, *Coherent analytic sheaves*, *Grundlehren der mathematischen Wissenschaften*, vol. 265, Springer-Verlag, Berlin, 1984.
- [Gre78] Werner Greub, *Multilinear algebra*, second ed., Universitext, Springer-Verlag, New York, 1978.
- [Gri68] Phillip A. Griffiths, *Periods of integrals on algebraic manifolds. I*, *American Journal of Mathematics* **90** (1968), 568–626.

- [Gri69] ———, *On the periods of certain rational integrals. I, II*, Annals of Mathematics. Second Series **90** (1969), 460–495, 496–541.
- [Gro66] Alexander Grothendieck, *On the de Rham cohomology of algebraic varieties*, Publications Mathématiques de l’IHÉS **29** (1966), 95–103.
- [Gro69a] ———, *Hodge’s general conjecture is false for trivial reasons*, Topology **8** (1969), 299–303.
- [Gro69b] ———, *Standard conjectures on algebraic cycles*, Algebraic Geometry (Internat. Colloq., Tata Inst. Fund. Res., Bombay, 1968), Oxford University Press, London, 1969, pp. 193–199.
- [Gro71] ———, *Revêtements étales et groupe fondamental (SGA 1)*, Lecture Notes in Mathematics, vol. 224, Springer-Verlag, Berlin, 1971.
- [Hal60] Paul R. Halmos, *Naïve set theory*, D. Van Nostrand, Princeton, NJ, 1960.
- [Hal74] ———, *Finite-dimensional vector spaces*, second ed., Undergraduate Texts in Mathematics, Springer-Verlag, New York, 1974.
- [Har77] Robin Hartshorne, *Algebraic geometry*, Graduate Texts in Mathematics, vol. 52, Springer-Verlag, New York-Heidelberg, 1977.
- [Har02] Philip Hartman, *Ordinary differential equations*, Classics in Applied Mathematics, vol. 38, Society for Industrial and Applied Mathematics (SIAM), Philadelphia, PA, 2002, Corrected reprint of the second (1982) edition.
- [Hat02] Allen Hatcher, *Algebraic topology*, Cambridge University Press, Cambridge, 2002.
- [Hen49] Leon Henkin, *The completeness of the first-order functional calculus*, Journal of Symbolic Logic **14** (1949), 159–166.
- [Hir64] Heisuke Hironaka, *Resolution of singularities of an algebraic variety over a field of characteristic zero. I, II*, Annals of Mathematics. Second Series **79** (1964), 109–203, 205–326.
- [Hir66] Friedrich Hirzebruch, *Topological methods in algebraic geometry*, third ed., Grundlehren der mathematischen Wissenschaften, vol. 131, Springer-Verlag, New York, 1966.
- [Hir76] Morris W. Hirsch, *Differential topology*, Graduate Texts in Mathematics, vol. 33, Springer-Verlag, New York-Heidelberg, 1976.
- [Hod52] W. V. D. Hodge, *The topological invariants of algebraic varieties*, Proceedings of the International Congress of Mathematicians, Cambridge, Mass., 1950, vol. 1, Amer. Math. Soc., Providence, R. I., 1952, pp. 182–192.
- [Hör90a] Lars Hörmander, *The analysis of linear partial differential operators. I*, second ed., Grundlehren der mathematischen Wissenschaften, vol. 256, Springer-Verlag, Berlin, 1990.
- [Hör90b] ———, *An introduction to complex analysis in several variables*, third ed., North-Holland Mathematical Library, vol. 7, North-Holland Publishing Co., Amsterdam, 1990.

- [Hum72] James E. Humphreys, *Introduction to lie algebras and representation theory*, Graduate Texts in Mathematics, vol. 9, Springer-Verlag, New York, 1972.
- [Huy05] Daniel Huybrechts, *Complex geometry: An introduction*, Universitext, Springer-Verlag, Berlin, 2005.
- [Ima76] Hideo Imai, *On the Hodge groups of some abelian varieties*, Kodai Mathematical Seminar Reports **27** (1976), no. 3, 367–372.
- [Ive86] Birger Iversen, *Cohomology of sheaves*, Universitext, Springer-Verlag, Berlin, 1986.
- [Jan92] Uwe Jannsen, *Motives, numerical equivalence, and semi-simplicity*, *Inventiones Mathematicae* **107** (1992), no. 3, 447–452.
- [Jan94] ———, *Motivic sheaves and filtrations on Chow groups*, *Motives* (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 245–302.
- [Jec03] Thomas Jech, *Set theory*, the third millennium ed., Springer Monographs in Mathematics, Springer-Verlag, Berlin, 2003.
- [Joy00] Dominic D. Joyce, *Compact manifolds with special holonomy*, Oxford Mathematical Monographs, Oxford University Press, Oxford, 2000.
- [Kan58] Daniel M. Kan, *Adjoint functors*, *Transactions of the American Mathematical Society* **87** (1958), 294–329.
- [Kel50] John L. Kelley, *The Tychonoff product theorem implies the axiom of choice*, *Fundamenta Mathematicae* **37** (1950), 75–76.
- [Kim05] Shun-Ichi Kimura, *Chow groups are finite dimensional, in some sense*, *Mathematische Annalen* **331** (2005), no. 1, 173–201.
- [Kle68] Steven L. Kleiman, *Algebraic cycles and the Weil conjectures*, *Dix exposés sur la cohomologie des schémas*, North-Holland, Amsterdam, 1968, pp. 359–386.
- [Kle94] ———, *The standard conjectures*, *Motives* (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 3–20.
- [KM74] Nicholas M. Katz and William Messing, *Some consequences of the Riemann hypothesis for varieties over finite fields*, *Inventiones Mathematicae* **23** (1974), 73–77.
- [KN63] Shoshichi Kobayashi and Katsumi Nomizu, *Foundations of differential geometry. Vol. I*, Interscience Publishers, New York-London, 1963.
- [Kna02] Anthony W. Knaapp, *Lie groups beyond an introduction*, second ed., Progress in Mathematics, vol. 140, Birkhäuser Boston, Boston, MA, 2002.
- [KO68] Nicholas M. Katz and Tadao Oda, *On the differentiation of de Rham cohomology classes with respect to parameters*, *Journal of Mathematics of Kyoto University* **8** (1968), 199–213.
- [Kod54] Kunihiko Kodaira, *On Kähler varieties of restricted type (an intrinsic characterization of algebraic varieties)*, *Annals of Mathematics. Second Series* **60** (1954), 28–48.

- [Kol92] János Kollár, *Trento examples*, Classification of Irregular Varieties (Trento, 1990), Lecture Notes in Mathematics, vol. 1515, Springer-Verlag, Berlin, 1992.
- [KS53] Kunihiro Kodaira and Donald C. Spencer, *Divisor class groups on algebraic varieties*, Proceedings of the National Academy of Sciences of the USA **39** (1953), 872–877.
- [KS67] Michio Kuga and Ichirô Satake, *Abelian varieties attached to polarized K3-surfaces*, Mathematische Annalen **169** (1967), 239–242.
- [KS90] Masaki Kashiwara and Pierre Schapira, *Sheaves on manifolds*, Grundlehren der mathematischen Wissenschaften, vol. 292, Springer-Verlag, Berlin, 1990.
- [KS06] ———, *Categories and sheaves*, Grundlehren der mathematischen Wissenschaften, vol. 332, Springer-Verlag, Berlin, 2006.
- [Kun11] Kenneth Kunen, *Set theory*, Studies in Logic, vol. 34, College Publications, London, 2011.
- [Lan30] Edmund Landau, *Grundlagen der Analysis*, Akademische Verlagsgesellschaft, Leipzig, 1930.
- [Lan93] Serge Lang, *Real and functional analysis*, third ed., Graduate Texts in Mathematics, vol. 142, Springer-Verlag, New York, 1993.
- [Lan02] ———, *Algebra*, revised third ed., Graduate Texts in Mathematics, vol. 211, Springer-Verlag, New York, 2002.
- [Laz04] Robert Lazarsfeld, *Positivity in algebraic geometry. I*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 48, Springer-Verlag, Berlin, 2004.
- [Lee13] John M. Lee, *Introduction to smooth manifolds*, second ed., Graduate Texts in Mathematics, vol. 218, Springer, New York, 2013.
- [Lee18] ———, *Introduction to riemannian manifolds*, second ed., Graduate Texts in Mathematics, vol. 176, Springer, Cham, 2018.
- [Lew99] James D. Lewis, *A survey of the Hodge conjecture*, second ed., CRM Monograph Series, vol. 10, American Mathematical Society, Providence, RI, 1999.
- [Lie68] David I. Lieberman, *Numerical and homological equivalence of algebraic cycles on Hodge manifolds*, American Journal of Mathematics **90** (1968), 366–374.
- [LM89] H. Blaine Lawson and Marie-Louise Michelsohn, *Spin geometry*, Princeton Mathematical Series, vol. 38, Princeton University Press, Princeton, NJ, 1989.
- [LS77] Roger C. Lyndon and Paul E. Schupp, *Combinatorial group theory*, Ergebnisse der Mathematik und ihrer Grenzgebiete, vol. 89, Springer-Verlag, Berlin-New York, 1977.
- [Mar23] Eyal Markman, *The monodromy of generalized Kummer varieties and algebraic cycles on their intermediate Jacobians*, Journal of the European Mathematical Society **25** (2023), no. 1, 231–321.
- [Mat86] Hideyuki Matsumura, *Commutative ring theory*, Cambridge Studies in Advanced Mathematics, vol. 8, Cambridge University Press, Cambridge, 1986.

- [Mau14] Daves Maulik, *Supersingular K3 surfaces for large primes*, Duke Mathematical Journal **163** (2014), no. 13, 2357–2425.
- [McC01] John McCleary, *A user's guide to spectral sequences*, second ed., Cambridge Studies in Advanced Mathematics, vol. 58, Cambridge University Press, Cambridge, 2001.
- [Mil63] John Milnor, *Morse theory*, Annals of Mathematics Studies, vol. 51, Princeton University Press, Princeton, NJ, 1963.
- [Mil80] James S. Milne, *étale cohomology*, Princeton Mathematical Series, vol. 33, Princeton University Press, Princeton, NJ, 1980.
- [Mil97] John W. Milnor, *Topology from the differentiable viewpoint*, Princeton Landmarks in Mathematics, Princeton University Press, Princeton, NJ, 1997.
- [Mil02] James S. Milne, *Polarizations and Grothendieck's standard conjectures*, Annals of Mathematics. Second Series **155** (2002), no. 2, 599–610.
- [Mil17] ———, *Algebraic groups: The theory of group schemes of finite type over a field*, Cambridge Studies in Advanced Mathematics, vol. 170, Cambridge University Press, Cambridge, 2017.
- [ML98] Saunders Mac Lane, *Categories for the working mathematician*, second ed., Graduate Texts in Mathematics, vol. 5, Springer-Verlag, New York, 1998.
- [MP15] Keerthi Madapusi Pera, *The Tate conjecture for K3 surfaces in odd characteristic*, Inventiones Mathematicae **201** (2015), no. 2, 625–668.
- [MS74] John W. Milnor and James D. Stasheff, *Characteristic classes*, Annals of Mathematics Studies, vol. 76, Princeton University Press, Princeton, NJ, 1974.
- [MT68] Robert E. Mosher and Martin C. Tangora, *Cohomology operations and applications in homotopy theory*, Harper & Row, New York, 1968.
- [Mum68] David Mumford, *Rational equivalence of 0-cycles on surfaces*, Journal of Mathematics of Kyoto University **9** (1968), 195–204.
- [Mum69] ———, *A note of Shimura's paper "Discontinuous groups and abelian varieties"*, Mathematische Annalen **181** (1969), 345–351.
- [Mum70] ———, *Abelian varieties*, Tata Institute of Fundamental Research Studies in Mathematics, vol. 5, Oxford University Press, London, 1970.
- [Mun00] James R. Munkres, *Topology*, second ed., Prentice Hall, Upper Saddle River, NJ, 2000.
- [Mur84] V. Kumar Murty, *Exceptional Hodge classes on certain abelian varieties*, Mathematische Annalen **268** (1984), no. 2, 197–206.
- [Mur93] Jacob P. Murre, *On a conjectural filtration on the Chow groups of an algebraic variety. I. The general conjectures and some examples*, Indagationes Mathematicae. New Series **4** (1993), no. 2, 177–188.
- [MZ99] Ben Moonen and Yuri G. Zarhin, *Hodge classes on abelian varieties of low dimension*, Mathematische Annalen **315** (1999), no. 4, 711–733.

- [NO85] Niels Nygaard and Arthur Ogus, *Tate's conjecture for K3 surfaces of finite height*, Annals of Mathematics. Second Series **122** (1985), no. 3, 461–507.
- [Nyg83] Niels O. Nygaard, *The Tate conjecture for ordinary K3 surfaces over finite fields*, Inventiones Mathematicae **74** (1983), no. 2, 213–237.
- [Par88] Kapil Paranjape, *Abelian varieties associated to certain K3 surfaces*, Compositio Mathematica **68** (1988), no. 1, 11–22.
- [Pin98] Richard Pink, *l -adic algebraic monodromy groups, cocharacters, and the Mumford–Tate conjecture*, Journal für die reine und angewandte Mathematik **495** (1998), 187–237.
- [PS08] Chris A. M. Peters and Joseph H. M. Steenbrink, *Mixed hodge structures*, Ergebnisse der Mathematik und ihrer Grenzgebiete. 3. Folge, vol. 52, Springer-Verlag, Berlin, 2008.
- [Qui69] Daniel Quillen, *On the formal group laws of unoriented and complex cobordism theory*, Bulletin of the American Mathematical Society **75** (1969), 1293–1298.
- [Rav86] Douglas C. Ravenel, *Complex cobordism and stable homotopy groups of spheres*, Pure and Applied Mathematics, vol. 121, Academic Press, Orlando, FL, 1986.
- [Ray78] Michel Raynaud, *Contre-exemple au “vanishing theorem” en caractéristique $p > 0$* , C. P. Ramanujam—a tribute, Tata Institute of Fundamental Research Studies in Mathematics, vol. 8, Springer, Berlin-New York, 1978, pp. 273–278.
- [Rie16] Emily Riehl, *Category theory in context*, Dover Publications, Mineola, NY, 2016.
- [Ros36] Barkley Rosser, *Extensions of some theorems of Gödel and Church*, Journal of Symbolic Logic **1** (1936), 87–91.
- [Rot95] Joseph J. Rotman, *An introduction to the theory of groups*, fourth ed., Graduate Texts in Mathematics, vol. 148, Springer-Verlag, New York, 1995.
- [RS80] Michael Reed and Barry Simon, *Methods of modern mathematical physics. I. functional analysis*, revised and enlarged ed., Academic Press, New York, 1980.
- [Rud69] Mary Ellen Rudin, *A new proof that metric spaces are paracompact*, Proceedings of the American Mathematical Society **20** (1969), 603.
- [Rud76] Walter Rudin, *Principles of mathematical analysis*, third ed., McGraw-Hill, New York, 1976.
- [Rud87] ———, *Real and complex analysis*, third ed., McGraw-Hill Book Co., New York, 1987.
- [Rud91] ———, *Functional analysis*, second ed., International Series in Pure and Applied Mathematics, McGraw-Hill, New York, 1991.
- [Sch88] Chad Schoen, *Hodge classes on self-products of a variety with an automorphism*, Compositio Mathematica **65** (1988), no. 1, 3–32.
- [Sch94] Anthony J. Scholl, *Classical motives*, Motives (Seattle, WA, 1991), Proceedings of Symposia in Pure Mathematics, vol. 55, Part 1, American Mathematical Society, Providence, RI, 1994, pp. 163–187.

- [SE62] N. E. Steenrod and D. B. A. Epstein, *Cohomology operations*, Annals of Mathematics Studies, vol. 50, Princeton University Press, Princeton, NJ, 1962.
- [Ser53] Jean-Pierre Serre, *Cohomologie modulo 2 des complexes d'Eilenberg-MacLane*, Commentarii Mathematici Helvetici **27** (1953), 198–232.
- [Ser56] ———, *Géométrie algébrique et géométrie analytique*, Annales de l'Institut Fourier **6** (1956), 1–42.
- [Ser64] ———, *Exemples de variétés projectives conjuguées non homéomorphes*, Comptes Rendus de l'Académie des Sciences de Paris **258** (1964), 4194–4196.
- [Ser72] ———, *Propriétés galoisiennes des points d'ordre fini des courbes elliptiques*, Inventiones Mathematicae **15** (1972), no. 4, 259–331.
- [Ser77] ———, *Linear representations of finite groups*, Graduate Texts in Mathematics, vol. 42, Springer-Verlag, New York, 1977.
- [Ser80] ———, *Trees*, Springer-Verlag, Berlin-New York, 1980.
- [Shi79] Tetsuji Shioda, *The Hodge conjecture for Fermat varieties*, Mathematische Annalen **245** (1979), no. 2, 175–184.
- [Sho67] Joseph R. Shoenfield, *Mathematical logic*, Addison-Wesley, Reading, MA, 1967.
- [Smi13] Peter Smith, *An introduction to Gödel's theorems*, second ed., Cambridge University Press, Cambridge, 2013.
- [Spa66] Edwin H. Spanier, *Algebraic topology*, McGraw-Hill Book Co., New York, 1966.
- [Spi65] Michael Spivak, *Calculus on manifolds*, W. A. Benjamin, New York-Amsterdam, 1965.
- [SS03] Elias M. Stein and Rami Shakarchi, *Complex analysis*, Princeton Lectures in Analysis, vol. 2, Princeton University Press, Princeton, NJ, 2003.
- [Ste15] Ian Stewart, *Galois theory*, fourth ed., CRC Press, Boca Raton, FL, 2015.
- [Sto48] A. H. Stone, *Paracompactness and product spaces*, Bulletin of the American Mathematical Society **54** (1948), 977–982.
- [Sto68] Robert E. Stong, *Notes on cobordism theory*, Mathematical Notes, Princeton University Press, Princeton, NJ; University of Tokyo Press, Tokyo, 1968.
- [SV05] Christophe Soulé and Claire Voisin, *Torsion cohomology classes and algebraic cycles on complex projective manifolds*, Advances in Mathematics **198** (2005), no. 1, 107–127.
- [Sza09] Tamás Szamuely, *Galois groups and fundamental groups*, Cambridge Studies in Advanced Mathematics, vol. 117, Cambridge University Press, Cambridge, 2009.
- [Tan82] Sergei G. Tankeev, *Cycles on simple abelian varieties of prime dimension*, Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya **46** (1982), no. 1, 155–170.
- [Tat65] John T. Tate, *Algebraic cycles and poles of zeta functions*, Arithmetical Algebraic Geometry (Proc. Conf. Purdue Univ., 1963), Harper & Row, New York, 1965, pp. 93–110.

- [Tat66a] John Tate, *Endomorphisms of abelian varieties over finite fields*, *Inventiones Mathematicae* **2** (1966), 134–144.
- [Tat66b] ———, *On the conjectures of Birch and Swinnerton-Dyer and a geometric analog*, *Séminaire Bourbaki*, 18e année (1965/66), 1966, Exposé 306.
- [The] The Stacks project authors, *The Stacks project*, <https://stacks.math.columbia.edu>.
- [Tho54] René Thom, *Quelques propriétés globales des variétés différentiables*, *Commentarii Mathematici Helvetici* **28** (1954), 17–86.
- [Thu76] William P. Thurston, *Some simple examples of symplectic manifolds*, *Proceedings of the American Mathematical Society* **55** (1976), no. 2, 467–468.
- [Tia87] Gang Tian, *Smoothness of the universal deformation space of compact Calabi–Yau manifolds and its Petersson–Weil metric*, *Mathematical Aspects of String Theory* (San Diego, 1986), *Advanced Series in Mathematical Physics*, vol. 1, World Scientific, Singapore, 1987, pp. 629–646.
- [TMR53] Alfred Tarski, Andrzej Mostowski, and Raphael M. Robinson, *Undecidable theories*, *Studies in Logic and the Foundations of Mathematics*, North-Holland, Amsterdam, 1953.
- [Tod89] Andrey N. Todorov, *The Weil–Petersson geometry of the moduli space of $SU(n \geq 3)$ (Calabi–Yau) manifolds. I*, *Communications in Mathematical Physics* **126** (1989), no. 2, 325–346.
- [Tot97] Burt Totaro, *Torsion algebraic cycles and complex cobordism*, *Journal of the American Mathematical Society* **10** (1997), 467–493.
- [Tot17] ———, *Recent progress on the Tate conjecture*, *Bulletin of the American Mathematical Society. New Series* **54** (2017), no. 4, 575–590.
- [vD13] Dirk van Dalen, *Logic and structure*, fifth ed., Universitext, Springer, London, 2013.
- [Voi02a] Claire Voisin, *A counterexample to the Hodge conjecture extended to Kähler varieties*, *International Mathematics Research Notices* (2002), no. 20, 1057–1075.
- [Voi02b] ———, *Hodge theory and complex algebraic geometry. I*, *Cambridge Studies in Advanced Mathematics*, vol. 76, Cambridge University Press, Cambridge, 2002.
- [Voi03] ———, *Hodge theory and complex algebraic geometry. II*, *Cambridge Studies in Advanced Mathematics*, vol. 77, Cambridge University Press, Cambridge, 2003.
- [Voi04] ———, *On the homotopy types of compact Kähler and complex projective manifolds*, *Inventiones Mathematicae* **157** (2004), no. 2, 329–343.
- [Voi06] ———, *On integral Hodge classes on uniruled or Calabi–Yau threefolds*, *Moduli Spaces and Arithmetic Geometry*, *Advanced Studies in Pure Mathematics*, vol. 45, Mathematical Society of Japan, Tokyo, 2006, pp. 43–73.

- [Voi07] ———, *Hodge loci and absolute Hodge classes*, *Compositio Mathematica* **143** (2007), no. 4, 945–958.
- [War83] Frank W. Warner, *Foundations of differentiable manifolds and lie groups*, Graduate Texts in Mathematics, vol. 94, Springer-Verlag, New York-Berlin, 1983.
- [Wat79] William C. Waterhouse, *Introduction to affine group schemes*, Graduate Texts in Mathematics, vol. 66, Springer-Verlag, New York, 1979.
- [Wei79] André Weil, *Abelian varieties and the Hodge ring*, *Œuvres Scientifiques. Collected Papers*. Vol. III, Springer-Verlag, New York, 1979, pp. 421–429.
- [Wei94] Charles A. Weibel, *An introduction to homological algebra*, Cambridge Studies in Advanced Mathematics, vol. 38, Cambridge University Press, Cambridge, 1994.
- [Wel08] Raymond O. Wells, *Differential analysis on complex manifolds*, third ed., Graduate Texts in Mathematics, vol. 65, Springer, New York, 2008.
- [Yau78] Shing-Tung Yau, *On the Ricci curvature of a compact Kähler manifold and the complex Monge–Ampère equation. I*, *Communications on Pure and Applied Mathematics* **31** (1978), no. 3, 339–411.
- [Zar75] Yuri G. Zarhin, *Endomorphisms of abelian varieties over fields of finite characteristic*, *Izvestiya Akademii Nauk SSSR. Seriya Matematicheskaya* **39** (1975), no. 2, 272–277.
- [Zer08] Ernst Zermelo, *Untersuchungen über die Grundlagen der Mengenlehre. I*, *Mathematische Annalen* **65** (1908), 261–281.
- [Zor35] Max Zorn, *A remark on method in transfinite algebra*, *Bulletin of the American Mathematical Society* **41** (1935), 667–670.
- [Zuc77] Steven Zucker, *The Hodge conjecture for cubic fourfolds*, *Compositio Mathematica* **34** (1977), no. 2, 199–209.